

# Service Manual

Technical Instructions



## Chemiluminescence Analyzer

**LIAISON<sup>®</sup> XS (Ref. I0090)**

DiaSorin S.p.A.  
Strada per Crescentino snc  
13040 - Saluggia (VC) Italy

---

## **IMPORTANT**

**The information contained herein is based on experience and knowledge relating to the subject matter as acquired by DiaSorin S.p.A.**

**DiaSorin is not liable for any loss or damage, including consequential or special damages resulting from the use and/ or misuse of the contained information, be it negligence or other fault.**



DiaSorin S.p.A.

Strada per Crescentino - 13040 Saluggia (Vercelli) - Italy

[www.diasorin.com](http://www.diasorin.com)



|                                   |
|-----------------------------------|
| <b>LIAISON® XS</b> Service Manual |
| Rev. 3 - 12/2021                  |

**DiaSorin S.p.A. All rights reserved**

## Description of Changes

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b> | Chapter 1.5.3.1 Added description about laser positioning in the Sample bay.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>2</b> | Chapter 2.5.3.1 Added description about laser positioning in the Sample bay.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>5</b> | <p>Chapter 5 Updated Technical Notes codes according to SmartSolve nomenclature.</p> <p>Chapter 5.3.2 Corrected typo.</p> <p>Chapter 5.3.4.4 Added to check the proper Ferrite snap installation.</p> <p>Chapter 5.3.5 Added picture of the Sample rack ID label.</p> <p>Chapter 5.3.6 Replaced picture of the Control rack ID label.</p> <p>Chapter 5.4.4 Added preliminary steps. Added a note about the country code.</p> <p>Chapter 5.4.6 Addition of warning for the last revision of the IFTP spreadsheet to be used.</p>                                                                                                                          |
| <b>6</b> | <p>Chapter 6 Updated Technical Notes codes according to SmartSolve nomenclature.</p> <p>Chapter 6.3.2 Corrected typo.</p> <p>Chapter 6.3.4.4 Added to check the proper Ferrite snap installation.</p> <p>Chapter 6.3.5 Added picture of the Sample rack ID label.</p> <p>Chapter 6.3.6 Replaced picture of the Control rack ID label.</p> <p>Chapter 6.3.8.3.2 New OLV verification to update the OLV calibration.</p> <p>Chapter 6.4.4 Added preliminary steps. Added a note about the country code.</p> <p>Chapter 6.4.6 Addition of warning for the last revision of the IFTP spreadsheet to be used.</p>                                             |
| <b>9</b> | <p>Chapter 9 Updated Technical Notes codes according to SmartSolve nomenclature.</p> <p>Chapter 9.2.2 Exchanged point 2 with point 3. Added insertion of a new EASY Waste box at the beginning of the PM. Disinfection Procedure task replaced with the new Service Decontamination task.</p> <p>Chapter 9.2.3 Removed creation of the Troubleshooting Complete backup from the PM.</p> <p>Chapter 9.2.7 Reworded point 4.</p> <p>Chapter 9.2.10 Added check of completion of the Antivirus scan.</p> <p>Chapter 9.2.11 Added RLU check as for the IFTP and added summary table for the acceptance criteria.</p> <p>Chapter 9.4.2 Updated Table 9–4.</p> |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>10</b> | <p>Chapter 10 Updated Technical Notes codes according to SmartSolve nomenclature.</p> <p>Chapter 10.2.2 Exchanged point 2 with point 3. Added insertion of a new EASY Waste box at the beginning of the PM. Disinfection Procedure task replaced with the new Service Decontamination task</p> <p>Chapter 10.2.3 Removed creation of the Troubleshooting Complete backup from the PM.</p> <p>Chapter 10.2.7 Reworded point 4.</p> <p>Chapter 10.2.10 Added check of completion of the Antivirus scan.</p> <p>Chapter 10.2.11 Added RLU check as for the IFTP and added summary table for the acceptance criteria.</p> <p>Chapter 10.4.2 Updated Table 10–4.</p> |
| <b>12</b> | Chapter 12 Updated according to SSW version 1.6.2111.4001.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>13</b> | <p>Chapter 13.4.1.1 Added default step size of 1 in Table 13–5.</p> <p>Chapter 13.8.1 Added new automatic functionality for teaching of Cuvette Tip Loading module.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>14</b> | Chapter 14.3.4.3 Added Ferrite snap installation and points about SW installation and teaching file / logfiles / databases restoring.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>16</b> | Chapter 16.4.2.1 Addition of caution note.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>17</b> | <p>Chapter 17.3.1.3 Added missing service instruction.</p> <p>Chapter 17.3.3 Added new replacement procedure about the SIPAL plastic parts.</p> <p>Chapter 17.3.9.3 Updated picture of the Washer rotor.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>18</b> | Chapter 18.3 Replaced the LiquidLength cut off value of 440 with the new value of 550.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>19</b> | <p>Chapter 19.3.3.2 Updated picture.</p> <p>Chapter 19.3.4.2 Updated picture.</p> <p>Chapter 19.3.14 Added the OLV calibration after SPOLV needle replacement.</p> <p>Chapter 19.3.15.3 Removed a service instruction not used.</p> <p>Chapter 19.3.23 New Service instruction.</p> <p>Chapter 19.3.24 New Service instruction.</p> <p>Chapter 19.3.25 New Service instruction.</p>                                                                                                                                                                                                                                                                             |



|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>20</b> | <p>Chapter 20.3.3.2 Updated picture.</p> <p>Chapter 20.3.4.2 Updated picture.</p> <p>Chapter 20.3.14 Added the OLV calibration after SPOLV needle replacement.</p> <p>Chapter 20.3.15.3 Removed a service instruction not used.</p> <p>Chapter 20.3.23 New Service instruction.</p> <p>Chapter 20.3.24 New Service instruction.</p> <p>Chapter 20.3.25 New Service instruction.</p> <p>Chapter 20.4.2 Updated and added new procedures.</p> <p>Chapter 20.5 Updated picture.</p> |
| <b>21</b> | <p>Chapter 21.1 Added laser safety warning already present in the Introduction chapter.</p> <p>Chapter 21.5.1 Updated picture.</p>                                                                                                                                                                                                                                                                                                                                               |
| <b>22</b> | <p>Chapter 22.3.2.3 Add reference to the new service instructions.</p> <p>Chapter 22.3.3.4 Add reference to the new service instructions.</p> <p>Chapter 22.3.5 Review of the service instructions for the new cooling module.</p> <p>Chapter 22.4.3 New procedure for the readjustment of the cam to the stirrer rod.</p>                                                                                                                                                       |
| <b>23</b> | <p>Chapter 23.2 Updated picture.</p> <p>Chapter 23.3 Removed a service instruction no more usable.</p> <p>Chapter 23.3.1 Updated the service instruction.</p>                                                                                                                                                                                                                                                                                                                    |
| <b>24</b> | <p>Chapter 24.2 Updated picture.</p> <p>Chapter 24.3 Removed a service instruction no more usable and updated picture.</p> <p>Chapter 24.3.1 Updated the service instruction.</p>                                                                                                                                                                                                                                                                                                |
| <b>25</b> | <p>Chapter 25.3.4 Updated Technical Notes codes according to SmartSolve nomenclature.</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>26</b> | <p>Chapter 26.3.4 Updated Technical Notes codes according to SmartSolve nomenclature.</p>                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>27</b> | <p>Chapter 27.3.2 Added new paragraphs about the replacement of the Starter Reagent flap.</p>                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>28</b> | <p>Chapter 28.2 Updated picture.</p> <p>Chapter 28.3 Removed a service instruction no more usable.</p> <p>Chapter 28.3.12 Updated the service instruction.</p> <p>Chapter 28.5.2 Correction of the height of Direct drain routing.</p>                                                                                                                                                                                                                                           |
| <b>29</b> | <p>Chapter 29.2 Updated picture.</p> <p>Chapter 29.3 Removed a service instruction no more usable.</p> <p>Chapter 29.3.13 Updated the service instruction.</p> <p>Chapter 29.5.2 Correction of the height of Direct drain routing.</p>                                                                                                                                                                                                                                           |

|           |                                                                                                                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>30</b> | Chapter 30.3 Removed a service instruction no more usable.<br>Chapter 30.3.8 Optimized the service instructions.<br>Chapter 30.4.1 Updated picture and table in order to correct wrong connection. |
| <b>31</b> | Chapter 31.3 Removed a service instruction no more usable.<br>Chapter 31.3.8 Optimized the service instructions.<br>Chapter 31.3.4 Updated picture and table in order to correct wrong connection. |
| <b>32</b> | Chapter 32.2 Updated Technical Notes codes according to SmartSolve nomenclature.<br>Chapter 32.3.6.5 Updated the DiaSorin file sharing system.                                                     |
| <b>33</b> | Chapter 33.4 Implemented all the changes on Chapter 9 and in Chapter 5.                                                                                                                            |
| <b>34</b> | Chapter 34.4 Implemented all the changes on Chapter 10 and in Chapter 6.                                                                                                                           |
| <b>35</b> | Chapter 35.2 Removed references to the handheld barcode scanner PN.<br>Added standard references as done in the User Manual.                                                                       |
| <b>36</b> | Chapter 36.2 Removed references to the handheld barcode scanner PN.<br>Added standard references as done in the User Manual.                                                                       |
| <b>39</b> | Table 39–11 Changed the required Y to N for H10.                                                                                                                                                   |
| <b>40</b> | Chapter 40 Removed FastObjects definition.                                                                                                                                                         |

# Table of Contents

|            |                                                                     |             |
|------------|---------------------------------------------------------------------|-------------|
| <b>1</b>   | <b>Introduction for internal-tanks configuration.....</b>           | <b>1-1</b>  |
| <b>1.1</b> | <b>Description.....</b>                                             | <b>1-1</b>  |
| 1.1.1      | Intended use .....                                                  | 1-1         |
| <b>1.2</b> | <b>Typographical conventions .....</b>                              | <b>1-2</b>  |
| 1.2.1      | Display of warnings and notes.....                                  | 1-2         |
| 1.2.2      | Used warning symbols.....                                           | 1-3         |
| 1.2.3      | Other symbols.....                                                  | 1-5         |
| 1.2.4      | Special types .....                                                 | 1-6         |
| <b>1.3</b> | <b>Metric system.....</b>                                           | <b>1-6</b>  |
| <b>1.4</b> | <b>Use of screw color codes .....</b>                               | <b>1-6</b>  |
| <b>1.5</b> | <b>Hazard instructions .....</b>                                    | <b>1-7</b>  |
| 1.5.1      | General safety .....                                                | 1-7         |
| 1.5.2      | Electrical hazards .....                                            | 1-8         |
| 1.5.3      | Physical Hazards .....                                              | 1-12        |
| 1.5.4      | Trip Hazards .....                                                  | 1-12        |
| 1.5.5      | Mechanical Hazards .....                                            | 1-14        |
| 1.5.6      | Biological Safety .....                                             | 1-15        |
| 1.5.7      | Chemical Hazards .....                                              | 1-17        |
| 1.5.8      | Spill clean-up .....                                                | 1-17        |
| 1.5.9      | Maintenance Work.....                                               | 1-17        |
| 1.5.10     | Disposal and Disinfection .....                                     | 1-20        |
| 1.5.11     | Caution on electromagnetic wave interference.....                   | 1-22        |
| 1.5.12     | Static magnetic field.....                                          | 1-23        |
| <b>1.6</b> | <b>Evaluation of Residual risk for Field Service Engineer .....</b> | <b>1-24</b> |
| <b>1.7</b> | <b>Instrument Labelling and Safety Labels.....</b>                  | <b>1-26</b> |
| 1.7.1      | Instrument Serial label .....                                       | 1-29        |
| 1.7.2      | Fuse / Power supply label.....                                      | 1-29        |
| 1.7.3      | ON/OFF label.....                                                   | 1-30        |
| 1.7.4      | USB and LAN label.....                                              | 1-30        |
| 1.7.5      | Disconnect main power label.....                                    | 1-31        |
| 1.7.6      | Ground label / Earth label .....                                    | 1-31        |
| 1.7.7      | Biohazard labels (4.5 cm) .....                                     | 1-32        |
| 1.7.8      | Biohazard labels (2.5 cm) .....                                     | 1-33        |
| 1.7.9      | Attention labels .....                                              | 1-34        |
| 1.7.10     | Hot surface labels .....                                            | 1-35        |
| 1.7.11     | Laser labels .....                                                  | 1-36        |
| 1.7.12     | Crush hazard labels.....                                            | 1-37        |
| 1.7.13     | Magnet labels .....                                                 | 1-37        |
| 1.7.14     | No heavy load labels .....                                          | 1-39        |

---

|            |                                                                     |             |
|------------|---------------------------------------------------------------------|-------------|
| 1.7.15     | Don't step on label labels.....                                     | 1-40        |
| 1.7.16     | Wash Buffer label .....                                             | 1-41        |
| 1.7.17     | Water label.....                                                    | 1-41        |
| 1.7.18     | Cleaning Solution label .....                                       | 1-42        |
| 1.7.19     | Waste label .....                                                   | 1-42        |
| 1.7.20     | Shockwatch, Tiltwatch and Temperature measuring strip .....         | 1-43        |
| <b>2</b>   | <b>Introduction for external-tanks configuration.....</b>           | <b>2-1</b>  |
| <b>2.1</b> | <b>Description .....</b>                                            | <b>2-1</b>  |
| 2.1.1      | Intended use .....                                                  | 2-1         |
| <b>2.2</b> | <b>Typographical conventions .....</b>                              | <b>2-2</b>  |
| 2.2.1      | Display of warnings and notes.....                                  | 2-2         |
| 2.2.2      | Used warning symbols.....                                           | 2-3         |
| 2.2.3      | Other symbols.....                                                  | 2-5         |
| 2.2.4      | Special types .....                                                 | 2-6         |
| <b>2.3</b> | <b>Metric system.....</b>                                           | <b>2-6</b>  |
| <b>2.4</b> | <b>Use of screw color codes .....</b>                               | <b>2-6</b>  |
| <b>2.5</b> | <b>Hazard instructions .....</b>                                    | <b>2-7</b>  |
| 2.5.1      | General safety .....                                                | 2-7         |
| 2.5.2      | Electrical hazards .....                                            | 2-8         |
| 2.5.3      | Physical Hazards .....                                              | 2-12        |
| 2.5.4      | Trip Hazards .....                                                  | 2-12        |
| 2.5.5      | Mechanical Hazards .....                                            | 2-14        |
| 2.5.6      | Biological Safety .....                                             | 2-15        |
| 2.5.7      | Chemical Hazards .....                                              | 2-17        |
| 2.5.8      | Spill clean-up .....                                                | 2-17        |
| 2.5.9      | Maintenance Work.....                                               | 2-17        |
| 2.5.10     | Disposal and Disinfection .....                                     | 2-20        |
| 2.5.11     | Caution on electromagnetic wave interference.....                   | 2-22        |
| 2.5.12     | Static magnetic field.....                                          | 2-23        |
| <b>2.6</b> | <b>Evaluation of Residual risk for Field Service Engineer .....</b> | <b>2-24</b> |
| <b>2.7</b> | <b>Instrument Labelling and Safety Labels.....</b>                  | <b>2-26</b> |
| 2.7.1      | Instrument Serial label .....                                       | 2-29        |
| 2.7.2      | Fuse / Power supply label.....                                      | 2-29        |
| 2.7.3      | ON/OFF label.....                                                   | 2-30        |
| 2.7.4      | USB and LAN label.....                                              | 2-30        |
| 2.7.5      | Disconnect main power label.....                                    | 2-31        |
| 2.7.6      | Ground label / Earth label .....                                    | 2-31        |
| 2.7.7      | Biohazard labels (4.5 cm) .....                                     | 2-32        |
| 2.7.8      | Biohazard labels (2.5 cm) .....                                     | 2-33        |
| 2.7.9      | Attention labels .....                                              | 2-34        |
| 2.7.10     | Hot surface labels .....                                            | 2-35        |
| 2.7.11     | Laser labels .....                                                  | 2-36        |

---

|            |                                                                                  |            |
|------------|----------------------------------------------------------------------------------|------------|
| 2.7.12     | Crush hazard labels.....                                                         | 2-37       |
| 2.7.13     | Magnet labels .....                                                              | 2-37       |
| 2.7.14     | No heavy load labels .....                                                       | 2-38       |
| 2.7.15     | Don't step on label labels.....                                                  | 2-40       |
| 2.7.16     | Connect/Disconnect both tubing and sensor label .....                            | 2-40       |
| 2.7.17     | Wash Buffer label .....                                                          | 2-41       |
| 2.7.18     | System Liquid label.....                                                         | 2-42       |
| 2.7.19     | Cleaning Solution label .....                                                    | 2-43       |
| 2.7.20     | Waste label .....                                                                | 2-43       |
| 2.7.21     | Shockwatch, Tiltwatch and Temperature measuring strip .....                      | 2-44       |
| <b>3</b>   | <b>Overview for internal-tanks configuration.....</b>                            | <b>3-1</b> |
| <b>3.1</b> | <b>Instrument overview.....</b>                                                  | <b>3-1</b> |
| 3.1.1      | Front view .....                                                                 | 3-2        |
| 3.1.2      | Left side inner view.....                                                        | 3-3        |
| 3.1.3      | Right side inner view.....                                                       | 3-4        |
| 3.1.4      | Left side view .....                                                             | 3-5        |
| 3.1.5      | Right side view.....                                                             | 3-6        |
| 3.1.6      | Fluidics overview .....                                                          | 3-7        |
| 3.1.7      | Electric Circuit overview.....                                                   | 3-8        |
| <b>4</b>   | <b>Overview for external-tanks configuration .....</b>                           | <b>4-1</b> |
| <b>4.1</b> | <b>Instrument overview.....</b>                                                  | <b>4-1</b> |
| 4.1.1      | Front view .....                                                                 | 4-2        |
| 4.1.2      | Left side inner view .....                                                       | 4-3        |
| 4.1.3      | Right side inner view.....                                                       | 4-4        |
| 4.1.4      | Left side view .....                                                             | 4-5        |
| 4.1.5      | Right side view.....                                                             | 4-6        |
| 4.1.6      | Liquid tanks view .....                                                          | 4-7        |
| 4.1.7      | Fluidics overview .....                                                          | 4-8        |
| 4.1.8      | Electric Circuit overview.....                                                   | 4-9        |
| <b>5</b>   | <b>LIAISON<sup>®</sup>XS Installation for internal-tanks configuration .....</b> | <b>5-1</b> |
| <b>5.1</b> | <b>Safety.....</b>                                                               | <b>5-1</b> |
| <b>5.2</b> | <b>Installation preparation .....</b>                                            | <b>5-3</b> |
| 5.2.1      | Standard supply with PN I0090 .....                                              | 5-3        |
| 5.2.2      | Additional material required for installation.....                               | 5-4        |
| 5.2.3      | Required tools.....                                                              | 5-4        |
| <b>5.3</b> | <b>Installation Qualification .....</b>                                          | <b>5-5</b> |
| 5.3.1      | Visual inspection before unpacking .....                                         | 5-5        |
| 5.3.2      | Unpacking, visual inspection and inventory .....                                 | 5-6        |
| 5.3.3      | Manual inspection.....                                                           | 5-14       |

|            |                                                                                                   |             |
|------------|---------------------------------------------------------------------------------------------------|-------------|
| 5.3.4      | System set-up .....                                                                               | 5-15        |
| 5.3.5      | Sample rack configuration .....                                                                   | 5-22        |
| 5.3.6      | Control rack configuration with Control rack A0243.....                                           | 5-29        |
| 5.3.7      | Pipettor and rack teaching .....                                                                  | 5-30        |
| 5.3.8      | Fluidic set-up .....                                                                              | 5-30        |
| <b>5.4</b> | <b>Operation Qualification .....</b>                                                              | <b>5-34</b> |
| 5.4.1      | Sample Bay .....                                                                                  | 5-34        |
| 5.4.2      | System.....                                                                                       | 5-34        |
| 5.4.3      | Backup of system configuration.....                                                               | 5-34        |
| 5.4.4      | <b>Manuals update.....</b>                                                                        | <b>5-34</b> |
| 5.4.5      | System Verification .....                                                                         | 5-35        |
| 5.4.6      | Installation Final Test.....                                                                      | 5-35        |
| 5.4.7      | Printout customization .....                                                                      | 5-38        |
| <b>6</b>   | <b>LIAISON®XS Installation for external-tanks configuration .....</b>                             | <b>6-1</b>  |
| <b>6.1</b> | <b>Safety .....</b>                                                                               | <b>6-1</b>  |
| <b>6.2</b> | <b>Installation preparation .....</b>                                                             | <b>6-3</b>  |
| 6.2.1      | Standard supply with PN I0090 .....                                                               | 6-3         |
| 6.2.2      | Additional material required for installation.....                                                | 6-4         |
| 6.2.3      | Required tools.....                                                                               | 6-4         |
| <b>6.3</b> | <b>Installation Qualification .....</b>                                                           | <b>6-5</b>  |
| 6.3.1      | Visual inspection before unpacking .....                                                          | 6-5         |
| 6.3.2      | Unpacking, visual inspection and inventory .....                                                  | 6-6         |
| 6.3.3      | Manual inspection.....                                                                            | 6-15        |
| 6.3.4      | System set-up.....                                                                                | 6-16        |
| 6.3.5      | Sample rack configuration .....                                                                   | 6-23        |
| 6.3.6      | Control rack configuration with Control rack A0243.....                                           | 6-30        |
| 6.3.7      | Pipettor and rack teaching .....                                                                  | 6-32        |
| 6.3.8      | Fluidic set-up .....                                                                              | 6-32        |
| <b>6.4</b> | <b>Operation Qualification .....</b>                                                              | <b>6-35</b> |
| 6.4.1      | Sample Bay .....                                                                                  | 6-35        |
| 6.4.2      | System.....                                                                                       | 6-36        |
| 6.4.3      | Backup of system configuration.....                                                               | 6-36        |
| 6.4.4      | <b>Manuals update.....</b>                                                                        | <b>6-36</b> |
| 6.4.5      | System Verification .....                                                                         | 6-36        |
| 6.4.6      | Installation Final Test.....                                                                      | 6-36        |
| 6.4.7      | Printout customization .....                                                                      | 6-39        |
| <b>7</b>   | <b>LIAISON®XS Removal for internal-tanks configuration .....</b>                                  | <b>7-1</b>  |
| <b>7.1</b> | <b>Safety .....</b>                                                                               | <b>7-1</b>  |
| <b>7.2</b> | <b>Preparation of sodium hypochlorite solution .....</b>                                          | <b>7-3</b>  |
| 7.2.1      | Instruction to prepare 1 L of solution at a concentration of active chlorine ranging from 0.5% to |             |

0.9% 7-3

**7.3 Data deleting and decontamination ..... 7-3**

|       |                                            |     |
|-------|--------------------------------------------|-----|
| 7.3.1 | Decontamination of Liquid Waste tank ..... | 7-3 |
| 7.3.2 | Removal of all consumable parts.....       | 7-4 |
| 7.3.3 | Cleaning procedures.....                   | 7-4 |
| 7.3.4 | Confirm decontamination.....               | 7-5 |

**7.4 Packaging of the system..... 7-5**

|       |                                                |      |
|-------|------------------------------------------------|------|
| 7.4.1 | Preparation .....                              | 7-5  |
| 7.4.2 | Packing for the accessories cardboard box..... | 7-21 |
| 7.4.3 | Packing of the system.....                     | 7-36 |
| 7.4.4 | Transport labeling.....                        | 7-43 |

**8 LIAISON<sup>®</sup> XS Removal for external-tanks configuration ..... 8-1****8.1 Safety ..... 8-1****8.2 Preparation of sodium hypochlorite solution ..... 8-3**

|       |                        |     |
|-------|------------------------|-----|
| 8.2.1 | Cleaning Solution..... | 8-3 |
|-------|------------------------|-----|

**8.3 Data deleting and decontamination ..... 8-3**

|       |                                            |     |
|-------|--------------------------------------------|-----|
| 8.3.1 | Decontamination of Liquid Waste tank ..... | 8-3 |
| 8.3.2 | Removal of all consumable parts.....       | 8-4 |
| 8.3.3 | Cleaning procedures.....                   | 8-4 |
| 8.3.4 | Confirm decontamination.....               | 8-5 |

**8.4 Packaging of the system..... 8-5**

|       |                                                |      |
|-------|------------------------------------------------|------|
| 8.4.1 | Preparation .....                              | 8-5  |
| 8.4.2 | Packing for the accessories cardboard box..... | 8-21 |
| 8.4.3 | Packing of the system.....                     | 8-36 |
| 8.4.4 | Transport labeling.....                        | 8-43 |

**9 Maintenance Schedule for internal-tank configuration ..... 9-1****9.1 Safety ..... 9-1****9.2 Maintenance ..... 9-3**

|        |                                                      |     |
|--------|------------------------------------------------------|-----|
| 9.2.1  | List of required material .....                      | 9-3 |
| 9.2.2  | System cleaning .....                                | 9-3 |
| 9.2.3  | PC & Accessories.....                                | 9-3 |
| 9.2.4  | Preparation for Maintenance of Internal Modules..... | 9-4 |
| 9.2.5  | Maintenance of the Washer and Reader Modules .....   | 9-4 |
| 9.2.6  | Maintenance of the Pumps and Tubings .....           | 9-5 |
| 9.2.7  | Maintenance of the Reagent Module.....               | 9-5 |
| 9.2.8  | Maintenance of the Pipettor.....                     | 9-5 |
| 9.2.9  | Cleaning of other surfaces.....                      | 9-6 |
| 9.2.10 | System Restore & Periodic Recalibrations .....       | 9-6 |
| 9.2.11 | Requalification test .....                           | 9-7 |

|             |                                                                            |              |
|-------------|----------------------------------------------------------------------------|--------------|
| <b>9.3</b>  | <b>Preventive Maintenance kit for Field Service Engineers .....</b>        | <b>9-8</b>   |
| 9.3.1       | Yearly Maintenance kit .....                                               | 9-8          |
| 9.3.2       | Replacing on demand parts.....                                             | 9-8          |
| <b>9.4</b>  | <b>Instrument requalification after service visit .....</b>                | <b>9-9</b>   |
| 9.4.1       | Background and rationales .....                                            | 9-9          |
| 9.4.2       | Service interventions .....                                                | 9-9          |
| <b>9.5</b>  | <b>Other activities during service visits .....</b>                        | <b>9-13</b>  |
| 9.5.1       | Instrument recovery and test .....                                         | 9-13         |
| 9.5.2       | User Manual last revision check .....                                      | 9-13         |
| <b>9.6</b>  | <b>System Shutdown and Wakeup for long term inactivity procedure .....</b> | <b>9-13</b>  |
| 9.6.1       | Procedure for system Shutdown and Wakeup .....                             | 9-14         |
| <b>10</b>   | <b>Maintenance Schedule for external-tanks configuration .....</b>         | <b>10-1</b>  |
| <b>10.1</b> | <b>Safety .....</b>                                                        | <b>10-1</b>  |
| <b>10.2</b> | <b>Maintenance .....</b>                                                   | <b>10-3</b>  |
| 10.2.1      | List of required material .....                                            | 10-3         |
| 10.2.2      | System cleaning .....                                                      | 10-3         |
| 10.2.3      | PC & Accessories .....                                                     | 10-3         |
| 10.2.4      | Preparation for Maintenance of Internal Modules.....                       | 10-4         |
| 10.2.5      | Maintenance of the Washer and Reader Modules .....                         | 10-5         |
| 10.2.6      | Maintenance of the Pumps and Tubings .....                                 | 10-5         |
| 10.2.7      | Maintenance of the Reagent Module.....                                     | 10-5         |
| 10.2.8      | Maintenance of the Pipettor.....                                           | 10-6         |
| 10.2.9      | Cleaning of other surfaces.....                                            | 10-6         |
| 10.2.10     | System Restore & Periodic Recalibrations .....                             | 10-6         |
| 10.2.11     | Requalification test .....                                                 | 10-7         |
| <b>10.3</b> | <b>Preventive Maintenance kit for Field Service Engineers .....</b>        | <b>10-9</b>  |
| 10.3.1      | Yearly Maintenance kit .....                                               | 10-9         |
| 10.3.2      | Replacing on demand parts.....                                             | 10-9         |
| <b>10.4</b> | <b>Instrument requalification after service visit .....</b>                | <b>10-10</b> |
| 10.4.1      | Background and rationales .....                                            | 10-10        |
| 10.4.2      | Service interventions .....                                                | 10-10        |
| <b>10.5</b> | <b>Other activities during service visits .....</b>                        | <b>10-14</b> |
| 10.5.1      | Instrument recovery and test .....                                         | 10-14        |
| 10.5.2      | User Manual last revision check .....                                      | 10-14        |
| <b>10.6</b> | <b>System Shutdown and Wakeup for long term inactivity procedure .....</b> | <b>10-14</b> |
| 10.6.1      | Procedure for system Shutdown and Wakeup .....                             | 10-15        |



|             |                                                               |              |
|-------------|---------------------------------------------------------------|--------------|
| <b>11</b>   | <b>Service Software for internal-tanks configuration.....</b> | <b>11-1</b>  |
| <b>11.1</b> | <b>Use of the Service Software .....</b>                      | <b>11-1</b>  |
| 11.1.1      | General Statement.....                                        | 11-1         |
| 11.1.2      | Before starting the Service Software .....                    | 11-1         |
| 11.1.3      | Use of the Service Software .....                             | 11-1         |
| 11.1.4      | General handling of the Service Software .....                | 11-3         |
| 11.1.5      | Interface Selection .....                                     | 11-8         |
| 11.1.6      | Errors .....                                                  | 11-8         |
| 11.1.7      | General description of Calibration .....                      | 11-9         |
| 11.1.8      | General description of Parameter .....                        | 11-11        |
| <b>11.2</b> | <b>COP Module .....</b>                                       | <b>11-14</b> |
| 11.2.1      | Information.....                                              | 11-14        |
| 11.2.2      | Parameter .....                                               | 11-17        |
| <b>11.3</b> | <b>Front and Loading .....</b>                                | <b>11-18</b> |
| 11.3.1      | General .....                                                 | 11-18        |
| 11.3.2      | Parameter .....                                               | 11-20        |
| 11.3.3      | Sample Bay Scanner .....                                      | 11-21        |
| <b>11.4</b> | <b>Incubator .....</b>                                        | <b>11-24</b> |
| 11.4.1      | General .....                                                 | 11-24        |
| 11.4.2      | Parameter .....                                               | 11-25        |
| 11.4.3      | Teach & Calibration .....                                     | 11-26        |
| <b>11.5</b> | <b>Washer.....</b>                                            | <b>11-28</b> |
| 11.5.1      | General.....                                                  | 11-28        |
| 11.5.2      | Parameter .....                                               | 11-30        |
| 11.5.3      | Teach & Calibration .....                                     | 11-31        |
| <b>11.6</b> | <b>Reader.....</b>                                            | <b>11-35</b> |
| 11.6.1      | General .....                                                 | 11-35        |
| 11.6.2      | Parameter .....                                               | 11-37        |
| 11.6.3      | Teach & Calibration .....                                     | 11-38        |
| <b>11.7</b> | <b>Pipettor .....</b>                                         | <b>11-43</b> |
| 11.7.1      | General .....                                                 | 11-43        |
| 11.7.2      | Parameter .....                                               | 11-44        |
| 11.7.3      | Move .....                                                    | 11-45        |
| 11.7.4      | Test.....                                                     | 11-49        |
| 11.7.5      | Teach & Calibration .....                                     | 11-54        |
| <b>11.8</b> | <b>System .....</b>                                           | <b>11-58</b> |
| 11.8.1      | General .....                                                 | 11-58        |
| 11.8.2      | Cuvette Tray State.....                                       | 11-66        |
| 11.8.3      | Loading Report .....                                          | 11-67        |
| 11.8.4      | System Parameter .....                                        | 11-68        |

---

|             |                                                                |              |
|-------------|----------------------------------------------------------------|--------------|
| <b>12</b>   | <b>Service Software for external-tanks configuration .....</b> | <b>12-1</b>  |
| <hr/>       |                                                                |              |
| <b>12.1</b> | <b>Use of the Service Software .....</b>                       | <b>12-1</b>  |
| 12.1.1      | General Statement.....                                         | 12-1         |
| 12.1.2      | Before starting the Service Software .....                     | 12-1         |
| 12.1.3      | Use of the Service Software .....                              | 12-1         |
| 12.1.4      | General handling of the Service Software .....                 | 12-3         |
| 12.1.5      | Interface Selection .....                                      | 12-8         |
| 12.1.6      | Errors .....                                                   | 12-8         |
| 12.1.7      | General description of Calibration .....                       | 12-9         |
| 12.1.8      | General description of Parameter .....                         | 12-11        |
| <b>12.2</b> | <b>COP Module .....</b>                                        | <b>12-13</b> |
| 12.2.1      | General .....                                                  | 12-14        |
| 12.2.2      | Parameter .....                                                | 12-17        |
| <b>12.3</b> | <b>Front and Loading .....</b>                                 | <b>12-18</b> |
| 12.3.1      | General .....                                                  | 12-18        |
| 12.3.2      | Parameter .....                                                | 12-20        |
| 12.3.3      | Sample Bay Scanner .....                                       | 12-21        |
| <b>12.4</b> | <b>Incubator .....</b>                                         | <b>12-24</b> |
| 12.4.1      | General .....                                                  | 12-24        |
| 12.4.2      | Parameter .....                                                | 12-25        |
| 12.4.3      | Teach & Calibration .....                                      | 12-26        |
| <b>12.5</b> | <b>Washer .....</b>                                            | <b>12-29</b> |
| 12.5.1      | General .....                                                  | 12-29        |
| 12.5.2      | Parameter .....                                                | 12-31        |
| 12.5.3      | Teach & Calibration .....                                      | 12-32        |
| <b>12.6</b> | <b>Reader .....</b>                                            | <b>12-37</b> |
| 12.6.1      | General .....                                                  | 12-37        |
| 12.6.2      | Parameter .....                                                | 12-39        |
| 12.6.3      | Teach & Calibration .....                                      | 12-40        |
| <b>12.7</b> | <b>Pipettor .....</b>                                          | <b>12-46</b> |
| 12.7.1      | General .....                                                  | 12-46        |
| 12.7.2      | Parameter .....                                                | 12-47        |
| 12.7.3      | Move .....                                                     | 12-48        |
| 12.7.4      | Test .....                                                     | 12-52        |
| 12.7.5      | Teach & Calibration .....                                      | 12-57        |
| <b>12.8</b> | <b>System .....</b>                                            | <b>12-60</b> |
| 12.8.1      | General .....                                                  | 12-60        |
| 12.8.2      | Cuvette Tray State .....                                       | 12-69        |
| 12.8.3      | Loading Report .....                                           | 12-70        |
| 12.8.4      | System Parameter .....                                         | 12-71        |

---

|              |                                           |              |
|--------------|-------------------------------------------|--------------|
| <b>13</b>    | <b>Teaching of the Pipettor .....</b>     | <b>13-1</b>  |
| <b>13.1</b>  | <b>General Statement .....</b>            | <b>13-1</b>  |
| <b>13.2</b>  | <b>Safety .....</b>                       | <b>13-1</b>  |
| <b>13.3</b>  | <b>Overview .....</b>                     | <b>13-2</b>  |
| 13.3.1       | Overview Teacher menu .....               | 13-2         |
| 13.3.2       | Teaching sequence .....                   | 13-5         |
| <b>13.4</b>  | <b>Teaching the pipettor module .....</b> | <b>13-6</b>  |
| 13.4.1       | General information .....                 | 13-7         |
| 13.4.2       | Rack files .....                          | 13-12        |
| 13.4.3       | Bottle Shapes .....                       | 13-14        |
| 13.4.4       | Start of Teacher software .....           | 13-15        |
| <b>13.5</b>  | <b>Reagent probe reference .....</b>      | <b>13-16</b> |
| 13.5.1       | Teaching sequence .....                   | 13-16        |
| <b>13.6</b>  | <b>Sample probe reference .....</b>       | <b>13-18</b> |
| 13.6.1       | Teaching sequence .....                   | 13-18        |
| <b>13.7</b>  | <b>Gripper offset .....</b>               | <b>13-20</b> |
| 13.7.1       | Teaching sequence .....                   | 13-20        |
| <b>13.8</b>  | <b>Cuvette tray .....</b>                 | <b>13-22</b> |
| 13.8.1       | Teaching X, Y, Zmax .....                 | 13-22        |
| <b>13.9</b>  | <b>Tip tray .....</b>                     | <b>13-24</b> |
| 13.9.1       | Teaching X, Y, Zmax .....                 | 13-24        |
| <b>13.10</b> | <b>Washer .....</b>                       | <b>13-25</b> |
| 13.10.1      | Teaching X, Y, Zmax .....                 | 13-25        |
| <b>13.11</b> | <b>Sample bay .....</b>                   | <b>13-27</b> |
| 13.11.1      | Teaching X, Y, Zmax .....                 | 13-27        |
| <b>13.12</b> | <b>Incubator sample probe .....</b>       | <b>13-29</b> |
| 13.12.1      | Teaching X, Y, Zmax .....                 | 13-29        |
| <b>13.13</b> | <b>Reader .....</b>                       | <b>13-30</b> |
| 13.13.1      | Teaching X, Y, Zmax .....                 | 13-30        |
| <b>13.14</b> | <b>Incubator reagent probe .....</b>      | <b>13-31</b> |
| 13.14.1      | Teaching X, Y, Zmax .....                 | 13-31        |
| <b>13.15</b> | <b>Wash station .....</b>                 | <b>13-32</b> |
| 13.15.1      | Teaching X, Y, Zmax .....                 | 13-32        |
| <b>13.16</b> | <b>Ancillary bay .....</b>                | <b>13-35</b> |
| 13.16.1      | Teaching X, Y, Zmax .....                 | 13-35        |

---

|              |                                                                  |              |
|--------------|------------------------------------------------------------------|--------------|
| <b>13.17</b> | <b>Reagent bay .....</b>                                         | <b>13-36</b> |
| 13.17.1      | Teaching X, Y, Zmax .....                                        | 13-36        |
| <b>13.18</b> | <b>Sample Rack Teaching.....</b>                                 | <b>13-38</b> |
| 13.18.1      | Case 1: cylindrical tubes with standard inner diameter .....     | 13-38        |
| 13.18.2      | Case 2: cylindrical tubes with not standard inner diameter ..... | 13-39        |
| 13.18.3      | Case 3: not standard conical tubes .....                         | 13-41        |
| 13.18.4      | BottleShapes.xml setting .....                                   | 13-42        |
| <b>13.19</b> | <b>Transfer sequence.....</b>                                    | <b>13-44</b> |
| <b>14</b>    | <b>Covers .....</b>                                              | <b>14-1</b>  |

---

|             |                                                           |              |
|-------------|-----------------------------------------------------------|--------------|
| <b>14.1</b> | <b>Safety.....</b>                                        | <b>14-1</b>  |
| <b>14.2</b> | <b>Overview.....</b>                                      | <b>14-1</b>  |
| <b>14.3</b> | <b>Replacements and repairs .....</b>                     | <b>14-4</b>  |
| 14.3.1      | Replacement of the Left Side Cover.....                   | 14-4         |
| 14.3.2      | Replace of the Bulk Fluid Door .....                      | 14-6         |
| 14.3.3      | Replacement of the Right Side Cover .....                 | 14-7         |
| 14.3.4      | Replacement of the Panel PC .....                         | 14-9         |
| 14.3.5      | Replacement of the Left Top Cover.....                    | 14-12        |
| 14.3.6      | Replacement of the Right Top Cover .....                  | 14-13        |
| 14.3.7      | Replacement of the Top Cover.....                         | 14-14        |
| 14.3.8      | Replacement of the Front Module .....                     | 14-16        |
| 14.3.9      | Replacement of the Gas Pressure Spring at Top Cover ..... | 14-16        |
| <b>14.4</b> | <b>Alignment of Left and Right Top Covers.....</b>        | <b>14-20</b> |
| <b>14.5</b> | <b>Electric connections, switches and signals.....</b>    | <b>14-21</b> |
| 14.5.1      | Soft Power Button PCB .....                               | 14-21        |

---

|           |                                          |             |
|-----------|------------------------------------------|-------------|
| <b>15</b> | <b>Cuvette - Tip Loading Module.....</b> | <b>15-1</b> |
|-----------|------------------------------------------|-------------|

---

|             |                                                                            |             |
|-------------|----------------------------------------------------------------------------|-------------|
| <b>15.1</b> | <b>Safety.....</b>                                                         | <b>15-1</b> |
| <b>15.2</b> | <b>Overview.....</b>                                                       | <b>15-2</b> |
| <b>15.3</b> | <b>Replacement and repairs .....</b>                                       | <b>15-3</b> |
| 15.3.1      | Replacement of the Cuvette - Tip Loading Module .....                      | 15-3        |
| 15.3.2      | Replacement of the Cuvette - Tip drawer switch / snap-in lock.....         | 15-4        |
| 15.3.3      | Alignment of the Cuvette - Tip Loading Module handle.....                  | 15-5        |
| <b>15.4</b> | <b>Adjustment and calibration .....</b>                                    | <b>15-7</b> |
| 15.4.1      | Adjustment of the Cuvette - Tip Loading Module switch / snap-in lock ..... | 15-7        |

|             |                                                                    |              |
|-------------|--------------------------------------------------------------------|--------------|
| <b>16</b>   | <b>Incubator Module .....</b>                                      | <b>16-1</b>  |
| <b>16.1</b> | <b>Safety .....</b>                                                | <b>16-1</b>  |
| <b>16.2</b> | <b>Overview .....</b>                                              | <b>16-2</b>  |
| <b>16.3</b> | <b>Replacement and repairs .....</b>                               | <b>16-4</b>  |
| 16.3.1      | Replacement of the Incubator cover .....                           | 16-4         |
| 16.3.2      | Replacement of the Incubator Module .....                          | 16-5         |
| 16.3.3      | Replacement of the Incubator rotor .....                           | 16-7         |
| 16.3.4      | Replacement of the bearings of the Incubator frame and cover ..... | 16-9         |
| 16.3.5      | Replacement of the Incubator motor .....                           | 16-11        |
| 16.3.6      | Replacement of the Incubator isolation foam .....                  | 16-13        |
| 16.3.7      | Replacement of the Incubator CU PCBA .....                         | 16-15        |
| 16.3.8      | Replacement of the Incubator light guide .....                     | 16-17        |
| <b>16.4</b> | <b>Adjustment and calibration .....</b>                            | <b>16-19</b> |
| 16.4.1      | Initialization routine .....                                       | 16-19        |
| 16.4.2      | Calibration of the Incubator Module .....                          | 16-19        |
| <b>16.5</b> | <b>Electric connections, switches and signals .....</b>            | <b>16-21</b> |
| 16.5.1      | INCUBATOR CU PCBA .....                                            | 16-21        |
| <b>17</b>   | <b>Washer Module .....</b>                                         | <b>17-1</b>  |
| <b>17.1</b> | <b>Safety .....</b>                                                | <b>17-1</b>  |
| <b>17.2</b> | <b>Overview .....</b>                                              | <b>17-1</b>  |
| 17.2.1      | SIPAL overview .....                                               | 17-4         |
| <b>17.3</b> | <b>Replacements and repairs .....</b>                              | <b>17-6</b>  |
| 17.3.1      | Replacement of the spindle aspiration probe .....                  | 17-6         |
| 17.3.2      | Replacement of the SIPAL .....                                     | 17-12        |
| 17.3.3      | Replacement of SIPAL plastic parts .....                           | 17-16        |
| 17.3.4      | Replacement of the valve .....                                     | 17-19        |
| 17.3.5      | Replacement of the dispense manifold .....                         | 17-22        |
| 17.3.6      | Replacement of the Washer plate .....                              | 17-24        |
| 17.3.7      | Replacement of the Washer CU PCB .....                             | 17-27        |
| 17.3.8      | Replacement of the Washer light barrier .....                      | 17-30        |
| 17.3.9      | Replacement of the Washer rotor .....                              | 17-32        |
| 17.3.10     | Replacement of the Washer Module .....                             | 17-35        |
| 17.3.11     | Replacement of the Washer Motor .....                              | 17-37        |
| 17.3.12     | Replacement of the aspiration lift PCBA .....                      | 17-40        |
| 17.3.13     | Replacement of the spindle drive of SIPAL .....                    | 17-42        |
| <b>17.4</b> | <b>Adjustment and calibration .....</b>                            | <b>17-44</b> |
| 17.4.1      | Initialization routine .....                                       | 17-44        |
| 17.4.2      | Calibration of the Washer Module .....                             | 17-45        |

|             |                                                        |              |
|-------------|--------------------------------------------------------|--------------|
| <b>17.5</b> | <b>Electric connections, switches and signals.....</b> | <b>17-48</b> |
| 17.5.1      | Washer CU PCBA board .....                             | 17-48        |
| 17.5.2      | Aspiration lift PCBA board .....                       | 17-49        |
| <b>18</b>   | <b>Reader Module .....</b>                             | <b>18-1</b>  |
| <b>18.1</b> | <b>Safety .....</b>                                    | <b>18-1</b>  |
| <b>18.2</b> | <b>Overview .....</b>                                  | <b>18-2</b>  |
| <b>18.3</b> | <b>Replacements and repairs .....</b>                  | <b>18-4</b>  |
| 18.3.1      | Replacement of the spindle aspiration probe .....      | 18-4         |
| 18.3.2      | Replacement of the Reader cover and SIPAL .....        | 18-9         |
| 18.3.3      | Replacement of the aspiration lift PCBA .....          | 18-13        |
| 18.3.4      | Replacement of the spindle drive of SIPAL .....        | 18-15        |
| 18.3.5      | Replacement of the Reader rotor .....                  | 18-17        |
| 18.3.6      | Replacement of the Reader Module .....                 | 18-19        |
| 18.3.7      | Replacement of the Reader light barrier .....          | 18-22        |
| 18.3.8      | Replacement of the Reader rotor motor .....            | 18-24        |
| 18.3.9      | Replacement of the PCB Luminometer CU .....            | 18-26        |
| 18.3.10     | Replacement of the PMT .....                           | 18-29        |
| <b>18.4</b> | <b>Adjustment and calibration .....</b>                | <b>18-31</b> |
| 18.4.1      | Initialization routine .....                           | 18-31        |
| 18.4.2      | In case of a new PMT .....                             | 18-31        |
| 18.4.3      | Calibration of the Reader Module .....                 | 18-32        |
| <b>18.5</b> | <b>Electric connections, switches and signals.....</b> | <b>18-34</b> |
| 18.5.1      | Luminometer CU PCBA board.....                         | 18-34        |
| 18.5.2      | Reader SIPAL CU PCBA board.....                        | 18-35        |
| <b>19</b>   | <b>Pipettor for internal-tanks configuration .....</b> | <b>19-1</b>  |
| <b>19.1</b> | <b>Safety .....</b>                                    | <b>19-1</b>  |
| <b>19.2</b> | <b>Overview .....</b>                                  | <b>19-2</b>  |
| 19.2.1      | Pipettor Overview .....                                | 19-2         |
| 19.2.2      | Left Arm overview .....                                | 19-3         |
| 19.2.3      | Right Arm overview .....                               | 19-5         |
| 19.2.4      | Sample Pipettor overview .....                         | 19-7         |
| 19.2.5      | Gripper overview .....                                 | 19-8         |
| 19.2.6      | Reagent Pipettor overview .....                        | 19-9         |
| <b>19.3</b> | <b>Replacements and repairs .....</b>                  | <b>19-10</b> |
| 19.3.1      | Replacement of the Sample Pipettor .....               | 19-10        |
| 19.3.2      | Replacement of the Dual Sledge CU PCB .....            | 19-15        |
| 19.3.3      | Replacement of the Gripper Sensor PCB .....            | 19-18        |
| 19.3.4      | Replacement of the Gripper module .....                | 19-20        |
| 19.3.5      | Replacement of the Gripper CU PCB .....                | 19-23        |

|             |                                                                    |              |
|-------------|--------------------------------------------------------------------|--------------|
| 19.3.6      | Replacement of the Gripper slide bearings .....                    | 19-25        |
| 19.3.7      | Replacement of the Sample Pipettor Z-Motor .....                   | 19-27        |
| 19.3.8      | Replacement of the Gripper Z-Motor .....                           | 19-30        |
| 19.3.9      | Replacement of the X-axis toothed belt of Left Pipettor Arm .....  | 19-33        |
| 19.3.10     | Replacement of the X-Motor of Left Pipettor Arm .....              | 19-36        |
| 19.3.11     | Replacement of the Y-axis toothed belt of Left Pipettor Arm .....  | 19-38        |
| 19.3.12     | Replacement of the Y-Motor of Left Pipettor Arm .....              | 19-42        |
| 19.3.13     | Replacement of the Left Pipettor Arm .....                         | 19-45        |
| 19.3.14     | Replacement of the SPOLV Needle .....                              | 19-50        |
| 19.3.15     | Replacement of the SPOLV .....                                     | 19-52        |
| 19.3.16     | Replacement of the Single Sledge CU PCB .....                      | 19-55        |
| 19.3.17     | Replacement of the Reagent Pipettor Z-Motor .....                  | 19-57        |
| 19.3.18     | Replacement of the X-axis toothed belt of Right Pipettor Arm ..... | 19-59        |
| 19.3.19     | Replacement of the X-Motor of Right Pipettor Arm .....             | 19-61        |
| 19.3.20     | Replacement of the Y-axis toothed belt of Right Pipettor Arm ..... | 19-63        |
| 19.3.21     | Replacement of the Y-Motor of Right Pipettor Arm .....             | 19-66        |
| 19.3.22     | Replacement of the Right Pipettor Arm .....                        | 19-69        |
| 19.3.23     | Replacement of the Left arm Y-Home sensor .....                    | 19-73        |
| 19.3.24     | Replacement of the Right arm Y-Home sensor .....                   | 19-75        |
| 19.3.25     | Replacement of the SPOLV Z-Home sensor .....                       | 19-76        |
| <b>19.4</b> | <b>Adjustment and calibration .....</b>                            | <b>19-78</b> |
| 19.4.1      | Initialization routine .....                                       | 19-78        |
| 19.4.2      | Calibration of the Pipettor Module .....                           | 19-78        |
| <b>19.5</b> | <b>Priming of Pipettor .....</b>                                   | <b>19-80</b> |
| 19.5.1      | Initialization routine .....                                       | 19-80        |
| 19.5.2      | Initial rinsing with DI water .....                                | 19-80        |
| 19.5.3      | Rinsing with Wash Buffer .....                                     | 19-81        |
| <b>19.6</b> | <b>Pipettor Tests .....</b>                                        | <b>19-82</b> |
| 19.6.1      | LLD Endurance test .....                                           | 19-82        |
| 19.6.2      | User LLD Endurance test .....                                      | 19-82        |
| 19.6.3      | Pipettor test .....                                                | 19-82        |
| <b>19.7</b> | <b>Cuvette Endurance Run .....</b>                                 | <b>19-82</b> |
| <b>19.8</b> | <b>Electric connections, switches and signals .....</b>            | <b>19-83</b> |
| 19.8.1      | X-Sledge-Dual-Z EC CU PCBA board .....                             | 19-83        |
| 19.8.2      | Gripper CU PCBA board .....                                        | 19-85        |
| 19.8.3      | Gripper sensor PCBA board .....                                    | 19-86        |
| 19.8.4      | X-Z-Sledge EC CU PCBA board .....                                  | 19-87        |
| <b>20</b>   | <b>Pipettor for external-tanks configuration .....</b>             | <b>20-1</b>  |
| <b>20.1</b> | <b>Safety .....</b>                                                | <b>20-1</b>  |
| <b>20.2</b> | <b>Overview .....</b>                                              | <b>20-2</b>  |
| 20.2.1      | Pipettor Overview .....                                            | 20-2         |
| 20.2.2      | Left Arm overview .....                                            | 20-3         |

|             |                                                                    |              |
|-------------|--------------------------------------------------------------------|--------------|
| 20.2.3      | Right Arm overview.....                                            | 20-5         |
| 20.2.4      | Sample Pipettor overview .....                                     | 20-7         |
| 20.2.5      | Gripper overview.....                                              | 20-8         |
| 20.2.6      | Reagent Pipettor overview.....                                     | 20-9         |
| <b>20.3</b> | <b>Replacements and repairs .....</b>                              | <b>20-10</b> |
| 20.3.1      | Replacement of the Sample Pipettor .....                           | 20-10        |
| 20.3.2      | Replacement of the Dual Sledge CU PCB .....                        | 20-15        |
| 20.3.3      | Replacement of the Gripper Sensor PCB.....                         | 20-18        |
| 20.3.4      | Replacement of the Gripper module.....                             | 20-20        |
| 20.3.5      | Replacement of the Gripper CU PCB.....                             | 20-23        |
| 20.3.6      | Replacement of the Gripper slide bearings .....                    | 20-25        |
| 20.3.7      | Replacement of the Sample Pipettor Z-Motor .....                   | 20-27        |
| 20.3.8      | Replacement of the Gripper Z-Motor .....                           | 20-30        |
| 20.3.9      | Replacement of the X-axis toothed belt of Left Pipettor Arm.....   | 20-33        |
| 20.3.10     | Replacement of the X-Motor of Left Pipettor Arm.....               | 20-36        |
| 20.3.11     | Replacement of the Y-axis toothed belt of Left Pipettor Arm.....   | 20-38        |
| 20.3.12     | Replacement of the Y-Motor of Left Pipettor Arm.....               | 20-42        |
| 20.3.13     | Replacement of the Left Pipettor Arm.....                          | 20-45        |
| 20.3.14     | Replacement of the SPOLV Needle .....                              | 20-50        |
| 20.3.15     | Replacement of the SPOLV.....                                      | 20-52        |
| 20.3.16     | Replacement of the Single Sledge CU PCB.....                       | 20-56        |
| 20.3.17     | Replacement of the Reagent Pipettor Z-Motor .....                  | 20-58        |
| 20.3.18     | Replacement of the X-axis toothed belt of Right Pipettor Arm ..... | 20-60        |
| 20.3.19     | Replacement of the X-Motor of Right Pipettor Arm .....             | 20-62        |
| 20.3.20     | Replacement of the Y-axis toothed belt of Right Pipettor Arm ..... | 20-64        |
| 20.3.21     | Replacement of the Y-Motor of Right Pipettor Arm .....             | 20-67        |
| 20.3.22     | Replacement of the Right Pipettor Arm .....                        | 20-70        |
| 20.3.23     | Replacement of the Left arm Y-Home sensor .....                    | 20-74        |
| 20.3.24     | Replacement of the Right arm Y-Home sensor .....                   | 20-76        |
| 20.3.25     | Replacement of the SPOLV Z-Home sensor.....                        | 20-77        |
| <b>20.4</b> | <b>Adjustment and calibration .....</b>                            | <b>20-79</b> |
| 20.4.1      | Initialization routine.....                                        | 20-79        |
| 20.4.2      | Calibration of the Pipettor Module .....                           | 20-79        |
| <b>20.5</b> | <b>Priming of Pipettor .....</b>                                   | <b>20-81</b> |
| 20.5.1      | Initialization routine.....                                        | 20-81        |
| 20.5.2      | Rinsing with System Liquid.....                                    | 20-81        |
| <b>20.6</b> | <b>Pipettor Tests .....</b>                                        | <b>20-83</b> |
| 20.6.1      | LLD Endurance test.....                                            | 20-83        |
| 20.6.2      | User LLD Endurance test .....                                      | 20-83        |
| 20.6.3      | Pipettor test .....                                                | 20-83        |
| <b>20.7</b> | <b>Cuvette Endurance Run .....</b>                                 | <b>20-83</b> |
| <b>20.8</b> | <b>Electric connections, switches and signals.....</b>             | <b>20-84</b> |
| 20.8.1      | X-Sledge-Dual-Z EC CU PCBA board.....                              | 20-84        |
| 20.8.2      | Gripper CU PCBA board.....                                         | 20-86        |
| 20.8.3      | Gripper sensor PCBA board .....                                    | 20-87        |



|        |                                   |       |
|--------|-----------------------------------|-------|
| 20.8.4 | X-Z-Sledge EC CU PCBA board ..... | 20-88 |
|--------|-----------------------------------|-------|

## **21 Sample Module.....21-1**

### **21.1 Safety .....21-1**

### **21.2 Overview.....21-2**

|        |                    |      |
|--------|--------------------|------|
| 21.2.1 | Sample Module..... | 21-2 |
| 21.2.2 | Sample rack.....   | 21-4 |
| 21.2.3 | Control rack ..... | 21-4 |

### **21.3 Replacement and repairs .....21-5**

|        |                                             |       |
|--------|---------------------------------------------|-------|
| 21.3.1 | Replacement of the Sample Bay cover .....   | 21-5  |
| 21.3.2 | Replacement of the Sample Module.....       | 21-7  |
| 21.3.3 | Replacement of the push/push mechanism..... | 21-9  |
| 21.3.4 | Replacement of the Sample Bay Guides.....   | 21-11 |
| 21.3.5 | Replacement of the barcode scanner.....     | 21-13 |
| 21.3.6 | Replacement of the Sample Bay PCBA .....    | 21-17 |
| 21.3.7 | Replacement of the Temperature sensor ..... | 21-19 |

### **21.4 Adjustment and calibration .....21-21**

|        |                                                  |       |
|--------|--------------------------------------------------|-------|
| 21.4.1 | Initialization routine.....                      | 21-21 |
| 21.4.2 | Check and Verification of the Sample Module..... | 21-21 |

### **21.5 Electric connections, switches and signals.....21-23**

|        |                      |       |
|--------|----------------------|-------|
| 21.5.1 | Sample Bay PCBA..... | 21-23 |
|--------|----------------------|-------|

## **22 Reagent Module .....22-1**

### **22.1 Safety .....22-1**

### **22.2 Overview.....22-2**

### **22.3 Replacement and repairs .....22-4**

|         |                                                           |       |
|---------|-----------------------------------------------------------|-------|
| 22.3.1  | Replacement of the Reagent Bay cover .....                | 22-4  |
| 22.3.2  | Replacement of the Stirrer Rod and Stirrer Bearings ..... | 22-5  |
| 22.3.3  | Replacement of the Stirrer Motor.....                     | 22-8  |
| 22.3.4  | Replacement of the Reagent Module .....                   | 22-12 |
| 22.3.5  | Replacement of the Cooling Module.....                    | 22-14 |
| 22.3.6  | Replacement of the Reagent Loading Bay CU PCB .....       | 22-18 |
| 22.3.7  | Replacement of the Main Insulation .....                  | 22-20 |
| 22.3.8  | Replacement of the PCB, Antenna 10 channel .....          | 22-23 |
| 22.3.9  | Replacement of the Quad RFID Antenna PCB.....             | 22-25 |
| 22.3.10 | Replacement of the Temperature Sensor.....                | 22-27 |

### **22.4 Adjustment and calibration .....22-30**

|        |                                                    |       |
|--------|----------------------------------------------------|-------|
| 22.4.1 | Initialization routine.....                        | 22-30 |
| 22.4.2 | Check and Verification of the Reagent Module ..... | 22-30 |
| 22.4.3 | Readjustment of the cam to the stirrer rod .....   | 22-31 |

---

|             |                                                                      |              |
|-------------|----------------------------------------------------------------------|--------------|
| <b>22.5</b> | <b>Electric connections, switches and signals.....</b>               | <b>22-33</b> |
| 22.5.1      | Reagent Loading Bay CU PCB.....                                      | 22-33        |
| 22.5.2      | Quad RFID Antenna PCB.....                                           | 22-34        |
| 22.5.3      | 10-Channel RFID Antenna PCB .....                                    | 22-34        |
| <b>23</b>   | <b>Starter Reagent Module for internal-tanks configuration .....</b> | <b>23-1</b>  |
| <hr/>       |                                                                      |              |
| <b>23.1</b> | <b>Safety .....</b>                                                  | <b>23-1</b>  |
| <b>23.2</b> | <b>Overview.....</b>                                                 | <b>23-2</b>  |
| <b>23.3</b> | <b>Replacement and repairs .....</b>                                 | <b>23-4</b>  |
| 23.3.1      | Replacement of the Starter Reagent Module .....                      | 23-4         |
| 23.3.2      | Replacement of the Starter 1 sensor .....                            | 23-7         |
| 23.3.3      | Replacement of the Starter 2 sensor .....                            | 23-11        |
| 23.3.4      | Replacement of the Dual antenna PCBA board .....                     | 23-14        |
| <b>23.4</b> | <b>Adjustment and calibration .....</b>                              | <b>23-16</b> |
| 23.4.1      | Initialization routines .....                                        | 23-16        |
| 23.4.2      | Calibration of the Starter Reagent Module .....                      | 23-16        |
| 23.4.3      | Check of Starters bubble sensors.....                                | 23-17        |
| 23.4.4      | Verifying RFID .....                                                 | 23-18        |
| <b>23.5</b> | <b>Electric connections, switches and signals.....</b>               | <b>23-19</b> |
| 23.5.1      | Dual antenna PCBA board .....                                        | 23-19        |
| <b>24</b>   | <b>Starter Reagent Module for external-tanks configuration .....</b> | <b>24-1</b>  |
| <hr/>       |                                                                      |              |
| <b>24.1</b> | <b>Safety .....</b>                                                  | <b>24-1</b>  |
| <b>24.2</b> | <b>Overview.....</b>                                                 | <b>24-2</b>  |
| <b>24.3</b> | <b>Replacement and repairs .....</b>                                 | <b>24-4</b>  |
| 24.3.1      | Replacement of the Starter Reagent Module .....                      | 24-4         |
| 24.3.2      | Replacement of the Starter 1 sensor .....                            | 24-7         |
| 24.3.3      | Replacement of the Starter 2 sensor .....                            | 24-11        |
| 24.3.4      | Replacement of the Dual antenna PCBA board .....                     | 24-14        |
| <b>24.4</b> | <b>Adjustment and calibration .....</b>                              | <b>24-16</b> |
| 24.4.1      | Initialization routines .....                                        | 24-16        |
| 24.4.2      | Calibration of the Starter Reagent Module .....                      | 24-16        |
| 24.4.3      | Check of Starters bubble sensors.....                                | 24-17        |
| 24.4.4      | Verifying RFID .....                                                 | 24-18        |
| <b>24.5</b> | <b>Electric connections, switches and signals.....</b>               | <b>24-19</b> |
| 24.5.1      | Dual antenna PCBA board .....                                        | 24-19        |

---

|             |                                                                                  |              |
|-------------|----------------------------------------------------------------------------------|--------------|
| <b>25</b>   | <b>Pumps for internal-tanks configuration.....</b>                               | <b>25-1</b>  |
| <b>25.1</b> | <b>Safety.....</b>                                                               | <b>25-1</b>  |
| <b>25.2</b> | <b>Overview.....</b>                                                             | <b>25-2</b>  |
| 25.2.1      | Liquid Distribution Overview .....                                               | 25-3         |
| <b>25.3</b> | <b>Replacement of Pumps .....</b>                                                | <b>25-4</b>  |
| 25.3.1      | Replacement of the LWP (Liquid Waste Pump, 3-channels peristaltic pump) .....    | 25-4         |
| 25.3.2      | Replacement of the WAP (Waste Aspiration Pump, 5-channels peristaltic pump)..... | 25-7         |
| 25.3.3      | Replacement of the Peristaltic Pumps Tubings.....                                | 25-12        |
| 25.3.4      | Replacement of the SDP (Starter Dispense Pumps).....                             | 25-16        |
| 25.3.5      | Replacement of the PRP (Probe Rinsing Pump).....                                 | 25-20        |
| 25.3.6      | Replacement of the WBDP (Wash Buffer Dispense Pump) .....                        | 25-25        |
| 25.3.7      | Replacement of the WBDP Cartridge .....                                          | 25-29        |
| 25.3.8      | Replacement of the WBDP Cartridge valve unit.....                                | 25-33        |
| <b>25.4</b> | <b>Adjustment and Calibration.....</b>                                           | <b>25-35</b> |
| 25.4.1      | Initialization routines .....                                                    | 25-35        |
| 25.4.2      | Calibration of the SDPs .....                                                    | 25-35        |
| 25.4.3      | Calibration of the WBDP .....                                                    | 25-35        |
| <b>25.5</b> | <b>Electrical connections, switches and signals.....</b>                         | <b>25-36</b> |
| 25.5.1      | LWP and WAP PCBA .....                                                           | 25-36        |
| 25.5.2      | WBDP PCBA .....                                                                  | 25-38        |
| 25.5.3      | SDP PCBA.....                                                                    | 25-40        |
| <b>26</b>   | <b>Pumps for external-tanks configuration.....</b>                               | <b>26-1</b>  |
| <b>26.1</b> | <b>Safety.....</b>                                                               | <b>26-1</b>  |
| <b>26.2</b> | <b>Overview.....</b>                                                             | <b>26-2</b>  |
| 26.2.1      | Liquid Distribution Overview .....                                               | 26-3         |
| <b>26.3</b> | <b>Replacement of Pumps .....</b>                                                | <b>26-4</b>  |
| 26.3.1      | Replacement of the LWP (Liquid Waste Pump, 3-channels peristaltic pump) .....    | 26-4         |
| 26.3.2      | Replacement of the WAP (Waste Aspiration Pump, 5-channels peristaltic pump)..... | 26-7         |
| 26.3.3      | Replacement of the Peristaltic Pumps Tubings.....                                | 26-12        |
| 26.3.4      | Replacement of the SDP (Starter Dispense Pumps).....                             | 26-16        |
| 26.3.5      | Replacement of the PRP (Probe Rinsing Pump).....                                 | 26-20        |
| 26.3.6      | Replacement of the WBDP (Wash Buffer Dispense Pump) .....                        | 26-25        |
| 26.3.7      | Replacement of the WBDP Cartridge .....                                          | 26-29        |
| 26.3.8      | Replacement of the WBDP Cartridge valve unit.....                                | 26-33        |
| <b>26.4</b> | <b>Adjustment and Calibration.....</b>                                           | <b>26-35</b> |
| 26.4.1      | Initialization routines .....                                                    | 26-35        |
| 26.4.2      | Calibration of the SDPs .....                                                    | 26-35        |
| 26.4.3      | Calibration of the WBDP .....                                                    | 26-35        |

---

|             |                                                                  |              |
|-------------|------------------------------------------------------------------|--------------|
| <b>26.5</b> | <b>Electrical connections, switches and signals.....</b>         | <b>26-36</b> |
| 26.5.1      | LWP and WAP PCBA .....                                           | 26-36        |
| 26.5.2      | WBDP PCBA .....                                                  | 26-38        |
| 26.5.3      | SDP PCBA.....                                                    | 26-40        |
| <b>27</b>   | <b>Front Module .....</b>                                        | <b>27-1</b>  |
| <hr/>       |                                                                  |              |
| <b>27.1</b> | <b>Safety .....</b>                                              | <b>27-1</b>  |
| <b>27.2</b> | <b>Overview .....</b>                                            | <b>27-2</b>  |
| <b>27.3</b> | <b>Replacement and repairs .....</b>                             | <b>27-4</b>  |
| 27.3.1      | Replacement of the Front Module .....                            | 27-4         |
| 27.3.2      | Replacement of the Starter Reagent flap .....                    | 27-6         |
| 27.3.3      | Replacement of the Gas Spring for Starter Reagent Flap .....     | 27-12        |
| 27.3.4      | Replacement of the Gas Spring for Sample Area Flap.....          | 27-14        |
| 27.3.5      | Replacement of the Gas Spring of Reagent Area flap.....          | 27-16        |
| 27.3.6      | Replacement of the Front Cover CU PCB .....                      | 27-17        |
| 27.3.7      | Replacement of the Reed Switch .....                             | 27-20        |
| 27.3.8      | Replacement of the Snap-in plate .....                           | 27-22        |
| <b>27.4</b> | <b>Adjustment and calibration .....</b>                          | <b>27-24</b> |
| 27.4.1      | Initialization routine.....                                      | 27-24        |
| 27.4.2      | Adjustment and calibration of the Front Module .....             | 27-24        |
| <b>27.5</b> | <b>Electric connections, switches and signals.....</b>           | <b>27-25</b> |
| 27.5.1      | Front cover CU PCBA .....                                        | 27-25        |
| <b>27.6</b> | <b>Magnetic Particles Resuspension Area.....</b>                 | <b>27-27</b> |
| <b>28</b>   | <b>Tubing for internal-tanks configuration.....</b>              | <b>28-1</b>  |
| <hr/>       |                                                                  |              |
| <b>28.1</b> | <b>Safety .....</b>                                              | <b>28-1</b>  |
| 28.1.1      | Liquid Distribution Overview .....                               | 28-2         |
| <b>28.2</b> | <b>Overview .....</b>                                            | <b>28-3</b>  |
| 28.2.1      | Overview tubing .....                                            | 28-3         |
| 28.2.2      | Overview Waste tubing at Left Energy Chain (LWP).....            | 28-5         |
| 28.2.3      | Overview Waste tubing at Right Energy Chain (WAP).....           | 28-5         |
| <b>28.3</b> | <b>Replacements and repairs .....</b>                            | <b>28-7</b>  |
| 28.3.1      | Replacement of the Liquid distribution valve (LDV).....          | 28-7         |
| 28.3.2      | Replacement of the LDV-PRP tubing .....                          | 28-10        |
| 28.3.3      | Replacement of the DI WATER TANK-LDV .....                       | 28-12        |
| 28.3.4      | Replacement of the PRP-SPOLV tubing .....                        | 28-15        |
| 28.3.5      | Replacement of the WB TANK-WBDP-LDV .....                        | 28-17        |
| 28.3.6      | Replacement of the Peristaltic pumps rack .....                  | 28-20        |
| 28.3.7      | Replacement of the Waste tubing at Left Energy Chain (LWP).....  | 28-23        |
| 28.3.8      | Replacement of the Waste tubing at Right Energy Chain (WAP)..... | 28-27        |

---

|             |                                                                  |              |
|-------------|------------------------------------------------------------------|--------------|
| 28.3.9      | Replacement of the Waste interface-Waste tank tubing.....        | 28-32        |
| 28.3.10     | Replacement of the Liquid Waste level sensor.....                | 28-34        |
| 28.3.11     | Replacement of the WBDP-Washer tubings.....                      | 28-38        |
| 28.3.12     | Replacement of the SDP-Starters tubings.....                     | 28-40        |
| 28.3.13     | Replacement of the Reader-SDP tubings.....                       | 28-43        |
| 28.3.14     | Replacement of the Wash station cover.....                       | 28-47        |
| 28.3.15     | Replacement of the Wash station.....                             | 28-49        |
| <b>28.4</b> | <b>Verifying of tank presence sensors and level sensors.....</b> | <b>28-51</b> |
| 28.4.1      | Initialization routine.....                                      | 28-51        |
| 28.4.2      | Verifying of the DI water tank presence sensor.....              | 28-51        |
| 28.4.3      | Verifying of the DI water tank level sensor.....                 | 28-51        |
| 28.4.4      | Verifying of the Wash Buffer tank presence sensor.....           | 28-52        |
| 28.4.5      | Verifying of the Wash Buffer tank level sensor.....              | 28-52        |
| 28.4.6      | Verifying of the Liquid Waste tank presence sensor.....          | 28-52        |
| 28.4.7      | Verifying of the Liquid Waste tank level sensor.....             | 28-52        |
| <b>28.5</b> | <b>Direct Drain Configuration.....</b>                           | <b>28-53</b> |
| 28.5.1      | Preliminary steps.....                                           | 28-54        |
| 28.5.2      | Installation.....                                                | 28-54        |
| 28.5.3      | Parameter setting.....                                           | 28-54        |
| 28.5.4      | Requalification.....                                             | 28-55        |
| <b>29</b>   | <b>Tubing for external-tanks configuration.....</b>              | <b>29-1</b>  |
| <b>29.1</b> | <b>Safety.....</b>                                               | <b>29-1</b>  |
| 29.1.1      | Liquid Distribution Overview.....                                | 29-2         |
| <b>29.2</b> | <b>Overview.....</b>                                             | <b>29-3</b>  |
| 29.2.1      | Overview tubing.....                                             | 29-3         |
| 29.2.2      | Overview Waste tubing at Left Energy Chain (LWP).....            | 29-5         |
| 29.2.3      | Overview Waste tubing at Right Energy Chain (WAP).....           | 29-5         |
| <b>29.3</b> | <b>Replacements and repairs.....</b>                             | <b>29-7</b>  |
| 29.3.1      | Replacement of the MANN filter.....                              | 29-7         |
| 29.3.2      | Replacement of the S.....ystem Liquid tank sensor cable.....     | 29-10        |
| 29.3.3      | Replacement of the SYSTEM LIQUID TANK-MANN FILTER tubing.....    | 29-12        |
| 29.3.4      | Replacement of the PRP-SPOLV tubing.....                         | 29-15        |
| 29.3.5      | Replacement of the WB TANK-WBDP tubing.....                      | 29-17        |
| 29.3.6      | Replacement of the Wash Buffer tank sensor cable.....            | 29-19        |
| 29.3.7      | Replacement of the Peristaltic pumps rack.....                   | 29-22        |
| 29.3.8      | Replacement of the Waste tubing at Left Energy Chain (LWP).....  | 29-25        |
| 29.3.9      | Replacement of the Waste tubing at Right Energy Chain (WAP)..... | 29-29        |
| 29.3.10     | Replacement of the Waste interface-Waste tank tubing.....        | 29-34        |
| 29.3.11     | Replacement of the Liquid Waste level sensor.....                | 29-36        |
| 29.3.12     | Replacement of the WBDP-Washer tubings.....                      | 29-40        |
| 29.3.13     | Replacement of the SDP-Starters tubings.....                     | 29-42        |
| 29.3.14     | Replacement of the Reader-SDP tubings.....                       | 29-45        |
| 29.3.15     | Replacement of the Wash station cover.....                       | 29-48        |

|             |                                                                     |              |
|-------------|---------------------------------------------------------------------|--------------|
| 29.3.16     | Replacement of the Wash station .....                               | 29-50        |
| <b>29.4</b> | <b>Verifying of tank presence sensors and level sensors .....</b>   | <b>29-52</b> |
| 29.4.1      | Initialization routine .....                                        | 29-52        |
| 29.4.2      | Verifying of the System Liquid tank presence sensor .....           | 29-52        |
| 29.4.3      | Verifying of the System Liquid tank level sensor .....              | 29-52        |
| 29.4.4      | Verifying of the Wash Buffer tank presence sensor .....             | 29-53        |
| 29.4.5      | Verifying of the Wash Buffer tank level sensor .....                | 29-53        |
| 29.4.6      | Verifying of the Liquid Waste tank presence sensor .....            | 29-53        |
| 29.4.7      | Verifying of the Liquid Waste tank level sensor .....               | 29-53        |
| <b>29.5</b> | <b>Direct Drain Configuration .....</b>                             | <b>29-54</b> |
| 29.5.1      | Preliminary steps .....                                             | 29-55        |
| 29.5.2      | Installation .....                                                  | 29-55        |
| 29.5.3      | Parameter setting .....                                             | 29-55        |
| 29.5.4      | Requalification .....                                               | 29-56        |
| <b>30</b>   | <b>Electrical Components for internal-tanks configuration .....</b> | <b>30-1</b>  |
| <b>30.1</b> | <b>Safety .....</b>                                                 | <b>30-1</b>  |
| <b>30.2</b> | <b>Overview .....</b>                                               | <b>30-2</b>  |
| <b>30.3</b> | <b>Replacement and repair .....</b>                                 | <b>30-3</b>  |
| 30.3.1      | Replacement of the Liquid Support PCBA .....                        | 30-3         |
| 30.3.2      | Replacement of the Pipettor CU PCBA .....                           | 30-5         |
| 30.3.3      | Replacement of PWR CAN EXT fuses .....                              | 30-6         |
| 30.3.4      | Replacement of the PWR CAN EXT. CU PCBA .....                       | 30-9         |
| 30.3.5      | Replacement of the COP EUS2CAN PCBA .....                           | 30-10        |
| 30.3.6      | Replacement of Fuses of the Main Switch .....                       | 30-12        |
| 30.3.7      | Replacement of the User Access Panel CU PCB .....                   | 30-14        |
| 30.3.8      | Replacement of the Power Supply, 24V, 350W .....                    | 30-17        |
| 30.3.9      | <b>Replacement of the Remote Start Button PCB .....</b>             | <b>30-20</b> |
| <b>30.4</b> | <b>Electric connections, switches and signals .....</b>             | <b>30-21</b> |
| 30.4.1      | PWR CAN EXT. CU PCB .....                                           | 30-21        |
| 30.4.2      | Pipettor CU PCB .....                                               | 30-23        |
| 30.4.3      | COP EUS2CAN PCB .....                                               | 30-25        |
| 30.4.4      | Liquid Support PCB .....                                            | 30-27        |
| 30.4.5      | User access panel CU PCBA .....                                     | 30-28        |
| 30.4.6      | Remote Start Button PCB .....                                       | 30-29        |

|             |                                                                            |              |
|-------------|----------------------------------------------------------------------------|--------------|
| <b>31</b>   | <b>Electrical Components for external-tanks configuration .....</b>        | <b>31-1</b>  |
| <b>31.1</b> | <b>Safety .....</b>                                                        | <b>31-1</b>  |
| <b>31.2</b> | <b>Overview .....</b>                                                      | <b>31-2</b>  |
| <b>31.3</b> | <b>Replacement and repair .....</b>                                        | <b>31-3</b>  |
| 31.3.1      | Replacement of the Liquid Support PCBA .....                               | 31-3         |
| 31.3.2      | Replacement of the Pipettor CU PCBA .....                                  | 31-5         |
| 31.3.3      | Replacement of PWR CAN EXT fuses .....                                     | 31-6         |
| 31.3.4      | Replacement of the PWR CAN EXT. CU PCBA .....                              | 31-9         |
| 31.3.5      | Replacement of the COP EUS2CAN PCBA .....                                  | 31-10        |
| 31.3.6      | Replacement of Fuses of the Main Switch .....                              | 31-12        |
| 31.3.7      | Replacement of the User Access Panel CU PCB .....                          | 31-14        |
| 31.3.8      | Replacement of the Power Supply, 24V, 350W .....                           | 31-17        |
| 31.3.9      | <b>Replacement of the Remote Start Button PCB .....</b>                    | <b>31-20</b> |
| <b>31.4</b> | <b>Electric connections, switches and signals .....</b>                    | <b>31-21</b> |
| 31.4.1      | PWR CAN EXT. CU PCB .....                                                  | 31-21        |
| 31.4.2      | Pipettor CU PCB .....                                                      | 31-23        |
| 31.4.3      | COP EUS2CAN PCB .....                                                      | 31-25        |
| 31.4.4      | Liquid Support PCB .....                                                   | 31-27        |
| 31.4.5      | User access panel CU PCBA .....                                            | 31-28        |
| 31.4.6      | Remote Start Button PCB .....                                              | 31-29        |
| <b>32</b>   | <b>Software and Firmware Topics .....</b>                                  | <b>32-1</b>  |
| <b>32.1</b> | <b>Safety .....</b>                                                        | <b>32-1</b>  |
| <b>32.2</b> | <b>LIAISON® XS PC System .....</b>                                         | <b>32-2</b>  |
| 32.2.1      | Handheld Barcode Reader configuration .....                                | 32-2         |
| 32.2.2      | Touchscreen calibration .....                                              | 32-3         |
| <b>32.3</b> | <b>LIAISON® XS Software .....</b>                                          | <b>32-4</b>  |
| 32.3.1      | Recover after HDD loss (Manufacturer Default - all data lost) .....        | 32-4         |
| 32.3.2      | Change system Date and Time on Operating System .....                      | 32-5         |
| 32.3.3      | Software packages .....                                                    | 32-5         |
| 32.3.4      | Software and Firmware Installation/Update .....                            | 32-6         |
| 32.3.5      | Software structure .....                                                   | 32-7         |
| 32.3.6      | Files and information required to perform troubleshooting activities ..... | 32-9         |
| <b>32.4</b> | <b>LIAISON® XS Antivirus .....</b>                                         | <b>32-10</b> |
| 32.4.1      | Antivirus Installation .....                                               | 32-10        |
| 32.4.2      | Antivirus Update .....                                                     | 32-10        |
| 32.4.3      | Antivirus functioning and Virus detection .....                            | 32-11        |
| 32.4.4      | Antivirus management .....                                                 | 32-12        |
| <b>32.5</b> | <b>Software FSE functionalities .....</b>                                  | <b>32-13</b> |
| 32.5.1      | Main UI FSE buttons .....                                                  | 32-14        |

---

|             |                                                                 |              |
|-------------|-----------------------------------------------------------------|--------------|
| 32.5.2      | LIS Details .....                                               | 32-16        |
| 32.5.3      | Sub Category System - Deleted .....                             | 32-19        |
| 32.5.4      | Sub Category System - Service.....                              | 32-20        |
| 32.5.5      | Sub Category Diagnostic - Options .....                         | 32-23        |
| <b>32.6</b> | <b>LIS connection .....</b>                                     | <b>32-25</b> |
| 32.6.1      | LIS configuration.....                                          | 32-26        |
| 32.6.2      | Connection scheme .....                                         | 32-26        |
| 32.6.3      | Troubleshooting .....                                           | 32-27        |
| <b>32.7</b> | <b>Remote access tool .....</b>                                 | <b>32-27</b> |
| 32.7.1      | Remote access tool setup .....                                  | 32-27        |
| 32.7.2      | Correct usage of the remote access tool and restrictions.....   | 32-27        |
| <b>33</b>   | <b>Service Check List for internal-tanks configuration.....</b> | <b>33-1</b>  |
| <b>33.1</b> | <b>LIAISON® XS Installation Qualification (IQ).....</b>         | <b>33-1</b>  |
| <b>33.2</b> | <b>LIAISON® XS Operational Qualification (OQ).....</b>          | <b>33-4</b>  |
| <b>33.3</b> | <b>LIAISON® XS Uninstallation .....</b>                         | <b>33-7</b>  |
| <b>33.4</b> | <b>LIAISON® XS Yearly Maintenance .....</b>                     | <b>33-10</b> |
| <b>34</b>   | <b>Service Check List for external-tanks configuration.....</b> | <b>34-1</b>  |
| <b>34.1</b> | <b>LIAISON® XS Installation Qualification (IQ).....</b>         | <b>34-1</b>  |
| <b>34.2</b> | <b>LIAISON® XS Operational Qualification (OQ).....</b>          | <b>34-5</b>  |
| <b>34.3</b> | <b>LIAISON® XS Uninstallation .....</b>                         | <b>34-8</b>  |
| <b>34.4</b> | <b>LIAISON® XS Yearly Maintenance .....</b>                     | <b>34-11</b> |
| <b>35</b>   | <b>Technical data for internal-tanks configuration.....</b>     | <b>35-1</b>  |
| <b>35.1</b> | <b>Power requirements .....</b>                                 | <b>35-1</b>  |
| <b>35.2</b> | <b>Laser barcode scanner .....</b>                              | <b>35-2</b>  |
| 35.2.1      | Handheld barcode reader .....                                   | 35-2         |

---



|              |                                                             |             |
|--------------|-------------------------------------------------------------|-------------|
| <b>35.3</b>  | <b>Computer and connections .....</b>                       | <b>35-2</b> |
| <b>35.4</b>  | <b>Dimensions and weight.....</b>                           | <b>35-3</b> |
| <b>35.5</b>  | <b>System clearance .....</b>                               | <b>35-4</b> |
| <b>35.6</b>  | <b>Environmental conditions.....</b>                        | <b>35-5</b> |
| <b>35.7</b>  | <b>Noise emission .....</b>                                 | <b>35-5</b> |
| <b>35.8</b>  | <b>Temperature Ranges .....</b>                             | <b>35-5</b> |
| <b>35.9</b>  | <b>Water quality .....</b>                                  | <b>35-6</b> |
| <b>35.10</b> | <b>Packaging.....</b>                                       | <b>35-7</b> |
| <b>36</b>    | <b>Technical data for external-tanks configuration.....</b> | <b>36-1</b> |
| <b>36.1</b>  | <b>Power requirements .....</b>                             | <b>36-1</b> |
| <b>36.2</b>  | <b>Laser barcode scanner .....</b>                          | <b>36-2</b> |
| 36.2.1       | Handheld barcode reader .....                               | 36-2        |
| <b>36.3</b>  | <b>Computer and connections .....</b>                       | <b>36-2</b> |
| <b>36.4</b>  | <b>Dimensions and weight.....</b>                           | <b>36-3</b> |
| <b>36.5</b>  | <b>System clearance .....</b>                               | <b>36-4</b> |
| <b>36.6</b>  | <b>Environmental conditions.....</b>                        | <b>36-6</b> |
| <b>36.7</b>  | <b>Noise emission .....</b>                                 | <b>36-6</b> |
| <b>36.8</b>  | <b>Temperature Ranges .....</b>                             | <b>36-6</b> |
| <b>36.9</b>  | <b>Water quality .....</b>                                  | <b>36-7</b> |
| <b>36.10</b> | <b>Packaging.....</b>                                       | <b>36-8</b> |
| <b>37</b>    | <b>Packing List for internal-tanks configuration .....</b>  | <b>37-1</b> |
| <b>37.1</b>  | <b>Packing list.....</b>                                    | <b>37-1</b> |
| 37.1.1       | Instrument cardboard box.....                               | 37-1        |
| 37.1.2       | Accessories cardboard box .....                             | 37-1        |
| 37.1.3       | Cover cardboard box .....                                   | 37-1        |
| 37.1.4       | Small accessories cardboard box.....                        | 37-1        |
| 37.1.5       | Panel PC cardboard box.....                                 | 37-2        |
| 37.1.6       | <b>LIAISON® XS</b> accessories set .....                    | 37-2        |

|              |                                                              |              |
|--------------|--------------------------------------------------------------|--------------|
| <b>38</b>    | <b>Packing List for external-tanks configuration.....</b>    | <b>38-1</b>  |
| <b>38.1</b>  | <b>Packing list.....</b>                                     | <b>38-1</b>  |
| 38.1.1       | Instrument cardboard box.....                                | 38-1         |
| 38.1.2       | Accessories cardboard box .....                              | 38-1         |
| 38.1.3       | Cover cardboard box .....                                    | 38-1         |
| 38.1.4       | Small accessories cardboard box.....                         | 38-1         |
| 38.1.5       | Panel PC cardboard box.....                                  | 38-2         |
| 38.1.6       | <b>LIAISON®</b> XS accessories set .....                     | 38-2         |
| <b>39</b>    | <b>LIS Interface Protocol.....</b>                           | <b>39-1</b>  |
| <b>39.1</b>  | <b>Purpose .....</b>                                         | <b>39-1</b>  |
| <b>39.2</b>  | <b>Related documents.....</b>                                | <b>39-1</b>  |
| <b>39.3</b>  | <b>Overview.....</b>                                         | <b>39-1</b>  |
| <b>39.4</b>  | <b>General Information and Specification.....</b>            | <b>39-1</b>  |
| 39.4.1       | Transfers initiated from <b>LIAISON®</b> XS .....            | 39-2         |
| 39.4.2       | LIS initiated transfers .....                                | 39-2         |
| <b>39.5</b>  | <b>Limitations.....</b>                                      | <b>39-3</b>  |
| 39.5.1       | Compatibility with LXL .....                                 | 39-3         |
| <b>39.6</b>  | <b>Result transmission configuration .....</b>               | <b>39-4</b>  |
| <b>39.7</b>  | <b>Logical Structure of the Message Level Protocol .....</b> | <b>39-4</b>  |
| <b>39.8</b>  | <b>Incoming transmission examples .....</b>                  | <b>39-5</b>  |
| 39.8.1       | Test orders.....                                             | 39-5         |
| 39.8.2       | Test Results.....                                            | 39-9         |
| <b>39.9</b>  | <b>Data Record Usage.....</b>                                | <b>39-12</b> |
| 39.9.1       | Message Header Record.....                                   | 39-12        |
| 39.9.2       | Patient Information Record .....                             | 39-13        |
| 39.9.3       | Test Order Record .....                                      | 39-17        |
| 39.9.4       | Result Record .....                                          | 39-22        |
| 39.9.5       | Result Comment Record .....                                  | 39-27        |
| 39.9.6       | Query Record .....                                           | 39-30        |
| 39.9.7       | Message Terminator Record .....                              | 39-32        |
| 39.9.8       | Scientific Record.....                                       | 39-33        |
| 39.9.9       | Manufacturer Information Record .....                        | 39-33        |
| <b>39.10</b> | <b>Tools for Troubleshooting.....</b>                        | <b>39-35</b> |
| 39.10.1      | ASTM 1381 Layer.....                                         | 39-35        |
| 39.10.2      | ASTM 1394 Layer.....                                         | 39-35        |

---

|              |                                                                                               |              |
|--------------|-----------------------------------------------------------------------------------------------|--------------|
| <b>39.11</b> | <b>Contention Case .....</b>                                                                  | <b>39-35</b> |
| <b>39.12</b> | <b>Differences between LIAISON, LIAISON® XL and LIAISON® XS configuration parameters39-37</b> |              |
| <b>40</b>    | <b>Glossary.....</b>                                                                          | <b>40-1</b>  |

---



# 1 Introduction for internal-tanks configuration

This manual was created to provide all technical instructions to allow a correct installation, maintenance, repair and uninstallation of the **LIAISON<sup>®</sup> XS** Analyzer.

This document is intended for Field Service Engineers (FSEs) for training and performing field service. FSEs shall follow all the instructions included in this Service Manual together with those provided in the User Manual.

## 1.1 Description

The system consists of a robotic instrument, a Panel-PC and a software that runs on the Panel-PC. The software provides the user-interface that gives the operator the ability to control the system, and the software performs data reduction on the measurement results (measured by the instrument).

The instrument contains a transport system for plastic cuvettes, a pipetting system with two pipetting arms (one with a steel needle and one with disposable tips), an incubator, a washer and a luminescence measurement chamber. In addition to that, the instrument contains modules for loading samples, reagents, tips and cuvettes into the instrument.

PC and instrument are connected together to exchange data.

### 1.1.1 Intended use

The **LIAISON<sup>®</sup> XS** is a fully automated chemiluminescence analyzer performing the complete sample processing (sample pre-dilutions, sample and reagent dispensing, incubations, wash processes, cuvette transports) as well as the measurement and evaluation. The instrument is controlled via the Windows PC **LIAISON<sup>®</sup> XS** software. This software, which was specifically designed for this purpose, allows the user to process the pre-defined assays. The clear structure with intuitive user-guidance allows simple and quick operation of daily routine jobs.

---

**NOTE**

**LIAISON<sup>®</sup> XS** system was developed and tested according to the regulations of the IvD Directive. It is strictly intended for professional In-vitro-Diagnostic use. It has to be used only with Chemiluminescence Immunoassays, authorized by DiaSorin S.p.A. for the **LIAISON<sup>®</sup> XS** Analyzer.

---

Any changes to the instrument that are not authorized by the manufacturer will lead to the loss of the conformity to the applicable regulations the manufacturer has declared. In this case, the customer is responsible for the fulfillment of the applicable regulations.

## 1.2 Typographical conventions

The warnings, notes and symbols described hereafter are used in the current manual, on the instrument and on its packaging.

### 1.2.1 Display of warnings and notes



---

**The signal word “Danger” and a related symbol point to imminent dangers.**

The non-observance of a danger warning is imminent and will probably result in death or at least serious irreversible injuries. A damage of the system or an adverse effect on the system function cannot be excluded.

---



---

**The signal word “Warning” and a related symbol point to potential dangers.**

The non-observance of a warning is potential and can result in death or at least serious irreversible injuries. A damage of the system or an adverse effect on the system function cannot be excluded.

---



---

**The signal word “Caution” and a related symbol point to potential dangers/problems.**

The non-observance of safety instructions can result in minor injuries. A damage of the system or an adverse effect on the system function cannot be excluded.

---



---

**The signal word “Caution” points to potential problems.**

The non-observance of a safety instruction can result in damage of the system or an adverse effect on the system function.

---



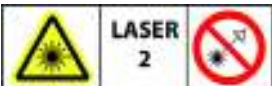
---

**The signal word “Note” points to potential problems.**

The non-observance of notes can result in an adverse effect on the system function (result deterioration).

---

## 1.2.2 Used warning symbols

|                                                                                     |                                                                                            |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|    | Caution, risk of danger to person or damage to equipment!<br>Consult instructions for use! |
|    | Biohazard!                                                                                 |
|    | Electrical hazard!                                                                         |
|    | Laser hazard!                                                                              |
|    | Laser hazard and laser information label                                                   |
|  | Caution, hot surface!                                                                      |
|  | Crush hazard!                                                                              |
|  | Magnetic field                                                                             |
|  | Cut injury hazard!                                                                         |
|  | No heavy load                                                                              |
|  | Disconnect main power connector before servicing!                                          |

## 1. Introduction for internal-tanks configuration

|                                                                                     |                                                   |
|-------------------------------------------------------------------------------------|---------------------------------------------------|
|    | Disconnect main power connector before servicing! |
|    | Power Supply characteristics                      |
|    | Grounding label                                   |
|    | LAN connection                                    |
|    | USB label                                         |
|    | Waste                                             |
|   | Shockwatch label                                  |
|  | Tiltwatch label                                   |
|  | DI Water Tank                                     |
|  | Wash Buffer Tank                                  |
|  | Liquid Waste Tank                                 |
|  | Cleaning Tank                                     |
|  | Main Switch                                       |
|  | Don't step on                                     |



Table 1–1: Warning symbols

## 1.2.3 Other symbols

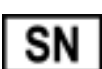
|                                                                                     |                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Manufactured by                                                                                                                                                                                                                                                                                |
|    | Manufactured date                                                                                                                                                                                                                                                                              |
|    | In Vitro Diagnostic                                                                                                                                                                                                                                                                            |
|    | CE mark                                                                                                                                                                                                                                                                                        |
|    | ID number                                                                                                                                                                                                                                                                                      |
|   | Serial number                                                                                                                                                                                                                                                                                  |
|  | Restriction of certain Hazardous Substances (RoHS) in electronic equipment                                                                                                                                                                                                                     |
|  | Certification label of the Nemko GmbH & Co. KG                                                                                                                                                                                                                                                 |
|  | Consult instructions for use!                                                                                                                                                                                                                                                                  |
| <a href="http://www.diasorin.com">www.diasorin.com</a>                              | DiaSorin website                                                                                                                                                                                                                                                                               |
| EU: ☎ 00 800 60 61 62 63                                                            | Toll free number                                                                                                                                                                                                                                                                               |
|  | <b>Disposal of Electrical and Electronic Equipment</b><br>In the European Union, electrical and electronic equipment must not be disposed with other household-type waste. It must be collected separately. Please observe the relevant legal regulations effective in the respective country. |
|  | Caution                                                                                                                                                                                                                                                                                        |

Table 1–2: Other symbols

### 1.2.4 Special types

| Typology              | Description                                                               | Example                   |
|-----------------------|---------------------------------------------------------------------------|---------------------------|
| LEDs and signal lamps | LEDs (Light Emitting Diode) and signal lamps are printed in special type. | Power LED, Error LED      |
| Menu items and fields | Menu items and fields are printed in bold and italic type.                | <b><i>File</i></b> Menu   |
| Buttons               | Buttons are printed in bold type.                                         | <b>Open</b> button.       |
| Keys                  | Keys are printed in italic type.                                          | Press <b><i>Enter</i></b> |
| File examples         | File examples are printed in typewriter font.                             | DRIVER=C:\SERVICE\DRIVERS |

Table 1–3: Typology

## 1.3 Metric system

The instruments are developed and built in the metric system.

All fasteners as screws and nuts used in this instrument have metrical dimensions, therefore use only metrical tools as wrenches and screwdrivers.

Do not use screws and nuts of a non-metric system.

Exceptions:

If it is explicitly mentioned in the instructions, then the ANSI-standard parts shall be used.

## 1.4 Use of screw color codes

Red thread-locker lacquer is used:

- Red thread-locker lacquer on a screw head shall be not opened by a field service engineer, except he has all the necessary tools to adjust and re-assemble the part.

Green thread-locker lacquer is used:

- Green thread-locker lacquer on a screw head marks this one as a critical connection, but the field service engineer can open this one and lock it again with green thread-locker lacquer.

Not marked screws:

- A screw without color lacquer code is uncritical.

## 1.5 Hazard instructions

The following hazard instructions must be observed at all times, both before and during operation and during maintenance.

---

**WARNING****Handling of the Service Manual**

The Service Manual is provided for field service engineers safety and gives important instructions for the handling of the system described.

Read all instructions!

---

The **LIAISON® XS** is designed and manufactured in accordance with the safety requirements for electronic and medical systems. It is the operators responsibility to comply with local and national laws, regulations and laboratory procedures for installation and operation of the instrument.

### 1.5.1 General safety

---

**WARNING****Non-Observance of Hazard Instructions**

The non-observance of hazard instructions can result in serious personal injuries and material damages.

- Follow all hazard instructions included in this manual.
  - Follow all warnings marked on the instrument.
- 

---

**WARNING****Function of the Instrument**

Please follow closely the steps contained in the individual instructions to achieve a correct functioning of the instrument.

---

## 1. Introduction for internal-tanks configuration

---

**WARNING****Use of the System according to Intended Use only**

Improper use of the system can produce erroneous results, produce damage to the system and cause personal injuries.

- The handling and maintenance of the system must only be performed by trained and authorized personnel.
- Before servicing the system, the Instruction for Use Manual and the Service Manual must be completely read and understood.
- The instrument must only be used in accordance with the intended use as described in this manual.
- Use only the consumables and accessories specifically produced for the instrument (e.g. disposable tips, cuvettes, integrals).
- For maintaining safety, do not modify the **LIAISON® XS** Diagnostic System, do not use parts other than the specified ones and do not remove the safety devices, unless strictly necessary.

The manufacturer assumes no liability for any damages, including those to third parties, caused by improper use or handling of the system.

---

**WARNING****Danger due to Overheating**

Improper place of installation/operation of the system may cause fire or serious system damage in case of overheating.

- Select the place of installation of the system so that the ventilation slots are not blocked or covered.
  - Select the place of installation of the system so that air can circulate.
  - Never cover ventilation slots.
- 

### 1.5.2 Electrical hazards

The **LIAISON® XS** Diagnostic System does not pose uncommon electrical hazards to operators if it is installed and operated without alteration and is connected to a power source that meets required specifications, see Chapter 35.1 on page 35-1.

---

**DANGER****Electrocution/Fire Hazard!**

Non-observance of rules and regulations can cause serious personal injuries with deadly consequences and material damages.

- National rules and legal regulations for the safe electrical operation of the system must be observed.
- 

**DANGER****Electrocution/Fire Hazard!**

The wrong operation voltage can cause serious personal injuries with deadly consequences and material damages (e.g. fire).

- Before connecting the system to the mains supply, check the operating voltage is correct.
  - The instrument must operate with the type of power source indicated on the dedicated label positioned near the switch. If there are doubts regarding the type of power available, consult an authorized technical person of the laboratory or the local electric power company.
-

**⚠ DANGER****Electrocution/Fire Hazard!**

Improper connection of the system and the peripheral devices to mains supply can cause serious personal injuries with deadly consequences and material damages (e.g. fire).

- Use only connection and extension cables with a protective earth conductor and sufficient capacity (performance, power) to connect the system and the peripheral devices to the mains supply.
  - Use a properly grounded electrical outlet of correct voltage and current handling capability.
  - Never interrupt the grounding contacts.
  - Grounding of the system and its peripheral devices to the same protective earth potential must be ensured.
  - The use of a power strip shall be limited to DiaSorin approved configurations, see Chapter 35.1 on page 35-1.
  - Where appropriate, the Analyzer is installed with additional external switch for residual current circuit-breakers with over-current protection.
  - Where appropriate, the Analyzer is integrated in UPS device.
- 

**⚠ DANGER****Electrocution/Fire Hazard!**

Damaged connecting cables can cause serious personal injuries with deadly consequences and material damages (e.g. fire).

- Damaged connecting cables must be replaced immediately!
  - No objects may be placed on the connecting cables.
  - Connecting cables must be laid so that they cannot be squeezed or damaged.
  - Connecting cables must be laid so that they do not lay in accessible or drivable areas.
- 

**⚠ DANGER****High Voltage!**

The touching of high voltage parts or cables can result in serious personal injuries with deadly consequences.

- Never touch high voltage parts with instrument on.
  - Separate the system from the mains supply before maintaining or cleaning it.
  - Be careful! The following parts can be under high voltage during operation:
    - UPS device (100-240V, high current)
    - Main supply unit (100-240V, high current)
    - Enclosure metal part in Single Fault conditions (132V r.m.s., low current)
    - Photomultiplier (PMT) (~ 1000 Volts, low current)
- 

**⚠ DANGER****Electrocution/Fire Hazard!**

Defective systems can result in serious injuries with deadly consequences and material damages (e.g. fire).

- Separate immediately the defective system from the mains supply, if a safe usage is no longer possible.
  - Secure the defective system against reconnection.
  - Label the defective system clearly as being defective.
-

## 1. Introduction for internal-tanks configuration

---

** DANGER****Electrocution!**

Working with electrical devices on wet floors can cause serious injuries with deadly consequences and material damages due to electrocution.

- Surfaces (e.g. floors) must be dry when working with the system.
  - Keep liquids away from all connectors of electrical or communication components.
  - Do not touch any switches or outlets with wet hands.
- 

** DANGER****Danger of Electrocution**

If the system is not separated from the mains supply before maintenance works, this can cause serious injuries with deadly consequences due to electrocution. Additionally, there is the danger that the system starts and causes injuries (e.g. contusion, cuts etc.) to the person working with the system.

- Determine and correct the cause of a blown fuse or thrown circuit-breaker before attempting to resume operations of the system.
  - Separate the system from the mains supply and secure it against restarting before replacing fuses.
  - Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.
  - Never repair or bridge blown fuses.
  - Never short-circuit the fuse holder.
- 

** DANGER****Danger of Electrocution or Mechanical Injuries**

If the system is not separated from the mains supply before maintenance works, this can cause serious injuries with deadly consequences due to electrocution. Additionally, there is the danger that the system starts and causes injuries (e.g. contusion, cuts etc.) to the person working with the system.

- Switch off the system, separate it from the mains supply and protect it against restarting. When system is secured start cleaning, disinfection, maintenance or repair works.
  - Make sure that nobody works on the system and that all covers are attached and closed before reconnecting the system to the mains supply.
  - Do not disconnect any electrical connection while the power is on.
  - Disconnect the power cord before cleaning major liquid spills. Clean spilled fluids immediately.
-

**WARNING****Danger due to Improper Place of Installation**

Improper place of installation of the system can cause accidents with serious injuries with deadly consequences, fire or serious system damages, because the system cannot be switched off or be separated from the mains supply.

- Ensure that the power supply and mains switch/ emergency stop are easily accessible in the place of installation of the system.
  - Ensure UPS proper ventilation.
  - The Analyzer is not designed and constructed to be operated in a potentially explosive atmosphere, so it must not be installed and used in a laboratory with potentially explosive atmosphere. Field Service Engineer must verify with the end user that the place chosen for installing th Analyzer fulfils such requirements. The responsibility of such check is in charge of the end user.
-

### 1.5.3 Physical Hazards

#### 1.5.3.1 Laser Hazards

**WARNING**

---

**Eye Injuries due to Laser Radiation**

Laser radiation causes eye irritations when the laser beam is looked for a long period of time.

- Never look directly into the laser beam!
- Do not introduce optical devices into the system.
- Remove watches and reflective jewellery before operating the laser.
- Caution during operation and testing of the laser (barcode scanner and hand held barcode reader) must be taken due to laser class (class 2).
- The barcode reader is positioned on the right side of the Sample loading bay. The laser beam exits from an opening positioned on the internal right side of the Sample loading bay. The laser beam is orthogonal to the internal right side surface.

Wrong usage of operating elements or adjustments or the non-observance of processes can cause a dangerous emission of laser radiation.

---

#### 1.5.3.2 Heavy Objects

Follow supplied instructions for a safe unpacking and placement of the Analyzer during Installation, see Chapter 5 on page 5-1, as well as Uninstallation, see Chapter 7 on page 7-1.

Use proper lifting techniques when handling full tanks (DI water, Wash Buffer solution and waste). Use care when handling the containers to reduce the risk of injury.

**WARNING**

---

**High Weight**

Improper lifting/carrying of the system may cause the system to fall down. Falling down may cause serious injuries to the personnel or serious material damages.

- Observe the technical specifications of the system to avoid misjudgments.
- 

**WARNING**

---

**Never lift the system by yourself!**

Manual handling of heavy loads may cause injuries.

The **LIAISON<sup>®</sup> XS** instrument shall NOT be lifted and carried by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

---

### 1.5.4 Trip Hazards

The **LIAISON<sup>®</sup> XS** Diagnostic System is equipped with power cord and various connectors. To avoid tripping hazards, ensure cords are properly stowed.



**NOTE**

---

In the event of a damage, prejudice or personal injury occurred in connection with the use of the **LIAISON® XS** Diagnostic System, the Field Service Engineer shall immediately contact the local DiaSorin Representative.

---

### **1.5.5 Mechanical Hazards**

The **LIAISON® XS** Diagnostic System is an automated system that operates under computer control. As with most automated equipment, there is potential for injury and bodily harm from moving mechanical components whenever the system is in operation.

The **LIAISON® XS** Diagnostic System minimizes mechanical hazards by providing guards to protect against accidental contact with moving components, and encoding the software with safety features.

During servicing of the Analyzer, a Field Service Engineer is potentially exposed to the following moving mechanical components:

- Pipettor arms and probes
- Cuvette gripper arm
- Incubator
- Washer
- Reader

Basic elements of mechanical safety include, but are not limited to:

- Wear protective gloves during cleaning, maintenance or service in general.
- Avoid contact with the sharp metal edges.
- Never bypass or override a safety device.
- Never allow any part of your body to enter a range of mechanical movement during system operation.
- Do not wear articles of clothing or accessories that could catch on the system.
- Keep pockets free of items that could fall into the system.
- The right pipettor probe, the washer probes and the reader probe are sharp and potentially contaminated with infectious material. Adopt all safety precautions required when servicing such areas.
- Use caution when performing cleaning, maintenance or service in general.
- Be aware that, in the event of a system malfunction or an unexpected sequence of movements, reflex actions could occur, causing injury.

### 1.5.6 Biological Safety

Performing the following activities, Field Service Engineers may be exposed to potentially infectious materials:

- Handling samples, reagents, calibrators, and controls;
- Cleaning spills;
- Handling and disposing of waste;
- Performing maintenance procedures;
- Performing cleaning procedures.

During servicing it is important to pay particular attention to the potentially contaminated parts listed below:

- Liquid waste tank and liquid waste basin;
- Solid waste bin and solid waste box;
- Liquid waste tubes (from pumps to waste tank);
- Left arm edge;
- Right arm needle;
- Right arm washing station;
- Washer and reader aspiration needles and tubes;
- Washer Aspiration Pump (WAP) tubes;
- Liquid Waste Pump (LWP) tubes;
- Incubator cover (especially near dispensing points);
- Incubator rotor and closer area;
- Washer and reader assemblies (especially internal parts);
- Sample area, sample and control racks (in case of spillage);
- Reagent loading area (in case of spillage);
- Reagent cover (in case of spillage);
- Any surfaces in case of spillage.

The following information help the technicians in minimizing the impact of the exposure.



Field Service Engineers should consider all clinical samples, reagents, calibrators, controls, and used reaction vessels and consumables (e.g. tips) that contain human-sourced material as potentially infectious. They should also consider all system surfaces or components that have come in contact with human-sourced material as potentially infectious. No known test method can offer complete assurance that products derived from human-sourced material will not transmit infection. Therefore, all products derived from human-sourced materials and system components exposed to human-sourced materials should be considered potentially infectious.

---

## 1. Introduction for internal-tanks configuration

---

**⚠ DANGER****Risk of infection!**

Parts, in contact to potentially infectious liquid, must be treated as potentially infectious. Improper handling of infectious parts can cause skin irritations, illnesses and possible death.

- Observe local and national provisions, legislation and laboratory regulations.
  - Use appropriate disposable gloves.
  - Use an appropriate lab coat.
  - Use an appropriate eye protection.
  - Avoid contact between skin/mucous membrane and samples/test reagents or parts of the instrument.
  - Do not pipette by mouth.
  - Do not eat, drink, smoke, apply cosmetics or handle contact lenses when handling human-sourced material or contaminated system components.
  - Clean and disinfect the system immediately if potentially infectious material has been spilled.
  - Any reagent spills should be washed with a chemical use sodium hypochlorite solution with 0.5% active chlorine (e.g. dilution 1:10 of a solution at 5% active chlorine) and disposed as though potentially infectious.
  - All samples, biological reagents and materials used in the assays must be considered potentially able to transmit infectious agents. They should therefore be disposed in accordance with the prevailing regulations and guidelines of the agencies holding jurisdiction over the laboratory, and the regulations of each Country.
  - Liquid waste must be decontaminated with a chemical use sodium hypochlorite solution with 0.1% active chlorine (e.g. dilution 1:50 of a solution at 5% active chlorine) for at least half a hour.
  - Do not use broken or chipped tubes or bottles.
  - Observe the instructions in the package inserts for a correct use of the reagents.
- 

**CAUTION****Organic solvents**

Reagent containers and hoses for system liquid and waste can be seriously damaged by organic solvents and become unusable.

- Never use organic solvents.
- 

**WARNING****Handling of Cleaning/ Disinfection Products**

Pay attention when handling cleaning/ disinfection products, because they are harmful as indicated on the bottle.

- Read the instructions of the products before use.
  - Do not use products undiluted!
-

### 1.5.7 Chemical Hazards

Field Service Engineer may be exposed to hazardous chemicals when handling reagents, calibrators and controls.

Exposure to hazardous chemicals is minimized by following instructions provided in the assay-specific documentation (such as a package insert), on product-specific labels and product-specific MSDS (Material Safety Data Sheets).

In general, observe the following precautions when handling chemicals:

- Wear suitable protective gloves.
- Consult MSDS for safe use instructions and precautions.
- Avoid contact with skin and eyes. If contact with material is anticipated, wear impervious gloves, protective eye wear, and clothing.
- Maintain good housekeeping. Do not eat, drink, or store food and beverages in areas where chemicals are used.
- Seek medical attention if irritation or signs of toxicity occur after exposure.

### 1.5.8 Spill clean-up

Clean spills in accordance with established biosafety practices and follow instructions in the MSDS (Material Safety Data Sheets). In general, safe work practices for cleaning spills include:

- Wear appropriate personal protective equipment, such as gloves, eye wear and lab coat.
- Absorb the spill with absorbent material.
- Wipe the spill area with detergent solution.
- Wipe the area clean with an appropriate disinfectant such as a chemical use sodium hypochlorite solution with 0.5% active chlorine (e.g. dilution 1:10 of a solution at 5% active chlorine).

### 1.5.9 Maintenance Work

---

**WARNING**

---

**Unapproved Maintenance Works**

Unapproved or inaccurately performed maintenance works can result in serious personal injuries and material damages.

- Only perform maintenance procedures described in this manual.
  - Follow closely the steps contained in the individual instructions.
  - For the maintenance works, only use original parts approved by DiaSorin.
  - Tests and maintenance works specified by the manufacturer must be performed to ensure the safe operation of the system and the proper functioning of the system.
  - All service and maintenance works which are not described in this instruction must be performed by qualified and authorized service technicians.
  - Any changes made to the instrument that are not authorized by the manufacturer will lead to the loss of guarantee.
-

## 1. Introduction for internal-tanks configuration

---

**WARNING****Danger of Electrocution due to not correct replacement of instrument parts**

Not correct replacement of instrument parts may cause spillage of liquid on PCB or electrical cables.

- Only perform maintenance procedures described in this manual.
  - Follow closely the steps contained in the individual instructions.
  - For the maintenance works, only use original parts approved by DiaSorin.
  - Tests and maintenance works specified by the manufacturer must be performed to ensure the safe operation of the system and the proper functioning of the system.
  - All service and maintenance works which are not described in this instruction must be performed by qualified and authorized service technicians.
  - Any changes made to the instrument that are not authorized by the manufacturer will lead to the loss of guarantee.
- 

**WARNING****Danger of Electrocution or Mechanical Injuries during Required Maintenance Works on the Voltage-Carrying System**

If the system cannot be separated from the mains supply during required maintenance works, additional precautions must be taken to avoid serious injuries with deadly consequences due to electrocution or injuries by the system (e.g. contusion, cuts etc.) due to electrocution.

- Only switch on the system if this is explicitly required for the minimum necessary time only.
  - Perform the maintenance works with highest caution.
  - Never touch electrical connecting contacts.
  - Please note that mechanical components (e.g. pipettor) can move unexpectedly.
  - Assure other personnel shall not access the instrument during the service activities and do not leave the instrument unattended.
- 

**WARNING****Danger of Electrocution or Mechanical Injuries by Missing or Opened Protective Covers**

To avoid serious injuries with deadly consequences due to electrocution or injuries by the system (e.g. contusion, cuts etc.), protective covers may only be opened or removed for certain maintenance procedures and with the highest level of caution.

- Only perform maintenance procedures described in this manual.
  - Make sure that nobody works on the system and that all covers are attached and closed before reconnecting the system to the mains supply.
  - Make sure that all covers are attached before switching on the instrument.
  - Switch off the system, separate it from the mains supply and protect the system against restarting, if protective covers/gears are missing.
  - Make sure that the pipettors have stopped before opening covers and/or accessing the working area of the instrument.
  - Avoid touching the pipettors and other moving parts while the system is in operation.
  - Perform all maintenance procedures with the highest level of caution.
  - Assure other personnel shall not access the instrument during the service activities and do not leave the instrument unattended.
-

**WARNING****Sharp Edges!**

Sheet metal parts and PCBs located behind protective covers might have sharp edges. Contact might lead to injuries.

- Always wear the suitable protective gloves during the service activities.
  - Use caution at corners and edges!
- 

**WARNING****Liquid in the System**

Liquid which gets into the system can cause illnesses with deadly consequences in case of contact. The system can be damaged by liquids.

- Switch off the system.
  - Separate the system from the mains supply.
  - Wear suitable protective clothing.
  - Clean, disinfect and dry the system according to the applicable local and national provisions, legislation and laboratory procedures.
  - Assure the system is completely dry before switch it on.
- 

**WARNING****Defects in the Liquid System**

Defect or leaky tubes, syringes, valves or pumps lead to deterioration of the pipetting results and consequently corruption of final results. Furthermore incorrectly flushed tips can cause mixing up of sample material.

- Check the system on drops and ponding.
  - Check tubes, syringes, valves or pumps periodically.
-

## 1. Introduction for internal-tanks configuration

---

### 1.5.10 Disposal and Disinfection

**CAUTION****Unsuitable Sterilisation Method**

Containers and components for liquids or waste are seriously damaged at autoclaving. Disinfect the containers and components with a suitable disinfection method.

---

**CAUTION****Cleaning or Disinfection**

Observe the following aspects during cleaning or disinfection because otherwise breakdowns or damages can be the result.

- Always wear the suitable protective gloves during the service activities.
  - Liquid cleaning or disinfection solutions may only be used with a moistened cleaning tissue. They may not be poured into or sprayed into the system.
  - Cleaning or disinfection solutions and procedures must be approved.
  - Cleaning or disinfection solutions must not come into contact with bearings and guides, as otherwise the gr film may dissolve!
  - Cleaning or disinfection solutions must not be used in the proximity of circuit boards, light barriers and perspex surfaces!
- 

**CAUTION****Handling/Cleaning of Optic Surfaces**

Optic surfaces (e. g. scanners, lenses, sensors) must be free of dust and fat.

- Always wear the suitable protective gloves during the service activities.
  - Do not touch any optic surfaces.
  - Only clean the optic surfaces with a softy and lint-free cloth.
  - Do not use any aggressive detergents (e.g. acetone).
- 

**WARNING****Disposal of Infectious Waste**

Potential infectious material and all parts that may come in contact with potential infectious material must be disposed according to the local and national provisions, legislation and laboratory procedures.

---

**WARNING****Disposal**

- The instrument, the packaging material and all parts that have been replaced must be disposed according to the applicable local and national provisions, legislation and laboratory procedures.

For more information about disposal, contact the manufacturer.

---



**CAUTION****Touch Screen Cleaning**

Improper cleaning could damage the touch screen surface.

- Always wear the suitable protective gloves during the service activities.
  - Use soft clothes with neutral detergent or with ethanol to clean the touch screen.
  - Do not use any chemical solvent, acidic or alkali solution.
  - Do not allow liquid from soaking into the joint of film and glass which may result in peeling or malfunctioning.
- 

**WARNING****Battery Handling**

The product may contain an internal lithium manganese dioxide, vanadium pentoxide, or alkaline battery or battery pack. There is risk of fire and burns if the battery pack is not handled properly. To reduce the risk of personal injuries:

- Do not attempt to recharge the battery.
  - Do not expose to temperatures higher than 60°C (140°F).
  - Do not disassemble, crush, puncture, short external contacts, or dispose of in fire or water.
  - Risk of explosion if battery is replaced by an incorrect type. Dispose of used batteries according to the instructions.
  - Replace only with the spare designated for this product.
-

### **1.5.11 Caution on electromagnetic wave interference**

#### **1.5.11.1 Electromagnetic wave interference given by LIAISON® XS to other equipment**

The **LIAISON® XS** instrument was designed and manufactured in compliance to the applicable EMC standards.

Use the cables attached at the installation for connection between the devices in the system. The proper use of the specified cables minimizes electromagnetic wave interference.

Installation and service of the system or changes in the installation may never be performed by non-authorized persons not approved by DiaSorin S.p.A. Movable multiple plug sockets different from the one installed by DiaSorin S.p.A. personnel are not allowed.

However, there is no guarantee that the **LIAISON® XS** instrument will not cause electromagnetic wave interference.

- a. When **LIAISON® XS** instrument may be the cause, turn off the power of the **LIAISON® XS** and check the functions of the other equipment. **LIAISON® XS**
- b. If the functioning is improved, the **LIAISON® XS** instrument probably is the cause.

#### **1.5.11.2 Electromagnetic wave interference that may be given to the LIAISON® XS instrument**

If the **LIAISON® XS** instrument is used near equipment that generates strong electric and magnetic field, noises may enter the Instrument and affect the performances and functions.

Use the cables provided at the installation for connection between the devices in the system. The proper use of the specified cables minimizes electromagnetic wave interference.

Installation and service of the system or changes in the installation shall be performed only by authorized persons approved by DiaSorin S.p.A. Movable multiple plug sockets different from the one installed by DiaSorin S.p.A. personnel are not allowed.

However, there is no guarantee that the **LIAISON® XS** instrument will not be affected by electromagnetic wave interference.

- a. When other equipment may be the cause, turn off the power of the equipment and check the functions of the **LIAISON® XS** instrument.
- b. If the functioning is improved, interference from the equipment probably is the cause.

Try the following to correct the issues.

- a. Move the **LIAISON® XS** instrument further away from the equipment that may be the cause.
- b. Connect the power cord of the **LIAISON® XS** instrument to an outlet that is on a different circuit from the equipment that may be the cause.
- c. Check that the electromagnetic wave interference does not affect the other equipment which is connected with this instrument.

**NOTE****Transient Emissions and Interference Resistance**

This instrument meets all requirements described in standard IEC 61326 on transient emissions and interference resistance.

- This instrument was developed and tested according to CISPR11 Class A. It can cause radio interference in domestic environments. In this case it may be required to take action to eliminate such interference.
  - Before setup and operation of the instrument, the electromagnetic environment shall be evaluated.
  - Do not use the instrument in the proximity of sources with excessive electromagnetic radiation (e.g. unshielded, deliberately operated high frequency sources) since they could interfere with the proper operation of the instrument.
- 

**1.5.11.3 Radio Interferences****CAUTION****Interference by Mobile Phones**

Mobile phones can affect the correct function of the instrument.

- Do not use mobile phones next to a running instrument.
- 

**Federal Communications Commission (FCC) Notice**

This equipment was tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the Instructions for Use and Service Manual, may cause harmful interferences to radio communications. Operation of this equipment in a residential area is likely to cause harmful interferences. In this case, the user will be required to correct the interferences at his own expense.

**Canadian Department of Communications Compliance Statement**

This equipment does not exceed Class A limits per radio noise emissions for digital apparatus set out in the Radio Interference Regulations of the Canadian Department of Communications. Operation in a residential area may cause unacceptable interference to radio and TV reception, requiring the owner or operator to take whatever steps are necessary to correct the interference.

**1.5.12 Static magnetic field**

Magnets are used in the instrument and clearly identified with labels (see Chapter 1.2.2 on page 1-3).

Concerning precautionary limits of exposure to magnetic fields, international standards establish 0.5 mT (5 gauss) as the magnetic flux density threshold level for static magnetic fields to prevent inadvertent harm to persons with implanted electronic medical devices and implants containing ferromagnetic material.

All magnets in the instruments are either completely within the 5 gauss line or are exceeding it to an extent which is not expected to have impact on the user during normal operations.

## 1. Introduction for internal-tanks configuration

---

**CAUTION****Handling of module labeled with magnetic field warning**

Precautions have to be taken during the handling of the module labeled with magnetic field warning (e.g. Washer module, Resuspension tool, Tanks...) since magnet could exceed the 5 gauss limit and may come in contact with the body of the FSE and potential cause interference with implanted medical devices.

---

**CAUTION****Handling of containers**

Precautions have to be taken during the handling of 5.5 liters and 1.5 liters containers since magnet of the level sensor exceeds the 5 gauss limit near the handle and may come in contact with the body of the user.

---

## 1.6 Evaluation of Residual risk for Field Service Engineer

The aim of this section is to bring to Field Service Engineer attention to the status of residual risks, despite the inherent safe design measures, safeguarding and complementary protective measures adopted by the Manufacturer to lower the risks to an acceptable level.

Field Service Engineer must read carefully the section “Safety residual risks for User” contained in **LIAISON® XS** User Manual as well as the present chapter in order to understand the residual risks and learn the recommended instructions to work and service safely on the **LIAISON® XS** Diagnostic System.

If further clarification should be needed, it is recommended to contact the local DiaSorin representative.

**WARNING**

**The FSE credentials, user name and password, must be kept confidential at all times and they should never be shared or exposed to others.**

---

**NOTE****Design of the Analyzer to facilitate handling**

The **LIAISON® XS** Diagnostic System and its component parts are capable of being handled, transported, packaged and stored safely, without damage.

The Analyzer is either wheels and/ or handles equipped to facilitate its handling during unpacking and positioning at customer site.

---

**NOTE****Protection against mechanical hazards**

The **LIAISON® XS** Diagnostic System and its components and fittings are stable enough to avoid overturning, falling or uncontrolled movements during transportation, assembly, dismantling and any other action involving the machinery.

The various parts of the Analyzer and their linkages are able to withstand the stresses to which they are subject when used.

The durability of the materials used is adequate for the nature of the working environment foreseen by DiaSorin S.p.A., in particular as regards the phenomena of fatigue, ageing, corrosion and abrasion. To date, there is no evidence of parts showed to be weak.

A remaining mechanical hazard is related to moving parts.

---

**WARNING****Handling of left top cover**

When the technician is logged in the software with a FSE access right and the “Allow open cover” checkbox is selected, the monitoring of the left top cover presence sensor is disabled, so the Analyzer does not stop moving when the cover is opened. Field Service Engineer must pay particular attention when interacting with the Analyzer for servicing and maintenance purposes.

- Left top cover shall be opened only when required for particular service activity for the minimum necessary time only.
  - Never insert hands or head inside instrument when pipettors or other instrument parts are running.
  - Move in a safe position pipettor arms before operating on internal instrument parts, as possible far away from working area.
  - Pay attention to internal potentially cutting parts (pipettor arms, right pipettor needle, sharp angles and edges).
  - Assure other personnel shall not access the instrument during the service activities and do not leave the instrument unattended.
-

## 1.7 Instrument Labelling and Safety Labels

This section provides information on the instrument labelling, with particular focus on the safety labels.

Safety labels are used on the **LIAISON® XS** system and in the documentation, to identify potentially dangerous conditions. Some safety labels are not accessible to the user but are, in some circumstances, visible to the Field Service Engineer.

### WARNING



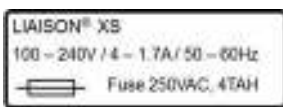
#### Missing Warnings

Missing or unreadable warning labels or type labels can result in non-identified dangers. This can cause serious personal injuries and material damages.

- Check the system on missing or unreadable warning labels and type labels.
- Missing or unreadable warning labels or type labels must be replaced.

|    |  |                                                                                                                                                                                                                                                                                                                                               |
|----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. |  | <p>Attention label:</p> <ul style="list-style-type: none"> <li>• Left Pipettor arm</li> <li>• Right Pipettor arm</li> <li>• Starter Reagent flap</li> <li>• Bulk Fluid area</li> </ul>                                                                                                                                                        |
| 2. |  | <p>Biological hazard label:</p> <ul style="list-style-type: none"> <li>• Left top cover, inside</li> <li>• Right top cover, inside</li> <li>• Sample Module flap, inside</li> <li>• Solid Waste bin, inside</li> <li>• Reagent Module flap, inside</li> <li>• Liquid Waste tank</li> <li>• Liquid Waste Basin</li> </ul>                      |
| 3. |  | <p>Laser warning pictogram label:</p> <ul style="list-style-type: none"> <li>• Sample Module</li> </ul> <p>Class 2 Laser warning information label:</p> <ul style="list-style-type: none"> <li>• Sample Module flap, inside</li> </ul> <p>Laser hazard label:</p> <ul style="list-style-type: none"> <li>• External barcode reader</li> </ul> |
| 4. |  | <p>Hot surface label:</p> <ul style="list-style-type: none"> <li>• Left Pipettor arm</li> <li>• Right Pipettor arm</li> </ul>                                                                                                                                                                                                                 |

## 1. Introduction for internal-tanks configuration

|     |                                                                                     |                                                                                                                                                                                                                                |
|-----|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.  |    | Crush hazard <ul style="list-style-type: none"> <li>On top of the Front Module, near the Sample Module</li> <li>On top of the Front Module, near the Reagent Module</li> </ul>                                                 |
| 6.  |    | Magnet label: <ul style="list-style-type: none"> <li>Washer Module</li> <li>DI Water tank</li> <li>Wash Buffer tank</li> <li>Cleaning Solution tank</li> <li>Resuspension tool</li> <li>Rear side of the instrument</li> </ul> |
| 7.  |    | No heavy load <ul style="list-style-type: none"> <li>Starter Reagent flap</li> <li>Sample Module flap</li> <li>Reagent Module flap</li> </ul>                                                                                  |
| 8.  |    | ON/OFF label <ul style="list-style-type: none"> <li>Right side of the instrument</li> </ul>                                                                                                                                    |
| 9.  |  | Disconnect main power before servicing: <ul style="list-style-type: none"> <li>Right side of the instrument, under the Right side cover</li> </ul>                                                                             |
| 10. |  | Fuse/Power Supply: <ul style="list-style-type: none"> <li>Right side of the instrument, above the main switch</li> </ul>                                                                                                       |
| 11. |  | LAN label <ul style="list-style-type: none"> <li>Right side of the instrument, above the LAN connector</li> </ul>                                                                                                              |
| 12. |  | USB label <ul style="list-style-type: none"> <li>Right side of the instrument, above the USB connector</li> </ul>                                                                                                              |
| 13. |  | Grounding label: <ul style="list-style-type: none"> <li>Bottom of the instrument, underneath of the main housing</li> </ul>                                                                                                    |
| 14. |  | Instrument Serial label (illustration only, for demonstrating label position): <ul style="list-style-type: none"> <li>Rear side of instrument</li> </ul>                                                                       |
| 15. |  | Shockwatch label: <ul style="list-style-type: none"> <li>Rear side of the instrument</li> </ul>                                                                                                                                |

## 1. Introduction for internal-tanks configuration

|     |                                                                                     |                                                                        |
|-----|-------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| 16. |    | Tiltwatch label:<br>• Rear side of the instrument                      |
| 17. |    | Temperature measuring strip<br>• Rear side of the instrument           |
| 18. |    | Wash label<br>• On top of the wash container                           |
| 19. |   | Waste label<br>• At the front of the waste container.                  |
| 20. |  | H <sub>2</sub> O label<br>• On top of the DI water container           |
| 21. |  | Cleaning solution label<br>• On top of the cleaning solution container |
| 22. |  | Don't step on<br>• Liquid Waste Basin                                  |

Table 1–4: Labels



### 1.7.1 Instrument Serial label

Instrument Serial label is positioned on:

- Rear side of the instrument



Figure 1–1: Instrument Serial label positioning

### 1.7.2 Fuse / Power supply label

Fuse / Power supply label is positioned on:

- Right side of the instrument, above the main switch

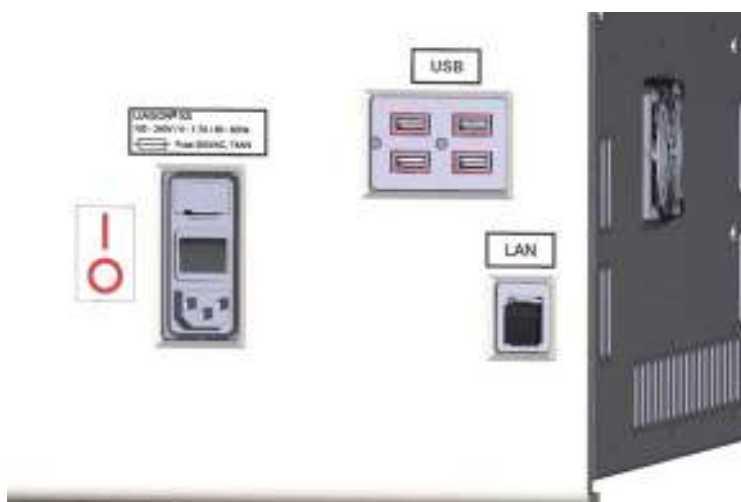
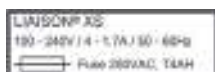


Figure 1–2: Fuse / Power supply label positioning

## 1. Introduction for internal-tanks configuration

### 1.7.3 ON/OFF label



ON/OFF label is positioned on:

- Right side of the instrument, at the left of the main switch

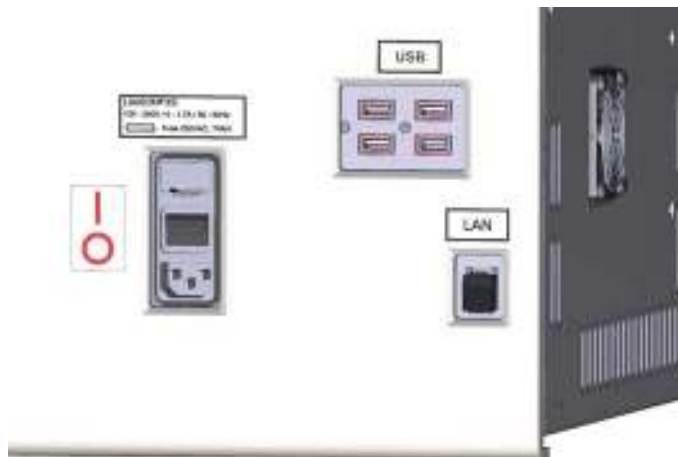


Figure 1–3: ON/OFF label positioning

### 1.7.4 USB and LAN label



USB and LAN labels are positioned on:

- Right side of the instrument,
  - USB label is positioned above the USB connectors
  - LAN label is positioned above the LAN connector

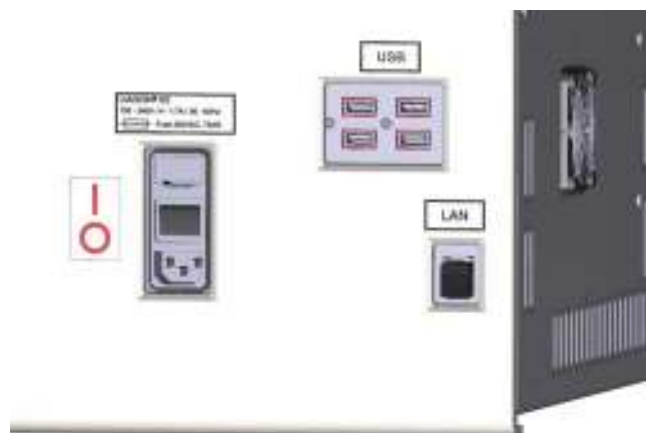


Figure 1–4: USB and LAN label positioning

### 1.7.5 Disconnect main power label

Disconnect main power label is positioned on:

- Right side of the instrument, under the Right Side cover



Figure 1-5: Disconnect main power label positioning

### 1.7.6 Ground label / Earth label



Ground label is positioned on:

- Bottom of the instrument, underneath the main housing, close to the User Access Panel PCB.

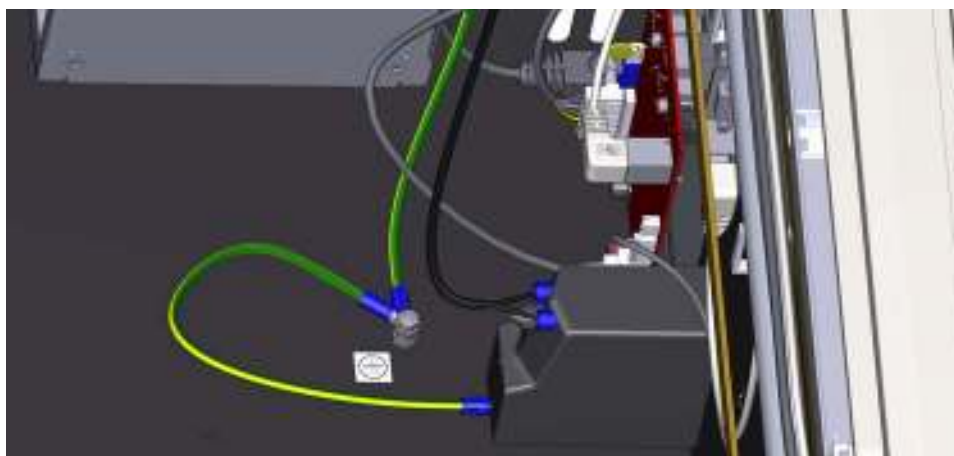


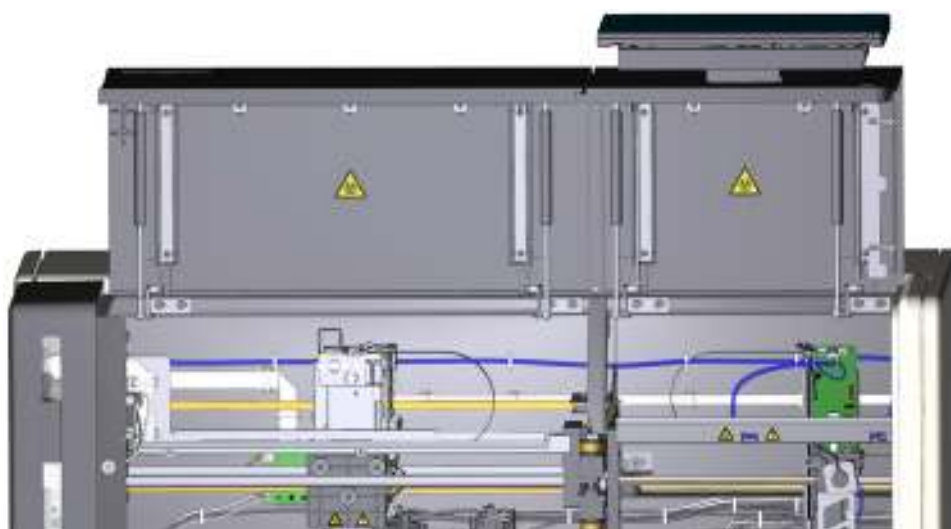
Figure 1-6: Ground label positioning

### 1.7.7 Biohazard labels (4.5 cm)



Biological hazard labels are positioned on:

- Left top cover, inside
- Right top cover, inside
- Liquid Waste tank
- Liquid Waste Basin



*Figure 1–7: Biohazard label, left and right top cover*



*Figure 1–8: Biohazard label, Liquid Waste tank*



Figure 1-9: Biohazard label, Liquid Waste Basin

### 1.7.8 Biohazard labels (2.5 cm)



Biohazard labels are positioned on:

- Sample Module, flap
- Reagent Module, flap
- Solid Waste Bin, inside



Figure 1-10: Biohazard label positioning

### 1.7.9 Attention labels



Attention labels are positioned on:

- Left Pipettor arm
- Right Pipettor arm
- Starter Reagent flap
- Bulk Fluid area

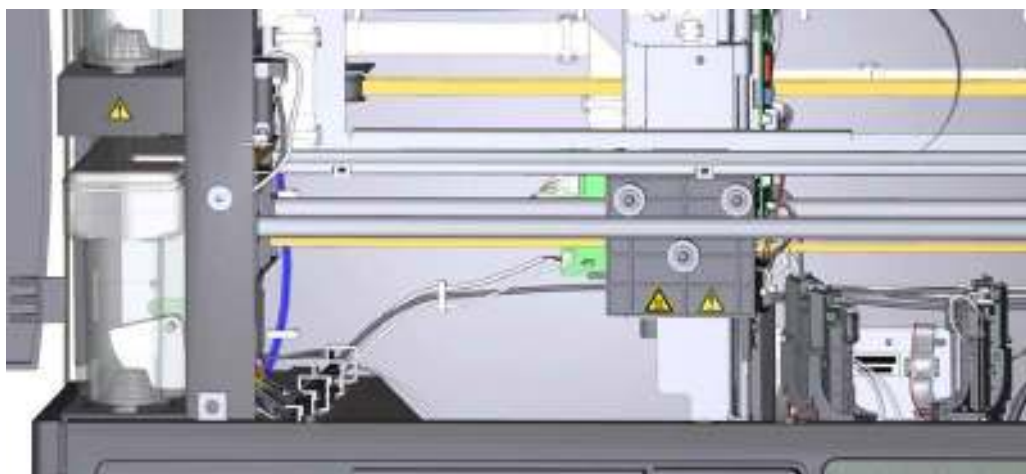


Figure 1-11: Attention label, Left Pipettor arm and Bulk Fluid area

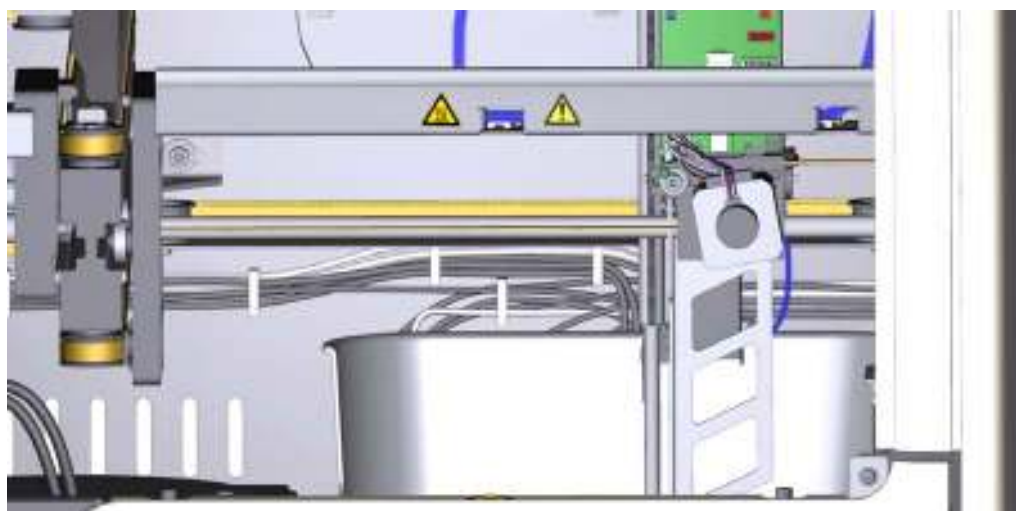


Figure 1-12: Attention label, Right Pipettor arm



Figure 1–13: Attention label, Starter Reagent flap

### 1.7.10 Hot surface labels



Hot surface labels are positioned on:

- Left Pipettor arm
- Right Pipettor arm

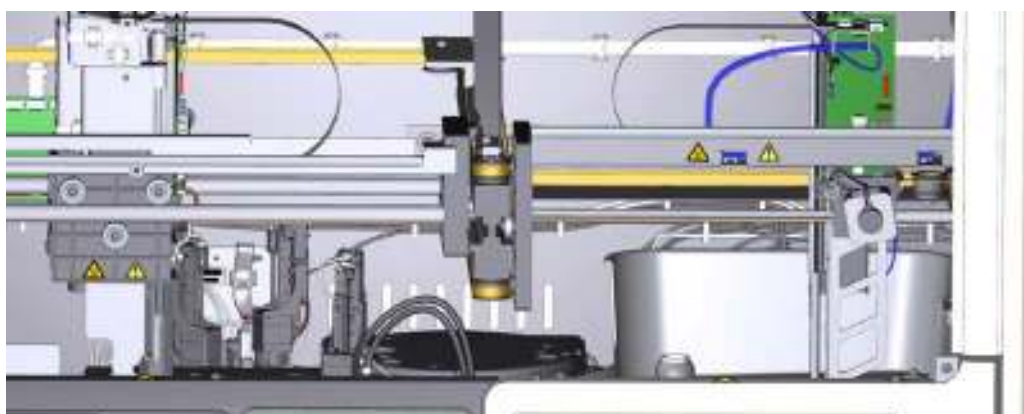


Figure 1–14: Hot surface label, Left and Right Pipettor arms



## 1. Introduction for internal-tanks configuration

### 1.7.11 Laser labels



Laser warning pictogram label is positioned on:

- Sample module, front view



Class 2 Laser warning information label is positioned on:

- Sample Module flap



Laser warning label is positioned on:

- External barcode reader



Figure 1–15: Laser warning labels



Figure 1–16: Laser label, External barcode reader



### 1.7.12 Crush hazard labels



Crush hazard labels are positioned on:

- On top of the Front Module, where the Left and Right top covers close



Figure 1–17: Crush hazard label positioning

### 1.7.13 Magnet labels



Magnet labels are positioned on:

- Washer Module, on top
- DI Water tank
- Wash Buffer tank
- Cleaning Solution tank
- Resuspension Tool
- Rear side of the instrument

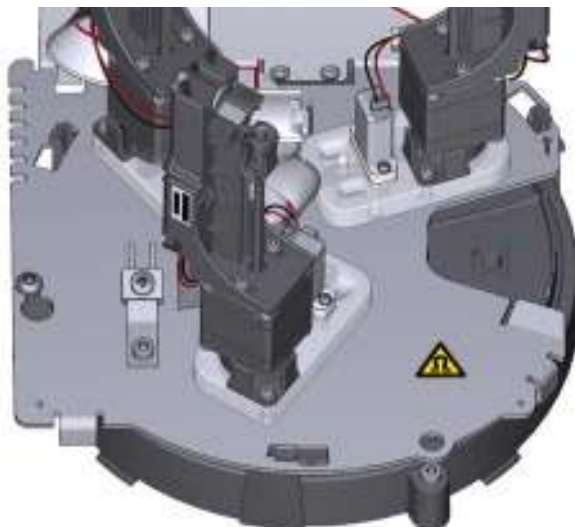


Figure 1–18: Washer Module, magnet label

## 1. Introduction for internal-tanks configuration



Figure 1-19: Magnet label, DI Water tank



Figure 1-20: Magnet label, Cleaning Solution tank



Figure 1-21: Magnet label, Wash Buffer container



Figure 1-22: Magnet label, Resuspension Tool

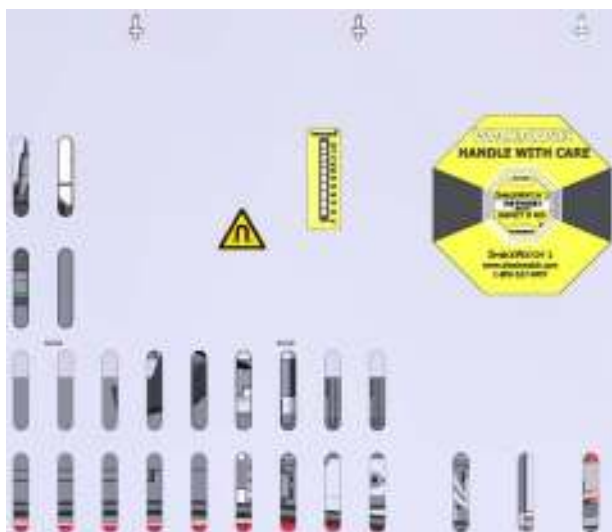


Figure 1–23: Magnet label, Instrument rear view

#### 1.7.14 No heavy load labels



No heavy load labels are positioned on:

- Starter Reagent flap
- Sample Module flap
- Reagent Module flap



Figure 1–24: No heavy load label, Starter Reagent flap

## 1. Introduction for internal-tanks configuration



Figure 1-25: No heavy load label, Sample Module flap



Figure 1-26: No heavy load label, Reagent Module flap

### 1.7.15 Don't step on label labels



Don't step on labels are positioned on:

- Liquid Waste Basin



Figure 1-27: Liquid Waste Basin, Don't step on labels

### 1.7.16 Wash Buffer label



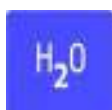
Wash label is positioned on:

- On top of the Wash Buffer tank.



*Figure 1–28: Wash Buffer tank*

### 1.7.17 Water label



Water label is positioned on:

- On top of the DI Water tank.



*Figure 1–29: DI Water tank*

### 1.7.18 Cleaning Solution label

Cleaning  
Solution

Cleaning solution label is positioned on:

- On top of the Cleaning Solution tank.



Figure 1–30: Cleaning Solution tank

### 1.7.19 Waste label

WASTE

Waste label is positioned on:

- On the front of the Liquid Waste tank.



Figure 1–31: Liquid Waste tank

### 1.7.20 Shockwatch, Tiltwatch and Temperature measuring strip

Shockwatch, Tiltwatch and Measuring strip labels are positioned on:

- Rear side of the instrument

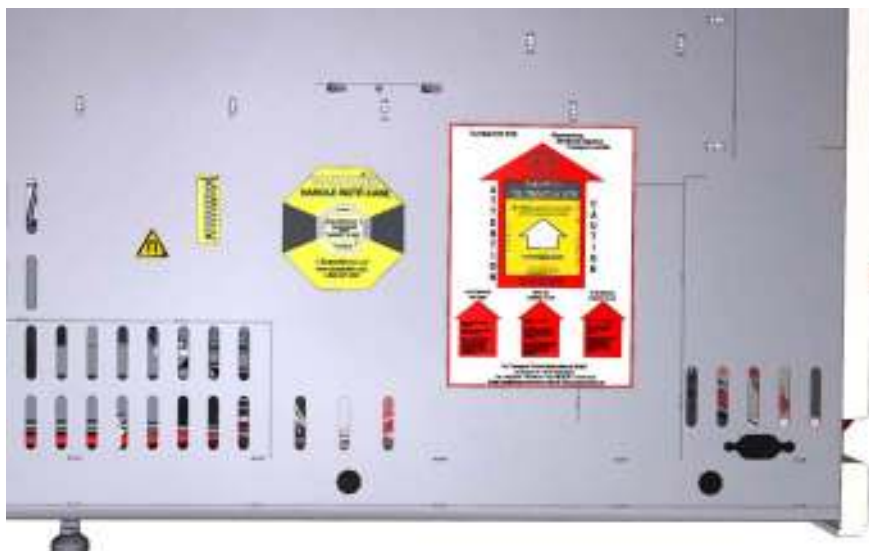


Figure 1–32: Shockwatch, Tiltwatch and Temperature measuring strip labels





## 2 Introduction for external-tanks configuration

This manual was created to provide all technical instructions to allow a correct installation, maintenance, repair and uninstallation of the **LIAISON<sup>®</sup> XS** Analyzer.

This document is intended for Field Service Engineers (FSEs) for training and performing field service. FSEs shall follow all the instructions included in this Service Manual together with those provided in the User Manual.

### 2.1 Description

The system consists of a robotic instrument, a Panel-PC and a software that runs on the Panel-PC. The software provides the user-interface that gives the operator the ability to control the system, and the software performs data reduction on the measurement results (measured by the instrument).

The instrument contains a transport system for plastic cuvettes, a pipetting system with two pipetting arms (one with a steel needle and one with disposable tips), an incubator, a washer and a luminescence measurement chamber. In addition to that, the instrument contains modules for loading samples, reagents, tips and cuvettes into the instrument.

PC and instrument are connected together to exchange data.

#### 2.1.1 Intended use

The **LIAISON<sup>®</sup> XS** is a fully automated chemiluminescence analyzer performing the complete sample processing (sample pre-dilutions, sample and reagent dispensing, incubations, wash processes, cuvette transports) as well as the measurement and evaluation. The instrument is controlled via the Windows PC **LIAISON<sup>®</sup> XS** software. This software, which was specifically designed for this purpose, allows the user to process the pre-defined assays. The clear structure with intuitive user-guidance allows simple and quick operation of daily routine jobs.

---

**NOTE**

**LIAISON<sup>®</sup> XS** system was developed and tested according to the regulations of the IvD Directive. It is strictly intended for professional In-vitro-Diagnostic use. It has to be used only with Chemiluminescence Immunoassays, authorized by DiaSorin S.p.A. for the **LIAISON<sup>®</sup> XS** Analyzer.

---

Any changes to the instrument that are not authorized by the manufacturer will lead to the loss of the conformity to the applicable regulations the manufacturer has declared. In this case, the customer is responsible for the fulfillment of the applicable regulations.

## 2.2 Typographical conventions

The warnings, notes and symbols described hereafter are used in the current manual, on the instrument and on its packaging.

### 2.2.1 Display of warnings and notes

**DANGER**

---

**The signal word “Danger” and a related symbol point to imminent dangers.**

The non-observance of a danger warning is imminent and will probably result in death or at least serious irreversible injuries. A damage of the system or an adverse effect on the system function cannot be excluded.

---

**WARNING**

---

**The signal word “Warning” and a related symbol point to potential dangers.**

The non-observance of a warning is potential and can result in death or at least serious irreversible injuries. A damage of the system or an adverse effect on the system function cannot be excluded.

---

**CAUTION**

---

**The signal word “Caution” and a related symbol point to potential dangers/problems.**

The non-observance of safety instructions can result in minor injuries. A damage of the system or an adverse effect on the system function cannot be excluded.

---

**CAUTION**

---

**The signal word “Caution” points to potential problems.**

The non-observance of a safety instruction can result in damage of the system or an adverse effect on the system function.

---

**NOTE**

---

**The signal word “Note” points to potential problems.**

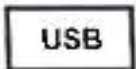
The non-observance of notes can result in an adverse effect on the system function (result deterioration).

---

## 2.2.2 Used warning symbols

|                                                                                     |                                                                                            |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
|    | Caution, risk of danger to person or damage to equipment!<br>Consult instructions for use! |
|    | Biohazard!                                                                                 |
|    | Electrical hazard!                                                                         |
|    | Laser hazard!                                                                              |
|    | Laser hazard and laser information label                                                   |
|  | Caution, hot surface!                                                                      |
|  | Crush hazard!                                                                              |
|  | Magnetic field                                                                             |
|  | Cut injury hazard!                                                                         |
|  | No heavy load                                                                              |
|  | Disconnect main power connector before servicing!                                          |

## 2. Introduction for external-tanks configuration

|                                                                                     |                                                   |
|-------------------------------------------------------------------------------------|---------------------------------------------------|
|    | Disconnect main power connector before servicing! |
|    | Power Supply characteristics                      |
|    | Grounding label                                   |
|    | LAN connection                                    |
|    | USB label                                         |
|    | Waste                                             |
|   | Shockwatch label                                  |
|  | Tiltwatch label                                   |
|  | Connect/disconnect BOTH tubing and sensor         |
|  | System liquid Tank                                |
|  | Wash Buffer Tank                                  |
|  | Liquid Waste Tank                                 |
|  | Cleaning Tank                                     |
|  | Main Switch                                       |



Don't step on

Table 2–1: Warning symbols

### 2.2.3 Other symbols

|                                                                                     |                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Manufactured by                                                                                                                                                                                                                                                                                |
|    | Manufactured date                                                                                                                                                                                                                                                                              |
|    | In Vitro Diagnostic                                                                                                                                                                                                                                                                            |
|   | CE mark                                                                                                                                                                                                                                                                                        |
|  | ID number                                                                                                                                                                                                                                                                                      |
|  | Serial number                                                                                                                                                                                                                                                                                  |
|  | Restriction of certain Hazardous Substances (RoHS) in electronic equipment                                                                                                                                                                                                                     |
| C N US                                                                              | Certification label of the Nemko GmbH & Co. KG                                                                                                                                                                                                                                                 |
|  | Consult instructions for use!                                                                                                                                                                                                                                                                  |
| www.diasorin.com                                                                    | DiaSorin website                                                                                                                                                                                                                                                                               |
| EU: ☎ 00 800 60 61 62 63                                                            | Toll free number                                                                                                                                                                                                                                                                               |
|  | <b>Disposal of Electrical and Electronic Equipment</b><br>In the European Union, electrical and electronic equipment must not be disposed with other household-type waste. It must be collected separately. Please observe the relevant legal regulations effective in the respective country. |
|  | Caution                                                                                                                                                                                                                                                                                        |

Table 2–2: Other symbols

### 2.2.4 Special types

| Typology              | Description                                                               | Example                   |
|-----------------------|---------------------------------------------------------------------------|---------------------------|
| LEDs and signal lamps | LEDs (Light Emitting Diode) and signal lamps are printed in special type. | Power LED, Error LED      |
| Menu items and fields | Menu items and fields are printed in bold and italic type.                | <b><i>File</i></b> Menu   |
| Buttons               | Buttons are printed in bold type.                                         | <b>Open</b> button.       |
| Keys                  | Keys are printed in italic type.                                          | Press <i>Enter</i>        |
| File examples         | File examples are printed in typewriter font.                             | DRIVER=C:\SERVICE\DRIVERS |

Table 2-3: Typology

## 2.3 Metric system

The instruments are developed and built in the metric system.

All fasteners as screws and nuts used in this instrument have metrical dimensions, therefore use only metrical tools as wrenches and screwdrivers.

Do not use screws and nuts of a non-metric system.

Exceptions:

If it is explicitly mentioned in the instructions, then the ANSI-standard parts shall be used.

## 2.4 Use of screw color codes

Red thread-locker lacquer is used:

- Red thread-locker lacquer on a screw head shall be not opened by a field service engineer, except he has all the necessary tools to adjust and re-assemble the part.

Green thread-locker lacquer is used:

- Green thread-locker lacquer on a screw head marks this one as a critical connection, but the field service engineer can open this one and lock it again with green thread-locker lacquer.

Not marked screws:

- A screw without color lacquer code is uncritical.

## 2.5 Hazard instructions

The following hazard instructions must be observed at all times, both before and during operation and during maintenance.

---

**WARNING****Handling of the Service Manual**

The Service Manual is provided for field service engineers safety and gives important instructions for the handling of the system described.

Read all instructions!

---

The **LIAISON® XS** is designed and manufactured in accordance with the safety requirements for electronic and medical systems. It is the operators responsibility to comply with local and national laws, regulations and laboratory procedures for installation and operation of the instrument.

### 2.5.1 General safety

---

**WARNING****Non-Observance of Hazard Instructions**

The non-observance of hazard instructions can result in serious personal injuries and material damages.

- Follow all hazard instructions included in this manual.
  - Follow all warnings marked on the instrument.
- 

---

**WARNING****Function of the Instrument**

Please follow closely the steps contained in the individual instructions to achieve a correct functioning of the instrument.

---

## 2. Introduction for external-tanks configuration

---

**WARNING****Use of the System according to Intended Use only**

Improper use of the system can produce erroneous results, produce damage to the system and cause personal injuries.

- The handling and maintenance of the system must only be performed by trained and authorized personnel.
- Before servicing the system, the Instruction for Use Manual and the Service Manual must be completely read and understood.
- The instrument must only be used in accordance with the intended use as described in this manual.
- Use only the consumables and accessories specifically produced for the instrument (e.g. disposable tips, cuvettes, integrals).
- For maintaining safety, do not modify the **LIAISON<sup>®</sup> XS** Diagnostic System, do not use parts other than the specified ones and do not remove the safety devices, unless strictly necessary.

The manufacturer assumes no liability for any damages, including those to third parties, caused by improper use or handling of the system.

---

**WARNING****Danger due to Overheating**

Improper place of installation/operation of the system may cause fire or serious system damage in case of overheating.

- Select the place of installation of the system so that the ventilation slots are not blocked or covered.
  - Select the place of installation of the system so that air can circulate.
  - Never cover ventilation slots.
- 

### 2.5.2 Electrical hazards

The **LIAISON<sup>®</sup> XS** Diagnostic System does not pose uncommon electrical hazards to operators if it is installed and operated without alteration and is connected to a power source that meets required specifications, see Chapter 36.1 on page 36-1.

---

**DANGER****Electrocution/Fire Hazard!**

Non-observance of rules and regulations can cause serious personal injuries with deadly consequences and material damages.

- National rules and legal regulations for the safe electrical operation of the system must be observed.
- 

**DANGER****Electrocution/Fire Hazard!**

The wrong operation voltage can cause serious personal injuries with deadly consequences and material damages (e.g. fire).

- Before connecting the system to the mains supply, check the operating voltage is correct.
  - The instrument must operate with the type of power source indicated on the dedicated label positioned near the switch. If there are doubts regarding the type of power available, consult an authorized technical person of the laboratory or the local electric power company.
-



**⚠ DANGER****Electrocution/Fire Hazard!**

Improper connection of the system and the peripheral devices to mains supply can cause serious personal injuries with deadly consequences and material damages (e.g. fire).

- Use only connection and extension cables with a protective earth conductor and sufficient capacity (performance, power) to connect the system and the peripheral devices to the mains supply.
  - Use a properly grounded electrical outlet of correct voltage and current handling capability.
  - Never interrupt the grounding contacts.
  - Grounding of the system and its peripheral devices to the same protective earth potential must be ensured.
  - The use of a power strip shall be limited to DiaSorin approved configurations, see Chapter 36.1 on page 36-1.
  - Where appropriate, the Analyzer is installed with additional external switch for residual current circuit-breakers with over-current protection.
  - Where appropriate, the Analyzer is integrated in UPS device.
- 

**⚠ DANGER****Electrocution/Fire Hazard!**

Damaged connecting cables can cause serious personal injuries with deadly consequences and material damages (e.g. fire).

- Damaged connecting cables must be replaced immediately!
  - No objects may be placed on the connecting cables.
  - Connecting cables must be laid so that they cannot be squeezed or damaged.
  - Connecting cables must be laid so that they do not lay in accessible or drivable areas.
- 

**⚠ DANGER****High Voltage!**

The touching of high voltage parts or cables can result in serious personal injuries with deadly consequences.

- Never touch high voltage parts with instrument on.
  - Separate the system from the mains supply before maintaining or cleaning it.
  - Be careful! The following parts can be under high voltage during operation:
    - UPS device (100-240V, high current)
    - Main supply unit (100-240V, high current)
    - Enclosure metal part in Single Fault conditions (132V r.m.s., low current)
    - Photomultiplier (PMT) (~ 1000 Volts, low current)
- 

**⚠ DANGER****Electrocution/Fire Hazard!**

Defective systems can result in serious injuries with deadly consequences and material damages (e.g. fire).

- Separate immediately the defective system from the mains supply, if a safe usage is no longer possible.
  - Secure the defective system against reconnection.
  - Label the defective system clearly as being defective.
-

## 2. Introduction for external-tanks configuration

---

** DANGER****Electrocution!**

Working with electrical devices on wet floors can cause serious injuries with deadly consequences and material damages due to electrocution.

- Surfaces (e.g. floors) must be dry when working with the system.
  - Keep liquids away from all connectors of electrical or communication components.
  - Do not touch any switches or outlets with wet hands.
- 

** DANGER****Danger of Electrocution**

If the system is not separated from the mains supply before maintenance works, this can cause serious injuries with deadly consequences due to electrocution. Additionally, there is the danger that the system starts and causes injuries (e.g. contusion, cuts etc.) to the person working with the system.

- Determine and correct the cause of a blown fuse or thrown circuit-breaker before attempting to resume operations of the system.
  - Separate the system from the mains supply and secure it against restarting before replacing fuses.
  - Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.
  - Never repair or bridge blown fuses.
  - Never short-circuit the fuse holder.
- 

** DANGER****Danger of Electrocution or Mechanical Injuries**

If the system is not separated from the mains supply before maintenance works, this can cause serious injuries with deadly consequences due to electrocution. Additionally, there is the danger that the system starts and causes injuries (e.g. contusion, cuts etc.) to the person working with the system.

- Switch off the system, separate it from the mains supply and protect it against restarting. When system is secured start cleaning, disinfection, maintenance or repair works.
  - Make sure that nobody works on the system and that all covers are attached and closed before reconnecting the system to the mains supply.
  - Do not disconnect any electrical connection while the power is on.
  - Disconnect the power cord before cleaning major liquid spills. Clean spilled fluids immediately.
-

**WARNING****Danger due to Improper Place of Installation**

Improper place of installation of the system can cause accidents with serious injuries with deadly consequences, fire or serious system damages, because the system cannot be switched off or be separated from the mains supply.

- Ensure that the power supply and mains switch/ emergency stop are easily accessible in the place of installation of the system.
  - Ensure UPS proper ventilation.
  - The Analyzer is not designed and constructed to be operated in a potentially explosive atmosphere, so it must not be installed and used in a laboratory with potentially explosive atmosphere. Field Service Engineer must verify with the end user that the place chosen for installing th Analyzer fulfils such requirements. The responsibility of such check is in charge of the end user.
-

## 2.5.3 Physical Hazards

### 2.5.3.1 Laser Hazards

**WARNING**

---

**Eye Injuries due to Laser Radiation**

Laser radiation causes eye irritations when the laser beam is looked for a long period of time.

- Never look directly into the laser beam!
- Do not introduce optical devices into the system.
- Remove watches and reflective jewellery before operating the laser.
- Caution during operation and testing of the laser (barcode scanner and hand held barcode reader) must be taken due to laser class (class 2).
- The barcode reader is positioned on the right side of the Sample loading bay. The laser beam exits from an opening positioned on the internal right side of the Sample loading bay. The laser beam is orthogonal to the internal right side surface.

Wrong usage of operating elements or adjustments or the non-observance of processes can cause a dangerous emission of laser radiation.

---

### 2.5.3.2 Heavy Objects

Follow supplied instructions for a safe unpacking and placement of the Analyzer during Installation, see Chapter 5 on page 5-1, as well as Uninstallation, see Chapter 7 on page 7-1.

Use proper lifting techniques when handling full tanks (System Liquid, Wash Buffer and Liquid Waste). Use care when handling the containers to reduce the risk of injury.

**WARNING**

---

**High Weight**

Improper lifting/carrying of the system may cause the system to fall down. Falling down may cause serious injuries to the personnel or serious material damages.

- Observe the technical specifications of the system to avoid misjudgments.
- 

**WARNING**

---

**Never lift the system by yourself!**

Manual handling of heavy loads may cause injuries.

The **LIAISON<sup>®</sup> XS** instrument shall NOT be lifted and carried by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

---

## 2.5.4 Trip Hazards

The **LIAISON<sup>®</sup> XS** Diagnostic System is equipped with power cord and various connectors. To avoid tripping hazards, ensure cords are properly stowed.

**NOTE**

In the event of a damage, prejudice or personal injury occurred in connection with the use of the **LIAISON<sup>®</sup> XS** Diagnostic System, the Field Service Engineer shall immediately contact the local DiaSorin Representative.

---

### **2.5.5 Mechanical Hazards**

The **LIAISON® XS** Diagnostic System is an automated system that operates under computer control. As with most automated equipment, there is potential for injury and bodily harm from moving mechanical components whenever the system is in operation.

The **LIAISON® XS** Diagnostic System minimizes mechanical hazards by providing guards to protect against accidental contact with moving components, and encoding the software with safety features.

During servicing of the Analyzer, a Field Service Engineer is potentially exposed to the following moving mechanical components:

- Pipettor arms and probes
- Cuvette gripper arm
- Incubator
- Washer
- Reader

Basic elements of mechanical safety include, but are not limited to:

- Wear protective gloves during cleaning, maintenance or service in general.
- Avoid contact with the sharp metal edges.
- Never bypass or override a safety device.
- Never allow any part of your body to enter a range of mechanical movement during system operation.
- Do not wear articles of clothing or accessories that could catch on the system.
- Keep pockets free of items that could fall into the system.
- The right pipettor probe, the washer probes and the reader probe are sharp and potentially contaminated with infectious material. Adopt all safety precautions required when servicing such areas.
- Use caution when performing cleaning, maintenance or service in general.
- Be aware that, in the event of a system malfunction or an unexpected sequence of movements, reflex actions could occur, causing injury.

### 2.5.6 Biological Safety

Performing the following activities, Field Service Engineers may be exposed to potentially infectious materials:

- Handling samples, reagents, calibrators, and controls;
- Cleaning spills;
- Handling and disposing of waste;
- Performing maintenance procedures;
- Performing cleaning procedures.

During servicing it is important to pay particular attention to the potentially contaminated parts listed below:

- Liquid waste tank and liquid waste basin;
- Solid waste bin and solid waste box;
- Liquid waste tubes (from pumps to waste tank);
- Left arm edge;
- Right arm needle;
- Right arm washing station;
- Washer and reader aspiration needles and tubes;
- Washer Aspiration Pump (WAP) tubes;
- Liquid Waste Pump (LWP) tubes;
- Incubator cover (especially near dispensing points);
- Incubator rotor and closer area;
- Washer and reader assemblies (especially internal parts);
- Sample area, sample and control racks (in case of spillage);
- Reagent loading area (in case of spillage);
- Reagent cover (in case of spillage);
- Any surfaces in case of spillage.

The following information help the technicians in minimizing the impact of the exposure.



Field Service Engineers should consider all clinical samples, reagents, calibrators, controls, and used reaction vessels and consumables (e.g. tips) that contain human-sourced material as potentially infectious. They should also consider all system surfaces or components that have come in contact with human-sourced material as potentially infectious. No known test method can offer complete assurance that products derived from human-sourced material will not transmit infection. Therefore, all products derived from human-sourced materials and system components exposed to human-sourced materials should be considered potentially infectious.

---

## 2. Introduction for external-tanks configuration

---

**⚠ DANGER****Risk of infection!**

Parts, in contact to potentially infectious liquid, must be treated as potentially infectious. Improper handling of infectious parts can cause skin irritations, illnesses and possible death.

- Observe local and national provisions, legislation and laboratory regulations.
  - Use appropriate disposable gloves.
  - Use an appropriate lab coat.
  - Use an appropriate eye protection.
  - Avoid contact between skin/mucous membrane and samples/test reagents or parts of the instrument.
  - Do not pipette by mouth.
  - Do not eat, drink, smoke, apply cosmetics or handle contact lenses when handling human-sourced material or contaminated system components.
  - Clean and disinfect the system immediately if potentially infectious material has been spilled.
  - Any reagent spills should be washed with a chemical use sodium hypochlorite solution with 0.5% active chlorine (e.g. dilution 1:10 of a solution at 5% active chlorine) and disposed as though potentially infectious.
  - All samples, biological reagents and materials used in the assays must be considered potentially able to transmit infectious agents. They should therefore be disposed in accordance with the prevailing regulations and guidelines of the agencies holding jurisdiction over the laboratory, and the regulations of each Country.
  - Liquid waste must be decontaminated with a chemical use sodium hypochlorite solution with 0.1% active chlorine (e.g. dilution 1:50 of a solution at 5% active chlorine) for at least half a hour.
  - Do not use broken or chipped tubes or bottles.
  - Observe the instructions in the package inserts for a correct use of the reagents.
- 

**CAUTION****Organic solvents**

Reagent containers and hoses for system liquid and waste can be seriously damaged by organic solvents and become unusable.

- Never use organic solvents.
- 

**WARNING****Handling of Cleaning/ Disinfection Products**

Pay attention when handling cleaning/ disinfection products, because they are harmful as indicated on the bottle.

- Read the instructions of the products before use.
  - Do not use products undiluted!
-



### 2.5.7 Chemical Hazards

Field Service Engineer may be exposed to hazardous chemicals when handling reagents, calibrators and controls.

Exposure to hazardous chemicals is minimized by following instructions provided in the assay-specific documentation (such as a package insert), on product-specific labels and product-specific MSDS (Material Safety Data Sheets).

In general, observe the following precautions when handling chemicals:

- Wear suitable protective gloves.
- Consult MSDS for safe use instructions and precautions.
- Avoid contact with skin and eyes. If contact with material is anticipated, wear impervious gloves, protective eye wear, and clothing.
- Maintain good housekeeping. Do not eat, drink, or store food and beverages in areas where chemicals are used.
- Seek medical attention if irritation or signs of toxicity occur after exposure.

### 2.5.8 Spill clean-up

Clean spills in accordance with established biosafety practices and follow instructions in the MSDS (Material Safety Data Sheets). In general, safe work practices for cleaning spills include:

- Wear appropriate personal protective equipment, such as gloves, eye wear and lab coat.
- Absorb the spill with absorbent material.
- Wipe the spill area with detergent solution.
- Wipe the area clean with an appropriate disinfectant such as a chemical use sodium hypochlorite solution with 0.5% active chlorine (e.g. dilution 1:10 of a solution at 5% active chlorine).

### 2.5.9 Maintenance Work

---

**WARNING**

---

**Unapproved Maintenance Works**

Unapproved or inaccurately performed maintenance works can result in serious personal injuries and material damages.

- Only perform maintenance procedures described in this manual.
  - Follow closely the steps contained in the individual instructions.
  - For the maintenance works, only use original parts approved by DiaSorin.
  - Tests and maintenance works specified by the manufacturer must be performed to ensure the safe operation of the system and the proper functioning of the system.
  - All service and maintenance works which are not described in this instruction must be performed by qualified and authorized service technicians.
  - Any changes made to the instrument that are not authorized by the manufacturer will lead to the loss of guarantee.
-

## 2. Introduction for external-tanks configuration

---

**WARNING****Danger of Electrocution due to not correct replacement of instrument parts**

Not correct replacement of instrument parts may cause spillage of liquid on PCB or electrical cables.

- Only perform maintenance procedures described in this manual.
  - Follow closely the steps contained in the individual instructions.
  - For the maintenance works, only use original parts approved by DiaSorin.
  - Tests and maintenance works specified by the manufacturer must be performed to ensure the safe operation of the system and the proper functioning of the system.
  - All service and maintenance works which are not described in this instruction must be performed by qualified and authorized service technicians.
  - Any changes made to the instrument that are not authorized by the manufacturer will lead to the loss of guarantee.
- 

**WARNING****Danger of Electrocution or Mechanical Injuries during Required Maintenance Works on the Voltage-Carrying System**

If the system cannot be separated from the mains supply during required maintenance works, additional precautions must be taken to avoid serious injuries with deadly consequences due to electrocution or injuries by the system (e.g. contusion, cuts etc.) due to electrocution.

- Only switch on the system if this is explicitly required for the minimum necessary time only.
  - Perform the maintenance works with highest caution.
  - Never touch electrical connecting contacts.
  - Please note that mechanical components (e.g. pipettor) can move unexpectedly.
  - Assure other personnel shall not access the instrument during the service activities and do not leave the instrument unattended.
- 

**WARNING****Danger of Electrocution or Mechanical Injuries by Missing or Opened Protective Covers**

To avoid serious injuries with deadly consequences due to electrocution or injuries by the system (e.g. contusion, cuts etc.), protective covers may only be opened or removed for certain maintenance procedures and with the highest level of caution.

- Only perform maintenance procedures described in this manual.
  - Make sure that nobody works on the system and that all covers are attached and closed before reconnecting the system to the mains supply.
  - Make sure that all covers are attached before switching on the instrument.
  - Switch off the system, separate it from the mains supply and protect the system against restarting, if protective covers/gears are missing.
  - Make sure that the pipettors have stopped before opening covers and/or accessing the working area of the instrument.
  - Avoid touching the pipettors and other moving parts while the system is in operation.
  - Perform all maintenance procedures with the highest level of caution.
  - Assure other personnel shall not access the instrument during the service activities and do not leave the instrument unattended.
-

**WARNING****Sharp Edges!**

Sheet metal parts and PCBs located behind protective covers might have sharp edges. Contact might lead to injuries.

- Always wear the suitable protective gloves during the service activities.
  - Use caution at corners and edges!
- 

**WARNING****Liquid in the System**

Liquid which gets into the system can cause illnesses with deadly consequences in case of contact. The system can be damaged by liquids.

- Switch off the system.
  - Separate the system from the mains supply.
  - Wear suitable protective clothing.
  - Clean, disinfect and dry the system according to the applicable local and national provisions, legislation and laboratory procedures.
  - Assure the system is completely dry before switch it on.
- 

**WARNING****Defects in the Liquid System**

Defect or leaky tubes, syringes, valves or pumps lead to deterioration of the pipetting results and consequently corruption of final results. Furthermore incorrectly flushed tips can cause mixing up of sample material.

- Check the system on drops and ponding.
  - Check tubes, syringes, valves or pumps periodically.
-

### 2.5.10 Disposal and Disinfection

**CAUTION****Unsuitable Sterilisation Method**

Containers and components for liquids or waste are seriously damaged at autoclaving. Disinfect the containers and components with a suitable disinfection method.

---

**CAUTION****Cleaning or Disinfection**

Observe the following aspects during cleaning or disinfection because otherwise breakdowns or damages can be the result.

- Always wear the suitable protective gloves during the service activities.
  - Liquid cleaning or disinfection solutions may only be used with a moistened cleaning tissue. They may not be poured into or sprayed into the system.
  - Cleaning or disinfection solutions and procedures must be approved.
  - Cleaning or disinfection solutions must not come into contact with bearings and guides, as otherwise the gr film may dissolve!
  - Cleaning or disinfection solutions must not be used in the proximity of circuit boards, light barriers and perspex surfaces!
- 

**CAUTION****Handling/Cleaning of Optic Surfaces**

Optic surfaces (e. g. scanners, lenses, sensors) must be free of dust and fat.

- Always wear the suitable protective gloves during the service activities.
  - Do not touch any optic surfaces.
  - Only clean the optic surfaces with a softy and lint-free cloth.
  - Do not use any aggressive detergents (e.g. acetone).
- 

**WARNING****Disposal of Infectious Waste**

Potential infectious material and all parts that may come in contact with potential infectious material must be disposed according to the local and national provisions, legislation and laboratory procedures.

---

**WARNING****Disposal**

- The instrument, the packaging material and all parts that have been replaced must be disposed according to the applicable local and national provisions, legislation and laboratory procedures.

For more information about disposal, contact the manufacturer.

---

**CAUTION****Touch Screen Cleaning**

Improper cleaning could damage the touch screen surface.

- Always wear the suitable protective gloves during the service activities.
  - Use soft clothes with neutral detergent or with ethanol to clean the touch screen.
  - Do not use any chemical solvent, acidic or alkali solution.
  - Do not allow liquid from soaking into the joint of film and glass which may result in peeling or malfunctioning.
- 

**WARNING****Battery Handling**

The product may contain an internal lithium manganese dioxide, vanadium pentoxide, or alkaline battery or battery pack. There is risk of fire and burns if the battery pack is not handled properly. To reduce the risk of personal injuries:

- Do not attempt to recharge the battery.
  - Do not expose to temperatures higher than 60°C (140°F).
  - Do not disassemble, crush, puncture, short external contacts, or dispose of in fire or water.
  - Risk of explosion if battery is replaced by an incorrect type. Dispose of used batteries according to the instructions.
  - Replace only with the spare designated for this product.
-

### **2.5.11 Caution on electromagnetic wave interference**

#### **2.5.11.1 Electromagnetic wave interference given by LIAISON® XS to other equipment**

The **LIAISON® XS** instrument was designed and manufactured in compliance to the applicable EMC standards.

Use the cables attached at the installation for connection between the devices in the system. The proper use of the specified cables minimizes electromagnetic wave interference.

Installation and service of the system or changes in the installation may never be performed by non-authorized persons not approved by DiaSorin S.p.A. Movable multiple plug sockets different from the one installed by DiaSorin S.p.A. personnel are not allowed.

However, there is no guarantee that the **LIAISON® XS** instrument will not cause electromagnetic wave interference.

- a. When **LIAISON® XS** instrument may be the cause, turn off the power of the **LIAISON® XS** and check the functions of the other equipment. **LIAISON® XS**
- b. If the functioning is improved, the **LIAISON® XS** instrument probably is the cause.

#### **2.5.11.2 Electromagnetic wave interference that may be given to the LIAISON® XS instrument**

If the **LIAISON® XS** instrument is used near equipment that generates strong electric and magnetic field, noises may enter the Instrument and affect the performances and functions.

Use the cables provided at the installation for connection between the devices in the system. The proper use of the specified cables minimizes electromagnetic wave interference.

Installation and service of the system or changes in the installation shall be performed only by authorized persons approved by DiaSorin S.p.A. Movable multiple plug sockets different from the one installed by DiaSorin S.p.A. personnel are not allowed.

However, there is no guarantee that the **LIAISON® XS** instrument will not be affected by electromagnetic wave interference.

- a. When other equipment may be the cause, turn off the power of the equipment and check the functions of the **LIAISON® XS** instrument.
- b. If the functioning is improved, interference from the equipment probably is the cause.

Try the following to correct the issues.

- a. Move the **LIAISON® XS** instrument further away from the equipment that may be the cause.
- b. Connect the power cord of the **LIAISON® XS** instrument to an outlet that is on a different circuit from the equipment that may be the cause.
- c. Check that the electromagnetic wave interference does not affect the other equipment which is connected with this instrument.

**NOTE****Transient Emissions and Interference Resistance**

This instrument meets all requirements described in standard IEC 61326 on transient emissions and interference resistance.

- This instrument was developed and tested according to CISPR11 Class A. It can cause radio interference in domestic environments. In this case it may be required to take action to eliminate such interference.
  - Before setup and operation of the instrument, the electromagnetic environment shall be evaluated.
  - Do not use the instrument in the proximity of sources with excessive electromagnetic radiation (e.g. unshielded, deliberately operated high frequency sources) since they could interfere with the proper operation of the instrument.
- 

**2.5.11.3 Radio Interferences****CAUTION****Interference by Mobile Phones**

Mobile phones can affect the correct function of the instrument.

- Do not use mobile phones next to a running instrument.
- 

**Federal Communications Commission (FCC) Notice**

This equipment was tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the Instructions for Use and Service Manual, may cause harmful interferences to radio communications. Operation of this equipment in a residential area is likely to cause harmful interferences. In this case, the user will be required to correct the interferences at his own expense.

**Canadian Department of Communications Compliance Statement**

This equipment does not exceed Class A limits per radio noise emissions for digital apparatus set out in the Radio Interference Regulations of the Canadian Department of Communications. Operation in a residential area may cause unacceptable interference to radio and TV reception, requiring the owner or operator to take whatever steps are necessary to correct the interference.

**2.5.12 Static magnetic field**

Magnets are used in the instrument and clearly identified with labels (see Chapter 2.2.2 on page 2-3).

Concerning precautionary limits of exposure to magnetic fields, international standards establish 0.5 mT (5 gauss) as the magnetic flux density threshold level for static magnetic fields to prevent inadvertent harm to persons with implanted electronic medical devices and implants containing ferromagnetic material.

All magnets in the instruments are either completely within the 5 gauss line or are exceeding it to an extent which is not expected to have impact on the user during normal operations.

**CAUTION****Handling of module labeled with magnetic field warning**

Precautions have to be taken during the handling of the module labeled with magnetic field warning (e.g. Washer module, Resuspension tool...) since magnet could exceed the 5 gauss limit and may come in contact with the body of the FSE and potential cause interference with implanted medical devices.

---

## 2.6 Evaluation of Residual risk for Field Service Engineer

The aim of this section is to bring to Field Service Engineer attention to the status of residual risks, despite the inherent safe design measures, safeguarding and complementary protective measures adopted by the Manufacturer to lower the risks to an acceptable level.

Field Service Engineer must read carefully the section "Safety residual risks for User" contained in **LIAISON® XS** User Manual as well as the present chapter in order to understand the residual risks and learn the recommended instructions to work and service safely on the **LIAISON® XS** Diagnostic System.

If further clarification should be needed, it is recommended to contact the local DiaSorin representative.

**WARNING**

**The FSE credentials, user name and password, must be kept confidential at all times and they should never be shared or exposed to others.**

---

**NOTE****Design of the Analyzer to facilitate handling**

The **LIAISON® XS** Diagnostic System and its component parts are capable of being handled, transported, packaged and stored safely, without damage.

The Analyzer is either wheels and/ or handles equipped to facilitate its handling during unpacking and positioning at customer site.

---

**NOTE****Protection against mechanical hazards**

The **LIAISON® XS** Diagnostic System and its components and fittings are stable enough to avoid overturning, falling or uncontrolled movements during transportation, assembly, dismantling and any other action involving the machinery.

The various parts of the Analyzer and their linkages are able to withstand the stresses to which they are subject when used.

The durability of the materials used is adequate for the nature of the working environment foreseen by DiaSorin S.p.A., in particular as regards the phenomena of fatigue, ageing, corrosion and abrasion. To date, there is no evidence of parts showed to be weak.

A remaining mechanical hazard is related to moving parts.

---



**WARNING****Handling of left top cover**

When the technician is logged in the software with a FSE access right and the “Allow open cover” checkbox is selected, the monitoring of the left top cover presence sensor is disabled, so the Analyzer does not stop moving when the cover is opened. Field Service Engineer must pay particular attention when interacting with the Analyzer for servicing and maintenance purposes.

- Left top cover shall be opened only when required for particular service activity for the minimum necessary time only.
  - Never insert hands or head inside instrument when pipettors or other instrument parts are running.
  - Move in a safe position pipettor arms before operating on internal instrument parts, as possible far away from working area.
  - Pay attention to internal potentially cutting parts (pipettor arms, right pipettor needle, sharp angles and edges).
  - Assure other personnel shall not access the instrument during the service activities and do not leave the instrument unattended.
-

## 2.7 Instrument Labelling and Safety Labels

This section provides information on the instrument labelling, with particular focus on the safety labels.

Safety labels are used on the **LIAISON® XS** system and in the documentation, to identify potentially dangerous conditions. Some safety labels are not accessible to the user but are, in some circumstances, visible to the Field Service Engineer.

### WARNING



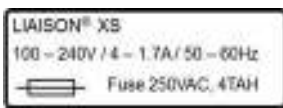
#### Missing Warnings

Missing or unreadable warning labels or type labels can result in non-identified dangers. This can cause serious personal injuries and material damages.

- Check the system on missing or unreadable warning labels and type labels.
- Missing or unreadable warning labels or type labels must be replaced.

|    |          |                                                                                                                                                                                                                                                                                                                                               |
|----|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. |          | <p>Attention label:</p> <ul style="list-style-type: none"> <li>• Left Pipettor arm</li> <li>• Right Pipettor arm</li> <li>• Starter Reagent flap</li> <li>• Bulk Fluid area</li> </ul>                                                                                                                                                        |
| 2. |          | <p>Biological hazard label:</p> <ul style="list-style-type: none"> <li>• Left top cover, inside</li> <li>• Right top cover, inside</li> <li>• Sample Module flap, inside</li> <li>• Solid Waste bin, inside</li> <li>• Reagent Module flap, inside</li> <li>• Liquid Waste tank</li> <li>• Liquid Waste Basin</li> </ul>                      |
| 3. | <br><br> | <p>Laser warning pictogram label:</p> <ul style="list-style-type: none"> <li>• Sample Module</li> </ul> <p>Class 2 Laser warning information label:</p> <ul style="list-style-type: none"> <li>• Sample Module flap, inside</li> </ul> <p>Laser hazard label:</p> <ul style="list-style-type: none"> <li>• External barcode reader</li> </ul> |
| 4. |          | <p>Hot surface label:</p> <ul style="list-style-type: none"> <li>• Left Pipettor arm</li> <li>• Right Pipettor arm</li> </ul>                                                                                                                                                                                                                 |

## 2. Introduction for external-tanks configuration

|     |                                                                                     |                                                                                                                                                                                |
|-----|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.  |    | Crush hazard <ul style="list-style-type: none"> <li>On top of the Front Module, near the Sample Module</li> <li>On top of the Front Module, near the Reagent Module</li> </ul> |
| 6.  |    | Magnet label: <ul style="list-style-type: none"> <li>Washer Module</li> <li>Resuspension tool</li> <li>Rear side of the instrument</li> </ul>                                  |
| 7.  |    | No heavy load <ul style="list-style-type: none"> <li>Starter Reagent flap</li> <li>Sample Module flap</li> <li>Reagent Module flap</li> </ul>                                  |
| 8.  |    | ON/OFF label <ul style="list-style-type: none"> <li>Right side of the instrument</li> </ul>                                                                                    |
| 9.  |    | Disconnect main power before servicing: <ul style="list-style-type: none"> <li>Right side of the instrument, under the Right side cover</li> </ul>                             |
| 10. |  | Fuse/Power Supply: <ul style="list-style-type: none"> <li>Right side of the instrument, above the main switch</li> </ul>                                                       |
| 11. |  | LAN label <ul style="list-style-type: none"> <li>Right side of the instrument, above the LAN connector</li> </ul>                                                              |
| 12. |  | USB label <ul style="list-style-type: none"> <li>Right side of the instrument, above the USB connector</li> </ul>                                                              |
| 13. |  | Grounding label: <ul style="list-style-type: none"> <li>Bottom of the instrument, underneath of the main housing</li> </ul>                                                    |
| 14. |  | Instrument Serial label (illustration only, for demonstrating label position): <ul style="list-style-type: none"> <li>Rear side of instrument</li> </ul>                       |
| 15. |  | Shockwatch label: <ul style="list-style-type: none"> <li>Rear side of the instrument</li> </ul>                                                                                |

## 2. Introduction for external-tanks configuration

|     |                                                                                     |                                                                                                                                                                                                              |
|-----|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16. |    | Tiltwatch label:<br><ul style="list-style-type: none"> <li>Rear side of the instrument</li> </ul>                                                                                                            |
| 17. |    | Temperature measuring strip<br><ul style="list-style-type: none"> <li>Rear side of the instrument</li> </ul>                                                                                                 |
| 18. |    | Connect/disconnect BOTH tubing and sensor<br><ul style="list-style-type: none"> <li>On top of the Wash Buffer tank</li> <li>On top of the System Liquid tank</li> <li>On top of the Cleaning tank</li> </ul> |
| 19. |   | Wash label<br><ul style="list-style-type: none"> <li>At the front of the Wash Buffer tank</li> </ul>                                                                                                         |
| 20. |  | Waste label<br><ul style="list-style-type: none"> <li>At the front of the Liquid Waste tank</li> </ul>                                                                                                       |
| 21. |  | System Liquid label<br><ul style="list-style-type: none"> <li>At the front of the System Liquid tank</li> </ul>                                                                                              |
| 22. |  | Cleaning solution label<br><ul style="list-style-type: none"> <li>At the front of the Cleaning tank</li> </ul>                                                                                               |
| 23. |  | Don't step on<br><ul style="list-style-type: none"> <li>Liquid Waste Basin</li> </ul>                                                                                                                        |

Table 2-4: Labels

### 2.7.1 Instrument Serial label

Instrument Serial label is positioned on:

- Rear side of the instrument



Figure 2–1: Instrument Serial label positioning

### 2.7.2 Fuse / Power supply label

Fuse / Power supply label is positioned on:

- Right side of the instrument, above the main switch

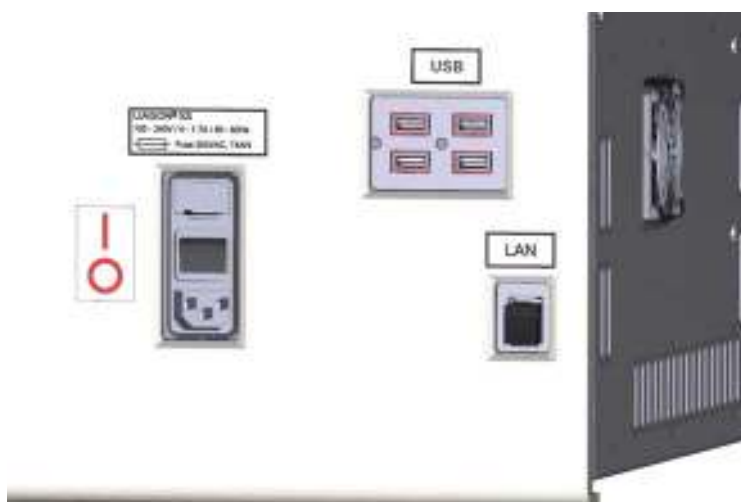
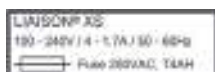


Figure 2–2: Fuse / Power supply label positioning

## 2. Introduction for external-tanks configuration

### 2.7.3 ON/OFF label



ON/OFF label is positioned on:

- Right side of the instrument, at the left of the main switch

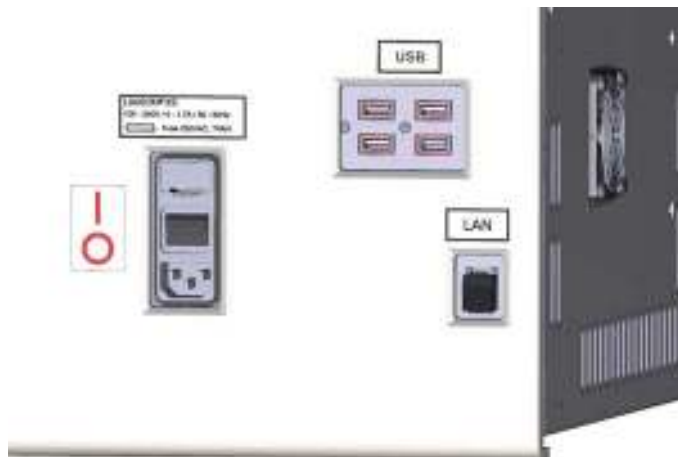


Figure 2–3: ON/OFF label positioning

### 2.7.4 USB and LAN label



USB and LAN labels are positioned on:

- Right side of the instrument,
  - USB label is positioned above the USB connectors
  - LAN label is positioned above the LAN connector

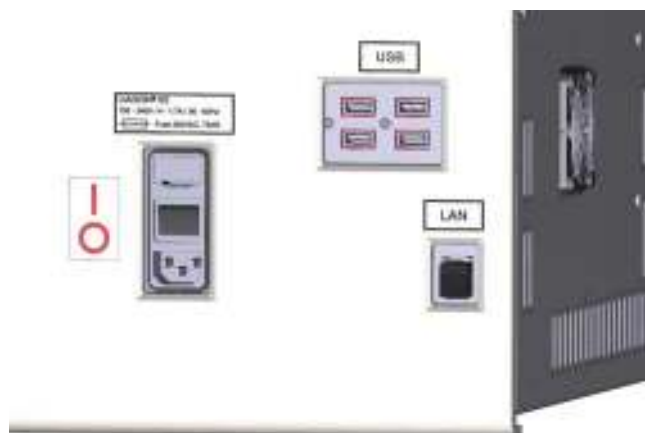


Figure 2–4: USB and LAN label positioning

### 2.7.5 Disconnect main power label

Disconnect main power label is positioned on:

- Right side of the instrument, under the Right Side cover



Figure 2-5: Disconnect main power label positioning

### 2.7.6 Ground label / Earth label



Ground label is positioned on:

- Bottom of the instrument, underneath the main housing, close to the User Access Panel PCB.

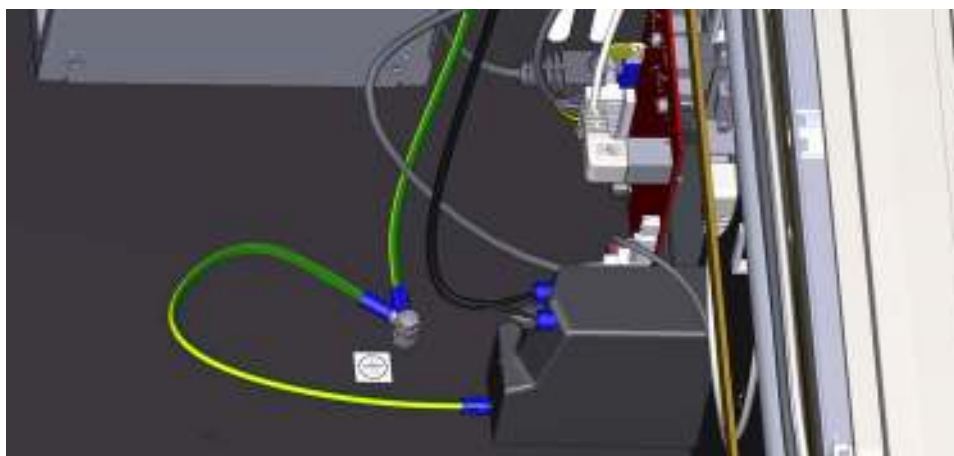


Figure 2-6: Ground label positioning

### 2.7.7 Biohazard labels (4.5 cm)



Biological hazard labels are positioned on:

- Left top cover, inside
- Right top cover, inside
- Liquid Waste tank
- Liquid Waste Basin

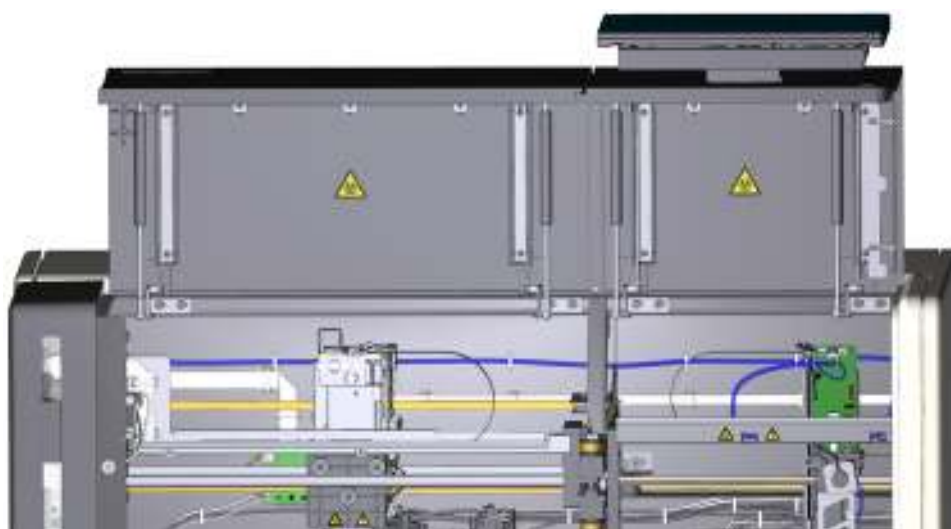


Figure 2–7: Biohazard label, left and right top cover



Figure 2–8: Biohazard label, Liquid Waste tank





Figure 2-9: Biohazard label, Liquid Waste Basin

### 2.7.8 Biohazard labels (2.5 cm)



Biohazard labels are positioned on:

- Sample Module, flap
- Reagent Module, flap
- Solid Waste Bin, inside



Figure 2-10: Biohazard label positioning

### 2.7.9 Attention labels



Attention labels are positioned on:

- Left Pipettor arm
- Right Pipettor arm
- Starter Reagent flap
- Bulk Fluid area

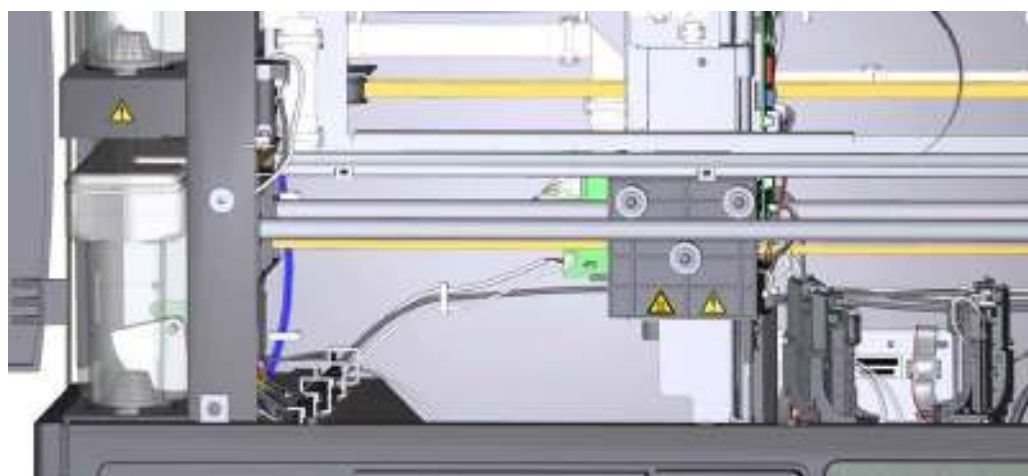


Figure 2-11: Attention label, Left Pipettor arm and Bulk Fluid area

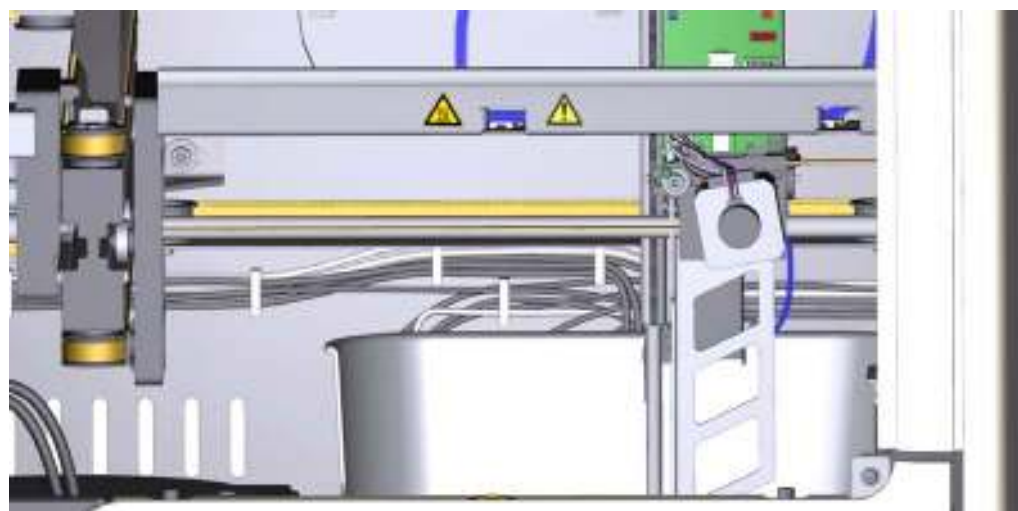


Figure 2-12: Attention label, Right Pipettor arm



Figure 2–13: Attention label, Starter Reagent flap

### 2.7.10 Hot surface labels



Hot surface labels are positioned on:

- Left Pipettor arm
- Right Pipettor arm

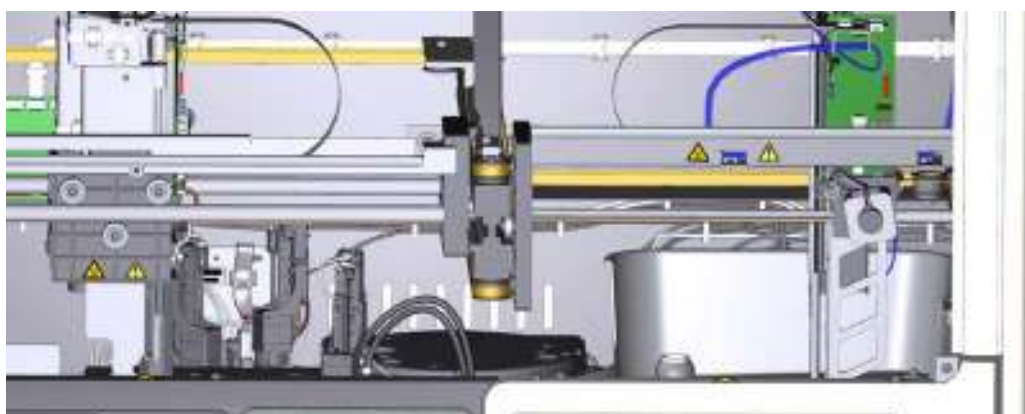


Figure 2–14: Hot surface label, Left and Right Pipettor arms

## 2. Introduction for external-tanks configuration

### 2.7.11 Laser labels



Laser warning pictogram label is positioned on:

- Sample module, front view



Class 2 Laser warning information label is positioned on:

- Sample Module flap



Laser warning label is positioned on:

- External barcode reader



Figure 2–15: Laser warning labels



Figure 2–16: Laser label, External barcode reader

### 2.7.12 Crush hazard labels



Crush hazard labels are positioned on:

- On top of the Front Module, where the Left and Right top covers close



Figure 2–17: Crush hazard label positioning

### 2.7.13 Magnet labels



Magnet labels are positioned on:

- Washer Module, on top
- Resuspension Tool
- Rear side of the instrument

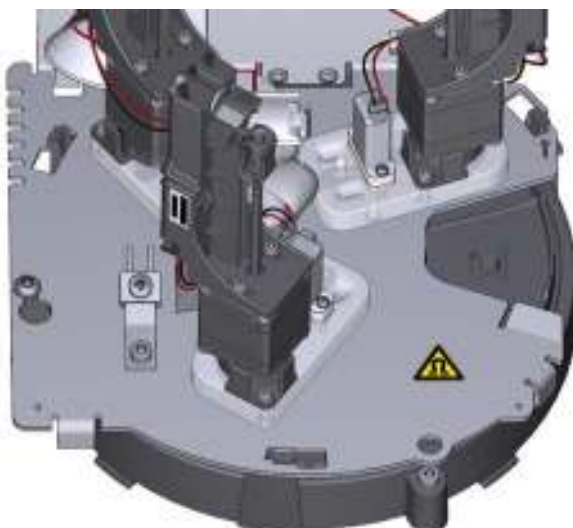


Figure 2–18: Washer Module, magnet label

## 2. Introduction for external-tanks configuration



Figure 2-19: Magnet label, Resuspension Tool

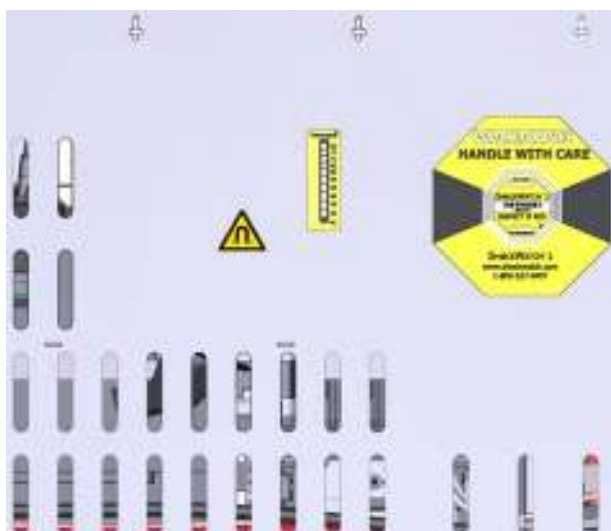


Figure 2-20: Magnet label, Instrument rear view

### 2.7.14 No heavy load labels



No heavy load labels are positioned on:

- Starter Reagent flap
- Sample Module flap
- Reagent Module flap



Figure 2-21: No heavy load label, Starter Reagent flap



Figure 2-22: No heavy load label, Sample Module flap



Figure 2-23: No heavy load label, Reagent Module flap



## 2. Introduction for external-tanks configuration

### 2.7.15 Don't step on label labels



Don't step on labels are positioned on:

- Liquid Waste Basin



Figure 2–24: Liquid Waste Basin, Don't step on labels

### 2.7.16 Connect/Disconnect both tubing and sensor label



Connect/disconnect BOTH tubing and sensor label is positioned on:

- On top of the Wash Buffer tank
- On top of the System Liquid tank
- On top of the Cleaning tank.



Figure 2–25: Connect/disconnect both tubing and sensor label on the Wash Buffer tank



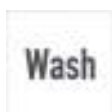


Figure 2-26: Connect/disconnect both tubing and sensor label on the System Liquid tank



Figure 2-27: Connect/disconnect both tubing and sensor label on the Cleaning tank

### 2.7.17 Wash Buffer label



Wash label is positioned on:

- At the front of the Wash Buffer tank.



Figure 2–28: Wash Buffer tank

### 2.7.18 System Liquid label



Water label is positioned on:

- At the front of the System Liquid tank.



Figure 2–29: System Liquid tank

### 2.7.19 Cleaning Solution label

Cleaning  
Solution

Cleaning solution label is positioned on:

- At the front of the Cleaning Solution tank.



Figure 2–30: Cleaning Solution tank

### 2.7.20 Waste label

WASTE

Waste label is positioned on:

- On the front of the Liquid Waste tank.



Figure 2–31: Liquid Waste tank

### 2.7.21 Shockwatch, Tiltwatch and Temperature measuring strip

Shockwatch, Tiltwatch and Measuring strip labels are positioned on:

- Rear side of the instrument

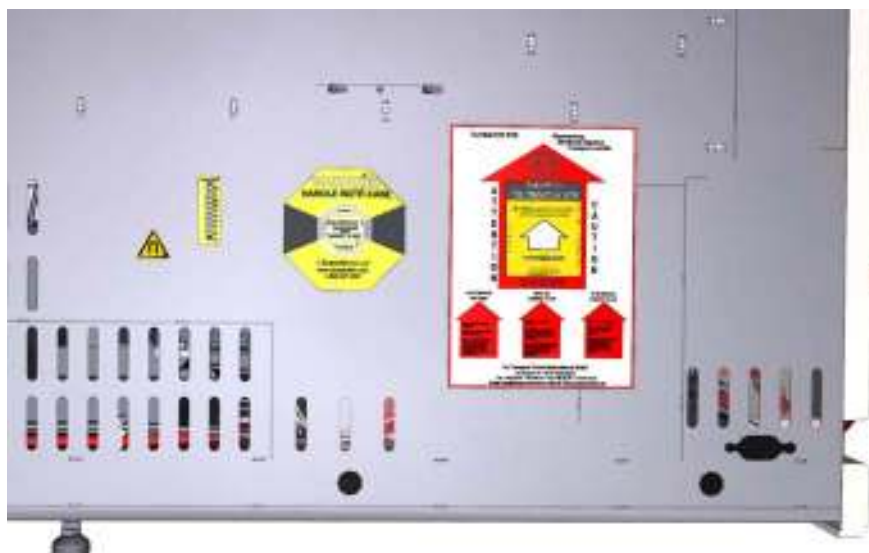


Figure 2–32: Shockwatch, Tiltwatch and Temperature measuring strip labels

## 3 Overview for internal-tanks configuration

The purpose of this section is to provide a summary of the instrument components.

### 3.1 Instrument overview

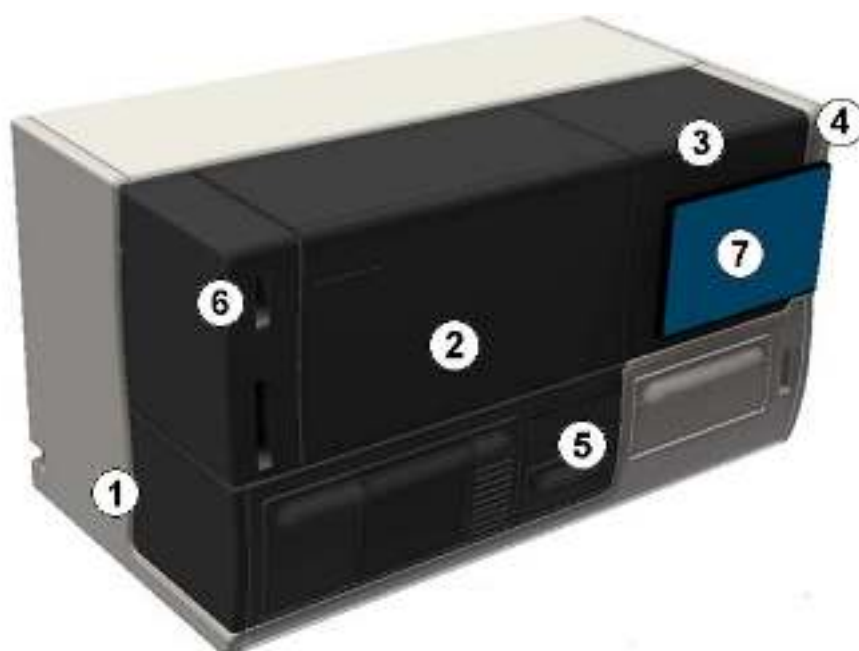


Figure 3-1: System overview

#### CAUTION

Opening the cover during system operations may result in operation interruption and not reported results.

| No. | Description                |
|-----|----------------------------|
| 1)  | Left Side Cover            |
| 2)  | Left Top Cover             |
| 3)  | Right Top Cover            |
| 4)  | Right Side Cover           |
| 5)  | Front Module               |
| 6)  | Bulk Fluid Door            |
| 7)  | Panel PC with Touch Screen |

Table 3-1: System overview

## 3. Overview for internal-tanks configuration

## 3.1.1 Front view

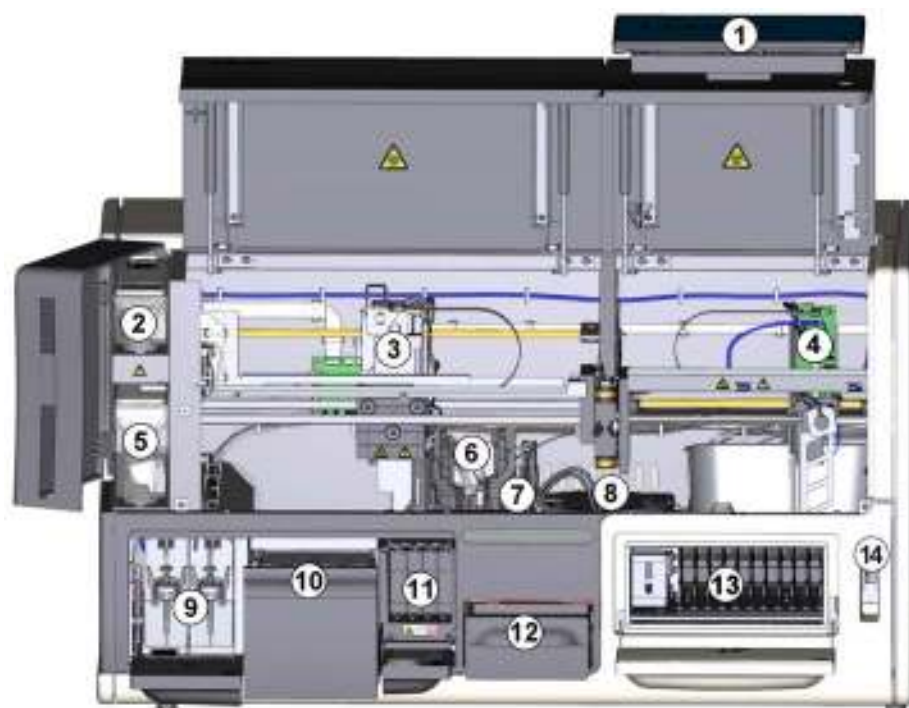


Figure 3-2: Instrument front view

| No. | Description                                           |
|-----|-------------------------------------------------------|
| 1)  | Panel PC with touch screen                            |
| 2)  | DI Water 1.5L tank                                    |
| 3)  | Left Pipettor Arm (Sample Pipettor & Cuvette Gripper) |
| 4)  | Right Pipettor Arm (Reagent Pipettor)                 |
| 5)  | Wash Buffer 5.5L tank                                 |
| 6)  | Washer Module                                         |
| 7)  | Reader Module                                         |
| 8)  | Incubator Module                                      |
| 9)  | Starter Reagent Module                                |
| 10) | Cuvette-Tip Loading Module                            |
| 11) | Sample Module                                         |
| 12) | Solid Waste Bin                                       |
| 13) | Reagent Module                                        |
| 14) | Resuspension Tool                                     |

Table 3-2: Instrument front view

### 3.1.2 Left side inner view

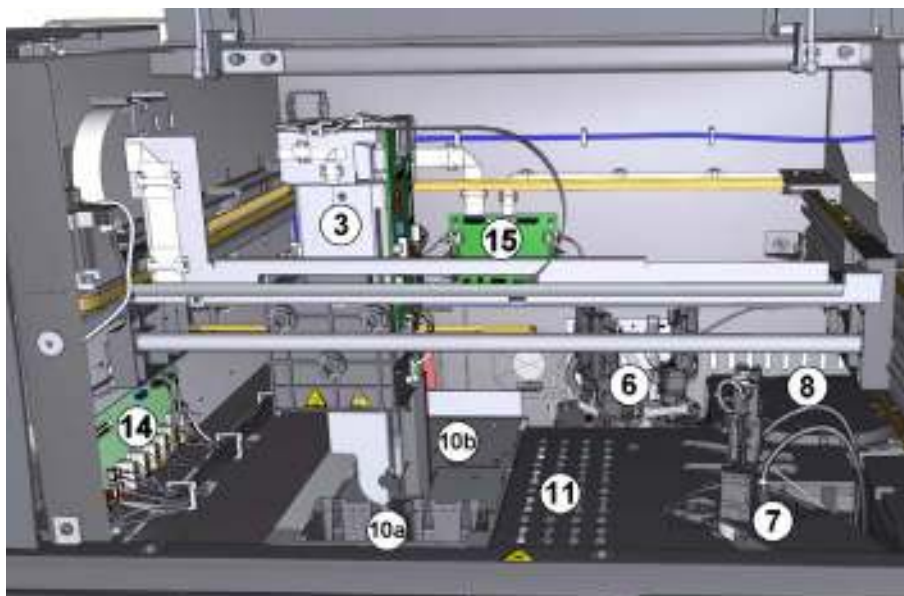


Figure 3-3: Instrument left side inner view

| No.  | Description                                           |
|------|-------------------------------------------------------|
| 3)   | Left Pipettor Arm (Sample Pipettor & Cuvette Gripper) |
| 6)   | Washer Module                                         |
| 7)   | Reader Module                                         |
| 8)   | Incubator Module                                      |
| 10a) | Tip Trays Area                                        |
| 10b) | Cuvette Trays Area                                    |
| 11)  | Sample Module                                         |
| 14)  | Liquid Support CU PCBA                                |
| 15)  | Pipettor CU PCBA                                      |

Table 3-3: Instrument left side inner view

### 3. Overview for internal-tanks configuration

#### 3.1.3 Right side inner view

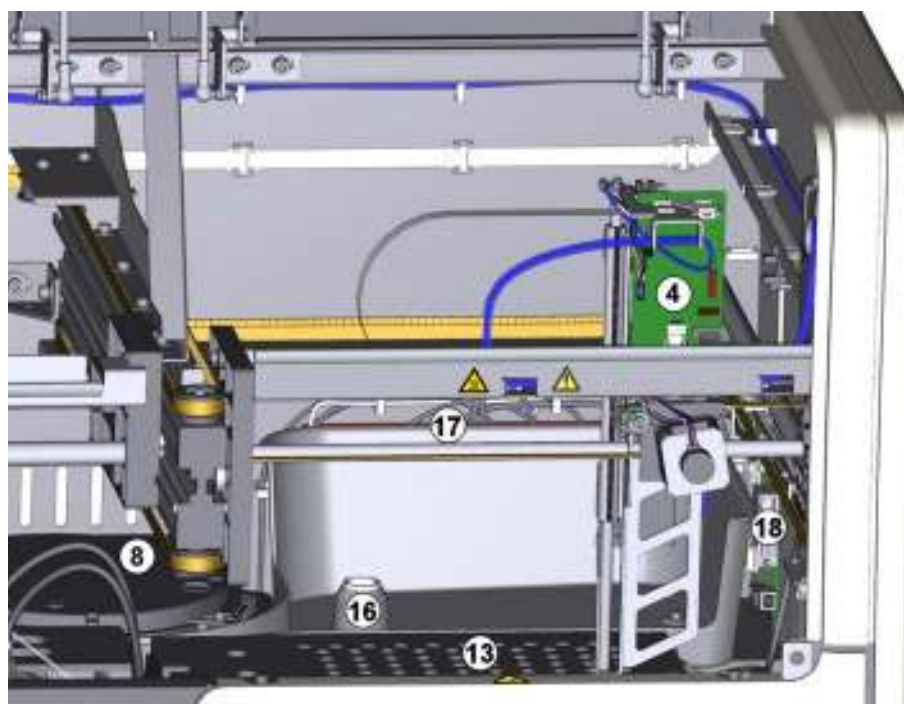


Figure 3-4: Instrument right side inner view

| No. | Description                           |
|-----|---------------------------------------|
| 4)  | Right Pipettor Arm (Reagent Pipettor) |
| 8)  | Incubator Module                      |
| 13) | Reagent Module                        |
| 16) | Pipettor Wash Station                 |
| 17) | Wash Station Cover                    |
| 18) | COP EUS2CAN PCBA                      |

Table 3-4: Instrument right side inner view



### 3.1.4 Left side view

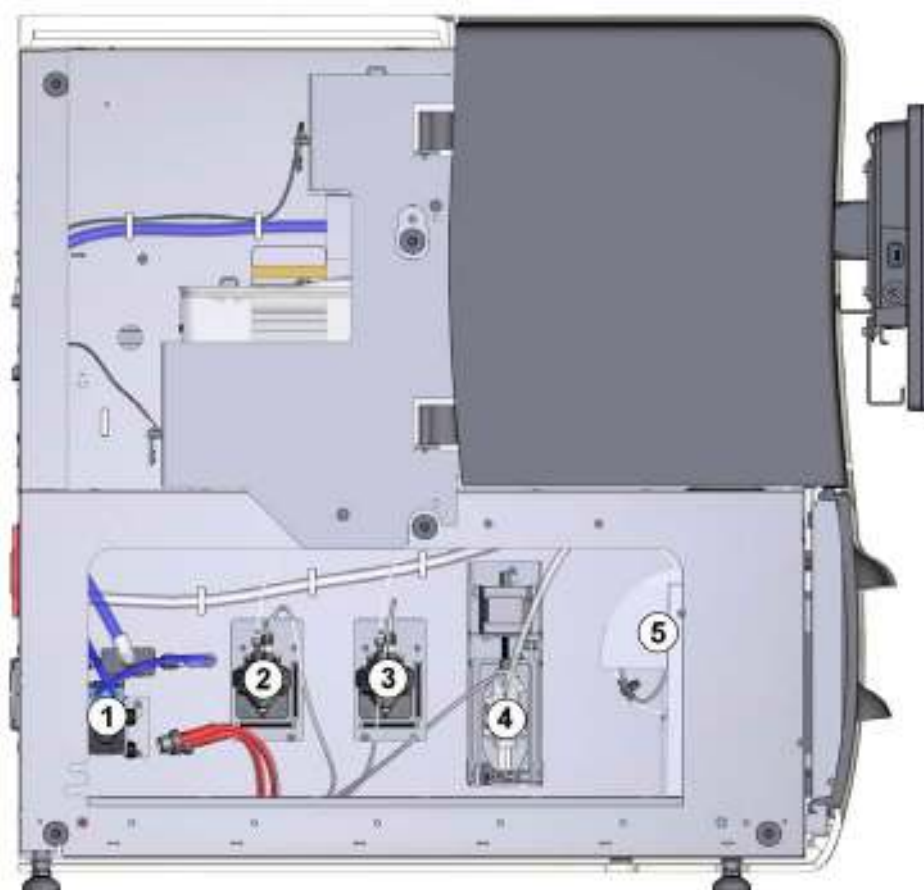


Figure 3–5: Left side view

| No. | Description                      |
|-----|----------------------------------|
| 1)  | PRP (Probe Rinsing Pump)         |
| 2)  | SDP 1 (Starter Dispense Pump)    |
| 3)  | SDP 2 (Starter Dispense Pump)    |
| 4)  | WBDP (Wash Buffer Dispense Pump) |
| 5)  | Starter Reagent Module           |

Table 3–5: Left side view

### 3. Overview for internal-tanks configuration

#### 3.1.5 Right side view



Figure 3–6: Right side view

| No. | Description       |
|-----|-------------------|
| 1)  | Soft Power Button |
| 2)  | Power Main Switch |
| 3)  | USB Ports         |
| 4)  | LAN Connector     |

Table 3–6: Right side view

## 3.1.6 Fluidics overview

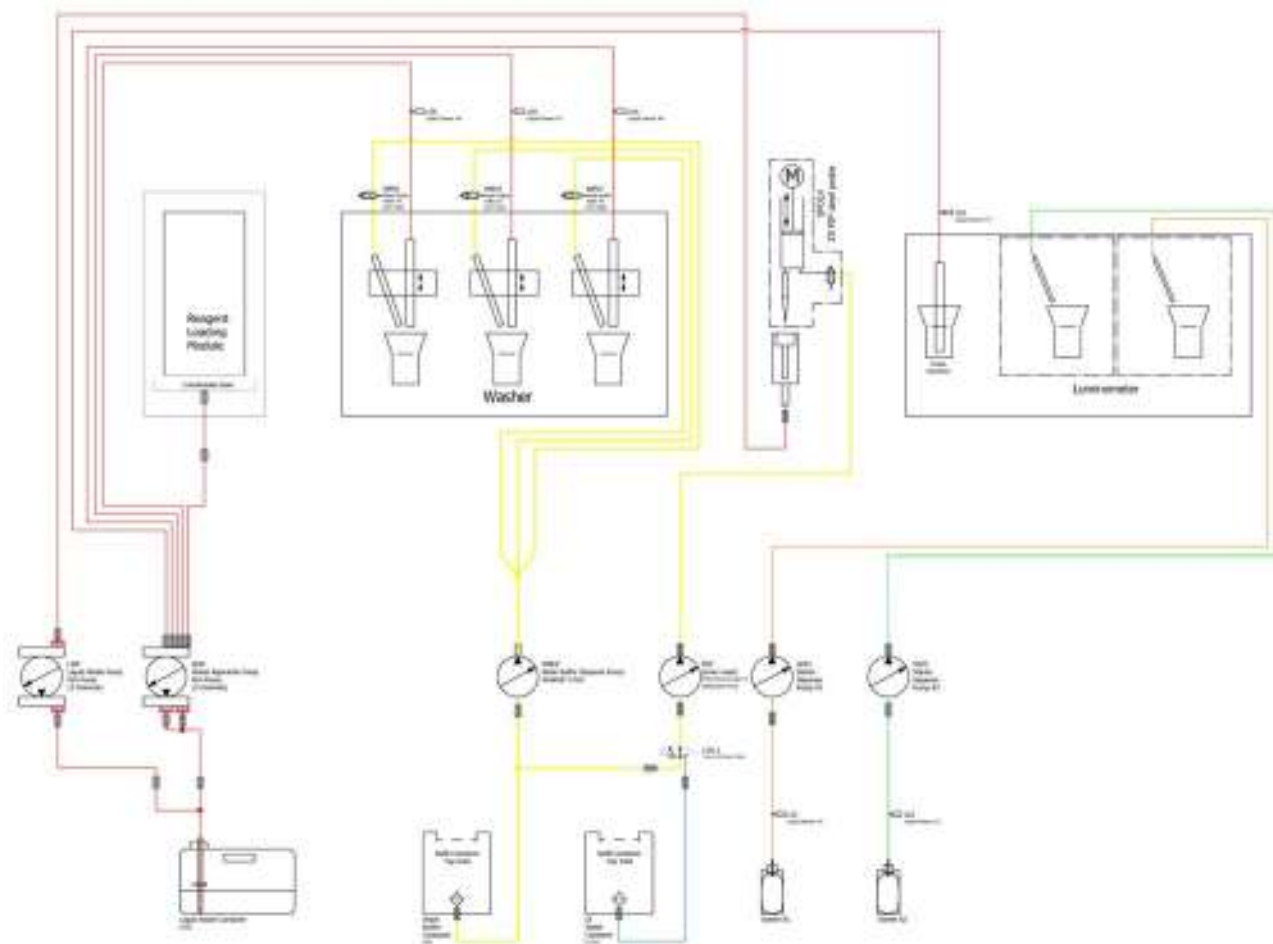


Figure 3–7: Fluidics overview

### 3. Overview for internal-tanks configuration

#### 3.1.7 Electric Circuit overview

#### NOTE

See module chapters for detailed information of the electric connections (boards, pumps, valves, sensors, switches and LEDs).

#### LIAISON XS -- 2290 -- Electronic Architecture Sheet

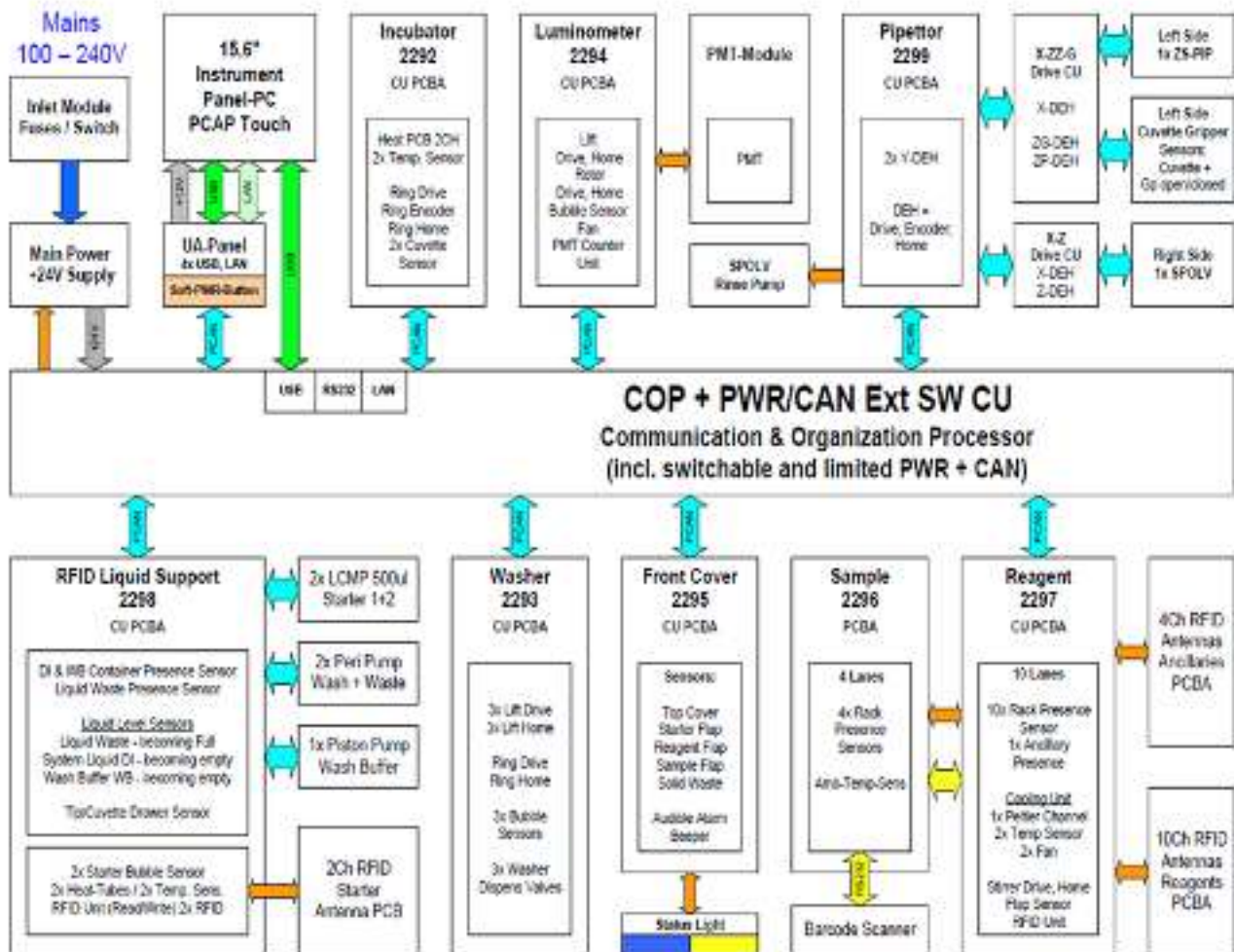


Figure 3–8: Electric Circuit overview

## 4 Overview for external-tanks configuration

The purpose of this section is to provide a summary of the instrument components.

### 4.1 Instrument overview

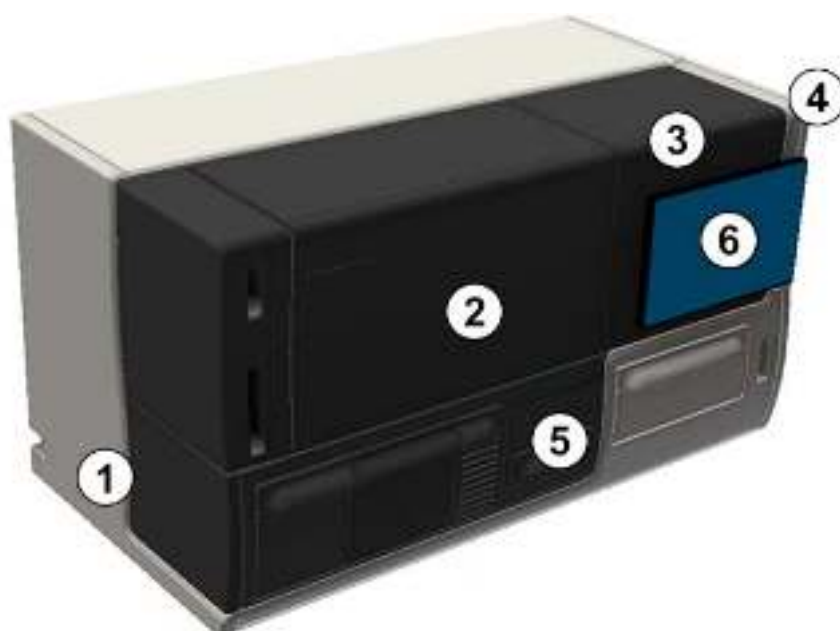


Figure 4-1: System overview

#### CAUTION

Opening the cover during system operations may result in operation interruption and not reported results.

| No. | Description                |
|-----|----------------------------|
| 1)  | Left Side Cover            |
| 2)  | Left Top Cover             |
| 3)  | Right Top Cover            |
| 4)  | Right Side Cover           |
| 5)  | Front Module               |
| 6)  | Panel PC with Touch Screen |

Table 4-1: System overview

## 4. Overview for external-tanks configuration

## 4.1.1 Front view

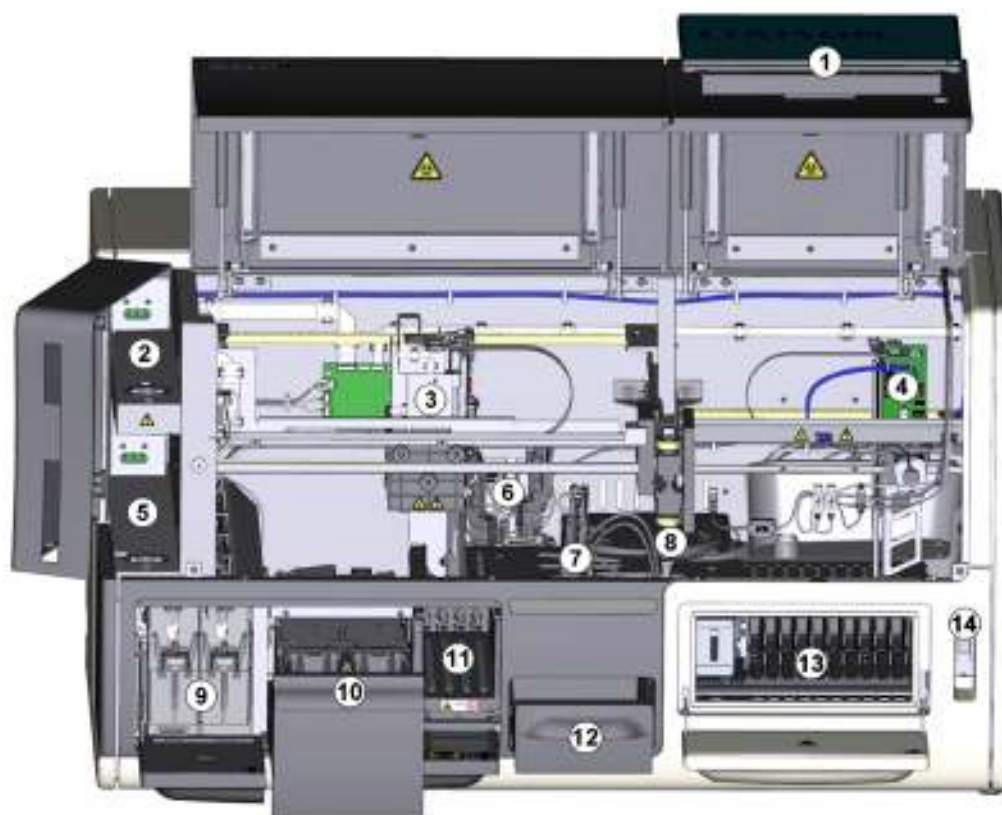


Figure 4-2: Instrument front view

| No. | Description                                           |
|-----|-------------------------------------------------------|
| 1)  | Panel PC with touch screen                            |
| 2)  | Not used                                              |
| 3)  | Left Pipettor Arm (Sample Pipettor & Cuvette Gripper) |
| 4)  | Right Pipettor Arm (Reagent Pipettor)                 |
| 5)  | Not used                                              |
| 6)  | Washer Module                                         |
| 7)  | Reader Module                                         |
| 8)  | Incubator Module                                      |
| 9)  | Starter Reagent Module                                |
| 10) | Cuvette-Tip Loading Module                            |
| 11) | Sample Module                                         |
| 12) | Solid Waste Bin                                       |
| 13) | Reagent Module                                        |
| 14) | Resuspension Tool                                     |

Table 4-2: Instrument front view

### 4.1.2 Left side inner view

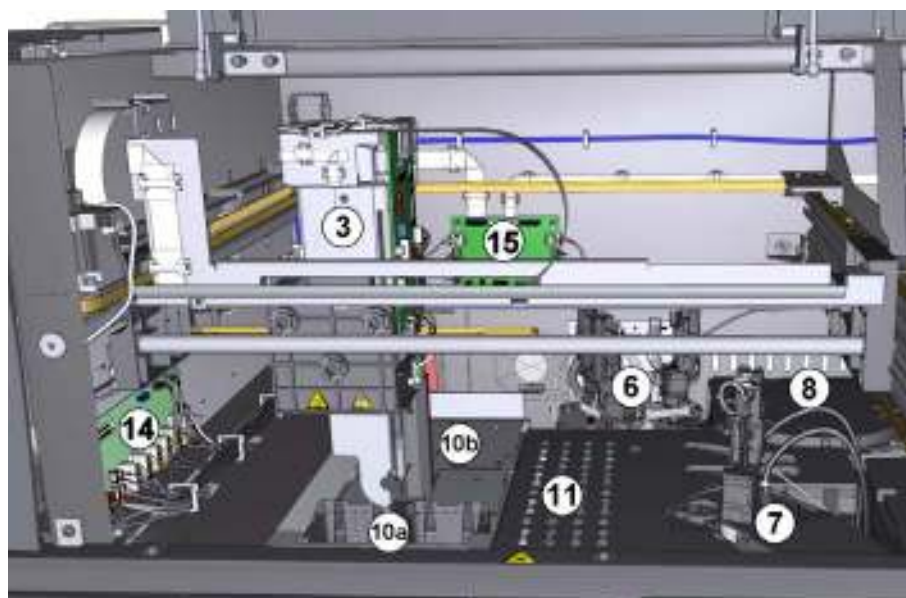


Figure 4-3: Instrument left side inner view

| No.  | Description                                           |
|------|-------------------------------------------------------|
| 3)   | Left Pipettor Arm (Sample Pipettor & Cuvette Gripper) |
| 6)   | Washer Module                                         |
| 7)   | Reader Module                                         |
| 8)   | Incubator Module                                      |
| 10a) | Tip Trays Area                                        |
| 10b) | Cuvette Trays Area                                    |
| 11)  | Sample Module                                         |
| 14)  | Liquid Support CU PCBA                                |
| 15)  | Pipettor CU PCBA                                      |

Table 4-3: Instrument left side inner view



## 4. Overview for external-tanks configuration

### 4.1.3 Right side inner view

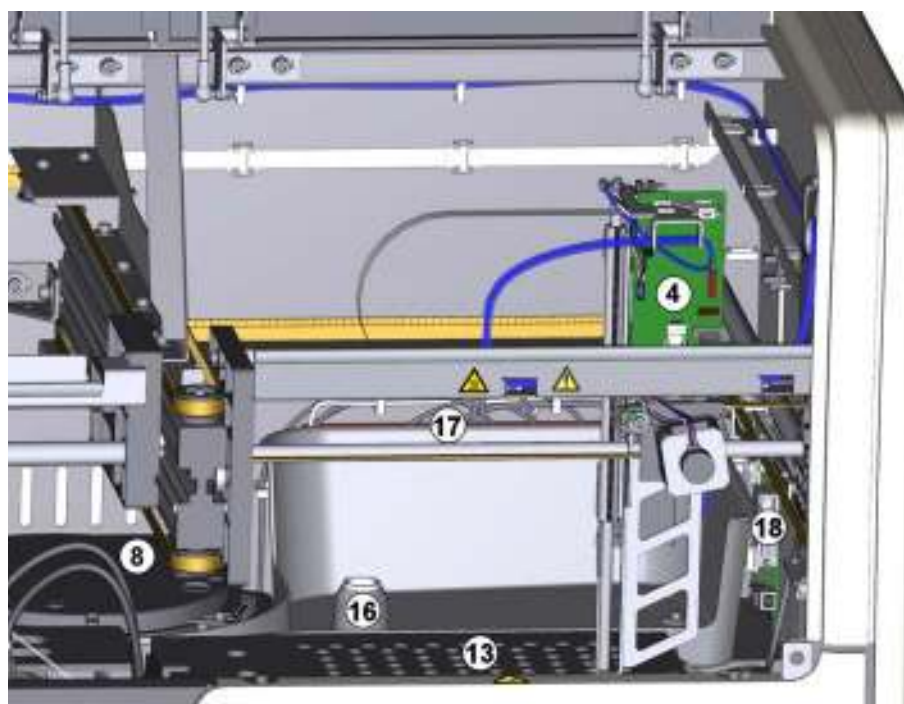


Figure 4–4: Instrument right side inner view

| No. | Description                           |
|-----|---------------------------------------|
| 4)  | Right Pipettor Arm (Reagent Pipettor) |
| 8)  | Incubator Module                      |
| 13) | Reagent Module                        |
| 16) | Pipettor Wash Station                 |
| 17) | Wash Station Cover                    |
| 18) | COP EUS2CAN PCBA                      |

Table 4–4: Instrument right side inner view



#### 4.1.4 Left side view

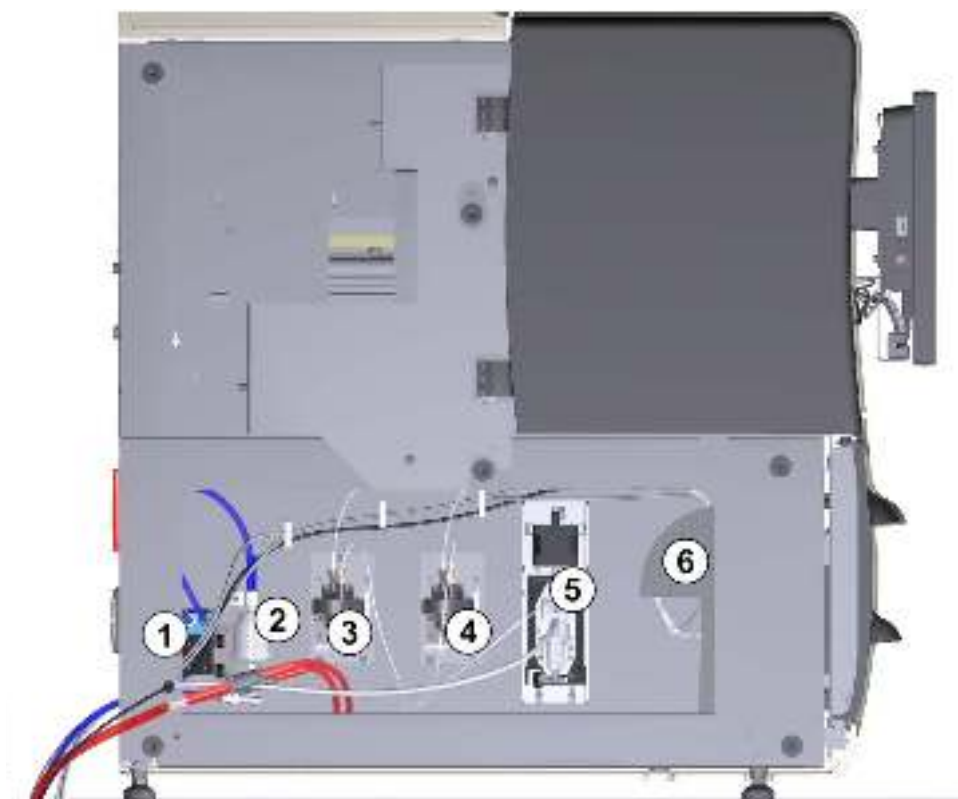


Figure 4–5: Left side view

| No. | Description                      |
|-----|----------------------------------|
| 1)  | PRP (Probe Rinsing Pump)         |
| 2)  | Mann filter                      |
| 3)  | SDP 1 (Starter Dispense Pump)    |
| 4)  | SDP 2 (Starter Dispense Pump)    |
| 5)  | WBDP (Wash Buffer Dispense Pump) |
| 6)  | Starter Reagent Module           |

Table 4–5: Left side view

4. Overview for external-tanks configuration

4.1.5 Right side view



Figure 4–6: Right side view

| No. | Description       |
|-----|-------------------|
| 1)  | Soft Power Button |
| 2)  | Power Main Switch |
| 3)  | USB Ports         |
| 4)  | LAN Connector     |

Table 4–6: Right side view

### 4.1.6 Liquid tanks view

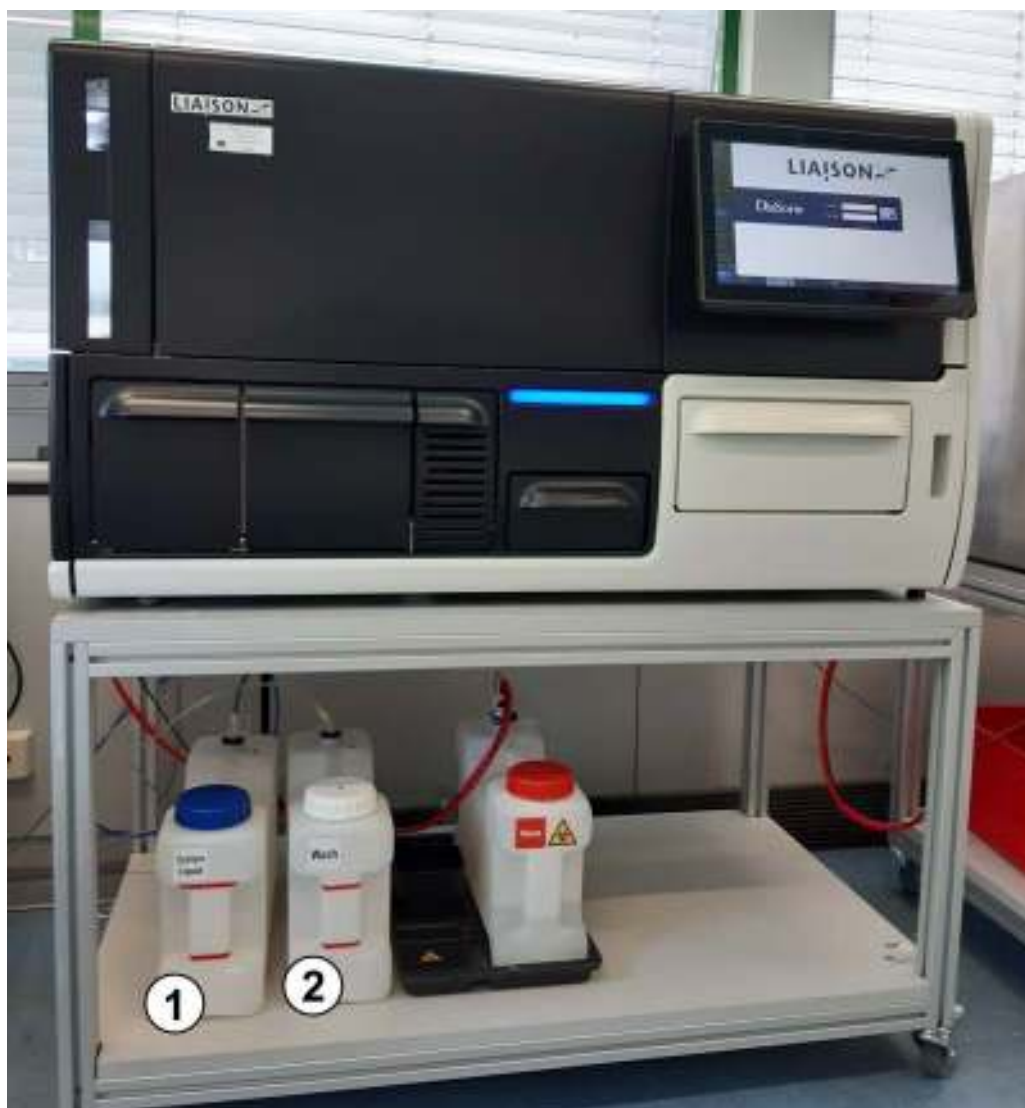


Figure 4-7: Liquid tanks overview

| No. | Description        |
|-----|--------------------|
| 1)  | System Liquid tank |
| 2)  | Wash Buffer tank   |

Table 4-7: Liquid tanks view

## 4. Overview for external-tanks configuration

## 4.1.7 Fluidics overview

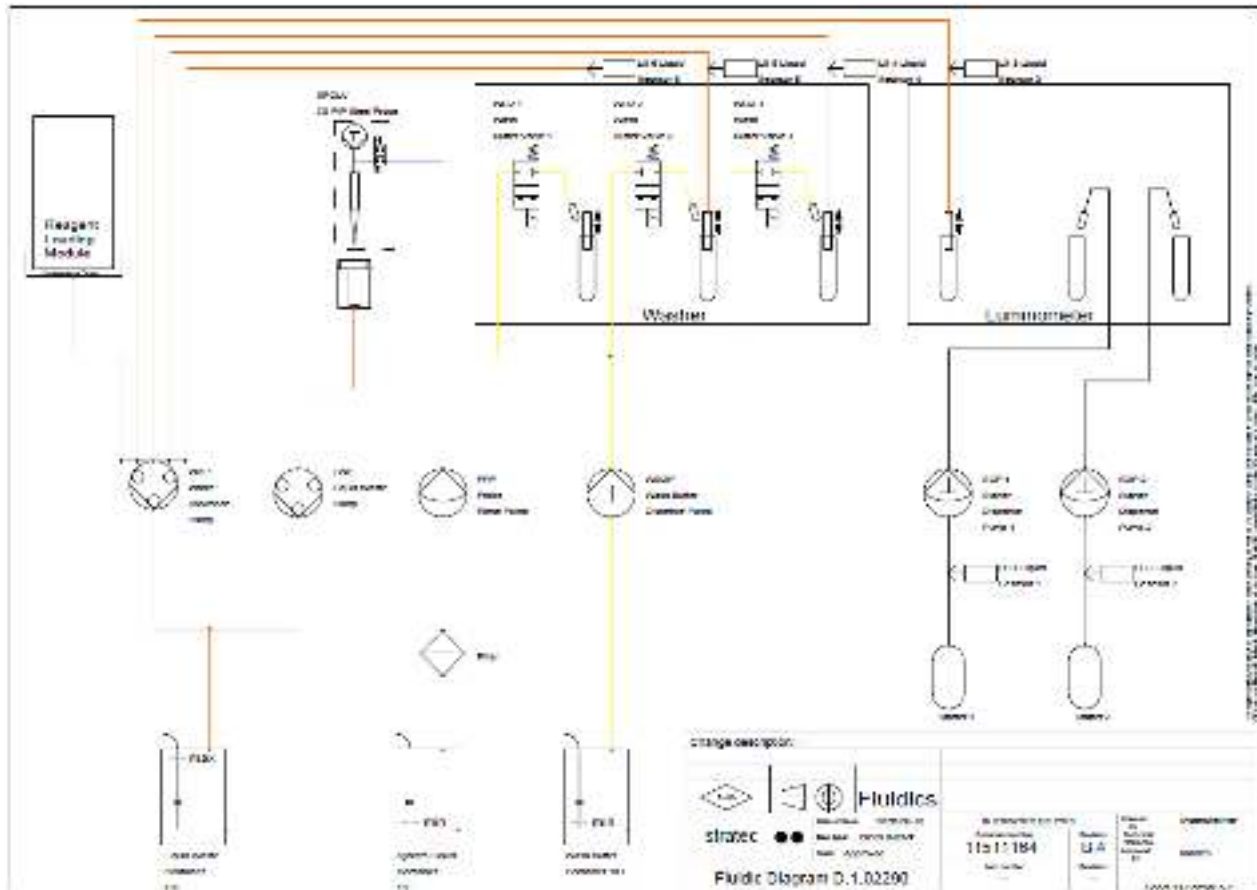


Figure 4–8: Fluidics overview

## 4.1.8 Electric Circuit overview

**NOTE**

See module chapters for detailed information of the electric connections (boards, pumps, valves, sensors, switches and LEDs).

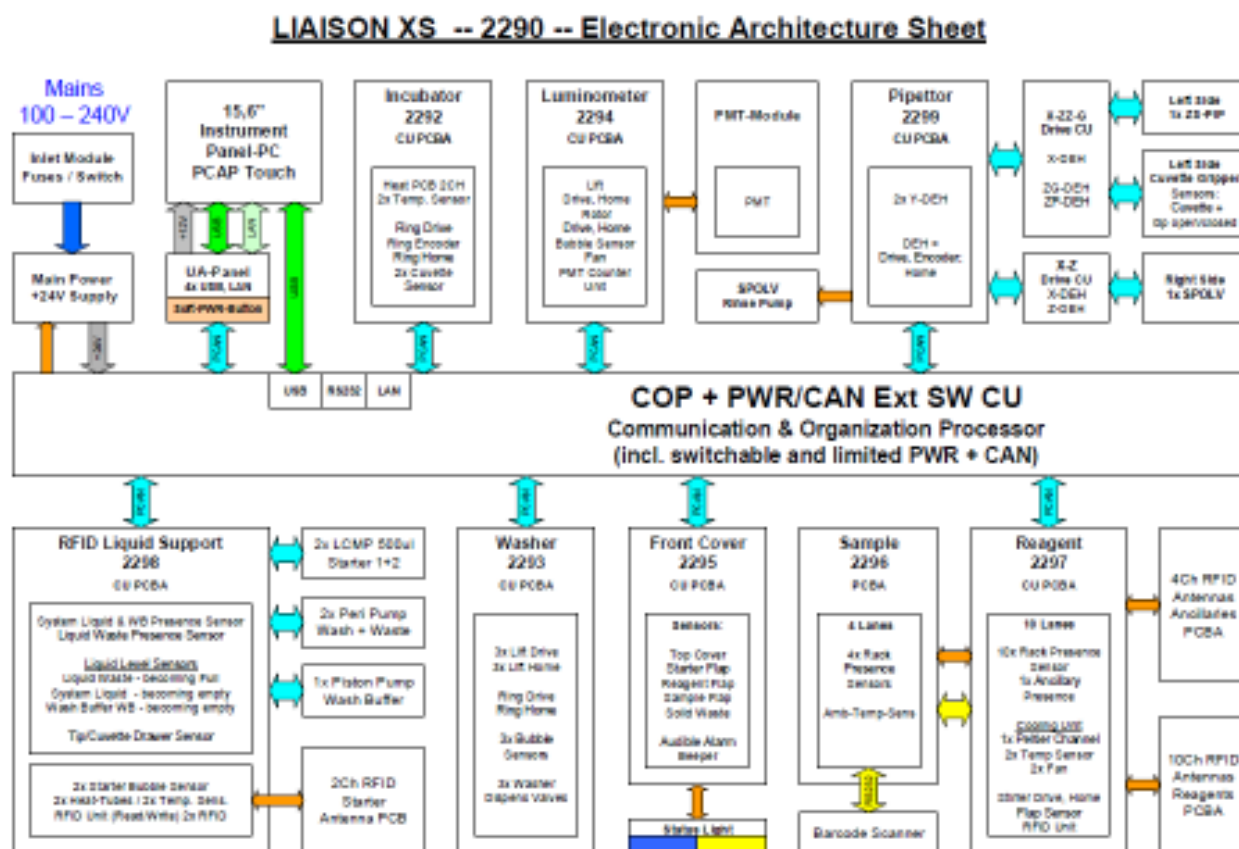


Figure 4–9: Electric Circuit overview

#### 4. Overview for external-tanks configuration

---

# 5 LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

As the **LIAISON<sup>®</sup>XS** instrument is a complex instrument and contains delicate electrical and mechanical components, it must be unpacked and installed by a qualified Service Engineer.

The procedures reported in this chapter allow to complete the installation of **LIAISON<sup>®</sup>XS** analyzers successfully.

## 5.1 Safety

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7.
  - Follow all working instructions included in this manual.
  - Follow all warnings marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
- 

### WARNING



#### Falling down of the system

Due to improper place of installation of the system, the system can fall down. If the system falls down, the user can be seriously injured or the system can be seriously damaged. Position the system on a stable platform, cart, stand or table, see Chapter 35.4 on page 35-3.

- Observe the technical specifications of the system, see Chapter 35 on page 35-1.
- 

### WARNING



#### Never lift the system by yourself!

Manual handling of heavy loads may cause injuries.

The **LIAISON<sup>®</sup>XS** instrument shall NOT be lifted and carried by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

---

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

---

**CAUTION****Check System**

During the installation of the system, always follow the steps indicated in **LIAISON<sup>®</sup>XS** Installation Qualification (IQ, see Chapter 33.1 on page 33-1) and Operational Qualification (OQ, see Chapter 33.2 on page 33-4) check lists. The proper execution and compilation of these check lists will ensure the proper installation of the instruments.

---



## 5.2 Installation preparation

**NOTE****Pre-Site Inspection visit for installation**

A Pre-Site Inspection form has to be used during the preliminary visit at the customer site in order to evaluate the customer facilities and the laboratory accessibility and requirement for a proper instrument installation. See Corp-TN-000294 (former LXS TN 011) for details.

---

### 5.2.1 Standard supply with PN I0090

- Instrument with cables (schuko plug 220 Volt)
- N° 1 Front Module
- N° 1 Left Top Cover
- N° 1 Right Top Cover
- N° 1 Left Side Cover
- N° 1 Right Side Cover
- N° 1 Bulk Fluid Door
- N° 1 Top Cover
- Panel PC
- N° 1 DI Water 1.5L tank
- N° 2 Wash Buffer solution 5.5L tanks
- N° 1 Cleaning solution 1.5L tank
- N° 1 Liquid waste tank
- N° 1 Solid waste bin
- N° 1 Cuvette - Tip Loading Module handle
- N° 3 Pump caps for Peristaltic pumps
- N° 1 Ancillary tray
- N° 5 Sample racks
- N°1 Label Set for Sample Rack
- Sample Rack Front Handle Labels
- N° 1 Velcro strip for USB stick
- N° 1 PN 10136280 - TANK FILLING STAND
- N° 1 PN A0179 - EXTERNAL BARCODE SCANNER or N° 1 PN A0256 - EXTERNAL BARCODE SCANNER
- N° 1 PN A0128 - INTEGRAL RACK HOLDER
- N° 1 PN A0120 - SAMPLE RACK HOLDER LIAISON XL/XS (1 NU)
- N° 1 PN A0055 - SAMPLE TUBE TOOL
- N° 1 PN A0118 - SAMPLE RACK ADAPTER - SET 60 (1 NC)
- N° 1 PN A0107 - ANCILLARY ADAPTOR FOR LC (SET OF 2)
- N° 1 PN A0243 - XS CONTROL RACK 8 POSITIONS (QTY 2)
- N° 1 PN X0054 - LIAISON<sup>®</sup> EASY Waste (QTY 10)
- N° 1 PN A0244 - XS LIQUID WASTE BASIN

## 5. LIAISON®XS Installation for internal-tanks configuration

- N° 1 PN A0251 - XS BACKUP USB STICK 16GB

### NOTE

Other accessories, such as:

- Printer (PN A0240, or newer, or locally validated)
- UPS Global 30 (PN I0078)
- XS Direct Drain kit (10127604)
- Smart Table (PN A0245)

are NOT delivered by standard supply and, if required, need to be ordered and managed separately (see Corp-TN-000294 - former LXS TN 011).

### 5.2.2 Additional material required for installation

| Product Code                              | Quantity | Description               |
|-------------------------------------------|----------|---------------------------|
| 319150                                    | 1        | LIAISON Light Check 12    |
| X0053                                     | 1        | LIAISON Cuvettes on trays |
| X0055                                     | 1        | LIAISON Disposable Tips   |
| 319300                                    | 1        | LIAISON EASY Starter Kit  |
| 319301                                    | 1        | LIAISON EASY Wash Buffer  |
| see Corp-TN-000292<br>(former LXS TN 009) | 1        | LIAISON Assay             |
| see Corp-TN-000292<br>(former LXS TN 009) | 1        | LIAISON Control for Assay |

Table 5–1: Additional material required for installation

### NOTE

The above listed material is sufficient for the Field Service Engineer installation only and MUST be ordered separately.

Anything required for the Customer Training, Start up, Routine (Assays, Controls, additional Consumables, etc.), MUST be ordered separately as well.

The Assay and related control to be ordered depends on the Country where the instrument will be installed (refer to Corp-TN-000292 - former LXS TN 009).

### 5.2.3 Required tools

The following tools are required to perform installation:

- Set of hexagon socket wrench (Allen key)
- Set of cross slotted screw driver
- Set of flat screw driver
- Set of torx screw driver
- Set of torx L-keys

- Adjustable wrench
- Cutter
- Magnetic work light
- Ruler (at least 200mm-8inches)
- Keyboard (handheld suggested)

## 5.3 Installation Qualification

### NOTE

---

#### Packaging

Keep the packaging to allow for safe transportation in case the system must be shipped in the future.

---

#### 5.3.1 Visual inspection before unpacking

1. Verify the transport box is without damages.
2. Verify the indication of shock-watch (1) and thermometer sensor (2).

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration



Figure 5-1: Shock-watch and thermometer sensors

### 5.3.2 Unpacking, visual inspection and inventory

1. Check the completeness of delivery by means of the packing list, see Chapter 37.1 on page 37-1.
2. Remove Accessories and Covers box from the top of instrument box.
3. Open the Accessory Box.
4. Check the Accessory Box parts according to the packing list, see Chapter 37.1 on page 37-1.
5. Visual inspect that the Covers are without damages or gaps.
6. Open the instrument box, removing its lid.
7. Lift cardboard box up to the top and remove it.
8. Remove the C-foam piece from both sides around the bottom of the instrument.
9. Remove the shipping foil.
10. Remove the foam cover from the top of the instrument.
11. Visually inspect that the chassis of the **LIAISON<sup>®</sup> XS** instrument is without damages or gaps.
12. Visually inspect that no visible cuts are present in the system or pipetting tubing.

---

**5. LIAISON<sup>®</sup> XS Installation for internal-tanks configuration**

---

13. Raise the adjustable feet by 1 cm using the nuts, before moving the system from the shipping palette.

**WARNING****Never lift the system by yourself!**

Manual handling of heavy loads may cause injuries.

The **LIAISON<sup>®</sup> XS** instrument shall NOT be lifted and carried by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

---

14. Move the instrument from the shipping palette and position it on a stable platform, cart, stand or table, see Chapter 35.4 on page 35-3.
15. Verify the indication of the tilt- and shock-watch at the rear chassis.



*Figure 5-2: Unpacking of instrument*

**NOTE****System clearance**

Allow ample clearance between the system and its surroundings as shown below.

---

## 5. LIAISON<sup>®</sup> XS Installation for internal-tanks configuration

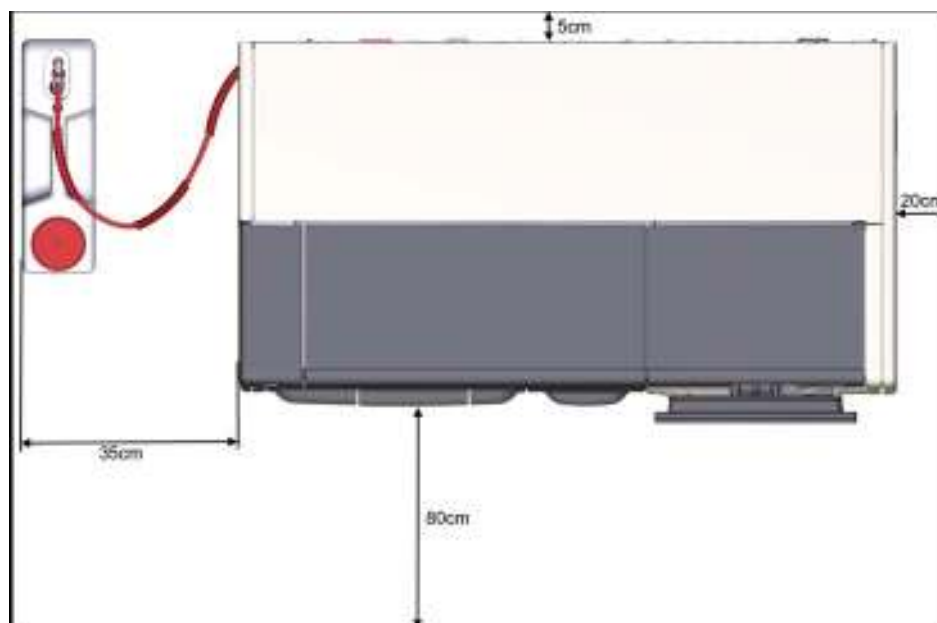


Figure 5-3: System clearance, top view

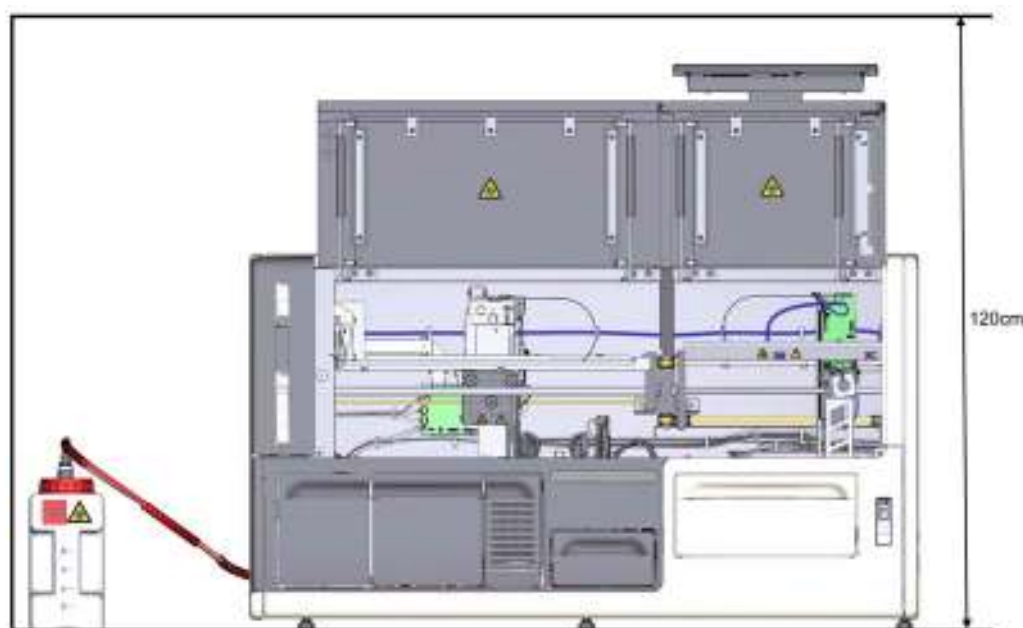


Figure 5-4: System clearance, front view

### 5.3.2.1 System leveling and stabilization

1. Adjust the instrument by leveling the adjustable feet (2) using the nuts until the level (1) at the Reagent Module is in adjustment.



Figure 5-5: Instrument level at the Reagent Module

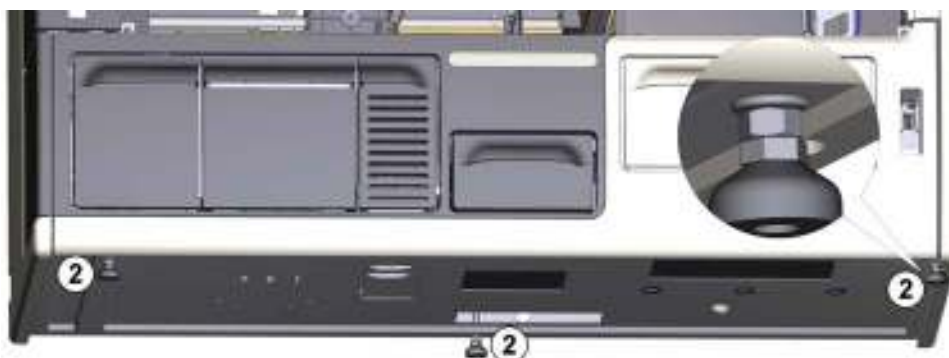


Figure 5-6: Adjustable feet of the instrument

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

---

### 5.3.2.2 Remove securing foam tubes and cable ties at Pipettor Arms

1. Carefully cut and remove the cable tie (1) at the Gripper Module.



*Figure 5–7: Cable tie at the Gripper Module*



## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

2. Remove the foam tubes at the Left and Right Pipettor Arms.



Figure 5-8: Left Pipettor Arm, front y-axis



Figure 5-9: Left Pipettor Arm, rear y-axis

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

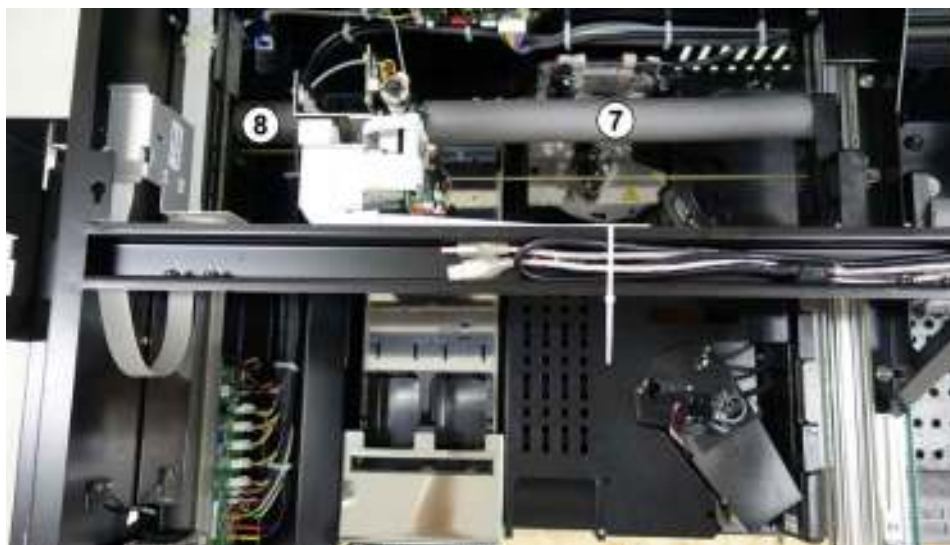


Figure 5-10: Left Pipettor Arm, right x-axis

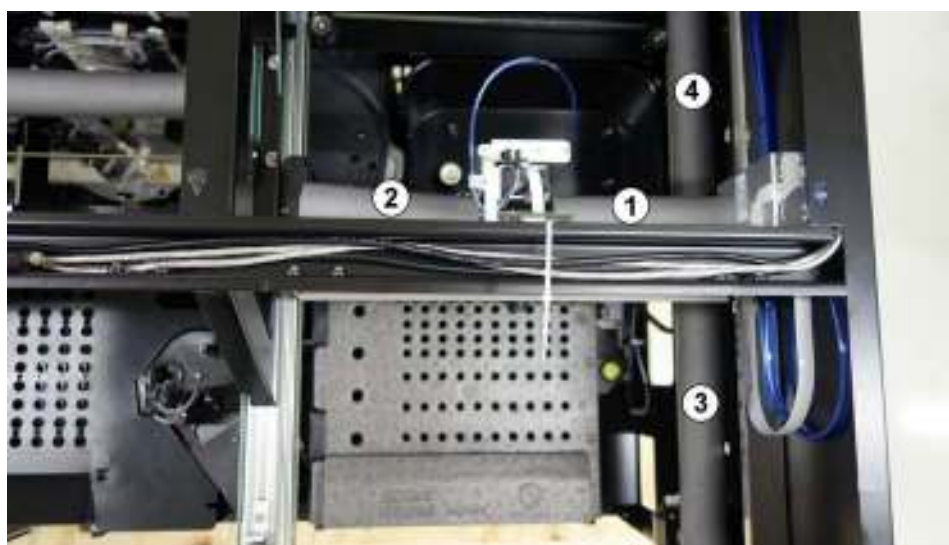


Figure 5-11: Right Pipettor Arm, x-axis and y-axis

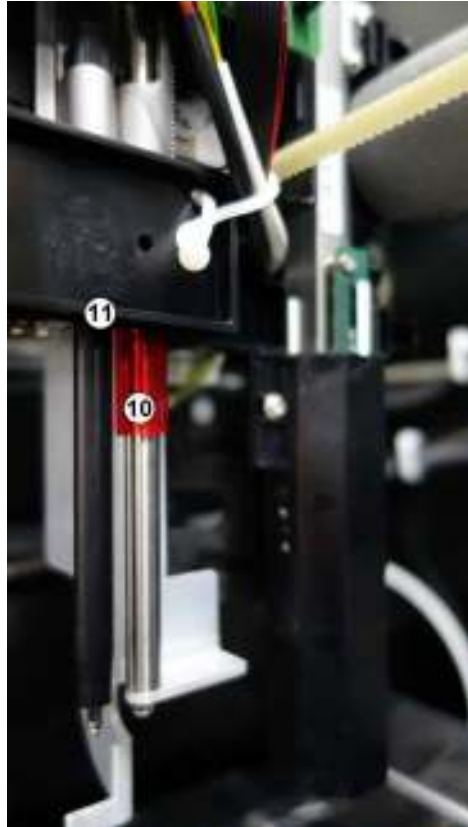
3. Remove securing tube (10) at z-axis of the Sample pipettor (11).

---

**5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration**

---

4. Remove securing tube (12) at z-axis of the SPOLV (13).



*Figure 5-12: Sample pipettor*

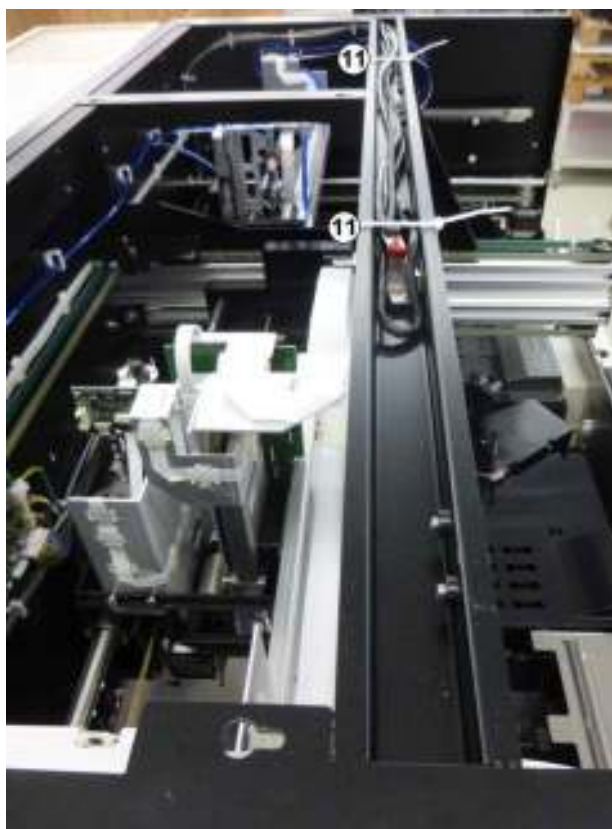


*Figure 5-13: SPOLV*

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

---

5. Cut the cable tie (11) and unwind the cables of the Panel PC at the support bar.



*Figure 5-14: Support bar*

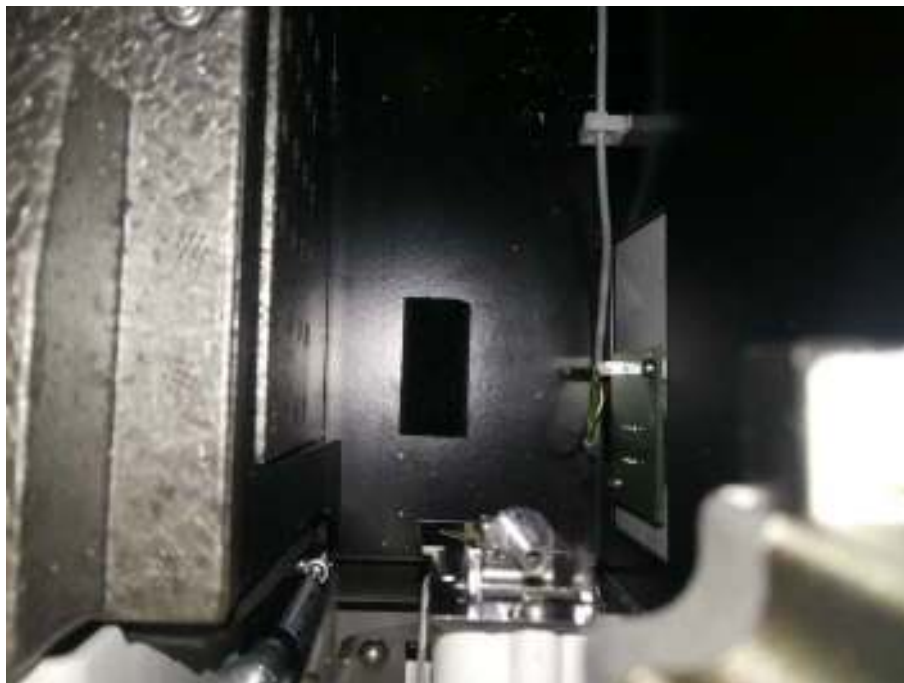
### 5.3.3 Manual inspection

1. Verify that both the Pipetting Arms are gliding smooth and without force in x, y and z direction.

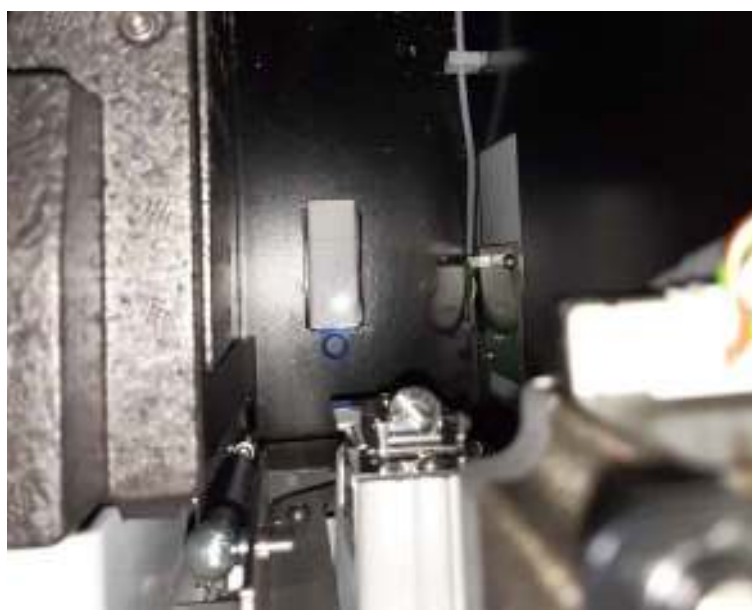
### 5.3.4 System set-up

#### 5.3.4.1 Velcro strip for USB stick set up

1. Place the velcro strip for the USB stick at the bottom of the chassis, next to the right side of the Reagent Module.
2. Place the USB stick.



*Figure 5-15: Velcro strip positioning*



*Figure 5-16: USB stick positioning*



## 5. LIAISON®XS Installation for internal-tanks configuration

---

### 5.3.4.2 LWP and WAP set up

1. Plug the peristaltic tubes into the LWP and WAP pumps.
2. Set and close the caps onto the LWP (1) and WAP (2) pumps.



*Figure 5–17: Peristaltic pumps without caps*

**5.3.4.3 Tanks set up**

1. Rinse all the tanks with DI water (DI water tank, Wash Buffer tank, Cleaning solution tank).
2. Fill DI water tank with 1.5L of DI water and put it into the instrument.
3. Fill Wash Buffer tank according to the IFU and put it into the instrument.
4. Unpack Liquid Waste tubing and slide the tube and the cable (1) into the cutout (rear left side of instrument).

**NOTE**

Be aware that both the cable and the tubing have a dedicated housing in the frame.

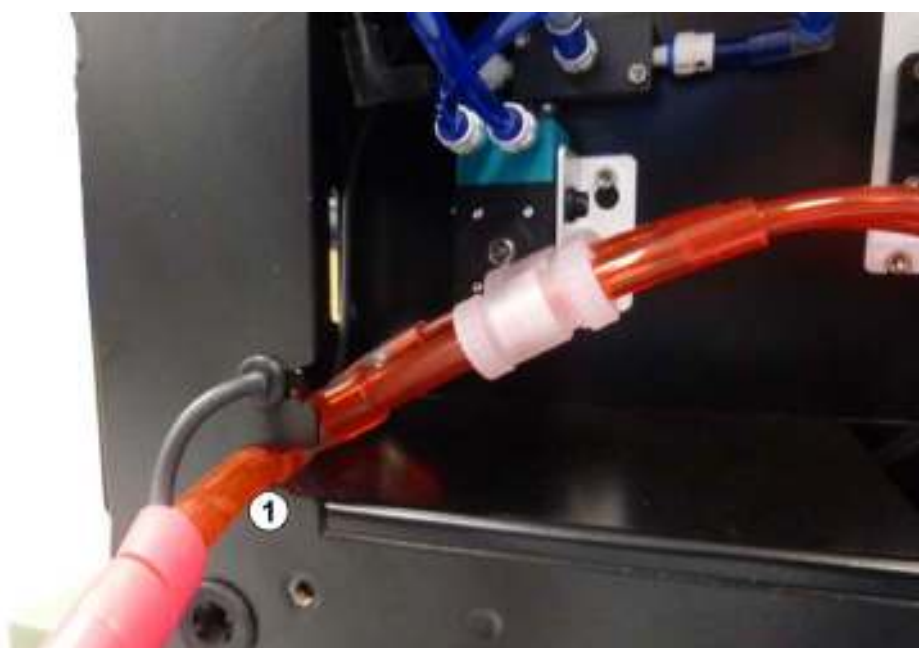
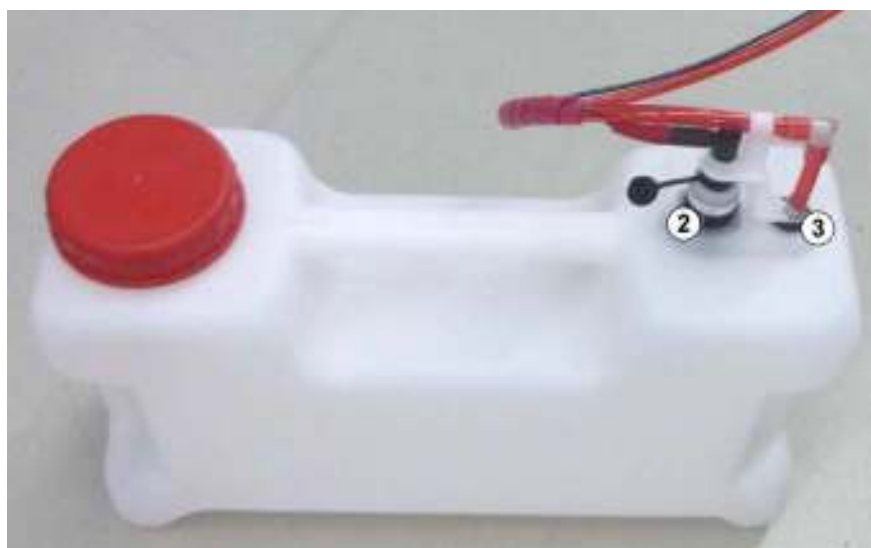


Figure 5–18: Waste tube layout

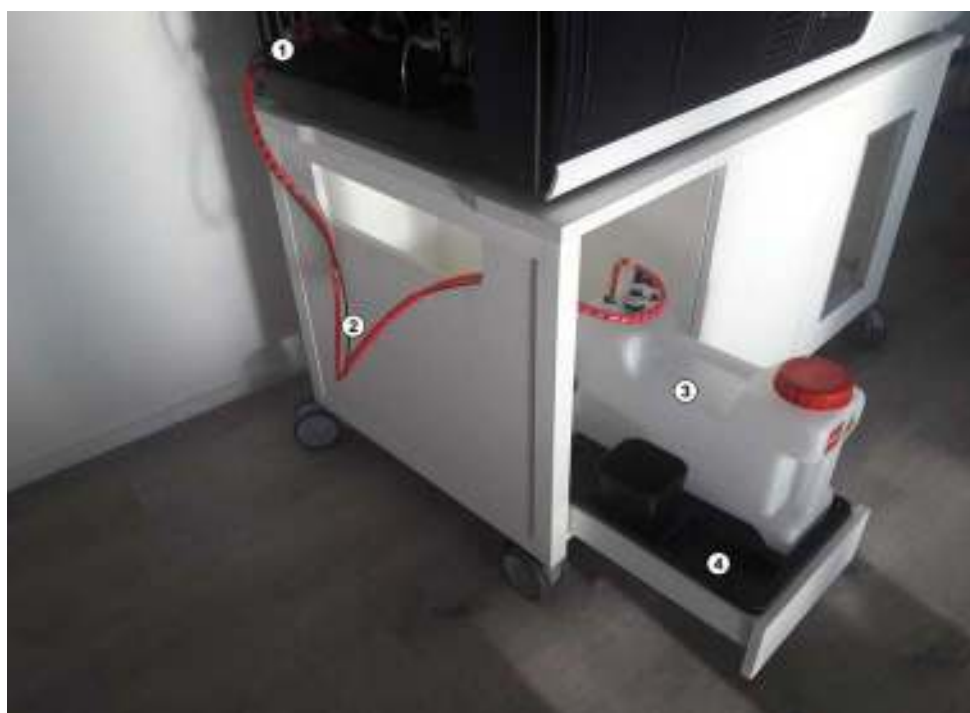
## 5. LIAISON®XS Installation for internal-tanks configuration

5. Connect the tubing (2) and the sensor cable (3) to the Liquid waste tank.



*Figure 5-19: Liquid Waste tank connection*

6. Place the Liquid waste tank (3) in the Liquid waste basin (4).



*Figure 5-20: Liquid Waste tubing connected to the Liquid Waste tank*



**Wrong placement of the Liquid waste tank**

Too high installation of the Liquid waste tank can lead to a liquid drawback.

- It is preferred to place the Liquid waste tank below the instrument.

**5.3.4.4 Assemble Covers**

1. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
2. Assemble the Top cover, see Chapter 14.3.7.3 on page 14-16.
3. Assemble the Left Top cover, see Chapter 14.3.5.3 on page 14-12.
4. Assemble the Right top cover, see Chapter 14.3.6.3 on page 14-14.
5. Assemble the Panel PC, see Chapter 14.3.4.3 on page 14-11.
6. Check the proper position and installation of the Ferrite snap, see Chapter 14.3.4.3 on page 14-11.

**NOTE**

**Ferrite snap is delivered in a box inside the Panel PC box.**



Figure 5-21: Box with the Ferrite snap inside the Panel PC box



Figure 5-22: Open Ferrite snap

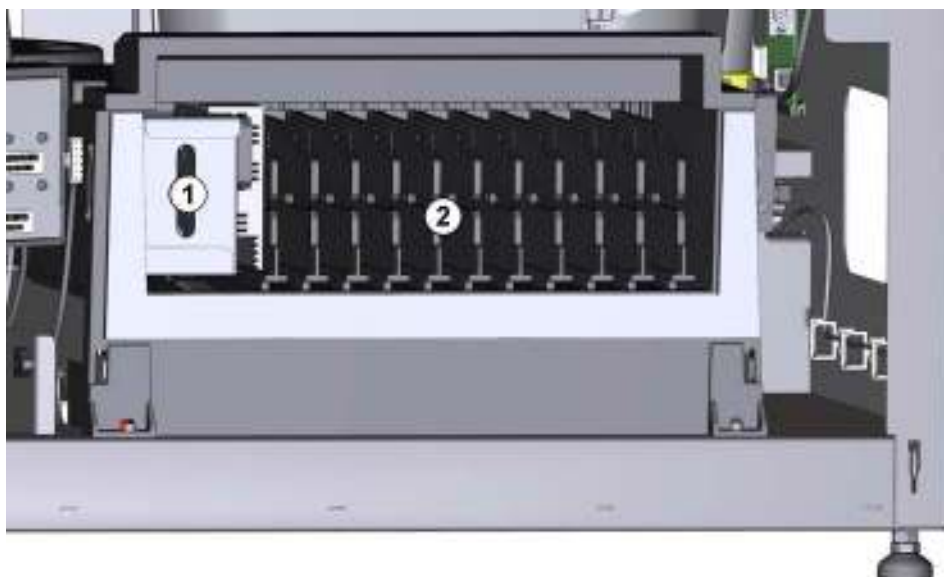
7. Check the alignment of the Covers, see Chapter 14.4 on page 14-20.
8. Assemble the Bulk Fluid Door, see Chapter 14.3.2.3 on page 14-6.
9. Assemble the Left Side cover, see Chapter 14.3.1.4 on page 14-5.
10. Assemble the Right Side cover, see Chapter 14.3.3.4 on page 14-9.
11. Check the system leveling, see Chapter 5.3.2.1 on page 5-9.
12. Re-tension the Y belt of both Left and Right pipettor Arms (see Chapter 19.3.11.4 on page 19-41 and see Chapter 19.3.20.4 on page 19-65).
  - a. For the Left Pipettor Arm: loosen the screw of the toothed belt wheel, let the spring expand to tension the belt and then fasten the screw. Move the Left

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

- Pipettor Arm forward and back a couple of times, to check that the pipettor arm is parallel. Fasten the screw at the belt holder.
- b. For the Right Pipettor Arm: loosen the screw of the toothed belt wheel, let the spring expand to tension the belt and then fasten the screw. Move the Right Pipettor Arm forward and back a couple of times, to check that the pipettor arm is parallel.
13. Remove the protective foil from the Status light.

### 5.3.4.5 Insert tray and set the Cuvette - Tip Loading Module

1. Insert Ancillary tray (1) into the Reagent Module (2).

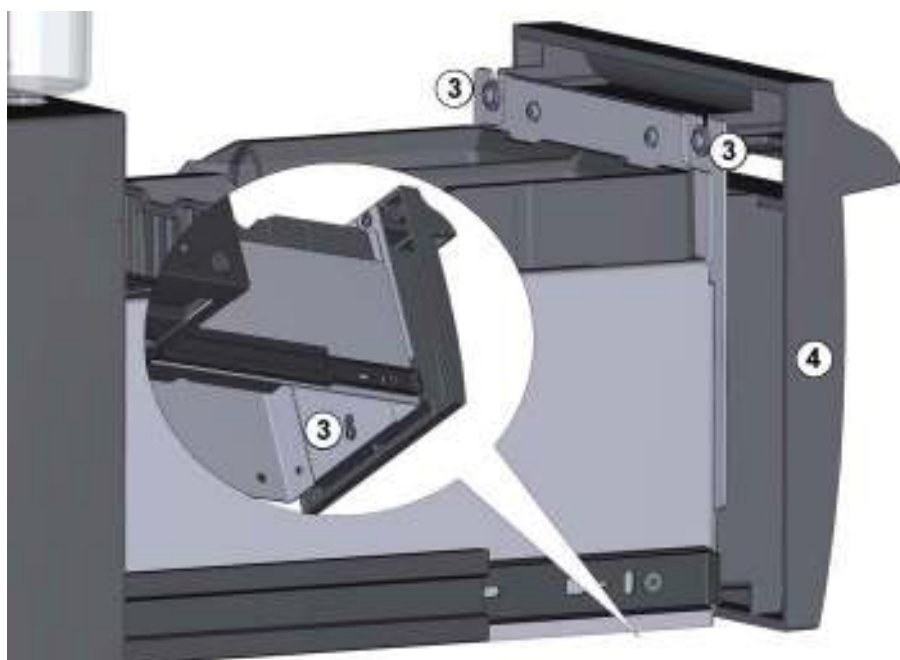


*Figure 5–23: Ancillary tray inside the Reagent Module*

2. Install the handle (4) of the Cuvette - Tip Loading Module and place the screws (3).

**5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration**

3. Align the Cuvette - Tip Loading Module handle, see Chapter 15.3.3 on page 15-5.



*Figure 5-24: Cuvette - Tip Loading Module Handle*

4. Insert cuvette trays in the Cuvette - Tip Loading Module.
5. Insert tip trays in the Cuvette - Tip Loading Module.
6. Insert Solid waste bin (5).



*Figure 5-25: Solid waste bin*

7. Insert the LIAISON EASY Waste inside the Solid Waste bin.

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

8. Insert the Starter reagents bottles into the dedicated module.

### 5.3.4.6 Switch On

1. Plug in the system.
2. Switch on the system.
3. Wait for the Soft Power Button to stop the blinking and press it.

### 5.3.5 Sample rack configuration

15 mm spacers (PN A0118) can be assembled together in order to realize adapters with different heights that shall be used with different typologies of Sample tubes.

#### NOTE

1 bag of 15 mm spacers (containing 60 pieces each) is provided by default in the Accessory Box of each instrument.

In case more spacers are needed to complete the proper configuration of customer Sample racks, service technicians shall order them separately.

The number of spacers needed to complete the configuration of customer Sample racks shall be estimated by service technicians during the pre-site inspection (see Corp-TN-000294 - former LXS TN 011).

The table below reports the different allowed configurations, with the associated total heights that can be obtained; for each configuration, the information regarding the heights of the supported tubes are also indicated. For tubes higher than 85 mm no spacers shall be used.

| Number of spacers | Picture                                                                             | Adapter height [mm] | Adapter height [steps] | Supported tube heights [mm] |
|-------------------|-------------------------------------------------------------------------------------|---------------------|------------------------|-----------------------------|
| 1                 |  | 15 mm               | 300                    | 76 mm – 85 mm               |
| 2                 |  | 25 mm               | 500                    | 66 mm – 75 mm               |

5. LIAISON<sup>®</sup> XS Installation for internal-tanks configuration

|   |                                                                                   |       |     |               |
|---|-----------------------------------------------------------------------------------|-------|-----|---------------|
| 3 |  | 35 mm | 700 | 56 mm – 65 mm |
| 4 |  | 45 mm | 900 | up to 55 mm   |

Table 5–2: Height of different spacer configurations

## 5.3.5.1 Configuration of Sample racks

Adapters with different heights (i.e. assembled using a different number of spacers) shall be used to customize the different rack typologies supported by LIAISON<sup>®</sup> XS system. The table below reports the configuration associated to each standard rack typology.

| Rack ID | Inner Diameter [mm]                                                                                           | Tube Height [mm] | Number of spacers                       |
|---------|---------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------|
| A       | 10 mm                                                                                                         | 86 mm - 100 mm   | 0                                       |
| B       | 15 mm                                                                                                         | 86 mm - 100 mm   | 0                                       |
| C       | Use only for DiaSorin S.p.A. special large vials (i.e. some types of control bottles or external calibrators) |                  | 0<br>(Dedicated Rack needed - PN A0243) |
| E       | 13 mm                                                                                                         | 86 mm - 100 mm   | 0                                       |
| F       | 12 mm                                                                                                         | 86 mm - 100 mm   | 0                                       |

## 5. LIAISON®XS Installation for internal-tanks configuration

|   |                                                                                                        |                |                                                                                           |
|---|--------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------|
| H | LIAISON® Stool extraction device (X0034)                                                               |                | 0<br>(Dedicated Adaptor needed - PN A0214)                                                |
| I | 11 mm                                                                                                  | 86 mm - 100 mm | 0                                                                                         |
| J | 9 mm                                                                                                   | 56 mm - 65 mm  | 3                                                                                         |
| K | 8 mm                                                                                                   | ≤ 55 mm        | 4                                                                                         |
| L | Used only for DiaSorin S.p.A. glass vials (i.e. some types of control bottles or external calibrators) | 55 ±0.5 mm     | 4                                                                                         |
| M | Calprotectin extraction device (X0043)                                                                 | 100 mm         | 0                                                                                         |
| N | 7 mm                                                                                                   | ≤ 55 mm        | 4                                                                                         |
| P | Used only for pediatric tubes                                                                          |                | 4<br>(Might need dedicated adapters, i.e. for Eppendorf tubes - PN 9450106)               |
| Q | 6 mm                                                                                                   | ≤ 55 mm        | 4                                                                                         |
| W | 14 mm                                                                                                  | 86 mm - 100 mm | 0                                                                                         |
| X | 10 mm                                                                                                  | 75 mm          | 2<br>(QuantiFERON®-TB Gold Plus Blood Collection Tubes shall be used with this rack type) |

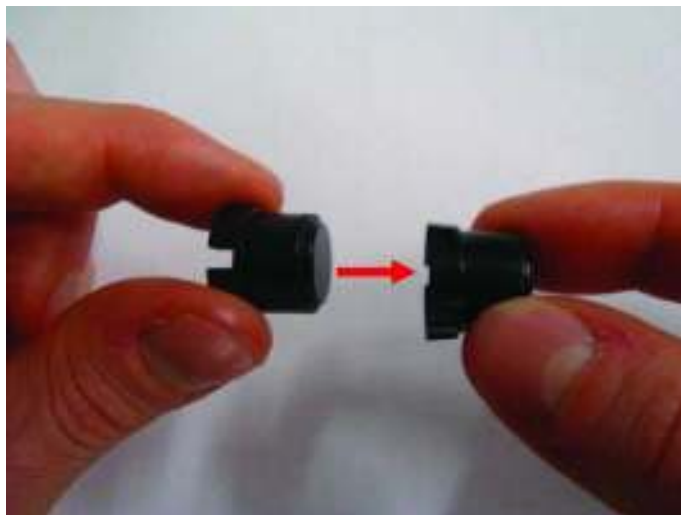
Table 5–3: Number of spacers associated to each standard rack typology

**LIAISON® XS** system supports also 4 additional rack typologies (G, O, Y and Z) that shall be properly configured in case the customer requirements are not satisfied by the characteristics (in term of tube height and inner diameter) of the standard rack typologies reported in Table 5–3. In this situation the proper customization of the sample racks shall be performed measuring the height of the customer tubes and configuring the proper adapter according to the rules presented in Table 5–2.

In any case it is important to consider that **the maximum number of spacers that can be assembled together is always equal to 4.**

### 5.3.5.2 Configuration procedure

1. Push the top part of a spacer into another one, see Figure 5–26: on page 5-25.



*Figure 5–26: Assembly of 2 spacers*

2. If needed, repeat point 1 in order to assemble 1 or 2 additional spacers, see Figure 5–27: on page 5-25.

---

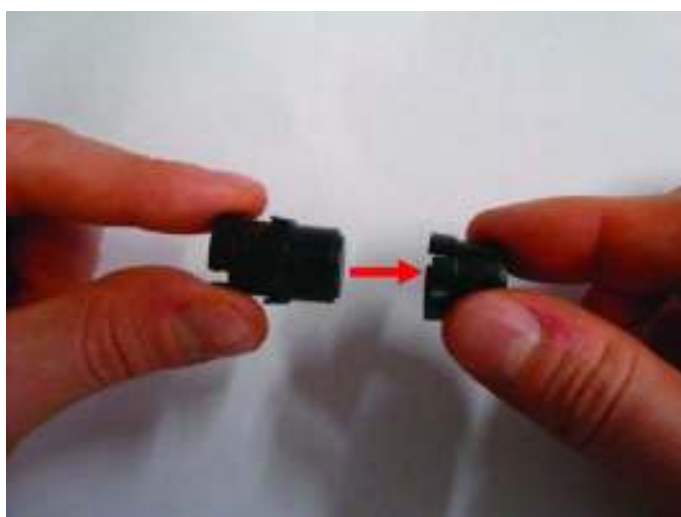
**NOTE**

To identify the proper number of spacers to assemble:

- For standard racks, refer to Table 5–3.

For customized racks, refer to Table 5–2.

---



*Figure 5–27: Assembly of additional spacers*

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

---

3. Verify the tightness of the assembly.
4. Push the adapters configured with the proper number of spacers into each position of the sample rack.
5. Verify that the bottom border of the lower spacer is in contact with the bottom of the sample rack (refer to the red circle of Figure 5–28). If needed, an Allen key can be used to push the spacers to their final position.

### NOTE

If a blockage is verified during the insertion, move the spacers a little further up, then push them down again.

---



*Figure 5–28: 15 mm spacers positioning*

6. Complete the customization of the Sample racks by applying the proper barcode ID labels, see also Figure 5–29, and rack ID labels, see also Figure 5–30.



## 5. LIAISON<sup>®</sup> XS Installation for internal-tanks configuration

7. The additional Sample rack should be labeled as L rack.



*Figure 5–29: Barcode ID positioning*

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

---



*Figure 5–30: Rack ID positioning*

### 5.3.6 Control rack configuration with Control rack A0243

Control rack configuration is intended for C rack.

The rack ID "C" shall be used with control bottles of the kits packed in Italy/ Germany and with the control bottles of the kits packed in USA.

**NOTE**

If more control racks are needed, it is possible to order them as additional spare parts (PN. A0243).



Figure 5-31: Barcode ID of rack C



Figure 5-32: Control rack ID positioning

### 5.3.7 Pipettor and rack teaching

1. Perform the Pipettor Arms teaching via **Teacher Software**, see Chapter 13.4 on page 13-6.
2. If needed, perform the teaching of customized Sample racks (G, O, Y and Z racks, see Chapter 13.18 on page 13-38).
3. Close **Teacher Software**.

### 5.3.8 Fluidic set-up

**CAUTION**

Before rinsing, make sure that all tube connections and fittings are properly assembled and that all containers are filled (DI water tank, Wash Buffer tank) and the Starters bottles are present. During the complete rinsing process, take care that there are no leaks.

**NOTE**

Before rinsing, make sure that the Solid Waste bin is inserted (with the dedicated LIAISON EASY Waste) and the Liquid Waste tank is connected.

#### 5.3.8.1 Prime SPOLV

##### 5.3.8.1.1 System Initialization

- a. Close any open software and start the **Service Software**.
- b. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- c. Click onto the **Instrument** button and choose the **Full initialization via COP** button to start the initialization sequence.
- d. The process is successful when the **Processing** window closes without failure message.

##### 5.3.8.1.2 Initial rinsing with DI water

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Liquid** button and choose the **Prime Pipettor** button to open the dedicated window.
- c. At the window **Prime Pipettor** for the initial rinsing with DI water, use following values and start with the button **Execute**.
  - Cycles: 22
  - Wash time: 18
  - OLV Calibration: Disabled
  - Liquid: DI Water
- d. The process is successful when the **Processing** window closes without failure message.

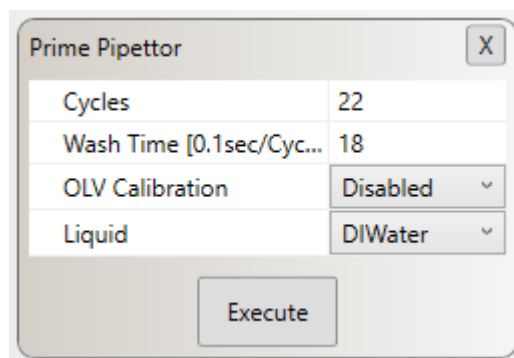


Figure 5–33: Pipettor rinsing with DI water

**CAUTION****Be aware of aspiration at the Wash station**

Check that the Wash station does not overflow during the rinsing of the SPOLV.

**5.3.8.1.3 Rinsing with Wash Buffer**

- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Liquid** button and choose the **Prime Pipettor** button to open the dedicated window.
- At the window **Prime Pipettor** for the rinsing with Wash Buffer use following values and start with **Execute**.
  - Cycles: 22
  - Wash time: 18
  - OLV Calibration: Enabled
  - Liquid: Wash Buffer

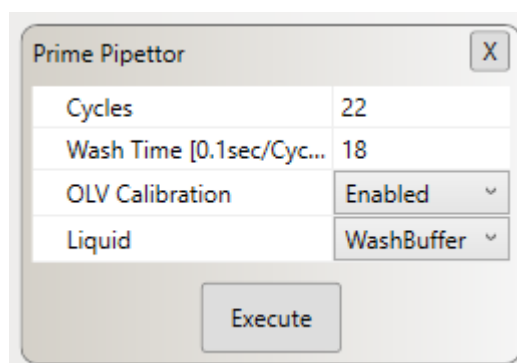


Figure 5–34: Pipettor rinsing with Wash Buffer

- The process is successful when the **Processing** window closes without failure message.

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

### 5.3.8.2 Prime Washer Module

#### 5.3.8.2.1 Assign cuvettes

- Open the Cuvette - Tip Loading Module.
- Place the cuvette trays in the dedicated position of the Cuvette - Tip Loading Module and close the it.
- In the **Service Software** at the navigation window, choose **System** and the **Cuvette Tray State** bracket.
- Fill the Cuvette Trays showed on the screen according to the actual situation and then press the button **Set Cuvette State**.

#### CAUTION

##### Be aware of aspiration

Check that at the last cycles the aspiration at all three Washer needles aspirates the fluid that was previously dispensed.

The tubing to the manifolds shall be without air bubbles and tight.

#### CAUTION

##### Be aware of initialization

Be aware that the system initialization will delete any Cuvette State setting. After a system initialization, always check the Cuvette State.

#### 5.3.8.2.2 Rinsing of Washer Module

- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Liquid** button and choose the **Prime Washer** button to open the dedicated window.
- At the window **Prime Washer** use following values and start with **Execute**.
  - Cycles Port 1: 22
  - Cycles Port 2: 11
  - Cycles Port 3: 11
  - Liquid Control: Enabled

| Prime Washer  |         |
|---------------|---------|
| Cycles Port 1 | 22      |
| Cycles Port 2 | 11      |
| Cycles Port 3 | 11      |
| LiquidControl | Enabled |

Execute

Figure 5–35: Rinsing of Washer Module

- The process is successful when the **Processing** window closes without failure message.

### 5.3.8.3 Prime Reader Module

#### 5.3.8.3.1 Assign cuvettes

- Open the Cuvette - Tip Loading Module.
- Place the cuvette trays in the dedicated position of the Cuvette - Tip Loading Module and close the it.
- In the **Service Software** at the navigation window, choose **System** and the **Cuvette Tray State** bracket.
- Fill the Cuvette Trays showed on the screen according to the actual situation and then press the button **Set Cuvette State**.

#### CAUTION

##### Be aware of aspiration

Check that at the last cycles the aspiration at the needle aspirates the fluid that was previously dispensed.

The tubing to the manifolds shall be without air bubbles and tight.

#### CAUTION

##### Be aware of initialization

Be aware that the system initialization will delete any Cuvette State setting. After a system initialization, always check the Cuvette State.

#### 5.3.8.3.2 Rinsing of Reader Module

- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Liquid** button and choose the **Prime Reader** button to open the dedicated window.
- At the window **Prime Reader** use following values and start with **Execute**.
  - Cycles: 40
  - Liquid Control Enabled

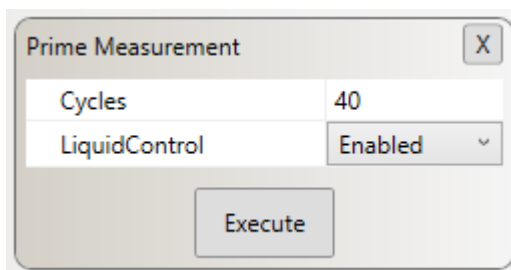


Figure 5–36: Rinsing of Reader Module

- The process is successful when the **Processing** window closes without failure message.

## 5.4 Operation Qualification

### 5.4.1 Sample Bay

1. Via **Service Software**, initialize the Sample Module, see Chapter 21.4.1 on page 21-21.
2. Check barcode scanner configuration, see Chapter 21.4.2.1 on page 21-21.
3. Perform a barcode reading test using customer barcoded vials, see Chapter 21.4.2.2 on page 21-21.
4. If needed, plug in the external handheld barcode reader and configure it, see Chapter 32.2.1 on page 32-2.

### 5.4.2 System

1. Via **Service Software**, initialize the System, see Chapter 11.8.1 on page 11-58.
2. Perform the Cuvette Endurance Run, see Chapter 11.8.1.2.1 on page 11-61.

---

**CAUTION**

**During the Cuvette Endurance Run, check the cuvette transfer from the Incubator to the Washer and vice-versa**

Check that cuvette insertion by the Gripper inside the Incubator is smooth and without noises.

Check that cuvette pickup from the Incubator and insertion into the Washer is smooth and without noises.

Check that cuvette pickup from the Washer and insertion back into the Incubator is smooth and without noises.

A not proper cuvette transfer is usually caused by a misalignment between the gripper and the Incubator rotor. If needed, change the Incubator rotor calibration (see Chapter 11.4.3.1 on page 11-26) such to compensate the misalignment; an adjustment of 3-4 steps clockwise could be sufficient.

---

### 5.4.3 Backup of system configuration

1. Perform backup of `Coordinate.coo` file.
2. Perform backup of “.rac” files.

### 5.4.4 Manuals update

1. Open the **LIAISON<sup>®</sup> XS Main UI**.
2. Ensure that the country code is set accordingly to the user needs to download/import appropriate IFU for the specific language/country, see Chapter 32.5.4 on page 32-20.
3. Install the most updated version of the Instrument Manuals, corresponding to the Instrument SW version and customer's language for the specific country. This can be done:
  - using the **Download** functionality available in the **Online Help** section of the **Main UI** (only if the system was previously connected to lot);



- using the **Acquire** functionality available in the **Online Help** section of the **Main UI** in case the appropriate instrument package was previously downloaded from **Dialog**.

### 5.4.5 System Verification

1. Open the **LIAISON<sup>®</sup> XS Main UI**.
2. Log-in as Service user and initialize the instrument.
3. Check the **Status - Consumables** toggle and assign the Consumables (cuvettes and tips) in the **Status - Plastic** toggle.
4. Load cuvette in the Incubator Module.

### 5.4.6 Installation Final Test

#### 5.4.6.1 Instrument setup

Before testing:

- All units must be properly installed and checked for the HW performances;
- The instrument shall reach a stable thermal condition;
- The instrument shall be fully primed with approved and valid Starters reagents and Wash Buffer.

#### 5.4.6.2 Materials

See Chapter 5.2.2 on page 5-4 for details.

Last revision of `Installation_Final_Test_Liaison_XS_[Assay].xls` spreadsheet shall be also available to collect all data. This file is attached to Corp-TN-000292 (former LXS TN 009) document. Last revision shall be downloaded and used.

#### 5.4.6.3 Background Wash (BGW)

The Background Wash (BGW) measures the noise of the system inherent the washing and signal acquisition steps, mainly dependent on the involved assemblies (i.e Washer and Reader modules, while Pipettors are not involved).

Ten cuvettes are loaded in the Incubator and moved to the Washer; during the washing phase, different Wash Buffer volumes are injected and aspirated in each cuvette at the 3 Washer ports:

- aspiration, 700µl injection, 700µl aspiration, 1200µl injection (at the first injector);
- 1200µl aspiration, 700µl injection, 700µl aspiration, 700µl injection (at the second injector);
- 700µl aspiration, 700µl injection, 700µl aspiration (at the third port).

Then, the cuvettes are moved to the Reader where Starters pumps inject 200 µl each, in each cuvette.

The readings of the 10 replicates provide 10 RLU values, whose CV% is used in order to assess the precision of the washing and of the measurement.

Three series of ten replicates of Background Wash have to be performed.

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

### 5.4.6.4 Light Check Right and Left (LCR/LCL)

Light Check (LC) assesses the performance of Pipettors, both left and right, and of the Reader module.

Ten replicates are run; for each replicate, the relevant Pipettor arm aspirates the LC solution and dispenses it into the cuvette; then the cuvette moves to the Reader, where Starters are injected (200µl for each Starter Pump).

The readings of the 10 replicates provide 10 RLUs values, whose CV% is used in order to assess the precision of the pipetting and of the measurement.

Six series of ten replicates of Light Check Left has to be performed.

Six series of ten replicates of Light Check Right has to be performed.

### 5.4.6.5 IFTP BGW & LCs Procedure

The following procedure applies:

1. Prepare 2 bottles of Light Check solution, following the Instructions for Use inserted in the package;
2. Put one bottle in the dedicated holder of the Ancillary tray and load the Ancillary tray in the Reagent bay;
3. Put the second bottle in the dedicated sample rack (L rack);
4. Insert the sample rack and assign the Light Check Product Number (319101) as SID;
5. Run **Service - 3BGW 6LCL 6LCR** task from the dedicated window of **Main UI (Maintenance, Tasks)**;
6. Remember to double **Accept** the Light Check Left and Right insertion in the **Service - 3BGW 6LCL 6LCR** task;
7. Export the results from **Maintenance, Logs** and check the results;
8. Collect and process the sets of data with the dedicated spreadsheet.

### Acceptance criteria

Refer to Corp-TN-000292 (former LXS TN 009) for details.

### NOTE

The execution of **Service - 3BGW 6LCL 6LCR** maintenance task represents the easiest way to run Background Wash and Light Check Installation Final Test.

Anyway, it is possible to run the test also using other maintenance tasks:

- Service - 3 series Background Wash, Service - 6 series LightCheck Left and Service - 6 series LightCheck Left;
- Background Wash + Background Wash + Background Wash, LightCheck Left + LightCheck Left + ..., LightCheck Right + LightCheck Right + ...

In any cases, the successful execution of:

- 3 consecutive Background,
- 6 consecutive Light Check Left,
- 6 consecutive Light Check Right,

shall be always achieved.

#### 5.4.6.6 Assay performance testing

The Assay kit should be performed in order to verify the performance of the instrument to perform an Assay Routine, with the contribution of Pipettors, Incubator, Washer and Reader modules.

Calibration and twenty replicates of an Assay kit has to be performed. The Assay kit depends upon the Country the instrument is installed in (see Corp-TN-000292 - former LXS TN 009 for details.)

#### 5.4.6.7 IFTP Assay Procedure

The following procedure applies:

1. Unpack, handle and prepare Reagent and Control according to the related Instruction For Use;
2. Define the specified Control scanning the DiaSorin Control COA (Certificate of Analysis) by using the Handheld Barcode Reader or by inserting manually the data from the COA;
3. Set the Control replicates to 20;
4. Load the Assay kit in the Reagent Area, if the integral is not correctly detected load the correct Assay Protocol;
5. Wait the agitation time and start the Assay kit calibration;
6. Place Control vial on its proper rack and load it in Sample Area;
7. Run the Assay Control (20 replicates are needed);
8. Check that the calibration of the Assay kit is valid;
9. Check that the average control result dose lies inside the COA specified range,
10. Collect the dose values (from the 20 replicates) and process data with the dedicated spreadsheet (attached to Corp-TN-000292 - former LXS TN 009) to ensure the CV%;
11. Restore the number of Control replicates to the original value.

#### NOTE

If the Control lies outside the expected range, calibration should be repeated and Control retested. If Control value obtained after successful calibration lies repeatedly outside the predefined range, the test should be repeated using an unopened Control vial. If Control values lie outside the expected ranges, execute troubleshooting.

#### Acceptance criteria

Refer to Corp-TN-000292 (former LXS TN 009) for details.

#### 5.4.6.8 Data collection

1. Perform raw data and calibration export and backup.
2. Verify that `Installation_Final_Test_Liaison_XS_[Assay].xls` file has been completed.

## 5. LIAISON<sup>®</sup>XS Installation for internal-tanks configuration

---

**NOTE**

The results of the installation activities (BGW, LCL, LCR, Calibration and Control run) must be attached to the Service records (for Subsidiaries only: SGAT WS), together with a detailed description of the Service activities performed on the analyzer.

---

### 5.4.7 Printout customization

1. Install the printer (if necessary, see Corp-TN-000293 - former LXS TN 010 for details).
2. Perform printout customization (if required by the customer).

# 6 LIAISON<sup>®</sup>XS Installation for external-tanks configuration

As the **LIAISON<sup>®</sup>XS** instrument is a complex instrument and contains delicate electrical and mechanical components, it must be unpacked and installed by a qualified Service Engineer.

The procedures reported in this chapter allow to complete the installation of **LIAISON<sup>®</sup>XS** analyzers successfully.

## 6.1 Safety

---

**WARNING****Non-observance of safety instructions**

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 2.5 on page 2-7.
  - Follow all working instructions included in this manual.
  - Follow all warnings marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
- 

---

**WARNING****Falling down of the system**

Due to improper place of installation of the system, the system can fall down. If the system falls down, the user can be seriously injured or the system can be seriously damaged. Position the system on a stable platform, cart, stand or table, see Chapter 36.4 on page 36-3.

- Observe the technical specifications of the system, see Chapter 36 on page 36-1.
- 

---

**WARNING****Never lift the system by yourself!**

Manual handling of heavy loads may cause injuries.

The **LIAISON<sup>®</sup>XS** instrument shall NOT be lifted and carried by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

---

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

---

**CAUTION****Check System**

During the installation of the system, always follow the steps indicated in **LIAISON<sup>®</sup>XS** Installation Qualification (IQ, see Chapter 34.1 on page 34-1) and Operational Qualification (OQ, see Chapter 34.2 on page 34-5) check lists. The proper execution and compilation of these check lists will ensure the proper installation of the instruments.

---

## 6.2 Installation preparation

### NOTE

#### Pre-Site Inspection visit for installation

A Pre-Site Inspection form has to be used during the preliminary visit at the customer site in order to evaluate the customer facilities and the laboratory accessibility and requirement for a proper instrument installation. See Corp-TN-000294 (former LXS TN 011) for details.

### 6.2.1 Standard supply with PN I0090

- Instrument with cables (schuko plug 220 Volt)
- N° 1 Front Module
- N° 1 Left Top Cover
- N° 1 Right Top Cover
- N° 1 Left Side Cover
- N° 1 Right Side Cover
- N° 1 Bulk Fluid Door
- N° 1 Top Cover
- Panel PC
- N° 1 System Liquid 10L tank
- N° 2 Wash Buffer solution 10L tanks
- N° 1 Cleaning solution 10L tank
- N° 1 Liquid waste tank
- N° 1 Solid waste bin
- N° 1 Cuvette - Tip Loading Module handle
- N° 3 Pump caps for Peristaltic pumps
- N° 1 Ancillary tray
- N° 5 Sample racks
- N°1 Label Set for Sample Rack
- Sample Rack Front Handle Labels
- N° 1 Velcro strip for USB stick
- N° 1 PN 10136280 - TANK FILLING STAND
- N° 1 PN A0179 - EXTERNAL BARCODE SCANNER or N° 1 PN A0256 - EXTERNAL BARCODE SCANNER
- N° 1 PN A0128 - INTEGRAL RACK HOLDER
- N° 1 PN A0120 - SAMPLE RACK HOLDER LIAISON XL/XS (1 NU)
- N° 1 PN A0055 - SAMPLE TUBE TOOL
- N° 1 PN A0118 - SAMPLE RACK ADAPTER - SET 60 (1 NC)
- N° 1 PN A0107 - ANCILLARY ADAPTOR FOR LC (SET OF 2)
- N° 1 PN A0243 - XS CONTROL RACK 8 POSITIONS (QTY 2)
- N° 1 PN X0054 - LIAISON<sup>®</sup> EASY Waste (QTY 10)
- N° 1 PN A0244 - XS LIQUID WASTE BASIN

## 6. LIAISON®XS Installation for external-tanks configuration

- N° 1 PN A0251 - XS BACKUP USB STICK 16GB

### NOTE

Other accessories, such as:

- Printer (PN A0240, or newer, or locally validated)
- UPS Global 30 (PN I0078)
- XS Direct Drain kit (10127604)
- Smart Table (PN A0245, starting from SN 00053)

are NOT delivered by standard supply and, if required, need to be ordered and managed separately (see Corp-TN-000294 - former LXS TN 011).

### 6.2.2 Additional material required for installation

| Product Code                              | Quantity | Description                |
|-------------------------------------------|----------|----------------------------|
| 319150                                    | 1        | LIAISON Light Check 12     |
| X0053                                     | 1        | LIAISON Cuvettes on trays  |
| X0055                                     | 1        | LIAISON Disposable Tips    |
| 319300                                    | 1        | LIAISON EASY Starter Kit   |
| 319301                                    | 1        | LIAISON EASY Wash Buffer   |
| 319302                                    | 1        | LIAISON EASY System Liquid |
| see Corp-TN-000292<br>(former LXS TN 009) | 1        | LIAISON Assay              |
| see Corp-TN-000292<br>(former LXS TN 009) | 1        | LIAISON Control for Assay  |

Table 6–1: Additional material required for installation

### NOTE

The above listed material is sufficient for the Field Service Engineer installation only and MUST be ordered separately.

Anything required for the Customer Training, Start up, Routine (Assays, Controls, additional Consumables, etc.), MUST be ordered separately as well.

The Assay and related control to be ordered depends on the Country where the instrument will be installed (refer to Corp-TN-000292 - former LXS TN 009).

### 6.2.3 Required tools

The following tools are required to perform installation:

- Set of hexagon socket wrench (Allen key)
- Set of cross slotted screw driver
- Set of flat screw driver



- Set of torx screw driver
- Set of torx L-keys
- Adjustable wrench
- Cutter
- Magnetic work light
- Ruler (at least 200mm-8inches)
- Keyboard (handheld suggested)

## 6.3 Installation Qualification

---

**NOTE****Packaging**

Keep the packaging to allow for safe transportation in case the system must be shipped in the future.

---

### 6.3.1 Visual inspection before unpacking

1. Verify the transport box is without damages.
2. Verify the indication of shock-watch (1) and thermometer sensor (2).

## 6. LIAISON®XS Installation for external-tanks configuration



Figure 6-1: Shock-watch and thermometer sensors

### 6.3.2 Unpacking, visual inspection and inventory

1. Check the completeness of delivery by means of the packing list, see Chapter 38.1 on page 38-1.
2. Remove Accessories and Covers box from the top of instrument box.
3. Open the Accessory Box.
4. Check the Accessory Box parts according to the packing list, see Chapter 38.1 on page 38-1.
5. Visual inspect that the Covers are without damages or gaps.
6. Open the instrument box, removing its lid.
7. Lift cardboard box up to the top and remove it.
8. Remove the C-foam piece from both sides around the bottom of the instrument.
9. Remove the shipping foil.
10. Remove the foam cover from the top of the instrument.
11. Visually inspect that the chassis of the **LIAISON® XS** instrument is without damages or gaps.
12. Visually inspect that no visible cuts are present in the system or pipetting tubing.

---

**6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration**

---

13. Raise the adjustable feet by 1 cm using the nuts, before moving the system from the shipping palette.

**WARNING****Never lift the system by yourself!**

Manual handling of heavy loads may cause injuries.

The **LIAISON<sup>®</sup> XS** instrument shall NOT be lifted and carried by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

- 
14. Move the instrument from the shipping palette and position it on a stable platform, cart, stand or table, see Chapter 36.4 on page 36-3.
  15. Verify the indication of the tilt- and shock-watch at the rear chassis.



*Figure 6-2: Unpacking of instrument*

**NOTE****System clearance**

Allow ample clearance between the system and its surroundings as shown below.

---

## 6. LIAISON<sup>®</sup> XS Installation for external-tanks configuration

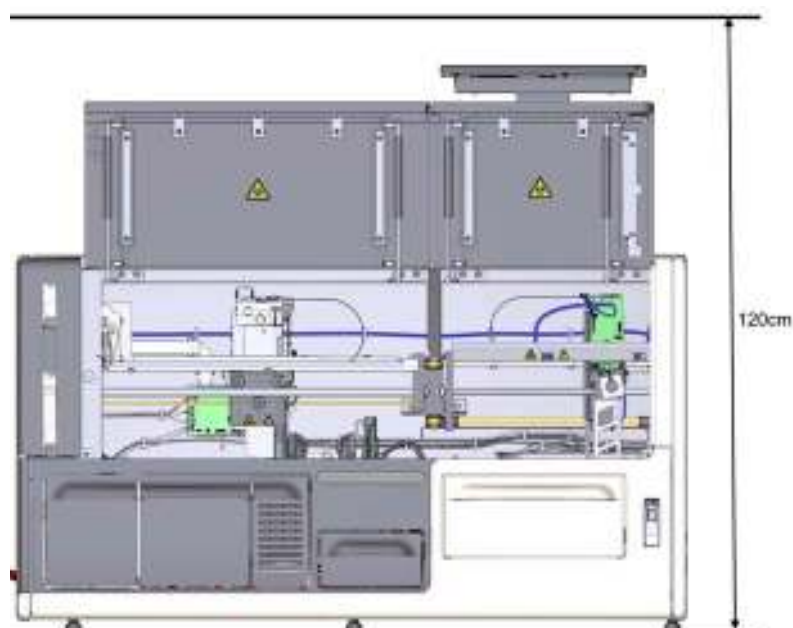


Figure 6–3: System clearance at the top



Figure 6–4: System clearance, side view

## 6. LIAISON®XS Installation for external-tanks configuration

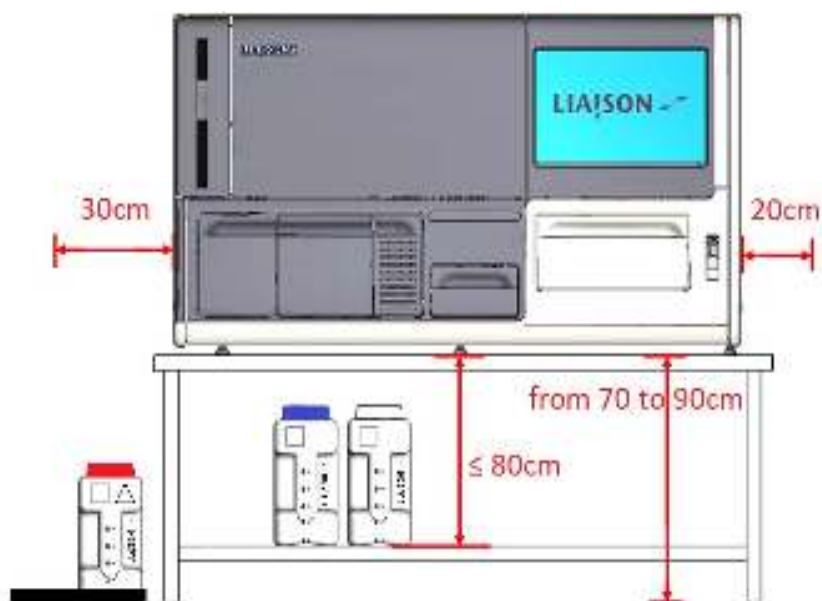


Figure 6–5: System clearance, front view, in case of tanks placed below the system

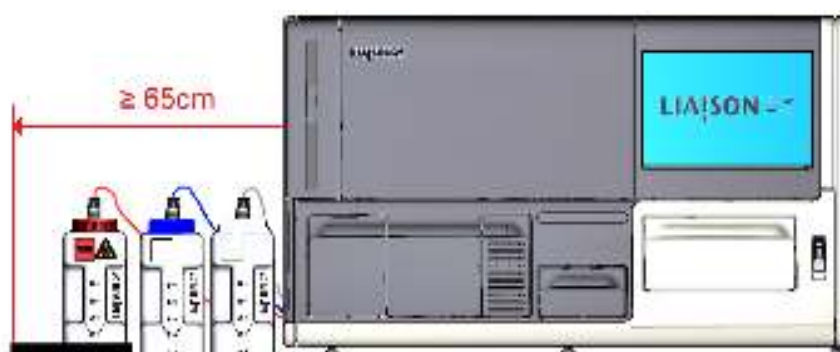


Figure 6–6: Left clearance, in case of tanks placed on the bench  
(right clearance unchanged)

## 6. LIAISON<sup>®</sup> XS Installation for external-tanks configuration

### 6.3.2.1 System leveling and stabilization

1. Adjust the instrument by leveling the adjustable feet (2) using the nuts until the level (1) at the Reagent Module is in adjustment.



Figure 6-7: Instrument level at the Reagent Module

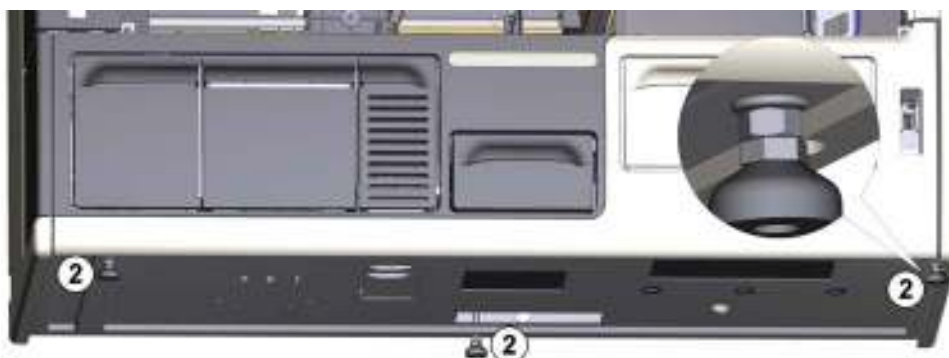


Figure 6-8: Adjustable feet of the instrument

**6.3.2.2 Remove securing foam tubes and cable ties at Pipettor Arms**

1. Carefully cut and remove the cable tie (1) at the Gripper Module.



*Figure 6–9: Cable tie at the Gripper Module*



## 6. LIAISON®XS Installation for external-tanks configuration

2. Remove the foam tubes at the Left and Right Pipettor Arms.



Figure 6-10: Left Pipettor Arm, front y-axis



Figure 6-11: Left Pipettor Arm, rear y-axis



## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

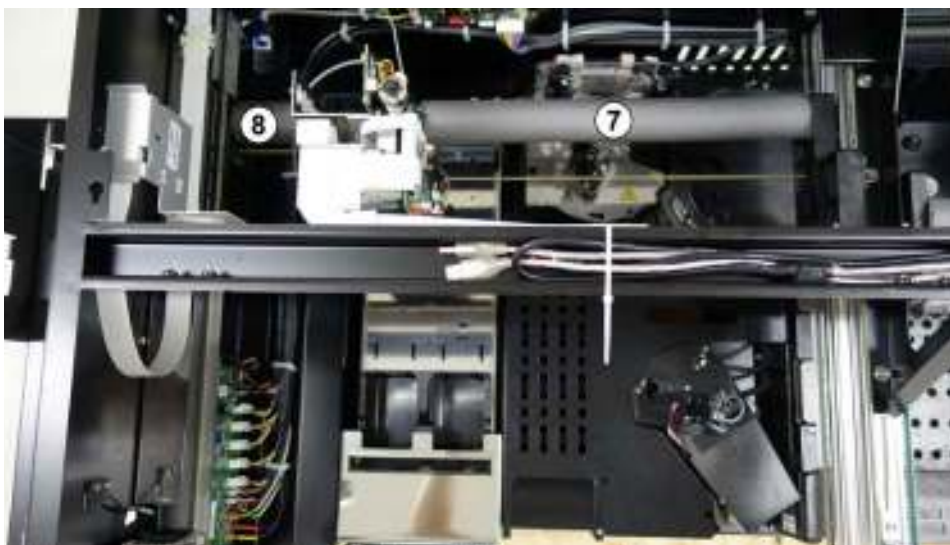


Figure 6-12: Left Pipettor Arm, right x-axis

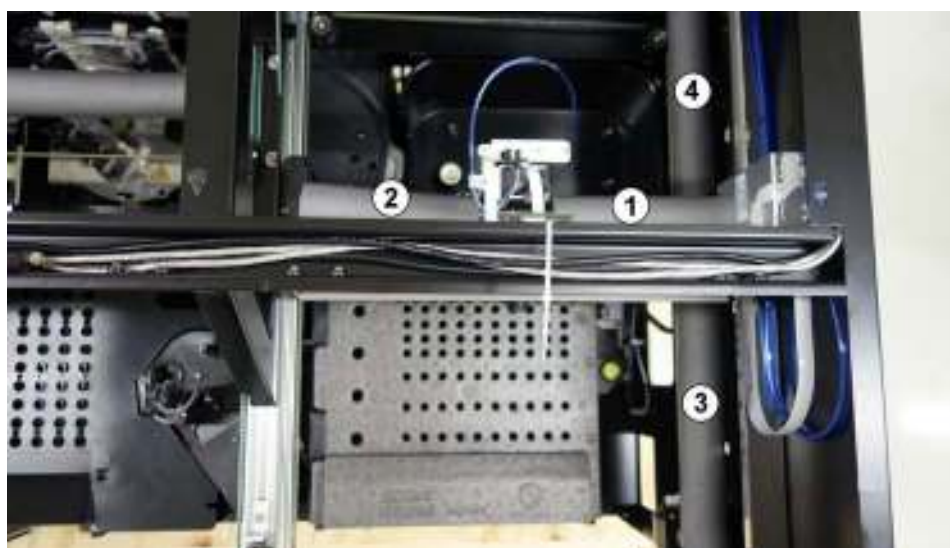


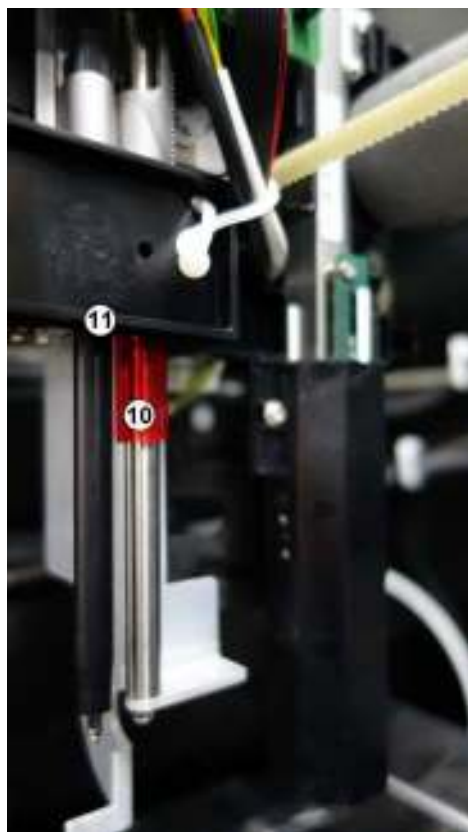
Figure 6-13: Right Pipettor Arm, x-axis and y-axis

3. Remove securing tube (10) at z-axis of the Sample pipettor (11).

**6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration**

---

4. Remove securing tube (12) at z-axis of the SPOLV (13).



*Figure 6-14: Sample pipettor*



*Figure 6-15: SPOLV*

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

5. Cut the cable tie (11) and unwind the cables of the Panel PC at the support bar.

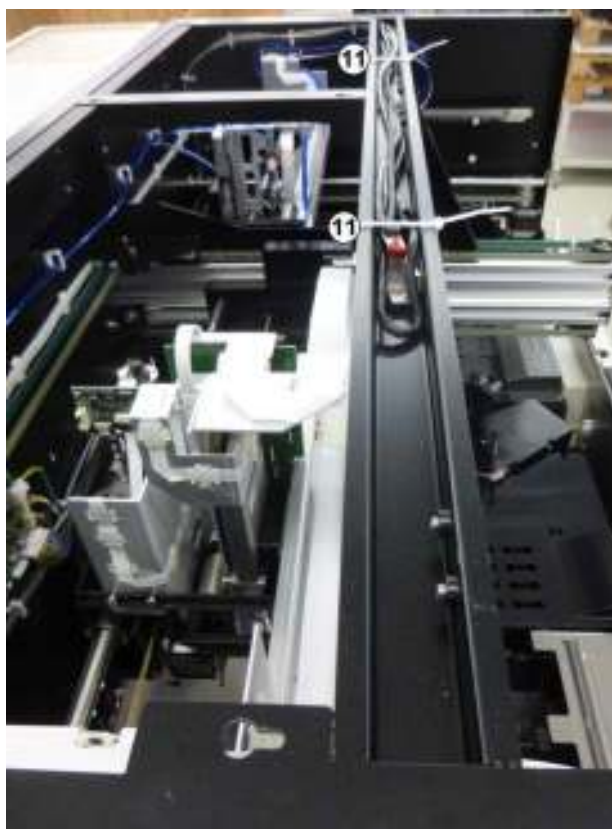


Figure 6-16: Support bar

### 6.3.3 Manual inspection

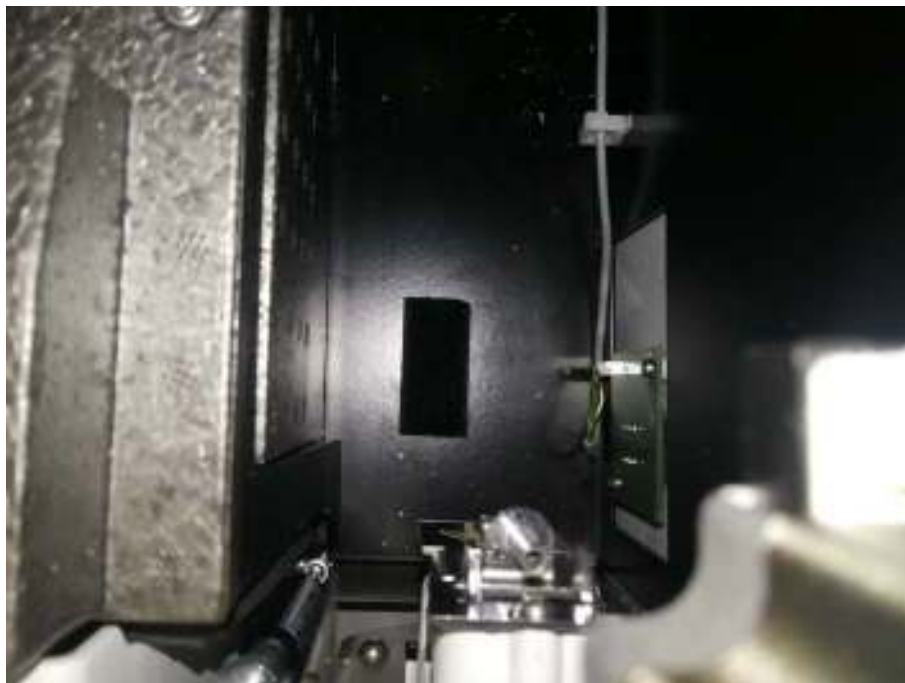
1. Verify that both the Pipetting Arms are gliding smooth and without force in x, y and z direction.

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

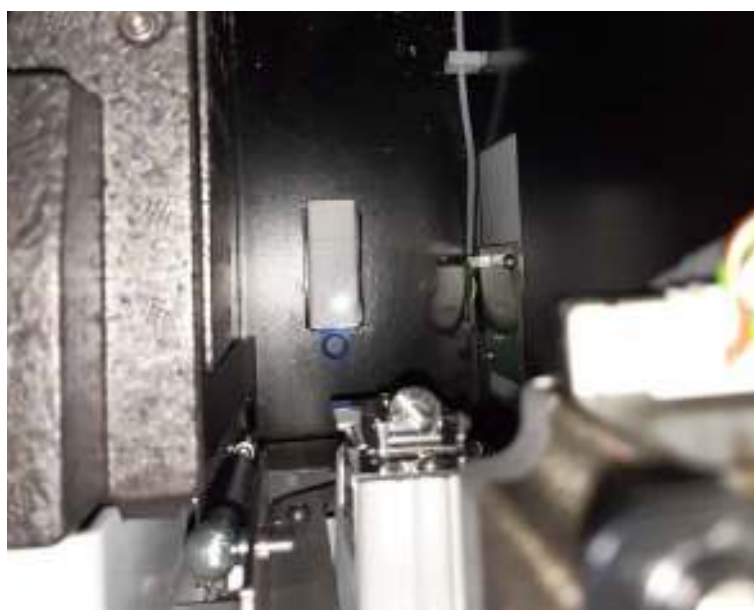
### 6.3.4 System set-up

#### 6.3.4.1 Velcro strip for USB stick set up

1. Place the velcro strip for the USB stick at the bottom of the chassis, next to the right side of the Reagent Module.
2. Place the USB stick.



*Figure 6-17: Velcro strip positioning*



*Figure 6-18: USB stick positioning*

**6.3.4.2 LWP and WAP set up**

1. Plug the peristaltic tubes into the LWP and WAP pumps.
2. Set and close the caps onto the LWP (1) and WAP (2) pumps.



*Figure 6–19: Peristaltic pumps without caps*



## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

### 6.3.4.3 Tanks set up

1. Rinse all the tanks with DI water (Wash Buffer tank, System Liquid tank, Cleaning solution tank).
2. Fill System Liquid tank according to the IFU.
3. Fill Wash Buffer tank according to the IFU.
4. Unpack the System Liquid tubing (1) and slide the tubing and the cable into the cutout at the rear left corner of the instrument.
5. Unpack the Wash Buffer tubing (2) and slide the tubing and the cable into the cutout at the rear left corner of the instrument.
6. Unpack Liquid Waste tubing (3) and slide the tube and the cable into the cutout at the rear left side of instrument.

**NOTE**

Be aware that the cables and the tubings have a dedicated housing in the frame.



Figure 6–20: Wash Buffer, System Liquid and Waste tube layout

7. Connect the tubing (4) and the sensor cable (5) to the System Liquid tank, see chapter “Load and Unload the System Liquid tank” of the User Manual.

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

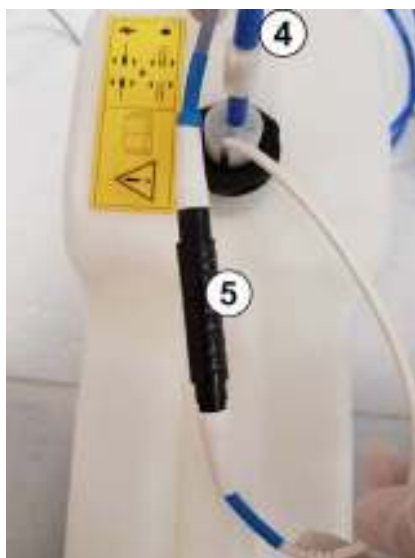


Figure 6-21: System Liquid tank connection

8. Connect the tubing (6) and the cable (7) to the Wash Buffer tank, see chapter “Load and Unload the Wash Buffer tank” of the User Manual.



Figure 6-22: Wash Buffer tank connection



### Wrong placement of the System Liquid and Wash Buffer tanks

Too low installation of the tanks can lead to a not proper performance of the pumps.

- The tanks shall be positioned at a maximum distance of 800mm from the bench/table where the system is placed on. For the allowed positioning, see Chapter 6.3.2 on page 6-6.

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

9. Connect the tubing (8) and the sensor cable (9) to the Liquid waste tank.

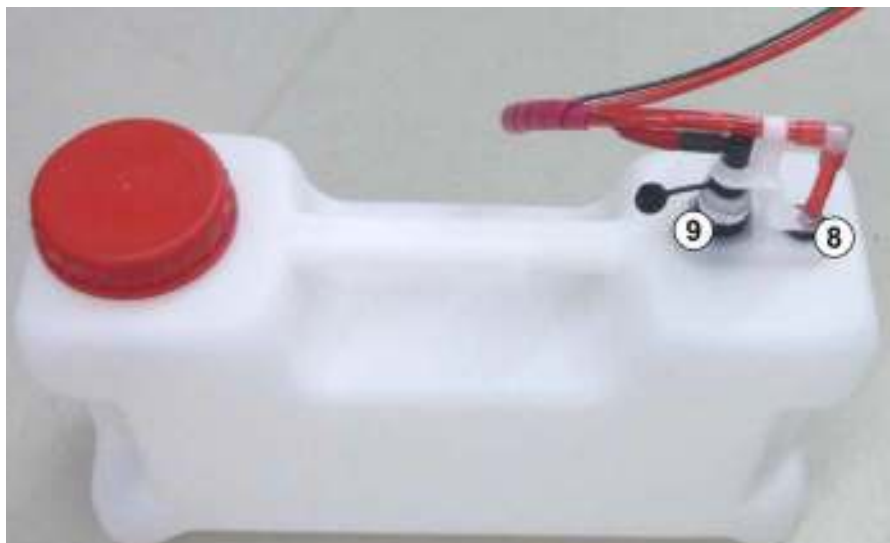


Figure 6-23: Liquid Waste tank connection

10. Place the Liquid waste tank in the Liquid waste basin (10).

**DANGER**



### Wrong placement of the Liquid waste tank

Too high installation of the Liquid waste tank can lead to a liquid drawback.

- It is preferred to place the Liquid waste tank below the instrument.



Figure 6-24: Liquid waste tank and basin, System Liquid and Wash Buffer tanks in Smart Table (PN A0245, from SN 00053)



#### 6.3.4.4 Assemble Covers

1. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
2. Assemble the Top cover, see Chapter 14.3.7.3 on page 14-16.
3. Assemble the Left Top cover, see Chapter 14.3.5.3 on page 14-12.
4. Assemble the Right top cover, see Chapter 14.3.6.3 on page 14-14.
5. Assemble the Panel PC, see Chapter 14.3.4.3 on page 14-11.
6. Check the proper position and installation of the Ferrite snap, see Chapter 14.3.4.3 on page 14-11.



Figure 6-26: Ferrite snap connected to the power cable of the Panel PC

#### NOTE

Ferrite snap is delivered in a box inside the Panel PC box.



Figure 6-27: Box with the Ferrite snap inside the Panel PC box



Figure 6-28: Open Ferrite snap

7. Check the alignment of the Covers, see Chapter 14.4 on page 14-20.
8. Assemble the Bulk Fluid Door, see Chapter 14.3.2.3 on page 14-6.
9. Assemble the Left Side cover, see Chapter 14.3.1.4 on page 14-5.

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

10. Assemble the Right Side cover, see Chapter 14.3.3.4 on page 14-9.
11. Check the system leveling, see Chapter 6.3.2.1 on page 6-10.
12. Re-tension the Y belt of both Left and Right pipettor Arms (see Chapter 20.3.11.4 on page 20-41 and see Chapter 20.3.20.4 on page 20-66).
  - a. For the Left Pipettor Arm: loosen the screw of the toothed belt wheel, let the spring expand to tension the belt and then fasten the screw. Move the Left Pipettor Arm forward and back a couple of times, to check that the pipettor arm is parallel. Fasten the screw at the belt holder.
  - b. For the Right Pipettor Arm: loosen the screw of the toothed belt wheel, let the spring expand to tension the belt and then fasten the screw. Move the Right Pipettor Arm forward and back a couple of times, to check that the pipettor arm is parallel.
13. Remove the protective foil from the Status light.

### 6.3.4.5 Insert tray and set the Cuvette - Tip Loading Module

1. Insert Ancillary tray (1) into the Reagent Module (2).

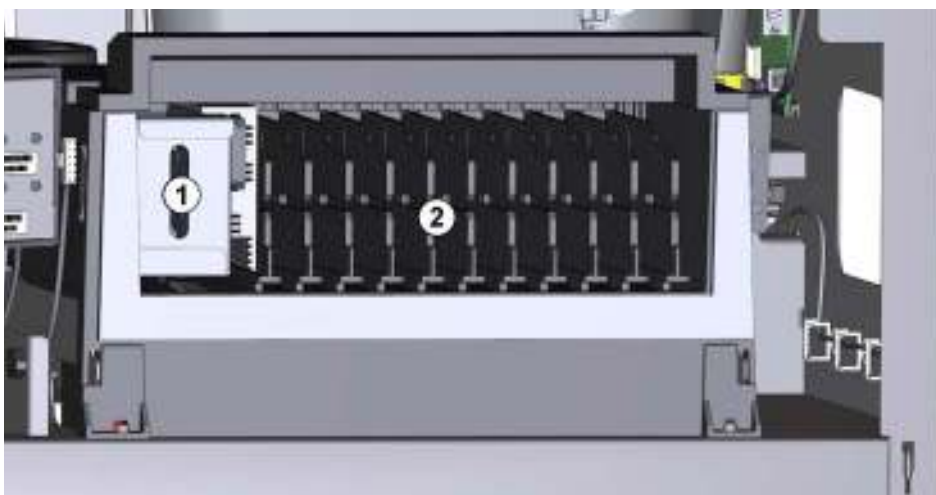


Figure 6-29: Ancillary tray inside the Reagent Module

2. Install the handle (4) of the Cuvette - Tip Loading Module and place screws (3).

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

3. Align the Cuvette - Tip Loading Module handle, see Chapter 15.3.3 on page 15-5.

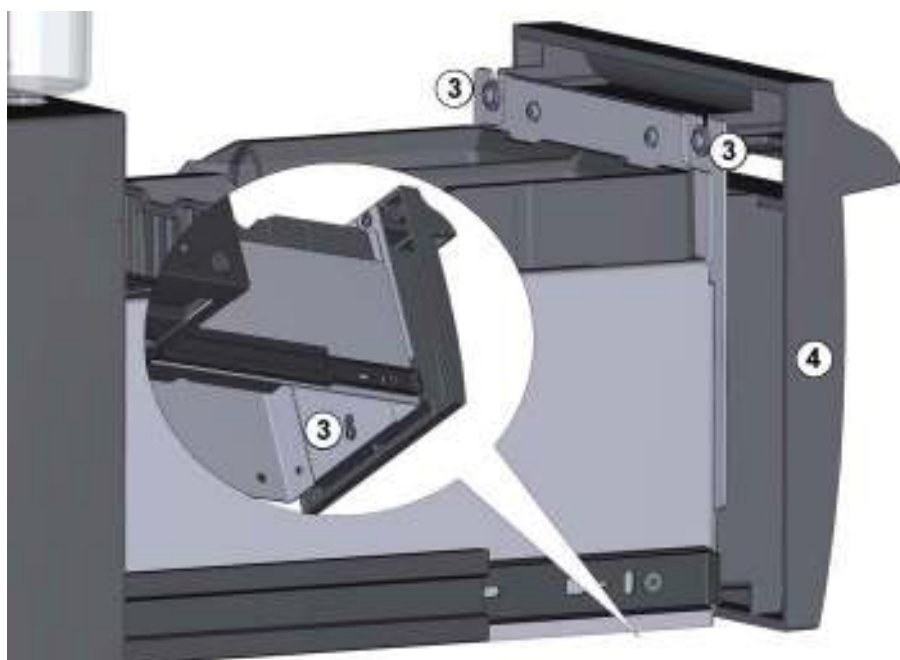


Figure 6–30: Cuvette - Tip Loading Module Handle

4. Insert cuvette trays in the Cuvette - Tip Loading Module.
5. Insert tip trays in the Cuvette - Tip Loading Module.
6. Insert Solid waste bin (5).



Figure 6–31: Solid waste bin

7. Insert the LIAISON EASY Waste box inside the Solid Waste bin.

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

8. Insert the Starter reagents bottles into the dedicated module.

### 6.3.4.6 Switch On

1. Plug in the system.
2. Switch on the system.
3. Wait for the Soft Power Button to stop the blinking and press it.

### 6.3.5 Sample rack configuration

15 mm spacers (PN A0118) can be assembled together in order to realize adapters with different heights that shall be used with different typologies of Sample tubes.

#### NOTE

1 bag of 15 mm spacers (containing 60 pieces each) is provided by default in the Accessory Box of each instrument.

In case more spacers are needed to complete the proper configuration of customer Sample racks, service technicians shall order them separately.

The number of spacers needed to complete the configuration of customer Sample racks shall be estimated by service technicians during the pre-site inspection (see see Corp-TN-000294 - former LXS TN 011).

The table below reports the different allowed configurations, with the associated total heights that can be obtained; for each configuration, the information regarding the heights of the supported tubes are also indicated. For tubes higher than 85 mm no spacers shall be used.

| Number of spacers | Picture                                                                             | Adapter height [mm] | Adapter height [steps] | Supported tube heights [mm] |
|-------------------|-------------------------------------------------------------------------------------|---------------------|------------------------|-----------------------------|
| 1                 |  | 15 mm               | 300                    | 76 mm – 85 mm               |
| 2                 |  | 25 mm               | 500                    | 66 mm – 75 mm               |

6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

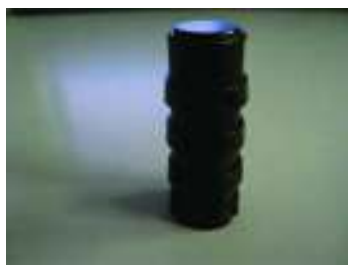
|   |                                                                                   |       |     |               |
|---|-----------------------------------------------------------------------------------|-------|-----|---------------|
| 3 |  | 35 mm | 700 | 56 mm – 65 mm |
| 4 |  | 45 mm | 900 | up to 55 mm   |

Table 6–2: Height of different spacer configurations

## 6.3.5.1 Configuration of Sample racks

Adapters with different heights (i.e. assembled using a different number of spacers) shall be used to customize the different rack typologies supported by LIAISON<sup>®</sup> XS system. The table below reports the configuration associated to each standard rack typology.

| Rack ID | Inner Diameter [mm]                                                                                           | Tube Height [mm] | Number of spacers                       |
|---------|---------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------|
| A       | 10 mm                                                                                                         | 86 mm - 100 mm   | 0                                       |
| B       | 15 mm                                                                                                         | 86 mm - 100 mm   | 0                                       |
| C       | Use only for DiaSorin S.p.A. special large vials (i.e. some types of control bottles or external calibrators) |                  | 0<br>(Dedicated Rack needed - PN A0243) |
| E       | 13 mm                                                                                                         | 86 mm - 100 mm   | 0                                       |
| F       | 12 mm                                                                                                         | 86 mm - 100 mm   | 0                                       |

## 6. LIAISON® XS Installation for external-tanks configuration

|   |                                                                                                        |                |                                                                                           |
|---|--------------------------------------------------------------------------------------------------------|----------------|-------------------------------------------------------------------------------------------|
| H | LIAISON® Stool extraction device (X0034)                                                               |                | 0<br>(Dedicated Adaptor needed - PN A0214)                                                |
| I | 11 mm                                                                                                  | 86 mm - 100 mm | 0                                                                                         |
| J | 9 mm                                                                                                   | 56 mm - 65 mm  | 3                                                                                         |
| K | 8 mm                                                                                                   | ≤ 55 mm        | 4                                                                                         |
| L | Used only for DiaSorin S.p.A. glass vials (i.e. some types of control bottles or external calibrators) | 55 ±0.5 mm     | 4                                                                                         |
| M | Calprotectin extraction device (X0043)                                                                 | 100 mm         | 0                                                                                         |
| N | 7 mm                                                                                                   | ≤ 55 mm        | 4                                                                                         |
| P | Used only for pediatric tubes                                                                          |                | 4<br>(Might need dedicated adapters, i.e. for Eppendorf tubes - PN 9450106)               |
| Q | 6 mm                                                                                                   | ≤ 55 mm        | 4                                                                                         |
| W | 14 mm                                                                                                  | 86 mm - 100 mm | 0                                                                                         |
| X | 10 mm                                                                                                  | 75 mm          | 2<br>(QuantiFERON®-TB Gold Plus Blood Collection Tubes shall be used with this rack type) |

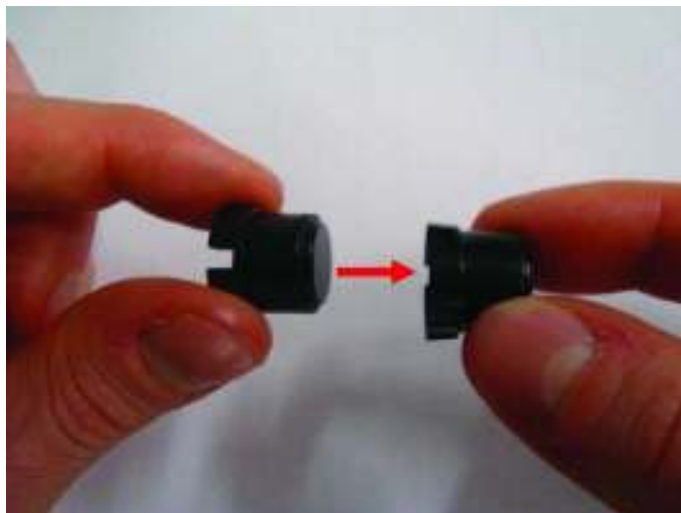
Table 6–3: Number of spacers associated to each standard rack typology

**LIAISON® XS** system supports also 4 additional rack typologies (G, O, Y and Z) that shall be properly configured in case the customer requirements are not satisfied by the characteristics (in term of tube height and inner diameter) of the standard rack typologies reported in Table 6–3. In this situation the proper customization of the sample racks shall be performed measuring the height of the customer tubes and configuring the proper adapter according to the rules presented in Table 6–2.

In any case it is important to consider that **the maximum number of spacers that can be assembled together is always equal to 4.**

### 6.3.5.2 Configuration procedure

1. Push the top part of a spacer into another one, see Figure 6–30: on page 6-26.



*Figure 6–32: Assembly of 2 spacers*

2. If needed, repeat point 1 in order to assemble 1 or 2 additional spacers, see Figure 6–31: on page 6-27.

---

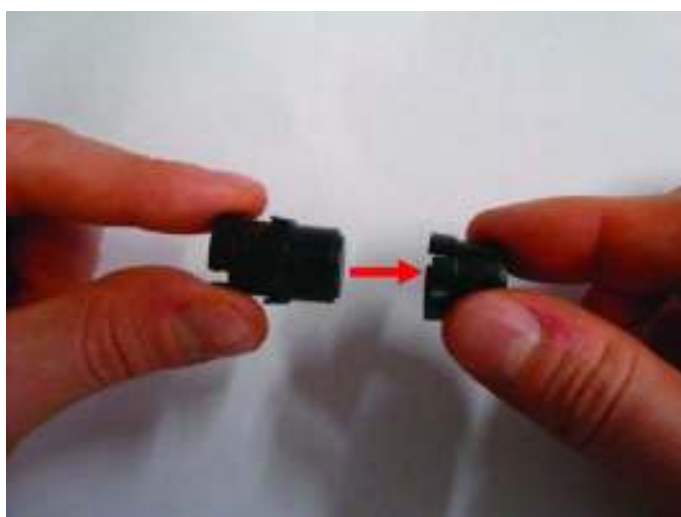
**NOTE**

To identify the proper number of spacers to assemble:

- For standard racks, refer to Table 6–3.

For customized racks, refer to Table 6–2.

---



*Figure 6–33: Assembly of additional spacers*

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

3. Verify the tightness of the assembly.
4. Push the adapters configured with the proper number of spacers into each position of the sample rack.
5. Verify that the bottom border of the lower spacer is in contact with the bottom of the sample rack (refer to the red circle of Figure 6–32). If needed, an Allen key can be used to push the spacers to their final position.

### NOTE

If a blockage is verified during the insertion, move the spacers a little further up, then push them down again.



Figure 6–34: 15 mm spacers positioning

6. Complete the customization of the Sample racks by applying the proper barcode ID labels, see also Figure 6–33, and rack ID labels, see also Figure 6–34.
7. The additional Sample rack should be labeled as L rack.



## 6. LIAISON<sup>®</sup> XS Installation for external-tanks configuration



Figure 6–35: Barcode ID positioning



Figure 6–36: Rack ID positioning

## 6. LIAISON® XS Installation for external-tanks configuration

### 6.3.6 Control rack configuration with Control rack A0243

Control rack configuration is intended for C rack.

The rack ID “C” shall be used with control bottles of the kits packed in Italy/ Germany and with the control bottles of the kits packed in USA.

#### NOTE

If more control racks are needed, it is possible to order them as additional spare parts (PN. A0243).



Figure 6-37: Barcode ID of rack C



Figure 6-38: Control rack ID positioning



## 6. LIAISON®XS Installation for external-tanks configuration

---

### 6.3.7 Pipettor and rack teaching

1. Perform the Pipettor Arms teaching via **Teacher Software**, see Chapter 13.4 on page 13-6.
2. If needed, perform the teaching of customized Sample racks (G, O, Y and Z racks, see Chapter 13.18 on page 13-38).
3. Close **Teacher Software**.

### 6.3.8 Fluidic set-up

**CAUTION**

Before rinsing, make sure that all tube connections and fittings are properly assembled and that all containers are filled (System liquid tank, Wash Buffer tank) and the Starters bottles are present. During the complete rinsing process, take care that there are no leaks.

**NOTE**

Before rinsing, make sure that the Solid Waste bin is inserted (with the dedicated LIAISON EASY Waste) and the Liquid Waste tank is connected.

#### 6.3.8.1 Prime SPOLV

##### 6.3.8.1.1 System Initialization

- a. Close any open software and start the **Service Software**.
- b. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- c. Click onto the **Instrument** button and choose the **Full initialization via COP** button to start the initialization sequence.
- d. The process is successful when the **Processing** window closes without failure message.

##### 6.3.8.1.2 Rinsing of SPOLV module

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Liquid** button and choose the **Prime Pipettor** button to open the dedicated window.
- c. At the window **Prime Pipettor** for the rinsing with System Liquid use following values and start with **Execute**.
  - Cycles: 24
  - Wash time: 18
  - OLV Verification: Enabled

6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

Figure 6–39: Pipettor rinsing

- d. The process is successful when the **Processing** window closes without failure message.

**CAUTION****Be aware of aspiration at the Wash station**

Check that the Wash station does not overflow during the rinsing of the SPOLV.

**WARNING****Air inside the MANN filter**

While the system is priming the SPOLV, please observe the liquid flow from the System liquid tank through the MANN filter: pay attention that the filter is not full of air. If needed, tap on it to remove air as much as possible.

**6.3.8.2 Prime Washer Module****6.3.8.2.1 Assign cuvettes**

- Open the Cuvette - Tip Loading Module.
- Place the cuvette trays in the dedicated position of the Cuvette - Tip Loading Module and close the it.
- In the **Service Software** at the navigation window, choose **System** and the **Cuvette Tray State** bracket.
- Fill the Cuvette Trays showed on the screen according to the actual situation and then press the button **Set Cuvette State**.

**CAUTION****Be aware of aspiration**

Check that at the last cycles the aspiration at all three Washer needles aspirates the fluid that was previously dispensed.

The tubing to the manifolds shall be without air bubbles and tight.

**CAUTION****Be aware of initialization**

Be aware that the system initialization will delete any Cuvette State setting. After a system initialization, always check the Cuvette State.

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

### 6.3.8.2.2 Rinsing of Washer Module

- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Liquid** button and choose the **Prime Washer** button to open the dedicated window.
- At the window **Prime Washer** use following values and start with **Execute**.
  - Cycles Port 1: 38
  - Cycles Port 2: 11
  - Cycles Port 3: 11
  - Liquid Control: Enabled

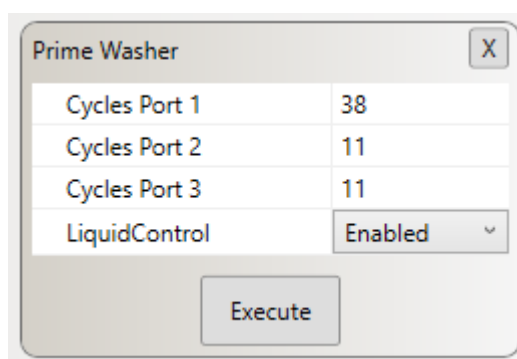


Figure 6–40: Rinsing of Washer Module

- The process is successful when the **Processing** window closes without failure message.

### 6.3.8.3 Prime Reader Module

#### 6.3.8.3.1 Assign cuvettes

- Open the Cuvette - Tip Loading Module.
- Place the cuvette trays in the dedicated position of the Cuvette - Tip Loading Module and close the it.
- In the **Service Software** at the navigation window, choose **System** and the **Cuvette Tray State** bracket.
- Fill the Cuvette Trays showed on the screen according to the actual situation and then press the button **Set Cuvette State**.

#### CAUTION

##### Be aware of aspiration

Check that at the last cycles the aspiration at the needle aspirates the fluid that was previously dispensed.

The tubing to the manifolds shall be without air bubbles and tight.

**CAUTION****Be aware of initialization**

Be aware that the system initialization will delete any Cuvette State setting. After a system initialization, always check the Cuvette State.

**6.3.8.3.2 Rinsing of Reader Module**

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Liquid** button and choose the **Prime Reader** button to open the dedicated window.
- c. At the window **Prime Reader** use following values and start with **Execute**.
  - Cycles: 40
  - Liquid Control Enabled

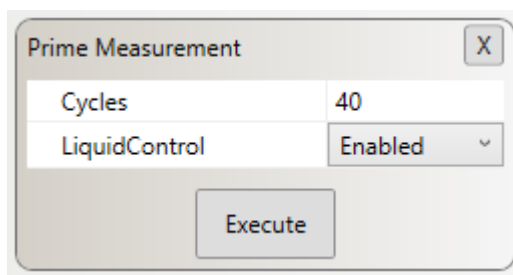


Figure 6–41: Rinsing of Reader Module

- d. The process is successful when the **Processing** window closes without failure message.

## 6.4 Operation Qualification

### 6.4.1 Sample Bay

1. Via **Service Software**, initialize the Sample Module, see Chapter 21.4.1 on page 21-21.
2. Check barcode scanner configuration, see Chapter 21.4.2.1 on page 21-21.
3. Perform a barcode reading test using customer barcoded vials, see Chapter 21.4.2.2 on page 21-21.
4. If needed, plug in the external handheld barcode reader and configure it, see Chapter 32.2.1 on page 32-2.

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

---

### 6.4.2 System

1. Via **Service Software**, initialize the System, see Chapter 12.8.1 on page 12-60.
2. Perform the Cuvette Endurance Run, see Chapter 12.8.1.2.1 on page 12-63.

### 6.4.3 Backup of system configuration

1. Perform backup of `Coordinate.coo` file.
2. Perform backup of “.rac” files.

### 6.4.4 Manuals update

1. Open the **LIAISON<sup>®</sup> XS Main UI**.
2. Ensure that the country code is set accordingly to the user needs to download/import appropriate IFU for the specific language/country, see Chapter 32.5.4 on page 32-20.
3. Install the most updated version of the Instrument Manuals, corresponding to the Instrument SW version and customer's language for the specific country. This can be done:
  - using the **Download** functionality available in the **Online Help** section of the **Main UI** (only if the system was previously connected to lot);
  - using the **Acquire** functionality available in the **Online Help** section of the **Main UI** in case the appropriate instrument package was previously downloaded from **Dialog**.

### 6.4.5 System Verification

1. Open the **LIAISON<sup>®</sup> XS Main UI**.
2. Log-in as Service user and initialize the instrument.
3. Check the **Status - Consumables** toggle and assign the Consumables (cuvettes and tips) in the **Status - Plastic** toggle.
4. Load cuvette in the Incubator Module.

### 6.4.6 Installation Final Test

#### 6.4.6.1 Instrument setup

Before testing:

- All units must be properly installed and checked for the HW performances;
- The instrument shall reach a stable thermal condition;
- The instrument shall be fully primed with approved and valid Starters reagents, Wash Buffer and System Liquid.

#### 6.4.6.2 Materials

See Chapter 6.2.2 on page 6-4 for details.



Last revision of `Installation_Final_Test_Liaison_XS_[Assay].xls` spreadsheet shall be also available to collect all data. This file is attached to Corp-TN-000292 (former LXS TN 009) document. Last revision shall be downloaded and used.

#### 6.4.6.3 Background Wash (BGW)

The Background Wash (BGW) measures the noise of the system inherent the washing and signal acquisition steps, mainly dependent on the involved assemblies (i.e Washer and Reader modules, while Pipettors are not involved).

Ten cuvettes are loaded in the Incubator and moved to the Washer; during the washing phase, different Wash Buffer volumes are injected and aspirated in each cuvette at the 3 Washer ports:

- aspiration, 700µl injection, 700µl aspiration, 1200µl injection (at the first injector);
- 1200µl aspiration, 700µl injection, 700µl aspiration, 700µl injection (at the second injector);
- 700µl aspiration, 700µl injection, 700µl aspiration (at the third port).

Then, the cuvettes are moved to the Reader where Starters pumps inject 200 µl each, in each cuvette.

The readings of the 10 replicates provide 10 RLU values, whose CV% is used in order to assess the precision of the washing and of the measurement.

Three series of ten replicates of Background Wash have to be performed.

#### 6.4.6.4 Light Check Right and Left (LCR/LCL)

Light Check (LC) assesses the performance of Pipettors, both left and right, and of the Reader module.

Ten replicates are run; for each replicate, the relevant Pipettor arm aspirates the LC solution and dispenses it into the cuvette; then the cuvette moves to the Reader, where Starters are injected (200µl for each Starter Pump).

The readings of the 10 replicates provide 10 RLUs values, whose CV% is used in order to assess the precision of the pipetting and of the measurement.

Six series of ten replicates of Light Check Left has to be performed.

Six series of ten replicates of Light Check Right has to be performed.

#### 6.4.6.5 IFTP BGW & LCs Procedure

The following procedure applies:

1. Prepare 2 bottles of Light Check solution, following the Instructions for Use inserted in the package;
2. Put one bottle in the dedicated holder of the Ancillary tray and load the Ancillary tray in the Reagent bay;
3. Put the second bottle in the dedicated sample rack (L rack);
4. Insert the sample rack and assign the Light Check Product Number (319101) as SID;
5. Run **Service - 3BGW 6LCL 6LCR** task from the dedicated window of **Main UI (Maintenance, Tasks)**;
6. Remember to double **Accept** the Light Check Left and Right insertion in the **Service - 3BGW 6LCL 6LCR** task;
7. Export the results from **Maintenance, Logs** and check the results;

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

---

8. Collect and process the sets of data with the dedicated spreadsheet.

### Acceptance criteria

Refer to Corp-TN-000292 (former LXS TN 009) for details.

#### NOTE

The execution of **Service - 3BGW 6LCL 6LCR** maintenance task represents the easiest way to run Background Wash and Light Check Installation Final Test.

Anyway, it is possible to run the test also using other maintenance tasks:

- Service - 3 series Background Wash, Service - 6 series LightCheck Left and Service - 6 series LightCheck Left;
- Background Wash + Background Wash + Background Wash, LightCheck Left + LightCheck Left + ..., LightCheck Right + LightCheck Right + ...

In any cases, the successful execution of:

- 3 consecutive Background,
- 6 consecutive Light Check Left,
- 6 consecutive Light Check Right,

shall be always achieved.

---

### 6.4.6.6 Assay performance testing

The Assay kit should be performed in order to verify the performance of the instrument to perform an Assay Routine, with the contribution of Pipettors, Incubator, Washer and Reader modules.

Calibration and twenty replicates of an Assay kit has to be performed. The Assay kit depends upon the Country the instrument is installed in (see Corp-TN-000292 - former LXS TN 009 for details).

### 6.4.6.7 IFTP Assay Procedure

The following procedure applies:

1. Unpack, handle and prepare Reagent and Control according to the related Instruction For Use;
2. Define the specified Control scanning the DiaSorin Control COA (Certificate of Analysis) by using the Handheld Barcode Reader or by inserting manually the data from the COA;
3. Set the Control replicates to 20;
4. Load the Assay kit in the Reagent Area, if the integral is not correctly detected load the correct Assay Protocol;
5. Wait the agitation time and start the Assay kit calibration;
6. Place Control vial on its proper rack and load it in Sample Area;
7. Run the Assay Control (20 replicates are needed);
8. Check that the calibration of the Assay kit is valid;
9. Check that the average control result dose lies inside the COA specified range,
10. Collect the dose values (from the 20 replicates) and process data with the dedicated spreadsheet (attached to Corp-TN-000292 - former LXS TN 009) to ensure the CV%;
11. Restore the number of Control replicates to the original value.

**NOTE**

If the Control lies outside the expected range, calibration should be repeated and Control retested. If Control value obtained after successful calibration lies repeatedly outside the predefined range, the test should be repeated using an unopened Control vial. If Control values lie outside the expected ranges, execute troubleshooting.

---

**Acceptance criteria**

Refer to Corp-TN-000292 (former LXS TN 009) for details.

**6.4.6.8 Data collection**

1. Perform raw data and calibration export and backup.
2. Verify that `Installation_Final_Test_Liaison_XS_[Assay].xls` file has been completed.

**NOTE**

The results of the installation activities (BGW, LCL, LCR, Calibration and Control run) must be attached to the Service records (for Subsidiaries only: SGAT WS), together with a detailed description of the Service activities performed on the analyzer.

---

**6.4.7 Printout customization**

1. Install the printer (if necessary, see Corp-TN-000293 - former LXS TN 010 for details).
2. Perform printout customization (if required by the customer).

## 6. LIAISON<sup>®</sup>XS Installation for external-tanks configuration

---

# 7 LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

As LIAISON<sup>®</sup> XS instrument is a complex system and contains delicate electrical and mechanical components, it must be packed by qualified service engineers.

The procedures reported in this chapter allows to successfully complete the removal of LIAISON<sup>®</sup> XS analyzer.

## 7.1 Safety

### WARNING



#### Non-Observance of Safety instructions

The non-observance of safety instruction can result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7.
- Follow all working instruction included in this manual.
- Follow all warning marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

### WARNING



#### Risk of injury due to high weight

Improper lifting/carrying of the instrument can cause serious injuries to the personnel and/or material damages.

- Use suitable transport aids (e.g. fork lift, crane, lift truck) for lifting/carrying the instrument.
- Observe the technical specifications of the instrument, see Chapter 35 on page 35-1, to avoid misjudgments.
- Observe the valid work safety directives and use auxiliary means, if necessary.

### WARNING



#### Never lift the system by yourself!

Manual handling of heavy loads may cause injuries.

The LIAISON<sup>®</sup> XS instrument shall NOT be lifted and carries by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

---

**CAUTION****Instrument reinstallation**

If a removed instrument shall be reinstalled, always replace the Washer and Reader aspiration needles and all tubings of DI water and Wash Buffer lines (for details see the dedicated chapters).

---

**CAUTION****Check System**

During the removal of the system, always follow the steps indicated in **LIAISON<sup>®</sup> XS** Uninstallation check list (see Chapter 33.3 on page 33-7). The proper execution and compilation of this check list will ensure the proper removal of the instruments.

---

## 7.2 Preparation of sodium hypochlorite solution

The execution of some procedures to perform during instrument removal foresees the use of a sodium hypochlorite solution that shall be prepared according to the procedures reported below.

### 7.2.1 Instruction to prepare 1 L of solution at a concentration of active chlorine ranging from 0.5% to 0.9%

1. Fill the cleaning solution tank with 0.9 L of water.
2. Add 100 mL of a sodium hypochlorite solution (e.g. bleach) with a content of active chlorine ranging from 5% to 9%. In case the available sodium hypochlorite should have a different titer, please adjust the dilution accordingly.
3. Place the proper cap and briskly mix the container.

## 7.3 Data deleting and decontamination

1. Delete patient database.
2. Empty Liquid Waste tank, add hypochlorite solution (see chapter 5.4.6 “Connecting and separating the Liquid Waste” of the User Manual) and load it back into the system.
3. Using the solution prepared in Chapter 7.2.1, run **System Shutdown** task from the dedicated window of **Main User Interface** (Main category **Maintenance**, sub-category **Task**), skipping the last step. See also chapter 5.11.1 “Procedure for System Shutdown and Wakeup” of the User Manual).
4. Remove all tanks from the system.

### 7.3.1 Decontamination of Liquid Waste tank

---

**CAUTION**

The procedure reported in this section shall be applied only when liquid waste tank shall be shipped with the analyzer. Otherwise, it is possible to dispose the tank without performing this additional decontamination.

---

**CAUTION**

The below reported procedure shall be applied only once the Liquid waste tank has been emptied according to the procedure described in chapter 5.4.6 “Connecting and separating the Liquid Waste” of the User Manual.

In particular it is necessary to verify that at least 200 mL of commercial hypochlorite or bleach are loaded in the waste tank at the beginning of the procedure.

---

1. Disconnect the sensor connector together with the liquid connector of the liquid waste tank.
2. Unscrew the cap of the tank.
3. Add DI water until tank is completely filled.
4. Screw the cap back and wait for at least 30 minutes (during this period it is possible to continue with the decontamination of the other parts of the instrument).
5. Empty the tank and rinse it thoroughly with DI water.
6. Repeat the previous operation at least two more times.

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

---

**NOTE**

To ensure an adequate rinsing, it is recommended to fill the tank at least at half volume.

---

### 7.3.2 Removal of all consumable parts

1. Close any open software and start the **Service Software**.
2. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
3. Click onto the **Instrument** button and choose the **Full initialization via COP** button to start the initialization sequence.
4. The process is successful when the **Processing** window closes without failure message.
5. Click onto the **Cuvette Handling** button and choose the **Clear all** button to automatically empty the Washer Module, the Incubator Module and the Reader Module.
6. Close the **Service Software**.
7. Remove all the samples tubes out of the Sample racks.
8. Remove all the integral kits out of the Reagent Module.
9. Remove all the ancillaries and holders out of the Ancillary rack.
10. Remove all the disposable tips trays out of the Cuvette Tip drawer.
11. Remove all the cuvette trays out of the Cuvette Tip drawer.
12. Close the EASY Waste and remove it from the Solid Waste bin.

### 7.3.3 Cleaning procedures

**NOTE**

Tanks shall be cleaned using the procedures reported in chapter 7.5 “Clean the wash buffer and DI water tanks” of the User Manual, but using sodium hypochlorite solution at 0.5%-0.9% of active chlorine.

---

1. Clean the pipettors, the instrument interior, the tanks and the covers (see chapters 7.5 “Clean the wash buffer and DI water tanks” and 7.6 “Manual Cleaning of the instrument” of the User Manual).
2. Using tissues soaked with a solution of sodium hypochlorite clean:
  - a. Incubator cover (inner and outer);
  - b. Reader cover (inner and outer);
  - c. Liquid waste basin;
  - d. Liquid waste tank (externally, if it is necessary to ship it with the instrument);
  - e. Tank areas;
  - f. Solid waste bin;
  - g. Sample bay cover;
  - h. PCB Anti-splash cover.
3. Using tissues soaked with an alcoholic/disinfectant solution clean:



- a. Reagent Module cover;
  - b. Cuvette Tip drawer;
  - c. Starter Reagents Module;
  - d. Sample racks;
  - e. Ancillary tray;
  - f. Panel-PC;
  - g. Top covers (inner parts);
  - h. Covers (outer parts).
4. Remove and dispose the washer and reader needles.

#### 7.3.4 Confirm decontamination

1. Complete the certificate of the decontamination (included in DiaSorin Technical GOP 23.7201).
2. Add the certificate of decontamination to the transportation documents.

### 7.4 Packaging of the system

#### 7.4.1 Preparation

---

**NOTE**

At least two persons are required for packing the instrument.

---

##### 7.4.1.1 Removal of the Panel PC

1. Open the Top covers.
2. Loose two screws (3) of cable carrier (2).
3. Slide the cable carrier (2) out of the keyholes and set cable carrier aside.

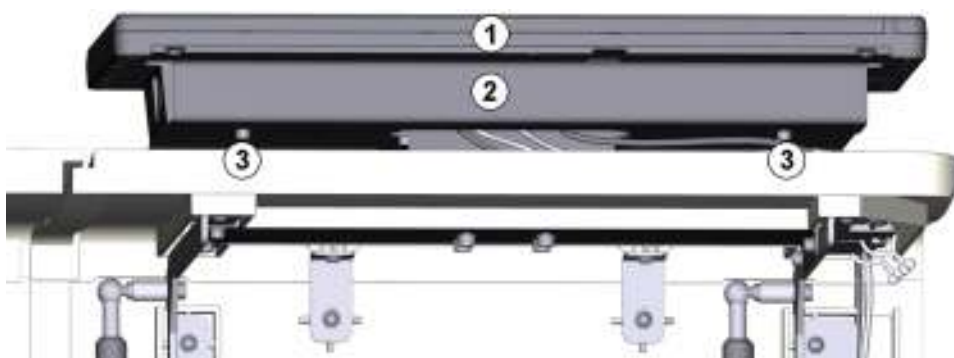


Figure 7–1: Panel PC with Right Top Cover opened, front view

## 7. LIAISON®XS Removal for internal-tanks configuration

4. Disconnect Panel PC connections power cable (4), USB connection (5 & 6) and LAN cable (7).

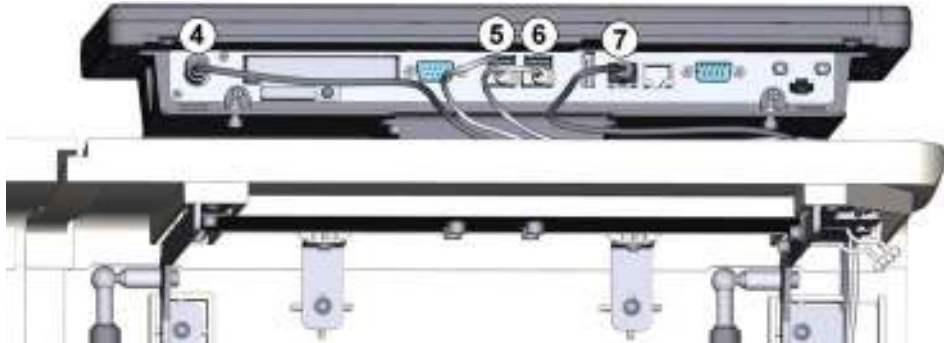


Figure 7-2: Panel PC with Right Top Cover opened, connections

5. Loosen four screws (9) of Panel PC. Lift Panel PC (1) out of the screw holes and set it aside.

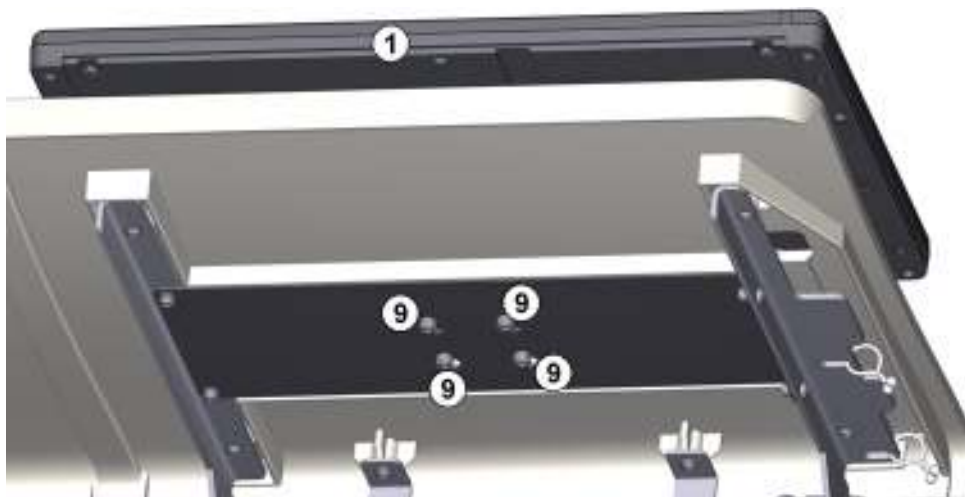


Figure 7-3: Panel PC screws

### 7.4.1.2 Remove Left and Right Top covers

1. Hold the Left Top Cover (1) and unscrew four screws (2) from the cover brackets.

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

2. Lift Left Top Cover up and set it aside.

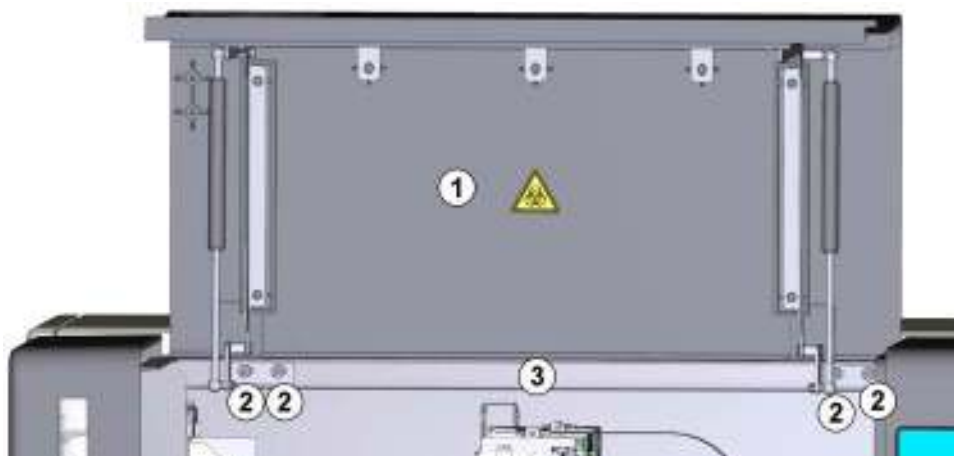


Figure 7-4: Left Top Cover opened, with details

3. Fasten the four screws (2) back into the support bar (3).
4. Move to the Right Top Cover and remove Panel PC cables (1) out of the cable holders (2).

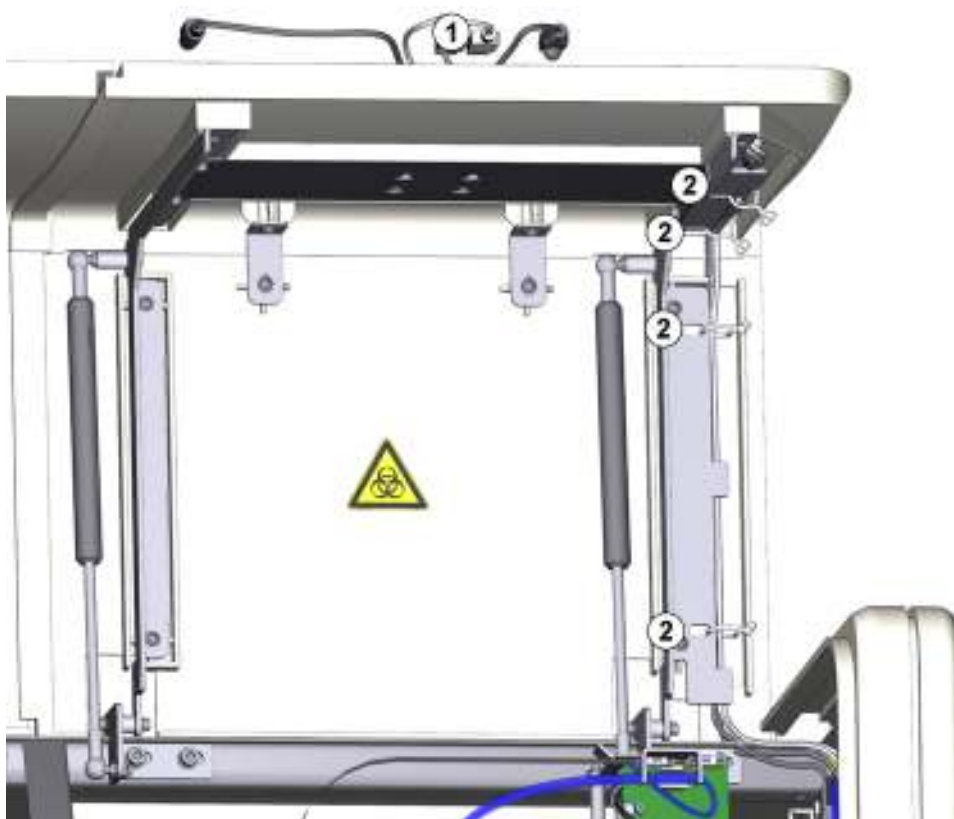


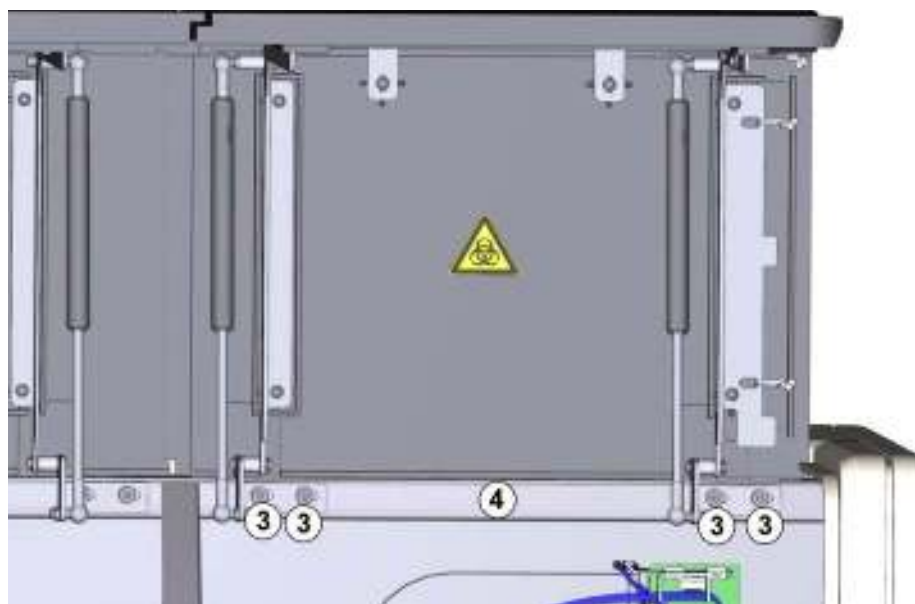
Figure 7-5: Right Top Cover with Panel PC cable

5. Hold the Right Top Cover and unscrew four screws (3) from the cover brackets.
6. Lift Right Top Cover up and set it aside.

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

---

7. Fasten the four screws (3) back into the support bar (4).



*Figure 7-6: Right Top Cover with detail*

### 7.4.1.3 Removal of the Right Side Cover

1. Unscrew cross-slotted screw (3).
2. Unplug the snap locks of the Right Side Cover and unplug the cable from the Soft Power Button PCBA (4).
3. Set Right Side Cover to the side.

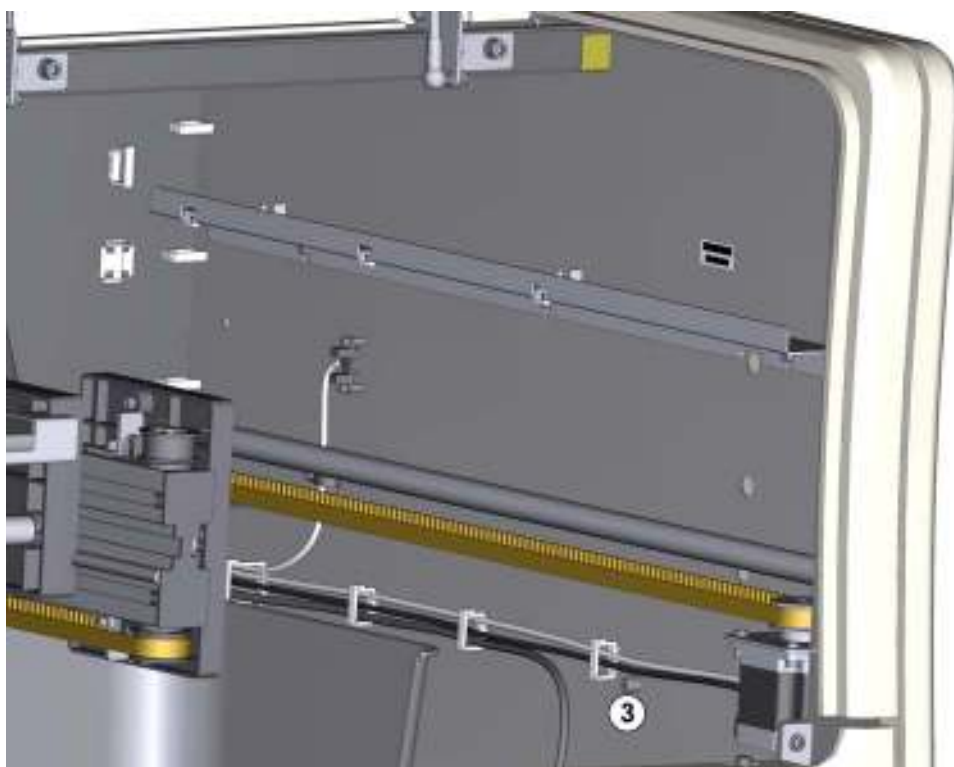


Figure 7-7: Fixing screw of the Right Side Cover, inside the instrument

#### 7.4.1.4 Removal of the Left Side Cover and Bulk Fluid Door

1. Unscrew cross-slotted screw (1).
2. Unplug the snap locks of the Left Side Cover and set the cover to the side.

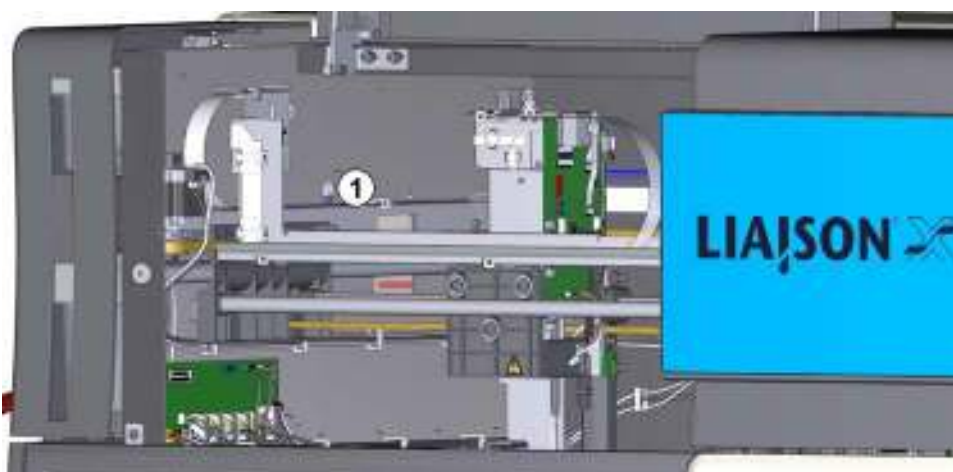


Figure 7-8: Fixing screw of the Left Side Cover, inside the instrument

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

3. Lift the Bulk Fluid door (1) out of the hinges (2) and set it aside.

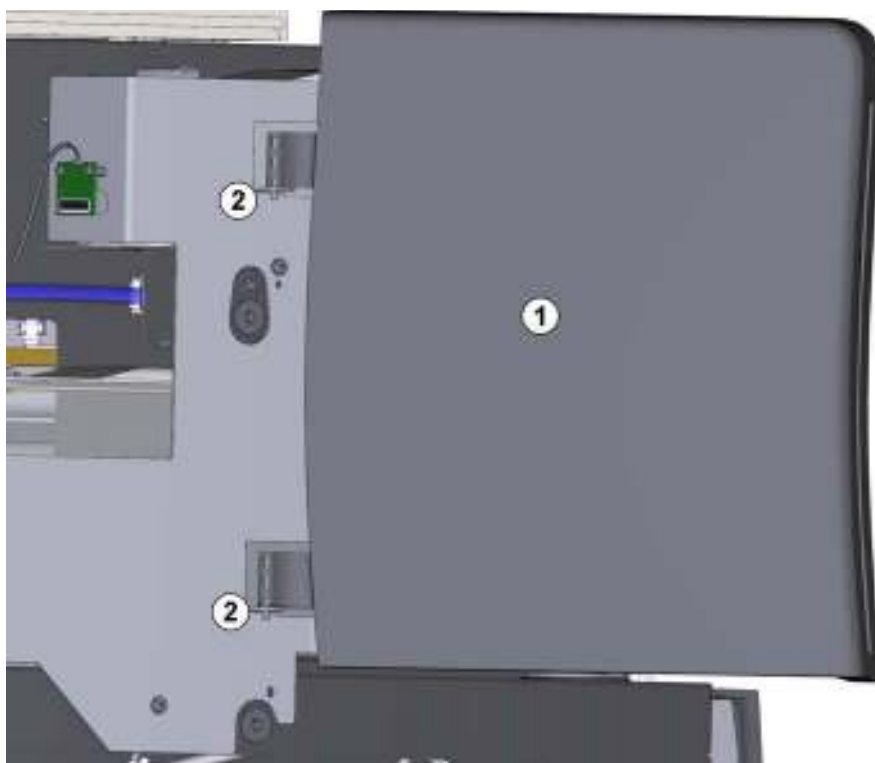


Figure 7-9: Bulk Fluid door

### 7.4.1.5 Removal of the Top Cover

1. Loosen two screws (2) at right side of the Top Cover (1).

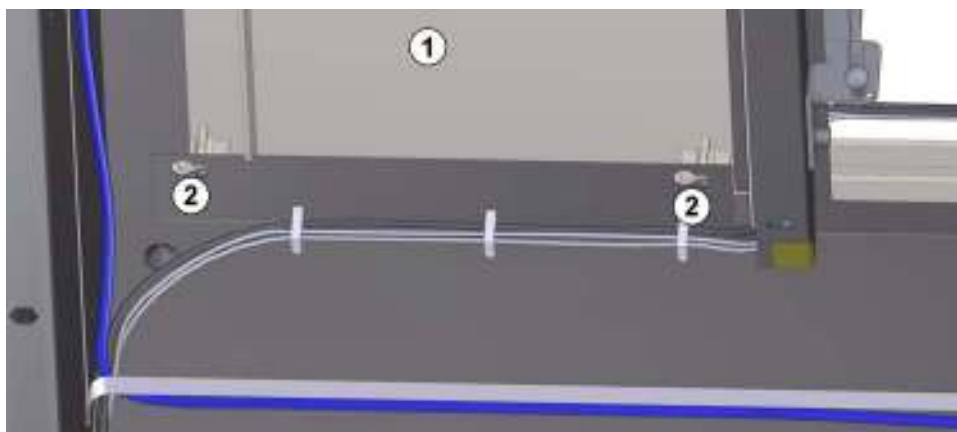


Figure 7-10: Top Cover, right side view inside the instrument

## 7. LIAISON<sup>®</sup> XS Removal for internal-tanks configuration

2. Loosen screw (2) at middle of the Top Cover (1).

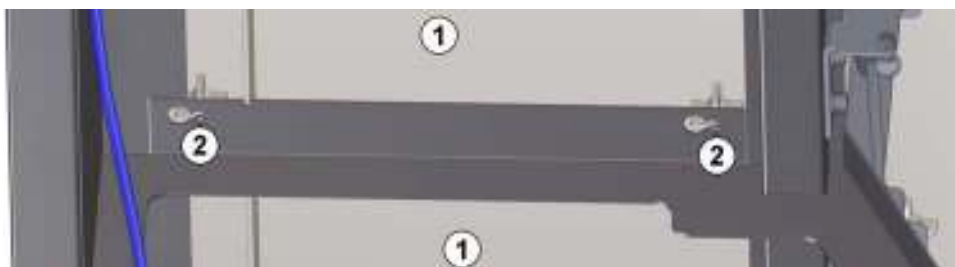


Figure 7-11: Top Cover, middle view inside the instrument

3. Loosen screw (2) at left of the Top Cover (1).

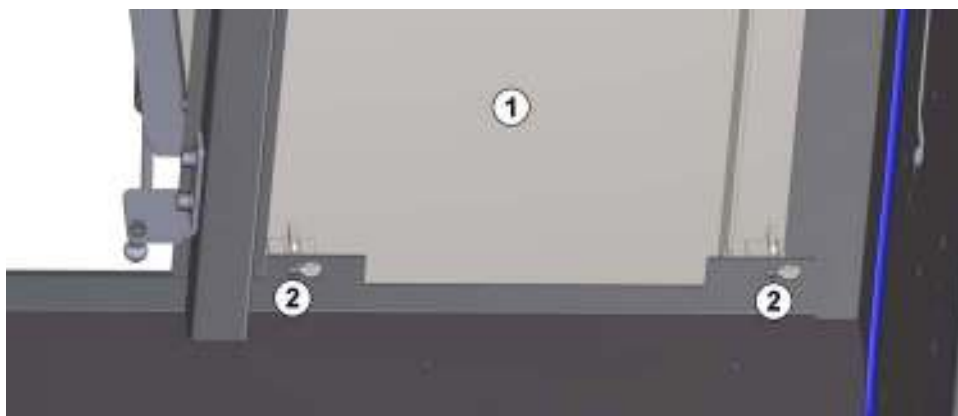


Figure 7-12: Top Cover, left side view inside the instrument

4. Slide Top Cover out of the key holes. Lift Top Cover out and set it aside.
5. Fasten six screws (2) of the Top Cover (1).

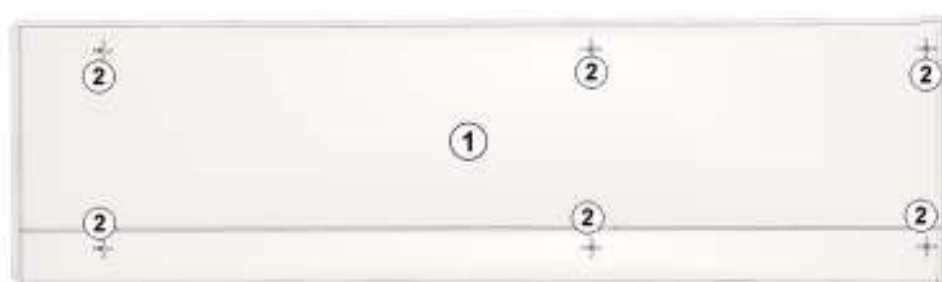


Figure 7-13: Top cover, bottom view



## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

6. Lay the Panel PC cable (3) into the gap of the support bar and secure the cables with two cable ties (4).

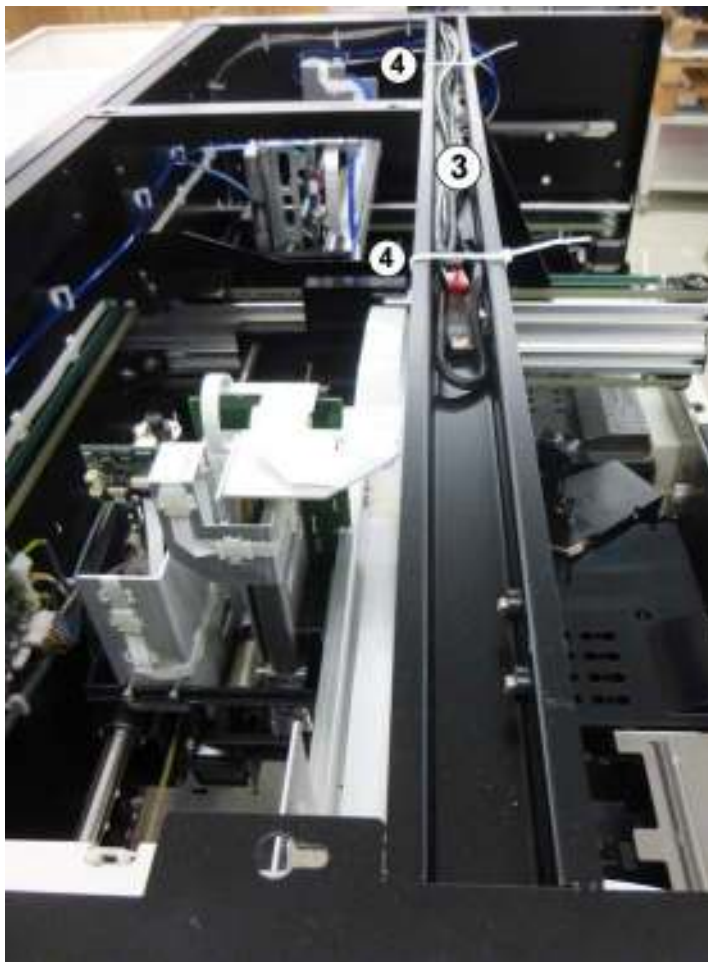


Figure 7-14: Gap of support bar



#### 7.4.1.6 Removal of the Front Module

1. Loose screws (5) and remove the handle (4) of the Cuvette - Tip Loading Module. Then, close the Cuvette - Tip Loading Module.

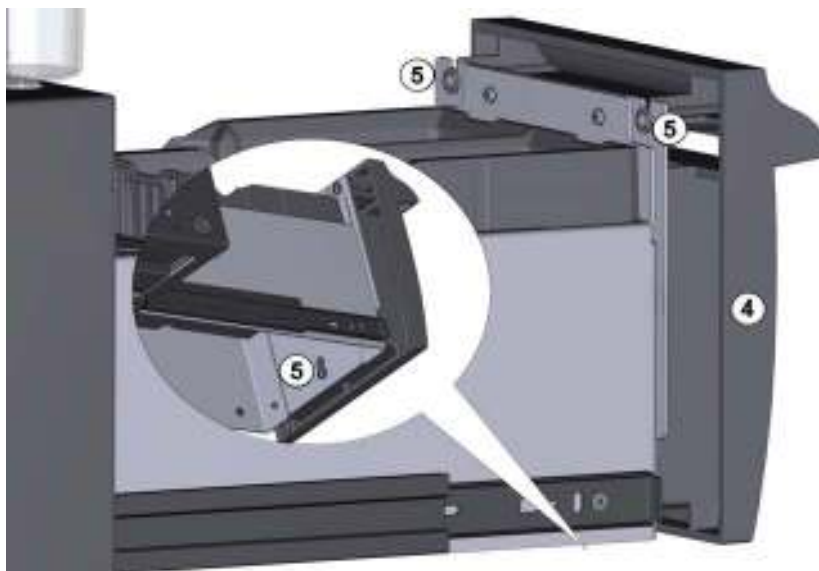


Figure 7-15: Cuvette - Tip Loading Module

2. Lift and pull Solid Waste bin out (3).

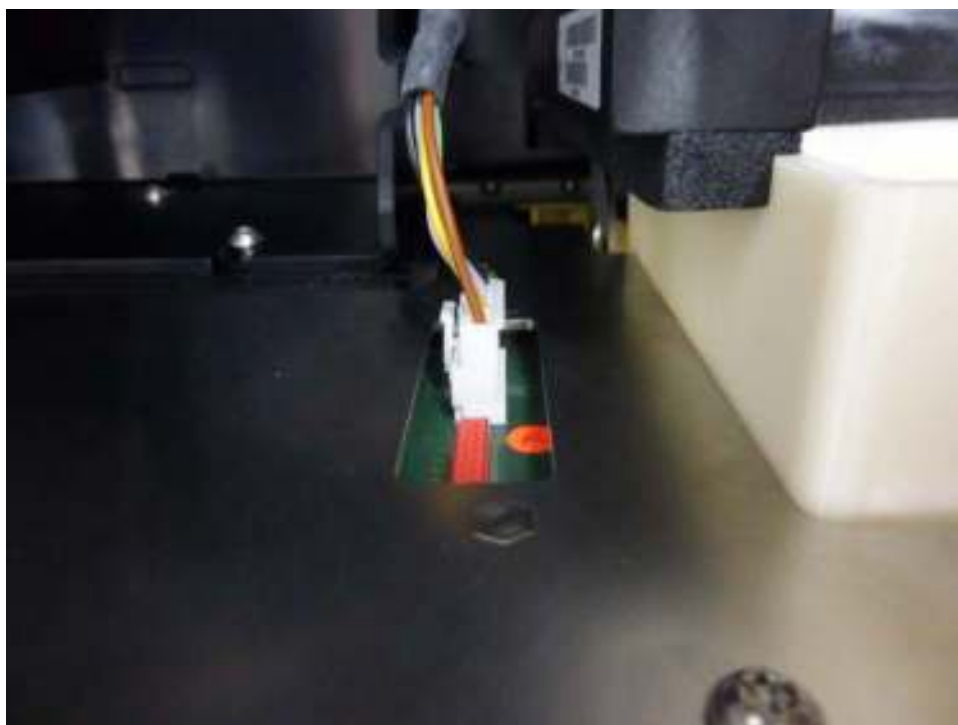


Figure 7-16: Solid waste bin handle

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

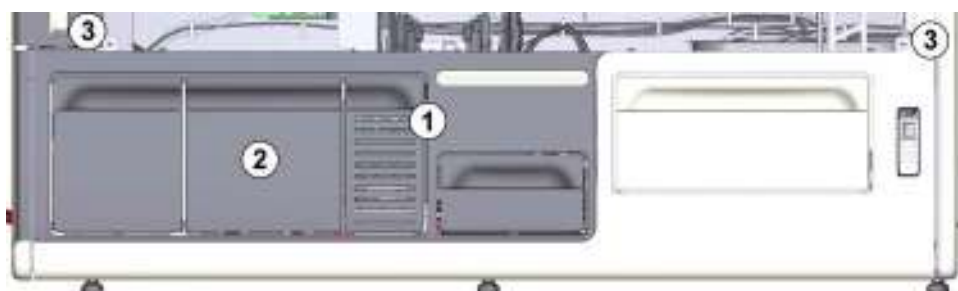
---

3. Unplug the Front Module PWR/Can cable.



*Figure 7–17: Front Module PWR/Can cable*

4. Unscrew two screws (3), tilt the Front Module (1) to the front, lift it up and set it aside.



*Figure 7–18: Front Module*

#### 7.4.1.7 Removal of Ancillary tray

1. Remove Ancillary tray (7) out of Reagent Module (6).

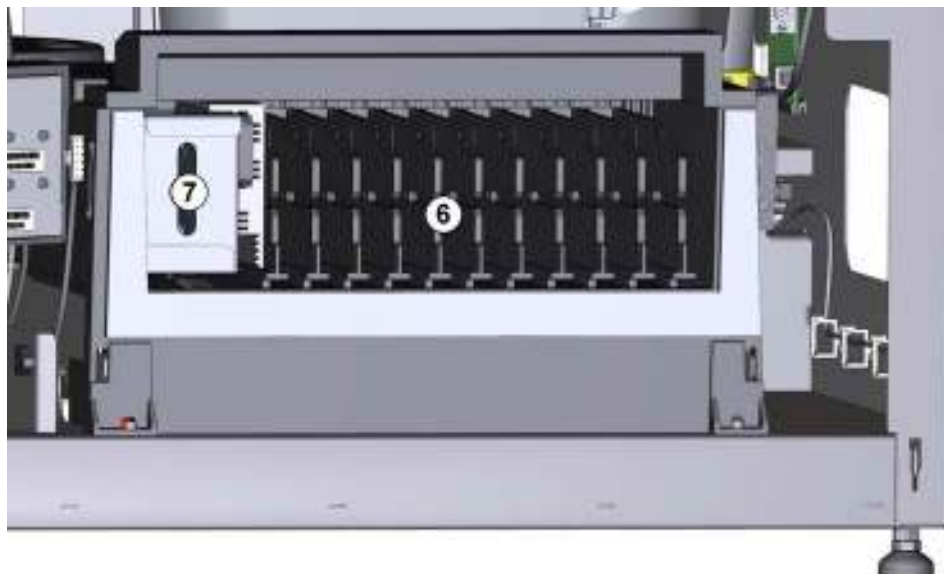


Figure 7-19: Reagent Module, front view without Front Module

#### 7.4.1.8 Secure Sample racks

1. Insert the empty Sample racks (8) into the Sample Module.

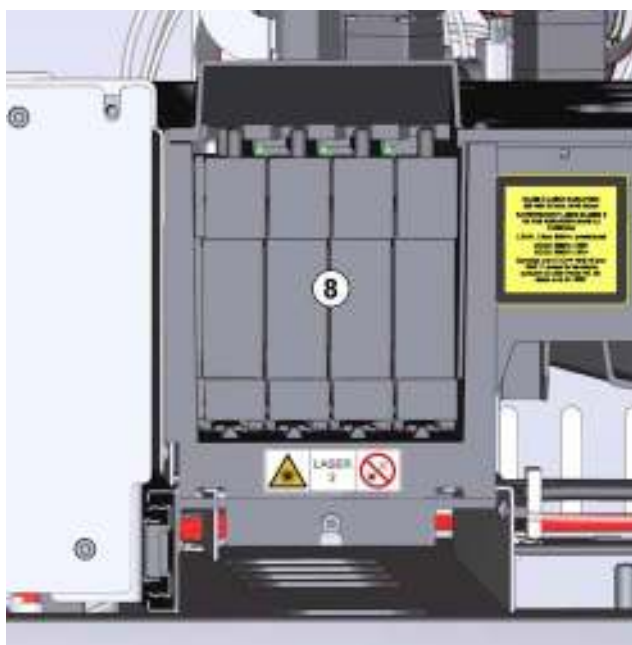


Figure 7-20: Sample Module, front view without Front Module

## 7. LIAISON®XS Removal for internal-tanks configuration

### 7.4.1.9 Secure Pipettors and Gripper

1. Place the securing tube (2) over the z-axis bar of the Sample Pipettor (1).
2. Place the securing tube (4) over the z-axis bar of the SPOLV (3).

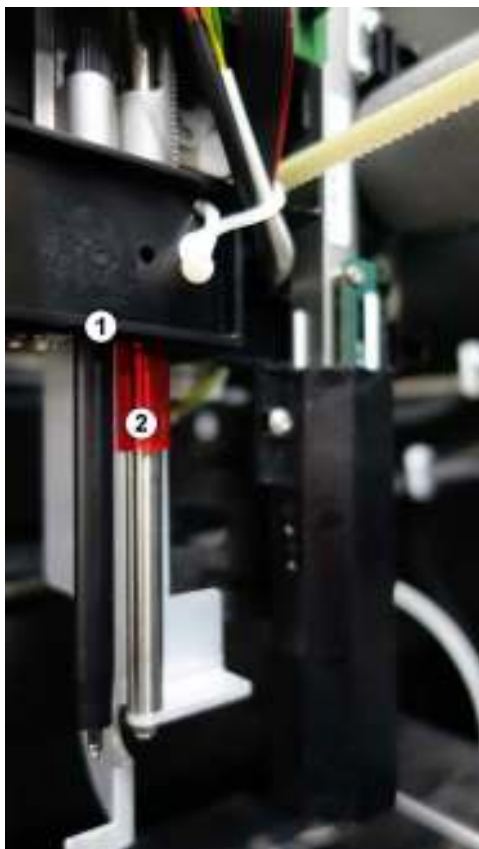


Figure 7-21: Sample pipettor



Figure 7-22: SPOLV

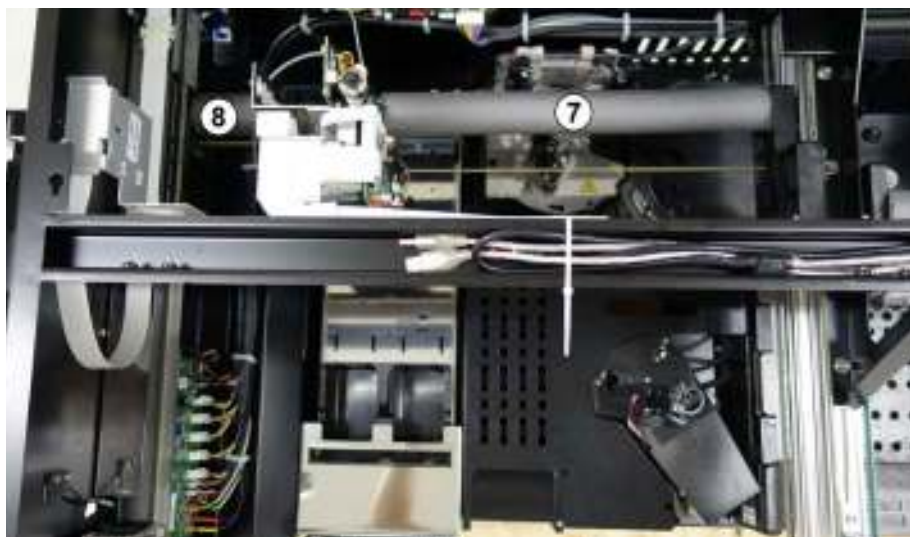
## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

3. Secure Gripper (5) in z-axis with a cable tie (6).



*Figure 7-23: Gripper with secure cable tie*

4. Place the securing foam (7) and (8) over the x-axis of the Left Pipettor Arm.

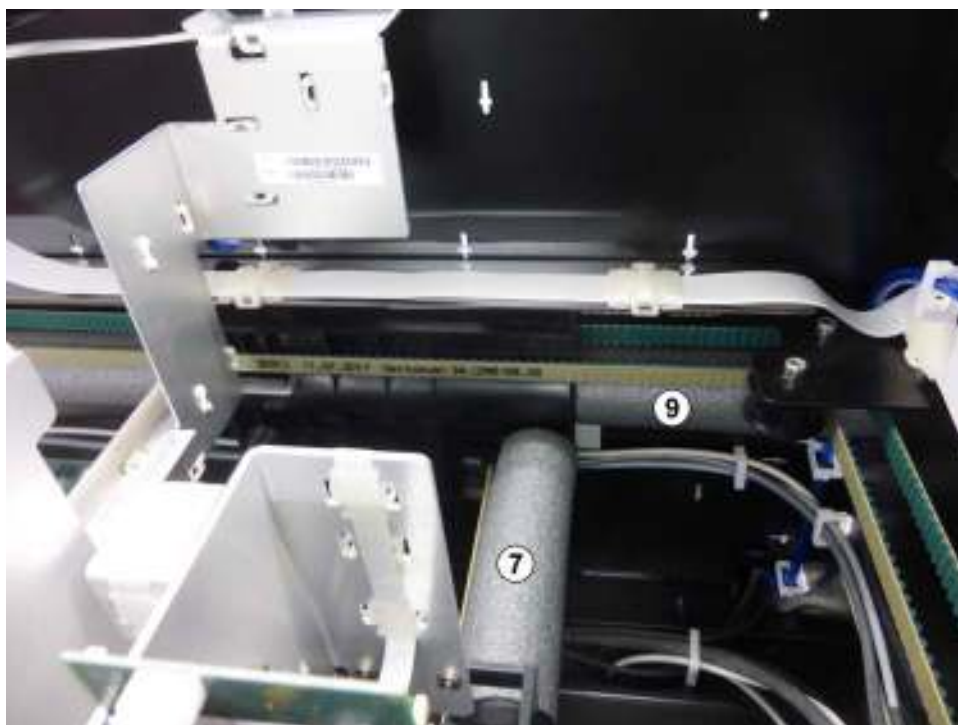


*Figure 7-24: Left Pipettor Arm with securing foam over the x-axis*



**7. LIAISON®XS Removal for internal-tanks configuration**

5. Place the securing foam (9) over the rear left y-axis of the Left Pipettor Arm.



*Figure 7-25: Left Pipettor Arm with securing foam on y-axis rear side*

6. Place the securing foam (10) over the front left y-axis of the Left Pipettor Arm.



*Figure 7-26: Left Pipettor Arm with securing foam on y-axis front side*

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

7. Place the securing foam (11) and (12) over x-axis of the Right Pipettor Arm.
8. Place the securing foam (13) and (14) over the front and rear y-axis of the Right Pipettor Arm.

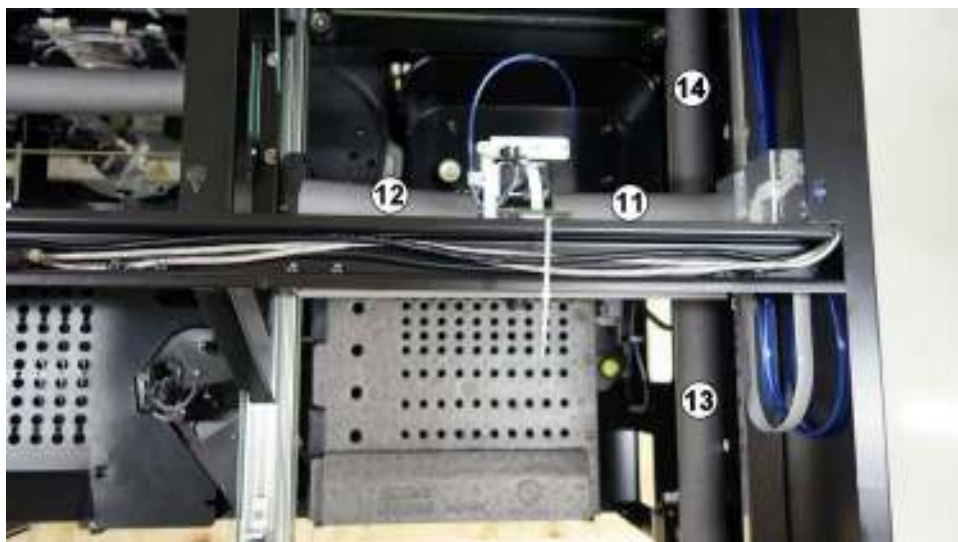


Figure 7-27: Right arm, securing foam on x- and y-axis

### 7.4.1.10 Removal of the Liquid Waste tubing

1. Carefully remove the liquid waste tubing and cable out of the key hole (3) in the instrument Left Side.



Figure 7-28: Liquid waste tubing

## 7. LIAISON<sup>®</sup> XS Removal for internal-tanks configuration

2. Wrap the end of the liquid waste tubing and cable with a lint free cloth and secure it with an adhesive tape, placing them in the instrument Left Side, under the Starters pumps



Figure 7-29: Instrument Left Side

### 7.4.1.11 Removal of the Peristaltic pumps caps (WAP and LWP)

1. Remove the caps of the LWP (10) and of the WAP (11).
2. Remove on one side each of the Peristaltic tubes from the two pumps.



Figure 7-30: WAP and LWP pumps without the caps and with the tubes on one side removed



## 7.4.2 Packing for the accessories cardboard box

### 7.4.2.1 Packing of the Panel PC

1. Insert the Panel PC (2) into the bag.
2. Set the Panel PC into the modeled foam (3) in the original Panel PC cardboard box (1).



Figure 7-31: Packing Panel PC

3. Set the modeled foam (4) onto the Panel PC.



Figure 7-32: Packing Panel PC

4. Insert the main power cable (5) into the cardboard box.

## 7. LIAISON®XS Removal for internal-tanks configuration

5. Close the cardboard box and secure with adhesive packing tape.



*Figure 7-33: Packing Panel PC*

### 7.4.2.2 Packing of the Side Covers

1. Insert the bottom foam plate (2) into the cardboard box (1).



*Figure 7-34: Cardboard box for the covers*

## 7. LIAISON<sup>®</sup> XS Removal for internal-tanks configuration

2. Lay the Right Side Cover (3) onto the foam plate.



Figure 7-35: Cardboard box for the covers

3. Insert the foam plates (4) into the Right Side Cover.



Figure 7-36: Cardboard box for the covers

**7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration**

---

4. Insert the Left Side Cover (5).



*Figure 7–37: Cardboard box for the covers*

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

5. Insert two foam strips (6) onto the Left Side Cover.
6. Close the cardboard box with adhesive packing tape.



Figure 7-38: Cardboard box for the covers

### 7.4.2.3 Packing of the small accessories

1. Insert the foam frame (1) into the cardboard box.



Figure 7-39: Cardboard box for the small accessories

## 7. LIAISON®XS Removal for internal-tanks configuration

2. Insert the Bulk Fluidic Door (2) into the foam frame.



Figure 7-40: Cardboard box for the small accessories

3. Insert the Solid Waste Bin (3) into two foam blocks (4) and (5).



Figure 7-41: Cardboard box for the small accessories

4. Insert the Cuvette - Tip Loading Module handle (6) into two foam blocks (7) and (8) and insert it in the cardboard box.
5. Insert the Ancillary tray (9) and the Barcode Label set (10) into the pocket of the foam block.
6. Pack the Peristaltic pumps caps (11) into bubble wrap and insert into the box.

7. Close the cardboard box with adhesive packing tape.



Figure 7-42: Cardboard box for the small accessories

#### 7.4.2.4 Packing of the other accessories

1. If necessary, clean and decontaminate the following accessories and pack them all together in the dedicated box:
  - Sample tube tool;
  - Ancillary adapters;
  - Two Control Racks;
  - Sample Rack;
  - Sample rack holder;
  - Integral rack holder;
  - External Barcode Scanner;
  - Liquid Waste Basin.

## 7. LIAISON®XS Removal for internal-tanks configuration

### 7.4.2.5 Packing of the Accessories Box

1. Insert the foam block (2) for the DI water tank into the Accessory box (1).

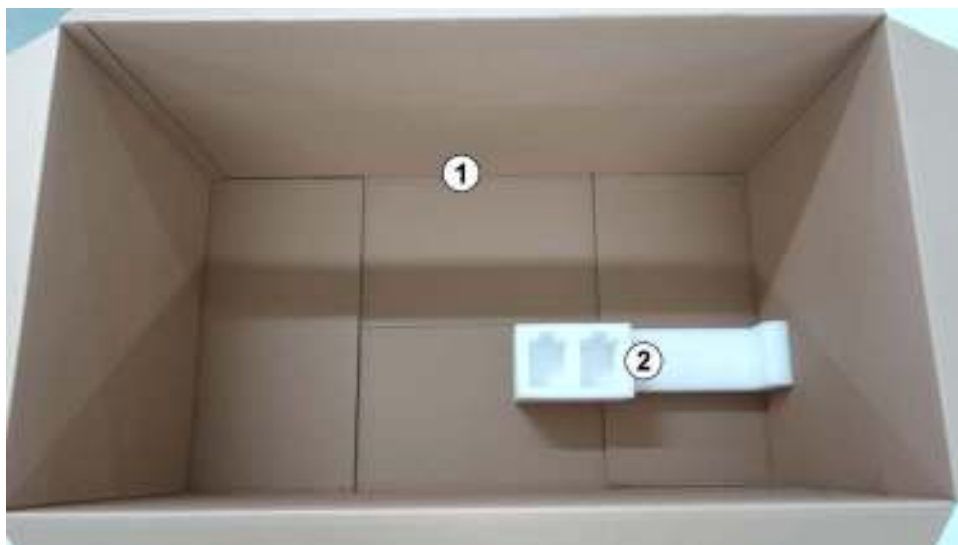


Figure 7-43: Accessory box

2. Lay the packed Side Covers (3) into the box (1).



Figure 7-44: Accessory box



## 7. LIAISON<sup>®</sup> XS Removal for internal-tanks configuration

3. Insert the foam piece (4) onto the box of the Side Covers.



Figure 7-45: Accessory box

4. Lay the Left Top Cover (5) onto the foam piece.
5. Secure the sheet metal corners with two open foam tubes (6).



Figure 7-46: Accessory box

**7. LIAISON®XS Removal for internal-tanks configuration**

---

6. Insert the cardboard box with small accessories (7).



*Figure 7-47: Accessory box*

7. Insert the foam piece for the Right Top Cover (8) onto the cardboard box.



*Figure 7-48: Accessory box*

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

8. Lay the Right Top Cover (9) onto the foam piece.



Figure 7-49: Accessory box

9. Insert the Front Module (10) into the foam blocks (11).



Figure 7-50: Front Module, left side



Figure 7-51: Front Module, right side

**7. LIAISON®XS Removal for internal-tanks configuration**

10. Slide the Front Module (10) with the foam parts (11) into the Accessory box.



*Figure 7-52: Accessory box*

11. Slide the Top Cover (12) into the foam blocks (11).



*Figure 7-53: Accessory box*

## 7. LIAISON<sup>®</sup> XS Removal for internal-tanks configuration

12. Insert the foam piece (13) above the Right Top Cover.



*Figure 7-54: Accessory box*

13. Insert Instrument accessories (Panel PC accessories and Power cables Europe and US) inside the dedicated box.



*Figure 7-55: Panel PC accessories & Power cables*

## 7. LIAISON®XS Removal for internal-tanks configuration

---

14. Lay the Panel PC box (14) onto the Right Top Cover.



*Figure 7-56: Accessory box*

15. Insert the two Wash Buffer tanks (in horizontal) and DI water and Cleaning solution tanks (in vertical) into the foam blocks.



*Figure 7-57: Wash buffer tanks and DI water and Cleaning solution tanks*



---

**7. LIAISON<sup>®</sup> XS Removal for internal-tanks configuration**

---

16. Add the foam piece onto the tanks.



*Figure 7-58: Accessory box*

17. Insert the other small accessory box.
18. Insert the Liquid Waste Tank (if saved and decontaminated).



*Figure 7-59: Accessory box*

19. Lay on top of the Side Covers crushed packing paper.
20. Close the Accessory box with adhesive tape.

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

---

### 7.4.3 Packing of the system

**WARNING****Never lift the system by yourself!**

Manual handling of heavy loads may cause injuries.

The **LIAISON<sup>®</sup> XS** instrument shall NOT be lifted and carried by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

---

**WARNING****Risk of injury due to high weight**

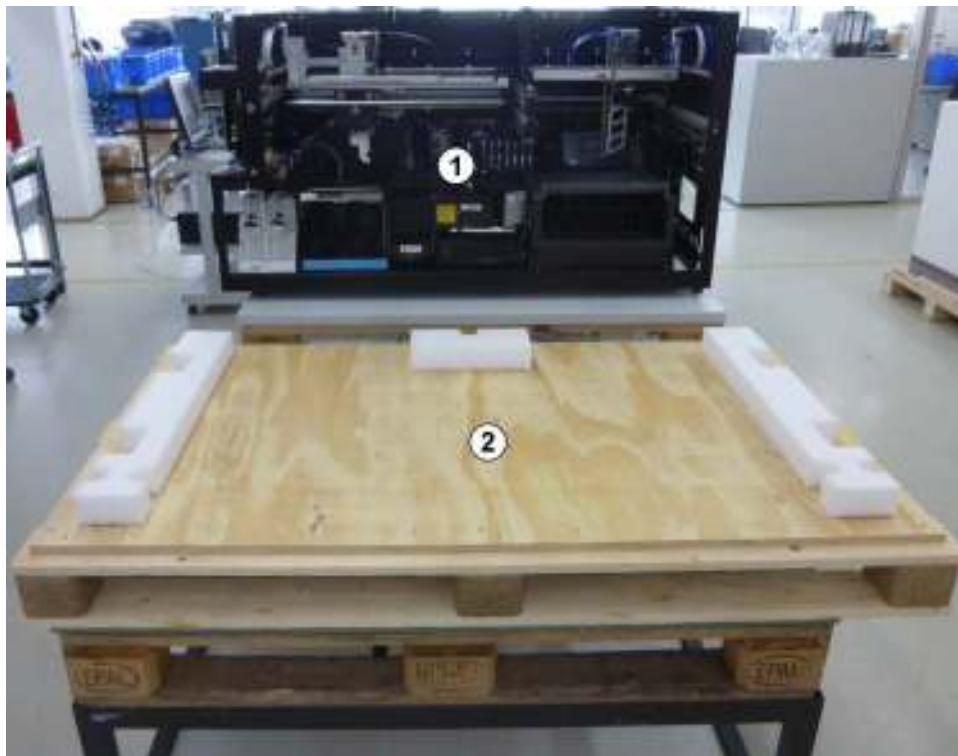
Improper lifting/carrying of the instrument can cause serious injuries to the personnel and/or material damages.

- Use suitable transport aids (e.g. fork lift, crane, lift truck) for lifting/carrying the instrument.
  - Observe the technical specifications of the instrument (see Chapter 35.4 on page 35-3) to avoid misjudgments.
  - Observe the valid work safety directives and use auxiliary means, if necessary.
-



## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

1. Place the instrument (1) onto the shipping pallet (2), taking care to match the pallet housing for the instrument feet.



*Figure 7–60: Instrument and shipping pallet*

## 7. LIAISON®XS Removal for internal-tanks configuration

2. Slide the foam cover (3 and 4) on top of the instrument.



*Figure 7-61: Instrument foam on the Left Side*



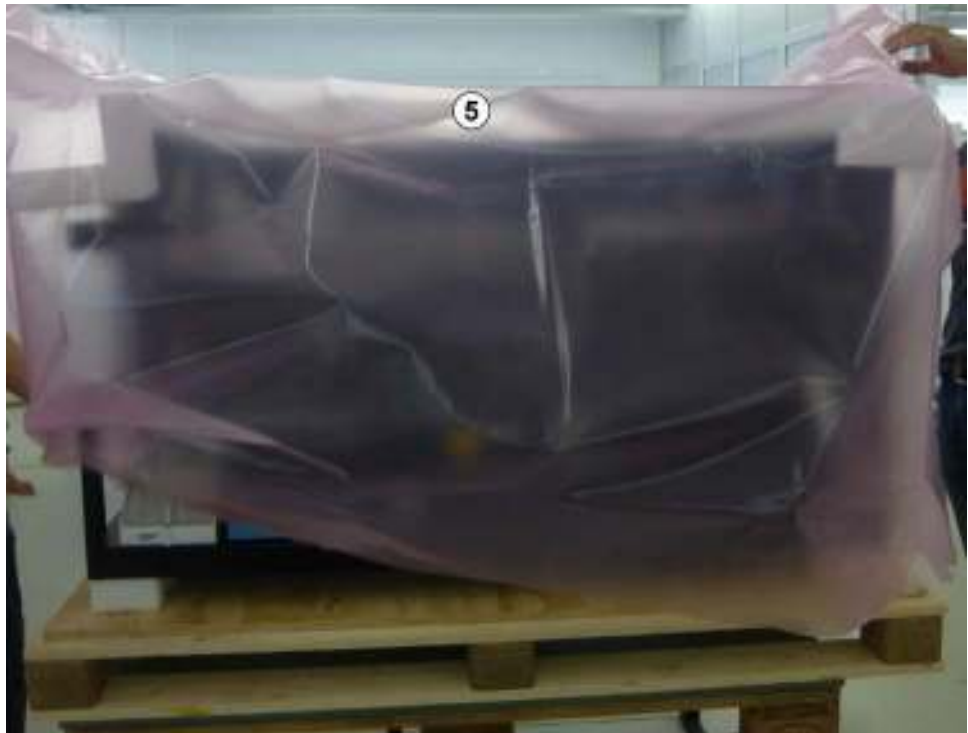
*Figure 7-62: Instrument foam on the Right Side*

---

**7. LIAISON<sup>®</sup> XS Removal for internal-tanks configuration**

---

3. Cover the system with the plastic bag (5).



*Figure 7–63: Instrument covered with plastic bag*

**7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration**

---

4. Lay the C-foam piece (6 and 7) on both sides around the bottom of the instrument.



*Figure 7-64: Instrument C-foam on the Right Side*



*Figure 7-65: Instrument C-foam on the Left Side*

**7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration**

5. Slide the outer cardboard cover (8) over the instrument and over the bottom foam.



*Figure 7-66: Outer cardboard cover*

6. Place the cardboard lid over the outer cardboard cover.



*Figure 7-67: Outer cardboard lid*

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

7. Lift the accessories cardboard box on top of the instrument.
8. Secure accessories cardboard box with packaging tape.



Figure 7-68: Final packaging of the instrument



#### 7.4.4 Transport labeling

1. Add the Shock-watch (1) and the Thermometer indicator (2) onto the packaging cover.



Figure 7–69: Shock-watch and Thermometer indicator

## 7. LIAISON<sup>®</sup>XS Removal for internal-tanks configuration

---



## 8 LIAISON<sup>®</sup> XS Removal for external-tanks configuration

As **LIAISON<sup>®</sup> XS** instrument is a complex system and contains delicate electrical and mechanical components, it must be packed by qualified service engineers.

The procedures reported in this chapter allows to successfully complete the removal of **LIAISON<sup>®</sup> XS** analyzer.

### 8.1 Safety

**WARNING**

---

**Non-Observance of Safety instructions**

The non-observance of safety instruction can result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 2.5 on page 2-7.
  - Follow all working instruction included in this manual.
  - Follow all warning marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
- 

**WARNING**

---

**Risk of injury due to high weight**

Improper lifting/carrying of the instrument can cause serious injuries to the personnel and/or material damages.

- Use suitable transport aids (e.g. fork lift, crane, lift truck) for lifting/carrying the instrument.
  - Observe the technical specifications of the instrument, see Chapter 36 on page 36-1, to avoid misjudgments.
  - Observe the valid work safety directives and use auxiliary means, if necessary.
- 

**WARNING**

---

**Never lift the system by yourself!**

Manual handling of heavy loads may cause injuries.

The **LIAISON<sup>®</sup> XS** instrument shall NOT be lifted and carries by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

---

## 8. LIAISON®XS Removal for external-tanks configuration

---

**CAUTION****Instrument reinstallation**

If a removed instrument shall be reinstalled, always replace the Washer and Reader aspiration needles and all tubings of System Liquid and Wash Buffer lines (for details see the dedicated chapters).

---

**CAUTION****Check System**

During the removal of the system, always follow the steps indicated in **LIAISON® XS** Uninstallation check list (see Chapter 34.3 on page 34-8). The proper execution and compilation of this check list will ensure the proper removal of the instruments.

---

## 8.2 Preparation of sodium hypochlorite solution

The execution of some procedures to perform during instrument removal foresees the use of a sodium hypochlorite solution that shall be prepared according to the procedures reported below.

### 8.2.1 Cleaning Solution

Follow the instruction to prepare the Cleaning Solution at a concentration from 0.5 to 0.9% of active chlorine.

1. Fill the Cleaning Solution tank with 1800mL of distilled water.
2. Add 200 mL of a sodium hypochlorite solution (e.g. bleach) with a content of active chlorine ranging from 5% to 9%. In case the available sodium hypochlorite should have a different titer, please adjust the dilution accordingly.
3. Place the proper cap and shake the tank gently to remove residuals.

## 8.3 Data deleting and decontamination

1. Delete patient database.
2. Empty Liquid Waste tank, add hypochlorite solution (see Chapter “Connecting and separating the Liquid Waste” of the User Manual) and load it back into the system.
3. Using the solution prepared in Chapter 8.2.1, run **System Shutdown** task from the dedicated window of **Main User Interface** (Main category **Maintenance**, sub-category **Task**), skipping the last step, see Chapter “Shutdown/End of Day Activities” of the User Manual.
4. Disconnect all tanks from the system.

### 8.3.1 Decontamination of Liquid Waste tank

#### CAUTION

The procedure reported in this section shall be applied only when liquid waste tank shall be shipped with the analyzer. Otherwise, it is possible to dispose the tank without performing this additional decontamination.

#### CAUTION



The below reported procedure shall be applied only once the Liquid waste tank has been emptied according to the procedure described in chapter “Connecting and separating the Liquid Waste” of the User Manual.

In particular it is necessary to verify that at least 200 mL of commercial hypochlorite or bleach are loaded in the waste tank at the beginning of the procedure.

1. Disconnect the sensor connector together with the liquid connector of the liquid waste tank.
2. Unscrew the cap of the tank.
3. Add DI water until tank is completely filled.
4. Screw the cap back and wait for at least 30 minutes (during this period it is possible to continue with the decontamination of the other parts of the instrument).
5. Empty the tank and rinse it thoroughly with DI water.

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

---

6. Repeat the previous operation at least two more times.

**NOTE**

To ensure an adequate rinsing, it is recommended to fill the tank at least at half volume.

---

### 8.3.2 Removal of all consumable parts

1. Close any open software and start the **Service Software**.
2. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
3. Click onto the **Instrument** button and choose the **Full initialization via COP** button to start the initialization sequence.
4. The process is successful when the **Processing** window closes without failure message.
5. Click onto the **Cuvette Handling** button and choose the **Clear all** button to automatically empty the Washer Module, the Incubator Module and the Reader Module.
6. Close the **Service Software**.
7. Remove all the samples tubes out of the Sample racks.
8. Remove all the integral kits out of the Reagent Module.
9. Remove all the ancillaries and holders out of the Ancillary rack.
10. Remove all the disposable tips trays out of the Cuvette Tip drawer.
11. Remove all the cuvette trays out of the Cuvette Tip drawer.
12. Close the EASY Waste and remove it from the Solid Waste bin.

### 8.3.3 Cleaning procedures

**NOTE**

Tanks shall be cleaned using the procedures reported in Chapter “Cleaning the wash buffer and system liquid tanks” of the User Manual but using sodium hypochlorite solution at 0.5%-0.9% of active chlorine.

---

1. Clean the pipettors, the instrument interior, the tanks and the covers (see Chapter “Cleaning the wash buffer and system liquid tanks” and Chapter “Manual Cleaning of the instrument” of the User Manual).
2. Using tissues soaked with a solution of sodium hypochlorite clean:
  - a. Incubator cover (inner and outer);
  - b. Reader cover (inner and outer);
  - c. Liquid waste basin;
  - d. Liquid waste tank (externally, if it is necessary to ship it with the instrument);
  - e. Tank areas;
  - f. Solid waste bin;
  - g. Sample bay cover;
  - h. PCB Anti-splash cover.
3. Using tissues soaked with an alcoholic/disinfectant solution clean:

- a. Reagent Module cover;
  - b. Cuvette Tip drawer;
  - c. Starter Reagents Module;
  - d. Sample racks;
  - e. Ancillary tray;
  - f. Panel-PC;
  - g. Top covers (inner parts);
  - h. Covers (outer parts).
4. Remove and dispose the Washer and Reader needles.

#### 8.3.4 Confirm decontamination

1. Complete the certificate of the decontamination (included in DiaSorin Technical GOP 23.7201).
2. Add the certificate of decontamination to the transportation documents.

### 8.4 Packaging of the system

#### 8.4.1 Preparation

**NOTE**

At least two persons are required for packing the instrument.

##### 8.4.1.1 Removal of the Panel PC

1. Open the Top covers.
2. Loose two screws (3) of cable carrier (2).
3. Slide the cable carrier (2) out of the keyholes and set cable carrier aside.

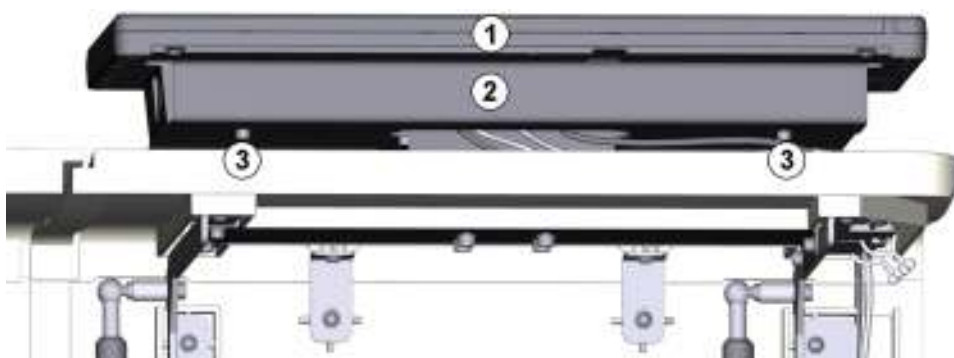


Figure 8–1: Panel PC with Right Top Cover opened, front view

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

4. Disconnect Panel PC connections power cable (4), USB connection (5 & 6) and LAN cable (7).

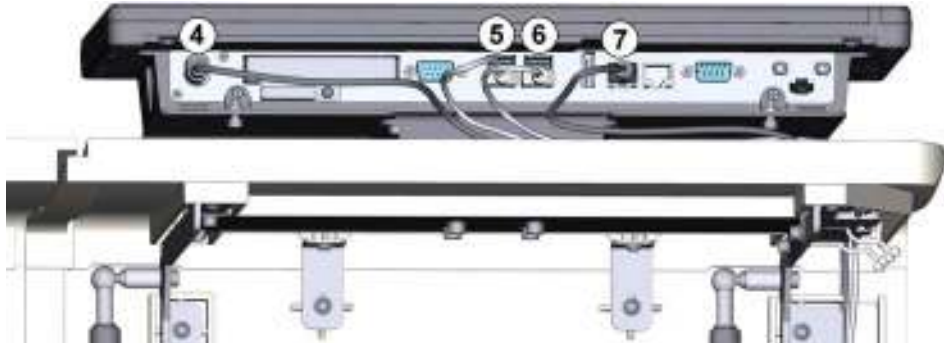


Figure 8-2: Panel PC with Right Top Cover opened, connections

5. Loosen four screws (9) of Panel PC. Lift Panel PC (1) out of the screw holes and set it aside.

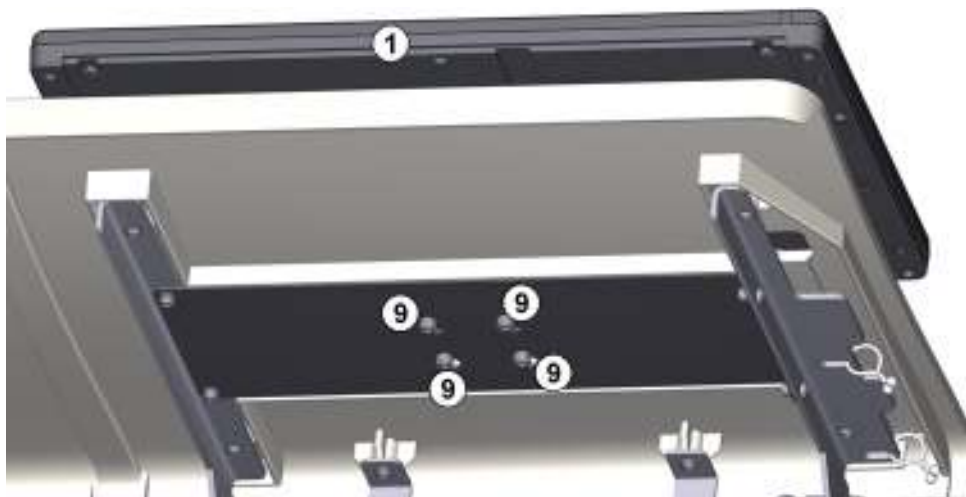


Figure 8-3: Panel PC screws

### 8.4.1.2 Remove Left and Right Top covers

1. Hold the Left Top Cover (1) and unscrew four screws (2) from the cover brackets.

## 8. LIAISON<sup>®</sup> XS Removal for external-tanks configuration

2. Lift Left Top Cover up and set it aside.

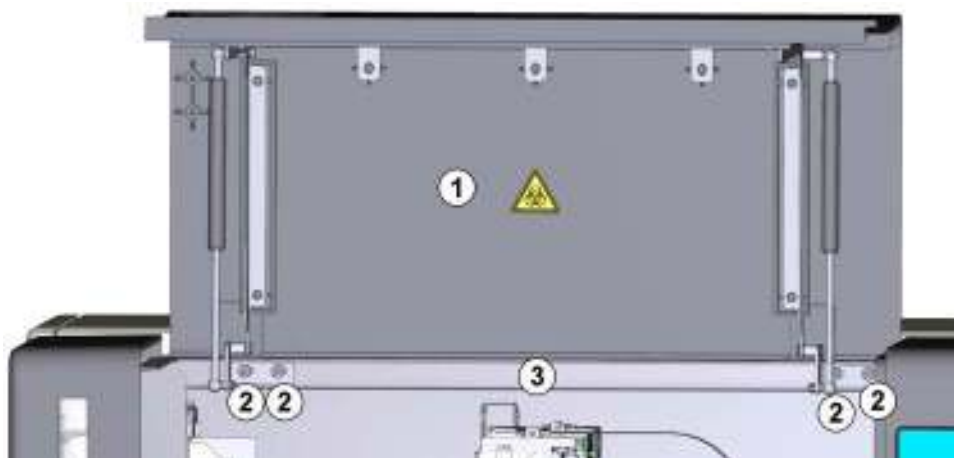


Figure 8-4: Left Top Cover opened, with details

3. Fasten the four screws (2) back into the support bar (3).
4. Move to the Right Top Cover and remove Panel PC cables (1) out of the cable holders (2).

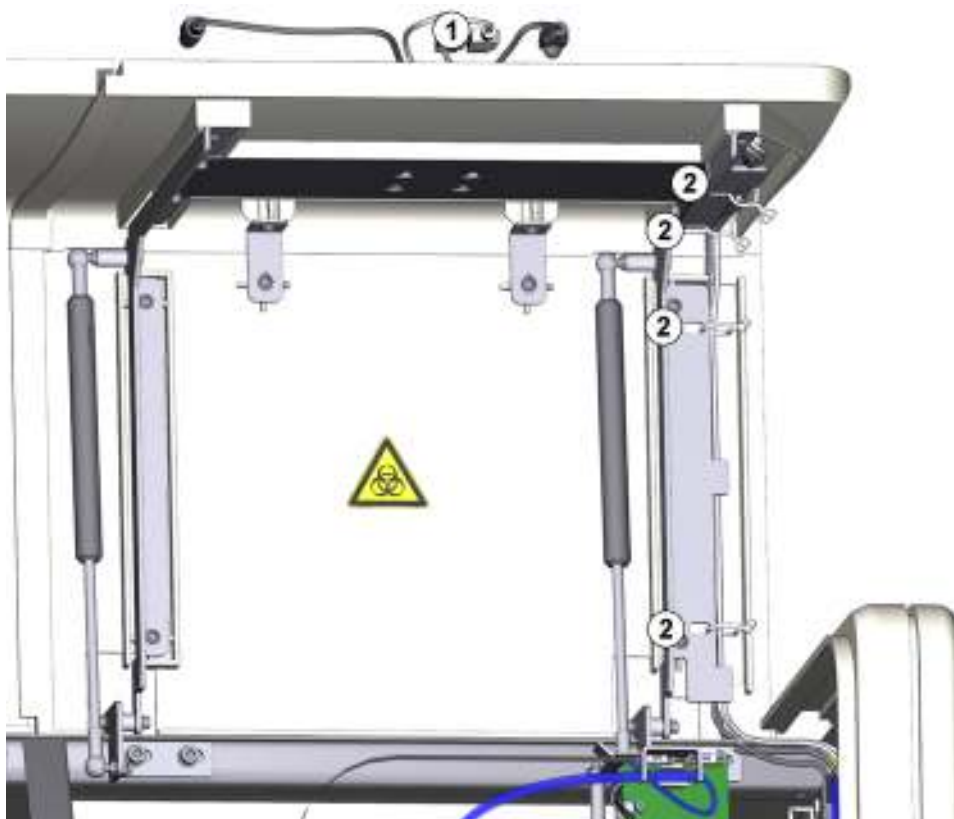


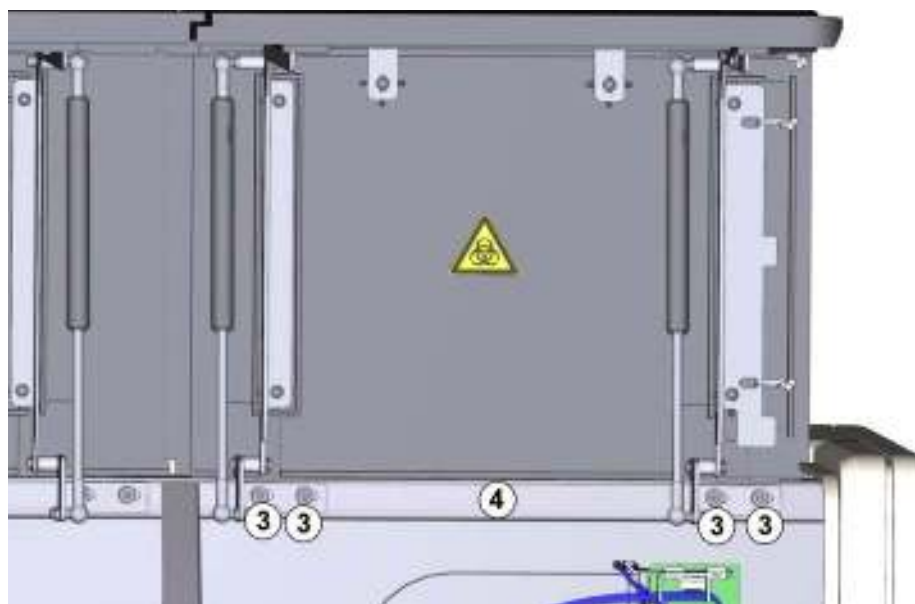
Figure 8-5: Right Top Cover with Panel PC cable

5. Hold the Right Top Cover and unscrew four screws (3) from the cover brackets.
6. Lift Right Top Cover up and set it aside.

## 8. LIAISON®XS Removal for external-tanks configuration

---

7. Fasten the four screws (3) back into the support bar (4).



*Table 8–1: Right Top Cover with detail*

### 8.4.1.3 Removal of the Right Side Cover

1. Unscrew cross-slotted screw (3).
2. Unplug the snap locks of the Right Side Cover and unplug the cable from the Soft Power Button PCBA (4).
3. Set Right Side Cover to the side.



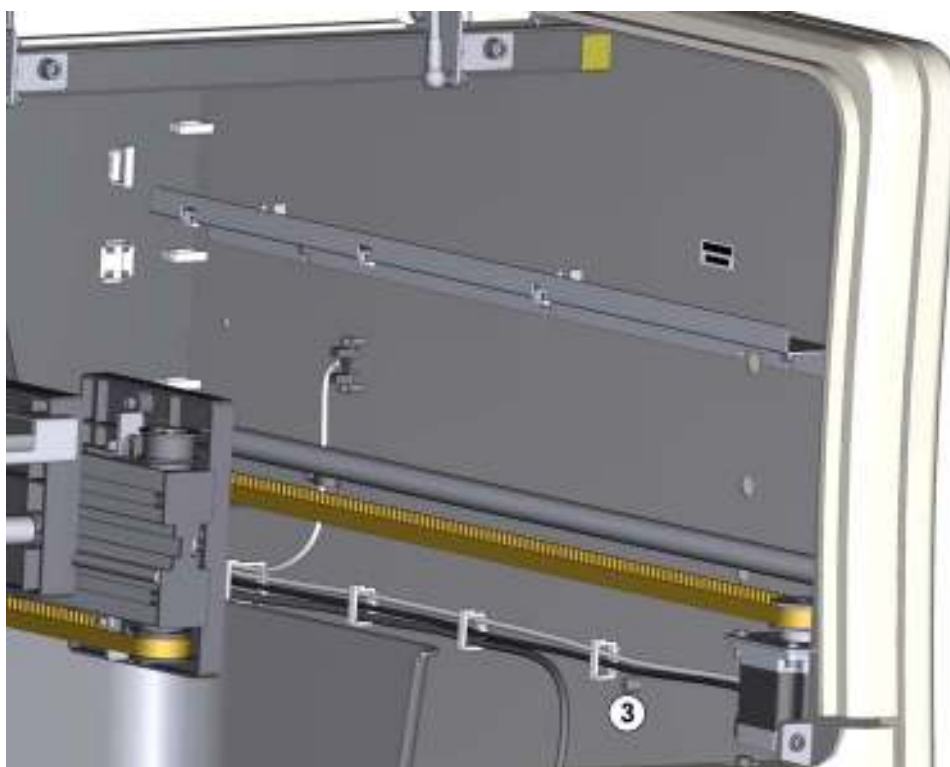


Figure 8-6: Fixing screw of the Right Side Cover, inside the instrument

#### 8.4.1.4 Removal of the Left Side Cover and Bulk Fluid Door

1. Unscrew cross-slotted screw (1).
2. Unplug the snap locks of the Left Side Cover and set the cover to the side.

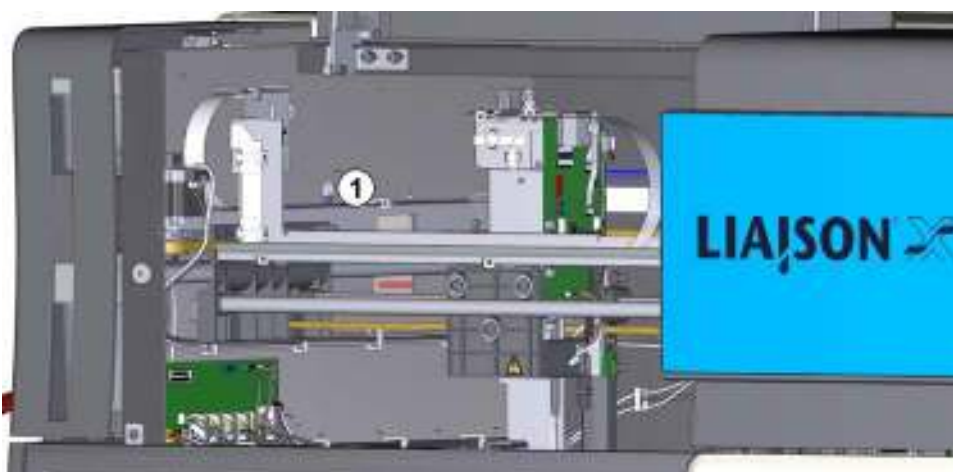


Figure 8-7: Fixing screw of the Left Side Cover, inside the instrument

## 8. LIAISON®XS Removal for external-tanks configuration

3. Lift the Bulk Fluid door (1) out of the hinges (2) and set it aside.

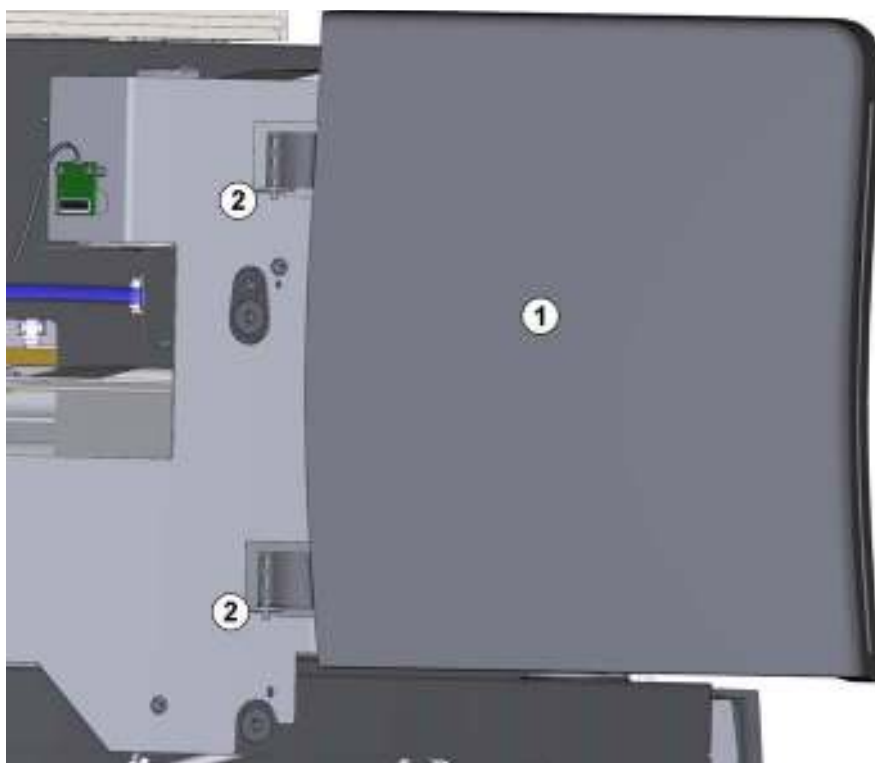


Figure 8-8: Bulk Fluid door

### 8.4.1.5 Removal of the Top Cover

1. Loosen two screws (2) at right side of the Top Cover (1).

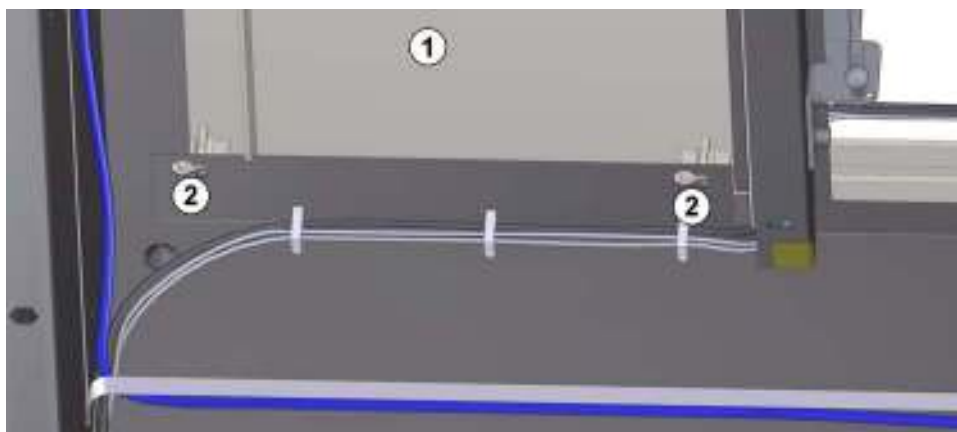


Figure 8-9: Top Cover, right side view inside the instrument

## 8. LIAISON<sup>®</sup> XS Removal for external-tanks configuration

2. Loosen screw (2) at middle of the Top Cover (1).

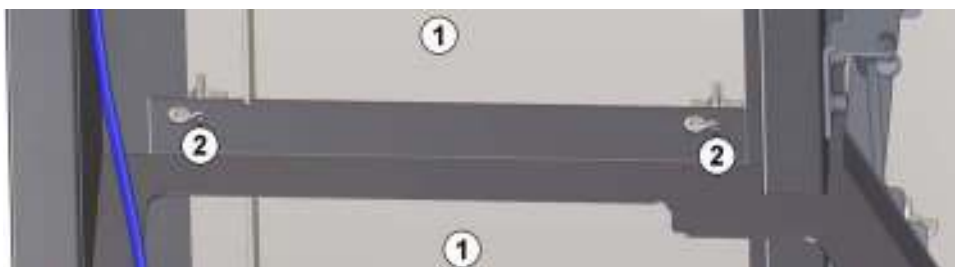


Figure 8-10: Top Cover, middle view inside the instrument

3. Loosen screw (2) at left of the Top Cover (1).

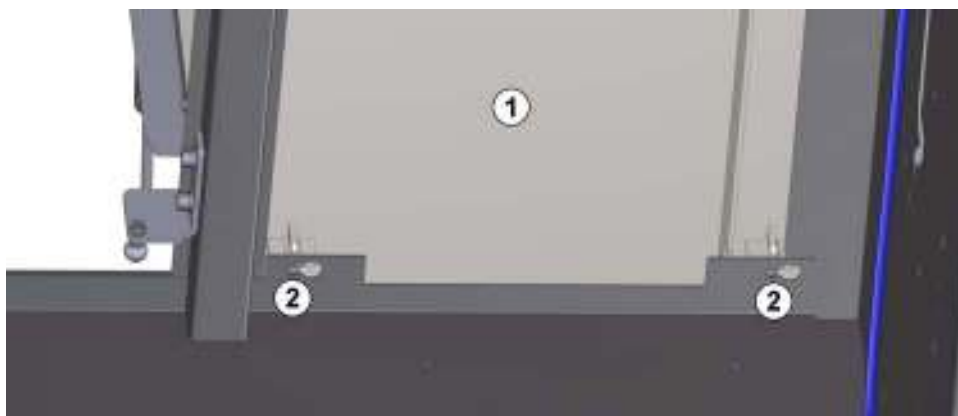


Figure 8-11: Top Cover, left side view inside the instrument

4. Slide Top Cover out of the key holes. Lift Top Cover out and set it aside.
5. Fasten six screws (2) of the Top Cover (1).

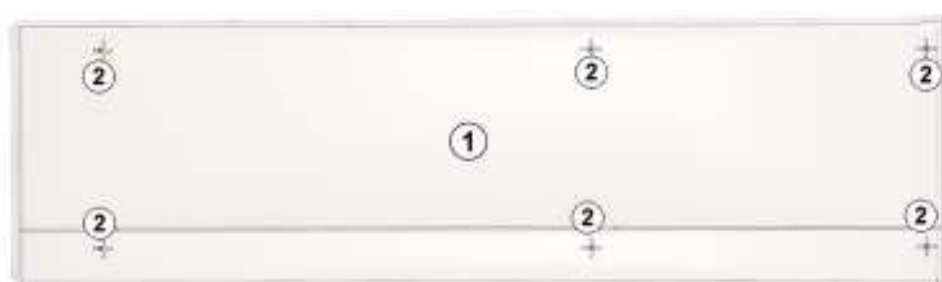


Figure 8-12: Top cover, bottom view

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

6. Lay the Panel PC cable (3) into the gap of the support bar and secure the cables with two cable ties (4).

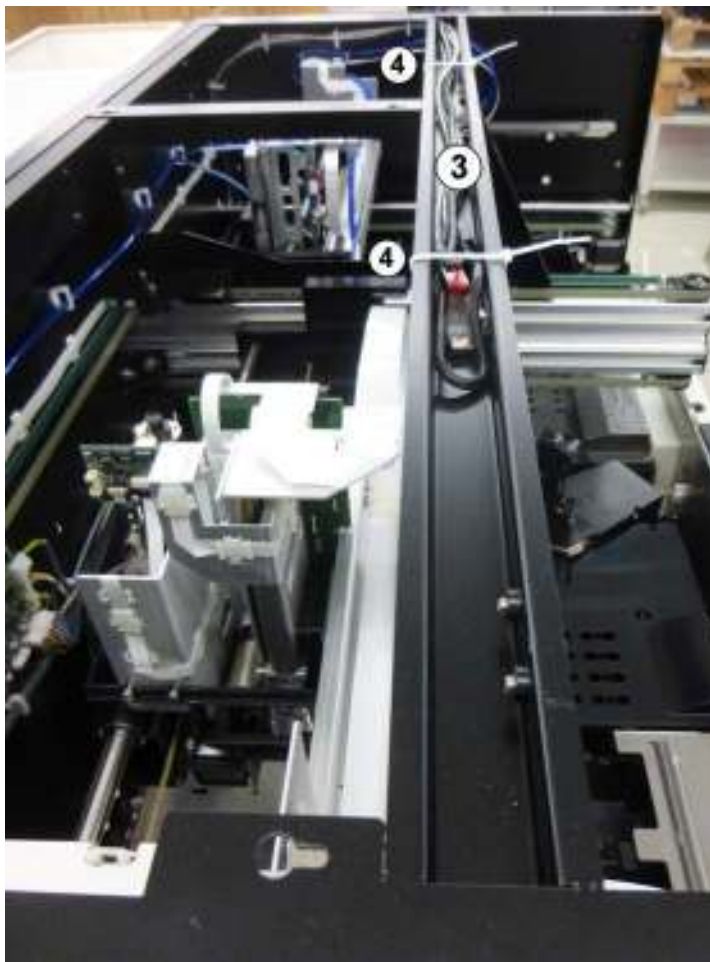


Figure 8-13: Gap of support bar

#### 8.4.1.6 Removal of the Front Module

1. Loose screws (5) and remove the handle (4) of the Cuvette - Tip Loading Module. Then, close the Cuvette - Tip Loading Module.

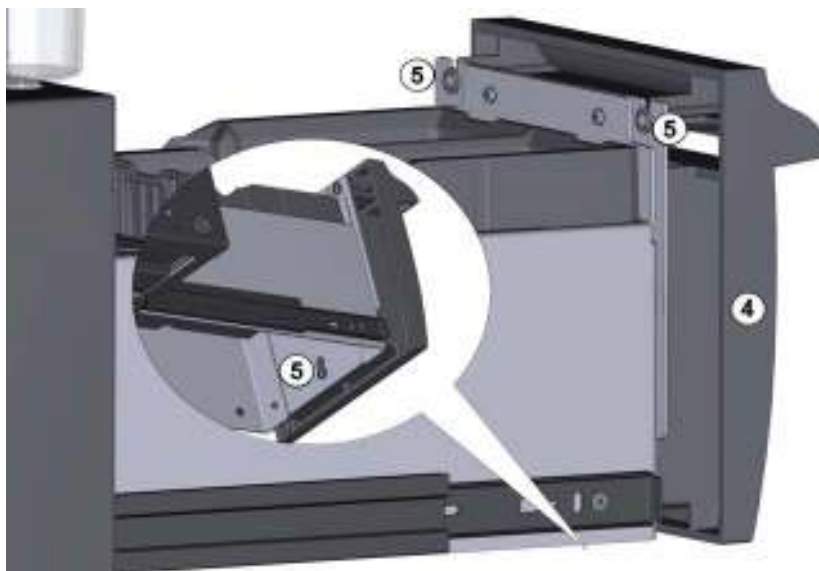


Figure 8–14: Cuvette - Tip Loading Module

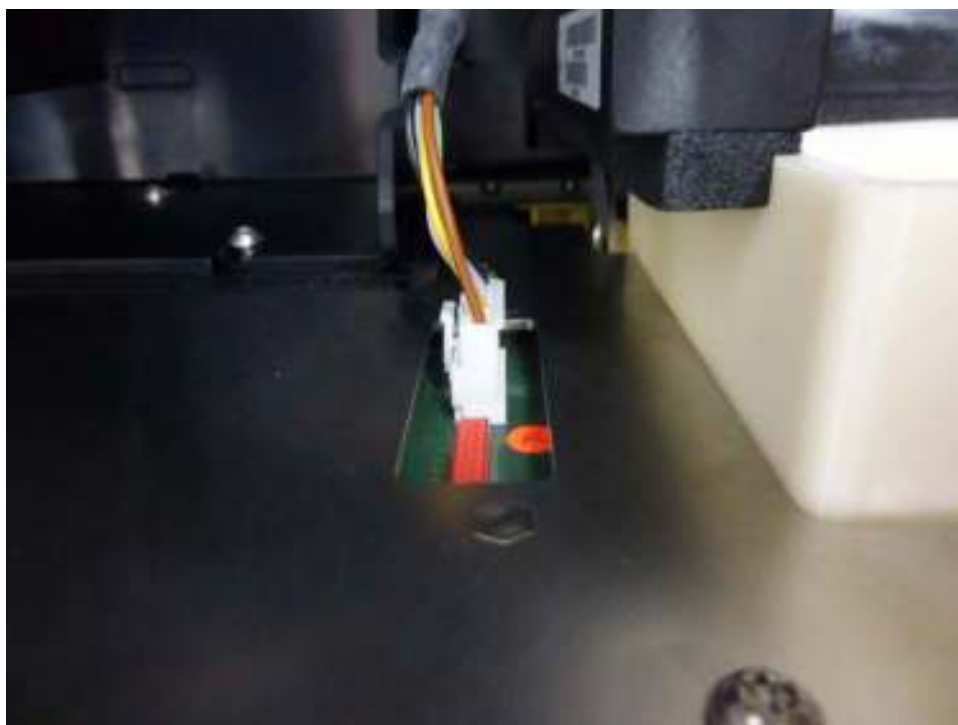
2. Lift and pull Solid Waste bin out (3).



Figure 8–15: Solid waste bin handle

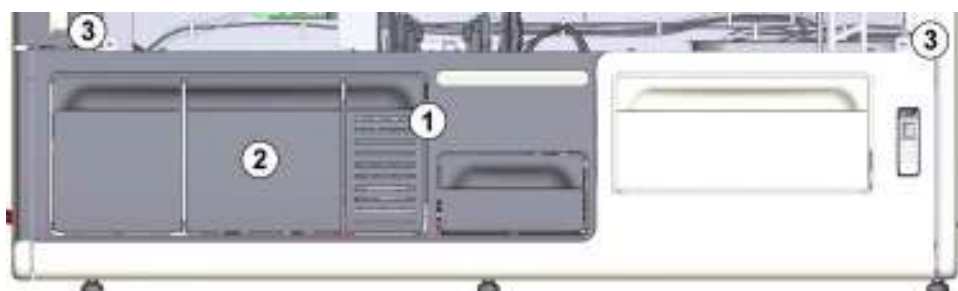
## 8. LIAISON®XS Removal for external-tanks configuration

3. Unplug the Front Module PWR/Can cable.



*Figure 8–16: Front Module PWR/Can cable*

4. Unscrew two screws (3), tilt the Front Module (1) to the front, lift it up and set it aside.



*Figure 8–17: Front Module*

**8.4.1.7 Removal of Ancillary tray**

1. Remove Ancillary tray (7) out of Reagent Module (6).

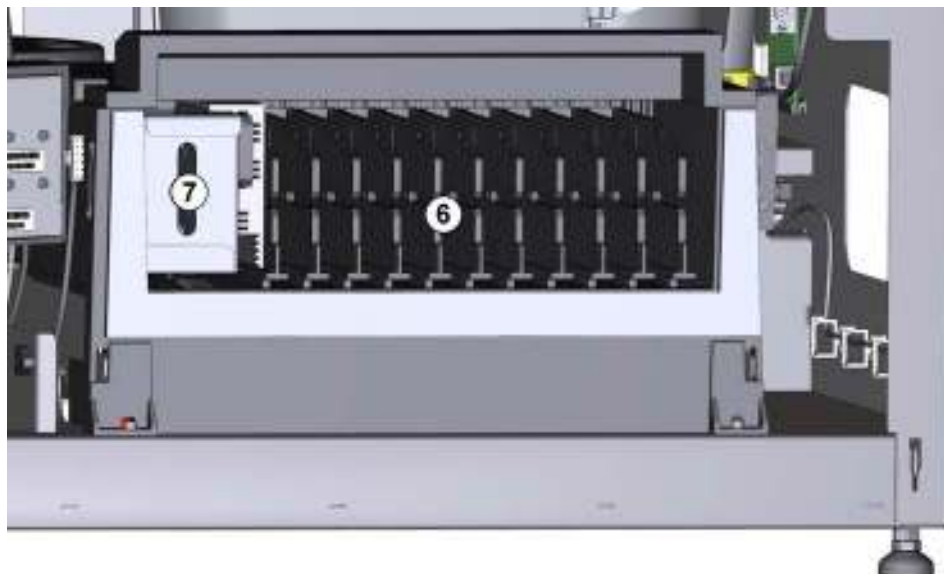


Figure 8-18: Reagent Module, front view without Front Module

**8.4.1.8 Secure Sample racks**

1. Insert the empty Sample racks (8) into the Sample Module.

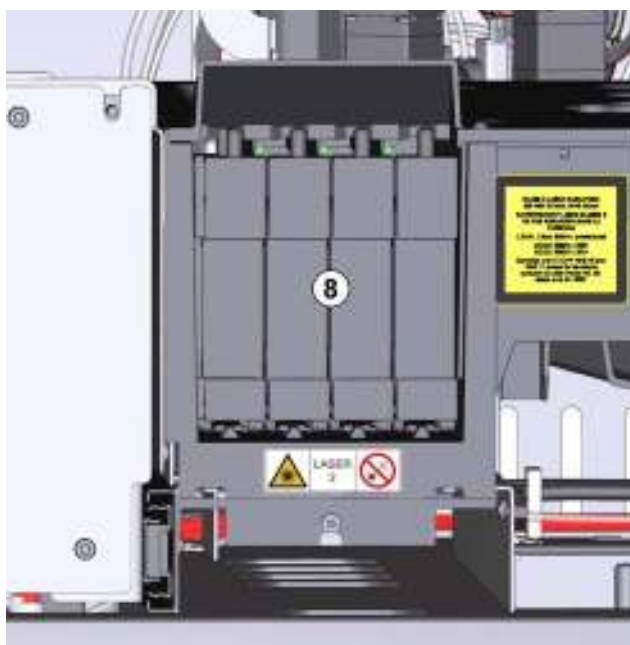


Figure 8-19: Sample Module, front view without Front Module



## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

### 8.4.1.9 Secure Pipettors and Gripper

1. Place the securing tube (2) over the z-axis bar of the Sample Pipettor (1).
2. Place the securing tube (4) over the z-axis bar of the SPOLV (3).

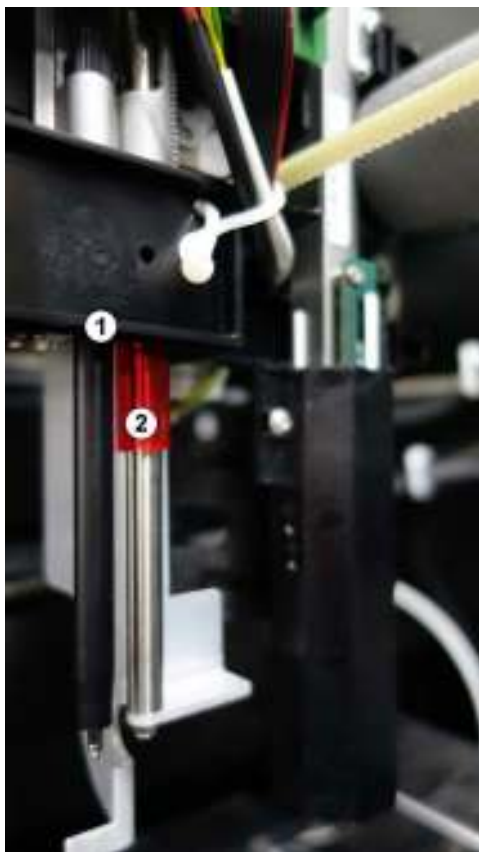


Figure 8-20: Sample pipettor



Figure 8-21: SPOLV



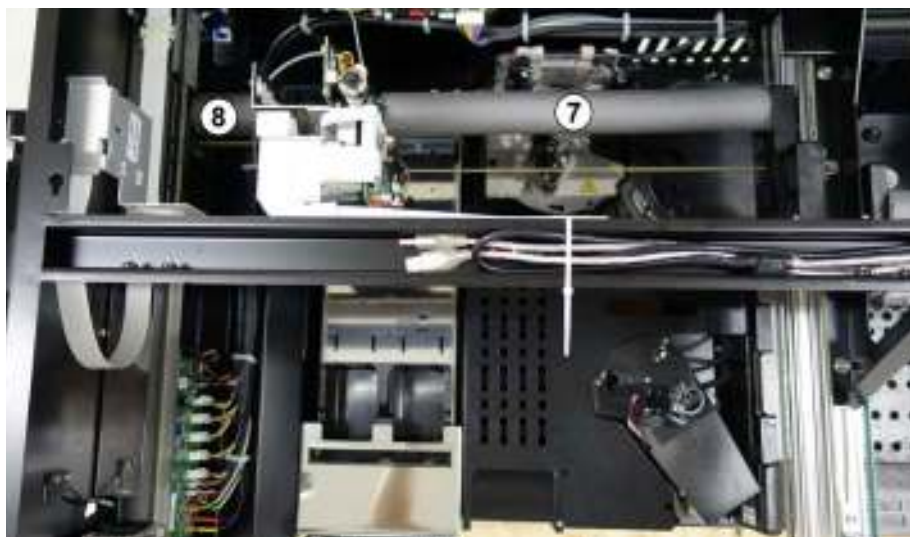
## 8. LIAISON<sup>®</sup> XS Removal for external-tanks configuration

3. Secure Gripper (5) in z-axis with a cable tie (6).



*Figure 8-22: Gripper with secure cable tie*

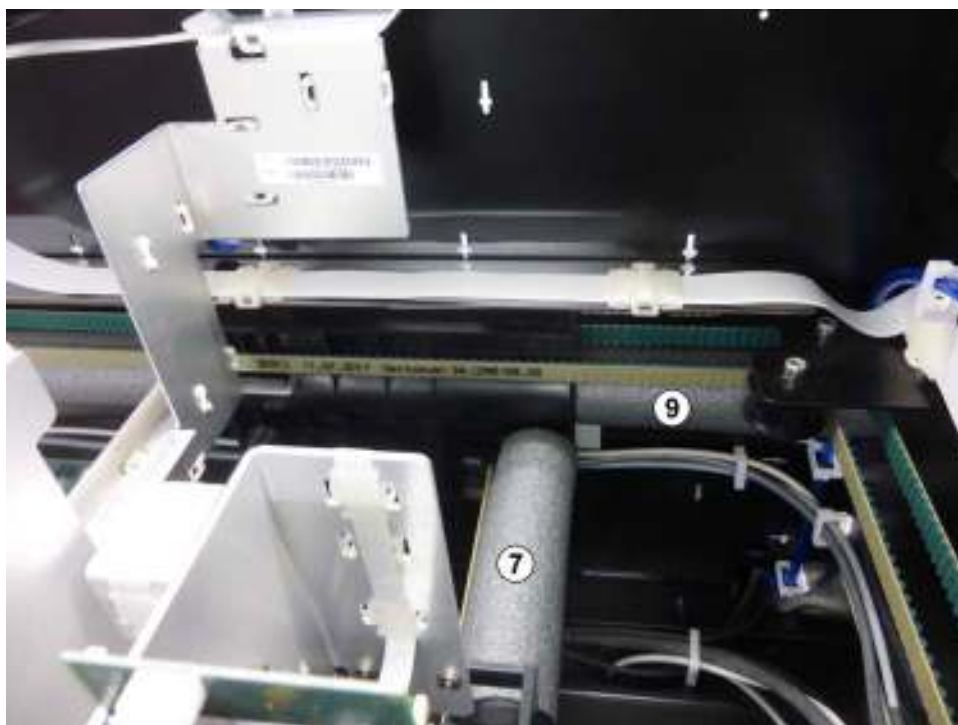
4. Place the securing foam (7) and (8) over the x-axis of the Left Pipettor Arm.



*Figure 8-23: Left Pipettor Arm with securing foam over the x-axis*

**8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration**

5. Place the securing foam (9) over the rear left y-axis of the Left Pipettor Arm.



*Figure 8-24: Left Pipettor Arm with securing foam on y-axis rear side*

6. Place the securing foam (10) over the front left y-axis of the Left Pipettor Arm.



*Figure 8-25: Left Pipettor Arm with securing foam on y-axis front side*

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

7. Place the securing foam (11) and (12) over x-axis of the Right Pipettor Arm.
8. Place the securing foam (13) and (14) over the front and rear y-axis of the Right Pipettor Arm.

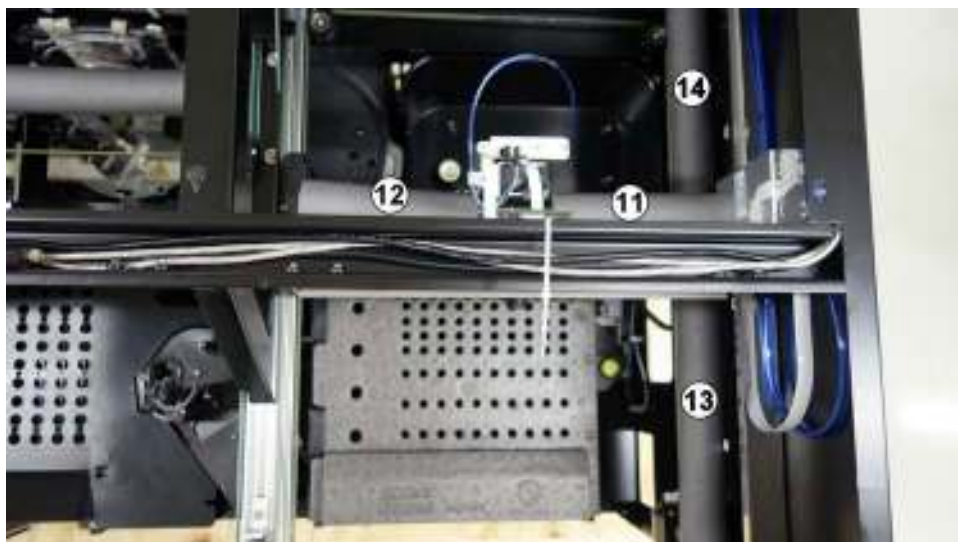


Figure 8-26: Right arm, securing foam on x- and y-axis

### 8.4.1.10 Removal of the System Liquid, Wash Buffer and Liquid Waste tubings

1. Carefully remove the System Liquid tubing (1) and cable from the cutout at the rear left corner of the instrument.

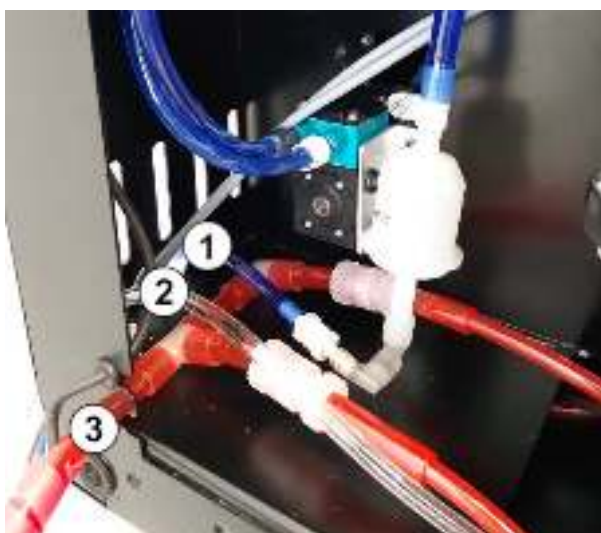


Figure 8-27: Tubing layout

2. Carefully remove the Wash Buffer tubing (2) and cable from the cutout at the rear left corner of the instrument.

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

3. Carefully remove the Liquid Waste tubing (3) and cable from the keyhole at the left side of the instrument.
4. Wrap the end of the liquid waste tubing and cable with a lint free cloth (4) and secure it with an adhesive tape, placing them in the instrument Left Side, under the Starters pumps.

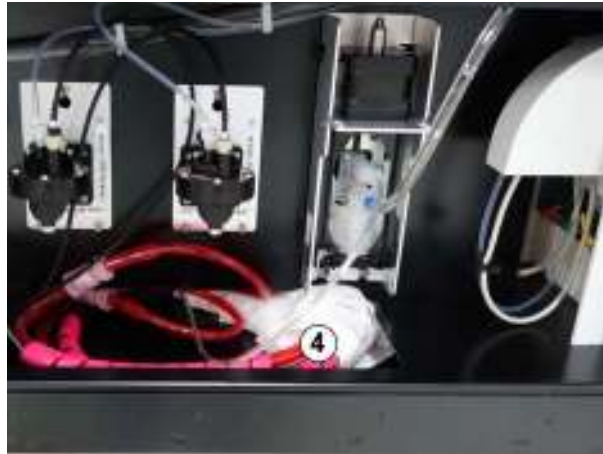


Figure 8-28: Instrument Left Side, liquid waste tubing secured

5. Repeat the previous step also for the System Liquid tubing and cable and the Wash Buffer tubing and cable.

### 8.4.1.11 Removal of the Peristaltic pumps caps (WAP and LWP)

1. Remove the caps of the LWP (10) and WAP (11).
2. Remove on one side each of the Peristaltic tubes from the two pumps.



Figure 8-29: WAP and LWP pumps without caps and with the tubes removed on one side



## 8.4.2 Packing for the accessories cardboard box

### 8.4.2.1 Packing of the Panel PC

1. Insert the Panel PC (2) into the bag.
2. Set the Panel PC into the modeled foam (3) in the original Panel PC cardboard box (1).



Figure 8-30: Packing Panel PC

3. Set the modeled foam (4) onto the Panel PC.



Figure 8-31: Packing Panel PC

4. Insert the main power cable (5) into the cardboard box.

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

5. Close the cardboard box and secure with adhesive packing tape.



*Figure 8-32: Packing Panel PC*

### 8.4.2.2 Packing of the Side Covers

1. Insert the bottom foam plate (2) into the cardboard box (1).



*Figure 8-33: Cardboard box for the covers*

## 8. LIAISON<sup>®</sup> XS Removal for external-tanks configuration

2. Lay the Right Side Cover (3) onto the foam plate.



Figure 8-34: Cardboard box for the covers

3. Insert the foam plates (4) into the Right Side Cover.



Figure 8-35: Cardboard box for the covers

**8. LIAISON®XS Removal for external-tanks configuration**

---

4. Insert the Left Side Cover (5).



*Figure 8-36: Cardboard box for the covers*

5. Insert two foam strips (6) onto the Left Side Cover.
6. Close the cardboard box with adhesive packing tape.



*Figure 8-37: Cardboard box for the covers*



#### 8.4.2.3 Packing of the small accessories

1. Insert the foam frame (1) into the cardboard box.



Figure 8-38: Cardboard box for the small accessories

2. Insert the Bulk Fluidic Door (2) into the foam frame.



Figure 8-39: Cardboard box for the small accessories

**8. LIAISON®XS Removal for external-tanks configuration**

3. Insert the Solid Waste Bin (3) into two foam blocks (4) and (5).



*Figure 8-40: Cardboard box for the small accessories*

4. Insert the Cuvette - Tip Loading Module handle (6) into two foam blocks (7) and (8) and insert it in the cardboard box.
5. Insert the Ancillary tray (9) and the Barcode Label set (10) into the pocket of the foam block.
6. Pack the Peristaltic pumps caps (11) into bubble wrap and insert into the box.
7. Close the cardboard box with adhesive packing tape.



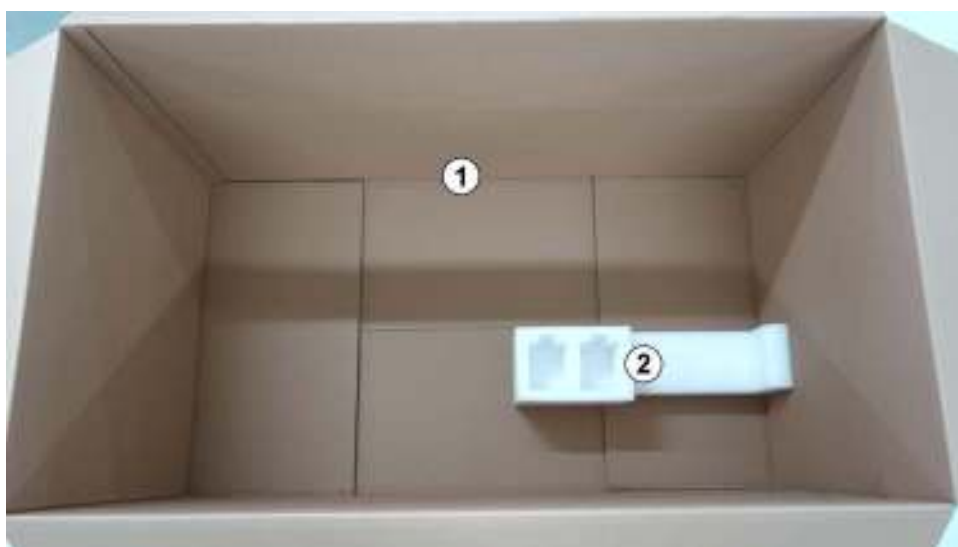
*Figure 8-41: Cardboard box for the small accessories*

**8.4.2.4 Packing of the other accessories**

1. If necessary, clean and decontaminate the following accessories and pack them all together in the dedicated box:
  - Sample tube tool;
  - Ancillary adapters;
  - Two Control Racks;
  - Sample Rack;
  - Sample rack holder;
  - Integral rack holder;
  - External Barcode Scanner;
  - Liquid Waste Basin.

**8.4.2.5 Packing of the Accessories Box**

1. Insert the foam block (2) into the Accessory box (1).



*Figure 8-42: Accessory box*

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

2. Lay the packed Side Covers (3) into the box (1).



Figure 8-43: Accessory box

3. Insert the foam piece (4) onto the box of the Side Covers.



Figure 8-44: Accessory box

4. Lay the Left Top Cover (5) onto the foam piece.

## 8. LIAISON<sup>®</sup> XS Removal for external-tanks configuration

5. Secure the sheet metal corners with two open foam tubes (6).



Figure 8-45: Accessory box

6. Insert the cardboard box with small accessories (7).



Figure 8-46: Accessory box

**8. LIAISON®XS Removal for external-tanks configuration**

---

7. Insert the foam piece for the Right Top Cover (8) onto the cardboard box.



*Figure 8-47: Accessory box*

8. Lay the Right Top Cover (9) onto the foam piece.



*Figure 8-48: Accessory box*

9. Insert the Front Module (10) into the foam blocks (11).

## 8. LIAISON<sup>®</sup> XS Removal for external-tanks configuration



Figure 8-49: Front Module, left side



Figure 8-50: Front Module, right side

10. Slide the Front Module (10) with the foam parts (11) into the Accessory box.



Figure 8-51: Accessory box



## 8. LIAISON®XS Removal for external-tanks configuration

11. Slide the Top Cover (12) into the foam blocks (11).



*Figure 8-52: Accessory box*

12. Insert the foam piece (13) above the Right Top Cover.



*Figure 8-53: Accessory box*



## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

13. Insert Instrument accessories (Panel PC accessories and Power cables Europe and US) inside the dedicated box.



*Figure 8-54: Panel PC accessories & Power cables*

14. Lay the Panel PC box (14) onto the Right Top Cover.



*Figure 8-55: Accessory box*

## 8. LIAISON®XS Removal for external-tanks configuration

---

15. Insert the other small accessory box (15).



*Figure 8-56: Accessory box*

16. Insert the Liquid Waste tank (16) (if saved and decontaminated).



*Figure 8-57: Accessory box*

17. Add the two Wash Buffer tanks (17), the System Liquid tank (18) and the Cleaning tank (19).

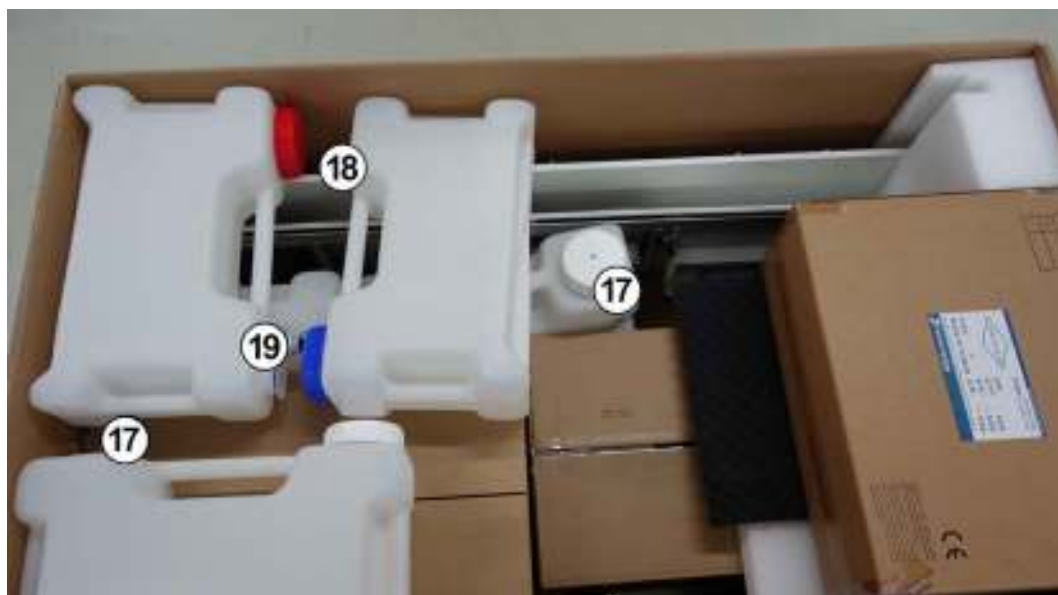


Figure 8-58: Accessory box (for illustration only: labeling and color-coding of the tanks may not reflect the real ones)

18. Add the foam piece (20).

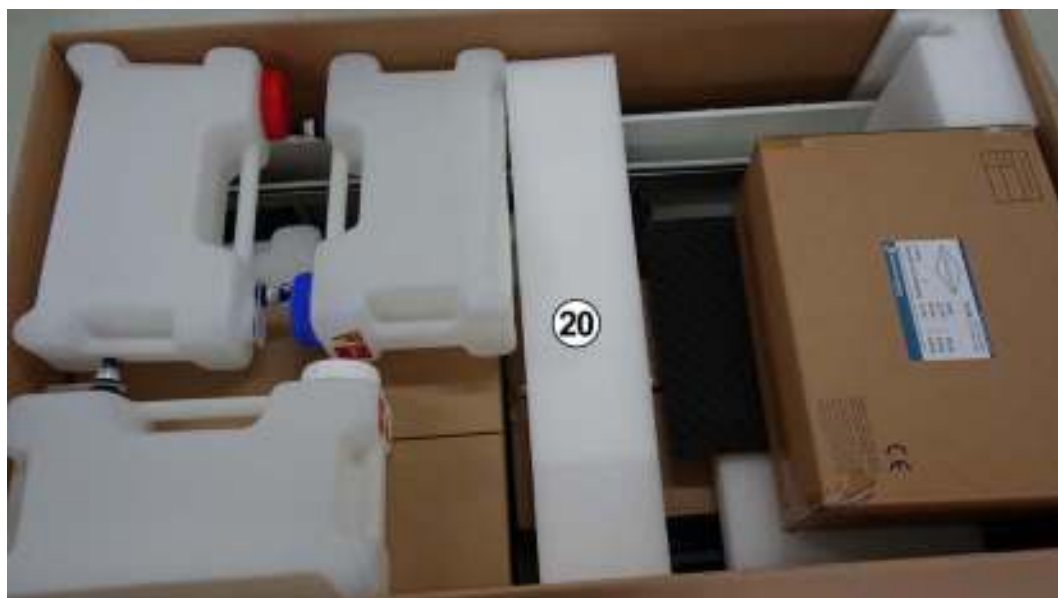


Figure 8-59: Accessory box (for illustration only: labeling and color-coding of the tanks may not reflect the real ones)

19. Lay on top the crushed packing paper.
20. Close the Accessory box with adhesive tape.

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

---

### 8.4.3 Packing of the system

**WARNING****Never lift the system by yourself!**

Manual handling of heavy loads may cause injuries.

The **LIAISON<sup>®</sup> XS** instrument shall NOT be lifted and carried by FSE.

Transportation, movement and positioning of the instrument shall be done only by qualified personnel, trained and authorized for the handling of heavy loads and by using suitable aid devices.

For any additional need concerning the movement of the instrument, contact DiaSorin S.p.A.

---

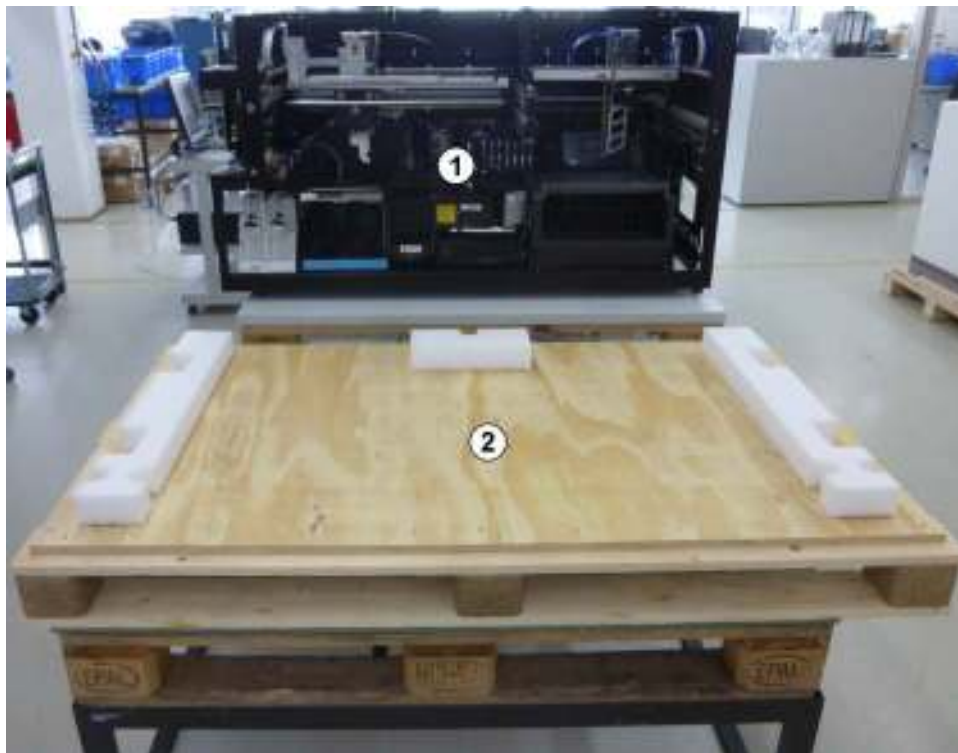
**WARNING****Risk of injury due to high weight**

Improper lifting/carrying of the instrument can cause serious injuries to the personnel and/or material damages.

- Use suitable transport aids (e.g. fork lift, crane, lift truck) for lifting/carrying the instrument.
  - Observe the technical specifications of the instrument (see Chapter 36.4 on page 36-3) to avoid misjudgments.
  - Observe the valid work safety directives and use auxiliary means, if necessary.
-

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

1. Place the instrument (1) onto the shipping pallet (2), taking care to match the pallet housing for the instrument feet.



*Figure 8–60: Instrument and shipping pallet*

## 8. LIAISON®XS Removal for external-tanks configuration

---

2. Slide the foam cover (3 and 4) on top of the instrument.



*Figure 8-61: Instrument foam on the Left Side*



*Figure 8-62: Instrument foam on the Right Side*

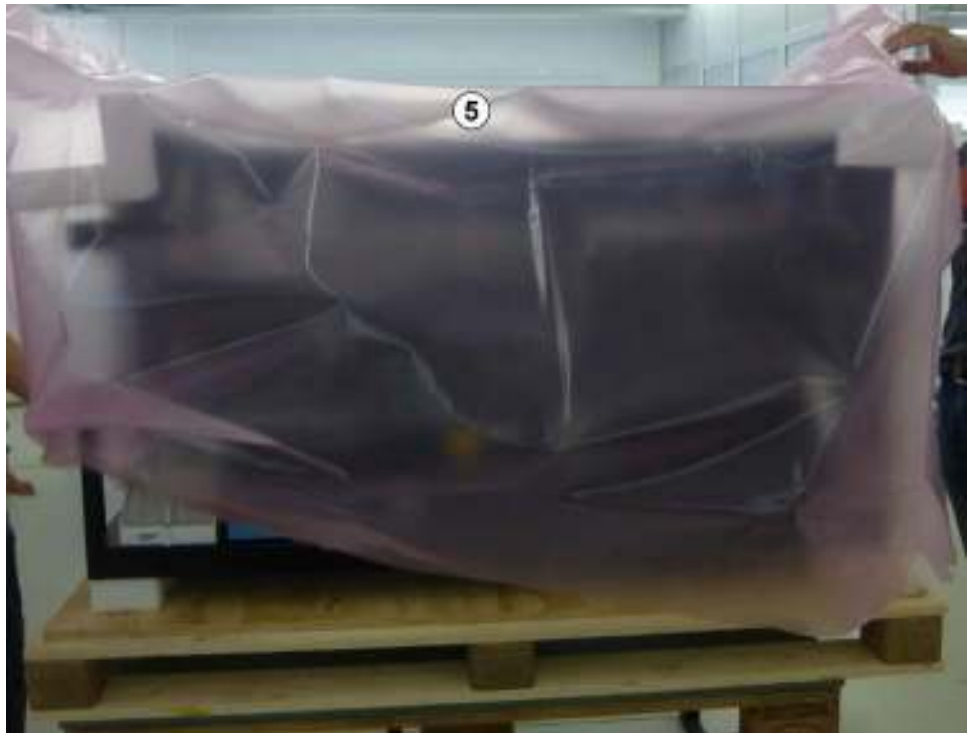


---

**8. LIAISON<sup>®</sup> XS Removal for external-tanks configuration**

---

3. Cover the system with the plastic bag (5).



*Figure 8–63: Instrument covered with plastic bag*

**8. LIAISON®XS Removal for external-tanks configuration**

4. Lay the C-foam piece (6 and 7) on both sides around the bottom of the instrument.



*Figure 8-64: Instrument C-foam on the Right Side*



*Figure 8-65: Instrument C-foam on the Left Side*



**8. LIAISON<sup>®</sup> XS Removal for external-tanks configuration**

5. Slide the outer cardboard cover (8) over the instrument and over the bottom foam.



*Figure 8-66: Outer cardboard cover*

6. Place the cardboard lid over the outer cardboard cover.



*Figure 8-67: Outer cardboard lid*

## 8. LIAISON®XS Removal for external-tanks configuration

7. Lift the accessories cardboard box on top of the instrument.
8. Secure accessories cardboard box with packaging tape.



Figure 8-68: Final packaging of the instrument

#### 8.4.4 Transport labeling

1. Add the Shock-watch (1) and the Thermometer indicator (2) onto the packaging cover.



Figure 8–69: Shock-watch and Thermometer indicator

## 8. LIAISON<sup>®</sup>XS Removal for external-tanks configuration

---

## 9 Maintenance Schedule for internal-tank configuration

The **LIAISON® XS** instrument was designed to be as maintenance free as possible. However, given the functional, precision and reliability requirements of the complex applications the instruments are used for, regular care and attention are necessary to maintain the high quality of instrument performance. For some components regular maintenance procedures could not be avoided in the design process.

The maintenance intervals given below assume a working time of the instrument of 8 h per day, 5 days a week, 180 results/day, in a normal laboratory environment. Upon more intense usage of the instrument or usage in demanding environmental conditions (dust, heat, humidity) the maintenance intervals should be shortened accordingly.

The description of requalification, cleaning and lubrication procedures to be performed during service visits are also reported in this chapter.

### 9.1 Safety

#### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

#### WARNING



#### Improper Maintenance work

Improper maintenance work can result in serious personal injuries and material damages.

- Take off watches and jewelry before performing any maintenance works.

#### WARNING



#### Defects in the liquid system

Defect or leaking tubes, syringes, valves or pumps lead to deterioration of the pipetting results and consequently corruption of final results. Furthermore incorrectly flushed tips can cause mixing up of sample material.

- Check the instrument for drops and pooling of liquid on surfaces.
- Check tubes, syringes, valves or pumps periodically.

## 9. Maintenance Schedule for internal-tank configuration

---

**CAUTION****Maintenance responsibility**

Defective parts that result from the non-fulfillment of proper maintenance intervals are the sole responsibility of country specific distributors. DiaSorin S.p.A. is not liable for defective parts due to non-performance of required maintenance intervals.

---

**CAUTION****Check System**

During the periodic maintenance of the system, always follow the steps indicated in **LIAISON® XS** Yearly Maintenance (see Chapter 33.4 on page 33-10) check list. The proper execution and compilation of these check list will ensure the completion of the proper periodic maintenance of the instruments.

---

## 9.2 Maintenance

### 9.2.1 List of required material

- Set of hexagon socket wrench (Allen key).
- Set of torx screw driver.
- Set of cross slotted screw driver.
- Flat screw driver.

### 9.2.2 System cleaning



---

#### Wrong placement of the Liquid Waste tank

Too high installation of the Liquid waste tank can lead to a liquid drawback.

- It is preferred to place the Liquid waste tank below the instrument.

- 
1. Empty Liquid Waste tank, add hypochlorite solution (see Chapter 5.4.6 "Connecting and separating the Liquid Waste" of the User Manual) and load it back into the system. This does not apply to those systems connected to direct drain.
  2. Remove the cuvettes from the system in the Washer, Reader and Incubator performing the **Empty Incubator** from Main category **Diagnostics**.
  3. Run **Service Decontamination** task from the dedicated window of **Main User Interface** (Main category **Maintenance**, sub-category **Task**).
  4. Check the tanks cleaning: if needed, clean them according to Long Term Maintenance, see the User Manual.
  5. Remove all the samples tubes out of the Sample racks.
  6. Remove all the integral kits out of the Reagent Module.
  7. Remove all the ancillaries and holders out of the Ancillary rack.
  8. Remove all the disposable tips trays out of the Cuvette Tip drawer.
  9. Remove all the cuvette trays out of the Cuvette Tip drawer.
  10. Close the LIAISON<sup>®</sup> EASY Waste box with its cap and remove it from the Solid Waste bin.
  11. Insert a new LIAISON<sup>®</sup> EASY Waste box in the solid waste bin.

### 9.2.3 PC & Accessories

1. Clean PC.
2. Check that databases (**Results/ All**, **Results/ Archived**, **Results/ Deleted**, **Results/ Calibrations**, **Maintenance/ Log**) have not reached dimensions that compromise the proper browsing of the database. In case this condition is verified, clean up the databases:
  - Reduce as much as possible the items in the **Results/ All** page (i.e. delete unnecessary results according to customer feedback);
  - Reduce as much as possible the items in the **Results/ Archived** page (i.e. delete unnecessary results according to customer feedback);

## 9. Maintenance Schedule for internal-tank configuration

---

- Reduce as much as possible the items in the **Results/ Calibrations** page (i.e. remove all Unused calibrations);
  - Delete all items from **Results/ Deleted** page;
  - Delete all unused group definitions (according to customer feedback);
  - Delete all unused control definitions (according to customer feedback);
  - Delete **Maintenance/ Log** (according to customer feedback).
3. Check printer functionalities (if present).
  4. Check the presence of the most updated version of the Manuals, corresponding to the Instrument SW version and customer's language for the specific country. If needed, update the Manual as written in the Installation procedure, see Chapter 5.4.4 on page 5-34.
  5. Close any open software and log-in as Administrator.
  6. Check that at the least 20% of C:\ partition of the hard disk is free. If this condition is not verified, delete the items present in C:\LiaisonXS\Data\Temp\archive.
  7. Check UPS functionalities (if present).
  8. Run Antivirus application as per following instructions:
    - Launch the "av.bat" file from the path: C:\LiaisonXS\Tools\AV\
    - While the Virus Scan is ongoing do not reboot the PC or logoff from the current account, to avoid the interruption of the scansion
    - A message informs that the scansion is completed
    - The results are appended in "C:\LiaisonXS\Tools\AV\clamlog.txt"

### 9.2.4 Preparation for Maintenance of Internal Modules

1. Start the **Service Software**.
2. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
3. Click onto the **Instrument** button and choose the **Power Off** button to operate in safe condition.
4. The process is successful when the **Processing** window closes without failure message.
5. Open top covers.
6. Move Pipettor arms to a safe location.

### 9.2.5 Maintenance of the Washer and Reader Modules

**NOTE**

The cleanings described below shall be performed using lint free cloth and isopropanol; as a final rinsing DI water shall be used.

1. Replace all the Washer and Reader Aspiration needles, see Chapter 17.3.1 on page 17-6 and see Chapter 18.3.1 on page 18-4.
2. Perform a visual check for crystallization in the Reader rotor and cover. Turn the Reader rotor and, if necessary, remove the Reader cover and clean both the



---

**9. Maintenance Schedule for internal-tank configuration**

---

Reader rotor and the Reader cover, see Chapter 18.3.2 on page 18-9 and see Chapter 18.3.5 on page 18-17.

3. Perform a visual check for crystallization in the Washer rotor and plate. Turn the Washer rotor and, if necessary, remove the Washer plate and clean the Washer rotor, plate and dispense nozzles, see Chapter 17.3.6 on page 17-24, see Chapter 17.3.9 on page 17-32 and see Chapter 17.3.2 on page 17-12.

**9.2.6 Maintenance of the Pumps and Tubings**

1. Replace the WBDP cartridge, see Chapter 25.3.7 on page 25-29.
2. Replace the WAP Peristaltic tubings, see Chapter 25.3.3 on page 25-12.
3. Replace the LWP Peristaltic tubings, see Chapter 25.3.3 on page 25-12.
4. Inspect and, if necessary, replace Waste tubing at Left and Right Energy Chain, see Chapter 28.3.7 on page 28-23 and see Chapter 28.3.8 on page 28-27.

**9.2.7 Maintenance of the Reagent Module****NOTE**

The cleanings described below shall be performed using lint free cloth and isopropanol; as a final rinsing DI water shall be used.

**NOTE**

During the cleaning of the Reagent Module, be sure to remove as many residual of dust between the fins and in the holes hosting the sliding bearings as possible.

1. Remove the Reagent Module cover, see Chapter 22.3.1.2 on page 22-4.
2. Clean the Reagent Module (inner and outer, cover included) using an alcohol/disinfectant solution.
3. Assemble the Reagent Module cover, see Chapter 22.3.1.3 on page 22-4.
4. If needed, clean the crank drive. Then, lubricate it, see Chapter 22.3.3.5 on page 22-10.

**9.2.8 Maintenance of the Pipettor****NOTE**

The maintenance of the pipettor shall be performed using lint free cloth, isopropanol and oil (PN 72229616).

**CAUTION**

Before using the oil, the isopropanol must be evaporated or wiped off from the rods. Do not apply too much oil on the rods, because of potential system contamination.

1. Wipe the whole length of the X and Y Pipettor rods with a lint free cloth soaked with isopropanol.

## 9. Maintenance Schedule for internal-tank configuration

---

2. Wipe all Pipettor rods with a lint free cloth soaked with oil (PN 72229616). A thin film of oil shall remain.
3. Verify the proper behavior moving both Pipettor modules 5 times in axial X-direction and 5 times in axial Y-direction.
4. Inspect and, if necessary, replace the Gripper slide bearings, see Chapter 19.3.6 on page 19-25.

### 9.2.9 Cleaning of other surfaces

**NOTE**

The cleanings described below shall be performed using an alcohol/ disinfectant solution.

1. Clean the Cuvette - Tip Loading Module.
2. Clean inner parts of Left and Right Top Covers.

### 9.2.10 System Restore & Periodic Recalibrations

1. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
2. Click onto the **Instrument** button and choose the **Power On** button to restore the system power source.
3. The process is successful when the **Processing** window closes without failure message.
4. Close any open software and start the **Teacher Software** from the Windows Administrator account.
5. Check Pipettor teaching, see Chapter 13 on page 13-1.
6. Perform backup of `Coordinate.coo` file.
7. Perform backup of ".rac" files.
8. Close any open software and start the **Service Software** from the Windows Administrator account.
9. Initialize the system, see Chapter 11.8.1 on page 11-58.
10. Perform the Washer SIPAL Lift Calibration, see Chapter 11.5.3.2 on page 11-32.
11. Perform the Washer Aspiration Monitoring Calibration, see Chapter 11.5.3.3 on page 11-33.
12. Perform the WBDP Calibration, see Chapter 11.5.3.4 on page 11-34.
13. Perform the Reader SIPAL Lift Calibration, see Chapter 11.6.3.2 on page 11-39.
14. Perform the Reader Aspiration Monitoring Calibration, see Chapter 11.6.3.3 on page 11-40.
15. Insert at least two cuvette trays in the Cuvette - Tip Loading Module and set them in the System - Cuvette Tray State, see Chapter 11.8.2 on page 11-66.
16. Perform the Cuvette Endurance Run, see Chapter 11.8.1.2.1 on page 11-61.
17. Prime the fluidics, see Chapter 11.8.1.3.1 on page 11-63.
18. Check that the Antivirus scan has been completed.
19. Close any open software and restart the PC.
20. Log into the **Main UI Software** and initialize the system.
21. Reset the counter of the Solid Waste.

## 9. Maintenance Schedule for internal-tank configuration

22. Insert cuvette trays and disposable tips trays in the Cuvette - Tip Loading Module.
23. Check the **Status - Consumables** toggle and assign the Consumables (cuvettes and tips) in the **Status - Plastic** toggle in **Main UI Software**.
24. Load the cuvette in the Incubator Module.

### 9.2.11 Requalification test

1. Run **Service - 3 Series Background Wash** task from the dedicated window of **Main User Interface** (Main Category **Maintenance**, sub-category **Task**).
2. For any given series, verify that CV observed is less than or equal to 10.00% and the average signal developed is equal to or smaller than 1000 RLU.
3. Run **Service - 3 Series LightCheck Right** task from the dedicated window of **Main User Interface** (Main Category **Maintenance**, sub-category **Task**).
4. For any given series, verify that CV observed is less than or equal to 4.00%, that the minimum average signal developed is higher than 160000 RLU and lower than 245000 RLU.
5. Run **Service - 3 Series LightCheck Left** task from the dedicated window of **Main User Interface** (Main Category **Maintenance**, sub-category **Task**).
6. For any given series, verify that CV observed is less than or equal to 4.00%, that the minimum average signal developed is higher than 345000 RLU and lower than 520000 RLU.

|            | Max CV% | Min RLU | Max RLU |
|------------|---------|---------|---------|
| <b>BGW</b> | 10.00%  | 0       | 1000    |
| <b>LCR</b> | 4.00%   | 160000  | 245000  |
| <b>LCL</b> | 4.00%   | 345000  | 520000  |

Table 9-1: Summary of the Acceptance Criteria

7. Perform the calibration and the control of one of the kits available at customer site.
8. Verify that the obtained values are in range.
9. Perform raw data export and backup.
10. Inform the customer that all assays in use must be recalibrated.

#### NOTE

For the re-run rules in case of failures of tests, please refer to Corp-TN-000292 (former LXS TN 009).

#### NOTE

The successful execution of 3 consecutive BGW, 3 consecutive LCL and 3 consecutive LCR shall be always achieved.

#### NOTE

The results of the requalification activities (BGW, LCL, LCR, Calibration and Control run) must be attached to the Service records (for Subsidiaries only: SGAT WS), together with a detailed description of the Service activities performed on the analyzer.

## 9.3 Preventive Maintenance kit for Field Service Engineers

### 9.3.1 Yearly Maintenance kit

| Module                    | Description                  | Action  | Link to procedure                                                  |
|---------------------------|------------------------------|---------|--------------------------------------------------------------------|
| Reader and Washer         | Set of 4 Aspiration Needle   | Replace | see Chapter 18.3.1 on page 18-4<br>see Chapter 17.3.1 on page 17-6 |
| Waste Aspiration Pump     | Peristaltic tubing 2-channel | Replace | see Chapter 25.3.3 on page 25-12                                   |
|                           | Peristaltic tubing 3-channel | Replace | see Chapter 25.3.3 on page 25-12                                   |
| Liquid Waste Pump         | Peristaltic tubing 3-channel | Replace | see Chapter 25.3.3 on page 25-12                                   |
| Wash Buffer Dispense Pump | Cartridge, piston and screw  | Replace | see Chapter 25.3.7 on page 25-29                                   |

Table 9–2: Yearly Maintenance

### 9.3.2 Replacing on demand parts

| Module        | Description                                                                                                     | Action                     | Link to procedure                |
|---------------|-----------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------|
| Waste tubings | Waste tubing at Left Energy Chain:<br>Wash station-LWP<br>LWP-Waste interface                                   | Upon inspection, as needed | see Chapter 28.3.7 on page 28-23 |
|               | Waste tubing at Right Energy Chain:<br>WAP-Waste interface,<br>Reagent Module-WAP,<br>Reader-WAP,<br>Washer-WAP | Upon inspection, as needed | see Chapter 28.3.8 on page 28-27 |
| Gripper       | Slide bearing set                                                                                               | Upon inspection, as needed | see Chapter 19.3.6 on page 19-25 |

Table 9–3: Maintenance on demand

## 9.4 Instrument requalification after service visit

During a Field Service intervention some parts of the instrument (assembly, sensor, motor, pump, tubing, etc.) may be replaced to troubleshoot and solve the problem on the instrument. Depending on the replaced spare part, before releasing the instrument to run patient samples, a final re-qualification may be required, as well as the reagent integral re-calibration. These spare parts are specified in Table 9–4.

In case some parts of the instrument are removed and mounted back for troubleshooting purposes, refer to the relevant dedicated paragraphs in order to assess the need of mechanical recalibration/teaching.

In case the replaced spare part is not present in the Table 9–4, refer to the assembly the replaced spare part belongs to.

Screws, washers, nuts, clips, fuses and other minor parts are deemed not to require any re-qualification/re-calibration.

### 9.4.1 Background and rationales

The Spare Part replacements detailed in Table 9–4, see Chapter 9.4.2 on page 9-9:

- may involve critical instrument components for the core characteristics of assay performance (i.e.: pipettor, incubator, washer, reader) and, due to this, may require a specific re-qualification related to the area of interest;
- for the same reason may require re-calibration of the assays used by the Final User, being potentially impacting the assay calibration RLU levels.

### 9.4.2 Service interventions

| Assembly    | Spare Part                        | BGW | LCL | LCR | Recal |
|-------------|-----------------------------------|-----|-----|-----|-------|
| Covers      | Front Cover GAS spring            | -   | -   | -   | -     |
| Covers      | Front Cover PCB                   | -   | -   | -   | -     |
| Covers      | Starter Reagent flap              | -   | -   | -   | -     |
| Electronics | Pipettor PCB                      | -   | -   | -   | -     |
| Electronics | Liquid Support PCB                | -   | -   | -   | -     |
| Electronics | COP                               | 1   | 1   | 1   | -     |
| Electronics | PWR CAN PCB                       | -   | -   | -   | -     |
| Electronics | Power Supply                      | -   | -   | -   | -     |
| Electronics | Fuse, 3.15A, 5.08 MM (Set of 14)  | -   | -   | -   | -     |
| Electronics | Fuse, 5.0 A, 6.3x32MM (Set of 10) | -   | -   | -   | -     |
| Electronics | Fuse, 4.0 A, 5x20MM (Set of 10)   | -   | -   | -   | -     |
| Electronics | Soft Power button PCB             | -   | -   | -   | -     |
| Incubator   | Incubator Assembly                | -   | 3   | 3   | Y     |
| Incubator   | Incubator PCB                     | -   | 3   | 3   | Y     |
| Incubator   | Incubator Cover w/ Bearings       | -   | -   | -   | -     |
| Incubator   | Incubator Rotor                   | -   | 1   | 1   | -     |
| Incubator   | Incubator Motor w/ Cables         | -   | 1   | 1   | -     |
| Incubator   | Light Guide                       | -   | 1   | 1   | -     |
| Incubator   | Incubator Base Bearing            | -   | -   | -   | -     |

## 9. Maintenance Schedule for internal-tank configuration

|                   |                                     |   |   |   |                     |
|-------------------|-------------------------------------|---|---|---|---------------------|
| Miscellaneous     | Flat Cable Pipettor Clips 17 mm     | - | - | - | -                   |
| Miscellaneous     | Flat Cable Pipettor Clips 26,5 mm   | - | - | - | -                   |
| Miscellaneous     | Flat Cable Pipettor Clips 12 mm     | - | - | - | -                   |
| Miscellaneous     | Consumable Drawer Switch            | - | - | - | -                   |
| Miscellaneous     | Sample rack labels                  | - | - | - | -                   |
| PC                | Panel Control Unit                  | - | - | - | -                   |
| Pipettor Left     | Pipettor Left Gantry                | 3 | 3 | - | -                   |
| Pipettor Left     | Sample Pipettor                     | - | 3 | - | Y* (see note below) |
| Pipettor Left     | Gripper                             | - | - | - | -                   |
| Pipettor Left     | Flat Cable Pipettor Left Arm        | - | - | - | -                   |
| Pipettor Left     | Gripper Z-Motor                     | - | - | - | -                   |
| Pipettor Left     | Pipettors Z-Motor                   | - | 1 | - | -                   |
| Pipettor Left     | Pipettors X-Motor w Cables          | - | 1 | - | -                   |
| Pipettor Left     | Gripper PCB                         | - | - | - | -                   |
| Pipettor Left     | Flat Cable Gripper                  | - | - | - | -                   |
| Pipettor Left     | Flat Cable Sample Pipettor          | - | - | - | -                   |
| Pipettor Left     | Dual Sledge PCB                     | - | 1 | - | -                   |
| Pipettor Left     | Gripper bearing L shape             | - | - | - | -                   |
| Pipettor Left     | Gripper Floating Sliding Plate      | - | - | - | -                   |
| Pipettor Left     | Slide Bearing Gripper               | - | - | - | -                   |
| Pipettor Left     | Gripper Sensor                      | - | - | - | -                   |
| Pipettor Left     | Left Pipettor X-Timing Belt         | - | 1 | - | -                   |
| Pipettor Left     | Left Pipettor Y-Timing Belt         | - | 1 | - | -                   |
| Pipettor Left     | Pipettors Y-Motor w Cables          | - | 1 | - | -                   |
| Pipettor Left     | Pipettor Timing Belt Holder         | - | 1 | - | -                   |
| Pipettor Left     | Left Arm Y-axis home sensor         | - | - | - | -                   |
| Pipettor Left     | Mechanical stop for Y-belt          | - | - | - | -                   |
| Pipettor Right    | Pipettor Right Gantry               | - | - | 3 | Y                   |
| Pipettor Right    | SPOLV                               | - | - | 3 | Y                   |
| Pipettor Right    | SPOLV Needle                        | - | - | 3 | Y                   |
| Pipettor Right    | Flat Cable Pipettor Right Arm       | - | - | - | -                   |
| Pipettor Right    | Flat Cable SPOLV                    | - | - | - | -                   |
| Pipettor Right    | Single Sledge PCB                   | - | - | 1 | -                   |
| Pipettor Right    | Pipettor Wash Station               | - | - | 1 | -                   |
| Pipettor Right    | Right Pipettor X-Timing Belt        | - | - | 1 | -                   |
| Pipettor Right    | Right Pipettor Y-Timing Belt        | - | - | 1 | -                   |
| Pipettor Right    | Pipettors Y-Motor w Cables          | - | - | 1 | -                   |
| Pipettor Right    | Pipettors Z-Motor                   | - | - | 1 | -                   |
| Pipettor Right    | Pipettors X-Motor w Cables          | - | - | 1 | -                   |
| Pipettor Right    | Pipettor Timing Belt Holder         | - | - | 1 | -                   |
| Pipettor Right    | Right Arm Y-axis home sensor        | - | - | - | -                   |
| Pipettor Right    | SPOLV Z-axis home sensor            | - | - | - | -                   |
| Pipettor Right    | Mechanical stop for Y-belt          | - | - | - | -                   |
| Pumps and Tubings | PRP-SPOLV Tubing                    | - | - | - | -                   |
| Pumps and Tubings | Reader-Started Dispense Pump tubing | 3 | - | 3 | -                   |

## 9. Maintenance Schedule for internal-tank configuration

|                   |                                      |   |   |   |   |
|-------------------|--------------------------------------|---|---|---|---|
| Pumps and Tubings | WAP Waste Aspiration Pump            | 3 | - | - | Y |
| Pumps and Tubings | LWP Liquid Waste Pump                | - | - | 3 | Y |
| Pumps and Tubings | WBDP Wash Buffer Dispense Pump       | 3 | - | - | Y |
| Pumps and Tubings | WBDP cartridge                       | 3 | - | - | Y |
| Pumps and Tubings | WBDP Valve                           | 3 | - | - | Y |
| Pumps and Tubings | WBDP-Washer tubing                   | - | - | - | - |
| Pumps and Tubings | Starter Reagent Pump                 | 3 | - | 3 | Y |
| Pumps and Tubings | Liquid Distribution Valve            | - | - | 1 | - |
| Pumps and Tubings | LDV-PRP tubing                       | - | - | - | - |
| Pumps and Tubings | KNF 60 Pump                          | - | - | 3 | Y |
| Pumps and Tubings | Tubing Energy Chain Right            | 3 | - | - | - |
| Pumps and Tubings | Tubing Energy Chain Left             | - | - | 1 | - |
| Pumps and Tubings | WAP Peristaltic Tubes (2+3 Channel)  | 3 | - | - | Y |
| Pumps and Tubings | LWP Peristaltic Tubes                | - | - | 1 | Y |
| Pumps and Tubings | Pressure Shoe WAP and LWP (set of 6) | 1 | - | 1 | - |
| Pumps and Tubings | Direct Drain Kit                     | - | - | - | - |
| Reader            | Reader Assembly                      | 3 | - | 3 | Y |
| Reader            | PMT Module                           | 3 | - | 3 | Y |
| Reader            | Aspiration Needle                    | - | - | 3 | - |
| Reader            | SIPAL                                | - | - | 3 | - |
| Reader            | SIPAL (Plastic Only)                 | - | - | 1 | - |
| Reader            | Washer/Reader Motor W/ Cable         | - | - | 1 | - |
| Reader            | Reader Cover                         | - | - | - | - |
| Reader            | Reader Rotor                         | 3 | - | - | - |
| Reader            | Reader Light Barrier                 | 1 | - | 1 | - |
| Reader            | Reader PCB                           | 3 | - | 3 | Y |
| Reader            | Washer/Reader Motor W/ Cable         | - | - | 1 | - |
| Reagent Area      | Reagent Area Assembly                | - | - | - | - |
| Reagent Area      | RFID Antenna PCB Integral 10 Channel | - | - | - | - |
| Reagent Area      | RFID Antenna PCB Ancillary           | - | - | - | - |
| Reagent Area      | Stirrer Bar                          | - | - | - | - |
| Reagent Area      | Reagent Motor                        | - | - | - | - |
| Reagent Area      | Cooling Module (QTY 1)               | - | - | - | - |
| Reagent Area      | Reagent Module PCB                   | - | - | - | - |
| Reagent Area      | Reagent Area Temp Sensor             | - | - | - | - |
| Reagent Area      | Reagent Stirrer Bearings             | - | - | - | - |
| Reagent Area      | Snap In Plate                        | - | - | - | - |
| Sample Area       | Sample Area Assembly                 | - | - | - | - |
| Sample Area       | Sample Module PCB                    | - | - | - | - |
| Sample Area       | Push-push Mechanism (Set of 10)      | - | - | - | - |
| Sample Area       | Sample Rack                          | - | - | - | - |
| Sample Area       | Barcode Scanner                      | - | - | - | - |
| Sample Area       | Sample Temperature Sensor            | - | - | - | - |
| Sample Area       | Sample Module Cover                  | - | - | - | - |
| Starter Area      | Starter Reagent Assembly             | - | - | - | - |

## 9. Maintenance Schedule for internal-tank configuration

|              |                                    |   |   |   |   |
|--------------|------------------------------------|---|---|---|---|
| Starter Area | RFID Antenna PCB Starters          | - | - | - | - |
| Starter Area | Starter bottle sensor clip         | - | - | - | - |
| Starter Area | Starter 1 sensor & tubing          | 1 | - | 1 | - |
| Starter Area | Starter 2 sensor & tubing          | 1 | - | 1 | - |
| Tanks Area   | System Liquid Tank                 | - | - | - | - |
| Tanks Area   | Cleaning Tank                      | - | - | - | - |
| Tanks Area   | System Liquid tank Presence Sensor | - | - | - | - |
| Tanks Area   | System liquid docking and tubing   | - | - | - | - |
| Tanks Area   | Waste Tank                         | - | - | - | - |
| Tanks Area   | Float Switch Waste tank            | - | - | - | - |
| Tanks Area   | Liquid Waste Sensor Cable          | - | - | - | - |
| Tanks Area   | DI water docking and tubing        | - | - | - | - |
| Tanks Area   | DI water Tank Presence Sensor      | - | - | - | - |
| Tanks Area   | DI-WB Led PCB cables               | - | - | - | - |
| Tanks Area   | Tank filling stand                 | - | - | - | - |
| Washer       | Washer Assembly                    | 3 | - | - | Y |
| Washer       | Washer PCB                         | 3 | - | - | Y |
| Washer       | Washer/Reader Motor W/ Cable       | 1 | - | - | - |
| Washer       | Washer Rotor                       | 1 | - | - | - |
| Washer       | Aspiration Needle                  | 3 | - | - | - |
| Washer       | SIPAL                              | 3 | - | - | - |
| Washer       | SIPAL (Plastic Only)               | 1 | - | - | - |
| Washer       | Dispense Manifold w/ Valve         | 3 | - | - | - |
| Washer       | Washer Light Barrier               | - | - | - | - |
| Washer       | Washer Valve                       | 3 | - | - | - |

Table 9–4: Requalification

**NOTE**

Table 9–4 lists all parts for which assays re-calibration is required upon conclusion of a service intervention (“Recal” column with “Y” value).

The FSE shall guarantee that at least the first kit re-calibration is successfully completed before the service call is closed.

Kit re-calibration is deemed not required when the object of the intervention is a part not listed in the “Recal” column. In such a case FSE may inform the end user that the kit recalibration is not needed. Anyway the end user has the right to decide if performing or not the kit recalibration.

**NOTE**

The results of the requalification activities must be attached to the Service records (for Subsidiaries only: SGAT WS), together with a detailed description of the Service activities performed on the analyzer.

**NOTE**

The kit recalibration upon Sample Pipettor replacement is mandatory only in case customer runs any Assay with external calibrator.



**NOTE**

3 repetitions mean “3 successful consecutive repetitions”. See Corp-TN-000292 (former LXS TN 009) for re-rerun rules.

---

## 9.5 Other activities during service visits

In order to check the system, at the end of each service visit, before performing the system requalification indicated in Chapter 9.4 on page 9-9, it is required to perform also the operations described in the following paragraphs.

### 9.5.1 Instrument recovery and test

After remounting of all dismantled assemblies, perform the following steps:

1. Close any open software and start the **Service Software** from the Windows Administrator account.
2. Initialize the system, see Chapter 11.8.1.1 on page 11-58.
3. Perform the Cuvette Endurance Run, see Chapter 11.8.1.2.1 on page 11-61.

### 9.5.2 User Manual last revision check

Check if the most updated version of the Instrument Manuals, corresponding to the Instrument SW version and customer's language for the specific country is available on the system.

If not, install the most updated version of the Instrument Manuals, see Chapter 5.4.4 on page 5-34. This can be done:

- using the **Download** functionality available in the **Online Help** section of the **Main UI**;
- using the **Acquire** functionality available in the **Online Help** section of the **Main UI** in case the appropriate instrument package was previously downloaded from **Dialog**.

## 9.6 System Shutdown and Wakeup for long term inactivity procedure

Aim of this section is to provide the instructions to shutdown the **LIAISON<sup>®</sup> XS** system in case of long period of inactivity (more than 2 weeks) and its subsequent restart.

Both activities could be managed by the user.

The following procedure can be applied on:

- new instruments (out of the crates) successfully installed that need to be switched off for a long period due to commercial aspect;
- operating instruments that need to be switched off for a long period due to user needs.

**NOTE**

Weekly and maintenance is not needed in case of long term instrument shutdown.

---

### 9.6.1 Procedure for system Shutdown and Wakeup

**NOTE**

Shutdown might be possible in case of long period of inactivity (more than 2 weeks). The required procedure for the shutdown and restart of the system from long period is described in the Chapter 5.11 “Shutdown/End of Day Activities” of the User Manual and also in the following paragraph.

---

**NOTE**

Before starting the System Shutdown Maintenance task, 1 L of Cleaning Solution shall be prepared according to Chapter 7.3 “Preparation of the Cleaning Solution in the cleaning tank” of the User Manual.

---

#### 9.6.1.1 System Shutdown Maintenance Task

1. Ensure that the instrument is not in “running” or “maintenance” status before starting the maintenance;
2. Tap on the **Maintenance** main category and select the **System Shutdown Maintenance** Task;
3. Tap on the “Perform” Button to start the System Shutdown Maintenance Task;
4. Follow the instructions described in the Maintenance Task.

#### 9.6.1.2 System Wakeup Maintenance Task

1. Switch on the instrument according to Chapter 5.3 “Start up” of the User Manual;
2. Ensure that the instrument is not in “running” or “maintenance” status before starting the maintenance;
3. Tap on the **Maintenance** main category and select the **System Wakeup Maintenance** Task;
4. Tap on the “Perform” Button to start the System Wakeup Maintenance Task;
5. Follow the instructions described in the Maintenance Task.

# 10 Maintenance Schedule for external-tanks configuration

The **LIAISON® XS** instrument was designed to be as maintenance free as possible. However, given the functional, precision and reliability requirements of the complex applications the instruments are used for, regular care and attention are necessary to maintain the high quality of instrument performance. For some components regular maintenance procedures could not be avoided in the design process.

The maintenance intervals given below assume a working time of the instrument of 8 h per day, 5 days a week, 180 results/day, in a normal laboratory environment. Upon more intense usage of the instrument or usage in demanding environmental conditions (dust, heat, humidity) the maintenance intervals should be shortened accordingly.

The description of requalification, cleaning and lubrication procedures to be performed during service visits are also reported in this chapter.

## 10.1 Safety

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 2.5 on page 2-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

### WARNING



#### Improper Maintenance work

Improper maintenance work can result in serious personal injuries and material damages.

- Take off watches and jewelry before performing any maintenance works.

### WARNING



#### Defects in the liquid system

Defect or leaking tubes, syringes, valves or pumps lead to deterioration of the pipetting results and consequently corruption of final results. Furthermore incorrectly flushed tips can cause mixing up of sample material.

- Check the instrument for drops and pooling of liquid on surfaces.
- Check tubes, syringes, valves or pumps periodically.

---

**10. Maintenance Schedule for external-tanks configuration**

---

**CAUTION**

---

**Maintenance responsibility**

Defective parts that result from the non-fulfillment of proper maintenance intervals are the sole responsibility of country specific distributors. DiaSorin S.p.A. is not liable for defective parts due to non-performance of required maintenance intervals.

---

**CAUTION**

---

**Check System**

During the periodic maintenance of the system, always follow the steps indicated in **LIAISON® XS** Yearly Maintenance (see Chapter 34.4 on page 34-11) check list. The proper execution and compilation of these check list will ensure the completion of the proper periodic maintenance of the instruments.

---

## 10.2 Maintenance

### 10.2.1 List of required material

- Set of hexagon socket wrench (Allen key).
- Set of torx screw driver.
- Set of cross slotted screw driver.
- Flat screw driver.

### 10.2.2 System cleaning



#### Wrong placement of the Liquid Waste tank

Too high installation of the Liquid waste tank can lead to a liquid drawback.

- It is preferred to place the Liquid waste tank below the instrument.



#### Wrong placement of the System Liquid and Wash Buffer tanks

Too low installation of the tanks can lead to a not proper performance of the pumps.

- The tanks shall be positioned at a maximum distance of 800mm from the bench/table where the system is placed on. For the allowed positioning, see Chapter 6.3.2 on page 6-6.

1. Empty Liquid Waste tank, add hypochlorite solution (see Chapter 5.4.6 “Connecting and separating the Liquid Waste” of the User Manual) and load it back into the system. This does not apply to those systems connected to direct drain.
2. Remove the cuvettes from the system in the Washer, Reader and Incubator performing the **Empty Incubator** from Main category **Diagnostics**.
3. Run **Service Decontamination** task from the dedicated window of **Main User Interface** (Main category **Maintenance**, sub-category **Task**).
4. Check the tanks cleaning: if needed, clean them according to Long Term Maintenance, see the User Manual.
5. Remove all the samples tubes out of the Sample racks.
6. Remove all the integral kits out of the Reagent Module.
7. Remove all the ancillaries and holders out of the Ancillary rack.
8. Remove all the disposable tips trays out of the Cuvette Tip drawer.
9. Remove all the cuvette trays out of the Cuvette Tip drawer.
10. Close the LIAISON® EASY Waste box with its cap and remove it from the Solid Waste bin.
11. Insert a new LIAISON® EASY Waste box in the solid waste bin.

### 10.2.3 PC & Accessories

1. Clean PC.

## 10. Maintenance Schedule for external-tanks configuration

---

2. Check that databases (**Results/ All**, **Results/ Archived**, **Results/ Deleted**, **Results/ Calibrations**, **Maintenance/ Log**) have not reached dimensions that compromise the proper browsing of the database. In case this condition is verified, clean up the databases:
  - Reduce as much as possible the items in the **Results/ All** page (i.e. delete unnecessary results according to customer feedback);
  - Reduce as much as possible the items in the **Results/ Archived** page (i.e. delete unnecessary results according to customer feedback);
  - Reduce as much as possible the items in the **Results/ Calibrations** page (i.e. remove all Unused calibrations);
  - Delete all items from **Results/ Deleted** page;
  - Delete all unused group definitions (according to customer feedback);
  - Delete all unused control definitions (according to customer feedback);
  - Delete **Maintenance/ Log** (according to customer feedback).
3. Check printer functionalities (if present).
4. Check the presence of the most updated version of the Manuals, corresponding to the Instrument SW version and customer's language for the specific country. If needed, update the Manual as written in the Installation procedure, see Chapter 6.4.4 on page 6-36.
5. Close any open software and log-in as Administrator.
6. Check that at the least 20% of C:\ partition of the hard disk is free. If this condition is not verified, delete the items present in C:\LiaisonXS\Data\Temp\archive.
7. Check UPS functionalities (if present).
8. Run Antivirus application as per following instructions:
  - Launch the "av.bat" file from the path: C:\LiaisonXS\Tools\AV\
  - While the Virus Scan is ongoing do not reboot the PC or logoff from the current account, to avoid the interruption of the scansion
  - A message informs that the scansion is completed
  - The results are appended in "C:\LiaisonXS\Tools\AV\clamlog.txt".

### 10.2.4 Preparation for Maintenance of Internal Modules

1. Start the **Service Software**.
2. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
3. Click onto the **Instrument** button and choose the **Power Off** button to operate in safe condition.
4. The process is successful when the **Processing** window closes without failure message.
5. Open top covers.
6. Move Pipettor arms to a safe location.

### 10.2.5 Maintenance of the Washer and Reader Modules

**NOTE**

The cleanings described below shall be performed using lint free cloth and isopropanol; as a final rinsing DI water shall be used.

---

1. Replace all the Washer and Reader Aspiration needles, see Chapter 17.3.1 on page 17-6 and see Chapter 18.3.1 on page 18-4.
2. Perform a visual check for crystallization in the Reader rotor and cover. Turn the Reader rotor and, if necessary, remove the Reader cover and clean both the Reader rotor and the Reader cover, see Chapter 18.3.2 on page 18-9 and see Chapter 18.3.5 on page 18-17.
3. Perform a visual check for crystallization in the Washer rotor and plate. Turn the Washer rotor and, if necessary, remove the Washer plate and clean the Washer rotor, plate and dispense nozzles, see Chapter 17.3.6 on page 17-24, see Chapter 17.3.9 on page 17-32 and see Chapter 17.3.2 on page 17-12.

### 10.2.6 Maintenance of the Pumps and Tubings

1. Replace the WBDP cartridge, see Chapter 26.3.7 on page 26-29.
2. Replace the WAP Peristaltic tubings, see Chapter 26.3.3 on page 26-12.
3. Replace the LWP Peristaltic tubings, see Chapter 26.3.3 on page 26-12.
4. Replace the MANN filter, see Chapter 29.3.1 on page 29-7.
5. Inspect and, if necessary, replace Waste tubing at Left and Right Energy Chain, see Chapter 29.3.8 on page 29-25 and see Chapter 29.3.9 on page 29-29.

### 10.2.7 Maintenance of the Reagent Module

**NOTE**

The cleanings described below shall be performed using lint free cloth and isopropanol; as a final rinsing DI water shall be used.

---

**NOTE**

During the cleaning of the Reagent Module, be sure to remove as many residual of dust between the fins and in the holes hosting the sliding bearings as possible.

---

1. Remove the Reagent Module cover, see Chapter 22.3.1.2 on page 22-4.
2. Clean the Reagent Module (inner and outer, cover included) using an alcohol/disinfectant solution.
3. Assemble the Reagent Module cover, see Chapter 22.3.1.3 on page 22-4
4. If needed, clean the crank drive. Then, lubricate it, see Chapter 22.3.3.5 on page 22-10.

### 10.2.8 Maintenance of the Pipettor

**NOTE**

The maintenance of the pipettor shall be performed using lint free cloth, isopropanol and oil (PN 72229616).

---

**CAUTION**

Before using the oil, the isopropanol must be evaporated or wiped off from the rods. Do not apply too much oil on the rods, because of potential system contamination.

---

1. Wipe the whole length of the X and Y Pipettor rods with a lint free cloth soaked with isopropanol.
2. Wipe all Pipettor rods with a lint free cloth soaked with oil (PN 72229616). A thin film of oil shall remain.
3. Verify the proper behavior moving both Pipettor modules 5 times in axial X-direction and 5 times in axial Y-direction.
4. Inspect and, if necessary, replace the Gripper slide bearings, see Chapter 20.3.6 on page 20-25.

### 10.2.9 Cleaning of other surfaces

**NOTE**

The cleanings described below shall be performed using an alcohol/ disinfectant solution.

---

1. Clean the Cuvette - Tip Loading Module.
2. Clean inner parts of Left and Right Top Covers.
3. Clean the stand-up for the tanks, if present.

### 10.2.10 System Restore & Periodic Recalibrations

1. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
2. Click onto the **Instrument** button and choose the **Power On** button to restore the system power source.
3. The process is successful when the **Processing** window closes without failure message.
4. Close any open software and start the **Teacher Software** from the Windows Administrator account.
5. Check Pipettor teaching, see Chapter 13 on page 13-1.
6. Perform backup of `Coordinate.coo` file.
7. Perform backup of ".rac" files.
8. Close any open software and start the **Service Software** from the Windows Administrator account.
9. Initialize the system, see Chapter 12.8.1 on page 12-60.
10. Perform the Washer SIPAL Lift Calibration, see Chapter 12.5.3.2 on page 12-34.
11. Perform the Washer Aspiration Monitoring Calibration, see Chapter 12.5.3.3 on page 12-35.



## 10. Maintenance Schedule for external-tanks configuration

12. Perform the WBDP Calibration, see Chapter 12.5.3.4 on page 12-36.
13. Perform the Reader SIPAL Lift Calibration, see Chapter 12.6.3.2 on page 12-41.
14. Perform the Reader Aspiration Monitoring Calibration, see Chapter 12.6.3.3 on page 12-42.
15. Insert at least two cuvette trays in the Cuvette - Tip Loading Module and set them in the System - Cuvette Tray State, see Chapter 12.8.2 on page 12-69.
16. Perform the Cuvette Endurance Run, see Chapter 12.8.1.2.1 on page 12-63.
17. Prime the fluidics, see Chapter 12.8.1.3.1 on page 12-66.
18. Check that the Antivirus scan has been completed.
19. Close any open software and restart the PC.
20. Log into the **Main UI Software** and initialize the system.
21. Reset the counter of the Solid Waste.
22. Insert cuvette trays and disposable tips trays in the Cuvette - Tip Loading Module.
23. Check the **Status - Consumables** toggle and assign the Consumables (cuvettes and tips) in the **Status - Plastic** toggle in **Main UI Software**.
24. Load the cuvette in the Incubator Module.

### 10.2.11 Requalification test

1. Run **Service - 3 Series Background Wash** task from the dedicated window of **Main User Interface** (Main Category **Maintenance**, sub-category **Task**).
2. For any given series, verify that CV observed is less than or equal to 10.00% and the average signal developed is equal to or smaller than 1000 RLU.
3. Run **Service - 3 Series LightCheck Right** task from the dedicated window of **Main User Interface** (Main Category **Maintenance**, sub-category **Task**).
4. For any given series, verify that CV observed is less than or equal to 4.00%, that the minimum average signal developed is higher than 160000 RLU and lower than 245000 RLU.
5. Run **Service - 3 Series LightCheck Left** task from the dedicated window of **Main User Interface** (Main Category **Maintenance**, sub-category **Task**).
6. For any given series, verify that CV observed is less than or equal to 4.00%, that the minimum average signal developed is higher than 345000 RLU and lower than 520000 RLU.

|            | Max CV% | Min RLU | Max RLU |
|------------|---------|---------|---------|
| <b>BGW</b> | 10.00%  | 0       | 1000    |
| <b>LCR</b> | 4.00%   | 160000  | 245000  |
| <b>LCL</b> | 4.00%   | 345000  | 520000  |

Table 10–1: Summary of the Acceptance Criteria

7. Perform the calibration and the control of one of the kits available at customer site.
8. Verify that the obtained values are in range.
9. Perform raw data export and backup.
10. Inform the customer that all assays in use must be recalibrated.

---

**10. Maintenance Schedule for external-tanks configuration**

---

**NOTE**

For the re-run rules in case of failures of tests, please refer to Corp-TN-000292 (former LXS TN 009).

---

**NOTE**

The successful execution of 3 consecutive BGW, 3 consecutive LCL and 3 consecutive LCR shall be always achieved.

---

**NOTE**

The results of the requalification activities (BGW, LCL, LCR, Calibration and Control run) must be attached to the Service records (for Subsidiaries only: SGAT WS), together with a detailed description of the Service activities performed on the analyzer.

---

## 10.3 Preventive Maintenance kit for Field Service Engineers

### 10.3.1 Yearly Maintenance kit

| Module                    | Description                  | Action  | Link to procedure                                                  |
|---------------------------|------------------------------|---------|--------------------------------------------------------------------|
| Reader and Washer         | Set of 4 Aspiration Needle   | Replace | see Chapter 18.3.1 on page 18-4<br>see Chapter 17.3.1 on page 17-6 |
| Waste Aspiration Pump     | Peristaltic tubing 2-channel | Replace | see Chapter 26.3.3 on page 26-12                                   |
|                           | Peristaltic tubing 3-channel | Replace | see Chapter 26.3.3 on page 26-12                                   |
| Liquid Waste Pump         | Peristaltic tubing 3-channel | Replace | see Chapter 26.3.3 on page 26-12                                   |
| Wash Buffer Dispense Pump | Cartridge, piston and screw  | Replace | see Chapter 26.3.7 on page 26-29                                   |
| Tubing                    | MANN filter                  | Replace | see Chapter 29.3.1 on page 29-7                                    |

Table 10–2: Yearly Maintenance

### 10.3.2 Replacing on demand parts

| Module        | Description                                                                                                     | Action                     | Link to procedure                |
|---------------|-----------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------------|
| Waste tubings | Waste tubing at Left Energy Chain:<br>Wash station-LWP<br>LWP-Waste interface                                   | Upon inspection, as needed | see Chapter 29.3.8 on page 29-25 |
|               | Waste tubing at Right Energy Chain:<br>WAP-Waste interface,<br>Reagent Module-WAP,<br>Reader-WAP,<br>Washer-WAP | Upon inspection, as needed | see Chapter 29.3.9 on page 29-29 |
| Gripper       | Slide bearing set                                                                                               | Upon inspection, as needed | see Chapter 20.3.6 on page 20-25 |

Table 10–3: Maintenance on demand

## 10.4 Instrument requalification after service visit

During a Field Service intervention some parts of the instrument (assembly, sensor, motor, pump, tubing, etc.) may be replaced to troubleshoot and solve the problem on the instrument. Depending on the replaced spare part, before releasing the instrument to run patient samples, a final re-qualification may be required, as well as the reagent integral re-calibration. These spare parts are specified in Table 10–4.

In case some parts of the instrument are removed and mounted back for troubleshooting purposes, refer to the relevant dedicated paragraphs in order to assess the need of mechanical recalibration/teaching.

In case the replaced spare part is not present in the Table 10–4, refer to the assembly the replaced spare part belongs to.

Screws, washers, nuts, clips, fuses and other minor parts are deemed not to require any re-qualification/re-calibration.

### 10.4.1 Background and rationales

The Spare Part replacements detailed in Table 10–4, see Chapter 10.4.2 on page 10-10:

- may involve critical instrument components for the core characteristics of assay performance (i.e.: pipettor, incubator, washer, reader) and, due to this, may require a specific re-qualification related to the area of interest;
- for the same reason may require re-calibration of the assays used by the Final User, being potentially impacting the assay calibration RLU levels.

### 10.4.2 Service interventions

| Assembly    | Spare Part                        | BGW | LCL | LCR | Recal |
|-------------|-----------------------------------|-----|-----|-----|-------|
| Covers      | Front Cover GAS spring            | -   | -   | -   | -     |
| Covers      | Front Cover PCB                   | -   | -   | -   | -     |
| Covers      | Starter Reagent flap              | -   | -   | -   | -     |
| Electronics | Pipettor PCB                      | -   | -   | -   | -     |
| Electronics | Liquid Support PCB                | -   | -   | -   | -     |
| Electronics | COP                               | 1   | 1   | 1   | -     |
| Electronics | PWR CAN PCB                       | -   | -   | -   | -     |
| Electronics | Power Supply                      | -   | -   | -   | -     |
| Electronics | Fuse, 3.15A, 5.08 MM (Set of 14)  | -   | -   | -   | -     |
| Electronics | Fuse, 5.0 A, 6.3x32MM (Set of 10) | -   | -   | -   | -     |
| Electronics | Fuse, 4.0 A, 5x20MM (Set of 10)   | -   | -   | -   | -     |
| Electronics | Soft Power button PCB             | -   | -   | -   | -     |
| Incubator   | Incubator Assembly                | -   | 3   | 3   | Y     |
| Incubator   | Incubator PCB                     | -   | 3   | 3   | Y     |
| Incubator   | Incubator Cover w/ Bearings       | -   | -   | -   | -     |
| Incubator   | Incubator Rotor                   | -   | 1   | 1   | -     |
| Incubator   | Incubator Motor w/ Cables         | -   | 1   | 1   | -     |
| Incubator   | Light Guide                       | -   | 1   | 1   | -     |
| Incubator   | Incubator Base Bearing            | -   | -   | -   | -     |

## 10. Maintenance Schedule for external-tanks configuration

|                   |                                   |   |   |   |                     |
|-------------------|-----------------------------------|---|---|---|---------------------|
| Miscellaneous     | Flat Cable Pipettor Clips 17 mm   | - | - | - | -                   |
| Miscellaneous     | Flat Cable Pipettor Clips 26,5 mm | - | - | - | -                   |
| Miscellaneous     | Flat Cable Pipettor Clips 12 mm   | - | - | - | -                   |
| Miscellaneous     | Consumable Drawer Switch          | - | - | - | -                   |
| Miscellaneous     | Sample rack labels                | - | - | - | -                   |
| PC                | Panel Control Unit                | - | - | - | -                   |
| Pipettor Left     | Pipettor Left Gantry              | 3 | 3 | - | -                   |
| Pipettor Left     | Sample Pipettor                   | - | 3 | - | Y* (see note below) |
| Pipettor Left     | Gripper                           | - | - | - | -                   |
| Pipettor Left     | Flat Cable Pipettor Left Arm      | - | - | - | -                   |
| Pipettor Left     | Gripper Z-Motor                   | - | - | - | -                   |
| Pipettor Left     | Pipettors Z-Motor                 | - | 1 | - | -                   |
| Pipettor Left     | Pipettors X-Motor w Cables        | - | 1 | - | -                   |
| Pipettor Left     | Gripper PCB                       | - | - | - | -                   |
| Pipettor Left     | Flat Cable Gripper                | - | - | - | -                   |
| Pipettor Left     | Flat Cable Sample Pipettor        | - | - | - | -                   |
| Pipettor Left     | Dual Sledge PCB                   | - | 1 | - | -                   |
| Pipettor Left     | Gripper bearing L shape           | - | - | - | -                   |
| Pipettor Left     | Gripper Floating Sliding Plate    | - | - | - | -                   |
| Pipettor Left     | Slide Bearing Gripper             | - | - | - | -                   |
| Pipettor Left     | Gripper Sensor                    | - | - | - | -                   |
| Pipettor Left     | Left Pipettor X-Timing Belt       | - | 1 | - | -                   |
| Pipettor Left     | Left Pipettor Y-Timing Belt       | - | 1 | - | -                   |
| Pipettor Left     | Pipettors Y-Motor w Cables        | - | 1 | - | -                   |
| Pipettor Left     | Pipettor Timing Belt Holder       | - | 1 | - | -                   |
| Pipettor Left     | Left Arm Y-axis home sensor       | - | - | - | -                   |
| Pipettor Left     | Mechanical stop for Y-belt        | - | - | - | -                   |
| Pipettor Right    | Pipettor Right Gantry             | - | - | 3 | Y                   |
| Pipettor Right    | SPOLV                             | - | - | 3 | Y                   |
| Pipettor Right    | SPOLV Needle                      | - | - | 3 | Y                   |
| Pipettor Right    | Flat Cable Pipettor Right Arm     | - | - | - | -                   |
| Pipettor Right    | Flat Cable SPOLV                  | - | - | - | -                   |
| Pipettor Right    | Single Sledge PCB                 | - | - | 1 | -                   |
| Pipettor Right    | Pipettor Wash Station             | - | - | 1 | -                   |
| Pipettor Right    | Right Pipettor X-Timing Belt      | - | - | 1 | -                   |
| Pipettor Right    | Right Pipettor Y-Timing Belt      | - | - | 1 | -                   |
| Pipettor Right    | Pipettors Y-Motor w Cables        | - | - | 1 | -                   |
| Pipettor Right    | Pipettors Z-Motor                 | - | - | 1 | -                   |
| Pipettor Right    | Pipettors X-Motor w Cables        | - | - | 1 | -                   |
| Pipettor Right    | Pipettor Timing Belt Holder       | - | - | 1 | -                   |
| Pipettor Right    | Right Arm Y-axis home sensor      | - | - | - | -                   |
| Pipettor Right    | SPOLV Z-axis home sensor          | - | - | - | -                   |
| Pipettor Right    | Mechanical stop for Y-belt        | - | - | - | -                   |
| Pumps and Tubings | PRP-SPOLV Tubing                  | - | - | - | -                   |
| Pumps and Tubings | MANN filter                       | - | - | 3 | -                   |

## 10. Maintenance Schedule for external-tanks configuration

|                   |                                      |   |   |   |   |
|-------------------|--------------------------------------|---|---|---|---|
| Pumps and Tubings | Reader-Started Dispense Pump tubing  | 3 | - | 3 | - |
| Pumps and Tubings | WAP Waste Aspiration Pump            | 3 | - | - | Y |
| Pumps and Tubings | LWP Liquid Waste Pump                | - | - | 3 | Y |
| Pumps and Tubings | WBDP Wash Buffer Dispense Pump       | 3 | - | - | Y |
| Pumps and Tubings | WBDP cartridge                       | 3 | - | - | Y |
| Pumps and Tubings | WBDP Valve                           | 3 | - | - | Y |
| Pumps and Tubings | WBDP-Washer tubing                   | - | - | - | - |
| Pumps and Tubings | Starter Reagent Pump                 | 3 | - | 3 | Y |
| Pumps and Tubings | KNF 60 Pump                          | - | - | 3 | Y |
| Pumps and Tubings | Tubing Energy Chain Right            | 3 | - | - | - |
| Pumps and Tubings | Tubing Energy Chain Left             | - | - | 1 | - |
| Pumps and Tubings | WAP Peristaltic Tubes (2+3 Channel)  | 3 | - | - | Y |
| Pumps and Tubings | LWP Peristaltic Tubes                | - | - | 1 | Y |
| Pumps and Tubings | Pressure Shoe WAP and LWP (set of 6) | 1 | - | 1 | - |
| Pumps and Tubings | Direct Drain Kit                     | - | - | - | - |
| Reader            | Reader Assembly                      | 3 | - | 3 | Y |
| Reader            | PMT Module                           | 3 | - | 3 | Y |
| Reader            | Aspiration Needle                    | - | - | 3 | - |
| Reader            | SIPAL                                | - | - | 3 | - |
| Reader            | SIPAL (Plastic Only)                 | - | - | 1 | - |
| Reader            | Washer/Reader Motor W/ Cable         | - | - | 1 | - |
| Reader            | Reader Cover                         | - | - | - | - |
| Reader            | Reader Rotor                         | 3 | - | - | - |
| Reader            | Reader Light Barrier                 | 1 | - | 1 | - |
| Reader            | Reader PCB                           | 3 | - | 3 | Y |
| Reader            | Washer/Reader Motor W/ Cable         | - | - | 1 | - |
| Reagent Area      | Reagent Area Assembly                | - | - | - | - |
| Reagent Area      | RFID Antenna PCB Integral 10 Channel | - | - | - | - |
| Reagent Area      | RFID Antenna PCB Ancillary           | - | - | - | - |
| Reagent Area      | Stirrer Bar                          | - | - | - | - |
| Reagent Area      | Reagent Motor                        | - | - | - | - |
| Reagent Area      | Cooling Module (QTY 1)               | - | - | - | - |
| Reagent Area      | Reagent Module PCB                   | - | - | - | - |
| Reagent Area      | Reagent Area Temp Sensor             | - | - | - | - |
| Reagent Area      | Reagent Stirrer Bearings             | - | - | - | - |
| Reagent Area      | Snap In Plate                        | - | - | - | - |
| Sample Area       | Sample Area Assembly                 | - | - | - | - |
| Sample Area       | Sample Module PCB                    | - | - | - | - |
| Sample Area       | Push-push Mechanism (Set of 10)      | - | - | - | - |
| Sample Area       | Sample Rack                          | - | - | - | - |
| Sample Area       | Barcode Scanner                      | - | - | - | - |
| Sample Area       | Sample Temperature Sensor            | - | - | - | - |
| Sample Area       | Sample Module Cover                  | - | - | - | - |
| Starter Area      | Starter Reagent Assembly             | - | - | - | - |
| Starter Area      | RFID Antenna PCB Starters            | - | - | - | - |

## 10. Maintenance Schedule for external-tanks configuration

|              |                                          |   |   |   |   |
|--------------|------------------------------------------|---|---|---|---|
| Starter Area | Starter bottle sensor clip               | - | - | - | - |
| Starter Area | Starter 1 sensor & tubing                | 1 | - | 1 | - |
| Starter Area | Starter 2 sensor & tubing                | 1 | - | 1 | - |
| Tanks Area   | Wash Buffer Tank                         | - | - | - | - |
| Tanks Area   | System Liquid Tank                       | - | - | - | - |
| Tanks Area   | Cleaning Tank                            | - | - | - | - |
| Tanks Area   | System Liquid tank Presence Sensor cable | - | - | - | - |
| Tanks Area   | Wash buffer tank Presence Sensor cable   | - | - | - | - |
| Tanks Area   | Wash buffer tank-WBDP tubing             | - | - | - | - |
| Tanks Area   | System liquid tank-filter tubing         | - | - | - | - |
| Tanks Area   | Waste Tank                               | - | - | - | - |
| Tanks Area   | Float Switch Waste tank                  | - | - | - | - |
| Tanks Area   | Liquid Waste Sensor Cable                | - | - | - | - |
| Washer       | Washer Assembly                          | 3 | - | - | Y |
| Washer       | Washer PCB                               | 3 | - | - | Y |
| Washer       | Washer/Reader Motor W/ Cable             | 1 | - | - | - |
| Washer       | Washer Rotor                             | 1 | - | - | - |
| Washer       | Aspiration Needle                        | 3 | - | - | - |
| Washer       | SIPAL                                    | 3 | - | - | - |
| Washer       | SIPAL (Plastic Only)                     | 1 | - | - | - |
| Washer       | Dispense Manifold w/ Valve               | 3 | - | - | - |
| Washer       | Washer Light Barrier                     | - | - | - | - |
| Washer       | Washer Valve                             | 3 | - | - | - |

Table 10–4: Requalification

**NOTE**

Table 10–4 lists all parts for which assays re-calibration is required upon conclusion of a service intervention (“Recal” column with “Y” value).

The FSE shall guarantee that at least the first kit re-calibration is successfully completed before the service call is closed.

Kit re-calibration is deemed not required when the object of the intervention is a part not listed in the “Recal” column. In such a case FSE may inform the end user that the kit recalibration is not needed. Anyway the end user has the right to decide if performing or not the kit recalibration.

**NOTE**

The results of the requalification activities must be attached to the Service records (for Subsidiaries only: SGAT WS), together with a detailed description of the Service activities performed on the analyzer.

**NOTE**

The kit recalibration upon Sample Pipettor replacement is mandatory only in case customer runs any Assay with external calibrator.

**NOTE**

3 repetitions mean “3 successful consecutive repetitions”. See Corp-TN-000292 (former LXS TN 009) for re-rerun rules.

---

## 10.5 Other activities during service visits

In order to check the system, at the end of each service visit, before performing the system requalification indicated in Chapter 10.4 on page 10-10, it is required to perform also the operations described in the following paragraphs.

### 10.5.1 Instrument recovery and test

After remounting of all dismantled assemblies, perform the following steps:

1. Close any open software and start the **Service Software** from the Windows Administrator account.
2. Initialize the system, see Chapter 12.8.1.1 on page 12-60.
3. Perform the Cuvette Endurance Run, see Chapter 12.8.1.2.1 on page 12-63.

### 10.5.2 User Manual last revision check

Check if the most updated version of the Instrument Manuals, corresponding to the Instrument SW version and customer's language for the specific country is available on the system.

If not, install the most updated version of the Instrument Manuals, see Chapter 6.4.4 on page 6-36. This can be done:

- using the **Download** functionality available in the **Online Help** section of the **Main UI**;
- using the **Acquire** functionality available in the **Online Help** section of the **Main UI** in case the appropriate instrument package was previously downloaded from **Dialog**.

## 10.6 System Shutdown and Wakeup for long term inactivity procedure

Aim of this section is to provide the instructions to shutdown the **LIAISON® XS** system in case of long period of inactivity (more than 2 weeks) and its subsequent restart.

Both activities could be managed by the user.

The following procedure can be applied on:

- new instruments (out of the crates) successfully installed that need to be switched off for a long period due to commercial aspect;
- operating instruments that need to be switched off for a long period due to user needs.

**NOTE**

Weekly and maintenance is not needed in case of long term instrument shutdown.

---



### 10.6.1 Procedure for system Shutdown and Wakeup

**NOTE**

Shutdown might be possible in case of long period of inactivity (more than 2 weeks). The required procedure for the shutdown and restart of the system from long period is described in the Chapter “Shutdown/End of Day Activities” of the User Manual and also in the following paragraph.

**NOTE**

Before starting the System Shutdown Maintenance task, 1 L of Cleaning Solution shall be prepared according to Chapter “Preparation of the Cleaning Solution in the cleaning tank” of the User Manual.

#### 10.6.1.1 System Shutdown Maintenance Task

1. Ensure that the instrument is not in “running” or “maintenance” status before starting the maintenance;
2. Tap on the **Maintenance** main category and select the **System Shutdown Maintenance** Task;
3. Tap on the “Perform” Button to start the System Shutdown Maintenance Task;
4. Follow the instructions described in the Maintenance Task.

#### 10.6.1.2 System Wakeup Maintenance Task

1. Switch on the instrument according to User Manual chapter 5.3;
2. Ensure that the instrument is not in “running” or “maintenance” status before starting the maintenance;
3. Tap on the **Maintenance** main category and select the **System Wakeup Maintenance** Task;
4. Tap on the “Perform” Button to start the System Wakeup Maintenance Task;
5. Follow the instructions described in the Maintenance Task.

## 10. Maintenance Schedule for external-tanks configuration

---

# 11 Service Software for internal-tanks configuration

This chapter describes basic functions of the **Service Software**.

## 11.1 Use of the Service Software

### 11.1.1 General Statement

This chapter contains descriptive information of the windows, fields and buttons of **Service Software**. For related calibration procedures, see the calibration sections of the individual modules.

The software allows to control necessary functions of the instrument and to perform adjustments of the function parameters.

The **Service Software** is used to perform a part of the final testing of the instruments during installation. In addition, the service engineer can use it as a tool to check in-field instruments and adjust them if required.

### 11.1.2 Before starting the Service Software

This chapter refers to **Service Software version 1.6.1906.28001** applied with **Main Software version 1.3.5**.

If it is necessary to switch from the **Main User Interface (Main UI)** or the **Teacher Software** to the **Service Software**, shut down the previously used software.

Then, log-in as Administrator in the Windows account before open the **Service Software**.

### 11.1.3 Use of the Service Software

---

**WARNING**

---

**Possible Instrument Damages**

It is easily possible to damage parts of the instrument with the **Service Software**.

- Make sure, that the movement course is free, before you start the movement of a module.
-

## 11. Service Software for internal-tanks configuration

---

**NOTE****Close other programs**

If the **Teacher Software** or the **Main UI** was running before, these programs have to be closed!

---

**NOTE****Log-in as Administrator**

Before open the **Service Software**, be sure to be logged-in as Administrator.

---

### 11.1.3.1 Start



1. To start the **Service Software**, click on the icon on the desktop twice or select the menu item in the Windows start menu.
2. Enter the password.
3. Click on **Login**.

**WARNING**

The FSE credentials, user name and password, must be kept confidential at all times and they should never be shared or exposed to others.

---

### 11.1.3.2 Interface Selection configuration (first start)

The first time you open the **Service Software**, there is the need to perform configuration of the **Interface Selection**.



1. Click on the **Application Configuration** button and select the **Interface Selection** menu entry, see Chapter 11.1.5 on page 11-8.
2. Click on the **LAN/USB Port** category tab.
3. Select the **WCF1** row, **Dispatcher** (setting only for **Service Software** at first run).
4. Click on the **Select** button.



## 11. Service Software for internal-tanks configuration

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <p><b>Navigation tree:</b></p> <p>The <b>Navigation tree</b> on the left-hand side of the <b>Service Software</b> window displays all modules with controller boards. You can work with it in the same manner as you work with the Windows Explorer. A module entry includes further categories with several functions. Click on the arrow in front of the module name or on the module name to display all available categories. Click on the arrow again to close the module entry.</p> <p>Click on a category to open it as <b>Category tab</b> on the right-hand side.</p> |
| 2  | <p><b>Category tab:</b></p> <p>To obtain a better overview the functions have been grouped into categories. These groups are opened by clicking on the corresponding category in the <b>Navigation tree</b> and displayed as a separate <b>Category tab</b> on the right-hand side. Each <b>Category tab</b> will display the appropriate features. By clicking on a <b>Category tab</b> you can switch between the different tabs. It is also possible to close unused <b>Category tabs</b>.</p>                                                                              |
| 3  | <p><b>Trace (log) area:</b></p> <p>The <b>Trace (log) area</b> displays the communication between the <b>Service Software</b> and the instrument modules. Errors will be displayed as red entries. A slider on the right side enables you to move the displayed text.</p>                                                                                                                                                                                                                                                                                                      |
| 4  | <p><b>Group:</b></p> <p>A <b>Group</b> combines functions which belong together.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5  | <p><b>Selection list:</b></p> <p>A <b>Selection list</b> displays all possible values/entries for this function. Select one of the displayed values/entries. The selected value/entry will be highlighted.</p>                                                                                                                                                                                                                                                                                                                                                                 |
| 6  | <p><b>Information box:</b></p> <p>An <b>Information box</b> displays information about a performed function (e.g. status of a sensor).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7  | <p><b>Button:</b> Non Volatile Memory</p> <p>Reload all Parameter's from Non-Volatile Memory to RAM.</p> <p>Overwrite ALL Parameter's in Non-Volatile Memory</p>                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8  | <p><b>Button:</b> File</p> <p>Read from Device and save.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 9  | <p><b>Check box:</b></p> <p>A <b>Check box</b> allows you to activate/deactivate an option of a function. Click on the <b>Check box</b> to activate (a checkmark is displayed) or deactivate the option.</p>                                                                                                                                                                                                                                                                                                                                                                   |
| 10 | <p><b>Number field:</b></p> <p>A <b>Number field</b> displays all possible number values for this function. Click on the small up/down arrows to increase/decrease the displayed number or click into the <b>Number field</b> and enter a number.</p> <p>Click on the big down arrow to get a calculator.</p>                                                                                                                                                                                                                                                                  |
| 11 | <p><b>Selection box:</b></p> <p>A <b>Selection box</b> displays the selected value/entry on the top. Click on the selection box to display all possible values/entries. Select one of the displayed values/entries.</p>                                                                                                                                                                                                                                                                                                                                                        |
| 12 | <p>Deactivated buttons, fields, selection boxes etc. are gray. It is not possible to use it in this situation. The software controls their use.</p>                                                                                                                                                                                                                                                                                                                                                                                                                            |

## 11. Service Software for internal-tanks configuration

|    |                                                                                                                                                                                                                                                                                                                                                                                            |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | <p><b>Input box:</b></p> <p><b>Input boxes</b> allow to enter values into it. Click on the field and enter the value with the (on screen) keyboard.</p>                                                                                                                                                                                                                                    |
| 14 | <p><b>Table:</b></p> <p><b>Tables</b> provide an overview of a large amount of data. Some tables allow you to change/enter values or to perform a function by clicking on a table cell/row.</p>                                                                                                                                                                                            |
| 15 | <p><b>Option box:</b></p> <p>An <b>Option box</b> to select an option of a function. Click on the <b>Option box</b> to select the option.</p> <p><b>Option boxes</b> are organized normally in groups. Only one <b>Option box</b> of these group can be selected.</p>                                                                                                                      |
| 16 | <p><b>Filter Messages and error levels:</b></p> <p>Choose with the check boxes the <b>Event Trace</b> windows display text:</p> <ul style="list-style-type: none"> <li>• Grey - Displays message events</li> <li>• Yellow - Displays warning events</li> <li>• Orange - Displays critical events</li> <li>• Red - Displays fatals events</li> <li>• White - Flash auto messages</li> </ul> |
| 17 | <p><b>Module box of the instrument layout:</b></p> <p>A <b>module box</b> can be selected to show a window on the right hand side with a closer look of the module. There you can read out the position coordinates of each single field position.</p>                                                                                                                                     |
|    | <p><b>Tool tip:</b></p> <p>Some buttons, fields, selection boxes etc. display a <b>Tool tip</b> with additional information about it. To display it move your mouse pointer to the button, field, selection box etc. and wait a short time.</p>                                                                                                                                            |

Table 11–1: Service Software control elements

## 11.1.4.2 General buttons

General buttons which are displayed on the top of **Service Software** and apply on dialogs or tabs:

| Button                                                                              | Description                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | The <b>Application Configuration</b> button opens the <b>Interface Selection</b> dialog.                                                                                                 |
|  | File / Folder Operation: <ul style="list-style-type: none"> <li>• Zip Logging Files</li> <li>• Open Logging Folder</li> <li>• Open Report Folder</li> <li>• Open Image Folder</li> </ul> |
|  | Power state of instrument.                                                                                                                                                               |

## 11. Service Software for internal-tanks configuration

| Button                                                                            | Description                                                                                                                                                         |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | When <b>Stop</b> button is pressed, the Device stops immediately without finishing the last action or command. The device will terminate in a not initialized mode. |
|  | Shows the Online Help.                                                                                                                                              |
|  | Shows the copyright list for used software modules.                                                                                                                 |
|  | The <b>Exit</b> button quits the <b>Service Software</b> .                                                                                                          |
|  | Repeats the last used operation (e.g. updating the last query result of sensors, repeating the last command).                                                       |
|  | Repeats the last used operation automatically (delay time 2 s).                                                                                                     |

Table 11-2: General Buttons

11.1.4.3 **Device is Busy** window

The **Device is Busy** window is displayed if a **Service Software** operation starts an instrument operation (e.g. movement of the Pipettor) and the instrument needs time to perform it.

| Button                                                                              | Description                                                                                                                     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Cancel</b>                                                                       | The Device stops one finished the last action or command. The Device will terminate in a ready (initialized) condition.         |
|  | The Device stops immediately without finishing the last action or command. The device will terminate in a not initialized mode. |

Table 11-3: Functions of the **Device is Busy** window11.1.4.4 **Keyboard Control**

It is also possible to teach and calibrate the modules with a keyboard. The keyboard must be connected and the **Keyboard Control** check box in the module driver must be activated.

| Key                 | Description         |
|---------------------|---------------------|
| -                   | Decrease step size  |
| +                   | Increase step size  |
| <b>Cursor down</b>  | Y to front          |
| <b>Cursor left</b>  | X left or close lid |
| <b>Cursor right</b> | X right or open lid |



---

11. Service Software for internal-tanks configuration

---

| Key       | Description |
|-----------|-------------|
| Cursor up | Y to rear   |
| Page Up   | Z up        |
| Page down | Z down      |

Table 11–4: Functions of **Keys**

## 11. Service Software for internal-tanks configuration

## 11.1.5 Interface Selection



Click on **Application Configuration** button to open the **Interface Selection** window.

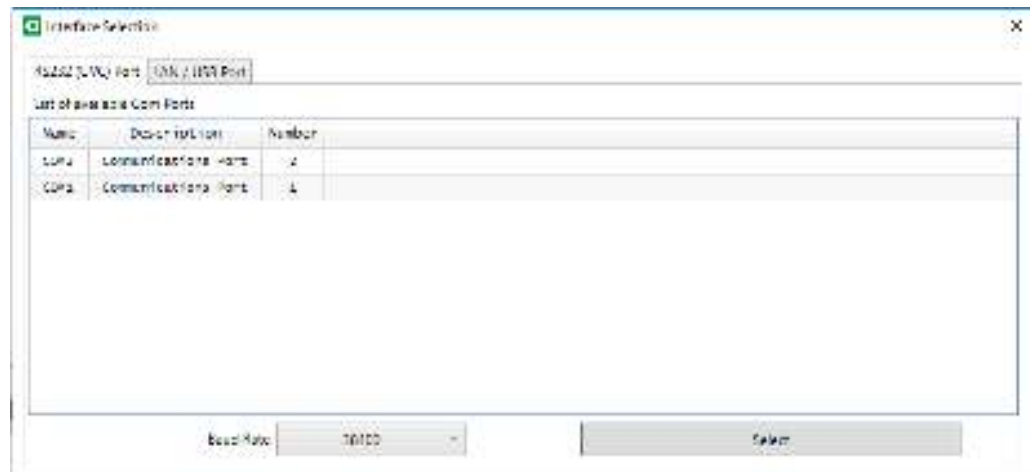


Figure 11–3: Interface Selection

| Tab or field                                                                        | Description                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RS232 (UVC) Port</b>                                                             | Tab button for RS232 (UVC) Port.                                                                                                                                                                 |
| <b>LAN / USB port</b>                                                               | Tab button for LAN / USB port.                                                                                                                                                                   |
| <b>List of available</b>                                                            | Com Ports: <ul style="list-style-type: none"> <li>• COM2</li> <li>• COM1</li> </ul> CopProt Stack Devices: <ul style="list-style-type: none"> <li>• USB1</li> <li>• WCF1 (Dispatcher)</li> </ul> |
| <b>Baud Rate</b>                                                                    | Choose between 9600, 38400, 115200.                                                                                                                                                              |
|  | Update the CopProt window.                                                                                                                                                                       |
| <b>Select</b>                                                                       | Choose the selected protocol.                                                                                                                                                                    |

Table 11–5: Description for **Interface Selection** window

## 11.1.6 Errors

Messages and errors will be displayed in the Event trace window as:

- Grey - Displays fatals events
- Yellow - Displays warning events

- Orange - Displays critical events
- Red - Displays fatal events.
- White - Flash auto messages.

### 11.1.7 General description of Calibration

#### NOTE

The following description is only a theoretical description. See the specific module for all the calibration procedures.

#### 11.1.7.1 Teach & Calibration

The **Teach & Calibration** window of a specific module in the **Navigation tree** displays a list with all possible teachable areas of the module.

##### 11.1.7.1.1 Buttons for Rotor Init Offset

The following table displays a list with all the **Buttons** used to calibrate the rotor (i.e. Incubator rotor, Washer rotor and Reader rotor).

| Column and function  |                                                                                     | Button name                 | Description                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Relative Move</b> |  | <b>Arrow buttons</b>        | Rotates the rotor one step or more steps clockwise. The number of steps are displayed in the field below the buttons.         |
|                      |  |                             | Rotates the rotor one step or more steps counter clockwise. The number of steps are displayed in the field below the buttons. |
|                      |  | <b>Get current position</b> | Gets the current position and display it in the field.                                                                        |
| <b>Move to</b>       |  | <b>Move to</b>              | Moves the rotor to the entered position.                                                                                      |
| <b>Stored</b>        |  | <b>Stored position</b>      | Copies the stored teach position (displayed in the field) into the <b>Move to</b> field.                                      |
| <b>Option</b>        |  | <b>Option</b>               | Displays optional functions for the sub module (e.g. initialize, unpower).                                                    |
| <b>Key-board</b>     |  | <b>Keyboard</b>             | Activates the keyboard control for this axis.                                                                                 |
| <b>Save</b>          |                                                                                     | <b>Save</b>                 | Saves the current teach position into the module RAM.<br>Note that the RAM will be lost after power off the instrument.       |

## 11. Service Software for internal-tanks configuration

| Column and function |  | Button name                        | Description                                                                                     |
|---------------------|--|------------------------------------|-------------------------------------------------------------------------------------------------|
| <i>Write to NVM</i> |  | <b>Write ALL Parameters to NVM</b> | All teach positions will be written into the non-volatile memory (NVM) of the affected modules. |
| <i>Write to NVM</i> |  | <b>Affected Modules</b>            | Shows the affected modules.                                                                     |
| <i>Summarize</i>    |  |                                    | The summarize list shows the results of the calibration steps.                                  |

Table 11–6: Button for Rotor Init Offset

## 11.1.7.1.2 Buttons for Cuvette Gripper Open Position

The following table displays a list with all the **Buttons** used to calibrate the Open Position of the Gripper Module.

| Column and function  |                                                                                     | Button name                        | Description                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <i>Relative Move</i> |  | <b>Arrow buttons</b>               | Moves the Gripper claws to close one step or more steps. The number of steps are displayed in the field below the buttons. |
|                      |  |                                    | Moves the Gripper claws to open one step or more steps. The number of steps are displayed in the field below the buttons.  |
|                      |  | <b>Get current position</b>        | Get the current position and display it in the field.                                                                      |
| <i>Move to</i>       |  | <b>Move to</b>                     | Moves the Gripper claws to the entered position.                                                                           |
| <i>Stored</i>        |  | <b>Stored position</b>             | Copies the stored teach position (displayed in the field) into the <i>Move to</i> field.                                   |
| <i>Option</i>        |  | <b>Option</b>                      | Displays optional functions for the sub module (e.g. initialize, unpower).                                                 |
| <i>Key-board</i>     |  | <b>Keyboard</b>                    | Activates the keyboard control for this axis.                                                                              |
| <i>Save</i>          |                                                                                     | <b>Save</b>                        | Saves the current teach position into the module RAM.<br>Note that the RAM will be lost after power off the instrument.    |
| <i>Write to NVM</i>  |                                                                                     | <b>Write ALL Parameters to NVM</b> | All teach positions will be written into the non-volatile memory (NVM) of the affected modules.                            |

| Column and function          |  | Button name                      | Description                                                    |
|------------------------------|--|----------------------------------|----------------------------------------------------------------|
| <a href="#">Write to NVM</a> |  | <a href="#">Affected Modules</a> | Shows the affected modules.                                    |
| <a href="#">Summarize</a>    |  |                                  | The summarize list shows the results of the calibration steps. |

Table 11–7: Button for Cuvette Gripper Open Position

### 11.1.8 General description of Parameter

The [Parameter](#) window of a specific module in the [Navigation tree](#) displays a list with all parameters of the module.

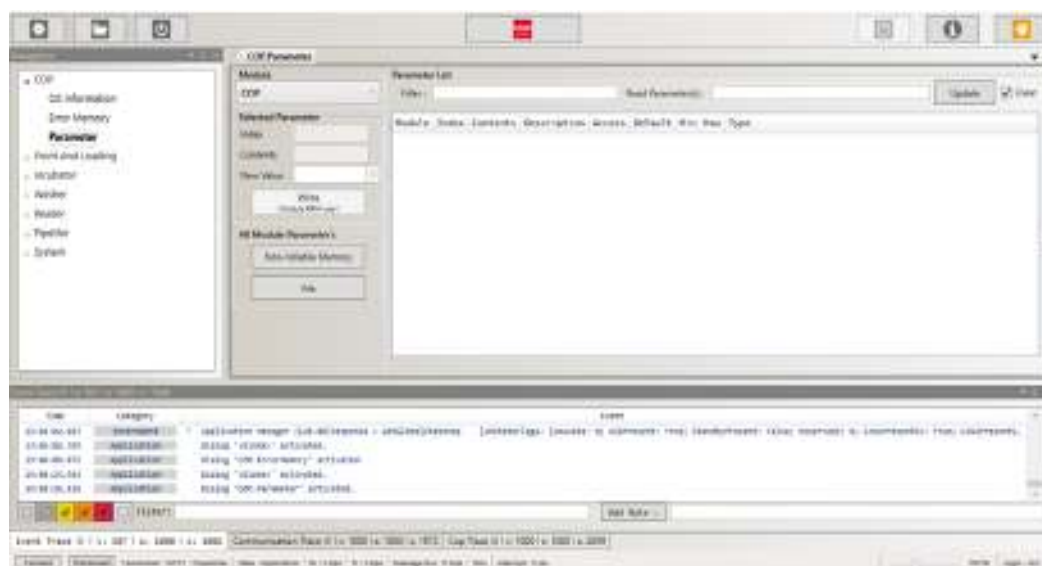


Figure 11–4: Example of COP - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

#### 11.1.8.1 [Module](#) group

| Function/<br>Button         | Description                          |
|-----------------------------|--------------------------------------|
| <a href="#">Module list</a> | List of all modules with parameters. |

Table 11–8: Functions of the **Module** group11.1.8.2 **Selected Parameter** group

| Function/<br>Button | Description                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Index</b>        | Shows the index number of the parameter.                                                                                                                 |
| <b>Contents</b>     | Displays the selected content/value of the parameter.                                                                                                    |
| <b>New Value</b>    | Enter the new value of the selected parameter.                                                                                                           |
| <b>Write</b>        | Writes the new parameter value into the <b>Service Software</b> module memory (RAM).<br>When the instrument is switched off, the parameter will be lost. |

Table 11–9: Functions of the **Selected Parameter** group11.1.8.3 **All Module Parameter's** group

| Function/<br>Button        | Description                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Non-Volatile Memory</b> | Use of current parameters of a instrument module. <ul style="list-style-type: none"> <li><b>Reload</b>: Reloads the parameters from the Non-Volatile Memory (NVM) of the module into the module RAM.</li> <li><b>Overwrite</b>: All parameters from the module RAM will be written into the Non-Volatile Memory (NVM) of the module.</li> </ul> |
| <b>File</b>                | Use of a parameter file of a instrument module. <ul style="list-style-type: none"> <li><b>Read from Device and Save</b>: loads a saved parameter file and send it to an instrument module.</li> </ul>                                                                                                                                           |

Table 11–10: Functions of the **All Module Parameter's** group11.1.8.4 **Parameter List** group

| Function/<br>Button      | Description                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Filter</b>            | Filters the parameters table by entered text.                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Read Parameter(s)</b> | Allows to show all parameters or only selected parameters. <ul style="list-style-type: none"> <li>Show all parameters: Let the field empty.</li> <li>Show only selected parameters: Examples: <ul style="list-style-type: none"> <li>5,6,20: shows only parameters 5, 6, and 20</li> <li>2-10: shows parameters 2 to 10</li> <li>5,9-12,15,20,30-60: shows parameters 5, 9 to 12, 15, 20, and 30 to 60</li> </ul> </li> </ul> |

| Function/<br>Button | Description                                                         |
|---------------------|---------------------------------------------------------------------|
| <b>Update</b>       | Reads the parameters of the module and updates the parameter table. |
| <b>Clear</b>        | Deletes all entries before updating the table.                      |

Table 11–11: Functions of the **Parameter List** group

| Table Column       | Description                                        |
|--------------------|----------------------------------------------------|
| <b>Module</b>      | Module name.                                       |
| <b>Index</b>       | Parameter number.                                  |
| <b>Contents</b>    | Parameter value.                                   |
| <b>Description</b> | Parameter name.                                    |
| <b>Access</b>      | Access level (Normal, Teach, ReadOnly).            |
| <b>Default</b>     | Default value of the parameter.                    |
| <b>Min</b>         | Allowed minimum value.                             |
| <b>Max</b>         | Allowed maximum value.                             |
| <b>Type</b>        | Parameter type (e.g. Int16: only positive values). |

Table 11–12: Columns of the table in the **Parameter List** group

## 11. Service Software for internal-tanks configuration

### 11.2 COP Module

The COP (Communication and Organization Processor) is built into the instrument and controls all the other modules. The COP in the **Service Software** inquires information about the internal status (Worklist, COP drivers, CAN-bus system) of the COP and the PC communication.

#### 11.2.1 Information

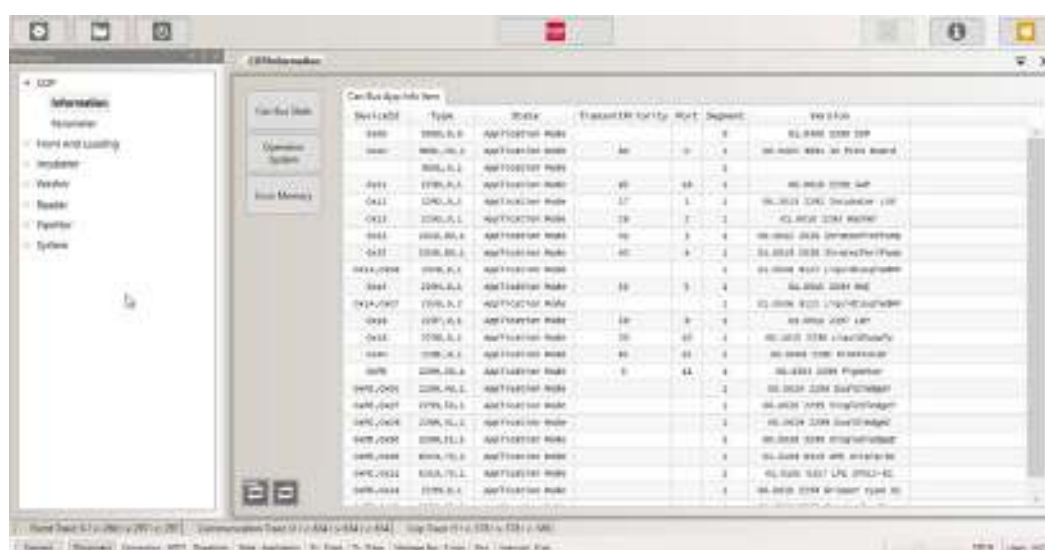


Figure 11–5: COP - Information

| Function/<br>Button     | Description                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Can Bus State</b>    | Displays all CAN modules with device address, CAN address and if they registered or not.                                            |
| <b>Operation System</b> | Displays all operating system related CAN information (e.g. the number of accepted CAN messages for dispatching to the assemblies). |
| <b>Error Memory</b>     | Displays Error Messages.                                                                                                            |

Table 11–13: Functions of COP - Information



## 11. Service Software for internal-tanks configuration



Figure 11–6: COP - Operation system

| Function/<br>Button        | Description                                                                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Can Bus Statistics</b>  | Displays all CAN modules with device address, CAN address and if they registered or not.                                                                                      |
| <b>Driver State</b>        | Displays COP internals (e.g. the number of hosted processes, the task calls per second and the state of work list processor).                                                 |
| <b>Messaging State</b>     | Displays IPC internals (e.g. the number of messages from one process sent to another).                                                                                        |
| <b>Process Information</b> | Displays all COP drivers, applications and some of their internal information (e.g. device address, process priority, number of task calls and the assigned Worklist queues). |
| <b>System Monitor</b>      | Gives information about CAN and communication voltages.                                                                                                                       |
| <b>Worklist State</b>      | Displays the work list..                                                                                                                                                      |

Table 11–14: Functions of COP - Operation System

## 11. Service Software for internal-tanks configuration

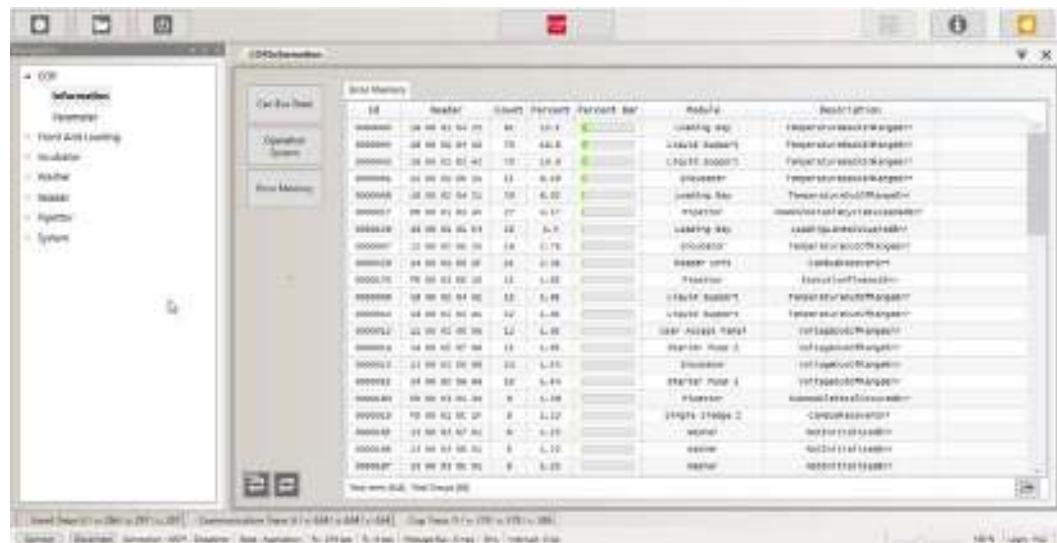


Figure 11-7: COP - Error Memory

| Function/<br>Button | Description                         |
|---------------------|-------------------------------------|
| <b>ID</b>           | Identification number of the error. |
| <b>Header</b>       | Hex numbers of the error.           |
| <b>Count</b>        | Error occurrence.                   |
| <b>Percent</b>      | Error rate in percent.              |
| <b>Percent bar</b>  | Displays the error rate in a bar.   |
| <b>Module</b>       | Shows the module name.              |
| <b>Description</b>  | Displays event name.                |

Table 11-15: Functions of COP - Error Memory

### 11.2.2 Parameter



Figure 11–8: COP - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 11-11.

In the **Module** group, it is possible to select COP module.

## 11.3 Front and Loading

### 11.3.1 General

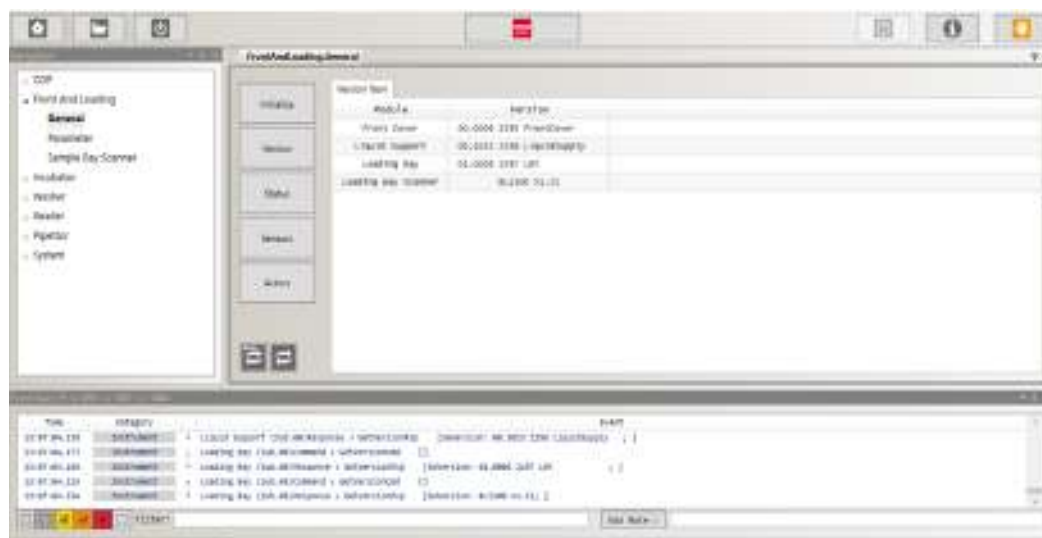


Figure 11–9: Front and Loading - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                                                                                             |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                                                                                                 |
| <b>Status</b>       | Displays information about the state of the module in the table: <ul style="list-style-type: none"> <li>• Front Cover</li> <li>• Loading Bay</li> <li>• Liquid Support</li> </ul> Select the desired part in the button menu.                                                  |
| <b>Sensors</b>      | Displays information about the state of the module sensors in the table: <ul style="list-style-type: none"> <li>• Front Cover</li> <li>• Loading Bay All</li> <li>• Loading Bay Temp. only</li> <li>• Liquid Support</li> </ul> Select the desired sensors in the button menu. |

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>       | <p>Activates, deactivates etc. the actors of the module:</p> <ul style="list-style-type: none"> <li>• Cooling <ul style="list-style-type: none"> <li>○ On</li> <li>○ Off</li> </ul> </li> <li>• Stirrer <ul style="list-style-type: none"> <li>○ On</li> <li>○ Off</li> <li>○ Auto</li> </ul> </li> <li>• Buzzer <ul style="list-style-type: none"> <li>○ On</li> <li>○ Off</li> <li>○ Parametrized On, which opens the Buzzer On window, where there is the possibility to choose the Buffer sound type and repetition to perform</li> </ul> </li> <li>• WAP <ul style="list-style-type: none"> <li>○ Start at Routine Speed</li> <li>○ Start at Prime Speed</li> <li>○ Start Parameterized, which opens the Start WAP Timed window, where there is the possibility to choose the WAP speed and time to run</li> <li>○ Stop</li> </ul> </li> <li>• Status Light Yellow or Blue <ul style="list-style-type: none"> <li>○ On / Off</li> <li>○ Flashing slow / fast</li> </ul> </li> </ul> <p>Select the desired actor in the button menu.</p> |

Table 11–16: Functions of Front and Loading - General

### 11.3.2 Parameter

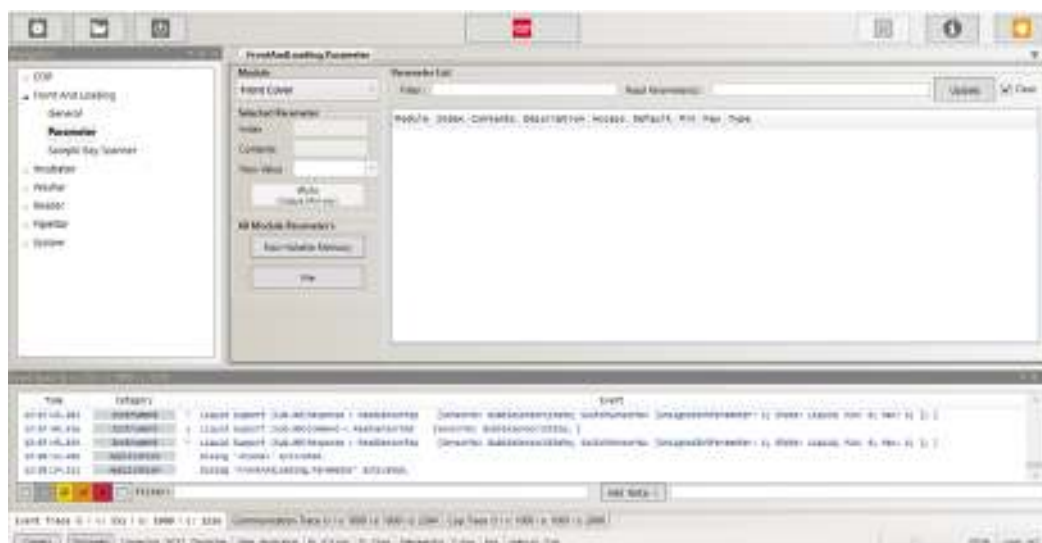


Figure 11–10: Front and Loading - Parameter

### NOTE

## Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 11-11.

In the **Module** group, it is possible to select:

- Front Cover
- Loading Bay
- Liquid Support
- Wash Aspiration Pump
- User Access Panel

### 11.3.3 Sample Bay Scanner

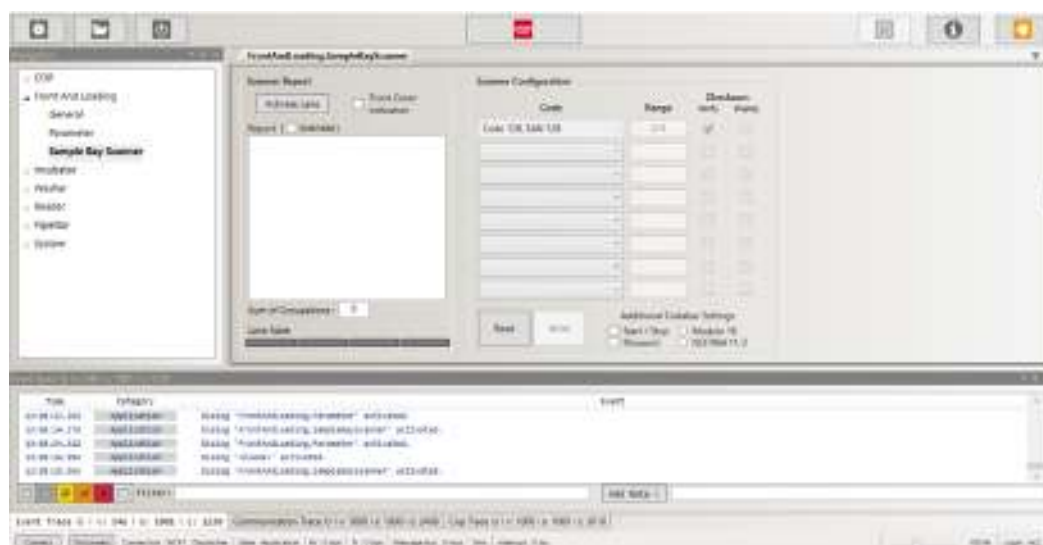


Figure 11–11: Front and Loading - Sample Bay Scanner

#### 11.3.3.1 Scanner Report group

| Function/Button               | Description                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Activate Lane</b>          | Activates the selected sample lane to read the barcodes of an inserted rack.                                     |
| <b>Front Cover indication</b> | Check box to activate / deactivate front module LED indication of activated rack lane.                           |
| <b>Report</b>                 | Shows the read barcodes.<br>If you activate the <b>Extended</b> check box, the report shows further information. |
| <b>Sum of Occupations</b>     | Displays sum of detected racks                                                                                   |
| <b>Lane State</b>             | Shows the state of the lane.                                                                                     |

Table 11–17: Functions of the **Scanner Report** group

## 11. Service Software for internal-tanks configuration

11.3.3.2 *Scanner Configuration* group

| Function/<br>Button                | Description                                                                                                                                                                                                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Code</b>                        | Select the barcode formats from the selection box that the customer will use: <ul style="list-style-type: none"> <li>• 2/5 Interleaved</li> <li>• Code 39</li> <li>• Code 93</li> <li>• Code 128, EAN 128</li> <li>• EAN 8 / 13</li> <li>• Codabar</li> <li>• UPCA, UPCE</li> <li>• Not Supported Code</li> </ul> |
| <b>Range</b>                       | Use this field to enter the bar code character length range value(s) for the selected bar code type. The acceptable bar code range is from 3 to 25 characters.                                                                                                                                                    |
| <b>Checksum</b>                    | If checksum is used with the code type selected, select either <b>Verify</b> , <b>Display</b> , or both.                                                                                                                                                                                                          |
| <b>Read</b>                        | Select this button to display the code types information stored on the EEPROM of the barcode scanner.                                                                                                                                                                                                             |
| <b>Write</b>                       | Select this button to write the code types information onto the EEPROM of the bar ode scanner.                                                                                                                                                                                                                    |
| <b>Additional Codabar Settings</b> | Select this to read monarch codabar symbologies as well as other types of codabar bar code symbologies. De-select this to better read nonmonarch codabar symbologies of poor quality.                                                                                                                             |

Table 11–18: Functions of the *Scanner Configuration* group

## 11.3.3.3 Barcode scanner configuration

In the **Service Software** at the navigation window, choose **Front and Loading** and the **Sample Bay Scanner** bracket.

- Click onto the **Read** button to read out the actual configuration.
  - Code 39 Range: 4 - 22
  - Code 128, EAN 128 Range: 4 - 22
  - Codabar Range: 4 - 22
- Clear all the check boxes that are not grayed out at the Checksum (**Verify** & **Display**).
- Click onto the **Write** button to save the settings.

## 11.3.3.4 Barcode reading test

In the **Service Software** at the navigation window choose **Front and Loading** and the **Sample Bay Scanner** bracket.

- Click onto the **Activate Lane - Activate** button to start the scanner.
- Insert a sample rack and check the correct reading of barcodes.



## 11. Service Software for internal-tanks configuration

---

- c. Test each Sample Bay lane. For each lane, the barcode scanner has to be reactivated.

At the beginning, the background of the Lane State is gray.

When a rack is moved to the end stop, the status of the lane switches to green.

When a rack is removed, the status of the lane switches from green to red.

## 11.4 Incubator

### 11.4.1 General

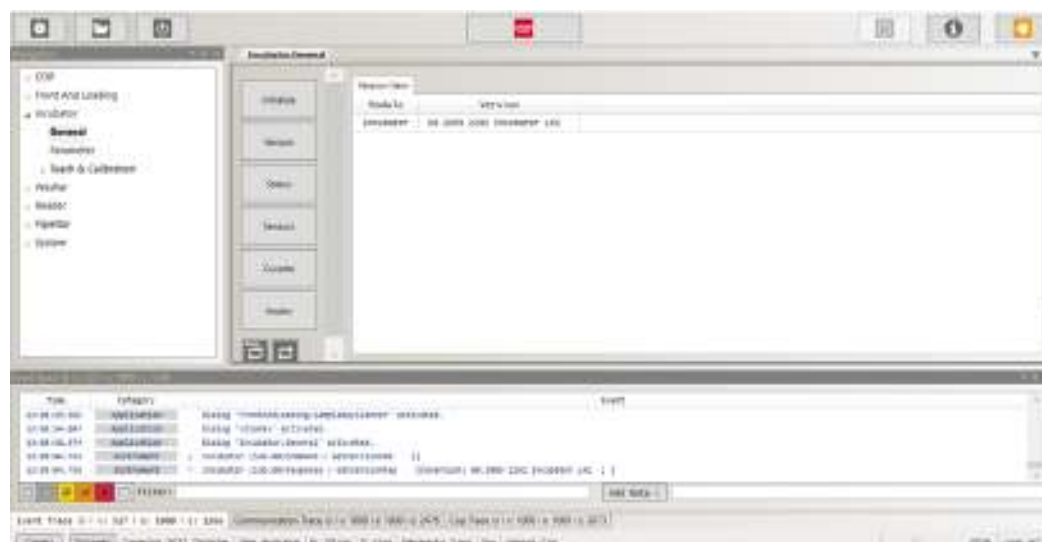


Figure 11–12: Incubator - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Status</b>       | Displays information about the state in the table.<br>Select the desired part in the button menu.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Sensors</b>      | Displays information about the state of the incubator sensors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Cuvette</b>      | <p>Move:</p> <ul style="list-style-type: none"> <li>Cuvette 1..25 (Inner), moves the selected cuvettes to the chosen positions</li> <li>Cuvette 26..57 (outer), moves the selected cuvettes to the chosen positions</li> <li>Parameterized, opens the window where there is the possibility to choose the cuvette and in which position moving it</li> <li>Fake move On</li> <li>Fake move Off</li> </ul> <p>Check:</p> <ul style="list-style-type: none"> <li>All cuvettes</li> <li>Cuvette 1..25 (Inner), checks the chosen cuvettes.</li> <li>Cuvette 26..57 (Outer), checks the chosen cuvettes.</li> <li>Parameterized, opens the window Check cuvette, where there is the possibility to choose the cuvette position to check</li> </ul> |

| Function/<br>Button | Description                            |
|---------------------|----------------------------------------|
| <b>Heater</b>       | Switch the Incubator Heater On or Off. |

Table 11–19: Functions of Incubator - General

### 11.4.2 Parameter

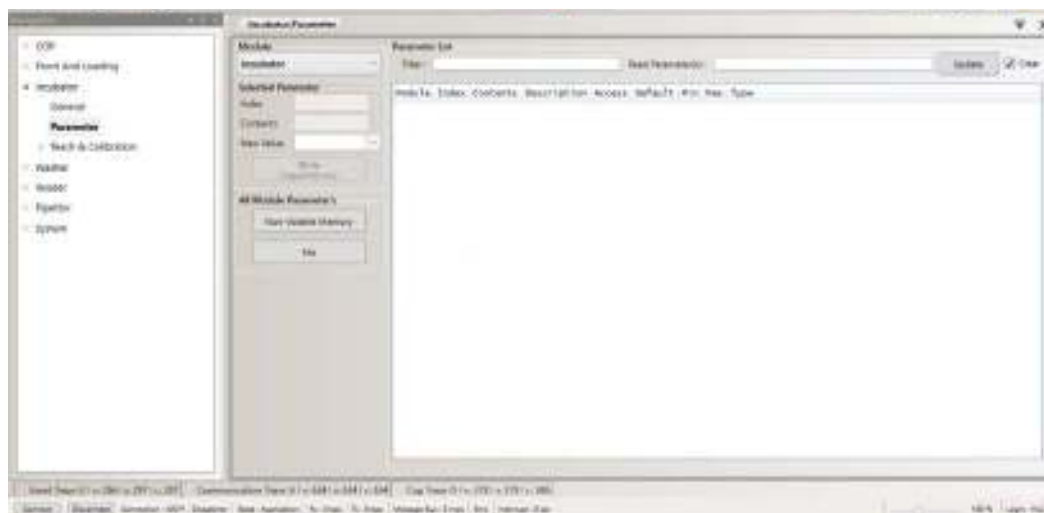


Figure 11–13: Incubator - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 11-11.

In the **Module** group, it is possible to select:

- Incubator
- COP

## 11. Service Software for internal-tanks configuration

### 11.4.3 Teach & Calibration

#### 11.4.3.1 Incubator Rotor Calibration



Figure 11–14: Incubator - Teach & Calibration - Ring

The Incubator Rotor Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

About the function of all the buttons, see Chapter 11.1.7.1.1 on page 11-9.

In the **Service Software** at the navigation window, choose **Incubator** then **Teach & Calibration** bracket and select **Ring**.

#### Step 1

- a. Follow the instructions and then press the **Done** button.

#### Step 2

- b. Move the Incubator rotor by the arrow buttons to align the slot on the rotor to the reference position of the Incubator frame.



Figure 11–15: Incubator rotor, calibration position

- c. Press **Save** button to save the position.

#### Step 3

- d. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

#### Step 4

- e. The process is successful when, at the **Summarize** panel, all steps are in green.

### CAUTION

**A mis-calibration of the Incubator rotor might be necessary in order to have a proper cuvette transfer from the Incubator to the Washer and vice-versa.**

Perform a Cuvette Endurance Run (see Chapter 11.8.1.2.1 on page 11-61) and check that the cuvette transfer is smooth and without noises.

If needed, change the Incubator rotor calibration such to compensate the observed mis-alignment between the Gripper and Incubator rotor; an adjustment of 3-4 steps clockwise could be sufficient.

## 11.4.3.2 Incubator Cuvette Sensors Calibration



Figure 11–16: Incubator - Teach &amp; Calibration - Cuvette Sensors

The Incubator Cuvette Sensors Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window, choose **Incubator** then **Teach & Calibration** bracket and select **Cuvette Sensors**.

- Step 1** a. Follow the instruction: remove the cuvettes from the gripper loading position (i.e. at positions 1 and 26), if any, and dispose of them; then press the **Done** button.
- Step 2** b. Follow the instruction by pressing **Done** button. The Incubator will move Cuvette 1 & 26 positions to the sensor.
- Step 3** c. Follow the instruction by pressing **Start** button to start the calibration procedure.  
d. Check the results and proceed by pressing **Done** button.
- Step 4** e. Store the taught positions in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 5** f. Follow the instruction: insert new cuvettes at the gripper loading position (i.e. at positions 1 and 26), if were present, and press the **Done** button.
- Step 6** g. The process is successful when, at the **Summarize** panel, all steps are in green.

## 11.5 Washer

### 11.5.1 General

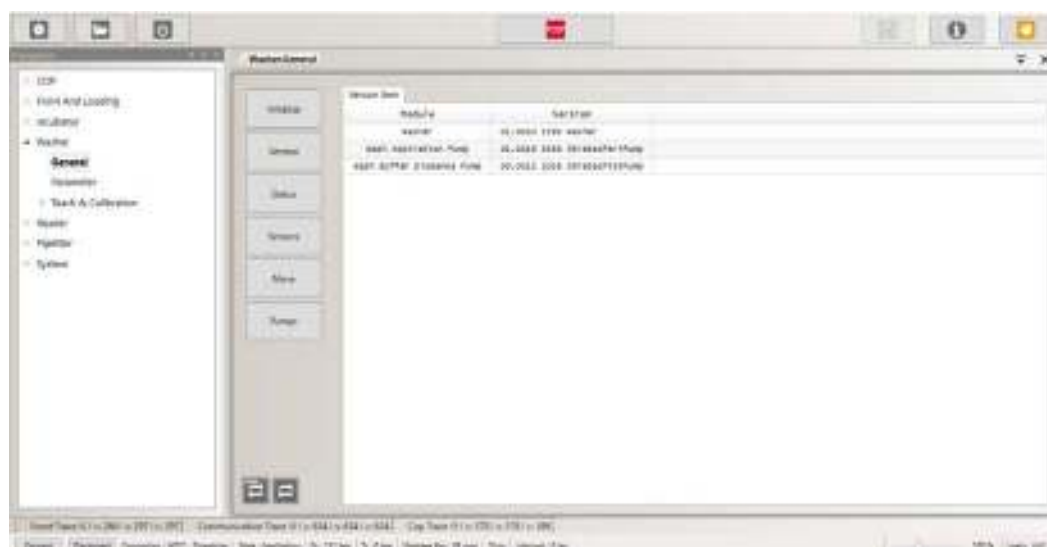


Figure 11–17: Washer - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                          |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                              |
| <b>Status</b>       | Displays information about the state in the table. Select the desired part in the button menu.                                                                                                              |
| <b>Sensors</b>      | Displays information about the state of the chosen module sensors: <ul style="list-style-type: none"> <li>• Washer</li> <li>• WAP</li> <li>• WBDP</li> </ul> Select the desired sensors in the button menu. |

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Move</b>         | <p>Ring moves the Washer rotor of:</p> <ul style="list-style-type: none"> <li>• 1 Step</li> <li>• 3 Steps</li> <li>• Parameterized opens the window Move Washer Ring where there is the possibility to choose of how many steps move the Washer rotor</li> <li>• Random Move</li> </ul> <p>Lift 1, 2 or 3 moves the Washer selected aspiration needle to the following positions:</p> <ul style="list-style-type: none"> <li>• to Home</li> <li>• to Bottom</li> <li>• Parameterized opens the window Move Lift 1, 2 or 3 where there is the possibility to choose at which position move the Lift and at which speed</li> </ul> <p>Lift Cycles opens the window Lifts Cycle Move where there is the possibility to parametrize all the movement of the Lift together or not. Set 0 cycles to perform a never-ending cycles movement.</p>                                       |
| <b>Pumps</b>        | <p>WAP:</p> <ul style="list-style-type: none"> <li>• Start at Routine Speed</li> <li>• Start at Prime Speed</li> <li>• Start Parameterized opens the Start WAP Timed window where there is the possibility to choose the WAP speed and time to run</li> <li>• Stop</li> </ul> <p>WBDP:</p> <ul style="list-style-type: none"> <li>• Load: <ul style="list-style-type: none"> <li>○ 1900 µl</li> <li>○ 2100 µl</li> <li>○ Parameterized opens the window Load WBDP where there is the possibility to choose the volume to load in uL</li> </ul> </li> <li>• Dispense Port 1, 2 or 3: <ul style="list-style-type: none"> <li>○ 700 µl</li> <li>○ 1200 µl</li> <li>○ Parameterized opens the window Dispense WBDP Port 1, 2 or 3 where there is the possibility to choose the volume to dispense in uL and if activate or not the tracking</li> </ul> </li> <li>• Empty</li> </ul> |

Table 11–20: Functions of Washer - General

### 11.5.2 Parameter

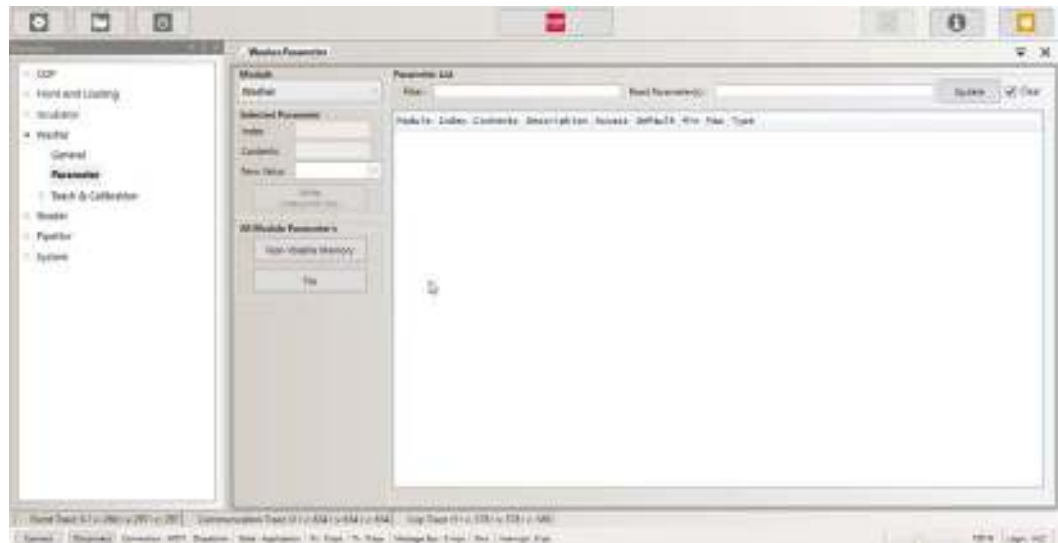


Figure 11–18: Washer - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 11-11.

In the **Module** group, it is possible to select:

- Washer
- Wash Aspiration Pump
- Wash Buffer Dispense Pump
- COP



### 11.5.3 Teach & Calibration

#### 11.5.3.1 Washer Rotor Calibration



Figure 11–19: Washer - Teach & Calibration - Ring

The Washer Rotor Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

About the function of all the buttons, see Chapter 11.1.7.1.1 on page 11-9.

In the **Service Software** at the navigation window, choose **Washer** then the **Teach & Calibration** bracket and select **Ring**.

**Step 1**

- Follow the instructions and then press the **Done** button.

**Step 2**

- Move the Washer rotor by the arrow buttons, to align the slot on the rotor to the reference position on the Washer base.



Figure 11–20: Washer rotor, calibration position

- Press **Save** button to save the position.

**Step 3**

- Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

**Step 4**

- The process is successful when, at the **Summarize** panel, all steps are in green.

## 11. Service Software for internal-tanks configuration

### 11.5.3.2 Washer SIPAL Lifts Calibration

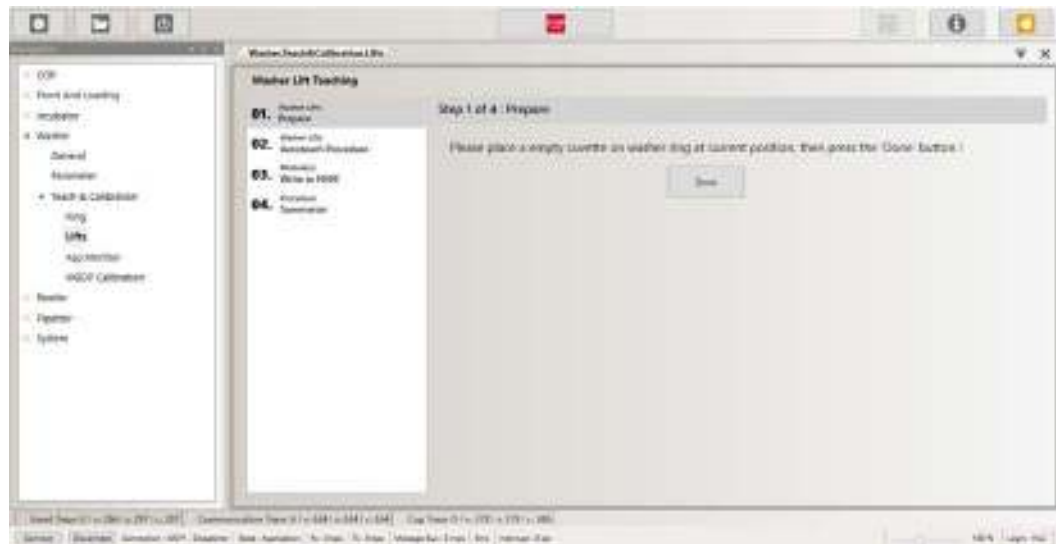


Figure 11–21: Washer - Teach & Calibration - Lifts

The Washer SIPAL Lift Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software**, at the navigation window, choose Washer then the **Teach & Calibration** bracket and select **Lifts**..

- Step 1**
  - a. Follow the instructions and then press the **Done** button.
- Step 2**
  - b. Press the **Start** button; the calibration procedure starts automatically.
  - c. When the **Processing** window closes without failure message, check the results and press the **Done** button.
- Step 3**
  - d. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 4**
  - e. The process is successful when, at the **Summarize** panel, all steps are in green.

## 11.5.3.3 Washer Aspiration Monitoring Sensors Calibration

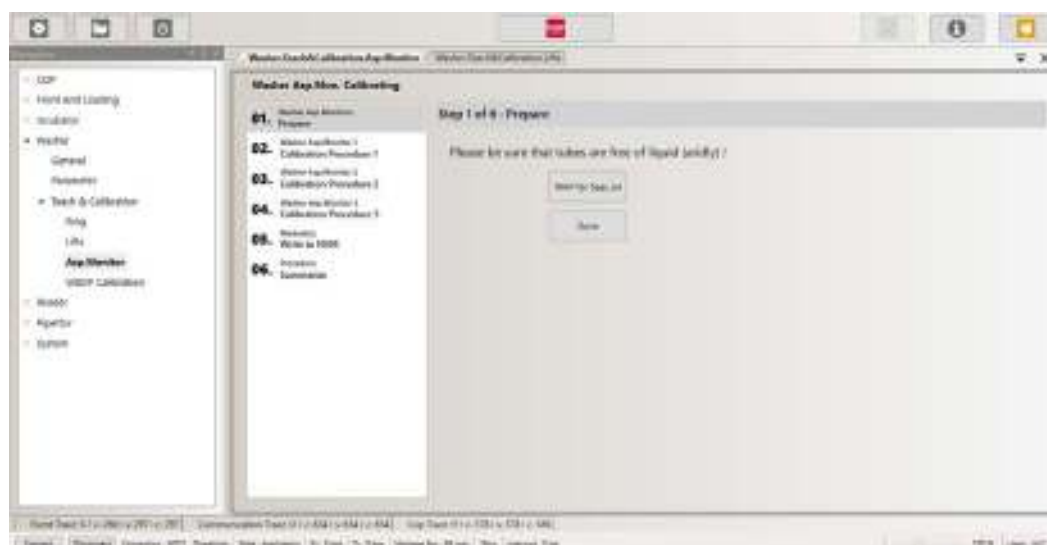


Figure 11–22: Washer - Teach &amp; Calibration - Asp. monitor

The Washer Aspiration Monitoring Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software**, at the navigation window, choose **Washer** then the **Teach & Calibration** bracket and select the **Washer Asp. Mon.** bracket.:

- Step 1**
  - a. Press the **WAP for 5sec** on button, to ensure that tubes are free of liquid.
  - b. When the Processing window closes, press the **Done** button.
- Step 2**
  - c. Press the **Start** button; the calibration procedure for aspiration monitor 1 starts automatically.
  - d. When the **Processing** window closes without failure message, check the result and press the **Done** button.
- Step 3**
  - e. Press the **Start** button; the calibration procedure for aspiration monitor 2 starts automatically.
  - f. When the **Processing** window closes without failure message, check the result and press the **Done** button.
- Step 4**
  - g. Press the **Start** button; the calibration procedure for aspiration monitor 3 starts automatically.
  - h. When the **Processing** window closes without failure message, check the result and press the **Done** button.
- Step 5**
  - i. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 6**
  - j. The process is successful when, at the **Summarize** panel, all steps are in green.

## 11. Service Software for internal-tanks configuration

### 11.5.3.4 WBDP Calibration

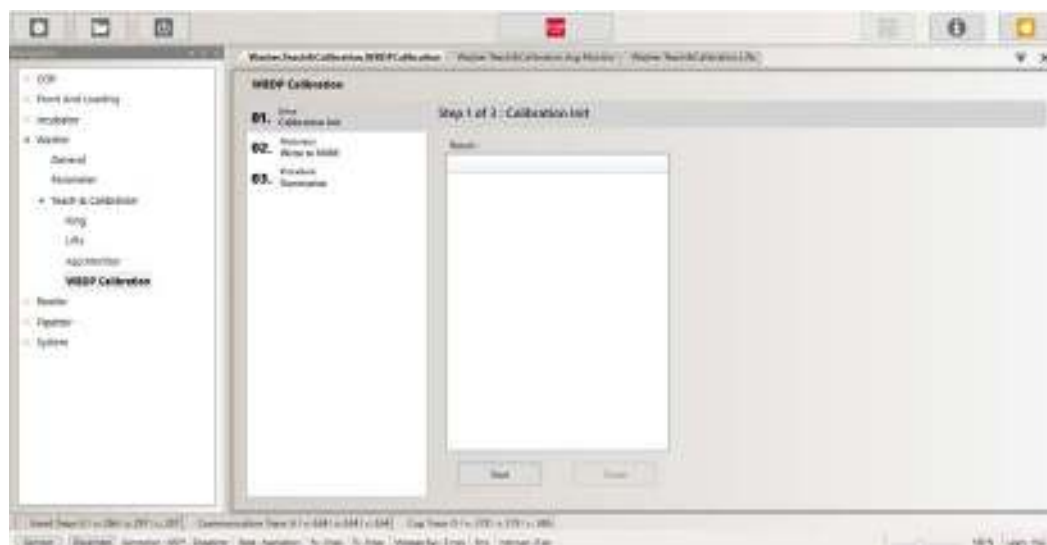


Figure 11–23: Washer - Teach & Calibration - WBDP Calibration

The WBDP Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window choose **Washer** and the **Teach & Calibration** bracket and select **WBDP Calibrating** bracket.

#### Step 1

- Press the **Start** button; the calibration procedure for WBDP starts automatically.
- When the **Processing** window closes without failure message, press the **Done** button.

#### Step 2

- Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.

#### Step 3

- The process is successful when, at the **Summarize** panel, all steps are in green.

## 11.6 Reader

### 11.6.1 General

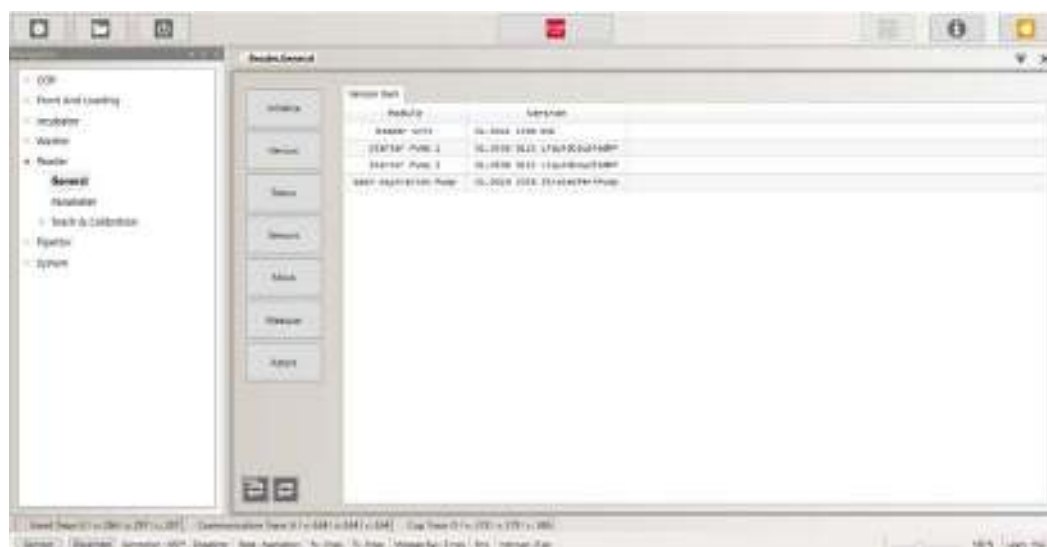


Figure 11-24: Reader - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                                                                                                                                                                                                                                                     |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Status</b>       | Displays information about the state in the table. Select the desired part in the button menu.                                                                                                                                                                                                                                                                                                                                         |
| <b>Sensors</b>      | Read displays information about the state of the module sensors in the table: <ul style="list-style-type: none"> <li>• Reader Unit</li> <li>• Starter 1 Pump</li> <li>• Starter 2 Pump</li> </ul> Check displays instruction to perform the check of the module sensors: <ul style="list-style-type: none"> <li>• Starter 1 Bottle Sensor</li> <li>• Starter 2 Bottle Sensor</li> </ul> Select the desired sensors in the button menu. |

## 11. Service Software for internal-tanks configuration

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Move</b>         | <p>Rotor to moves the Reader rotor to the following positions:</p> <ul style="list-style-type: none"> <li>• Load / Unload</li> <li>• Dark Count</li> <li>• Starter 1</li> <li>• Starter 2 / RLU</li> <li>• Aspirate</li> </ul> <p>Lift moves the Reader aspiration needle to the following positions:</p> <ul style="list-style-type: none"> <li>• to Home</li> <li>• to Bottom</li> <li>• Cycles opens the window Lifts Cycle, where there is the possibility to cycle the movement of the Lift</li> </ul>                                                                                                                                                                                                                                                                                                                                              |
| <b>Measure</b>      | <p>Choose the Dark Count reading length and in which position perform the reading:</p> <ul style="list-style-type: none"> <li>• At Dark Count Position <ul style="list-style-type: none"> <li>◦ 1 to 5 sec</li> </ul> </li> <li>• At Current Position <ul style="list-style-type: none"> <li>◦ 1 to 5 sec</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Actors</b>       | <p>Select the desired actor in the button menu.</p> <p>Tube Heating switches the heating of the Starters tubings:</p> <ul style="list-style-type: none"> <li>• Both On / Off</li> <li>• Starter 1 On / Off</li> <li>• Starter 2 On / Off</li> </ul> <p>High voltage:</p> <ul style="list-style-type: none"> <li>• Switch On or Off</li> </ul> <p>WAP:</p> <ul style="list-style-type: none"> <li>• Start at Routine Speed</li> <li>• Start at Prime Speed</li> <li>• Start Parameterized opens the Start WAP Timed window, where there is the possibility to choose the WAP speed and time to run</li> <li>• Stop</li> </ul> <p>Starter Dispense Pump 1 and 2 (SP1 and SP2):</p> <ul style="list-style-type: none"> <li>• Fill to 400 µl or Parameterized</li> <li>• Dispense 200 µl or 400 µl or Parameterized</li> <li>• Empty SP 1 or SP 2</li> </ul> |

Table 11–21: Functions of Reader - General

### 11.6.2 Parameter

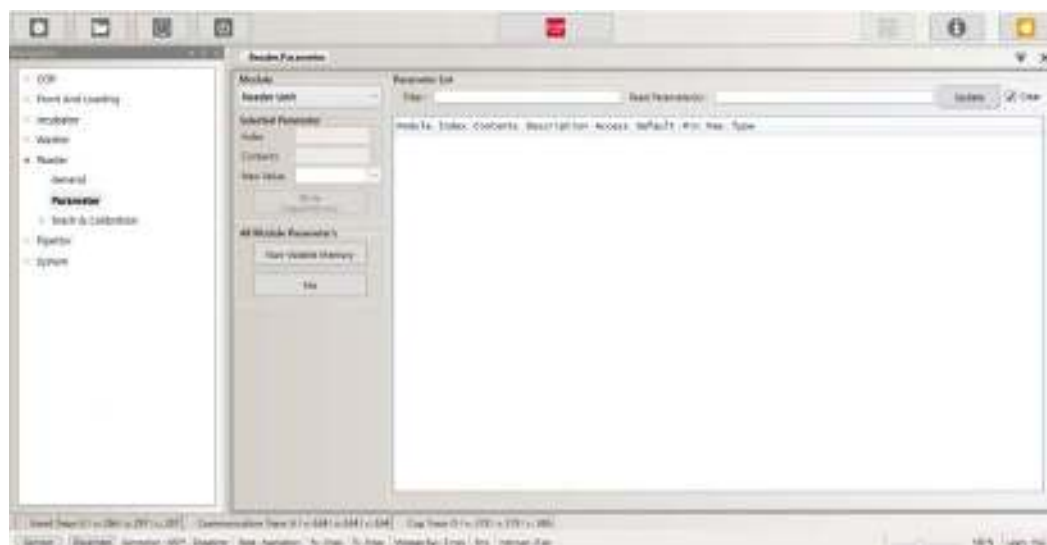


Figure 11–25: Reader - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 11-11.

In the **Module** group, it is possible to select:

- Reader Unit
- Starter Pump 1
- Starter Pump 2
- Wash Aspiration Pump
- Liquid Support
- COP

## 11. Service Software for internal-tanks configuration

### 11.6.3 Teach & Calibration

#### 11.6.3.1 Reader Rotor Calibration



Figure 11–26: Reader - Teach & Calibration - Rotor

The Reader Rotor Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

About the function of all the buttons, see Chapter 11.1.7.1.1 on page 11-9.

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select **Rotor**.

#### Step 1

- a. Press the **Done** button to perform the init.

#### Step 2

- b. Move the Reader rotor by the arrow buttons to align the teaching point to the reference position in the Reader cover.



Figure 11–27: Reader rotor, calibration position

- c. Press **Save** button to save the position.

#### Step 3

- d. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

#### Step 4

- e. The process is successful when, at the **Summarize** panel, all steps are in green.



## 11.6.3.2 Reader SIPAL Lift Calibration

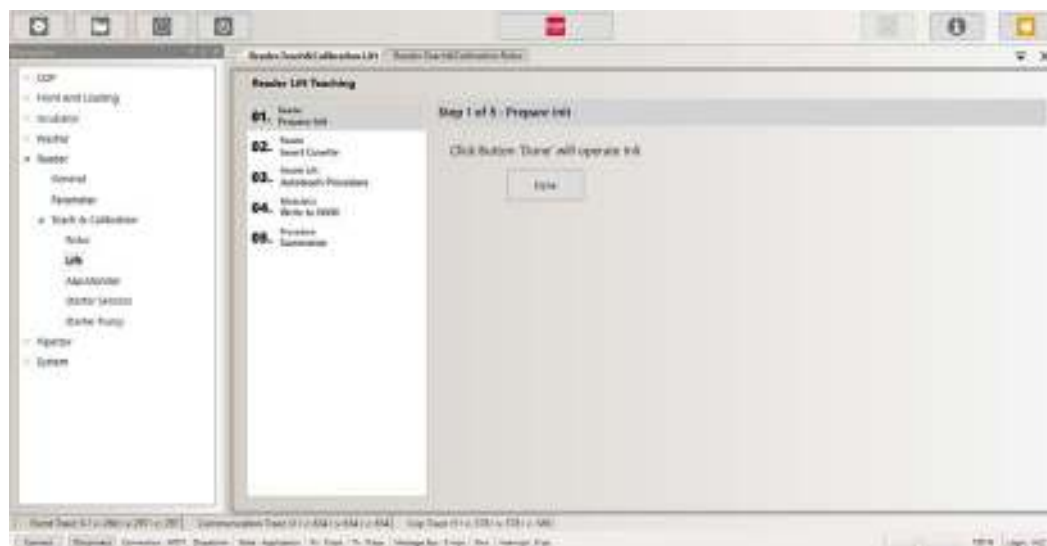


Figure 11–28: Reader - Teach &amp; Calibration - Lift

The Reader SIPAL Lift Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window, choose **Reader** then **Teach & Calibration** bracket and select **Lift**.

- Step 1** a. Follow the instructions and press the **Done** button.
- Step 2** b. Place an empty cuvette in the Reader loading position, then press the **Done** button.
- Step 3** c. Press the **Start** button; the calibration procedure starts automatically.  
d. When the **Processing** window closes without failure message, press the **Done** button.
- Step 4** e. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 5** f. The process is successful when, at the **Summarize** panel, all steps are in green.

## 11. Service Software for internal-tanks configuration

## 11.6.3.3 Reader Aspiration Monitoring Sensors Calibration



Figure 11–29: Reader - Teach &amp; Calibration - Asp. monitor

The Reader Aspiration Monitoring Sensors Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select the **Asp.Monitor**.

- Step 1**
  - a. Press the **WAP for 5sec** on button, to ensure that tubes are free of liquid.
  - b. When the **Processing** window closes, press the **Done** button.
- Step 2**
  - c. Press the **Start** button; the calibration procedure starts automatically.
  - d. When the **Processing** window closes without failure message, press the **Done** button.
- Step 3**
  - e. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 4**
  - f. The process is successful when, at the **Summarize** panel, all steps are in green.

## 11.6.3.4 Starter Sensors Calibration



Figure 11–30: Reader - Teach &amp; Calibration - Starter Sensors

The Starter Sensors Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select the **Starter Sensors**.

**Step 1**

- a. Press the **Start** button, to start the procedure.
- b. Follow the instruction at the pop-up and press **OK** button.



Figure 11–31: Sensor calibration pop-up for Starter 1

- c. When the **Processing** window closes, press the **Done** button.

**Step 2**

- d. Press the **Start** button, to start the procedure.
- e. Follow the instruction at the pop-up and press **OK** button.



Figure 11–32: Sensor calibration pop-up for Starter 2

- f. When the **Processing** window closes without failure message, press the **Done** button.

## 11. Service Software for internal-tanks configuration

**Step 3** g. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

**Step 4** h. The process is successful when, at the **Summarize** panel, all steps are in green.

### 11.6.3.5 Starter Pumps Calibration

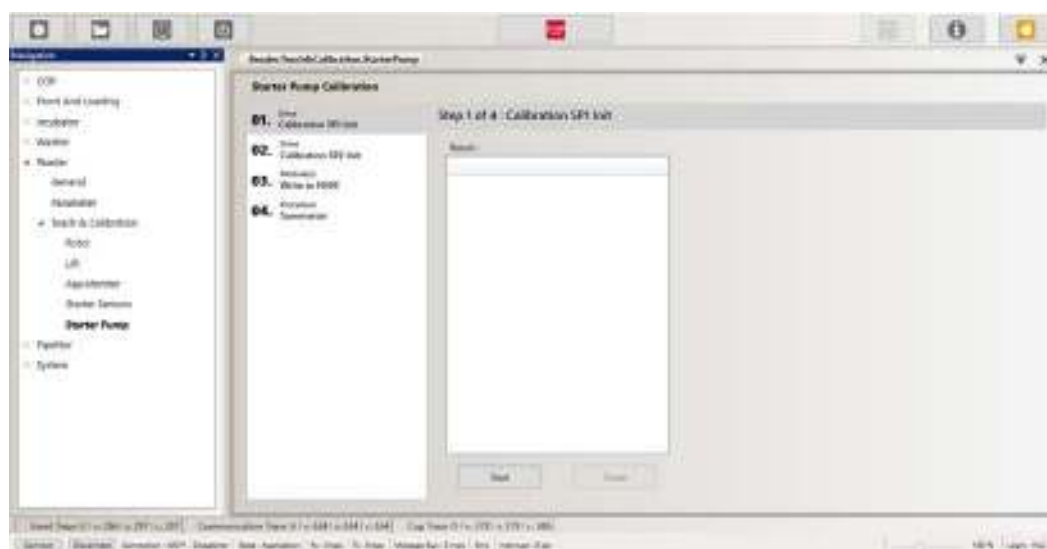


Figure 11–33: Reader - Teach & Calibration - Starter Pump

The Starter Pumps Sensors Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select the **Starter Pump**.

**Step 1** a. Click on the **Start** button to start the automatic calibration of the SDP1.  
b. Check the results and proceed by pressing **Done** button.

**Step 2** c. Click on the **Start** button to start the automatic calibration of the SDP2.  
d. Follow the instruction by pressing **Done** button.

**Step 3** e. Store the taught positions in the EEPROM by selecting **Write ALL Parameters to NVM** button.

**Step 4** f. The process is successful when, at the **Summarize** panel, all steps are in green.

## 11.7 Pipettor

### 11.7.1 General

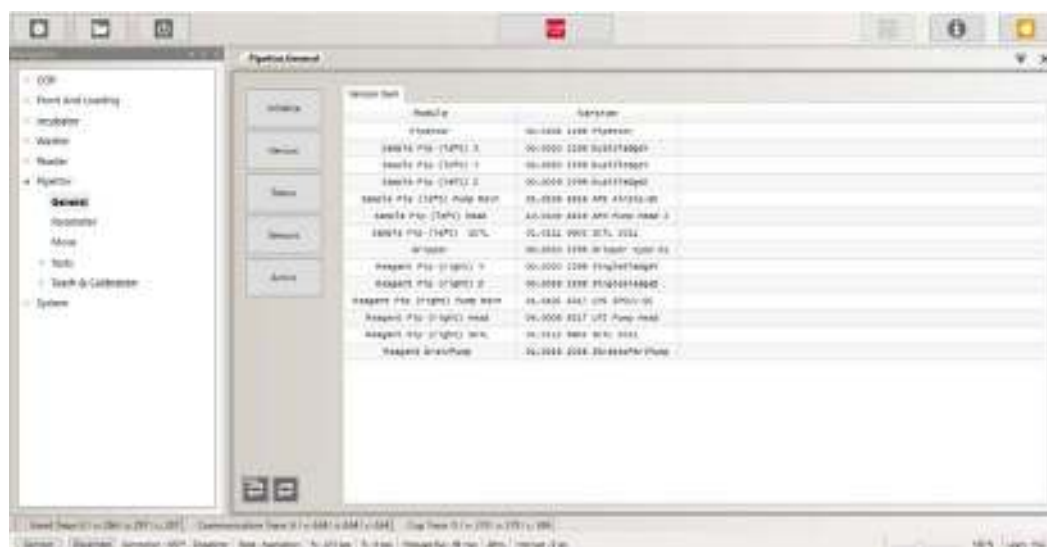


Figure 11–34: Pipettor - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                                    |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                                        |
| <b>Status</b>       | Displays information about the state in the table. Select the desired part in the button menu.                                                                                                                        |
| <b>Sensors</b>      | Displays information about the state of the Pipettor Module sensors in the table.                                                                                                                                     |
| <b>Actors</b>       | LWP: <ul style="list-style-type: none"> <li>• Switch On or Off</li> </ul> Cuvette Gripper: <ul style="list-style-type: none"> <li>• Open or Close the claws</li> </ul> Select the desired sensors in the button menu. |

*Table 11-22: Functions of Pipettor - General*

### 11.7.2 Parameter

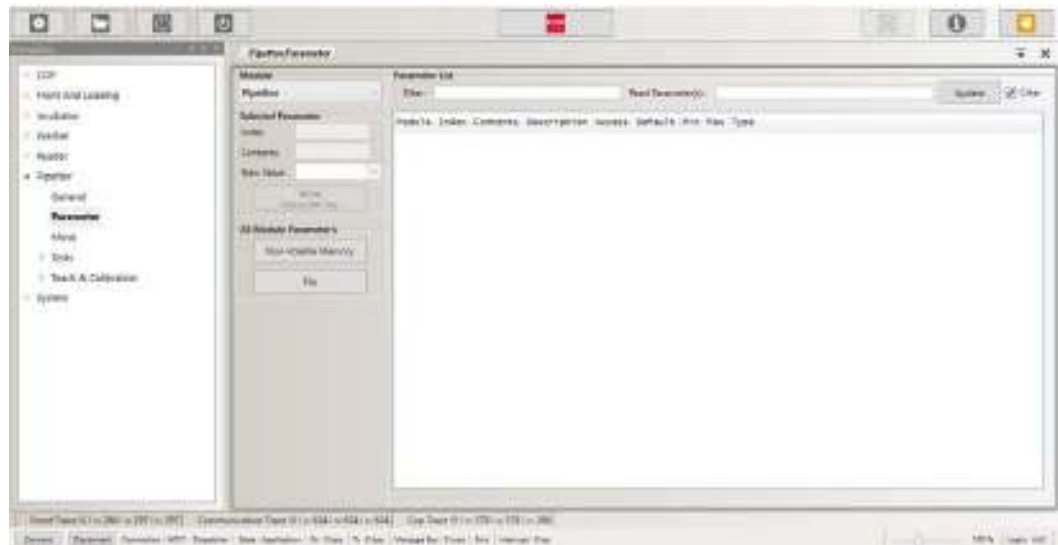


Figure 11–35: Pipettor - Parameter

### NOTE

## Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 11-11.

In the **Module** group, it is possible to select:

- Pipettor
- Liquid Waste Pump
- Gripper Module
- AirPip Sample Probe
- Single Sledge Y
- Single Sledge Z
- SPOLV Reagent Probe
- Dual Sledge Y
- Dual Sledge Z

### 11.7.3 Move

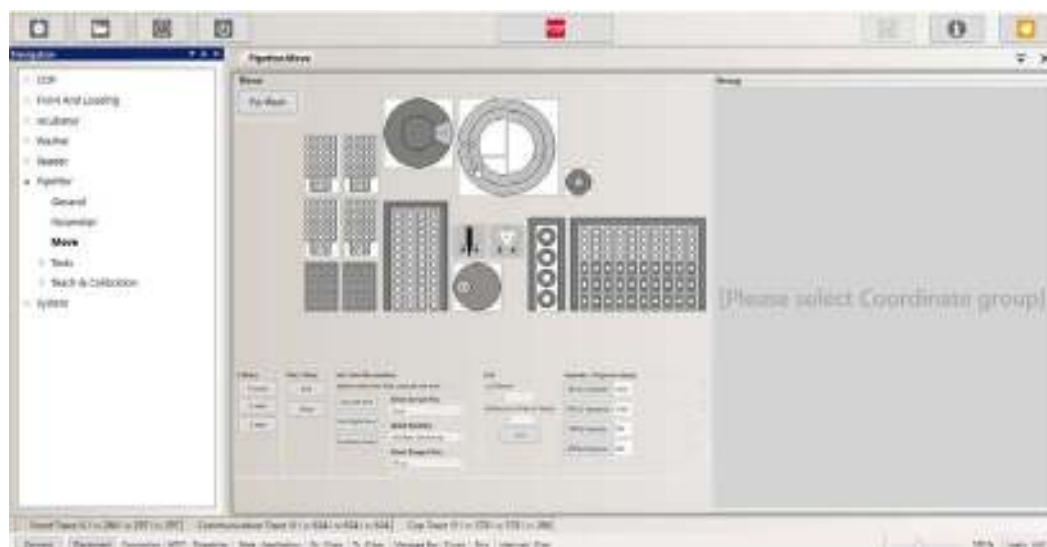


Figure 11–36: Pipettor - Move

#### NOTE

Remember to Initialize the Pipettors before open the **Move** tab.

The **Move** tab shows all the positions, which can be reach by Sample pipettor, SPOLV ad Gripper Module.

Follow the steps to move the Pipettor arms to a desired position:

- Tap on the desired instrument position (e.g. Sample bay, Incubator pipetting position etc.). All possible positions will shown as detailed figure on the right side.
- Double tap on the desired position of the detailed figure on the right side. The specific module moves to the required position.
- Perform an action (Pick / Place cuvette or disposable tip, Aspirate / Dispense liquid, LLD test ect).

#### 11.7.3.1 Rinse group

| Function/<br>Button | Description               |
|---------------------|---------------------------|
| <b>Pip Wash</b>     | Rinses / washes the SPOLV |

Table 11–23: Functions of the **Rinse** group

## 11. Service Software for internal-tanks configuration

11.7.3.2 **Z Move** group**NOTE****Collision of Pipettor**

Be aware that the Z-move will happen at the last XY-position of the Pipettor.

| Function/<br>Button | Description                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------|
| <b>Z travel</b>     | Moves the Pipettor arms to its Z travel position (Z move happens at last XY-position of the Pipettor). |
| <b>Z start</b>      | Moves the Pipettor arms to its Z start position.                                                       |
| <b>Z max</b>        | Moves the Pipettor arms to its Z max position.                                                         |

Table 11–24: Functions of the **Z Move** group11.7.3.3 **Pick / Place** group

| Function/<br>Button | Description                                                             |
|---------------------|-------------------------------------------------------------------------|
| <b>Pick</b>         | Picks up cuvette or disposable tip, depending on the Left Arm position. |
| <b>Place</b>        | Places a cuvette or disposable tip to a desired position.               |

Table 11–25: Functions of the **Pick / Place** group11.7.3.4 **Init / Rack file selection** group

| Function/<br>Button   | Description                             |
|-----------------------|-----------------------------------------|
| <b>Init Left Arm</b>  | Initializes Left arm.                   |
| <b>Init Right Arm</b> | Initializes Right arm.                  |
| <b>Init Both Arms</b> | Initializes both arms to the same time. |

Table 11–26: Functions of the **Init** group

| Function/<br>Button       | Description                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Select Sample Rack</b> | Select registered .rac file for Sample bay (to open the drop down field, a init of the left arm must be carried out). |



## 11. Service Software for internal-tanks configuration

| Function/<br>Button        | Description                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Select Ancillary</b>    | Select registered .rac file for Ancillary bay (to open the drop down field, a init of the right arm must be carried out). |
| <b>Select Reagent Rack</b> | Select registered .rac file for Reagent bay (to open the drop down field, a init of the right arm must be carried out).   |

Table 11–27: Functions of the **Rack file selection** group11.7.3.5 **LLD** group

| Function/<br>Button     | Description                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LLD Result</b>       | Shows the results of the LLD test performed, depending on the Pipettor arm used: <ul style="list-style-type: none"> <li>Left arm position, the Sample pipettor performs a barometric check (B-LLD)</li> <li>Right arm position, the SPOLV Module performs a capacitive check (C-LLD)</li> </ul> |
| <b>Distance to Zmax</b> | Shows the Liquid Level relative distance to Zmax expressed in steps, after the LLD test performed.                                                                                                                                                                                              |
| <b>LLD</b>              | Starts the Liquid Level Detection with the last Pipettor arm moved into the dedicated position.                                                                                                                                                                                                 |

Table 11–28: Functions of the **LLD** group11.7.3.6 **Aspirate / Dispense** group

| Function/Button       | Description                                                                                                                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SPOLV Aspirate</b> | Insert the amount of steps to aspirate by the SPOLV in the box, or moving the arrow left (to decrease) or right (to increase).<br>Press the button <b>SPOLV Aspirate</b> to start the aspiration of the entered steps at the SPOLV.                     |
| <b>SPOLV Dispense</b> | Insert the amount of steps to dispense by the SPOLV in the box, or moving the arrow left (to decrease) or right (to increase).<br>Press the button <b>SPOLV Dispense</b> to start the dispensation of the entered steps at the SPOLV.                   |
| <b>ZSPip Aspirate</b> | Insert the amount of steps to aspirate by the Sample pipettor in the box, or moving the arrow left (to decrease) or right (to increase).<br>Press the button <b>ZSPip Aspirate</b> to start the aspiration of the entered steps at the Sample pipettor. |

## 11. Service Software for internal-tanks configuration

---

| Function/Button       | Description                                                                                                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ZSPip Dispense</b> | Insert the amount of steps to dispense by the ZSPip in the box, or moving the arrow left (to decrease) or right (to increase).<br>Press the button <b>ZSPip Dispense</b> to start the dispensation of the entered steps at the Sample pipettor. |

Table 11–29: Functions of the **Aspirate / Dispense** group

## 11.7.4 Test

### 11.7.4.1 LLD Endurance

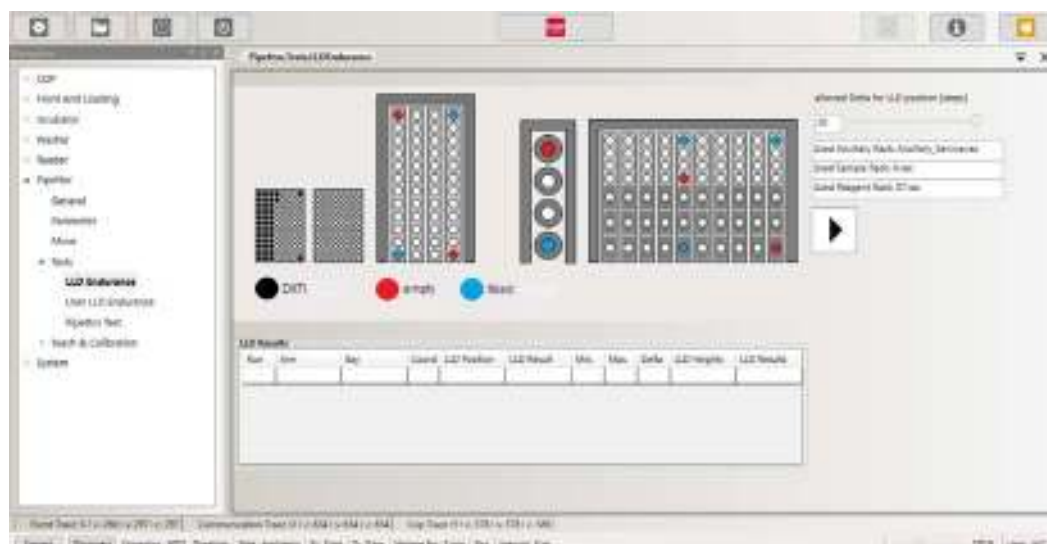


Figure 11–37: Pipettor - Test - LLD Endurance

| Function/Button                                                                     | Description                    |
|-------------------------------------------------------------------------------------|--------------------------------|
|  | Start the LLD test customized. |

Table 11–30: Function of the **LLD Endurance**

#### 11.7.4.1.1 Precondition

- The pre-set amount of a LLD Endurance runs is 10.
- The positions for disposable tips and LLD positions are pre-defined (marked with colors: black means disposable tips required, red means Empty, blue means Filled).
- After the test starting, an initialization is always carried out.
- After the test finishing, a report will be generated.
- Always the same .rac files have to be used (Ancillary\_Service.rac, A.rac and 07.rac).

#### 11.7.4.1.2 Preparation

- Insert a full-filled disposable tip tray in the left tray.
- Insert a empty tube in the first position and a filled tube in the last position of a Sample rack A. Insert the rack in the first lane of the Sample bay.
- Insert a empty tube in the last position and a filled tube in the first position of a Sample rack A. Insert the rack in the last lane of the Sample bay.

## 11. Service Software for internal-tanks configuration

- Insert a empty bottle in the first position and a filled bottle in the last position of the Ancillary tray (in the dedicated holder). Insert the Ancillary tray in its lane on the Reagent bay.
- Insert a reagent kit in the fifth line of the Reagent bay with the following configuration: first and last (seventh) vial filled, forth vial empty
- Insert a reagent kit in the last lane of the Reagent bay with the following configuration: first vial filled, last vial (seventh) empty.

### 11.7.4.1.3 General procedure of LLD Endurance test

- The Sample pipettor picks up a disposable tip (only needed for LLD test of the Sample bay).
- The LLD test starts the first run at defined coordinate.
- The results are displayed in a table.
- The system verifies if the results of the LLD heights are in the predefined range (allowed delta for LLD position).
- The Sample pipettor ejects the disposable tip (if present).
- The system repeats the LLD Endurance test for the number of runs (10).

### 11.7.4.2 User LLD Endurance

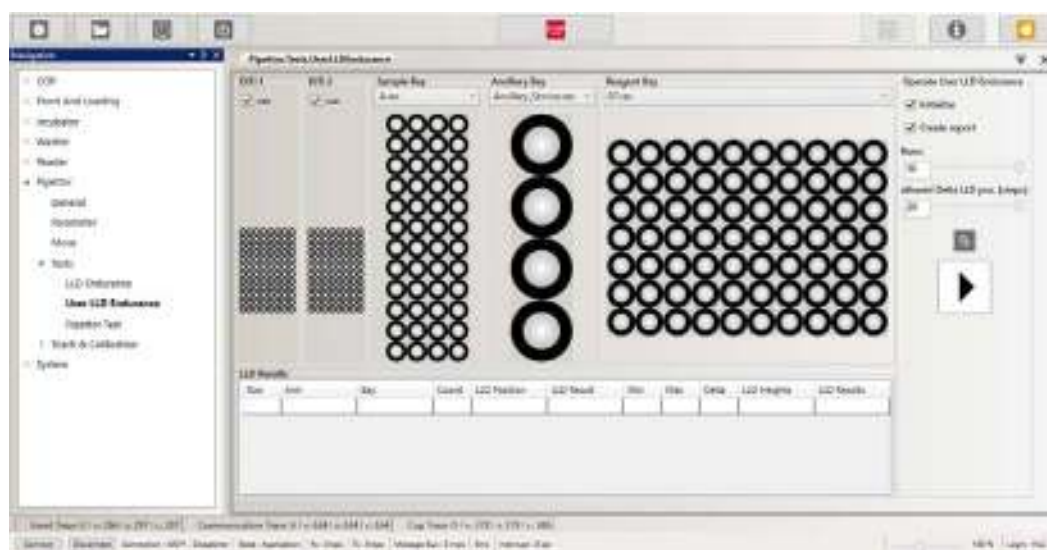


Figure 11–38: Pipettor - Test - User LLD Endurance

#### 11.7.4.2.1 DiTi 1 and DiTi 2 group

| Function/<br>Button      | Description                                                                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DiTi 1 and DiTi 2</b> | Check box to enable the Disposable tips tray usage.<br>In the selected Disposable tip trays, the Disposable tips needed will be marked automatically. |

Table 11–31: Function of **DiTi 1** and **DiTi 2** group**11.7.4.2.2 Sample Bay group**

| Function/<br>Button                                                               | Description                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sample Bay</b>                                                                 | Dropdown menu to select the Sample rack file.                                                                                                                                                                                                                                          |
|  | Positions of the Sample bay to select by clicking on them.<br>The selected positions will be marked.<br>In the selected Sample bay positions, the Left Pipettor will perform the LLD.<br>The Disposable tips needed will be marked automatically after the Sample bay positions chose. |

Table 11–32: Function of **Sample Bay** group**11.7.4.2.3 Ancillary Bay group**

| Function/<br>Button                                                                 | Description                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ancillary Bay</b>                                                                | Dropdown menu to select the Ancillary rack file.                                                                                                                                             |
|  | Positions of the Ancillary bay to select by clicking on them.<br>The selected positions will be marked.<br>In the selected Ancillary bay positions, the Right Pipettor will perform the LLD. |

Table 11–33: Function of **Ancillary Bay** group**11.7.4.2.4 Reagent Bay group**

| Function/<br>Button                                                                 | Description                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reagent Bay</b>                                                                  | Dropdown menu to select the Reagent rack file.                                                                                                                                           |
|  | Positions of the Reagent bay to select by clicking on them.<br>The selected positions will be marked.<br>In the selected Reagent bay positions, the Right Pipettor will perform the LLD. |

Table 11–34: Function of **Reagent Bay** group

## 11. Service Software for internal-tanks configuration

11.7.4.2.5 *Operate User LLD Endurance group*

| Function/<br>Button                                                                | Description                                                                                                                                                                 |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Initialize</i>                                                                  | Check box to select the initialization of the instrument at the beginning of the LLD test.                                                                                  |
| <i>Create report</i>                                                               | Check box to select the Report creation at the end of the LLD test.                                                                                                         |
| <i>Runs</i>                                                                        | Runs to be performed by the system during the LLD test (default is set to 10 run).<br>Select from 1 to 10 runs, moving the arrow left (to decrease) or right (to increase). |
| <i>allowed Delta LLD pos.</i>                                                      | Allowed delta for LLD positions, expressed in steps (default is set to 20 steps).<br>Select from 1 to 20 steps, moving the arrow left (to decrease) or right (to increase). |
|   | Clear list button.                                                                                                                                                          |
|  | Start the LLD test customized.                                                                                                                                              |

Table 11–35: Function of *Operate User LLD Endurance group*11.7.4.2.6 *Preparation*

- a. Customize on the screen the LLD test:
  - select the .rac files to be used;
  - select the Disposable tip tray/trays to be used;
  - select the Sample bay, Ancillary tray and Reagent bay positions where the Pipettors will perform the LLD test;
  - select the number of test runs to be performed;
  - select the allowed delta for LLD positions.
- b. Insert all the needed consumables, sample racks, reagent kit etc in the system.
- c. Press the **Play** button.

## 11.7.4.3 Pipettor Test

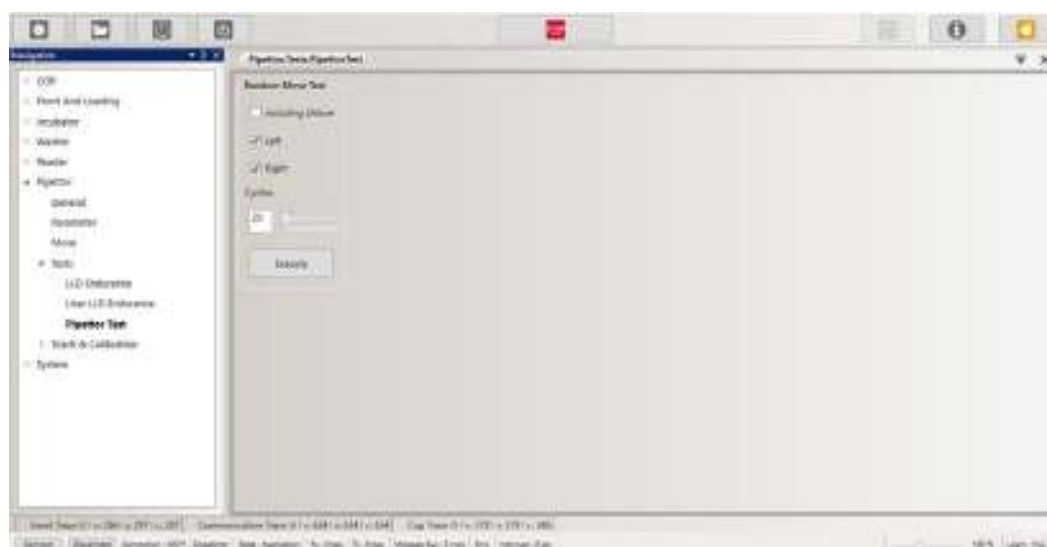


Figure 11–39: Pipettor - Tests - Pipettor Test

## 11.7.4.3.1 Random Move Test group

| Function/<br>Button     | Description                                                                                                                                                                             |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Including Z-Move</b> | Check box to enable the Pipettors Z movements in the random cycle.                                                                                                                      |
| <b>Left / Right</b>     | Check box to choose the arm / arms which will perform the test.                                                                                                                         |
| <b>Cycles</b>           | Cycles to be performed by the arms during the Pipettor test (default is set to 20 cycles).<br>Select from 1 to 1000 cycles, moving the arrow left (to decrease) or right (to increase). |
| <b>Execute</b>          | Starts the random movements.                                                                                                                                                            |

Table 11–36: Random Move Test

## 11. Service Software for internal-tanks configuration

## 11.7.5 Teach &amp; Calibration

## 11.7.5.1 Gripper Claws Calibration

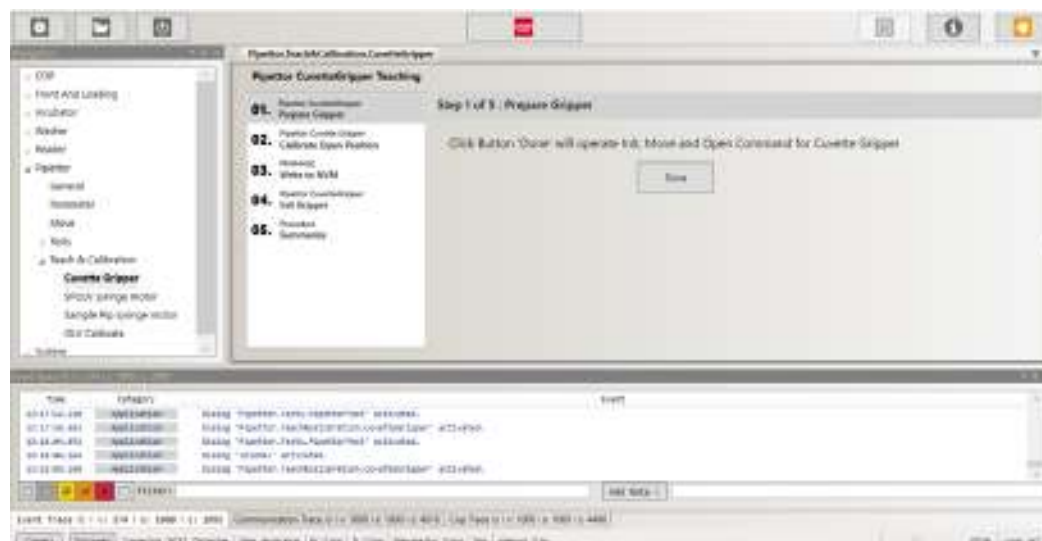


Figure 11–40: Pipettor - Teach &amp; Calibration - Cuvette Gripper

The Gripper Claws Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

About the function of all the buttons, see Chapter 11.1.7.1.1 on page 11-9.

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **Cuvette Gripper**

- Step 1**
  - a. Remove the front right cuvette tray in the Cuvette - Tip drawer. Read the instructions and then press the **Done** button to initialize the gripper.
- Step 2**
  - b. Take a cuvette and manually hold it between the gripper claws.
  - c. Use the buttons to close the gripper claws in steps of 10, until the gripper holds on to the cuvette by itself.
  - d. Use the buttons to open the gripper claws in steps of 10, until the cuvette falls down.
  - e. Use the buttons to further open the gripper claws by **30** steps.
  - f. Press the **Save** button to save the adjustment.
- Step 3**
  - g. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 4**
  - h. Read the instructions and then press the **Done** button to initialize the gripper.
- Step 5**
  - i. The process is successful when, at the **Summarize** panel, all steps are in green.





## 11. Service Software for internal-tanks configuration

### 11.7.5.3 Sample Pip syringe motor

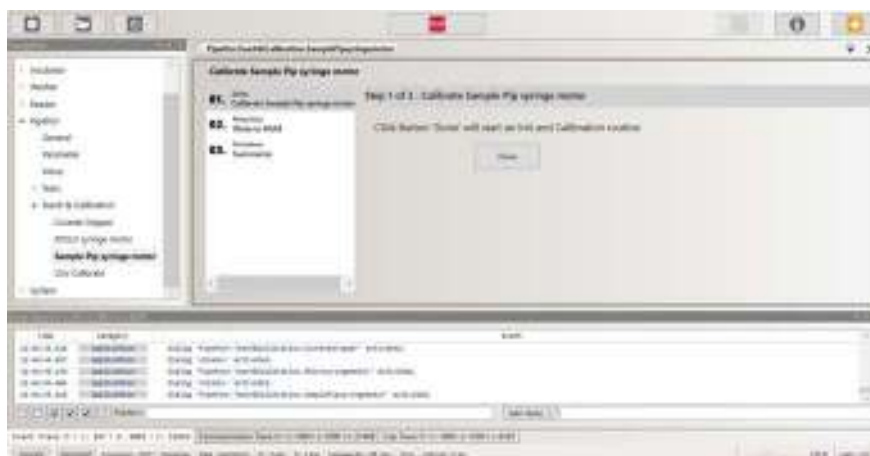


Figure 11-42: Pipettor - Teach & Calibration - Sample Pip syringe motor

The Sample Pip syringe motor Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **Sample Pip syringe motor**.

- Step 1**
  - a. Read the instructions and then press the **Done** button; the calibration procedure starts automatically.
  - b. The procedure is successful when the **Processing** window closes without failure message.
- Step 2**
  - c. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 3**
  - d. The process is successful when, at the **Summarize** panel, all steps are in green.

## 11.7.5.4 OLV Calibrate

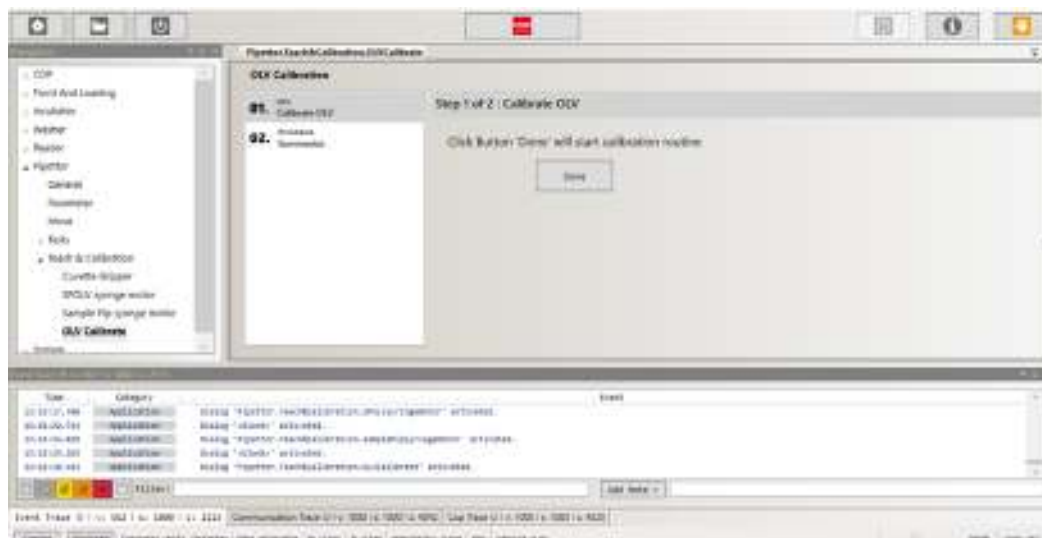


Figure 11–43: Pipettor - Teach &amp; Calibration - OLV calibrate

The OLV Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **OLV Calibrate**.

**Step 1**

- Read the instructions and then press the **Done** button; the calibration procedure starts automatically.
- The procedure is successful when the **Processing** window closes without failure message.

**Step 2**

- The process is successful when, at the **Summarize** panel, all steps are in green.

## 11.8 System

### 11.8.1 General

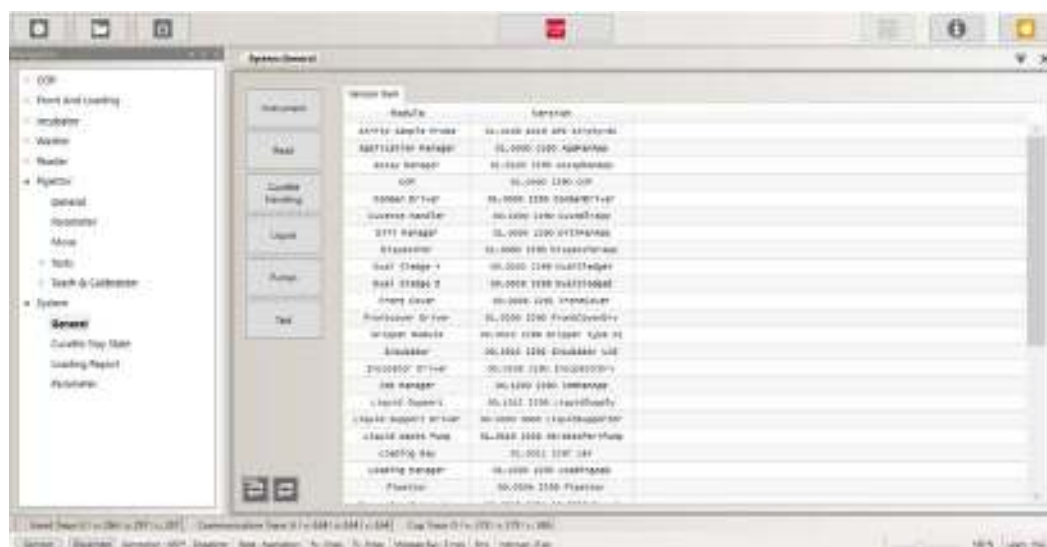


Figure 11-44: System - General - Version

#### 11.8.1.1 System General

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instrument</b>   | Full initialization via COP: <ul style="list-style-type: none"> <li>Initialize the instrument</li> </ul> Power On: <ul style="list-style-type: none"> <li>Powers on the instrument</li> </ul> Power Off: <ul style="list-style-type: none"> <li>Powers off the instrument.</li> </ul>                                                                                                                                                                                                                                                                  |
| <b>Read</b>         | Displays information about the Versions,<br>Temperature: <ul style="list-style-type: none"> <li>Displays information about the state of the instrument temperature sensors</li> </ul> Liquid State: <ul style="list-style-type: none"> <li>Displays information about the state of the instrument liquid sensors</li> </ul> Counter: <ul style="list-style-type: none"> <li>Reads the cycle counter from the selected modules.</li> </ul> Serial: <ul style="list-style-type: none"> <li>Reads the Serial number from the selected modules.</li> </ul> |

| Function/<br>Button     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cuvette Handling</b> | <p>Clear All:</p> <ul style="list-style-type: none"> <li>The gripper picks up all the cuvettes in the instrument (from Washer, Incubator and Reader) and dispose them via the Reader Module.</li> </ul> <p>Clear Incubator:</p> <ul style="list-style-type: none"> <li>The gripper picks up all the cuvettes in the Incubator and dispose them via the Reader Module.</li> </ul> <p>Clear Washer:</p> <ul style="list-style-type: none"> <li>The gripper picks up all the cuvettes in the Washer and dispose them via the Reader Module.</li> </ul> <p>Clear Measurement:</p> <ul style="list-style-type: none"> <li>Cleans the cuvette in the Reader Module.</li> </ul> <p>Fill Incubator:</p> <ul style="list-style-type: none"> <li>The gripper picks up the cuvettes from the trays in order to fill the Incubator (the cuvette trays must be set in the <b>Cuvette Tray State</b>).</li> </ul> <p>Cuvette Endurance Run open the Cuvette Endurance Run window where there is the possibility to choose the setting for the test, see Chapter 11.8.1.2 on page 11-61.</p> |
| <b>Liquid</b>           | <p>Prime all, see Chapter 11.8.1.3 on page 11-62.</p> <p>Prime Pipettor, see Chapter 11.8.1.4 on page 11-64.</p> <p>Prime Washer, see Chapter 11.8.1.5 on page 11-65.</p> <p>Prime Measurement, see Chapter 11.8.1.6 on page 11-65.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## 11. Service Software for internal-tanks configuration

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pumps</b>        | <p>WAP:</p> <ul style="list-style-type: none"> <li>Start at Routine Speed</li> <li>Start at Prime Speed</li> <li>Start Parameterized opens the Start WAP Timed window, where there is the possibility to choose the WAP speed and time to run</li> <li>Stop</li> </ul> <p>WBDP:</p> <ul style="list-style-type: none"> <li>Load: <ul style="list-style-type: none"> <li>1900 µl</li> <li>2100 µl</li> <li>Parameterized opens the window Load WBDP where there is the possibility to choose the volume to load in uL</li> </ul> </li> <li>Dispense Port 1, 2 or 3: <ul style="list-style-type: none"> <li>700 µl</li> <li>1200 µl</li> <li>Parameterized opens the window Dispense WBDP Port 1, 2 or 3 where there is the possibility to choose the volume to dispense in uL and if activate or not the tracking</li> </ul> </li> <li>Empty the pump</li> </ul> <p>Starter Dispense Pump 1 and 2 (SP1 and SP2):</p> <ul style="list-style-type: none"> <li>Fill to 400 µl or Parameterized</li> <li>Dispense 200 µl or 400 µl or Parameterized</li> <li>Empty the pump</li> </ul> <p>LWP:</p> <p>Switch On or Off</p> |
| <b>Test</b>         | <p>Cuvette Movement Test</p> <ul style="list-style-type: none"> <li>Follow arising instructions.</li> <li>The cuvette trays must be set in the <b>Cuvette Tray State</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Table 11–37: Functions of System - General

## 11.8.1.2 Cuvette Endurance Run

| Function/<br>Button                | Description                                                                                                                                                                                     |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialization</b>              | Check box to perform the system initialization before the start of the Cuvette Endurance Run.                                                                                                   |
| <b>Quantity</b>                    | Quantity of cuvette to test in the Cuvette Endurance Run (default value is set to 57).<br>Choose a value from 1 to 57.                                                                          |
| <b>StartWith</b>                   | Incubator rotor ring from which the Cuvette Endurance Run starts: <ul style="list-style-type: none"> <li>• Inner (cuvette positions 1-25)</li> <li>• Outer (cuvette positions 26-57)</li> </ul> |
| <b>EndWith</b>                     | Position in which the Cuvette Endurance Run finishes: <ul style="list-style-type: none"> <li>• Incubator</li> <li>• Waste</li> </ul>                                                            |
| <b>Inc. to Washer cycles</b>       | Number of cycles of the cuvette inside the Washer Module.<br>Set 0 to avoid the Washer Module.                                                                                                  |
| <b>PresenceCheck</b>               | Check box to activate the check of the cuvette presence inside the Incubator Module.                                                                                                            |
| <b>LiftMove</b>                    | Check box to activate the Up and Down move of the Washer aspiration needles.                                                                                                                    |
| <b>Execute</b>                     | Button to start the Cuvette Endurance Run.                                                                                                                                                      |
| <b>Cuvette Endurance Info Item</b> | Reports the actions of the system during the Cuvette Endurance Run.                                                                                                                             |

Table 11–38: Functions of the **Cuvette Endurance Run** window

## 11.8.1.2.1 Standard Cuvette Endurance Run

a. Set the cuvette trays in the **Cuvette Tray State**.



b. Customize on the screen the **Cuvette Endurance Run** test:

- select the initialization of the instrument;
- select 10 cuvette to test;
- select Outer to start;
- select Waste to end;
- select 2 cycles inside the Washer;
- select the cuvette presence check in the Incubator;
- uncheck the Lift movement;

c. Press the **Execute** button.

d. Check if the placing of the cuvette by the Gripper inside the Incubator and the Washer module is smooth and without noises.

e. Perform again the Cuvette Endurance Run, selecting Inner to start.

**CAUTION**

**During the Cuvette Endurance Run, check the cuvette transfer from the Incubator to the Washer and vice-versa**

Check that cuvette insertion by the Gripper inside the Incubator is smooth and without noises.

Check that cuvette pickup from the Incubator and insertion into the Washer is smooth and without noises.

Check that cuvette pickup from the Washer and insertion back into the Incubator is smooth and without noises.

A not proper cuvette transfer is usually caused by a misalignment between the gripper and the Incubator rotor. If needed, change the Incubator rotor calibration (see Chapter 11.4.3.1 on page 11-26) such to compensate the misalignment; an adjustment of 3-4 steps clockwise could be sufficient.

### 11.8.1.3 Prime all

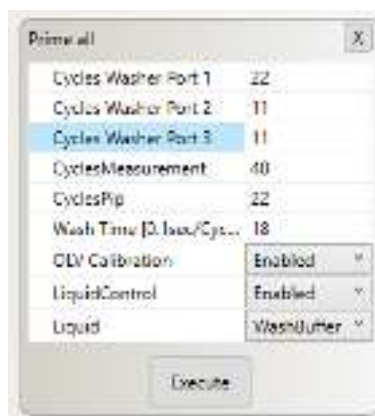
| Function/Button                      | Description                                                                                                                                                                                             |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cycles Washer Port 1, 2 and 3</b> | Select the Wash Buffer prime cycles to perform at each Washer SIPAL.                                                                                                                                    |
| <b>Cycles Measurement</b>            | Select the Starter reagents prime cycles to perform at the Reader Module.                                                                                                                               |
| <b>Cycles Pip</b>                    | Select the prime cycles to perform at the SPOLV.                                                                                                                                                        |
| <b>Wash Time</b>                     | Select the wash time during the SPOLV prime (0.1second/cycles).                                                                                                                                         |
| <b>OLV Calibration</b>               | Enables or disables the OLV Calibration.                                                                                                                                                                |
| <b>Liquid Control</b>                | Enables or disables the Aspiration monitoring check on the three Washer SIPAL and on the Reader SIPAL.                                                                                                  |
| <b>Liquid</b>                        | Select the liquid to prime the SPOLV: <ul style="list-style-type: none"> <li>• DIWater (the system primes using the 1.5L tank)</li> <li>• WashBuffer (the system primes using the 5.5L tank)</li> </ul> |
| <b>Execute</b>                       | Button to start the Cuvette Endurance Run.                                                                                                                                                              |
| <b>Bubble Measure Result Item</b>    | If the Liquid Control and/or the OLV Calibration are enabled, displays the results.                                                                                                                     |

Table 11–39: Functions of the **Prime all** window



**11.8.1.3.1 Standard Prime all**

- a. Insert all needed consumables and liquids.
- b. Set the cuvette trays in the **Cuvette Tray State**.
- c. Customize on the screen the **Prime all** window with the following values:
  - Cycles Washer Port 1 : 22
  - Cycles Washer Port 2: 11
  - Cycles Washer Port 3: 11
  - CyclesMeasurement: 40
  - CyclesPip: 22
  - Wash time: 18
  - OLV Calibration: Enabled
  - LiquidControl: Enabled
  - Liquid: WashBuffer



- d. Press the **Execute** button.
- e. The process is successful when the **Processing** window closes without failure message.

## 11. Service Software for internal-tanks configuration

### 11.8.1.4 Standard Prime SPOLV procedure

#### 11.8.1.4.1 Initial rinsing with DI water

In the **Service Software** at the navigation window, choose **System** and the **General** bracket. Click onto the **Liquid** button and choose the **Prime Pipettor** button to open the dedicated window.

- a. At the window **Prime Pipettor**, use following values:
  - Cycles: 22
  - Wash time: 18
  - OLV Calibration: Disabled
  - Liquid: DI Water

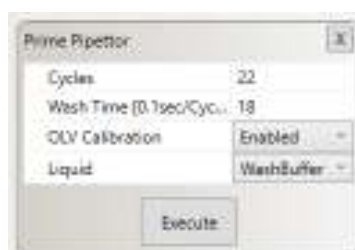


- b. Press the **Execute** button.
- c. The process is successful when the **Processing** window closes without failure message.

#### 11.8.1.4.2 Rinsing with Wash Buffer

In the **Service Software** at the navigation window, choose **System** and the **General** bracket. Click onto the **Liquid** button and choose the **Prime Pipettor** button to open the dedicated window.

- a. At the window **Prime Pipettor**, use following values:
  - Cycles: 22
  - Wash time: 18
  - OLV Calibration: Disabled
  - Liquid: Wash Buffer



- b. Press the **Execute** button.
- c. The process is successful when the **Processing** window closes without failure message.

### 11.8.1.5 Standard Prime Washer Module procedure

#### 11.8.1.5.1 Rinsing of Washer Module

In the **Service Software** at the navigation window, choose **System** and the **General** bracket. Click onto the **Liquid** button and choose the **Prime Washer** button to open the dedicated window.

- a. Set the cuvette trays in the **Cuvette Tray State**.
- b. At the window **Prime Washer** use following values:
  - Cycles Port 1: 22
  - Cycles Port 2: 11
  - Cycles Port 3: 11
  - Liquid Control: Enabled



- c. Press the **Execute** button.
- d. The process is successful when the **Processing** window closes without failure message.

### 11.8.1.6 Standard Prime Reader Module procedure

#### 11.8.1.6.1 Rinsing of Reader Module

In the **Service Software** at the navigation window, choose **System** and the **General** bracket. Click onto the **Liquid** button and choose the **Prime Reader** button to open the dedicated window.

- a. Set the cuvette trays in the **Cuvette Tray State**.
- b. At the window **Prime Reader** use following values and start with **Execute**.
  - Cycles: 40
  - Liquid Control Enabled



- c. The process is successful when the **Processing** window closes without failure message.

### 11.8.2 Cuvette Tray State

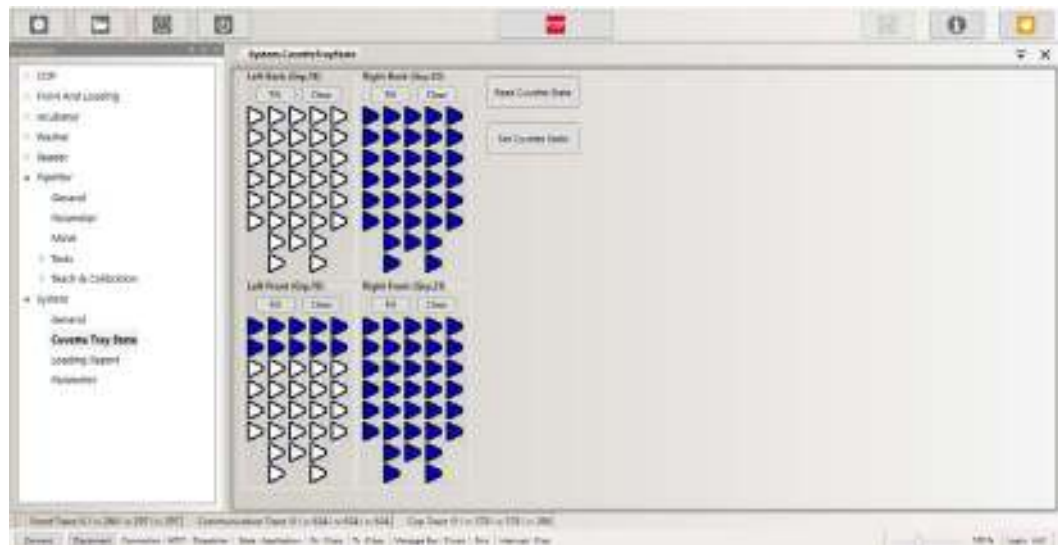


Figure 11-45: System - Cuvette Tray State

| Function/Button                                                                     | Description                                                                                         |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Fill, Clear                                                                         | Fill or clear the chosen cuvette tray only at the display.                                          |
|  | Fill or clear the chosen cuvette only at the display.<br>The cuvette will be marked in blue if set. |
| Read Cuvette State                                                                  | Reads and shows the last set cuvette status from the cuvette tray.                                  |
| Set Cuvette State                                                                   | Takes over the chosen setting of the display into the cuvette tray.                                 |

*Table 11–40: Functions of the Cuvette Tray State*

### 11.8.3 Loading Report

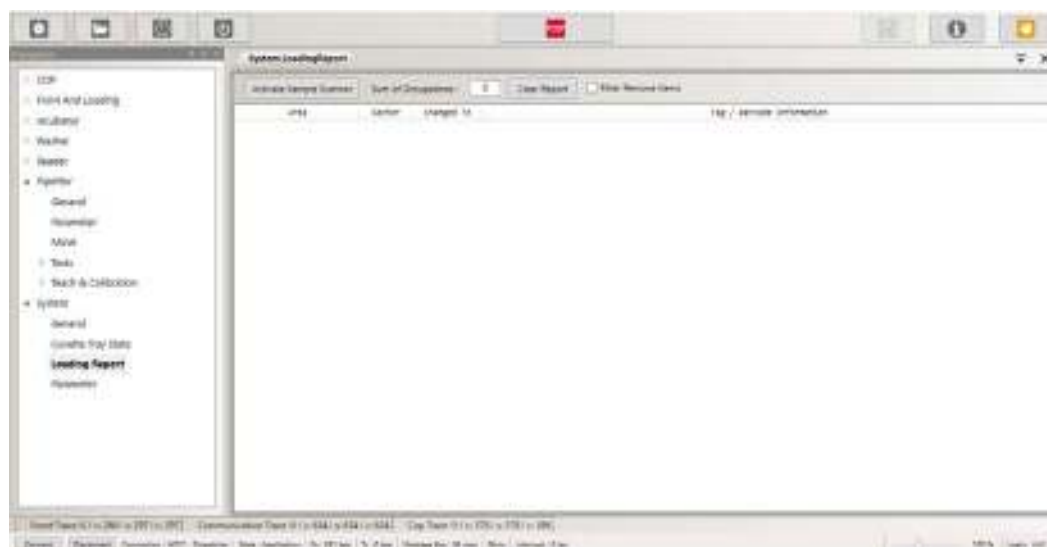


Figure 11–46: System - Loading Report

| Function/Button                | Description                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------|
| <b>Activate Sample Scanner</b> | Switch on the sample scanner to scan the sample rack.                                      |
| <b>Sum of Occupations</b>      | Sum of all loading changes.                                                                |
| <b>Clear Report</b>            | Report will be deleted.                                                                    |
| <b>Filter Remove Items</b>     | If check box is marked, then the loading changes for removing of racks will be suppressed. |

Table 11–41: Functions of the Loading Report

#### 11.8.3.1 Verifying RFID

In the **Service Software** at the navigation window choose **System** and the **Loading Report** bracket.

The changes of the materials to be loaded are shown after loading or unloading of the materials.

- When inserting the Starter bottles or the Assay reagent kits in the dedicated area, after a defined time interval the presence of the bottles and of the kits will be prompted.
- The RFID will be read and displayed as a dialog.

### 11.8.4 System Parameter

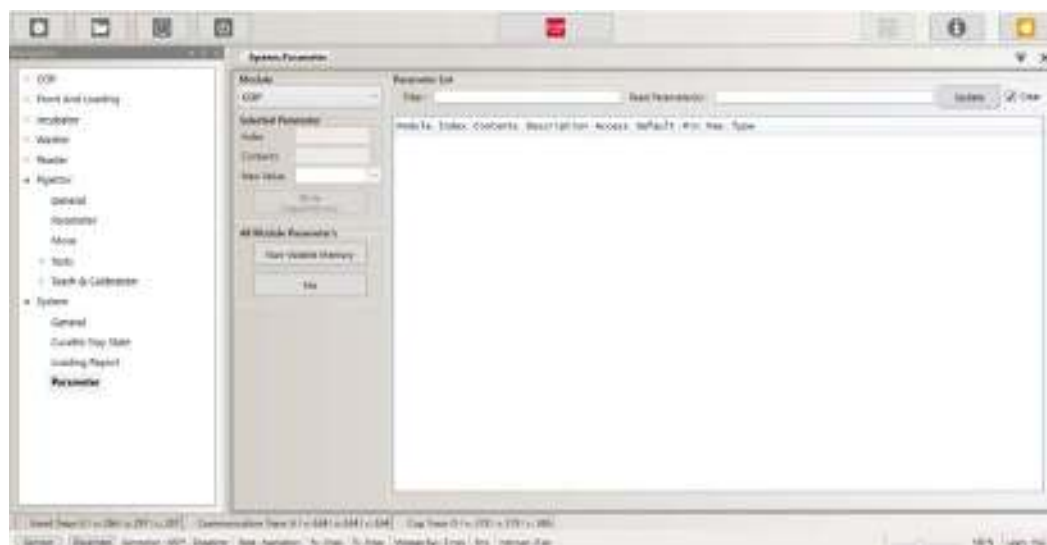


Figure 11–47: System - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 11-11.

In the **Module** group, it is possible to select:

- COP
- Power Can Extension
- Front Cover
- Loading Bay
- Liquid Support
- Wash Aspiration Pump
- Incubator
- Reader Unit
- Starter Pump 1 & 2
- Pipettor
- Liquid Waste Pump
- Gripper Module
- AirPip Sample Probe
- Single Sledge Y
- Single Sledge Z
- SPOLV Reagent Probe
- Dual Sledge Y
- Dual Sledge Z

## 11. Service Software for internal-tanks configuration

---

- Washer
- Wash Buffer Dispense Pump





# 12 Service Software for external-tanks configuration

This chapter describes basic functions of the **Service Software**.

## 12.1 Use of the Service Software

### 12.1.1 General Statement

This chapter contains descriptive information of the windows, fields and buttons of **Service Software**. For related calibration procedures, see the calibration sections of the individual modules.

The software allows to control necessary functions of the instrument and to perform adjustments of the function parameters.

The **Service Software** is used to perform a part of the final testing of the instruments during installation. In addition, the service engineer can use it as a tool to check in-field instruments and adjust them if required.

### 12.1.2 Before starting the Service Software

This chapter is applicable from **Service Software version 1.6.2111.4001** and **Main Software version 1.4.9**.

If it is necessary to switch from the **Main User Interface (Main UI)** or the **Teacher Software** to the **Service Software**, shut down the previously used software.

Then, log-in as Administrator in the Windows account before open the **Service Software**.

### 12.1.3 Use of the Service Software

#### WARNING



---

#### Possible Instrument Damages

It is easily possible to damage parts of the instrument with the **Service Software**.

- Make sure, that the movement course is free, before you start the movement of a module.
-

**NOTE**

**Close other programs**

If the **Teacher Software** or the **Main UI** was running before, these programs have to be closed!

---

**NOTE**

**Log-in as Administrator**

Before open the **Service Software**, be sure to be logged-in as Administrator.

---

**12.1.3.1 Start**



1. To start the **Service Software**, click on the icon on the desktop twice or select the menu item in the Windows start menu.
2. Enter the password.
3. Click on **Login**.

**WARNING**



The FSE credentials, user name and password, must be kept confidential at all times and they should never be shared or exposed to others.

---

**12.1.3.2 Interface Selection configuration (first start)**

The first time you open the **Service Software**, there is the need to perform configuration of the **Interface Selection**.



1. Click on the **Application Configuration** button and select the **Interface Selection** menu entry, see Chapter 12.1.5 on page 12-8.
2. Click on the **LAN/USB Port** category tab.
3. Select the **WCF1** row, **Dispatcher** (setting only for **Service Software** at first run).
4. Click on the **Select** button.

## 12.1.4 General handling of the Service Software

Several parts and buttons are identical on all module drivers. This chapter will describe their function.

### 12.1.4.1 Overview

Overview of the **Service Software** control elements.

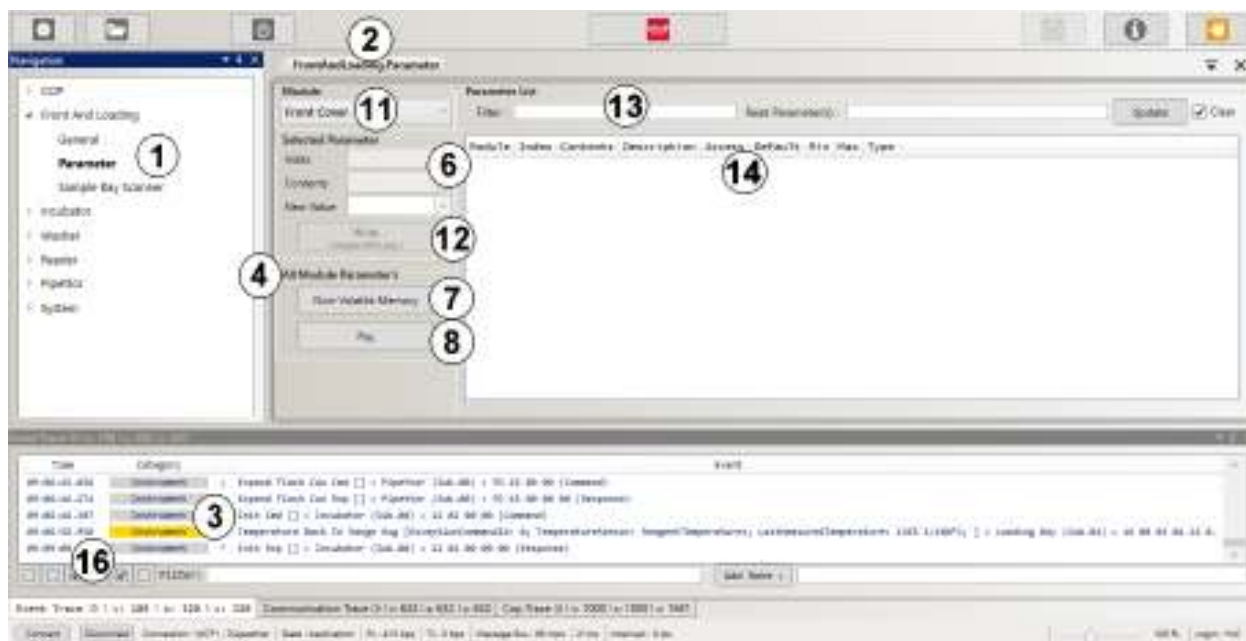


Figure 12-1: Service Software Control Elements I

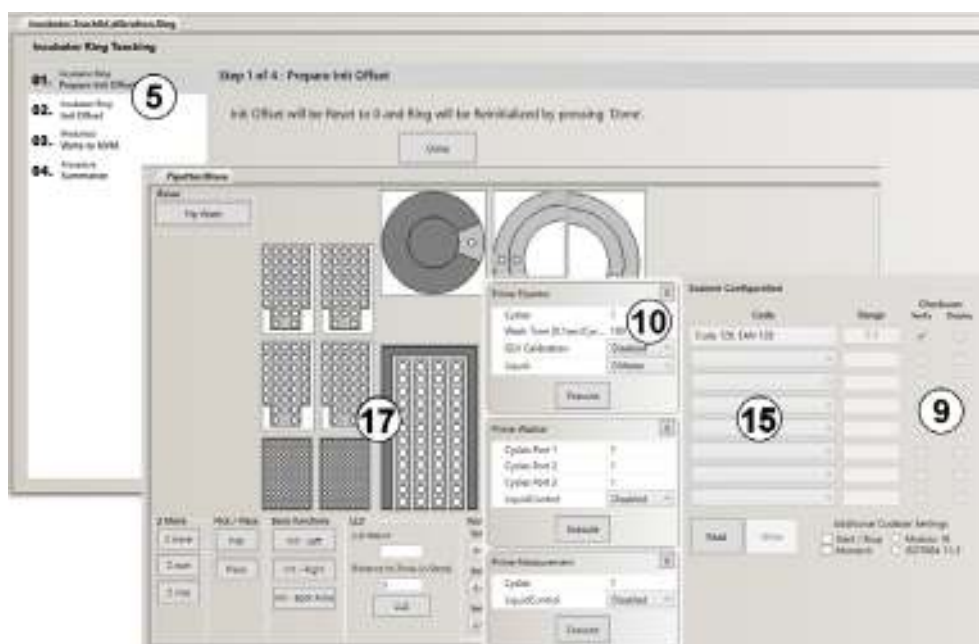


Figure 12-2: Service Software Control Elements II

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <p><b>Navigation tree:</b></p> <p>The <b>Navigation tree</b> on the left-hand side of the <b>Service Software</b> window displays all modules with controller boards. You can work with it in the same manner as you work with the Windows Explorer. A module entry includes further categories with several functions. Click on the arrow in front of the module name or on the module name to display all available categories. Click on the arrow again to close the module entry.</p> <p>Click on a category to open it as <b>Category tab</b> on the right-hand side.</p> |
| 2  | <p><b>Category tab:</b></p> <p>To obtain a better overview the functions have been grouped into categories. These groups are opened by clicking on the corresponding category in the <b>Navigation tree</b> and displayed as a separate <b>Category tab</b> on the right-hand side. Each <b>Category tab</b> will display the appropriate features. By clicking on a <b>Category tab</b> you can switch between the different tabs. It is also possible to close unused <b>Category tabs</b>.</p>                                                                              |
| 3  | <p><b>Trace (log) area:</b></p> <p>The <b>Trace (log) area</b> displays the communication between the <b>Service Software</b> and the instrument modules. Errors will be displayed as red entries. A slider on the right side enables you to move the displayed text.</p>                                                                                                                                                                                                                                                                                                      |
| 4  | <p><b>Group:</b></p> <p>A <b>Group</b> combines functions which belong together.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 5  | <p><b>Selection list:</b></p> <p>A <b>Selection list</b> displays all possible values/entries for this function. Select one of the displayed values/entries. The selected value/entry will be highlighted.</p>                                                                                                                                                                                                                                                                                                                                                                 |
| 6  | <p><b>Information box:</b></p> <p>An <b>Information box</b> displays information about a performed function (e.g. status of a sensor).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 7  | <p><b>Button:</b> Non Volatile Memory</p> <p>Reload all Parameter's from Non-Volatile Memory to RAM.</p> <p>Overwrite ALL Parameter's in Non-Volatile Memory</p>                                                                                                                                                                                                                                                                                                                                                                                                               |
| 8  | <p><b>Button:</b> File</p> <p>Read from Device and save.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 9  | <p><b>Check box:</b></p> <p>A <b>Check box</b> allows you to activate/deactivate an option of a function. Click on the <b>Check box</b> to activate (a checkmark is displayed) or deactivate the option.</p>                                                                                                                                                                                                                                                                                                                                                                   |
| 10 | <p><b>Number field:</b></p> <p>A <b>Number field</b> displays all possible number values for this function. Click on the small up/down arrows to increase/decrease the displayed number or click into the <b>Number field</b> and enter a number.</p> <p>Click on the big down arrow to get a calculator.</p>                                                                                                                                                                                                                                                                  |
| 11 | <p><b>Selection box:</b></p> <p>A <b>Selection box</b> displays the selected value/entry on the top. Click on the selection box to display all possible values/entries. Select one of the displayed values/entries.</p>                                                                                                                                                                                                                                                                                                                                                        |
| 12 | <p>Deactivated buttons, fields, selection boxes etc. are gray. It is not possible to use it in this situation. The software controls their use.</p>                                                                                                                                                                                                                                                                                                                                                                                                                            |

|    |                                                                                                                                                                                                                                                                                                                                                                                            |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | <p><b>Input box:</b></p> <p><b>Input boxes</b> allow to enter values into it. Click on the field and enter the value with the (on screen) keyboard.</p>                                                                                                                                                                                                                                    |
| 14 | <p><b>Table:</b></p> <p><b>Tables</b> provide an overview of a large amount of data. Some tables allow you to change/enter values or to perform a function by clicking on a table cell/row.</p>                                                                                                                                                                                            |
| 15 | <p><b>Option box:</b></p> <p>An <b>Option box</b> to select an option of a function. Click on the <b>Option box</b> to select the option.</p> <p><b>Option boxes</b> are organized normally in groups. Only one <b>Option box</b> of these group can be selected.</p>                                                                                                                      |
| 16 | <p><b>Filter Messages and error levels:</b></p> <p>Choose with the check boxes the <b>Event Trace</b> windows display text:</p> <ul style="list-style-type: none"> <li>• Grey - Displays message events</li> <li>• Yellow - Displays warning events</li> <li>• Orange - Displays critical events</li> <li>• Red - Displays fatals events</li> <li>• White - Flash auto messages</li> </ul> |
| 17 | <p><b>Module box of the instrument layout:</b></p> <p>A module box can be selected to show a window on the right hand side with a closer look of the module. There you can read out the position coordinates of each single field position.</p>                                                                                                                                            |
|    | <p><b>Tool tip:</b></p> <p>Some buttons, fields, selection boxes etc. display a <b>Tool tip</b> with additional information about it. To display it move your mouse pointer to the button, field, selection box etc. and wait a short time.</p>                                                                                                                                            |

Table 12–1: Service Software control elements

#### 12.1.4.2 General buttons

General buttons which are displayed on the top of **Service Software** and apply on dialogs or tabs:

| Button                                                                              | Description                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | The <b>Application Configuration</b> button opens the <b>Interface Selection</b> dialog.                                                                                                 |
|  | File / Folder Operation: <ul style="list-style-type: none"> <li>• Zip Logging Files</li> <li>• Open Logging Folder</li> <li>• Open Report Folder</li> <li>• Open Image Folder</li> </ul> |
|  | Power state of instrument.                                                                                                                                                               |

| Button                                                                            | Description                                                                                                                                                         |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | When <b>Stop</b> button is pressed, the Device stops immediately without finishing the last action or command. The device will terminate in a not initialized mode. |
|  | Shows the Online Help.                                                                                                                                              |
|  | Shows the copyright list for used software modules.                                                                                                                 |
|  | The <b>Exit</b> button quits the <b>Service Software</b> .                                                                                                          |
|  | Repeats the last used operation (e.g. updating the last query result of sensors, repeating the last command).                                                       |
|  | Repeats the last used operation automatically (delay time 2 s).                                                                                                     |
|  | Export of the General Dialog into a PDF.                                                                                                                            |

Table 12–2: General Buttons

#### 12.1.4.3 **Device is Busy** window

The **Device is Busy** window is displayed if a **Service Software** operation starts an instrument operation (e.g. movement of the Pipettor) and the instrument needs time to perform it.

| Button                                                                              | Description                                                                                                                     |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| <b>Cancel</b>                                                                       | The Device stops one finished the last action or command. The Device will terminate in a ready (initialized) condition.         |
|  | The Device stops immediately without finishing the last action or command. The device will terminate in a not initialized mode. |

Table 12–3: Functions of the **Device is Busy** window

#### 12.1.4.4 **Keyboard Control**

It is also possible to teach and calibrate the modules with a keyboard. The keyboard must be connected and the **Keyboard Control** check box in the module driver must be activated.

| Key                | Description        |
|--------------------|--------------------|
| -                  | Decrease step size |
| +                  | Increase step size |
| <b>Cursor down</b> | Y to front         |

| Key          | Description         |
|--------------|---------------------|
| Cursor left  | X left or close lid |
| Cursor right | X right or open lid |
| Cursor up    | Y to rear           |
| Page Up      | Z up                |
| Page down    | Z down              |

Table 12–4: Functions of **Keys**

12.1.5 Interface Selection



Click on **Application Configuration** button to open the **Interface Selection** window.

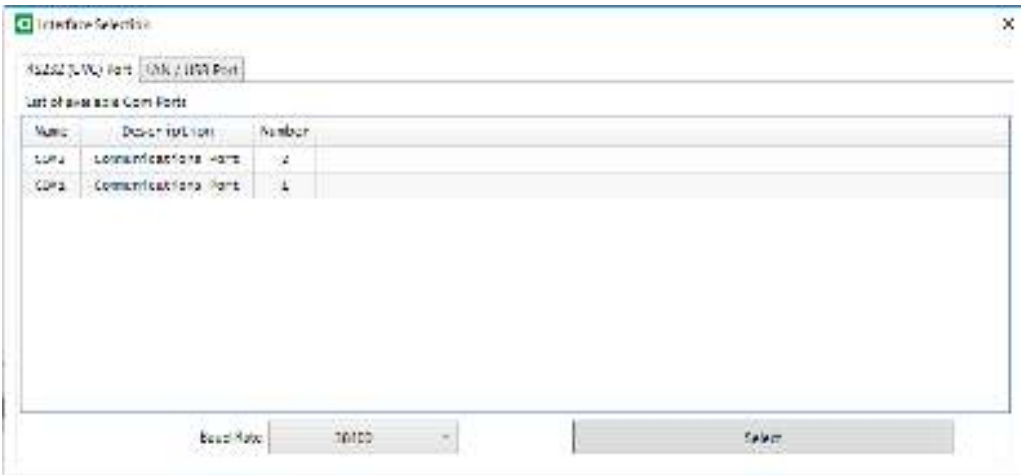


Figure 12–3: Interface Selection

| Tab or field                                                                        | Description                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RS232 (UVC) Port</b>                                                             | Tab button for RS232 (UVC) Port.                                                                                                                                                                 |
| <b>LAN / USB port</b>                                                               | Tab button for LAN / USB port.                                                                                                                                                                   |
| <b>List of available</b>                                                            | Com Ports: <ul style="list-style-type: none"> <li>• COM2</li> <li>• COM1</li> </ul> CopProt Stack Devices: <ul style="list-style-type: none"> <li>• USB1</li> <li>• WCF1 (Dispatcher)</li> </ul> |
| <b>Baud Rate</b>                                                                    | Choose between 9600, 38400, 115200.                                                                                                                                                              |
|  | Update the CopProt window.                                                                                                                                                                       |
| <b>Select</b>                                                                       | Choose the selected protocol.                                                                                                                                                                    |

Table 12–5: Description for **Interface Selection** window

12.1.6 Errors

Messages and errors will be displayed in the Event trace window as:

- Grey - Displays fatals events
- Yellow - Displays warning events



- Orange - Displays critical events
- Red - Displays fatal events.
- White - Flash auto messages.

### 12.1.7 General description of Calibration

#### NOTE

The following description is only a theoretical description. See the specific module for all the calibration procedures.

#### 12.1.7.1 Teach & Calibration

The **Teach & Calibration** window of a specific module in the **Navigation tree** displays a list with all possible teachable areas of the module.

##### 12.1.7.1.1 Buttons for Rotor Init Offset

The following table displays a list with all the **Buttons** used to calibrate the rotor (i.e. Incubator rotor, Washer rotor and Reader rotor).

| Column and function  |                                                                                     | Button name                 | Description                                                                                                                   |
|----------------------|-------------------------------------------------------------------------------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Relative Move</b> |  | <b>Arrow buttons</b>        | Rotates the rotor one step or more steps clockwise. The number of steps are displayed in the field below the buttons.         |
|                      |  |                             | Rotates the rotor one step or more steps counter clockwise. The number of steps are displayed in the field below the buttons. |
|                      |  | <b>Get current position</b> | Gets the current position and display it in the field.                                                                        |
| <b>Move to</b>       |  | <b>Move to</b>              | Moves the rotor to the entered position.                                                                                      |
| <b>Stored</b>        |  | <b>Stored position</b>      | Copies the stored teach position (displayed in the field) into the <b>Move to</b> field.                                      |
| <b>Option</b>        |  | <b>Option</b>               | Displays optional functions for the sub module (e.g. initialize, unpower).                                                    |
| <b>Key-board</b>     |  | <b>Keyboard</b>             | Activates the keyboard control for this axis.                                                                                 |
| <b>Save</b>          |                                                                                     | <b>Save</b>                 | Saves the current teach position into the module RAM.<br>Note that the RAM will be lost after power off the instrument.       |

## 12. Service Software for external-tanks configuration

| Column and function |  | Button name                        | Description                                                                                     |
|---------------------|--|------------------------------------|-------------------------------------------------------------------------------------------------|
| <i>Write to NVM</i> |  | <b>Write ALL Parameters to NVM</b> | All teach positions will be written into the non-volatile memory (NVM) of the affected modules. |
| <i>Write to NVM</i> |  | <b>Affected Modules</b>            | Shows the affected modules.                                                                     |
| <i>Summary</i>      |  |                                    | The summary list shows the results of the calibration steps.                                    |

Table 12–6: Button for Rotor Init Offset

### 12.1.7.1.2 Buttons for Cuvette Gripper Open Position

The following table displays a list with all the **Buttons** used to calibrate the Open Position of the Gripper Module.

| Column and function  |                                                                                     | Button name                        | Description                                                                                                                |
|----------------------|-------------------------------------------------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| <i>Relative Move</i> |  | <b>Arrow buttons</b>               | Moves the Gripper claws to close one step or more steps. The number of steps are displayed in the field below the buttons. |
|                      |  |                                    | Moves the Gripper claws to open one step or more steps. The number of steps are displayed in the field below the buttons.  |
|                      |  | <b>Get current position</b>        | Get the current position and display it in the field.                                                                      |
| <i>Move to</i>       |  | <b>Move to</b>                     | Moves the Gripper claws to the entered position.                                                                           |
| <i>Stored</i>        |  | <b>Stored position</b>             | Copies the stored teach position (displayed in the field) into the <i>Move to</i> field.                                   |
| <i>Option</i>        |  | <b>Option</b>                      | Displays optional functions for the sub module (e.g. initialize, unpower).                                                 |
| <i>Key-board</i>     |  | <b>Keyboard</b>                    | Activates the keyboard control for this axis.                                                                              |
| <i>Save</i>          |                                                                                     | <b>Save</b>                        | Saves the current teach position into the module RAM.<br>Note that the RAM will be lost after power off the instrument.    |
| <i>Write to NVM</i>  |                                                                                     | <b>Write ALL Parameters to NVM</b> | All teach positions will be written into the non-volatile memory (NVM) of the affected modules.                            |

| Column and function          |  | Button name             | Description                                                  |
|------------------------------|--|-------------------------|--------------------------------------------------------------|
| <a href="#">Write to NVM</a> |  | <b>Affected Modules</b> | Shows the affected modules.                                  |
| <a href="#">Summary</a>      |  |                         | The summary list shows the results of the calibration steps. |

Table 12–7: Button for Cuvette Gripper Open Position

### 12.1.8 General description of Parameter

The **Parameter** window of a specific module in the **Navigation tree** displays a list with all parameters of the module.

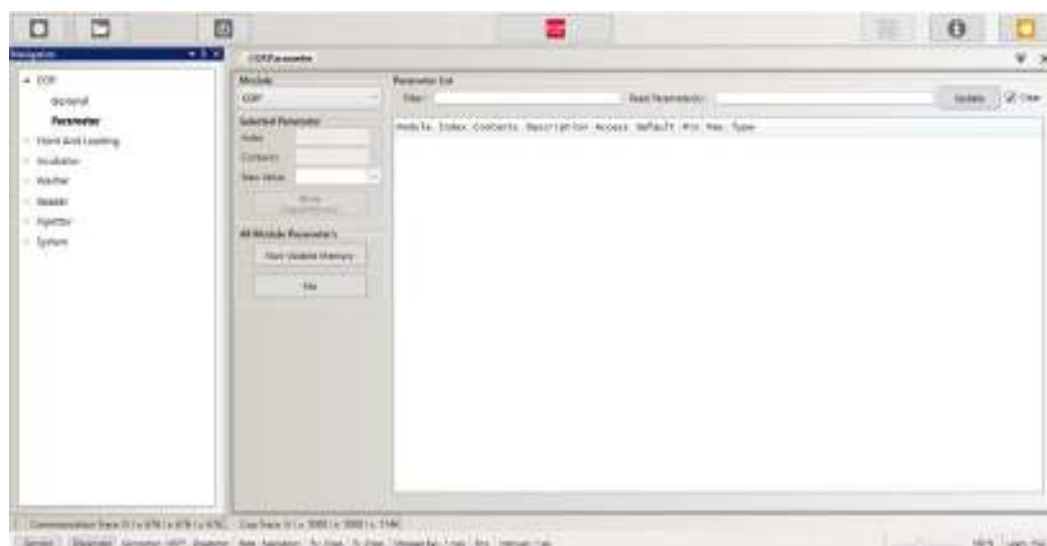


Figure 12–4: Example of COP - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

#### 12.1.8.1 **Module** group

| Function/<br>Button         | Description                          |
|-----------------------------|--------------------------------------|
| <a href="#">Module list</a> | List of all modules with parameters. |

Table 12–8: Functions of the **Module** group

### 12.1.8.2 *Selected Parameter* group

| Function/<br>Button | Description                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Index</i>        | Shows the index number of the parameter.                                                                                                                 |
| <i>Contents</i>     | Displays the selected content/value of the parameter.                                                                                                    |
| <i>New Value</i>    | Enter the new value of the selected parameter.                                                                                                           |
| <i>Write</i>        | Writes the new parameter value into the <b>Service Software</b> module memory (RAM).<br>When the instrument is switched off, the parameter will be lost. |

Table 12–9: Functions of the *Selected Parameter* group

### 12.1.8.3 *All Module Parameter's* group

| Function/<br>Button        | Description                                                                                                                                                                                                                                                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Non-Volatile Memory</b> | Use of current parameters of a instrument module. <ul style="list-style-type: none"> <li><b>Reload</b>: Reloads the parameters from the Non-Volatile Memory (NVM) of the module into the module RAM.</li> <li><b>Overwrite</b>: All parameters from the module RAM will be written into the Non-Volatile Memory (NVM) of the module.</li> </ul> |
| <b>File</b>                | Use of a parameter file of a instrument module. <ul style="list-style-type: none"> <li><b>Read from Device and Save</b>: loads a saved parameter file and send it to an instrument module.</li> </ul>                                                                                                                                           |

Table 12–10: Functions of the *All Module Parameter's* group

### 12.1.8.4 *Parameter List* group

| Function/<br>Button      | Description                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Filter</i>            | Filters the parameters table by entered text.                                                                                                                                                                                                                                                                                                                                                                                 |
| <i>Read Parameter(s)</i> | Allows to show all parameters or only selected parameters. <ul style="list-style-type: none"> <li>Show all parameters: Let the field empty.</li> <li>Show only selected parameters: Examples: <ul style="list-style-type: none"> <li>5,6,20: shows only parameters 5, 6, and 20</li> <li>2-10: shows parameters 2 to 10</li> <li>5,9-12,15,20,30-60: shows parameters 5, 9 to 12, 15, 20, and 30 to 60</li> </ul> </li> </ul> |
| <i>Update</i>            | Reads the parameters of the module and updates the parameter table.                                                                                                                                                                                                                                                                                                                                                           |

| Function/<br>Button | Description                                    |
|---------------------|------------------------------------------------|
| <b>Clear</b>        | Deletes all entries before updating the table. |

Table 12–11: Functions of the **Parameter List** group

| Table Column       | Description                                        |
|--------------------|----------------------------------------------------|
| <b>Module</b>      | Module name.                                       |
| <b>Index</b>       | Parameter number.                                  |
| <b>Contents</b>    | Parameter value.                                   |
| <b>Description</b> | Parameter name.                                    |
| <b>Access</b>      | Access level (Normal, Teach, ReadOnly).            |
| <b>Default</b>     | Default value of the parameter.                    |
| <b>Min</b>         | Allowed minimum value.                             |
| <b>Max</b>         | Allowed maximum value.                             |
| <b>Type</b>        | Parameter type (e.g. Int16: only positive values). |

Table 12–12: Columns of the table in the **Parameter List** group

## 12.2 COP Module

The COP (Communication and Organization Processor) is built into the instrument and controls all the other modules. The COP in the **Service Software** inquires information about the internal status (Worklist, COP drivers, CAN-bus system) of the COP and the PC communication.

## 12.2.1 General

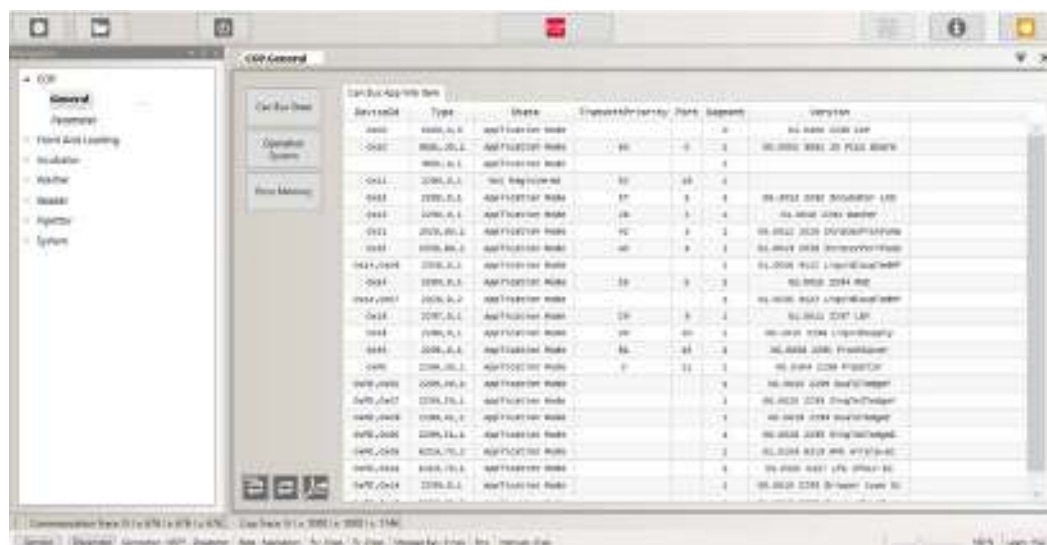


Figure 12–5: COP - General - Can Bus State

| Function/<br>Button     | Description                                                                                                                         |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <b>Can Bus State</b>    | Displays all CAN modules with device address, CAN address and if they registered or not.                                            |
| <b>Operation System</b> | Displays all operating system related CAN information (e.g. the number of accepted CAN messages for dispatching to the assemblies). |
| <b>Error Memory</b>     | Displays Error Messages.                                                                                                            |

Table 12–13: Functions of COP - Information

## 12. Service Software for external-tanks configuration

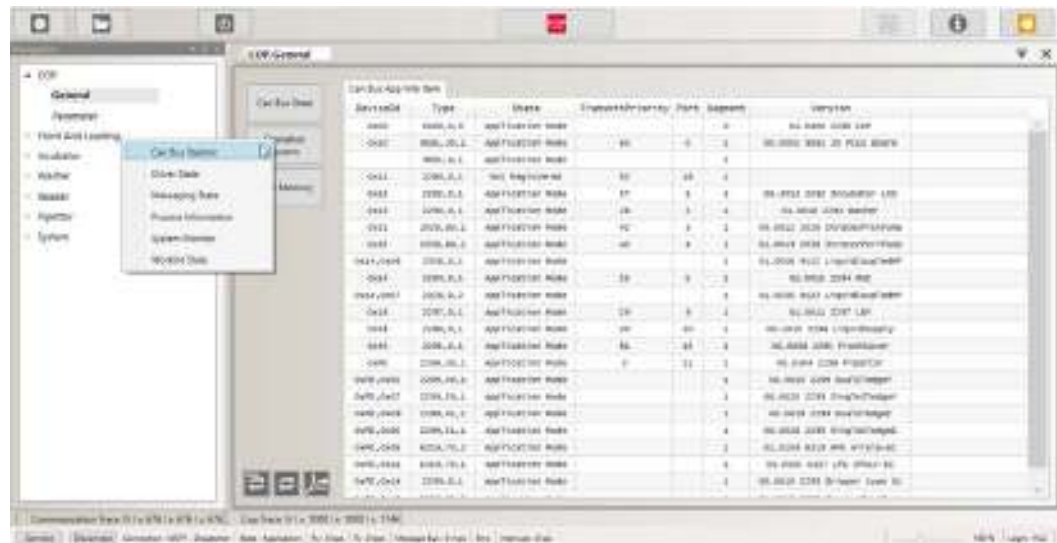


Figure 12–6: COP - General - Operation System

| Function/<br>Button        | Description                                                                                                                                                                   |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Can Bus Statistics</b>  | Displays all CAN modules with device address, CAN address and if they registered or not.                                                                                      |
| <b>Driver State</b>        | Displays COP internals (e.g. the number of hosted processes, the task calls per second and the state of work list processor).                                                 |
| <b>Messaging State</b>     | Displays IPC internals (e.g. the number of messages from one process sent to another).                                                                                        |
| <b>Process Information</b> | Displays all COP drivers, applications and some of their internal information (e.g. device address, process priority, number of task calls and the assigned Worklist queues). |
| <b>System Monitor</b>      | Gives information about CAN and communication voltages.                                                                                                                       |
| <b>Worklist State</b>      | Displays the work list.                                                                                                                                                       |

Table 12–14: Functions of COP - Operation System

## 12. Service Software for external-tanks configuration

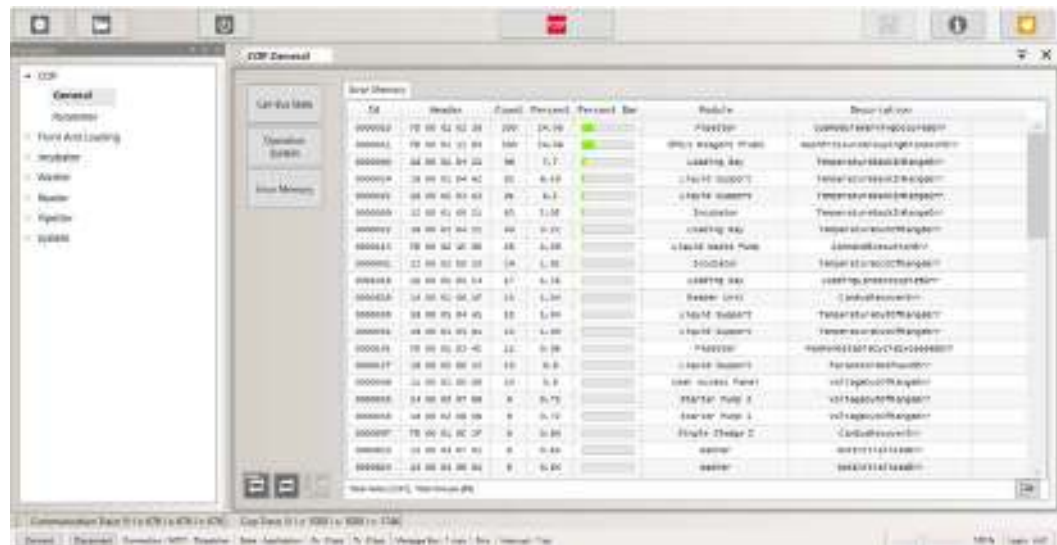


Figure 12-7: COP - Error Memory

| Function/<br>Button | Description                         |
|---------------------|-------------------------------------|
| <b>ID</b>           | Identification number of the error. |
| <b>Header</b>       | Hex numbers of the error.           |
| <b>Count</b>        | Error occurrence.                   |
| <b>Percent</b>      | Error rate in percent.              |
| <b>Percent bar</b>  | Displays the error rate in a bar.   |
| <b>Module</b>       | Shows the module name.              |
| <b>Description</b>  | Displays event name.                |

Table 12-15: Functions of COP - Error Memory



### 12.2.2 Parameter

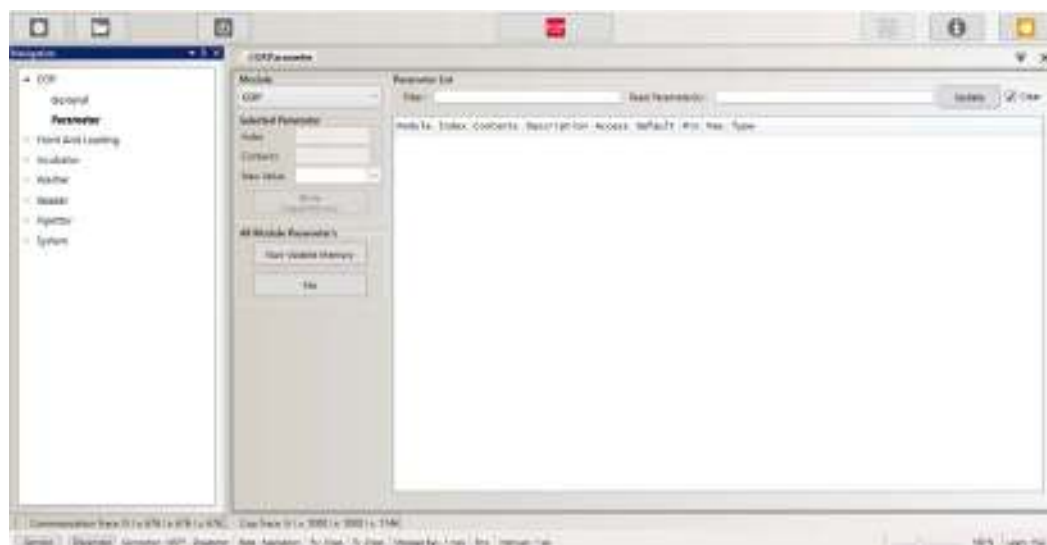


Figure 12–8: COP - Parameter

#### NOTE

##### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 12-11.

In the **Module** group, it is possible to select COP module.

## 12.3 Front and Loading

### 12.3.1 General



Figure 12–9: Front and Loading - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                                                                                             |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                                                                                                 |
| <b>Status</b>       | Displays information about the state of the module in the table: <ul style="list-style-type: none"> <li>• Front Cover</li> <li>• Loading Bay</li> <li>• Liquid Support</li> </ul> Select the desired part in the button menu.                                                  |
| <b>Sensors</b>      | Displays information about the state of the module sensors in the table: <ul style="list-style-type: none"> <li>• Front Cover</li> <li>• Loading Bay All</li> <li>• Loading Bay Temp. only</li> <li>• Liquid Support</li> </ul> Select the desired sensors in the button menu. |

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Actors</b>       | <p>Activates, deactivates etc. the actors of the module:</p> <ul style="list-style-type: none"> <li>• Cooling <ul style="list-style-type: none"> <li>○ On</li> <li>○ Off</li> </ul> </li> <li>• Stirrer <ul style="list-style-type: none"> <li>○ On</li> <li>○ Off</li> <li>○ Auto</li> </ul> </li> <li>• Buzzer <ul style="list-style-type: none"> <li>○ On</li> <li>○ Off</li> <li>○ Parametrized On, which opens the Buzzer On window, where there is the possibility to choose the Buffer sound type and repetition to perform</li> </ul> </li> <li>• WAP <ul style="list-style-type: none"> <li>○ Start at Routine Speed</li> <li>○ Start at Prime Speed</li> <li>○ Start Parameterized, which opens the Start WAP Timed window, where there is the possibility to choose the WAP speed and time to run</li> <li>○ Stop</li> </ul> </li> <li>• Status Light Yellow or Blue <ul style="list-style-type: none"> <li>○ On / Off</li> <li>○ Flashing slow / fast</li> </ul> </li> </ul> <p>Select the desired actor in the button menu.</p> |

Table 12–16: Functions of Front and Loading - General

### 12.3.2 Parameter

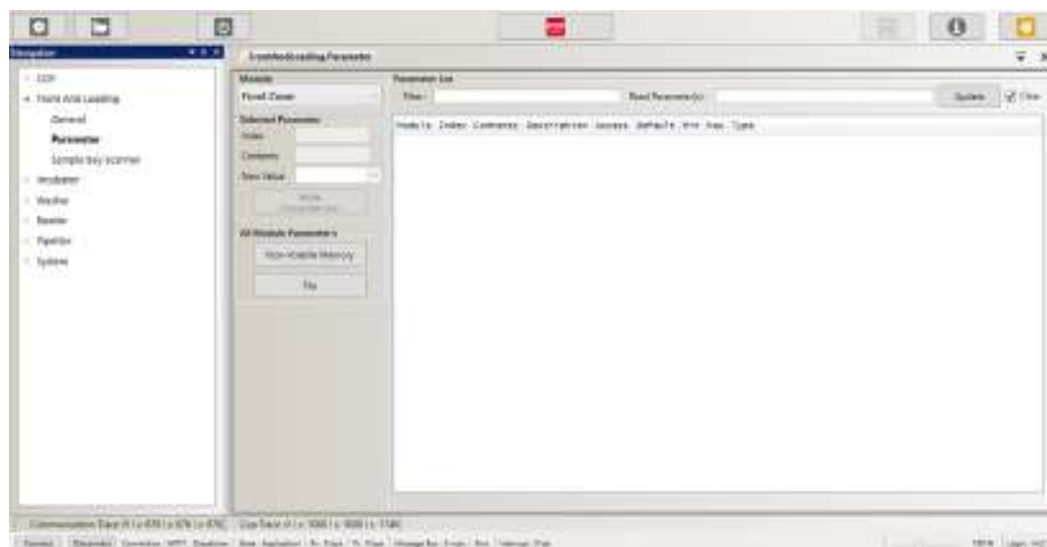


Figure 12–10: Front and Loading - Parameter

#### NOTE

##### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 12-11.

In the **Module** group, it is possible to select:

- Front Cover
- Loading Bay
- Liquid Support
- Wash Aspiration Pump
- User Access Panel

### 12.3.3 Sample Bay Scanner



Figure 12–11: Front and Loading - Sample Bay Scanner

#### 12.3.3.1 Scanner Report group

| Function/Button               | Description                                                                                                      |
|-------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>Activate Lane</b>          | Activates the selected sample lane to read the barcodes of an inserted rack.                                     |
| <b>Front Cover indication</b> | Check box to activate / deactivate front module LED indication of activated rack lane.                           |
| <b>Report</b>                 | Shows the read barcodes.<br>If you activate the <b>Extended</b> check box, the report shows further information. |
| <b>Sum of Occupations</b>     | Displays sum of detected racks                                                                                   |
| <b>Lane State</b>             | Shows the state of the lane.                                                                                     |

Table 12–17: Functions of the **Scanner Report** group

### 12.3.3.2 *Scanner Configuration* group

| Function/<br>Button                | Description                                                                                                                                                                                                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Code</b>                        | Select the barcode formats from the selection box that the customer will use: <ul style="list-style-type: none"> <li>• 2/5 Interleaved</li> <li>• Code 39</li> <li>• Code 93</li> <li>• Code 128, EAN 128</li> <li>• EAN 8 / 13</li> <li>• Codabar</li> <li>• UPCA, UPCE</li> <li>• Not Supported Code</li> </ul> |
| <b>Range</b>                       | Use this field to enter the bar code character length range value(s) for the selected bar code type. The acceptable bar code range is from 3 to 25 characters.                                                                                                                                                    |
| <b>Checksum</b>                    | If checksum is used with the code type selected, select either <b>Verify</b> , <b>Display</b> , or both.                                                                                                                                                                                                          |
| <b>Read</b>                        | Select this button to display the code types information stored on the EEPROM of the barcode scanner.                                                                                                                                                                                                             |
| <b>Write</b>                       | Select this button to write the code types information onto the EEPROM of the bar ode scanner.                                                                                                                                                                                                                    |
| <b>Additional Codabar Settings</b> | Select this to read monarch codabar symbologies as well as other types of codabar bar code symbologies. De-select this to better read nonmonarch codabar symbologies of poor quality.                                                                                                                             |

Table 12–18: Functions of the *Scanner Configuration* group

### 12.3.3.3 Barcode scanner configuration

In the **Service Software** at the navigation window, choose *Front and Loading* and the *Sample Bay Scanner* bracket.

- Click onto the **Read** button to read out the actual configuration.
  - Code 39 Range: 4 - 22
  - Code 128, EAN 128 Range: 4 - 22
  - Codabar Range: 4 - 22
- Clear all the check boxes that are not grayed out at the Checksum (**Verify** & **Display**).
- Click onto the **Write** button to save the settings.

### 12.3.3.4 Barcode reading test

In the **Service Software** at the navigation window choose *Front and Loading* and the *Sample Bay Scanner* bracket.

## 12. Service Software for external-tanks configuration

---

- a. Click onto the **Activate Lane - Activate** button to start the scanner.
- b. Insert a sample rack and check the correct reading of barcodes.
- c. Test each Sample Bay lane. For each lane, the barcode scanner has to be reactivated.

At the beginning, the background of the Lane State is gray.

When a rack is moved to the end stop, the status of the lane switches to green.

When a rack is removed, the status of the lane switches from green to red.

## 12.4 Incubator

### 12.4.1 General

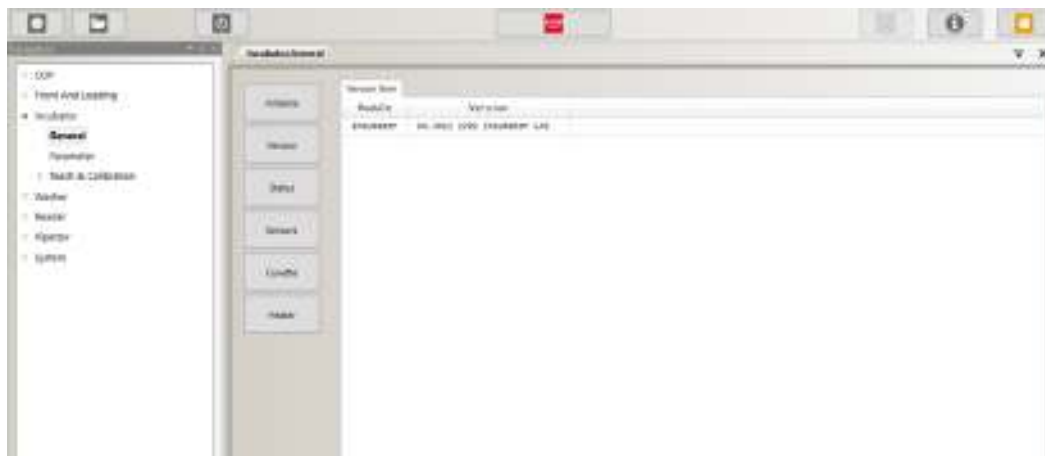


Figure 12–12: Incubator - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Status</b>       | Displays information about the state in the table.<br>Select the desired part in the button menu.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Sensors</b>      | Displays information about the state of the incubator sensors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Cuvette</b>      | <p>Move:</p> <ul style="list-style-type: none"> <li>• Cuvette 1..25 (Inner), moves the selected cuvettes to the chosen positions</li> <li>• Cuvette 26..57 (outer), moves the selected cuvettes to the chosen positions</li> <li>• Parameterized, opens the window Incubator cuvette move, where it is possible to choose the cuvette and in which position moving it</li> <li>• Fake move On</li> <li>• Fake move Off</li> </ul> <p>Check:</p> <ul style="list-style-type: none"> <li>• All cuvettes</li> <li>• Cuvette 1..25 (Inner), checks the chosen cuvettes.</li> <li>• Cuvette 26..57 (Outer), checks the chosen cuvettes.</li> <li>• Parameterized, opens the window Check cuvette, where there is the possibility to choose the cuvette position to check</li> </ul> |
| <b>Heater</b>       | Switch the Incubator Heater On or Off.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

Table 12–19: Functions of Incubator - General



### 12.4.2 Parameter

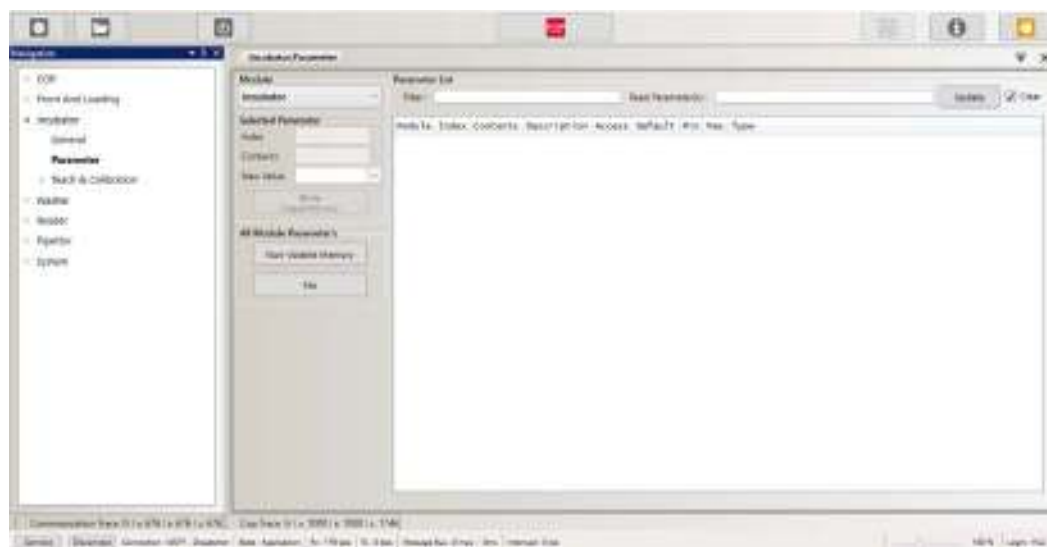


Figure 12–13: Incubator - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 12-11.

In the **Module** group, it is possible to select:

- Incubator
- COP

### 12.4.3 Teach & Calibration

#### 12.4.3.1 Incubator Rotor Calibration

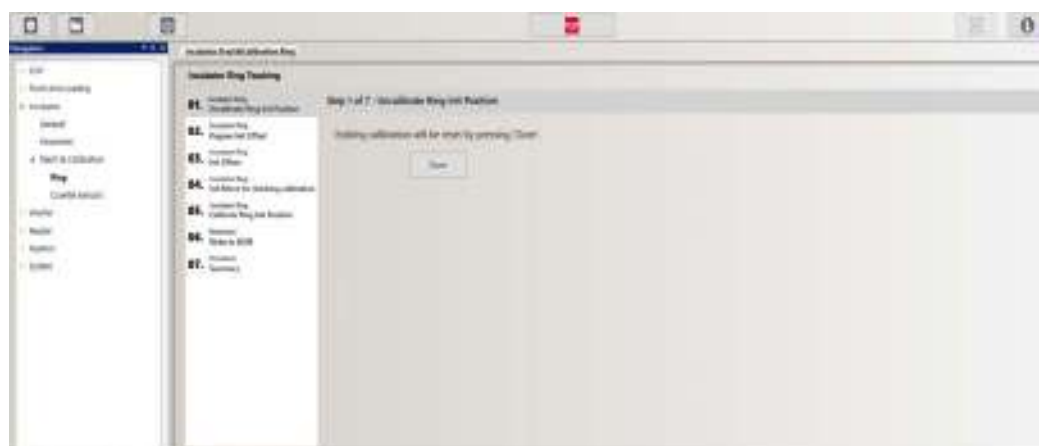


Figure 12–14: Incubator - Teach & Calibration - Ring

The Incubator Rotor Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

About the function of all the buttons, see Chapter 12.1.7.1.1 on page 12-9.

In the **Service Software** at the navigation window, choose **Incubator** then **Teach & Calibration** bracket and select **Ring**.

- Step 1**
  - a. Follow the instructions and then press the **Done** button to reset the calibration.
- Step 2**
  - b. Follow the instructions and then press the **Done** button to initialize the module.
- Step 3**
  - c. Move the Incubator rotor by the arrow buttons to align the slot on the rotor to the reference position of the Incubator frame.



**Figure 12–15: Incubator rotor, calibration position**

- d. Press **Save** button to save the position.
  - e. Follow the instructions and then press the **Done** button to initialize again the module.
- Step 5**
- f. Follow the instructions and then press the **Done** button to perform automatic calibration of the module.

**Step 6** g. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

**Step 7** h. The process is successful when, at the **Summarize** panel, all steps are in green.

### CAUTION

**A mis-calibration of the Incubator rotor might be necessary in order to have a proper cuvette transfer from the Incubator to the Washer and vice-versa.**

Perform a Cuvette Endurance Run (see Chapter 12.8.1.2.1 on page 12-63) and check that the cuvette transfer is smooth and without noises.

If needed, change the Incubator rotor calibration such to compensate the observed mis-alignment between the Gripper and Incubator rotor; an adjustment of 3-4 steps clockwise could be sufficient.

#### 12.4.3.2 Incubator Cuvette Sensors Calibration

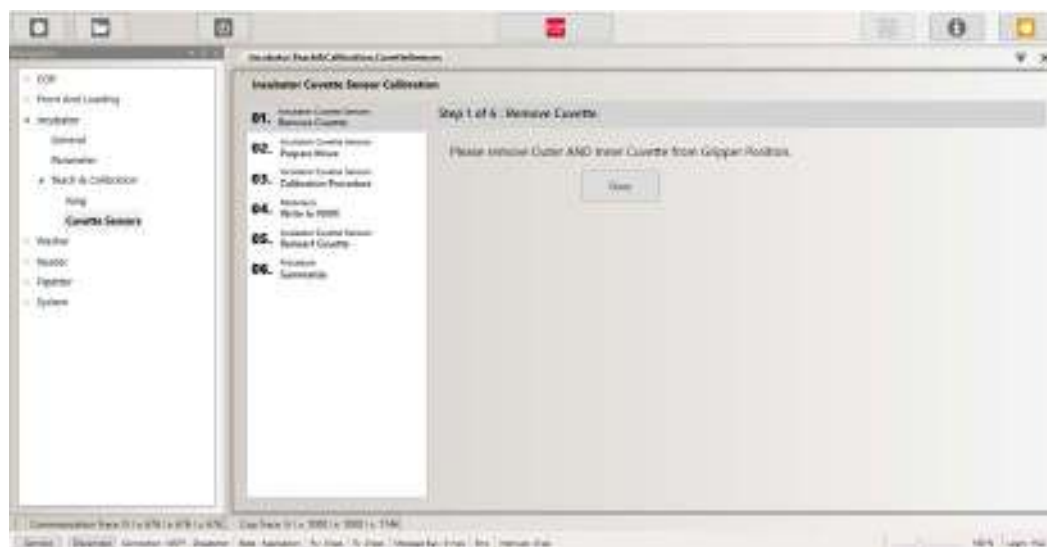


Figure 12–16: Incubator - Teach & Calibration - Cuvette Sensors

The Incubator Cuvette Sensors Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window, choose **Incubator** then **Teach & Calibration** bracket and select **Cuvette Sensors**.

**Step 1** a. Follow the instruction: remove the cuvettes from the gripper loading position (i.e. at positions 1 and 26), if any, and dispose of them; then press the **Done** button.

**Step 2** b. Follow the instruction by pressing **Done** button. The Incubator will move Cuvette 1 & 26 positions to the sensor.

- Step 3**
- c. Follow the instruction by pressing **Start** button to start the calibration procedure.
  - d. Check the results and proceed by pressing **Done** button.
- Step 4**
- e. Store the taught positions in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 5**
- f. Follow the instruction: insert new cuvettes at the gripper loading position (i.e. at positions 1 and 26), if were present, and press the **Done** button.
- Step 6**
- g. The process is successful when, at the **Summarize** panel, all steps are in green.

## 12.5 Washer

### 12.5.1 General

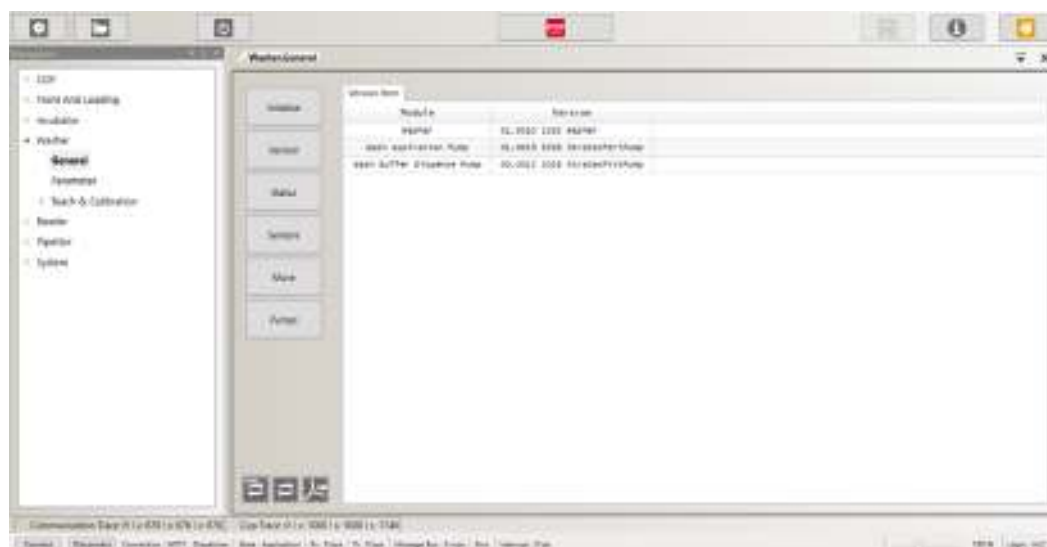


Figure 12–17: Washer - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                 |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                          |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                              |
| <b>Status</b>       | Displays information about the state in the table. Select the desired part in the button menu.                                                                                                              |
| <b>Sensors</b>      | Displays information about the state of the chosen module sensors: <ul style="list-style-type: none"> <li>• Washer</li> <li>• WAP</li> <li>• WBDP</li> </ul> Select the desired sensors in the button menu. |

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Move</b>         | <p>Ring moves the Washer rotor of:</p> <ul style="list-style-type: none"> <li>• 1 Step</li> <li>• 3 Steps</li> <li>• Parameterized opens the window Move Washer Ring where there is the possibility to choose of how many steps move the Washer rotor</li> <li>• Random Move</li> </ul> <p>Lift 1, 2 or 3 moves the Washer selected aspiration needle to the following positions:</p> <ul style="list-style-type: none"> <li>• to Home</li> <li>• to Bottom</li> <li>• Parameterized opens the window Move Lift 1, 2 or 3 where there is the possibility to choose at which position move the Lift and at which speed</li> </ul> <p>Lift Cycles opens the window Lifts Cycle Move where there is the possibility to parametrize all the movement of the Lift together or not. Set 0 cycles to perform a never-ending cycles movement.</p>                                       |
| <b>Pumps</b>        | <p>WAP:</p> <ul style="list-style-type: none"> <li>• Start at Routine Speed</li> <li>• Start at Prime Speed</li> <li>• Start Parameterized opens the Start WAP Timed window where there is the possibility to choose the WAP speed and time to run</li> <li>• Stop</li> </ul> <p>WBDP:</p> <ul style="list-style-type: none"> <li>• Load: <ul style="list-style-type: none"> <li>○ 1900 µl</li> <li>○ 2100 µl</li> <li>○ Parameterized opens the window Load WBDP where there is the possibility to choose the volume to load in uL</li> </ul> </li> <li>• Dispense Port 1, 2 or 3: <ul style="list-style-type: none"> <li>○ 700 µl</li> <li>○ 1200 µl</li> <li>○ Parameterized opens the window Dispense WBDP Port 1, 2 or 3 where there is the possibility to choose the volume to dispense in uL and if activate or not the tracking</li> </ul> </li> <li>• Empty</li> </ul> |

Table 12–20: Functions of Washer - General

### 12.5.2 Parameter

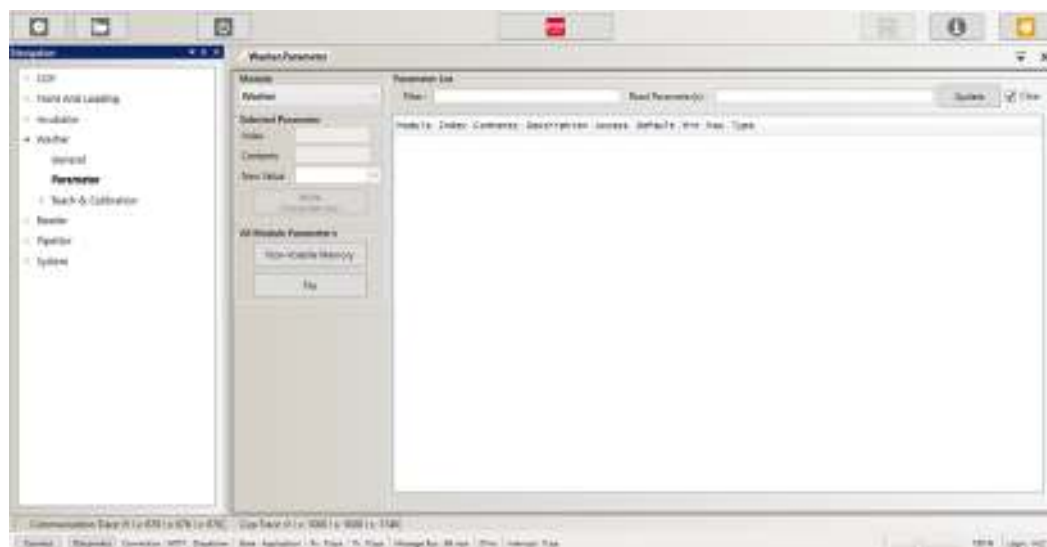


Figure 12–18: Washer - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 12-11.

In the **Module** group, it is possible to select:

- Washer
- Wash Aspiration Pump
- Wash Buffer Dispense Pump
- COP

### 12.5.3 Teach & Calibration

#### 12.5.3.1 Washer Rotor Calibration

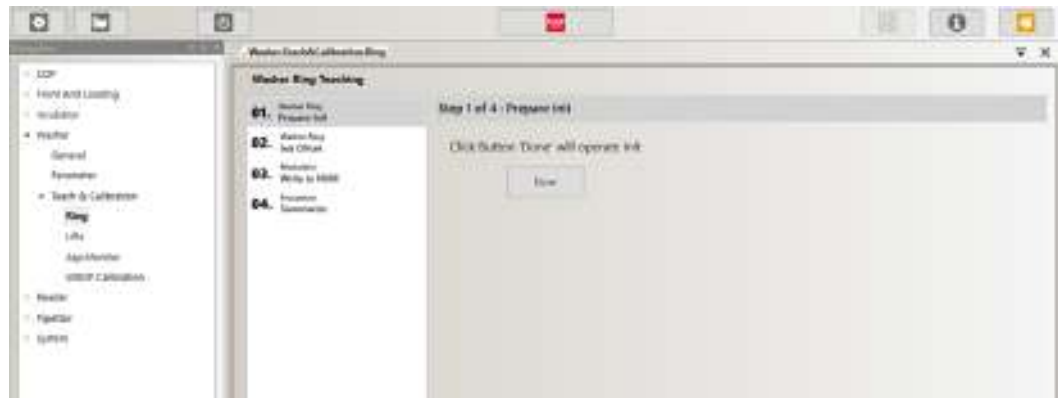


Figure 12–19: Washer - Teach & Calibration - Ring

The Washer Rotor Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

About the function of all the buttons, see Chapter 12.1.7.1.1 on page 12-9.

In the **Service Software** at the navigation window, choose **Washer** then the **Teach & Calibration** bracket and select **Ring**.

#### Step 1

- Follow the instructions and then press the **Done** button.

#### Step 2

- Move the Washer rotor by the arrow buttons, to align the slot on the rotor to the reference position on the Washer base.



Figure 12–20: Washer rotor, calibration position

- Press **Save** button to save the position.



**Step 3**

- d. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

**Step 4**

- e. The process is successful when, at the **Summarize** panel, all steps are in green.

## 12.5.3.2 Washer SIPAL Lifts Calibration

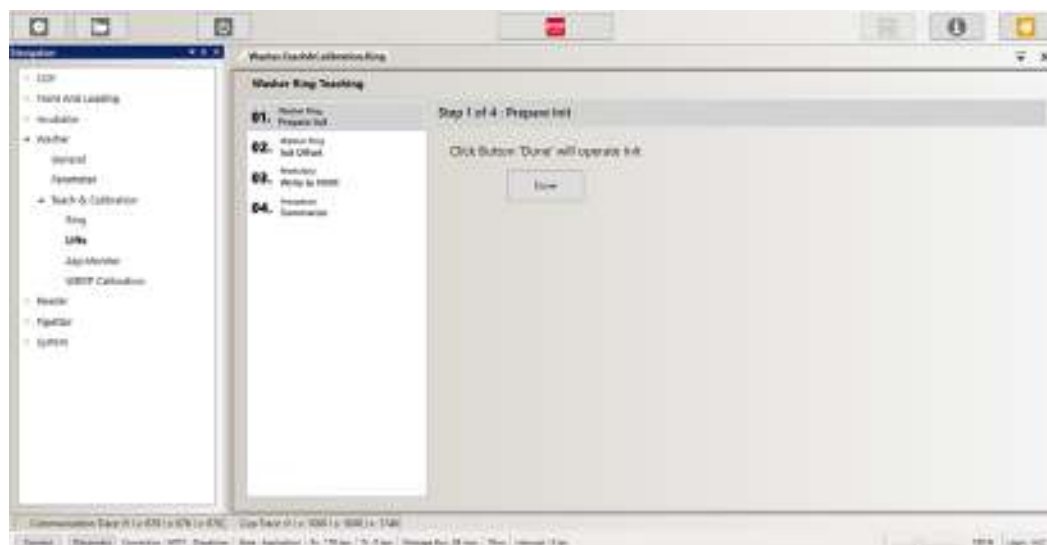


Figure 12–21: Washer - Teach &amp; Calibration - Lifts

The Washer SIPAL Lift Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software**, at the navigation window, choose Washer then the **Teach & Calibration** bracket and select **Lifts**.

- Step 1**
  - a. Follow the instructions and then press the **Done** button.
- Step 2**
  - b. Press the **Start** button; the calibration procedure starts automatically.
  - c. When the **Processing** window closes without failure message, check the results and press the **Done** button.
- Step 3**
  - d. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 4**
  - e. The process is successful when, at the **Summarize** panel, all steps are in green.

## 12.5.3.3 Washer Aspiration Monitoring Sensors Calibration

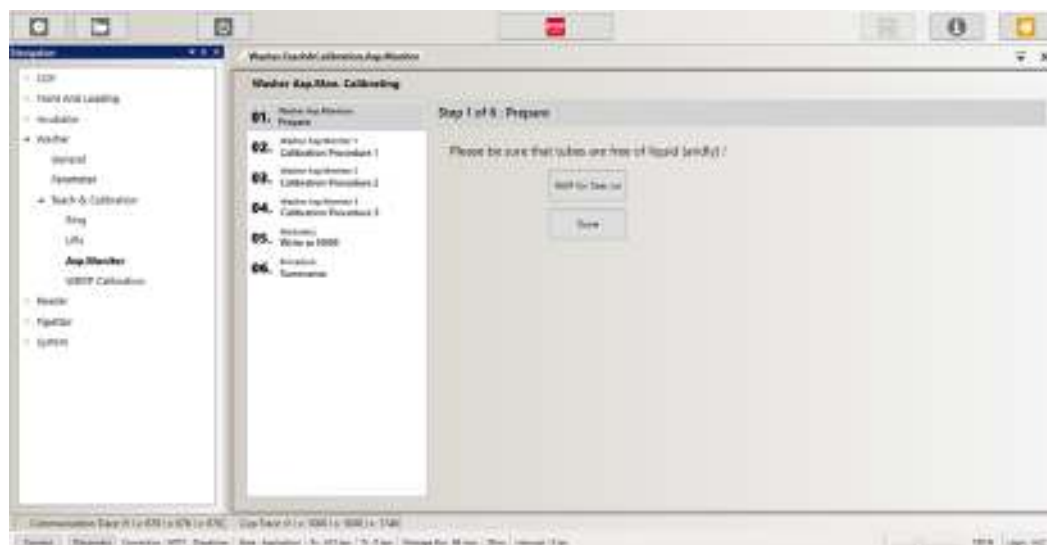


Figure 12–22: Washer - Teach &amp; Calibration - Asp. monitor

The Washer Aspiration Monitoring Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software**, at the navigation window, choose **Washer** then the **Teach & Calibration** bracket and select the **Washer Asp. Mon.** bracket.:

- Step 1**
  - a. Press the **WAP for 5sec** on button, to ensure that tubes are free of liquid.
  - b. When the Processing window closes, press the **Done** button.
- Step 2**
  - c. Press the **Start** button; the calibration procedure for aspiration monitor 1 starts automatically.
  - d. When the **Processing** window closes without failure message, check the result and press the **Done** button.
- Step 3**
  - e. Press the **Start** button; the calibration procedure for aspiration monitor 2 starts automatically.
  - f. When the **Processing** window closes without failure message, check the result and press the **Done** button.
- Step 4**
  - g. Press the **Start** button; the calibration procedure for aspiration monitor 3 starts automatically.
  - h. When the **Processing** window closes without failure message, check the result and press the **Done** button.
- Step 5**
  - i. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 6**
  - j. The process is successful when, at the **Summarize** panel, all steps are in green.

#### 12.5.3.4 WBDP Calibration

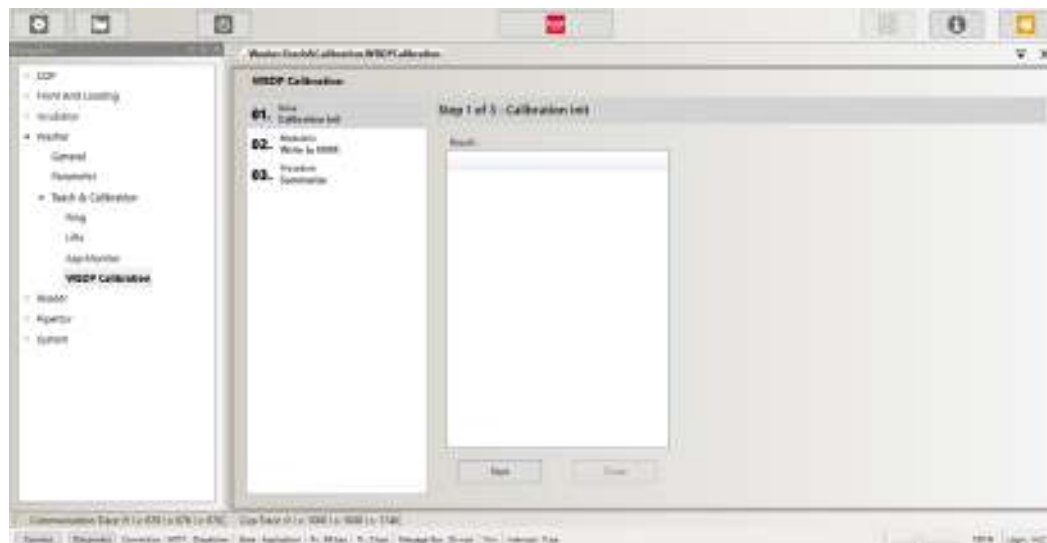


Figure 12–23: Washer - Teach & Calibration - WBDP Calibration

The WBDP Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window choose **Washer** and the **Teach & Calibration** bracket and select **WBDP Calibrating** bracket.

##### Step 1

- a. Press the **Start** button; the calibration procedure for WBDP starts automatically.
- b. When the **Processing** window closes without failure message, press the **Done** button.

##### Step 2

- c. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.

##### Step 3

- d. The process is successful when, at the **Summarize** panel, all steps are in green.

## 12.6 Reader

### 12.6.1 General

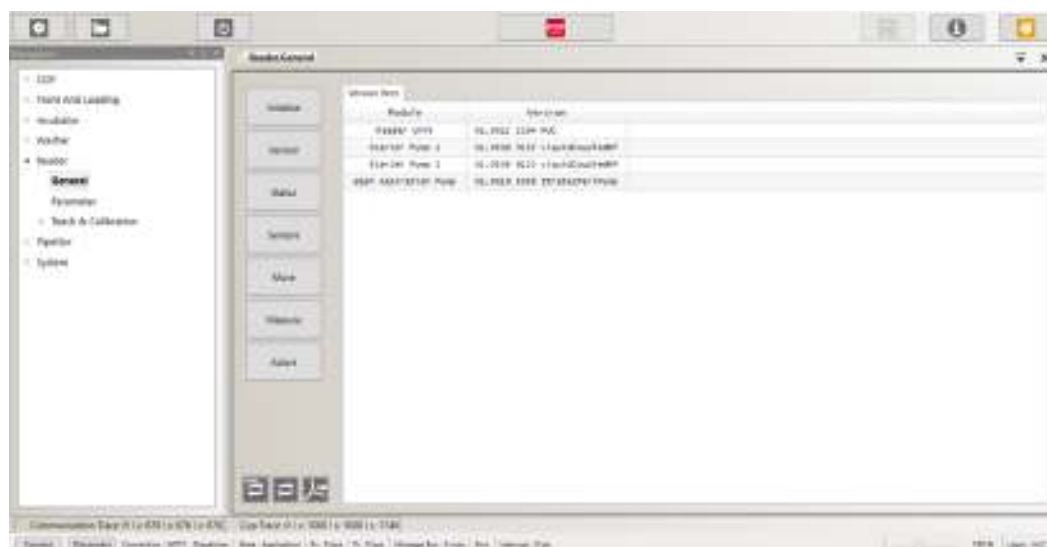


Figure 12-24: Reader - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                                                                                                                                                                                                                                                     |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Status</b>       | Displays information about the state in the table. Select the desired part in the button menu.                                                                                                                                                                                                                                                                                                                                         |
| <b>Sensors</b>      | Read displays information about the state of the module sensors in the table: <ul style="list-style-type: none"> <li>• Reader Unit</li> <li>• Starter 1 Pump</li> <li>• Starter 2 Pump</li> </ul> Check displays instruction to perform the check of the module sensors: <ul style="list-style-type: none"> <li>• Starter 1 Bottle Sensor</li> <li>• Starter 2 Bottle Sensor</li> </ul> Select the desired sensors in the button menu. |

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Move</b>         | <p>Rotor to moves the Reader rotor to the following positions:</p> <ul style="list-style-type: none"> <li>• Load / Unload</li> <li>• Dark Count</li> <li>• Starter 1</li> <li>• Starter 2 / RLU</li> <li>• Aspirate</li> </ul> <p>Lift moves the Reader aspiration needle to the following positions:</p> <ul style="list-style-type: none"> <li>• to Home</li> <li>• to Bottom</li> <li>• Cycles opens the window Lifts Cycle, where there is the possibility to cycle the movement of the Lift</li> </ul>                                                                                                                                                                                                                                                                                                                                              |
| <b>Measure</b>      | <p>Choose the Dark Count reading length and in which position perform the reading:</p> <ul style="list-style-type: none"> <li>• At Dark Count Position <ul style="list-style-type: none"> <li>◦ 1 to 5 sec</li> </ul> </li> <li>• At Current Position <ul style="list-style-type: none"> <li>◦ 1 to 5 sec</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>Actors</b>       | <p>Select the desired actor in the button menu.</p> <p>Tube Heating switches the heating of the Starters tubings:</p> <ul style="list-style-type: none"> <li>• Both On / Off</li> <li>• Starter 1 On / Off</li> <li>• Starter 2 On / Off</li> </ul> <p>High voltage:</p> <ul style="list-style-type: none"> <li>• Switch On or Off</li> </ul> <p>WAP:</p> <ul style="list-style-type: none"> <li>• Start at Routine Speed</li> <li>• Start at Prime Speed</li> <li>• Start Parameterized opens the Start WAP Timed window, where there is the possibility to choose the WAP speed and time to run</li> <li>• Stop</li> </ul> <p>Starter Dispense Pump 1 and 2 (SP1 and SP2):</p> <ul style="list-style-type: none"> <li>• Fill to 400 µl or Parameterized</li> <li>• Dispense 200 µl or 400 µl or Parameterized</li> <li>• Empty SP 1 or SP 2</li> </ul> |

Table 12–21: Functions of Reader - General



### 12.6.3 Teach & Calibration

### 12.6.3.1 Reader Rotor Calibration

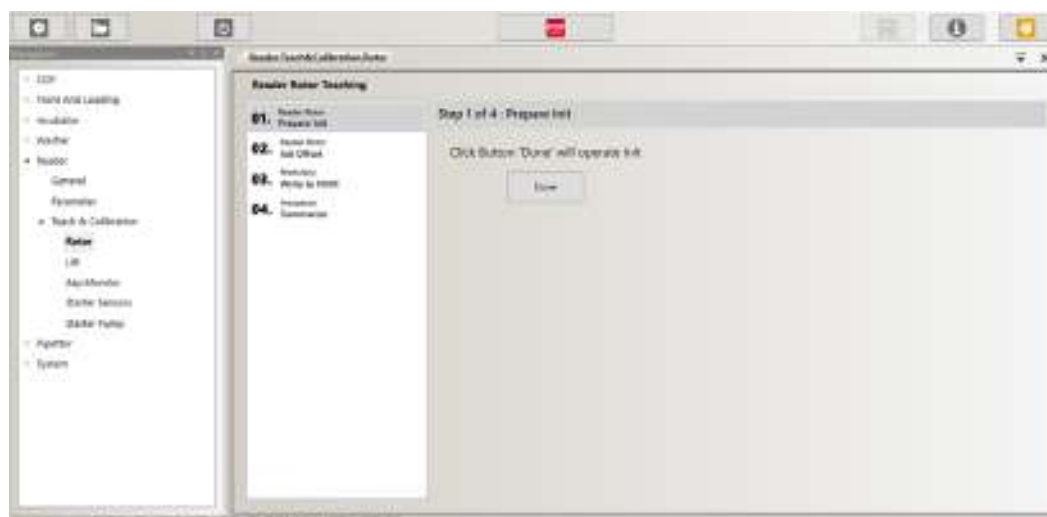


Figure 12-26: Reader - Teach & Calibration - Rotor

The Reader Rotor Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

About the function of all the buttons, see Chapter 12.1.7.1.1 on page 12-9.

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select **Rotor**.

### Step 1

- a. Press the **Done** button to perform the init.

## Step 2

- b. Move the Reader rotor by the arrow buttons to align the teaching point to the reference position in the Reader cover.



Figure 12-27: Reader rotor, calibration position

- c. Press **Save** button to save the position.



**Step 3** d. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

**Step 4** e. The process is successful when, at the **Summarize** panel, all steps are in green.

### 12.6.3.2 Reader SIPAL Lift Calibration

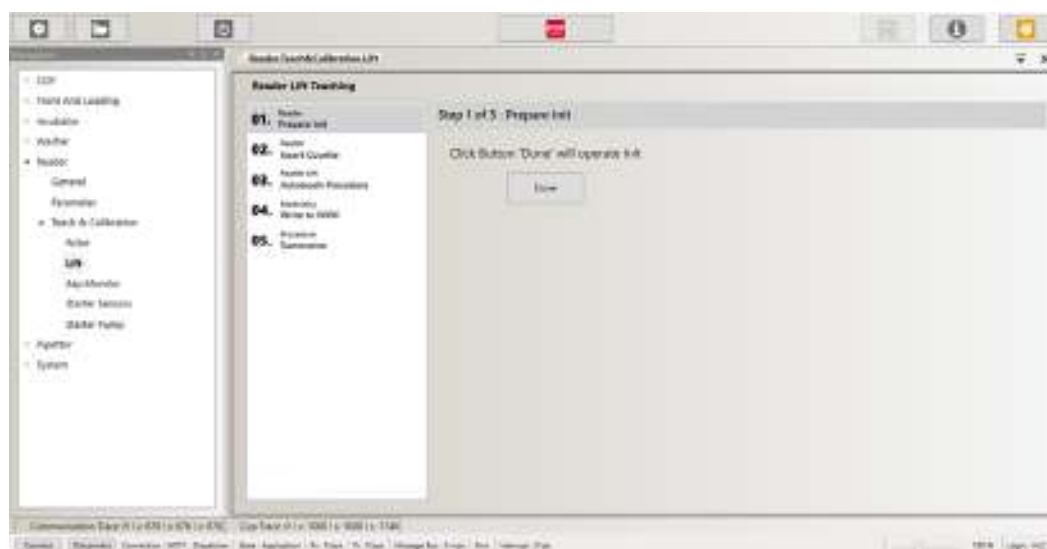


Figure 12–28: Reader - Teach & Calibration - Lift

The Reader SIPAL Lift Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window, choose **Reader** then **Teach & Calibration** bracket and select **Lift**.

**Step 1** a. Follow the instructions and press the **Done** button.

**Step 2** b. Place an empty cuvette in the Reader loading position, then press the **Done** button.

**Step 3** c. Press the **Start** button; the calibration procedure starts automatically.  
d. When the **Processing** window closes without failure message, press the **Done** button.

**Step 4** e. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

**Step 5** f. The process is successful when, at the **Summarize** panel, all steps are in green.

## 12.6.3.3 Reader Aspiration Monitoring Sensors Calibration

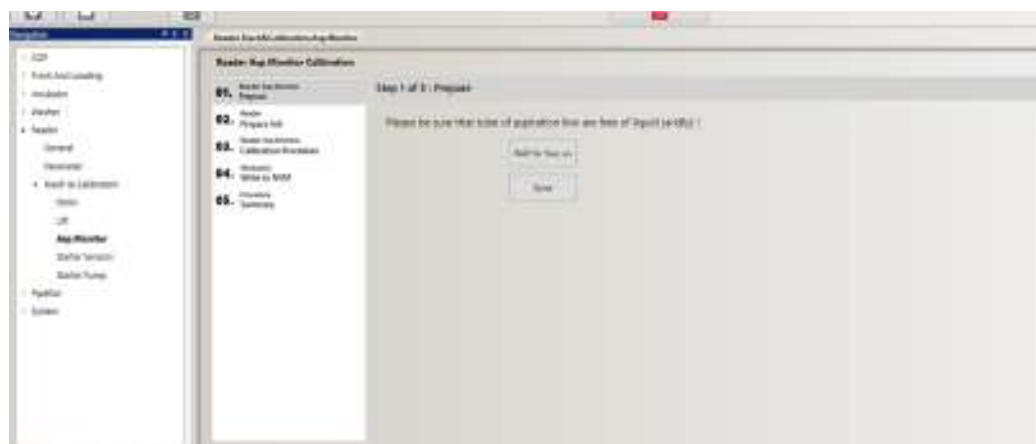


Figure 12–29: Reader - Teach &amp; Calibration - Asp. monitor

The Reader Aspiration Monitoring Sensors Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select the **Asp.Monitor**.

- Step 1**
  - a. Press the **WAP for 5sec** on button, to ensure that tubes are free of liquid.
  - b. When the **Processing** window closes, press the **Done** button.
- Step 2**
  - c. Press the **Done** button, to ensure the system will initialize the Reader.
- Step 3**
  - d. Press the **Start** button; the calibration procedure starts automatically.
  - e. When the **Processing** window closes without failure message, press the **Done** button.
- Step 4**
  - f. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 5**
  - g. The process is successful when, at the **Summarize** panel, all steps are in green.

## 12.6.3.4 Starter Sensors Calibration

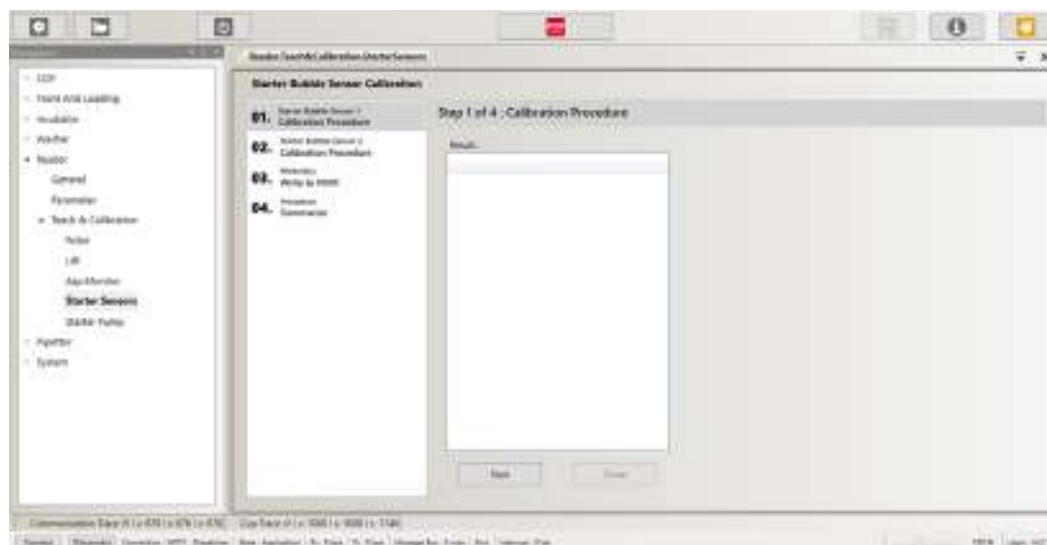


Figure 12–30: Reader - Teach &amp; Calibration - Starter Sensors

The Starter Sensors Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select the **Starter Sensors**.

**Step 1**

- a. Press the **Start** button, to start the procedure.
- b. Follow the instruction at the pop-up and press **OK** button.



Figure 12–31: Sensor calibration pop-up for Starter 1

- c. When the **Processing** window closes, press the **Done** button.

## 12. Service Software for external-tanks configuration

### Step 2

- d. Press the **Start** button, to start the procedure.
- e. Follow the instruction at the pop-up and press **OK** button.

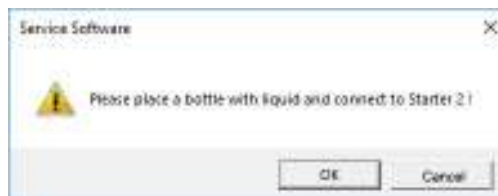


Figure 12-32: Sensor calibration pop-up for Starter 2

- f. When the **Processing** window closes without failure message, press the **Done** button.

### Step 3

- g. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

### Step 4

- h. The process is successful when, at the **Summarize** panel, all steps are in green.

### 12.6.3.5 Starter Pumps Calibration

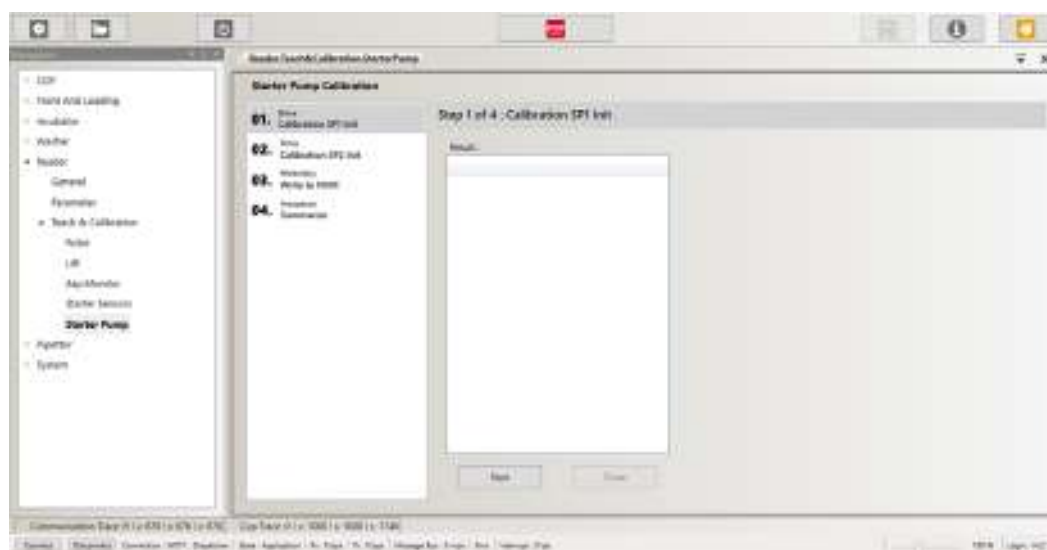


Figure 12-33: Reader - Teach & Calibration - Starter Pump

The Starter Pumps Sensors Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select the **Starter Pump**.

### Step 1

- a. Click on the **Start** button to start the automatic calibration of the SDP1.
- b. Check the results and proceed by pressing **Done** button.

**Step 2**

- c. Click on the **Start** button to start the automatic calibration of the SDP2.
- d. Follow the instruction by pressing **Done** button.

**Step 3**

- e. Store the taught positions in the EEPROM by selecting **Write ALL Parameters to NVM** button.

**Step 4**

- f. The process is successful when, at the **Summarize** panel, all steps are in green.

## 12.7 Pipettor

### 12.7.1 General

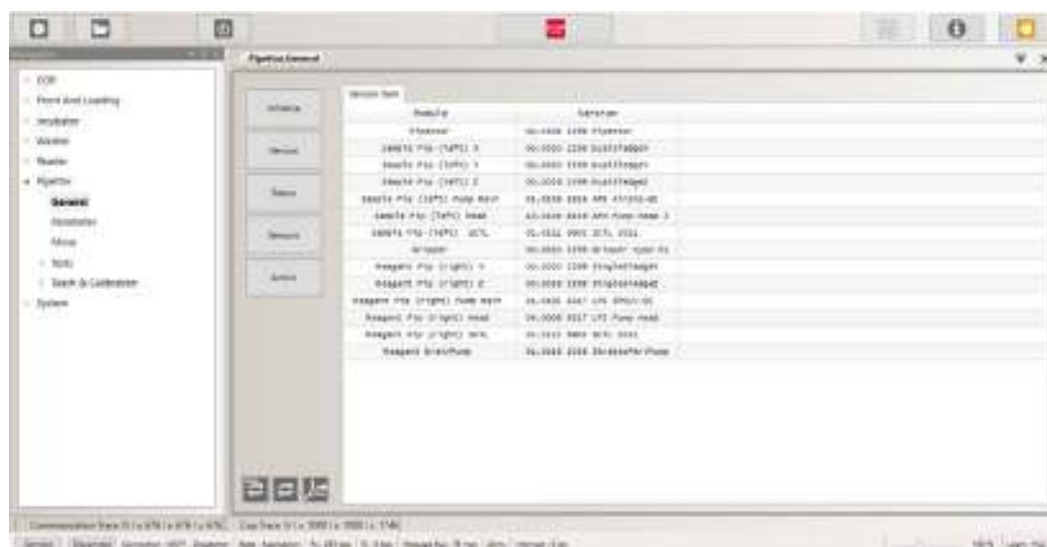


Figure 12–34: Pipettor - General - Version

| Function/<br>Button | Description                                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialize</b>   | Initializes the modules or controlled modules.<br>Select all sub module or a single sub module in the button menu.                                                                                                      |
| <b>Version</b>      | Displays information about the firmware versions in the table.                                                                                                                                                          |
| <b>Status</b>       | Displays information about the state in the table. Select the desired part in the button menu.                                                                                                                          |
| <b>Sensors</b>      | Displays information about the state of the Pipettor Module sensors in the table.                                                                                                                                       |
| <b>Actors</b>       | LWP:<br><ul style="list-style-type: none"> <li>Switch On or Off</li> </ul> Cuvette Gripper:<br><ul style="list-style-type: none"> <li>Open or Close the claws</li> </ul> Select the desired sensors in the button menu. |

Table 12–22: Functions of Pipettor - General

### 12.7.2 Parameter

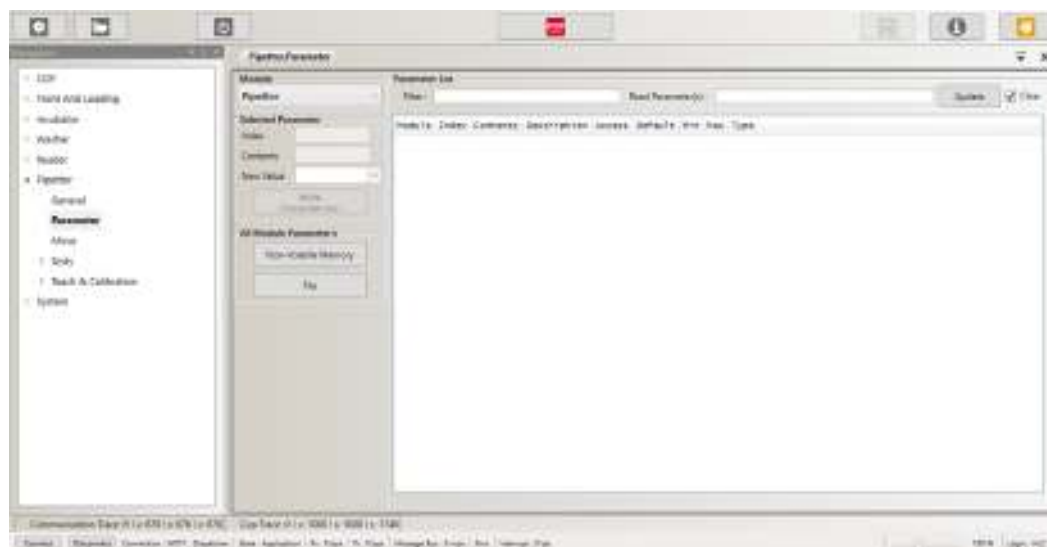


Figure 12-35: Pipettor - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 12-11.

In the **Module** group, it is possible to select:

- Pipettor
- Liquid Waste Pump
- Gripper Module
- AirPip Sample Probe
- Single Sledge Y
- Single Sledge Z
- SPOLV Reagent Probe
- Dual Sledge Y
- Dual Sledge Z

### 12.7.3 Move

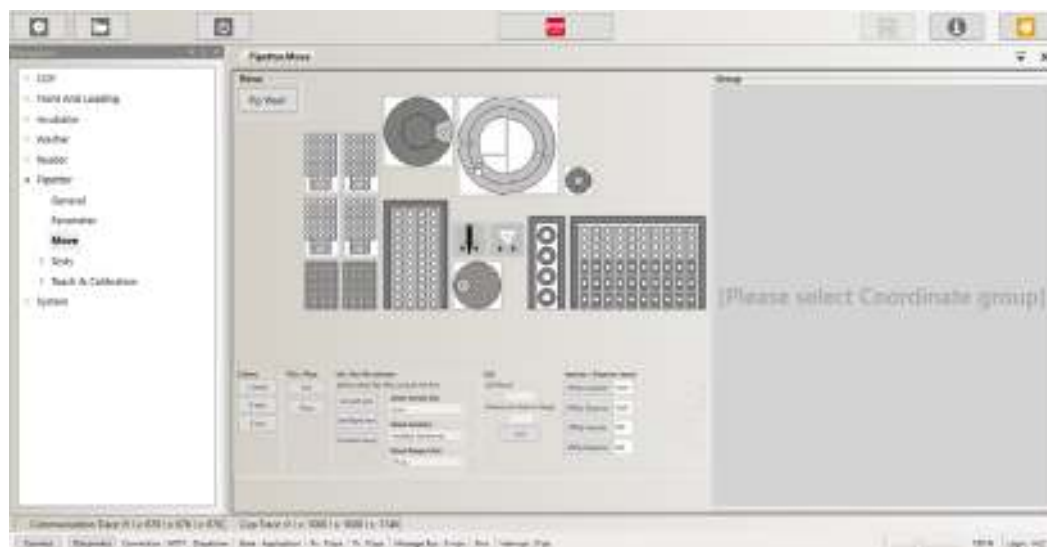


Figure 12–36: Pipettor - Move

#### NOTE

Remember to Initialize the Pipettors before open the **Move** tab.

The **Move** tab shows all the positions, which can be reach by the Sample pipettor, SPOLV and Gripper Module.

Follow the steps to move the Pipettor arms to a desired position:

- Tap on the desired instrument position (e.g. Sample bay, Incubator pipetting position etc.). All possible positions will shown as detailed figure on the right side.
- Double tap on the desired position of the detailed figure on the right side. The specific module moves to the required position.
- Perform an action (Pick / Place cuvette or disposable tip, Aspirate / Dispense liquid, LLD test ect).

#### 12.7.3.1 Rinse group

| Function/<br>Button | Description               |
|---------------------|---------------------------|
| <b>Pip Wash</b>     | Rinses / washes the SPOLV |

Table 12–23: Functions of the **Rinse** group



12.7.3.2 **Z Move** group**NOTE****Collision of Pipettor**

Be aware that the Z-move will happen at the last XY-position of the Pipettor.

| Function/<br>Button | Description                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------|
| <b>Z travel</b>     | Moves the Pipettor arms to its Z travel position (Z move happens at last XY-position of the Pipettor). |
| <b>Z start</b>      | Moves the Pipettor arms to its Z start position.                                                       |
| <b>Z max</b>        | Moves the Pipettor arms to its Z max position.                                                         |

Table 12–24: Functions of the **Z Move** group12.7.3.3 **Pick / Place** group

| Function/<br>Button | Description                                                             |
|---------------------|-------------------------------------------------------------------------|
| <b>Pick</b>         | Picks up cuvette or disposable tip, depending on the Left Arm position. |
| <b>Place</b>        | Places a cuvette or disposable tip to a desired position.               |

Table 12–25: Functions of the **Pick / Place** group12.7.3.4 **Init / Rack file selection** group

| Function/<br>Button   | Description                             |
|-----------------------|-----------------------------------------|
| <b>Init Left Arm</b>  | Initializes Left arm.                   |
| <b>Init Right Arm</b> | Initializes Right arm.                  |
| <b>Init Both Arms</b> | Initializes both arms to the same time. |

Table 12–26: Functions of the **Init** group

| Function/<br>Button       | Description                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>Select Sample Rack</b> | Select registered .rac file for Sample bay (to open the drop down field, a init of the left arm must be carried out). |

## 12. Service Software for external-tanks configuration

| Function/<br>Button        | Description                                                                                                               |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Select Ancillary</b>    | Select registered .rac file for Ancillary bay (to open the drop down field, a init of the right arm must be carried out). |
| <b>Select Reagent Rack</b> | Select registered .rac file for Reagent bay (to open the drop down field, a init of the right arm must be carried out).   |

Table 12–27: Functions of the **Rack file selection** group

### 12.7.3.5 **LLD** group

| Function/<br>Button     | Description                                                                                                                                                                                                                                                                                     |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LLD Result</b>       | Shows the results of the LLD test performed, depending on the Pipettor arm used: <ul style="list-style-type: none"> <li>Left arm position, the Sample pipettor performs a barometric check (B-LLD)</li> <li>Right arm position, the SPOLV Module performs a capacitive check (C-LLD)</li> </ul> |
| <b>Distance to Zmax</b> | Shows the Liquid Level relative distance to Zmax expressed in steps, after the LLD test performed.                                                                                                                                                                                              |
| <b>LLD</b>              | Starts the Liquid Level Detection with the last Pipettor arm moved into the dedicated position.                                                                                                                                                                                                 |

Table 12–28: Functions of the **LLD** group

### 12.7.3.6 **Aspirate / Dispense** group

| Function/Button       | Description                                                                                                                                                                                                                                             |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>SPOLV Aspirate</b> | Insert the amount of steps to aspirate by the SPOLV in the box, or moving the arrow left (to decrease) or right (to increase).<br>Press the button <b>SPOLV Aspirate</b> to start the aspiration of the entered steps at the SPOLV.                     |
| <b>SPOLV Dispense</b> | Insert the amount of steps to dispense by the SPOLV in the box, or moving the arrow left (to decrease) or right (to increase).<br>Press the button <b>SPOLV Dispense</b> to start the dispensation of the entered steps at the SPOLV.                   |
| <b>ZSPip Aspirate</b> | Insert the amount of steps to aspirate by the Sample pipettor in the box, or moving the arrow left (to decrease) or right (to increase).<br>Press the button <b>ZSPip Aspirate</b> to start the aspiration of the entered steps at the Sample pipettor. |

| Function/Button       | Description                                                                                                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ZSPip Dispense</b> | Insert the amount of steps to dispense by the ZSPip in the box, or moving the arrow left (to decrease) or right (to increase).<br>Press the button <b>ZSPip Dispense</b> to start the dispensation of the entered steps at the Sample pipettor. |

Table 12–29: Functions of the **Aspirate / Dispense** group

### 12.7.4 Test

#### 12.7.4.1 LLD Endurance

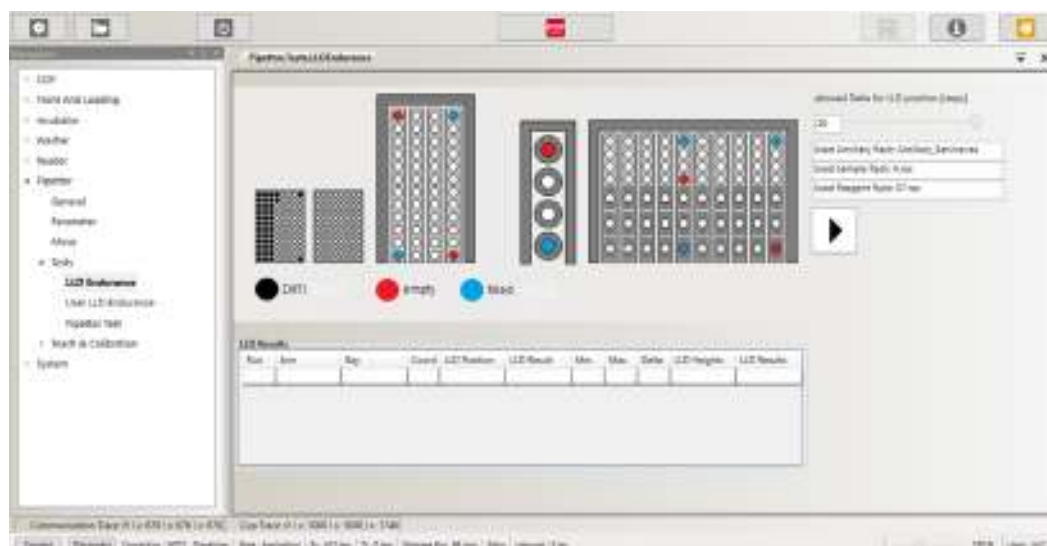


Figure 12-37: Pipettor - Test - LLD Endurance

| Function/Button                                                                     | Description                    |
|-------------------------------------------------------------------------------------|--------------------------------|
|  | Start the LLD test customized. |

Table 12–30: Function of the **LLD Endurance**

#### 12.7.4.1.1 Precondition

- The pre-set amount of a LLD Endurance runs is 10.
- The positions for disposable tips and LLD positions are pre-defined (marked with colors: black means disposable tips required, red means Empty, blue means Filled).
- After the test starting, an initialization is always carried out.
- After the test finishing, a report will be generated.
- Always the same .rac files have to be used (Ancillary\_Service.rac, A.rac and 07.rac).

#### 12.7.4.1.2 Preparation

- Insert a full-filled disposable tip tray in the left tray.
- Insert a empty tube in the first position and a filled tube in the last position of a Sample rack A. Insert the rack in the first lane of the Sample bay.
- Insert a empty tube in the last position and a filled tube in the first position of a Sample rack A. Insert the rack in the last lane of the Sample bay.

## 12. Service Software for external-tanks configuration

- d. Insert a empty bottle in the first position and a filled bottle in the last position of the Ancillary tray (in the dedicated holder). Insert the Ancillary tray in its lane on the Reagent bay.
- e. Insert a reagent kit in the fifth line of the Reagent bay with the following configuration: first and last (seventh) vial filled, forth vial empty
- f. Insert a reagent kit in the last lane of the Reagent bay with the following configuration: first vial filled, last vial (seventh) empty.

## 12.7.4.1.3 General procedure of LLD Endurance test

- a. The Sample pipettor picks up a disposable tip (only needed for LLD test of the Sample bay).
- b. The LLD test starts the first run at defined coordinate.
- c. The results are displayed in a table.
- d. The system verifies if the results of the LLD heights are in the predefined range (allowed delta for LLD position).
- e. The Sample pipettor ejects the disposable tip (if present).
- f. The system repeats the LLD Endurance test for the number of runs (10).

## 12.7.4.2 User LLD Endurance

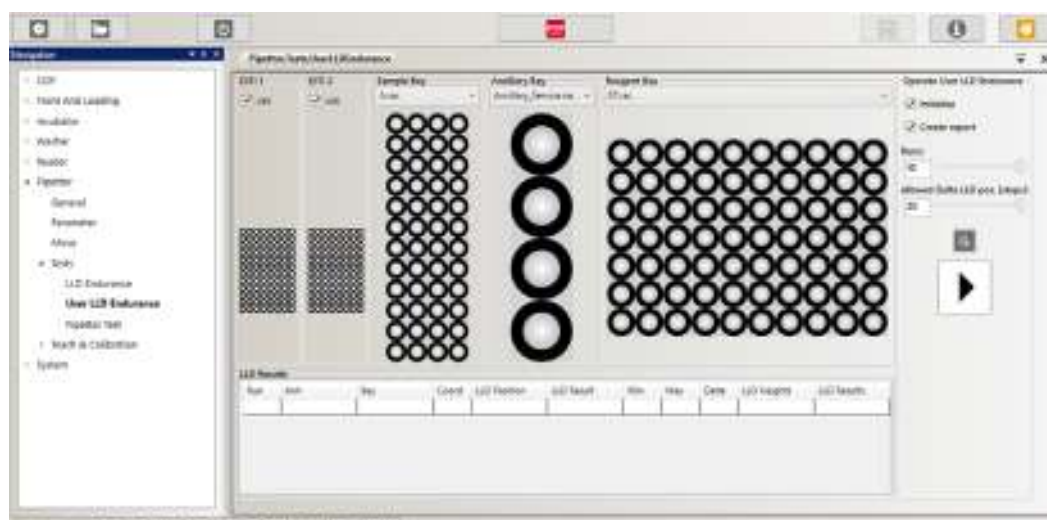


Figure 12–38: Pipettor - Test - User LLD Endurance

## 12.7.4.2.1 DiTi 1 and DiTi 2 group

| Function/<br>Button      | Description                                                                                                                                           |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>DiTi 1 and DiTi 2</b> | Check box to enable the Disposable tips tray usage.<br>In the selected Disposable tip trays, the Disposable tips needed will be marked automatically. |

Table 12–31: Function of **DiTi 1** and **DiTi 2** group

#### 12.7.4.2.2 *Sample Bay group*

| Function/<br>Button                                                               | Description                                                                                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Sample Bay</i>                                                                 | Dropdown menu to select the Sample rack file.                                                                                                                                                                                                                                          |
|  | Positions of the Sample bay to select by clicking on them.<br>The selected positions will be marked.<br>In the selected Sample bay positions, the Left Pipettor will perform the LLD.<br>The Disposable tips needed will be marked automatically after the Sample bay positions chose. |

Table 12–32: Function of *Sample Bay* group

#### 12.7.4.2.3 *Ancillary Bay group*

| Function/<br>Button                                                                 | Description                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Ancillary Bay</i>                                                                | Dropdown menu to select the Ancillary rack file.                                                                                                                                             |
|  | Positions of the Ancillary bay to select by clicking on them.<br>The selected positions will be marked.<br>In the selected Ancillary bay positions, the Right Pipettor will perform the LLD. |

Table 12–33: Function of *Ancillary Bay* group

#### 12.7.4.2.4 *Reagent Bay group*

| Function/<br>Button                                                                 | Description                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Reagent Bay</i>                                                                  | Dropdown menu to select the Reagent rack file.                                                                                                                                           |
|  | Positions of the Reagent bay to select by clicking on them.<br>The selected positions will be marked.<br>In the selected Reagent bay positions, the Right Pipettor will perform the LLD. |

Table 12–34: Function of *Reagent Bay* group

12.7.4.2.5 *Operate User LLD Endurance group*

| Function/<br>Button                                                                | Description                                                                                                                                                                 |
|------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Initialize</i>                                                                  | Check box to select the initialization of the instrument at the beginning of the LLD test.                                                                                  |
| <i>Create report</i>                                                               | Check box to select the Report creation at the end of the LLD test.                                                                                                         |
| <i>Runs</i>                                                                        | Runs to be performed by the system during the LLD test (default is set to 10 run).<br>Select from 1 to 10 runs, moving the arrow left (to decrease) or right (to increase). |
| <i>allowed Delta LLD pos.</i>                                                      | Allowed delta for LLD positions, expressed in steps (default is set to 20 steps).<br>Select from 1 to 20 steps, moving the arrow left (to decrease) or right (to increase). |
|   | Clear list button.                                                                                                                                                          |
|  | Start the LLD test customized.                                                                                                                                              |

Table 12–35: Function of *Operate User LLD Endurance group*12.7.4.2.6 *Preparation*

- a. Customize on the screen the LLD test:
  - select the .rac files to be used;
  - select the Disposable tip tray/trays to be used;
  - select the Sample bay, Ancillary tray and Reagent bay positions where the Pipettors will perform the LLD test;
  - select the number of test runs to be performed;
  - select the allowed delta for LLD positions.
- b. Insert all the needed consumables, sample racks, reagent kit etc in the system.
- c. Press the **Play** button.

### 12.7.4.3 Pipettor Test

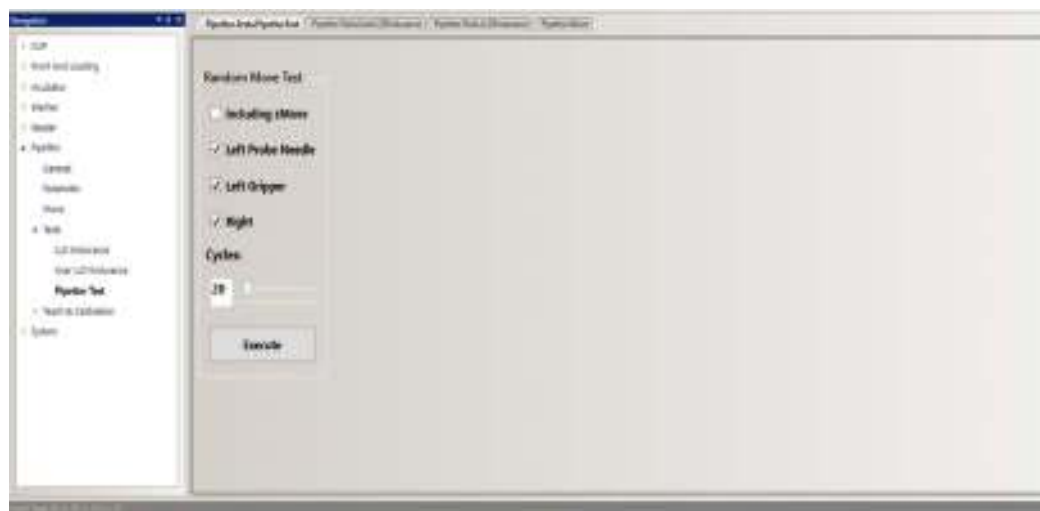


Figure 12–39: Pipettor - Tests - Pipettor Test

#### 12.7.4.3.1 Random Move Test group

| Function/<br>Button                             | Description                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Including Z-Move</b>                         | Check box to enable the Pipettors Z movements in the random cycle.                                                                                                                                                                                                                                                        |
| <b>Left Gripper / Left Probe Needle / Right</b> | Check box to choose the arm / arms which will perform the test. In addition, the Left Arm could move only to the routing points of the Sample pipettor (Left Probe Needle), only to the routing points of the Gripper module (Left Gripper) or to both routing points during the same test, according to the checked box. |
| <b>Cycles</b>                                   | Cycles to be performed by the arms during the Pipettor test (default is set to 20 cycles).<br>Select from 1 to 1000 cycles, moving the arrow left (to decrease) or right (to increase).                                                                                                                                   |
| <b>Execute</b>                                  | Starts the random movements.                                                                                                                                                                                                                                                                                              |

Table 12–36: Random Move Test



## 12.7.5 Teach & Calibration

### 12.7.5.1 Gripper Claws Calibration

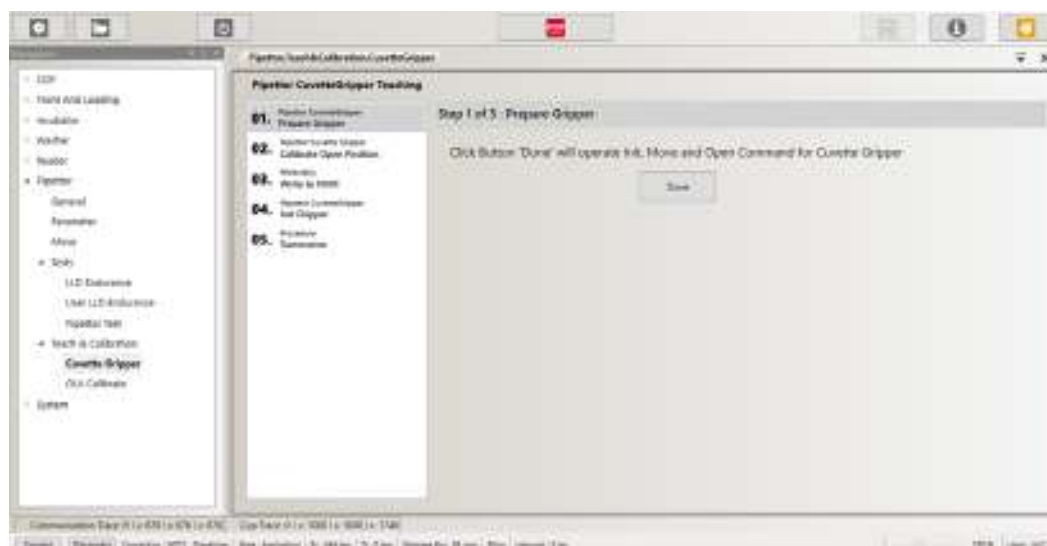


Figure 12–40: Pipettor - Teach & Calibration - Cuvette Gripper

The Gripper Claws Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

About the function of all the buttons, see Chapter 12.1.7.1.1 on page 12-9.

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **Cuvette Gripper**

- Step 1**
  - a. Remove the front right cuvette tray in the Cuvette - Tip drawer. Read the instructions and then press the **Done** button to initialize the gripper.
- Step 2**
  - b. Take a cuvette and manually hold it between the gripper claws.
  - c. Use the buttons to close the gripper claws in steps of 10, until the gripper holds on to the cuvette by itself.
  - d. Use the buttons to open the gripper claws in steps of 10, until the cuvette falls down.
  - e. Use the buttons to further open the gripper claws by **30** steps.
  - f. Press the **Save** button to save the adjustment.
- Step 3**
  - g. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 4**
  - h. Read the instructions and then press the **Done** button to initialize the gripper.
- Step 5**
  - i. The process is successful when, at the **Summarize** panel, all steps are in green.

### 12.7.5.2 OLV Verify

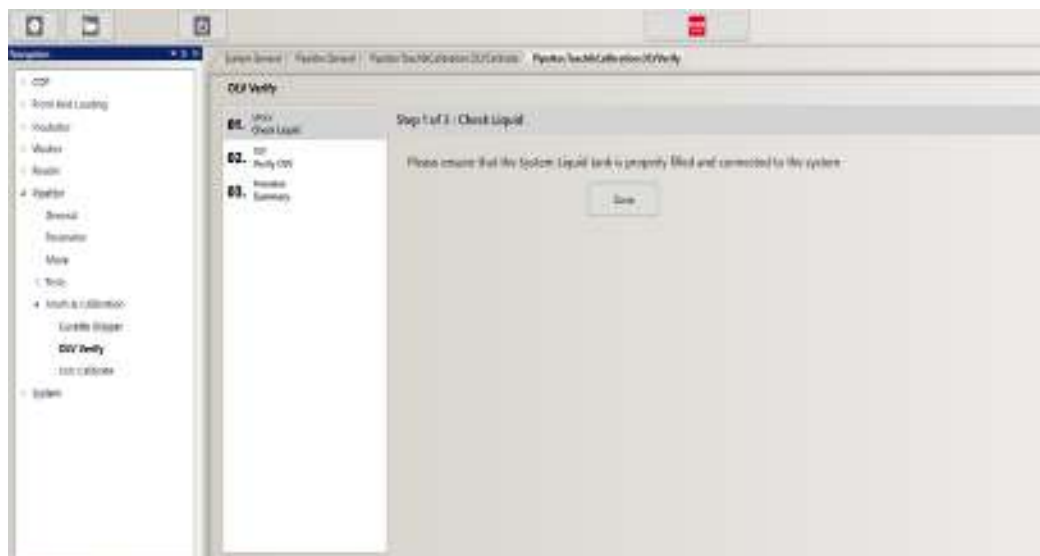


Figure 12-41: Pipettor - Teach & Calibration - OLV verify

The OLV Verification contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **OLV Verify**.

- Step 1**
  - a. Read the instructions to ensure the System Liquid tank is properly filled and connected to the system and then press the **Done** button.
- Step 2**
  - b. Press the **Do** button; the verification routine starts automatically.
  - c. The procedure is successful when the **Processing** window closes without failure message.
- Step 3**
  - d. The process is successful when, at the **Summarize** panel, all steps are in green. The Info column shows you the results.

## 12.7.5.3 OLV Calibrate

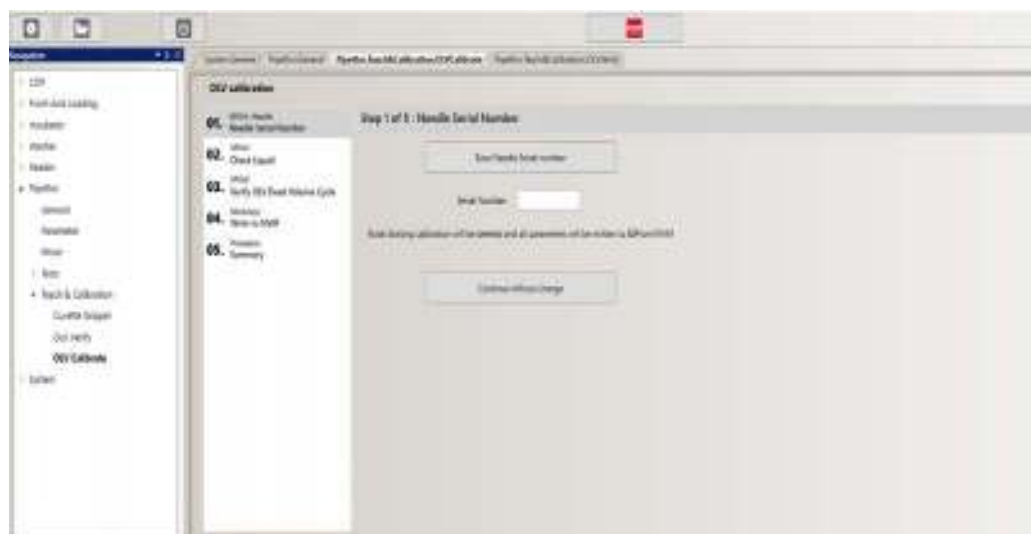


Figure 12–42: Pipettor - Teach &amp; Calibration - OLV calibrate

The OLV Calibration contains several steps. In order to obtain a correct result, it is necessary to keep the order of the steps.

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **OLV Calibrate**.

- Step 1**
  - a. In case the SPOLV needle has just been replaced, please enter in the Serial Number field, the new Serial Number of the needle, then continue by pressing the **Save Needle Serial number** button. Otherwise, press the **Continue without change** button
- Step 2**
  - b. Read the instructions to ensure the System Liquid tank is properly filled and connected to the system and then press the **Done** button; the calibration routine starts automatically.
- Step 3**
  - c. When the calibration is finished, the dialog **Verify OLV Dead Volume Cycle** will open automatically. The calibration perform 5 verification runs. In case of a failed calibration, one or more verification run results are highlighted in red. In case of a successful calibration, the total result is set to "Ok". If the total result is "Ok", press the **Done** button. Otherwise, return to **Step 2** or check hardware, i.e. the PRP-SPOLV tubing, the SPOLV module, the SPOLV needle, subsequently.
- Step 4**
  - d. Store the taught position in the EEPROM by selecting the **Store teaching values to NVM** button.
- Step 5**
  - e. The process is successful when, at the **Summarize** panel, all steps are in green.

## 12.8 System

### 12.8.1 General

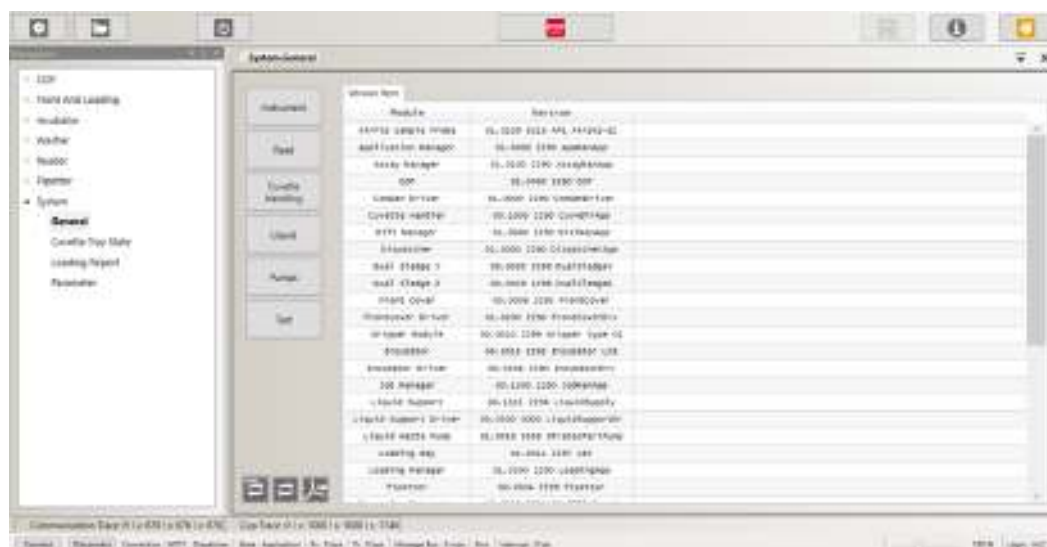


Figure 12-43: System - General - Read

### 12.8.1.1 System General

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Instrument</b>   | <p>Full initialization via COP:</p> <ul style="list-style-type: none"> <li>• Initialize the instrument</li> </ul> <p>Power On:</p> <ul style="list-style-type: none"> <li>• Powers on the instrument</li> </ul> <p>Power Off:</p> <ul style="list-style-type: none"> <li>• Powers off the instrument.</li> </ul>                                                                                                                                                                                                                                                                                   |
| <b>Read</b>         | <p>Displays information about the:</p> <p>Version</p> <p>Temperature:</p> <ul style="list-style-type: none"> <li>• Displays information about the state of the instrument temperature sensors</li> </ul> <p>Liquid State:</p> <ul style="list-style-type: none"> <li>• Displays information about the state of the instrument liquid sensors</li> </ul> <p>Counter</p> <ul style="list-style-type: none"> <li>• Reads the cycle counter from the selected modules.</li> </ul> <p>Serial</p> <ul style="list-style-type: none"> <li>• Reads the Serial number from the selected modules.</li> </ul> |

## 12. Service Software for external-tanks configuration

| Function/<br>Button     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Cuvette Handling</b> | <p>Clear All:</p> <ul style="list-style-type: none"> <li>The gripper picks up all the cuvettes in the instrument (from Washer, Incubator and Reader) and dispose them via the Reader Module.</li> </ul> <p>Clear Incubator:</p> <ul style="list-style-type: none"> <li>The gripper picks up all the cuvettes in the Incubator and dispose them via the Reader Module.</li> </ul> <p>Clear Washer:</p> <ul style="list-style-type: none"> <li>The gripper picks up all the cuvettes in the Washer and dispose them via the Reader Module.</li> </ul> <p>Clear Measurement:</p> <ul style="list-style-type: none"> <li>Cleans the cuvette in the Reader Module.</li> </ul> <p>Fill Incubator:</p> <ul style="list-style-type: none"> <li>The gripper picks up the cuvettes from the trays in order to fill the Incubator (the cuvette trays must be set in the <b>Cuvette Tray State</b>).</li> </ul> <p>Cuvette Endurance Run; opens the Cuvette Endurance Run window where there is the possibility to choose the setting for the test, see Chapter 12.8.1.2 on page 12-62.</p> |
| <b>Liquid</b>           | <p>Prime all, see Chapter 12.8.1.3 on page 12-65.</p> <p>Prime Pipettor, see Chapter 12.8.1.4 on page 12-67.</p> <p>Prime Washer, see Chapter 12.8.1.5 on page 12-67.</p> <p>Prime Reader see Chapter 12.8.1.6 on page 12-68.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

| Function/<br>Button | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pumps</b>        | <p>WAP:</p> <ul style="list-style-type: none"> <li>Start at Routine Speed</li> <li>Start at Prime Speed</li> <li>Start Parameterized opens the Start WAP Timed window, where there is the possibility to choose the WAP speed and time to run</li> <li>Stop</li> </ul> <p>WBDP:</p> <ul style="list-style-type: none"> <li>Load: <ul style="list-style-type: none"> <li>1900 µl</li> <li>2100 µl</li> <li>Parameterized opens the window Load WBDP where there is the possibility to choose the volume to load in uL</li> </ul> </li> <li>Dispense Port 1, 2 or 3: <ul style="list-style-type: none"> <li>700 µl</li> <li>1200 µl</li> <li>Parameterized opens the window Dispense WBDP Port 1, 2 or 3 where there is the possibility to choose the volume to dispense in uL and if activate or not the tracking</li> </ul> </li> <li>Empty the pump</li> </ul> <p>Starter Dispense Pump 1 and 2 (SP1 and SP2):</p> <ul style="list-style-type: none"> <li>Fill to 400 µl or Parameterized</li> <li>Dispense 200 µl or 400 µl or Parameterized</li> <li>Empty the pump</li> </ul> <p>LWP:</p> <p>Switch On or Off</p> |
| <b>Test</b>         | <p>Cuvette Movement Test</p> <ul style="list-style-type: none"> <li>Follow arising instructions.</li> <li>The cuvette trays must be set in the <b>Cuvette Tray State</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

Table 12–37: Functions of System - General

### 12.8.1.2 Cuvette Endurance Run

| Function/<br>Button   | Description                                                                                                                                                                                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Initialization</b> | Check box to perform the system initialization before the start of the Cuvette Endurance Run.                                                                                               |
| <b>Quantity</b>       | Quantity of cuvette to test in the Cuvette Endurance Run (default value is set to 57).<br>Choose a value from 1 to 57.                                                                      |
| <b>StartWith</b>      | Incubator rotor ring from which the Cuvette Endurance Run starts: <ul style="list-style-type: none"> <li>Inner (cuvette positions 1-25)</li> <li>Outer (cuvette positions 26-57)</li> </ul> |

| Function/<br>Button                | Description                                                                                                                          |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <b>EndWith</b>                     | Position in which the Cuvette Endurance Run finishes: <ul style="list-style-type: none"> <li>• Incubator</li> <li>• Waste</li> </ul> |
| <b>Inc. to Washer cycles</b>       | Number of cycles of the cuvette inside the Washer Module. Set 0 to avoid the Washer Module.                                          |
| <b>PresenceCheck</b>               | Check box to activate the check of the cuvette presence inside the Incubator Module.                                                 |
| <b>LiftMove</b>                    | Check box to activate the Up and Down move of the Washer aspiration needles.                                                         |
| <b>Execute</b>                     | Button to start the Cuvette Endurance Run.                                                                                           |
| <b>Cuvette Endurance Info Item</b> | Reports the actions of the system during the Cuvette Endurance Run.                                                                  |

Table 12–38: Functions of the **Cuvette Endurance Run** window

#### 12.8.1.2.1 Standard Cuvette Endurance Run

a. Set the cuvette trays in the **Cuvette Tray State**.



b. Customize on the screen the **Cuvette Endurance Run** test:

- select the initialization of the instrument;
- select 10 cuvette to test;
- select Outer to start;
- select Waste to end;
- select 2 cycles inside the Washer;
- select the cuvette presence check in the Incubator;
- uncheck the Lift movement;

c. Press the **Execute** button.

d. Check if the placing of the cuvette by the Gripper inside the Incubator and the Washer module is smooth and without noises.

e. Perform again the Cuvette Endurance Run, selecting Inner to start.

**CAUTION**

**During the Cuvette Endurance Run, check the cuvette transfer from the Incubator to the Washer and vice-versa**

Check that cuvette insertion by the Gripper inside the Incubator is smooth and without noises.

Check that cuvette pickup from the Incubator and insertion into the Washer is smooth and without noises.

Check that cuvette pickup from the Washer and insertion back into the Incubator is smooth and without noises.

A not proper cuvette transfer is usually caused by a misalignment between the gripper and the Incubator rotor. If needed, change the Incubator rotor calibration (see Chapter 12.4.3.1 on page 12-26) such to compensate the misalignment; an adjustment of 3-4 steps clockwise could be sufficient.

---



## 12.8.1.3 Prime all

| Function/Button                      | Description                                                                                            |
|--------------------------------------|--------------------------------------------------------------------------------------------------------|
| <i>Cycles Washer Port 1, 2 and 3</i> | Select the Wash Buffer prime cycles to perform at each Washer SIPAL.                                   |
| <i>Cycles Measurement</i>            | Select the Starter reagents prime cycles to perform at the Reader Module.                              |
| <i>Cycles Pip</i>                    | Select the prime cycles to perform at the SPOLV.                                                       |
| <i>Wash Time</i>                     | Select the wash time during the SPOLV prime (0.1second/cycles).                                        |
| <i>OLV Verification</i>              | Enables or disables the OLV Calibration.                                                               |
| <i>Liquid Control</i>                | Enables or disables the Aspiration monitoring check on the three Washer SIPAL and on the Reader SIPAL. |
| <i>Execute</i>                       | Button to start the Cuvette Endurance Run.                                                             |
| <i>Bubble Measure Result Item</i>    | If the Liquid Control and/or the OLV Verification are enabled, displays the results.                   |

Table 12–39: Functions of the *Prime all* window

### 12.8.1.3.1 *Standard Prime all*

- a. Insert all needed consumables and liquids.
- b. Set the cuvette trays in the *Cuvette Tray State*.
- c. Customize on the screen the *Prime all* window:
  - Cycles Washer Port 1: 38
  - Cycles Washer Port 2: 11
  - Cycles Washer Port 3: 11
  - CyclesMeasurement: 40
  - CyclesPip: 24
  - Wash time: 18
  - OLV Verification: Enabled
  - LiquidControl: Enabled

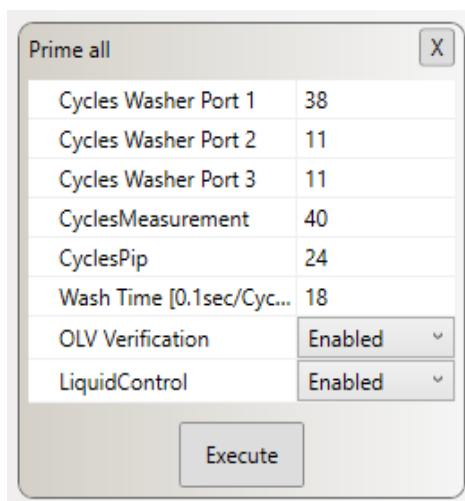


Figure 12–44: Standard Prime all

- d. Press the *Execute* button.
- e. The process is successful when the *Processing* window closes without failure message.

#### 12.8.1.4 Standard Prime SPOLV procedure

##### 12.8.1.4.1 Rinsing of SPOLV module

In the **Service Software** at the navigation window, choose **System** and the **General** bracket. Click onto the **Liquid** button and choose the **Prime Pipettor** button to open the dedicated window.

- a. At the window **Prime Pipettor** for the rinsing with System Liquid, use following values and start with the button **Execute**.
  - Cycles: 24
  - Wash time: 18
  - OLV Verification: Enabled

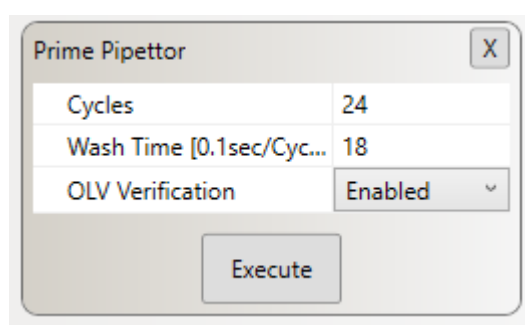


Figure 12–45: Pipettor rinsing

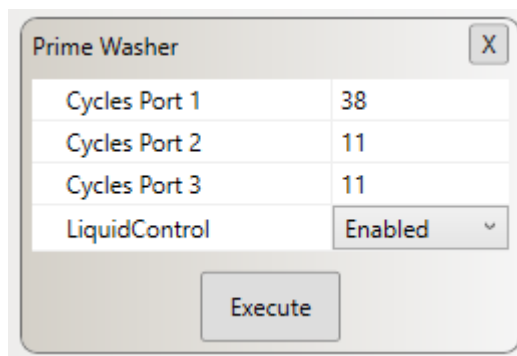
- b. The process is successful when the **Processing** window closes without failure message.

#### 12.8.1.5 Standard Prime Washer Module procedure

##### 12.8.1.5.1 Rinsing of Washer Module

In the **Service Software** at the navigation window, choose **System** and the **General** bracket. Click onto the **Liquid** button and choose the **Prime Washer** button to open the dedicated window.

- a. Set the cuvette trays in the **Cuvette Tray State**.
- b. At the window **Prime Washer** use following values and start with **Execute**.
  - Cycles Port 1: 38
  - Cycles Port 2: 11
  - Cycles Port 3: 11
  - Liquid Control: Enabled



| Prime Washer  |         |
|---------------|---------|
| Cycles Port 1 | 38      |
| Cycles Port 2 | 11      |
| Cycles Port 3 | 11      |
| LiquidControl | Enabled |

Execute

Figure 12–46: Rinsing of Washer Module

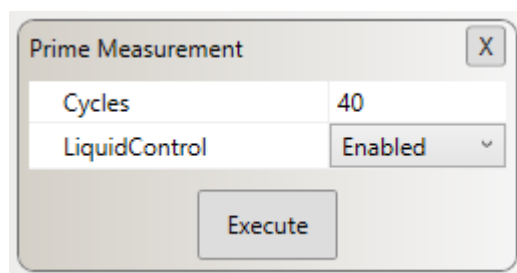
- c. The process is successful when the **Processing** window closes without failure message.

## 12.8.1.6 Standard Prime Reader Module procedure

### 12.8.1.6.1 Rinsing of Reader Module

In the **Service Software** at the navigation window, choose **System** and the **General** bracket. Click onto the **Liquid** button and choose the **Prime Reader** button to open the dedicated window.

- a. Set the cuvette trays in the **Cuvette Tray State**.
- b. At the window **Prime Reader** use following values and start with **Execute**.
  - Cycles: 40
  - Liquid Control Enabled



| Prime Measurement |         |
|-------------------|---------|
| Cycles            | 40      |
| LiquidControl     | Enabled |

Execute

Figure 12–47: Rinsing of Reader Module

- c. The process is successful when the **Processing** window closes without failure message.

## 12.8.2 Cuvette Tray State

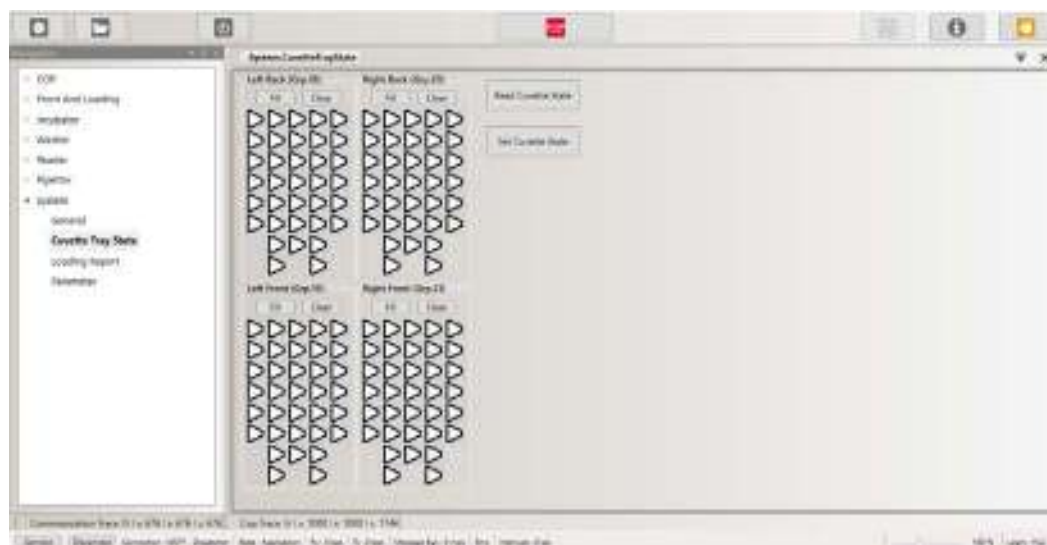


Figure 12–48: System - Cuvette Tray State

| Function/Button                                                                     | Description                                                                                         |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Fill, Clear</b>                                                                  | Fill or clear the chosen cuvette tray only at the display.                                          |
|  | Fill or clear the chosen cuvette only at the display.<br>The cuvette will be marked in blue if set. |
| <b>Read Cuvette State</b>                                                           | Reads and shows the last set cuvette status from the cuvette tray.                                  |
| <b>Set Cuvette State</b>                                                            | Takes over the chosen setting of the display into the cuvette tray.                                 |

Table 12–40: Functions of the Cuvette Tray State

### 12.8.3 Loading Report



Figure 12–49: System - Loading Report

| Function/Button                | Description                                                                                |
|--------------------------------|--------------------------------------------------------------------------------------------|
| <b>Activate Sample Scanner</b> | Switch on the sample scanner to scan the sample rack.                                      |
| <b>Sum of Occupations</b>      | Sum of all loading changes.                                                                |
| <b>Clear Report</b>            | Report will be deleted.                                                                    |
| <b>Filter Remove Items</b>     | If check box is marked, then the loading changes for removing of racks will be suppressed. |

Table 12–41: Functions of the Loading Report

#### 12.8.3.1 Verifying RFID

In the **Service Software** at the navigation window choose **System** and the **Loading Report** bracket.

The changes of the materials to be loaded are shown after loading or unloading of the materials.

- When inserting the Starter bottles or the Assay reagent kits in the dedicated area, after a defined time interval the presence of the bottles and of the kits will be prompted.
- The RFID will be read and displayed as a dialog.

### 12.8.4 System Parameter

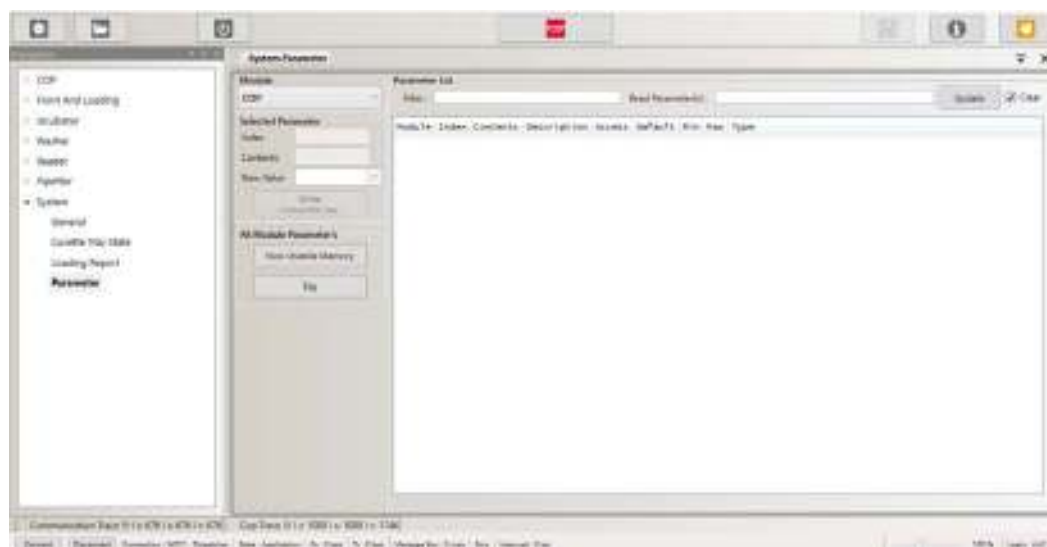


Figure 12–50: System - Parameter

#### NOTE

#### Improper change of parameters

Improper change of parameters can result in serious personal injuries, material damages and non functioning of the instrument. Consult the manufacturer before you change parameters.

For description about parameter, see Chapter on page 12-11.

In the **Module** group, it is possible to select:

- COP
- Power CAN Extension
- Front Cover
- Loading Bay
- Liquid Support
- Wash Aspiration Pump
- Incubator
- Reader Unit
- Starter Pump 1 & 2
- Pipettor
- Liquid Waste Pump
- Gripper Module
- AirPip Sample Probe
- Single Sledge Y
- Single Sledge Z
- SPOLV Reagent Probe
- Dual Sledge Y
- Dual Sledge Z

## 12. Service Software for external-tanks configuration

---

- Washer
- Wash Buffer Dispense Pump



# 13 Teaching of the Pipettor

This chapter describes the teaching for the **LIAISON<sup>®</sup> XS** instrument.

## 13.1 General Statement

If the **User Software** or the **Service Software** was running before, shut down the previously used software. Then login as Administrator and open the **Teacher Software**.

### NOTE

#### Close other programs

If the **Service Software** or the **User Software** was running before, these programs have to be closed!

### NOTE

#### Login as Administrator

The Teacher Software has to be opened from the Administrator Windows account!

### NOTE

#### External keyboard

It is necessary to use an external keyboard for teaching!

The keyboard has to be compatible with the operating system, see Chapter 35.3 on page 35-2 or see Chapter 36.3 on page 36-2.

## 13.2 Safety

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 or see Chapter 2.5 on page 2-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

### CAUTION

#### Use of the Software

Use **LIAISON<sup>®</sup> XS Teacher Software** only to perform described operations.

## 13. Teaching of the Pipettor

**WARNING**

### Function of the Instrument

Please follow closely the steps contained in the individual instructions to achieve a perfect function of the instrument.

## 13.3 Overview

### 13.3.1 Overview Teacher menu

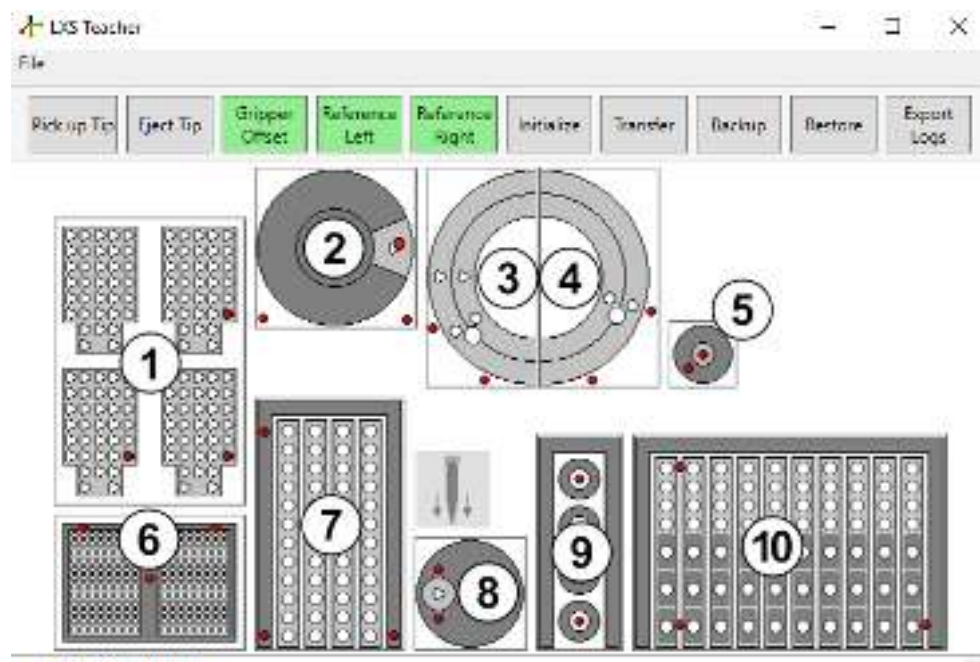


Figure 13–1: Teacher menu

Teach points of each module are shown with red dots.

Click on the desired position or button to teach the pipettor.

| Button / Position      | Description                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Pickup Tip</b>      | The Sample Pipettor move to the left DiTi tray and picks up a disposable tip from the left rearmost position. |
| <b>Eject Tip</b>       | The Sample Pipettor move to the eject position and releases the disposable tip.                               |
| <b>Gripper Offset</b>  | Allows teaching of the gripper respect to the Sample Pipettor. "Gripper offset" on page 20                    |
| <b>Reference Left</b>  | Allows teaching of the reference position for the left arm. "Sample probe reference" on page 18               |
| <b>Reference Right</b> | Allows teaching of the reference position for the right arm. "Reagent probe reference" on page 16             |
| <b>Initialize</b>      | Initializes the LIAISON <sup>®</sup> XS pipettors.                                                            |
| <b>Transfer</b>        | Transfers the coordinates to the EEPROM.                                                                      |
| <b>Backup</b>          | Creates a backup from several data files to a selected folder.                                                |
| <b>Restore</b>         | Restores a previously created backup.                                                                         |
| <b>Export Logs</b>     | Exports file CommunicationLogFile.txt or if available an error log file to a selected folder.                 |
| 1                      | "Cuvette tray" on page 22                                                                                     |
| 2                      | "Washer" on page 25                                                                                           |
| 3                      | "Incubator sample probe" on page 29                                                                           |
| 4                      | "Incubator reagent probe" on page 31                                                                          |
| 5                      | "Wash station" on page 32                                                                                     |
| 6                      | "Tip tray" on page 24                                                                                         |
| 7                      | "Sample bay" on page 27                                                                                       |
| 8                      | "Reader" on page 30                                                                                           |
| 9                      | "Ancillary bay" on page 35                                                                                    |
| 10                     | "Reagent bay" on page 36                                                                                      |

Table 13–1: Teacher overview

## 13. Teaching of the Pipettor

## Button overview

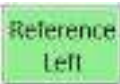
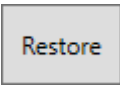
| Button                                                                              | Description                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Teach the reference position for the right arm.                                                                                                                                                           |
|    | Teach the reference position for the left arm.                                                                                                                                                            |
|    | A tip is picked up from the left rearmost position of the left DiTi tray.                                                                                                                                 |
|    | The tip is ejected at the drop-off position.                                                                                                                                                              |
|    | Teach the gripper respect to the sample pipettor.                                                                                                                                                         |
|  | Initialize the pipettors.                                                                                                                                                                                 |
|  | Transfers static coordinates to pipettor EEPROM                                                                                                                                                           |
|  | Saves following data: <ul style="list-style-type: none"><li>• Coordinate files,</li><li>• all rack files,</li><li>• BottleShapes.xml,</li><li>• PressureThresholds.xml</li></ul> to a preselected folder. |
|  | Restores only data from a previously created backup.<br>If no previous backup exists, a failure message appears.                                                                                          |
|  | Exports file CommunicationLogFile.txt and if there was a failure, an error log file to a pre-selected path.                                                                                               |

Table 13–2: Button overview

## File menu

| Menu      |                                                                                                      | Description                                                        |
|-----------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| File/New  |  Rack file          | Opens dialog to create a new rack file.<br>"Rack files" on page 12 |
| File/Open |  Coordinate file    | Opens window <b>Coordinate File</b> .                              |
|           |  Rack file          | Opens window to choose <b>Rack File</b> .                          |
|           |  Bottle shapes file | Opens <b>Bottle Shapes File</b> .                                  |
| File      |  Exit               | Closes the teach program.                                          |

Table 13–3: Menu overview

## 13.3.2 Teaching sequence

| Step | Description                          |
|------|--------------------------------------|
| 1.   | "Reagent probe reference" on page 16 |
| 2.   | "Sample probe reference" on page 18  |
| 3.   | "Gripper offset" on page 20          |
| 4.   | "Cuvette tray" on page 22            |
| 5.   | "Tip tray" on page 24                |
| 6.   | "Washer" on page 25                  |
| 7.   | "Sample bay" on page 27              |
| 8.   | "Incubator sample probe" on page 29  |
| 9.   | "Reader" on page 30                  |
| 10.  | "Incubator reagent probe" on page 31 |
| 11.  | "Wash station" on page 32            |
| 12.  | "Ancillary bay" on page 35           |
| 13.  | "Reagent bay" on page 36             |
| 14.  | "Sample Rack Teaching" on page 38    |

Table 13–4: Teaching sequence

## **13.4 Teaching the pipettor module**

**WARNING**

---

**Possible instrument damages**

It is easily possible to damage parts of the instrument with the teacher software.

- Make sure, that the movement course is free, before you start the pipettor movement to a module.
- 

**NOTE**

---

**External keyboard**

It is necessary to use an external keyboard for teaching!

The keyboard has to be compatible with the operating system, see Chapter 35.3 on page 35-2.

---

**NOTE**

---

**Cleaned and adjusted pipettor**

It is necessary to clean and adjust the pipettor mechanic before starting the teach process!

---

**NOTE**

---

**Reference positions and Gripper Offset**

All positions that have to be taught correlate to the Left and Right reference positions and to the offset between the gripper and the sample pipettor.

Therefore, it is important to teach the Left reference position, Right reference position and Gripper Offset before any other position is taught.

---

### 13.4.1 General information

#### 13.4.1.1 General handling

Clicking on a specific position of the teacher main window allows modification of the coordinates at the different positions. The pipettor moves to the currently set X- and Y-coordinates and the **Adjust Position** dialog appears. The Adjust positioning Dialog window comes always up when a position needs to be taught.

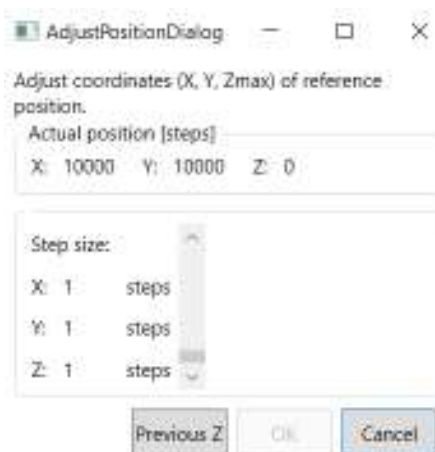


Figure 13–2: **Adjust Position** dialog

In order to avoid damage of the pipettors, the Z-axis does not move automatically.

In the **Adjust Position** dialog, the actual position of the pipettor is shown, in terms of number of steps currently moved from the Init position of each axis. Each time a movement is made, it is automatically added to the current value and may be viewed here in real time.

## 13. Teaching of the Pipettor

At the top of the dialog, the teach instruction is shown to inform which position has to be taught. The following keys of the keyboard or buttons can be used for adjusting the pipettor to the correct position:

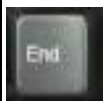
| Key/<br>Button:                                                                     | Action                                                                                                                                                                              | Key/<br>Button:                                                                                    | Action                                                                                                                                                      |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Moves the pipettor to the right (X-axis).                                                                                                                                           | <br><b>Shift</b> | Moves the pipettor down (Z-axis) half of the distance from the previously set Z position.                                                                   |
|    | Moves the pipettor to the left (X-axis).                                                                                                                                            |                  | Decreases the step size.<br>Alternative key <b>s</b> can be used from Version 1.0.7.0 on.                                                                   |
|    | Moves the pipettor to the rear (Y-axis).                                                                                                                                            |                   | Increases the step size.<br>Alternative key <b>a</b> can be used from Version 1.0.7.0 on.                                                                   |
|  | Moves the pipettor to the front (Y-axis).                                                                                                                                           |                | Moves the pipettor adapter to the stored Z-max position (Prev.Z).<br>See the safety note below!                                                             |
|  | Moves the pipettor up (Z-axis).                                                                                                                                                     | <b>Cancel</b>                                                                                      | If a failure has occurred, the procedure can be interrupted by pressing the <b>Cancel</b> button.<br>Rejects all changes for the taught position.           |
|  | Moves the pipettor down (Z-axis).                                                                                                                                                   | <b>OK</b>                                                                                          | If a position is adjusted correctly, then the step has to be finished with the <b>OK</b> button.<br>Saves the (changed) coordinates of the taught position. |
| <b>Step size bar</b>                                                                | The size of the movements can be adjusted by the step size bar. If the bar is adjusted to the bottom, the pipettor can be moved with minimum steps.<br>Default step size is 1 step. |                                                                                                    |                                                                                                                                                             |

Table 13–5: Use of the **Adjust Position** dialog



**NOTE****Destroying of the needle**

The **Prev. Z** button or **End** key should only be used if it is sure that the needle cannot hit the bottom! To reach the final position in X/Y/Z, small step sizes must be used!

---

**NOTE****Too much tolerance**

When checking the coordinates, be aware of the technical error of the axis in X-, Y- and Z-direction; a tolerance of 3 steps (minimal step size) can be accepted!

---

**NOTE****Position check**

If the positions are only checked (not changed in any direction), the teaching of this position (before saving the last position) must be finished by pressing the **Cancel** button. This has to be done by all different positions in the teacher!

---

**NOTE****Initializing after crash**

If the pipettor crashes (motors de-energized) while teaching any position, the pipettor must be initialized by pressing the **Init** button.

---

### 13.4.1.2 Resolution of Pipettor Steps

#### Step-size

In the “Step size” section the user may adjust the size of the steps in 3 ways:

1. The user may click on the arrows to raise or lower the step size. This can be achieved in increments of 1 for X,Y axis and 2 for Z axis.
2. The user may click and drag the slider to increase or decrease the step size.
3. The “+” or “a”, “-” or “s” buttons on the keyboard may be used to increase or decrease the size movement of the item to be taught. “+” or “a” = larger step sizes, “-” or “s” = lower step sizes. This can be achieved in double increments.

The “X”, “Y”, & “Z” symbols found on the left side of this section inform the user of the current step size chosen. Each time the step size is changed the user may view the new settings in this area.

Conversion to Teacher steps:

| Axis:                                                                                                                   | Steps/mm: | mm/steps: |
|-------------------------------------------------------------------------------------------------------------------------|-----------|-----------|
| At X-axis and Y-axis 10 Steps are equivalent to 1 mm, therefore 0,1 mm distance is the minimal travel range for 1 step. |           |           |
| X                                                                                                                       | 10        | 0,1       |
| Y                                                                                                                       | 10        | 0,1       |
| At Z-axis 20 Steps are equivalent to 1 mm, therefore 0,05 mm distance is the minimal travel range for 1 step.           |           |           |
| Z                                                                                                                       | 20        | 0,05      |

Table 13–6: Resolution of pipettor steps

### 13.4.1.3 Previous Z

The **Prev. Z** button moves the pipettor arm to the last taught “Z” position for that particular teach point. Pressing the “**Prev. Z**” does not affect any previously adjusted “X” or “Y” positions.

#### NOTE

The **Prev. Z** button should only be used if it is sure that the needle cannot hit the bottom! To reach the final position in X/Y/Z, small step sizes must be used!

#### Prev. Z

Before deciding to press the “**Prev. Z**” button, the following aspects shall be considered:

1. The last “Z” position taught is known by the user and will not cause a crash of the probe.
2. The “X” and the “Y” position are set so that the complete movement of the “**Prev. Z**” may be accomplished.

**Z-Adjustment****NOTE**

The “Z” setting for the pipettor travel along the X and Y axes will always be the same, given that all the assemblies are below this Z-travel setting and parallel to the pipettor sledges.

At first adjust the Xand-position. Then adjust the Z-position, so that the paper fits between the teach point and the tip adapter or needle in the positions where the paper is required.

**13.4.1.4 OK and Cancel**

When the technician has completed the teaching of a position, **OK** button must be pressed to confirm that the teaching has been completed.

**CAUTION**

Once **OK** button has been pressed, the taught coordinates are saved in `Coordinate.coo` file, so it is no more possible to restore the previous coordinates without reteaching the position.

If a mistake has been made, the technician may **Cancel** the last teaching step sequence.

## 13.4.2 Rack files

**CAUTION**

The standard “.rac” files loaded by default into the system shall not be modified: any modification of these files can cause possible crashes of pipettors and wrong behaviour of the system.

All functionalities described in this section shall be used only for the teaching of customized racks.

The teaching procedure of customized rack typologies is detailed in Chapter 13.18 on page 13-38.

## Overview Rack.rac

Figure 13–3: Overview Rack.rac

| Group  | Button / Field | Description                      |
|--------|----------------|----------------------------------|
| Header | File version   | Description of the file.         |
|        | Data Version   | For display only.                |
|        | Width          | Number of lanes the rack covers. |
|        | Loading Bay    | For display only.                |
|        | Positions      | Number of positions in the rack. |

Table 13–7: Overview Rack.rac

| Group                                         | Button / Field                        | Description                                                                                                                                                                                                                                                              |
|-----------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| List of rack positions with the current value |                                       |                                                                                                                                                                                                                                                                          |
|                                               | No.                                   |                                                                                                                                                                                                                                                                          |
|                                               | Ref. Pos.                             |                                                                                                                                                                                                                                                                          |
|                                               | dX                                    |                                                                                                                                                                                                                                                                          |
|                                               | dY                                    | For display only.                                                                                                                                                                                                                                                        |
|                                               | dZmax                                 |                                                                                                                                                                                                                                                                          |
|                                               | dZstart                               |                                                                                                                                                                                                                                                                          |
|                                               | Surface (mm <sup>2</sup> )            |                                                                                                                                                                                                                                                                          |
| Selected Positions                            |                                       |                                                                                                                                                                                                                                                                          |
|                                               | No.                                   |                                                                                                                                                                                                                                                                          |
|                                               | Ref. Pos.                             |                                                                                                                                                                                                                                                                          |
|                                               | dX                                    |                                                                                                                                                                                                                                                                          |
|                                               | dY                                    | Only for display, but the values can be taught.                                                                                                                                                                                                                          |
|                                               | dZmax                                 |                                                                                                                                                                                                                                                                          |
|                                               | dZstart                               |                                                                                                                                                                                                                                                                          |
|                                               | Surface (mm <sup>2</sup> )            | Surface in mm <sup>2</sup> of the vessel that is placed at this rack position.<br><b>Important!</b> This field and the associated button shall not be used. Tube sections shall be set using the <code>BottleShapes.xml</code> file (see Chapter 13.18.4 on page 13-42). |
|                                               | Apply to all                          | The settings in the display position can be copied into all following positions.                                                                                                                                                                                         |
| Teaching                                      |                                       |                                                                                                                                                                                                                                                                          |
|                                               | Area                                  | Drop down button, with possible choices: <ul style="list-style-type: none"> <li>• Reagent Area</li> <li>• Ancillary Area</li> <li>• Sample Area</li> </ul>                                                                                                               |
|                                               | Lane                                  | Drop down button: <ul style="list-style-type: none"> <li>• Lane selected for the teaching.</li> </ul>                                                                                                                                                                    |
|                                               | Interpolate from position to position | Used to teach a set of equidistant rack positions by only teaching the first and the last of that selection: coordinates for all intermediate positions are set automatically.                                                                                           |
| Buttons                                       |                                       |                                                                                                                                                                                                                                                                          |
|                                               | Teach                                 | Starts teach sequence for racks.                                                                                                                                                                                                                                         |
|                                               | Save                                  | Saves the settings of the current file.                                                                                                                                                                                                                                  |
|                                               | Save As                               | Saves the settings to a new file.                                                                                                                                                                                                                                        |
|                                               | Close                                 | Close window without saving.                                                                                                                                                                                                                                             |

Table 13–8: Overview Rack.rac

### 13.4.3 Bottle Shapes

BottleShapes.xml file shall be updated every time a new rack typology is configured on a LIAISON® XS analyzer; the setting of parameters available in BottleShapes.xml file allows to define the dimensions of the different cross sections of the rack vials.

The procedure for setting the BottleShapes.xml file is detailed in Chapter 13.18.4.



```

<?xml version="1.0" encoding="utf-8" ?>
<!--
This file contains the bottle shapes.
A bottle is described by a list of height/CrossSectionArea data pairs. The height is the vertical position
relative to Zmax (in mm).
- heights must be in descending order.
- The last height must be ZZERO.
- Note that there is usually a clearance between Zmax and bottom which must be taken into account
when calculating height/CrossSectionArea.
- BottleIndex (0..99)
- Height (Z-Zmax) (0..60000)
- CrossSectionArea (mm²) (1..10000)
- Resolution is 20[Zstep/mm]
-->
<BottleShapes xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:noNamespaceSchemaLocation="BottleShapes.xsd">

  <BottleShape BottleIndex="2" Description="Alkaline">
    <ShapeData Height="0" CrossSectionArea="0"/>
  </BottleShape>

  <BottleShape BottleIndex="3" Description="Backbone">
    <ShapeData Height="0" CrossSectionArea="120"/>
  </BottleShape>

```

Figure 13–4: Overview BottleShapesTemp.xml

| Button / Field   | Description                                                            |
|------------------|------------------------------------------------------------------------|
| Bottle Index     | ID of the bottle.                                                      |
| Description      | Name of the rack.                                                      |
| Height           | Vertical position calculated from $Z_{\max}$ and expressed in Z-steps. |
| CrossSectionArea | Cross section area at the defined height, expressed in $\text{mm}^2$ . |

Table 13–9: Overview BottleShapesTemp.xml

#### 13.4.4 Start of Teacher software

##### Tools

Keyboard

Sheet of paper 80g/m<sup>3</sup>

Ruler

### NOTE

#### Close other programs

If the service software or the User software is running, these programs have to be closed!

#### Teacher software

1. Login as Administrator.
2. Start the Teacher software.
3. Click on the **Log-On** button (no Password is needed).
4. The pipettor is running the initialization routine.
5. Teacher Menu appears.



Figure 13–5: Teacher Log-on window

## 13.5 Reagent probe reference

### 13.5.1 Teaching sequence

**NOTE**

The Reference Position shall be taught for both, left and right arms.

Due to the fact that the setting of all positions is related to the reference positions, in some cases, it is enough to teach the reference position. Therefore, it is also important to teach the reference position before all other positions.

**CAUTION**

When moving to Z-direction, be aware that the pipettor is already taught, if not, minimize the Step size.

Reference  
Right

1. To start the reference teaching sequence click onto **Reference Right** button.
2. The reagent arm moves to the X and Y coordinates of the reference position and the window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other.



Figure 13–6: Adjust reference position

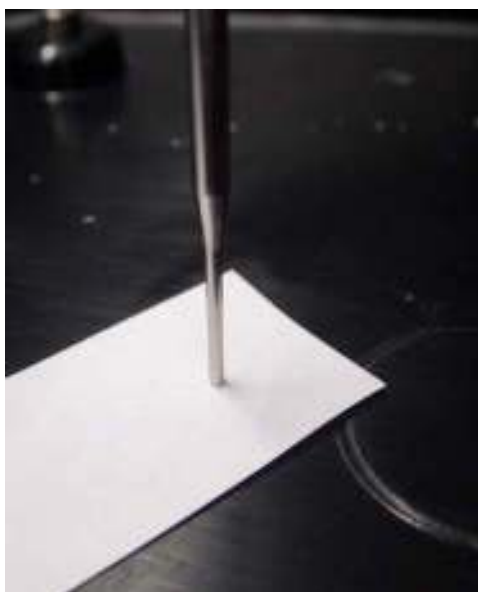


**Teaching of X- and Y- coordinates**

4. The point of the needle shall be taught in X- and Y-axis to be centered on the reference position bore.

**Teaching of Z-position**

5. Move the needle down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt, see also “Z-Adjustment” on page 11.



*Figure 13–7: Reagent reference position*

## 13.6 Sample probe reference

### 13.6.1 Teaching sequence

**NOTE**

The Reference Position shall be taught for both, left and right arms.

Due to the fact that the setting of all positions is related to the reference positions, in some cases it is enough to teach the reference position. Therefore, it is also important to teach the reference position before all other positions.

**CAUTION**

When moving to Z-direction, be aware that the Di-Ti drawer is completely open. Be also aware that pipettor is already taught, if not, minimize the Step size.

Reference  
Left

1. To start the reference teaching sequence click onto **Reference Left** button.
2. The sample arm moves to the X and Y coordinates of the reference position and the window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other.



Figure 13–8: Adjust reference position

**Teaching of X- and Y- coordinates**

4. The point of the tip adapter shall be taught in X- and Y-axis to be centered on the reference position bore. The reference position bore is in front of the round recess clearance.

**Teaching of Z-position**

5. Move the sample probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt, see also “Z-Adjustment” on page 11.



Figure 13–9: Sample reference position

## 13.7 Gripper offset

**NOTE**

The Gripper Offset shall be taught.

Due to the fact that the setting for all gripper positions is computed in relation to the left pipettor positions, that are in turn computed in relation to the reference position, in some cases it is enough to teach the reference position and the Gripper Offset. Therefore, it is important to teach the reference position and Gripper Offset before all other positions.

### 13.7.1 Teaching sequence

Gripper  
Offset

1. To start the gripper offset teaching sequence click onto **Gripper Offset** button.
2. The left arm moves to the X and Y coordinates of the first gripper teach position. The window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other. The teaching positions are at the consumable drawer, on the right front cuvette tray area.



Figure 13–10: Adjust reference position

#### Teaching of X- and Y- coordinates

4. The point of the tip adapter shall be taught to be centered on the little bore. The bore position is in the right front cuvette tray area.

**Teaching of Z-position**

5. Move the sample probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt. (see also “Z-Adjustment” on page 11)



Figure 13-11: Gripper offset, z-adjustment

**Teaching of Z-position**

6. The second position is taught with the gripper itself. Teach the Z-position, moving down the bottom edge of the gripper until the paper surface is almost touched. To move the paper, a slight resistance shall be felt. (see also “Z-Adjustment” on page 11)

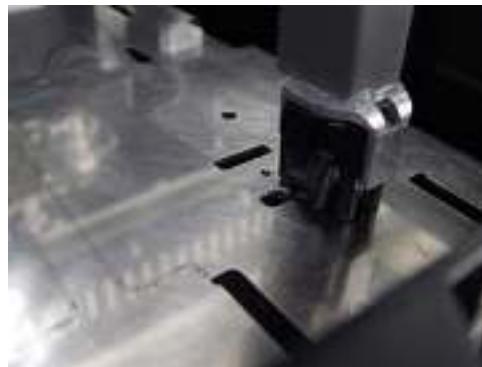


Figure 13-12: Gripper Offset Z

**Teaching of X- and Y- coordinates**

7. The third position is taught with the gripper itself. Teach the gripper in X- and Y-axis into the gripper-shaped bore. When the gripper is properly centered, it is possible to move the gripper into the bore.

### 13. Teaching of the Pipettor

---

8. The Z drive can be used for verification purpose, but will not change the Z-setting.

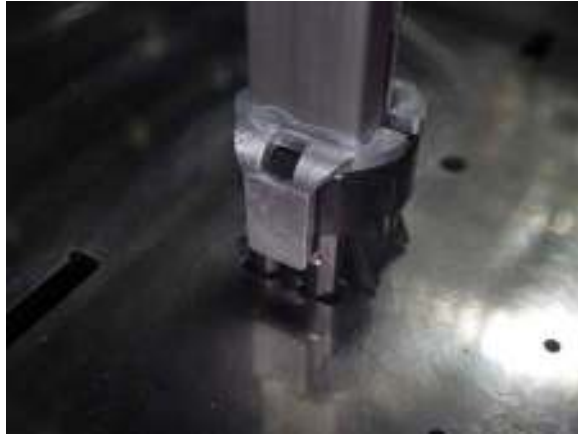
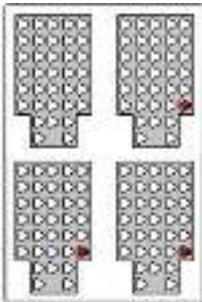


Figure 13-13: Gripper Offset

## 13.8 Cuvette tray

### 13.8.1 Teaching X, Y, Zmax



1. Ensure that the Cuvette-Tip Loading Module is completely pushed in and empty.

---

**WARNING****Empty the Cuvette-Tip Loading Module**

Remove all the cuvette trays before teaching the area.

This is needed to avoid collisions during the automatic teaching procedure.

---

2. To start the cuvette tray teaching sequence, click onto **Cuvette Tray** button.
3. The window **Adjust Position Dialog** opens.

4. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other. The three teaching positions are at the cuvette tray area.

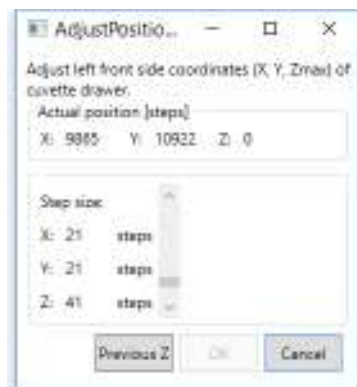


Figure 13–14: Adjust position dialog: Cuvette tray

#### Teaching of X- and Y- coordinates

5. For the three teaching positions, the point of the tip adapter shall be taught in X- and Y-axis to be centered on the bore position.

#### Teaching of Z-position

6. Move the sample probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt. (see also “Z-Adjustment” on page 11).
7. When the first position is taught successfully, press the **OK** button to acknowledge. The pipettor moves to the next teaching position.
8. Proceed as described in steps 6 and 7 also for the second and third teaching positions.

#### WARNING

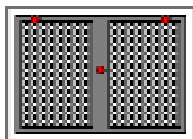


#### Automatic teaching procedure of the Cuvette-Tip Loading Module

After manual teaching of the three positions, an automatic procedure starts: Gripper detects the Z height of the drawer at three different positions for each cuvette tray. Pay attention to the Left Arm moving back and forth.

## 13.9 Tip tray

### 13.9.1 Teaching X, Y, Zmax



1. To start the tip tray teaching sequence, click onto the **Tip tray** button.
2. The window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other. The three teaching positions are at the DiTi tray area.



Figure 13–15: Adjust Position Dialog: 1<sup>st</sup> (left) tip tray position

#### Teaching of X- and Y- coordinates

4. The point of the tip adapter shall be taught in X- and Y-axis to be centered on the bore position.

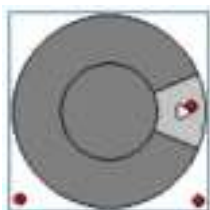
#### Teaching of Z-position

5. Move the sample probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt. (see also “Z-Adjustment” on page 11).
6. When the first position is taught successfully, press the **OK** button to acknowledge. The pipettor moves to the next teaching position.
7. Proceed as described in steps 4 to 6 also for the second and third teaching positions.



## 13.10 Washer

### 13.10.1 Teaching X, Y, Zmax



1. To start the washer teaching sequence, click onto the **Washer** button.
2. The window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other. The first two teaching positions are at the Washer metal plate. The third teaching position is directly on a cuvette which has to be inserted manually.



Figure 13–16: Adjust Position Dialog: washer position

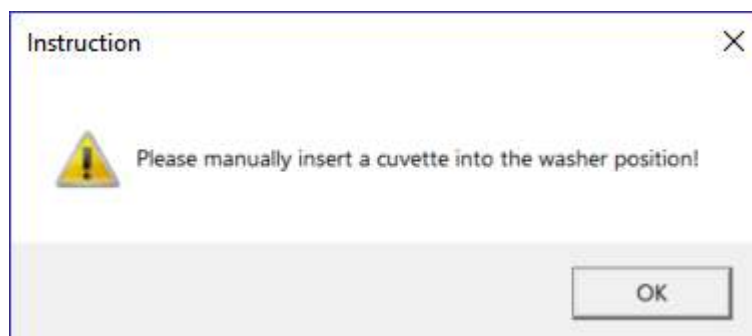


Figure 13–17: cuvette insertion for the third teaching position

#### Teaching of X- and Y- coordinates

4. The point of the tip adapter shall be taught in X- and Y-axis to be centered on the bore position of the first two teaching points.
5. The X- and Y- coordinates of the third teaching position - on the inserted cuvette - are automatically calculated and set by the **Teacher Software**.

### 13. Teaching of the Pipettor

---

#### Teaching of Z-position

6. For the first two teaching positions, move the sample probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt (see also “Z-Adjustment” on page 11).
7. When the first position is taught successfully, press the **OK** button to acknowledge. The pipettor moves to the second teaching position.
8. When the second position is taught successfully, press the **OK** button to acknowledge.
9. A popup opens, asking to manually insert a cuvette into the Washer. After having inserted the cuvette, press the **OK** button to acknowledge. The pipettor moves to the third teaching position on the cuvette.

---

**NOTE**

**Use of paper is optional for this step; if you prefer to use paper, you might take a longer piece of paper as access to the cuvette is limited!**

---

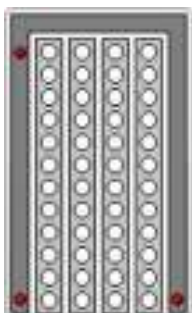
10. Move the Gripper down to the cuvette, decreasing the step size as you get close.
  - a. **Without using paper:** continue to move the Gripper down, until you observe a slight downward movement of the Washer rotor and cuvette. Then move the Gripper up, until you see and hear no more bouncing of the Washer rotor, meaning that the cuvette is barely touched.
  - b. **Using paper:** move the Gripper down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt (see also “Z-Adjustment” on page 11).
11. Press the **OK** button to acknowledge.



*Figure 13–18: Zmax teaching at third teaching position (case a: without using paper)*

## 13.11 Sample bay

### 13.11.1 Teaching X, Y, Zmax



1. To start the sample bay teaching sequence, click onto the **Sample bay** button.
2. The window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other. The three teaching positions are on the sample bay cover.



Figure 13–19: Adjust Position Dialog: sample bay position

#### Teaching of X- and Y- coordinates

4. The point of the tip adapter shall be taught in X- and Y-axis to be centered on the bore position.

#### Teaching of Z-position

5. Move the sample probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt. (see also “Z-Adjustment” on page 11).
6. When the first position is taught successfully, press the **OK** button to acknowledge. The pipettor moves to the second teaching position.
7. Proceed as described in steps 4 to 6 also for the second and third teach positions.
8. After the three teaching positions are taught, the sample arm moves to the tip-eject position.

#### Teaching of tip-eject Z-positions

9. The positions Z-max and Z-start need to be taught. These positions will be taught only in Z-axis: the X- and Y- positions are calculated automatically.

### 13. Teaching of the Pipettor

10. To teach the Z-max position, the sample probe has to be moved down until the eject mechanism is close to the metal sheet of the arm. The eject tube is about 1 mm above the metal sheet (can move down a little).



*Figure 13–20: Tip eject, Zmax position*

11. When Z-max position is taught, press the **OK** button to acknowledge. The pipettor moves up but keeps staying over the eject position.
12. To teach the Z-start position, the sample probe has to be moved down until the eject mechanism is 1 mm above the metal sheet when fully extended by hand (it does not touch the metal sheet).



*Figure 13–21: Tip eject, Z-start position*

## 13.12 Incubator sample probe

### 13.12.1 Teaching X, Y, Zmax



1. To start the incubator sample probe teaching sequence, click onto the **Sample pipetting** button.
2. The window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other.



Figure 13–22: Adjust Position Dialog: sample pipetting position

#### Teaching of X- and Y- coordinates

4. The point of the tip adapter shall be taught in X- and Y-axis to be centered on the bore position.

#### Teaching of Z-position

5. Move the sample probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt. (see also “Z-Adjustment” on page 11).
6. When the first position is taught successfully, press the **OK** button to acknowledge. The pipettor moves to the second teaching position.
7. Proceed as described in steps 4 to 6 also for the second teaching position.

## 13.13 Reader

### 13.13.1 Teaching X, Y, Zmax



1. To start the reader teaching sequence, click onto the **Reader** button.
2. The window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other.

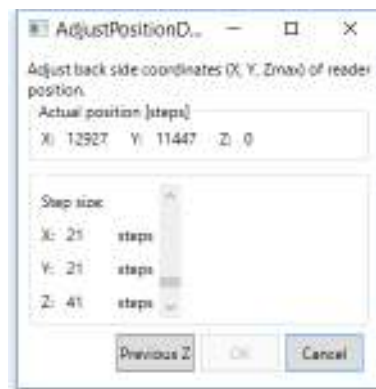


Figure 13–23: Adjust Position Dialog: reader position

#### Teaching of X- and Y- coordinates

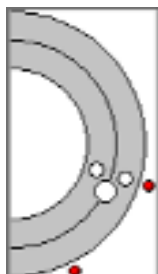
4. The point of the tip adapter shall be taught in X- and Y-axis to be centered on the bore position.

#### Teaching of Z-position

5. Move the sample probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt. (see also “Z-Adjustment” on page 11).
6. When the first position is taught successfully, press the **OK** button to acknowledge. The pipettor moves to the second teaching position.
7. Proceed as described in steps 4 to 6 also for the second teaching position.

## 13.14 Incubator reagent probe

### 13.14.1 Teaching X, Y, Zmax



1. To start the incubator reagent probe teaching sequence, click onto the **Incubator reagent probe** button.
2. The window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other.

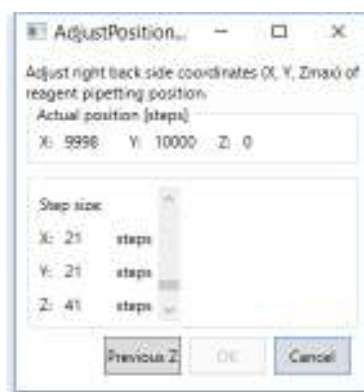


Figure 13–24: Adjust Position Dialog: reagent pipetting position

#### Teaching of X- and Y- coordinates

4. The point of the needle shall be taught in X- and Y-axis to be centered on the bore position.

#### Teaching of Z-position

5. Move the reagent probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt. (see also “Z-Adjustment” on page 11).
6. When the first position is taught successfully, press the **OK** button to acknowledge. The pipettor moves to the second teaching position.
7. Proceed as described in steps 4 to 6 also for the second teaching position.

## 13.15 Wash station

### 13.15.1 Teaching X, Y, Zmax



1. To start the wash station teaching sequence, click onto the **Wash station** button.
2. The window **Adjust Position Dialog** opens.
3. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other. The first two teaching positions are related to the wash position, while the third teaching position is related to the waste position.



Figure 13–25: Adjust Position Dialog: wash position

#### Teaching of X- and Y- coordinates

4. The point of the needle shall be taught in X- and Y-axis to be centered on the rinsing position of the wash station.

#### Teaching of 1<sup>st</sup> Z-position

5. To teach the first Z-position, move the needle carefully down until approaching the upper edge in the center of the wash station.
6. Move the needle into the wash station until the lower edge of the gear rack has a distance of 70mm to the upper edge of the wash station. In order to correctly set the distance of 70mm, place the ruler as shown in Figure 13–26 and level your eyes with the lower edge of the gear rack, in order to avoid parallax effect when reading the ruler scale. When Z-max is reached, the needle shall be in the middle of the bore.
7. When the position is taught successfully, press the **OK** button to acknowledge.





Figure 13-26: Wash station, Zmax



Figure 13-27: Wash station

#### Teaching of 2<sup>nd</sup> Z-position

8. The X- and Y- coordinates for the second teaching position are taken over from the first teach position. Move the reagent probe into the wash station until the lower edge of the gear rack has a distance of 135mm to the upper edge of the wash station.
9. In order to correctly set the distance of 135mm, place the ruler as shown in Figure 13-28 and level your eyes with the lower edge of the gear rack, in order to avoid parallax effect when reading the ruler scale.

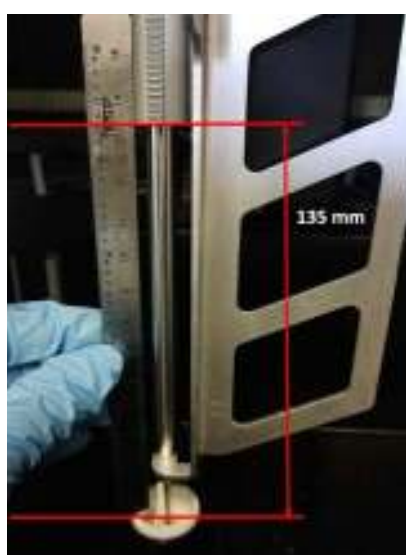


Figure 13-28: Wash station, Zstart

**Teaching of 3<sup>rd</sup> Z-position**

10. The X- coordinate from the 1<sup>st</sup> and 2<sup>nd</sup> positions can be used for the 3<sup>rd</sup> position.
11. Position the reagent probe over the deeper bore of the rinsing station. The Y-value is about 80 steps higher then the values of the 1<sup>st</sup> and 2<sup>nd</sup> positions.



Figure 13–29: Wash station, top view

12. Move the reagent probe down until the lower edge of the gear rack has a distance of 142mm to the upper edge of the wash station.
13. In order to correctly set the distance of 142mm, place the ruler as shown in Figure 13–30 and level your eyes with the lower edge of the gear rack, in order to avoid parallax effect when reading the ruler scale.



Figure 13–30: Wash station

## 13.16 Ancillary bay

### 13.16.1 Teaching X, Y, Zmax



1. Remove the Reagent Bay cover.
2. To start the ancillary bay teaching sequence, click onto the **Ancillary bay** button.
3. The window **Adjust Position Dialog** opens.
4. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other.

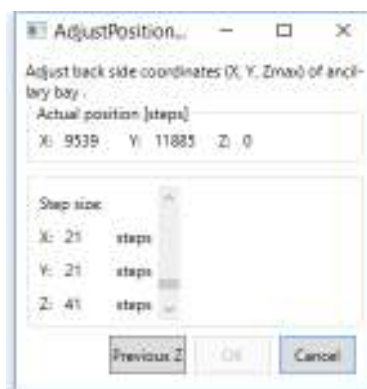


Figure 13–31: Adjust Position Dialog: ancillary bay position

#### Teaching of X- and Y- coordinates

5. The point of the needle shall be taught in X- and Y-axis to be centered on the bore position.

#### Teaching of Z-position

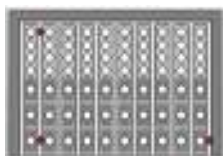
6. Move the reagent probe down until the paper surface is almost touched. To move the paper, a slight resistance shall be felt. (see also “Z-Adjustment” on page 11).
7. When the first teach position is taught successfully, press the **OK** button to acknowledge. The pipettor moves to the second teaching position.
8. Proceed as described in steps 4 to 6 also for the second teaching position.

#### Post-processing step

In case the Reagent Bay does not need to be taught, then reassemble the Reagent Bay cover.

## 13.17 Reagent bay

### 13.17.1 Teaching X, Y, Zmax



1. Remove the Reagent Bay cover.
2. To start the reagent bay teaching sequence, click onto the **Reagent bay** button.
3. The window **Adjust Position Dialog** opens.
4. Follow the commands shown in the Adjust Position Dialog window. The positions to be taught are displayed consecutively one after the other.



Figure 13–32: Adjust Position Dialog: reagent bay position

#### Teaching of X- and Y- coordinates

5. The point of the needle shall be taught in X- and Y-axis into the end of the notch, as close as possible to the notch edges but not touching them.

#### Teaching of Z-position

6. To adjust the Z- position, move the reagent probe to the right rear corner of the second cooling fin.
7. Move the reagent needle down, until the surface of the reference gap is barely touched, then raise it up of 2-3 steps. The needle is not allowed to touch the surface at all, it is always necessary to verify the possibility to move the edge freely.



Figure 13–33: Reagent bay, first teach position

8. When the first position is taught successfully, acknowledge with the **OK** button. The pipettor moves to the next teaching position.
9. Proceed as described in steps 4 to 7 also for the second and third teach positions.



*Figure 13–34: Reagent bay, second teach position*

#### **Post-processing steps**

10. Reassemble the Reagent Bay cover.

## 13.18 Sample Rack Teaching

### 13.18.1 Case 1: cylindrical tubes with standard inner diameter

#### NOTE

The procedure reported in this section can be applied in place of that described in chapter 13.18.2 every time the following criteria are verified:

- It is not possible to use one of the standard racks indicated in Chapter 6.3.5 on page 6-23.
- The inner diameter of the tubes to use is equal to one of those associated to standard rack typologies, Chapter 6.3.5 on page 6-23.

1. Using the information contained in Table 5–3 on page 24, identify the rack typology which has the same inner diameter of the tubes to use.
2. Open the **Teacher Software** and press the **Pick Tip** button; ensure that a tip is picked up.
3. In the **File Menu** of **Teacher Software**, choose **Open** and then **Rack file**.
4. Select the ".rac" file of the previously identified rack typology.
5. Click on **Save As** button and name it with one of the customized rack IDs: **G.rac**, **O.rac**, **Y.rac** and **Z.rac** are the options to choose among.
6. In **Area** drop-down menu, select **Sample Bay**.
7. In **Lane** drop-down menu, select **1**.
8. Verify that no racks are equipped in lane 1 of the Sample Area.
9. Check the **Interpolate from position ... to position ...** option, specifying the first and last positions of the rack to teach.
10. Press the **Teach** button.
11. In **Adjust Position** window that appears, press the **Prev. Z** button.
12. Measure the height of the customized tube and verify the number of spacers to use in each position (**Sp<sub>cust</sub>**), according to Table 5–2 on page 23.
13. Using the information reported in Table 5–2 on page 23, verify the number of steps (**step<sub>cust</sub>**) associated to the number of spacers **Sp<sub>cust</sub>** calculated at previous point.
14. Using the information reported in Table 5–3 on page 24, verify the standard number of spacers (**Sp<sub>st</sub>**) associated to the standard tube typology having the same inner diameter of the customized one.
15. Using the information reported in Table 5–2 on page 23, verify the number of steps (**step<sub>st</sub>**) associated to the number of spacers **Sp<sub>st</sub>** calculated at previous point.
16. Calculate the difference **Delta = step<sub>cust</sub> - step<sub>st</sub>**.
17. Using dedicated keys of the keyboard, move **Z<sub>max</sub>** coordinate of tube position 1 according to the following rules:
  - If **Delta > 0**, move left arm **up** for the number of steps calculated at point 16;
  - If **Delta ≤ 0**, move left arm **down** for the number of steps calculated at point 16.

**Example 1.** For the creation of a **Y.rac** file for tubes with 9 mm inner diameter and 75 mm height:

- A copy of the **J.rac**, named **Y.rac**, shall be created;

- 2 spacers ( $\text{Sp}_{\text{cust}}$ ) shall be used to configure each position;
  - The associated number of steps ( $\text{step}_{\text{cust}}$ ) is 500 steps;
  - The standard number of spacers ( $\text{Sp}_{\text{st}}$ ) is equal to 3 (i.e. the number associated to the  $\text{J.rac}$ );
  - The associated number of steps ( $\text{step}_{\text{st}}$ ) is 700 steps;
  - $\Delta = 500 \text{ steps} - 700 \text{ steps} = -200 \text{ steps}$ .
  - $Z_{\text{max}}$  coordinates of the  $\text{Y.rac}$  shall be **moved down** by 200 steps.
18. Press the **OK** button; **Adjust Position** window for teaching of  $Z_{\text{start}}$  appears.
  19. Press the **Prev. Z** button and then **OK** button, leaving the coordinate unchanged.
  20. Repeat points 11-19 to teach the last tube position.
  21. Press the **OK** button to save the taught coordinates.
  22. In lane 1, equip a sample rack customized with the needed number of spacers and with at least 1 tube at intermediate rack position.
  23. Uncheck the **Interpolate from position ... to position ...** option.
  24. Press the **Teach** button.
  25. Press the **Prev. Z** button and then **OK** button, leaving the coordinate unchanged.
  26. Repeat point 25 until the rack position where the tube is placed is reached.
  27. In **Adjust Position** window, press **Prev. Z** button.
  28. Moving the tube up and down, a gap shall be verified between the inner bottom of the tube and the edge of the tip.
  29. Repeat points 18 and 19.
  30. Press the **Save** button of **Rack** window.
  31. Press the **Close** button of **Rack** window.
  32. Press the **Eject Tip** button.
  33. Perform transfer of coordinates, see Chapter 13.19 on page 13-44.
  34. Perform the setting of `BottleShapes.xml` file, see Chapter 13.18.4 on page 13-42.

### 13.18.2 Case 2: cylindrical tubes with not standard inner diameter

1. Open the **Teacher Software** and press the **Pick Tip** button; ensure that a tip is picked up.
2. In the **File Menu** of **Teacher Software**, choose **New** and then **Rack file**.
3. In **Width** field, insert **1**.
4. In **Positions** field, insert **12**.
5. In **Surface** field, leave the default setting.
6. In **Area** drop-down menu, select **Sample Bay**.
7. In **Lane** drop-down menu, select **1**.
8. Verify that no racks are equipped in lane 1 of the Sample Area.
9. Check the **Interpolate from position ... to position ...** option, specifying the first and last positions of the rack to teach.
10. Calculate the cylindrical section (**S**) associated to the inner diameter of the tubes to use (**d**).

### 13. Teaching of the Pipettor

11. For the estimation of the  $Z_{\max}$  volume ( $V_{Z_{\max}}$ ) of the tubes, consider the following rules:
  - 150  $\mu\text{L}$  for cylindrical tubes with inner diameter equal or less than 10 mm;
  - 250  $\mu\text{L}$  for cylindrical tubes with inner diameter greater than 10 mm.
12. Using the previously identified volume, estimate the coordinate  $Z_{\max}$  to set and calculate the related number of steps to teach. Remember that 1 mm = 20 steps.

**Example 3.** Cylindrical Tube with 9.5 mm inner diameter:

- $V_{Z_{\max}}$ : 150  $\mu\text{L}$
- $d$ : 9.5 mm
- $S$ : 70.88  $\text{mm}^2$
- $Z_{\max}$  (in mm):  $150 \mu\text{L} / 70.88 \text{ mm}^2 = \sim 2.12 \text{ mm}$
- $Z_{\max}$  (in steps):  $2.12 \text{ mm} \cdot 20 \text{ steps/mm} = \sim 42 \text{ steps}$

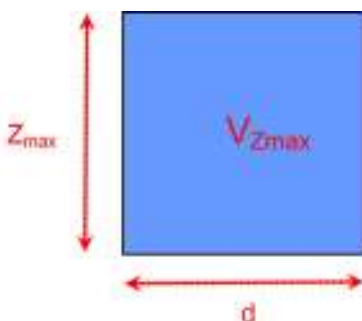


Figure 13–35: Cylindrical Tube Model

13. Press the **Teach** button.
14. Teach “X” and “Y” coordinates of rack position 1. The edge of the tip shall be aligned to the center of the tube.
15. Teach “ $Z_{\max}$ ” coordinate of rack position 1: move the left arm (with the tip) down, until the surface of the tube is barely touched. Then raise it up by the number of steps previously calculated.
16. Press the **OK** button; **Adjust Position** window for the teaching of  $Z_{\text{start}}$  coordinate appears.
17. Teach “ $Z_{\text{start}}$ ” coordinate. The edge of the tip shall be aligned to the upper border of the tube.
18. Press the **OK** button: **Adjust Position** window for the teaching of X, Y and  $Z_{\max}$  coordinates of last rack position appears.
19. Repeat points 14 to 17 for the last rack position.
20. Press the **OK** button to save the taught coordinates.
21. Press **Save As** button of **Rack** window.
22. Select the `C:\LiaisonXS\Data\Coordinates` folder; in the **File Name** field, insert the name of the rack to teach: `G.rac`, `O.rac`, `Y.rac` and `Z.rac` are the options to choose among.
23. Press the **Close** button of **Rack** window.



24. Press the **Eject Tip** button.
25. Perform transfer of coordinates, see Chapter 13.19 on page 13-44.
26. Perform the setting of `BottleShapes.xml` file, see Chapter 13.18.4 on page 13-42.

### 13.18.3 Case 3: not standard conical tubes

The following procedure shall be applied for the teaching of conical tubes different from Eppendorf 9.5 mm (inner diameter of cylindrical section) x 35 mm (height) ones. For this specific typology of tubes, standard `P.rac` is already available in the system.

1. Open the **Teacher Software** and press the **Pick Tip** button; ensure that a tip is picked up.
2. In the **File Menu** of **Teacher Software**, choose **New** and then **Rack file**.
3. In **Width** field, insert **1**.
4. In **Positions** field, insert **12**.
5. In **Surface** field, leave the default setting.
6. In **Area** drop-down menu, select **Sample Bay**.
7. In **Lane** drop-down menu, select **1**.
8. Verify that no racks are equipped in lane 1 of the Sample Area.
9. Check the **Interpolate from position ... to position ...** option, specifying the first and last positions of the rack to teach.
10. Press the **Teach** button.
11. Teach "X" and "Y" coordinates of rack position 1. The edge of the tip shall be aligned to the center of the tube.
12. Teach " $Z_{max}$ " coordinate of rack position 1: move the left arm down, until the tip is barely touching the tube bottom. Then raise it up by 3 mm (i.e. 60 steps).
13. Press the **OK** button; **Adjust Position** window for the teaching of  $Z_{start}$  coordinate appears.
14. Teach " $Z_{start}$ " coordinate. The edge of the tip shall be aligned to the upper border of the tube.
15. Press the **OK** button: **Adjust Position** window for the teaching of X, Y and  $Z_{max}$  coordinates of last rack position appears.
16. Repeat points 11 to 15 for the last rack position.
17. Press the **OK** button to save the taught coordinates.
18. Press **Save As** button of **Rack** window.
19. Select the `C:\LiaisonXS\Data\Coordinates` folder; in the **File Name** field, insert the name of the rack to teach: `G.rac`, `O.rac`, `Y.rac` and `Z.rac` are the options to choose among.
20. Press the **Close** button of **Rack** window.
21. Press the **Eject Tip** button.
22. Perform transfer of coordinates, see Chapter 13.19 on page 13-44.
23. Using a pipette, estimate the volume of liquid underneath  $Z_{max}$  coordinate ( $V_{Z_{max}}$ ).
24. Estimate the section ( $S_{Z_{max}}$ ) of  $Z_{max}$  cone (i.e. the "blue" cone of Figure 13-36). This section shall then be used for the setting of `BottleShapes.xml` file (see Chapter 13.18.4 on page 13-42).

## 13. Teaching of the Pipettor

25. Measure the height (**h**) of the conical portion.
26. Measure the inner diameter (**d**) of the cylindrical portion of the tube and calculate the associated section (**S**). Also this section shall be used for the setting of `BottleShapes.xml` file.
27. Perform the setting of `BottleShapes.xml` file, see Chapter 13.18.4 on page 13-42.
28. In order to prevent possible adverse effects due to mechanical tolerances and imprecision of the conical model, communicate to the customer a dead volume (**V<sub>d</sub>**) of 2 times **V<sub>Zmax</sub>**. In any case, a minimum **V<sub>d</sub>** of 50 µL shall be guaranteed.

**Example 4.** Conical Tube with  $V_{Zmax} = 30 \mu\text{L}$ :

- $V_{Zmax}$ : 30 µL
- $Z_{max}$ : 3 mm
- $S_{Zmax}$ :  $3 \cdot 30 \mu\text{L} / 3 \text{ mm} = 30 \text{ mm}^2$
- $V_d = 2 \cdot 30 \mu\text{L} = 60 \mu\text{L}$

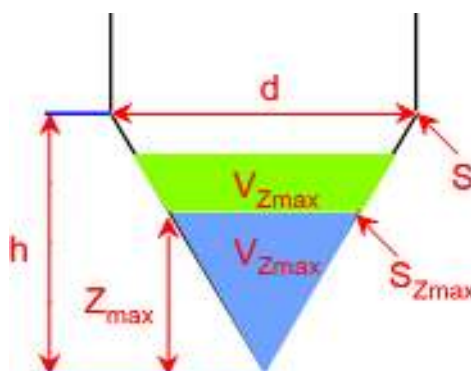


Figure 13–36: Conical Tube Model

### 13.18.4 BottleShapes.xml setting

`BottleShapes.xml` file shall be updated every time a new rack typology is configured on a **LIAISON® XS** analyzer; the setting of parameters available in `BottleShapes.xml` file allows to define the dimensions of the different sections of the rack vials.

The editing of `BottleShapes.xml` file shall be performed once completed the teaching with **Rack** window of **Teacher Software** (see previous chapters).

`BottleShapes.xml` file shall be opened from the **File Menu** of **Teacher Software**, by choosing **Open** and then **Bottle shapes file**.

Once opened the file, the information reported below shall be edited for each new rack typology:

```
<BottleShapes BottleIndex="N" Description="Name">
  <ShapeData Height="Step1" CrossSectionArea="Surface1"/>
  <ShapeData Height="Step2" CrossSectionArea="Surface2"/>
  ...
</BottleShape>
```

The meaning of the different fields to insert is described in Table 13–10.

| Function                  | Description                                                                                                                                                                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "N"                       | The bottle ID shall be indicated, according to the rule of Table 13–11.                                                                                                                                                                                                                                     |
| "Name"                    | The name of the rack (the same set also in <b>Rack</b> window of <b>Teacher Software</b> ) shall be indicated.<br>Possible options: "G", "O", "Y", "Z".                                                                                                                                                     |
| "StepN"<br>N= 1,2,3...    | Z coordinate (calculated from $Z_{\max}$ and expressed in <u>number of steps</u> ) of each section to define shall be indicated.<br>The coordinates shall be inserted from the highest to the lowest, until $Z_{\max}$ coordinate (0 steps) is reached.<br><b>Important!</b> Remember that 1 mm = 20 steps. |
| "SurfaceN"<br>N= 1,2,3... | The surface (in $\text{mm}^2$ ) of each section to define shall be indicated.                                                                                                                                                                                                                               |

Table 13–10: BottleShapes.xml Legenda

| Rack Name | Bottle ID |
|-----------|-----------|
| G         | 8         |
| O         | 16        |
| Y         | 26        |
| Z         | 27        |

Table 13–11: Bottle IDs

### CAUTION

The setting of sections in BottleShapes.xml file shall be always performed using integer values.

The definition of cylindrical vials (case 1, chapter 13.18.1 and case 2, chapter 13.18.2) requires only the definition of the section (**S**) at  $Z_{\max}$  coordinate (i.e. coordinate 0).

For the definition of conical vials (case 3, chapter 13.18.3) it is necessary to set at least 2 sections, one at the beginning of the cylindrical section, and one at  $Z_{\max}$ : the software is then able to use an interpolation method to complete the definition of the vial.

In case of more complex vials, it is necessary to indicate a parameter set (Z coordinate and the associated surface) for each sudden change of section (e.g. a step profile) detected.

**Example 5** – Configuration of BottleShapes.xml to create a Y.rac for the conical tube of Example 4 (see Figure 13–36):  $S = 78.5 \text{ mm}^2$ ,  $h = 10 \text{ mm}$ ,  $Z_{\max} = 3 \text{ mm}$ ,  $S_{Z_{\max}} = 30 \text{ mm}^2$ . In order to complete the activity, it is necessary to estimate the distance between  $Z_{\max}$  and the beginning of the cylindrical section:  $h - Z_{\max} = 7 \text{ mm}$  (140 steps).

```
<BottleShapes BottleIndex="26" Description="YRack">
  <ShapeData Height="140" CrossSectionArea="78"/>
  <ShapeData Height="0" CrossSectionArea="30"/>
</BottleShape>
```

## 13.19 Transfer sequence

**NOTE****Create a backup**

It is recommended to create a  as a precaution measure.

1. When all positions are taught, click onto the **Transfer** button to transfer the taught coordinates to the Pip CU. After completion of the transfer, the information window can be closed by clicking on the **OK** button.

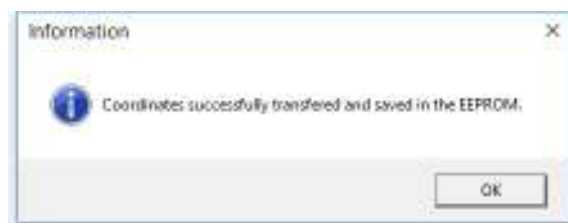


Figure 13–37: Transfer, information

2. When exiting the teacher software, a window comes up asking again if you want to transfer the coordinates. If you are sure that you have done the transfer before, then click onto the **No** button.



Figure 13–38: Close question

# 14 Covers

This chapter describes the outer Covers in detail. The first section of the chapter introduces a brief overview of the main outer covers, flaps and connections of the LIAISON<sup>®</sup> XS Analyzer.

Then the following section contains information about repair and calibration of these modules ("Replacement and Repair", section).

## 14.1 Safety

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 OR see Chapter 2.5 on page 2-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

## 14.2 Overview

This section shows an overview of the components of the LIAISON<sup>®</sup> XS outer Covers.

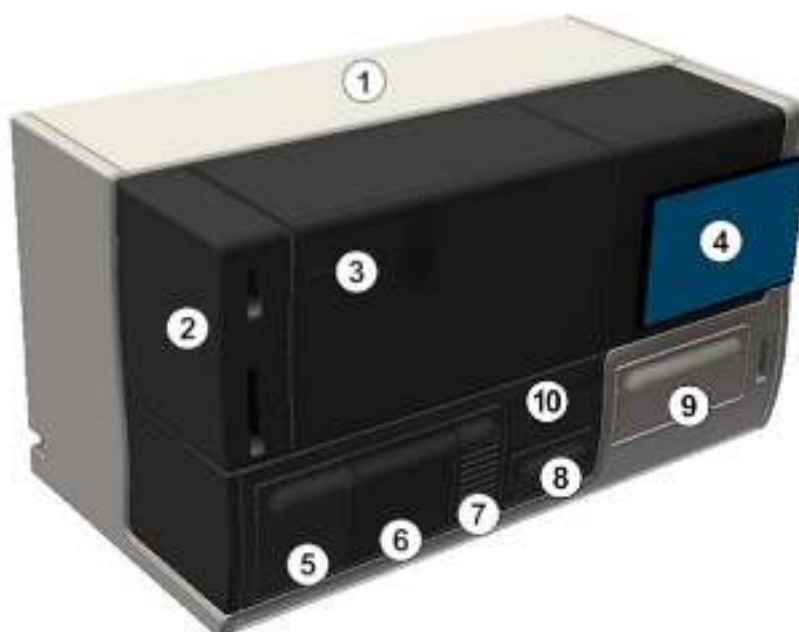
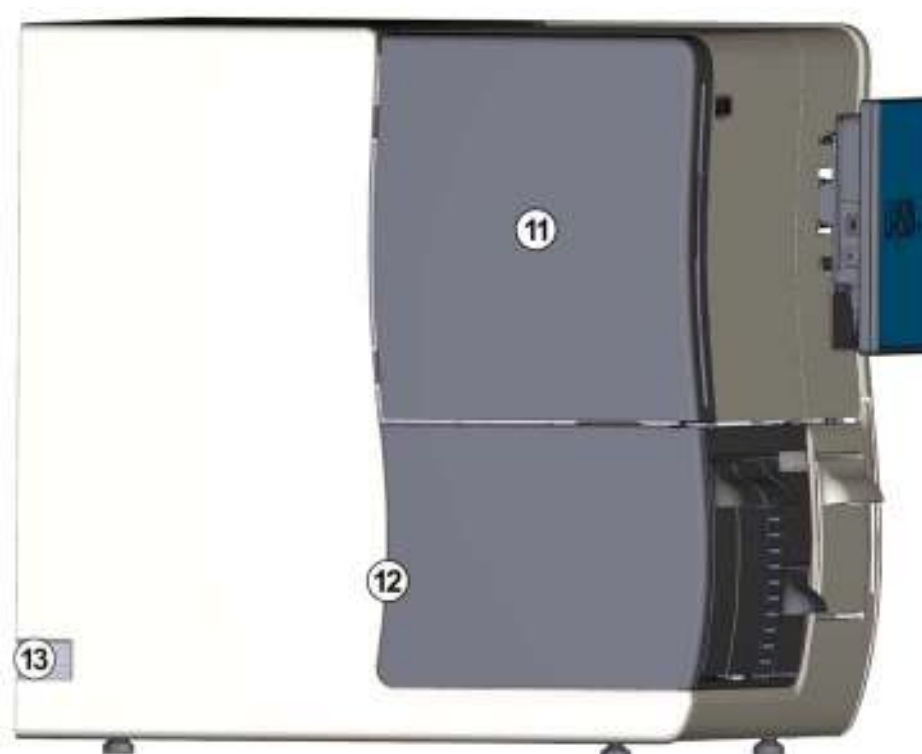


Figure 14–1: Instrument

## 14. Covers

|    |                                 |
|----|---------------------------------|
| 1  | Top cover                       |
| 2  | Bulk Fluid door                 |
| 3  | Left Top Cover                  |
| 4  | Right Top Cover (with Panel PC) |
| 5  | Starter Reagent Module flap     |
| 6  | Cuvette-Tip Loading Module      |
| 7  | Sample Module flap              |
| 8  | Solid waste bin                 |
| 9  | Reagent Module flap             |
| 10 | Front Module                    |

*Table 14–1: Instrument outer covers and flap overview*



*Figure 14–2: Instrument, left side view*

|    |                              |
|----|------------------------------|
| 11 | Bulk Fluid Door              |
| 12 | Left Side Cover              |
| 13 | Opening for tubes and cables |

*Table 14–2: Instrument Left Side Cover overview*

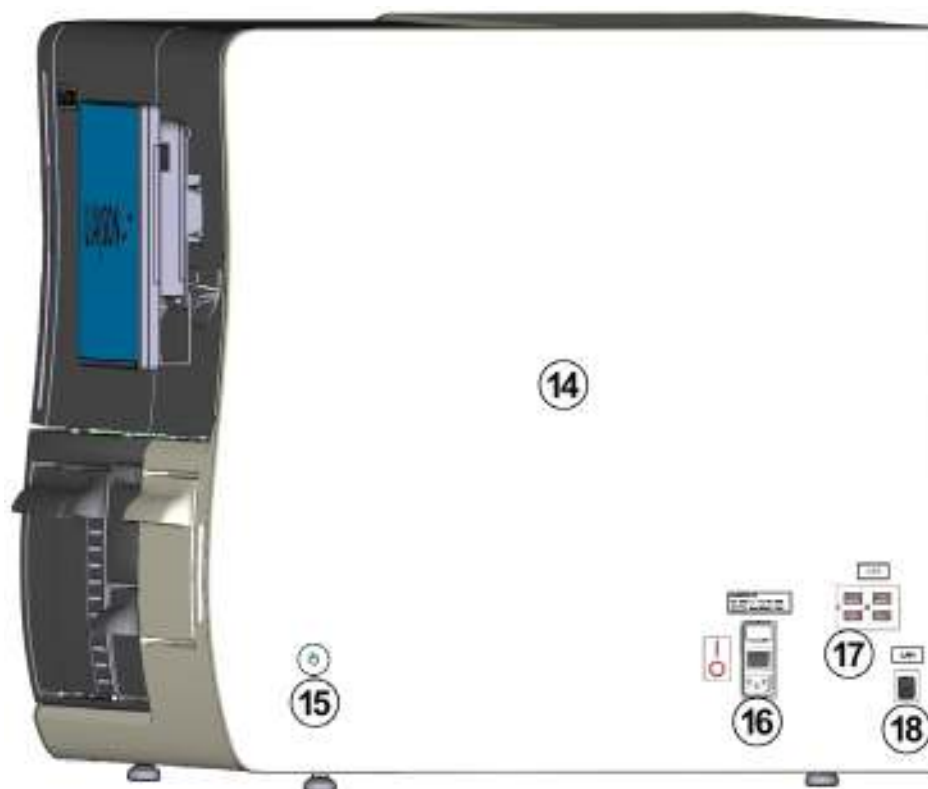


Figure 14-3: Instrument, right side view

|    |                            |
|----|----------------------------|
| 14 | Right Side Cover           |
| 15 | Soft Power Button          |
| 16 | Power Main Switch and fuse |
| 17 | USB Ports                  |
| 18 | LAN connector              |

Table 14-3: Instrument Right Side Cover and connection overview

### 14.3 Replacements and repairs

This section contains instructions for the repair of individual components of the instrument Covers. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

#### 14.3.1 Replacement of the Left Side Cover

##### 14.3.1.1 Inside view of Left Side Cover

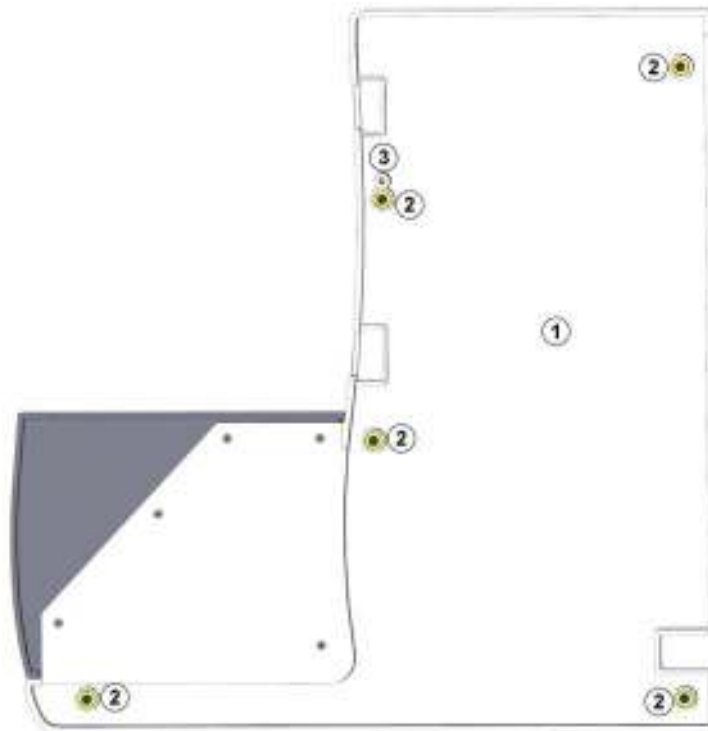


Figure 14-4: Left Side Cover, inside view

- |    |                                       |
|----|---------------------------------------|
| 1. | Left Side Cover                       |
| 2. | Snap lock pins                        |
| 3. | Thread for fixing cover to instrument |

Figure 14-5: Left Side Cover, overview



### 14.3.1.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Left Top Cover.
6. Move Sample pipettor arm to a safe location.

### 14.3.1.3 Removal of the Left Side Cover

1. Unscrew cross-slotted screw (1).
2. Unplug the snap locks of the Left Side Cover and set cover to the side.

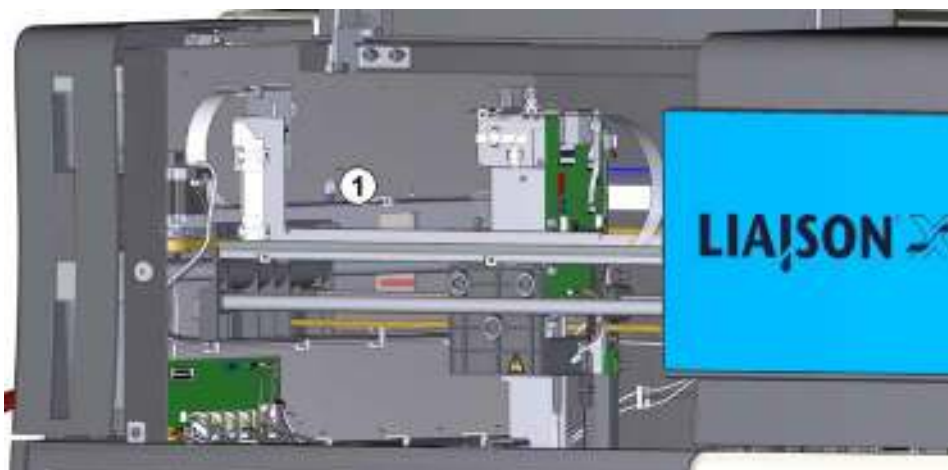


Figure 14–6: Fixing screw of the Left Side Cover, inside the instrument

### 14.3.1.4 Assembly of the Left Side Cover

1. Set Left Side Cover into the pins.
2. Press the Left Side Cover at the pins location till they snap in.
3. Fasten screw (1) to secure the Left Side Cover.

### 14.3.2 Replace of the Bulk Fluid Door

#### 14.3.2.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Left Top Cover.
6. Move Sample pipettor arm to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.

#### 14.3.2.2 Removal of the Bulk Fluid Door

1. Lift the Bulk Fluid door (1) out of the hinges (2) and set it aside.

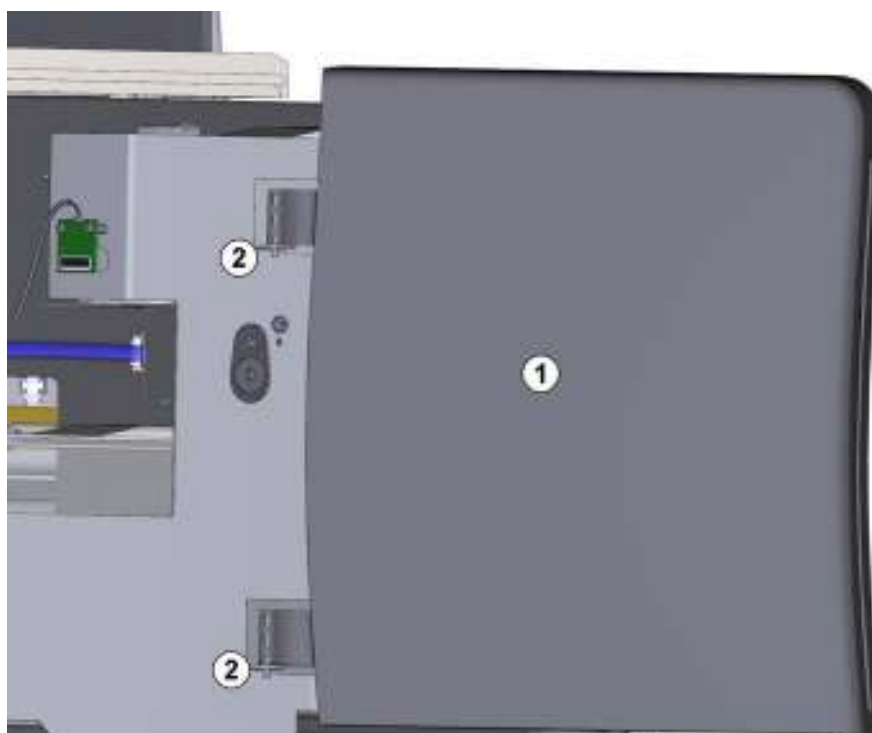


Figure 14–7: Bulk Fluid door

#### 14.3.2.3 Assembly of the Bulk Fluid Door

1. Place the Bulk Fluid Door (1) inside the hinges (2).
2. Assembly the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 14.3.3 Replacement of the Right Side Cover

#### 14.3.3.1 Inside view of Right Side Cover

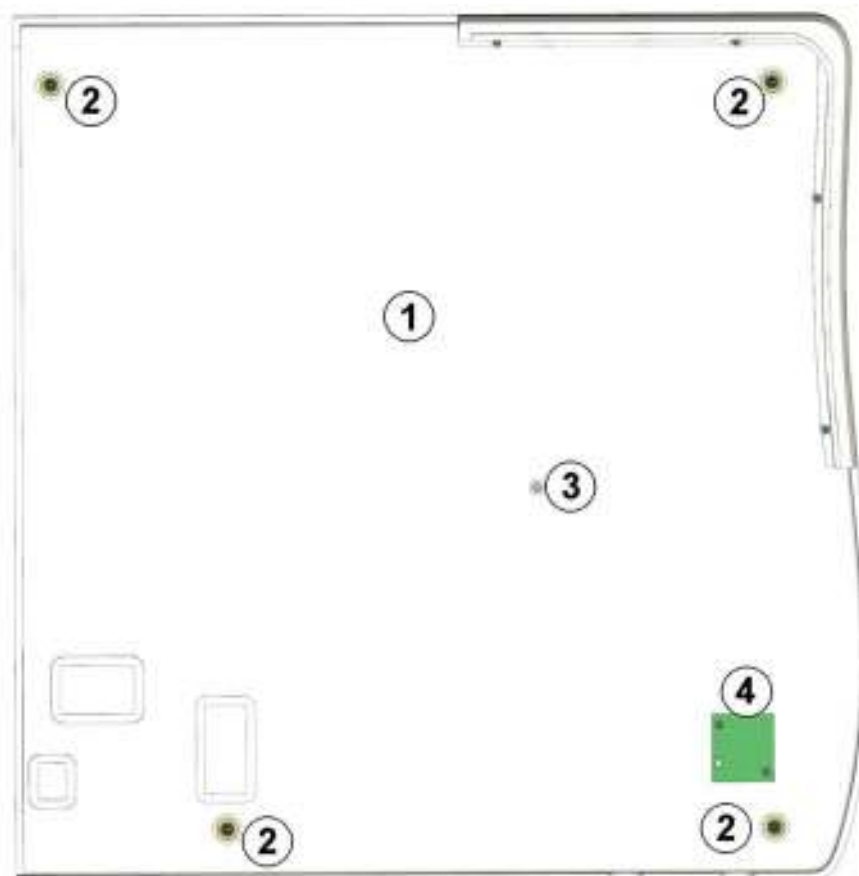


Figure 14-8: Right Side Cover, inside view

- |    |                                       |
|----|---------------------------------------|
| 1. | Right Side Cover                      |
| 2. | Snap lock pins                        |
| 3. | Thread for fixing cover to instrument |
| 4. | Soft Power Button PCBA                |

Figure 14-9: Right Side Cover, overview

**14.3.3.2 Preliminary steps**

1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Right Top Cover.
6. Move Right pipettor arm to a safe location.

**14.3.3.3 Removal of the Right Side Cover**

1. Unscrew cross-slotted screw (3).
2. Unplug the snap locks of the Right Side Cover and unplug the cable to the Soft Power Button PCBA (4).
3. Set Right Side Cover to the side.

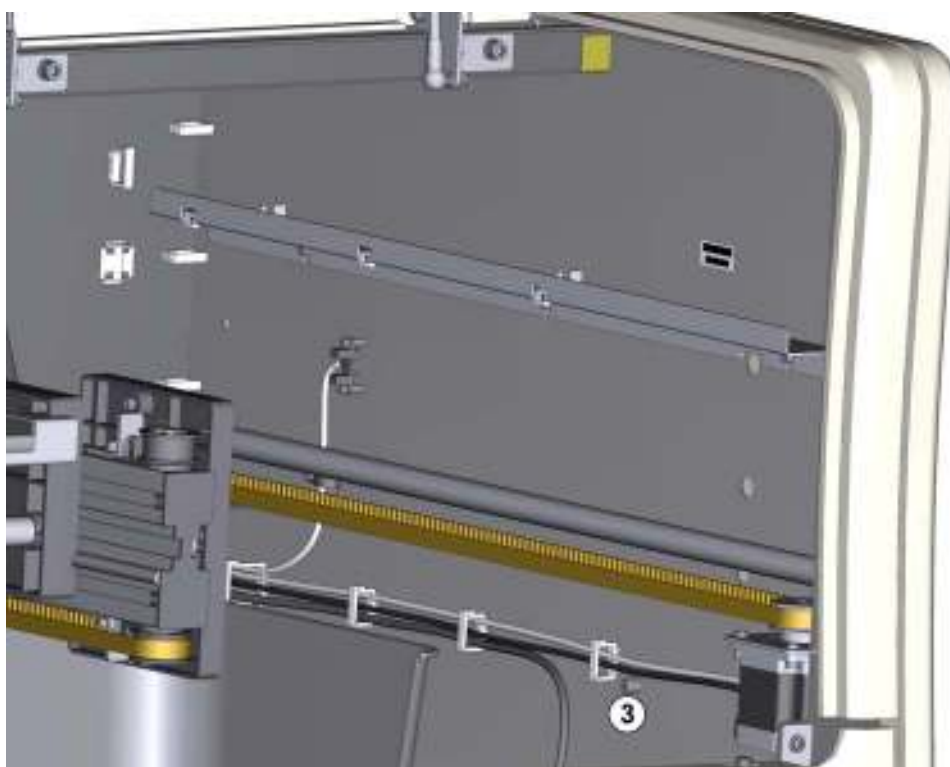


Figure 14–10: Fixing screw of the Right Side Cover, inside the instrument

#### 14.3.3.4 Assembly of the Right Side Cover

1. Connect cable onto Soft Power Button PCBA (4).
2. Set Right Side Cover into the pins.
3. Press the Right Side Cover at the pins location till they snap in.
4. Fasten screw (3) to secure the Right Side Cover.

#### 14.3.4 Replacement of the Panel PC

##### 14.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. If it is possible, save the Teaching coordinate file `Coordinate.coo` in the folder `C:\LiaisonXS\Data\Coordinates`.
3. If it is possible and the customer needs, save the folders containing logfile and database: `C:\LiaisonXS\Data\Temp` `C:\LiaisonXS\Data\Databases`.
4. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
5. On the right side of the system turn the power switch off.
6. Unplug the system power plug from the right side of the system.
7. Completely open the Top Covers.
8. Move Pipettor arms to a safe location.

##### 14.3.4.2 Removal of the Panel PC

1. Loose two screws (3) of cable carrier (2).
2. Slide the cable carrier (2) out of the keyholes and set cable carrier aside.

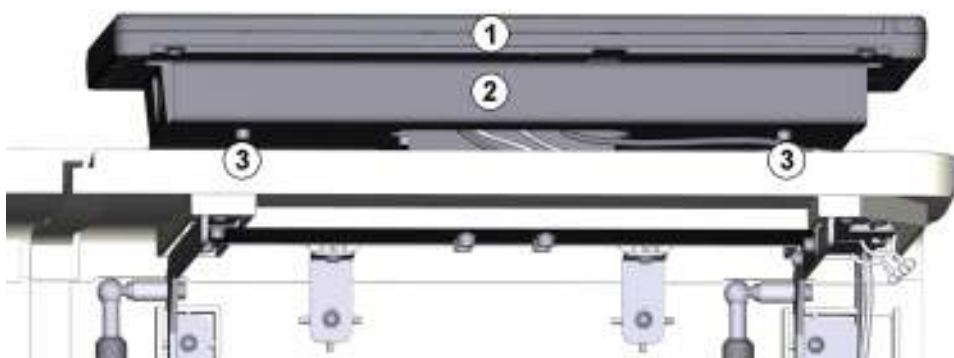
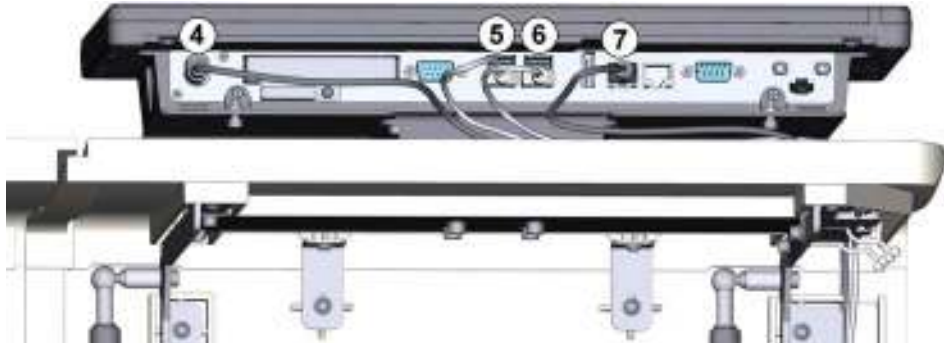


Figure 14–11: Panel PC with Right Top Cover opened, front view

3. Disconnect Panel PC connections power cable (4), USB connection (5 & 6) and LAN cable (7).

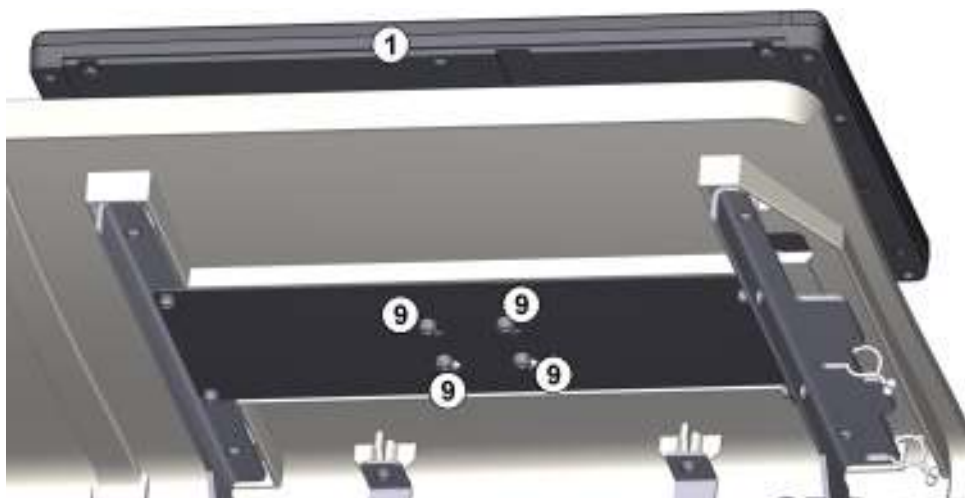


*Figure 14–12: Panel PC with Right Top Cover opened, connections*



*Figure 14–13: Ferrite snap connected to the power cable of the Panel PC*

4. Loosen four screws (9) of Panel PC. Lift Panel PC (1) out of the screw holes and set it aside.



*Figure 14–14: Panel PC screws*

#### 14.3.4.3 Assembly of the Panel PC

1. Set screw heads (9) of the Panel PC into the screw holes of the Right Top Cover.
2. Fasten four screws (9).
3. Connect the power cable (4), USB connection (5 & 6) and LAN cable (7) to the Panel PC (1).
4. Clip the Ferrite snap to the power cable: Ferrite must be kept open, put the cable in the Ferrite and close it with simple pressure by hand to reach clip closure. The position of the ferrite on the power cable is close to the connection to the Panel PC (about 35mm +/- 5mm from the power connector).

#### NOTE

Once the ferrite snap is first closed, it may be difficult to open it again without using a tool. Make sure to close the Ferrite snap once it is positioned as desired.



Figure 14–15: Position of the Ferrite snap on the power cable of the Panel PC

5. Slide the cable carrier (2) into the key holes and fasten two screws (3).
6. Switch on the instrument.
7. Switch on the Panel PC.
8. Install the newest Main Software version according to the dedicated TN.
9. If needed (and if they were saved), restore the Teaching Coordinate file and the logfiles/databases. Then, open the **Teacher Software** and press the button **Transfer** to transfer the coordinates.



### 14.3.5 Replacement of the Left Top Cover

#### 14.3.5.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Right and Left Top Cover.
6. Move Pipettor arms to a safe location.

#### 14.3.5.2 Removal of the Left Top Cover

1. Hold up to the Left Top Cover (1) and unscrew four screws (2) of the cover brackets.
2. Lift Left Top Cover up and set it aside.

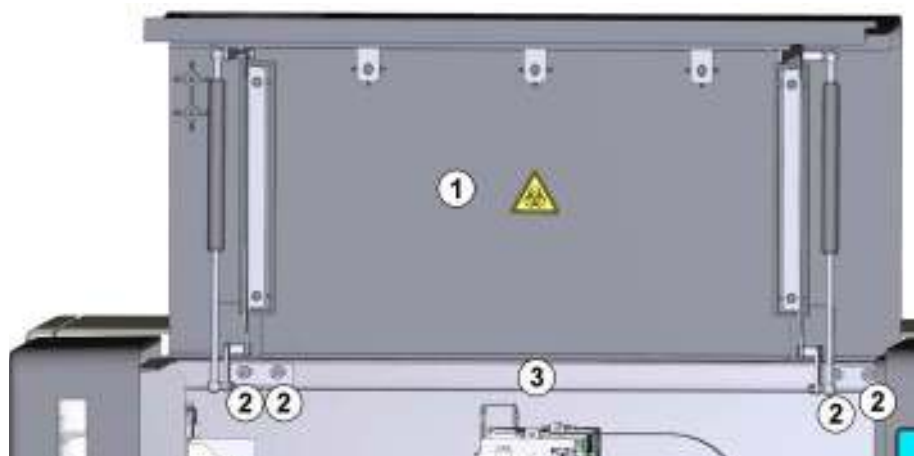


Figure 14–16: Left Top Cover opened, with details

3. Fasten the four screws (2) back into the support bar (3).

#### 14.3.5.3 Assembly of the Left Top Cover

1. Unscrew four screws (2) of the support bar (3).
2. Place the Left Top Cover into place and screw the cover bracket loose on.
3. Center and align Left Top Cover, see Chapter 14.4 on page 14-20.
4. Fasten four screws (2) at support bar.



### 14.3.6 Replacement of the Right Top Cover

#### 14.3.6.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Right and Left Top Cover.
6. Move Pipettor arms to a safe location.
7. Remove the Panel PC, see Chapter 14.3.4.2 on page 14-9.

#### 14.3.6.2 Removal of the Right Top Cover

1. Remove Panel PC cables (1) out of four cable holder (2).

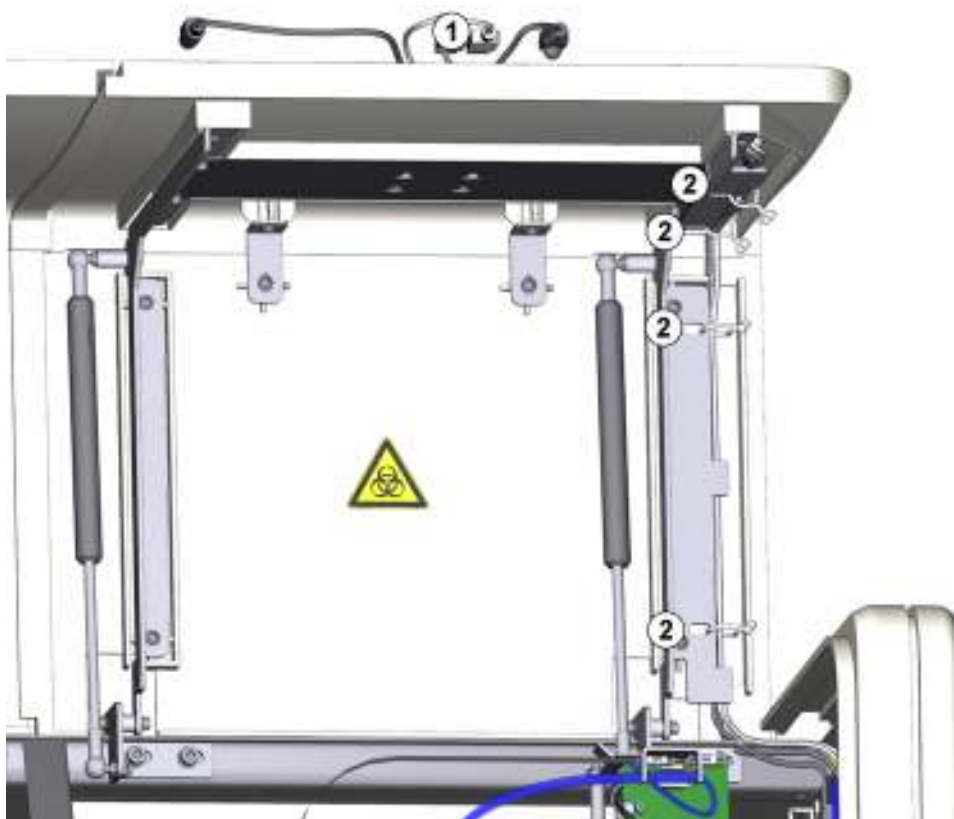


Figure 14–17: Right Top Cover with Panel PC cable

2. Hold up to the Right Top Cover and unscrew four screws (3) of the cover brackets.

3. Lift Right Top Cover up and set it aside.

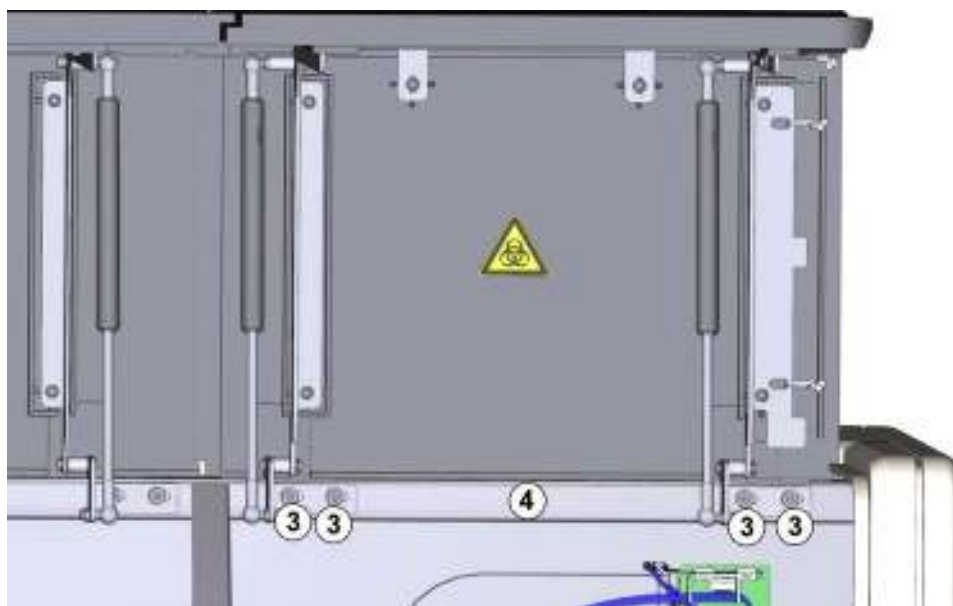


Table 14-4: Right Top Cover with detail

4. Fasten the four screws (3) back into the support bar (4).

#### 14.3.6.3 Assembly of the Right Top Cover

1. Unscrew four screws (3) of the support bar (4).
2. Lift the Right Top Cover into place and screw the cover bracket (8) loose on.
3. Center and align Right Top Cover, see Chapter 14.4 on page 14-20.
4. Fasten four screws (2).
5. Assembly of the Panel PC, see Chapter 14.3.4.3 on page 14-11.

### 14.3.7 Replacement of the Top Cover

#### 14.3.7.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Right and Left Top Cover.
6. Move Pipettor arms to a safe location.

### 14.3.7.2 Removal of the Top Cover

1. Loosen two screws (2) at right side of the Top Cover (1).

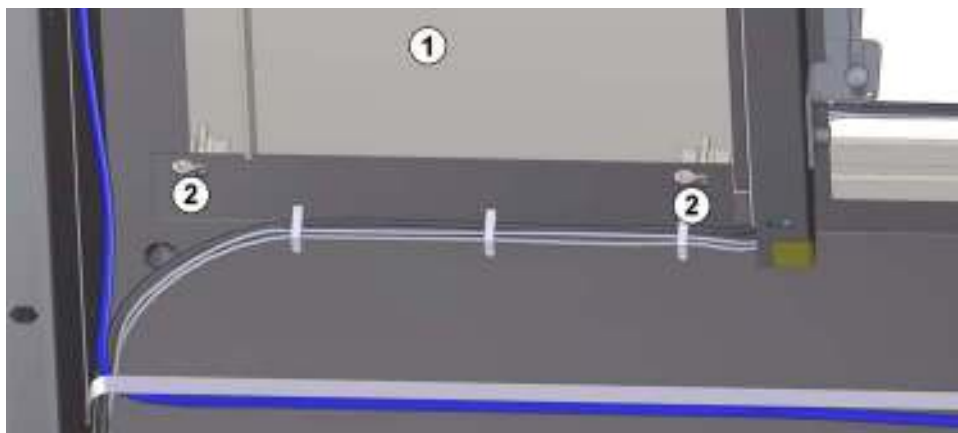


Figure 14-18: Top Cover, right side view inside the instrument

2. Loosen screw (2) at middle of the Top Cover (1).

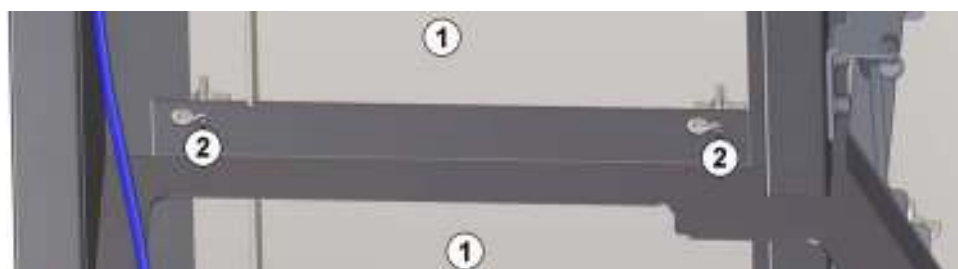


Figure 14-19: Top Cover, middle view inside the instrument

3. Loosen screw (2) at left of the Top Cover (1).

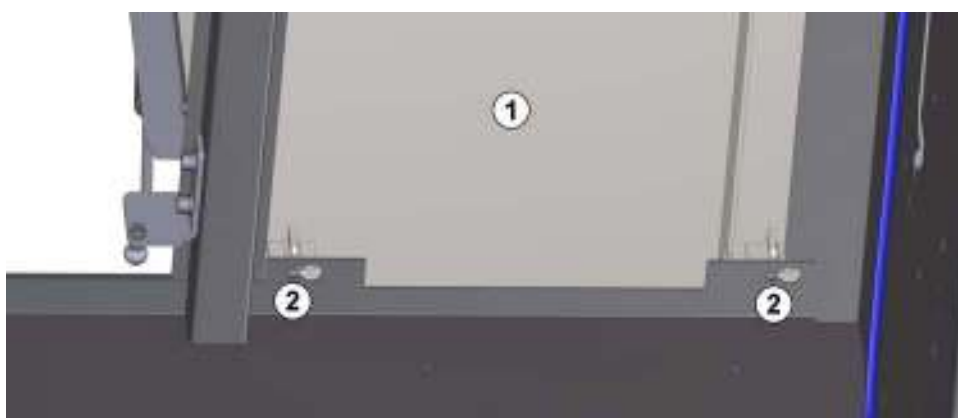


Figure 14-20: Top Cover, left side view inside the instrument

4. Slide Top Cover out of the key holes. Lift Top Cover out and set it aside.

5. Fasten six screws (2) of the Top Cover (1).



Figure 14–21: Top cover, bottom view

#### 14.3.7.3 Assembly of the Top Cover

1. Loosen screws (2) at Top Cover.
2. Set Top Cover onto the chassis and slide the six screw heads into the key holes.
3. Fasten six screws (2).

#### 14.3.8 Replacement of the Front Module

For the replacement and assembling of the Front Module, please see Chapter 27.3.1 on page 27-4.

#### 14.3.9 Replacement of the Gas Pressure Spring at Top Cover

##### 14.3.9.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Right and Left Top Cover.
6. Move Pipettor arms to a safe location.

##### 14.3.9.2 Removal of the Gas Pressure Spring at Top Cover

1. When the Top cover is fully opened, the gas pressure spring should be in a pressure free condition.

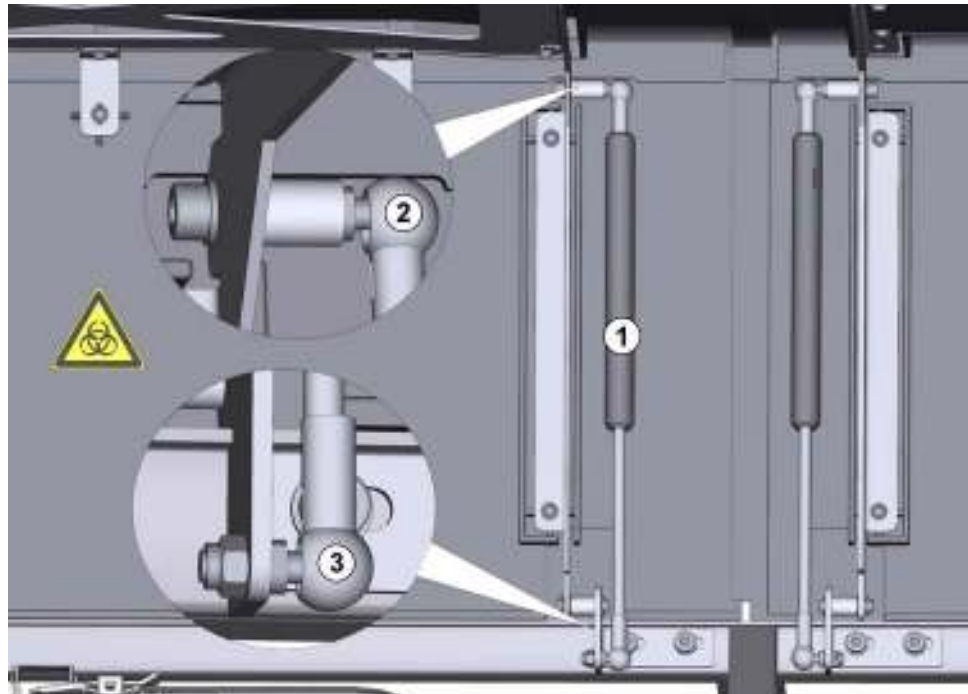


Figure 14–22: Left Top Cover

2. Hold up the Top cover, while removing the gas pressure springs.
3. Remove the securing spring (4) of the ball-hitch coupling (3) of the gas pressure spring.



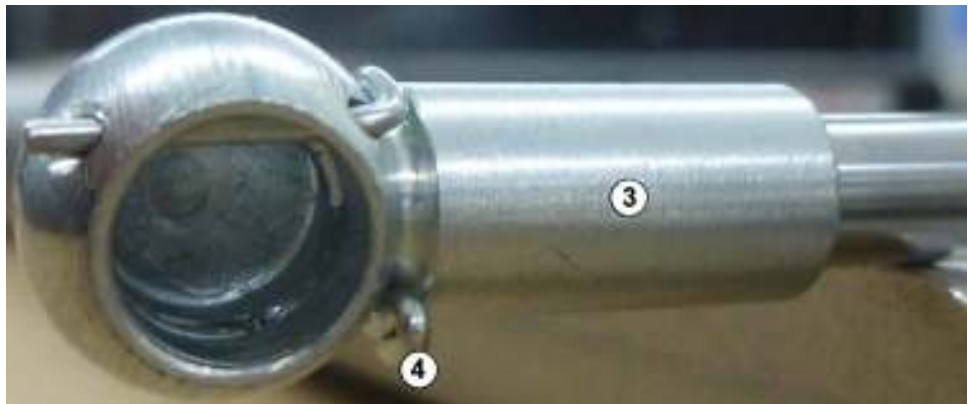
*Figure 14-23: Ball-hitch coupling of Gas Pressure Spring*

4. Unhook the ball-hitch coupling (3) out of the ball-head bolt (5) of the gas pressure spring.



*Figure 14-24: Ball-hitch coupling of Gas Pressure Spring*

5. Insert the securing ring (4) onto both ball-hitch coupling (3).



*Figure 14-25: Ball-hitch coupling with securing ring*

6. Proceed with the same sequence on the other side of the gas pressure spring as in step 2 to 3 was described.

#### **14.3.9.3 Assembly of the Gas Pressure Spring at Top Cover**

1. Remove the securing rings (4) out of the ball-hitch couplings (3) of both sides of the gas pressure spring.
2. Push the ball-head bolt (5) over the ball-hitch coupling (3) on both sides of the left top cover and insert the securing springs (4).

## 14.4 Alignment of Left and Right Top Covers

**NOTE**

The Left Top Cover has to overlap the right gradation on the left side of the Right Top Cover. As a result the Right Top Cover can only be opened if the Left Top Cover was opened before.



Figure 14-26: Instrument front view

1. Open the Left and Right Top Covers.
2. Loosen screws (4) of the Left and Right Top Covers.
3. Align the two Top Covers in order to have three equal gaps, see Figure 14-26: on page 14-20. As well the Left Top Cover overlaps (3) the Right Top Cover.
4. Fasten eight screws (4).



Figure 14-27: Instrument front view, Left and Right Top Covers opened



## 14.5 Electric connections, switches and signals

This section shows an overview of the electronic boards of the Covers.

### NOTE

see Chapter 3.1.7 on page 3-8 OR see Chapter 4.1.8 on page 4-9 for an overview of the complete electric circuit of the system.

### 14.5.1 Soft Power Button PCB



Figure 14–28: Front view



Figure 14–29: Rear view

Figure 14–30: Soft Power Button PCB, front and rear view

| No. | Connection           |
|-----|----------------------|
| J 1 | User Access Panel CU |

Table 14–5: Connections of the Soft Power Button PCB

| Button | Meaning/Function/Default Switch Setting |
|--------|-----------------------------------------|
| SW1    | Start Button, ON/OFF                    |

Table 14–6: Switch of the Soft Power Button PCB



# 15 Cuvette - Tip Loading Module

This chapter describes the Cuvette - Tip Loading Module in detail. The first section of the chapter introduces “Safety” notes concerning the Cuvette - Tip Loading Module, followed by a brief overview of the main parts composing this module.

Then the following section contains information about repair and calibration of this module (“Replacement and Repair”, “Adjustment and Calibration”).

- The module offers space for 2 tip trays and for 4 trays of cuvette, for a total amount of 192 disposable tips and 140 cuvettes.
- The Left Arm pipettor has complete access to that module: the Sample pipettor is able to pick up the disposable tips and the Gripper Module is able to pick up the cuvettes and to manage them inside the instrument.

## 15.1 Safety

---

**WARNING****Cross-contamination by multi-use**

Repeatedly use of the single-use cuvettes or tips will cause cross-contamination.

- Never reuse the single-use cuvettes or tips.
- 

---

**WARNING****Non-observance of safety instructions**

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 Or see Chapter 2.5 on page 2-7.
  - Follow all working instructions included in this manual.
  - Follow all warnings marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
-

## 15.2 Overview

This section shows an overview of the components of the Cuvette - Tip Loading Module.

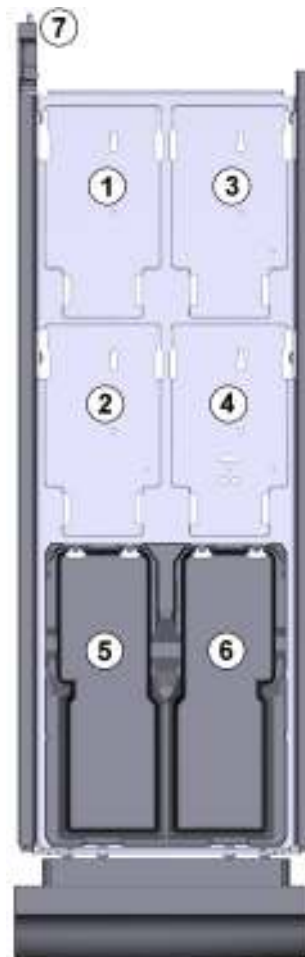


Figure 15–1: Cuvette Tip - Loading Module, top view

|   |                          |
|---|--------------------------|
| 1 | Cuvette tray 1           |
| 2 | Cuvette tray 2           |
| 3 | Cuvette tray 3           |
| 4 | Cuvette tray 4           |
| 5 | Disposable tip tray 1    |
| 6 | Disposable tip tray 2    |
| 7 | Locking mechanism switch |

Table 15–1: Cuvette Tip - Loading Module overview

## 15.3 Replacement and repairs

This section contains instructions for the repair of individual components of the module. After the repair of certain assemblies it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 15.3.1 Replacement of the Cuvette - Tip Loading Module

#### 15.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Left Top covers.
6. Move Left pipettor arm to a safe location.

#### 15.3.1.2 Removal of the Cuvette - Tip Loading Module

1. Pull the Cuvette - Tip drawer out.
2. Lift the hook on the drawer left side up, push the hook on the drawer right side down, and pull carefully the drawer out.
3. Push the linear guides inside the instrument.

#### 15.3.1.3 Assembly of the Cuvette - Tip Loading Module

1. Pull the linear guides out.
2. Slide the inner guides of the Cuvette - Tip drawer into the linear guides.
3. Push the drawer inside the instrument.

#### 15.3.1.4 Set of the Cuvette - Tip Loading Module

1. Perform the Cuvette - Tip Loading Module handle alignment, in case of Cuvette - Tip Loading Module replacement, see Chapter 15.3.3 on page 15-5.
2. Perform the Teaching of the Gripper offset, Cuvette Tray and Tip Tray in case of Cuvette - Tip Loading Module replacement, see Chapter 13.7 on page 13-20, see Chapter 13.8 on page 13-22, see Chapter 13.9 on page 13-24.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 15.3.2 Replacement of the Cuvette - Tip drawer switch / snap-in lock



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Left Top covers.
6. Move Left pipettor arm to a safe location.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3

#### 15.3.2.1 Removal of the switch / snap-in lock

1. Unscrew two screws (2) of the lock (1).
2. Disconnect the cable of the switch / snap-in lock.

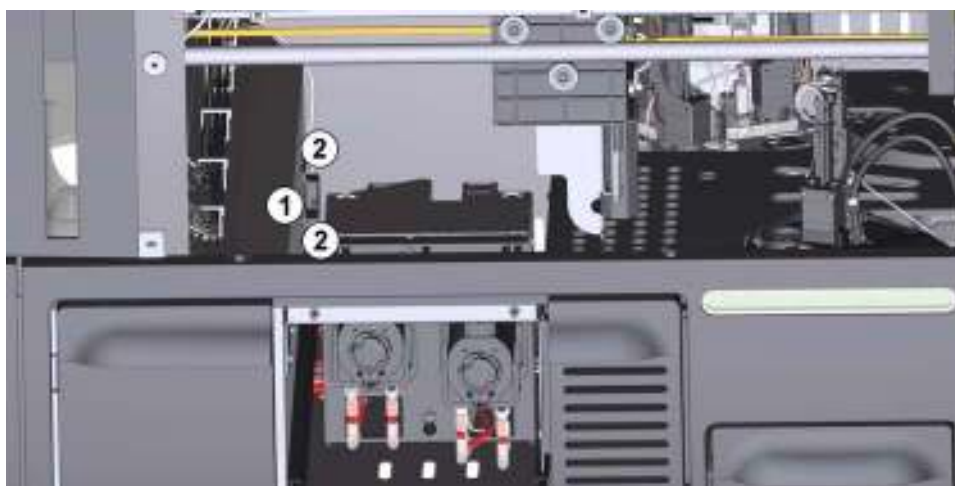


Figure 15-2: Switch / snap-in lock inside the instrument

### 15.3.2.2 Assembly of the switch / snap-in lock

1. Connect the two cables (3 - brown, 4 - black) to the snap lock.

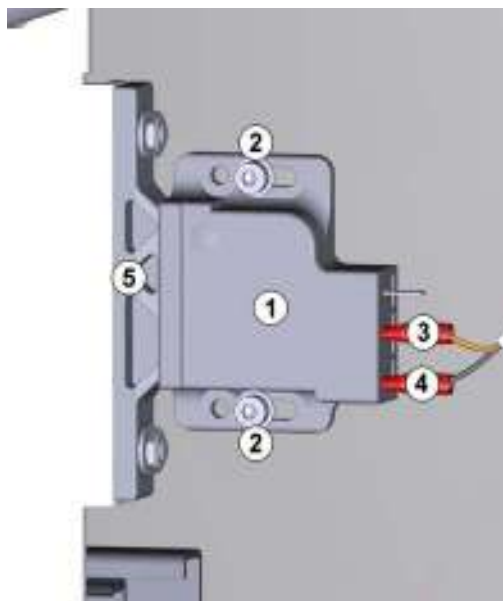


Figure 15-3: Switch / snap-in lock

2. Screw the snap lock loose onto the side wall.
3. Assemble the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.

### 15.3.2.3 Set of the switch / snap-in lock

1. Perform the alignment of the Cuvette - Tip Loading Module handle, see Chapter 15.3.3 on page 15-5.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 15.3.3 Alignment of the Cuvette - Tip Loading Module handle

1. Loosen three screws (2) of Cuvette - Tip Loading Module handle (1).

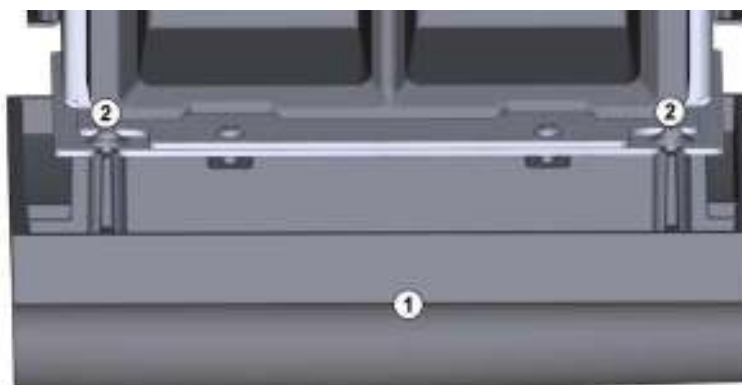


Figure 15-4: Cuvette - Tip Loading Module handle, top view

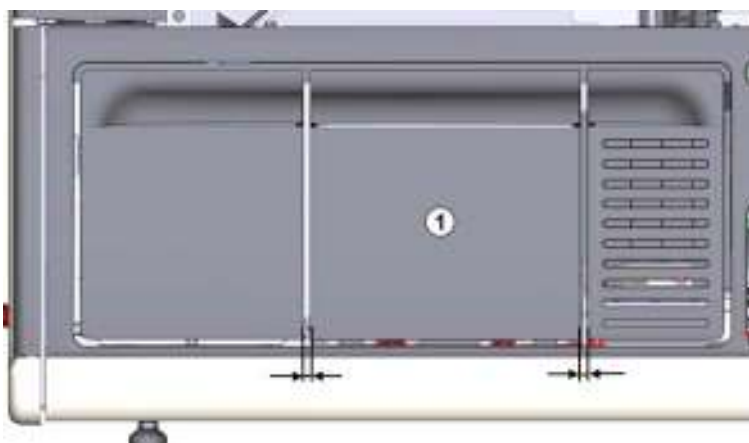
**15. Cuvette - Tip Loading Module**

---



*Figure 15-5: Cuvette - Tip Loading Module handle, bottom view*

2. The height of the Cuvette- Tip Loading Module Handle (1) shall be even and in the middle. When the Cuvette- Tip Loading Module Handle is leveled out, then fasten three screws (2).



*Figure 15-6: Front Module, front view*



## 15.4 Adjustment and calibration

### 15.4.1 Adjustment of the Cuvette - Tip Loading Module switch / snap-in lock

1. Set the screws of the snap lock loose onto the side wall.
2. Close the Cuvette - Tip drawer. Make sure that the enclosing angle (5) slides into the snap lock (1) and hook into it.
3. Align the Cuvette - Tip drawer handle with the Starter Reagent Module handle (7) and the Sample Module flap (8).
4. Fasten the screw and washer (2) of the snap-lock.
5. Check from **Service Software** the instrument is detecting correctly the status of the drawer selecting **Front and Loading** panel and then the button **Sensor - Liquid support**.

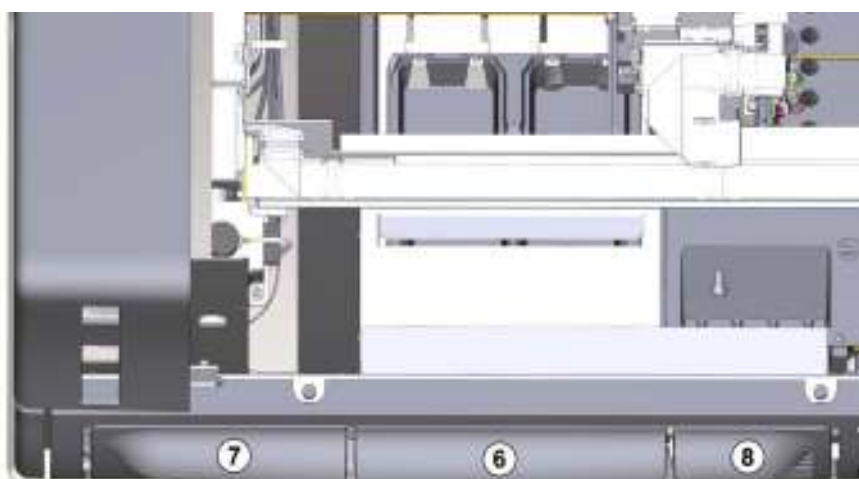


Figure 15-7: Alignment of Cuvette - Tip drawer, Starter Module and Sample Module handles



# 16 Incubator Module

This chapter describes the Incubator Module in detail. In the first section of the chapter “Safety” notes concerning Incubator, followed by a brief overview of the main parts composing this module.

The following section contains information about repair and calibration (“Replacement and Repair”, “Adjustment and Calibration” sections).

In the last section, Incubator CU PCBA layout and connections are described.

- The Incubator Module provides storage for the reaction cuvettes at 37°C, in order to facilitate the chemical reaction of the materials present in the cuvettes.
- It is mainly composed by a rotor, used in order to accommodate the cuvettes, moved by a stepper motor that is driven by a PCB (Incubator CU PCBA).
- The rotor has two concentric rings: the outer ring may host up to 32 cuvettes, used for the first incubation period (i.e. before the first washing step); the inner ring may host up to 25 cuvettes, used for the second and/or third incubation period, if any.
- It is the only assembly in the system that can automatically recognize the presence (or the absence) of a cuvette, by using 2 optical sensors (one for each ring).
- The cuvettes are moved by the motor to the relevant position along the rotor; they are loaded/unloaded by the Gripper in the loading/unloading position, while other positions are dedicated to the dispensing of the Sample Pipettor and of the Reagent Pipettor.

## 16.1 Safety

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 OR see Chapter 2.5 on page 2-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

### CAUTION



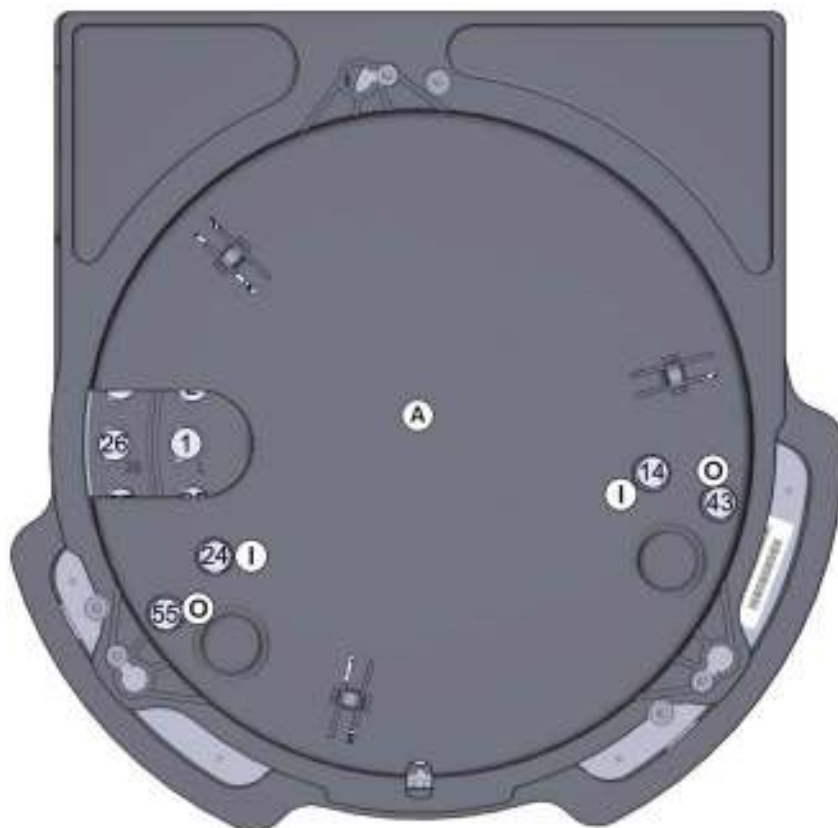
#### Risk of burn

The heating rings of the Incubator Module will reach high temperatures during operation and testing. Contact will cause injuries.

- Use appropriate gloves and PPE.
- Let the heating rings cool down before cleaning or maintenance.

## 16.2 Overview

This section shows an overview of the components of the Incubator Module.



*Figure 16–1: Incubator top view*

|    |                                                                      |
|----|----------------------------------------------------------------------|
| A  | Incubator cover                                                      |
| 1  | Cuvette load & unload position at inner ring (rotor positions 1-25)  |
| 26 | Cuvette load & unload position at outer ring (rotor positions 26-57) |
| I  | Pipetting positions at inner ring (24 not used, 14 for reagent)      |
| O  | Pipetting positions at outer ring (55 for sample, 43 for reagent)    |

*Table 16–1: Incubator Module overview*

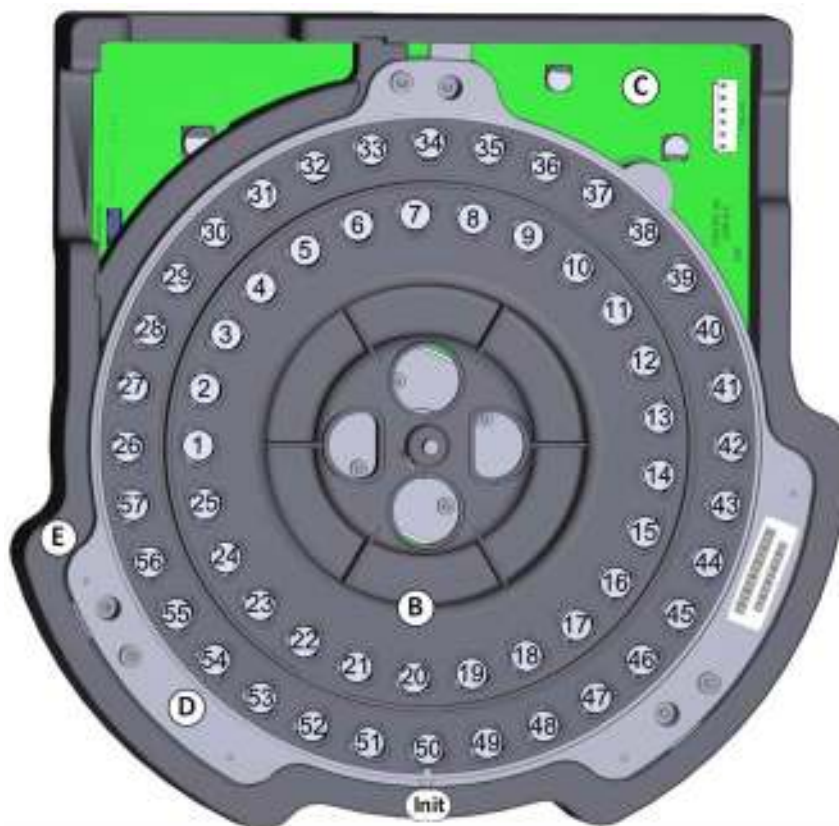


Figure 16–2: Incubator Module assembly and rotor position numbering.

|      |                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Init | Initializing position of the Incubator rotor                                                                                                            |
| B    | Incubator rotor <ul style="list-style-type: none"> <li>• 1-25 Cuvettes positions, inner ring</li> <li>• 26-57 Cuvettes positions, outer ring</li> </ul> |
| C    | Incubator CU PCBA                                                                                                                                       |
| D    | Incubator frame                                                                                                                                         |
| E    | Incubator Module protective foam                                                                                                                        |

Table 16–2: Incubator Module overview

## 16.3 Replacement and repairs

This section contains instructions for the repair of individual components of the Incubator Module. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 16.3.1 Replacement of the Incubator cover

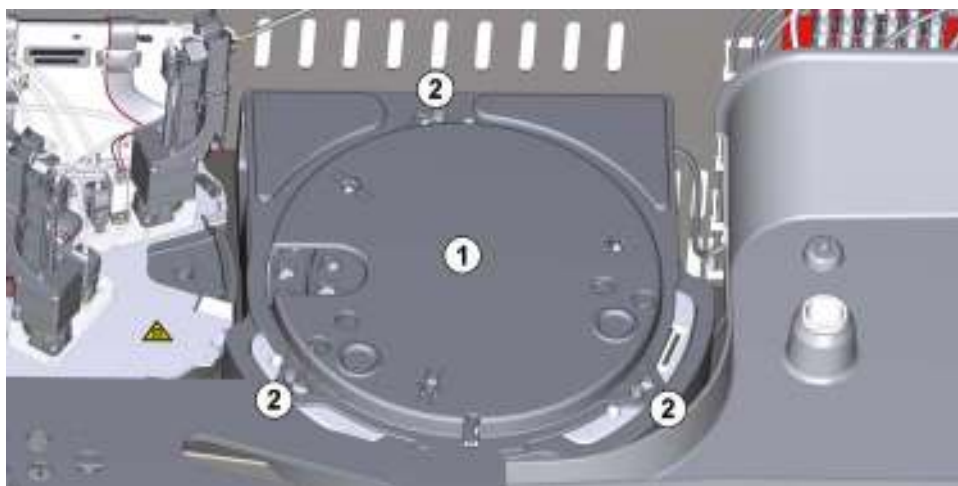
#### 16.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.

#### 16.3.1.2 Removal of the Incubator cover

1. Loose the three screws (2) of the Incubator cover (1).
2. Turn the Incubator cover (1) clockwise.
3. Lift the Incubator cover up and set it aside.



*Figure 16–3: Incubator cover*

#### 16.3.1.3 Assembly of the Incubator cover

1. Set the Incubator cover keyholes into the screws.
2. Turn the Incubator cover counter-clockwise to the end stop and tighten the three screws (2).

## 16.3.2 Replacement of the Incubator Module

### 16.3.2.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Incubator cover, see Chapter 16.3.1.2 on page 16-4.
8. Remove and dispose manually the cuvettes in the rotor, if any.

### 16.3.2.2 Removal of the Incubator Module

1. Unplug CAN connection (3).
2. Unscrew the three screws (4) of the Incubator Module frame.
3. Lift the Incubator Module (5) carefully out (see orange arrows for suggested grabbing position) and set it aside.

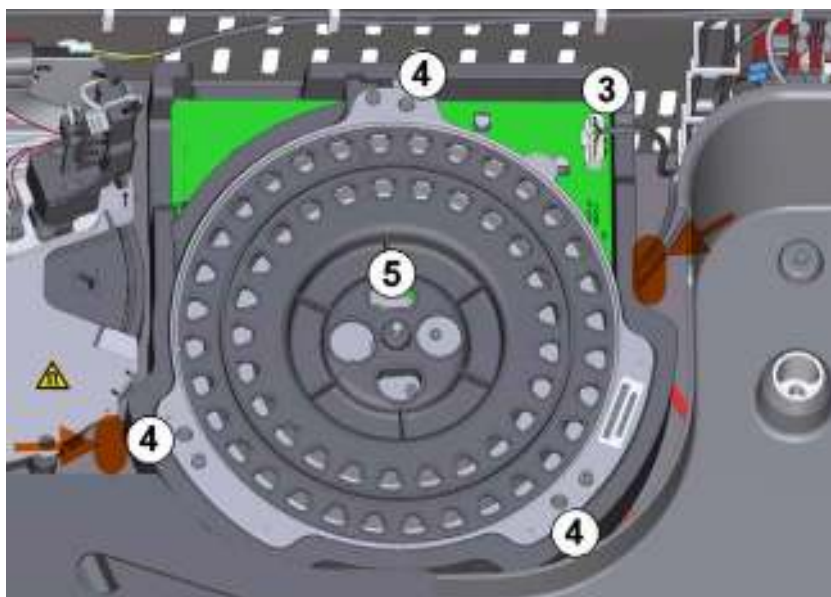


Figure 16-4: Incubator Module

**16.3.2.3 Assembly of the Incubator Module**

1. Set the Incubator Module (5) into place.
2. Tighten the three screws (4).
3. Plug the PWR CAN connector (3) into the Incubator CU PCBA.
4. Assemble the Incubator cover, see Chapter 16.3.1.3 on page 16-4.

**16.3.2.4 Set of the Incubator Module**

1. If necessary (new Incubator Module installed), run the Firmware Update of the complete system, see Chapter 32.3.4 on page 32-6.
2. Perform the calibration of the Incubator rotor, see Chapter 16.4.2.1 on page 16-19.
3. Perform the calibration of the cuvette presence sensors, see Chapter 16.4.2.2 on page 16-20.
4. Check the teaching of Incubator Module positions of both Left and Right Arms, see Chapter 13.12 on page 13-29 and see Chapter 13.14 on page 13-31.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.



### 16.3.3 Replacement of the Incubator rotor

#### 16.3.3.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Incubator cover, see Chapter 16.3.1.2 on page 16-4.
8. Remove and dispose manually the cuvettes in the rotor, if any.

#### 16.3.3.2 Removal of the Incubator rotor

1. Carefully pull the Incubator rotor out of the Incubator frame.

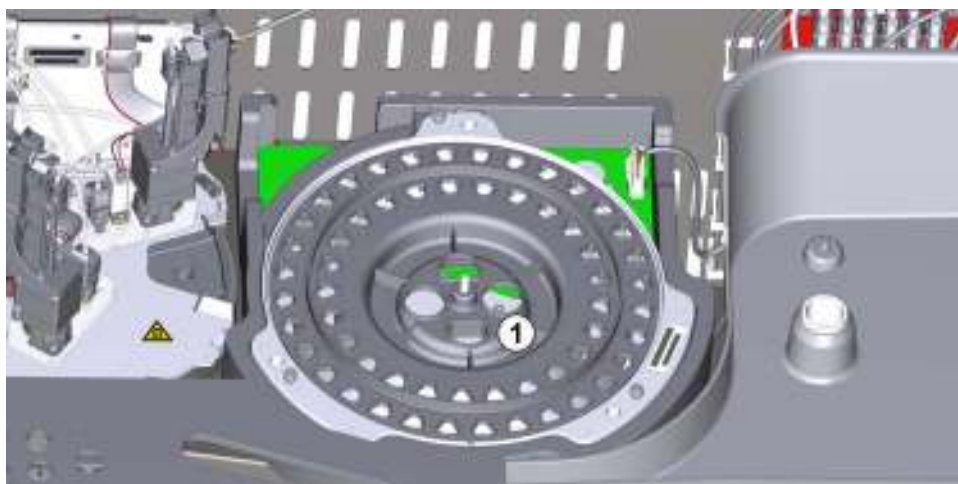


Figure 16–5: Incubator without cover

#### 16.3.3.3 Assembly of the Incubator rotor

1. Set the Incubator rotor onto the motor shaft (pay attention to the correct orientation, since they fit into each other only if properly aligned, due to the motor shaft shape) and slide the Incubator rotor down till it sits on the bearings of the Incubator frame.

#### NOTE

##### Break off of bearing holders

Be careful while removing the bearings (2) out of the Incubator cover.  
The bearing holder (3) can break off.

2. Assemble the Incubator cover, see Chapter 16.3.1.3 on page 16-4.

#### **16.3.3.4 Set the Incubator rotor**

1. Perform the calibration of the Incubator rotor, see Chapter 16.4.2.1 on page 16-19.
2. Perform the calibration of the cuvette presence sensors, see Chapter 16.4.2.2 on page 16-20.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 16.3.4 Replacement of the bearings of the Incubator frame and cover

#### 16.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Incubator cover, see Chapter 16.3.1.2 on page 16-4.

#### 16.3.4.2 Removal of the bearings of Incubator frame and cover

##### NOTE

##### Break off of bearing holders

Be careful while removing the bearings (2) out of the Incubator cover. The bearing holder (3) can break off.

1. Remove the bearings (2) carefully out of the Incubator cover (1).

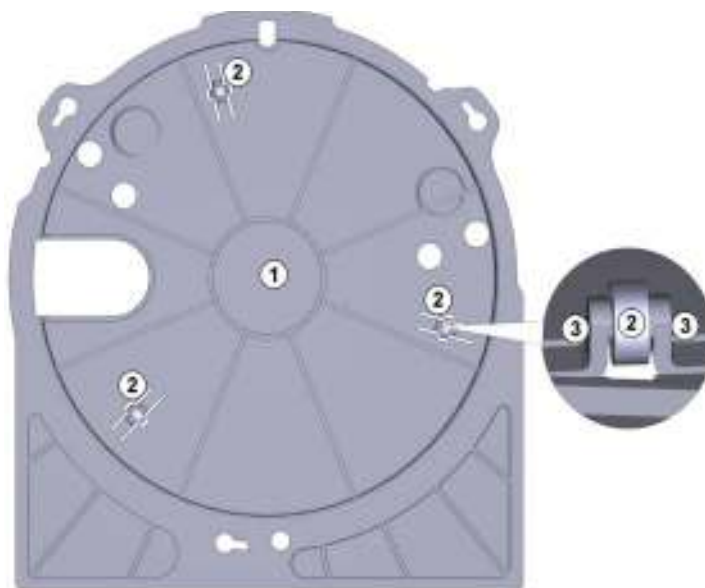
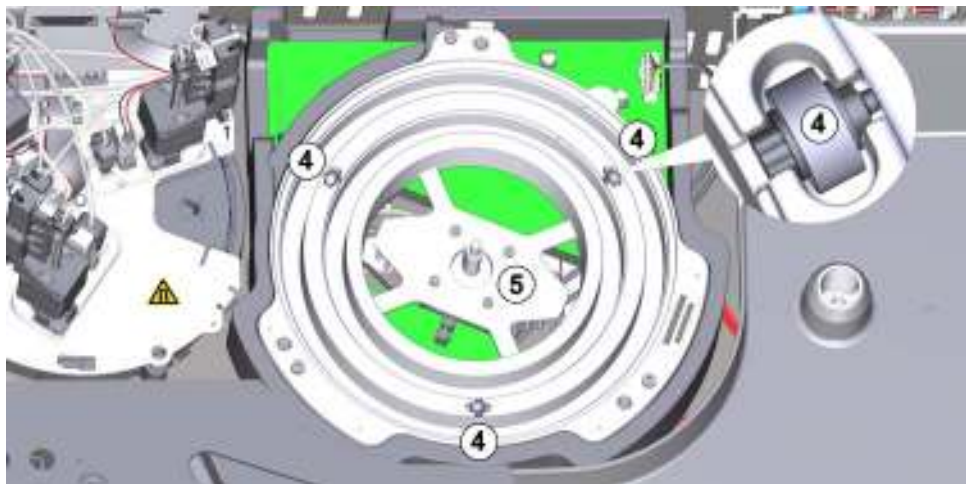


Figure 16–6: Incubator cover, bottom view

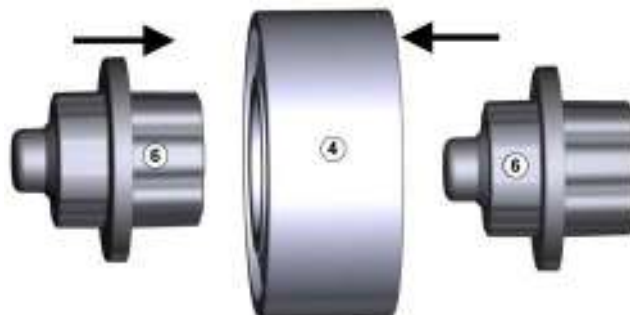
2. Remove the Incubator rotor, see Chapter 16.3.3.2 on page 16-7.
3. Remove the bearings (4) out of the Incubator frame (5).



*Figure 16-7: Incubator frame, with bearings and pins*

#### **16.3.4.3 Assembly of the bearings of Incubator frame and cover**

1. Set the two pins (6) inside of the bearing (4) together.



*Figure 16-8: Detail of the Incubator bearings*

2. Insert the pre-assembled bearings (4) into the Incubator frame (5) until they sit in their place. Pay attention to the orientation of the bearing (4), see Figure 16-7: on page 16-10.
3. Assemble the Incubator rotor, see Chapter 16.3.3.3 on page 16-7.
4. Insert the bearings into the Incubator cover until they click into place.
5. Assemble the Incubator cover, see Chapter 16.3.1.3 on page 16-4.

#### **16.3.4.4 Set the Incubator bearings**

1. Perform the calibration of the Incubator rotor, see Chapter 16.4.2.1 on page 16-19.
2. Perform the calibration of the cuvette presence sensors, see Chapter 16.4.2.2 on page 16-20.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 16.3.5 Replacement of the Incubator motor

#### 16.3.5.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Incubator cover, see Chapter 16.3.1.2 on page 16-4.
8. Remove the Incubator rotor, see Chapter 16.3.3.2 on page 16-7.
9. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.

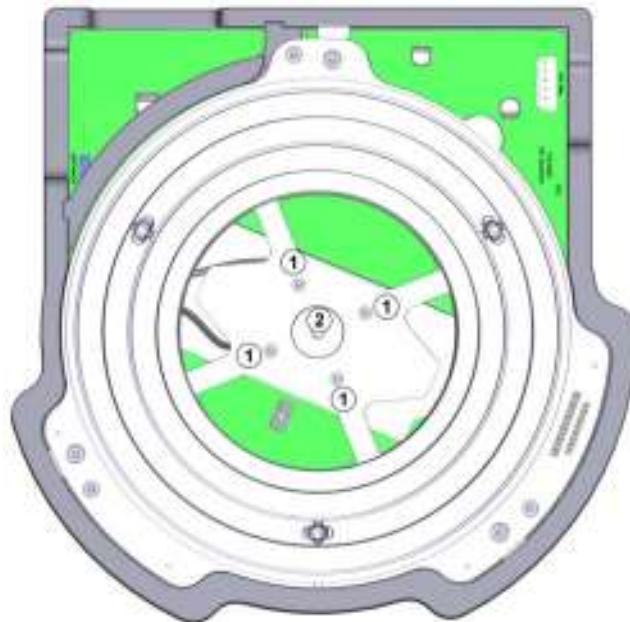
#### 16.3.5.2 Removal of the Incubator motor

1. Unplug motor cable (3) and encoder cable (4) out of Incubator CU PCBA (5).



Figure 16–9: Incubator Module, bottom view

2. Unscrew the four screws(1) of the Incubator motor (2).



*Figure 16–10: Incubator Module, top view without cover and rotor*

3. Remove the Incubator motor (2).

#### **16.3.5.3 Assembly of the Incubator motor**

1. Insert the Incubator motor (2) onto the Incubator Module and fasten with four screws (1).
2. Plug the motor cable (3) and the encoder cable (4) into the Incubator CU PCBA (5)
3. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
4. Assemble the Incubator rotor, see Chapter 16.3.3.3 on page 16-7.
5. Assemble the Incubator cover, see Chapter 16.3.1.3 on page 16-4.

#### **16.3.5.4 Set the Incubator motor**

1. Check the teaching of Incubator Module positions of both Left and Right Arms, see Chapter 13.12 on page 13-29 and see Chapter 13.14 on page 13-31.
2. Perform the calibration of the Incubator rotor, see Chapter 16.4.2.1 on page 16-19.
3. Perform the calibration of the cuvette presence sensors, see Chapter 16.4.2.2 on page 16-20.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

### 16.3.6 Replacement of the Incubator isolation foam

#### 16.3.6.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Incubator cover, see Chapter 16.3.1.2 on page 16-4.
8. Remove the Incubator rotor, see Chapter 16.3.3.2 on page 16-7.
9. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.

#### 16.3.6.2 Removal of the Incubator isolation foam

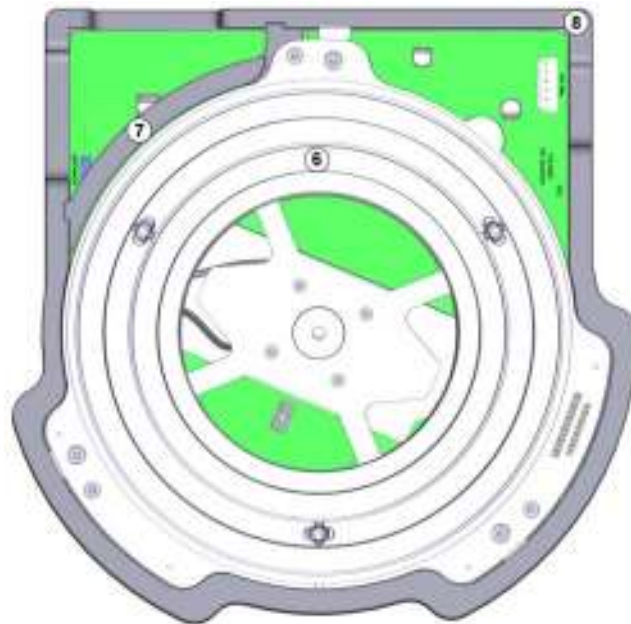
1. Unplug motor cable (3) and encoder cable (4) out of Incubator CU PCBA (5).



Figure 16–11: Incubator Module, bottom view

2. Lift the Incubator insulation bridge (7) out of the Incubator insulation (8).
3. Lift Incubator frame (6) along with motor and Incubator CU PCBA out of the insulation (8).





*Figure 16-12: Incubator Module, top view without cover and rotor*

#### **16.3.6.3 Assembly of the Incubator isolation foam**

1. Slide the Incubator insulation (8) over the Incubator frame (6).
2. Set the Incubator insulation bridge (7) into the Incubator insulation (8).
3. Plug the motor cable (3) and the encoder cable (4) into the Incubator CU PCBA (5).
4. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
5. Assemble the Incubator cover, see Chapter 16.3.1.3 on page 16-4.

#### **16.3.6.4 Set the Incubator isolation foam**

1. Check the teaching of Incubator Module positions of both Left and Right Arms, see Chapter 13.12 on page 13-29 and see Chapter 13.14 on page 13-31.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.



## 16.3.7 Replacement of the Incubator CU PCBA

### 16.3.7.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Incubator cover, see Chapter 16.3.1.2 on page 16-4.
8. Remove the Incubator rotor, see Chapter 16.3.3.2 on page 16-7.
9. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.
10. Remove the Incubator isolation foam, see Chapter 16.3.6.2 on page 16-13.

### 16.3.7.2 Removal of the Incubator CU PCBA

1. Unscrew six screws (9) of the Incubator CU PCBA (5) and remove the CU PCBA.

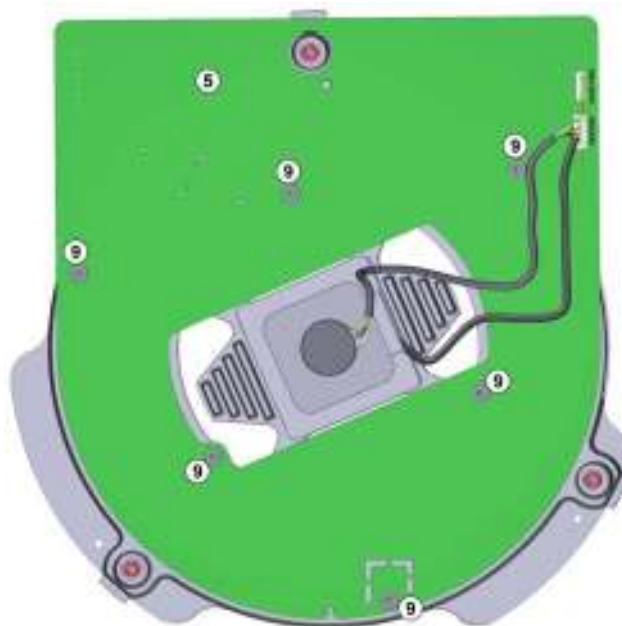


Figure 16–13: Incubator frame and Incubator CU PCBA, bottom view

### 16.3.7.3 Assembly of the Incubator CU PCBA

1. Carefully place the Incubator CU PCBA (5) on the Incubator frame (6).

#### NOTE

When placing the Incubator CU PCBA on the Incubator frame, ensure that the light guides are correctly positioned into their slots in the Incubator frame.

2. Fasten the Incubator CU PCBA (5) with six screws (9) onto the Incubator frame.
3. Assemble the Incubator isolation foam, see Chapter 16.3.6.3 on page 16-14.
4. Plug the motor cable (3) and the encoder cable (4) into the Incubator CU PCBA (5).
5. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
6. Assemble the Incubator rotor, see Chapter 16.3.3.3 on page 16-7.
7. Assemble the Incubator cover, see Chapter 16.3.1.3 on page 16-4.

#### **16.3.7.4 Set the Incubator CU PCBA**

1. If necessary (new Incubator CU PCBA installed), run the Firmware Update of the complete system, see Chapter 32.3.4 on page 32-6.
2. Check the teaching of Incubator Module positions of both Left and Right Arms, see Chapter 13.12 on page 13-29 and see Chapter 13.14 on page 13-31.
3. Perform the calibration of the Incubator rotor, see Chapter 16.4.2.1 on page 16-19.
4. Perform the calibration of the cuvette presence sensors, see Chapter 16.4.2.2 on page 16-20.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 16.3.8 Replacement of the Incubator light guide

### 16.3.8.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Incubator cover, see Chapter 16.3.1.2 on page 16-4.
8. Remove the Incubator rotor, see Chapter 16.3.3.2 on page 16-7.
9. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.
10. Remove the Incubator CU PCBA, see Chapter 16.3.7.2 on page 16-15.

### 16.3.8.2 Removal of the Incubator light guide

1. Remove the four Incubator light guides (10) out of the Incubator frame (6).

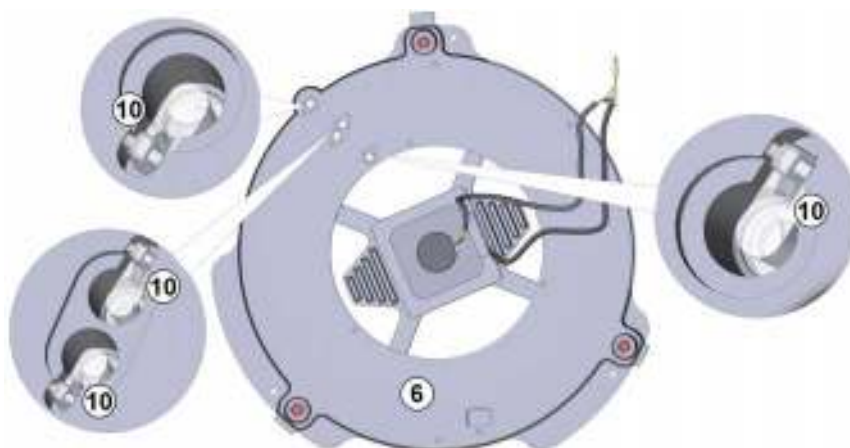


Figure 16–14: Incubator frame with motor and light guides, bottom view

### 16.3.8.3 Assembly of the light guide

1. Insert four light guides (10) into the bottom of the incubator frame (6).



*Figure 16–15: Incubator frame including motor, bottom view*

1. Assemble the Incubator CU PCBA, see Chapter 16.3.7.3 on page 16-15.
2. Assemble the Incubator isolation foam, see Chapter 16.3.6.3 on page 16-14.
3. Plug the motor cable (3) and the encoder cable (4) into the Incubator CU PCBA (5).
4. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
5. Assemble the Incubator rotor, see Chapter 16.3.3.3 on page 16-7.
6. Assemble the Incubator cover, see Chapter 16.3.1.3 on page 16-4.

#### **16.3.8.4 Set the Incubator light guide**

1. Check the teaching of Incubator Module positions of both Left and Right Arms, see Chapter 13.12 on page 13-29 and see Chapter 13.14 on page 13-31.
2. Perform the calibration of the Incubator rotor, see Chapter 16.4.2.1 on page 16-19.
3. Perform the calibration of the cuvette presence sensors, see Chapter 16.4.2.2 on page 16-20.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 16.4 Adjustment and calibration

This section contains information about the adjustment and calibration of the Incubator Module.

### NOTE

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).  
Do not run any of the software at the same time.

### 16.4.1 Initialization routine

- Close any open software and start the **Service Software** from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **Incubator** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog Controlled Modules** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 16.4.2 Calibration of the Incubator Module

#### 16.4.2.1 Calibration of the Incubator Rotor

In the **Service Software** at the navigation window, choose **Incubator** then **Teach & Calibration** bracket and select **Ring**.

#### Step 1

- Follow the instructions and then press the **Done** button.

#### Step 2

- Move the Incubator rotor by the arrow buttons to align the slot on the rotor to the reference position of the Incubator frame.



Figure 16–16: Incubator rotor, calibration position

#### Step 3

- Press **Save** button to save the position.
- Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

#### Step 4

- The process is successful when, at the **Summarize** panel, all steps are in green.

**CAUTION**

**A mis-calibration of the Incubator rotor might be necessary in order to have a proper cuvette transfer from the Incubator to the Washer and vice-versa.**

Perform a Cuvette Endurance Run (see Chapter 12.8.1.2.1 on page 12-63) and check that the cuvette transfer is smooth and without noises.

If needed, change the Incubator rotor calibration such to compensate the observed mis-alignment between the Gripper and Incubator rotor; an adjustment of 3-4 steps clockwise could be sufficient.

#### 16.4.2.2 Calibration of the Incubator Cuvette Sensors

In the **Service Software** at the navigation window, choose **Incubator** then **Teach & Calibration** bracket and select **Cuvette Sensors**.

- Step 1** a. Follow the instruction: remove the cuvettes from the gripper loading position (i.e. at positions 1 and 26), if any, and dispose of them; then press the **Done** button.
- Step 2** b. Follow the instruction by pressing **Done** button. The Incubator will move Cuvette 1 & 26 positions to the sensor.
- Step 3** c. Follow the instruction by pressing **Start** button to start the calibration procedure.  
d. Check the results and proceed by pressing **Done** button.
- Step 4** e. Store the taught positions in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 5** f. Follow the instruction: insert new cuvettes at the gripper loading position (i.e. at positions 1 and 26), if were present, and press the **Done** button.
- Step 6** g. The process is successful when, at the **Summarize** panel, all steps are in green.

## 16.5 Electric connections, switches and signals

This section shows an overview of the electronic boards of the Incubator Module.

### NOTE

see Chapter 3.1.7 on page 3-8 OR see Chapter 4.1.8 on page 4-9 for an overview of the complete electric circuit of the system.

### 16.5.1 INCUBATOR CU PCBA

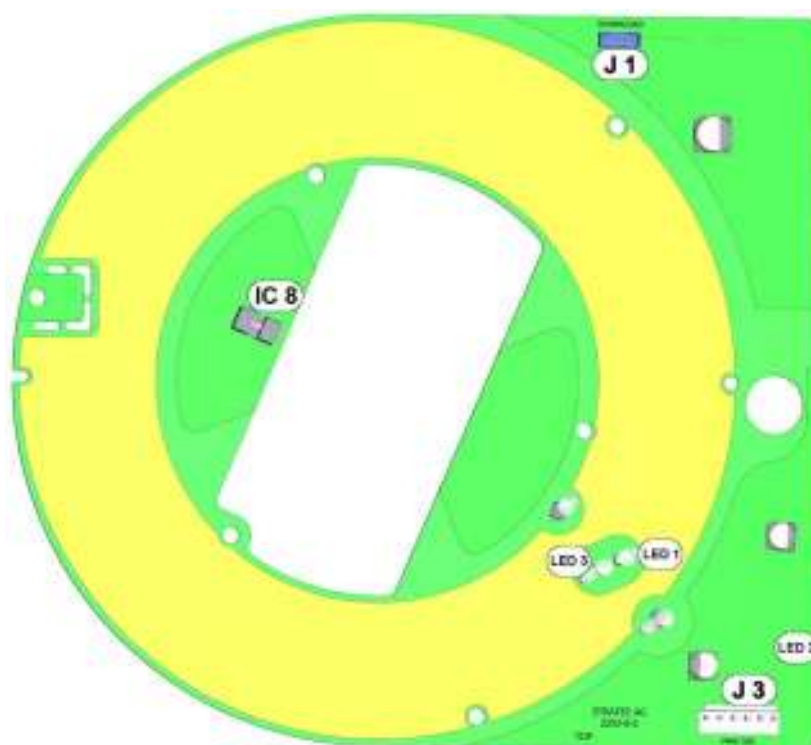


Figure 16–17: Incubator CU PCBA, top view

| Number | Connection |
|--------|------------|
| J1     | Download   |
| J3     | PWR/CAN    |

Table 16–3: Connections of the Incubator CU PCBA

| Number | Meaning/Function                           |
|--------|--------------------------------------------|
| LED 1  | Cathode (4Vmax) (transmitter and receiver) |
| LED 2  | Power                                      |
| LED 3  | Cathode (4Vmax) (transmitter and receiver) |

Table 16–4: LED of the Incubator CU PCBA

| Number | Meaning/Function |
|--------|------------------|
| IC 8   | Rotor init       |

Table 16–5: Light barrier of the Incubator CU PCBA

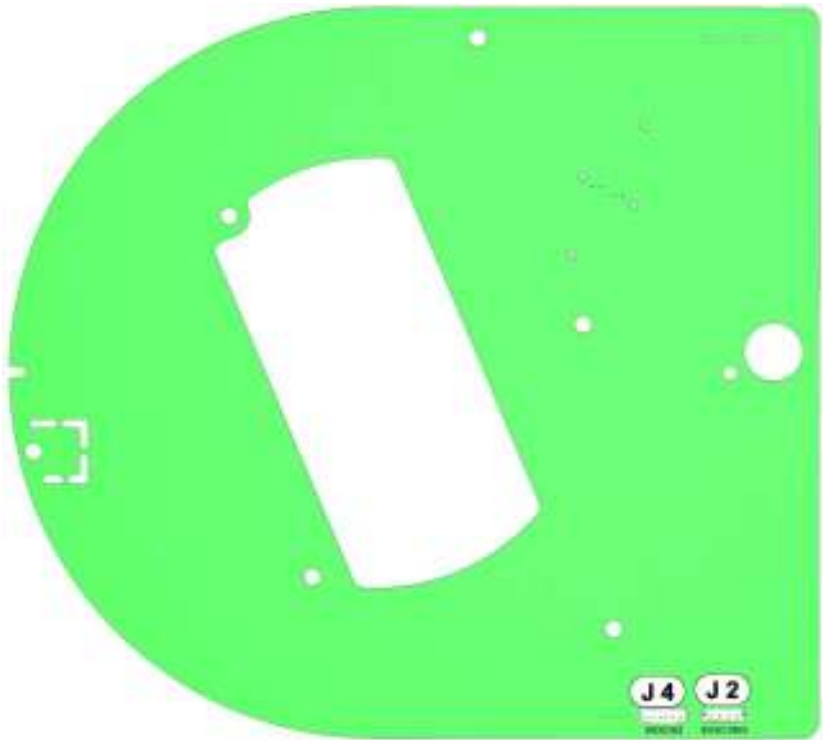


Figure 16–18: Incubator CU PCBA, bottom view

| Number | Connection            |
|--------|-----------------------|
| J2     | Incubator rotor drive |
| J4     | Encoder               |

Table 16–6: Connections of the Incubator CU PCBA



# 17 Washer Module

This chapter describes the Washer Module in detail. The first section of the chapter introduces “Safety” notes concerning Washer, followed by a brief overview of the main parts composing this module.

The following section contains information about repair and calibration (“Replacement and Repair” and “Adjustment and Calibration” sections).

In the last section, Washer CU PCBA layout, Aspiration Lift PCBA layout and connections are described.

- The Washer Module provides three physical wash positions. At each of these, one or more wash steps are performed, each represented by aspiration of liquid from reaction cuvettes and dispensation of Wash Buffer into the reaction cuvettes.
- The complete wash track consists of ten positions, including the three wash positions and one position for loading/unloading of cuvettes. Each cuvette passes all the ten positions during assay processing.
- Along the wash track, magnetic retention positions are located so that in each cuvette, particles that are magnetically attracted are not washed out.
- Cuvette transport along the wash track is achieved by a rotor, moved by a stepper motor that is driven by the Washer CU PCBA.
- Loading and unloading of cuvettes into/from the Washer rotor is performed by the gripper. The gripper brings cuvettes from the Incubator or from a cuvette tray. From the Washer, cuvettes are either returned to the Incubator, or delivered to the Reader (depending on assay type), or directly delivered to the solid waste.

## 17.1 Safety

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injury and material damage.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 Or see Chapter 2.5 on page 2-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

## 17.2 Overview

This section contains an overview of the components of the Washer Module.

The module consists of the following components:

- Washer base, with integrated retention/separation magnets.
- Cuvette rotor and drive.
- Washer plate, including three wash lifts and the Washer CU PCBA.

**17. Washer Module**

Each wash lift consists of the following components:

- Single Probe Aspiration Lift (SIPAL), for the aspiration of liquid from the cuvettes.
- Dispense manifold, for the dispensation of Wash Buffer into the cuvettes.



*Figure 17-1: Washer Module front view*

|    |                              |
|----|------------------------------|
| 1  | Washer base                  |
| 2  | Motor drive for Washer rotor |
| 3  | Washer rotor                 |
| 4  | Washer plate                 |
| 5  | SIPAL 1                      |
| 6  | SIPAL 2                      |
| 7  | SIPAL 3                      |
| 8  | Dispense manifold 1          |
| 9  | Dispense manifold 2          |
| 10 | Dispense manifold 3          |

*Table 17-1: Washer Module front view*



Figure 17–2: Washer Module back view

|    |                                     |
|----|-------------------------------------|
| 11 | Rotor light barrier - Init position |
| 12 | Washer CU PCBA                      |

Table 17–2: Washer Module back view

### 17.2.1 SIPAL overview

This section contains an overview of the SIPAL components.



Figure 17–3: SIPAL

|   |                          |
|---|--------------------------|
| 1 | Tube bayonet adapter     |
| 2 | Init light barrier       |
| 3 | Spindle aspiration probe |
| 4 | Spindle drive            |

Table 17–3: SIPAL



Figure 17–4: SIPAL, rear view

|   |                          |
|---|--------------------------|
| 5 | Liquid monitoring sensor |
| 6 | Aspiration lift PCBA     |
| 7 | Spindle drive connector  |

Table 17–4: SIPAL, rear view

## 17.3 Replacements and repairs

This section contains instructions for the repair of individual components of the Wash Module. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 17.3.1 Replacement of the spindle aspiration probe

Each spindle aspiration probe includes the aspiration needle.

#### 17.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.

#### 17.3.1.2 Removal of the Spindle Aspiration Probe

1. Unhook the tube (2) out of the hooks (3) of the spindle aspiration probe housing (1).
2. Unhook tube out of the guidance (4).

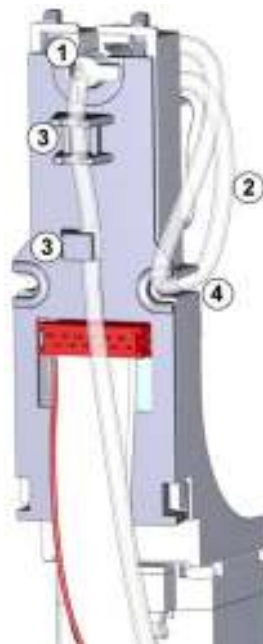


Figure 17–5: SIPAL 3, rear view

3. Insert the tube into the bayonet adapter.
4. Turn the bayonet adapter (5) counterclockwise a quarter turn.
5. Lift the tube out of the bayonet adapter (5).
6. Pull the tube (2) carefully out of the aspiration probe spindle and lay down the tube.

**NOTE**

In case more than one SIPAL will be exchanged or removed, label the individual tubes.

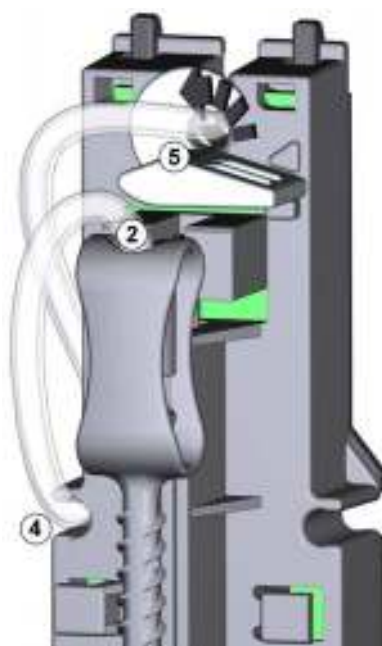


Figure 17-6: SIPAL 3, front view

7. Turn the bayonet adapter (6) so that the retainer lines up with the guide slot of the SIPAL.
8. Pull the spindle aspiration probe (7) out of the SIPAL.

**WARNING****Spindle aspiration probe is contaminated.**

The spindle aspiration probe shall not be left inside the instrument and/or in contact with instrument surfaces, to avoid a possible contamination.



*Figure 17-7: SIPAL, front view, removal of the aspiration needle*

#### **17.3.1.3 Assembly of the spindle aspiration probe**

1. Set the spindle aspiration probe (7) into the SIPAL guide and push the threaded part carefully into the threaded bushing of the motor.



*Figure 17-8: SIPAL, front view, assembly of the aspiration needle*

2. The tube length between the bayonet adapter and the spindle aspiration probe should measure 160mm.
3. Turn the bayonet adapter (6) clockwise a quarter turn to lock the tube.
4. Connect the tube with the spindle aspiration probe.



**NOTE**

The tube length between the spindle aspiration probe and the bayonet shall not be longer than 160mm to prevent aspiration monitoring malfunction.

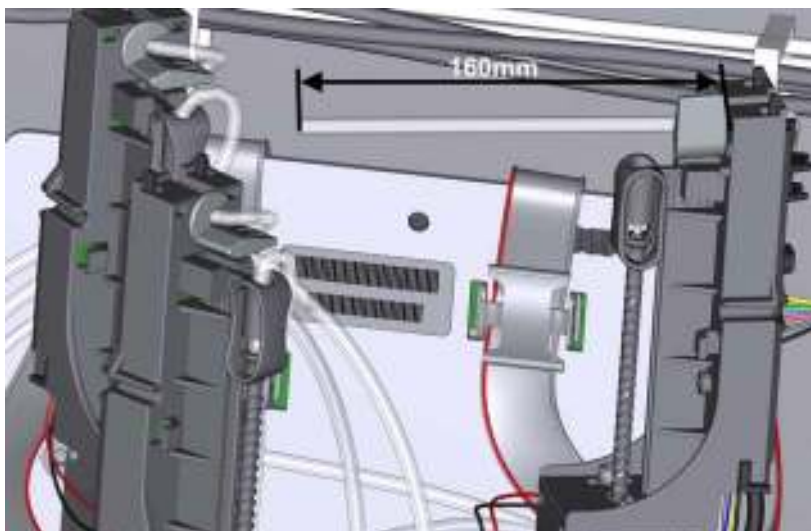


Figure 17-9: SIPAL 3, adjust tube length

5. Locate the tube in the hooks (3) and the guidance (4).

**NOTE**

Tubing shall fit straight into the bayonet.  
Tube shall not be kinked or twisted.

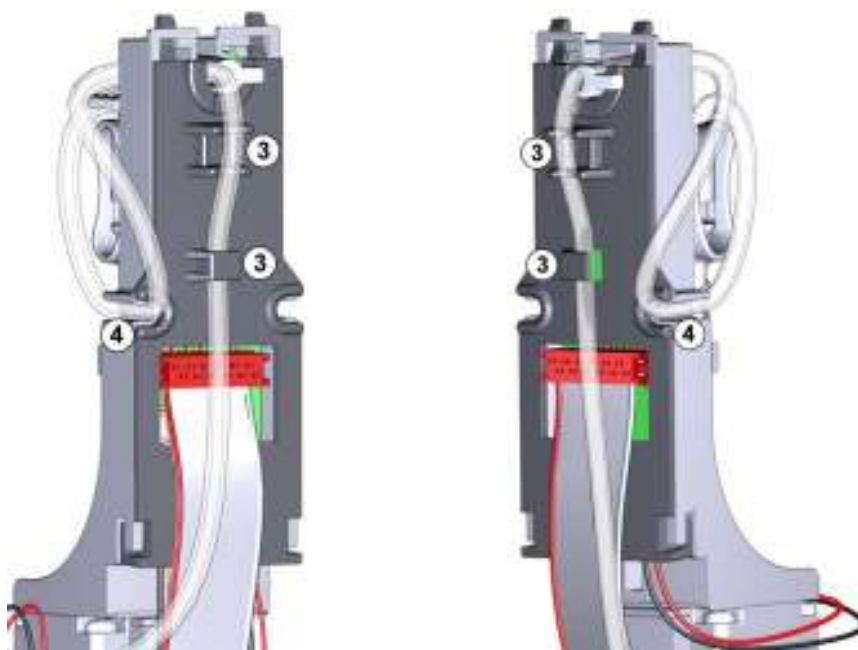


Figure 17-10: Connecting tubing on SIPAL 1 & 2, rear view

Figure 17-11: Connecting tubing on SIPAL 3, rear view

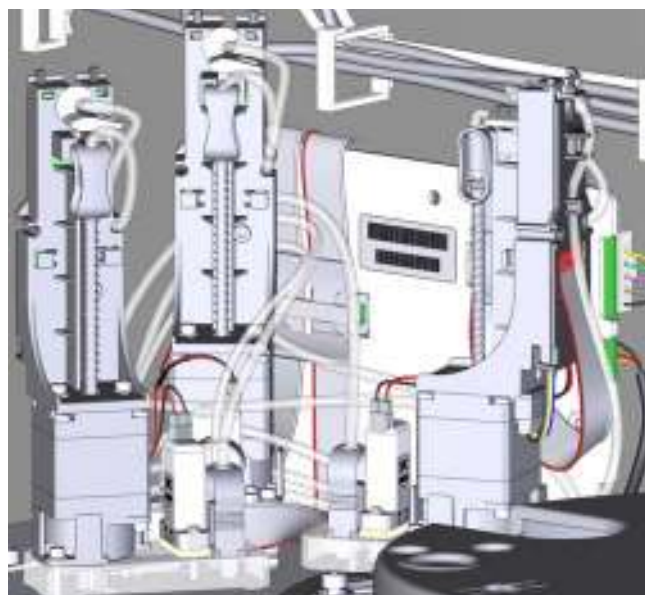


Figure 17-12: Tube routing on the SIPALs

#### 17.3.1.4 Set the spindle aspiration probes

1. Perform the calibration of the Washer SIPAL Lifts, see Chapter 17.4.2.2 on page 17-46.
2. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.

3. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

## 17.3.2 Replacement of the SIPAL

### 17.3.2.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the spindle aspiration probe, see Chapter 17.3.1.2 on page 17-6 from the SIPAL.

### 17.3.2.2 Removal of the SIPAL

1. Unplug flat ribbon cable connection (1) from the aspiration lift PCBA.
2. Remove bayonet tube adapter (4).
3. Unhook both hooks (3) and slide SIPAL protective cover (2) downwards out.

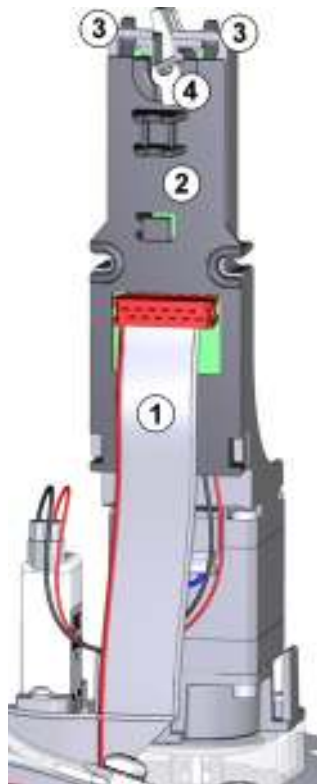


Figure 17–13: SIPAL, back view

4. Unplug the valve connector (J4) from the Probe Aspiration Lift PCBA (5).

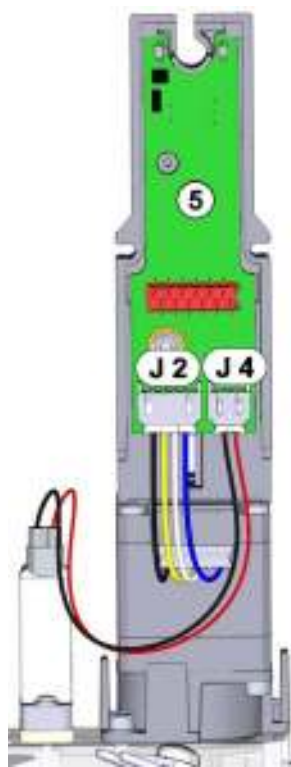


Figure 17–14: Probe Aspiration Lift PCBA

5. Unscrew the two screws (6) to remove the SIPAL and actuator flange (7) from the dispense manifold (8).

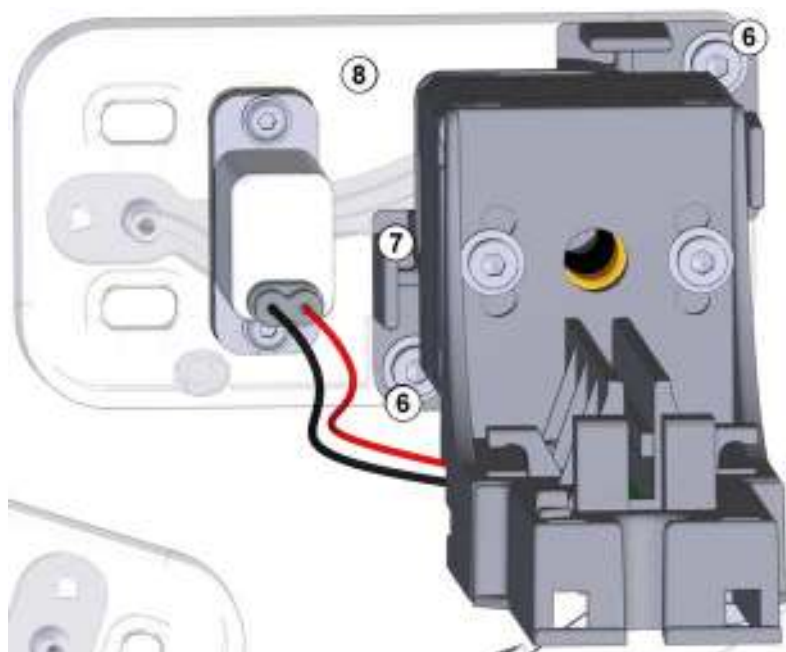
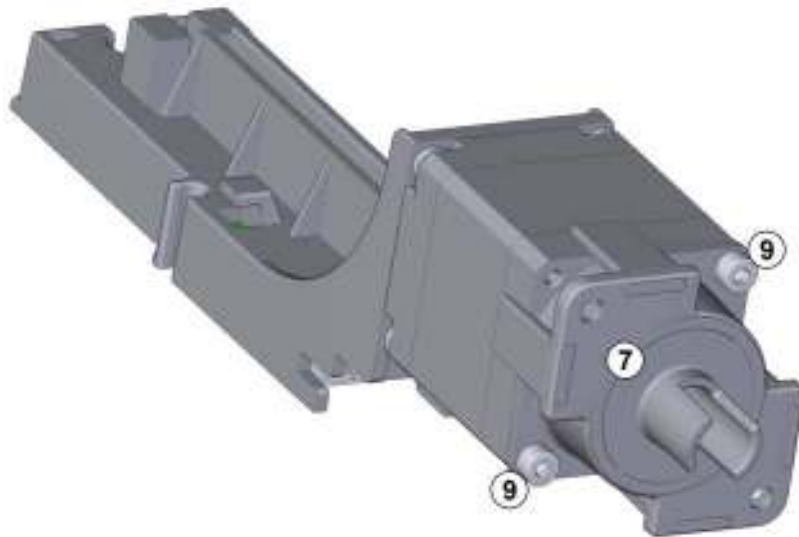


Figure 17–15: SIPAL top view

6. Unscrew two screws (9) to remove actuator flange (7).



*Figure 17-16: SIPAL, bottom view*

#### **17.3.2.3 Assembly of the SIPAL**

1. Set actuator flange (7) onto SIPAL motor and fasten with two screws (9).
2. Set SIPAL and actuator flange (7) onto the dispense manifold (8).
3. Holding the screw driver as shown in Figure 17-17, carefully turn the screw on the washer (6), just enough that neither the screw nor the whole lift assembly has any play. **Do not tighten the screw yet!** Do this consecutively for each of the two screws and washers (6).



*Figure 17-17: Carefully turn the screw*

4. Tighten the screws (6) by a half turn only.

**WARNING****Screws shall not be tightened too much.**

In case the screws (6) are tightened too much, the dispense manifold might be deformed and the cuvettes may collide with the injection nozzles during movement of the Washer rotor.

**NOTE**

To ensure that the SIPALs are correctly set, you can turn the Washer plate on a side and check the distance between the top edge of the injection nozzle and the Washer plate itself, using a caliber. The allowed distance is 2.9±0.4 mm.



Figure 17–18: Check distance between injection nozzle and Washer plate

5. Plug the valve connector (J4) onto the Probe Aspiration Lift PCBA (5).
6. Slide SIPAL protective cover (2) upwards in, till the both hooks (3) are snapping into place.
7. Insert bayonet tube adapter (4).
8. Plug the cable (1) onto the Probe Aspiration Lift PCBA.
9. Assemble the spindle aspiration probe, see Chapter 17.3.1.3 on page 17-8.

**17.3.2.4 Set the SIPAL**

1. Perform the calibration of the Washer SIPAL Lifts, see Chapter 17.4.2.2 on page 17-46.
2. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
3. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
4. Perform a standard Cuvette Endurance Run, see Chapter 11.8.1.2 on page 11-61 OR see Chapter 12.8.1.2 on page 12-62. During the run, check that no sequential noises are present.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.



### 17.3.3 Replacement of SIPAL plastic parts

There is a specific Part Number for the SIPAL plastic parts which combines together:

- Protective black cover of the aspiration lift PCBA (on the right of the picture);
- Bayonet lock adapter (on the bottom left of the picture);
- Plastic body of the SIPAL (in the middle of the picture);
- Black actuator flange of SIPAL motor (on the upper left of the picture).



*Figure 17–19: SIPAL plastics parts*

#### 17.3.3.1 Preliminary step

1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.

#### 17.3.3.2 Removal of the SIPAL plastic parts

1. Remove the Washer aspiration tubing from its fixations (1 and 2) on the SIPAL, see Figure 17–20.
2. Disconnect the tubing from the spindle aspiration probe.
3. Unplug the red cable (J11) from the aspiration lift PCBA by pulling horizontally (3).
4. To remove the protective black cover, simultaneously push gently the hooks (4) out at the top of the SIPAL while pulling the cover downwards.



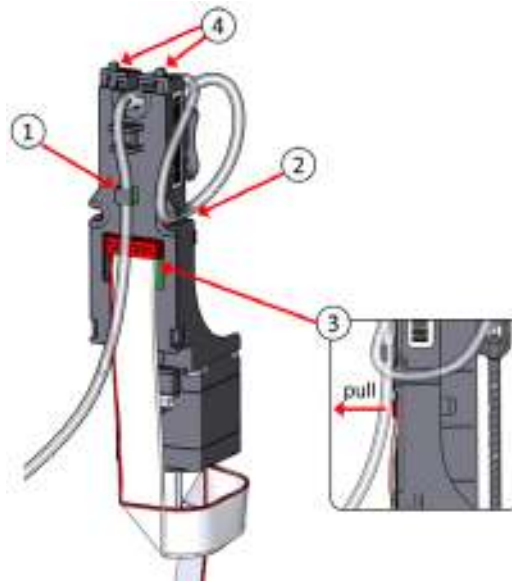


Figure 17–20: SIPAL plastic parts removal

5. The bayonet lock can now be removed.
6. From the aspiration lift PCBA, unplug the two cables (J2 and J4) by pulling downwards.
7. Remove the screws connecting the SIPAL body with the motor, see Figure 17–21.

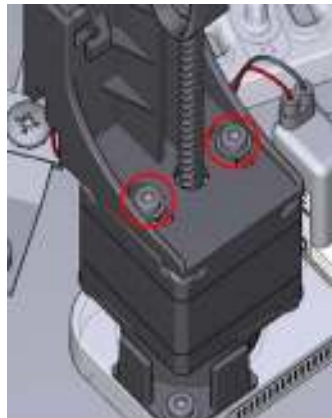
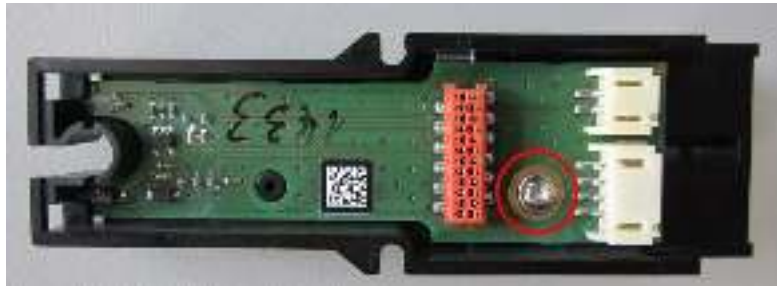


Figure 17–21: Screws connecting SIPAL body to its motor

8. Take out the SIPAL body from the system and remove the spindle aspiration probe.
9. Remove the screw holding the aspiration lift PCBA and exchange the SIPAL body. Install the screw again with the PCBA and make sure that the screw head is centered in the metallic circle. If not, align the PCBA properly with the plastic body, see Figure 17–22.



*Figure 17-22: Screw fixating the aspiration lift PCBA*

10. Remove the screws holding the actuator black flange on the transparent dispense manifold and take the motor and flange out of the system.
11. Exchange the actuator flange onto SIPAL motor and fasten with two screws. Set SIPAL and actuator flange onto the dispense manifold on the Washer module, see Chapter 17.3.2.3 on page 17-14.

#### **17.3.3.3 Assembly of the SIPAL plastic parts**

1. Reinstall the SIPAL body on its motor. Make sure to use the middle screw holes, see Figure 17-21.
2. Put the spindle aspiration probe back into the motor.
3. Reconnect the cables (J2 and J4) on the aspiration lift PCBA.
4. Install the bayonet lock and the protective black cover.
5. Reconnect the red cable (J11) on the aspiration lift PCBA.
6. The Washer aspiration tubing should be fixated on the same location as before in the bubble sensor. This is visible by the pressure marks in the tubing coming from the bayonet lock.
7. Reconnect it to the spindle aspiration probe and fix it on the plastic body.

#### **17.3.3.4 Set of the SIPAL plastic parts**

1. Perform the calibration of the Washer SIPAL Lifts, see Chapter 17.4.2.2 on page 17-46.
2. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
3. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

### 17.3.4 Replacement of the valve

#### 17.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the spindle aspiration probe, see Chapter 17.3.1.2 on page 17-6 from the SIPAL.

#### 17.3.4.2 Removal of the valve

1. Unplug the cable (1) from the Probe Aspiration Lift PCBA.
2. Unhook both hooks (3) and slide SIPAL protective cover (2) downwards out.

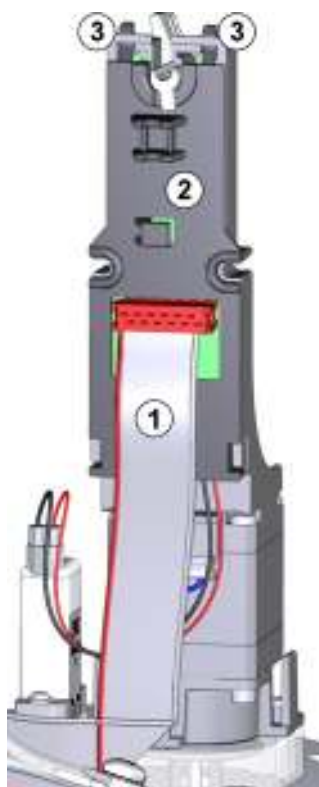
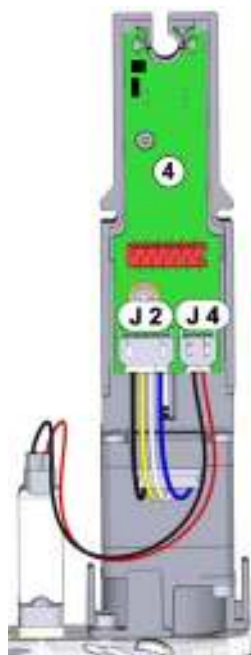


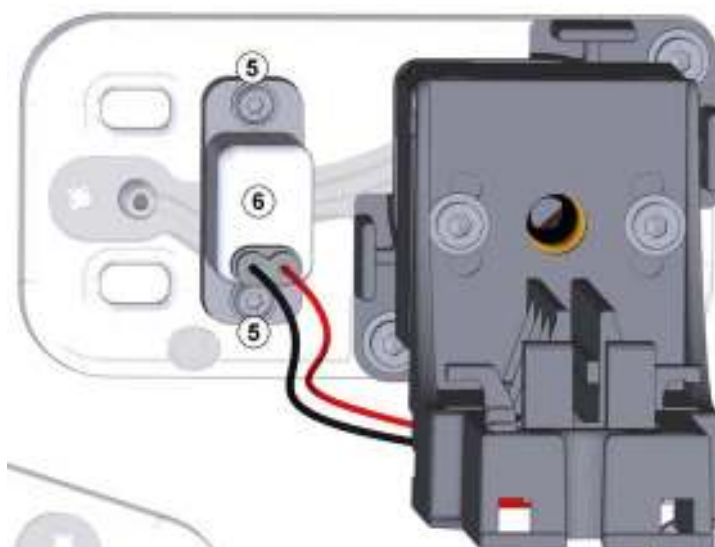
Figure 17–23: SIPAL, back view

3. Disconnect the valve connector (J4) from the Probe Aspiration Lift PCBA (4).



*Figure 17-24: Probe Aspiration Lift PCBA*

4. Unscrew the two screws (5) with TORX T6 and remove the valve (6).



*Figure 17-25: SIPAL with dispense manifold*

#### 17.3.4.3 Assembly of the valve

1. Set the valve (6) in the correct orientation onto the dispense base and fasten with two screws (5) with TORX T6.
2. Plug the valve connector (J4) onto the Probe Aspiration Lift PCBA (4).
3. Slide the SIPAL protective cover (2) upwards in, till the both hooks (3) are snapping into place.
4. Plug the cable (1) onto the Probe Aspiration Lift PCBA.
5. Assembly of the spindle aspiration probe, see Chapter 17.3.1.3 on page 17-8.

#### 17.3.4.4 Set the spindle aspiration probes

1. Perform the calibration of the Washer SIPAL Lift, see Chapter 17.4.2.2 on page 17-46.
2. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
3. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 Or see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 Or see Chapter 10.4 on page 10-10.

## 17.3.5 Replacement of the dispense manifold

### 17.3.5.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the spindle aspiration probe from the SIPAL, see Chapter 17.3.1.2 on page 17-6.
8. Remove the dispense valve, see Chapter 17.3.4.2 on page 17-19.
9. Remove the SIPAL, see Chapter 17.3.2.2 on page 17-12.

### 17.3.5.2 Removal of the dispense manifold

1. Pull out the manifold (6) from the washer plate.

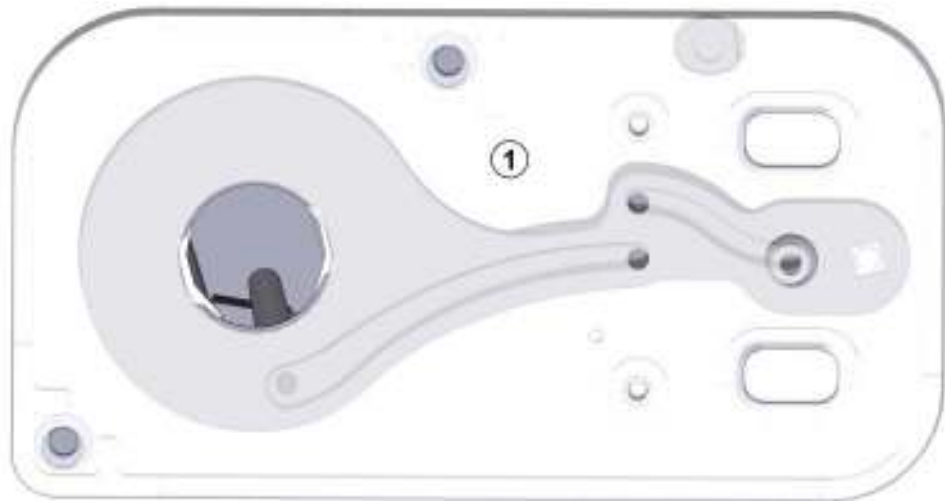


Figure 17–26: Dispense manifold

**17.3.5.3 Assembly of the dispense manifold**

1. Fit the dispense manifold (6) on the washer plate, align the bores (5) of the manifold to the threaded holes of the washer plate.
2. Assemble the SIPAL on the dispense manifold, see Chapter 17.3.2.3 on page 17-14.
3. Assemble the dispense valve on the dispense manifold, see Chapter 17.3.4.3 on page 17-21.
4. Assemble the spindle aspiration probe, see Chapter 17.3.1.3 on page 17-8.

**17.3.5.4 Set the spindle aspiration probes**

1. Perform the calibration of the Washer SIPAL Lifts ,see Chapter 17.4.2.2 on page 17-46.
2. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
3. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 Or see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 Or see Chapter 10.4 on page 10-10.

## 17.3.6 Replacement of the Washer plate

### 17.3.6.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the spindle aspiration probe from the SIPALs, see Chapter 17.3.1.2 on page 17-6.

### 17.3.6.2 Removal of the Washer plate

1. Disassemble the routing of tubes (1) and (2) along the washer plate (aspiration tubes from SIPAL 1 and SIPAL 2, respectively).
2. Unplug the SIPAL 3 aspiration tube (3) from the grounding connection (4) on the washer plate.

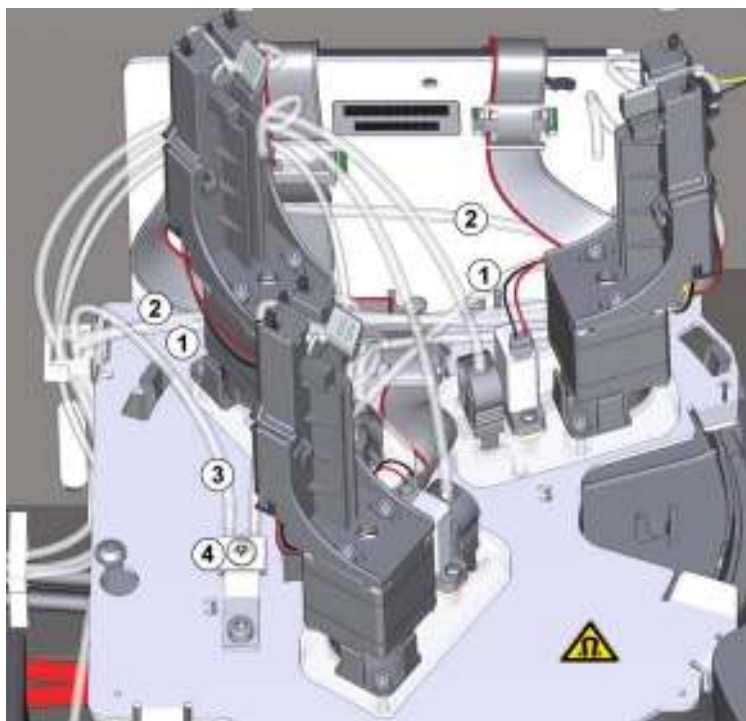


Figure 17–27: Disassembling the tube routing



3. Unplug the TA WBDP washers (5), (6) and (7) from the dispense manifolds of SIPAL 1, 2 and 3, respectively.
4. Remove all the tubes from the cable holder (8).
5. Loosen the screw (9) and turn the washer plate clockwise.
6. Disconnect the CAN connector (10) and the motor connector (11) from the Washer CU PCBA.
7. Lift the washer plate carefully out.

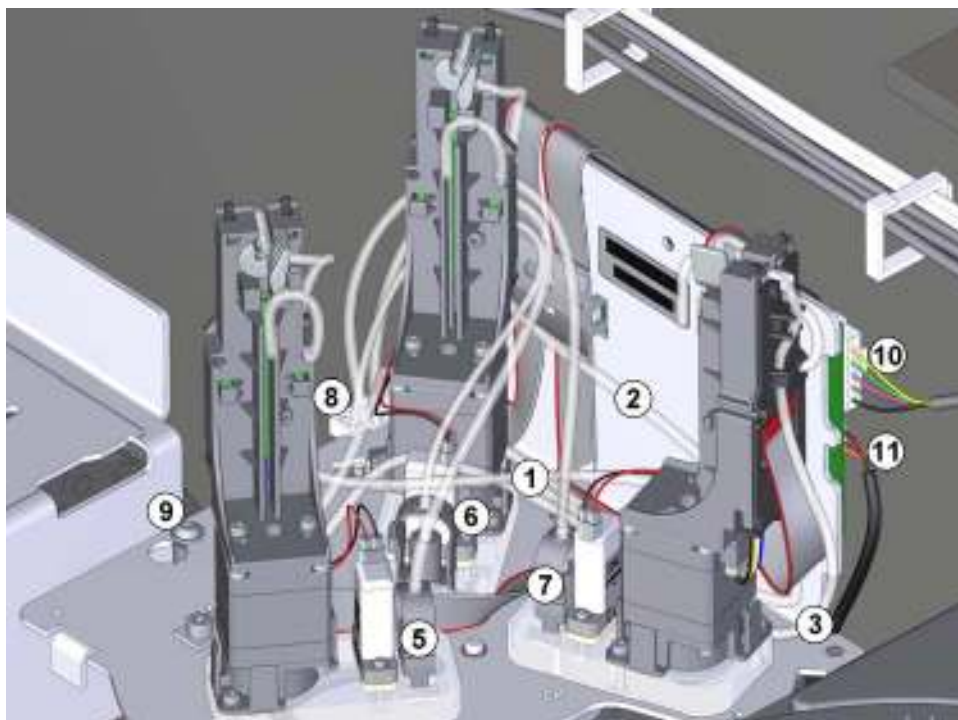


Figure 17–28: Unplugging connections of washer plate

**17.3.6.3 Assembly of the Washer plate**

1. Set the washer plate aligning properly the keyhole and the slots, to the washer base.
2. Connect the CAN connector (10) and the motor connector (11) onto the washer CU PCBA.
3. Turn the washer plate counterclockwise to the end stop.
4. Fasten the screw (9) to secure the washer plate.
5. Insert all the tubes into the cable holder (8).
6. Plug the TA WBDP washers (5), (6) and (7) onto the dispense manifolds of SIPAL 1, 2 and 3, respectively.
7. Re-assemble the routing of tubes (1) and (2) along the washer plate (aspiration tubes from SIPAL 1 and SIPAL 2, respectively).
8. Plug the SIPAL 3 aspiration tube (3) into the grounding connection (4) on the washer plate.
9. Assemble the spindle aspiration probes, see Chapter 17.3.1.3 on page 17-8 for all the SIPALs (1, 2 and 3).

**17.3.6.4 Set the Washer plate**

1. In case a new washer plate has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the calibration of the Washer SIPAL Lift, see Chapter 17.4.2.2 on page 17-46.
3. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
4. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
5. Check the teaching positions for the Left Pipettor on the Washer, see Chapter 13.10 on page 13-25.
6. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

## 17.3.7 Replacement of the Washer CU PCB

### 17.3.7.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the spindle aspiration probe from the SIPALs, see Chapter 17.3.1.2 on page 17-6.
8. Remove the washer plate, see Chapter 17.3.6.2 on page 17-24.

### 17.3.7.2 Removal of the Washer CU PCB

1. Unplug the following cables from Washer CU PCB (1):
  - Wash lift 1 (J3)
  - Wash lift 2+3 (J4)
  - Ring drive Init (J5)

2. Loosen the two screws (2) and slide down the Washer CU PCB (1) to remove it.

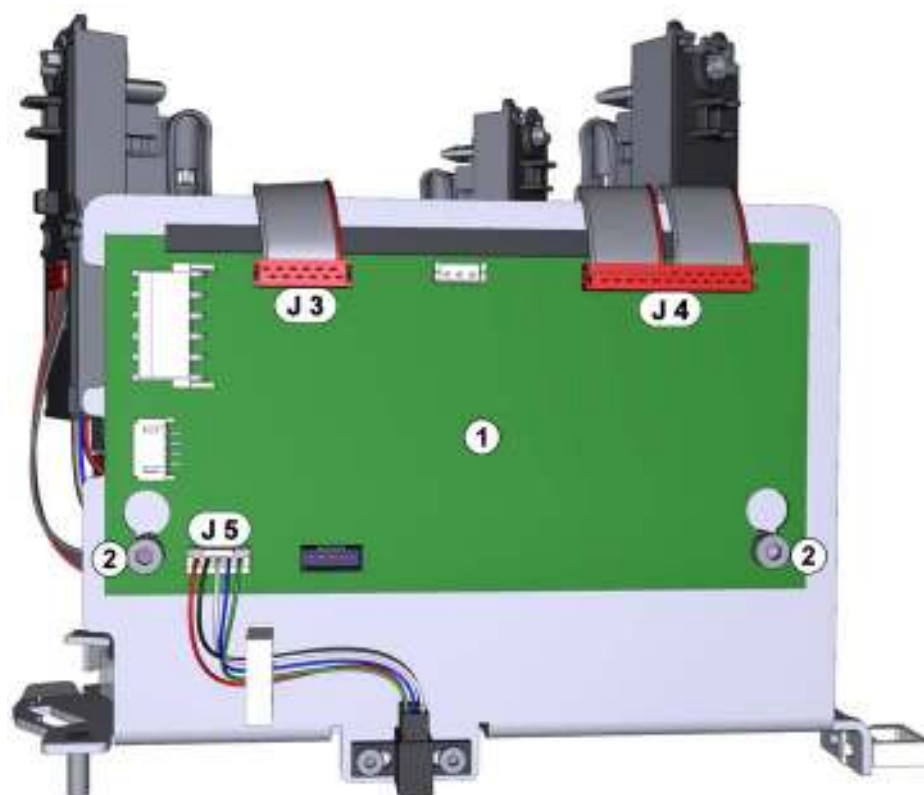


Figure 17–29: Washer CU PCBA

### 17.3.7.3 Assembly of the Washer CU PCB

1. Set the washer CU PCBA (1) into the keyholes and slide it up; fasten the two screws (2) to secure it.
2. Connect the following cables onto Washer CU PCB:
  - Wash lift 1 (J3)
  - Wash lift 2+3 (J4)
  - Ring drive init (J5)
3. Assemble the washer plate, see Chapter 17.3.6.3 on page 17-26.
4. Assemble the spindle aspiration probes, see Chapter 17.3.1.3 on page 17-8 for all the SIPALs.

### 17.3.7.4 Set the Washer CU PCB

1. In case a new PCB has been installed, proceed with the firmware update see Chapter 32.3.4 on page 32-6.
2. Perform the calibration of the Washer SIPAL Lift, see Chapter 17.4.2.2 on page 17-46.
3. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
4. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
5. Check the teaching positions for the Left Pipettor on the Washer, see Chapter 13.10 on page 13-25.
6. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

## 17.3.8 Replacement of the Washer light barrier

### 17.3.8.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the spindle aspiration probe from the SIPALs, see Chapter 17.3.1.2 on page 17-6.
8. Removal of the washer plate, see Chapter 17.3.6.2 on page 17-24.

### 17.3.8.2 Removal of the Washer light barrier

1. Unplug the connector at slot (J5).
2. Unscrew the two screws (2) and remove the light barrier (1).

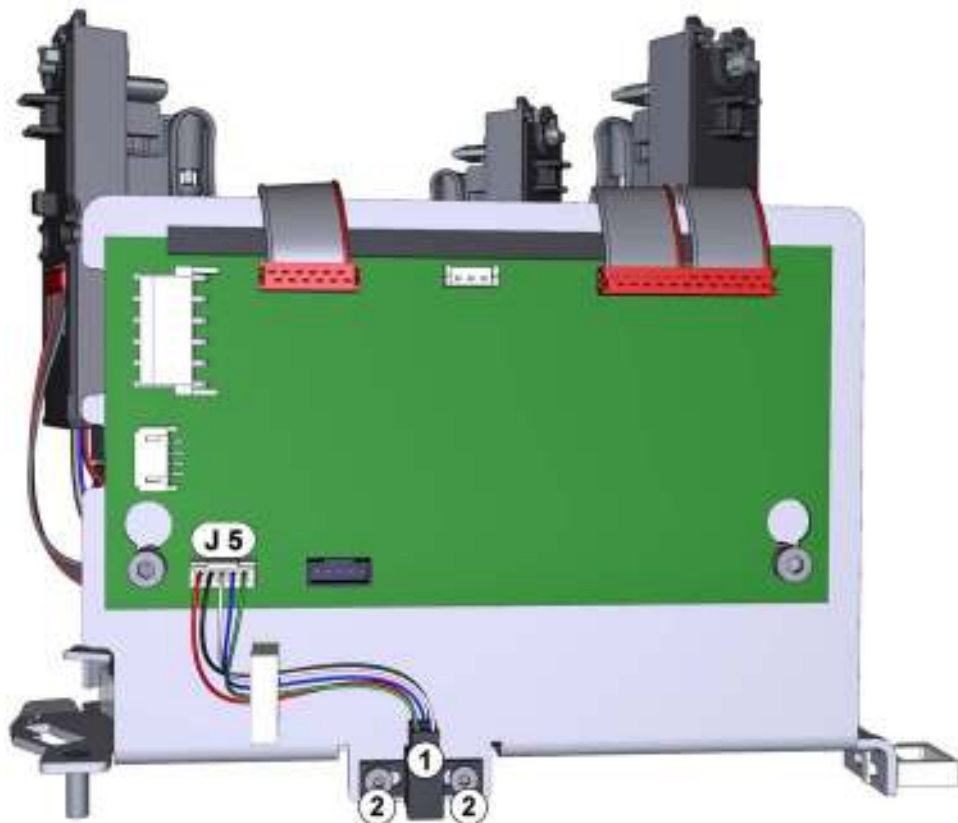


Figure 17–30: Washer plate, rear view

**17.3.8.3 Assembly of the Washer light barrier**

1. Locate the light barrier (1) and fasten with two screws (2).
2. Plug the light barrier cable connector into the dedicated slot (J5).
3. Assemble the washer plate, see Chapter 17.3.6.3 on page 17-26.
4. Assemble the spindle aspiration probe, see Chapter 17.3.1.3 on page 17-8 for all the SIPALs.

**17.3.8.4 Set of the Washer light barrier**

1. Perform the calibration of the Washer SIPAL Lift, see Chapter 17.4.2.2 on page 17-46.
2. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
3. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
4. Check the teaching positions for the Left Pipettor on the Washer, see Chapter 13.10 on page 13-25.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

## 17.3.9 Replacement of the Washer rotor

### 17.3.9.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the spindle aspiration probe from the SIPALs, see Chapter 17.3.1.2 on page 17-6.
8. Removal of the washer plate, see Chapter 17.3.6.2 on page 17-24.
9. Remove and dispose manually the cuvettes in the rotor, if any.

### 17.3.9.2 Removal of the Washer rotor

1. Pull the rotor (1) out of the washer base.

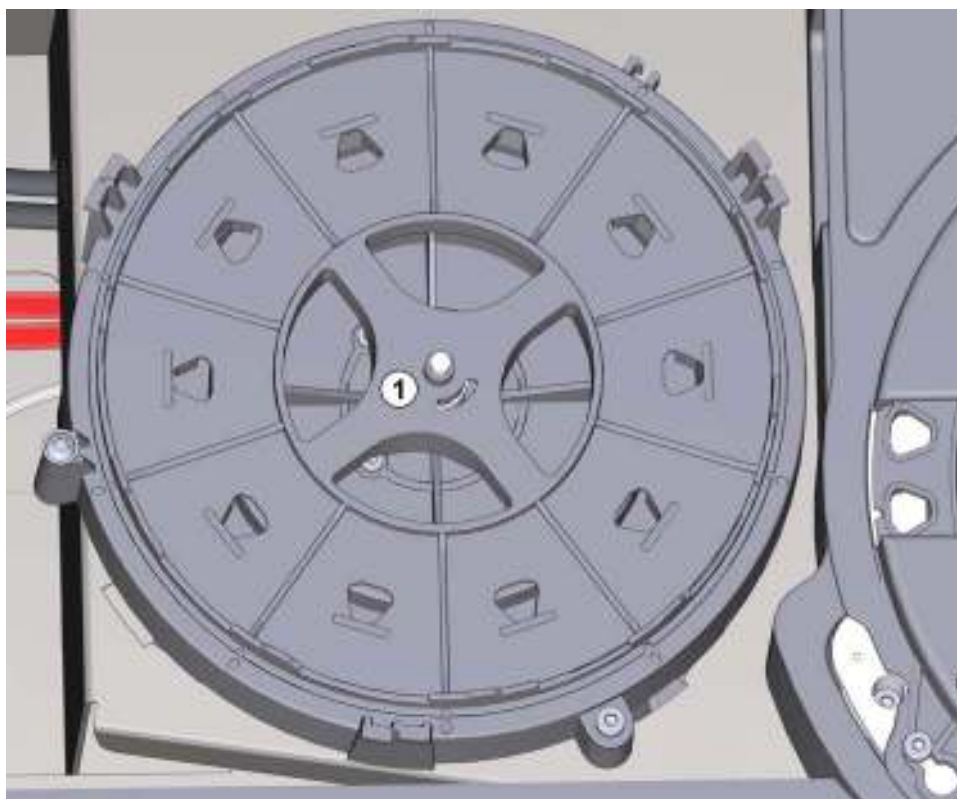


Figure 17–31: Washer rotor



### 17.3.9.3 Assembly of the Washer rotor

1. Check for correct alignment of the tube clamp:  
The tenterhooks (2) of the tube clamp shall sit in the middle of the lateral surface of the shaft support (3).

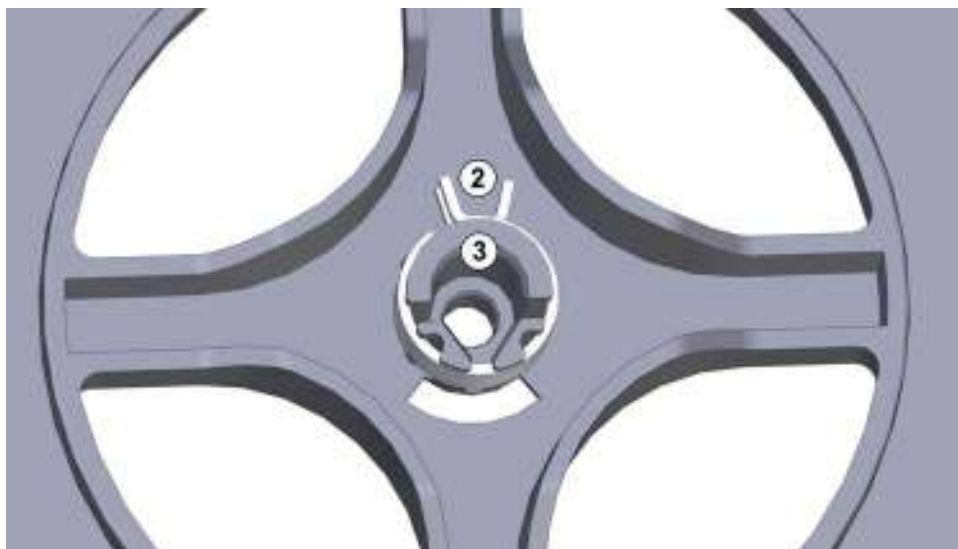


Figure 17-32: Washer rotor

2. Slide the washer rotor onto the motor shaft down to the end stop.



Figure 17-33: Washer rotor

3. Assemble the washer plate see Chapter 17.3.6.3 on page 17-26.
4. Assemble the spindle aspiration probe see Chapter 17.3.1.3 on page 17-8 for all the SIPALs.

**17.3.9.4 Set the Washer rotor**

1. Perform the calibration of the Washer Rotor, see Chapter 17.4.2.1 on page 17-45.
2. Perform the calibration of the Washer SIPAL Lift, see Chapter 17.4.2.2 on page 17-46.
3. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
4. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 or see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
5. Check the teaching positions for the Left Pipettor on the Washer, see Chapter 13.10 on page 13-25.
6. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 17.3.10 Replacement of the Washer Module

#### 17.3.10.1 Preliminary steps



1. Before replacing the wash module save the parameter file.
2. Exit the **LIAISON<sup>®</sup> XS** system software.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the spindle aspiration probe from the SIPALs, see Chapter 17.3.1.2 on page 17-6.
8. Remove the washer plate, see Chapter 17.3.6.2 on page 17-24.
9. Remove and dispose manually the cuvettes in the rotor, if any.

#### 17.3.10.2 Removal of the Washer Module

1. Remove the screw (1).
2. Turn the module base clockwise and lift it carefully out.

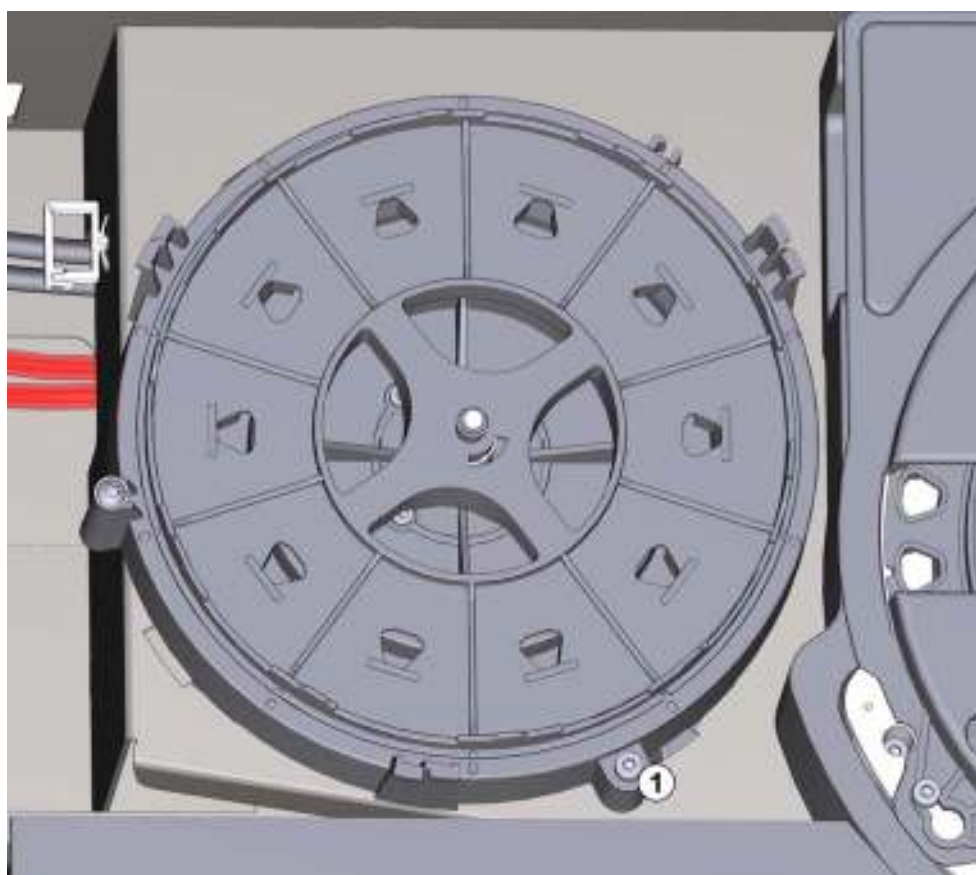


Figure 17–34: Washer module

**17.3.10.3 Assembly of the washer module**

1. Align the module to the guides and turn it counterclockwise to the end stop.
2. Fasten the screw (1).
3. Assemble the washer plate, see Chapter 17.3.6.3 on page 17-26.
4. Assemble the spindle aspiration probe, see Chapter 17.3.1.3 on page 17-8.

**17.3.10.4 Set of the Washer Module**

1. In case a new module has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the calibration of the Washer Rotor, see Chapter 17.4.2.1 on page 17-45.
3. Perform the calibration of the Washer SIPAL Lift, see Chapter 17.4.2.2 on page 17-46.
4. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
5. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
6. Check the teaching positions for the Left Pipettor on the Washer, see Chapter 13.10 on page 13-25.
7. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

### 17.3.11 Replacement of the Washer Motor

#### 17.3.11.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shutdown button in Windows, and wait until the shutdown is complete.
3. On the right side of the instrument, turn the power switch off.
4. Unplug the power cable from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the spindle aspiration probe from the SIPALs, see Chapter 17.3.1.2 on page 17-6.
8. Remove the Washer plate, see Chapter 17.3.6.2 on page 17-24.
9. Remove and dispose manually the cuvettes in the rotor, if any.
10. Remove the Washer rotor, see Chapter 17.3.9.2 on page 17-32.
11. Remove the Washer Module see Chapter 17.3.10.2 on page 17-35.

#### 17.3.11.2 Removal of the Washer Motor

1. Disconnect the motor cable from the motor.
2. Remove the four screws (1).



Figure 17–35: Washer motor screws

**17.3.11.3 Assembly of the Washer Motor**

1. Insert the motor (2) into the Washer Base. The Motor cable connector (3) shall be oriented toward the Washer Base cable holder (4).
2. Fasten the motor with four screws (1).



*Figure 17–36: Washer motor orientation*

3. Connect the motor cable onto the motor.
4. Assemble the Washer rotor onto the module, see Chapter 17.3.9.3 on page 17-33.
5. Assemble the Washer module, see Chapter 17.3.10.3 on page 17-36.
6. Assemble the Washer plate, see Chapter 17.3.6.3 on page 17-26.
7. Assemble the spindle aspiration probes, see Chapter 17.3.1.3 on page 17-8.

**17.3.11.4 Washer adjustment**

1. Perform the calibration of the Washer Rotor, see Chapter 17.4.2.1 on page 17-45.
2. Perform the calibration of the Washer SIPAL Lift, see Chapter 17.4.2.2 on page 17-46.
3. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
4. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.

5. Check the teaching positions for the Left Pipettor on the Washer, see Chapter 13.10 on page 13-25.
6. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 17.3.12 Replacement of the aspiration lift PCBA

### 17.3.12.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the spindle aspiration probe, see Chapter 17.3.1.2 on page 17-6.
8. Remove the SIPAL, see Chapter 17.3.2.2 on page 17-12.

### 17.3.12.2 Removal of the aspiration lift PCBA

1. Unhook the two hooks (2) of the cover (1) and slide the cover out.



Figure 17–37: SIPAL, remove cover

2. Unplug motor connector (J2).
3. Remove one screw (3) and lift aspiration lift PCBA (4) out of the SIPAL.

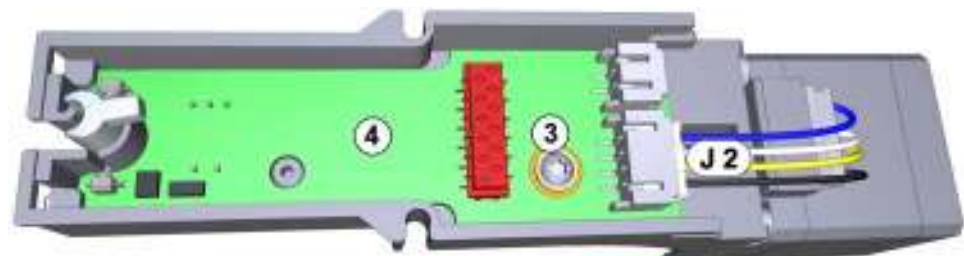


Figure 17–38: SIPAL, remove aspiration lift PCBA



**17.3.12.3 Assembly of the aspiration lift PCBA**

1. Set aspiration lift PCBA (4) into SIPAL and fasten with one screw (3).
2. Connect the motor connector (J2) onto the aspiration lift PCBA (4).
3. Set the cover onto the SIPAL and slide the cover till the hooks are snapping into position.
4. Assemble the SIPAL, see Chapter 17.3.2.3 on page 17-14.
5. Assemble the spindle aspiration probe, see Chapter 17.3.1.3 on page 17-8.

**17.3.12.4 Set the aspiration lift PCBA**

1. Perform the calibration of the Washer SIPAL Lifts, see Chapter 17.4.2.2 on page 17-46.
2. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
3. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

### 17.3.13 Replacement of the spindle drive of SIPAL

#### 17.3.13.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the spindle aspiration probe, see Chapter 17.3.1.2 on page 17-6.
8. Remove the SIPAL, see Chapter 17.3.2.2 on page 17-12.

#### 17.3.13.2 Removal of the spindle drive of SIPAL

1. Unplug motor connector (2) at the spindle drive (1).



Figure 17–39: SIPAL, remove cover

2. Remove two screws and washers (3).
3. Remove the spindle drive (1) out of the SIPAL.



Figure 17–40: SIPAL, remove spindle drive

### 17.3.13.3 Assembly of the spindle drive of SIPAL

**NOTE**

Be aware of the correct cable outlet (2) orientation of the spindle drive.

1. Set the spindle drive (1) onto the SIPAL chassis and fasten with two screws and washers (3).

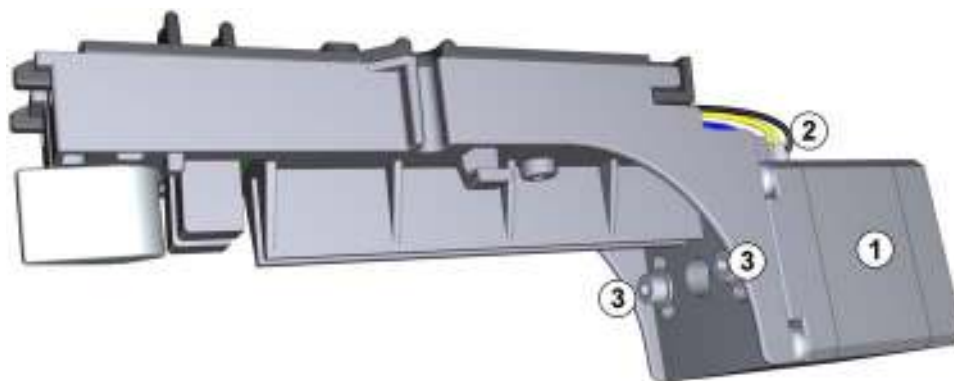


Figure 17-41: SIPAL, remove spindle drive, side view

2. Connect the motor connector onto the spindle drive (1).
3. Assemble the SIPAL, see Chapter 17.3.2.3 on page 17-14.
4. Assemble the spindle aspiration probe, see Chapter 17.3.1.3 on page 17-8.

### 17.3.13.4 Set of the spindle drive of SIPAL

1. Perform the calibration of the Washer SIPAL Lifts, see Chapter 17.4.2.2 on page 17-46.
2. Perform the calibration of Aspiration Monitoring Sensors, see Chapter 17.4.2.3 on page 17-46.
3. In the **Service Software**, perform a standard prime of the Washer Module, see Chapter 11.8.1.5 on page 11-65 OR see Chapter 12.8.1.5 on page 12-67.; in the result table, check that the LiquidLength value for each port exceeds the cut-off value of 900 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

## 17.4 Adjustment and calibration

This section contains information about the adjustment and calibration of the Washer Module.

### NOTE

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).  
Do not run any of the software at the same time.

### 17.4.1 Initialization routine

- Close any open software and start the **Service Software** from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **Washer** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog Controlled Modules** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

## 17.4.2 Calibration of the Washer Module

### 17.4.2.1 Calibration of the Washer Rotor

In the **Service Software** at the navigation window, choose **Washer** then the **Teach & Calibration** bracket and select **Ring**.

#### Step 1

- a. Follow the instructions and then press the **Done** button.

#### Step 2

- b. Move the Washer rotor by the arrow buttons, to align the slot on the rotor to the reference position on the Washer base.



Figure 17–42: Washer rotor, calibration position

#### Step 3

- c. Press **Save** button to save the position.
- d. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

#### Step 4

- e. The process is successful when, at the **Summarize** panel, all steps are in green.

**17.4.2.2 Calibration of the Washer SIPAL Lift**

In the **Service Software**, at the navigation window, choose Washer then the **Teach & Calibration** bracket and select **Lifts**.

- Step 1** a. Follow the instructions and then press the **Done** button.
- Step 2** b. Press the **Start** button; the calibration procedure starts automatically.  
c. When the **Processing** window closes without failure message, check the results and press the **Done** button.
- Step 3** d. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 4** e. The process is successful when, at the **Summarize** panel, all steps are in green.

**17.4.2.3 Calibration of the Washer Aspiration Monitoring Sensors**

In the **Service Software**, at the navigation window, choose **Washer** then the **Teach & Calibration** bracket and select the **Washer Asp. Mon.** bracket:

- Step 1** a. Press the **WAP for 5sec** on button, to ensure that tubes are free of liquid.  
b. When the **Processing** window closes, press the **Done** button.
- Step 2** c. Press the **Start** button; the calibration procedure for aspiration monitor 1 starts automatically.  
d. When the **Processing** window closes without failure message, check the result and press the **Done** button.
- Step 3** e. Press the **Start** button; the calibration procedure for aspiration monitor 2 starts automatically.  
f. When the **Processing** window closes without failure message, check the result and press the **Done** button.
- Step 4** g. Press the **Start** button; the calibration procedure for aspiration monitor 3 starts automatically.  
h. When the **Processing** window closes without failure message, check the result and press the **Done** button.
- Step 5** i. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 6** j. The process is successful when, at the **Summarize** panel, all steps are in green.

#### 17.4.2.4 WBDP calibration

In the **Service Software** at the navigation window choose **Washer** and the **Teach & Calibration** bracket and select **WBDP Calibrating** bracket.

##### Step 1

- a. Press the **Start** button; the calibration procedure for WBDP starts automatically.
- b. .When the **Processing** window closes without failure message, press the **Done** button.

##### Step 2

- c. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.

##### Step 3

- d. The process is successful when, at the **Summarize** panel, all steps are in green.

## 17.5 Electric connections, switches and signals

This section shows an overview of the electronic boards of the module.

### NOTE

see Chapter 3.1.7 on page 3-8 or see Chapter 4.1.8 on page 4-9 for an overview of the complete electric circuit of the system.

### 17.5.1 Washer CU PCBA board

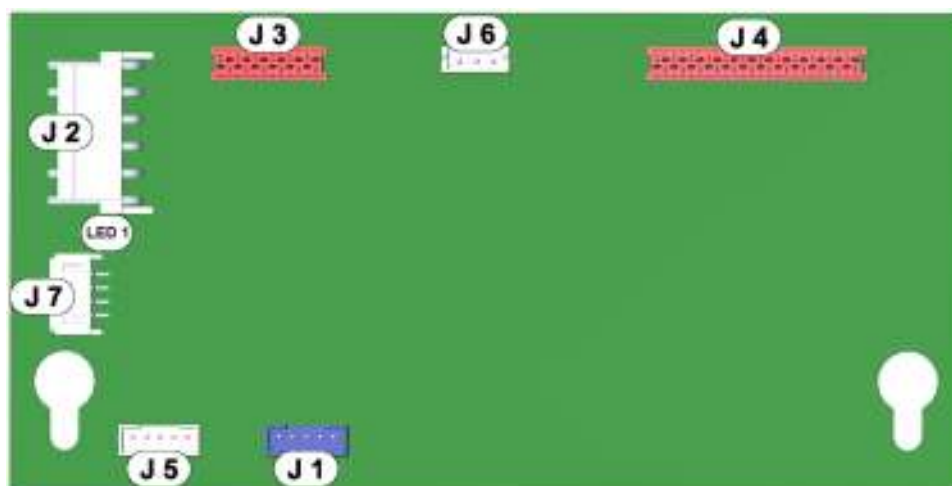


Figure 17–43: Washer CU PCBA board

| Number | Connection                              |
|--------|-----------------------------------------|
| J1     | Download                                |
| J2     | Power / CAN                             |
| J3     | Wash Lift 1                             |
| J4     | Wash Lift 2 + 3                         |
| J5     | Rotor position / Init (Ring drive/init) |
| J6     | not in use                              |
| J7     | Ring drive                              |

Table 17–5: Connections of the Washer CU PCBA board

| Number | Meaning/Function |
|--------|------------------|
| LED 1  | Power / CAN      |

Table 17–6: LED of the Washer CU PCBA board



### 17.5.2 Aspiration lift PCBA board

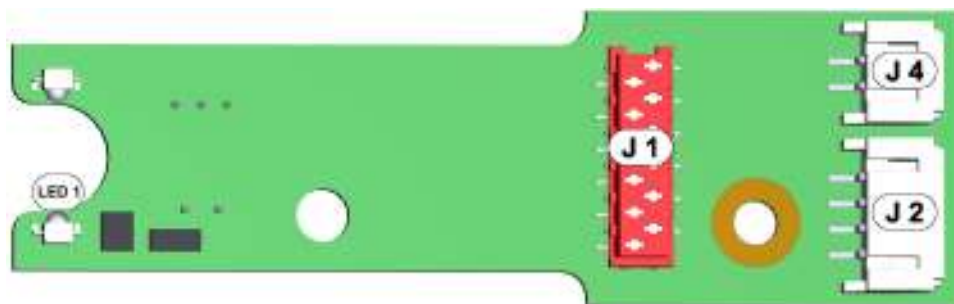


Figure 17-44: Aspiration lift PCBA board

| Number | Connection    |
|--------|---------------|
| J1     | Washer CU PCB |
| J2     | Needle MTR    |
| J4     | Valve         |

Table 17-7: Connections of the aspiration lift PCBA board

| Number | Meaning/Function |
|--------|------------------|
| LED 1  | Bubble detector  |

Table 17-8: LED of the aspiration lift PCBA board



Figure 17-45: Light barrier of the aspiration lift PCBA

| Number | Meaning/Function   |
|--------|--------------------|
| IC 2   | Init light barrier |

Table 17-9: Light barrier of the aspiration lift PCBA board



# 18 Reader Module

This chapter describes the Reader Module in detail. In the first section of the chapter “Safety” notes concerning Incubator are specified, while the next section gives a brief overview of the main parts composing this module.

This section is followed by a larger part containing information about repair and calibration of this module (sections “Replacement and Repair” and “Adjustment and Calibration”). In the last section Luminometer CU PCBA layout and connections are described.

- The Reader Module consists of a Measurement chamber, which can be rotated by a drive.
- The Reader Module controls this drive, that transports the cuvette through the Reader unit.
- The cuvette will be measured by a PMT module, a photo multiplier tube module.
- In the Measurement chamber, a defined amount of Starter reagents are dispensed into the cuvette, which causes a luminescent reaction.
- The PMT measures the light signals of the reaction.
- After the measurement, the Instrument aspirates liquid in the cuvette to a liquid waste container and disposes the cuvette to a solid waste container.

## 18.1 Safety

---

**⚠ DANGER****High voltage!**

Touching high voltage parts or cables will result in serious personal injury with deadly consequences.

- Never touch high voltage parts.
  - Separate the Instrument from the mains supply before working on or cleaning the Instrument.
  - Avoid contact with the following parts that can be under high voltage during operation: silver HV cable connectors and PMT.
- 

---

**⚠ WARNING****Non-observance of safety instructions**

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 OR see Chapter 2.5 on page 2-7.
  - Follow all working instructions included in this manual.
  - Follow all warnings marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
-

## 18.2 Overview

This section shows an overview of the components of the Reader Module.



Figure 18–1: Reader Module, side view

- |    |                                    |
|----|------------------------------------|
| 1. | Aspiration monitor (bubble sensor) |
| 2. | SIPAL                              |
| 3. | Cuvette loading                    |
| 4. | Starter Reagent 2                  |
| 5. | Starter Reagent 1                  |
| 6. | Reader Module                      |
| 7. | Motor drive for Reader rotor       |

Table 18–1: Reader Module, overview

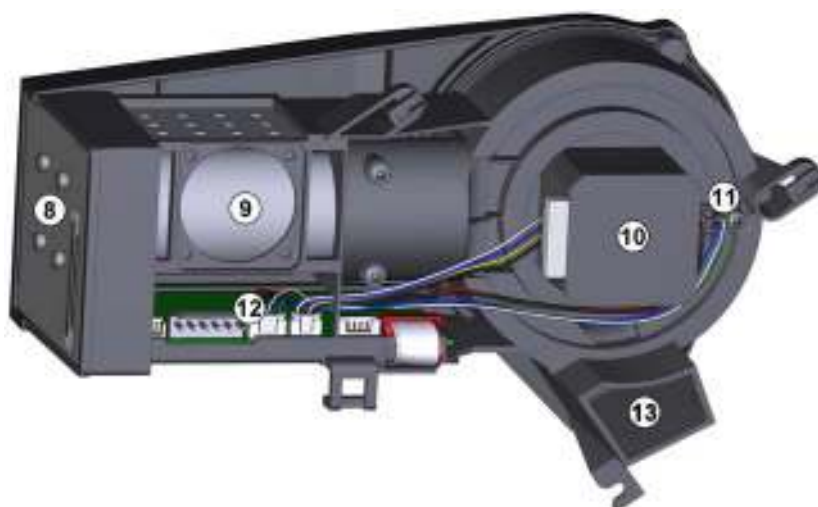


Figure 18–2: Reader Module, bottom view

- |     |                            |
|-----|----------------------------|
| 8.  | Back plate                 |
| 9.  | PMT, photo multiplier tube |
| 10. | Motor drive                |
| 11. | Init Sensor (rotary home)  |
| 12. | Luminometer CU PCBA        |
| 13. | Cuvette waste              |

Table 18–2: Reader Module, bottom view

## 18.3 Replacements and repairs

This section contains instructions for the repair of individual components of the Reader Module. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 18.3.1 Replacement of the spindle aspiration probe

The spindle aspiration probe includes the aspiration needle.

#### 18.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.

#### 18.3.1.2 Removal of the spindle aspiration probe

1. Unhook the tube (2) from the hooks (3) of the SIPAL housing (1).
2. Unhook tube out of the guidance (4).

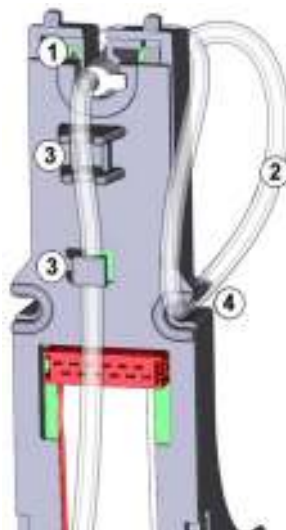


Figure 18–3: Reader SIPAL, rear view

3. Turn the bayonet adapter (5) counterclockwise a quarter turn.
4. Lift the tube out of the bayonet adapter (5). Pull the tube (2) carefully out of the aspiration probe and lay down the tube.



**18.3.1.3 Assembly of the spindle aspiration probe**

1. Set the spindle aspiration probe (7) into the slide and push the threaded part carefully into the threaded bushing of the motor.



*Figure 18–6: Reader SIPAL without tubing, front view, assembly of the aspiration needle*

2. The tube length from the bayonet adapter should measure about 160mm.
3. Turn the bayonet adapter (6) clockwise a quarter turn to lock the tube.
4. Connect the tube with the spindle aspiration probe.

**NOTE**

The tube length between the spindle aspiration probe and the bayonet shall not be longer than 160mm to prevent aspiration monitoring malfunction.



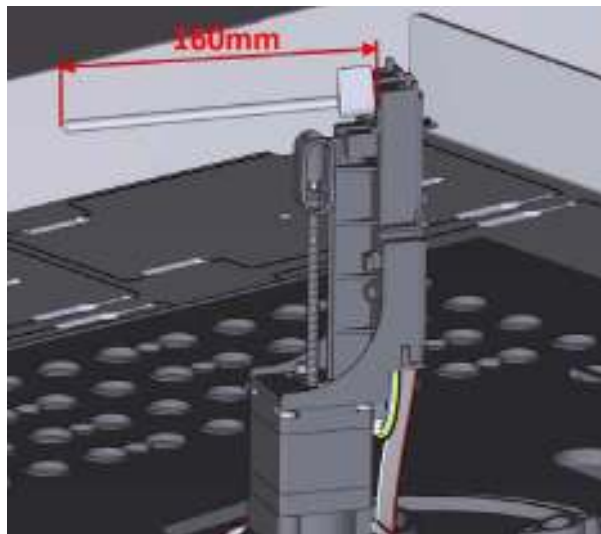


Figure 18-7: SIPAL

5. Locate the tube in the hooks (3) and the guidance (4).

**NOTE**

Tubing shall fit straight into the bayonet.

Tubes shall not be kinked or twisted.

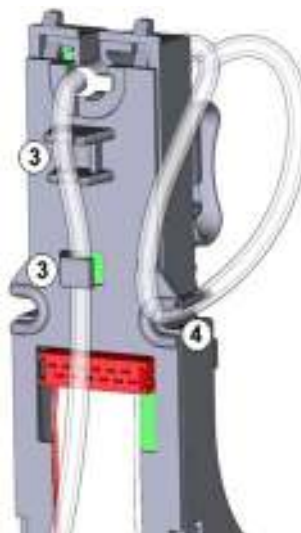


Figure 18-8: Reader SIPAL and aspiration tube routing, back view

**18.3.1.4 Set the spindle aspiration probe**

1. Perform the calibration of the lift, see Chapter 18.4.3.2 on page 18-33.
2. Perform the calibration of the aspiration monitor sensor, see Chapter 18.4.3.3 on page 18-33.

3. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 OR see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

## 18.3.2 Replacement of the Reader cover and SIPAL

### 18.3.2.1 Preliminary steps



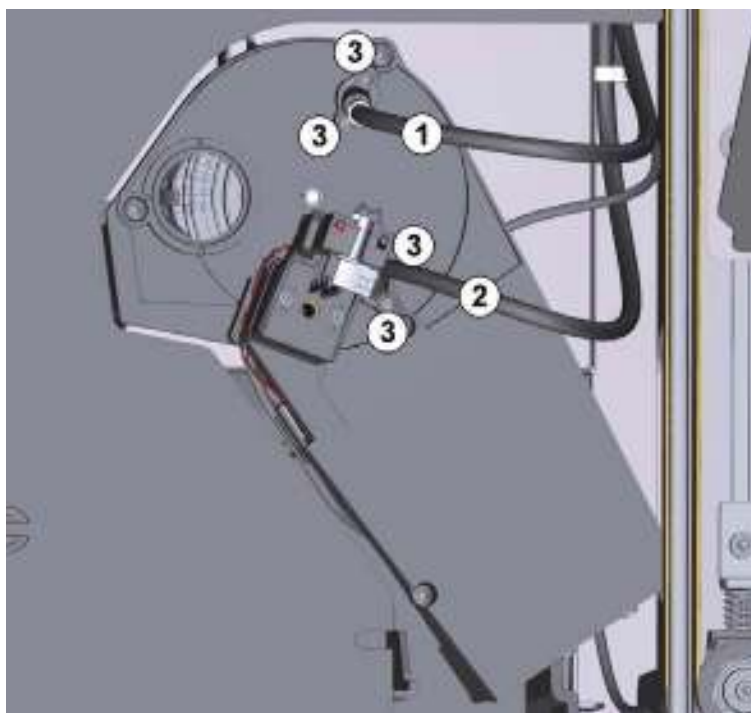
1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Removal of the spindle aspiration probe, see Chapter 18.3.1.2 on page 18-4.

### 18.3.2.2 Removal of the Reader cover and SIPAL

#### NOTE

**Disconnect both the Starter Reagent tubes before proceed with the removal of the reader cover.**

1. Unscrew four screws (3) of Starter Reagent tubes (1 and 2).
2. Put the Starter Reagent tube (1 and 2) aside.



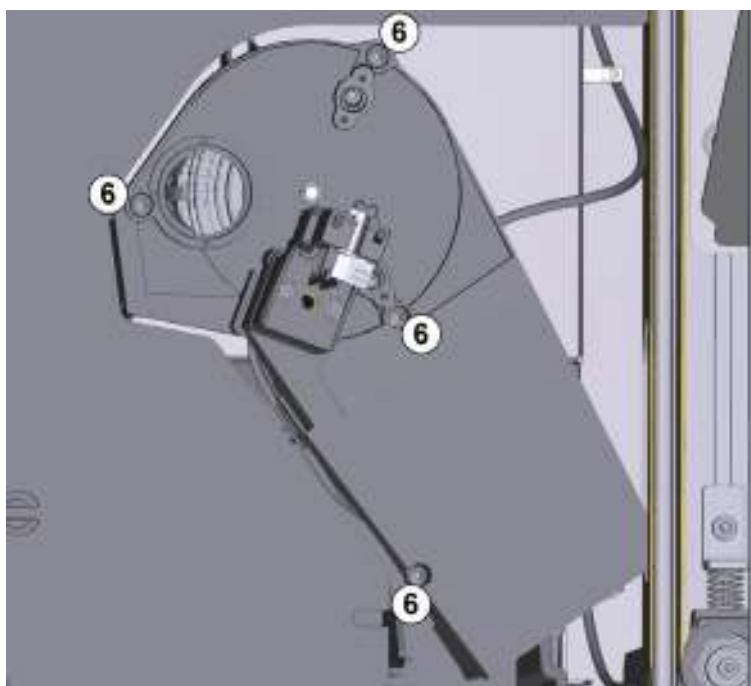
*Figure 18–9: Reader Module, inside the instrument, top view*

3. Unplug flat ribbon cable connection (J1) of aspiration lift PCBA and route it out of the guides.



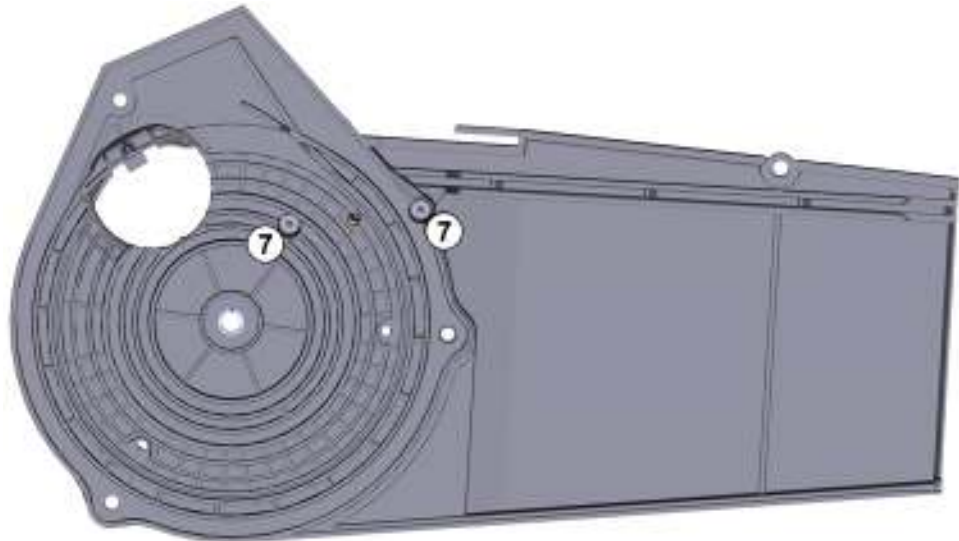
*Figure 18-10: Reader, SIPAL, back view*

4. Unscrew four screws (6) and lift Reader cover up.



*Figure 18-11: Reader Module, top view without tubing and needle*

5. Unscrew two screws (7) from the inner side of the Reader cover to release the SIPAL.



*Figure 18–12: Reader cover, bottom view*

6. Remove SIPAL and set it aside.

### 18.3.2.3 Assembly of the Reader cover and SIPAL

1. Set SIPAL (8) into the Reader cover and fasten with two screws (7).

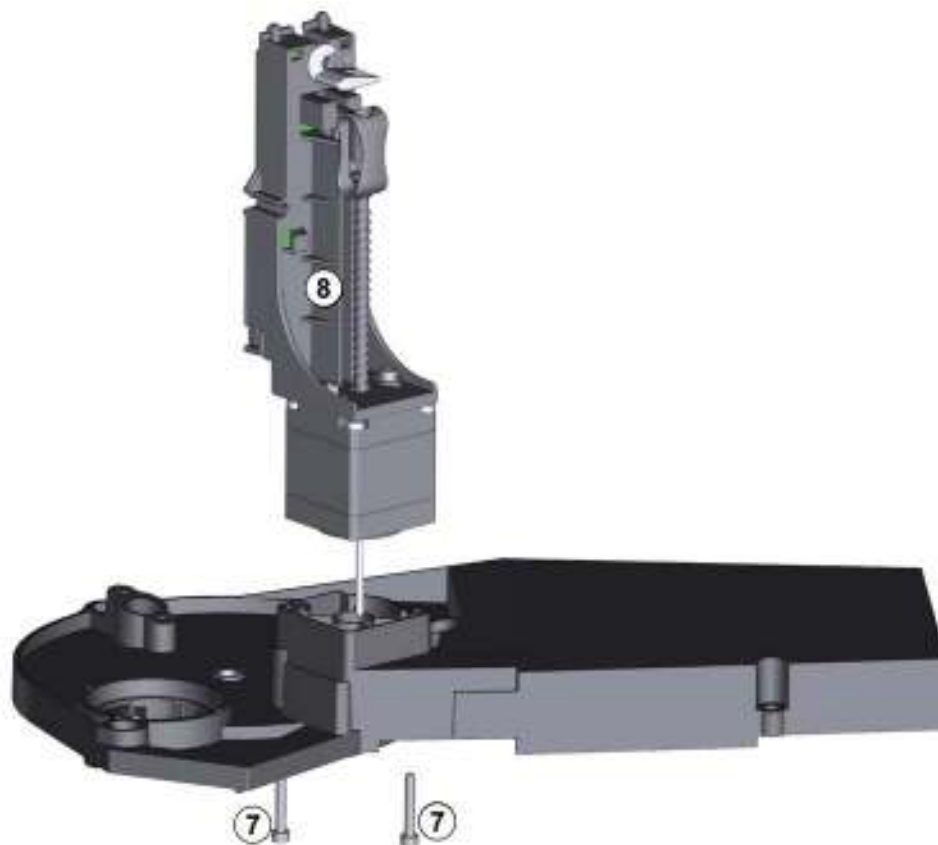


Figure 18–13: Reader cover and SIPAL

2. Set Reader cover onto the Reader Module and fasten four screws (6).
3. Connect flat ribbon cable connection (J1) onto aspiration lift PCBA.
4. Mount the two Starter Reagent tubes in the correct order onto the Reader Module.
5. Assemble the spindle aspiration probe, see Chapter on page 18-5.

### 18.3.2.4 Set the Reader cover and SIPAL

1. Perform the calibration of the Reader SIPAL lift, see Chapter 18.4.3.2 on page 18-33.
2. Perform the calibration of the Reader aspiration monitor, see Chapter 18.4.3.3 on page 18-33.
3. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 OR see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 18.3.3 Replacement of the aspiration lift PCBA

#### 18.3.3.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the spindle aspiration probe, see Chapter 18.3.1.2 on page 18-4.
8. Remove the Reader cover and SIPAL, see Chapter 18.3.2.2 on page 18-9.

#### 18.3.3.2 Removal of the aspiration lift PCBA

1. Unhook the two hooks (2) of the cover (1) and slide the cover out.

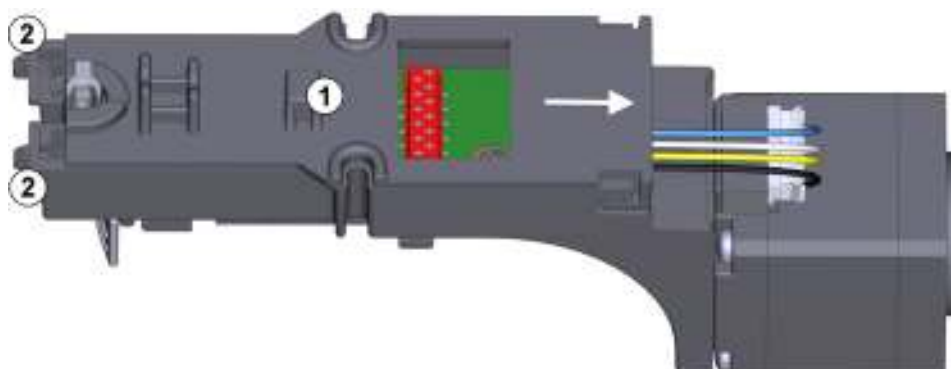


Figure 18–14: SIPAL, remove cover

2. Unplug motor connector (J2).
3. Remove one screw (3) and lift aspiration lift PCBA (4) out of the SIPAL.



Figure 18–15: SIPAL, remove aspiration lift PCBA

**18.3.3.3 Assembly of the aspiration lift PCBA**

1. Set aspiration lift PCBA (4) into SIPAL and fasten with one screw (3).
2. Connect the motor connector (J2) onto the aspiration lift PCBA (4).
3. Set the cover onto the SIPAL and slide the cover till the hooks are snapping into position.
4. Assemble the Reader cover and SIPAL, see Chapter 18.3.2.3 on page 18-12.
5. Assemble the spindle aspiration probe, see Chapter on page 18-5-

**18.3.3.4 Set the aspiration lift PCBA**

1. Perform the calibration of the Reader SIPAL lift, see Chapter 18.4.3.2 on page 18-33.
2. Perform the calibration of the Reader aspiration monitor, see Chapter 18.4.3.3 on page 18-33.
3. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 OR see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.



### 18.3.4 Replacement of the spindle drive of SIPAL

#### 18.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the spindle aspiration probe, see Chapter 18.3.1.2 on page 18-4.
8. Remove the Reader cover and SIPAL, see Chapter 18.3.2.2 on page 18-9.

#### 18.3.4.2 Removal of the spindle drive of SIPAL

1. Unplug motor connector (2) at the spindle drive (1).



Figure 18-16: SIPAL, remove cover

2. Remove two screws and washers (3).
3. Remove the spindle drive (1) out of the SIPAL.

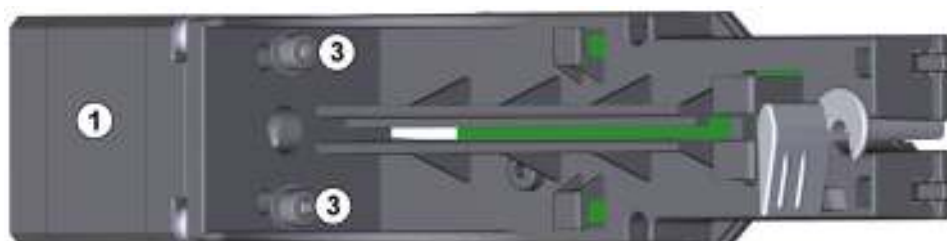


Figure 18-17: SIPAL, remove spindle drive

### 18.3.4.3 Assembly of the spindle drive of SIPAL

**NOTE**

Be aware of the correct cable outlet (2) orientation of the spindle drive.

1. Set the spindle drive (1) onto the SIPAL chassis and fasten with two screws and washers (3).



Figure 18-18: SIPAL, remove spindle drive, side view

2. Connect the motor connector onto the spindle drive (1).
3. Assemble the Reader cover and SIPAL, see Chapter 18.3.2.3 on page 18-12.
4. Assemble the spindle aspiration probe, see Chapter on page 18-5.

### 18.3.4.4 Set the spindle drive of SIPAL

1. Perform the calibration of the Reader SIPAL lift, see Chapter 18.4.3.2 on page 18-33.
2. Perform the calibration of the Reader aspiration monitor, see Chapter 18.4.3.3 on page 18-33.
3. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 OR see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 18.3.5 Replacement of the Reader rotor

#### 18.3.5.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Removal of the Reader cover and SIPAL, see Chapter 18.3.2.2 on page 18-9.

#### 18.3.5.2 Removal of the Reader rotor

1. Lift rotor (1) carefully out of the housing.

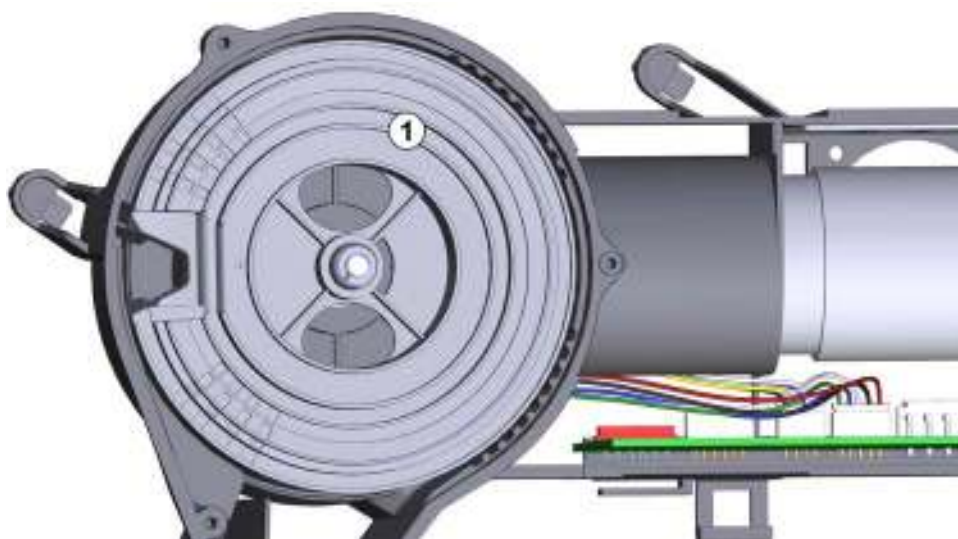
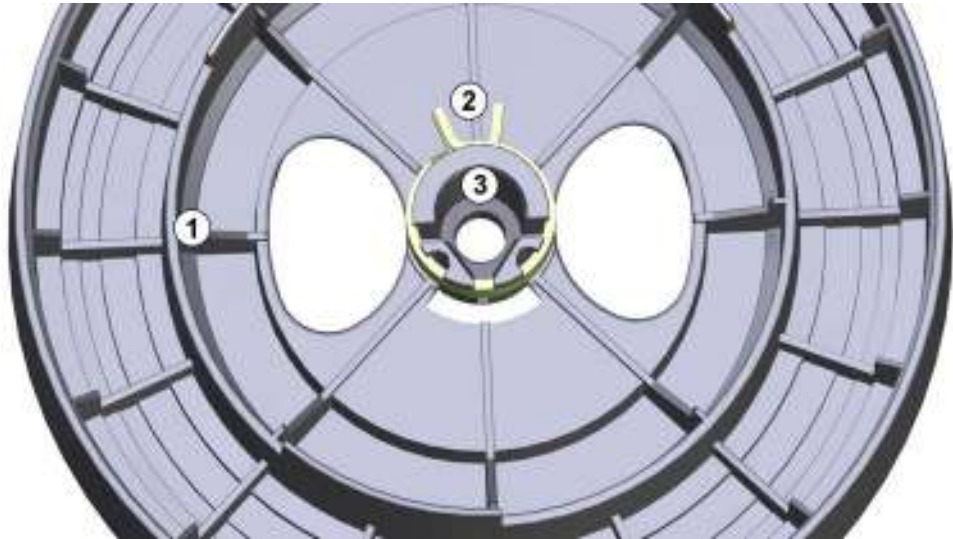


Figure 18–19: Reader Module, without cover, top view

### 18.3.5.3 Assembly of the Reader rotor

1. Check for correct alignment of the clamp (2):
  - The tenterhooks (2) of the clamp shall sit in the middle of the long surface (3) of the shaft support.



*Figure 18-20: Rotor*

2. Slide the Reader rotor onto the motor shaft down to the end stop.
3. Assemble the Reader cover and SIPAL, see Chapter 18.3.2.3 on page 18-12.
4. Perform the calibration of the Reader rotor, see Chapter 18.4.3.1 on page 18-32.

### 18.3.5.4 Set of the Reader rotor

1. Perform the calibration of the Reader rotor, see Chapter 18.4.3.1 on page 18-32.
2. Perform the calibration of the Reader SIPAL lift, see Chapter 18.4.3.2 on page 18-33.
3. Perform the calibration of the Reader aspiration monitor, see Chapter 18.4.3.3 on page 18-33.
4. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 Or see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 Or see Chapter 10.4 on page 10-10.

### 18.3.6 Replacement of the Reader Module

#### 18.3.6.1 Preliminary steps



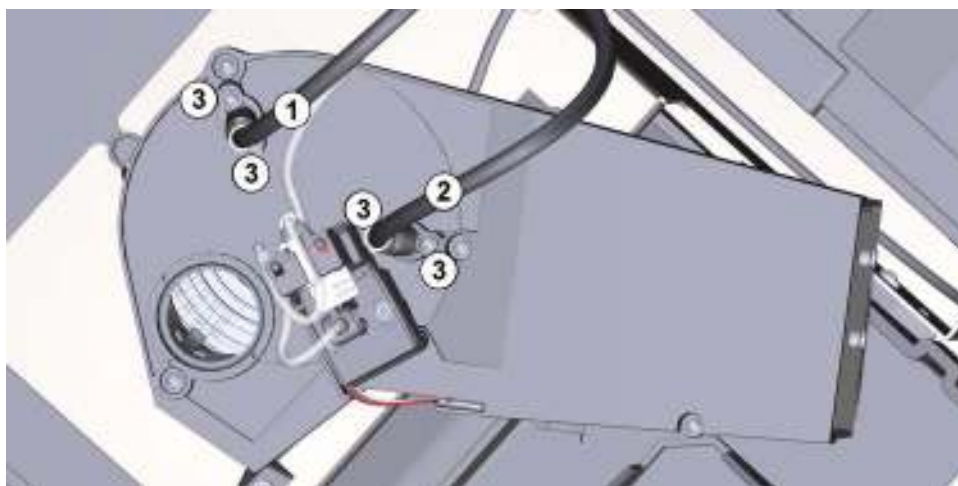
1. Before replacing the module save the parameter file.
2. Exit the **LIAISON<sup>®</sup> XS** system software.
3. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
4. On the right side of the system turn the power switch off.
5. Unplug the system power plug from the right side of the system.
6. Open top covers.
7. Move pipettor arms to a safe location.
8. Remove the Sample Module cover, see Chapter 21.3.1.2 on page 21-5.

#### 18.3.6.2 Removal of the Reader Module

**NOTE**

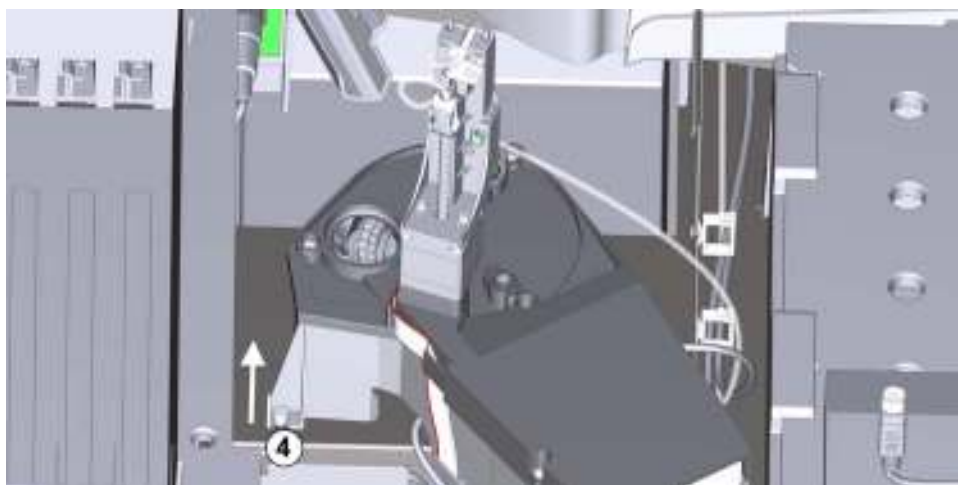
**First remove both the Starter Reagent tubes.**

1. Unscrew four screws (3) of Starter Reagent Tube (1 and 2).
2. Put the Starter Reagent Tube (1 and 2) aside.



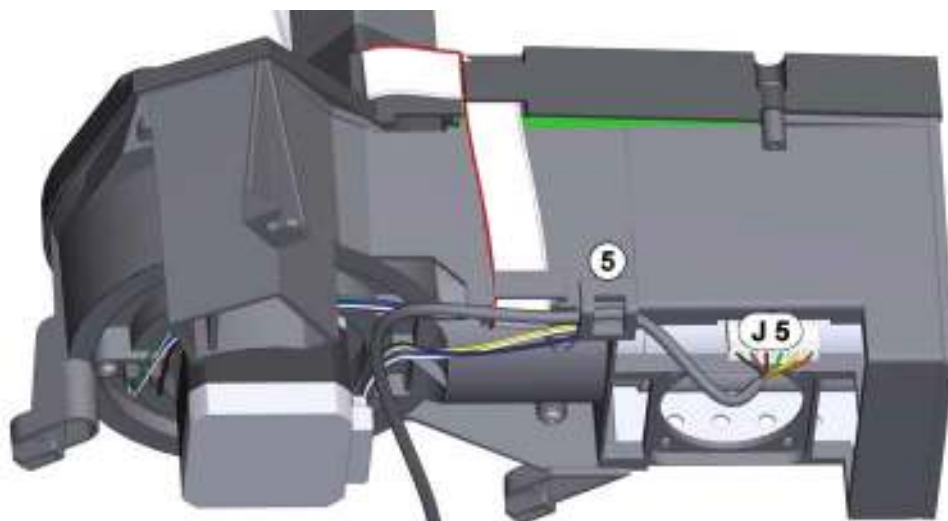
*Figure 18–21: Reader Module, top view*

3. Loosen screw (4) and push the Reader Module to the back.



*Figure 18-22: Reader Module inside the instrument*

4. Lift the Reader Module up and unplug CAN cable (J5).
5. Remove the PWR CAN connection from the Reader guiding hook (5).



*Figure 18-23: Reader Module, bottom view*

**18.3.6.3 Assembly of the Reader Module**

1. Connect PWR CAN connection (J5) onto Reader Module, fit the cable in the guiding hook (5).
2. Set Reader Module into the hooks of the frame, slide it in position and fasten screw (4).
  - Be aware to do not kink or twist cable or tubing; pay attention to the routing of the CAN cable (a wrong routing may affect the closing of the solid waste drawer or the normal fall of cuvettes into it).
3. Assemble the spindle aspiration probe together with tubing, see Chapter on page 18-5.
4. Assemble starter reagent tubing,
  - Pay attention to the correct installation of the starter 1 & starter 2 lines.
5. Assemble the Sample Module cover, see Chapter 21.3.1.3 on page 21-6.

**18.3.6.4 Set the Reader Module**

1. Run the Firmware Update of the complete system in case new Reader Module is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the calibration of the Reader rotor, see Chapter 18.4.3.1 on page 18-32.
3. Perform the calibration of the Reader SIPAL lift, see Chapter 18.4.3.2 on page 18-33.
4. Perform the calibration of the Reader aspiration monitoring sensor, see Chapter 18.4.3.3 on page 18-33.
5. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 OR see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
6. Check the teaching of Reader Module position of the Left Arm, see Chapter 13.13 on page 13-30.
7. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.



### 18.3.7 Replacement of the Reader light barrier

#### 18.3.7.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Sample Module cover, see Chapter 21.3.1.2 on page 21-5.
8. Remove the Reader cover and SIPAL, see Chapter 18.3.2.2 on page 18-9
9. Remove the Reader Module, see Chapter 18.3.6.2 on page 18-19.

#### 18.3.7.2 Removal of the Reader light barrier

1. Unplug light barrier connector of the Luminometer CU PCBA slot (J4).
2. Remove screw (2) and carefully pull out light barrier (1).

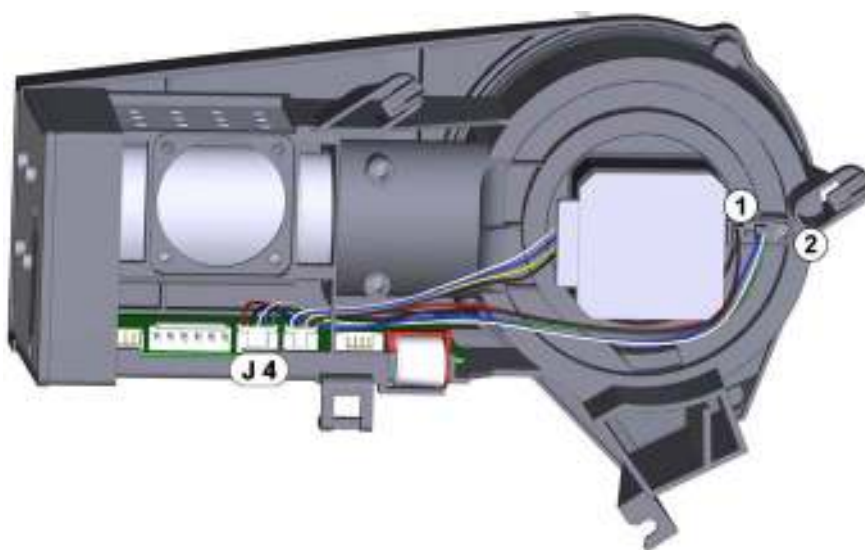


Figure 18–24: Reader Module, bottom view

#### 18.3.7.3 Assembly of the Reader light barrier

1. Set light barrier (1) into the Reader Module housing and fasten with screw (2).
2. Connect light barrier connector into slot (J4) at Luminometer CU PCBA.
3. Assemble the Reader Module, see Chapter 18.3.6.3 on page 18-21.
4. Assemble the Sample Module cover, see Chapter 21.3.1.3 on page 21-6.



**18.3.7.4 Set the Reader light barrier**

1. Run the Firmware Update of the complete system in case new Reader Module is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the calibration of the Reader rotor, see Chapter 18.4.3.1 on page 18-32.
3. Perform the calibration of the Reader SIPAL lift, see Chapter 18.4.3.2 on page 18-33.
4. Perform the calibration of the Reader aspiration monitoring sensor, see Chapter 18.4.3.3 on page 18-33.
5. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 OR see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
6. Check the teaching of Reader Module position of the Left Arm, see Chapter 13.13 on page 13-30.
7. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

## 18.3.8 Replacement of the Reader rotor motor

### 18.3.8.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Sample Module cover, see Chapter 21.3.1.2 on page 21-5.
8. Remove the Reader cover and SIPAL, see Chapter 18.3.2.2 on page 18-9.
9. Remove the Reader Module, see Chapter 18.3.6.2 on page 18-19.
10. Remove the Reader rotor, see Chapter 18.3.5.2 on page 18-17.

### 18.3.8.2 Removal of the Reader rotor motor

1. Unplug the motor connector (J2).

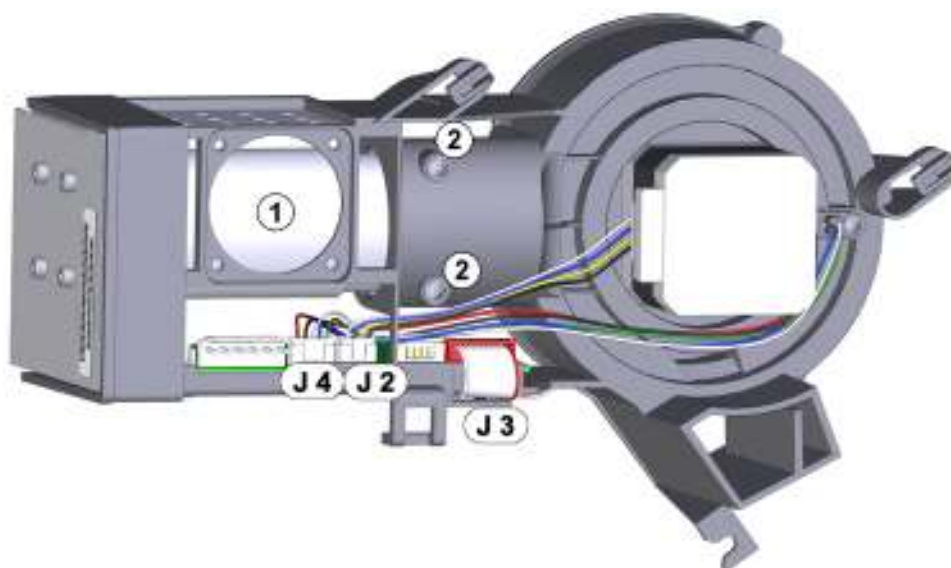


Figure 18–25: Reader Module bottom view

2. Unscrew the four screws (3) and remove the motor from the bottom of the module.

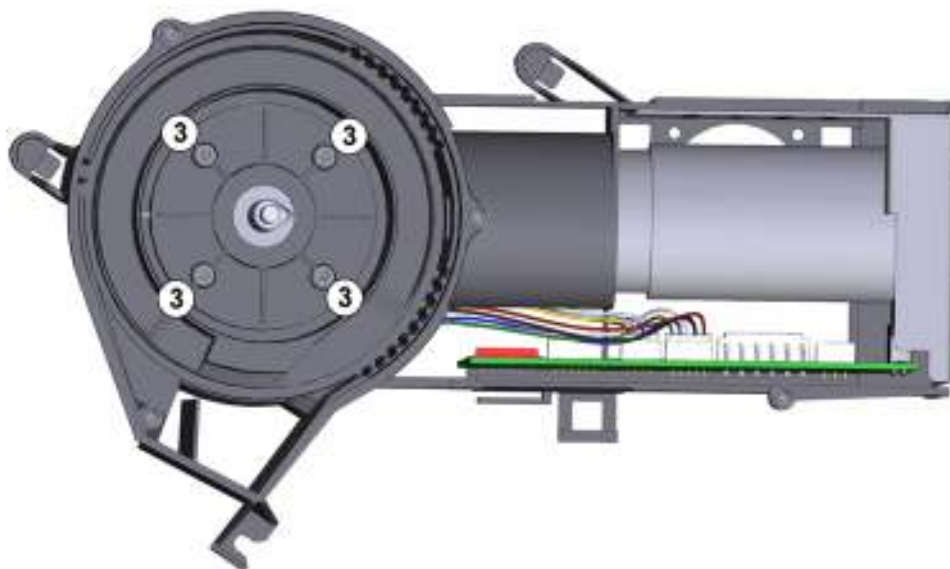


Figure 18–26: Reader Module without cover and rotor, top view

#### 18.3.8.3 Assembly of Reader rotor motor

1. Put the new Reader rotor motor in position.
2. Screw the four screws (3) of the Reader rotor motor.
3. Connect the motor connector (J2).
4. Assemble the Reader rotor, see Chapter 18.3.5.3 on page 18-18.
5. Assemble the Reader Module, see Chapter 18.3.6.3 on page 18-21.
6. Assemble the Reader cover and SIPAL, see Chapter 18.3.2.3 on page 18-12.
7. Assemble Sample Module cover, see Chapter 21.3.1.3 on page 21-6.

#### 18.3.8.4 Set the Reader rotor motor

1. Check the teaching of Reader Module position of the Left Arm, see Chapter 13.13 on page 13-30.
2. Perform the calibration of the Reader rotor, see Chapter 18.4.3.1 on page 18-32.
3. Perform the calibration of the lift, see Chapter 18.4.3.2 on page 18-33.
4. Perform the calibration of the Reader aspiration monitoring sensor, see Chapter 18.4.3.3 on page 18-33.
5. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 OR see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
6. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 OR see Chapter 10.4 on page 10-10.

### 18.3.9 Replacement of the PCB Luminometer CU

#### 18.3.9.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Sample Module cover, see Chapter 21.3.1.2 on page 21-5.
8. Remove the Reader Module, see Chapter 18.3.6.2 on page 18-19.
9. Remove the Reader cover and SIPAL, see Chapter 18.3.2.2 on page 18-9.

#### 18.3.9.2 Removal of the PCB Luminometer CU

1. Unplug all connections of the luminometer CU PCBA (J2, J3, J4).
2. Loosen two screws (2) and pull PMT Module with luminometer CU PCBA carefully out.

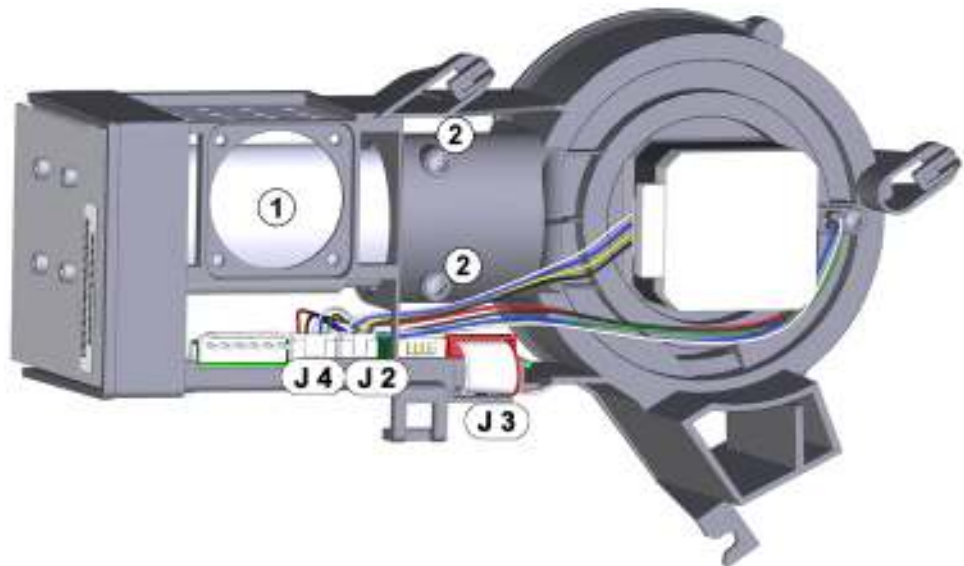


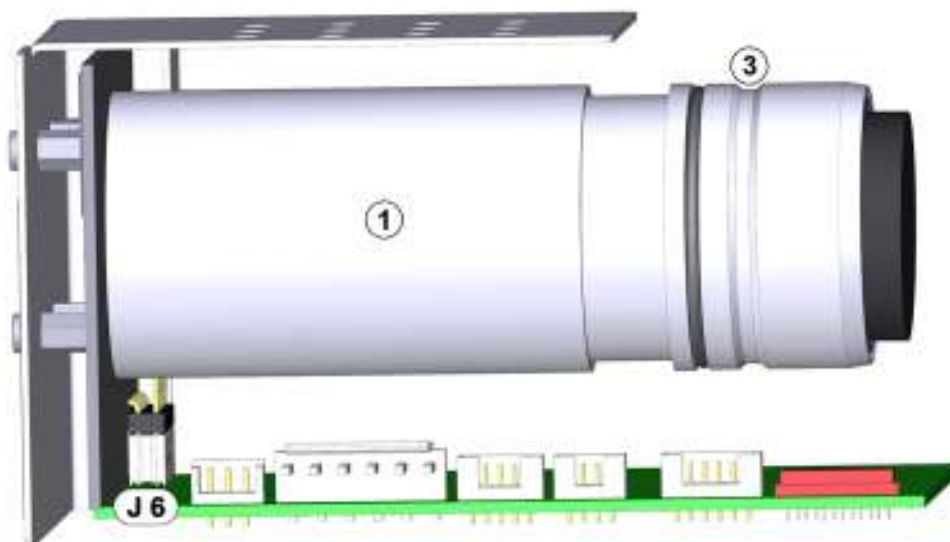
Figure 18–27: Reader Module, bottom view

3. Unplug luminometer CU PCBA of the PMT connector (J6).

**CAUTION**

**Do not expose the PMT Module to the sun light, cover lens of PMT could be damaged by light exposure.**

Cover PMT with a cover or a towel.



*Figure 18–28: PMT Module and luminometer CU PCBA*

### 18.3.9.3 Assembly of the PCB Luminometer CU PCB

**CAUTION**

**Do not remove PMT cover before mounting into the Reader Module.**

- Do not expose the PMT Module to the sun light, cover lens of PMT could be damaged by light exposure.
- Before inserting of the PMT Module, the PMT tube has to be clean and free of any residue.

1. Take note of the HV value handwritten onto the PMT module, in the following it will be needed for the setting.
2. Connect PMT Module connector (J6) of luminometer CU PCBA onto the PMT Module connector.

**NOTE**

**Carefully insert the screws (2) that slide into the notch (3).**

3. Slide the PMT Module with the luminometer CU PCBA carefully into the housing and carefully turn the two screws (2) into the notch of the PMT Module. Tighten the screws (2) slightly.
4. Connect Light barrier connector (J4), motor connector (J2) and flat ribbon cable measurement lift (J3) onto luminometer CU PCBA.
5. Assemble the Reader cover and SIPAL, see Chapter 18.3.2.3 on page 18-12.
6. Assemble the Reader Module, see Chapter 18.3.6.3 on page 18-21.

7. Assemble the Sample Module cover, see Chapter 21.3.1.3 on page 21-6.

#### **18.3.9.4 Set the PCB Luminometer CU PCB**

1. Run the Firmware Update of the complete system, see Chapter 32.3.4 on page 32-6.
2. In case of a new PMT, perform the parameter setting for the new PMT, see Chapter 18.4.2 on page 18-31.
3. Perform the calibration of the Reader rotor, see Chapter 18.4.3.1 on page 18-32.
4. Perform the calibration of the Reader SIPAL lift, see Chapter 18.4.3.2 on page 18-33.
5. Perform the calibration of the Reader aspiration monitoring sensor, see Chapter 18.4.3.3 on page 18-33.
6. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 or see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
7. Check the teaching of Reader Module position of the Left Arm, see Chapter 13.13 on page 13-30.
8. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 18.3.10 Replacement of the PMT

#### 18.3.10.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Sample Module cover, see Chapter 21.3.1.2 on page 21-5.
8. Remove the Reader Module, see Chapter 18.3.6.2 on page 18-19.
9. Remove the Reader cover and SIPAL, see Chapter 18.3.2.2 on page 18-9.
10. Remove the PCB Luminometer CU, see Chapter 18.3.9.2 on page 18-26.

#### 18.3.10.2 Removal of the PMT

1. Bar code label overview of the PMT (1).
  - (2) PMT bar code label
  - (3) Reader Module bar code label

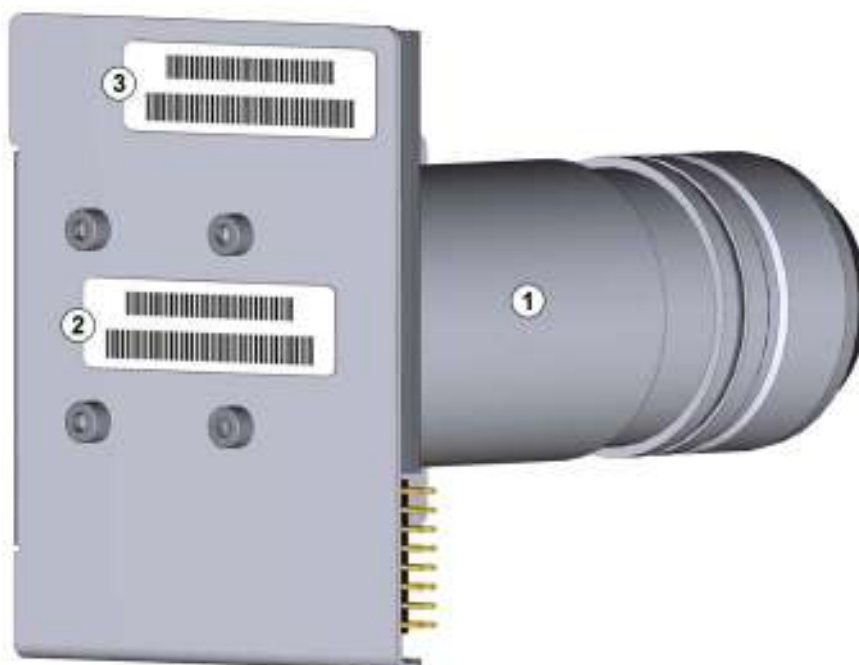


Figure 18–29: Back View of PMT Module

### 18.3.10.3 Assembly of the PMT

**CAUTION**

**Do not remove PMT cover before mounting into the Reader Module.**

- Do not expose the PMT Module to the sun light, cover lens of PMT could be damaged by light exposure.
- Before inserting of the PMT Module, the PMT tube has to be clean and free of any residue.

1. Take note of the HV value handwritten onto the PMT module, in the following it will be needed for the setting.
2. Assemble the PCB luminometer CU, see Chapter 18.3.9.3 on page 18-27.
3. Assemble the Reader cover and SIPAL, see Chapter 18.3.2.3 on page 18-12.
4. Assemble the Reader Module, see Chapter 18.3.6.3 on page 18-21.
5. Assemble the Sample Module cover, see Chapter 21.3.1.3 on page 21-6.

### 18.3.10.4 Set the PMT

1. Run the Firmware Update of the complete system, see Chapter 32.3.4 on page 32-6.
2. Check the teaching of Reader Module position of the Left Arm, see Chapter 13.13 on page 13-30
3. Perform the calibration of the Reader rotor, see Chapter 18.4.3.1 on page 18-32.
4. Perform the calibration of the Reader SIPAL lift, see Chapter 18.4.3.2 on page 18-33.
5. Perform the calibration of the Reader aspiration monitoring sensor, see Chapter 18.4.3.3 on page 18-33.
6. In the **Service Software**, perform a standard prime of the Reader Module, see Chapter 11.8.1.6 on page 11-65 Or see Chapter 12.8.1.6 on page 12-68.; in the result table, check that the LiquidLength value exceeds the cut-off value of 550 counts.
7. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 Or see Chapter 10.4 on page 10-10.



## 18.4 Adjustment and calibration

In this section information about the adjustment and calibration of Reader Module can be found.

### NOTE

---

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).

Do not run any of the software at the same time.

---

### 18.4.1 Initialization routine

- Close any open software and start the **Service Software** from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **Reader** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog Controlled Modules** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 18.4.2 In case of a new PMT

- At the navigation window, choose **Reader** and the **Parameter** bracket.
- Set **Authorization Code** to **1** and press **Set Authorization** button.
- Using **Read parameter(s)** choose 500, then press the button **Update**.
- Select the parameter #500; the value will be showed in **Contents** field.
- Enter the HV value handwritten onto the PMT module into the **New Value** field.
- Press **Write** button and then save the entered value by clicking onto **Non-Volatile Memory** button and then **Overwrite**.

### 18.4.3 Calibration of the Reader Module

#### 18.4.3.1 Calibration of the Reader Rotor

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select **Rotor**.

**Step 1**

- a. Press the **Done** button to perform the init.

**Step 2**

- b. Move the Reader rotor by the arrow buttons to align the teaching point to the reference position in the Reader cover.



*Figure 18–30: Reader rotor, calibration position*

- c. Press **Save** button to save the position.

**Step 3**

- d. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

**Step 4**

- e. The process is successful when, at the **Summarize** panel, all steps are in green.

#### 18.4.3.2 Calibration of the Reader SIPAL lift

In the **Service Software** at the navigation window, choose **Reader** then **Teach & Calibration** bracket and select **Lift**.

- Step 1**
  - a. Follow the instructions and place an empty cuvette in the Reader loading position, then press the **Done** button.
- Step 2**
  - b. Press the **Start** button; the calibration procedure starts automatically.
  - c. When the **Processing** window closes without failure message, press the **Done** button.
- Step 3**
  - d. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 4**
  - e. The process is successful when, at the **Summarize** panel, all steps are in green.

#### 18.4.3.3 Calibration of the Aspiration monitor

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select the **Asp.Monitor**.

- Step 1**
  - a. Press the **WAP for 5sec** on button, to ensure that tubes are free of liquid.
  - b. When the **Processing** window closes, press the **Done** button.
- Step 2**
  - c. Press the **Start** button; the calibration procedure starts automatically.
  - d. When the **Processing** window closes without failure message, press the **Done** button.
- Step 3**
  - e. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 4**
  - f. The process is successful when, at the **Summarize** panel, all steps are in green.

## 18.5 Electric connections, switches and signals

This section shows an overview of the electronic boards of the Reader Module.

### NOTE

see Chapter 3.1.7 on page 3-8 or see Chapter 4.1.8 on page 4-9 for an overview of the complete electric circuit of the system.

### 18.5.1 Luminometer CU PCBA board

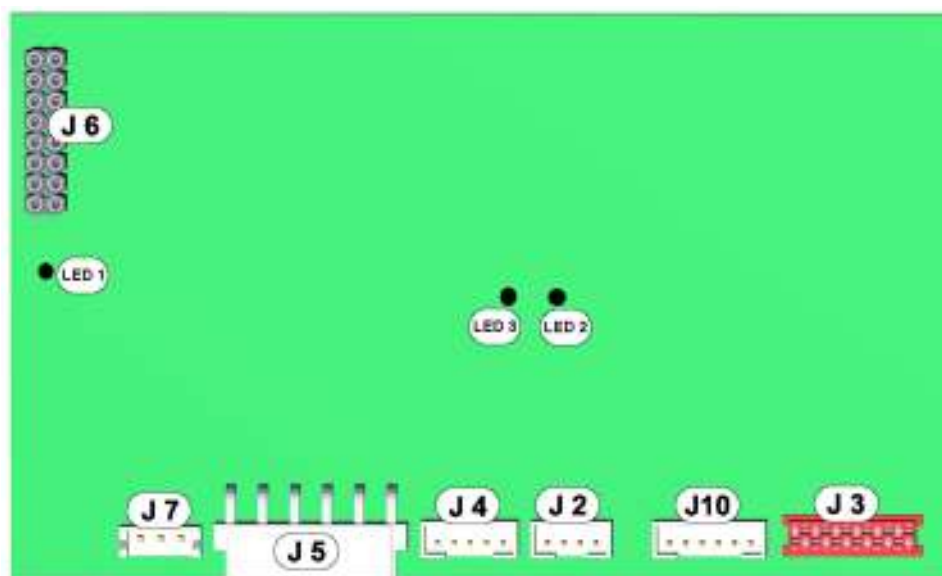


Figure 18–31: Luminometer CU PCBA board

| Number | Connection                          |
|--------|-------------------------------------|
| J2     | Motor drive                         |
| J3     | SIPAL (Lift and Aspiration Monitor) |
| J4     | Home sensor                         |
| J5     | Power / CAN                         |
| J6     | PMT module                          |
| J10    | Spare                               |

Table 18–3: Connections of the Luminometer CU PCBA board

| Number | Meaning/Function |
|--------|------------------|
| LED 1  | Power            |
| LED 2  | Status 1         |
| LED 3  | Status 2         |

Table 18–4: LED of the Luminometer CU PCBA board

## 18.5.2 Reader SIPAL CU PCBA board

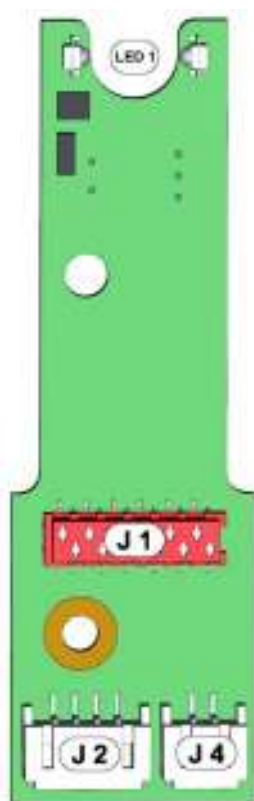


Figure 18-32: Reader SIPAL CU PCBA board, front view



Figure 18-33: Reader SIPAL CU PCBA board, rear view

| Number | Connection     |
|--------|----------------|
| J1     | Washer CU PCBA |
| J2     | Needle Motor   |
| J4     | Valve          |

Table 18-5: Connections of the SIPAL CU PCBA board

| Number | Meaning/Function |
|--------|------------------|
| LED 1  | Bubble detector  |

Table 18-6: LED of the SIPAL CU PCBA board

| Number | Meaning/Function   |
|--------|--------------------|
| IC 2   | Init light barrier |

Table 18-7: Sensor of the SIPAL CU PCBA board



# 19 Pipettor for internal-tanks configuration

This chapter describes the Pipettor in detail. The first section of the chapter introduces “Safety” notes concerning Pipettor, followed by a brief overview of the main parts composing this module.

The following sections contain information about repair, calibration and other service activities (“Replacement and Repair”, “Adjustment and Calibration”, “Priming of Pipettor” and “Pipettor Tests” sections).

In the last section, layout and connections of X-Sledge-Dual-Z EC CU PCBA, Gripper CU PCBA, Gripper Sensor PCBA and X-Z-Sledge EC CU PCBA are described.

The pipettor consists of a left and a right arm. They share a common XY-axis but are equipped with individual Z-axis.

The Left Arm carries the Sample Pipettor and the Gripper; the Right Arm carries the Reagent Pipettor, a non-magnetic stainless steel probe called SPOLV.

- The Gripper is used to transport any cuvette from the cuvette trays to the Incubator, Washer and Reader modules. It grabs a cuvette from the top.
- The Gripper shaft is driven together with the Sample Pipettor through a combined Y-sledge.
- The Sample Pipettor and the Gripper have an independent Z-axis for independent Z-travel.
- The SPOLV acts independently from the Left Arm, as there are no interference zones between the two arms.

## 19.1 Safety

### WARNING



#### Non-observance of safety instructions

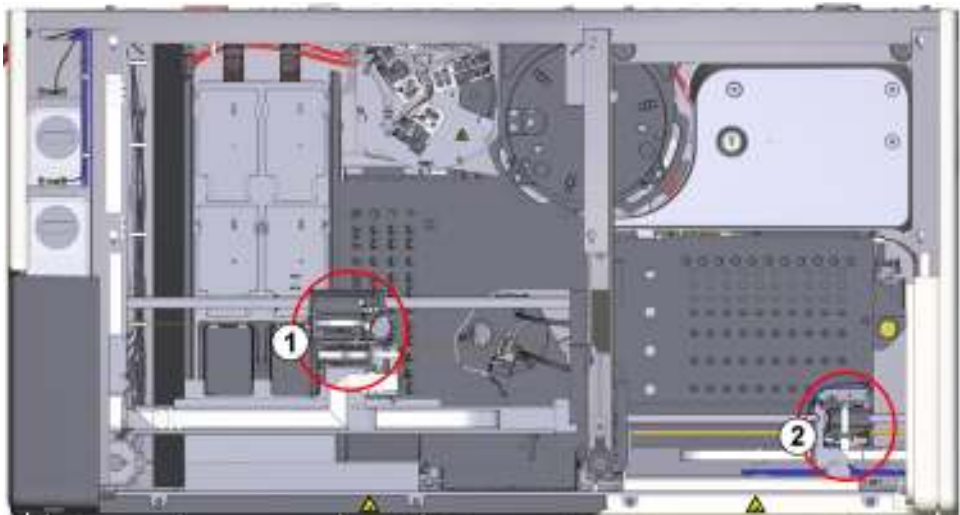
The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

**19.2 Overview**

This section contains an overview of the Pipettor components.

**19.2.1 Pipettor Overview**



*Figure 19–1: Instrument top view, without Top Covers*

|    |                                     |
|----|-------------------------------------|
| 1) | Left Arm (Sample Pipettor, Gripper) |
| 2) | Right Arm (SPOLV)                   |

*Table 19–1: Pipettor overview*



### 19.2.2 Left Arm overview

This section shows an overview of the Left Arm components.

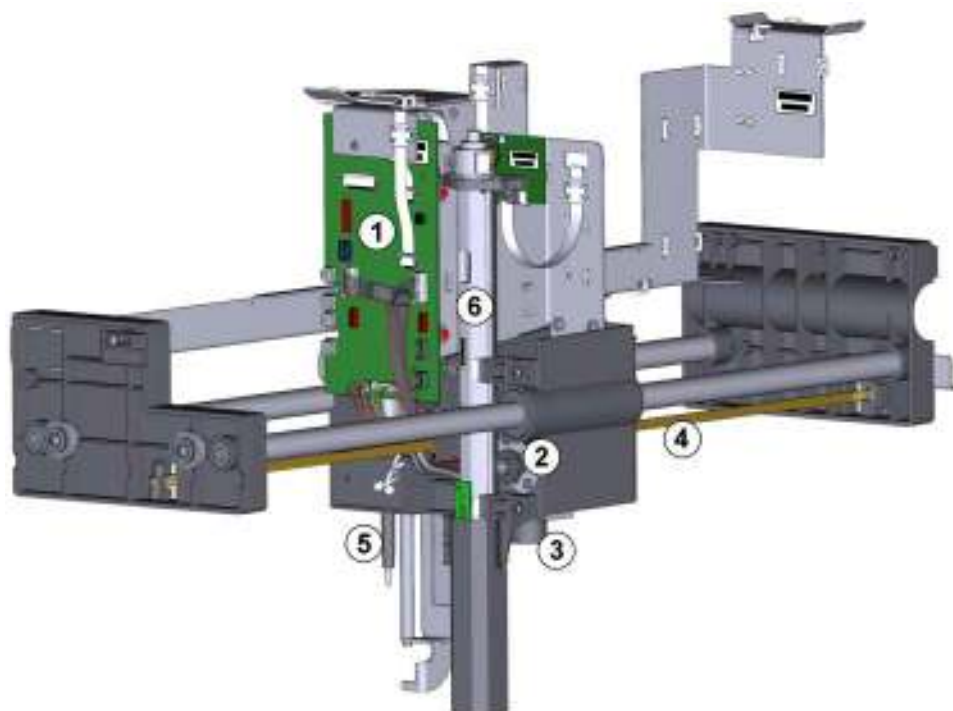


Figure 19–2: Left Arm overview

- |    |                            |
|----|----------------------------|
| 1) | X-Sledge-Dual-Z EC CU PCBA |
| 2) | Gripper Z-motor            |
| 3) | X-Sledge motor             |
| 4) | X-axis toothed belt        |
| 5) | Sample Pipettor            |
| 6) | Gripper                    |

Table 19–2: Left Arm overview

## 19. Pipettor for internal-tanks configuration

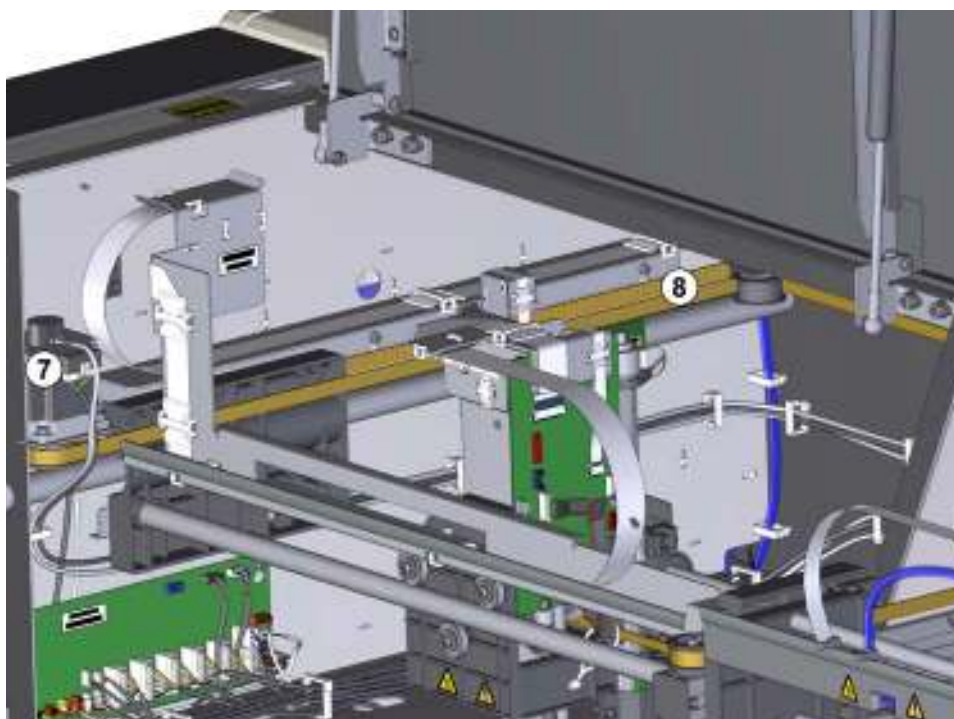


Figure 19-3: Left Arm overview

- |    |                     |
|----|---------------------|
| 7) | Y-Sledge motor      |
| 8) | Y-axis toothed belt |

Table 19-3: Left Arm overview

### 19.2.3 Right Arm overview

This section shows an overview of the Right Arm components.

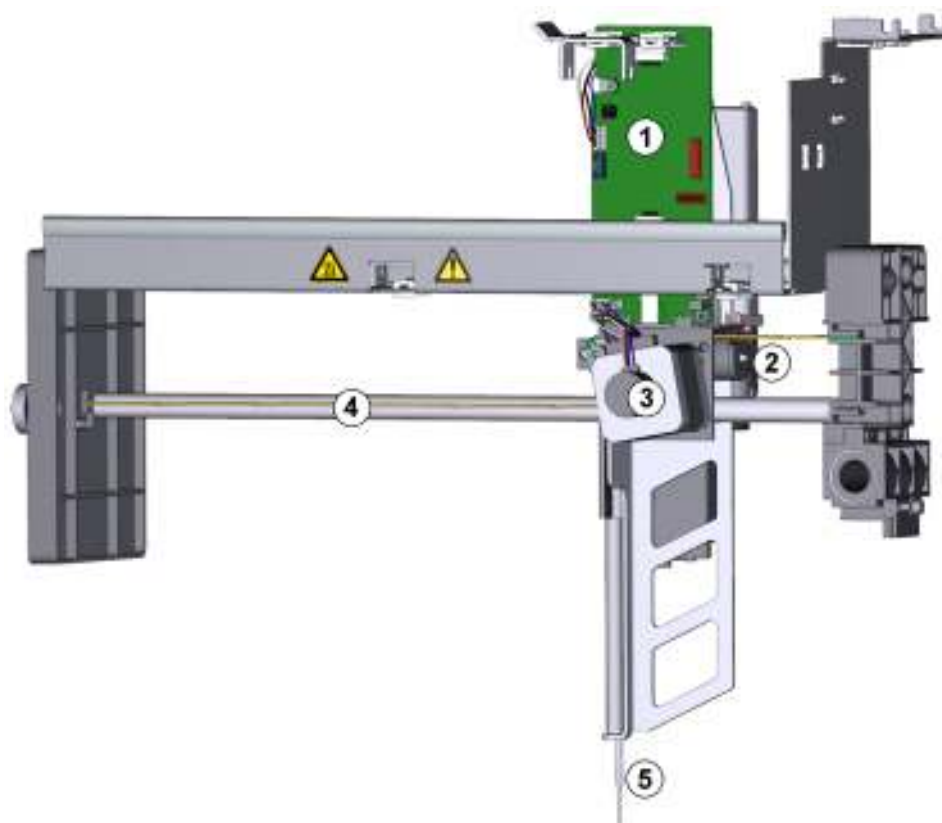
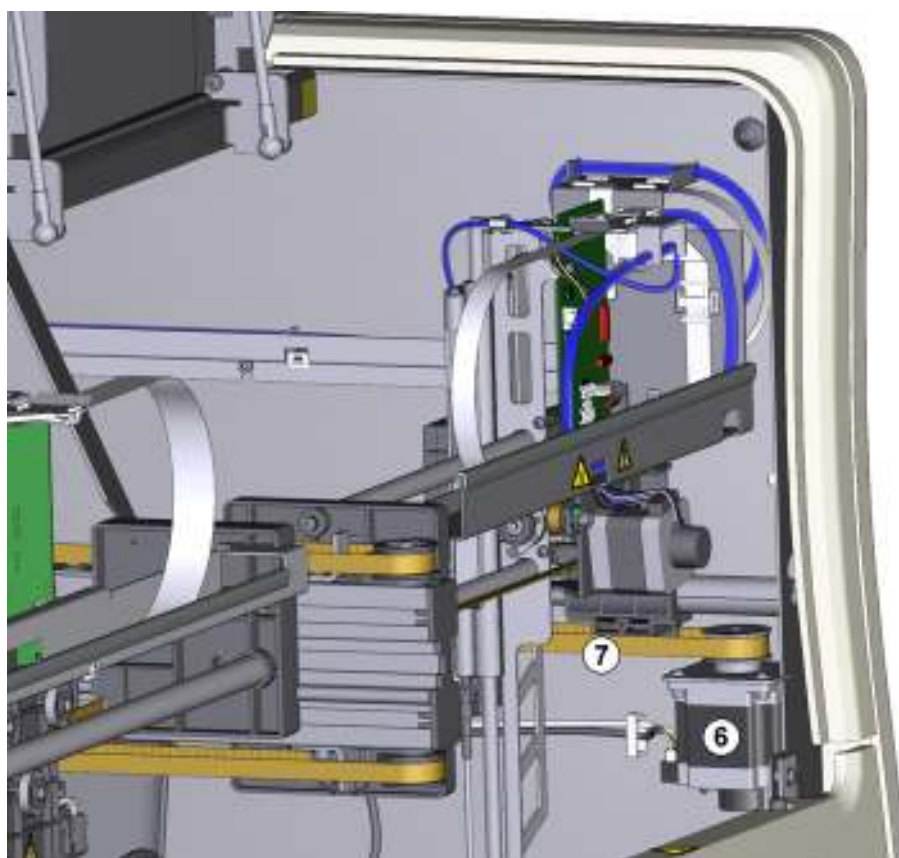


Figure 19–4: Right Arm overview

- |    |                       |
|----|-----------------------|
| 1) | X-Z-sledge EC CU PCBA |
| 2) | Z-motor               |
| 3) | X-sledge motor        |
| 4) | X-axis toothed belt   |
| 5) | SPOLV Needle          |

Table 19–4: Right Arm overview

## 19. Pipettor for internal-tanks configuration

*Figure 19–5: Right Arm overview*

- |    |                     |
|----|---------------------|
| 6) | Y-sledge motor      |
| 7) | Y-axis toothed belt |

*Table 19–5: Right Arm overview*

### 19.2.4 Sample Pipettor overview

This section shows an overview of the Sample Pipettor components.



*Figure 19–6: Sample Pipettor overview*

- |    |                     |
|----|---------------------|
| 1) | ZS-PIP ECXS CU PCBA |
| 2) | Tip adapter         |
| 3) | Guiding rod         |

*Table 19–6: Sample Pipettor overview*

### 19.2.5 Gripper overview

This section shows an overview of the Gripper components.

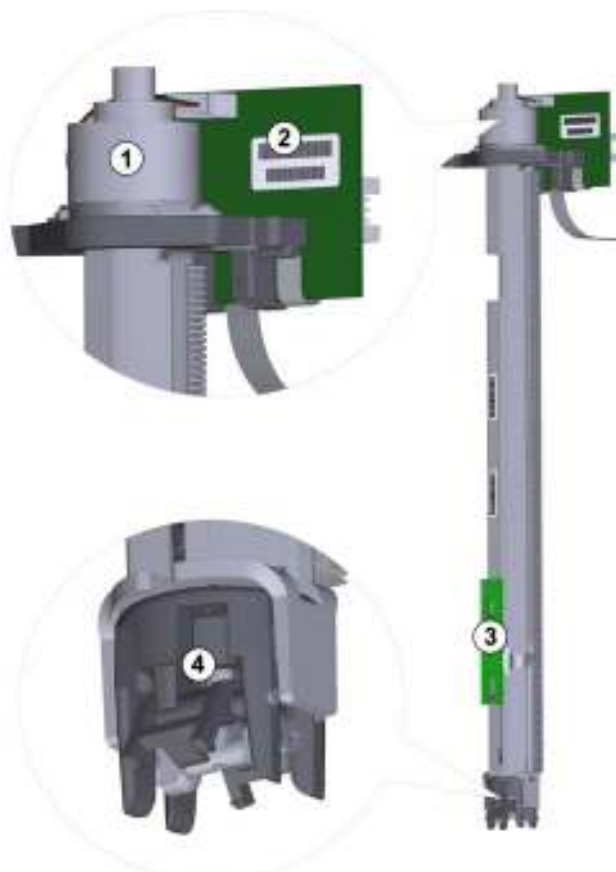


Figure 19–7: Gripper overview

- |    |                     |
|----|---------------------|
| 1) | Linear actuator     |
| 2) | Gripper CU PCBA     |
| 3) | Gripper Sensor PCBA |
| 4) | Finger claw         |

Table 19–7: Gripper overview

### 19.2.6 Reagent Pipettor overview

This section shows an overview of the Reagent Pipettor components.



Figure 19–8: Reagent Pipettor overview

|   |                                         |
|---|-----------------------------------------|
| 1 | Steel needle                            |
| 2 | Knurled screw (needle assembly)         |
| 3 | Fitting connection for PRP-SPOLV tubing |
| 4 | Motor                                   |
| 5 | SPOLV CU PCBA                           |

Table 19–8: Reagent Pipettor overview

## 19.3 Replacements and repairs

This section contains instructions for the repair of individual components of the Pipettor module. After several repair works, it is required that the individual components are re-adjusted. Follow the instructions in the individual sections.

### 19.3.1 Replacement of the Sample Pipettor

#### 19.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.

#### 19.3.1.2 Removal of the Sample Pipettor

1. Unlock the flat ribbon cables (connectors J2, J4 and J3).
2. Remove the cable holder (2).

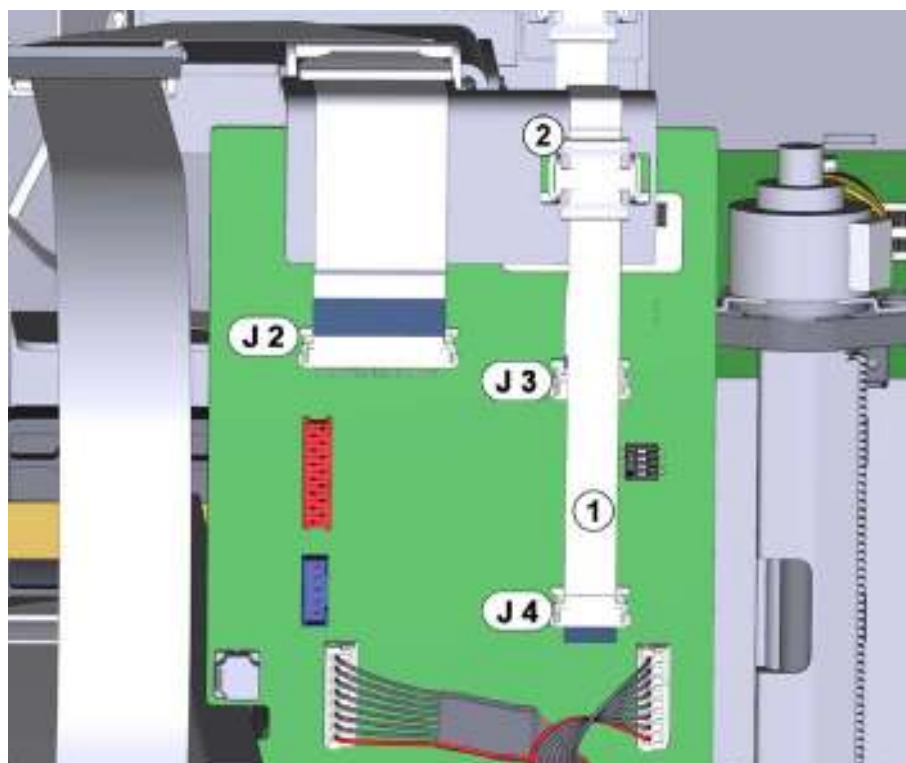


Figure 19–9: Left Arm, right side view



3. Remove the cable holder (2).

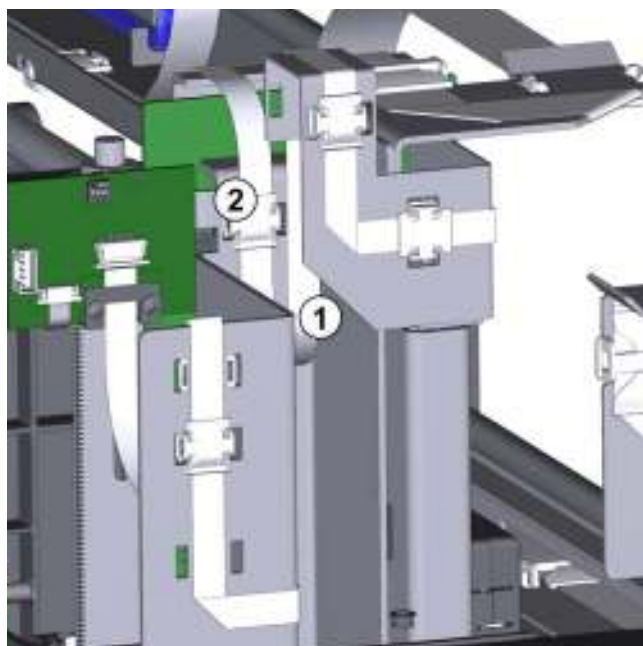


Figure 19-10: Left Arm, left side view

4. Remove the screw (4) on the rear of the X-Sledge-Dual-Z PCB.
5. Loosen the two screws (5) that anchor the top metal sheet (6).
6. Slide the top metal sheet (6) out, and set it aside to access the Sample Pipettor from top.

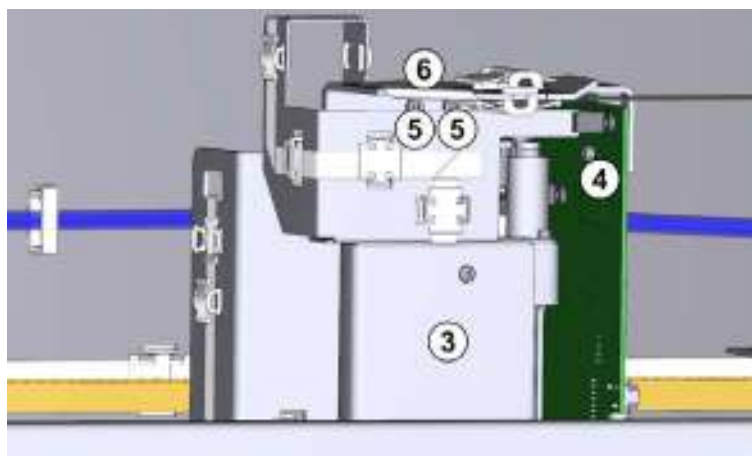
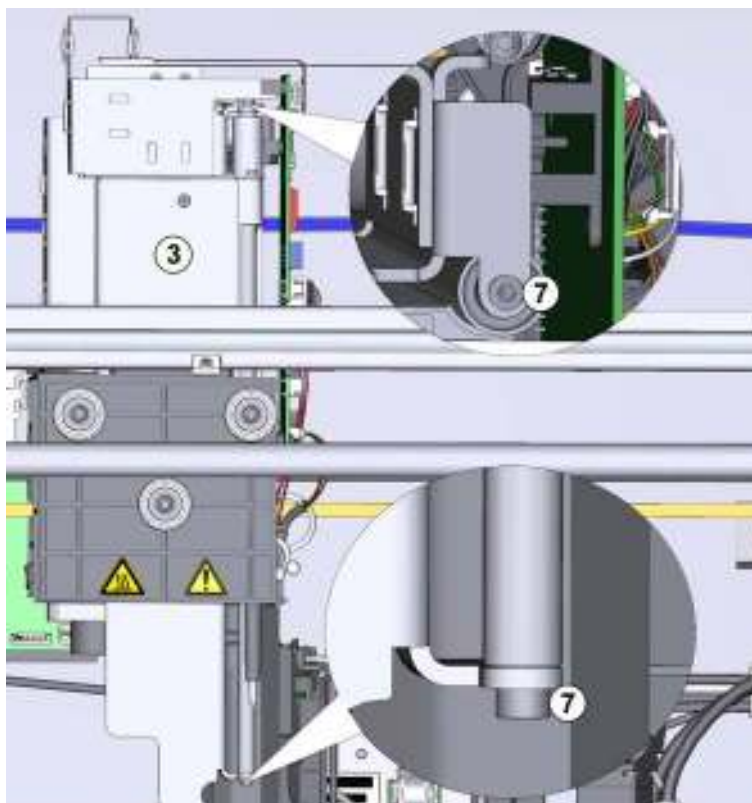


Figure 19-11: Left Arm, front view

7. Unscrew the top and bottom screws (7) holding on to the Sample Pipettor (3).
8. Slide the Sample Pipettor (3) out of the guide and lift it up until it can be removed. Be aware of the bearing, it may slip out!

## 19. Pipettor for internal-tanks configuration



*Figure 19-12: Sample Pipettor, front view*

9. Remove the five cable holders (9) to completely disassemble the routing of the Sample Pipettor flat ribbon cable.
10. Loosen the three screws (10) that anchor the cable support (11) to the Sample Pipettor.
11. Slide the cable support (11) out.

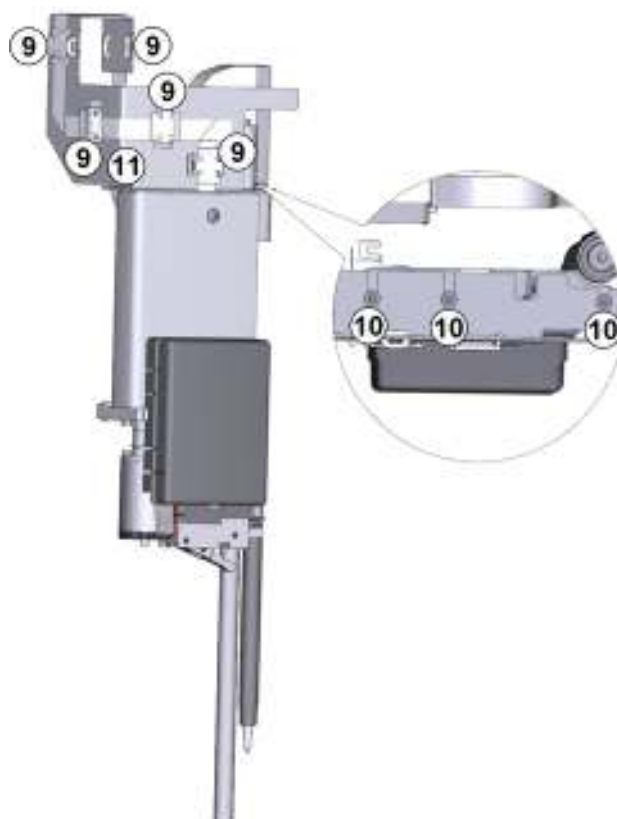


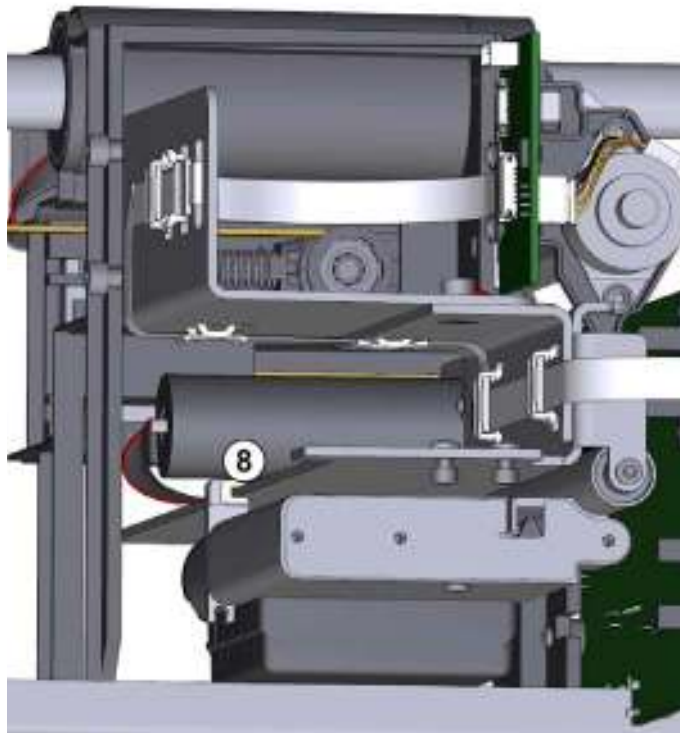
Figure 19–13: Sample Pipettor, side view

#### 19.3.1.3 Assembly of the Sample Pipettor

1. In case a new module has to be installed, check the proper setting of DIP-Switches onto the Sample Pipettor PCB (1000).
2. Lay the flat ribbon cable on the Sample Pipettor upper surface.
3. Place the cable support (11) on the Sample Pipettor upper surface.
4. Fasten the three screws (10) to lock the cable support.
5. Assemble the routing of the flat ribbon cable along the support, and lock it with the five cable holders (9).
6. Set the Z-guiding rod of the Sample Pipettor (3) into the dedicated position and fasten with the two screws (7). The bearing (8) must sit in the Z-bracket!

## 19. Pipettor for internal-tanks configuration

---



*Figure 19–14: Sample Pipettor alignment*

7. Slide the top metal sheet (6) into its dedicated position, and lock it fastening the two screws (5).
8. Screw the screw (4) to the rear of the X-Sledge-Dual-Z PCB.
9. Assemble the routing of Gripper and Sample Pipettor flat ribbon cables along the top metal sheet, and lock them with the two cable holders (2).
10. Lock the flat ribbon cables into the dedicated connectors (J2, J3 and J4).

### **19.3.1.4 Set of the Sample Pipettor**

1. In case a new module has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

## 19.3.2 Replacement of the Dual Sledge CU PCB

### 19.3.2.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.

### 19.3.2.2 Removal of the Dual Sledge CU PCB

1. Unplug all the cables from the Dual Sledge CU PCB (1), see Chapter 19.8.1 on page 19-83.

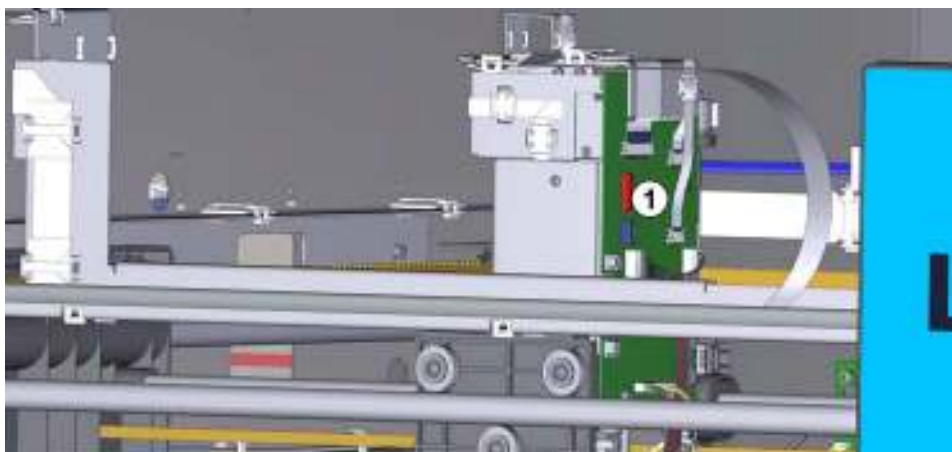
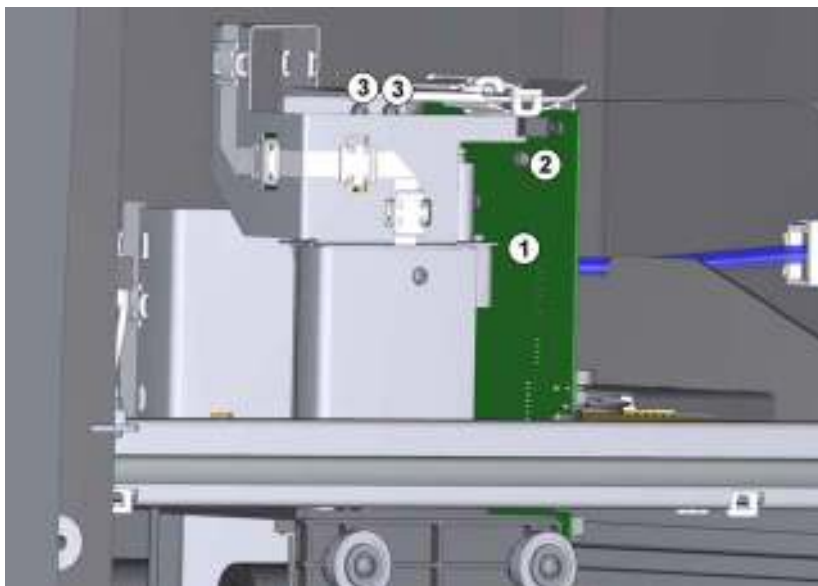


Figure 19–15: X-Sledge-Dual-Z CU PCB

**19. Pipettor for internal-tanks configuration**

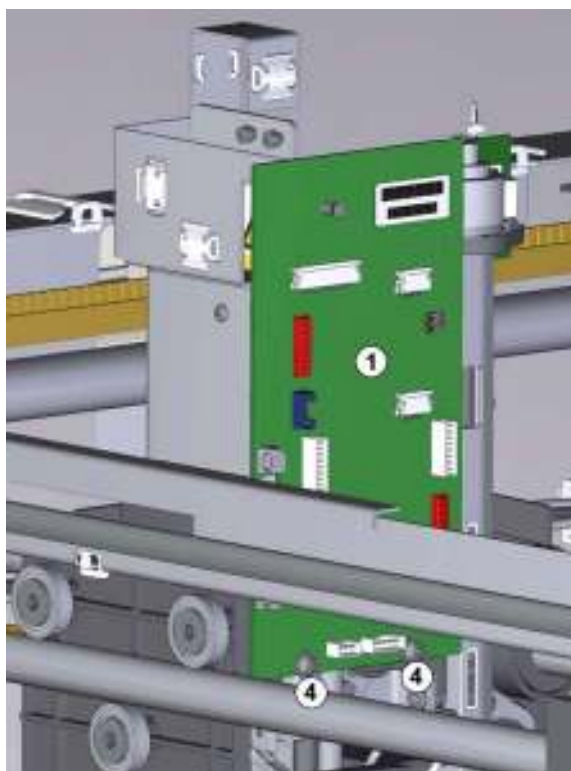
---

2. Unscrew the screw (2) at the rear side of the Dual Sledge CU PCB (1).
3. Loosen the two screws (3), lift up the top metal sheet and set it aside.



*Figure 19–16: X-Sledge-Dual-Z CU PCB, rear view*

4. Loosen the two screws (4) and slide out the Dual Sledge CU PCB (1).



*Figure 19–17: X-Sledge-Dual-Z CU PCB, front view*

**19.3.2.3 Assembly of the Dual Sledge CU PCB**

1. Set the Dual Sledge CU PCB (1) into place and fasten the two screws (4) at the bottom and the screw (2) at the rear side.
2. Plug all the cables onto the Dual Sledge CU PCB, see Chapter 19.8.1 on page 19-83.

**19.3.2.4 Set of the Dual Sledge CU PCB**

1. In case a new PCB has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

Dip switch setting:

| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | OFF                                     |
|            | 2 | OFF                                     |
|            | 3 | OFF                                     |
|            | 4 | OFF                                     |

*Figure 19–18: DIP-Switches of the PCB, Dual Sledge CU*



### 19.3.3 Replacement of the Gripper Sensor PCB

#### 19.3.3.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Cuvette-Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
6. Open the left top cover and approach the left pipettor arm.

#### 19.3.3.2 Removal of the Gripper Sensor PCB

1. Place the gripper over the empty space.
2. Remove the screw (2) and slide down the protective barrel (1) to remove it.

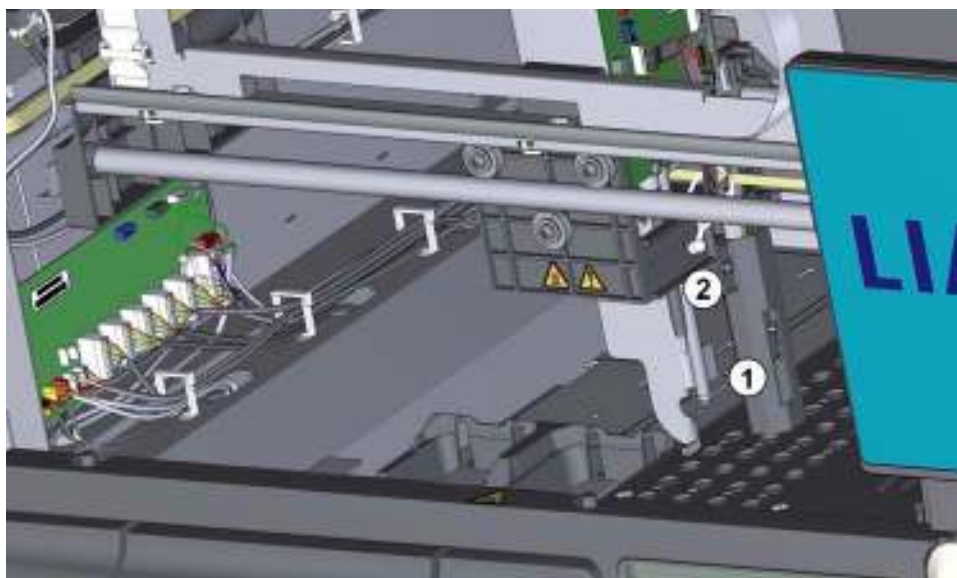


Figure 19–19: Pipettor Left Arm



3. Unplug the flat ribbon cable from the connector (J1) of Gripper Sensor PCB (3).
4. Unscrew the two screws (4).

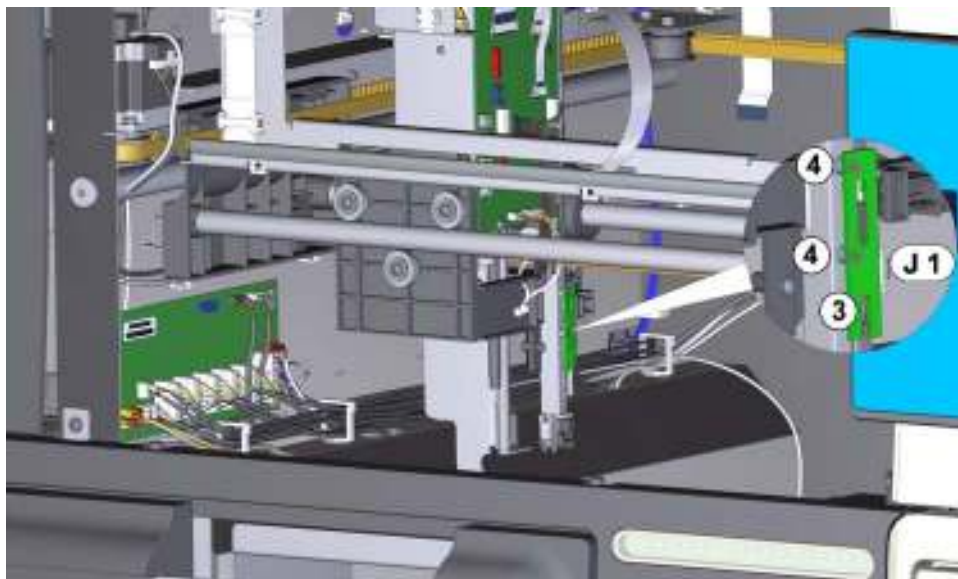


Figure 19–20: Gripper Sensor PCBA, detail

#### 19.3.3.3 Assembly of the Gripper Sensor PCB

1. Set the Gripper Sensor PCB (3) in with the steel spacer and fasten with the two screws (4).
2. Lock the flat ribbon cable into the connector (J1) of Gripper Sensor PCB.
3. Set the protective barrel (1) onto the Gripper and fasten the screw (2).

#### 19.3.3.4 Set the Gripper Sensor PCB

1. Assemble the Cuvette-Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.4 Replacement of the Gripper module

#### 19.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Cuvette-Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
6. Open the left top cover and approach the left pipettor arm.

#### 19.3.4.2 Removal of the Gripper module

1. Place the Gripper over the empty space.
2. Remove the screw (2) and slide down the protective barrel (1) to remove it.

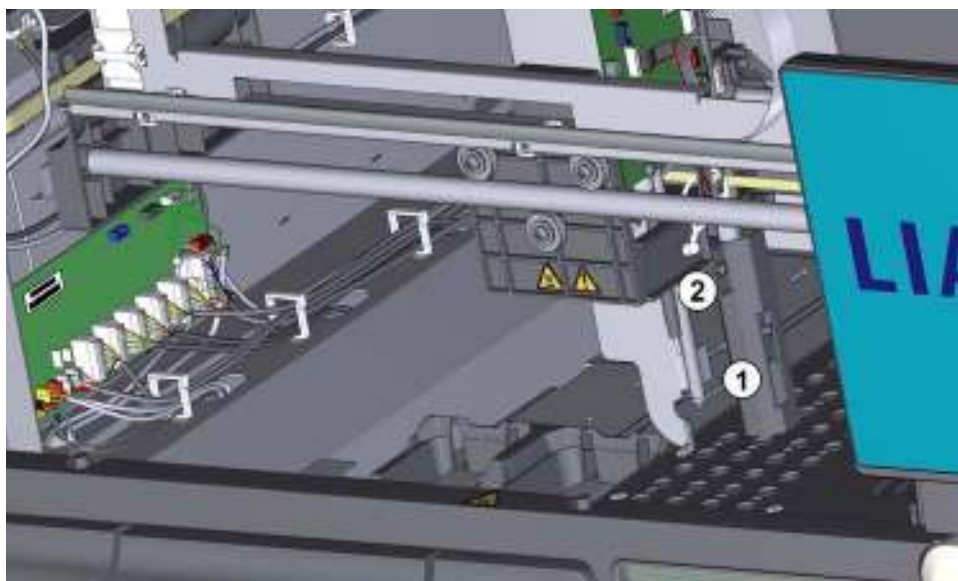


Figure 19–21: Pipettor Left Arm

3. Remove the two screws (4) from the cable holder.
4. Open the connector (5) and pull the flat ribbon cable (6) carefully out.

## 19. Pipettor for internal-tanks configuration

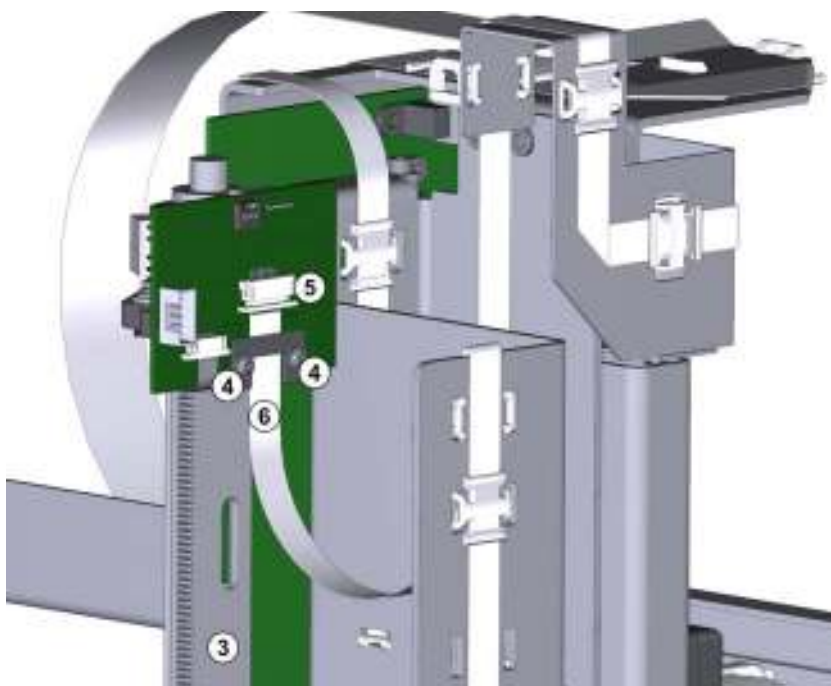


Figure 19-22: Gripper, rear view

5. Hold on to the gripper rack (3), loosen the screws (7) at the clamping levers, then lift the gripper rack (3) carefully out.
6. Watch for the slide bearings (8), they may fall down.

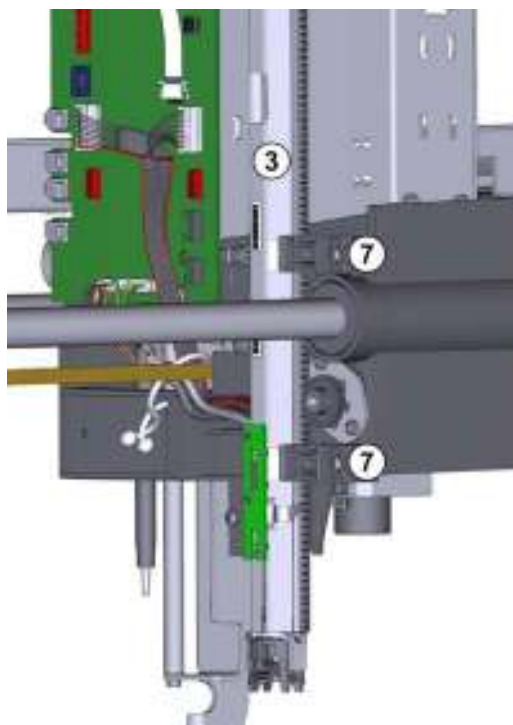


Figure 19-23: Gripper, left rear side view

#### 19.3.4.3 Assembly of the Gripper module

7. Slide the Gripper carefully in between the slide bearings (8) and clamping levers (9) and slide the toothed rack side into the toothed wheel (10).

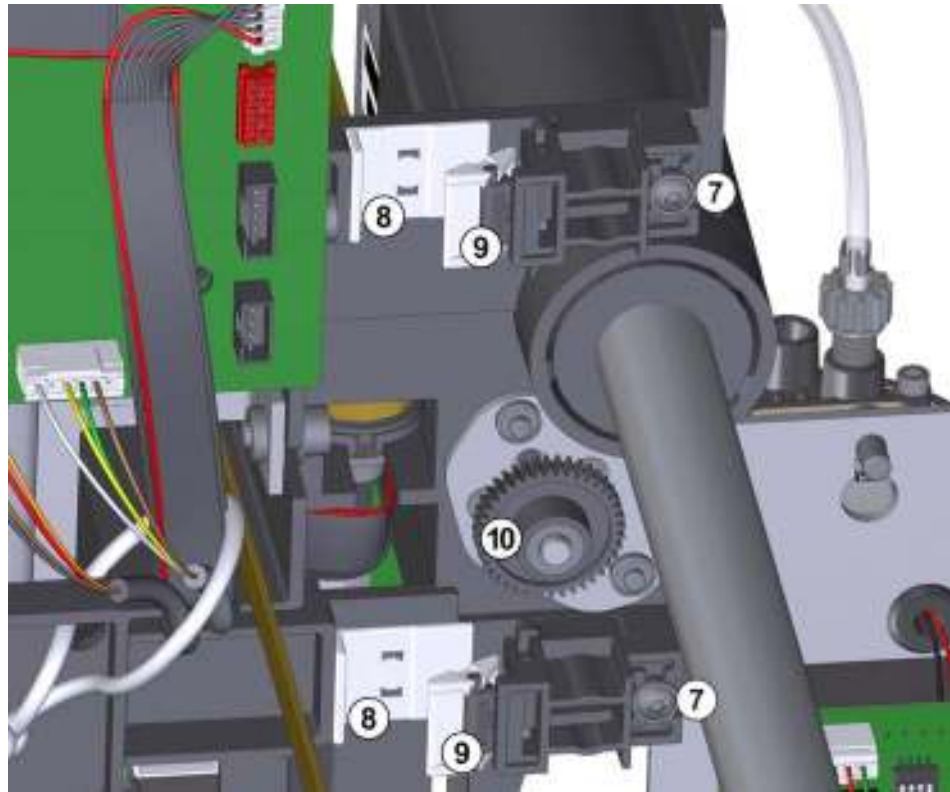


Figure 19–24: Gripper module, sliding bearings

8. Fasten the two screws (7) to clamp the gripper rack (3).
9. Insert the flat ribbon cable (6) under the cable holder.
10. Lock the flat ribbon cable (6) into the connector (5).
11. Fasten the two screws (4).
12. Set the protective barrel (1) onto the Gripper and fasten the screw (2).
13. Assemble the Cuvette-Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.

#### 19.3.4.4 Set of the Gripper module

1. In case a new module has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
3. In case a new module has been installed, calibrate the open position of the gripper claws, see Chapter 19.4.2.1 on page 19-78.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.5 Replacement of the Gripper CU PCB

#### 19.3.5.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.
6. Remove the Gripper Module, see Chapter 19.3.4.2 on page 19-20.

#### 19.3.5.2 Removal of the Gripper CU PCB

1. Unplug the motor cable at slot (J6) on the rear side of the Gripper CU PCB (1).

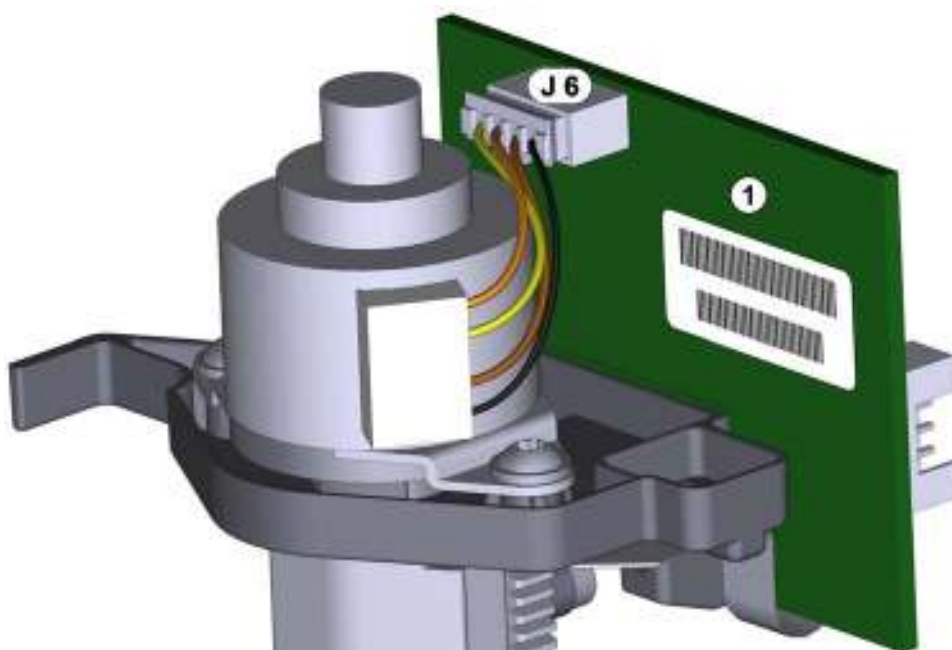


Figure 19–25: Gripper CU PCB, rear view

2. Disconnect the flat ribbon cable at slot (J2) of the Gripper CU PCB (1).
3. Remove the two screws (2).

## 19. Pipettor for internal-tanks configuration

4. Lift the Gripper CU PCB (1) out.

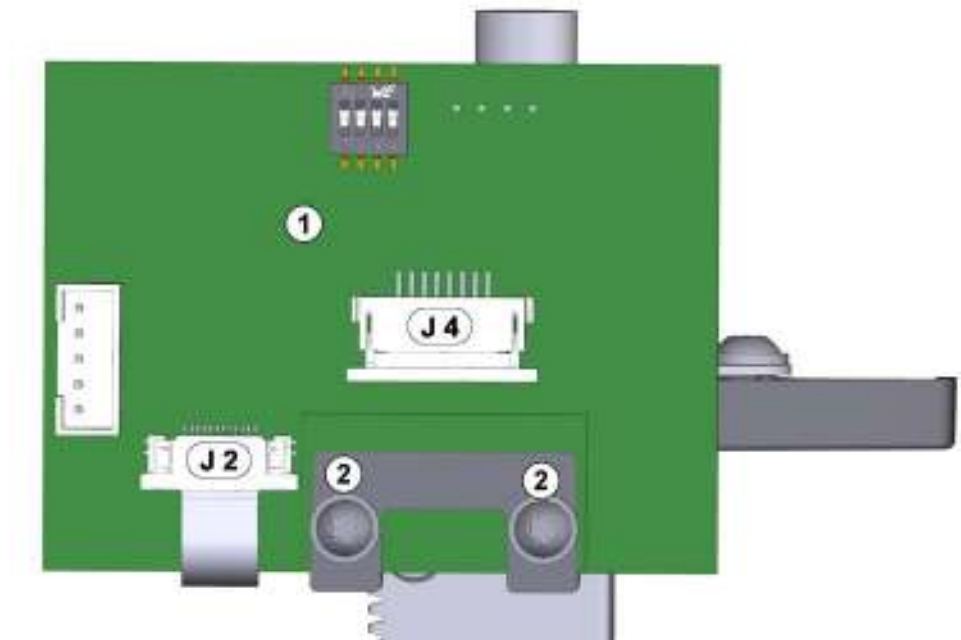


Figure 19–26: Gripper CU PCBA, front view

### 19.3.5.3 Assembly of the Gripper CU PCB

1. Set the Gripper CU PCB (1) into the holder and screw in the two screws (2).
2. Lock the flat ribbon cable into the sensor connector (J2).
3. Plug the motor cable into the connector (J6) on the rear side of the Gripper CU PCB (1).
4. Assemble the Gripper module, see Chapter 19.3.4.3 on page 19-22.

### 19.3.5.4 Set of the Gripper CU PCB

1. In case a new PCB has been installed, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the teaching of the Gripper Offset, see Chapter 13.4 on page 13-6.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 19.3.6 Replacement of the Gripper slide bearings

#### NOTE

The Gripper slide bearings are an add-on of yearly Preventive Maintenance kit.

The Field Service Engineer shall evaluate whether their replacement is needed or not, see Chapter 9.3.1 on page 9-8.

#### 19.3.6.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.
6. Remove the Gripper Module, see Chapter 19.3.4.2 on page 19-20.

#### 19.3.6.2 Removal of the Gripper slide bearings

1. Unhook the sliding plate (1) of the rocker clamp (2).
2. Remove the L-shaped sliding plate (3).

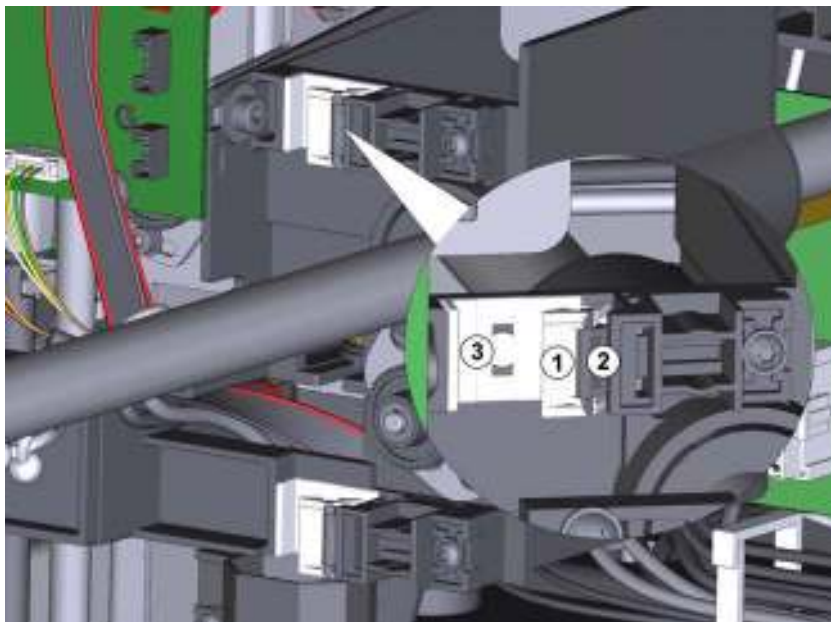


Figure 19–27: Sliding plate

#### 19.3.6.3 Assembly of the Gripper slide bearings

1. Insert the sliding plate (1) into the rocker clamp (2).
2. Set the L-shaped sliding plate in.

## **19. Pipettor for internal-tanks configuration**

---

3. Assemble the gripper module, see Chapter 19.3.4.3 on page 19-22.

### **19.3.6.4 Set of the Gripper slide bearings**

1. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 19.3.7 Replacement of the Sample Pipettor Z-Motor

#### 19.3.7.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.
6. Remove the Sample Pipettor, see Chapter 19.3.1.2 on page 19-10.
7. Remove the Gripper, see Chapter 19.3.4.2 on page 19-20.
8. Remove the Dual Sledge CU PCB, see Chapter 19.3.2.2 on page 19-15.

#### 19.3.7.2 Removal of the Sample Pipettor Z-Motor

**CAUTION**

Do not open or remove screws marked with locking lacquer.

1. Unscrew one screw (1) of the Z-bracket Assembly (2).

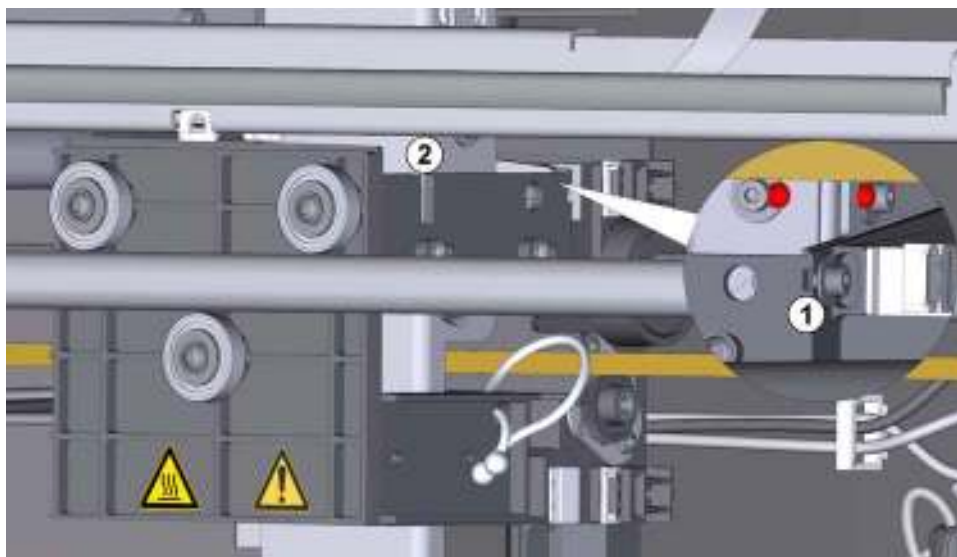


Figure 19–28: Z-bracket with Z-motor

## 19. Pipettor for internal-tanks configuration

2. Loosen the two screws (3) of the Z-bracket assembly (2) and lift the Z-bracket assembly carefully out of the XZ-sledge (4).

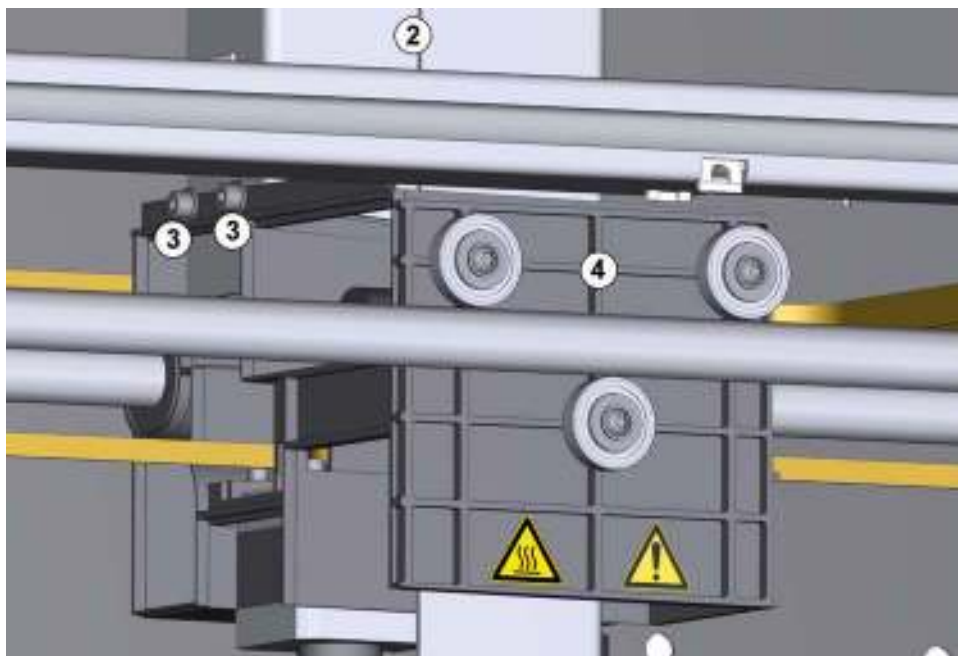


Figure 19-29: Removal of Z-bracket assembly from XZ-sledge

3. Remove two screws (5a and 5b) and slide the Z-motor (6) carefully out.

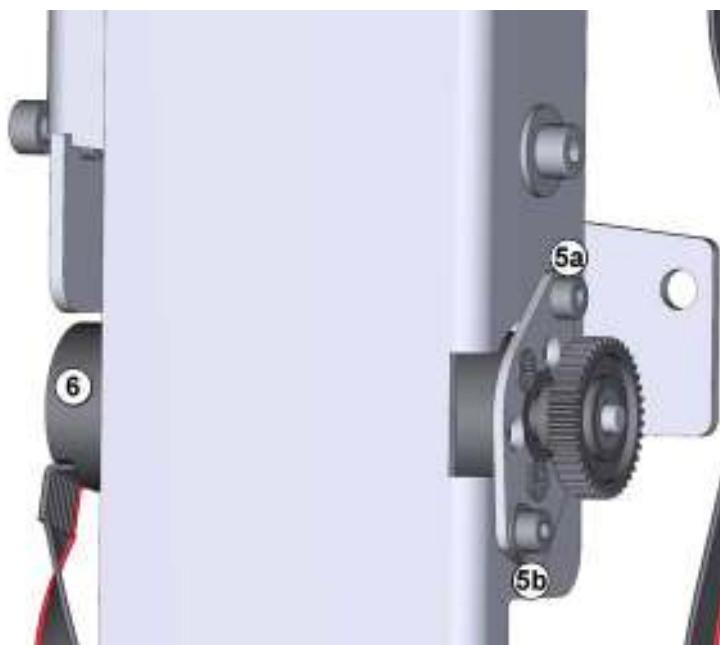


Figure 19-30: Removal of Z-motor from Z-bracket assembly

**19.3.7.3 Assembly of the Sample Pipettor Z-Motor**

1. Set the new Z-motor (6) into the Z-bracket assembly (2) and check for the correct position of the cable outlet. Set the Z-motor loose with one screw (5a) and with one screw and washer (5b).
2. Slide the Z-bracket assembly (2) into the XZ-sledge (4) from the top.
3. Fasten the Z-bracket assembly with one screw and washer (1) at the right side and with the two screws and washers (3) at the left side.
4. Assemble the Sample Pipettor, see Chapter 19.3.1.3 on page 19-13.
5. Adjust the clearance between the Z-motor gear and the Sample Pipettor toothed bar.
6. Fasten two screws (2) of Z-motor (6).
7. Assemble the PCB Dual Sledge CU, see Chapter 19.3.2.3 on page 19-17.
8. Assemble the gripper, see Chapter 19.3.4.3 on page 19-22.

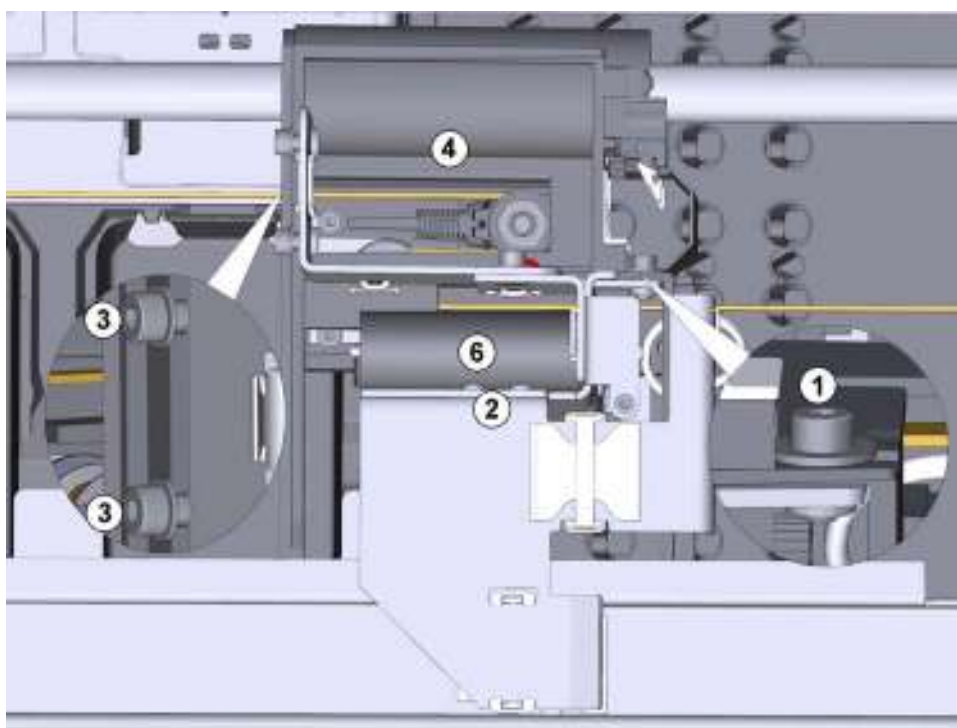


Figure 19–31: Z-bracket assembly, front and top view

**19.3.7.4 Set of the Sample Pipettor Z-Motor**

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.8 Replacement of the Gripper Z-Motor

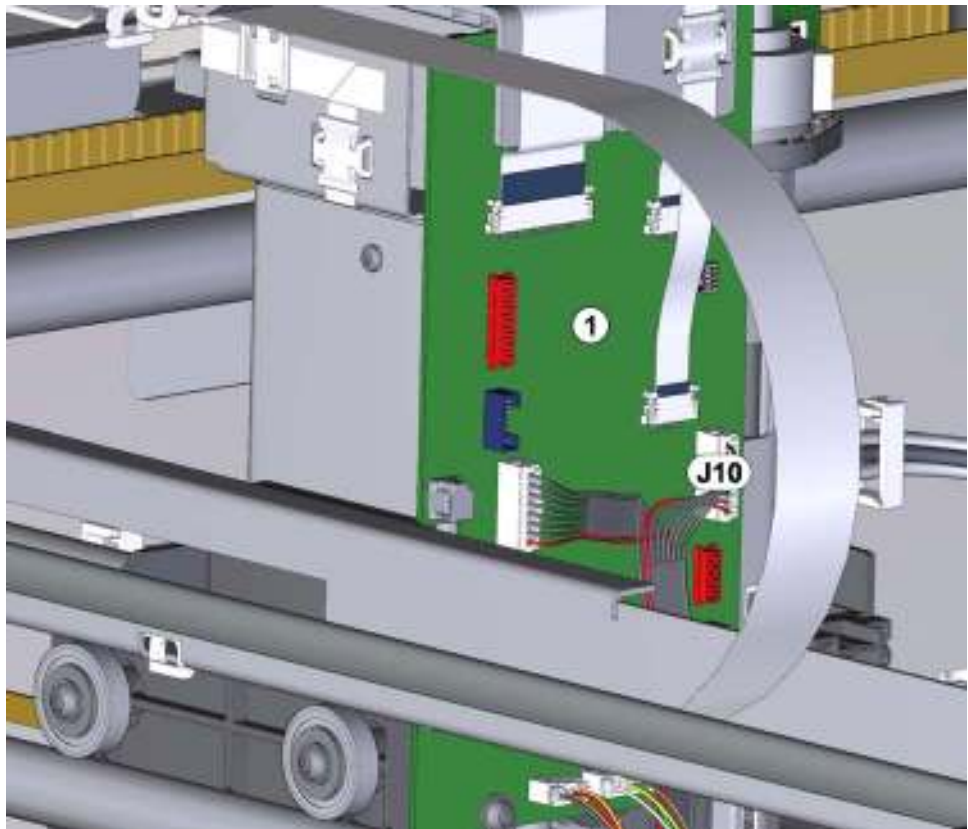
#### 19.3.8.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm
6. Remove the Gripper, see Chapter 19.3.4.2 on page 19-20.

#### 19.3.8.2 Removal of the Gripper Z-Motor

1. Unplug the motor cable at slot (J10) of Dual Sledge CU PCB (1).

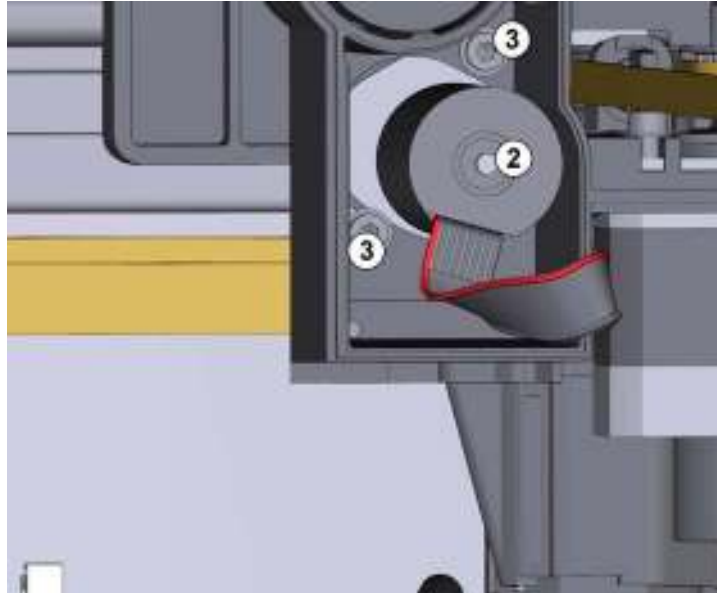


*Figure 19–32: PCB, Dual Sledge CU unplug Gripper Z-Motor*

**19. Pipettor for internal-tanks configuration**

---

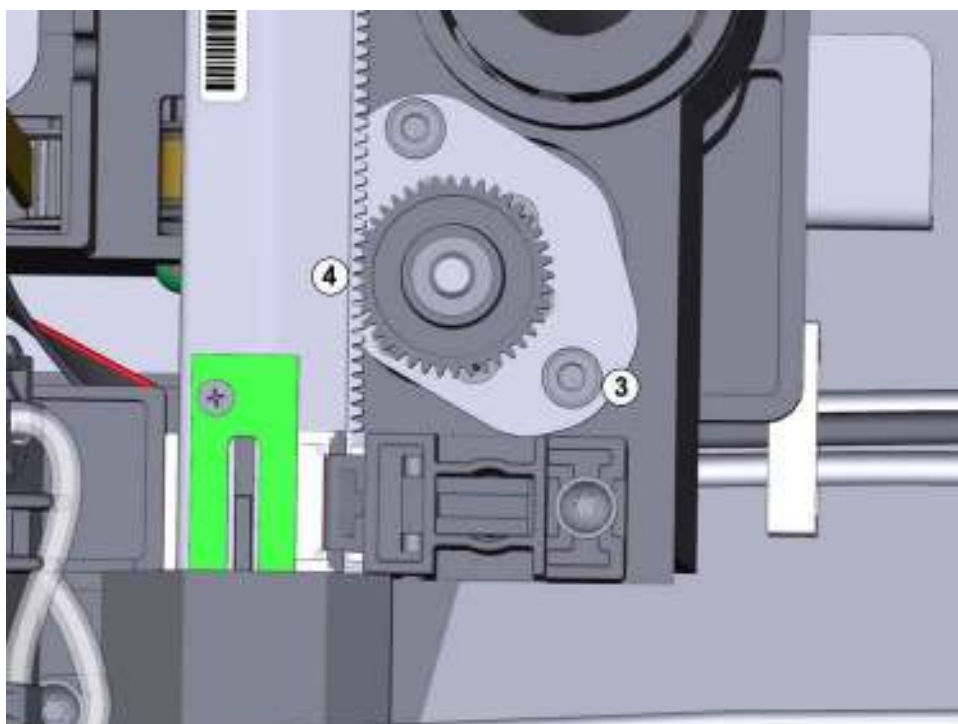
2. Unscrew two screws (3) and remove the Z-motor. Pull the motor cable carefully out.



*Figure 19–33: Gripper Z-Motor*

**19.3.8.3 Assembly of the Gripper Z-Motor**

1. Insert the new Z-Motor (2).
2. Set the Z-Motor loose with two screws (3).
3. Assemble the gripper, see Chapter 19.3.4.3 on page 19-22.
4. Adjust the clearance of the Z-Motor gear and the toothed bar (4).
5. Fasten the two screws (3).
6. Connect the motor cable onto slot (J10) of Dual Sledge CU PCB (1).



*Figure 19-34: Gripper Z-Motor gear clearance*

**19.3.8.4 Set the Gripper**

1. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 19.3.9 Replacement of the X-axis toothed belt of Left Pipettor Arm

#### 19.3.9.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.

#### 19.3.9.2 Removal of the X-Axis toothed belt of Left Pipettor Arm

1. Loosen the fixing screw (2) of guide pulley, push it towards the spring and fasten again the screw.

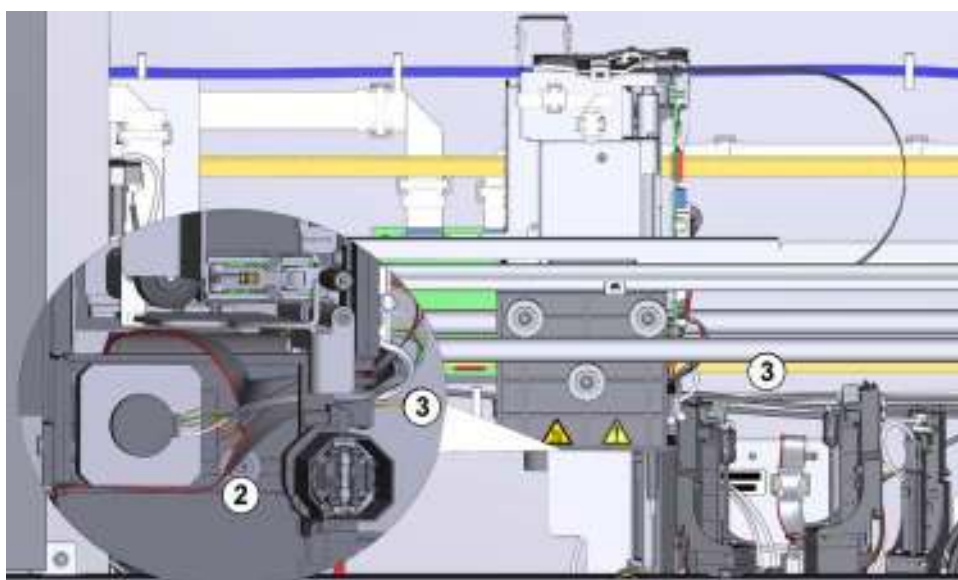


Figure 19–35: Left Pipettor arm, zoom window shows bottom view

## 19. Pipettor for internal-tanks configuration

2. Unhook both the belt holders (3) of X-Axis toothed belt (1) at left (4) and right (5) Y-Sledge of the left arm.

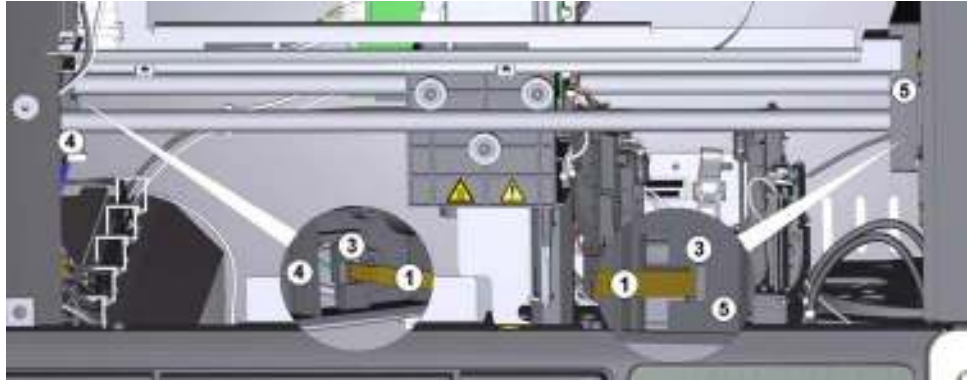


Figure 19-36: X-Axis toothed belt suspension

### 19.3.9.3 Assembly of the X-Axis toothed belt of Left Pipettor Arm

1. Check the toothed belt length, and if necessary cut it to a length of 770mm (the toothed belt has to be cut only along vales).
2. Thread the toothed belt (1) through the guide pulley (6) and the motor pulley (7).

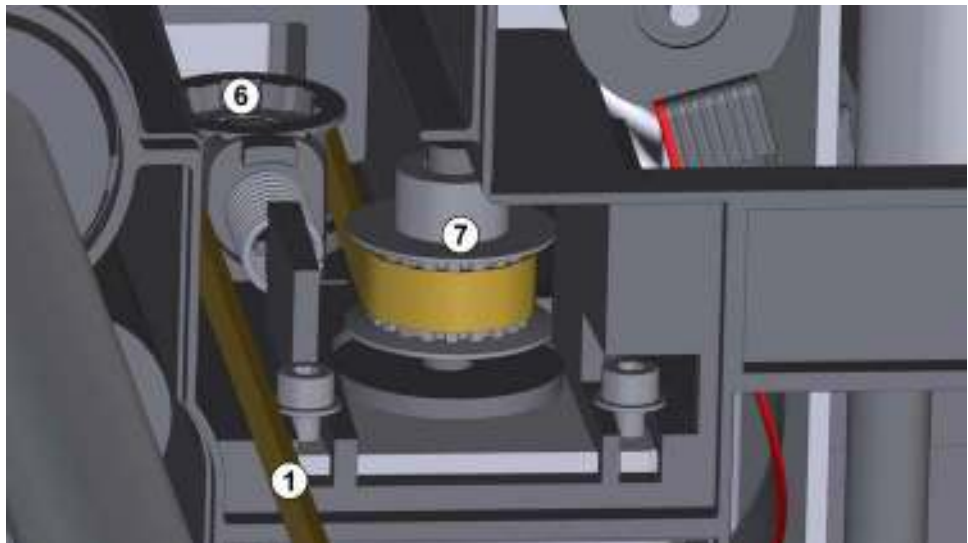


Figure 19-37: Guide pulleys of X-axis toothed belt

3. Insert the toothed belt into the belt holders (3) of X-Axis toothed belt (1).



Figure 19-38: Toothed belt holders



**NOTE****Belt adjustment**

By moving a tooth Inwards or outwards, the tension of the toothed belt can be adjusted.

---

4. Insert both the belt holders (3) at left (4) and right (5) Y-Sledge.
5. Slide the belt holders into place.
6. Loosen the fixing screw (2) of guide pulley, adjust the belt tension and then fasten the screw (2).
7. Add a drop of green locking lacquer onto the head of the fixing screw (2).

**19.3.9.4 Set the Left Pipettor Arm**

1. Perform the teaching of the Sample pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.10 Replacement of the X-Motor of Left Pipettor Arm

#### 19.3.10.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.
6. Remove the X-axis toothed belt of Left Pipettor Arm, see Chapter 19.3.9.2 on page 19-33.

#### 19.3.10.2 Removal of the X-Motor of Left Pipettor Arm

1. Unplug the motor cable (J5) and the encoder cable (J6) from Dual Sledge CU PCB (1).

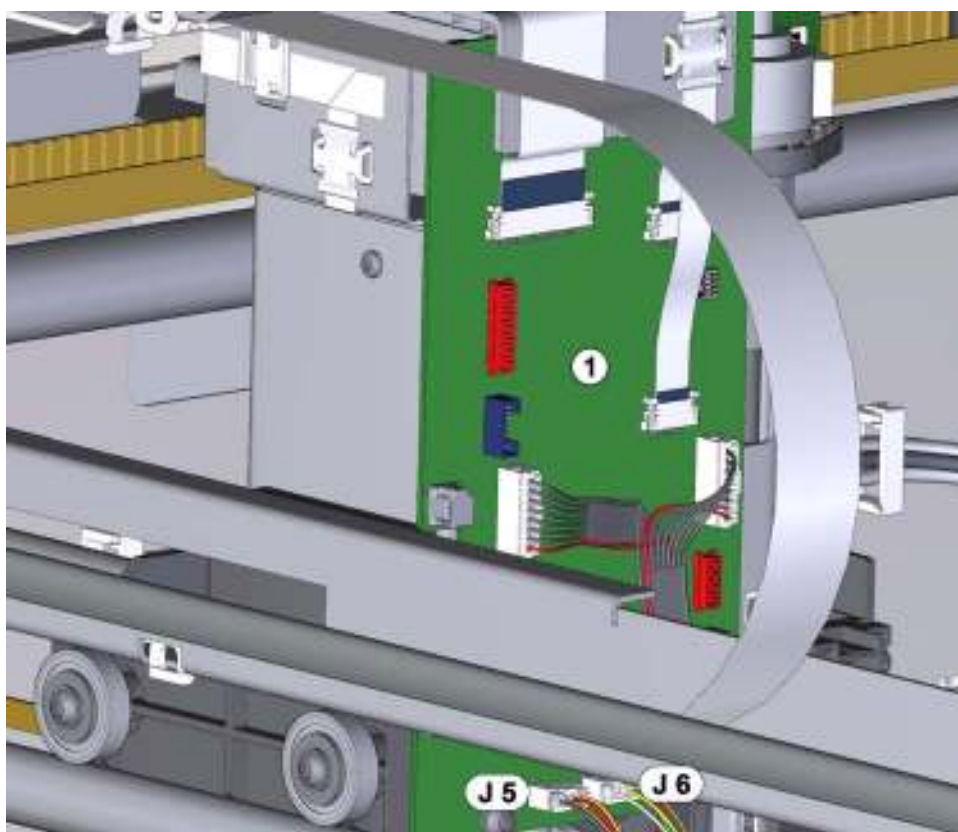


Figure 19–39: PCB, Dual Sledge CU

2. Hold on to the X-motor (5) and remove the two screws (4), then pull the X-motor (5) out.

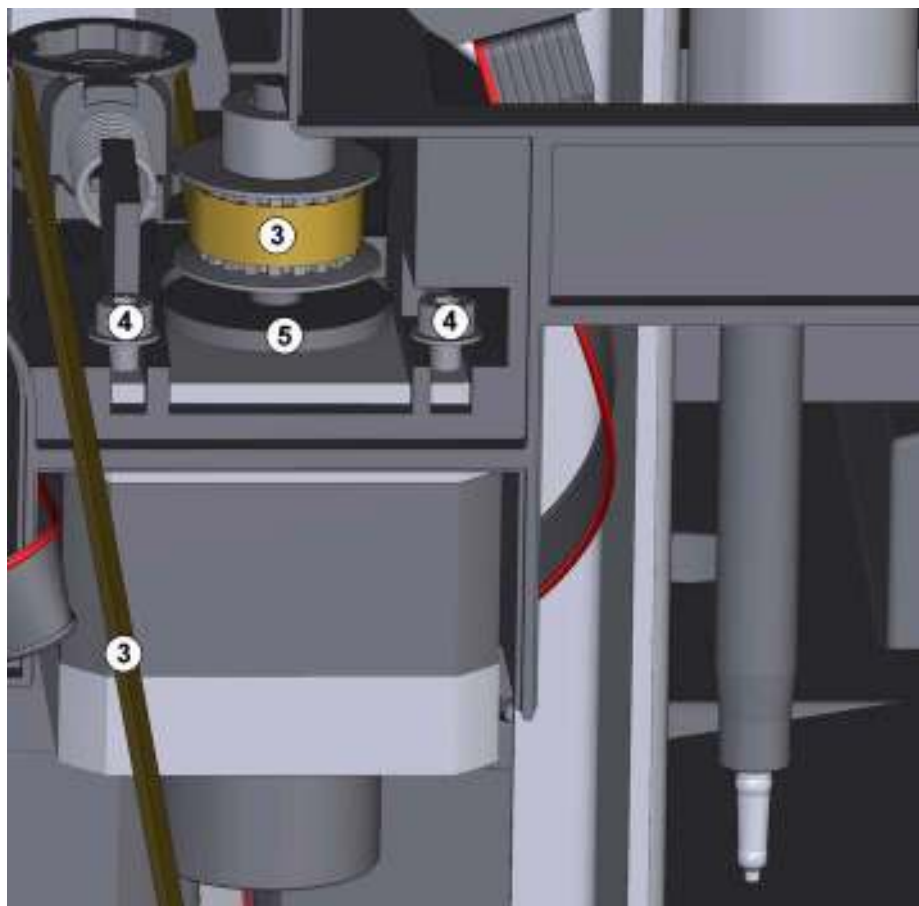


Figure 19-40: Left arm X-Motor, left side view

#### 19.3.10.3 Assembly of the X-Motor of Left Pipettor Arm

1. Insert the X-Motor into the frame and set it loose with the two screws (4).
2. Connect the motor cable (J5) and the encoder cable (J6) onto the Dual Sledge CU PCB (1).
3. Fasten the two screws (4).
4. Assemble the X-belt of the left pipettor arm, see Chapter 19.3.9.3 on page 19-34.

#### 19.3.10.4 Set the Left Pipettor Arm

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.11 Replacement of the Y-axis toothed belt of Left Pipettor Arm

#### 19.3.11.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the Top Covers.
6. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.

#### 19.3.11.2 Removal of the Y-axis toothed belt of Left Pipettor Arm

1. Loosen the screw of the toothed belt wheel (1).
2. Press the wheel against the end stop and fasten the screw to loosen the belt.

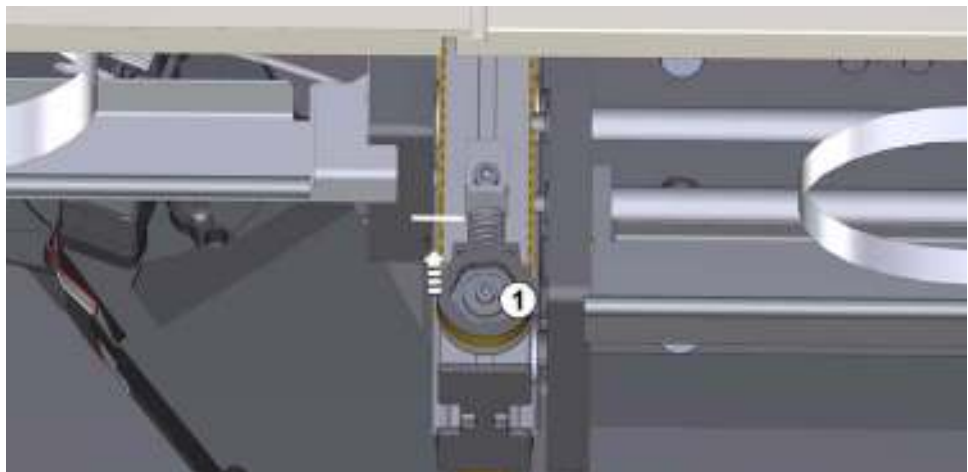


Figure 19-41: Belt tensioner

3. Move the Left Pipettor Arm to the back, until you see the front belt connection from the left side of the chassis.
4. Remove the two screws of the front belt connector (2) and slide the end of toothed belt out of the connector.



Figure 19-42: Chassis left side view, front belt connection

5. Move the Left Pipettor Arm to the front, until you see the rear belt connection from the left side of the chassis.
6. Remove the two screws of the rear belt connector (3) and slide the end of toothed belt out of the connector.

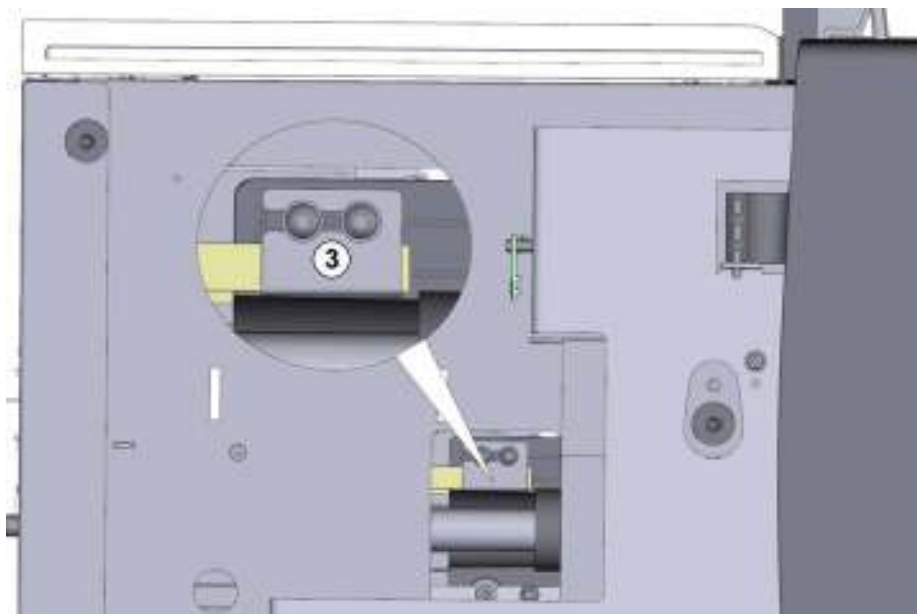


Figure 19-43: Chassis left side view, rear belt connection

## 19. Pipettor for internal-tanks configuration

7. Slide the toothed belt out of the belt holder (4).
8. Remove the toothed belt from the instrument.

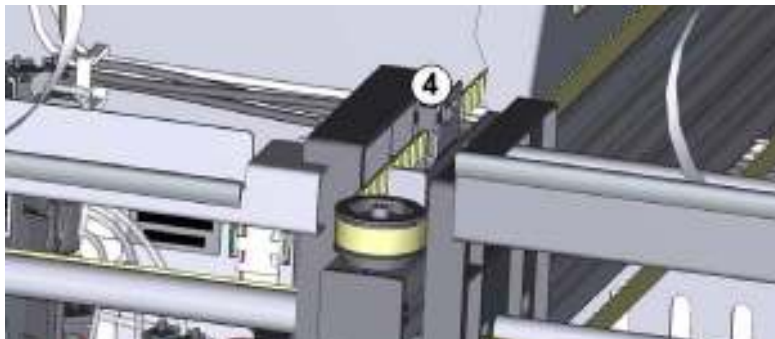


Figure 19-44: Inside view of the instrument

### 19.3.11.3 Assembly of the Y-axis toothed belt of Left Pipettor Arm

1. Check the toothed belt length, and if necessary cut it to a length of 3600mm (the toothed belt has to be cut only along vales).
2. Insert the toothed belt through the guide pulleys.

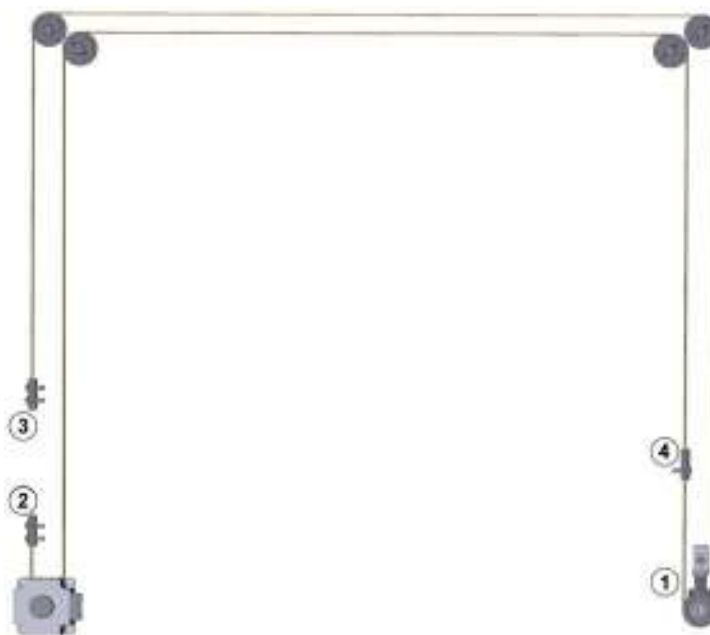


Figure 19-45: Layout of Left Arm toothed belt



Figure 19–46: Left Arm toothed belt

**NOTE**

**The toothed belt must not be twisted!**

**The toothed belt must sit correctly on the pulleys!**

3. Slide the toothed belt into the rear belt connector (3), set the connector into place and fasten to the Left Arm with the two screws.
4. Slide the toothed belt into the front belt connector (2), set the connector into place and fasten to the Left Arm with the two screws.
5. Slide the toothed belt into the belt holder (4). Loosen the screw at the belt holder.
6. Loosen the screw of the toothed belt wheel (1), let the spring expand to tension the belt and then fasten the screw.
7. Move the Left Pipettor Arm forward and back a couple of times.
8. Fasten the screw at the belt holder (4).
9. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

#### 19.3.11.4 Set the Y-axis toothed belt of Left Pipettor Arm

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.12 Replacement of the Y-Motor of Left Pipettor Arm

#### 19.3.12.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.

#### 19.3.12.2 Removal of the Y-Motor of Left Pipettor Arm

1. Unplug the motor cable (1) and the encoder cable (2) at the Y-motor (3).

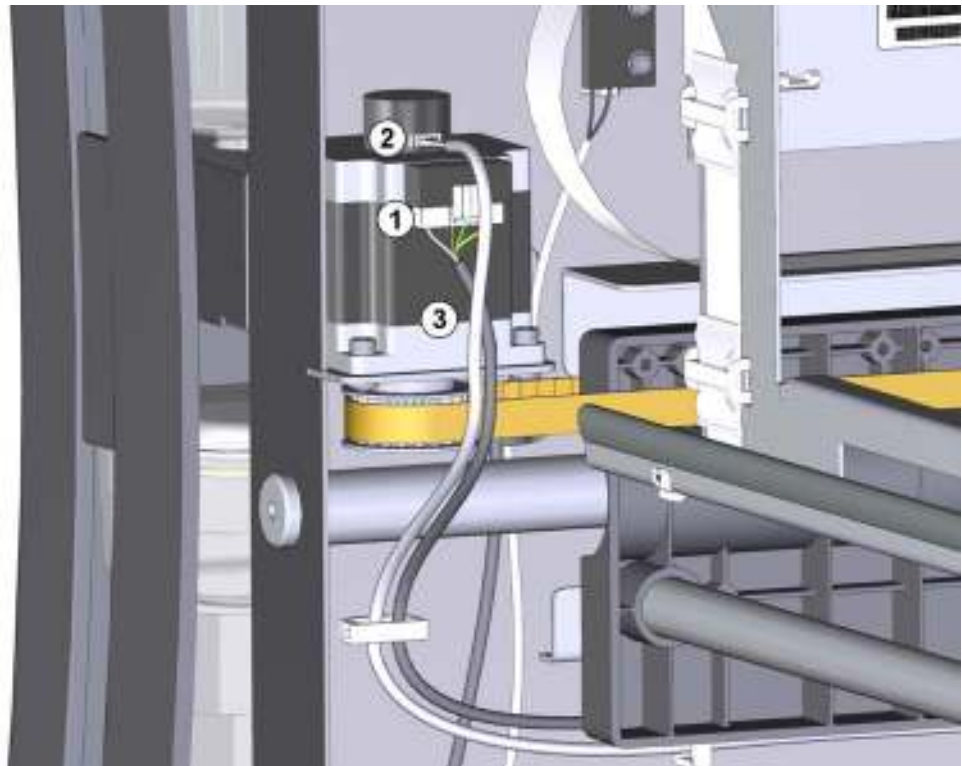


Figure 19–47: Y-Motor left pipettor arm



**19. Pipettor for internal-tanks configuration**

2. Loosen the fixing screw (4) of guide pulley (5), push it towards the spring and fasten again the screw (4).

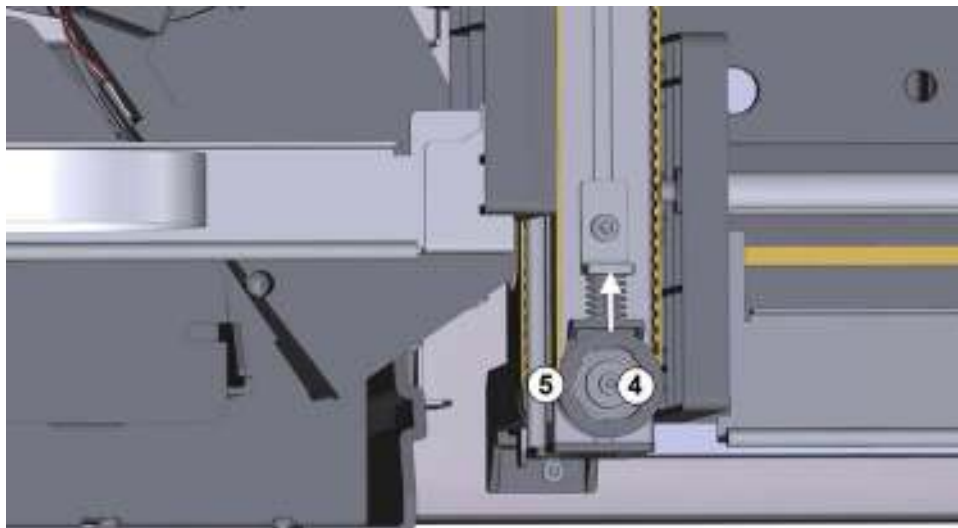


Figure 19-48: Left Pipettor arm

3. Remove the three screws (6) and lift the Y-Motor (3) out.

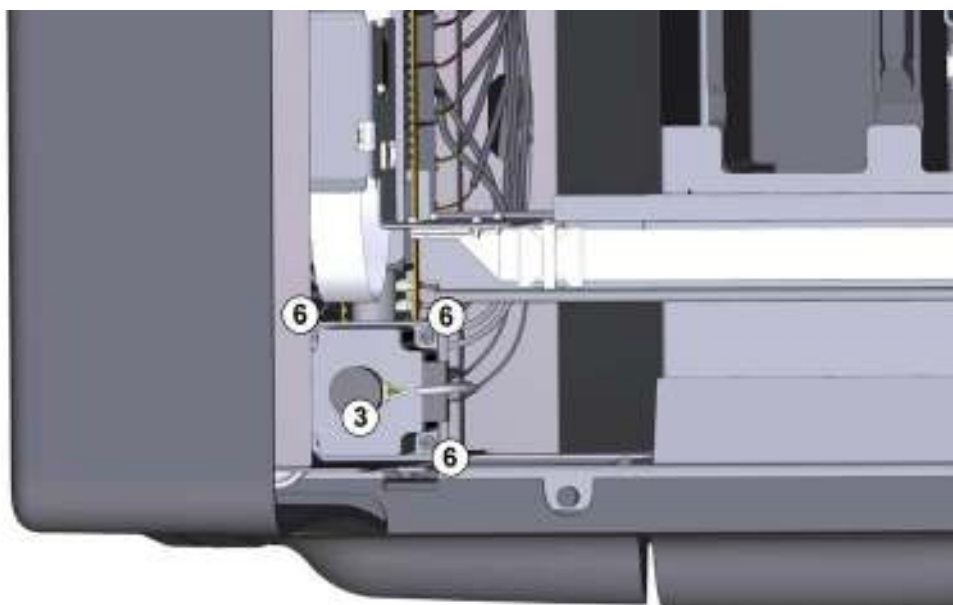


Figure 19-49: Y-Motor, left arm

**19.3.12.3 Assembly of the Y-Motor of Left Pipettor Arm**

1. Insert the Y-Motor (3) into the frame and thread the toothed belt over the motor pulley.
2. Fasten the three screws (6).
3. Loosen the fixing screw (4) of guide pulley (5) to adjust the belt tension, then fasten one screw (4).
4. Connect the motor cable (1) and the encoder cable (2) onto the Y-motor (3).

**19.3.12.4 Set the Left Pipettor Arm**

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.13 Replacement of the Left Pipettor Arm

#### 19.3.13.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the Top Covers.
6. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
7. Remove the Left Top Cover, see Chapter 14.3.5.2 on page 14-12.
8. Remove the Top Cover, see Chapter 14.3.7.2 on page 14-15.
9. Remove the Sample Pipettor, see Chapter 19.3.1.2 on page 19-10.
10. Remove the Gripper Module, see Chapter 19.3.4.2 on page 19-20.
11. Remove the toothed belt, see Chapter 19.3.11.2 on page 19-38.

## 19. Pipettor for internal-tanks configuration

### 19.3.13.2 Removal of the Left Pipettor Arm

1. Remove the cable holders (1).
2. Remove the flat ribbon cable (2) until the rear end of the cable support (3).

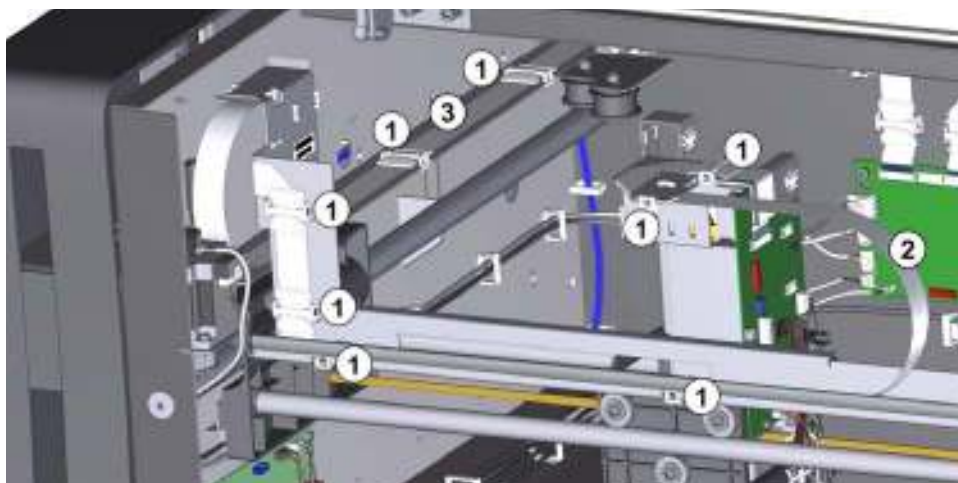


Figure 19-50: Left arm

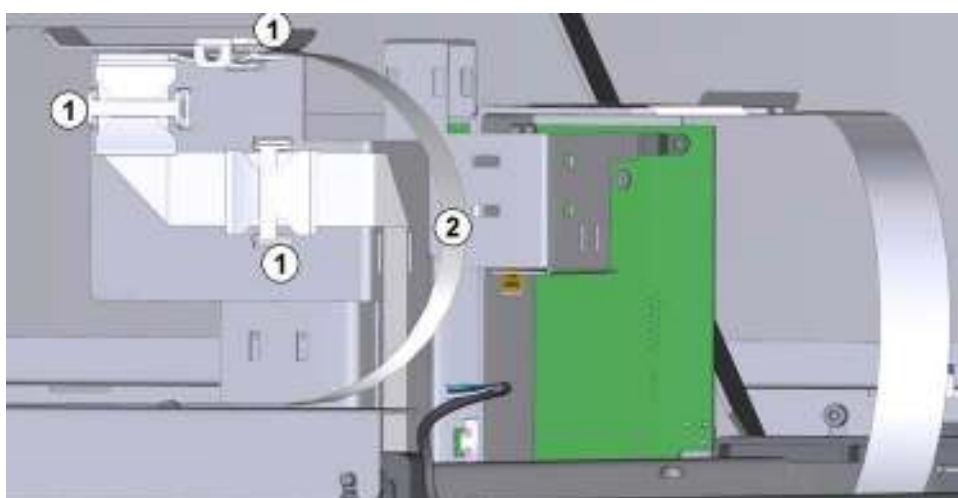


Figure 19-51: Flat ribbon cable routing, view without left side cover

3. Loosen the three screws (4) and slide the cable support (3) out of the chassis.

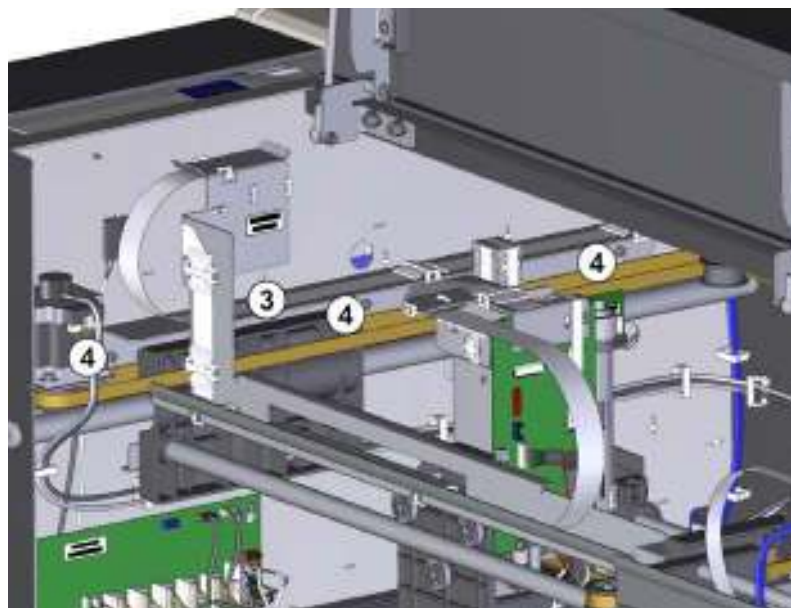


Figure 19-52: Left Arm cable support

4. Loosen the screw (5) of the Y-guide bar at the rear of the chassis.



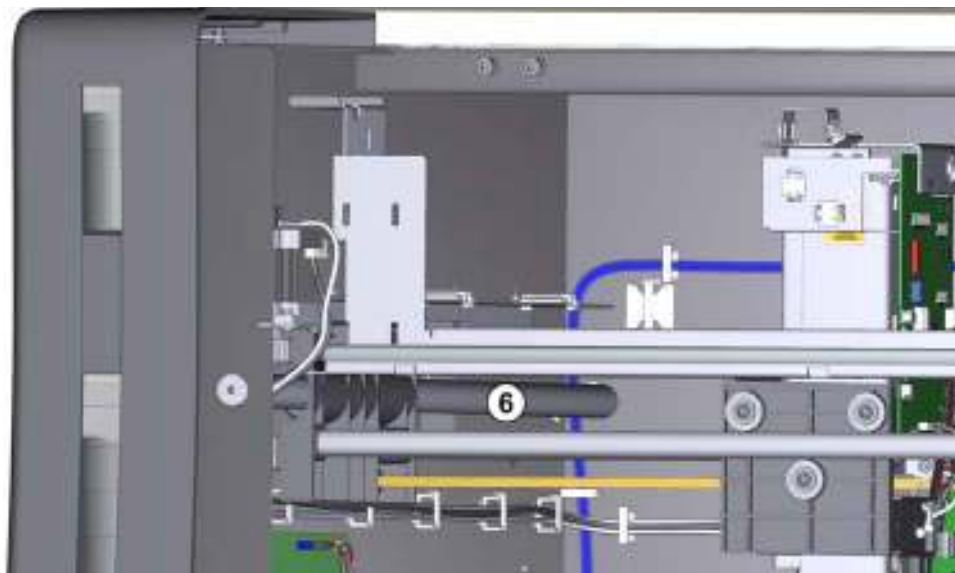
Figure 19-53: Instrument, rear view

5. Hold on to the Left Arm, it may slide to the front.
6. Lift the Y-guide bar (6) a bit up and slide it out of the keyhole at the rear wall.
7. Move the Left Pipettor Arm completely to the back, and hold on to it.
8. Pull the Y-guide bar (6) to the front and slide it out of the Left Arm.

## 19. Pipettor for internal-tanks configuration

---

9. Lift the Y-sledge left side up to slide the right side out of the guide.
10. Lift the Left Arm up and remove it from the top.



*Figure 19-54: Left Pipettor Arm*

### 19.3.13.3 Assembly of the Left Pipettor Arm

1. Place the Left Pipettor Arm into the rearmost area of the instrument from the top.
2. Slide the bearings (7) of the Y-sledge right side into the guide (8).
3. Slide the Y-guide bar (6) in and thread it into the Y-sledge left side.

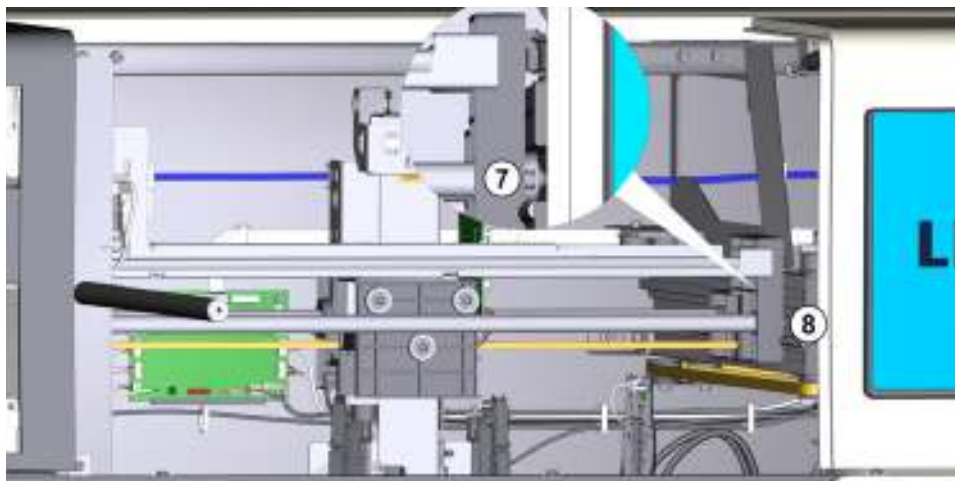


Figure 19-55: Left Pipettor Arm

4. Slide the screw head of Y-guide bar into the keyhole at the rear wall and fasten the screw (5).
5. Slide the cable support (3) into the chassis and fasten it with the screws (4).
6. Lay the flat ribbon cable (2) onto the cable support (3) and secure it with the cable holders (1).
7. Assemble the routing of the flat ribbon cable (2) and secure it with the cable holders (1).
8. Assemble the toothed belt, see Chapter 19.3.11.3 on page 19-40.
9. Assemble the Gripper module, see Chapter 19.3.4.3 on page 19-22.
10. Assemble the Sample Pipettor, see Chapter 19.3.1.3 on page 19-13.
11. Assemble the Top Cover, see Chapter 14.3.7.3 on page 14-16.
12. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 19.3.13.4 Set of the Left Pipettor Arm

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.14 Replacement of the SPOLV Needle

#### 19.3.14.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Reagent Bay cover, see Chapter 22.3.1.2 on page 22-4.
7. Move the Right Arm over the Ancillary lane.

**NOTE**

Ensure that any ancillary reagents have been unloaded from the instrument.

#### 19.3.14.2 Removal of the SPOLV Needle

1. Unplug the PRP-SPOLV tubing from the SPOLV.
2. Open the knurled screw (2).
3. Hold on to the SPOLV (1), pull the steel needle (3) down and remove it.

**NOTE**

Use a tissue to avoid drops.

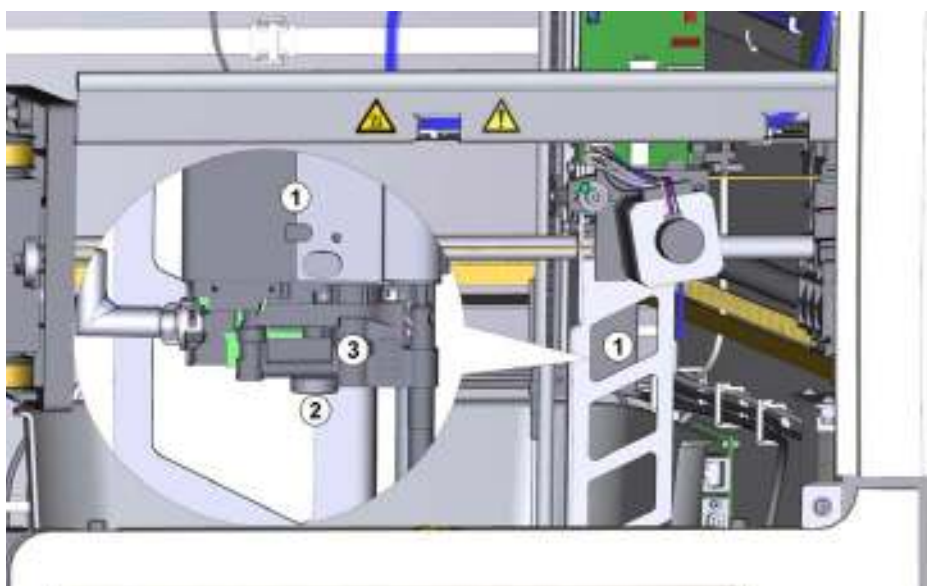


Figure 19–56: SPOLV Needle, zoom view from the rear



### 19.3.14.3 Assembly of the SPOLV Needle

#### NOTE

#### Need of SPOLV needle SN

The Serial Number of the installed SPOLV needle has to be recorded in the calibration dialogue of the Service Software. You can check the S/N on the needle screw; you can read it using a mirror and a light, without the need to disassemble the needle from the SPOLV.



1. Check the gasket (4) and O-Ring (5) for the proper installation.

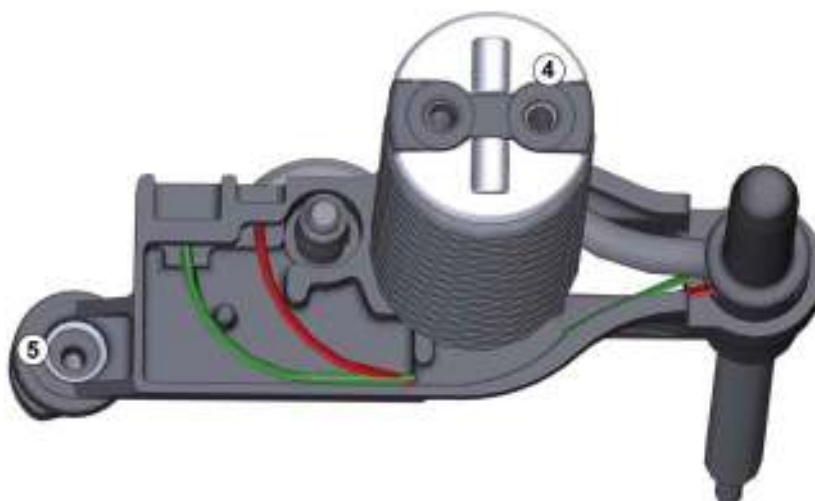


Figure 19-57: SPOLV Needle, inside view

2. Hold the SPOLV (1) and push the steel needle (3) into it.
3. Tighten the knurled screw (2) by hand.
4. Plug the PRP-SPOLV tubing.
5. Assemble the Reagent Bay cover, see Chapter 22.3.1.3 on page 22-4.

### 19.3.14.4 Set of the SPOLV

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the OLV calibration via Service Software, see Chapter 19.4.2.4 on page 19-79.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.15 Replacement of the SPOLV

#### 19.3.15.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers and approach the Right Pipettor Arm.

#### 19.3.15.2 Removal of the SPOLV

1. Unplug the PRP-SPOLV tube adapter (2) from the SPOLV (1).

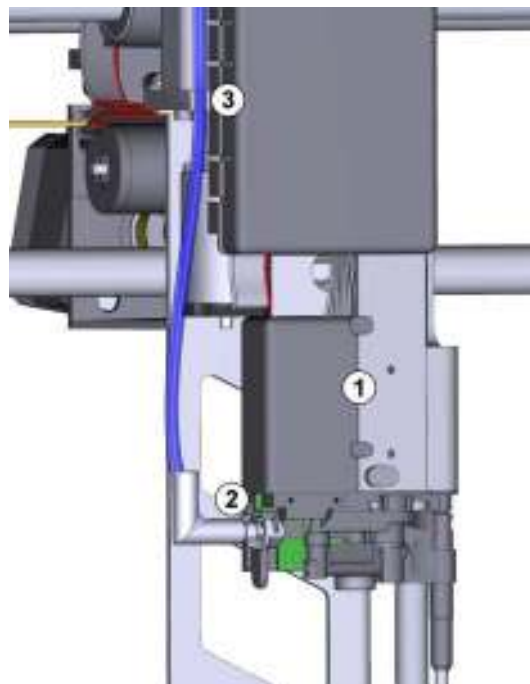


Figure 19–58: SPOLV rear view

#### NOTE

Use a tissue to avoid drops.

2. Unlock the PRP - SPOLV tube from the hook (3) on the rear of the SPOLV.
3. Remove the cable holder (connector J3) at the Single Sledge CU PCB (4).
4. Unlock the flat ribbon cable (connector J3) and pull it to the rear, out of the PCB.

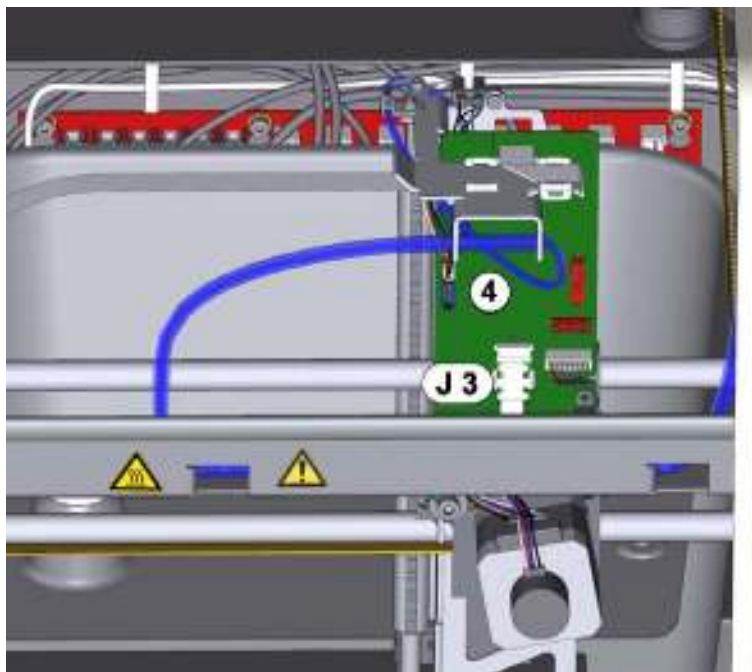


Figure 19-59: Single Sledge CU PCBA

5. Remove the top screw (5) and bottom screw (6) holding on to the SPOLV (1).
6. Slide the SPOLV out of the guide and remove it together with the Z-guiding rod (7). Be aware of the bearing (8), it may slip out!

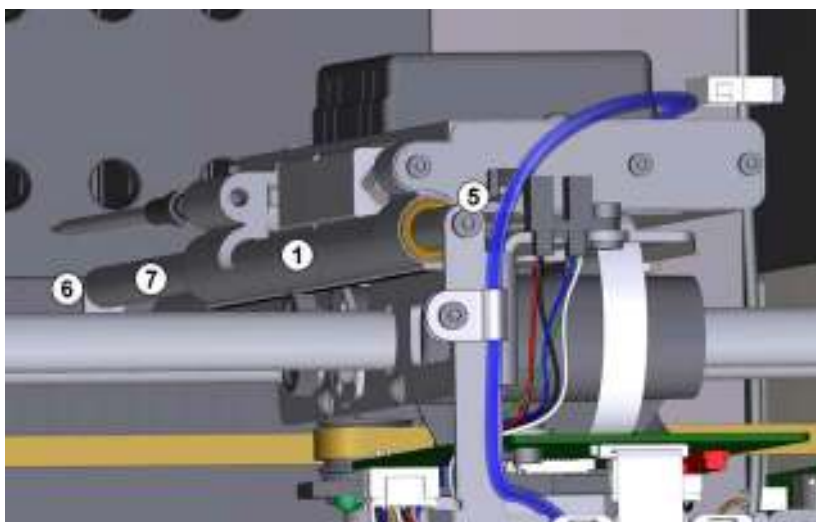


Figure 19-60: SPOLV, top view

**19.3.15.3 Assembly of the SPOLV**

1. Set the SPOLV Z-guiding rod (7) into the dedicated position and fasten with the bottom (6) and top (5) screws. The bearing (8) must sit in the Z-bracket!

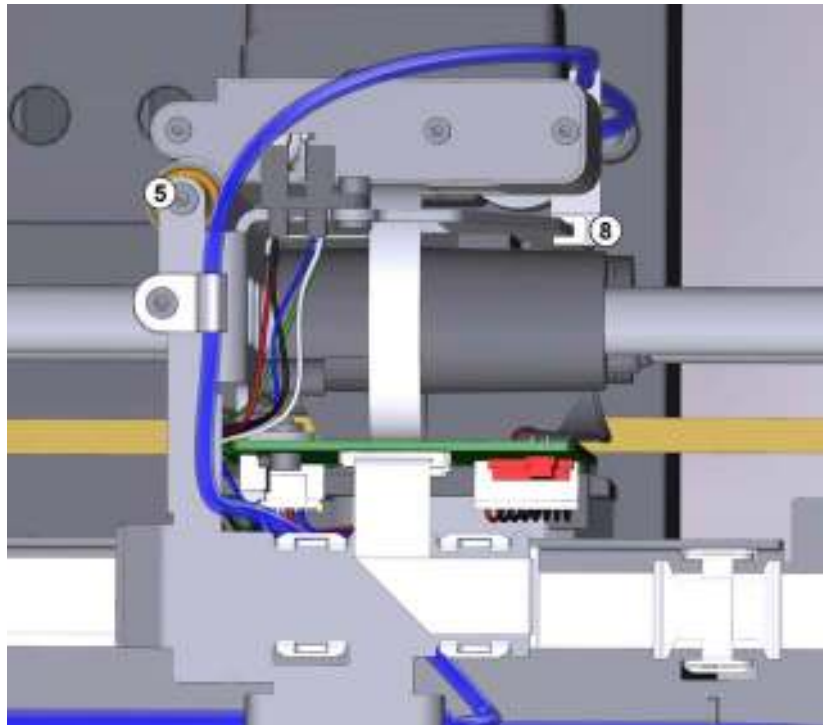


Figure 19–61: SPOLV, top view

2. Route the flat ribbon cable through the Single Sledge CU PCB (4) and lock it into the dedicated connector (J3).
3. Lock the flat ribbon cable with the cable holder.
4. Plug the PRP-SPOLV tube adapter (2) on the SPOLV.
5. Lock the PRP-SPOLV tube into the hook (3) on the rear of the SPOLV.

**19.3.15.4 Set of the Reagent Pipettor**

1. In case a new module has been installed, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the teaching of the Reagent Pipettor, see Chapter 13.5 on page 13-16.
3. Prime the SPOLV, see Chapter 19.5 on page 19-80.
4. Perform the OLV calibration, see Chapter 19.4.2.4 on page 19-79.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.16 Replacement of the Single Sledge CU PCB

#### 19.3.16.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers and approach the Right Pipettor Arm.

#### 19.3.16.2 Removal of the Single Sledge CU PCB

1. Unplug all cables from the Single Sledge CU PCB (1), see Chapter 19.8.4 on page 19-87.

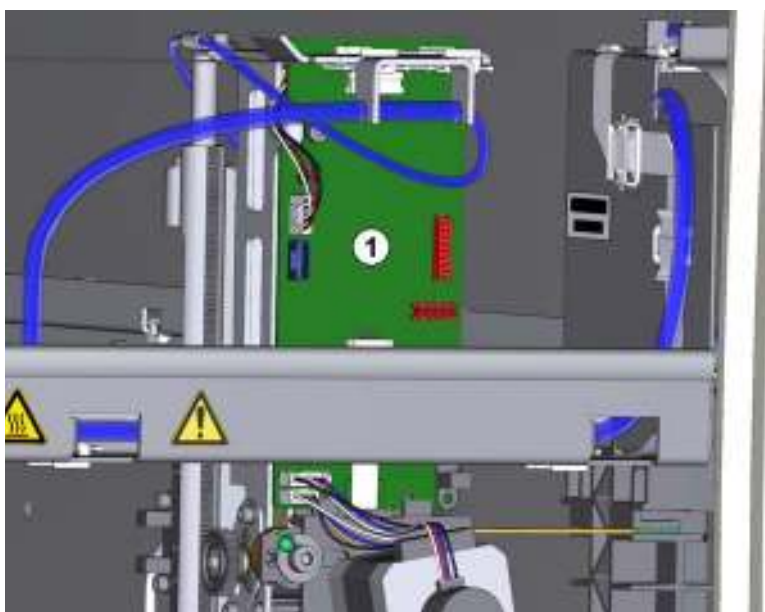


Figure 19–62: Single Sledge CU PCB

## 19. Pipettor for internal-tanks configuration

2. Unscrew the screw (2) and remove the Single Sledge CU PCB (1).

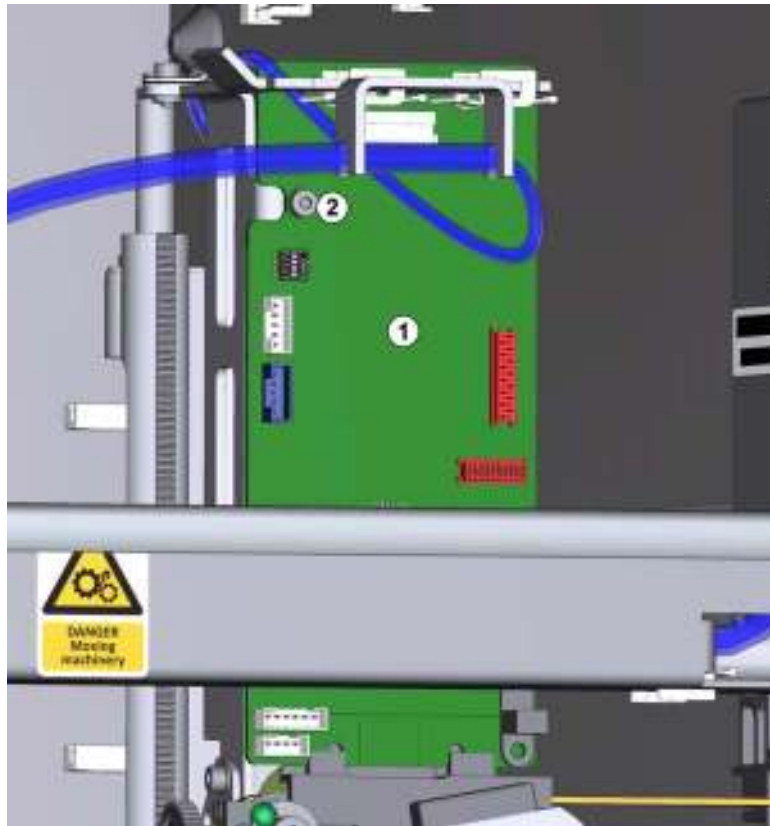


Figure 19-63: Single Sledge CU PCB removal

### 19.3.16.3 Assembly of the Single Sledge CU PCB

1. Insert the Single Sledge CU PCB (1) and fasten with the screw (2).
2. Plug all the cables onto the PCB, see Chapter 19.8.4 on page 19-87.

### 19.3.16.4 Set the Single Sledge CU PCB

1. In case a new PCB has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 19.3.17 Replacement of the Reagent Pipettor Z-Motor

#### 19.3.17.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers and approach the Right Pipettor Arm.
6. Remove the SPOLV, see Chapter 19.3.15.2 on page 19-52.

#### 19.3.17.2 Removal of the Reagent Pipettor Z-Motor

1. Unplug the Z-Motor cable (J8) from the Single Sledge CU PCB (1).

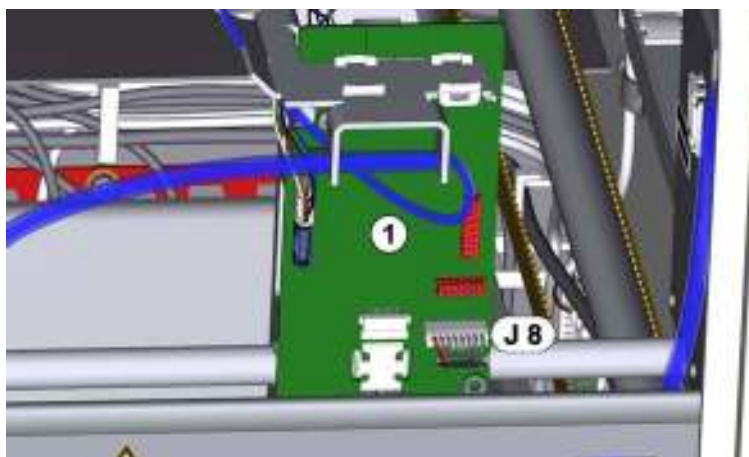


Figure 19-64: Reagent Pipettor, Single Sledge CU PCB

2. Unscrew two screws (2) and slide the Z-Motor (3) out.

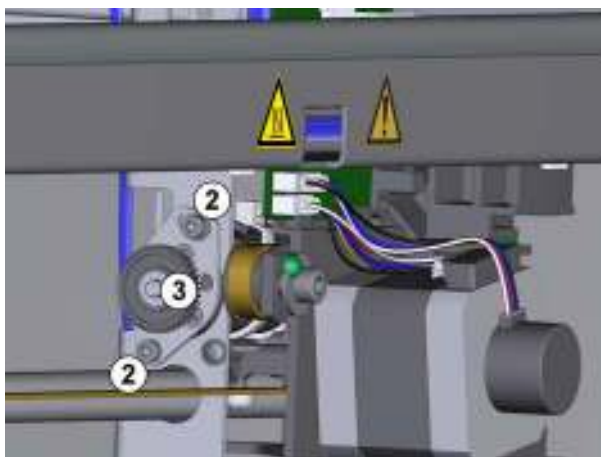
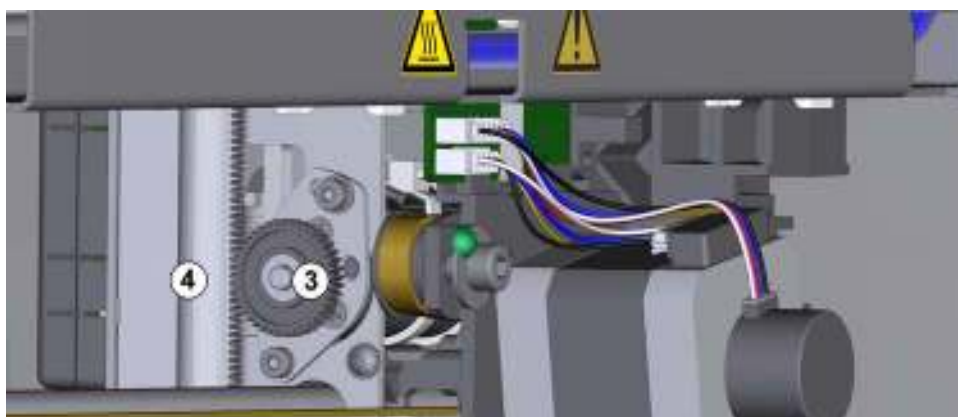


Figure 19-65: Reagent Pipettor, Z-Motor

**19.3.17.3 Assembly of the Reagent Pipettor Z-Motor**

1. Fold the flat ribbon cable of the Z-Motor and secure it with a cable tie.
2. Set the new Z-Motor (3) in.
3. Fasten the Z-Motor with two screws (2).
4. Assemble the SPOLV, see Chapter 19.3.15.3 on page 19-54.
5. Adjust the clearance between the Z-Motor gear (3) and the Reagent Pipettor toothed bar (4).
6. Fasten the Z-Motor with two screws (2).



*Figure 19-66: SPOLV Z-Motor gear clearance*

**19.3.17.4 Set the Reagent Pipettor**

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 19.3.18 Replacement of the X-axis toothed belt of Right Pipettor Arm

#### 19.3.18.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers and approach the Right Pipettor Arm.

#### 19.3.18.2 Removal of the X-Axis toothed belt of Right Pipettor Arm

1. Loosen the fixing screw (2) of the guide pulley (3), push the guide pulley onto the spring and fasten the screw to loosen the belt.

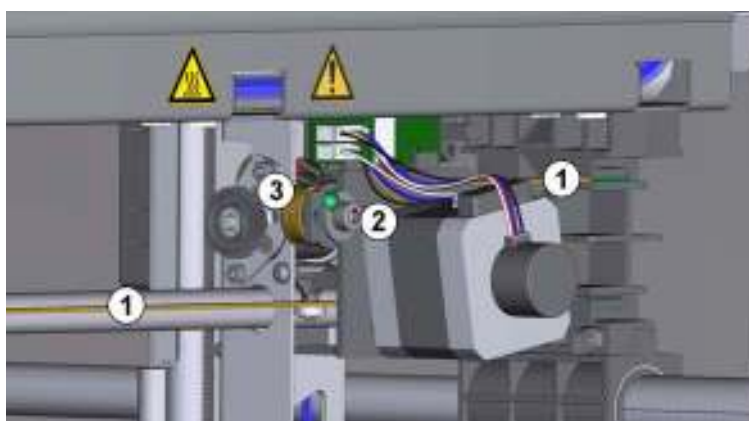


Figure 19–67: Right Pipettor arm

2. At the right side of Y-sledge (6), pull the toothed belt (1) out and then unhook the left belt holder (4) from the left side of Y-Sledge(5).

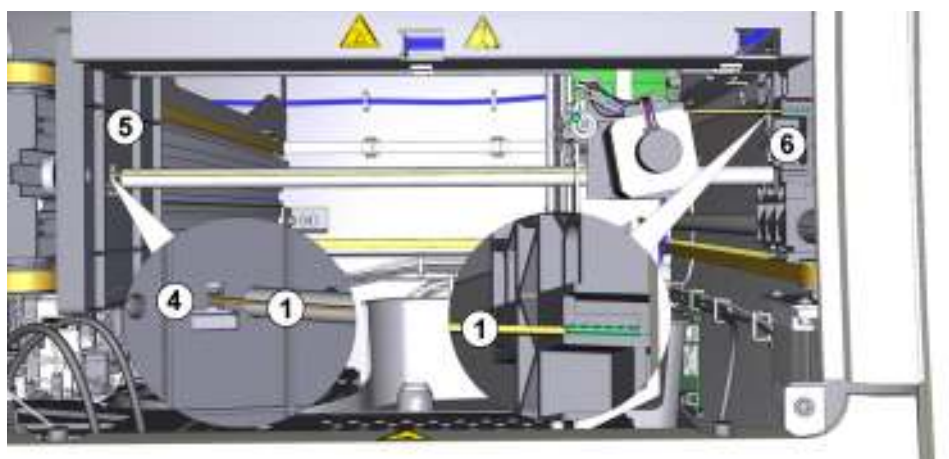
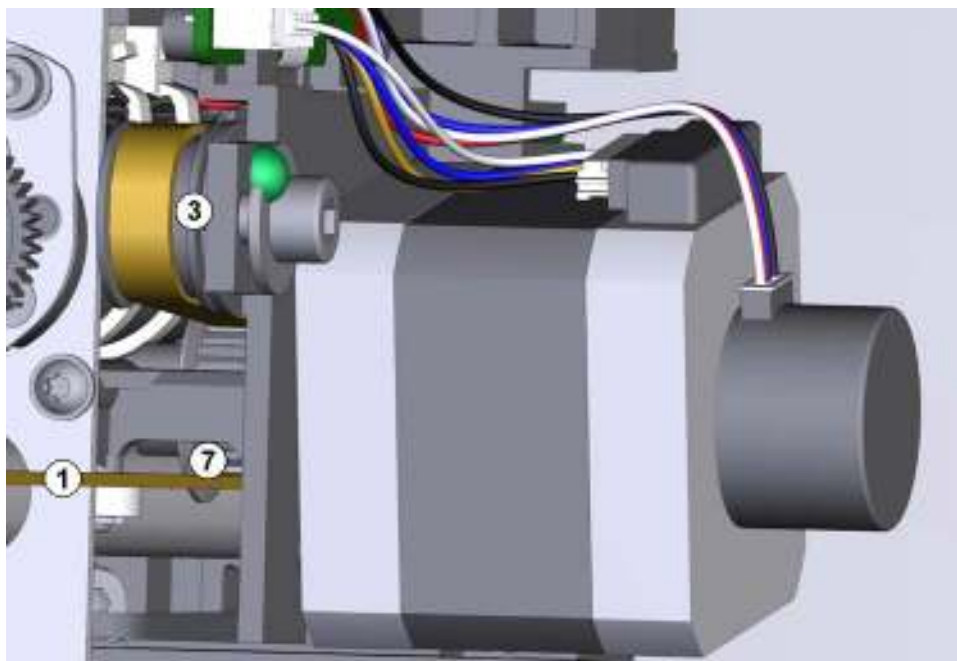


Figure 19–68: X-Axis toothed belt suspension

**19.3.18.3 Assembly of the X-Axis toothed belt of Right Pipettor Arm**

1. Thread the toothed belt (1) through the guide pulley (3) and the motor pulley (7).



*Figure 19–69: X-Axis Right Pipettor Arm*

2. Hang the left belt holder (4) into the left side of Y-Sledge (5).
3. Slide the toothed belt into the right upper part (6) of the Y-Sledge.
4. Loosen the fixing screw (2) of the guide pulley (3) to adjust the belt tension, then fasten the screw (2).

**19.3.18.4 Set the Right Pipettor Arm**

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.19 Replacement of the X-Motor of Right Pipettor Arm

#### 19.3.19.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.
7. Remove the SPOLV Needle, see Chapter 19.3.14.2 on page 19-50.
8. Remove the SPOLV, if needed, see Chapter 19.3.15.2 on page 19-52.
9. Remove the Y-axis toothed belt, see Chapter 19.3.20.2 on page 19-63.
10. Remove completely the Right Pipettor arm, see Chapter 19.3.22.2 on page 19-69.

#### 19.3.19.2 Removal of the X-Motor of Right Pipettor Arm

1. Loosen the fixing screw (2) of the guide pulley (3), push the guide pulley onto the spring and fasten the screw to loosen the belt.
2. Unplug the X-Motor cable (J4) and the X-Encoder cable (J5) from the Single Sledge CU PCB.

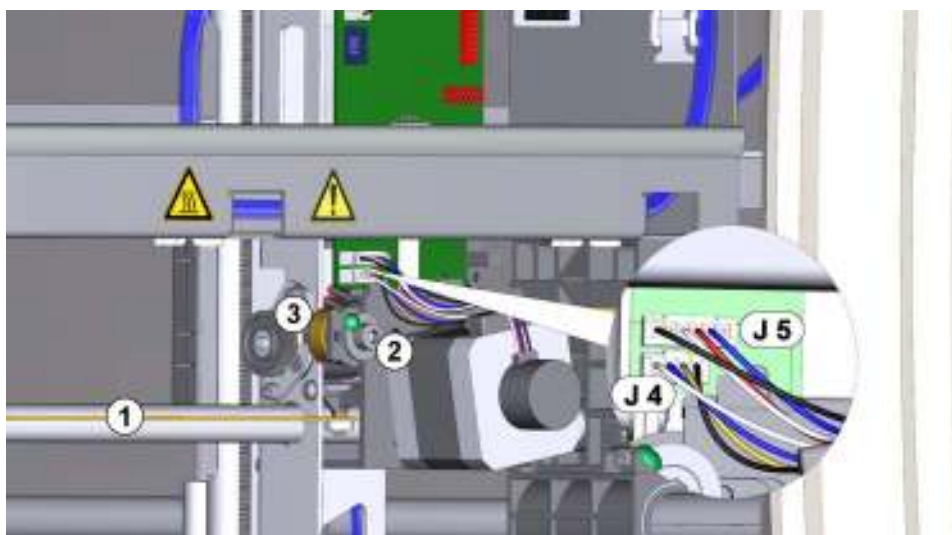


Figure 19–70: Right Pipettor arm

## 19. Pipettor for internal-tanks configuration

3. Loosen the three screws (4) of X-Motor (5).
4. Turn the X-Motor (5) counterclockwise and pull it out.

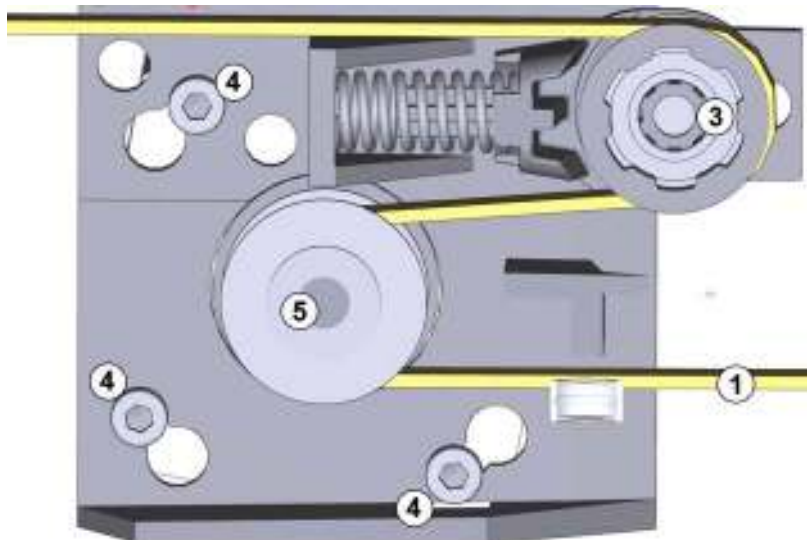


Figure 19-71: Inside view (cross section) of X-Motor mounting

### 19.3.19.3 Assembly of the X-Motor of Right Pipettor Arm

1. Insert the X-Motor (5) and lay the toothed belt around the motor pulley.
2. Turn the X-Motor clockwise and fasten the three screws (4).
3. Plug the X-Motor cable and the X-Encoder cable onto connectors (J4) and (J5) of the Single Sledge CU PCB, respectively.
4. Loosen the fixing screw (2) of the guide pulley (3) to tension the toothed belt (1), then fasten the fixing screw (2).
5. Assemble the Right Pipettor arm, see Chapter 19.3.22.3 on page 19-72.
6. Assemble the Y-axis toothed belt, see Chapter 19.3.20.3 on page 19-64.
7. Assemble the SPOLV, if needed, see Chapter 19.3.15.3 on page 19-54.
8. Assemble the SPOLV Needle, see Chapter 19.3.14.3 on page 19-51.
9. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.

### 19.3.19.4 Set the Right Pipettor Arm

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Prime the SPOLV, see Chapter 19.5 on page 19-80.
3. Perform the OLV calibration, see Chapter 19.4.2.4 on page 19-79.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.20 Replacement of the Y-axis toothed belt of Right Pipettor Arm

#### 19.3.20.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.

#### 19.3.20.2 Removal of the Y-axis toothed belt of Right Pipettor Arm

1. Loosen the screw of the toothed belt wheel (1).
2. Press the wheel against the end stop and fasten the screw to loosen the belt.

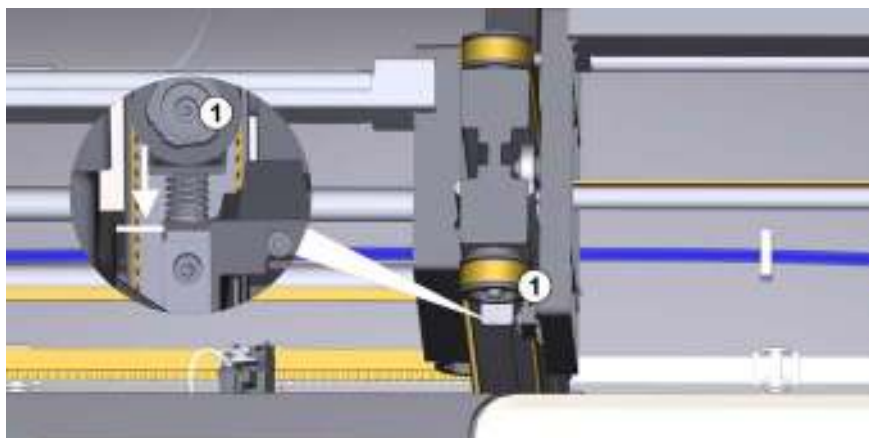


Figure 19-72: Right arm, belt tensioner

3. Remove the two screws (3) of the belt connector (2) and slide the ends of the toothed belt (4) out of the connector.

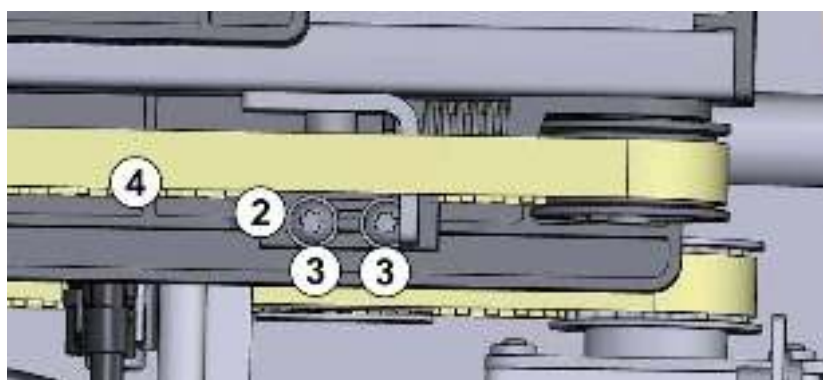


Figure 19-73: Right arm, toothed belt connector

## 19. Pipettor for internal-tanks configuration

4. Slide the toothed belt (4) out of the belt holder (5).
5. Remove the toothed belt from the instrument.

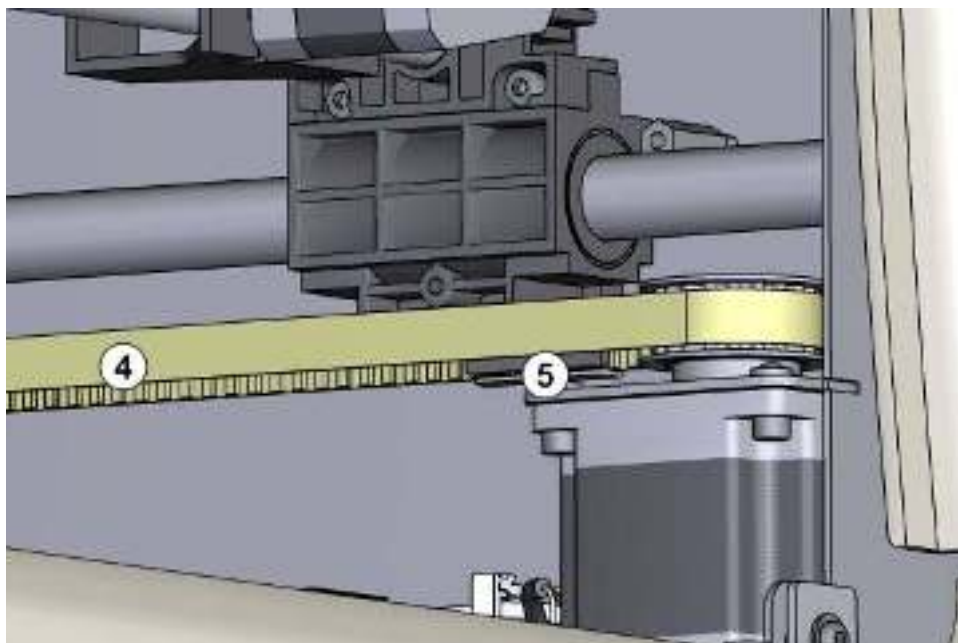


Figure 19–74: Toothed belt holder (right side)

### 19.3.20.3 Assembly of the Y-axis toothed belt of Right Pipettor Arm

1. Check the toothed belt length, and if necessary cut it to a length of 3265mm (the toothed belt has to be cut only along vales).
2. Insert the toothed belt through the guide pulleys.

#### NOTE

**The toothed belt must not be twisted!**

**The toothed belt must sit correctly on the pulleys!**



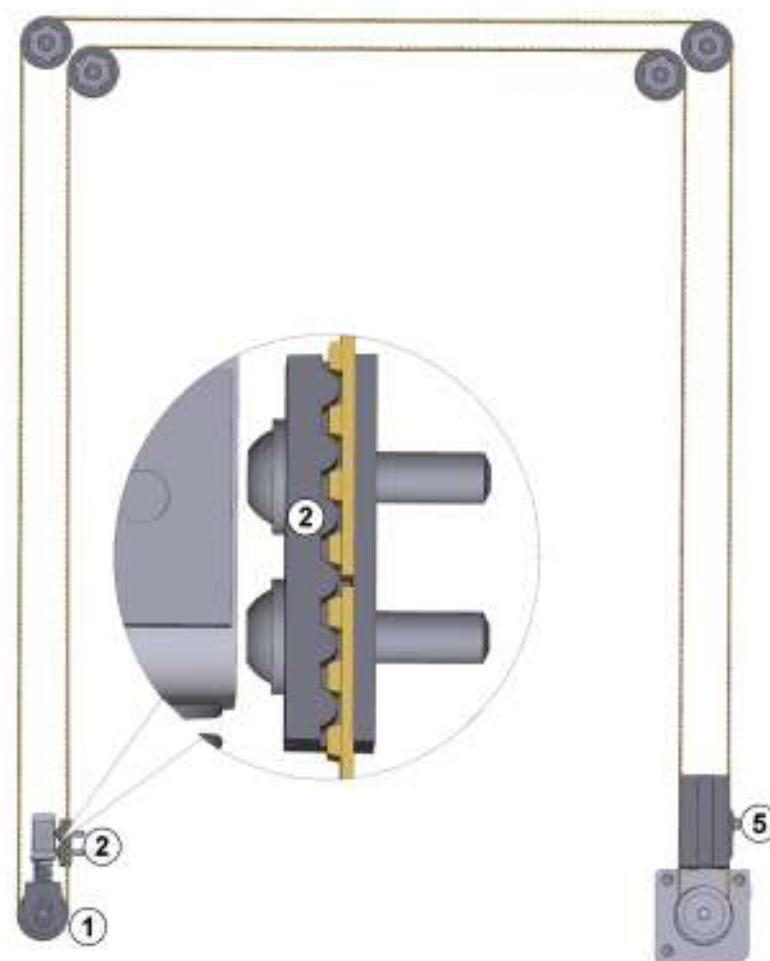


Figure 19–75: Layout of right arm toothed belt

3. Insert the two ends of the toothed belt (4) into the belt connector (2):
  - Slide the front end of the toothed belt with two teeth into the belt connector (2).
  - Slide the back end of the toothed belt with three teeth into the belt connector (2).
4. Set the connector (2) into place and fasten to the Right Arm with the screws (3).
5. Loosen the screw of the toothed belt wheel (1), let the spring expand to tension the belt and then fasten the screw.
6. Slide the toothed belt into the belt holder (5).
7. Move the Right Pipettor Arm forward and back a couple of times, to check that the pipettor arm is parallel. If necessary, loosen the belt at the belt wheel (1) and change the tooth insertion by one tooth forward or back into the belt holder (5).

#### 19.3.20.4 Set the Y-axis toothed belt of Right Pipettor Arm

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### **19.3.21 Replacement of the Y-Motor of Right Pipettor Arm**

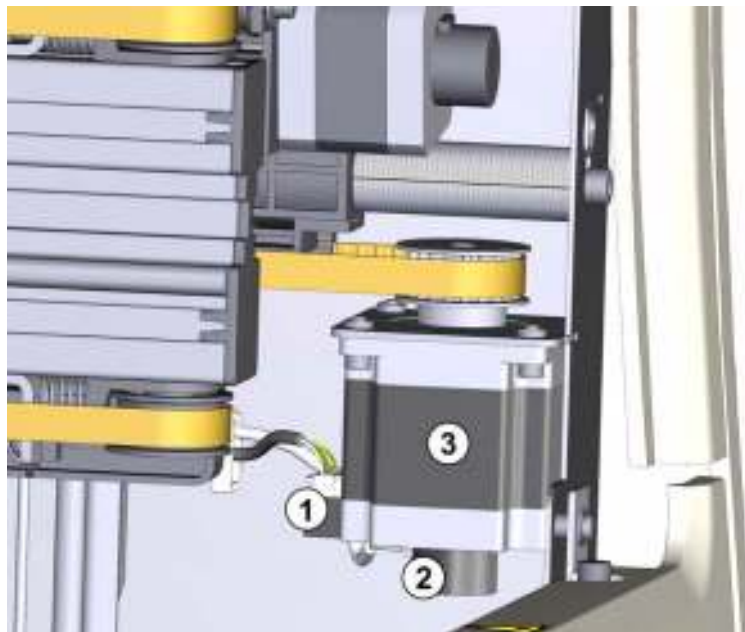
#### **19.3.21.1 Preliminary steps**



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.

#### **19.3.21.2 Removal of the Y-Motor of Right Pipettor Arm**

1. Unplug the motor cable (1) and the encoder cable (2) at the Y-motor (3).



*Figure 19–76: Y-Motor of Right Arm, side view*



**19. Pipettor for internal-tanks configuration**

2. Loosen the fixing screw (4) of guide pulley (5) and push it towards the spring to loosen the belt, then fasten the screw (4).

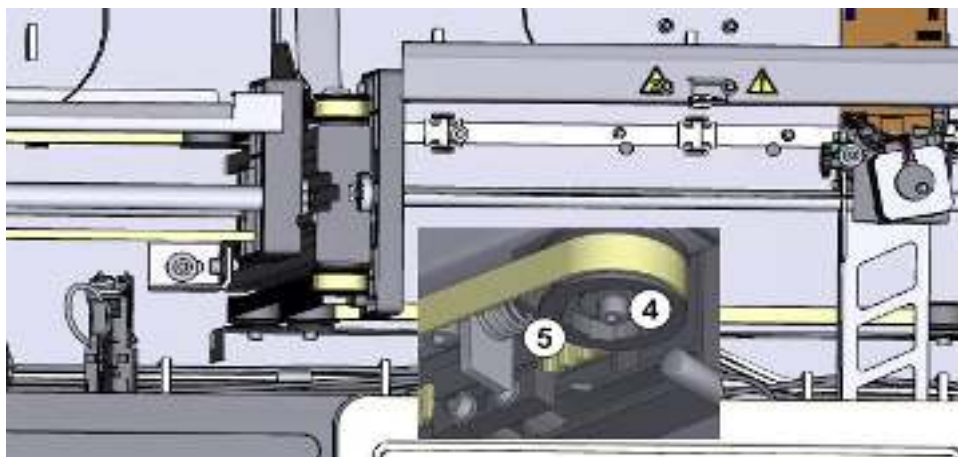


Figure 19-77: Right Pipettor Arm

3. Remove the three screws (6) and pull the Y-Motor (3) out.

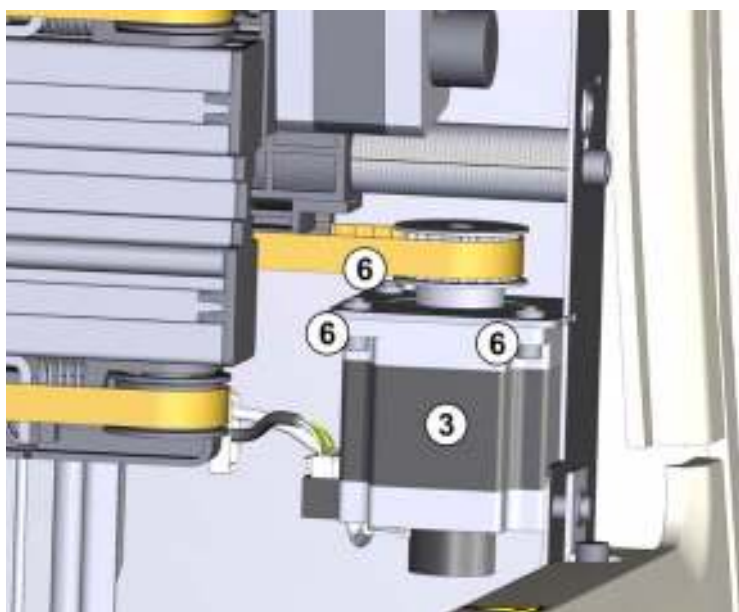


Figure 19-78: Y-Motor, Right arm

**19.3.21.3 Assembly of the Y-Motor of Right Pipettor Arm**

1. Insert the Y-Motor (3) into the frame and thread the toothed belt over the motor pulley.
2. Fasten the three screws (6).
3. Loosen the two fixing screws (4) of guide pulley (5) to adjust the belt tension, then fasten the two screws (4).
4. Plug the motor cable (1) and the encoder cable (2) onto the Y-motor (3).

**19.3.21.4 Set the Right Pipettor Arm**

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.22 Replacement of the Right Pipettor Arm

#### 19.3.22.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.
7. Remove the SPOLV Needle, see Chapter 19.3.14.2 on page 19-50.
8. Remove the SPOLV, see Chapter 19.3.15.2 on page 19-52.
9. Remove the toothed belt, see Chapter 19.3.20.2 on page 19-63.

#### 19.3.22.2 Removal of the Right Pipettor Arm

1. Remove the cable holders (2).
2. Unlock the flat ribbon cable (1) from the connector (3) and roll it up.
3. Unscrew the screw (5) of the tube clamp and unlock the PRP - SPOLV tube (4) from the hooks (6).

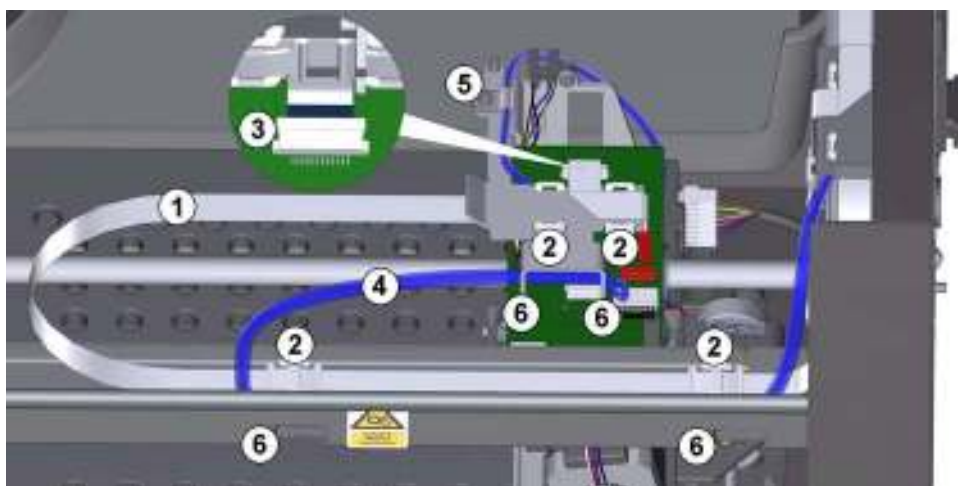


Figure 19–79: Right Arm

## 19. Pipettor for internal-tanks configuration

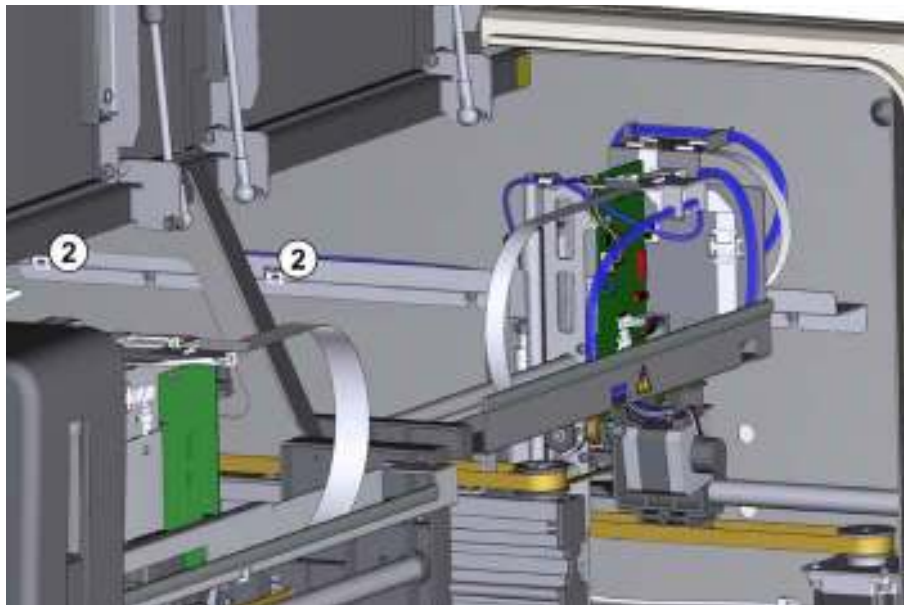


Figure 19-80: Right Arm

4. Remove the four cable holders (2) and roll up the flat ribbon cable (1).
5. Unlock the PRP - SPOLV tube (4) from the hooks (6).

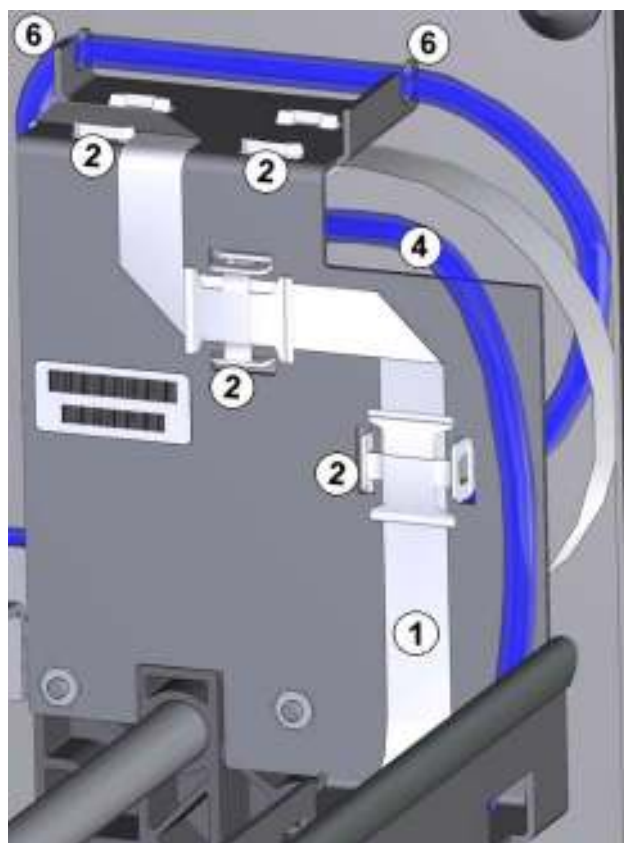
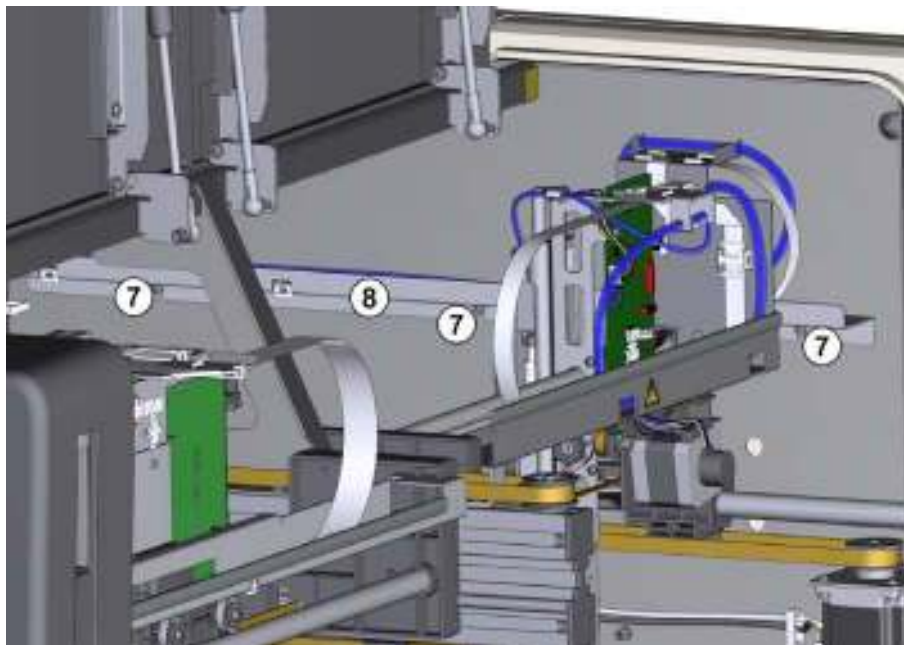


Figure 19-81: Pipettor right arm, flat ribbon cable routing

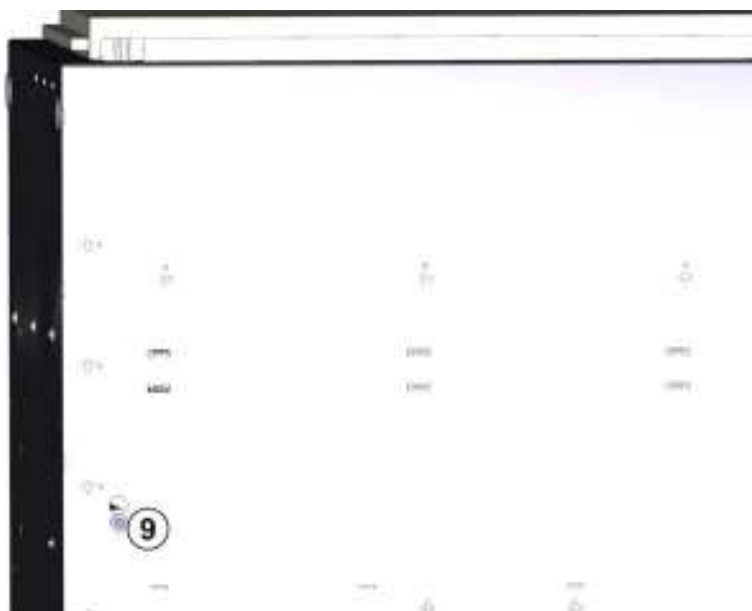
**19. Pipettor for internal-tanks configuration**

6. Loosen the three screws (7) and slide the cable support (8) out of the chassis.



*Figure 19-82: Right Arm cable support*

7. Loosen the screw (9) of the Y-guide bar at the rear of the chassis.



*Figure 19-83: Instrument rear view*

8. Loosen the screw (10) of the Y-guide bar at the front of the chassis.
9. Hold on to the Right Arm, it may slide to the front.

## 19. Pipettor for internal-tanks configuration

10. Lift the Y-guide bar (11) a bit up and slide it out of the keyhole at the rear wall.
11. Move the Right Pipettor Arm completely to the front, and hold on to it.
12. Pull the Y-guide bar (11) to the back and slide it out of the Right Arm.
13. Lift the Y-sledge right side up to slide the left side out of the guide, and remove the Right Arm from the instrument.

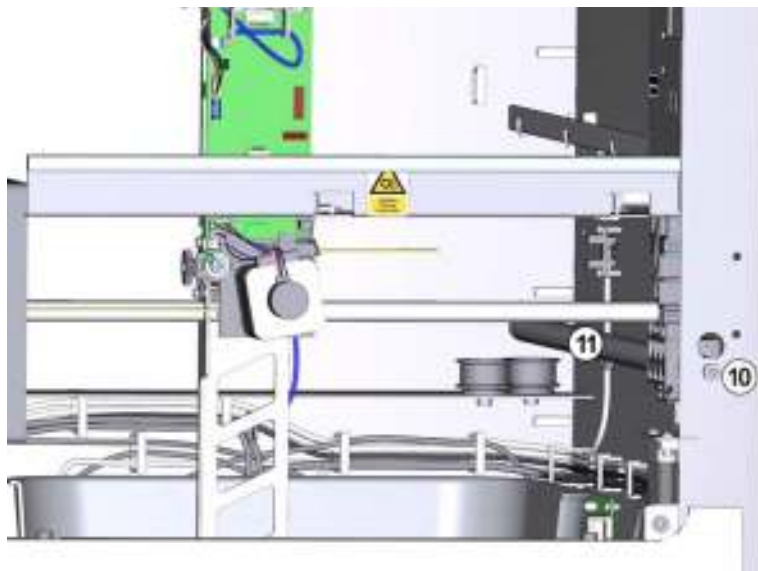


Figure 19-84: Instrument front view

### 19.3.22.3 Assembly of the Right Pipettor Arm

1. Slide the bearings (12) of the Y-sledge left side into the guide.
2. Slide the Y-guide bar in and thread it into the Y-sledge right side.

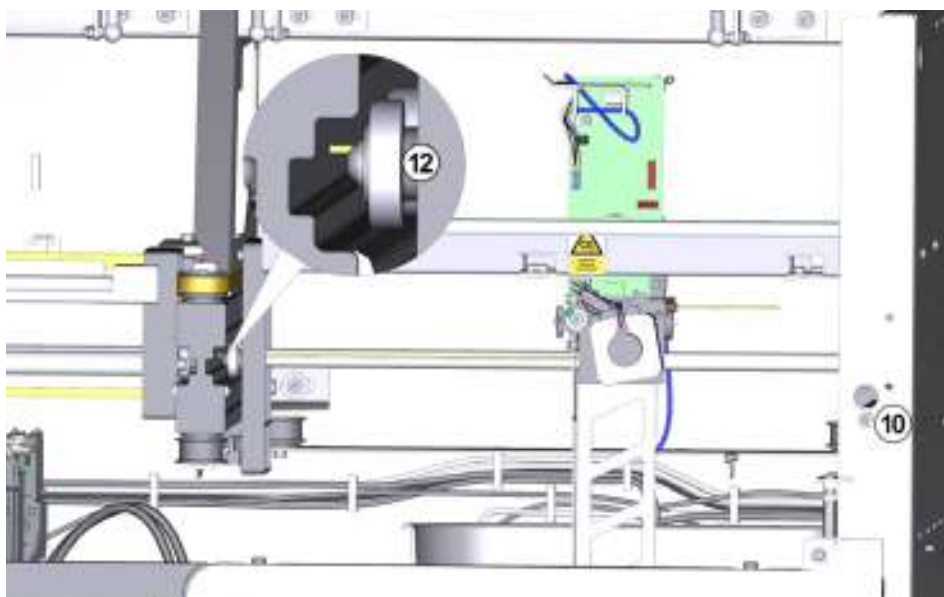


Figure 19-85: Instrument front view



3. Slide the screw heads of Y-guide bar into the keyholes and fasten the screws (9) and (10).
4. Slide the cable support (8) into the chassis and fasten it with the screws (7).
5. Assemble the routing of the flat ribbon cable (1) along the cable supports and secure it with the cable holders (2).
6. Lay the PRP- SPOLV tube (4) carefully in and hang it into the hooks (6).
7. Move the pipettor forward and back to check that the tube and the cable are properly positioned: ensure that the tube is not interfering with the flat ribbon cable onto the cable support.
8. Assemble the toothed belt, see Chapter 19.3.20.3 on page 19-64.
9. Assemble the SPOLV, see Chapter 19.3.15.3 on page 19-54.
10. Assemble the SPOLV Needle, see Chapter 19.3.14.3 on page 19-51.
11. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.

#### 19.3.22.4 Set the Right Pipettor Arm

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Prime the SPOLV, see Chapter 19.5 on page 19-80.
3. Perform the OLV calibration, see Chapter 19.4.2.4 on page 19-79.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 19.3.23 Replacement of the Left arm Y-Home sensor

#### 19.3.23.1 Preliminary step



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.

#### 19.3.23.2 Removal of the Left arm Y-Home sensor

1. Unplug the sensor Y-axis home sensor of left arm (connector J12 on the Pipettor CU PCBA), see Figure 19-87.
2. Unscrew the screws, which are holding the selected sensor, see Figure 19-88.
3. Open the cable holders.
4. Remove the old sensor from the instrument.

## 19. Pipettor for internal-tanks configuration

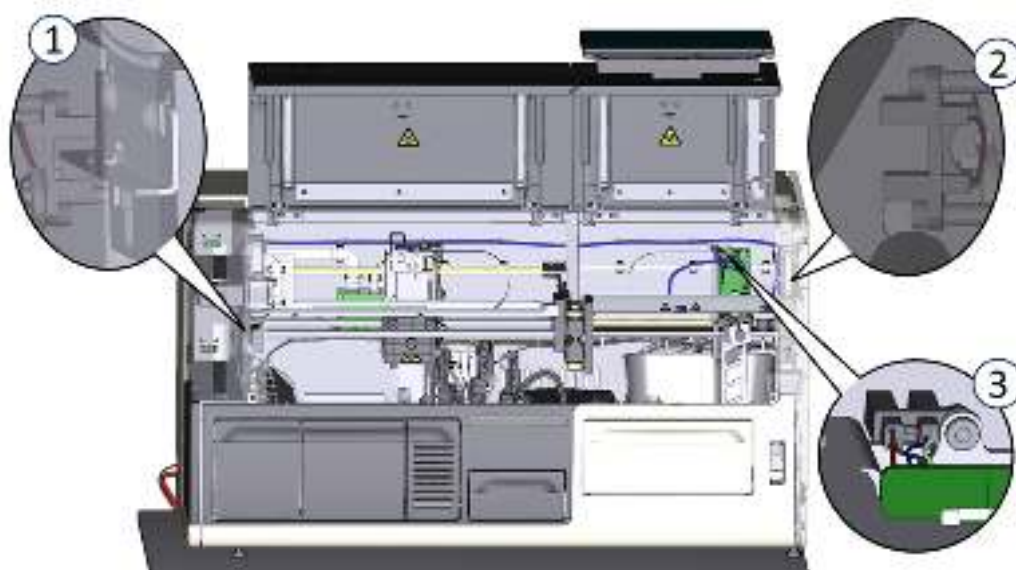


Figure 19–86: Pipettor Home sensors positioning inside the system. The circle 1 is the Left arm Y-Home sensor. The circle 2 is the Right arm Y-Home sensor. The circle 3 is the SPOLV Z-Home sensor

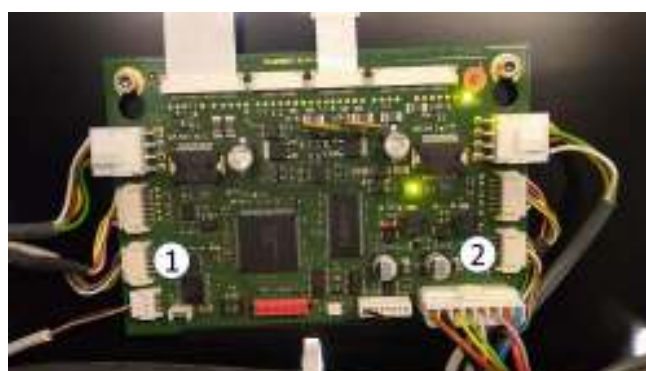


Figure 19–87: Pipettor CU PCBA



Figure 19–88: Left arm Y-Home sensor



### 19.3.23.3 Assembly of the Left arm Y-Home sensor

1. Install the new sensor according to the Figure 19–88 and fasten the screws.
2. Guide the cable through the holders and close them.
3. Connect the plug to the Pipettor CU PCBA (J12), see Figure 19–87.
4. Close the covers, turn on instrument, open the Main UI and initialize the system.

### 19.3.24 Replacement of the Right arm Y-Home sensor

#### 19.3.24.1 Preliminary step



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location.

#### 19.3.24.2 Removal of the Right arm Y-Home sensor

1. Unplug the sensor Y-axis home sensor of right arm (connector J8 on the Pipettor CU PCBA), see Figure 19–87.
2. Unscrew the screws, which are holding the selected sensor, see Figure 19–89.
3. Open the cable holders.
4. Remove the old sensor from the instrument.



Figure 19–89: Right arm Y-Home sensor

**19.3.24.3 Assembly of the Right arm Y-Home sensor**

1. Install the new sensor according to Figure 19–89 and fasten the screws.
2. Guide the cable through the holders and close them.
3. Connect the plug to the Pipettor CU PCBA (J8), see Figure 19–87.
4. Close the covers, turn on instrument, open the Main UI and initialize the system.

**19.3.25 Replacement of the SPOLV Z-Home sensor****19.3.25.1 Preliminary step**

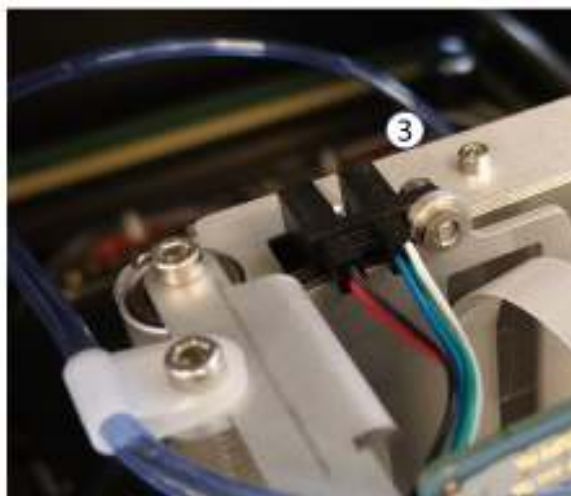
1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.

**19.3.25.2 Removal of the SPOLV Z-Home sensor**

1. Unplug the sensor Z-axis home sensor of SPOLV (connector J9 on the X-Z-sledge EC CU PCBA), see Figure 19–90.
2. Unscrew the screw which is holding the sensor, see Figure 19–91.
3. Remove the old sensor from the instrument.



*Figure 19–90: X-Z-sledge EC CU PCBA*



*Figure 19–91: SPOLV Z-Home sensor*

#### **19.3.25.3 Assembly of the SPOLV Z-Home sensor**

1. Install the new sensor according to Figure 19–91 and fasten the screw.
2. Connect the plug to the X-Z-sledge EC CU PCBA (J9), see Figure 19–90.
3. Close the covers, turn on instrument, open the Main UI and initialize the system.

## 19.4 Adjustment and calibration

This section contains information about the adjustment and calibration of the Pipettor Module.

### 19.4.1 Initialization routine

- Close any open software and start the **Service software** from the Windows Administrator account.
- In the **Service software** at the navigation window, choose **Pipettor** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog Controlled Modules** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 19.4.2 Calibration of the Pipettor Module

#### 19.4.2.1 Calibration of the Gripper Claws

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **Cuvette Gripper**.

- Step 1**
- Read the instructions and then press the **Done** button.
- Step 2**
- Take a cuvette and manually hold it between the gripper claws.
  - Use the buttons to close the gripper claws in steps of 10, until the gripper holds on to the cuvette.
  - Use the buttons to open the gripper claws in steps of 10, until the cuvette falls down.
  - Use the buttons to further open the gripper claws by **30** steps.
  - Press the **Save** button to save the adjustment.
- Step 3**
- Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 4**
- Read the instructions and then press the **Done** button.
- Step 5**
- The process is successful when, at the **Summarize** panel, all steps are in green.

#### 19.4.2.2 Calibration of the Sample Pipettor Syringe Motor

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **Sample Pip syringe motor**.

- Step 1**
- Read the instructions and then press the **Done** button; the calibration procedure starts automatically.
  - The procedure is successful when the **Processing** window closes without failure message.

**Step 2** c. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.

**Step 3** d. The process is successful when, at the **Summarize** panel, all steps are in green.

#### 19.4.2.3 Calibration of the SPOLV Syringe Motor

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **SPOLV syringe motor**.

**Step 1** a. Read the instructions and then press the **Done** button; the calibration procedure starts automatically.  
b. The procedure is successful when the **Processing** window closes without failure message.

**Step 2** c. Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.

**Step 3** d. The process is successful when, at the **Summarize** panel, all steps are in green.

#### 19.4.2.4 OLV Calibration

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **OLV Calibrate**.

**Step 1** a. Read the instructions and then press the **Done** button; the calibration procedure starts automatically.  
b. The procedure is successful when the **Processing** window closes without failure message.

**Step 2** c. The process is successful when, at the **Summarize** panel, all steps are in green.

## 19.5 Priming of Pipettor

**CAUTION**

Before rinsing, make sure that all tube connections and fittings are properly assembled. Ensure that all tanks are filled (DI Water, Wash Buffer and Starter Reagents) and the Liquid Waste tank is empty. During the complete rinsing process, ensure that there is no leakage (tube connections glued, inserted, rotated etc.).

**CAUTION**

Before rinsing, make sure that Solid Waste bin is inserted (with the dedicated EASY Waste) and the Liquid Waste tank is connected.

### 19.5.1 Initialization routine

- Close any open software and start the **Service software** from the Windows Administrator account.
- In the **Service software** at the navigation window, choose **System** and then the **General** bracket.
- Click onto the **Instrument** button and choose the **Full initialization via COP** to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 19.5.2 Initial rinsing with DI water

In the **Service software** at the navigation window, choose **System** and the **General** bracket.

- Click onto the **Liquid** button and choose the **Prime Pipettor** button to open the dedicated window.
- For initial rinsing, use the following values and start pressing the **Execute** button.
  - Cycles: 11
  - Wash time: 18
  - OLV Calibration: Disabled
  - Liquid: DI Water

| Prime Pipettor           |          |
|--------------------------|----------|
| Cycles                   | 11       |
| Wash Time [0.1sec/Cyc... | 18       |
| OLV Calibration          | Disabled |
| Liquid                   | DIWater  |
| <b>Execute</b>           |          |

## 19. Pipettor for internal-tanks configuration

- c. The process is successful when the **Processing** window closes without failure message.

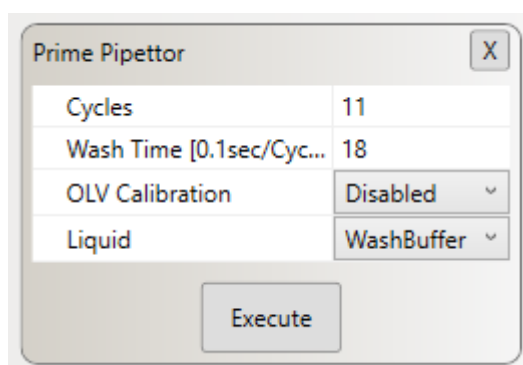
**CAUTION**

Check that the wash station does not overflow during rinsing of the SPOLV.

### 19.5.3 Rinsing with Wash Buffer

In the **Service software** at the navigation window, choose **System** and the **General** bracket.

- a. Click onto the **Liquid** button and choose the **Prime Pipettor** button to open the dedicated window.
- b. For initial rinsing, use the following values and start pressing the **Execute** button.
- Cycles: 11
  - Wash time: 18
  - OLV Calibration: Disabled
  - Liquid: Wash Buffer



- c. The process is successful when the **Processing** window closes without failure message.

**CAUTION**

Check that the wash station does not overflow during rinsing of the SPOLV.

## **19.6 Pipettor Tests**

### **19.6.1 LLD Endurance test**

For LLD Endurance test, see see Chapter 11.7.4.1 on page 11-49.

### **19.6.2 User LLD Endurance test**

For User LLD Endurance test, see Chapter 11.7.4.2 on page 11-50.

### **19.6.3 Pipettor test**

For Random movement test, see Chapter 11.7.4.3 on page 11-53.

## **19.7 Cuvette Endurance Run**

For Cuvette Endurance Run, see Chapter 11.8.1.2 on page 11-61.



## 19.8 Electric connections, switches and signals

This section shows an overview of the electronic boards of Left and Right Pipettor Arms.

### NOTE

see Chapter 3.1.7 on page 3-8 for an overview of the complete electric circuit of the system.

### 19.8.1 X-Sledge-Dual-Z EC CU PCBA board

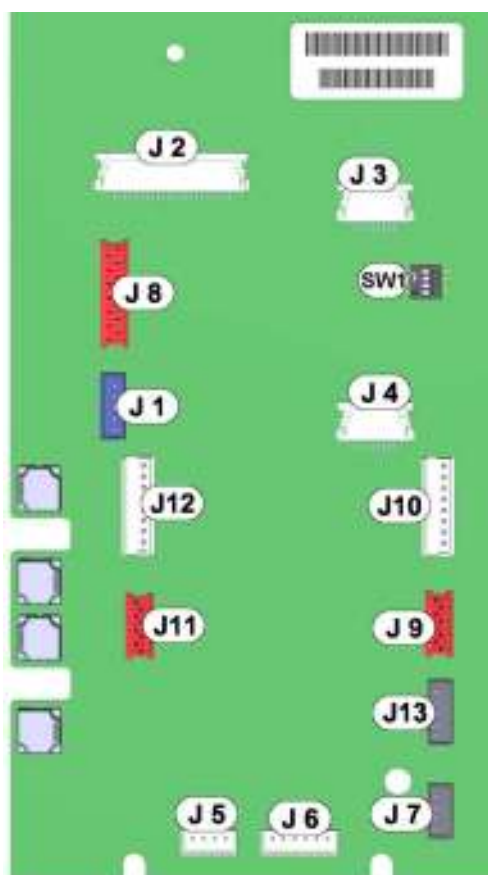


Figure 19–92: X-Sledge-Dual-Z EC CU PCBA board, front view

**19. Pipettor for internal-tanks configuration**

---

| Number | Connection            |
|--------|-----------------------|
| J1     | Download              |
| J2     | PWR / CAN             |
| J3     | Gripper CU PCBA       |
| J4     | Sample Pip EC CU PCBA |
| J5     | X-motor               |
| J6     | X-encoder             |
| J7     | not used              |
| J8     | Download (E1 SCTL)    |
| J9     | N/A                   |
| J10    | Gripper Z-motor       |
| J11    | N/A                   |
| J12    | Sample Pip Z-motor    |
| J13    | not used              |

*Table 19–9: Connections of the X-Sledge-Dual-Z EC CU PCBA board*

### 19.8.2 Gripper CU PCBA board

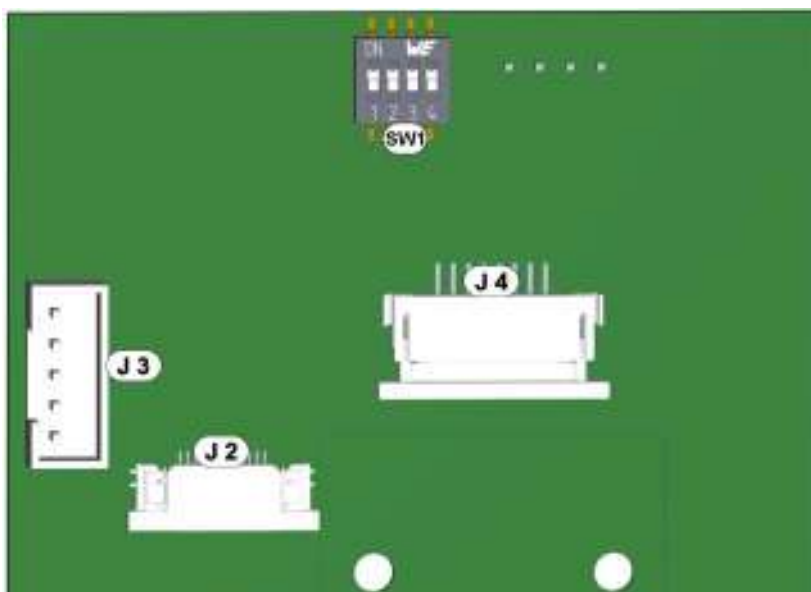


Figure 19–93: Gripper CU PCBA board, front view

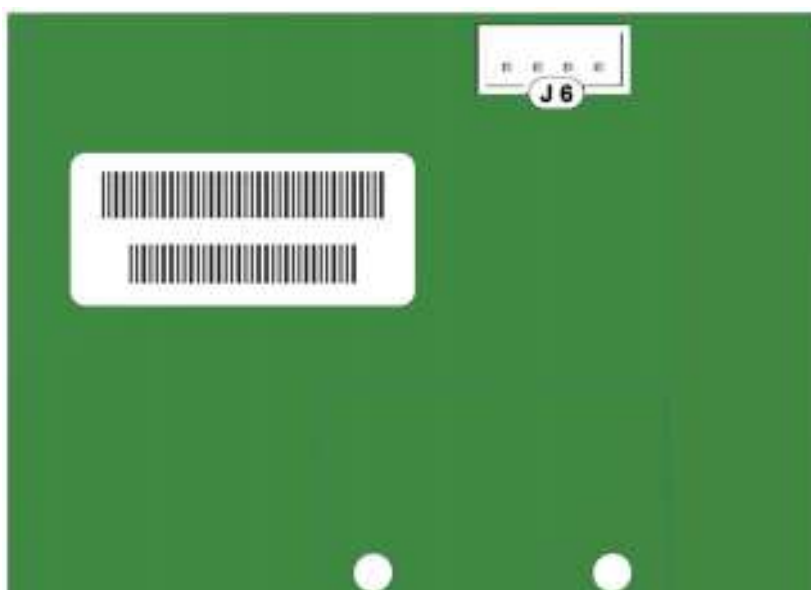


Figure 19–94: Gripper CU PCBA board, rear view

| Number | Connection       |
|--------|------------------|
| J2     | Front end Sensor |
| J3     | Down load        |
| J4     | PWR / CAN        |
| J6     | Gripper drive    |

Table 19–10: Connections of the Gripper CU PCBA board

## 19. Pipettor for internal-tanks configuration

| DIP-Switch |   |     | Meaning/Function/Default Switch Setting |
|------------|---|-----|-----------------------------------------|
| SW1        | 1 | OFF |                                         |
|            | 2 | OFF |                                         |
|            | 3 | OFF |                                         |
|            | 4 | OFF |                                         |

Table 19–11: DIP-Switches of the Gripper CU PCBA board

## 19.8.3 Gripper sensor PCBA board

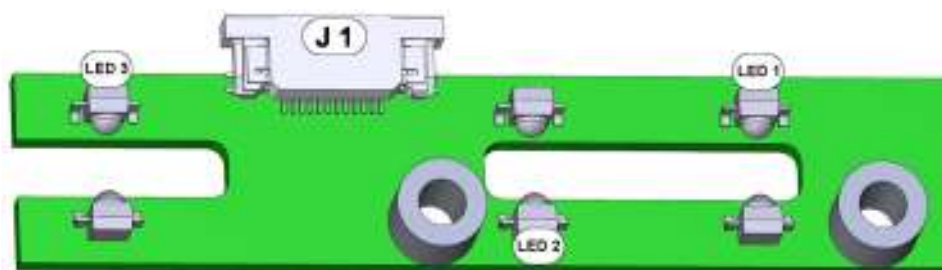


Figure 19–95: Gripper sensor PCBA board, front view

| Number | Connection                  |
|--------|-----------------------------|
| J1     | Front end sensor gripper CU |

Table 19–12: Connections of the Gripper sensor PCBA board

| Number   | Meaning/Function |
|----------|------------------|
| LED (Q1) |                  |
| LED (Q2) |                  |
| LED (Q3) |                  |

Table 19–13: LED of the Gripper sensor PCBA board

## 19.8.4 X-Z-Sledge EC CU PCBA board

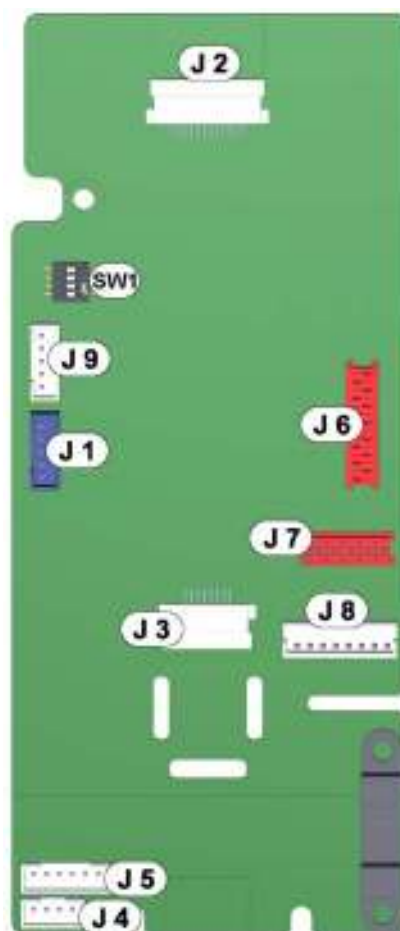


Figure 19–96: X-Z-Sledge EC CU PCBA board, front view

| Number | Connection          |
|--------|---------------------|
| J1     | Download            |
| J2     | PWR / CAN           |
| J3     | Connection to SPOLV |
| J4     | X-motor             |
| J5     | X-encoder           |
| J6     | Download (E1 SCTL)  |
| J7     | N/A                 |
| J8     | SPOLV Z-motor       |
| J9     | SPOLV Z-init        |

Table 19–14: Connections of the X-Z-Sledge EC CU PCBA board

## 19. Pipettor for internal-tanks configuration

---

| DIP-Switch |   |     | Meaning/Function/Default Switch Setting |
|------------|---|-----|-----------------------------------------|
| SW1        | 1 | OFF |                                         |
|            | 2 | OFF |                                         |
|            | 3 | OFF |                                         |
|            | 4 | OFF |                                         |

*Table 19–15: DIP-Switches of the X-Z-Sledge EC CU PCBA board*

## 20 Pipettor for external-tanks configuration

This chapter describes the Pipettor in detail. The first section of the chapter introduces “Safety” notes concerning Pipettor, followed by a brief overview of the main parts composing this module.

The following sections contain information about repair, calibration and other service activities (“Replacement and Repair”, “Adjustment and Calibration”, “Priming of Pipettor” and “Pipettor Tests” sections).

In the last section, layout and connections of X-Sledge-Dual-Z EC CU PCBA, Gripper CU PCBA, Gripper Sensor PCBA and X-Z-Sledge EC CU PCBA are described.

The pipettor consists of a left and a right arm. They share a common XY-axis but are equipped with individual Z-axis.

The Left Arm carries the Sample Pipettor and the Gripper; the Right Arm carries the Reagent Pipettor, a non-magnetic stainless steel probe called SPOLV.

- The Gripper is used to transport any cuvette from the cuvette trays to the Incubator, Washer and Reader modules. It grabs a cuvette from the top.
- The Gripper shaft is driven together with the Sample Pipettor through a combined Y-sledge.
- The Sample Pipettor and the Gripper have an independent Z-axis for independent Z-travel.
- The SPOLV acts independently from the Left Arm, as there are no interference zones between the two arms.

### 20.1 Safety

#### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 2.5 on page 2-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

## 20.2 Overview

This section contains an overview of the Pipettor components.

### 20.2.1 Pipettor Overview

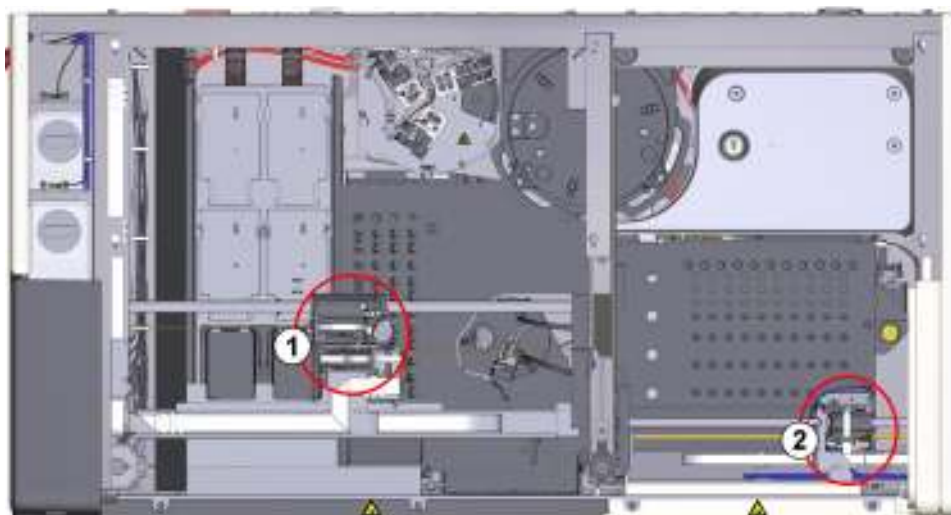


Figure 20–1: Instrument top view, without Top Covers

|    |                                     |
|----|-------------------------------------|
| 1) | Left Arm (Sample Pipettor, Gripper) |
| 2) | Right Arm (SPOLV)                   |

Table 20–1: Pipettor overview



### 20.2.2 Left Arm overview

This section shows an overview of the Left Arm components.

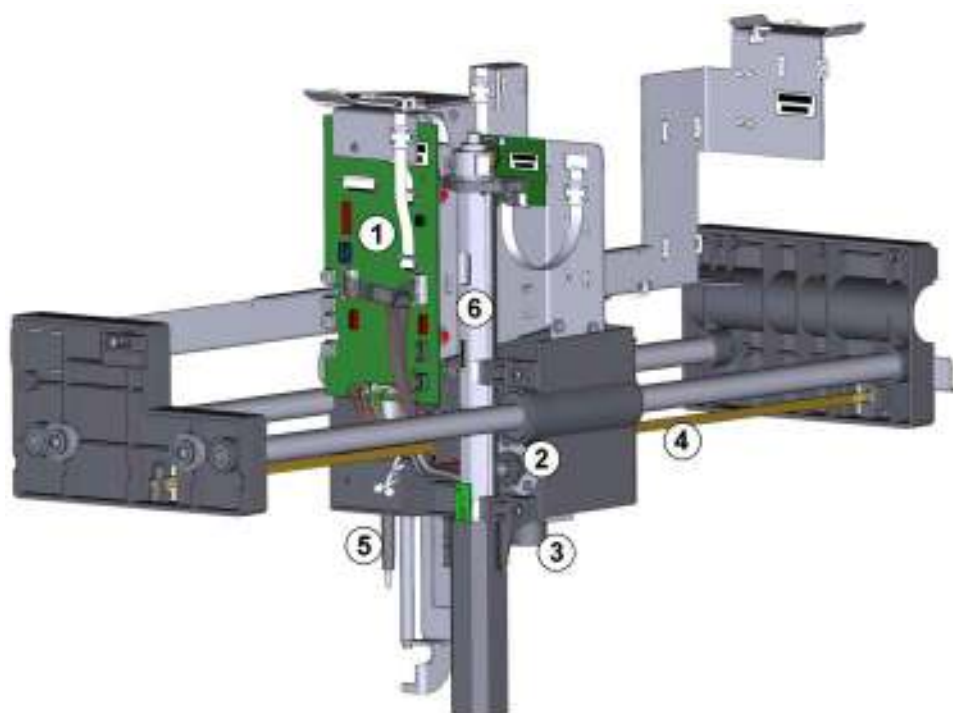


Figure 20–2: Left Arm overview

- |    |                            |
|----|----------------------------|
| 1) | X-Sledge-Dual-Z EC CU PCBA |
| 2) | Gripper Z-motor            |
| 3) | X-Sledge motor             |
| 4) | X-axis toothed belt        |
| 5) | Sample Pipettor            |
| 6) | Gripper                    |

Table 20–2: Left Arm overview

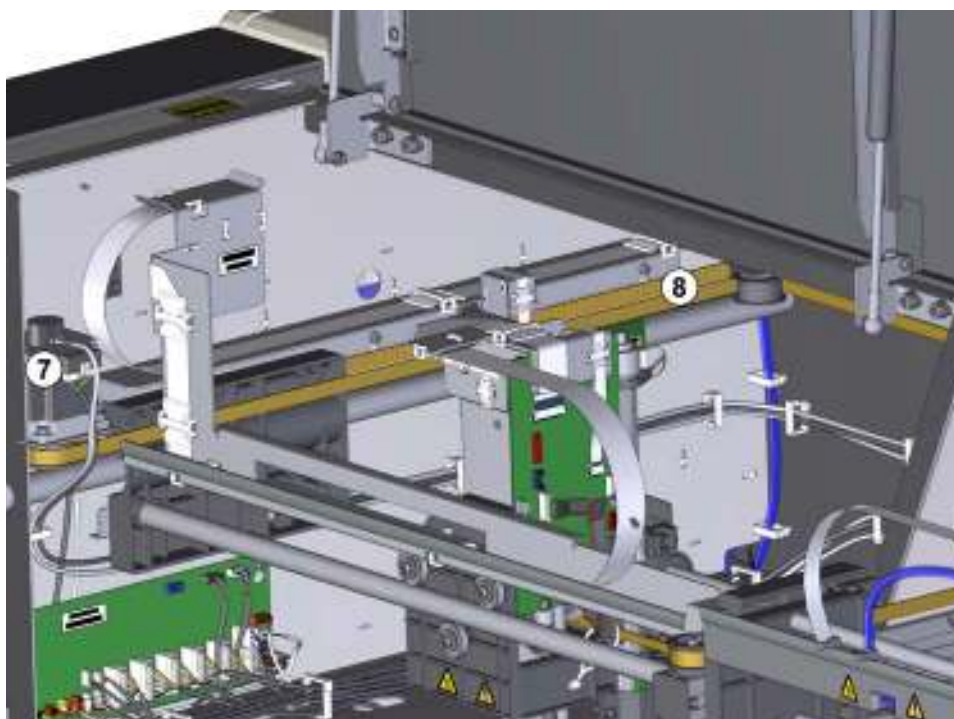


Figure 20-3: Left Arm overview

- |    |                     |
|----|---------------------|
| 7) | Y-Sledge motor      |
| 8) | Y-axis toothed belt |

Table 20-3: Left Arm overview

### 20.2.3 Right Arm overview

This section shows an overview of the Right Arm components.

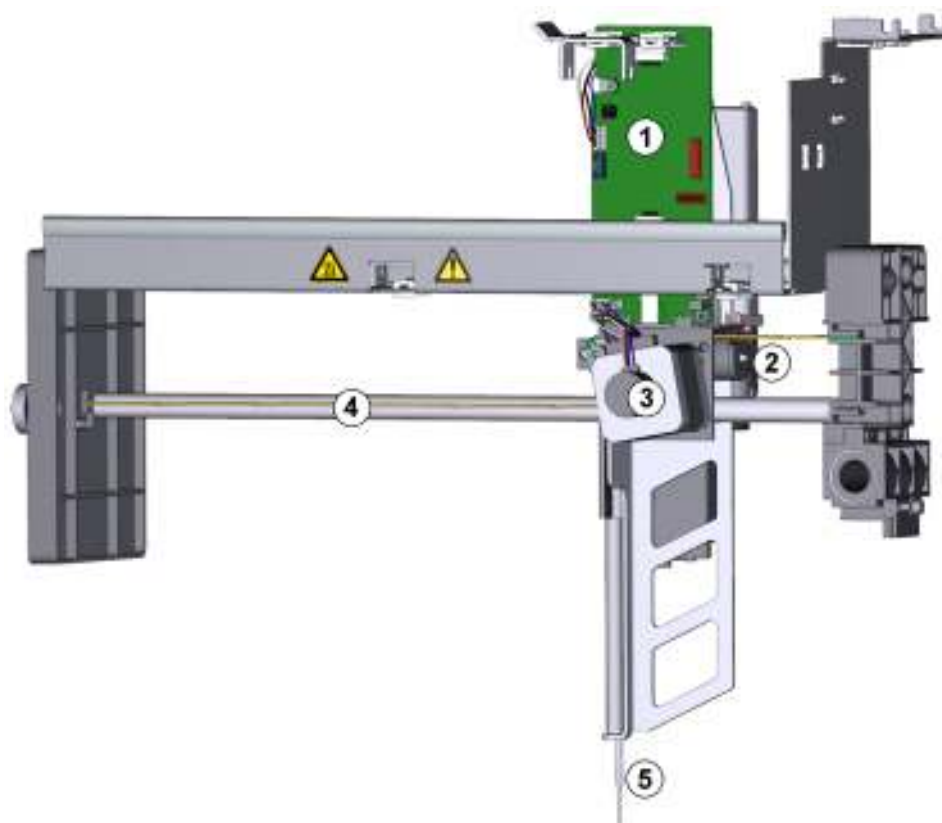
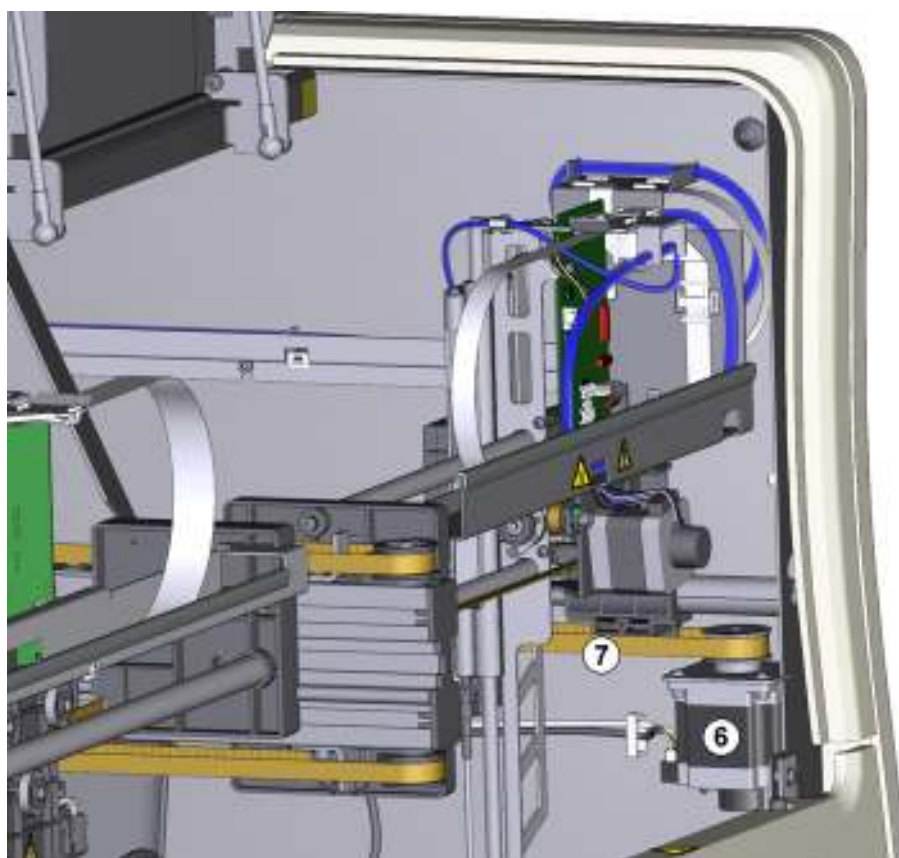


Figure 20–4: Right Arm overview

- |    |                       |
|----|-----------------------|
| 1) | X-Z-sledge EC CU PCBA |
| 2) | Z-motor               |
| 3) | X-sledge motor        |
| 4) | X-axis toothed belt   |
| 5) | SPOLV Needle          |

Table 20–4: Right Arm overview

## 20. Pipettor for external-tanks configuration

*Figure 20-5: Right Arm overview*

- |    |                     |
|----|---------------------|
| 6) | Y-sledge motor      |
| 7) | Y-axis toothed belt |

*Table 20-5: Right Arm overview*

### 20.2.4 Sample Pipettor overview

This section shows an overview of the Sample Pipettor components.



*Figure 20–6: Sample Pipettor overview*

- |    |                     |
|----|---------------------|
| 1) | ZS-PIP ECXS CU PCBA |
| 2) | Tip adapter         |
| 3) | Guiding rod         |

*Table 20–6: Sample Pipettor overview*

### 20.2.5 Gripper overview

This section shows an overview of the Gripper components.

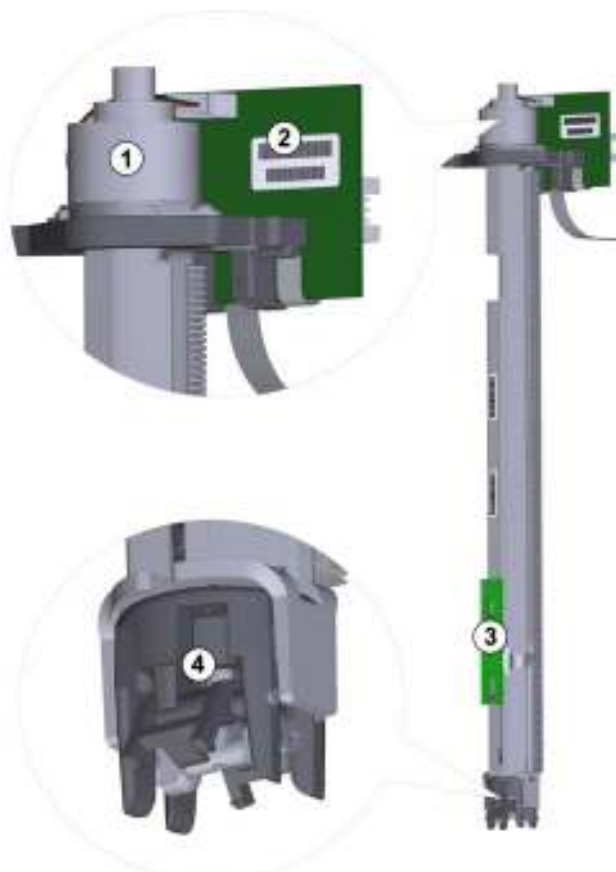


Figure 20–7: Gripper overview

- |    |                     |
|----|---------------------|
| 1) | Linear actuator     |
| 2) | Gripper CU PCBA     |
| 3) | Gripper Sensor PCBA |
| 4) | Finger claw         |

Table 20–7: Gripper overview

### 20.2.6 Reagent Pipettor overview

This section shows an overview of the Reagent Pipettor components.



Figure 20–8: Reagent Pipettor overview

|   |                                         |
|---|-----------------------------------------|
| 1 | Steel needle                            |
| 2 | Knurled screw (needle assembly)         |
| 3 | Fitting connection for PRP-SPOLV tubing |
| 4 | Motor                                   |
| 5 | SPOLV CU PCBA                           |

Table 20–8: Reagent Pipettor overview

## 20.3 Replacements and repairs

This section contains instructions for the repair of individual components of the Pipettor module. After several repair works, it is required that the individual components are re-adjusted. Follow the instructions in the individual sections.

### 20.3.1 Replacement of the Sample Pipettor

#### 20.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.

#### 20.3.1.2 Removal of the Sample Pipettor

1. Unlock the flat ribbon cables (connectors J2, J4 and J3).
2. Remove the cable holder (2).

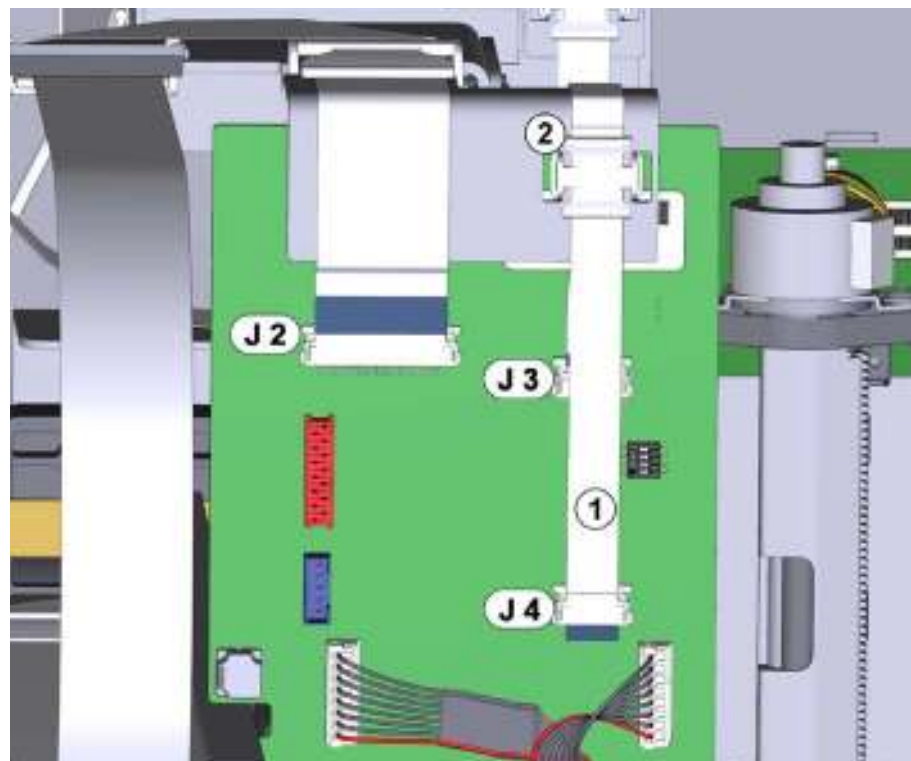


Figure 20–9: Left Arm, right side view



3. Remove the cable holder (2).

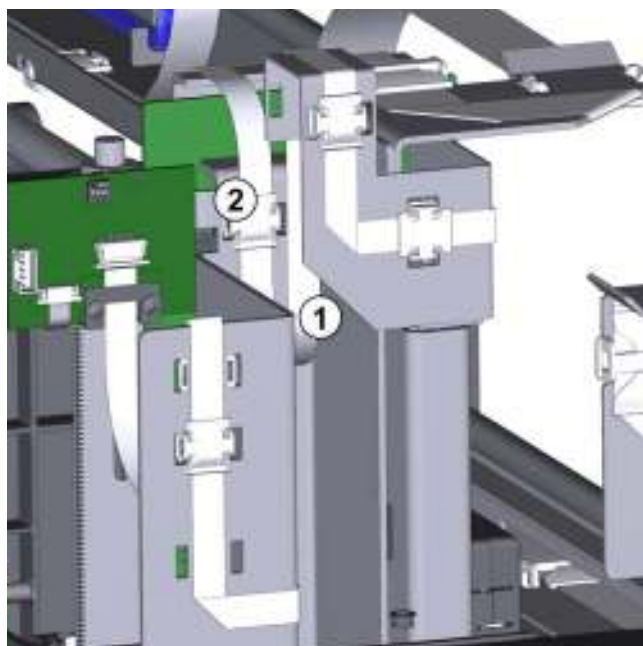


Figure 20-10: Left Arm, left side view

4. Remove the screw (4) on the rear of the X-Sledge-Dual-Z PCB.
5. Loosen the two screws (5) that anchor the top metal sheet (6).
6. Slide the top metal sheet (6) out, and set it aside to access the Sample Pipettor from top.

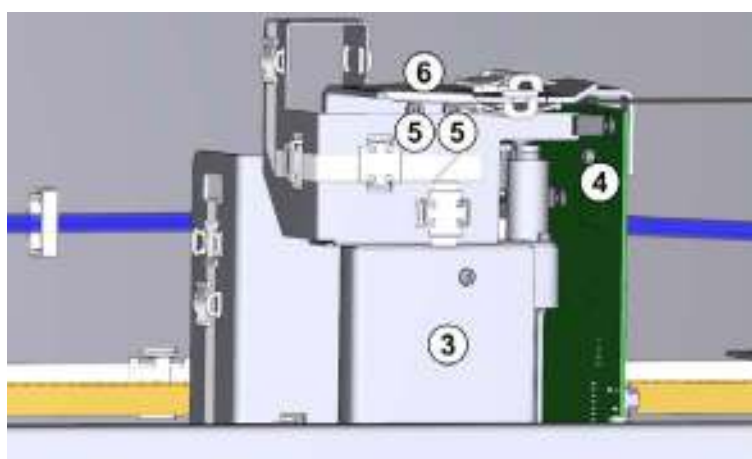
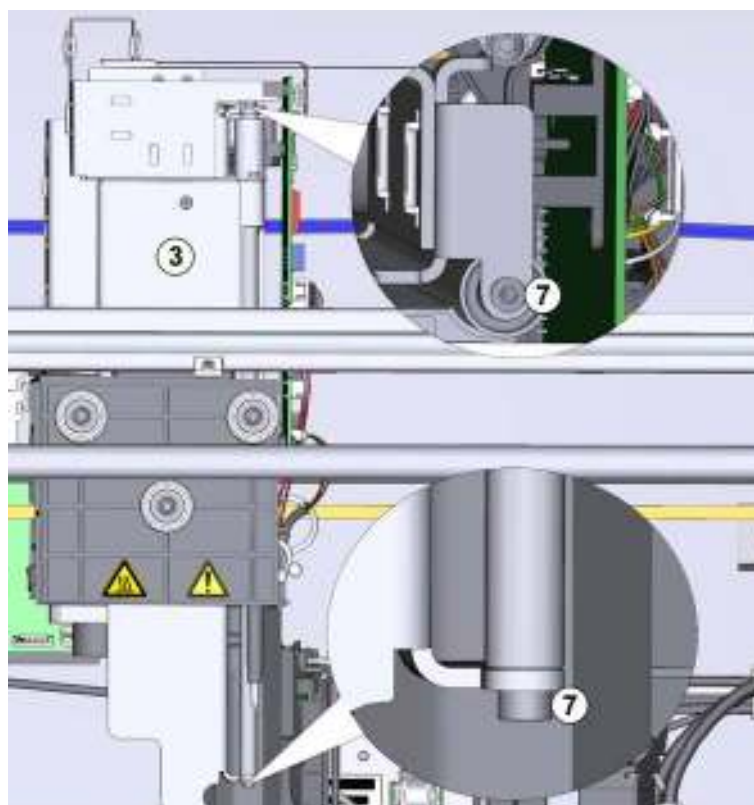


Figure 20-11: Left Arm, front view

7. Unscrew the top and bottom screws (7) holding on to the Sample Pipettor (3).
8. Slide the Sample Pipettor (3) out of the guide and lift it up until it can be removed. Be aware of the bearing, it may slip out!

## 20. Pipettor for external-tanks configuration



*Figure 20-12: Sample Pipettor, front view*

9. Remove the five cable holders (9) to completely disassemble the routing of the Sample Pipettor flat ribbon cable.
10. Loosen the three screws (10) that anchor the cable support (11) to the Sample Pipettor.
11. Slide the cable support (11) out.

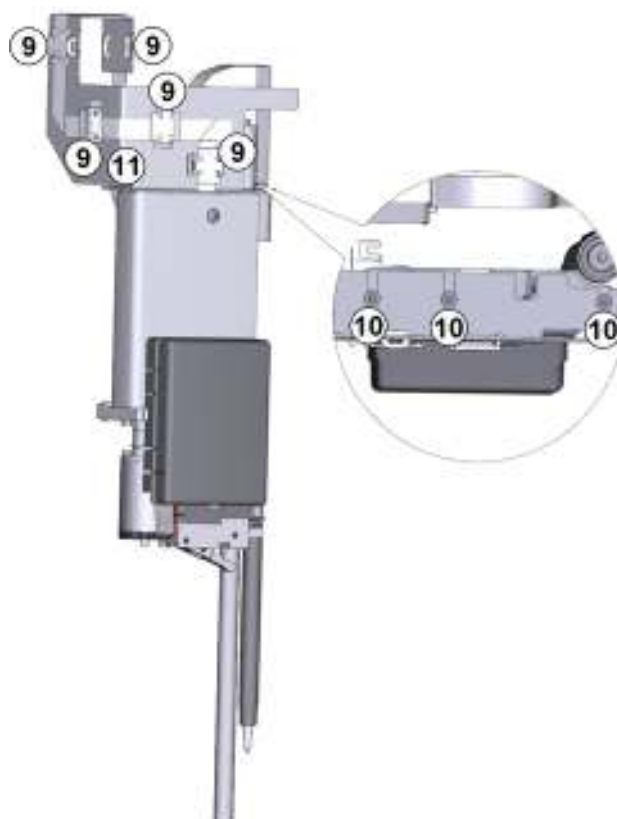
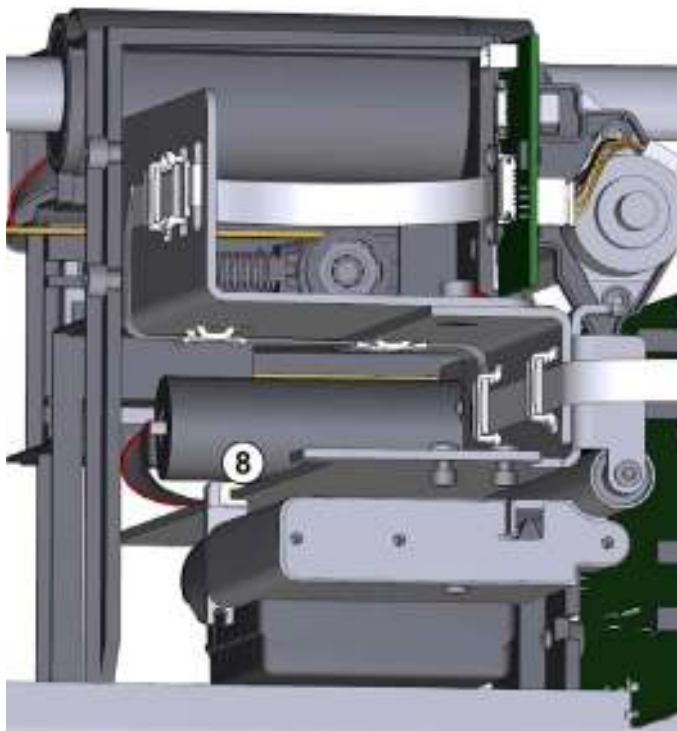


Figure 20–13: Sample Pipettor, side view

#### 20.3.1.3 Assembly of the Sample Pipettor

1. In case a new module has to be installed, check the proper setting of DIP-Switches onto the Sample Pipettor PCB (1000).
2. Lay the flat ribbon cable on the Sample Pipettor upper surface.
3. Place the cable support (11) on the Sample Pipettor upper surface.
4. Fasten the three screws (10) to lock the cable support.
5. Assemble the routing of the flat ribbon cable along the support, and lock it with the five cable holders (9).
6. Set the Z-guiding rod of the Sample Pipettor (3) into the dedicated position and fasten with the two screws (7). The bearing (8) must sit in the Z-bracket!



*Figure 20–14: Sample Pipettor alignment*

7. Slide the top metal sheet (6) into its dedicated position, and lock it fastening the two screws (5).
8. Screw the screw (4) to the rear of the X-Sledge-Dual-Z PCB.
9. Assemble the routing of Gripper and Sample Pipettor flat ribbon cables along the top metal sheet, and lock them with the two cable holders (2).
10. Lock the flat ribbon cables into the dedicated connectors (J2, J3 and J4).

#### **20.3.1.4 Set of the Sample Pipettor**

1. In case a new module has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

## 20.3.2 Replacement of the Dual Sledge CU PCB

### 20.3.2.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.

### 20.3.2.2 Removal of the Dual Sledge CU PCB

1. Unplug all the cables from the Dual Sledge CU PCB (1), see Chapter 20.8.1 on page 20-84.

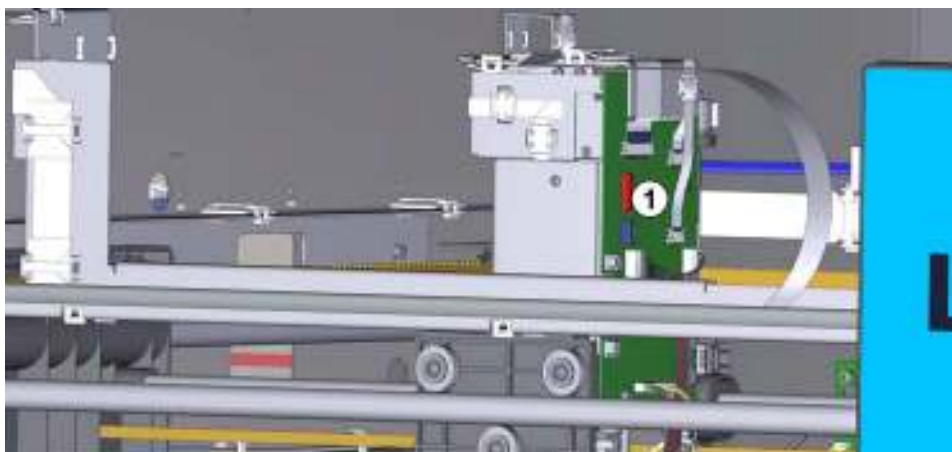


Figure 20–15: X-Sledge-Dual-Z CU PCB

## 20. Pipettor for external-tanks configuration

---

2. Unscrew the screw (2) at the rear side of the Dual Sledge CU PCB (1).
3. Loosen the two screws (3), lift up the top metal sheet and set it aside.

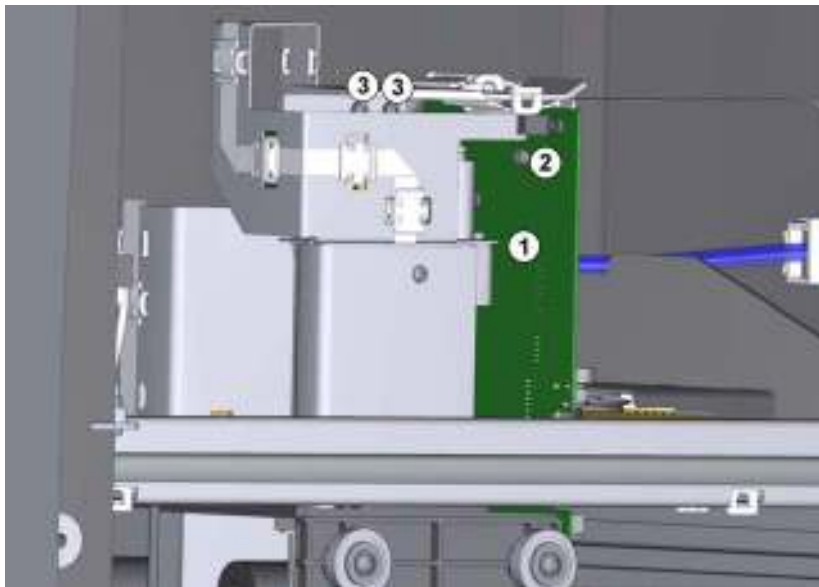


Figure 20–16: X-Sledge-Dual-Z CU PCB, rear view

4. Loosen the two screws (4) and slide out the Dual Sledge CU PCB (1).

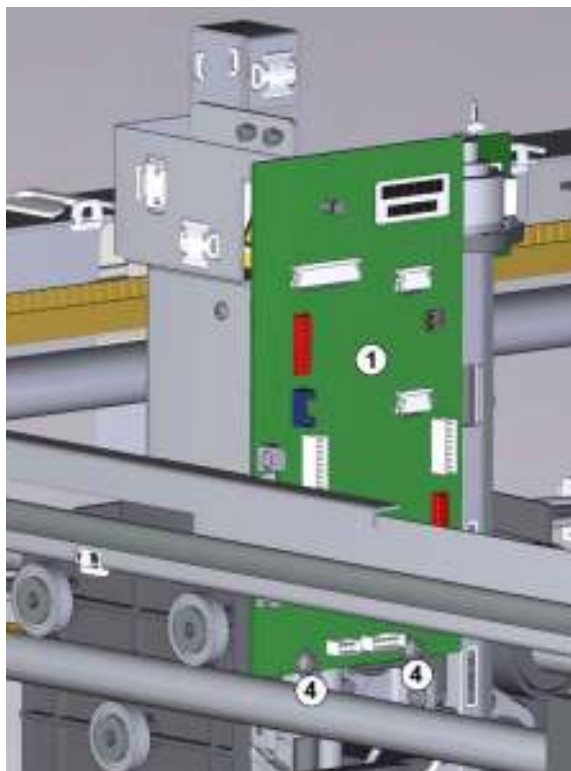


Figure 20–17: X-Sledge-Dual-Z CU PCB, front view

**20.3.2.3 Assembly of the Dual Sledge CU PCB**

1. Set the Dual Sledge CU PCB (1) into place and fasten the two screws (4) at the bottom and the screw (2) at the rear side.
2. Plug all the cables onto the Dual Sledge CU PCB, see Chapter 20.8.1 on page 20-84.

**20.3.2.4 Set of the Dual Sledge CU PCB**

1. In case a new PCB has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

Dip switch setting:

| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | OFF                                     |
|            | 2 | OFF                                     |
|            | 3 | OFF                                     |
|            | 4 | OFF                                     |

*Figure 20–18: DIP-Switches of the PCB, Dual Sledge CU*



### 20.3.3 Replacement of the Gripper Sensor PCB

#### 20.3.3.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Cuvette-Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
6. Open the left top cover and approach the left pipettor arm.

#### 20.3.3.2 Removal of the Gripper Sensor PCB

1. Place the gripper over the empty space.
2. Remove the screw (2) and slide down the protective barrel (1) to remove it.

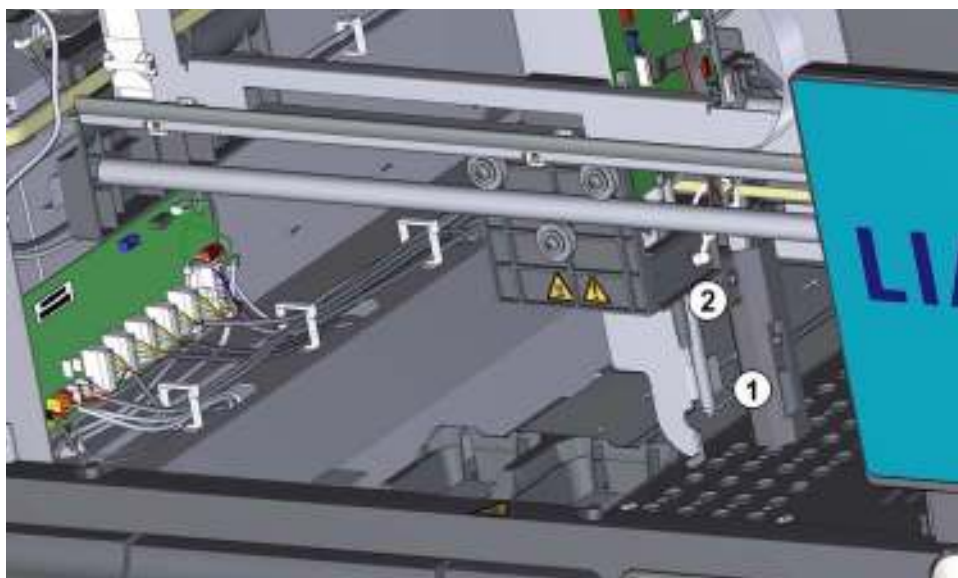


Figure 20–19: Pipettor Left Arm



3. Unplug the flat ribbon cable from the connector (J1) of Gripper Sensor PCB (3).
4. Unscrew the two screws (4).

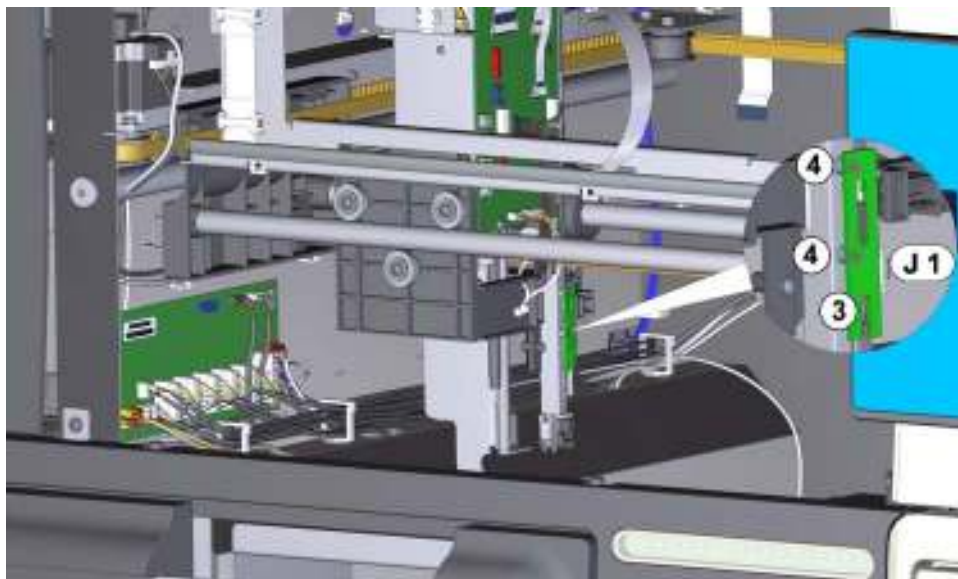


Figure 20–20: Gripper Sensor PCBA, detail

#### 20.3.3.3 Assembly of the Gripper Sensor PCB

1. Set the Gripper Sensor PCB (3) in with the steel spacer and fasten with the two screws (4).
2. Lock the flat ribbon cable into the connector (J1) of Gripper Sensor PCB.
3. Set the protective barrel (1) onto the Gripper and fasten the screw (2).

#### 20.3.3.4 Set the Gripper Sensor PCB

1. Assemble the Cuvette-Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.4 Replacement of the Gripper module

#### 20.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Cuvette-Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
6. Open the left top cover and approach the left pipettor arm.

#### 20.3.4.2 Removal of the Gripper module

1. Place the Gripper over the empty space.
2. Remove the screw (2) and slide down the protective barrel (1) to remove it.

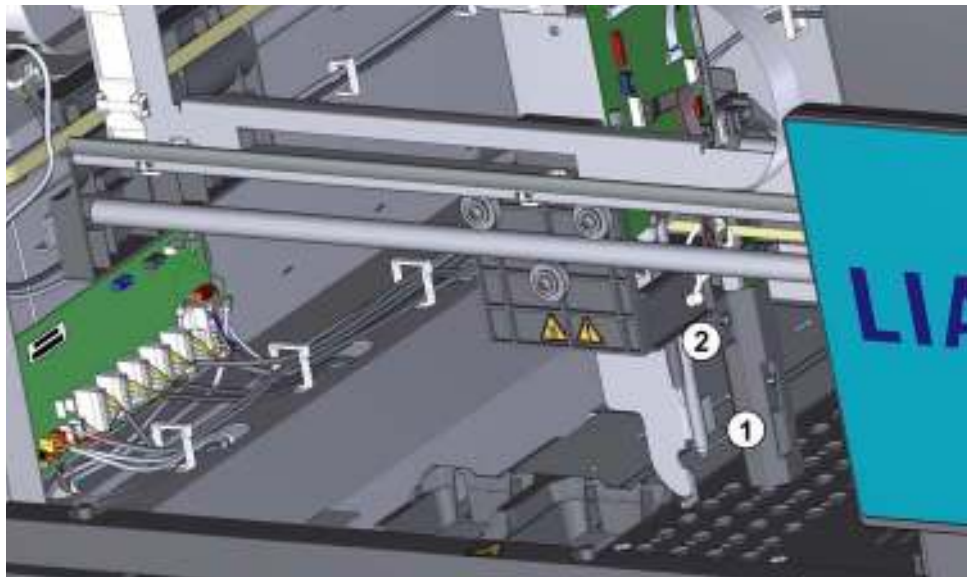


Figure 20–21: Pipettor Left Arm

3. Remove the two screws (4) from the cable holder.
4. Open the connector (5) and pull the flat ribbon cable (6) carefully out.

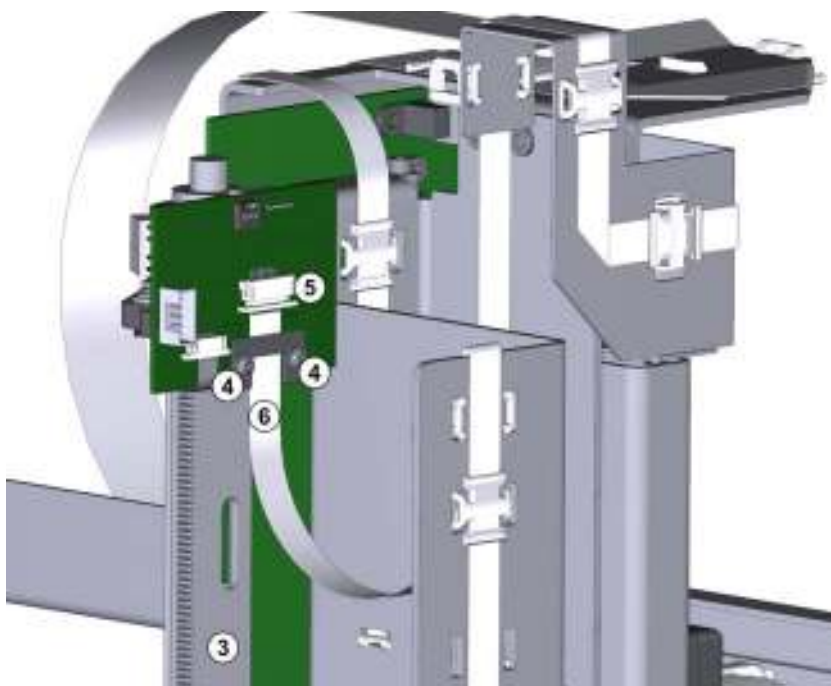


Figure 20-22: Gripper, rear view

5. Hold on to the gripper rack (3), loosen the screws (7) at the clamping levers, then lift the gripper rack (3) carefully out.
6. Watch for the slide bearings (8), they may fall down.

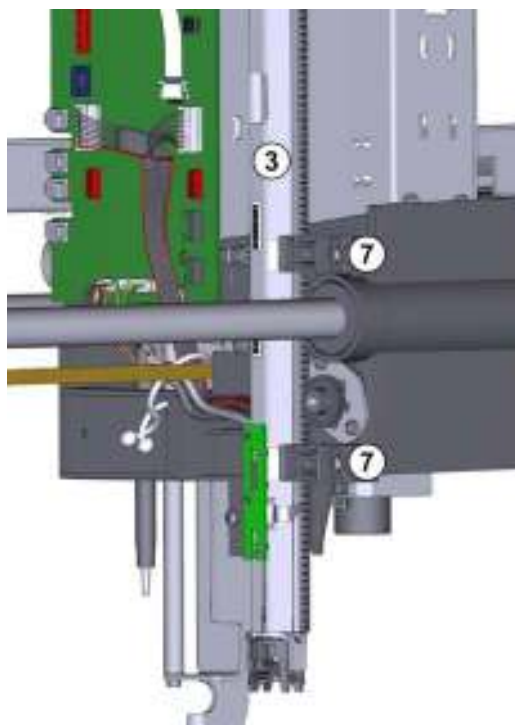


Figure 20-23: Gripper, left rear side view

#### 20.3.4.3 Assembly of the Gripper module

7. Slide the Gripper carefully in between the slide bearings (8) and clamping levers (9) and slide the toothed rack side into the toothed wheel (10).

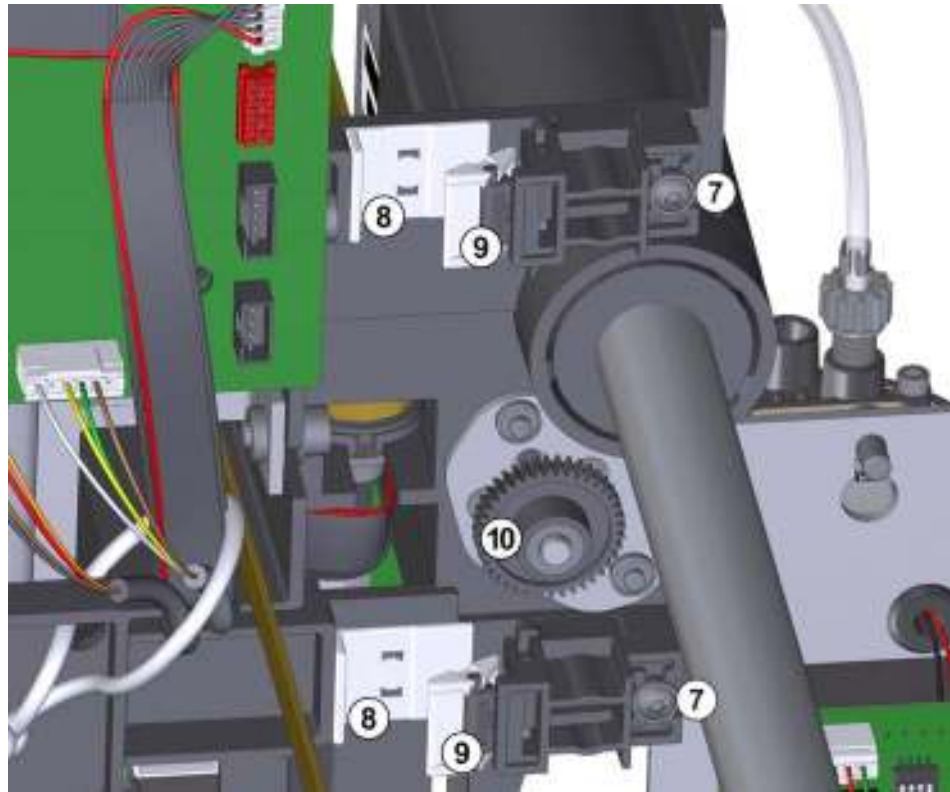


Figure 20–24: Gripper module, sliding bearings

8. Fasten the two screws (7) to clamp the gripper rack (3).
9. Insert the flat ribbon cable (6) under the cable holder.
10. Lock the flat ribbon cable (6) into the connector (5).
11. Fasten the two screws (4).
12. Set the protective barrel (1) onto the Gripper and fasten the screw (2).
13. Assemble the Cuvette-Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.

#### 20.3.4.4 Set of the Gripper module

1. In case a new module has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
3. In case a new module has been installed, calibrate the open position of the gripper claws, see Chapter 20.4.2.1 on page 20-79.
4. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.5 Replacement of the Gripper CU PCB

#### 20.3.5.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.
6. Remove the Gripper Module, see Chapter 20.3.4.2 on page 20-20.

#### 20.3.5.2 Removal of the Gripper CU PCB

1. Unplug the motor cable at slot (J6) on the rear side of the Gripper CU PCB (1).

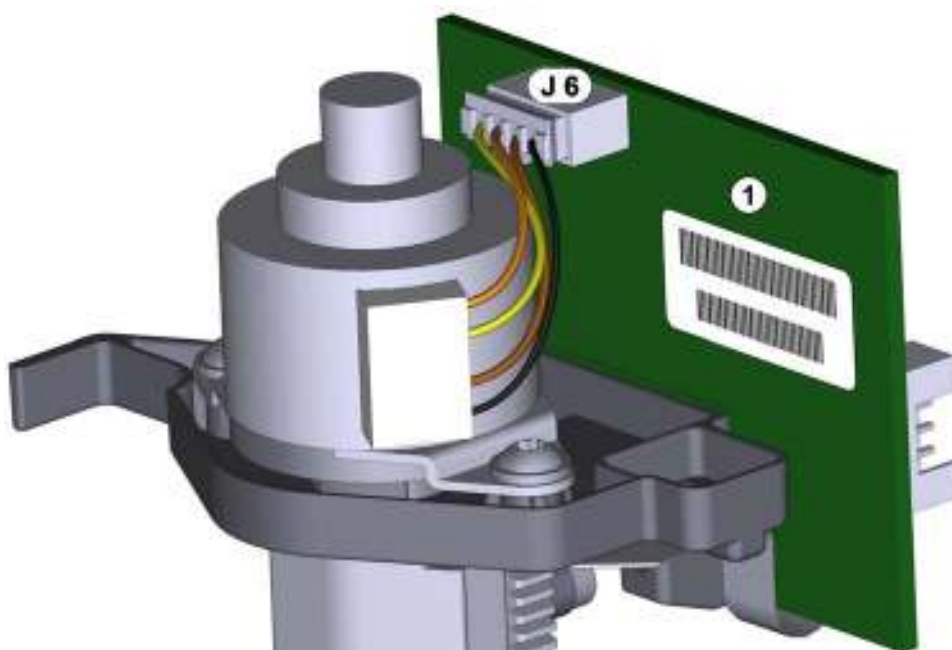


Figure 20–25: Gripper CU PCB, rear view

2. Disconnect the flat ribbon cable at slot (J2) of the Gripper CU PCB (1).
3. Remove the two screws (2).

## 20. Pipettor for external-tanks configuration

4. Lift the Gripper CU PCB (1) out.

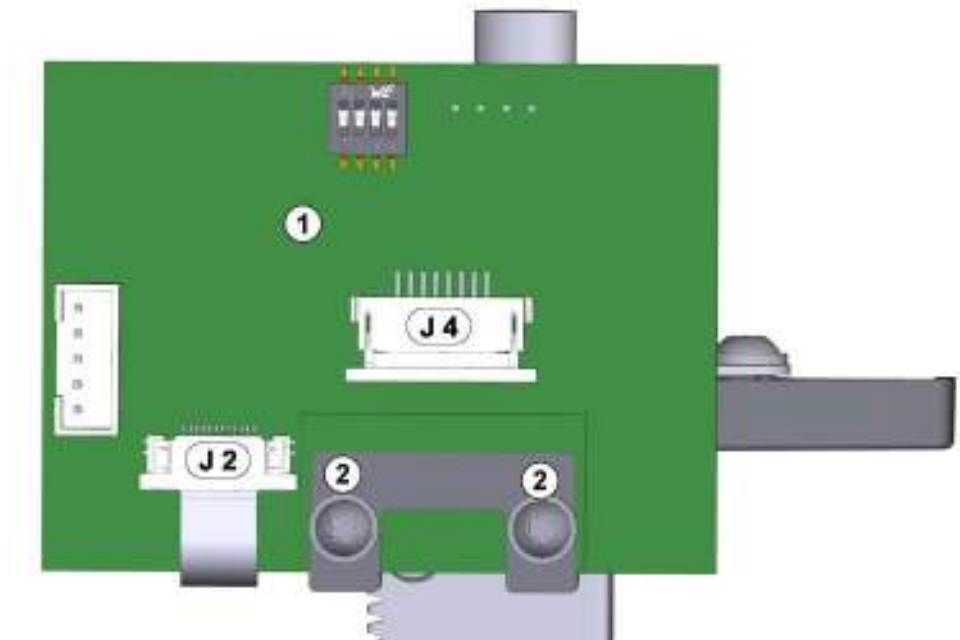


Figure 20–26: Gripper CU PCBA, front view

### 20.3.5.3 Assembly of the Gripper CU PCB

1. Set the Gripper CU PCB (1) into the holder and screw in the two screws (2).
2. Lock the flat ribbon cable into the sensor connector (J2).
3. Plug the motor cable into the connector (J6) on the rear side of the Gripper CU PCB (1).
4. Assemble the Gripper module, see Chapter 20.3.4.3 on page 20-22.

### 20.3.5.4 Set of the Gripper CU PCB

1. In case a new PCB has been installed, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 20.3.6 Replacement of the Gripper slide bearings

**NOTE**

The Gripper slide bearings are an add-on of yearly Preventive Maintenance kit.

The Field Service Engineer shall evaluate whether their replacement is needed or not, see Chapter 9.3.1 on page 9-8.

#### 20.3.6.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.
6. Remove the Gripper Module, see Chapter 20.3.4.2 on page 20-20.

#### 20.3.6.2 Removal of the Gripper slide bearings

1. Unhook the sliding plate (1) of the rocker clamp (2).
2. Remove the L-shaped sliding plate (3).

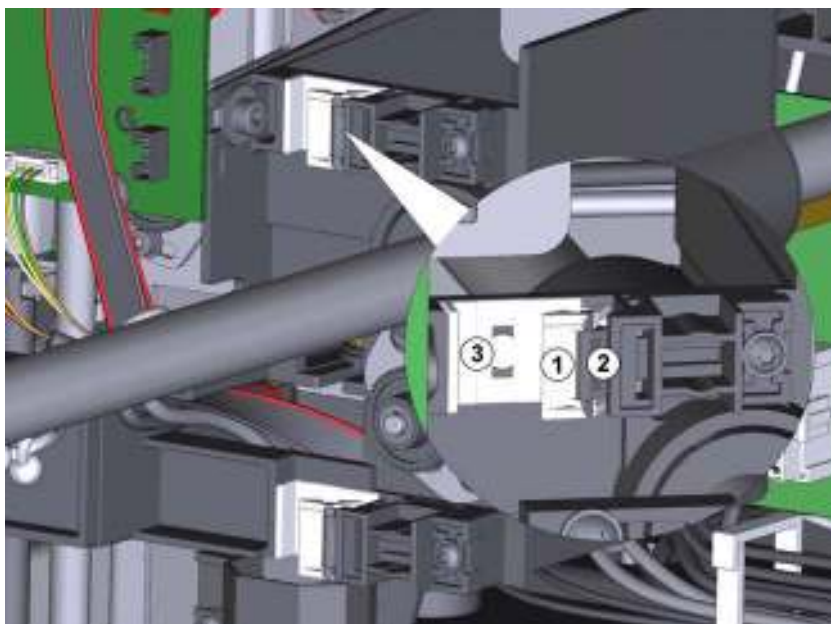


Figure 20–27: Sliding plate

#### 20.3.6.3 Assembly of the Gripper slide bearings

1. Insert the sliding plate (1) into the rocker clamp (2).
2. Set the L-shaped sliding plate in.

## **20. Pipettor for external-tanks configuration**

---

3. Assemble the gripper module, see Chapter 20.3.4.3 on page 20-22.

### **20.3.6.4 Set of the Gripper slide bearings**

1. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 20.3.7 Replacement of the Sample Pipettor Z-Motor

#### 20.3.7.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.
6. Remove the Sample Pipettor, see Chapter 20.3.1.2 on page 20-10.
7. Remove the Gripper, see Chapter 20.3.4.2 on page 20-20.
8. Remove the Dual Sledge CU PCB, see Chapter 20.3.2.2 on page 20-15.

#### 20.3.7.2 Removal of the Sample Pipettor Z-Motor

**CAUTION**

Do not open or remove screws marked with locking lacquer.

1. Unscrew one screw (1) of the Z-bracket Assembly (2).

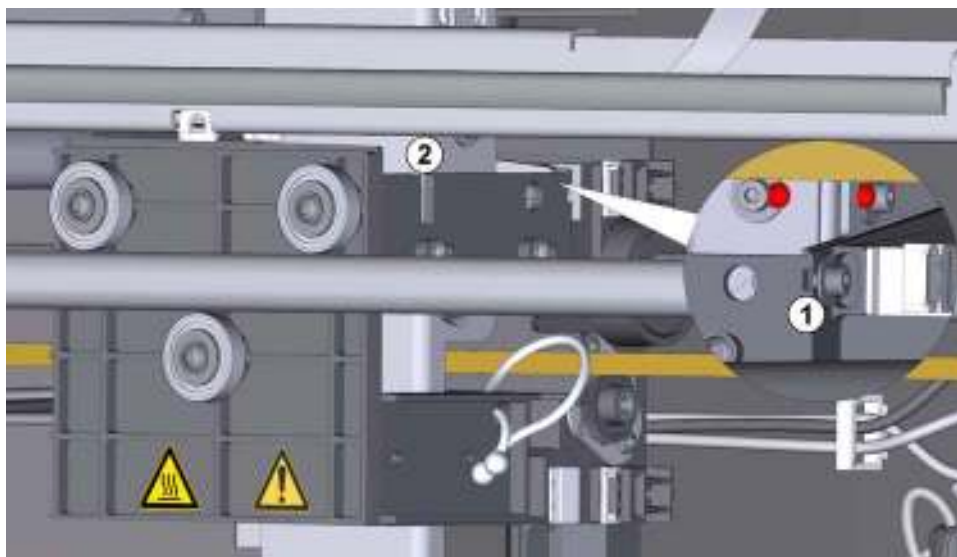
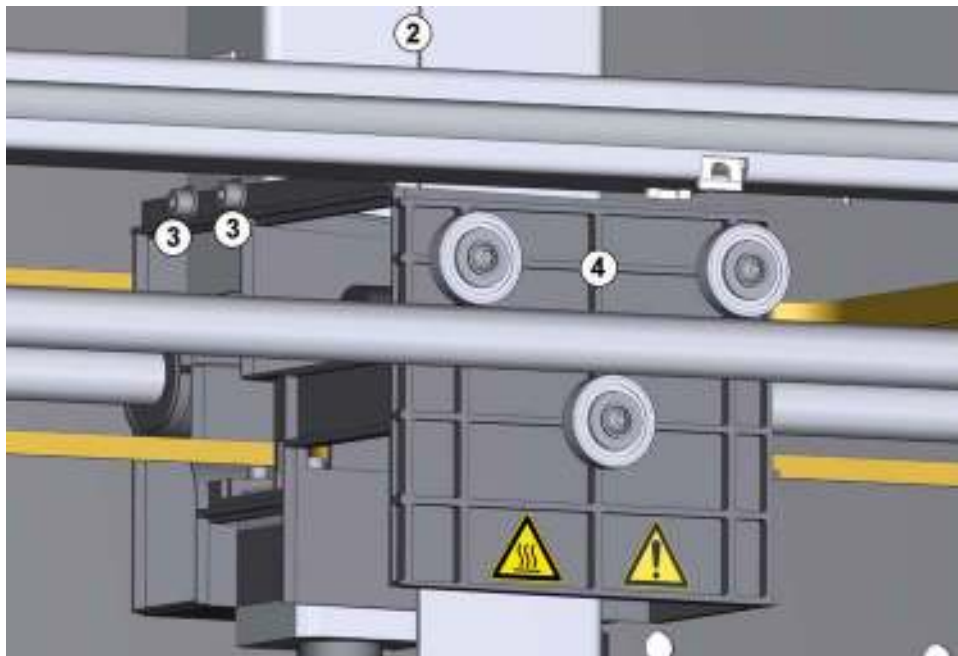


Figure 20–28: Z-bracket with Z-motor

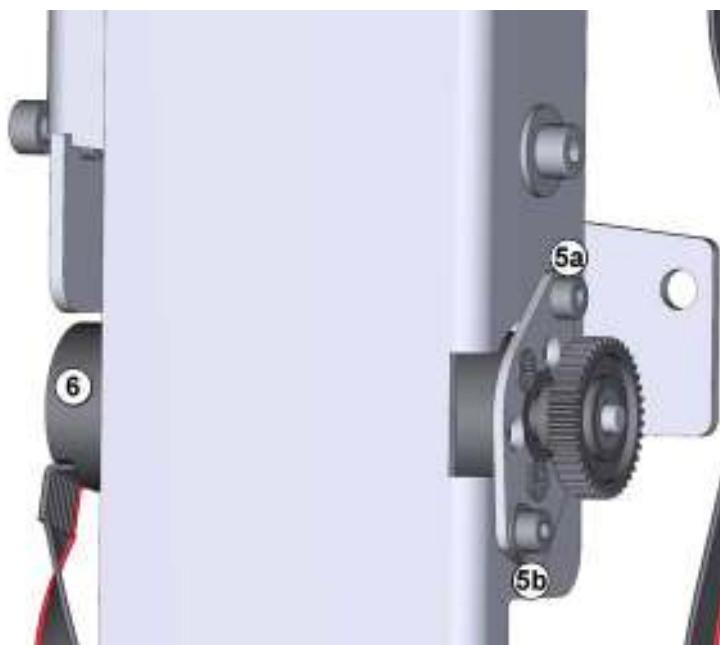
## 20. Pipettor for external-tanks configuration

2. Loosen the two screws (3) of the Z-bracket assembly (2) and lift the Z-bracket assembly carefully out of the XZ-sledge (4).



*Figure 20-29: Removal of Z-bracket assembly from XZ-sledge*

3. Remove two screws (5a and 5b) and slide the Z-motor (6) carefully out.



*Figure 20-30: Removal of Z-motor from Z-bracket assembly*

**20.3.7.3 Assembly of the Sample Pipettor Z-Motor**

1. Set the new Z-motor (6) into the Z-bracket assembly (2) and check for the correct position of the cable outlet. Set the Z-motor loose with one screw (5a) and with one screw and washer (5b).
2. Slide the Z-bracket assembly (2) into the XZ-sledge (4) from the top.
3. Fasten the Z-bracket assembly with one screw and washer (1) at the right side and with the two screws and washers (3) at the left side.
4. Assemble the Sample Pipettor, see Chapter 20.3.1.3 on page 20-13.
5. Adjust the clearance between the Z-motor gear and the Sample Pipettor toothed bar.
6. Fasten two screws (2) of Z-motor (6).
7. Assemble the PCB Dual Sledge CU, see Chapter 20.3.2.3 on page 20-17.
8. Assemble the gripper, see Chapter 20.3.4.3 on page 20-22.

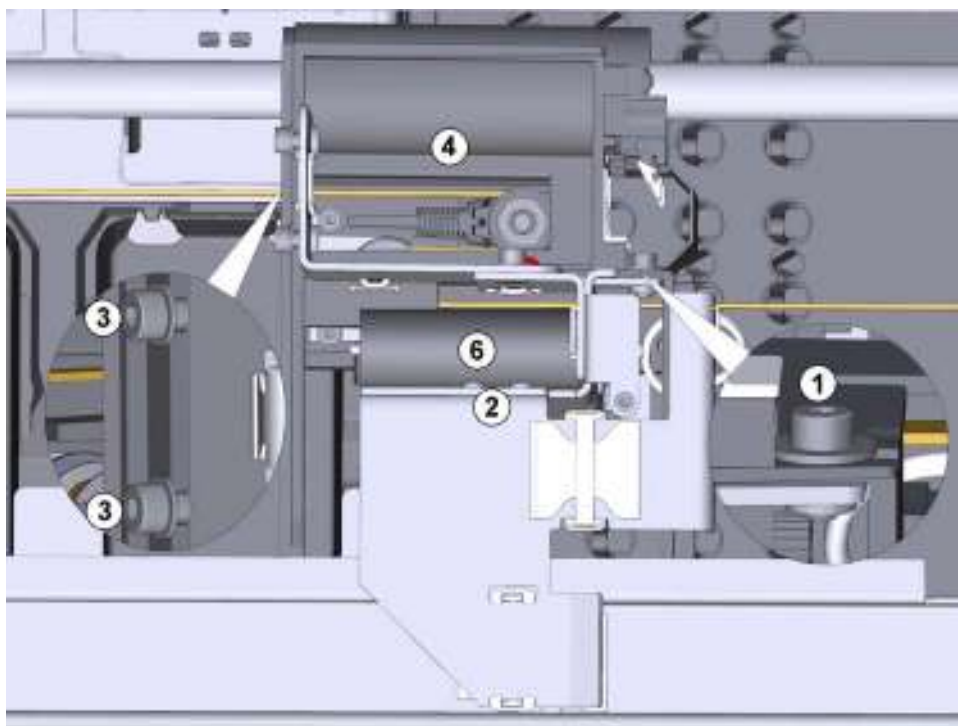


Figure 20-31: Z-bracket assembly, front and top view

**20.3.7.4 Set of the Sample Pipettor Z-Motor**

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### **20.3.8 Replacement of the Gripper Z-Motor**

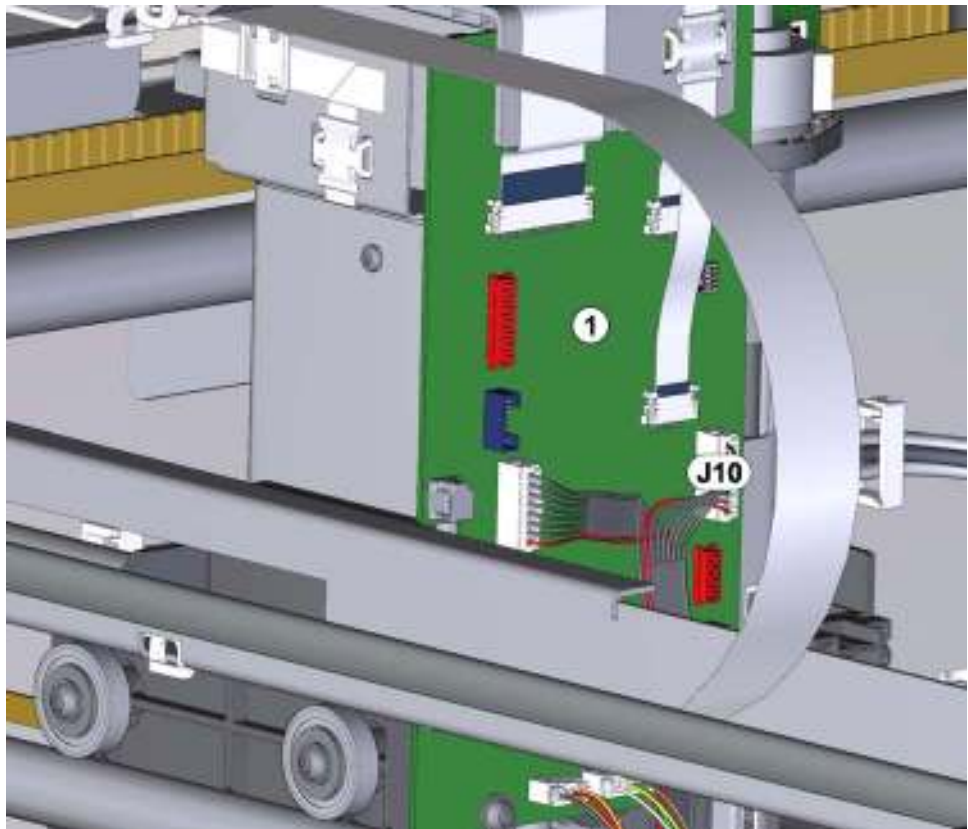
#### **20.3.8.1 Preliminary steps**



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm
6. Remove the Gripper, see Chapter 20.3.4.2 on page 20-20.

#### **20.3.8.2 Removal of the Gripper Z-Motor**

1. Unplug the motor cable at slot (J10) of Dual Sledge CU PCB (1).

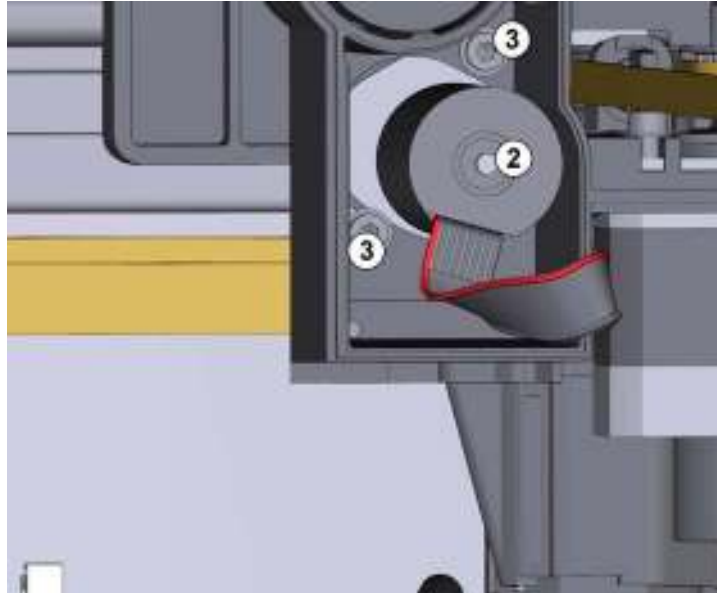


*Figure 20–32: PCB, Dual Sledge CU unplug Gripper Z-Motor*

**20. Pipettor for external-tanks configuration**

---

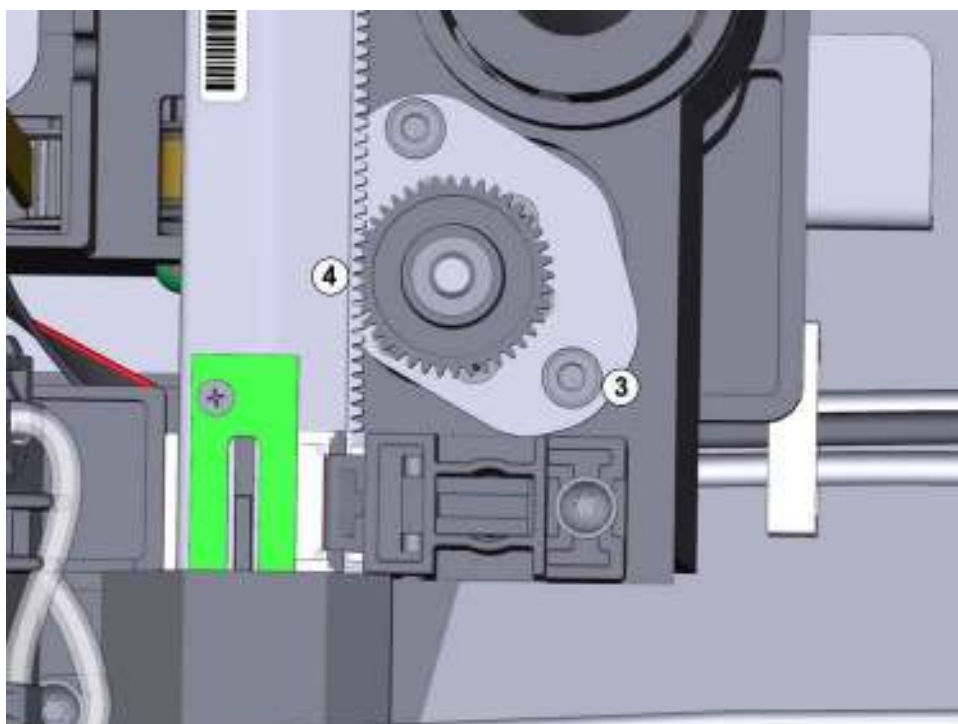
2. Unscrew two screws (3) and remove the Z-motor. Pull the motor cable carefully out.



*Figure 20–33: Gripper Z-Motor*

**20.3.8.3 Assembly of the Gripper Z-Motor**

1. Insert the new Z-Motor (2).
2. Set the Z-Motor loose with two screws (3).
3. Assemble the gripper, see Chapter 20.3.4.3 on page 20-22.
4. Adjust the clearance of the Z-Motor gear and the toothed bar (4).
5. Fasten the two screws (3).
6. Connect the motor cable onto slot (J10) of Dual Sledge CU PCB (1).



*Figure 20-34: Gripper Z-Motor gear clearance*

**20.3.8.4 Set the Gripper**

1. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 20.3.9 Replacement of the X-axis toothed belt of Left Pipettor Arm

#### 20.3.9.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.

#### 20.3.9.2 Removal of the X-Axis toothed belt of Left Pipettor Arm

1. Loosen the fixing screw (2) of guide pulley, push it towards the spring and fasten again the screw.

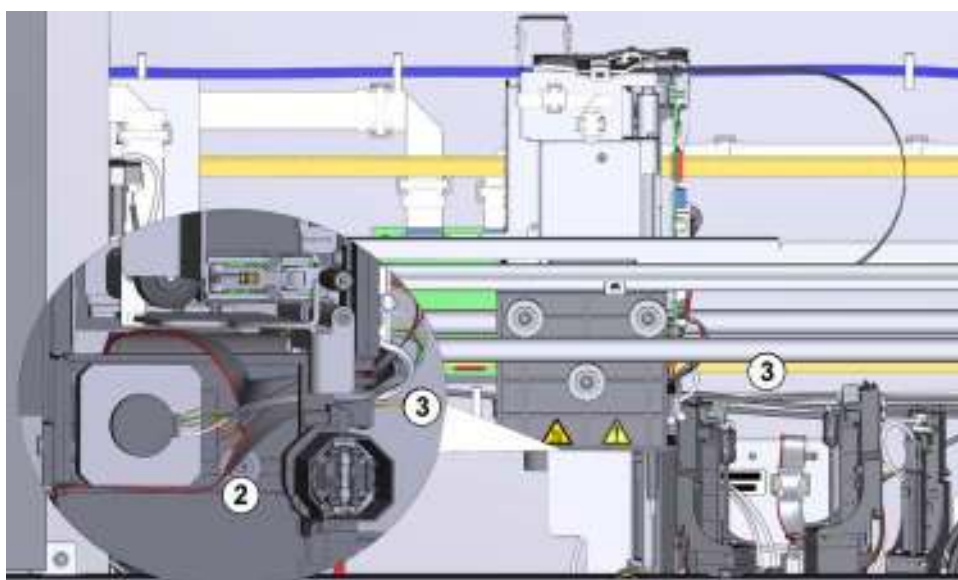


Figure 20–35: Left Pipettor arm, zoom window shows bottom view

## 20. Pipettor for external-tanks configuration

2. Unhook both the belt holders (3) of X-Axis toothed belt (1) at left (4) and right (5) Y-Sledge of the left arm.

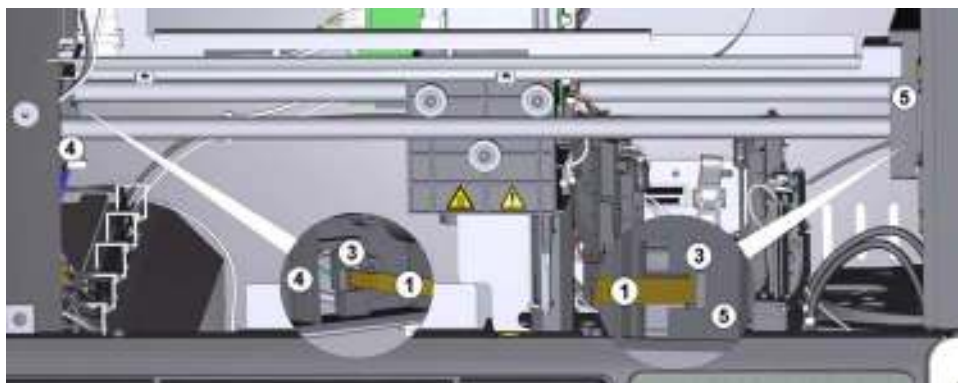


Figure 20-36: X-Axis toothed belt suspension

### 20.3.9.3 Assembly of the X-Axis toothed belt of Left Pipettor Arm

1. Check the toothed belt length, and if necessary cut it to a length of 770mm (the toothed belt has to be cut only along vales).
2. Thread the toothed belt (1) through the guide pulley (6) and the motor pulley (7).

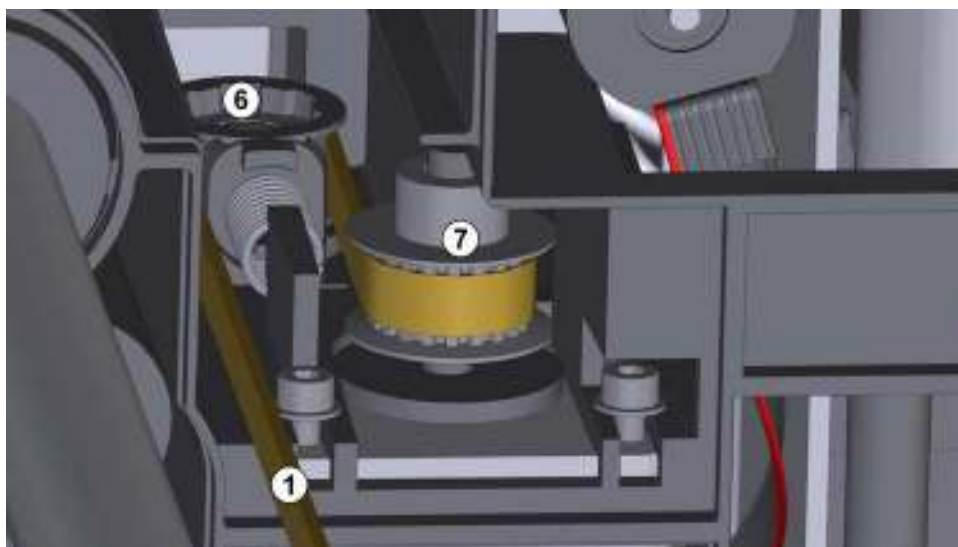


Figure 20-37: Guide pulleys of X-axis toothed belt

3. Insert the toothed belt into the belt holders (3) of X-Axis toothed belt (1).



Figure 20-38: Toothed belt holders



**NOTE****Belt adjustment**

By moving a tooth Inwards or outwards, the tension of the toothed belt can be adjusted.

---

4. Insert both the belt holders (3) at left (4) and right (5) Y-Sledge.
5. Slide the belt holders into place.
6. Loosen the fixing screw (2) of guide pulley, adjust the belt tension and then fasten the screw (2).
7. Add a drop of green locking lacquer onto the head of the fixing screw (2).

**20.3.9.4 Set the Left Pipettor Arm**

1. Perform the teaching of the Sample pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.10 Replacement of the X-Motor of Left Pipettor Arm

#### 20.3.10.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover and approach the left pipettor arm.
6. Remove the X-axis toothed belt of Left Pipettor Arm, see Chapter 20.3.9.2 on page 20-33.

#### 20.3.10.2 Removal of the X-Motor of Left Pipettor Arm

1. Unplug the motor cable (J5) and the encoder cable (J6) from Dual Sledge CU PCB (1).

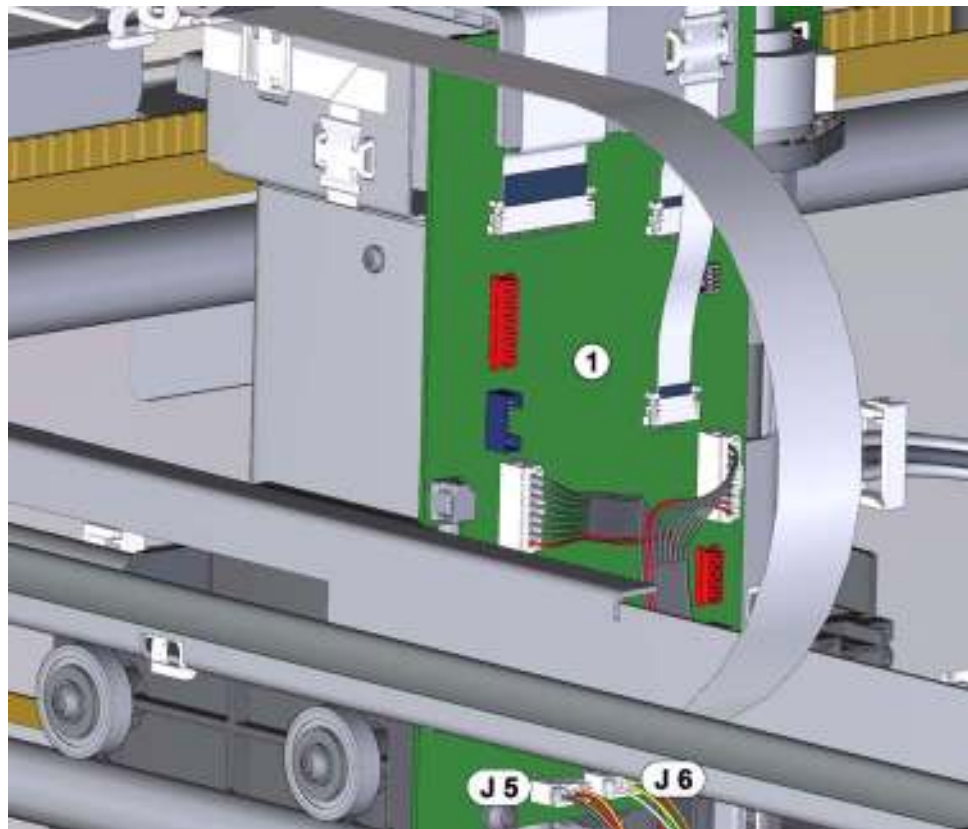


Figure 20–39: PCB, Dual Sledge CU

2. Hold on to the X-motor (5) and remove the two screws (4), then pull the X-motor (5) out.

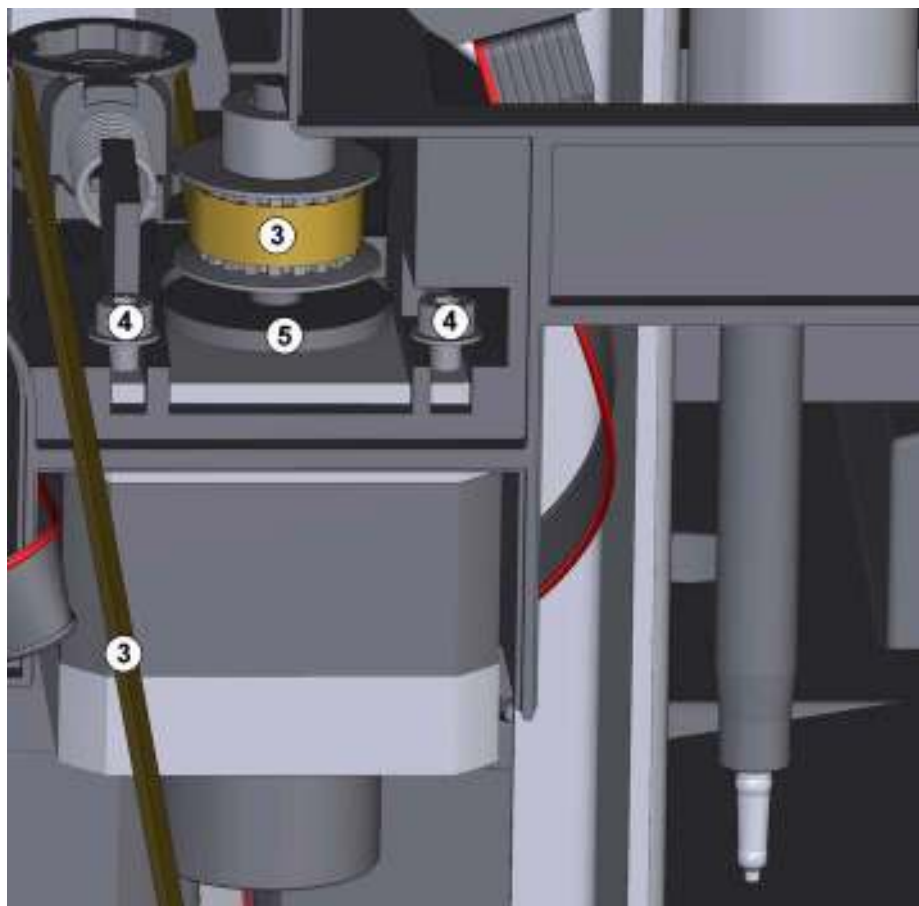


Figure 20-40: Left arm X-Motor, left side view

#### 20.3.10.3 Assembly of the X-Motor of Left Pipettor Arm

1. Insert the X-Motor into the frame and set it loose with the two screws (4).
2. Connect the motor cable (J5) and the encoder cable (J6) onto the Dual Sledge CU PCB (1).
3. Fasten the two screws (4).
4. Assemble the X-belt of the left pipettor arm, see Chapter 20.3.9.3 on page 20-34.

#### 20.3.10.4 Set the Left Pipettor Arm

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.11 Replacement of the Y-axis toothed belt of Left Pipettor Arm

#### 20.3.11.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the Top Covers.
6. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.

#### 20.3.11.2 Removal of the Y-axis toothed belt of Left Pipettor Arm

1. Loosen the screw of the toothed belt wheel (1).
2. Press the wheel against the end stop and fasten the screw to loosen the belt.

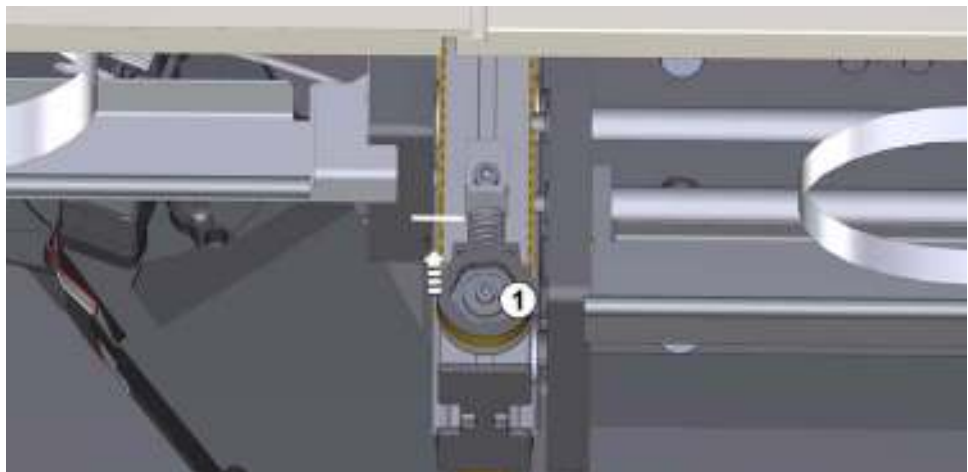


Figure 20–41: Belt tensioner

3. Move the Left Pipettor Arm to the back, until you see the front belt connection from the left side of the chassis.
4. Remove the two screws of the front belt connector (2) and slide the end of toothed belt out of the connector.

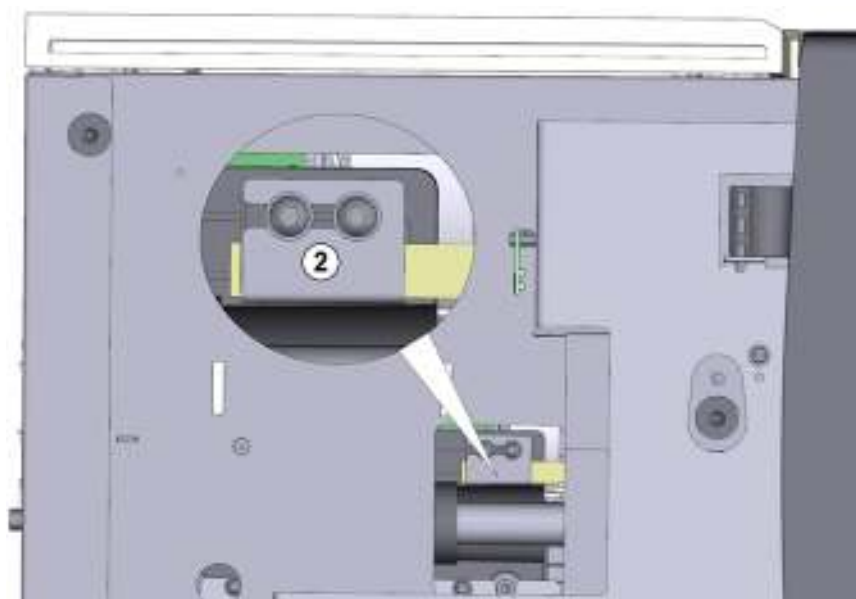


Figure 20-42: Chassis left side view, front belt connection

5. Move the Left Pipettor Arm to the front, until you see the rear belt connection from the left side of the chassis.
6. Remove the two screws of the rear belt connector (3) and slide the end of toothed belt out of the connector.

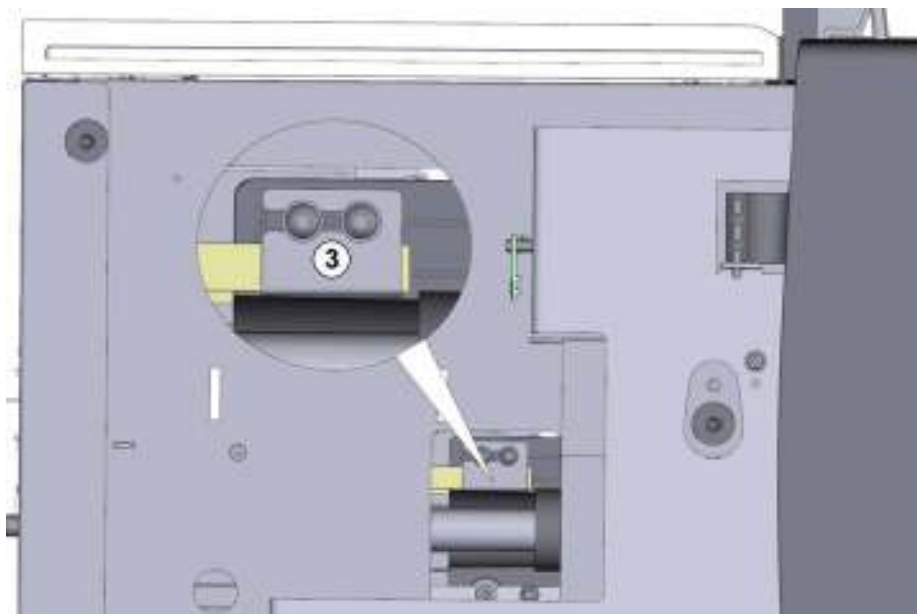


Figure 20-43: Chassis left side view, rear belt connection

## 20. Pipettor for external-tanks configuration

7. Slide the toothed belt out of the belt holder (4).
8. Remove the toothed belt from the instrument.



Figure 20-44: Inside view of the instrument

### 20.3.11.3 Assembly of the Y-axis toothed belt of Left Pipettor Arm

1. Check the toothed belt length, and if necessary cut it to a length of 3600mm (the toothed belt has to be cut only along vales).
2. Insert the toothed belt through the guide pulleys.

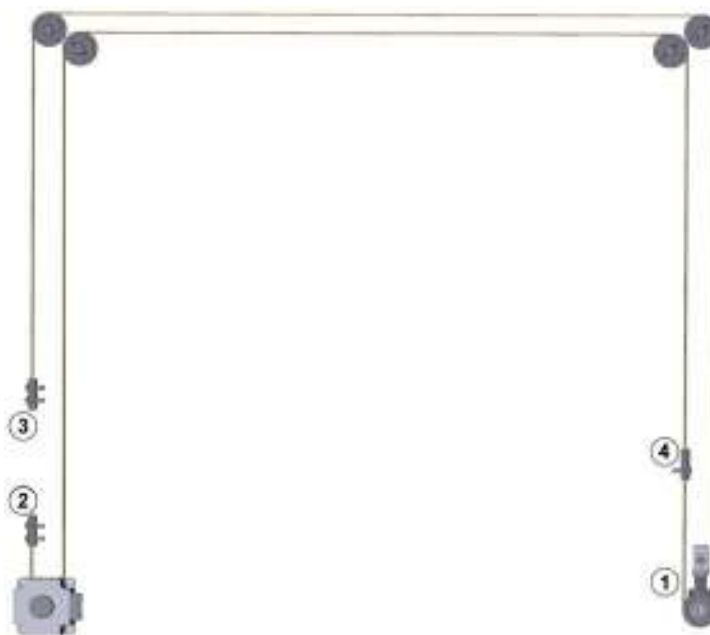


Figure 20-45: Layout of Left Arm toothed belt



Figure 20–46: Left Arm toothed belt

**NOTE**

**The toothed belt must not be twisted!**

**The toothed belt must sit correctly on the pulleys!**

3. Slide the toothed belt into the rear belt connector (3), set the connector into place and fasten to the Left Arm with the two screws.
4. Slide the toothed belt into the front belt connector (2), set the connector into place and fasten to the Left Arm with the two screws.
5. Slide the toothed belt into the belt holder (4). Loosen the screw at the belt holder.
6. Loosen the screw of the toothed belt wheel (1), let the spring expand to tension the belt and then fasten the screw.
7. Move the Left Pipettor Arm forward and back a couple of times.
8. Fasten the screw at the belt holder (4).
9. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

#### 20.3.11.4 Set the Y-axis toothed belt of Left Pipettor Arm

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### **20.3.12 Replacement of the Y-Motor of Left Pipettor Arm**

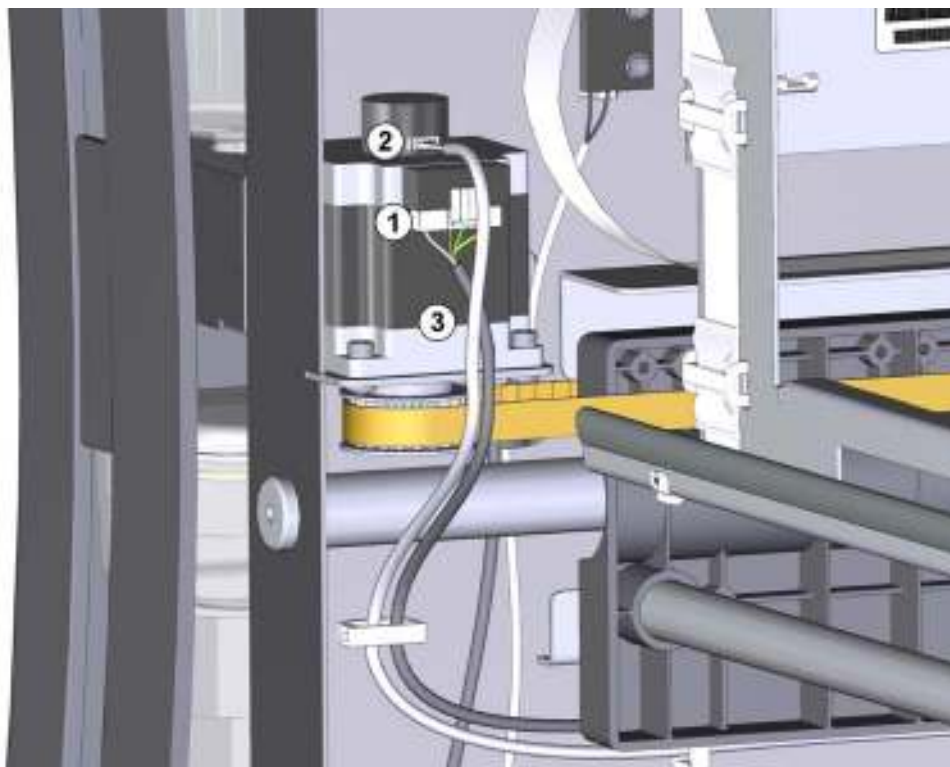
#### **20.3.12.1 Preliminary steps**



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.

#### **20.3.12.2 Removal of the Y-Motor of Left Pipettor Arm**

1. Unplug the motor cable (1) and the encoder cable (2) at the Y-motor (3).



*Figure 20–47: Y-Motor left pipettor arm*



**20. Pipettor for external-tanks configuration**

2. Loosen the fixing screw (4) of guide pulley (5), push it towards the spring and fasten again the screw (4).

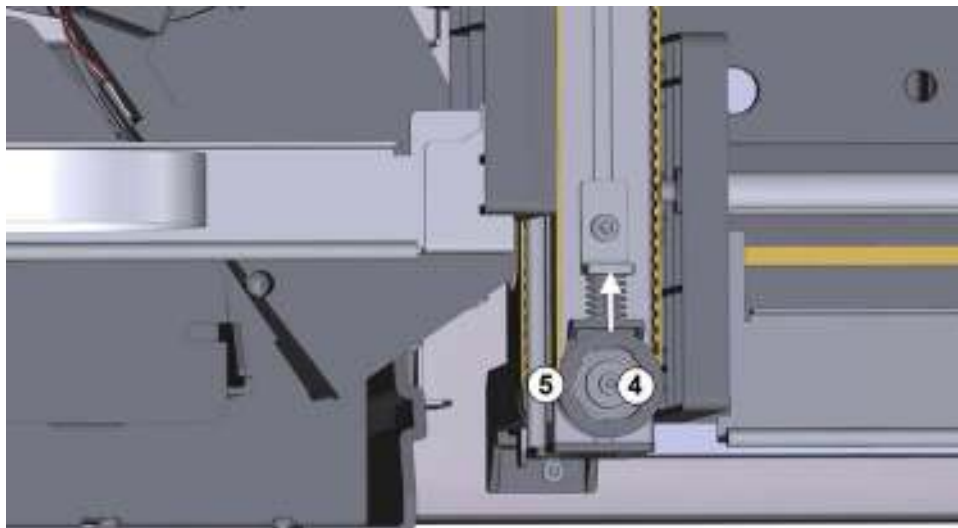


Figure 20-48: Left Pipettor arm

3. Remove the three screws (6) and lift the Y-Motor (3) out.

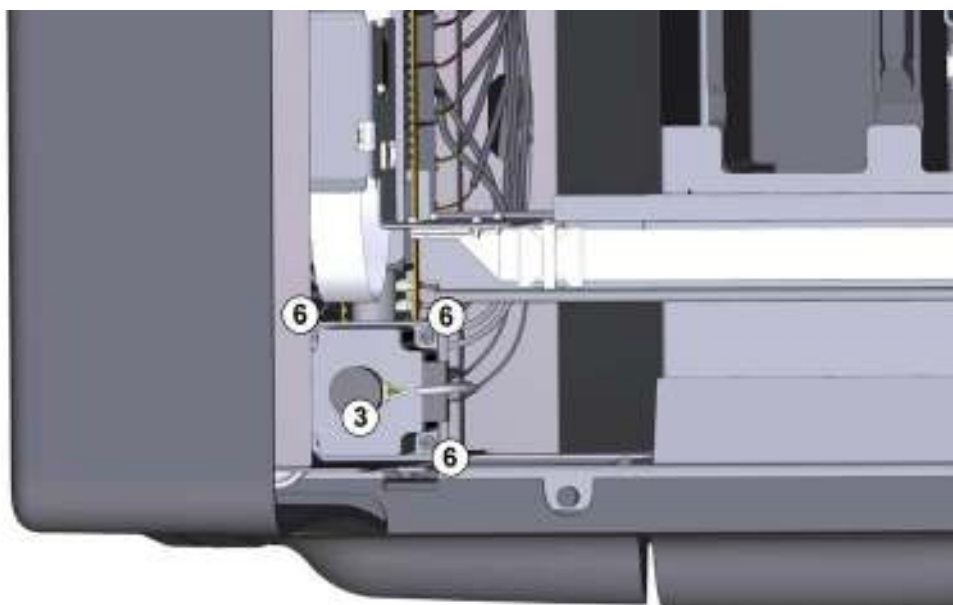


Figure 20-49: Y-Motor, left arm

**20. Pipettor for external-tanks configuration**

---

**20.3.12.3 Assembly of the Y-Motor of Left Pipettor Arm**

1. Insert the Y-Motor (3) into the frame and thread the toothed belt over the motor pulley.
2. Fasten the three screws (6).
3. Loosen the fixing screw (4) of guide pulley (5) to adjust the belt tension, then fasten one screw (4).
4. Connect the motor cable (1) and the encoder cable (2) onto the Y-motor (3).

**20.3.12.4 Set the Left Pipettor Arm**

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.13 Replacement of the Left Pipettor Arm

#### 20.3.13.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the Top Covers.
6. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
7. Remove the Left Top Cover, see Chapter 14.3.5.2 on page 14-12.
8. Remove the Top Cover, see Chapter 14.3.7.2 on page 14-15.
9. Remove the Sample Pipettor, see Chapter 20.3.1.2 on page 20-10.
10. Remove the Gripper Module, see Chapter 20.3.4.2 on page 20-20.
11. Remove the toothed belt, see Chapter 20.3.11.2 on page 20-38.

### 20.3.13.2 Removal of the Left Pipettor Arm

1. Remove the cable holders (1).
2. Remove the flat ribbon cable (2) until the rear end of the cable support (3).

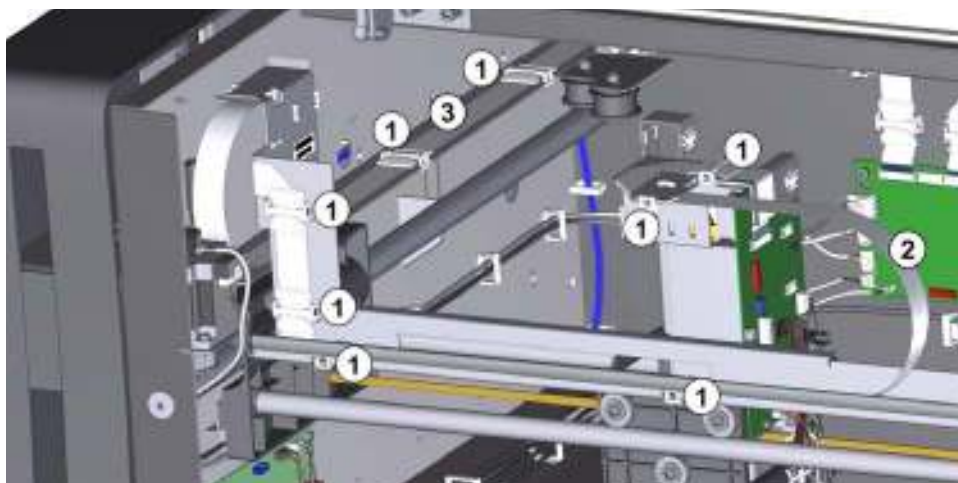


Figure 20-50: Left arm

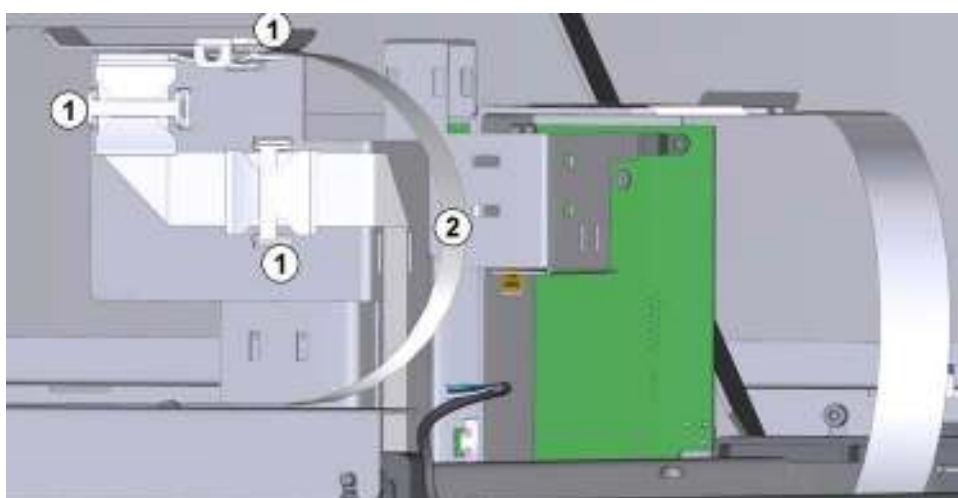


Figure 20-51: Flat ribbon cable routing, view without left side cover

3. Loosen the three screws (4) and slide the cable support (3) out of the chassis.

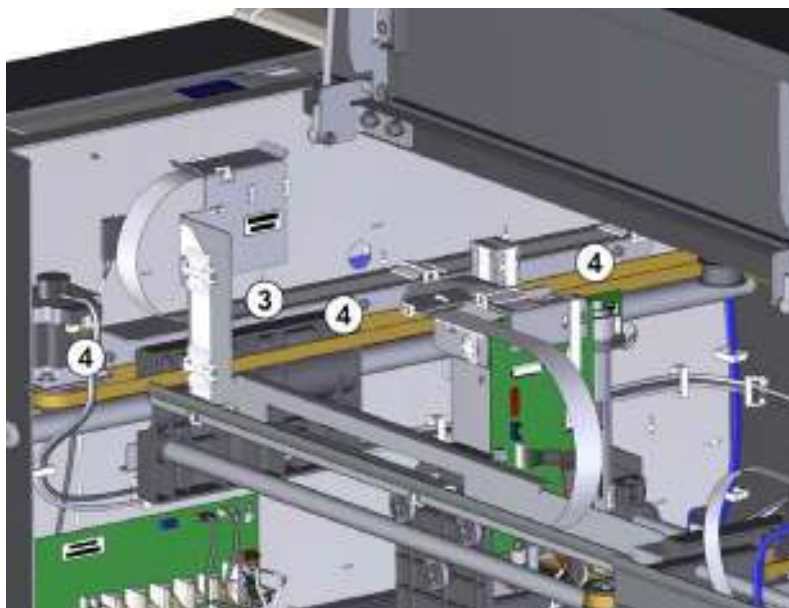


Figure 20-52: Left Arm cable support

4. Loosen the screw (5) of the Y-guide bar at the rear of the chassis.



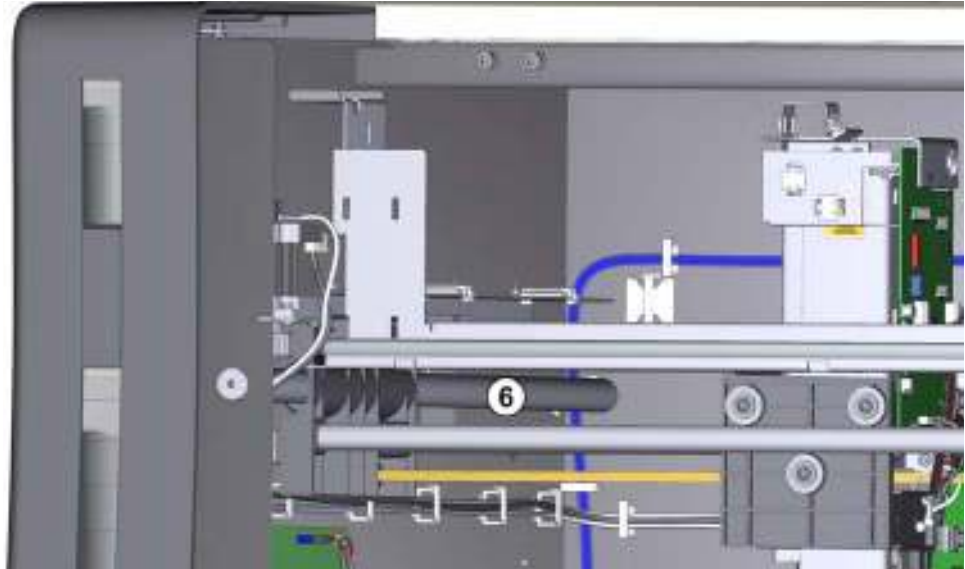
Figure 20-53: Instrument, rear view

5. Hold on to the Left Arm, it may slide to the front.
6. Lift the Y-guide bar (6) a bit up and slide it out of the keyhole at the rear wall.
7. Move the Left Pipettor Arm completely to the back, and hold on to it.
8. Pull the Y-guide bar (6) to the front and slide it out of the Left Arm.

**20. Pipettor for external-tanks configuration**

---

9. Lift the Y-sledge left side up to slide the right side out of the guide.
10. Lift the Left Arm up and remove it from the top.



*Figure 20-54: Left Pipettor Arm*

### 20.3.13.3 Assembly of the Left Pipettor Arm

1. Place the Left Pipettor Arm into the rearmost area of the instrument from the top.
2. Slide the bearings (7) of the Y-sledge right side into the guide (8).
3. Slide the Y-guide bar (6) in and thread it into the Y-sledge left side.

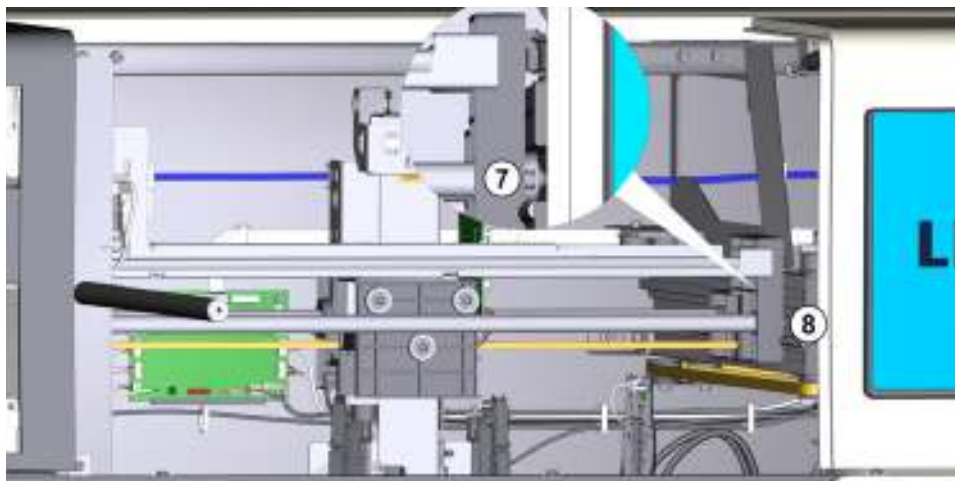


Figure 20-55: Left Pipettor Arm

4. Slide the screw head of Y-guide bar into the keyhole at the rear wall and fasten the screw (5).
5. Slide the cable support (3) into the chassis and fasten it with the screws (4).
6. Lay the flat ribbon cable (2) onto the cable support (3) and secure it with the cable holders (1).
7. Assemble the routing of the flat ribbon cable (2) and secure it with the cable holders (1).
8. Assemble the toothed belt, see Chapter 20.3.11.3 on page 20-40.
9. Assemble the Gripper module, see Chapter 20.3.4.3 on page 20-22.
10. Assemble the Sample Pipettor, see Chapter 20.3.1.3 on page 20-13.
11. Assemble the Top Cover, see Chapter 14.3.7.3 on page 14-16.
12. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 20.3.13.4 Set of the Left Pipettor Arm

1. Perform the teaching of the Sample Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the teaching of the Gripper Offset, see Chapter 13.7 on page 13-20.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 20.3.14 Replacement of the SPOLV Needle

#### 20.3.14.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Reagent Bay cover, see Chapter 22.3.1.2 on page 22-4.
7. Move the Right Arm over the Ancillary lane.

**NOTE**

Ensure that any ancillary reagents have been unloaded from the instrument.

#### 20.3.14.2 Removal of the SPOLV Needle

1. Unplug the PRP-SPOLV tubing from the SPOLV.
2. Open the knurled screw (2).
3. Hold on to the SPOLV (1), pull the steel needle (3) down and remove it.

**NOTE**

Use a tissue to avoid drops.

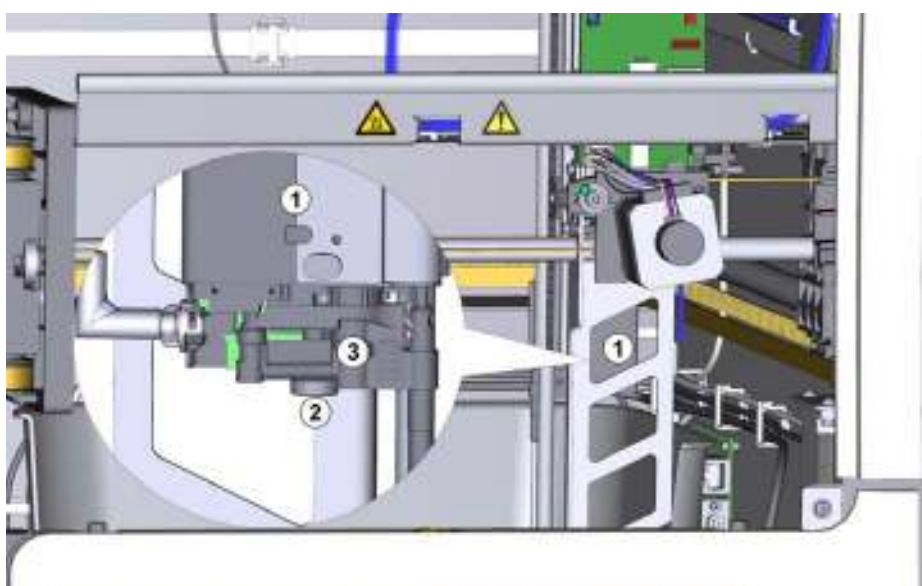


Figure 20–56: SPOLV Needle, zoom view from the rear



### 20.3.14.3 Assembly of the SPOLV Needle

#### NOTE

#### Need of SPOLV needle SN

The Serial Number of the installed SPOLV needle has to be recorded in the calibration dialogue of the Service Software. You can check the S/N on the needle screw; you can read it using a mirror and a light, without the need to disassemble the needle from the SPOLV.



1. Check the gasket (4) and O-Ring (5) for the proper installation.

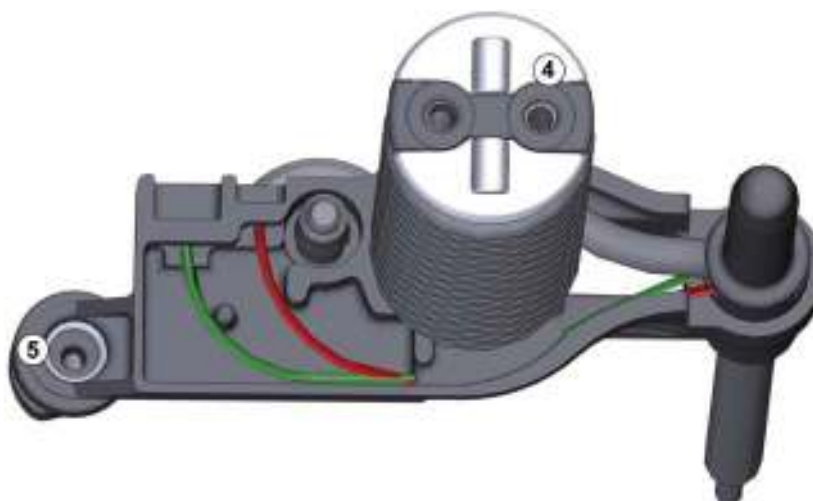


Figure 20–57: SPOLV Needle, inside view

2. Hold the SPOLV (1) and push the steel needle (3) into it.
3. Tighten the knurled screw (2) by hand.
4. Plug the PRP-SPOLV tubing.
5. Assemble the Reagent Bay cover, see Chapter 22.3.1.3 on page 22-4.

### 20.3.14.4 Set of the SPOLV

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the OLV calibration via Service Software, see Chapter 20.4.2.2 on page 20-79.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.15 Replacement of the SPOLV

#### 20.3.15.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers and approach the Right Pipettor Arm.

#### 20.3.15.2 Removal of the SPOLV

1. Unplug the PRP-SPOLV tube adapter (2) from the SPOLV (1).

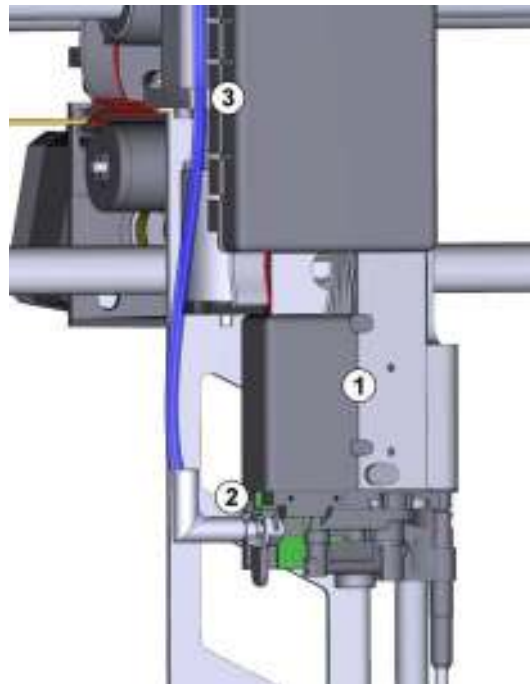


Figure 20–58: SPOLV rear view

#### NOTE

Use a tissue to avoid drops.

2. Unlock the PRP - SPOLV tube from the hook (3) on the rear of the SPOLV.
3. Remove the cable holder (connector J3) at the Single Sledge CU PCB (4).
4. Unlock the flat ribbon cable (connector J3) and pull it to the rear, out of the PCB.

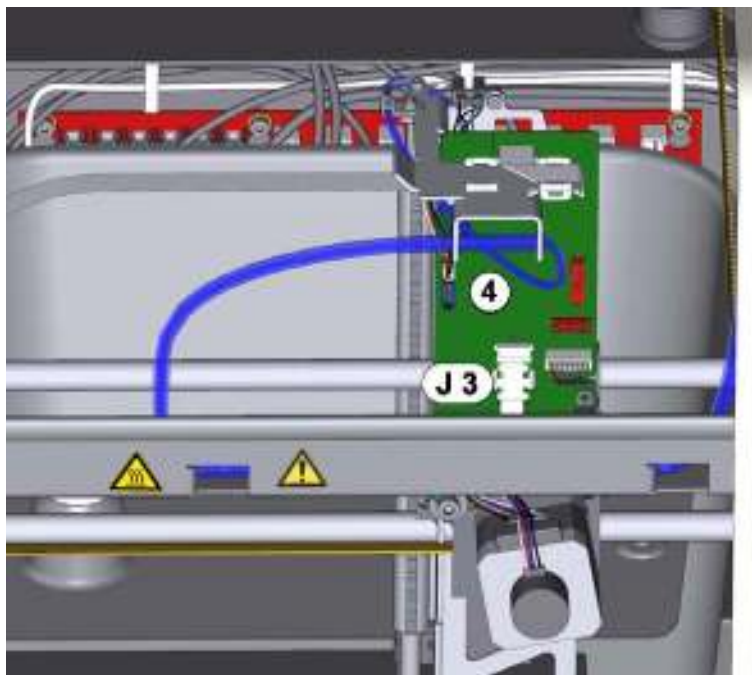


Figure 20-59: Single Sledge CU PCBA

5. Remove the top screw (5) and bottom screw (6) holding on to the SPOLV (1).
6. Slide the SPOLV out of the guide and remove it together with the Z-guiding rod (7). Be aware of the bearing (8), it may slip out!

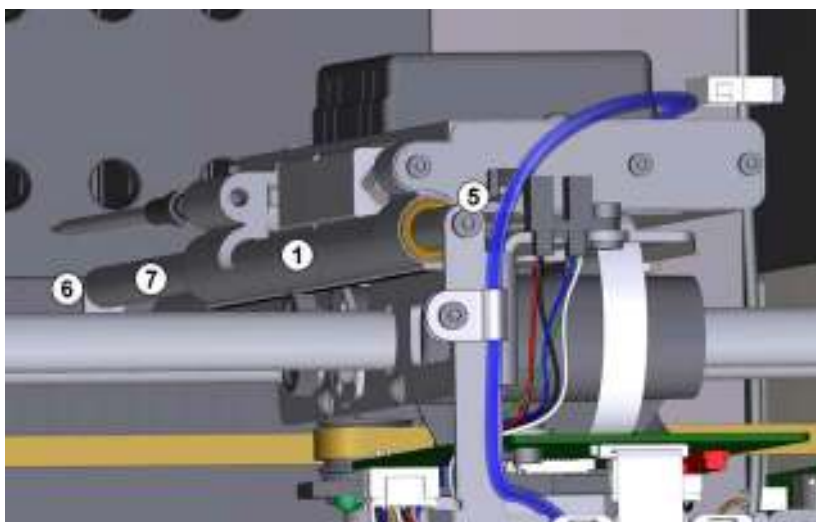


Figure 20-60: SPOLV, top view

## 20.3.15.3 Assembly of the SPOLV

**NOTE****Need of SPOLV needle SN**

The Serial Number of the installed SPOLV needle has to be recorded in the calibration dialogue of the Service Software. You can check the S/N on the needle screw; you can read it using a mirror and a light, without the need to disassemble the needle from the SPOLV.



1. Set the SPOLV Z-guiding rod (7) into the dedicated position and fasten with the bottom (6) and top (5) screws. The bearing (8) must sit in the Z-bracket!

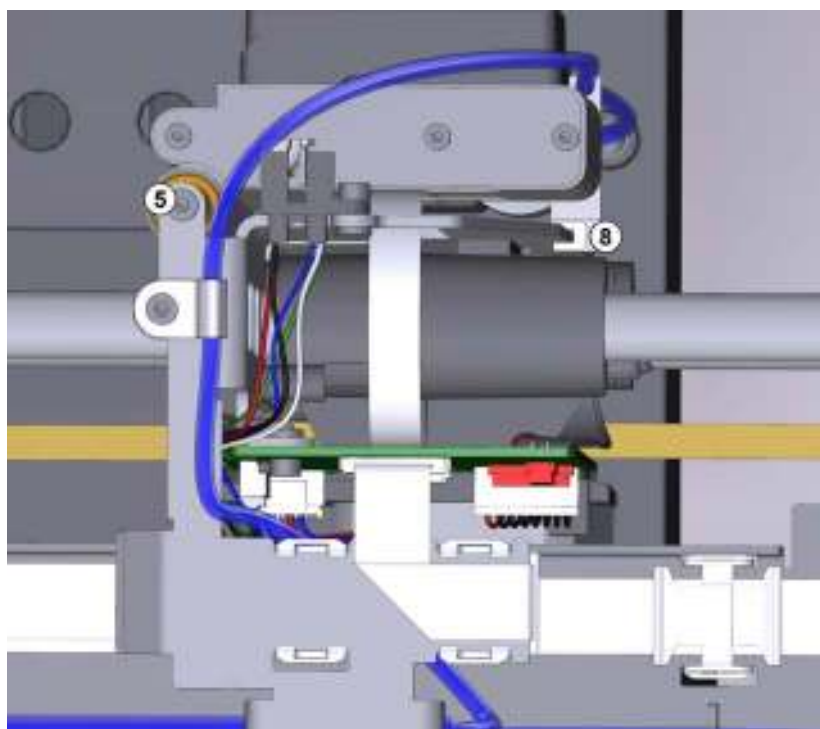


Figure 20–61: SPOLV, top view

2. Route the flat ribbon cable through the Single Sledge CU PCB (4) and lock it into the dedicated connector (J3).
3. Lock the flat ribbon cable with the cable holder.
4. Plug the PRP-SPOLV tube adapter (2) on the SPOLV.
5. Lock the PRP-SPOLV tube into the hook (3) on the rear of the SPOLV.

**20.3.15.4 Set of the Reagent Pipettor**

1. In case a new module has been installed, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
3. Prime the SPOLV, see Chapter 20.5 on page 20-81.
4. Perform the OLV calibration, see Chapter 20.4.2.2 on page 20-79.
5. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.16 Replacement of the Single Sledge CU PCB

#### 20.3.16.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers and approach the Right Pipettor Arm.

#### 20.3.16.2 Removal of the Single Sledge CU PCB

1. Unplug all cables from the Single Sledge CU PCB (1), see Chapter 20.8.4 on page 20-88.

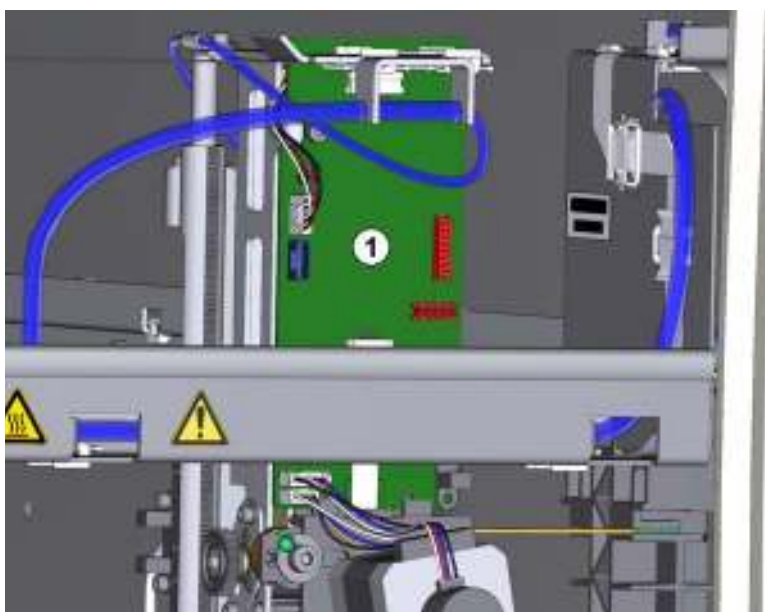


Figure 20–62: Single Sledge CU PCB

2. Unscrew the screw (2) and remove the Single Sledge CU PCB (1).

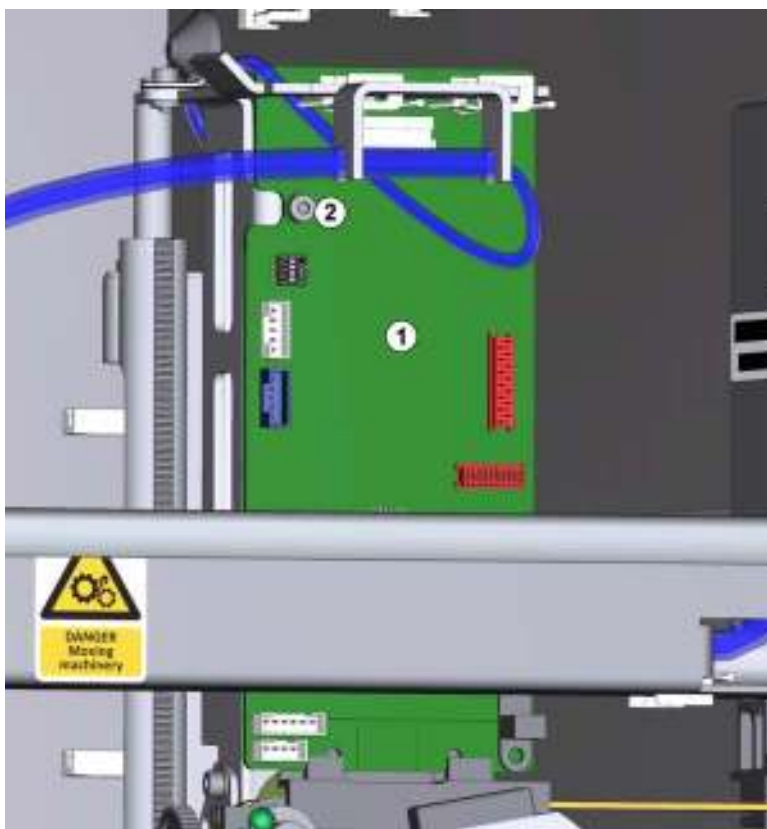


Figure 20-63: Single Sledge CU PCB removal

#### 20.3.16.3 Assembly of the Single Sledge CU PCB

1. Insert the Single Sledge CU PCB (1) and fasten with the screw (2).
2. Plug all the cables onto the PCB, see Chapter 20.8.4 on page 20-88.

#### 20.3.16.4 Set the Single Sledge CU PCB

1. In case a new PCB has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 20.3.17 Replacement of the Reagent Pipettor Z-Motor

#### 20.3.17.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers and approach the Right Pipettor Arm.
6. Remove the SPOLV, see Chapter 20.3.15.2 on page 20-52.

#### 20.3.17.2 Removal of the Reagent Pipettor Z-Motor

1. Unplug the Z-Motor cable (J8) from the Single Sledge CU PCB (1).

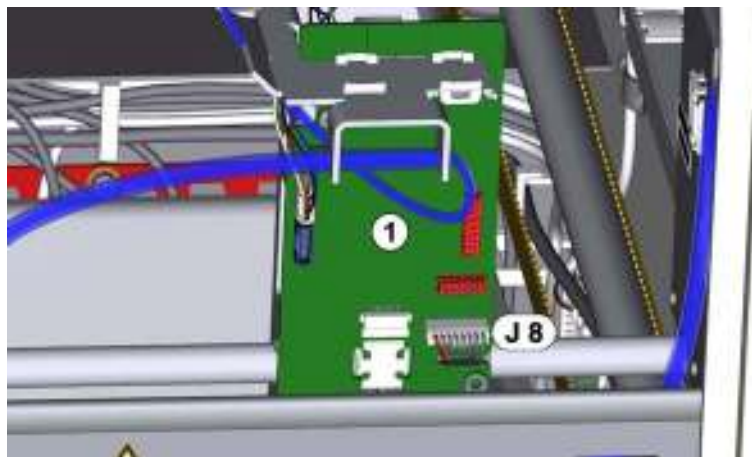


Figure 20-64: Reagent Pipettor, Single Sledge CU PCB

2. Unscrew two screws (2) and slide the Z-Motor (3) out.

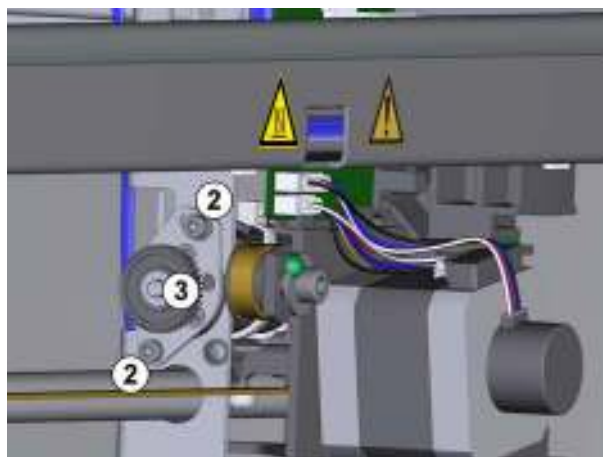


Figure 20-65: Reagent Pipettor, Z-Motor



### 20.3.17.3 Assembly of the Reagent Pipettor Z-Motor

1. Fold the flat ribbon cable of the Z-Motor and secure it with a cable tie.
2. Set the new Z-Motor (3) in.
3. Fasten the Z-Motor with two screws (2).
4. Assemble the SPOLV, see Chapter 20.3.15.3 on page 20-54.
5. Adjust the clearance between the Z-Motor gear (3) and the Reagent Pipettor toothed bar (4).
6. Fasten the Z-Motor with two screws (2).

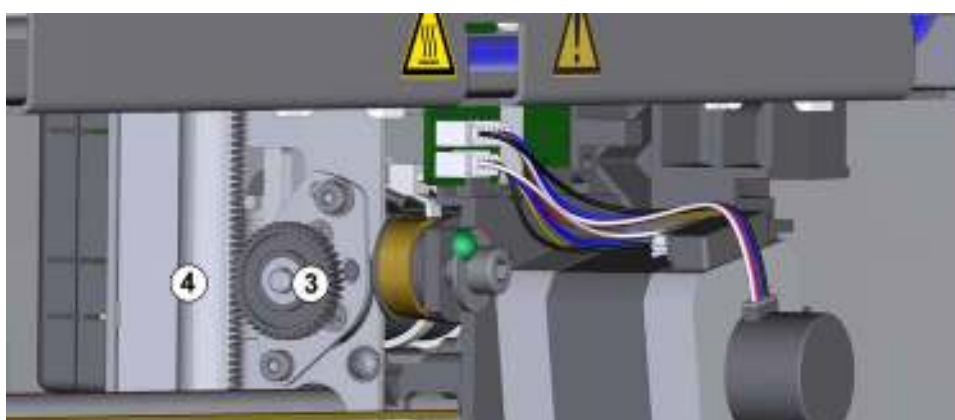


Figure 20–66: SPOLV Z-Motor gear clearance

### 20.3.17.4 Set the Reagent Pipettor

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.18 Replacement of the X-axis toothed belt of Right Pipettor Arm

#### 20.3.18.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers and approach the Right Pipettor Arm.

#### 20.3.18.2 Removal of the X-Axis toothed belt of Right Pipettor Arm

1. Loosen the fixing screw (2) of the guide pulley (3), push the guide pulley onto the spring and fasten the screw to loosen the belt.

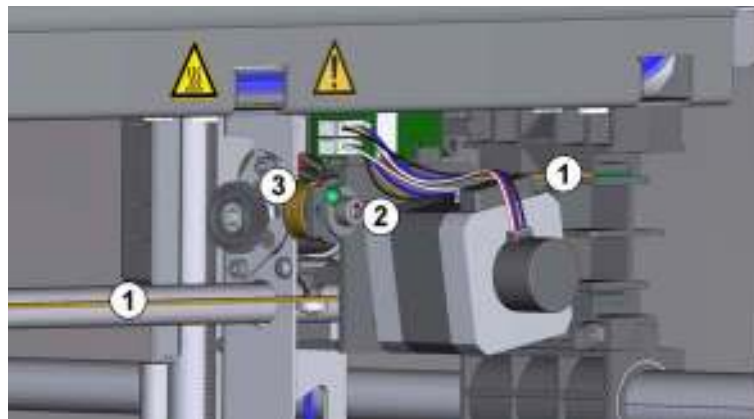


Figure 20–67: Right Pipettor arm

2. At the right side of Y-sledge (6), pull the toothed belt (1) out and then unhook the left belt holder (4) from the left side of Y-Sledge(5).

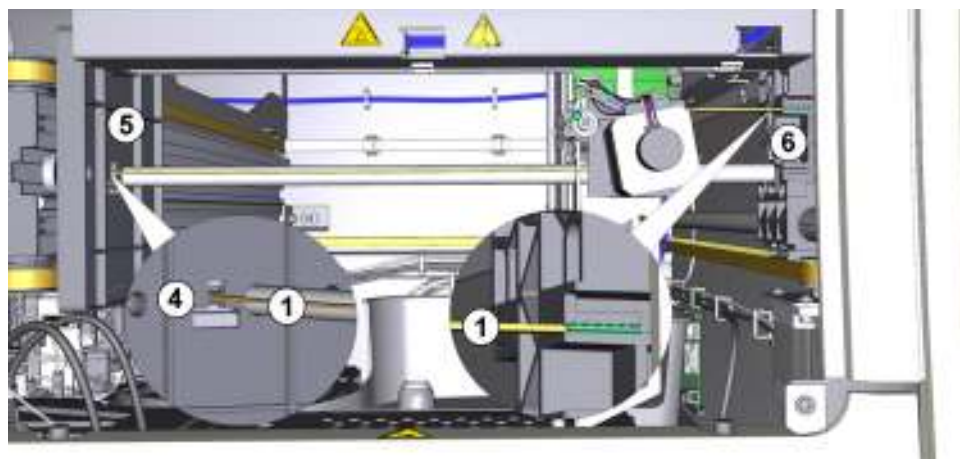
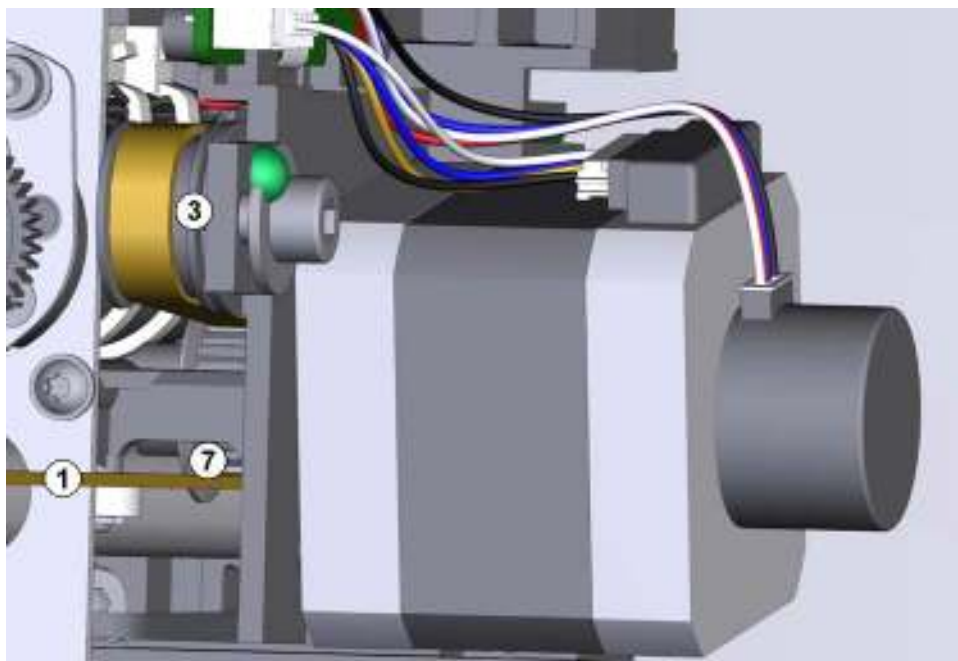


Figure 20–68: X-Axis toothed belt suspension

**20.3.18.3 Assembly of the X-Axis toothed belt of Right Pipettor Arm**

1. Thread the toothed belt (1) through the guide pulley (3) and the motor pulley (7).



*Figure 20–69: X-Axis Right Pipettor Arm*

2. Hang the left belt holder (4) into the left side of Y-Sledge (5).
3. Slide the toothed belt into the right upper part (6) of the Y-Sledge.
4. Loosen the fixing screw (2) of the guide pulley (3) to adjust the belt tension, then fasten the screw (2).

**20.3.18.4 Set the Right Pipettor Arm**

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.19 Replacement of the X-Motor of Right Pipettor Arm

#### 20.3.19.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.
7. Remove the SPOLV Needle, see Chapter 20.3.14.2 on page 20-50.
8. Remove the SPOLV, if needed, see Chapter 20.3.15.2 on page 20-52.
9. Remove the Y-axis toothed belt, see Chapter 20.3.20.2 on page 20-64.
10. Remove completely the Right Pipettor arm, see Chapter 20.3.22.2 on page 20-70.

#### 20.3.19.2 Removal of the X-Motor of Right Pipettor Arm

1. Loosen the fixing screw (2) of the guide pulley (3), push the guide pulley onto the spring and fasten the screw to loosen the belt.
2. Unplug the X-Motor cable (J4) and the X-Encoder cable (J5) from the Single Sledge CU PCB.

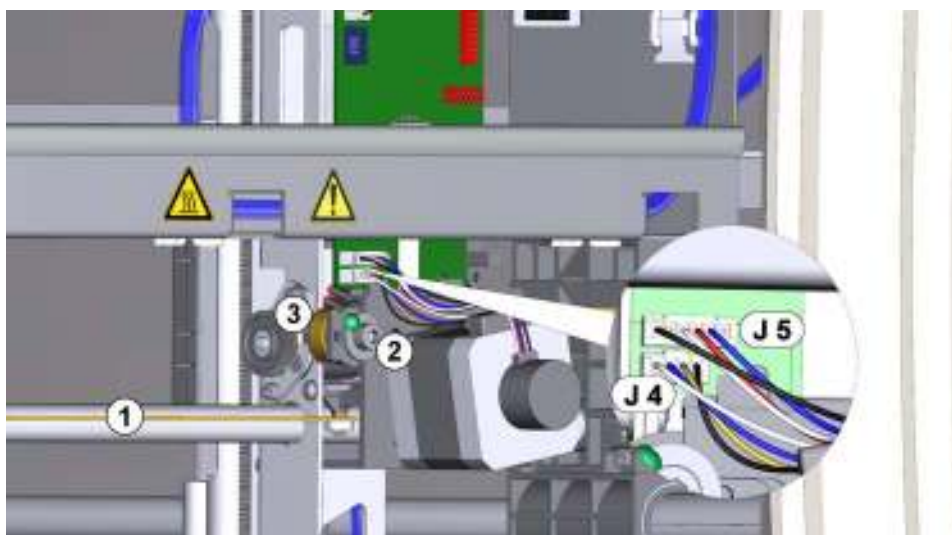


Figure 20–70: Right Pipettor arm

3. Loosen the three screws (4) of X-Motor (5).
4. Turn the X-Motor (5) counterclockwise and pull it out.

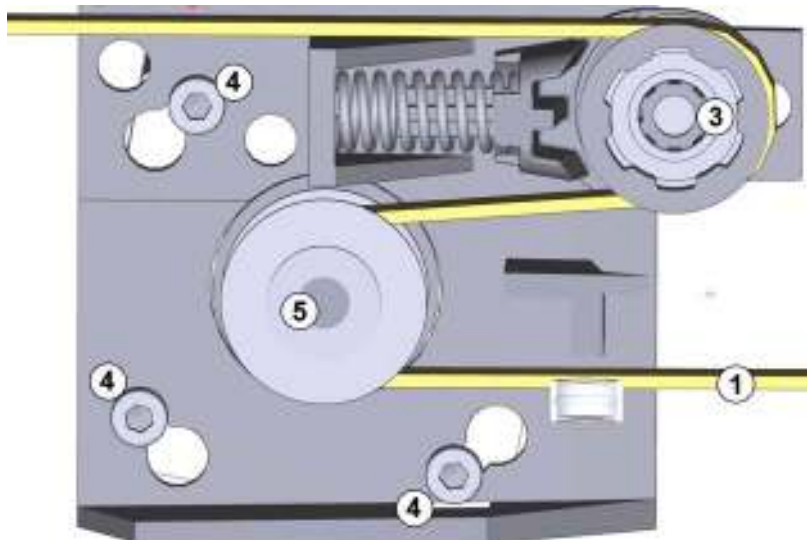


Figure 20-71: Inside view (cross section) of X-Motor mounting

#### 20.3.19.3 Assembly of the X-Motor of Right Pipettor Arm

1. Insert the X-Motor (5) and lay the toothed belt around the motor pulley.
2. Turn the X-Motor clockwise and fasten the three screws (4).
3. Plug the X-Motor cable and the X-Encoder cable onto connectors (J4) and (J5) of the Single Sledge CU PCB, respectively.
4. Loosen the fixing screw (2) of the guide pulley (3) to tension the toothed belt (1), then fasten the fixing screw (2).
5. Assemble the Right Pipettor arm, see Chapter 20.3.22.3 on page 20-73.
6. Assemble the Y-axis toothed belt, see Chapter 20.3.20.3 on page 20-65.
7. Assemble the SPOLV, if needed, see Chapter 20.3.15.3 on page 20-54.
8. Assemble the SPOLV Needle, see Chapter 20.3.14.3 on page 20-51.
9. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.

#### 20.3.19.4 Set the Right Pipettor Arm

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Prime the SPOLV, see Chapter 20.5 on page 20-81.
3. Perform the OLV calibration, see Chapter 20.4.2.2 on page 20-79.
4. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.20 Replacement of the Y-axis toothed belt of Right Pipettor Arm

#### 20.3.20.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.

#### 20.3.20.2 Removal of the Y-axis toothed belt of Right Pipettor Arm

1. Loosen the screw of the toothed belt wheel (1).
2. Press the wheel against the end stop and fasten the screw to loosen the belt.

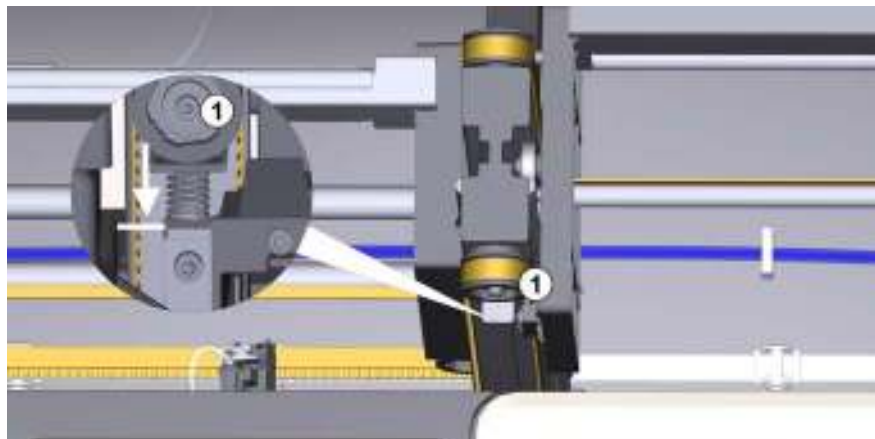


Figure 20-72: Right arm, belt tensioner

3. Remove the two screws (3) of the belt connector (2) and slide the ends of the toothed belt (4) out of the connector.

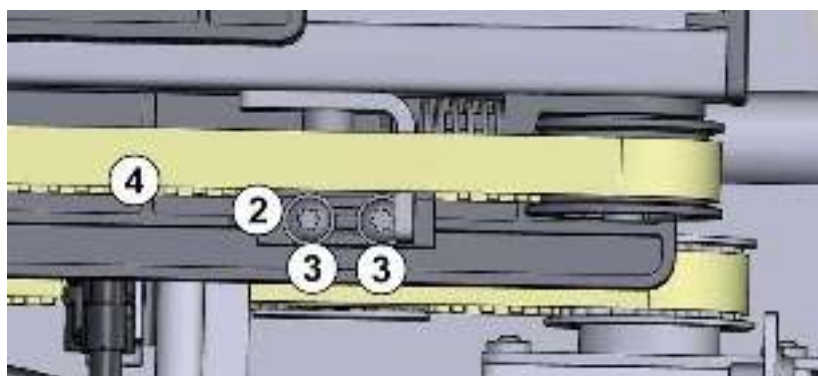
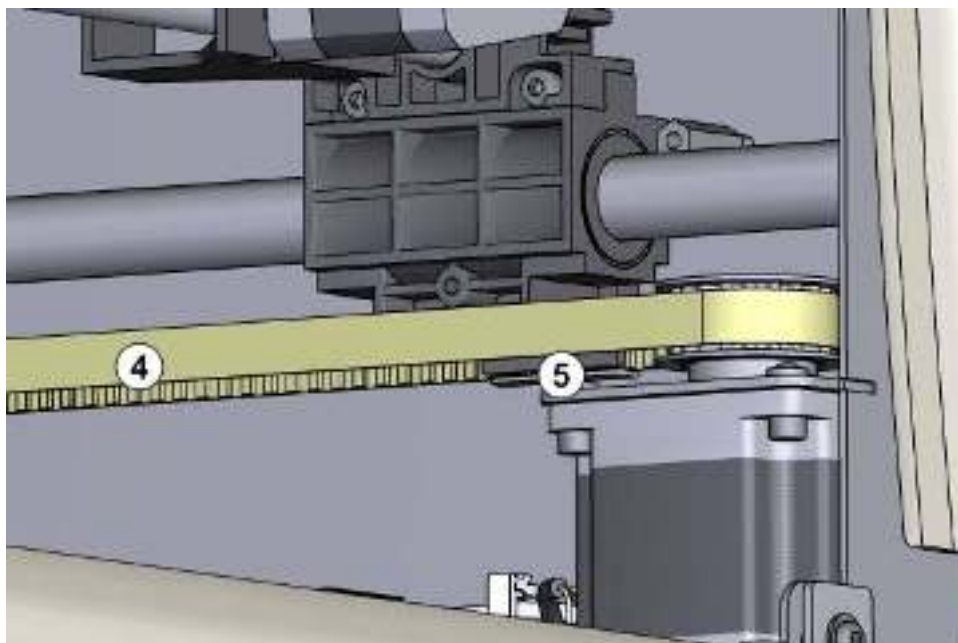


Figure 20-73: Right arm, toothed belt connector



4. Slide the toothed belt (4) out of the belt holder (5).
5. Remove the toothed belt from the instrument.



*Figure 20–74: Toothed belt holder (right side)*

#### **20.3.20.3 Assembly of the Y-axis toothed belt of Right Pipettor Arm**

1. Check the toothed belt length, and if necessary cut it to a length of 3265mm (the toothed belt has to be cut only along vales).
2. Insert the toothed belt through the guide pulleys.

---

**NOTE**

**The toothed belt must not be twisted!**

**The toothed belt must sit correctly on the pulleys!**

---

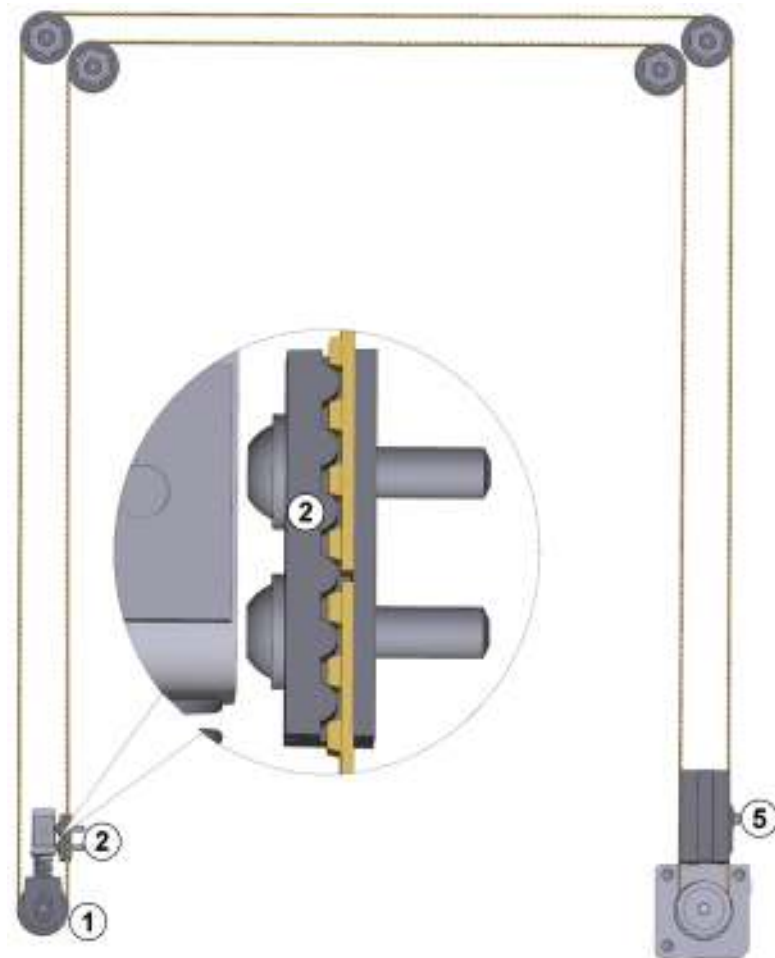


Figure 20–75: Layout of right arm toothed belt

3. Insert the two ends of the toothed belt (4) into the belt connector (2):
  - Slide the front end of the toothed belt with two teeth into the belt connector (2).
  - Slide the back end of the toothed belt with three teeth into the belt connector (2).
4. Set the connector (2) into place and fasten to the Right Arm with the screws (3).
5. Loosen the screw of the toothed belt wheel (1), let the spring expand to tension the belt and then fasten the screw.
6. Slide the toothed belt into the belt holder (5).
7. Move the Right Pipettor Arm forward and back a couple of times, to check that the pipettor arm is parallel. If necessary, loosen the belt at the belt wheel (1) and change the tooth insertion by one tooth forward or back into the belt holder (5).

#### 20.3.20.4 Set the Y-axis toothed belt of Right Pipettor Arm

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 20.3.21 Replacement of the Y-Motor of Right Pipettor Arm

#### 20.3.21.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.

#### 20.3.21.2 Removal of the Y-Motor of Right Pipettor Arm

1. Unplug the motor cable (1) and the encoder cable (2) at the Y-motor (3).

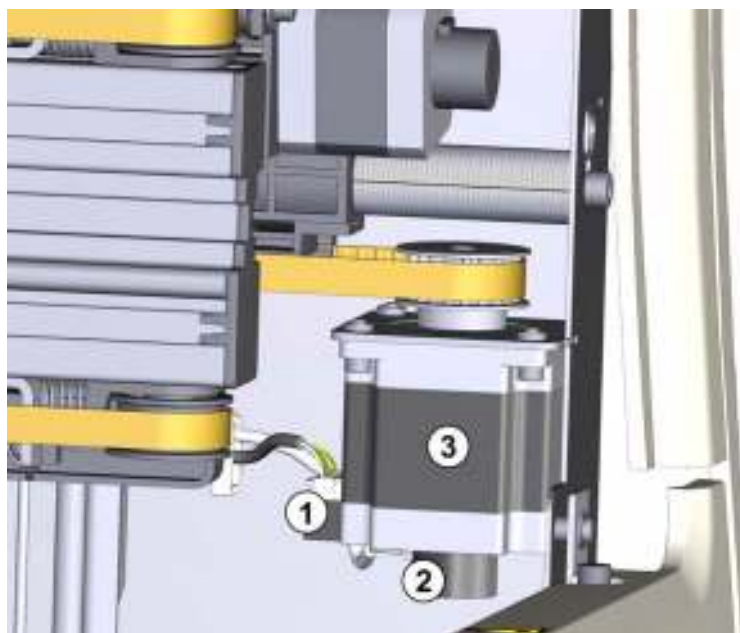


Figure 20–76: Y-Motor of Right Arm, side view

## 20. Pipettor for external-tanks configuration

2. Loosen the fixing screw (4) of guide pulley (5) and push it towards the spring to loosen the belt, then fasten the screw (4).

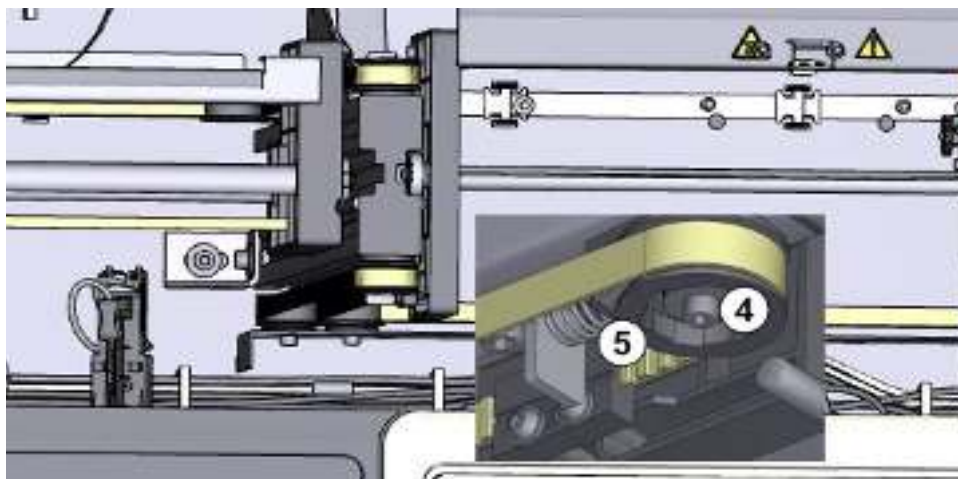


Figure 20-77: Right Pipettor Arm

3. Remove the three screws (6) and pull the Y-Motor (3) out.

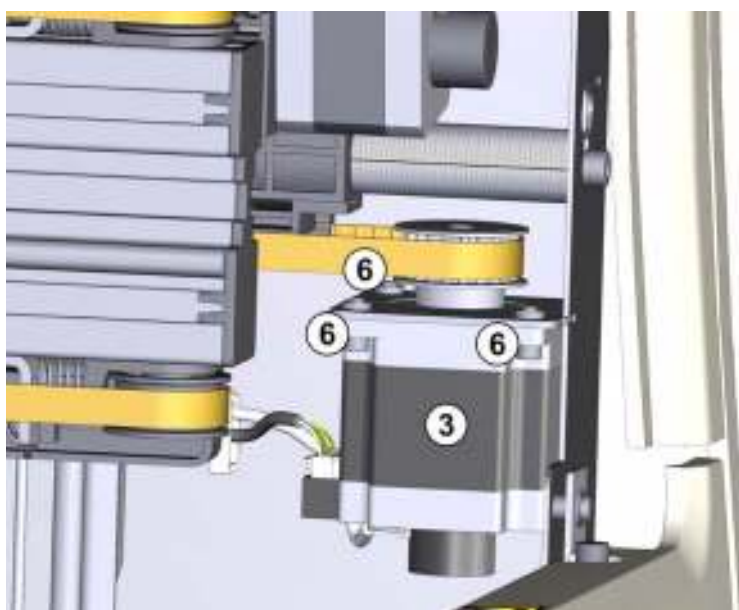


Figure 20-78: Y-Motor, Right arm

**20.3.21.3 Assembly of the Y-Motor of Right Pipettor Arm**

1. Insert the Y-Motor (3) into the frame and thread the toothed belt over the motor pulley.
2. Fasten the three screws (6).
3. Loosen the two fixing screws (4) of guide pulley (5) to adjust the belt tension, then fasten the two screws (4).
4. Plug the motor cable (1) and the encoder cable (2) onto the Y-motor (3).

**20.3.21.4 Set the Right Pipettor Arm**

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

## 20.3.22 Replacement of the Right Pipettor Arm

### 20.3.22.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.
7. Remove the SPOLV Needle, see Chapter 20.3.14.2 on page 20-50.
8. Remove the SPOLV, see Chapter 20.3.15.2 on page 20-52.
9. Remove the toothed belt, see Chapter 20.3.20.2 on page 20-64.

### 20.3.22.2 Removal of the Right Pipettor Arm

1. Remove the cable holders (2).
2. Unlock the flat ribbon cable (1) from the connector (3) and roll it up.
3. Unscrew the screw (5) of the tube clamp and unlock the PRP - SPOLV tube (4) from the hooks (6).

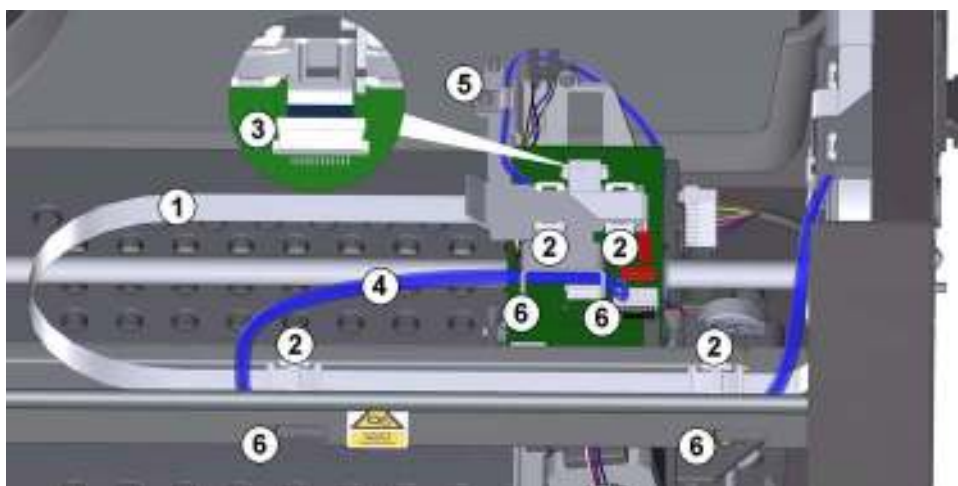


Figure 20–79: Right Arm

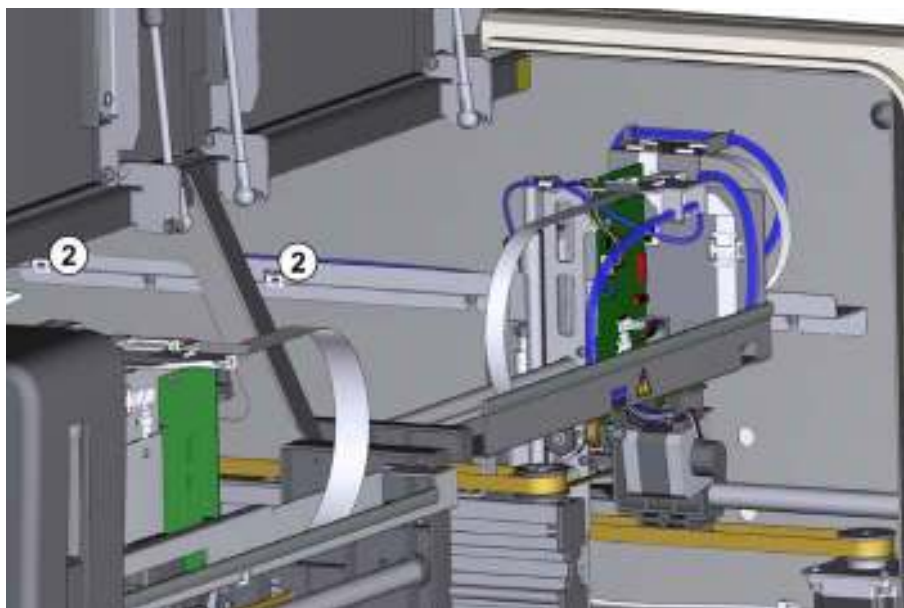


Figure 20-80: Right Arm

4. Remove the four cable holders (2) and roll up the flat ribbon cable (1).
5. Unlock the PRP - SPOLV tube (4) from the hooks (6).

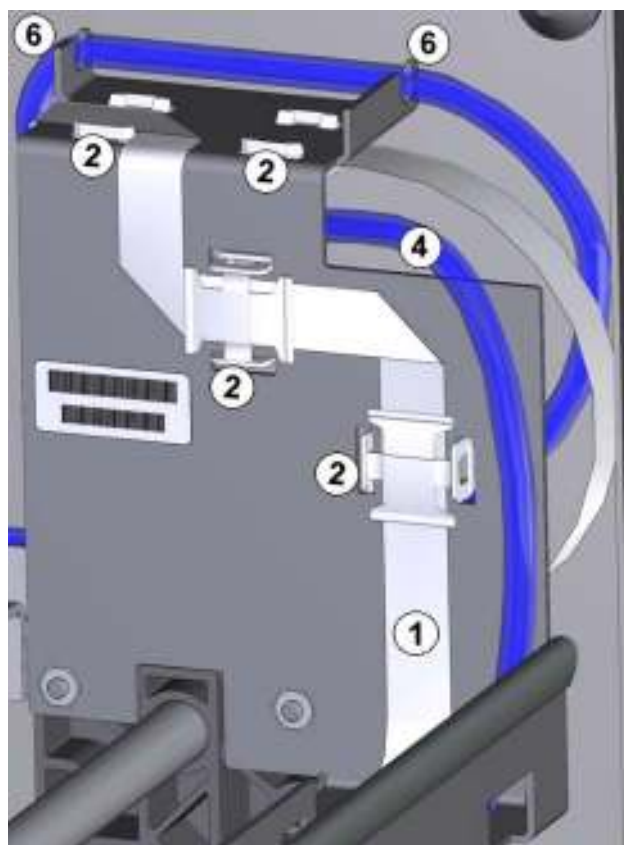
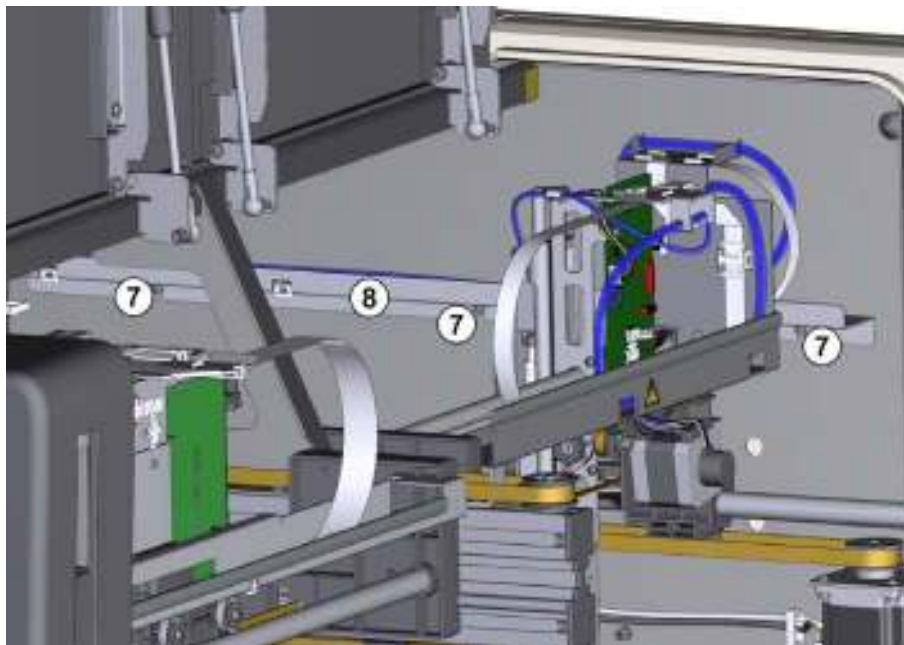


Figure 20-81: Pipettor right arm, flat ribbon cable routing

**20. Pipettor for external-tanks configuration**

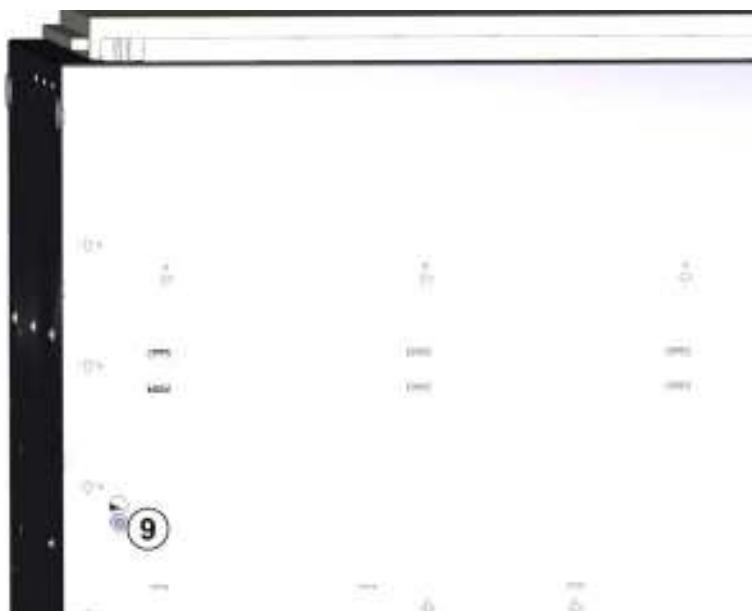
---

6. Loosen the three screws (7) and slide the cable support (8) out of the chassis.



*Figure 20-82: Right Arm cable support*

7. Loosen the screw (9) of the Y-guide bar at the rear of the chassis.



*Figure 20-83: Instrument rear view*

8. Loosen the screw (10) of the Y-guide bar at the front of the chassis.  
9. Hold on to the Right Arm, it may slide to the front.



**20. Pipettor for external-tanks configuration**

10. Lift the Y-guide bar (11) a bit up and slide it out of the keyhole at the rear wall.
11. Move the Right Pipettor Arm completely to the front, and hold on to it.
12. Pull the Y-guide bar (11) to the back and slide it out of the Right Arm.
13. Lift the Y-sledge right side up to slide the left side out of the guide, and remove the Right Arm from the instrument.

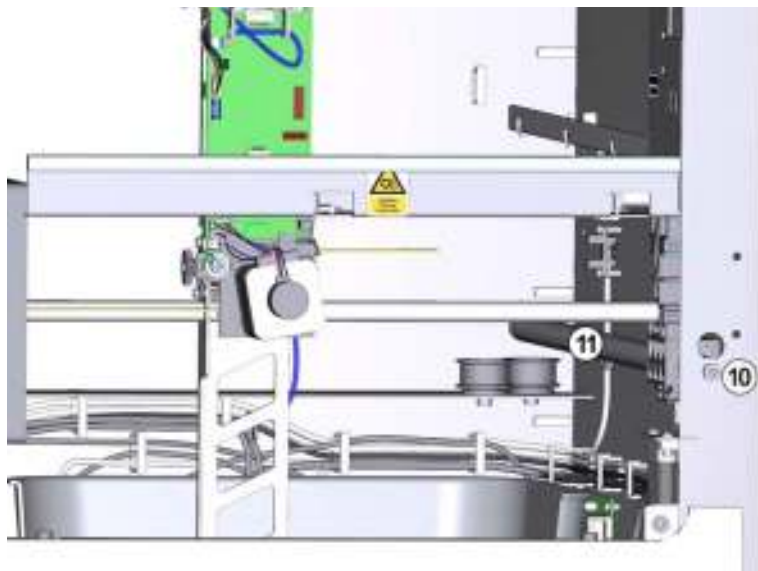


Figure 20-84: Instrument front view

**20.3.22.3 Assembly of the Right Pipettor Arm**

1. Slide the bearings (12) of the Y-sledge left side into the guide.
2. Slide the Y-guide bar in and thread it into the Y-sledge right side.

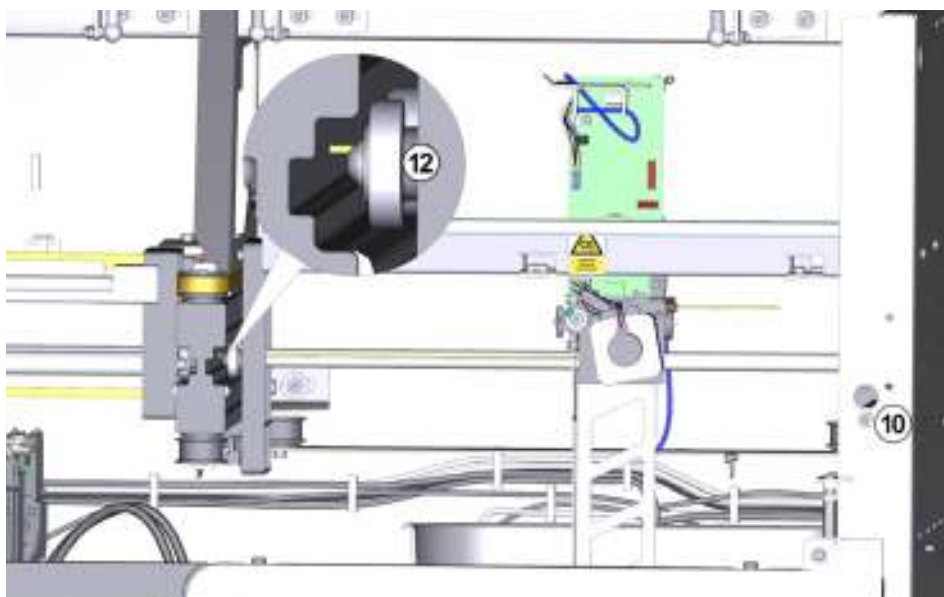


Figure 20-85: Instrument front view

## 20. Pipettor for external-tanks configuration

3. Slide the screw heads of Y-guide bar into the keyholes and fasten the screws (9) and (10).
4. Slide the cable support (8) into the chassis and fasten it with the screws (7).
5. Assemble the routing of the flat ribbon cable (1) along the cable supports and secure it with the cable holders (2).
6. Lay the PRP- SPOLV tube (4) carefully in and hang it into the hooks (6).
7. Move the pipettor forward and back to check that the tube and the cable are properly positioned: ensure that the tube is not interfering with the flat ribbon cable onto the cable support.
8. Assemble the toothed belt, see Chapter 20.3.20.3 on page 20-65.
9. Assemble the SPOLV, see Chapter 20.3.15.3 on page 20-54.
10. Assemble the SPOLV Needle, see Chapter 20.3.14.3 on page 20-51.
11. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.

### 20.3.22.4 Set the Right Pipettor Arm

1. Perform the teaching of the Reagent Pipettor, see Chapter 13.4 on page 13-6.
2. Prime the SPOLV, see Chapter 20.5 on page 20-81.
3. Perform the OLV calibration, see Chapter 20.4.2.2 on page 20-79.
4. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 20.3.23 Replacement of the Left arm Y-Home sensor

#### 20.3.23.1 Preliminary step



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the left pipettor arm to a safe location on the leftmost area of the instrument.

#### 20.3.23.2 Removal of the Left arm Y-Home sensor

1. Unplug the sensor Y-axis home sensor of left arm (connector J12 on the Pipettor CU PCBA), see Figure 20-87.
2. Unscrew the screws, which are holding the selected sensor, see Figure 20-88.
3. Open the cable holders.
4. Remove the old sensor from the instrument.



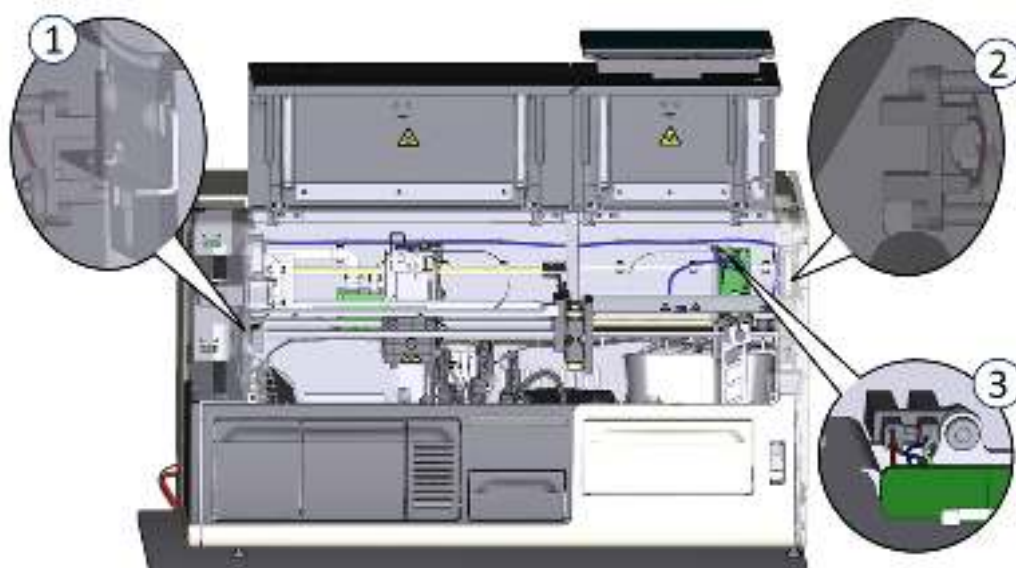


Figure 20–86: Pipettor Home sensors positioning inside the system. The circle 1 is the Left arm Y-Home sensor. The circle 2 is the Right arm Y-Home sensor. The circle 3 is the SPOLV Z-Home sensor

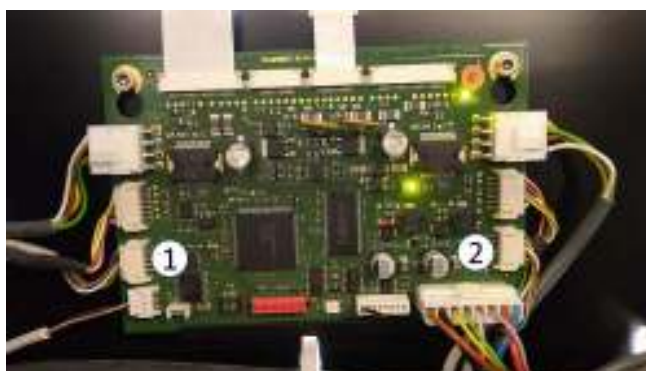


Figure 20–87: Pipettor CU PCBA



Figure 20–88: Left arm Y-Home sensor

**20.3.23.3 Assembly of the Left arm Y-Home sensor**

1. Install the new sensor according to the Figure 20–88 and fasten the screws.
2. Guide the cable through the holders and close them.
3. Connect the plug to the Pipettor CU PCBA (J12), see Figure 20–87.
4. Close the covers, turn on instrument, open the Main UI and initialize the system.

**20.3.24 Replacement of the Right arm Y-Home sensor****20.3.24.1 Preliminary step**

1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location.

**20.3.24.2 Removal of the Right arm Y-Home sensor**

1. Unplug the sensor Y-axis home sensor of right arm (connector J8 on the Pipettor CU PCBA), see Figure 20–87.
2. Unscrew the screws, which are holding the selected sensor, see Figure 20–89.
3. Open the cable holders.
4. Remove the old sensor from the instrument.



Figure 20–89: Right arm Y-Home sensor

### 20.3.24.3 Assembly of the Right arm Y-Home sensor

1. Install the new sensor according to Figure 20–89 and fasten the screws.
2. Guide the cable through the holders and close them.
3. Connect the plug to the Pipettor CU PCBA (J8), see Figure 20–87.
4. Close the covers, turn on instrument, open the Main UI and initialize the system.

### 20.3.25 Replacement of the SPOLV Z-Home sensor

#### 20.3.25.1 Preliminary step



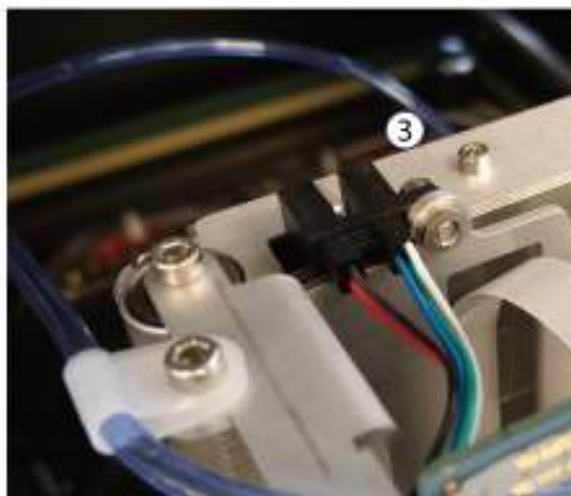
1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.

#### 20.3.25.2 Removal of the SPOLV Z-Home sensor

1. Unplug the sensor Z-axis home sensor of SPOLV (connector J9 on the X-Z-sledge EC CU PCBA), see Figure 20–90.
2. Unscrew the screw which is holding the sensor, see Figure 20–91.
3. Remove the old sensor from the instrument.



Figure 20–90: X-Z-sledge EC CU PCBA



*Figure 20–91: SPOLV Z-Home sensor*

#### **20.3.25.3 Assembly of the SPOLV Z-Home sensor**

1. Install the new sensor according to Figure 20–91 and fasten the screw.
2. Connect the plug to the X-Z-sledge EC CU PCBA (J9), see Figure 20–90.
3. Close the covers, turn on instrument, open the Main UI and initialize the system.

## 20.4 Adjustment and calibration

This section contains information about the adjustment and calibration of the Pipettor Module.

### 20.4.1 Initialization routine

- Close any open software and start the **Service software** from the Windows Administrator account.
- In the **Service software** at the navigation window, choose **Pipettor** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog Controlled Modules** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 20.4.2 Calibration of the Pipettor Module

#### 20.4.2.1 Calibration of the Gripper Claws

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **Cuvette Gripper**.

- Step 1**
- Read the instructions and then press the **Done** button.
- Step 2**
- Take a cuvette and manually hold it between the gripper claws.
  - Use the buttons to close the gripper claws in steps of 10, until the gripper holds on to the cuvette.
  - Use the buttons to open the gripper claws in steps of 10, until the cuvette falls down.
  - Use the buttons to further open the gripper claws by **30** steps.
  - Press the **Save** button to save the adjustment.
- Step 3**
- Store the taught position in the EEPROM by selecting the **Write ALL Parameters to NVM** button.
- Step 4**
- Read the instructions and then press the **Done** button to initialize the gripper.
- Step 5**
- The process is successful when, at the **Summarize** panel, all steps are in green.

#### 20.4.2.2 OLV Calibration

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **OLV Calibrate**.

- Step 1**
- In case the SPOLV needle has just been replaced, please enter in the Serial Number field, the new Serial Number of the needle, then continue by pressing the **Save Needle Serial number** button. Otherwise, press the **Continue without change** button

## 20. Pipettor for external-tanks configuration

---

- Step 2** b. Read the instructions to ensure the System Liquid tank is properly filled and connected to the system and then press the **Done** button; the calibration routine starts automatically.
- Step 3** c. When the calibration is finished, the dialog **Verify OLV Dead Volume Cycle** will open automatically. The calibration perform 5 verification runs. In case of a failed calibration, one or more verification run results are highlighted in red. In case of a successful calibration, the total result is set to "Ok". If the total result is "Ok", press the **Done** button. Otherwise, return to **Step 2** or check hardware, i.e. the PRP-SPOLV tubing, the SPOLV module, the SPOLV needle, subsequently.
- Step 4** d. Store the taught position in the EEPROM by selecting the **Store teaching values to NVM** button.
- Step 5** e. The process is successful when, at the **Summarize** panel, all steps are in green.

### 20.4.2.3 OLV Verify

In the **Service software**, at the navigation window, choose **Pipettor** then the **Teach & Calibration** bracket and select **OLV Verify**.

- Step 1** a. Read the instructions to ensure the System Liquid tank is properly filled and connected to the system and then press the **Done** button.
- Step 2** b. Press the **Do** button; the verification routine starts automatically.  
c. The procedure is successful when the **Processing** window closes without failure message.
- Step 3** d. The process is successful when, at the **Summarize** panel, all steps are in green. The Info column shows you the results.

## 20.5 Priming of Pipettor

### CAUTION

Before rinsing, make sure that all tube connections and fittings are properly assembled. Ensure that all tanks are filled (System Liquid, Wash Buffer and Starter Reagents) and the Liquid Waste tank is empty. During the complete rinsing process, ensure that there is no leakage (tube connections glued, inserted, rotated etc.).

### CAUTION

Before rinsing, make sure that Solid Waste bin is inserted (together with the dedicated EASY Waste) and the Liquid Waste tank is connected.

### 20.5.1 Initialization routine

- Close any open software and start the **Service software** from the Windows Administrator account.
- In the **Service software** at the navigation window, choose **System** and then the **General** bracket.
- Click onto the **Instrument** button and choose the **Full initialization via COP** to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 20.5.2 Rinsing with System Liquid

In the **Service software** at the navigation window, choose **System** and the **General** bracket.

- Click onto the **Liquid** button and choose the **Prime Pipettor** button to open the dedicated window
- For initial rinsing, use the following values and start pressing the **Execute** button.
  - Cycles: 12
  - Wash time: 18
  - OLV Verification: Disabled

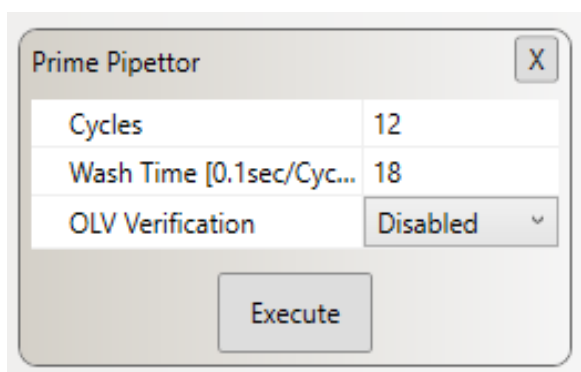


Figure 20–92: Pipettor rinsing

## 20. Pipettor for external-tanks configuration

---

- c. The process is successful when the *Processing* window closes without failure message.

---

**CAUTION**

Check that the wash station does not overflow during rinsing of the SPOLV.

---



## **20.6 Pipettor Tests**

### **20.6.1 LLD Endurance test**

For LLD Endurance test, see see Chapter 12.7.4.1 on page 12-52.

### **20.6.2 User LLD Endurance test**

For User LLD Endurance test, see Chapter 12.7.4.2 on page 12-53.

### **20.6.3 Pipettor test**

For Random movement test, see Chapter 12.7.4.3 on page 12-56.

## **20.7 Cuvette Endurance Run**

For Cuvette Endurance Run, see Chapter 12.8.1.2 on page 12-62.

## 20.8 Electric connections, switches and signals

This section shows an overview of the electronic boards of Left and Right Pipettor Arms.

**NOTE**

see Chapter 4.1.8 on page 4-9 for an overview of the complete electric circuit of the system.

### 20.8.1 X-Sledge-Dual-Z EC CU PCBA board

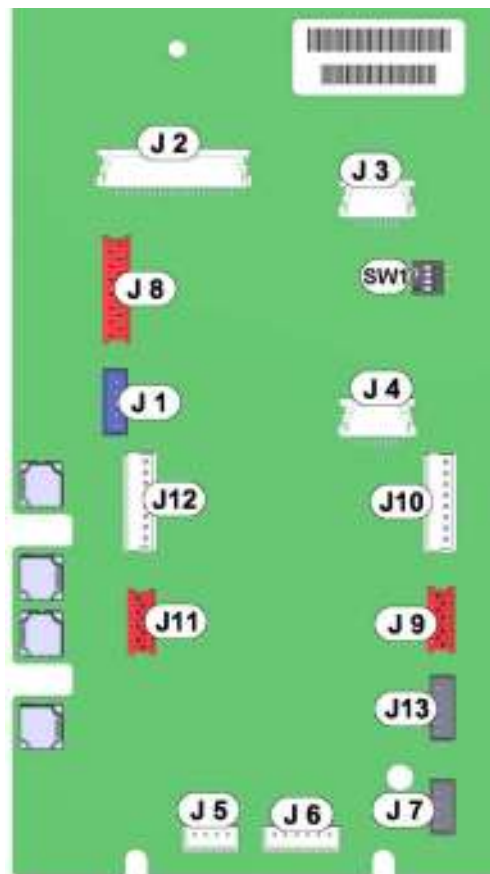


Figure 20–93: X-Sledge-Dual-Z EC CU PCBA board, front view

| Number | Connection            |
|--------|-----------------------|
| J1     | Download              |
| J2     | PWR / CAN             |
| J3     | Gripper CU PCBA       |
| J4     | Sample Pip EC CU PCBA |
| J5     | X-motor               |
| J6     | X-encoder             |
| J7     | not used              |
| J8     | Download (E1 SCTL)    |
| J9     | N/A                   |
| J10    | Gripper Z-motor       |
| J11    | N/A                   |
| J12    | Sample Pip Z-motor    |
| J13    | not used              |

*Table 20–9: Connections of the X-Sledge-Dual-Z EC CU PCBA board*

## 20.8.2 Gripper CU PCBA board

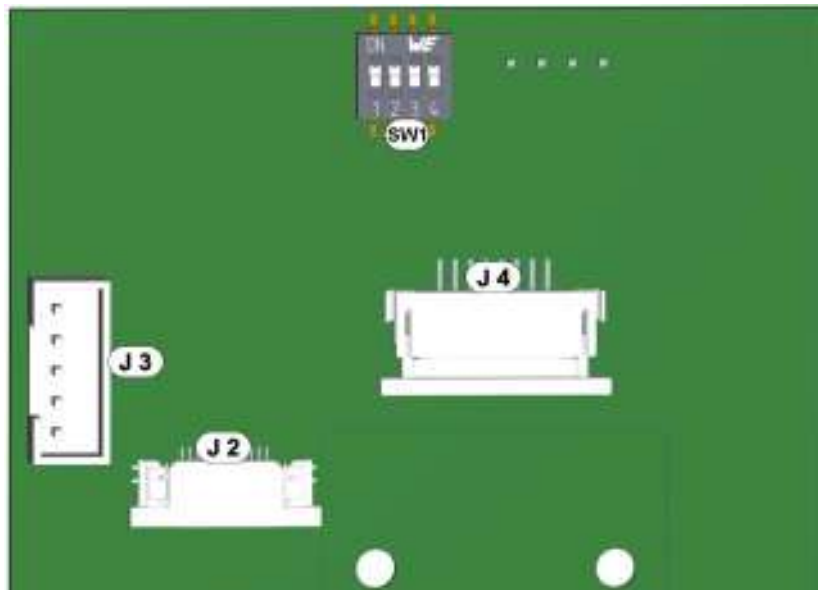


Figure 20–94: Gripper CU PCBA board, front view

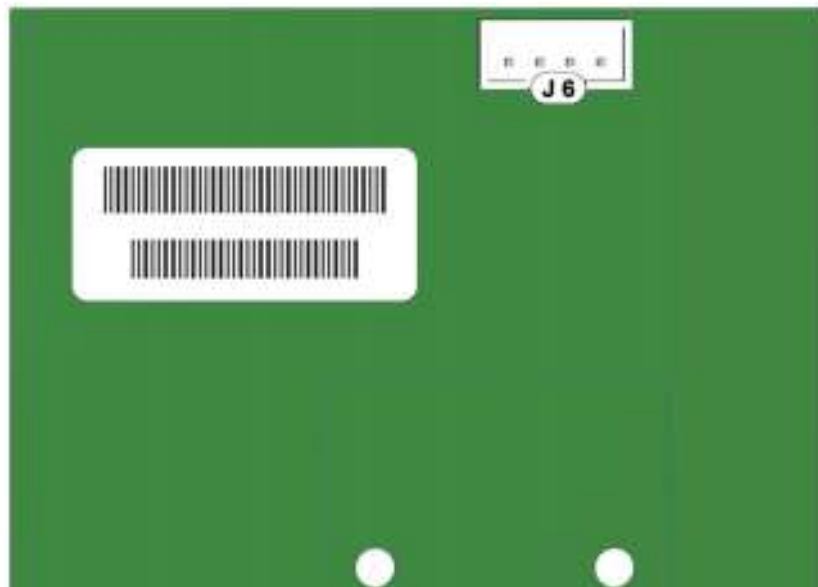


Figure 20–95: Gripper CU PCBA board, rear view

| Number | Connection       |
|--------|------------------|
| J2     | Front end Sensor |
| J3     | Down load        |
| J4     | PWR / CAN        |
| J6     | Gripper drive    |

Table 20–10: Connections of the Gripper CU PCBA board

| DIP-Switch |   |     | Meaning/Function/Default Switch Setting |
|------------|---|-----|-----------------------------------------|
| SW1        | 1 | OFF |                                         |
|            | 2 | OFF |                                         |
|            | 3 | OFF |                                         |
|            | 4 | OFF |                                         |

Table 20–11: DIP-Switches of the Gripper CU PCBA board

### 20.8.3 Gripper sensor PCBA board

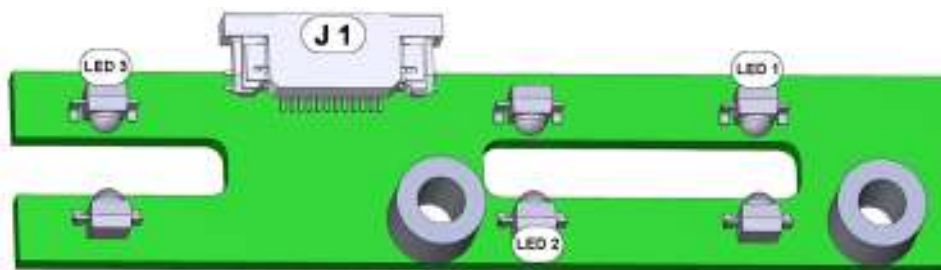


Figure 20–96: Gripper sensor PCBA board, front view

| Number | Connection                  |
|--------|-----------------------------|
| J1     | Front end sensor gripper CU |

Table 20–12: Connections of the Gripper sensor PCBA board

| Number   | Meaning/Function |
|----------|------------------|
| LED (Q1) |                  |
| LED (Q2) |                  |
| LED (Q3) |                  |

Table 20–13: LED of the Gripper sensor PCBA board

## 20.8.4 X-Z-Sledge EC CU PCBA board

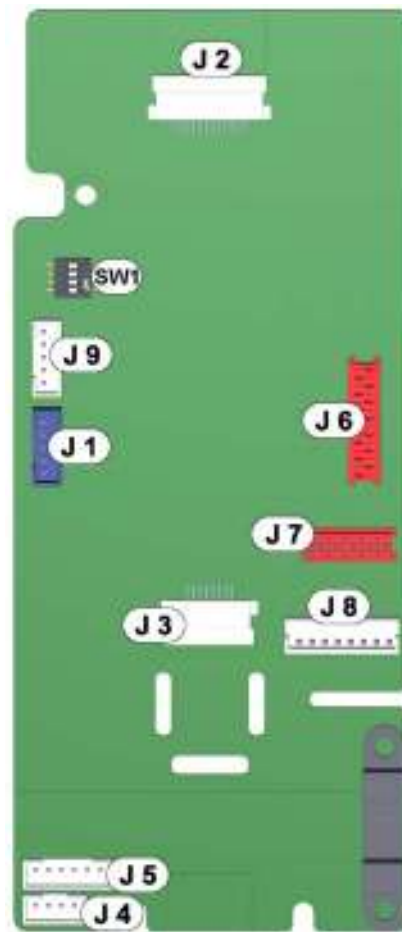


Figure 20–97: X-Z-Sledge EC CU PCBA board, front view

| Number | Connection          |
|--------|---------------------|
| J1     | Download            |
| J2     | PWR / CAN           |
| J3     | Connection to SPOLV |
| J4     | X-motor             |
| J5     | X-encoder           |
| J6     | Download (E1 SCTL)  |
| J7     | N/A                 |
| J8     | SPOLV Z-motor       |
| J9     | SPOLV Z-init        |

Table 20–14: Connections of the X-Z-Sledge EC CU PCBA board

| DIP-Switch |   |     | Meaning/Function/Default Switch Setting |
|------------|---|-----|-----------------------------------------|
| SW1        | 1 | OFF |                                         |
|            | 2 | OFF |                                         |
|            | 3 | OFF |                                         |
|            | 4 | OFF |                                         |

*Table 20–15: DIP-Switches of the X-Z-Sledge EC CU PCBA board*





# 21 Sample Module

This chapter describes the Sample Module in detail. The first section of the chapter introduces “Safety” notes concerning the Sample Bay, followed by a brief overview of the main parts composing this module.

The following section contains information about repair and verification (“Replacement and Repair” and “Check and Verification” sections).

In the last section, Sample Bay PCBA and connections are described.

The Sample Module supports loading/unloading of samples and controls in dedicated racks (sample racks and control racks, respectively).

- Tubes and bottles within racks are identified by a barcode scanner.
- The Sample Bay PCBA does not host a dedicated firmware; it is controlled by the RFID Loading Bay CU PCBA of the Reagent Module.

## 21.1 Safety

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 OR see Chapter 2.5 on page 2-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

### WARNING



#### Eye Injuries due to Laser Radiation

Laser radiation causes eye irritations when the laser beam is looked for a long period of time.

- Never look directly into the laser beam!
- Do not introduce optical devices into the system.
- Remove watches and reflective jewellery before operating the laser.
- Caution during operation and testing of the laser (barcode scanner and hand held barcode reader) must be taken due to laser class (class 2).
- The barcode reader is positioned on the right side of the Sample loading bay. The laser beam exits from an opening positioned on the internal right side of the Sample loading bay. The laser beam is orthogonal to the internal right side surface.

Wrong usage of operating elements or adjustments or the non-observance of processes can cause a dangerous emission of laser radiation.

21.2 Overview

21.2.1 Sample Module

This section contains an overview of the Sample Module components.



Figure 21–1: Sample Module, front view

|               |                  |
|---------------|------------------|
| 1             | Barcode scanner  |
| 2             | Sample Bay cover |
| 3             | Sample Bay PCBA  |
| Lane1 - Lane4 | Lane 1 to Lane 4 |

Table 21–1: Sample Module, front view

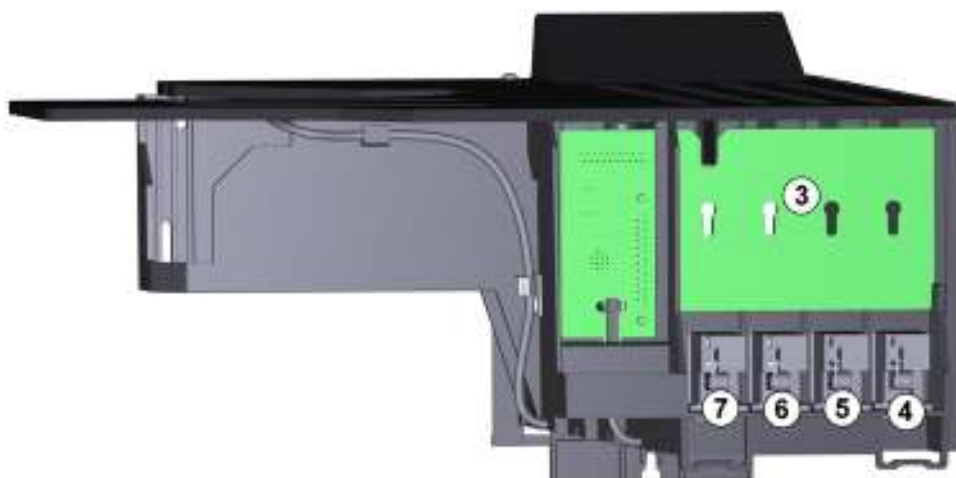


Figure 21-2: Sample Module, rear view

|   |                        |
|---|------------------------|
| 3 | Sample Bay PCBA        |
| 4 | Push Push lock, lane 1 |
| 5 | Push Push lock, lane 2 |
| 6 | Push Push lock, lane 3 |
| 7 | Push Push lock, lane 4 |

Table 21-2: Sample, Module, rear view

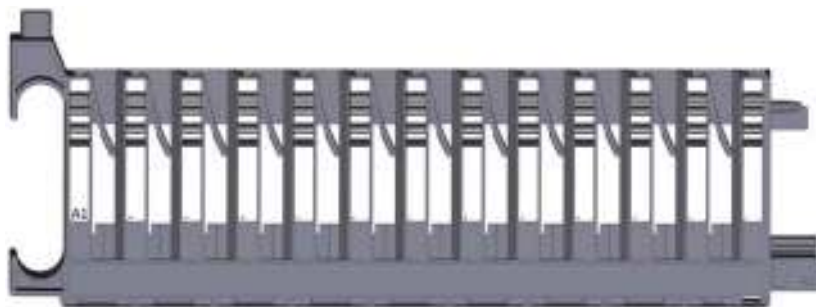


Figure 21-3: Sample Module, right side view

|   |                           |
|---|---------------------------|
| 1 | Barcode scanner           |
| 8 | Barcode scanner connector |
| 9 | Temperature sensor        |

Table 21-3: Sample Module, right side view

### 21.2.2 Sample rack



*Figure 21-4: Sample Rack*

### 21.2.3 Control rack



*Figure 21-5: Control Rack*

## 21.3 Replacement and repairs

This section contains instructions for the repair of individual components of the Sample Module. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 21.3.1 Replacement of the Sample Bay cover

#### 21.3.1.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** Main Software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the Top Covers and move the Left Pipettor arm completely to the left.

#### 21.3.1.2 Removal of the Sample Bay cover

1. Loosen the screw (1).
2. Lift the right side of Sample Bay cover (2) up to unlock the snap-in mechanism.
3. Slide the cover (2) to the right and remove it from the instrument.

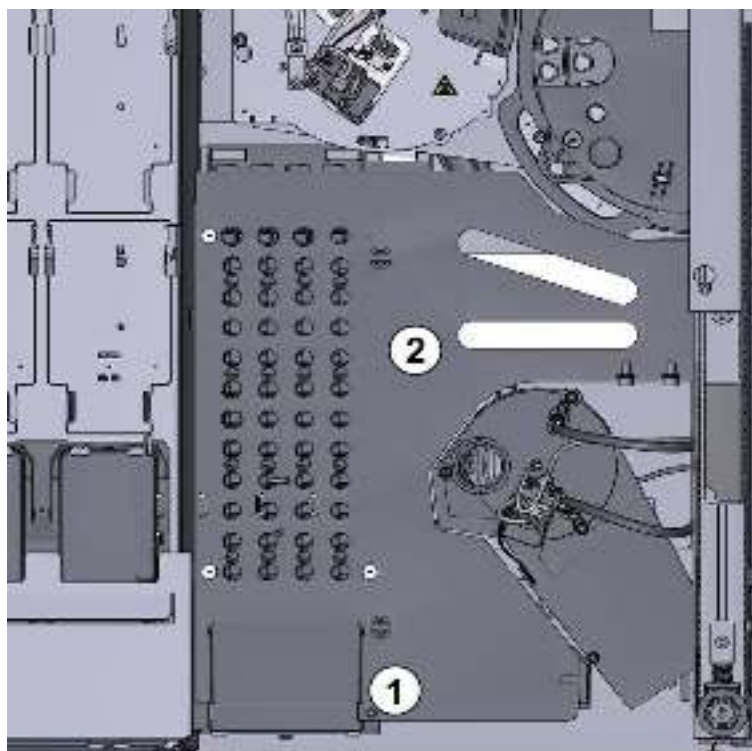


Figure 21–6: Sample Module, Sample Bay cover

**21.3.1.3 Assembly of the Sample Bay cover**

1. Insert the hooks of the Sample Bay cover (2) into the slots of the Sample Bay.
2. Lift down the right side of the cover (2), to lock the snap-in mechanism.
3. Fasten one screw (1).

## 21.3.2 Replacement of the Sample Module

### 21.3.2.1 Preliminary steps



4. Exit the **LIAISON<sup>®</sup> XS** Main Software.
5. Shut down the Panel PC from the Desk shield or from the Windows menu, and wait until the shutdown is complete.
6. On the right side of the instrument, turn off the Main Switch and unplug the power chord to completely cut the power to the instrument.
7. Open the Top Covers.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
9. Move the Left Pipettor arm to a safe location on the leftmost area of the instrument.
10. Remove the Sample Bay cover, see Chapter 21.3.1.2 on page 21-5.

### 21.3.2.2 Removal of the Sample Module

1. Unplug the cable from Loading Bay CU PCBA at the connection (J1).
2. Loosen the screw (2).
3. Pull the Sample Module (1) a bit to the front, until the hooks (3) are free.
4. Lift the Sample Module (1) out of the guides.

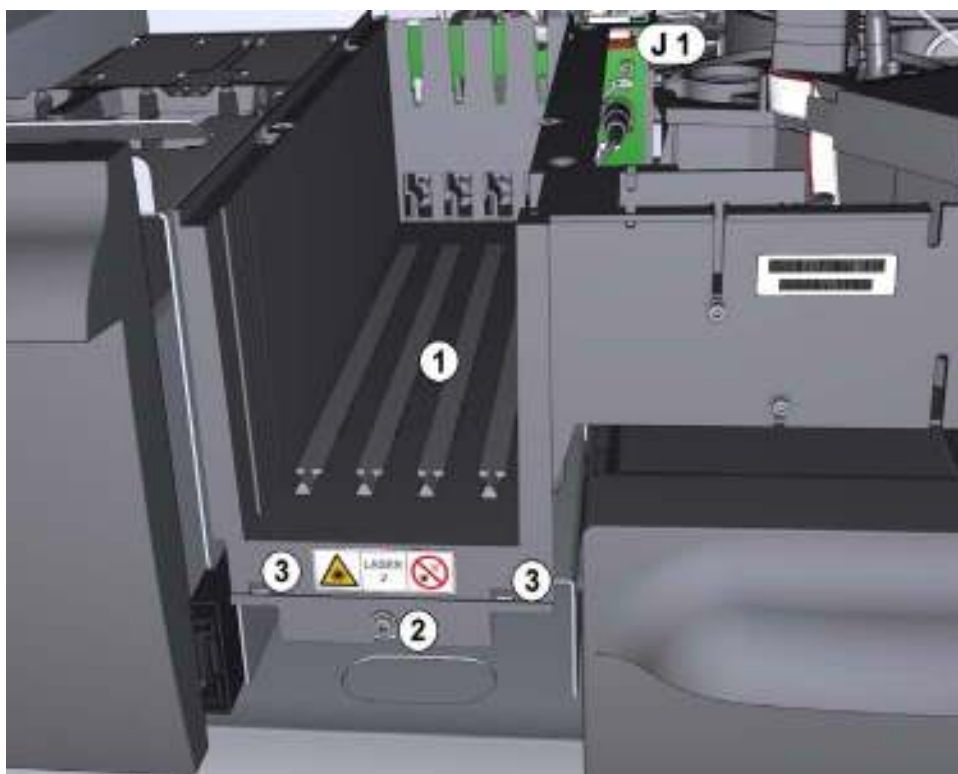


Figure 21–7: Sample Module

**21.3.2.3 Assembly of the Sample Module**

1. Set the Sample Module into the guides.
2. Push the Sample Module to the back, until hooks (3) are in position.
3. Fasten the screw (2).
4. Plug the cable from Loading Bay CU PCBA into the connector (J1).
5. Assemble the Sample Bay cover, see Chapter 21.3.1.3 on page 21-6.
6. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.

**21.3.2.4 Set of the Sample Module**

1. Check the teaching positions for the Left Pipettor, see Chapter 13.11 on page 13-27.
2. Perform the barcode reading test, see Chapter 21.4.2.2 on page 21-21.
3. Check the ambient temperature sensor, see Chapter 21.4.2.3 on page 21-22.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.



### 21.3.3 Replacement of the push/push mechanism

#### 21.3.3.1 Preliminary steps



5. Exit the **LIAISON<sup>®</sup> XS** Main Software.
6. Shut down the Panel PC from the Desk shield or from the Windows menu, and wait until the shutdown is complete.
7. On the right side of the instrument, turn off the Main Switch and unplug the power chord to completely cut the power to the instrument.
8. Open the Top Covers.
9. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
10. Move the Left Pipettor arm to a safe location on the leftmost area of the instrument.
11. Remove the Sample Bay cover, see Chapter 21.3.1.2 on page 21-5.
12. Remove the Sample Module, see Chapter 21.3.2.2 on page 21-7.

#### 21.3.3.2 Removal of the push-push mechanism

1. Lift the snap lock (2) of the push-push mechanism up.
2. Slide the push-push mechanism (1) to the back, lift it up and set it aside.

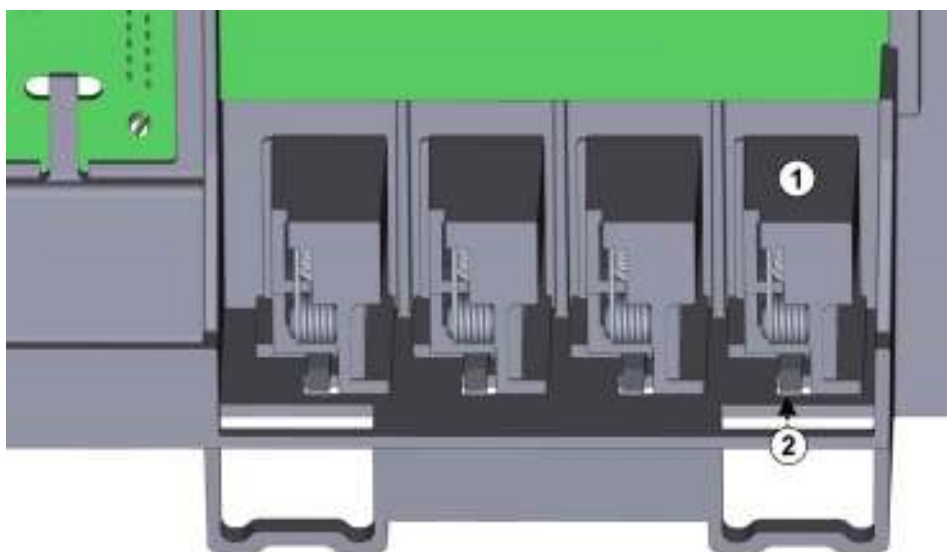
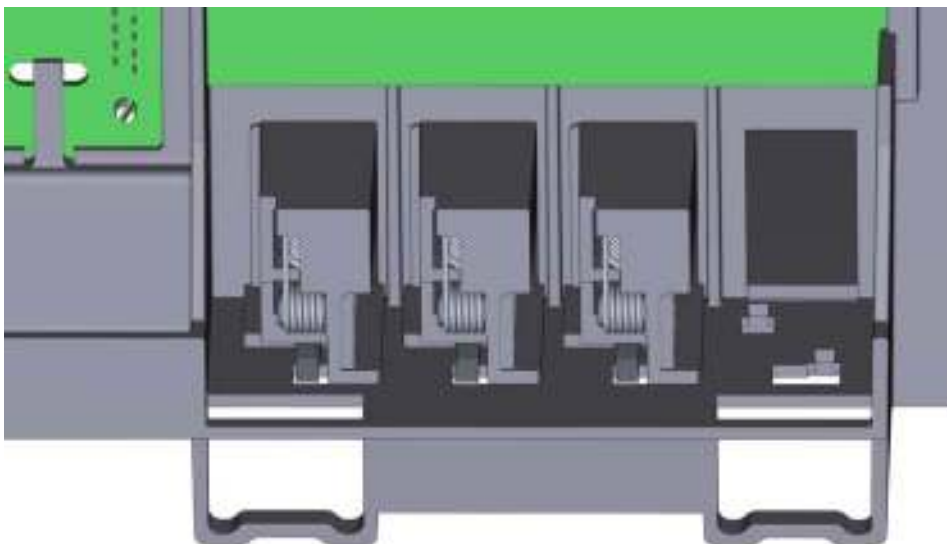


Figure 21–8: Sample Module, rear view

**21.3.3.3 Assembly of the push-push mechanism**

1. Insert the push-push mechanism (1) from the top into the cut-outs.
2. Slide the push-push mechanism to the front, until the snap lock is in position.
3. Assemble the Sample Module, see Chapter 21.3.2.3 on page 21-8.
4. Assemble the Sample Bay cover, see Chapter 21.3.1.3 on page 21-6.
5. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.



*Figure 21–9: Sample module, rear view, w/o one push push lock*

**21.3.3.4 Set of the Sample module**

1. Check the teaching positions for the Left Pipettor, see Chapter 13.11 on page 13-27.
2. Perform the barcode reading test, see Chapter 21.4.2.2 on page 21-21.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 21.3.4 Replacement of the Sample Bay Guides

### 21.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** Main Software.
2. Shut down the Panel PC from the Desk shield or from the Windows menu, and wait until the shutdown is complete.
3. On the right side of the instrument, turn off the Main Switch and unplug the power cable to completely cut the power to the instrument.
4. Open the Top Covers.
5. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
6. Move the Left Pipettor Arm to a safe location on the leftmost area of the instrument.
7. Remove the Sample Bay Cover, see Chapter 21.3.1.2 on page 21-5.

### 21.3.4.2 Removal of the Sample Bay Guides

1. Arch the sample bay guide in correspondence of the blocking pin (see gray little hole between the four sample bay insertion points of the second line showed in Figure 21–10).
2. Push the Sample bay guide to the back and lift it up.

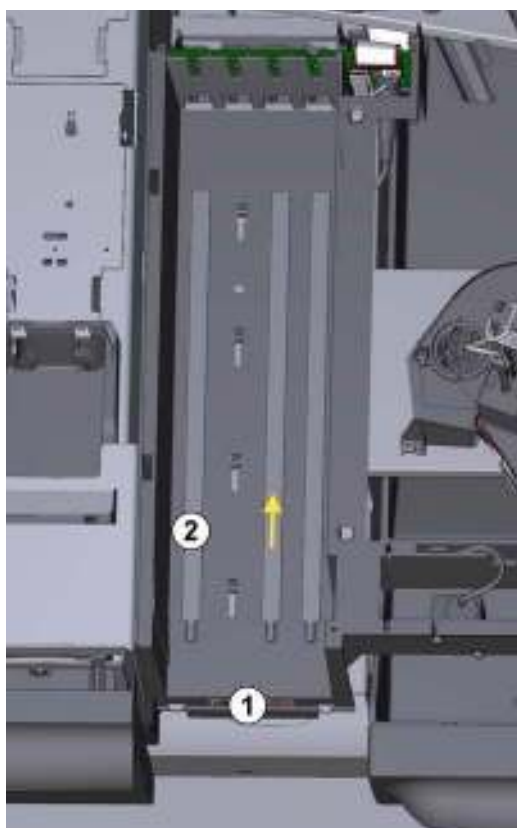


Figure 21–10: Sample Module, top view

**21.3.4.3 Assembly of the Sample Bay Guides**

1. Insert the Sample Bay guide and push it to the front until it is in position.
2. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
3. Assemble the Sample Bay Cover, see Chapter 21.3.1.3 on page 21-6.

**21.3.4.4 Set Sample Module**

1. Check the teaching positions for the Left Pipettor, see Chapter 13.11 on page 13-27.
2. Perform the barcode reading test, see Chapter 21.4.2.2 on page 21-21.

## 21.3.5 Replacement of the barcode scanner

### 21.3.5.1 Preliminary steps



3. Exit the **LIAISON<sup>®</sup> XS** Main Software.
4. Shut down the Panel PC from the Desk shield or from the Windows menu, and wait until the shutdown is complete.
5. On the right side of the instrument, turn off the Main Switch and unplug the power chord to completely cut the power to the instrument.
6. Switch off the instrument and separate it from main power supply.
7. Open the Top Covers.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
9. Move the Left Pipettor arm to a safe location on the leftmost area of the instrument.
10. Remove the Sample Bay cover, see Chapter 21.3.1.2 on page 21-5.
11. Remove the Sample Module, see Chapter 21.3.2.2 on page 21-7.

### 21.3.5.2 Removal of the barcode scanner

1. Unplug the barcode scanner connector (1) from the Sample Bay PCBA.
2. Unwire and unhook the barcode scanner cable from the brackets at the bottom of the Sample Bay frame.

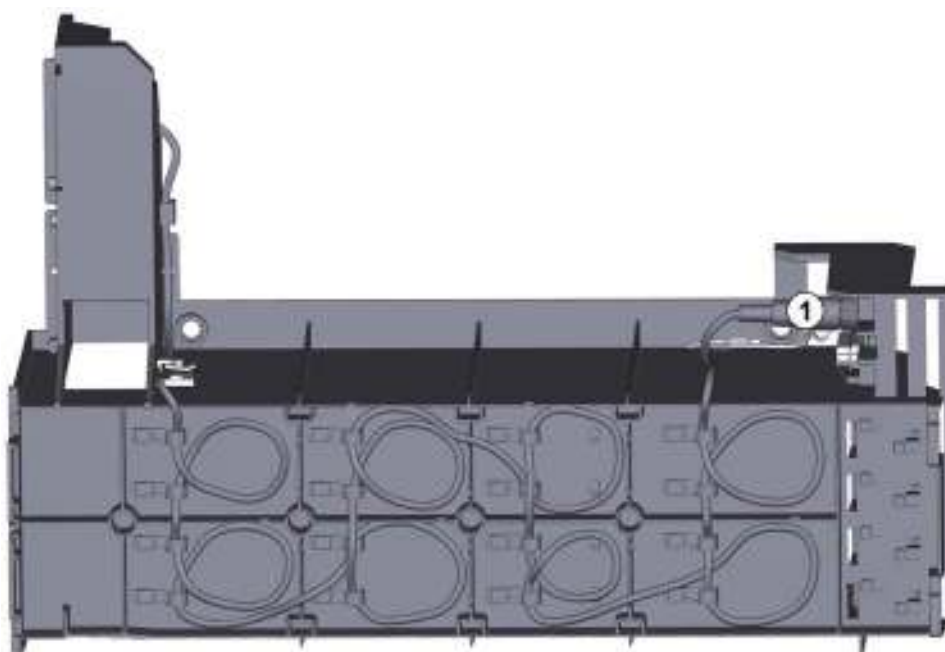


Figure 21–11: Sample Module, bottom view

3. Unscrew the two screws (2) and remove the barcode scanner.

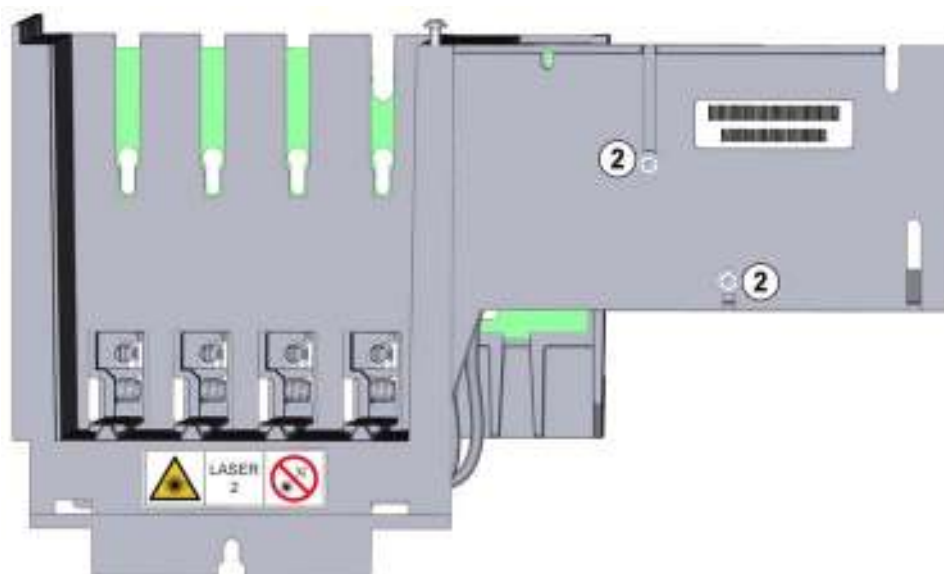


Figure 21-12: Sample Module, front view

#### 21.3.5.3 Assembly of the barcode scanner

1. Set the barcode scanner (3) into place and fasten it with washers and the two screws (2).

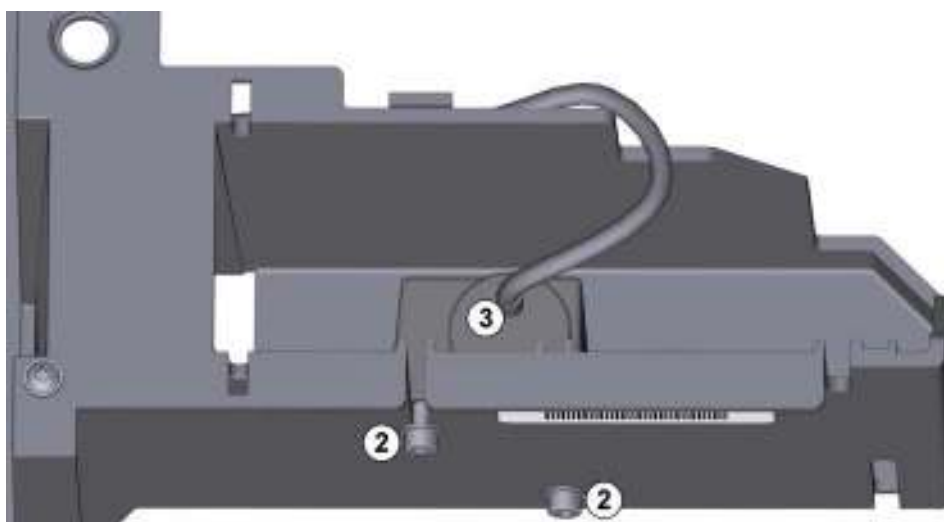
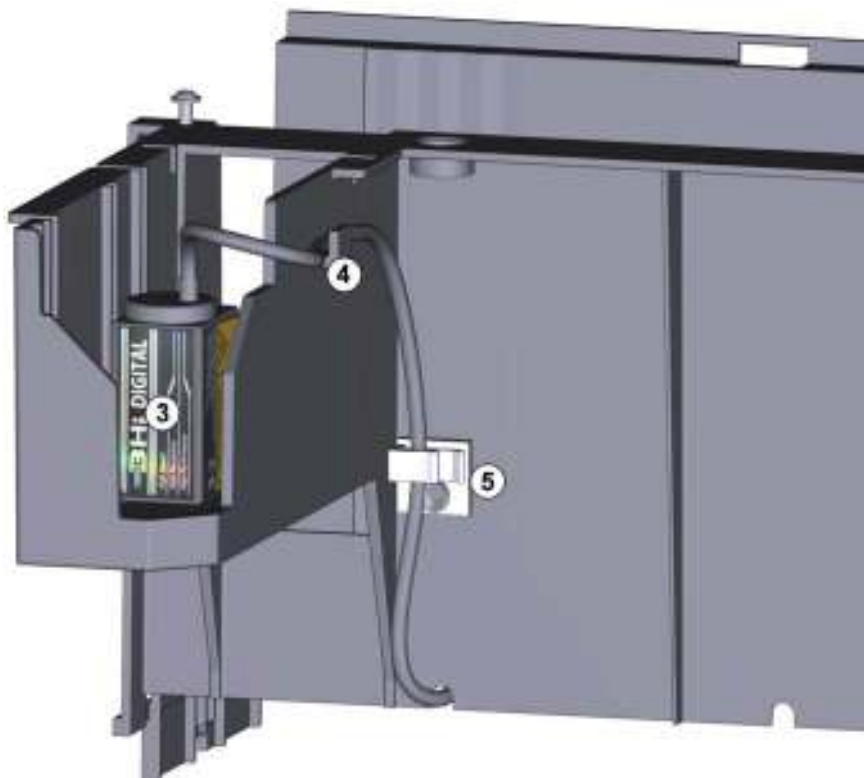


Figure 21-13: Barcode Scanner, top view

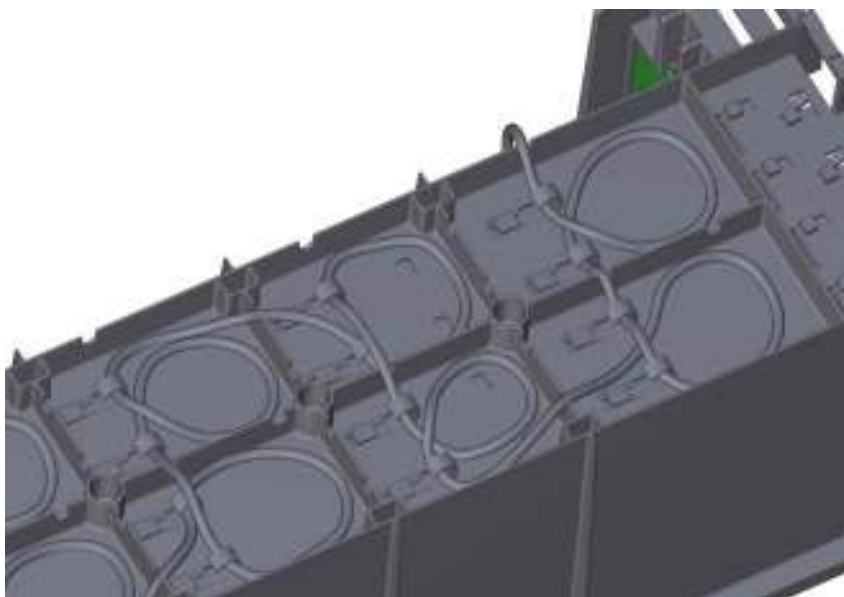
2. Lock the cable of the barcode scanner (3) into the fixing bracket (4) and the cable holder (5).



*Figure 21–14: Barcode scanner*

3. Lay the cable in loops (one loop per chamber) and insert the cable into the fixing brackets at the bottom.
4. Plug the barcode scanner connector (1) onto the Sample Bay PCBA.
5. Assemble the Sample Module, see Chapter 21.3.2.3 on page 21-8.
6. Assemble the Sample Bay cover, see Chapter 21.3.1.3 on page 21-6.

7. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.



*Figure 21–15: Barcode scanner cable layout*

#### **21.3.5.4 Set of the Sample Module**

1. Check the teaching positions for the Left Pipettor, see Chapter 13.11 on page 13-27.
2. Configure the barcode scanner, see Chapter 21.4.2.1 on page 21-21.
3. Perform the barcode reading test, see Chapter 21.4.2.2 on page 21-21.
4. Check the ambient temperature sensor, see Chapter 21.4.2.3 on page 21-22.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 Or see Chapter 10.4 on page 10-10.



### 21.3.6 Replacement of the Sample Bay PCBA

#### 21.3.6.1 Preliminary steps



6. Exit the **LIAISON<sup>®</sup> XS** Main Software.
7. Shut down the Panel PC from the Desk shield or from the Windows menu, and wait until the shutdown is complete.
8. On the right side of the instrument, turn off the Main Switch and unplug the power chord to completely cut the power to the instrument.
9. Open the Top Covers.
10. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
11. Move the Left Pipettor arm to a safe location on the leftmost area of the instrument.
12. Remove the Sample Bay cover, see Chapter 21.3.1.2 on page 21-5.
13. Remove the Sample Module, see Chapter 21.3.2.2 on page 21-7.

#### 21.3.6.2 Removal of the Sample Bay PCBA

1. Unplug the barcode scanner cable connector from slot (J2) and the temperature sensor connector from slot (J3).
2. Push the hook (3) in, lift the Sample Bay PCBA (1) carefully up and remove it.

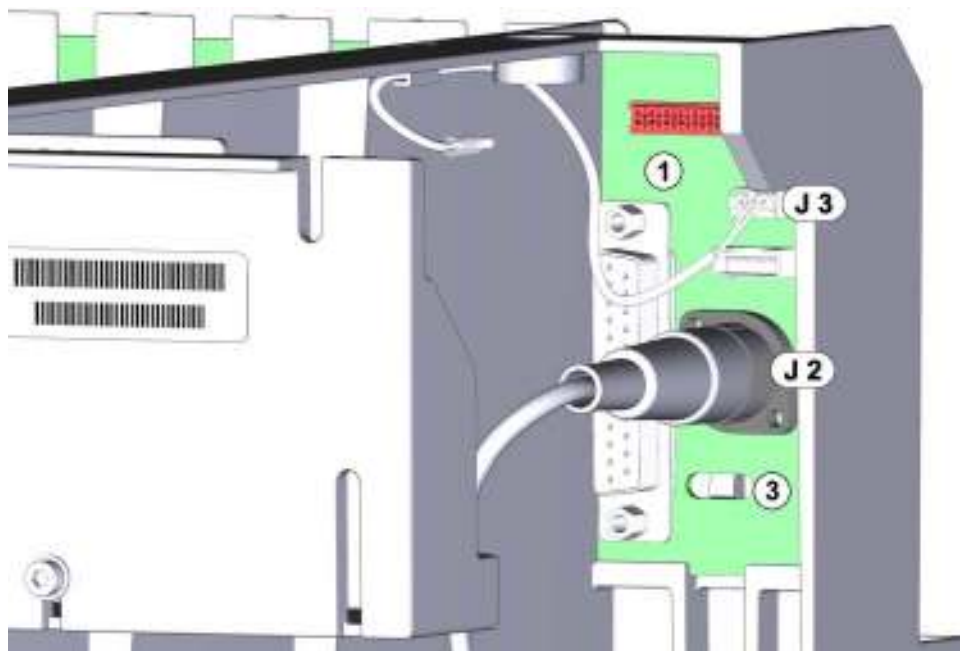


Figure 21–16: Sample Module

**21.3.6.3 Assembly of the Sample Bay PCBA**

1. Slide the sample bay PCBA (1) carefully into the guide of the Sample Bay, until the hook (3) snaps in.
2. Plug the barcode scanner cable connector into slot (J2) and the temperature sensor connector into slot (J3) of the Sample Bay PCBA.
3. Assemble the Sample Module, see Chapter 21.3.2.3 on page 21-8.
4. Assemble the Sample Bay cover, see Chapter 21.3.1.3 on page 21-6.
5. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.

**21.3.6.4 Set of the Sample Bay PCBA**

1. Check the teaching positions for the Left Pipettor, see Chapter 13.11 on page 13-27.
2. Perform the barcode reading test, see Chapter 21.4.2.2 on page 21-21.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 21.3.7 Replacement of the Temperature sensor

#### 21.3.7.1 Preliminary steps



4. Exit the **LIAISON<sup>®</sup> XS** Main Software.
5. Shut down the Panel PC from the Deskshield or from the Windows menu, and wait until the shutdown is complete.
6. On the right side of the instrument, turn off the Main Switch and unplug the power chord to completely cut the power to the instrument.
7. Open the Top Covers.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
9. Move the Left Pipettor arm to a safe location on the leftmost area of the instrument.
10. Remove the Sample Bay cover, see Chapter 21.3.1.2 on page 21-5.
11. Remove the Sample Module, see Chapter 21.3.2.2 on page 21-7.

#### 21.3.7.2 Removal of the Temperature sensor

1. Unplug the temperature sensor connector from slot (J3) of the Sample Bay PCBA.
2. Slide the temperature sensor (1) out of the fixing bracket (2).

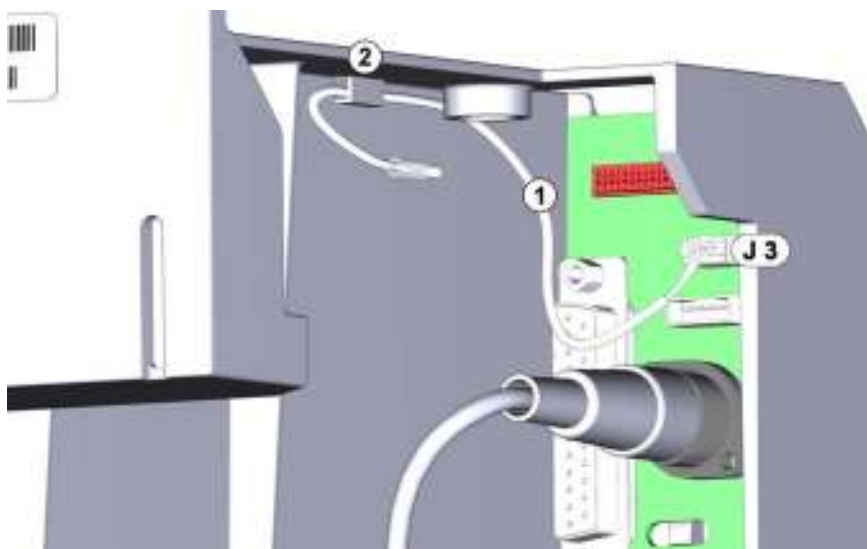


Figure 21–17: Temperature sensor

#### 21.3.7.3 Assembly of the Temperature sensor

1. Slide the temperature sensor (1) into the fixing bracket (2).
2. Plug the temperature sensor connector into slot (J3) of the Sample Bay PCBA.
3. Assemble the Sample Module, see Chapter 21.3.2.3 on page 21-8.
4. Assemble the Sample Bay cover, see Chapter 21.3.1.3 on page 21-6.
5. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.

**21.3.7.4 Set the Sample Module**

1. Check the teaching positions for the Left Pipettor, see Chapter 13.11 on page 13-27.
2. Check the ambient temperature sensor, see Chapter 21.4.2.3 on page 21-22.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 21.4 Adjustment and calibration

This section contains information about check and verification of the Sample Module.

### NOTE

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).

Do not run any of the software at the same time!

### 21.4.1 Initialization routine

- Close any open software and start the **Service Software** from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **Front and Loading** and the **General** bracket.
- Click onto the **Initialize** button and choose the **Loading Bay**. Press the **General** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 21.4.2 Check and Verification of the Sample Module

#### 21.4.2.1 Barcode scanner configuration

In the **Service Software** at the navigation window, choose **Front and Loading** and the **Sample Bay Scanner** bracket.

- Click onto the **Read** button to read out the actual configuration.
  - Code 39 Range: 4 - 22
  - Code 128, EAN 128 Range: 4 - 22
  - Codabar Range: 4 - 22
- Clear all the check boxes that are not grayed out at the Checksum (Verify & Display).
- Click onto the **Write** button to save the settings.

#### 21.4.2.2 Barcode reading test

In the **Service Software** at the navigation window choose **Front and Loading** and the **Sample Bay Scanner** bracket.

- Click onto the **Activate Lane - Activate** button to start the scanner.
- Insert a sample rack and check the correct reading of barcodes.
- Test each Sample Bay lane. For each lane, the barcode scanner has to be reactivated.

At the beginning, the background of the Lane State is gray.

When a rack is moved to the end stop, the status of the lane switches to green.

When a rack is removed, the status of the lane switches from green to red.

**21.4.2.3 Check of the Ambient Temperature Sensor**

- a. In the **Service Software** at the navigation window choose **Front and Loading** and the **General** bracket.
- b. Click onto the **Sensors** button and choose the **Loading Bay Temp. only** button.
- c. In the Temperature field, check that the Ambient Temperature showed is a value similar to room temperature.

## 21.5 Electric connections, switches and signals

This section shows an overview of the electronic boards of the Sample Module.

### NOTE

see Chapter 3.1.7 on page 3-8 or see Chapter 4.1.8 on page 4-9 for an overview of the complete electric circuit of the system.

### 21.5.1 Sample Bay PCBA

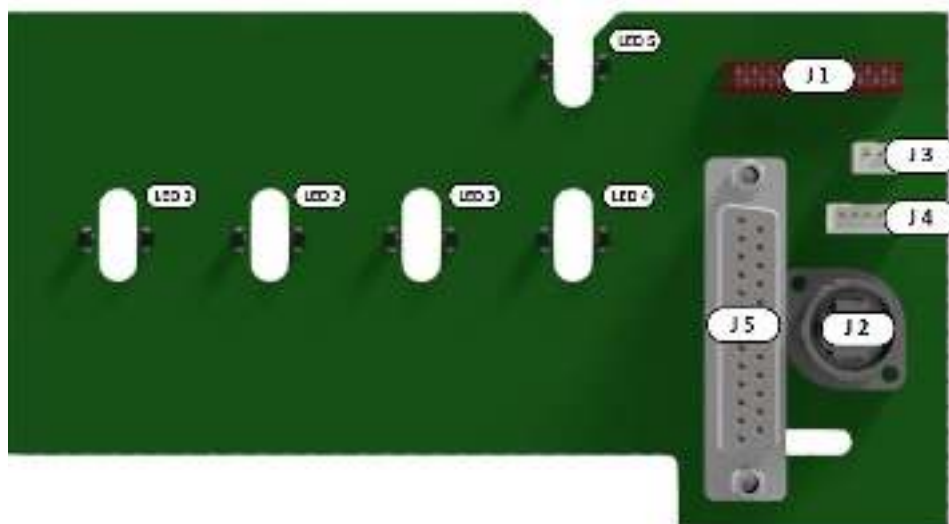


Figure 21-18: Sample bay PCBA

| Number | Connection                 |
|--------|----------------------------|
| J1     | Loading Bay CU PCBA        |
| J2     | Scanner (round plug)       |
| J3     | Ambient temperature sensor |
| J4     | not used                   |
| J5     | Scanner                    |

Table 21-4: Connections of the sample bay PCBA

| Number | Meaning/Function             |
|--------|------------------------------|
| LED 1  | Rack lane 1 (leftmost rack)  |
| LED 2  | Rack lane 2                  |
| LED 3  | Rack lane 3                  |
| LED 4  | Rack lane 4 (rightmost rack) |
| LED 5  | (optional)                   |

*Table 21–5: LED of the sample bay PCBA*



## 22 Reagent Module

This chapter describes the Reagent Module in detail. The first section of the chapter introduces “Safety” notes concerning the Reagent Bay, followed by a brief overview of the main parts composing this module.

The following section contains information about repair and calibration (“Replacement and Repair”, and “Adjustment and Calibration” sections).

In the last section, Reagent Loading Bay CU PCBA and connections are described.

- The Reagent Module stores reagents that are needed to process samples on the LXS system.
- Reagents are stored in a cooled environment, in order to prolong the on board stability of the chemicals.
- Reagents are agitated by a stirring rod, in order to ensure the suspension of the magnetic beads in the bead solution.
- The module supports loading/unloading of integrals and ancillaries via a flap, that is also insulating the reagent compartment.
- Integrals and ancillaries are identified via RFID technology.
- The Reagent Loading Bay CU PCBA also hosts the firmware to handle the Sample Bay.

### 22.1 Safety

---

**WARNING****Non-observance of safety instructions**

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 OR see Chapter 2.5 on page 2-7.
  - Follow all working instructions included in this manual.
  - Follow all warnings marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
-

## 22.2 Overview

This section shows an overview of the components of the Reagent Module.

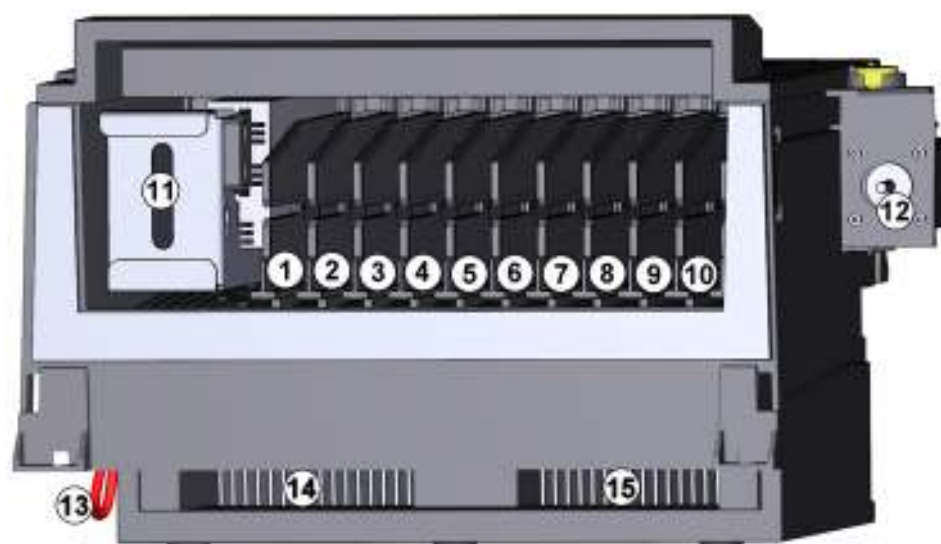


Figure 22–1: Reagent Module, front view

|        |                                 |
|--------|---------------------------------|
| 1 - 10 | Integral lanes                  |
| 11     | Ancillary tray                  |
| 12     | Stirrer drive motor             |
| 13     | Reagent Module-WAP drain tubing |
| 14     | Cooling module                  |
| 15     | Cooling module                  |

Table 22–1: Reagent Module, front view

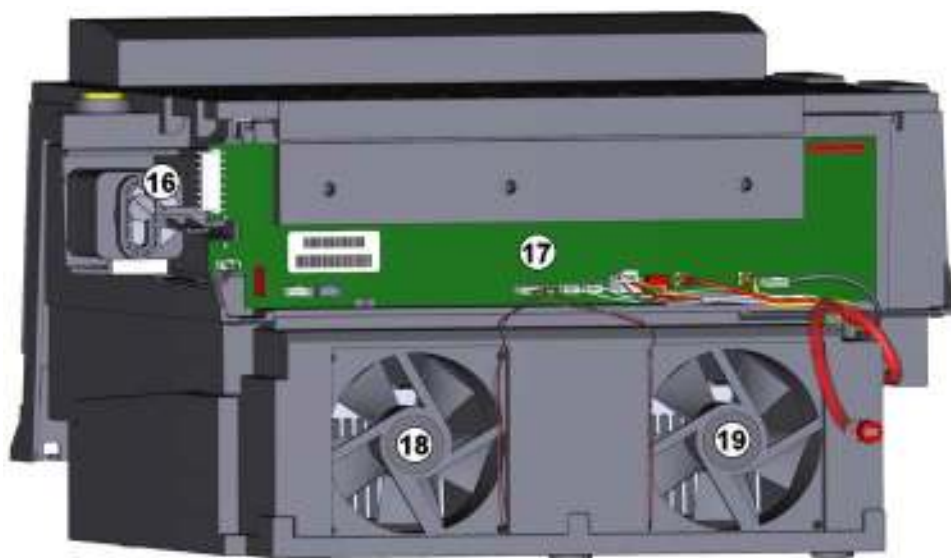


Figure 22–2: Reagent Module, rear view

|    |                            |
|----|----------------------------|
| 16 | Stirrer rod                |
| 17 | Reagent Loading Bay CU PCB |
| 18 | Cooling fan                |
| 19 | Cooling fan                |

Table 22–2: Reagent Module, rear view

## 22.3 Replacement and repairs

This section contains instructions for the repair of individual components of the Reagent Module. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 22.3.1 Replacement of the Reagent Bay cover

#### 22.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location on the rearmost area of the instrument.

#### 22.3.1.2 Removal of the Reagent Bay cover

1. Unscrew two screws (2), lift the Reagent Bay cover (1) up and set it aside.



Figure 22–3: Reagent Bay cover

#### 22.3.1.3 Assembly of the Reagent Bay cover

1. Slide the Reagent Bay cover (1) onto the Reagent Module and fasten with two screws (2).

## 22.3.2 Replacement of the Stirrer Rod and Stirrer Bearings

### 22.3.2.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location on the rearmost area of the instrument.
7. Remove the Reagent Bay cover, see Chapter 22.3.1.2 on page 22-4.
8. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.

### 22.3.2.2 Removal of the Stirrer Rod and Stirrer Bearings

#### NOTE

If the screw (1) is too much tightened and cannot be loosened, then the whole Reagent Module has to be removed, see Chapter 22.3.4.2 on page 22-12.

1. Unscrew the slotted screw (1) of the stirrer rod (2).
2. Slide the stirrer rod (2) carefully out.

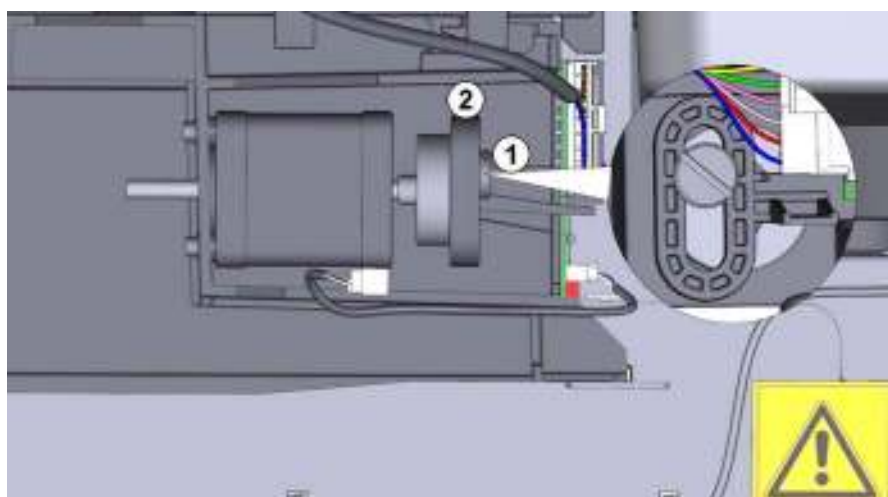


Figure 22–4: Remove stirrer rod

3. Push with a finger the stirrer bearings out of the cooling fins from the left side direction.

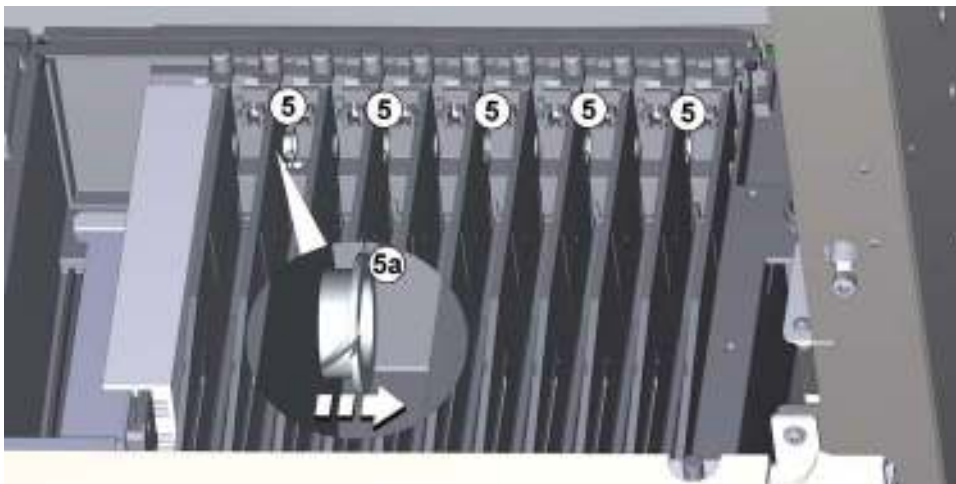


Figure 22-5: Reagent Module

### 22.3.2.3 Assembly of the Stirrer Rod and Stirrer Bearings

#### CAUTION

Pay attention to the bearing orientation, the bigger side of the stirrer bearing (5a) shall be positioned to the right side of the cooling fin.

1. Insert the new stirrer bearings (5) into the holes of the cooling fins.



Figure 22-6: Stirrer Bearing

2. Insert the stirrer rod (2) from the Right side opening of the Reagent Module.
3. Moisten the thread of the slotted screw (1) with a little bit of low-strength thread locker (Loctite 222).
4. Fasten the stirrer bar (2) with the moistened screw (1).
5. Remove any surplus of the thread locker.
6. If needed, perform the readjustment of the cam to the stirrer rod, see Section 22.4.3 on page 22-31.
7. If the Reagent Module was removed, then assemble the Reagent Module, see Chapter 22.3.4.3 on page 22-13.
8. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.

**22.3.2.4 Set of the Reagent Module**

1. If the Reagent Module was removed, check the teaching positions for the Right Pipettor, see Chapter 13.16 on page 13-35 and see Chapter 13.17 on page 13-36.
2. Assemble the Reagent Bay cover, see Chapter 22.3.1.3 on page 22-4.
3. Verify the stirring, see Chapter 22.4.2.2 on page 22-30.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 22.3.3 Replacement of the Stirrer Motor

#### 22.3.3.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location on the rearmost area of the instrument.
7. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.

#### 22.3.3.2 Removal of the Stirrer Motor

1. Unplug the motor cable connector (2) from the stirrer motor (1).
2. Unscrew the slotted screw (4) at the crank drive adapter (5).

#### NOTE

If the screw (4) is too much tightened and cannot be loosened, then the whole Reagent Module has to be removed, see Chapter 22.3.4.2 on page 22-12.

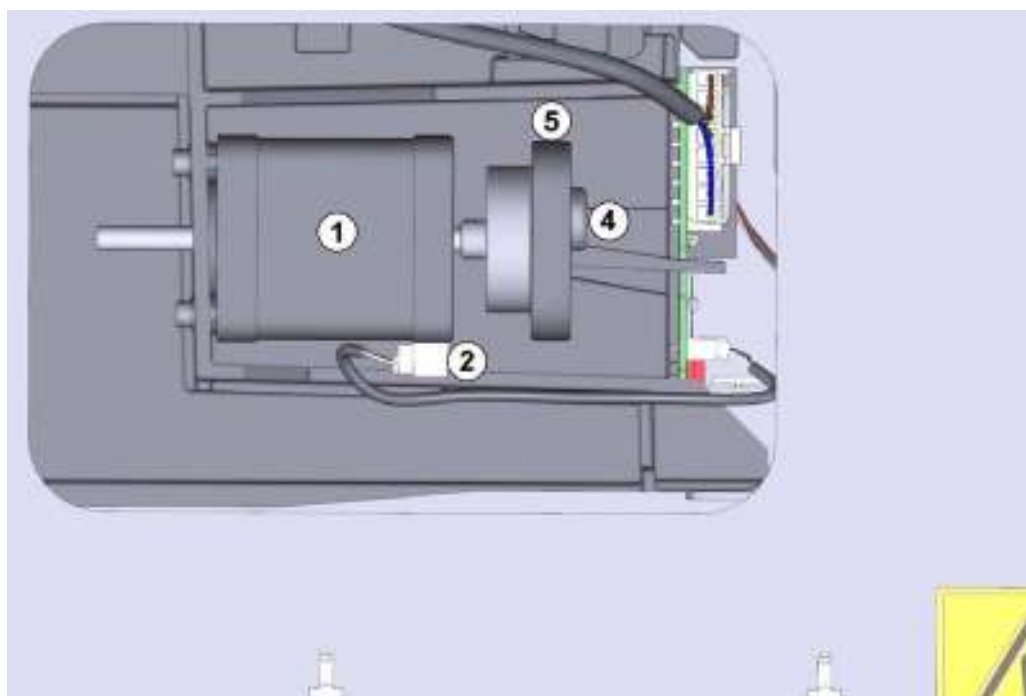


Figure 22–7: Unplug the stirrer motor cable and remove the slotted screw



3. Unscrew the four screws (3) and remove the stirrer motor (1).

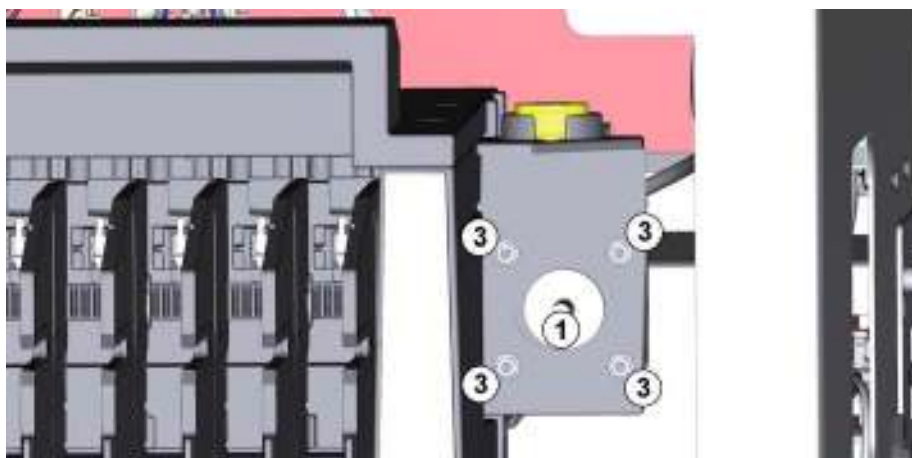


Figure 22–8: stirrer motor removal

#### 22.3.3.3 Assembly of the Stirrer Motor

1. Insert the new stirrer motor (1) orienting it properly (the motor cable outlet is positioned downwards) and fasten with four screws and washers (3).
2. Plug the motor cable (2) onto the stirrer motor (1).
3. Slide the crank drive onto the motor spindle. Do not fasten the two screws (6).
4. Moisten the slotted screw (4) with a little bit of low-strength thread locker (Loctite 222).
5. Fasten the slotted screw with the stirred rod (5) onto the crank drive.

#### 22.3.3.4 Setting of the Crank Drive

1. Completely rotate the crank drive two or three times to align the crank drive to the stirrer rod; then fasten the two set screws (6) of the crank drive.

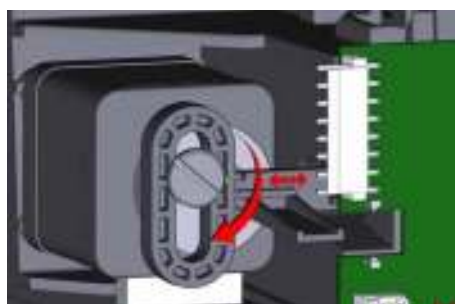


Figure 22–9: Crank Drive

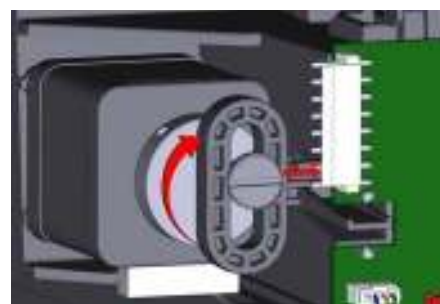
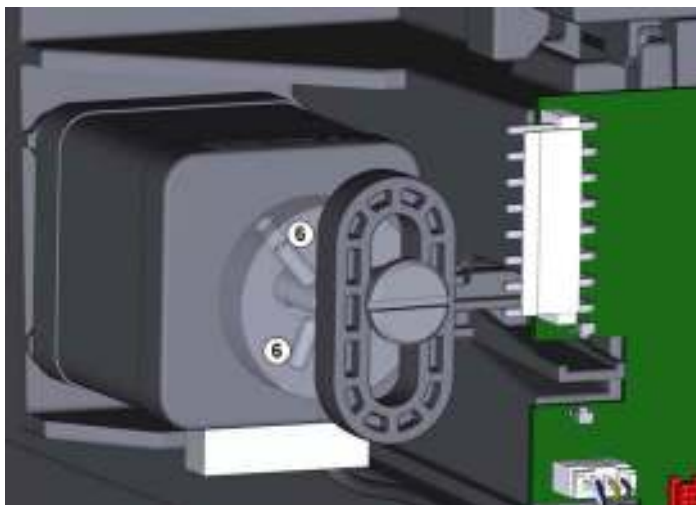


Figure 22–10: Crank Drive

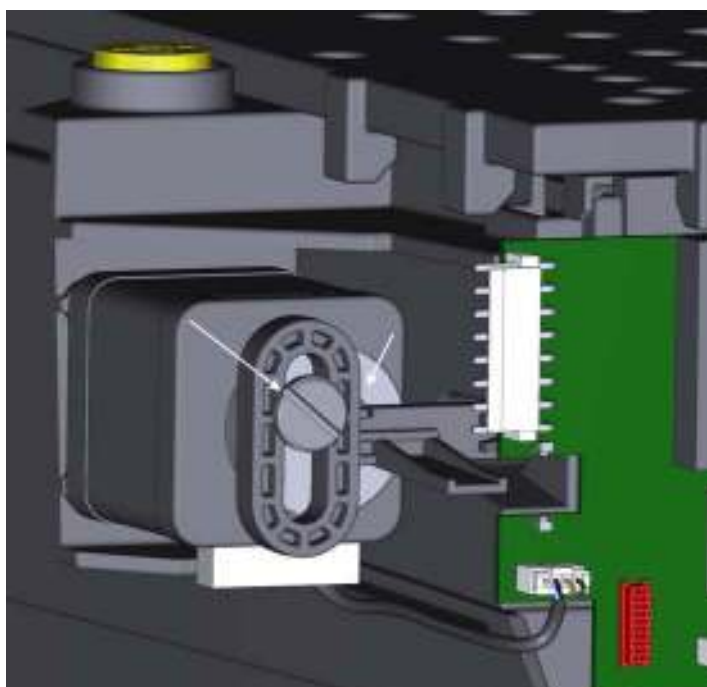


*Figure 22-11: Crank Drive, inner view*

2. If needed, perform the readjustment of the cam to the stirrer rod, see Section 22.4.3 on page 22-31.

#### **22.3.3.5 Lubrication of the Crank Drive**

1. Using a tissue soaked with grease, lubricate the crank drive of the motor and the stirrer rod cam.



*Figure 22-12: Crank Drive*

2. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.

### **22.3.3.6 Set of the Reagent Module**

1. If the Reagent Module was removed, check the teaching positions for the Right Pipettor, see Chapter 13.16 on page 13-35 and see Chapter 13.17 on page 13-36.
2. Verify the stirring, see Chapter 22.4.2.2 on page 22-30.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 22.3.4 Replacement of the Reagent Module

### 22.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm above the Incubator Module.
7. Remove the Wash Station cover, see Chapter 28.3.14.2 on page 28-47.
8. Remove the Front Cover, see Chapter 27.3.1.2 on page 27-4.

#### NOTE

Ensure that any reagent integrals and/or ancillary reagents have been unloaded from the instrument.

### 22.3.4.2 Removal of the Reagent Module

1. Unplug the following cables from the Reagent Loading Bay CU PCB:
  - PWR CAN cable connector (J14)
  - Sample Bay PCBA connector (J8)
2. Loosen the two screws (2) and carefully pull the Reagent Module (1) a bit out.

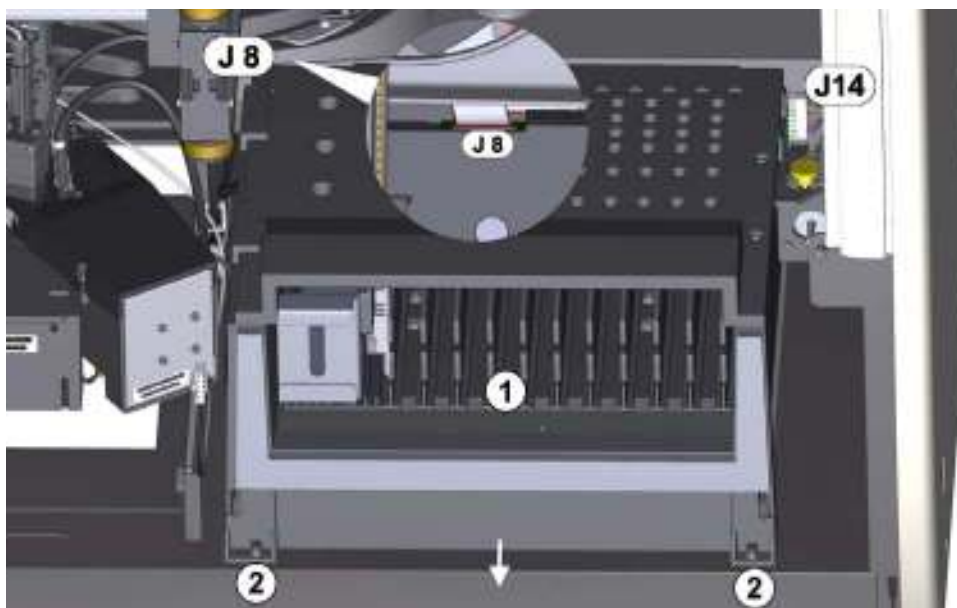


Figure 22–13: Reagent Module removal

3. Unscrew the connector of Reagent Module-WAP tubing (3).
4. Pull the module further to the front, lift it out and set it aside.

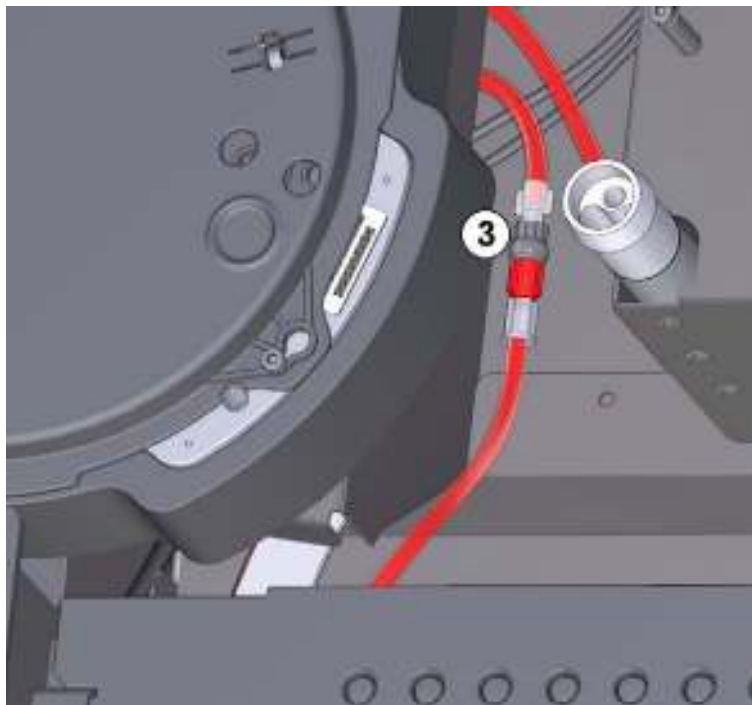


Figure 22-14: Reagent Module, disconnect waste tube

#### 22.3.4.3 Assembly of the Reagent Module

1. Partially insert the Reagent Module into the instrument.
2. Connect the Reagent Module-WAP tubing (3).
3. Plug the following cables onto the Reagent Loading Bay CU PCBA:
  - PWR CAN connector (J14)
  - Sample Bay PCBA connector (J8).
4. Push the module into the guides and fasten the two screws (2).
5. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
6. Assemble the Wash Station cover, see Chapter 28.3.14.3 on page 28-48.

#### 22.3.4.4 Set of the Reagent Module

1. In case a new module has been installed, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Check the teaching positions for the Right Pipettor, see Chapter 13.16 on page 13-35 and see Chapter 13.17 on page 13-36.
3. Check the condensation aspiration, see Chapter 22.4.2.1 on page 22-30.
4. Verify the stirring, see Chapter 22.4.2.2 on page 22-30.
5. Verify the Cooling Modules, see Chapter 22.4.2.3 on page 22-30.
6. Verify the microswitches and RFID, see Chapter 22.4.2.4 on page 22-31.
7. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## **22.3.5 Replacement of the Cooling Module**

### **22.3.5.1 Preliminary steps**



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm above the Incubator Module.
7. Remove the Reagent Module, see Chapter 22.3.4.2 on page 22-12.
8. Remove the Reagent Bay cover, see Chapter 22.3.1.2 on page 22-4.

### **22.3.5.2 Removal of the Cooling Module**

1. Turn the Reagent Module upside down.
2. Lift each side of the bottom insulation (3) a little bit up and remove it.



*Figure 22–15: Reagent Module, bottom insulation*

3. Unplug the power cables (4) of the cooling units at the clamping block (4).

4. Remove the cables from the insulation foam.

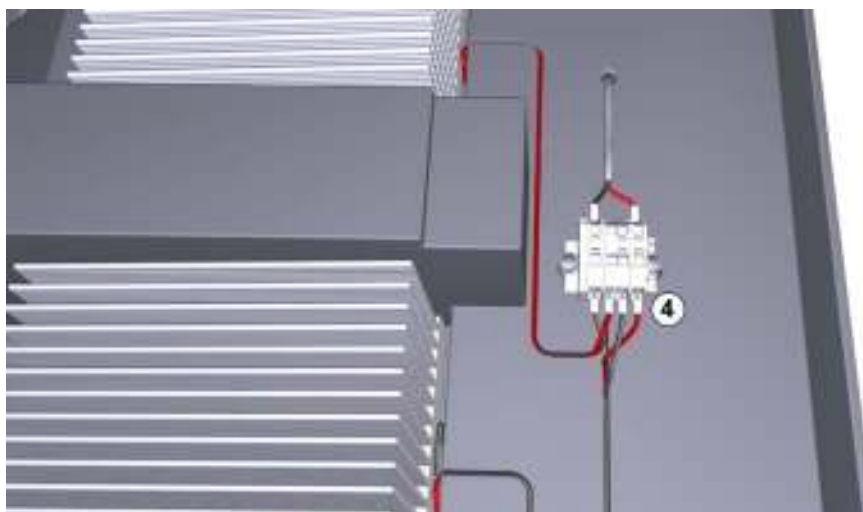


Figure 22-16: Reagent Module, bottom view

5. Unscrew four screws (5), paying attention to not lose the washers.

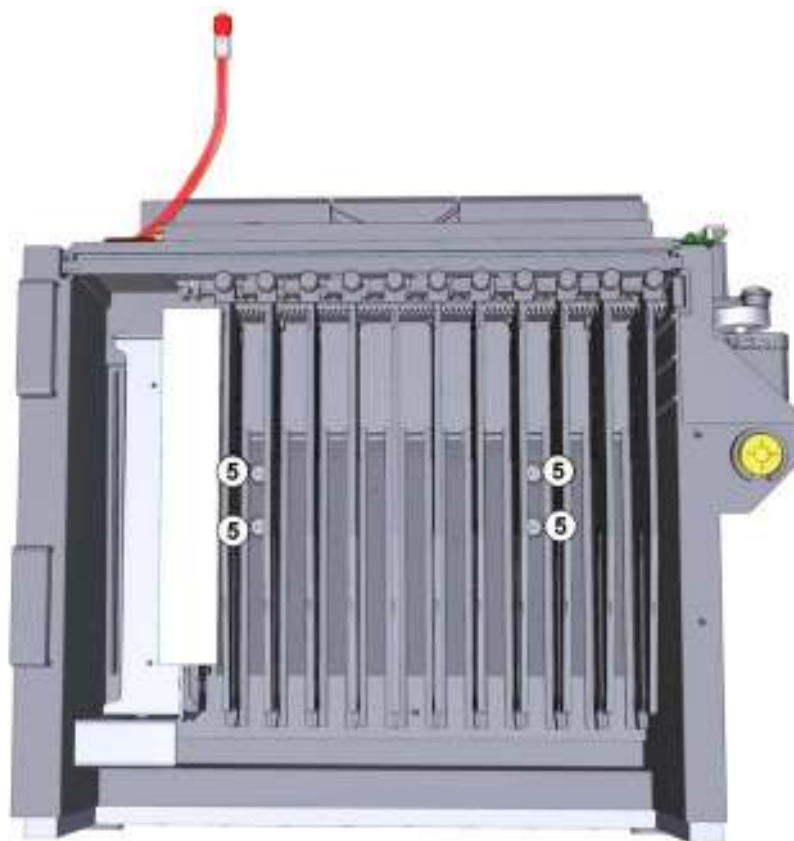
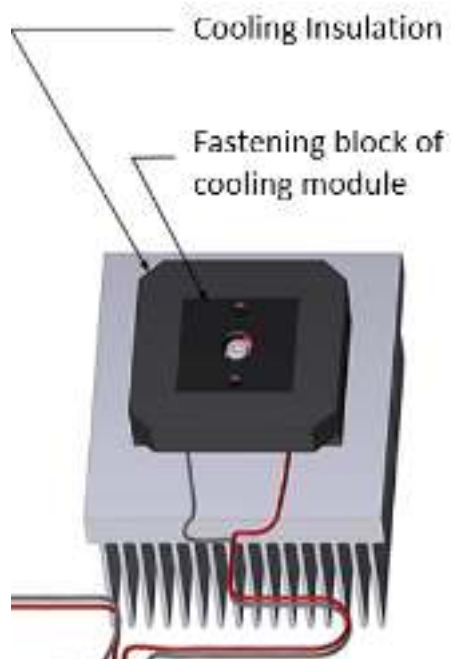


Figure 22-17: Reagent Module, top view

6. Remove the Cooling Modules.



*Figure 22–18: Cooling Module*

#### **22.3.5.3 Assembly of the Cooling Module**

1. Clean the top surfaces of the fastening block with isopropanol, make sure the isopropanol is evaporated (i.e. surface is not wet).
2. Paste the self adhesive heat transfer foil onto the fastening block of the new Cooling Module, see Figure 22–18.

---

**NOTE**

Do not squeeze or bend the cable of the Cooling Module.  
Take care of the cable outlet.

---

3. Set the Cooling Modules into the Reagent Module and fasten each with two screws (5) and two washers each.

---

**NOTE**

Avoid extra tightening force as the washers could be damaged by the screw head.

---

4. Lay the cables carefully into the grooves of the main insulation.
5. Connect the power cables of the cooling modules (10 and 11) to the clamping block (9).



| Position No. | Cable colour | Terminal connection | Cooling module |
|--------------|--------------|---------------------|----------------|
| 11a          | red          | rightmost           | 11             |
| 11b          | black        | middle right        | 11             |
| 10a          | red          | middle left         | 10             |
| 10b          | black        | leftmost            | 10             |

Table 22–3: Clamping block connection layout

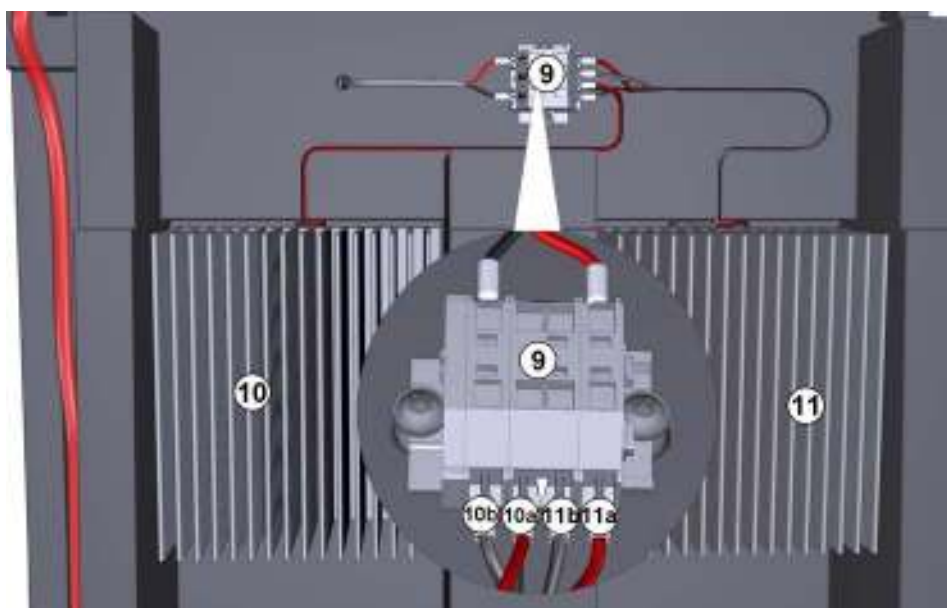


Figure 22–19: Cable layout for Cooling Module connection

6. Reconnect the plug to the connector on the PCBA.
7. Slide the bottom insulation (3) onto the Reagent Module.
8. Assemble the Reagent Module, see Chapter 22.3.4.3 on page 22-13.

#### 22.3.5.4 Set the Reagent Module

1. Check the teaching positions for the Right Pipettor, see Chapter 13.16 on page 13-35 and see Chapter 13.17 on page 13-36.
2. Assemble the Reagent Bay cover, see Chapter 22.3.1.3 on page 22-4.
3. Verify the Cooling Modules, see Chapter 22.4.2.3 on page 22-30.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## **22.3.6 Replacement of the Reagent Loading Bay CU PCB**

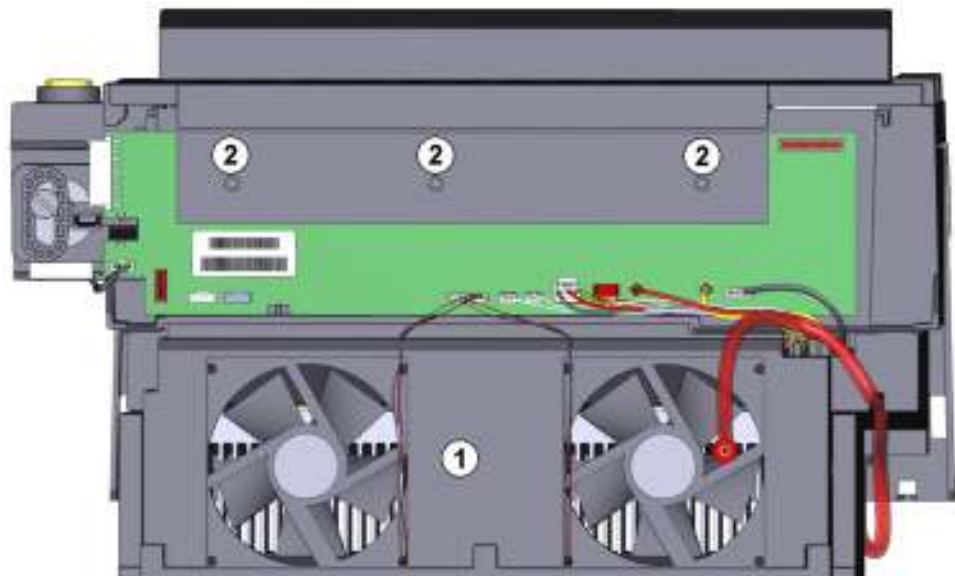
### **22.3.6.1 Preliminary steps**



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm above the Incubator Module.
7. Remove the Reagent Module, see Chapter 22.3.4.2 on page 22-12.

### **22.3.6.2 Removal of the Reagent Loading Bay CU PCB**

1. Unplug all cables from the Reagent Loading Bay CU PCB (1).
2. Unscrew three screws (2).
3. Carefully remove the Reagent Loading Bay CU PCB (1).



*Figure 22–20: Reagent Loading Bay CU PCB*

4. Slide the back silicone sealing (3) very carefully out of the micro switches (4).

#### **NOTE**

#### **Bending or breaking of contact spring of micro switch**

Be careful when removing the back silicone sealing.

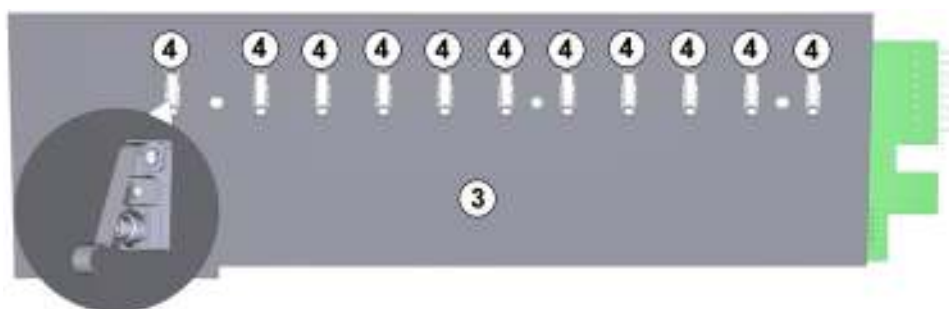


Figure 22–21: Reagent Loading Bay CU PCB, rear view (back silicone sealing)

### 22.3.6.3 Assembly of the Reagent Loading Bay CU PCB

#### NOTE

#### Bending or breaking of contact spring of micro switch

Be careful when assembling the back silicone sealing onto the Reagent Loading Bay CU PCB. Check each contact spring after sliding it through the silicone sealing.

1. Set the back silicone sealing (3) carefully onto the contact springs of the micro switches (4).
2. Carefully fasten the Reagent Loading Bay CU PCB (1) onto the Reagent Bay with three screws (2).
3. Assemble the Reagent Module, see Chapter 22.3.4.3 on page 22-13.

### 22.3.6.4 Set the Reagent Module

1. When the PCB is installed, then proceed with the firmware update.
2. Check the teaching positions for the Right Pipettor, see Chapter 13.16 on page 13-35 and see Chapter 13.17 on page 13-36.
3. Verify the microswitches, see Chapter 22.4.2.4 on page 22-31.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 22.3.7 Replacement of the Main Insulation

#### 22.3.7.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm above the Incubator Module.
7. Remove the Reagent Module, see Chapter 22.3.4.2 on page 22-12.
8. Remove the Cooling Module, see Chapter 22.3.5.2 on page 22-14.

#### 22.3.7.2 Removal of the Main Insulation

1. Unplug the fan cable (J13) from the Reagent Loading Bay CU PCB (1).
2. Remove carefully the cable of the fans (2) from the main insulation foam (3).

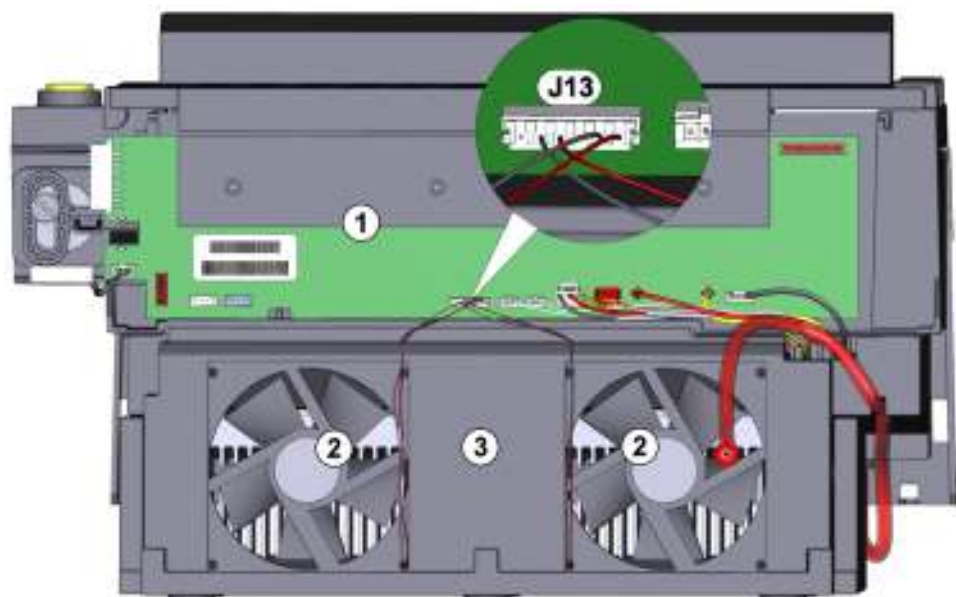


Figure 22–22: Reagent Module, remove fan cables

3. Turn the whole module upside down.
4. Lift the fans (2) carefully out of the main insulation foam.

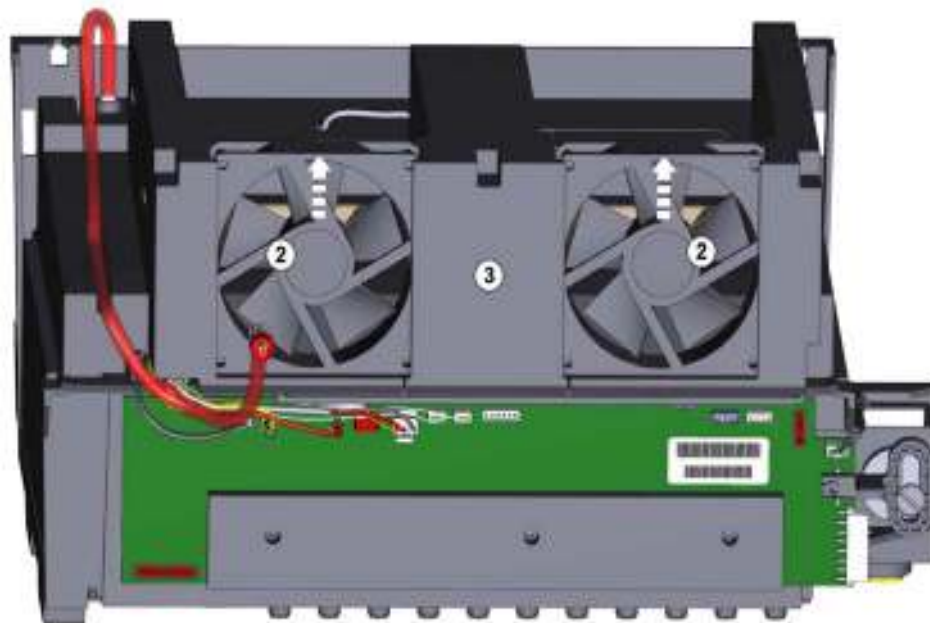


Figure 22-23: Remove fans

5. Unplug cables (5 and 6) at the clamping block (4).
6. Unscrew two screws (7) and remove the clamping block (4).

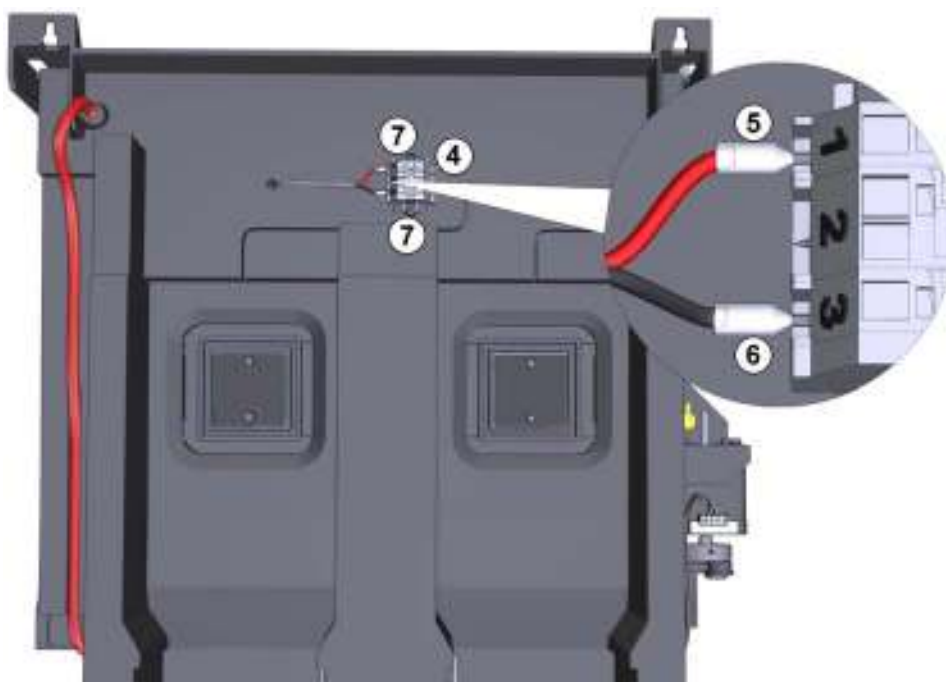
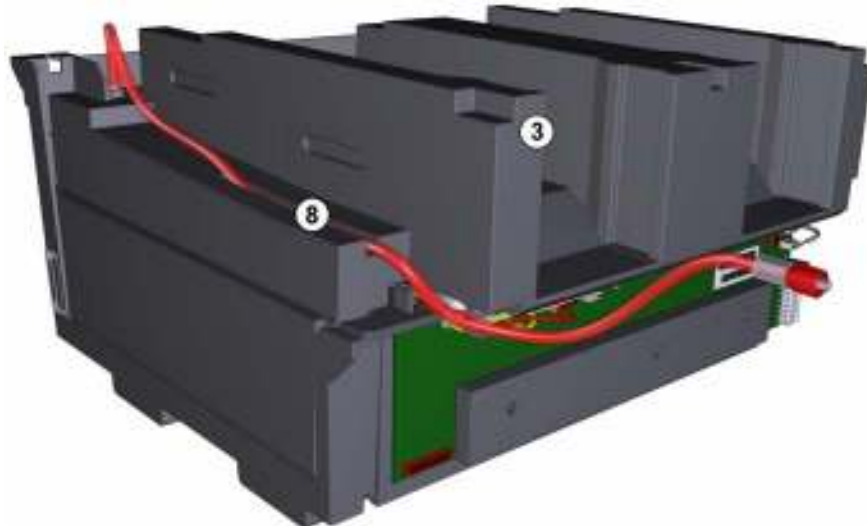


Figure 22-24: Clamping block removal

7. Remove the Reagent Module-WAP drain tubing (8) out of the main insulation.
8. Lift the main insulation (3) carefully up and set it aside.



*Figure 22-25: Main insulation*

#### **22.3.7.3 Assembly of the Main Insulation**

1. Hold the main insulation (3) over the Reagent Module and slide the power cables (5 and 6) through the main insulation.
2. Push the main insulation (3) carefully over the Reagent Module shell.

#### **CAUTION**

---

##### **Be careful while mounting the main insulation.**

Slide the main insulation carefully over the circuit board, connectors and cables. Do not damage or tear any cables and connectors.

---

3. Insert the Reagent Module-WAP tubing into the groove of the main insulation. Check that the tubing is not bent or squeezed.
4. Set the clamping block (4) and fasten with two screws (7).
5. Connect the power cables (5 and 6) to the clamping block (4).
6. Insert the fans (2) into the main insulation (3).
7. Plug the fan cables (J13) onto the Reagent Loading Bay CU PCB (1).
8. Assemble the Cooling Modules, see Chapter 22.3.5.3 on page 22-16.
9. Assemble the Reagent Module, see Chapter 22.3.4.3 on page 22-13.

#### **22.3.7.4 Set the Reagent Module**

1. Check the teaching positions for the Right Pipettor, see Chapter 13.16 on page 13-35 and see Chapter 13.17 on page 13-36.
2. Verify the Cooling Modules, see Chapter 22.4.2.3 on page 22-30.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9, or see Chapter 10.4 on page 10-10.



### 22.3.8 Replacement of the PCB, Antenna 10 channel

#### 22.3.8.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm above the Incubator Module.
7. Remove the Reagent Module, see Chapter 22.3.4.2 on page 22-12.
8. Remove the Cooling Module, see Chapter 22.3.5.2 on page 22-14.
9. Remove the Main Insulation, see Chapter 22.3.7.2 on page 22-20.

#### 22.3.8.2 Removal of the 10-Channel Antenna PCB

1. Unplug the antenna CTRL cable connector (J1) and the antenna cable connector (J2).
2. Remove the 10-channel antenna PCB (8) out of the positioning pins (9).

#### NOTE

##### Removal of the 10-Channel Antenna PCB

Be careful when removing the 10-channel antenna PCB. The PCB is fixed with a double-sided adhesive tape.

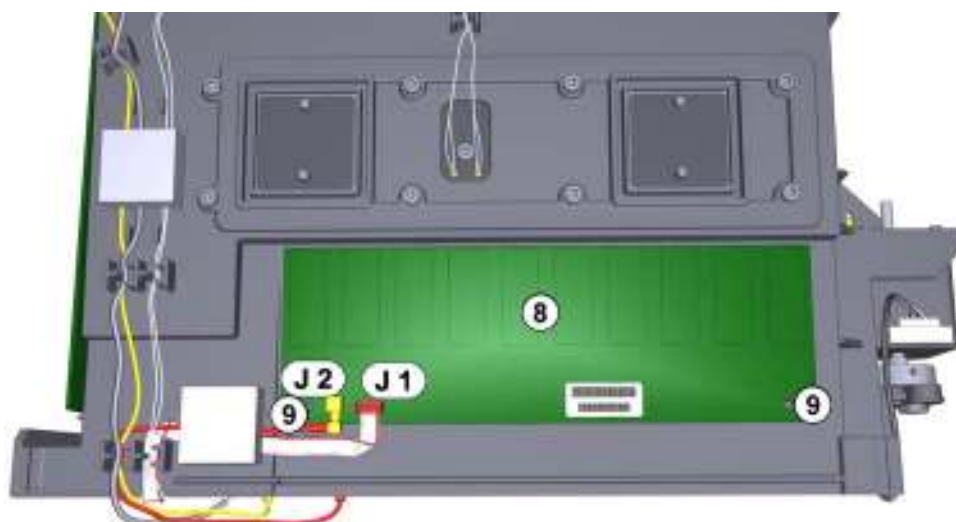


Figure 22–26: Remove 10-channel antenna PCB

**22.3.8.3 Assembly of the 10-Channel Antenna PCB**

1. At the Reagent Bay shell, clean the surface for the 10-channel antenna PCB.
2. Remove the protective foil of the adhesive tape on the rear of the 10-channel antenna PCB.
3. Attach the 10-channel antenna PCB aligning it to the positioning pins (9).
4. Plug the antenna CTRL cable (J1) and the antenna cable (J2) onto the 10-channel antenna PCB.
5. Assemble the Main Insulation, see Chapter 22.3.7.3 on page 22-22.
6. Assemble the Cooling Modules, see Chapter 22.3.5.3 on page 22-16.
7. Assemble the Reagent Module, see Chapter 22.3.4.3 on page 22-13.

**22.3.8.4 Set the Reagent Module**

1. Check the teaching positions for the Right Pipettor, see Chapter 13.16 on page 13-35 and see Chapter 13.17 on page 13-36.
2. Verify the Cooling Modules, see Chapter 22.4.2.3 on page 22-30.
3. Verify the RFID, see Chapter 22.4.2.4 on page 22-31.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.



### 22.3.9 Replacement of the Quad RFID Antenna PCB

#### 22.3.9.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm above the Incubator Module.
7. Remove the Reagent Module, see Chapter 22.3.4.2 on page 22-12.
8. Remove the Cooling Module, see Chapter 22.3.5.2 on page 22-14.
9. Remove the Main Insulation, see Chapter 22.3.7.2 on page 22-20.

#### 22.3.9.2 Removal of the Quad RFID Antenna PCB

1. Unplug the antenna cable connector (J1) and ancillaries cable connector (J2).
2. Remove the Quad RFID Antenna (1) carefully out of the positioning pins (2).

#### NOTE

##### Removal of the Quad RFID Antenna PCB

Be careful when removing the Quad RFID antenna PCB. The PCB is fixed with a double-sided adhesive tape.

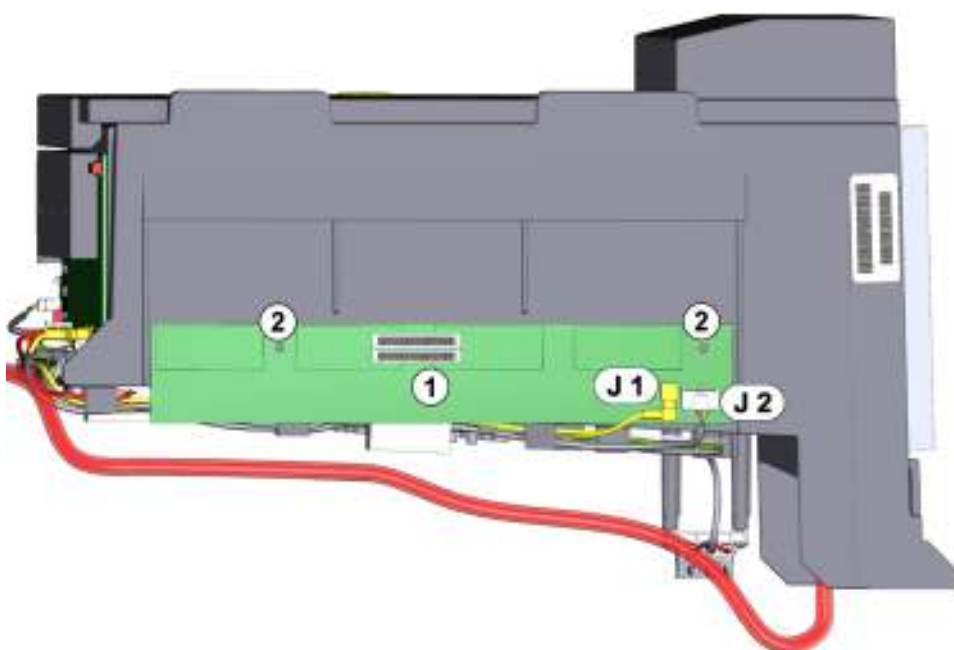


Figure 22–27: Quad RFID Antenna PCB removal

**22.3.9.3 Assembly of the Quad RFID Antenna PCB**

1. At the Reagent Bay shell, clean the surface for the Quad RFID antenna PCB.
2. Remove the protective foil of the adhesive tape on the rear of the Quad RFID antenna PCB.
3. Attach the Quad RFID antenna PCB aligning it to the positioning pins (2).
4. Plug the antenna cable (J1) and the ancillaries cable (J2) onto the Quad RFID antenna PCB.
5. Assemble the Main Insulation, see Chapter 22.3.7.3 on page 22-22.
6. Assemble the Cooling Modules, see Chapter 22.3.5.3 on page 22-16.
7. Assemble the Reagent Module, see Chapter 22.3.4.3 on page 22-13.

**22.3.9.4 Set the Reagent Module**

1. Check the teaching positions for the Right Pipettor, see Chapter 13.16 on page 13-35 and see Chapter 13.17 on page 13-36.
2. Verify the Cooling Modules, see Chapter 22.4.2.3 on page 22-30.
3. Verify the RFID, see Chapter 22.4.2.4 on page 22-31.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 Or see Chapter 10.4 on page 10-10.

## 22.3.10 Replacement of the Temperature Sensor

### 22.3.10.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm above the Incubator Module.
7. Remove the Reagent Module, see Chapter 22.3.4.2 on page 22-12.
8. Remove the Cooling Module, see Chapter 22.3.5.2 on page 22-14.
9. Remove the Main Insulation, see Chapter 22.3.7.2 on page 22-20.

### 22.3.10.2 Removal of the Temperature Sensor

1. Pull both temperature sensors (8) carefully out.

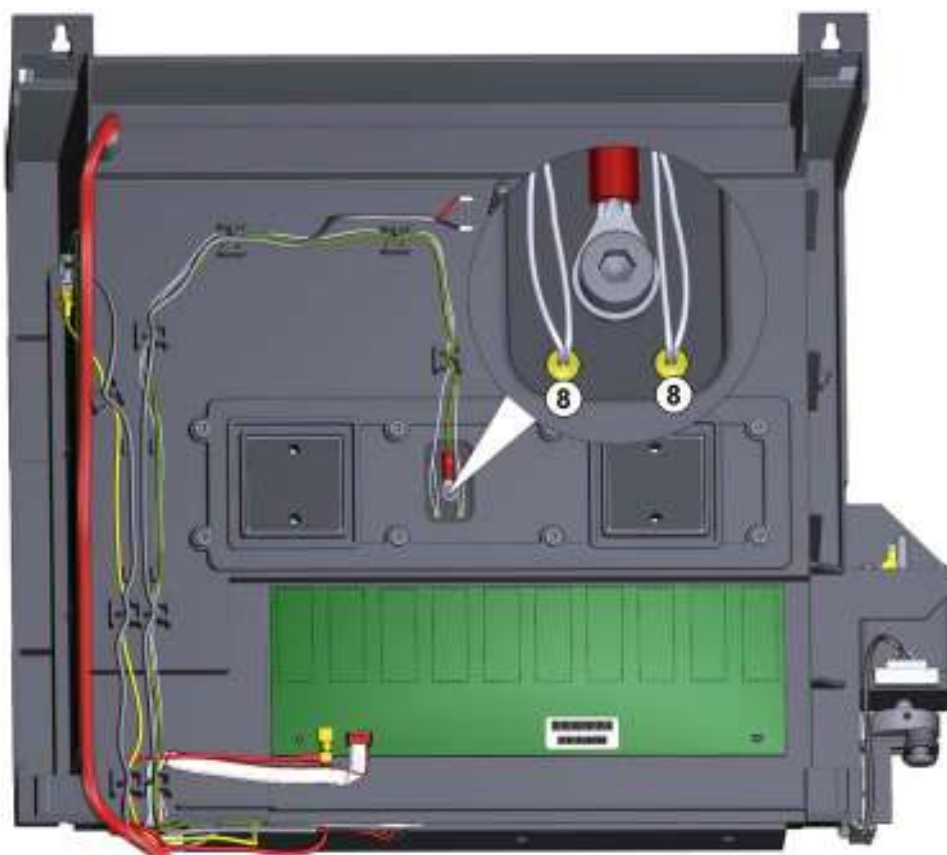


Figure 22–28: Remove Temperature Sensors

2. Unplug both temperature sensors (J10 and J11) from the Reagent Loading Bay CU PCB.

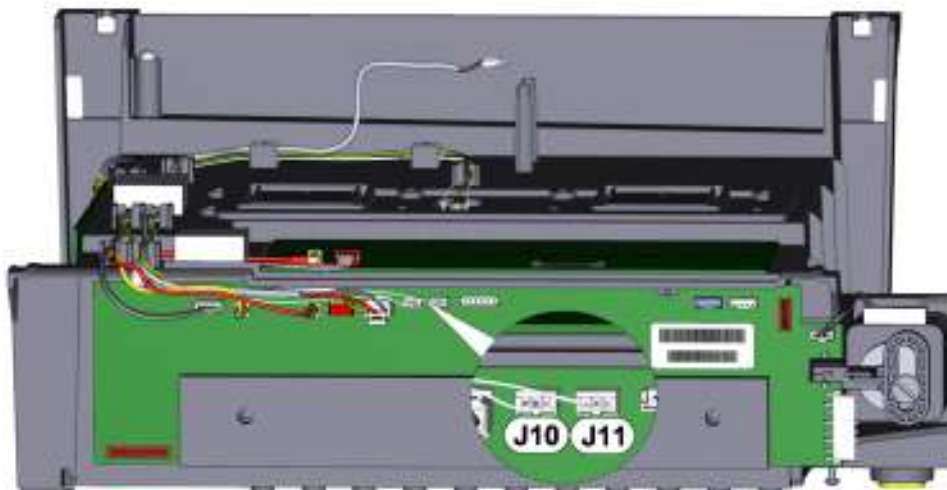


Figure 22-29: Reagent Module, unplug Temperature Sensors

#### 22.3.10.3 Assembly of the Temperature Sensor

1. Using a syringe, fill the drilled hole hosting the temperature sensor with the standard thermal grease (PN 3641010007) up to half of the hole height.
2. Insert both temperature sensors up to the hole end stop and remove any surplus of thermal grease.
3. Lay the temperatures sensor cables into the cable holders (10).
4. Connect both temperature sensors (J10 and J11) onto the Reagent Loading Bay CU PCB.

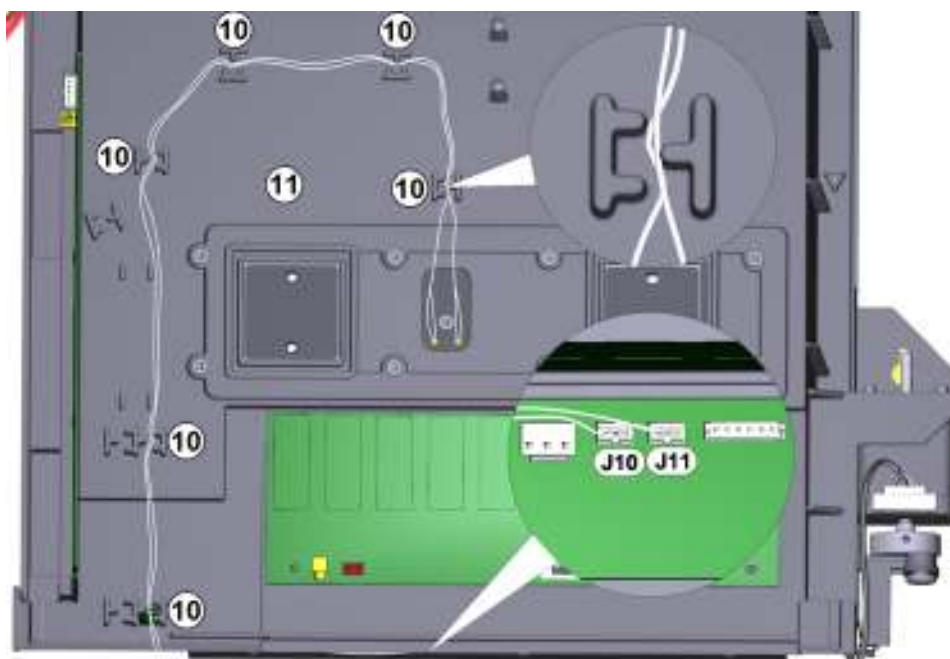


Figure 22-30: For better view, only the temperature Sensor cables are shown

5. Assemble the Main Insulation, see Chapter 22.3.7.3 on page 22-22.
6. Assemble the Cooling Modules, see Chapter 22.3.5.3 on page 22-16.
7. Assemble the Reagent Module, see Chapter 22.3.4.3 on page 22-13.

#### **22.3.10.4 Set the Reagent Module**

1. Check the teaching positions for the Right Pipettor, see Chapter 13.16 on page 13-35 and see Chapter 13.17 on page 13-36.
2. Verify the Cooling Modules, see Chapter 22.4.2.3 on page 22-30.
3. Verify the temperature sensors, see Chapter 22.4.2.5 on page 22-31.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 22.4 Adjustment and calibration

This section contains information about check and verification of the Reagent Module.

### NOTE

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).  
Do not run any of the software at the same time!

### 22.4.1 Initialization routine

- Close any open software and start the **Service Software** from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **Front and Loading** and the **General** bracket.
- Click onto the **Initialize** button and choose the **Loading Bay**. Press the **General** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 22.4.2 Check and Verification of the Reagent Module

#### 22.4.2.1 Check the condensation aspiration function

- Gently pour about 2-3 mL of DI water close to the drain of the Reagent Module, so that it is covered with liquid.
- In the **Service Software** at the navigation window, choose the **Washer - General** bracket.
- Click onto the **Initialize** button and choose **All Dialogue Controlled Modules** to start the initialization sequence.
- Press the **Pumps** button and choose **WAP - Start at routine speed**. The liquid present in the Reagent Module should be aspirated.
- Switch off the pump by pressing **WAP- Stop**.

#### 22.4.2.2 Verify the stirring

- In the **Service Software** at the navigation window, choose the **Front and Loading - General** bracket.
- Click onto the **Actors - Stirrer - On** button.
- Verify that the stirrer rod is activated.
- Switch off the stirrer rod by pressing the **Actors - Stirrer - Off** button.

#### 22.4.2.3 Verify the cooling

- In the **Service Software** at the navigation window, choose the **Front and Loading - General** bracket.
- Click onto the **Actors - Cooling - On** button.
- Click onto the **Sensors - Loading Bay All** button.
- Click onto the **Auto** update button:

- the Seebeck voltage display should show > 1000. And the temperatures of the Loading Bay 1 & 2 shall fall;
- the test should run minimum about 3 minutes.

#### 22.4.2.4 Verify the micro-switches and the RFIDs

- In the **Service Software** at the navigation window, choose the **System - General** bracket and press the **Instrument - Full initialization via COP** button.
- Choose the **System - Loading Report** bracket.
- While inserting the integrals and ancillaries, the presence of the kits and ancillaries will be prompted.
  - Test the single rows one after the other, checking the display of the report for plausibility.

#### 22.4.2.5 Check of the Reagent Bay Temperature Sensors

- In the **Service Software** at the navigation window, choose **Front and Loading** and the **General** bracket.
- Click onto the **Sensors** button and choose the **Loading Bay Temp. only** button.
- In the dedicated temperature fields, check that the Reagent Bay temperature is inside the required range.

### 22.4.3 Readjustment of the cam to the stirrer rod

#### NOTE

##### Remove the Reagent Area

The readjustment of the cam to the stirrer rod can be performed when the Reagent Area module is outside of the system.

1. Loosen the shaft screw on the cam that is holding the stirrer rod.

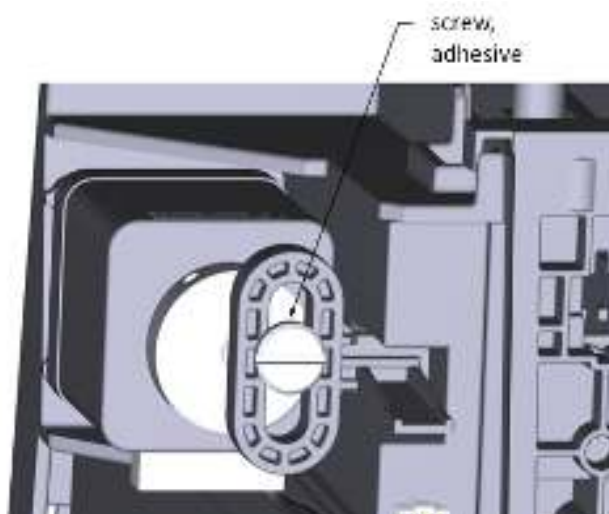


Figure 22–31: Screw on the cam holding the stirrer rod



2. Check the orientation of the cam to the stirrer rod interface: Both interfaces shall be flush to each other without tension or stress.
3. In case the face are not flush to each other, loosen the two adjusting screws (grub screw) on the cam, so the cam can be moved on the motor shaft, see Figure 22–32.

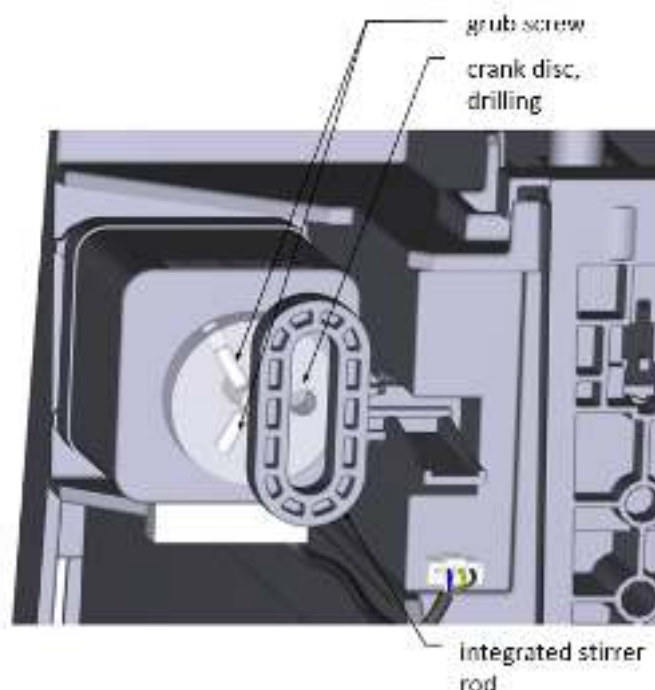


Figure 22–32: Position of the two adjusting screws (grub screw) in the cam

4. Adjust the cam to the interface of the stirrer rod. Both interfaces shall be flush to each other without tension or stress. Re-tighten the two adjusting screws.
5. Re-attach the shaft screw with a light film of low-strength thread locker (Loctite 222) to the cam. The cam and the stirrer rod must remain flush and free of tension or stress after this step.

**NOTE****Shaft screw shall be reused**

The shaft screw shall be reused and therefore shall have a remaining light film of Loctite 222 on it.

6. Do two to three turns with the cam and control the smooth running with the stirrer rod. If necessary, the two adjusting screws can be loosened again for correction and fixated after procedure.
7. Apply a thin film of lubricant to the cam to stirrer-rod interface in retracted and protruded position of the stirrer-rod. Remove excessive grease.



## 22.5 Electric connections, switches and signals

This section shows an overview of the electronic boards of the module.

### 22.5.1 Reagent Loading Bay CU PCB

RFID loading bay CU PCBA = Reagent loading bay CU PCBA = PCB, Loading Bay CU



Figure 22-33: Reagent Loading Bay CU PCB

| No. | Connection               | No. | Connection   |
|-----|--------------------------|-----|--------------|
| J 1 | Download                 | J 9 | Peltier      |
| J 2 | Ancillaries Antenna      | J10 | Temp 2       |
| J 3 | 10CH Antenna             | J11 | Temp 1       |
| J 4 | Stirrer drive            | J12 | not used     |
| J 5 | Debug (JTAG)             | J13 | Fan 1 + 2    |
| J 6 | 10CH Antenna CTRL        | J14 | PWR / CAN    |
| J 7 | Ancillaries Antenna CTRL | J15 | Reagent flap |
| J 8 | Sample PCBA              | J16 | Display      |

Table 22-4: Connections of the Reagent Loading Bay CU PCB

| Number | Meaning/Function |
|--------|------------------|
| LED 1  | Power            |
| LED 13 | Stirrer init     |

Table 22-5: LED of the Reagent Loading Bay CU PCB

### 22.5.2 Quad RFID Antenna PCB



Figure 22-34: Quad RFID antenna PCB

| No. | Connection           |
|-----|----------------------|
| J 1 | Ancillaries antennas |
| J 2 | Ancillaries          |

Table 22-6: Connections of the Quad RFID antenna PCB

### 22.5.3 10-Channel RFID Antenna PCB



Figure 22-35: 10-Channel RFID Antenna PCB

| No. | Connection    |
|-----|---------------|
| J 1 | Antennas CTRL |
| J 2 | Antenna       |

Table 22-7: Connections of the main 10-Channel RFID Antenna PCB

# 23 Starter Reagent Module for internal-tanks configuration

This chapter describes the Starter Reagent Module in detail.

After the overview of the module, the chapter continues with information about repair and calibration of the module (sections “Replacement and Repair”, “Adjustment and Calibration”). In the last section Dual antenna PCBA board is described.

- The Starter Reagent Module controls the Starter 1 and Starter 2 bottles.
- There are two bubble sensors which indicate an empty bottle and two RFID antenna which read the liquid level stored in the RFID tag of the bottles.

## 23.1 Safety

---

**WARNING****Non-observance of safety instructions**

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7.
  - Follow all working instructions included in this manual.
  - Follow all warnings marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
-

## 23.2 Overview

This section shows an overview of the components of the Starter Reagent Module.

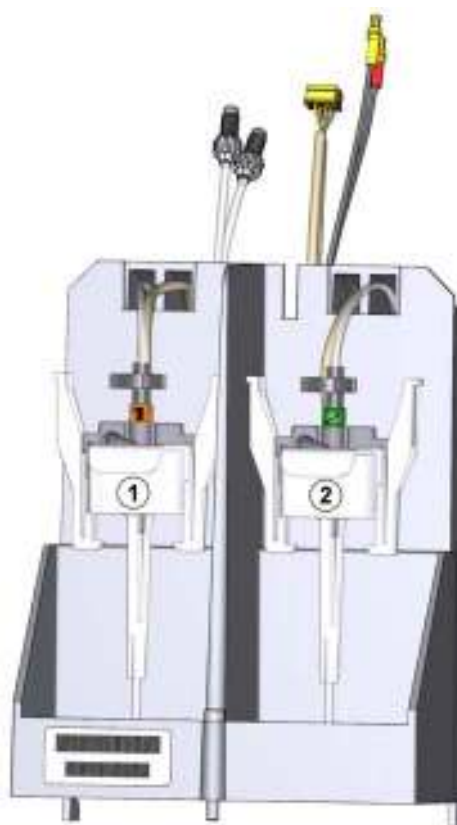


Figure 23–1: Starter Reagent Module, front view

- |    |                         |
|----|-------------------------|
| 1. | Starter 1 sensor & clip |
| 2. | Starter 2 sensor & clip |

Table 23–1: Overview Starter Reagent Module, front view

---

**23. Starter Reagent Module for internal-tanks configuration**

---



*Figure 23–2: Starter Reagent Module, rear view*

- |    |                               |
|----|-------------------------------|
| 3. | (J1) RFID antenna L1 (yellow) |
| 4. | (J2) RFID antenna L2 (red)    |

*Table 23–2: Overview Starter Reagent Module, rear view*

## 23.3 Replacement and repairs

This section contains instructions for the repair of individual components of the Starter Reagent Module. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 23.3.1 Replacement of the Starter Reagent Module

**NOTE**

Do not bend or damage any tubes while disassembly or assembly of the Starter Reagent Module.

#### 23.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.

#### 23.3.1.2 Removal of the Starter Reagent Module

1. Disconnect bubble sensor caps (2) of Starter reagent bottles (1).
2. Remove bottles out of Starter Reagent Module and dispose of them according to the IFU.

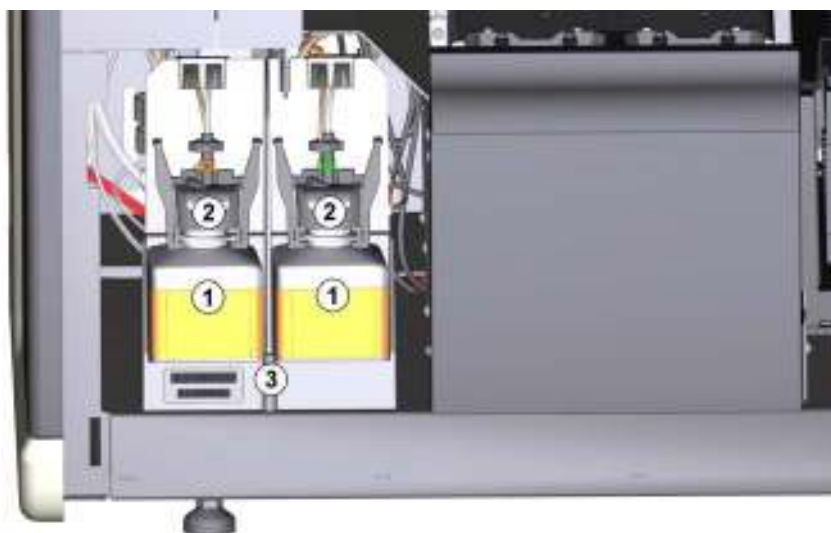


Figure 23–3: Starter Reagent Module inside the instrument

### 23. Starter Reagent Module for internal-tanks configuration

3. Unplug Starter Reagent cables from the Liquid support CU PCBA at slot J9 and J10.
4. Unplug RFID antenna cables at slot J13 and J14.
5. Remove the four cables out of the cable holder (4).

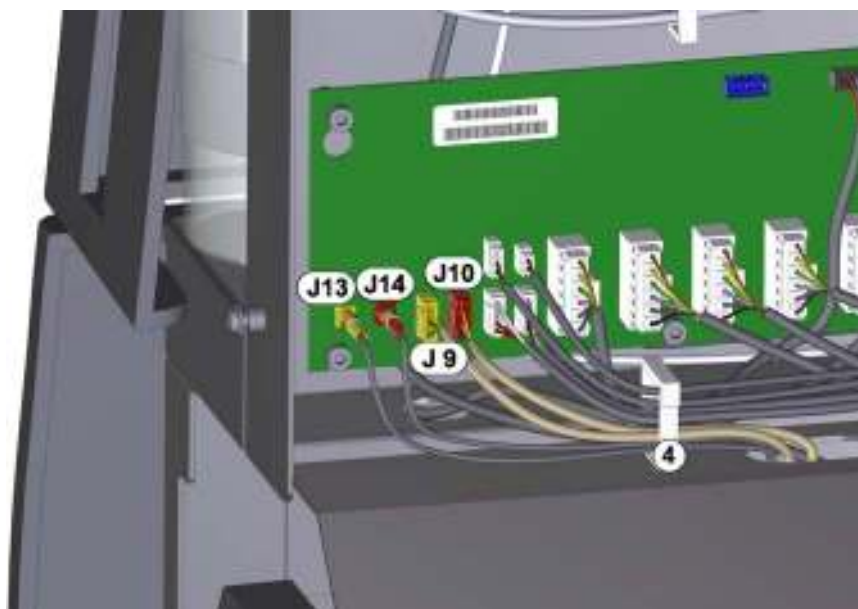


Figure 23-4: Liquid support CU PCBA

6. Unscrew fittings of the transparent tubes (5) (6) from the Starter Pump 1 (ST1) and from the Starter Pump 2 (ST2).

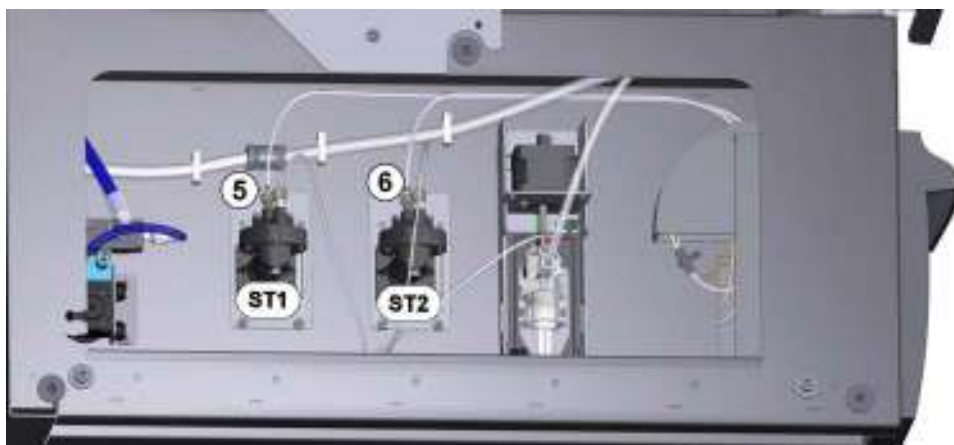


Figure 23-5: Starter Pumps

#### NOTE

Be aware: unscrewing the tubes fitting, dripping could occur at the end of starter tubing in the starter reagent area. Place a dry tissue in starter area to collect spilled liquid.

7. Unscrew the screw that fix the Starter Reagent Module (3) and lift the Module carefully out. Gently pull out the Starter Reagent module towards the Front

## 23. Starter Reagent Module for internal-tanks configuration

module collecting the tubings and cable sensors to avoid possible hooking of them to the system edges and chassis holes.

### 23.3.1.3 Assembly of the Starter Reagent Module

1. Set Starter Reagent Module, sliding the upper metallic guide through the crevice of the module and sliding hooks (7) the bottom plate of the instrument.
2. Fasten the Starter Reagent Module with one screw (3).

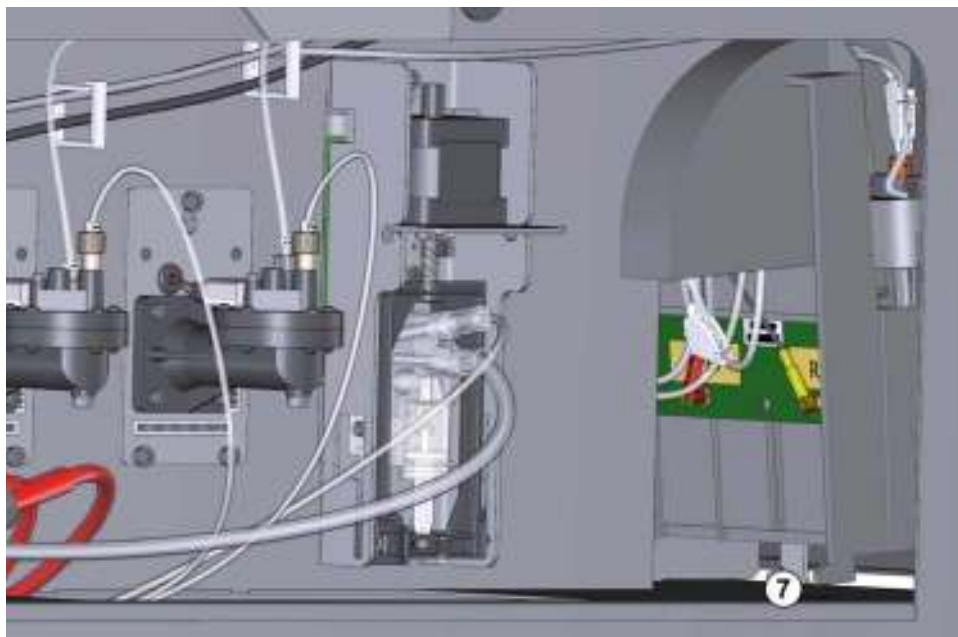


Figure 23-6: Starter Reagent Module back view



Figure 23-7: Upper crevice of the Starter Reagent module



## 23. Starter Reagent Module for internal-tanks configuration

- Slide the four cables in and connect it to Liquid support CU PCBA (J9, J10, J13 and J14). Insert the four sensor cables in the cable holder and close it.
- Screw the Starter transparent tubes fittings (5) (6) onto the Starter Reagent Pumps.

### NOTE

#### Before mounting the SP-Starters tubings.

Be aware of the correct installation of the Starter 1 & Starter 2 lines.

- Check the tube for proper installation.
- Tighten the front screw of the Starter Reagent Module.
- Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
- Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

#### 23.3.1.4 Set of the Starter Reagent Module

- Calibrate the Starters Bubble sensors, see Chapter 23.4.2.1 on page 23-16.
- If needed, prime the Reader, see Chapter 11.8.1.6 on page 11-65.
- Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 23.3.2 Replacement of the Starter 1 sensor

#### 23.3.2.1 Preliminary steps



- Exit the **LIAISON<sup>®</sup> XS** system software.
- Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
- On the right side of the system turn the power switch off.
- Unplug the system power plug from the right side of the system.
- Open top covers.
- Move pipettor arms to a safe location.
- Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
- Open the Starter Reagent Module flap.

## 23. Starter Reagent Module for internal-tanks configuration

### 23.3.2.2 Removal of the Starter 1 sensor

1. Disconnect bubble sensor cap (2a) of Starter 1 reagent bottle (1a).



Figure 23-8: Starter Reagent Module inside the instrument

2. Unplug Starter 1 reagent cable from the Liquid support CU PCBA at slot J9.
3. Remove the cable out of the cable holder (3).

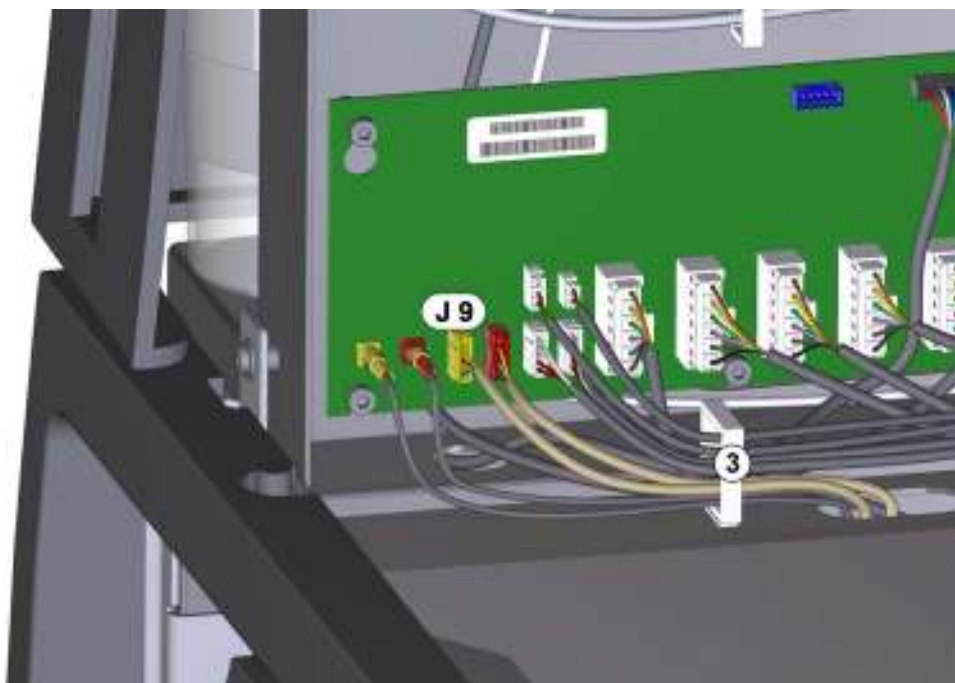
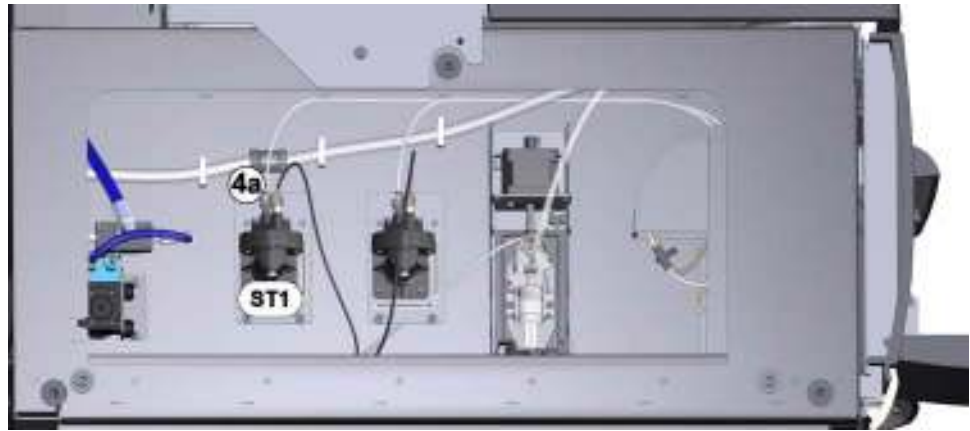


Figure 23-9: Liquid support CU PCBA

**23. Starter Reagent Module for internal-tanks configuration**

---

4. Unscrew fitting of the transparent tube (4a) from the Starter Pump 1 (ST1).



*Figure 23-10: Starter Pumps*

**NOTE**

Be aware: unscrewing the tube fitting, dripping could occur at the end of starter tubing in the starter reagent area. Place a dry tissue in starter area to collect spilled liquid.

5. Lift Starter 1 sensor with cable and tubing carefully out.

## 23. Starter Reagent Module for internal-tanks configuration

### 23.3.2.3 Assembly of the Starter 1 sensor

1. Insert the cable (5a) and the tube (6a) into the Starter Reagent Module left side.



Figure 23-11: Starter Reagent Module inside the instrument

2. Slide cable (5a) in and connect it to Liquid support CU PCBA at slot J9.
3. Fit the cable in the cable holder (3).
4. Slide tubing (6a) into the Starter Reagent Module and connect the fitting (4a) onto the Starter Reagent Pump (ST1).
5. Close the Starter Reagent Module flap.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 23.3.2.4 Set of the Starter 1 sensor

1. Calibrate the Starter 1 Bubble sensor, see Chapter 23.4.2.1 on page 23-16.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 23.3.3 Replacement of the Starter 2 sensor

#### 23.3.3.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Open the Starter Reagent Module flap.

#### 23.3.3.2 Removal of the Starter 2 sensor

1. Disconnect bubble sensor cap (2b) of right Starter 2 reagent bottle (1b).



Figure 23–12: Starter Reagent Module inside the instrument

2. Unplug Starter 2 Reagent cable from the Liquid support CU PCBA at slot J10.
3. Remove the cable out of the cable holder (3).

### 23. Starter Reagent Module for internal-tanks configuration

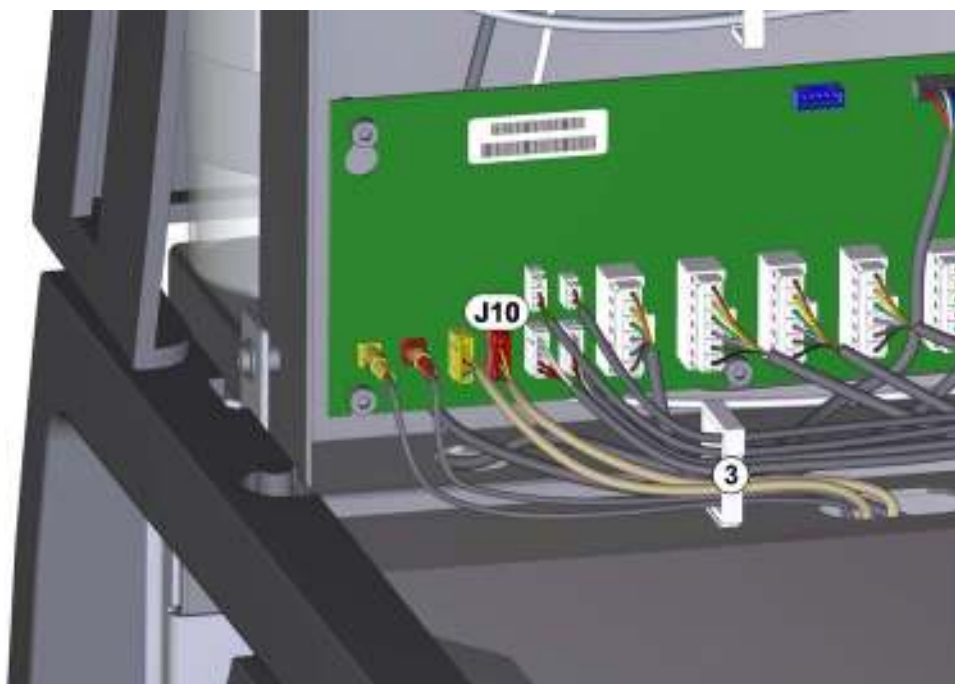


Figure 23-13: Liquid support CU PCBA

4. Unscrew fitting of the transparent tube (4a) out of Starter Pump 2 (ST2).

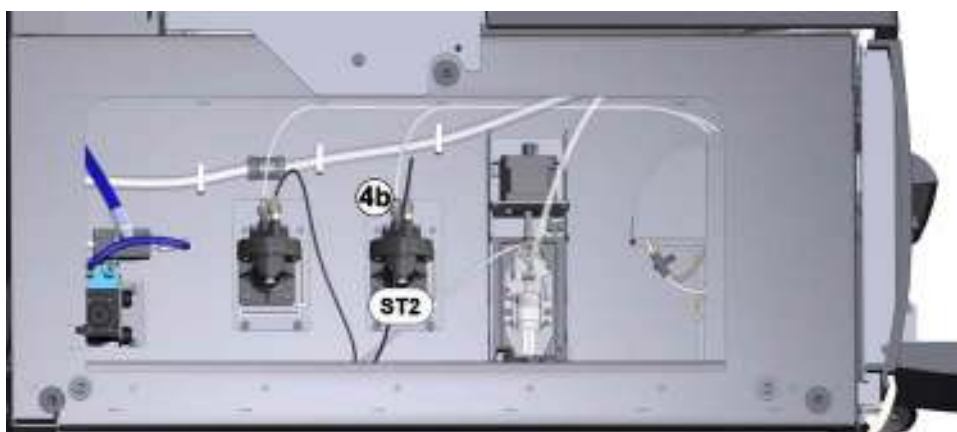


Figure 23-14: Starter Pumps

#### NOTE

Be aware: unscrewing the tube fitting, dripping could occur at the end of starter tubing in the starter reagent area. Place a dry tissue in starter area to collect spilled liquid.

5. Lift Starter 2 sensor with cable and tubing carefully out.

## 23. Starter Reagent Module for internal-tanks configuration

### 23.3.3.3 Assembly of the Starter 2 sensor

1. Insert the cable (5b) and the tube (6b) into the Starter Reagent Module right side.



Figure 23–15: Starter Reagent Module inside the instrument

2. Slide cable (5b) in and connect it to Liquid support CU PCBA at slot J10.
3. Fit the cable in the cable holder (3).
4. Slide tubing (6b) into the Starter Reagent Module and connect the fitting (4b) onto the Starter Reagent Pump (ST1).
5. Close the Starter Reagent Module flap.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 23.3.3.4 Set of the Starter 2 sensor

1. Calibrate the Starter 2 Bubble sensor, see Chapter 23.4.2.1 on page 23-16.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 23.3.4 Replacement of the Dual antenna PCBA board

#### 23.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
9. Remove the Starter Reagent Module, see Chapter 23.3.1.2 on page 23-4.

#### 23.3.4.2 Removal of the Dual antenna PCBA board

1. Disconnect antenna cables (J1 and J2) out of Dual antenna PCBA.
2. Unscrew one screw (1) and remove Dual antenna PCBA.

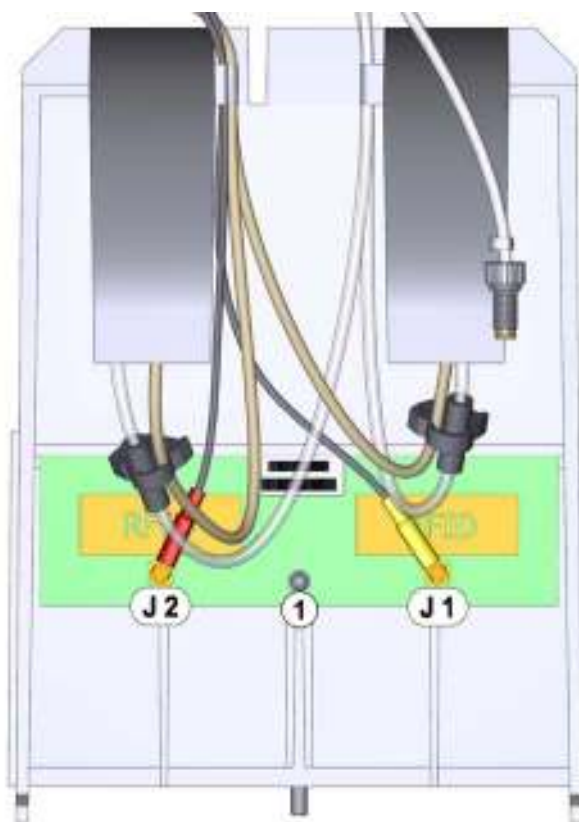


Figure 23–16: Starter Reagent Module, rear view



**23.3.4.3 Assembly of the Dual antenna PCBA board**

1. Set Dual antenna PCBA in and fasten one screw (1).
2. Connect the antenna cables onto the Dual antenna PCBA (J1) (J2).
3. Assemble the Starter Reagent Module, see Chapter 23.3.1.3 on page 23-6.
4. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
5. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
6. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

## 23.4 Adjustment and calibration

In this section information about the adjustment and calibration of Starter Reagent Module can be found.

### NOTE

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).  
Do not run any of the software at the same time.

### 23.4.1 Initialization routines

- Close any open software and start the **Service Software** from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **Reader** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog Controlled Modules** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 23.4.2 Calibration of the Starter Reagent Module

#### 23.4.2.1 Calibration of the Starter bubble sensors

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select **Starter Sensors**.

#### Step 1

- Press the **Start** button, to start the procedure.
- Follow the instruction at the pop-up and press **OK** button.



Figure 23–17: Sensor calibration pop-up for Starter 1

- When the **Processing** window closes, press the **Done** button.

## 23. Starter Reagent Module for internal-tanks configuration

### Step 2

- d. Press the **Start** button, to start the procedure.
- e. Follow the instruction at the pop-up and press **OK** button.

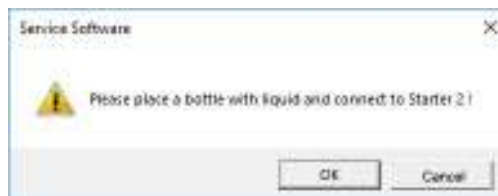


Figure 23–18: Sensor calibration pop-up for Starter 2

- f. When the **Processing** window closes without failure message, press the **Done** button.

### Step 3

- g. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

### Step 4

- h. The process is successful when, at the **Summarize** panel, all steps are in green.

### 23.4.2.2 Calibration of the Starter pumps

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select **Starter Pump**.

#### Step 1

- a. Press the **Start** button; the calibration procedure starts automatically.
- b. When the **Processing** window closes, press the **Done** button.

#### Step 2

- c. Press the **Start** button; the calibration procedure starts automatically.
- d. When the **Processing** window closes without failure message, press the **Done** button.

#### Step 3

- e. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

#### Step 4

- f. The process is successful when, at the **Summarize** panel, all steps are in green.

### 23.4.3 Check of Starters bubble sensors

In the **Service Software** at the navigation window choose **Reader** and the **General** bracket and select **Sensors** then **Check**.

#### Step 1

- a. Select the **Starter 1 bottle sensor** and follow the instructions.

#### Step 2

- b. Compare the TubeCondition result with the SensorState result (the status should be equal).

#### Step 3

- a. Select the **Starter 2 bottle sensor** and follow the instructions.

#### Step 4

- b. Compare the TubeCondition result with the SensorState result (the status should be equal).

#### **23.4.4 Verifying RFID**

In the **Service Software** at the navigation window choose **System** and the **Loading Report** bracket.

The changes of the materials to be loaded are shown after loading or unloading of the materials.

- a. When inserting the Starter bottles, after a defined time interval the presence of the bottles will be prompted.
- b. The RFID will be read and displayed as a dialog.

## 23.5 Electric connections, switches and signals

This section shows an overview of the electronic boards of the module.

**NOTE**

see Chapter 3.1.7 on page 3-8 for an overview of the complete electric circuit of the system.

### 23.5.1 Dual antenna PCBA board

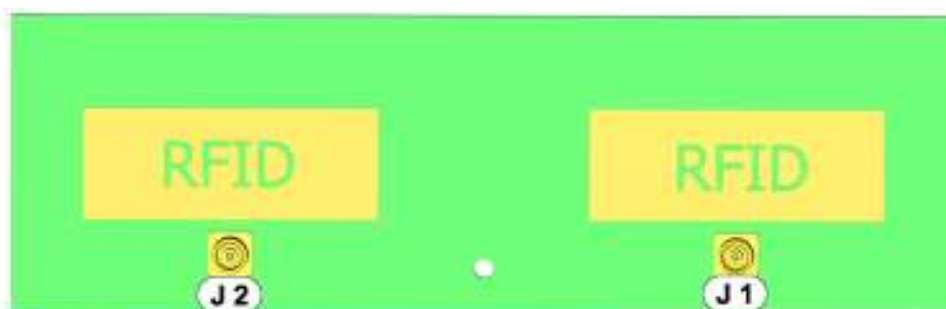


Figure 23-19: Dual antenna PCBA board

| Number |                                      | Connection |
|--------|--------------------------------------|------------|
| J1     | RFID antenna<br>(yellow shrink tube) |            |
| J2     | RFID antenna<br>(red shrink tube)    |            |

Table 23-3: Connections of the Dual antenna PCBA board

## 23. Starter Reagent Module for internal-tanks configuration

---

# 24 Starter Reagent Module for external-tanks configuration

This chapter describes the Starter Reagent Module in detail.

After the overview of the module, the chapter continues with information about repair and calibration of the module (sections “Replacement and Repair”, “Adjustment and Calibration”). In the last section Dual antenna PCBA board is described.

- The Starter Reagent Module controls the Starter 1 and Starter 2 bottles.
- There are two bubble sensors which indicate an empty bottle and two RFID antenna which read the liquid level stored in the RFID tag of the bottles.

## 24.1 Safety

---

**WARNING****Non-observance of safety instructions**

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 2.5 on page 2-7.
  - Follow all working instructions included in this manual.
  - Follow all warnings marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
-

## 24.2 Overview

This section shows an overview of the components of the Starter Reagent Module.

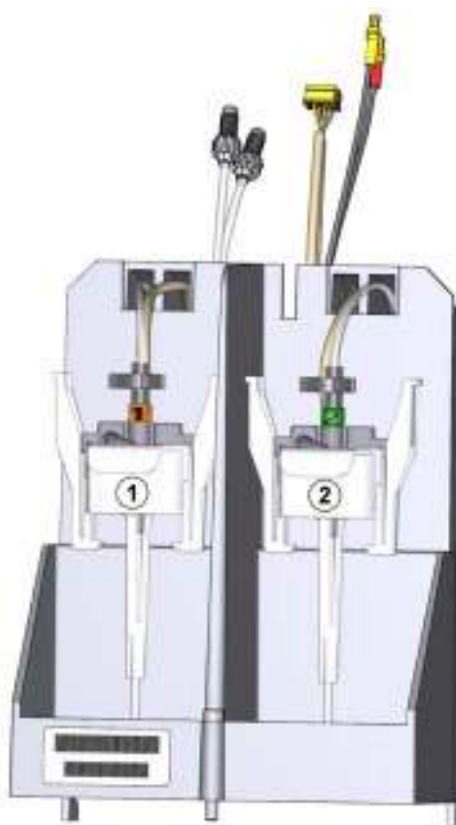


Figure 24–1: Starter Reagent Module, front view

- |    |                         |
|----|-------------------------|
| 1. | Starter 1 sensor & clip |
| 2. | Starter 2 sensor & clip |

Table 24–1: Overview Starter Reagent Module, front view



---

**24. Starter Reagent Module for external-tanks configuration**

---



*Figure 24–2: Starter Reagent Module, rear view*

- |    |                               |
|----|-------------------------------|
| 3. | (J1) RFID antenna L1 (yellow) |
| 4. | (J2) RFID antenna L2 (red)    |

*Table 24–2: Overview Starter Reagent Module, rear view*

## 24.3 Replacement and repairs

This section contains instructions for the repair of individual components of the Starter Reagent Module. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 24.3.1 Replacement of the Starter Reagent Module

**NOTE**

Do not bend or damage any tubes while disassembly or assembly of the Starter Reagent Module.

#### 24.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.

#### 24.3.1.2 Removal of the Starter Reagent Module

1. Disconnect bubble sensor caps (2) of Starter reagent bottles (1).
2. Remove bottles out of Starter Reagent Module and dispose of them according to the IFU.

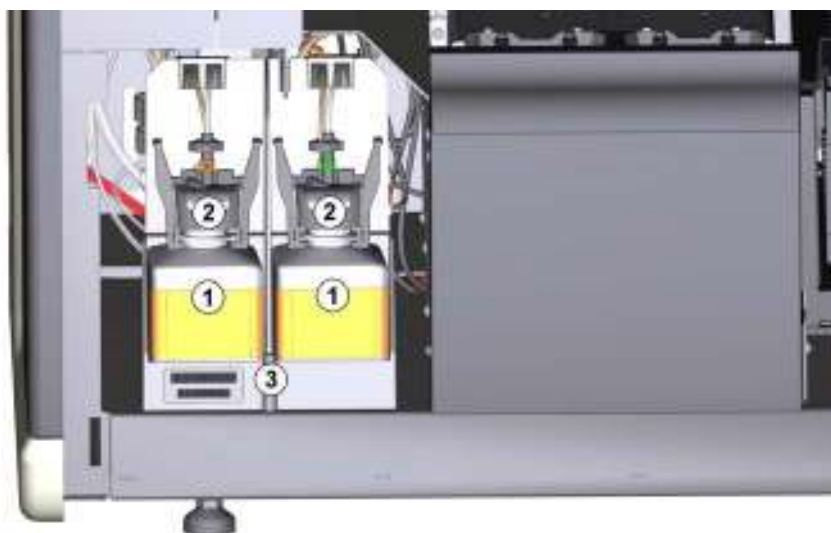


Figure 24–3: Starter Reagent Module inside the instrument

## 24. Starter Reagent Module for external-tanks configuration

3. Unplug Starter Reagent cables from the Liquid support CU PCBA at slot J9 and J10.
4. Unplug RFID antenna cables at slot J13 and J14.
5. Remove the four cables out of the cable holder (4).

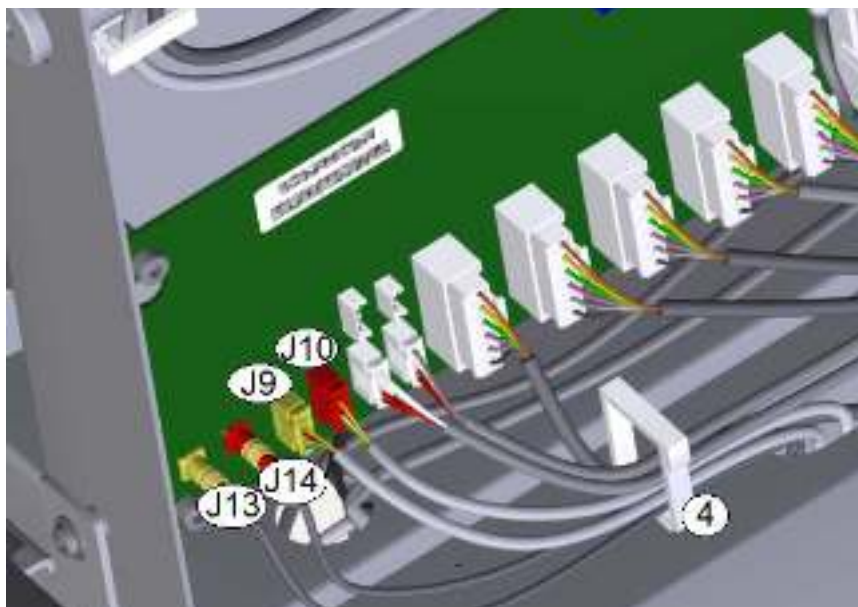


Figure 24-4: Liquid support CU PCBA

6. Unscrew fittings of the transparent tubes (5) (6) from the Starter Pump 1 (ST1) and from the Starter Pump 2 (ST2).

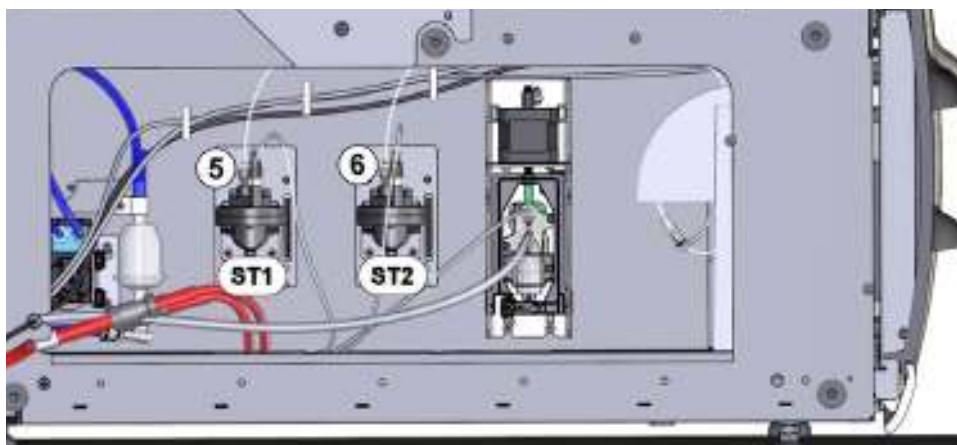


Figure 24-5: Starter Pumps

### NOTE

Be aware: unscrewing the tubes fitting, dripping could occur at the end of starter tubing in the starter reagent area. Place a dry tissue in starter area to collect spilled liquid.

7. Unscrew the screw that fix the Starter Reagent Module (3) and lift the Module carefully out. Gently pull out the Starter Reagent module towards the Front

## 24. Starter Reagent Module for external-tanks configuration

module collecting the tubings and cable sensors to avoid possible hooking of them to the system edges and chassis holes.

### 24.3.1.3 Assembly of the Starter Reagent Module

1. Set Starter Reagent Module, sliding the upper metallic guide through the crevice of the module and sliding hooks (7) the bottom plate of the instrument.
2. Fasten the Starter Reagent Module with one screw (3).

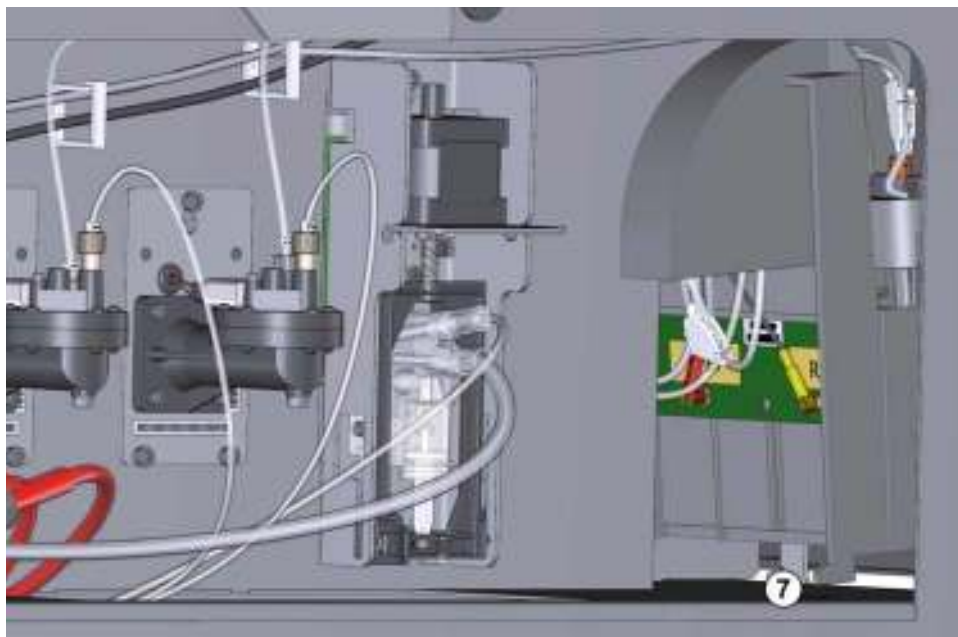


Figure 24-6: Starter Reagent Module back view



Figure 24-7: Upper crevice of the Starter Reagent module

## 24. Starter Reagent Module for external-tanks configuration

- Slide the four cables in and connect it to Liquid support CU PCBA (J9, J10, J13 and J14). Insert the four sensor cables in the cable holder and close it.
- Screw the Starter transparent tubes fittings (5) (6) onto the Starter Reagent Pumps.

### NOTE

#### Before mounting the SP-Starters tubings.

Be aware of the correct installation of the Starter 1 & Starter 2 lines.

- Check the tube for proper installation.
- Tighten the front screw of the Starter Reagent Module.
- Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
- Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

#### 24.3.1.4 Set of the Starter Reagent Module

- Calibrate the Starters Bubble sensors, see Chapter 24.4.2.1 on page 24-16.
- If needed, prime the Reader, see Chapter 12.8.1.6 on page 12-68.
- Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 24.3.2 Replacement of the Starter 1 sensor

#### 24.3.2.1 Preliminary steps



- Exit the **LIAISON<sup>®</sup> XS** system software.
- Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
- On the right side of the system turn the power switch off.
- Unplug the system power plug from the right side of the system.
- Open top covers.
- Move pipettor arms to a safe location.
- Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
- Open the Starter Reagent Module flap.

## 24. Starter Reagent Module for external-tanks configuration

### 24.3.2.2 Removal of the Starter 1 sensor

1. Disconnect bubble sensor cap (2a) of Starter 1 reagent bottle (1a).



Figure 24-8: Starter Reagent Module inside the instrument

2. Unplug Starter 1 reagent cable from the Liquid support CU PCBA at slot J9.
3. Remove the cable out of the cable holder (3).

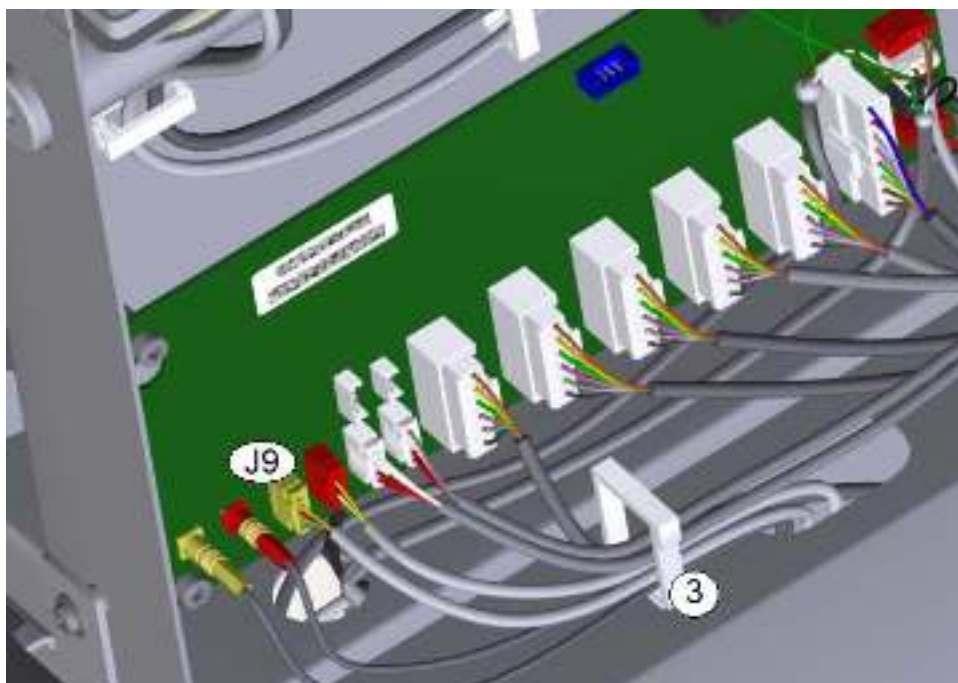


Figure 24-9: Liquid support CU PCBA

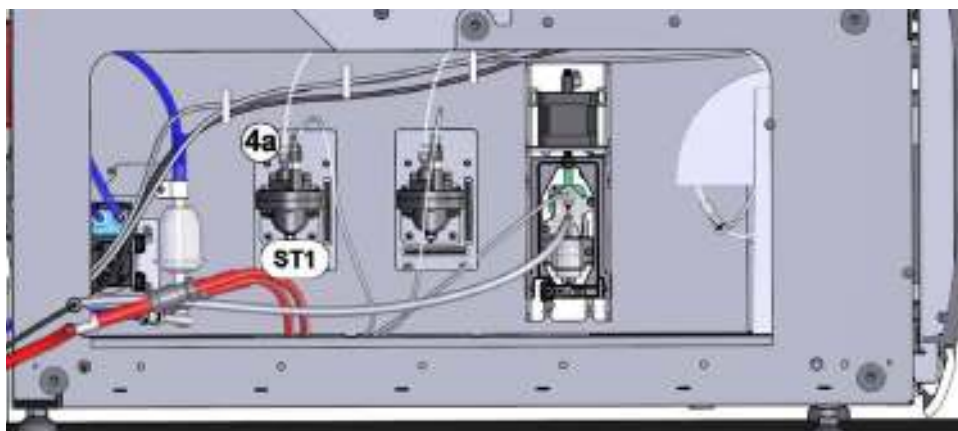


---

**24. Starter Reagent Module for external-tanks configuration**

---

4. Unscrew fitting of the transparent tube (4a) from the Starter Pump 1 (ST1).



*Figure 24–10: Starter Pumps*

**NOTE**

Be aware: unscrewing the tube fitting, dripping could occur at the end of starter tubing in the starter reagent area. Place a dry tissue in starter area to collect spilled liquid.

5. Lift Starter 1 sensor with cable and tubing carefully out.

## 24. Starter Reagent Module for external-tanks configuration

### 24.3.2.3 Assembly of the Starter 1 sensor

1. Insert the cable (5a) and the tube (6a) into the Starter Reagent Module left side.



Figure 24-11: Starter Reagent Module inside the instrument

2. Slide cable (5a) in and connect it to Liquid support CU PCBA at slot J9.
3. Fit the cable in the cable holder (3).
4. Slide tubing (6a) into the Starter Reagent Module and connect the fitting (4a) onto the Starter Reagent Pump (ST1).
5. Close the Starter Reagent Module flap.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 24.3.2.4 Set of the Starter 1 sensor

1. Calibrate the Starter 1 Bubble sensor, see Chapter 24.4.2.1 on page 24-16.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 24.3.3 Replacement of the Starter 2 sensor

#### 24.3.3.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Open the Starter Reagent Module flap.

#### 24.3.3.2 Removal of the Starter 2 sensor

1. Disconnect bubble sensor cap (2b) of right Starter 2 reagent bottle (1b).



Figure 24–12: Starter Reagent Module inside the instrument

2. Unplug Starter 2 Reagent cable from the Liquid support CU PCBA at slot J10.
3. Remove the cable out of the cable holder (3).

## 24. Starter Reagent Module for external-tanks configuration

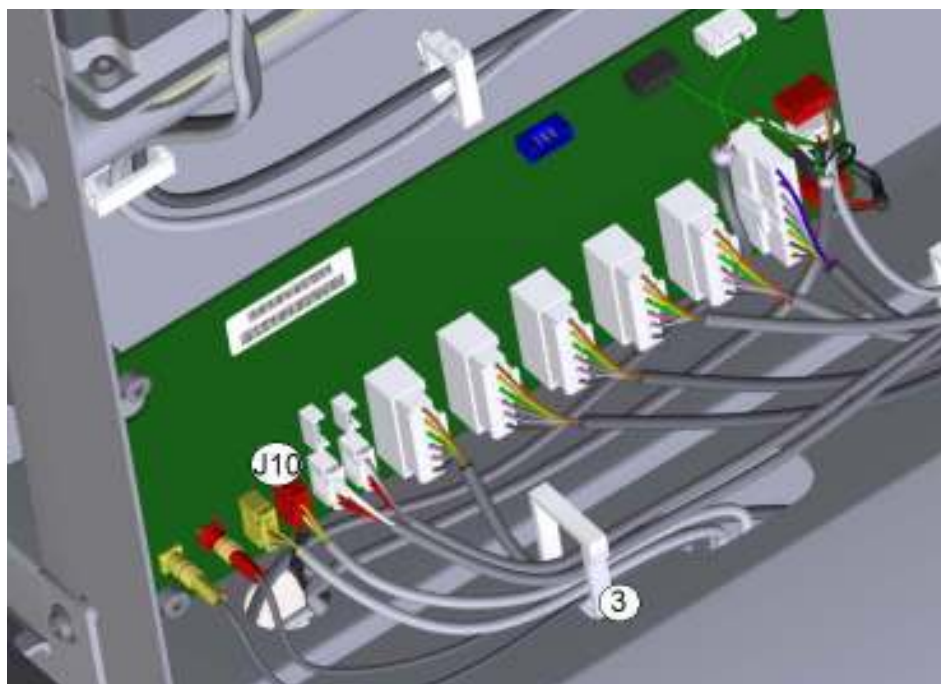


Figure 24-13: Liquid support CU PCBA

4. Unscrew fitting of the transparent tube (4a) out of Starter Pump 2 (ST2).

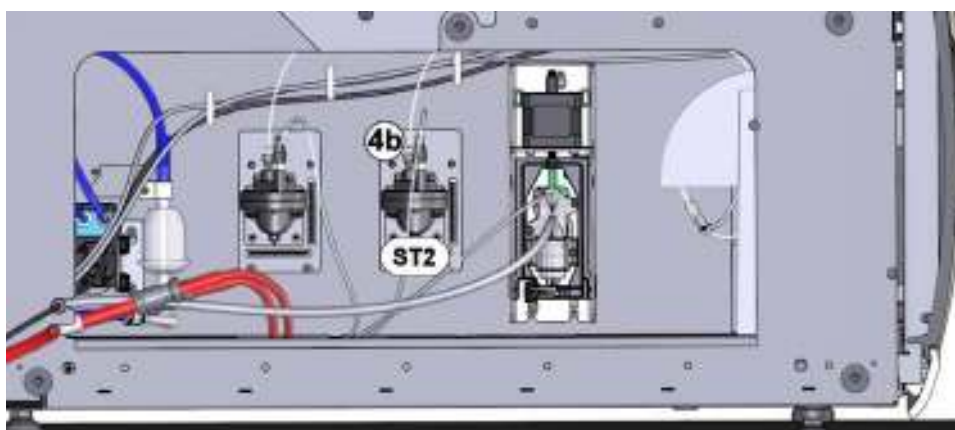


Figure 24-14: Starter Pumps

### NOTE

Be aware: unscrewing the tube fitting, dripping could occur at the end of starter tubing in the starter reagent area. Place a dry tissue in starter area to collect spilled liquid.

5. Lift Starter 2 sensor with cable and tubing carefully out.

## 24. Starter Reagent Module for external-tanks configuration

### 24.3.3.3 Assembly of the Starter 2 sensor

1. Insert the cable (5b) and the tube (6b) into the Starter Reagent Module right side.



Figure 24–15: Starter Reagent Module inside the instrument

2. Slide cable (5b) in and connect it to Liquid support CU PCBA at slot J10.
3. Fit the cable in the cable holder (3).
4. Slide tubing (6b) into the Starter Reagent Module and connect the fitting (4b) onto the Starter Reagent Pump (ST1).
5. Close the Starter Reagent Module flap.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 24.3.3.4 Set of the Starter 2 sensor

1. Calibrate the Starter 2 Bubble sensor, see Chapter 24.4.2.1 on page 24-16.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

## 24.3.4 Replacement of the Dual antenna PCBA board

### 24.3.4.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
9. Remove the Starter Reagent Module, see Chapter 24.3.1.2 on page 24-4.

### 24.3.4.2 Removal of the Dual antenna PCBA board

1. Disconnect antenna cables (J1 and J2) out of Dual antenna PCBA.
2. Unscrew one screw (1) and remove Dual antenna PCBA.

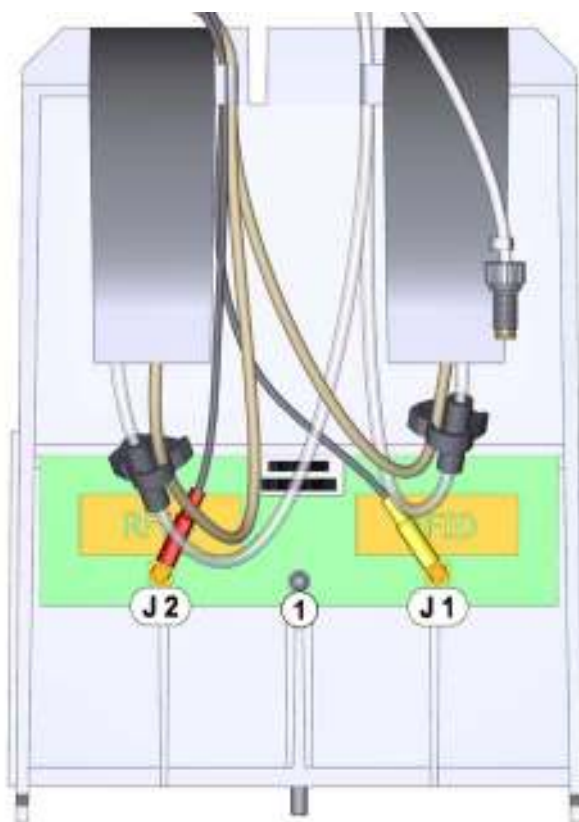


Figure 24–16: Starter Reagent Module, rear view

**24.3.4.3 Assembly of the Dual antenna PCBA board**

1. Set Dual antenna PCBA in and fasten one screw (1).
2. Connect the antenna cables onto the Dual antenna PCBA (J1) (J2).
3. Assemble the Starter Reagent Module, see Chapter 24.3.1.3 on page 24-6.
4. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
5. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
6. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

## 24.4 Adjustment and calibration

In this section information about the adjustment and calibration of Starter Reagent Module can be found.

### NOTE

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).  
Do not run any of the software at the same time.

### 24.4.1 Initialization routines

- Close any open software and start the **Service Software** from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **Reader** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog Controlled Modules** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 24.4.2 Calibration of the Starter Reagent Module

#### 24.4.2.1 Calibration of the Starter bubble sensors

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select **Starter Sensors**.

#### Step 1

- Press the **Start** button, to start the procedure.
- Follow the instruction at the pop-up and press **OK** button.



Figure 24–17: Sensor calibration pop-up for Starter 1

- When the **Processing** window closes, press the **Done** button.

## 24. Starter Reagent Module for external-tanks configuration

### Step 2

- d. Press the **Start** button, to start the procedure.
- e. Follow the instruction at the pop-up and press **OK** button.

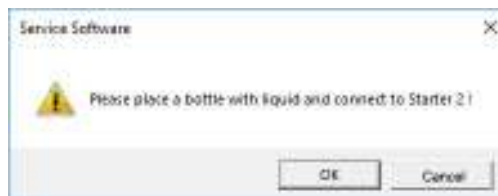


Figure 24–18: Sensor calibration pop-up for Starter 2

- f. When the **Processing** window closes without failure message, press the **Done** button.

### Step 3

- g. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

### Step 4

- h. The process is successful when, at the **Summarize** panel, all steps are in green.

### 24.4.2.2 Calibration of the Starter pumps

In the **Service Software** at the navigation window choose **Reader** and the **Teach & Calibration** bracket and select **Starter Pump**

### Step 1

- a. Press the **Start** button; the calibration procedure starts automatically.
- b. When the **Processing** window closes, press the **Done** button.

### Step 2

- c. Press the **Start** button; the calibration procedure starts automatically.
- d. When the **Processing** window closes without failure message, press the **Done** button.

### Step 3

- e. Store the taught position in the EEPROM by selecting **Write ALL Parameters to NVM** button.

### Step 4

- f. The process is successful when, at the **Summarize** panel, all steps are in green.

### 24.4.3 Check of Starters bubble sensors

In the **Service Software** at the navigation window choose **Reader** and the **General** bracket and select **Sensors** then **Check**.

### Step 1

- a. Select the **Starter 1 bottle sensor** and follow the instructions.

### Step 2

- b. Compare the TubeCondition result with the SensorState result (the status should be equal).

### Step 3

- a. Select the **Starter 2 bottle sensor** and follow the instructions.

### Step 4

- b. Compare the TubeCondition result with the SensorState result (the status should be equal).

#### **24.4.4 Verifying RFID**

In the **Service Software** at the navigation window choose **System** and the **Loading Report** bracket.

The changes of the materials to be loaded are shown after loading or unloading of the materials.

- a. When inserting the Starter bottles, after a defined time interval the presence of the bottles will be prompted.
- b. The RFID will be read and displayed as a dialog.



## 24.5 Electric connections, switches and signals

This section shows an overview of the electronic boards of the module.

**NOTE**

see Chapter 4.1.8 on page 4-9 for an overview of the complete electric circuit of the system.

### 24.5.1 Dual antenna PCBA board

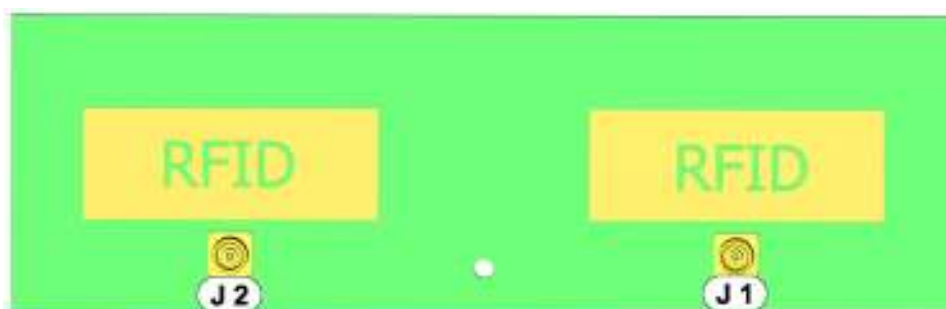


Figure 24-19: Dual antenna PCBA board

| Number |                                      | Connection |
|--------|--------------------------------------|------------|
| J1     | RFID antenna<br>(yellow shrink tube) |            |
| J2     | RFID antenna<br>(red shrink tube)    |            |

Table 24-3: Connections of the Dual antenna PCBA board

## 24. Starter Reagent Module for external-tanks configuration

---

# 25 Pumps for internal-tanks configuration

This chapter shows an overview of all pumps of the **LIAISON<sup>®</sup> XS** Analyzer.

In the first section of the chapter, “Safety” notes concerning pumps are specified, while the next section gives a brief overview of the pumps and where they are located inside the instrument.

This section is followed by a larger part containing informations about repair and calibration of all the pumps (sections “Replacement and Repair”, “Adjustment and Calibration”). In the last section, PCBA layout and connections are described.

## 25.1 Safety

### DANGER



#### Risk of infection!

The pumps and tubings, (parts with contact to potentially infectious liquid) shall be treated as potentially infectious. Improper handling of infectious parts will cause skin irritations, illnesses and possible death.

- Strictly follow the local and national provisions, legislation and laboratory regulations.
- Use appropriate gloves!
- Use an appropriate lab coat!
- Use an appropriate eye protection (e.g. protective glasses)!
- Avoid contact between skin/mucous membrane and samples/test reagents or parts of the instrument.
- Clean, disinfect and decontaminate the instrument immediately if potentially infectious material has been spilled.
- Do not use broken or chipped tubes or bottles.
- Observe the instructions in the package inserts for correct use of reagents.
- Observe the legal regulations for the handling of infectious material.
- Never use bio-hazardous liquids for testing the instrument!
- The instrument shall be cleaned, disinfected and decontaminated before servicing!

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

## 25.2 Overview

This section shows an overview of the pumps and components located on the Left Side of the instrument and below the Cuvette - Tip Loading Module.

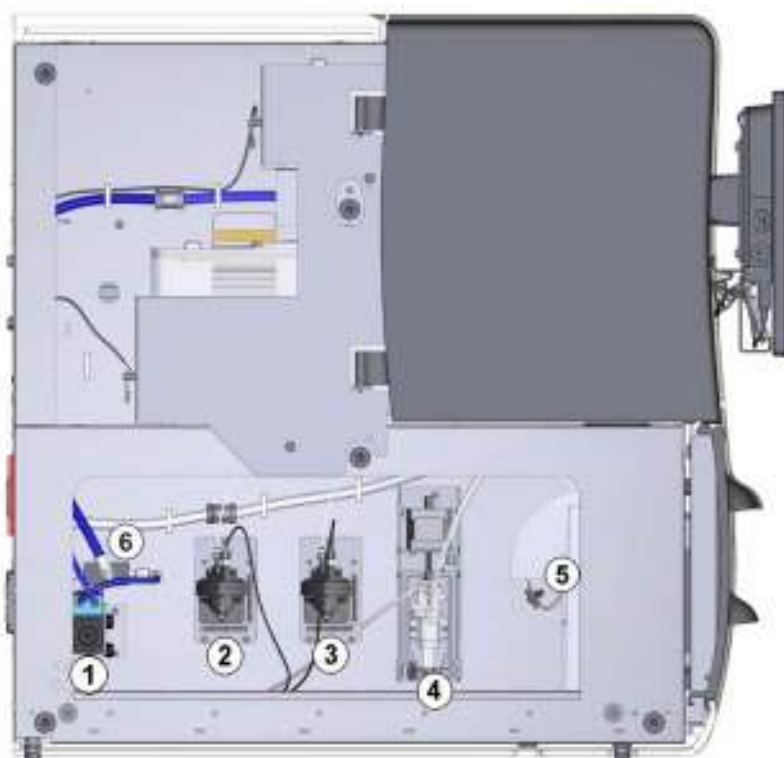


Figure 25-1: Instrument, left side view

| No. | Description                      |
|-----|----------------------------------|
| 1   | PRP (Probe Rinsing Pump)         |
| 2   | SDP1 (Starter Dispense Pump)     |
| 3   | SDP2 (Starter Dispense Pump)     |
| 4   | WBDP (Wash Buffer Dispense Pump) |
| 5   | Starter Reagent Module           |
| 6   | LDV (Liquid Distribution Valve)  |

Table 25-1: Pumps and components located behind the Left Side Cover of the instrument

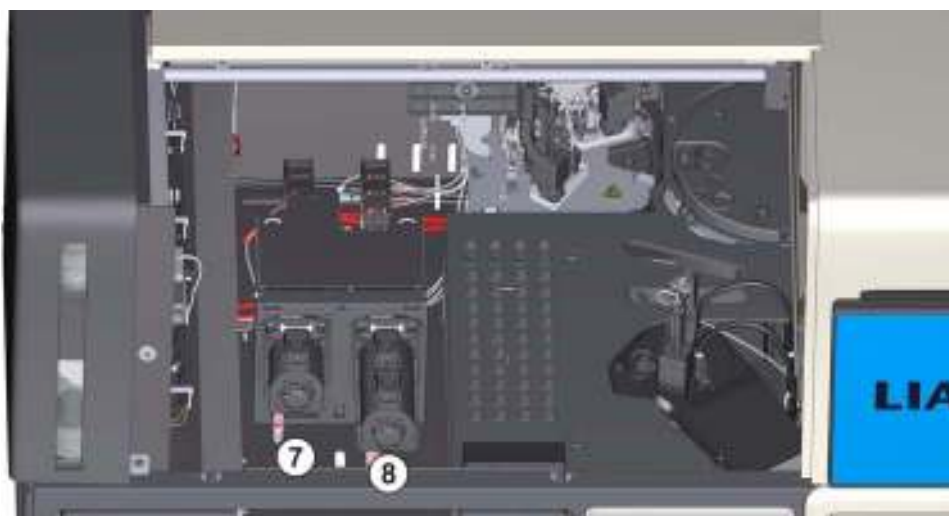


Figure 25-2: Instrument, with Left Top Cover and Cuvette - Tip drawer opened

| No. | Description                                                |
|-----|------------------------------------------------------------|
| 7   | LWP (Liquid Waste Pump) or 3-channels peristaltic pump     |
| 8   | WAP (Waste Aspiration Pump) or 5-channels peristaltic pump |

Table 25-2: Pumps located below the Cuvette - Tip Loading Module

### 25.2.1 Liquid Distribution Overview

#### NOTE

For an overview of the complete liquid distribution, see Chapter 3.1.6 on page 3-7.

## 25.3 Replacement of Pumps

### 25.3.1 Replacement of the LWP (Liquid Waste Pump, 3-channels peristaltic pump)

#### 25.3.1.1 Overview

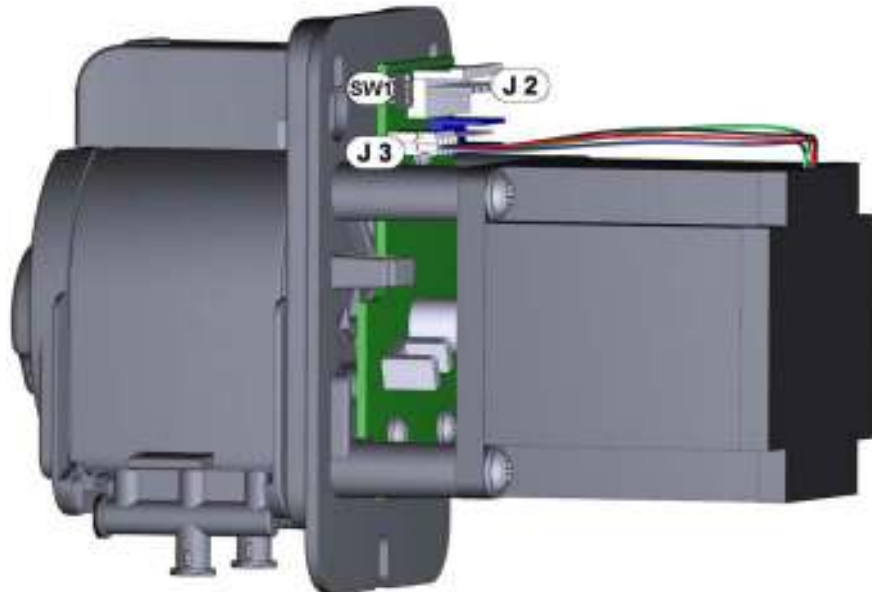


Figure 25–3: LWP

| Number |            | Connection |
|--------|------------|------------|
| J2     | PWR/CAN    |            |
| J3     | Pump motor |            |

Table 25–3: Connections of the LWP PCBA board

#### 25.3.1.2 Electrical Circuit Overview

**NOTE**

For the detail of the LWP connection and DIP-Switch setting, see Chapter 25.5.1 on page 25-36.

### 25.3.1.3 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.

### 25.3.1.4 Removal of the LWP (Liquid Waste Pump, 3-channels peristaltic pump)

1. Tilt peristaltic pumps rack up.
2. Unplug all tubing of LWP (A-B).

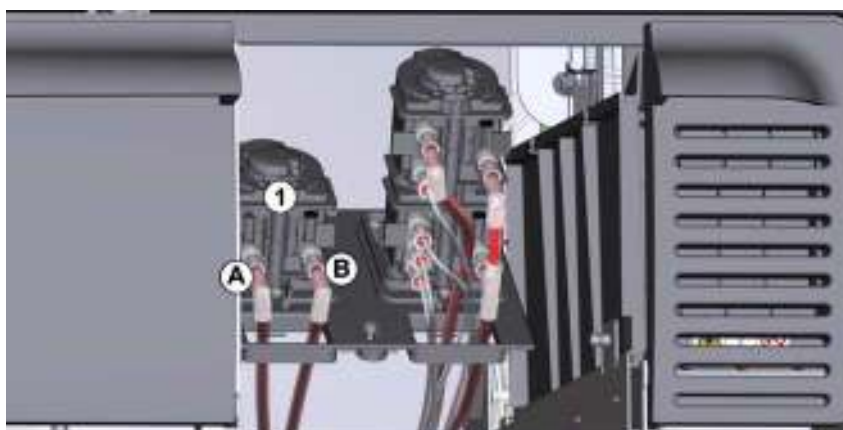


Figure 25-4: LWP, bottom view

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the LWP tubing.

3. Loosen three screws (2), lift the LWP (1) a bit up and carefully pull it out, paying attention that the PWR/CAN cable is still connected.

## 25. Pumps for internal-tanks configuration

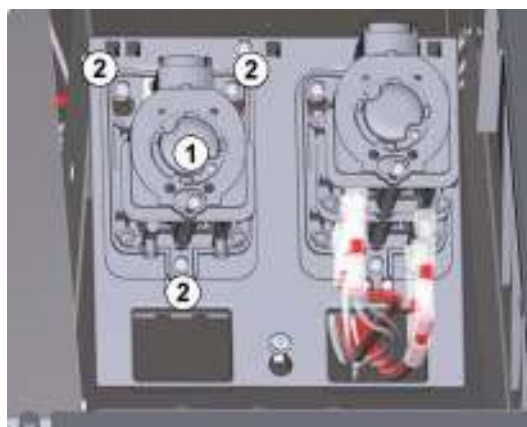


Figure 25–5: Peristaltic pumps rack, front view

4. Unplug PWR/CAN cable (J2) from LWP and remove the pump.



Figure 25–6: LWP, top view

### 25.3.1.5 Assembly of the LWP (Liquid Waste Pump, 3-channels peristaltic pump)

1. Check DIP-Switch on the LWP PCBA for correct setting, see Table 25–19.
2. Hold the LWP in front of the peristaltic pumps bracket and connect PWR/CAN cable onto LWP PCBA connector (J2).
3. Set keyholes into the screws and fasten three screws (2).
4. Connect following fitting onto LWP (1):



| No. | Connection at Pump | Tubing              | Connection to                    |
|-----|--------------------|---------------------|----------------------------------|
| A   | LWP                | Wash station-LWP    | Wash station                     |
| B   | LWP                | LWP-Waste interface | Waste interface<br>(Y-connector) |

Table 25–4: LWP, tube connection overview

5. Assemble the Cuvette - Tip drawer, see Chapter 15.3.1.3 on page 15-3.

#### 25.3.1.6 Set LWP

1. Run the Firmware Update of the complete system in case new LWP is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 25.3.2 Replacement of the WAP (Waste Aspiration Pump, 5-channels peristaltic pump)

#### 25.3.2.1 Overview

This section shows an overview of the connections of the Waste Aspiration Pump.

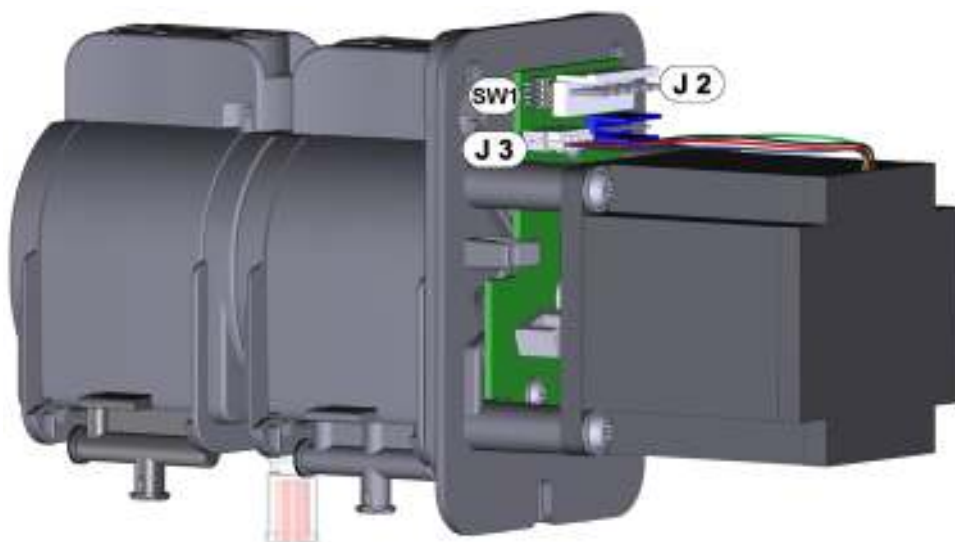


Figure 25–7: WAP, side view

| Number | Connection |
|--------|------------|
| J2     | PWR/CAN    |
| J3     | Pump motor |

## 25. Pumps for internal-tanks configuration

Table 25–5: Connections of the WAP PCBA

### 25.3.2.2 Electrical Circuit Overview

#### NOTE

For the detail of the WAP connection and DIP-Switch setting, see Chapter 25.5.1 on page 25-36.

### 25.3.2.3 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.

### 25.3.2.4 Removal of the WAP (5-channels peristaltic pump)

1. Tilt peristaltic pumps rack up.
2. Unplug all tubing of the WAP (C-I).

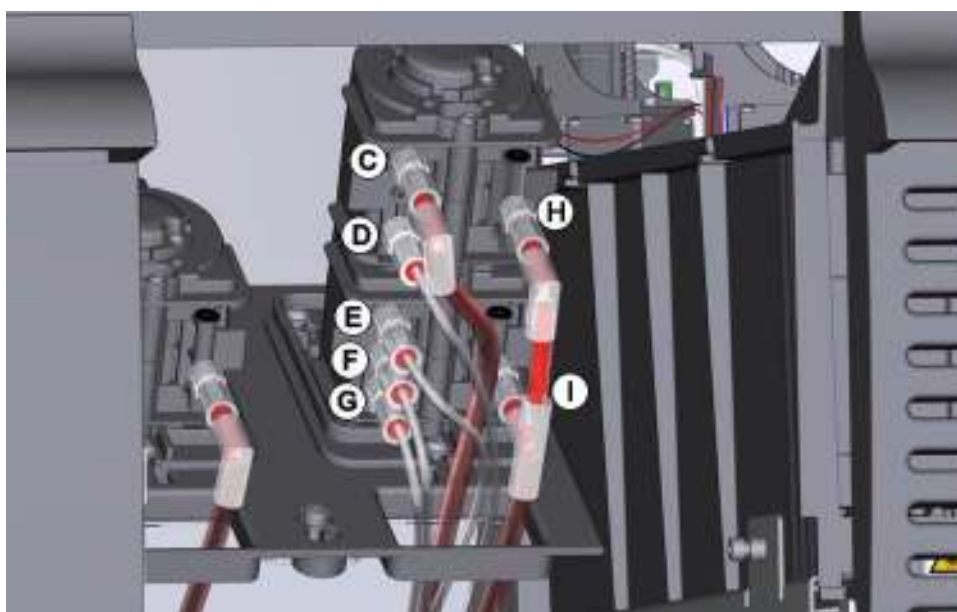


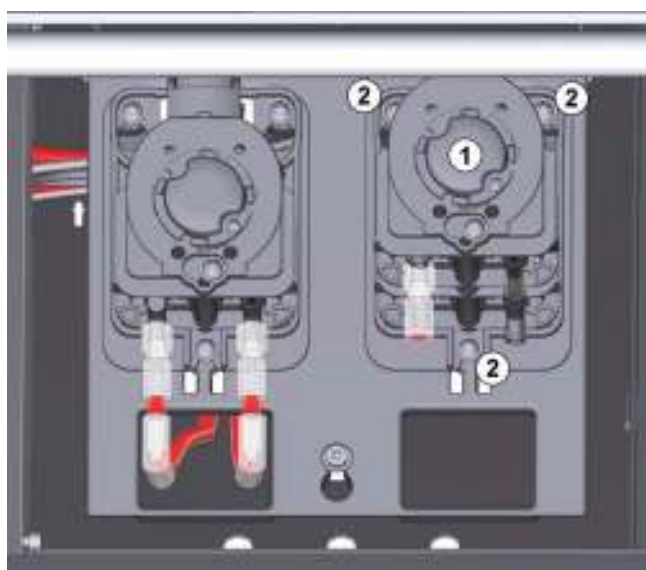
Figure 25–8: WAP, bottom view

**WARNING****Liquid Waste Spillage**

Please pay attention that liquid waste spillage is possible when unplugging the WAP tubing.

---

3. Loosen three screws (2) and lift the WAP (1) a bit up and carefully pull it out, paying attention that the PWR/CAN cable is still connected.



*Figure 25–9: Peristaltic pumps rack, front view*

## 25. Pumps for internal-tanks configuration

---

4. Unplug PWR/CAN cable (J2) from WAP and remove the pump.



*Figure 25–10: WAP, top view*

**25.3.2.5 Assembly of the WAP (Waste Aspiration Pump, 5-channels peristaltic pump)**

1. Check DIP-Switch on the WAP PCBA for correct setting, see Table 25–20.
2. Hold the WAP in front of the peristaltic pump bracket and connect PWR/CAN cable onto WAP PCBA connector (J2).
3. Set keyholes into the screws and fasten three screws (2).
4. Connect following fitting onto the WAP (1):

| No. | Connection at Pump | Tubing              | Connection to                     |
|-----|--------------------|---------------------|-----------------------------------|
| C   | WAP                | Reagent Module-WAP  | Drain of the Reagent Module       |
| D   | WAP                | Reader-WAP          | Reader SIPAL                      |
| E   | WAP                | Washer-WAP          | Washer SIPAL 1                    |
| F   | WAP                | Washer-WAP          | Washer SIPAL 2                    |
| G   | WAP                | Washer-WAP          | Washer SIAPL 3 and Grounding pipe |
| H   | WAP                | WAP-Waste interface | Waste interface (Y-connector)     |
| I   | WAP                | WAP-Waste interface | Waste interface (Y-connector)     |

*Table 25–6: WAP tube connection overview*

5. Assemble the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.

**25.3.2.6 Set WAP**

1. Run the Firmware Update of the complete system in case new WAP is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

## 25. Pumps for internal-tanks configuration

### 25.3.3 Replacement of the Peristaltic Pumps Tubings

#### 25.3.3.1 Overview



Figure 25-11: Peristaltic Pumps Tubings

| No. | Tubing for:                      |
|-----|----------------------------------|
| 1)  | LWP 3-channel Peristaltic Tubing |
| 2)  | WAP 3-channel Peristaltic Tubing |
| 3)  | WAP 2-channel Peristaltic Tubing |

Table 25-7: Peristaltic Pumps Tubings, overview



Figure 25-12: Prescribed position of Peristaltic Pumps Tubings

### 25.3.3.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Left Top Cover.
6. Move Sample pipettor to a safe location.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.

### 25.3.3.3 Removal of the Peristaltic Pumps Tubings

1. Unlock and remove the caps (1) of the Peristaltic Pumps (LWP and WAP) by pressing and turning the latches counterclockwise slightly less than 90°.



Figure 25–13: Peristaltic Pumps inside the instrument, top view

| No. | Description               |
|-----|---------------------------|
| 1)  | Caps                      |
| 2)  | Peristaltic Pumps Tubings |
| 3)  | Pump bracket              |

Table 25–8: Peristaltic Pumps, components



## 25. Pumps for internal-tanks configuration

2. Tilt the pump bracket (3) upwards.
3. Unscrew the tube fittings (5) from the peristaltic tubings.
4. Pull the Peristaltic Pumps Tubings (4) out from each side of the pumps, and remove them.

**WARNING****Liquid Waste Spillage**

Please pay attention that liquid waste spillage is possible when unplugging the WAP tubing.



Figure 25-14: Pump bracket tilted upwards

| No. | Description               |
|-----|---------------------------|
| 4)  | Peristaltic Pumps Tubings |
| 5)  | Tube fittings             |

Table 25-9: Pump bracket tilted upwards



#### 25.3.3.4 Assembly of the Peristaltic Pumps Tubings

**NOTE**

**The tubes must not be twisted inside of the pump.**

If the tubes are twisted, then check them for dents or kinks.

Make sure that the tubes are properly seated in the tabs.

When dents or kinks are found, the tubes shall be replaced.

**WARNING**

**The Luer connectors of the tubes have different colors:**

- Connect the tubes with red Luer connectors always to the right side of the pumps.
- Connect the tubes with white Luer connectors always to the left side of the pumps.

1. Make sure the correct tube is mounted in the proper position:
  - a. Liquid Waste Pump is on the left and hosts 3 tubes with one inlet on the left and one outlet on the right.
  - b. Waste Aspiration Pump is on the right and host two brackets:
    - i. the bracket on the rear of the instrument hosts 3 tubes with three inlets on the left and one outlet on the right,
    - ii. the bracket on the front of the instrument hosts 2 tubes with 2 inlets on the left and one outlet on the right.
2. Snap the tubes into the Liquid Waste Pump, by sliding each side into the proper guiding, until the tube handle clicks in both the front and rear brackets at each side of the pump.
3. Remount the cap of the Liquid Waste Pump: insert the cap, and while pushing it down, rotate the latches clockwise slightly less than 90°, finishing with the latch clicking at 12 hour.
4. Connect the white and red tube fittings to the Liquid Waste Pump.
5. Snap the tubes into the Waste Aspiration Pump, by sliding each side into the proper guiding, until the tube handle clicks in both in the front and rear brackets at each side of the pump.
6. Remount the caps of the Waste Aspiration Pump: insert the cap, and while pushing it down, rotate the latches clockwise slightly less than 90°, finishing with the latch clicking at 12 hour.
7. Connect the white and red tube fittings to the Waste Aspiration Pump.
8. Push the pump bracket downwards.
9. Place the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3
10. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 25.3.4 Replacement of the SDP (Starter Dispense Pumps)

#### 25.3.4.1 Overview

This section shows an overview of the SDP (Starter Dispense Pump).

The SDPs are LCM pump 500uL.

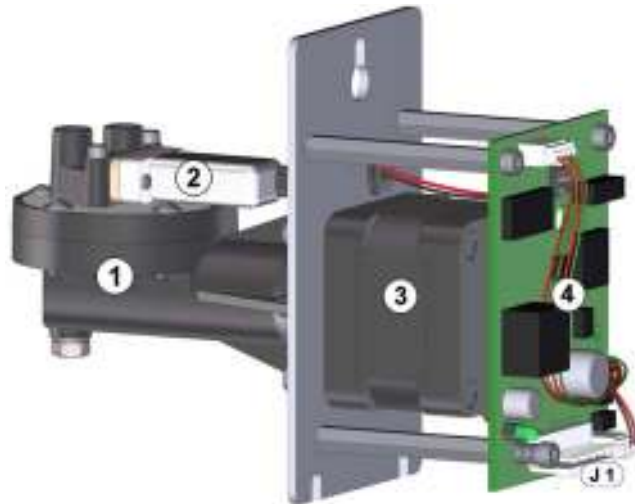


Figure 25-15: SDP

|   |                   |
|---|-------------------|
| 1 | Pump housing      |
| 2 | 2-way valve       |
| 3 | Pump motor        |
| 4 | SDP circuit board |

Table 25-10: SDP overview

#### NOTE

If the system is located at 2000mt. ASL or over you should install a Starter Pump shipped with a special package (custom vacuum package). For the detail, see Corp-TN-000529 (former LXS TN 028).

#### 25.3.4.2 Electrical Circuit Overview

#### NOTE

For the detail of the SDPs connection and DIP-Switch setting, see Chapter 25.5.3 on page 25-40.

### 25.3.4.3 Preliminary steps

#### NOTE

#### Priming in air

Before replacing the SDPs, remember to free the Starter lines priming in air by **Service Software**, see Chapter 11.8.1.6 on page 11-65.



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.

### 25.3.4.4 Removal of the SDPs

1. Unscrew fittings of tubings:
  - SDP1-Starters (Q)
  - Reader-SDP1 (S)
  - SDP1-Starters (R)
  - Reader-SDP1 (T)

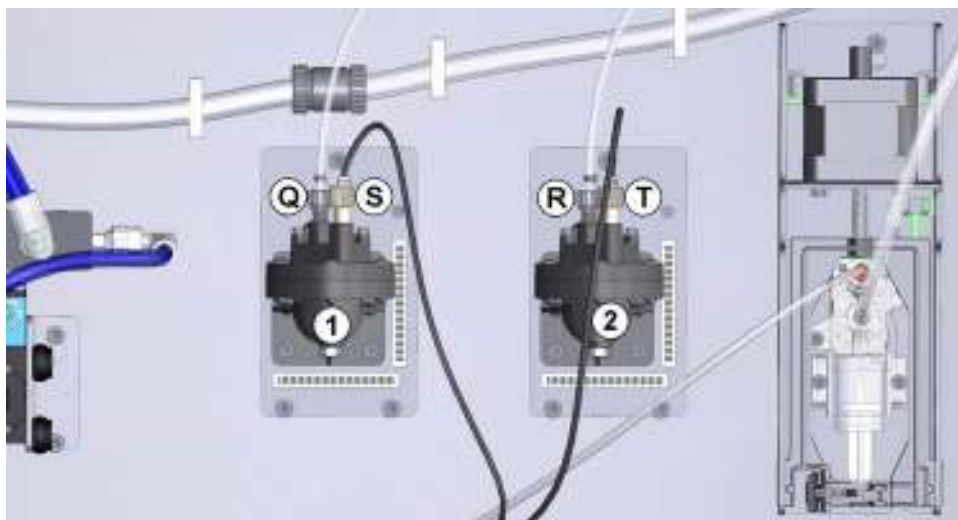


Figure 25–16: SDPs

## 25. Pumps for internal-tanks configuration

2. Loosen three screws (2), lift the SDP (1) a bit up and carefully pull it out, paying attention that the cable is still connected to the pump.

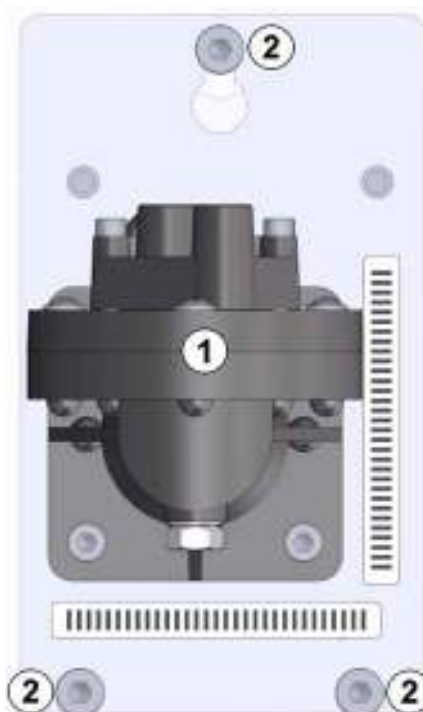


Figure 25-17: SDP, front view

3. Unplug PWR/CAN cable (J1) from the SDP and remove the pump.

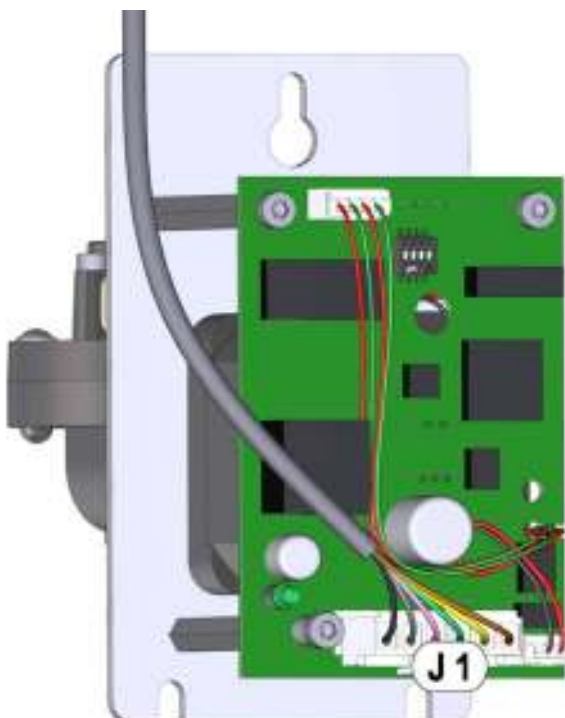


Figure 25-18: SDP, rear view

**25.3.4.5 Assembly of the SDPs**

1. Check DIP-Switch on the SDP PCBA for correct setting, see Table 25–27 and see Table 25–28.
2. Connect PWR/CAN cable onto pump connector (J1).
3. Hang the SDP into the three screws (2) and fasten screws.
4. Connect following fittings:

| No. | Connection at Pump | Tubing        | Connection to          |
|-----|--------------------|---------------|------------------------|
| Q   | SDP 1              | SDP1-Starters | Starter Reagent Module |
| S   | SDP 1              | Reader-SDP1   | Reader                 |
| R   | SDP 2              | SDP2-Starters | Starter Reagent Module |
| T   | SDP 2              | Reader-SDP2   | Reader                 |

*Table 25–11: SDP, tubings connections overview*

5. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**25.3.4.6 Set SDP**

1. Run the Firmware Update of the complete system in case new SDP is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the Starter Pump calibration, see Chapter 25.4.2.1 on page 25-35.
3. Perform Starter Reagents prime, see Chapter 11.8.1.6 on page 11-65.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

**NOTE****Priming in liquid**

Please, be sure that the Starter lines are primed with Starter Reagent before you leave the instrument.

### 25.3.5 Replacement of the PRP (Probe Rinsing Pump)

#### 25.3.5.1 Overview

This section shows an overview of the PRP (Probe Rinsing Pump).

The PRP is a membrane pump.



Figure 25–19: PRP

|   |            |
|---|------------|
| 1 | Pump motor |
| 2 | Valve      |

Table 25–12: PRP overview

#### 25.3.5.2 Preliminary steps

##### NOTE

##### Priming in air

Before replacing the PRP, remove the Wash Buffer tank and the DI water tank and remember to free the lines priming in air by **Service Software**, see Chapter 11.8.1.4 on page 11-64.



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
6. Remove the DI water tank and the Wash Buffer tank.

### 25.3.5.3 Removal of the PRP

1. Disconnect the tubings (N and P) of the PRP (1).

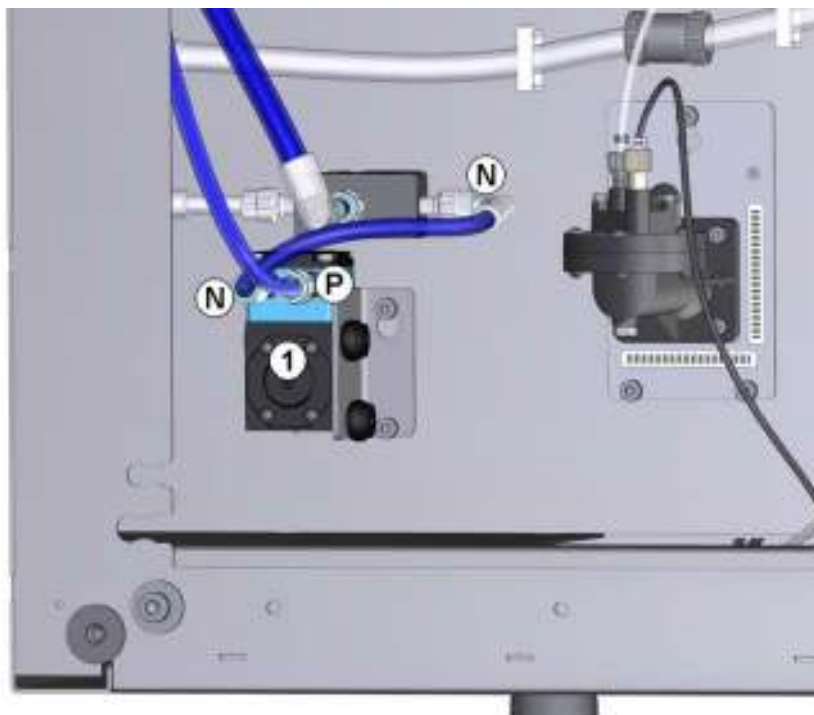
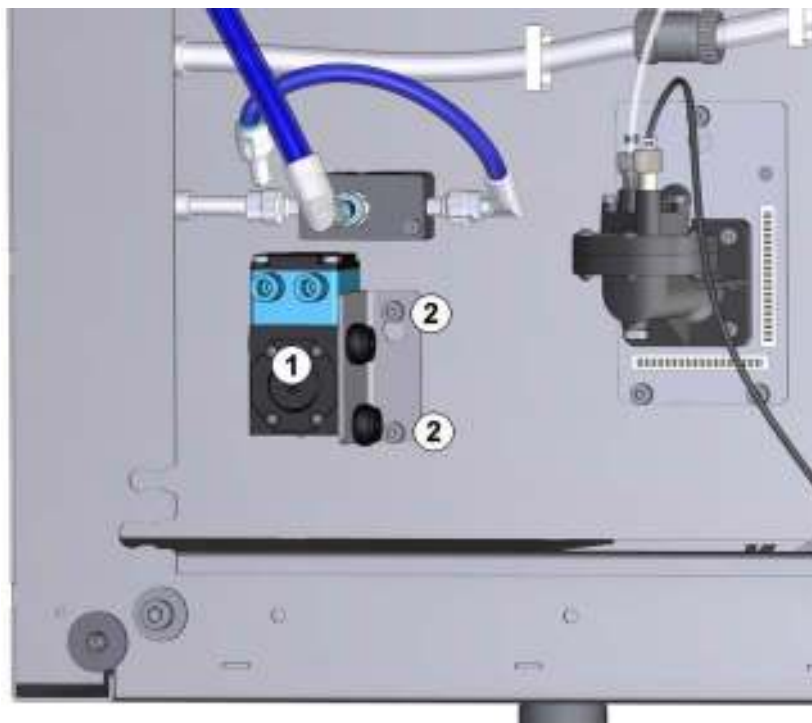


Figure 25–20: PRP inside the instrument

## 25. Pumps for internal-tanks configuration

---

2. Loosen two screws (2) of PRP (1), lift it a bit up and carefully pull it out, paying attention that the cables are still connected to the pump.



*Figure 25–21: PRP inside the instrument, without connected tubings*



3. Unplug cables of the PRP (1).



*Figure 25–22: PRP, top view*

## 25. Pumps for internal-tanks configuration

### 25.3.5.4 Assembly of the PRP

1. Connect the brown cable onto plus cable lug and the black cable onto minus cable lug of the PRP (1).



Figure 25-23: PRP, rear view

2. Hang the PRP into the two screws (2) and fasten screws.
3. Connect following fittings onto the PRP (1):

| No. | Connection at Pump | Tubing    | Connection to |
|-----|--------------------|-----------|---------------|
| P   | PRP                | PRP-SPOLV | SPOLV Module  |
| N   | PRP                | LDV-PRP   | LDV Valve     |

Table 25-13: PRP, tubings connections overview

4. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
5. Place back the DI water tank and the Wash Buffer tank.

### 25.3.5.5 Set PRP

1. Perform the SPOLV prime, see Chapter 11.8.1.4 on page 11-64.

#### NOTE

#### Priming in liquid

Please, be sure that the SPOLV line is primed before you leave the instrument.

2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 25.3.6 Replacement of the WBDP (Wash Buffer Dispense Pump)

#### 25.3.6.1 Overview

This section shows an overview of the WBDP (Wash Buffer Dispense Pump).

The WBDP is a piston pump.

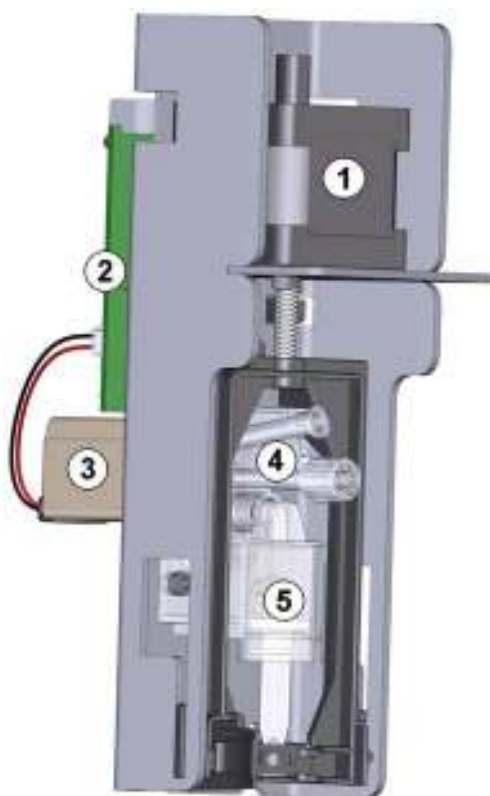


Figure 25–24: WBDP

|   |                                        |
|---|----------------------------------------|
| 1 | Pump motor                             |
| 2 | WBDP PCBA                              |
| 3 | 2 way valve                            |
| 4 | 3 way manifold                         |
| 5 | Cartridge (piston cylinder unit 2,5ml) |

Table 25–14: WBDP overview

## 25. Pumps for internal-tanks configuration

### 25.3.6.2 Electrical Circuit Overview

#### NOTE

For the detail of the WBDP connection and DIP-Switch setting, see Chapter Table 25–19: on page 25-37.

### 25.3.6.3 Preliminary steps

#### NOTE

#### Priming in air

Before replacing the WBDP, remove the Wash Buffer tank and remember to free the lines priming in air by **Service Software**, see Chapter 11.8.1.5 on page 11-65.



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
6. Remove the Wash Buffer tank.

### 25.3.6.4 Removal of the WBDP

1. Unscrew fittings of tubings of the WBDP:
  - WBDP-Washer (E-F-G)
  - WB TANK-WBDP-LDV (M)

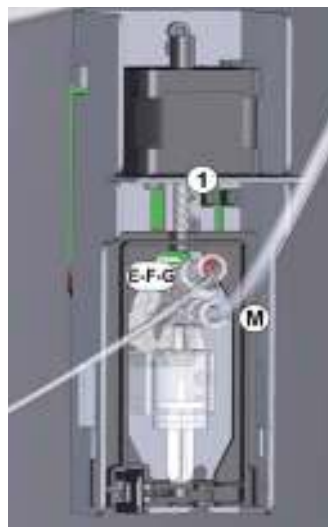


Figure 25–25: WBDP inside the instrument

**25. Pumps for internal-tanks configuration**

2. Loosen three screws (2), lift the WBDP (1) a bit up and carefully pull it out, paying attention that the cable is still connected to the WBDP.

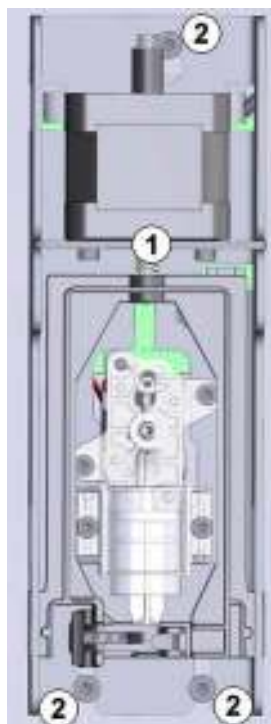


Figure 25–26: WBDP, front view

3. Unplug PWR/CAN cable (J3) from the WBDP and remove the pump.

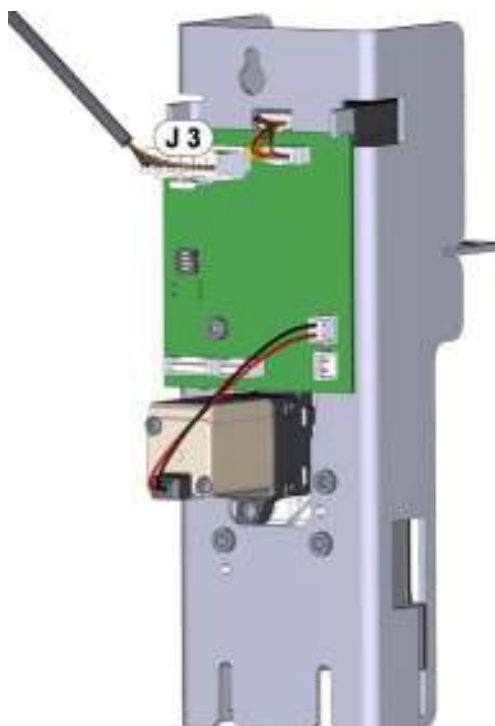


Figure 25–27: WBDP, rear view

## 25. Pumps for internal-tanks configuration

### 25.3.6.5 Assembly of the WBDP

1. Connect PWR/CAN cable onto the WBDP connector (J3).
2. Hang the WBDP into the three screws (2) and fasten screws.
3. Connect following fittings onto the WBDP (1):

| No.   | Connection at Pump | Tubing           | Connection to         |
|-------|--------------------|------------------|-----------------------|
| E-F-G | WBDP               | WBDP-Washer      | Washer Module         |
| M     | WBDP               | WB TANK-WBDP-LDV | Wash Buffer tank (5L) |

Table 25–15: WBDP, tubings connections overview

4. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
5. Place back the Wash Buffer tank.

### 25.3.6.6 Set WBDP

1. Run the Firmware Update of the complete system in case new WBDP is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the WBDP calibration, see Chapter 25.4.3.1 on page 25-35.
3. Perform the Washer prime, see Chapter 11.8.1.5 on page 11-65.

#### NOTE

#### Priming in liquid

Please, be sure that the Washer lines are primed with Wash Buffer before you leave the instrument.

4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

## 25.3.7 Replacement of the WBDP Cartridge

### 25.3.7.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
6. Remove the Wash Buffer tank.

---

**NOTE**

Before disconnecting the WBDP tubes, remove the Wash Buffer tank

---

### 25.3.7.2 Removal of the WBDP Cartridge

1. Disconnect the WBDP tubes, see Chapter 25.3.6.4 on page 25-26.
2. Turn motor shaft (2) to extend the Cartridge piston (3) to the maximum down.
3. Open piston holder (4).
4. Remove screws (5) out of the WBDP Cartridge.

## 25. Pumps for internal-tanks configuration

---

5. Pull the Cartridge up and out of the fitting connection (6).

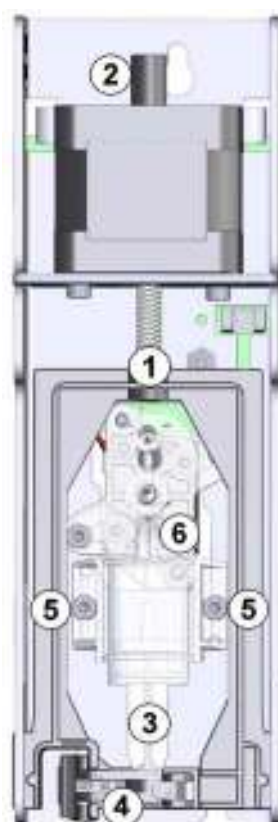


Figure 25-28: WBDP, front view



### 25.3.7.3 Assembly of the WBDP Cartridge

1. Check for correct sit of the sealing inside of the fitting connection.

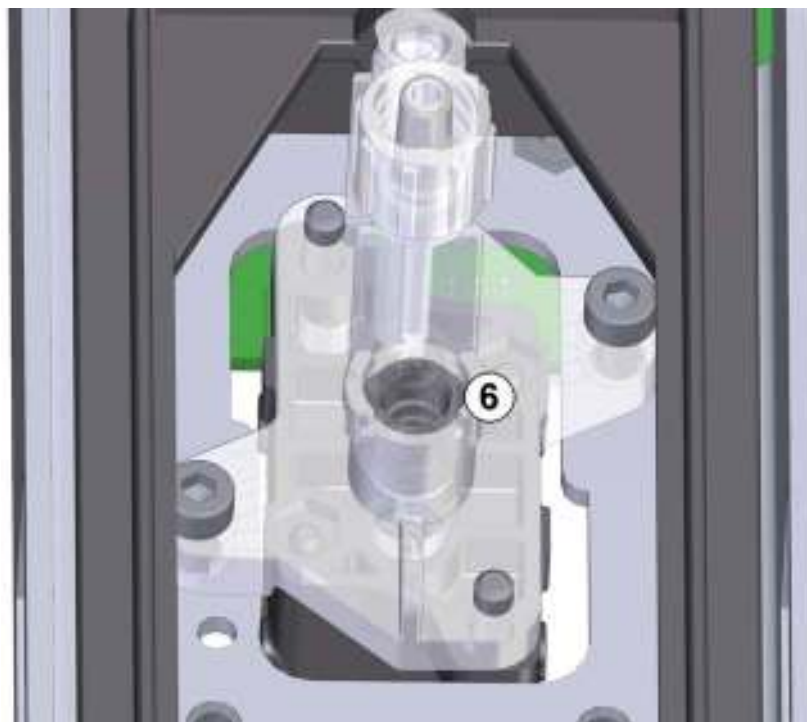


Figure 25-29: WBDP, 3-way manifold

2. Move the Cartridge fitting (7) carefully in the fitting connection (6) and be aware of the pins at the bottom of the cylinder housing that must sit correctly in the bore holes.

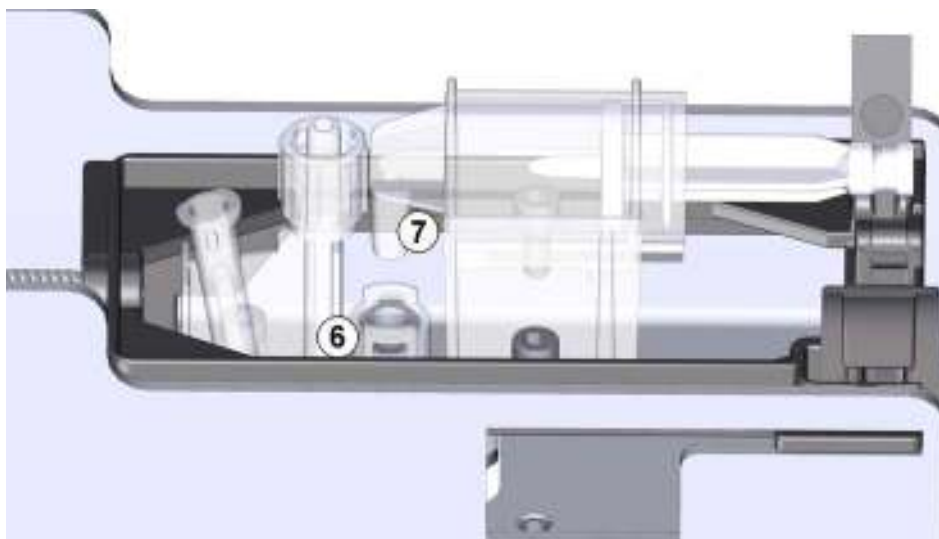


Figure 25-30: WBDP, side view

## 25. Pumps for internal-tanks configuration

---

3. The piston shaft (8) shall sit on the bottom of the pusher frame (9).
4. Then close the piston shaft holder (10).

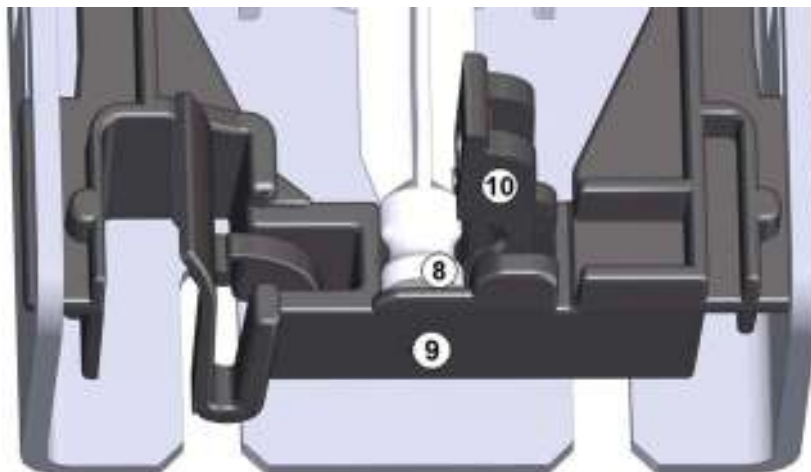


Figure 25–31: WBDP, piston bottom

5. Connect the WBDP tubes, see Chapter 25.3.6.5 on page 25-28.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
7. Place back the Wash Buffer tank.

### 25.3.7.4 Set WBDP cartridge

1. Perform the WBDP calibration, see Chapter 25.4.3.1 on page 25-35.
2. Perform the Washer prime, see Chapter 11.8.1.5 on page 11-65.

#### NOTE

#### Priming in liquid

Please, be sure that the Washer lines are primed with Wash Buffer before you leave the instrument.

3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

## 25.3.8 Replacement of the WBDP Cartridge valve unit

### 25.3.8.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
6. Remove the Wash Buffer tank.
7. Remove the WBDP, see Chapter 25.3.6.4 on page 25-26.
8. Remove the WBDP Cartridge, see Chapter 25.3.8.2 on page 25-33.

### 25.3.8.2 Removal of the WBDP Cartridge valve unit

---

**NOTE**

Before disconnecting the WBDP tubes, remove the Wash Buffer tank

---

1. Unplug the WBDP Cartridge valve cable at the WBDP PCBA.
2. Unscrew two screws (2).

## 25. Pumps for internal-tanks configuration

3. Turn the WBDP Cartridge valve unit (1) a bit counterclockwise and carefully lift it out.

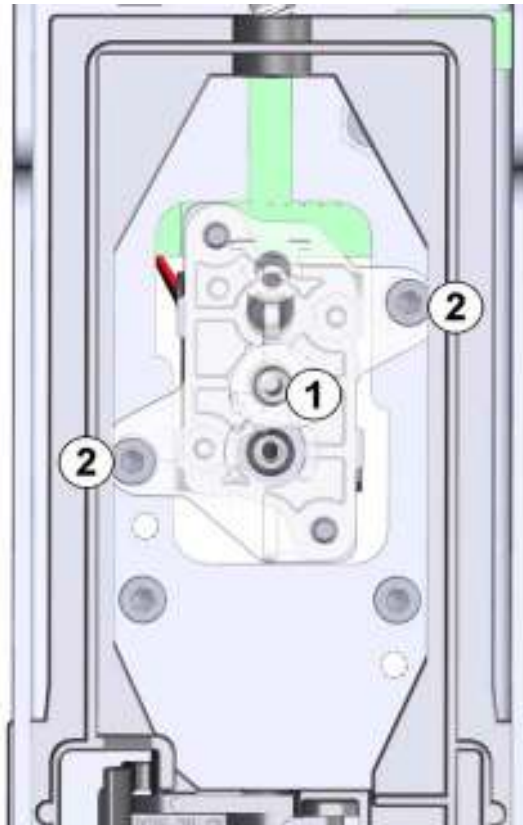


Figure 25–32: WBDP without Cartridge

### 25.3.8.3 Assembly of the WBDP Cartridge valve unit

1. Set the WBDP Cartridge valve unit (1) into the carrier for the pump and fasten with two screws (2).
2. Connect the WBDP Cartridge valve cable onto the WBDP PCBA.
3. Assemble the WBDP Cartridge, see Chapter 25.3.7.3 on page 25-31.
4. Assemble and set the WBDP, see Chapter 25.3.6.5 on page 25-28 and see Chapter 25.3.6.6 on page 25-28.
5. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
6. Place back the Wash Buffer tank.

### 25.3.8.4 Set WBDP Cartridge valve unit

1. Perform the WBDP calibration, see Chapter 25.4.3.1 on page 25-35.
2. Perform the Washer prime, see Chapter 11.8.1.5 on page 11-65.

#### NOTE

#### Priming in liquid

Please, be sure that the Washer lines are primed with Wash Buffer before you leave the instrument.

3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

## 25.4 Adjustment and Calibration

This section contains informations about the adjustment and calibration of the Pumps.

### 25.4.1 Initialization routines

- a. Close any open software and start the **Service Software**.
- b. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- c. Click on the **Instrument** button and choose the **Full initialization via COP** button to start the initialization sequence.
- d. The process is successful when the **Processing** window closes without failure message.

### 25.4.2 Calibration of the SDPs

#### 25.4.2.1 Calibration of the Starter Pump

In the **Service Software** at the navigation window choose **Reader** then **Teach & Calibration** bracket and select **Starter Pump**.

- Step 1**
- a. Click on the **Start** button to start the automatic calibration of the SDP1.
  - b. Check the results and proceed by pressing **Done** button.
- Step 2**
- c. Click on the **Start** button to start the automatic calibration of the SDP2.
  - d. Follow the instruction by pressing **Done** button.
- Step 3**
- e. Store the taught positions in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 4**
- f. The process is successful when, at the **Summarize** panel, all steps are in green.

### 25.4.3 Calibration of the WBDP

#### 25.4.3.1 Calibration of the WBDP

In the **Service Software** at the navigation window choose **Washer** then **Teach & Calibration** bracket and select **WBDP Calibration**.

- Step 1**
- a. Click on the **Start** button to start the automatic calibration of the WBDP.
  - b. Check the results and proceed by pressing **Done** button.
- Step 2**
- c. Store the taught positions in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 3**
- d. The process is successful when, at the **Summarize** panel, all steps are in green.

## 25.5 Electrical connections, switches and signals

This section shows an overview of the electronic boards of the pumps.

### NOTE

see Chapter 3.1.7 on page 3-8 for an overview of the complete electric circuit of the system.

### 25.5.1 LWP and WAP PCBA

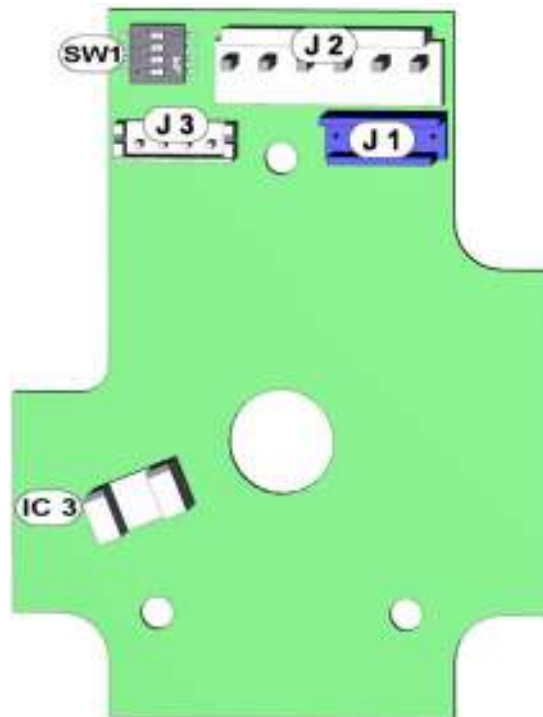


Figure 25–33: LWP and WAP PCBA

| Number | Connection |
|--------|------------|
| J1     | Download   |
| J2     | PWR/CAN    |
| J3     | Pump motor |

Table 25–16: Connections of the LWP and WAP PCBA

| Number | Meaning/Function     |
|--------|----------------------|
| IC 3   | Init (light barrier) |

Table 25–17: IC of the LWP and WAP PCBA

| Number | Meaning/Function     |
|--------|----------------------|
| SW 1   | Individual selection |

*Table 25–18: DIP-Switch of the LWP and WAP PCBA*

| DIP-Switch | Meaning/Function/Default Switch Setting |
|------------|-----------------------------------------|
| SW1 1      | ON                                      |
| 2          | OFF                                     |
| 3          | OFF                                     |
| 4          | OFF                                     |

*Table 25–19: DIP-Switch settings at LWP PCBA*

| DIP-Switch | Meaning/Function/Default Switch Setting |
|------------|-----------------------------------------|
| SW1 1      | OFF                                     |
| 2          | OFF                                     |
| 3          | OFF                                     |
| 4          | OFF                                     |

*Table 25–20: DIP-Switch settings at WAP PCBA*

25. Pumps for internal-tanks configuration

25.5.2 WBDP PCBA

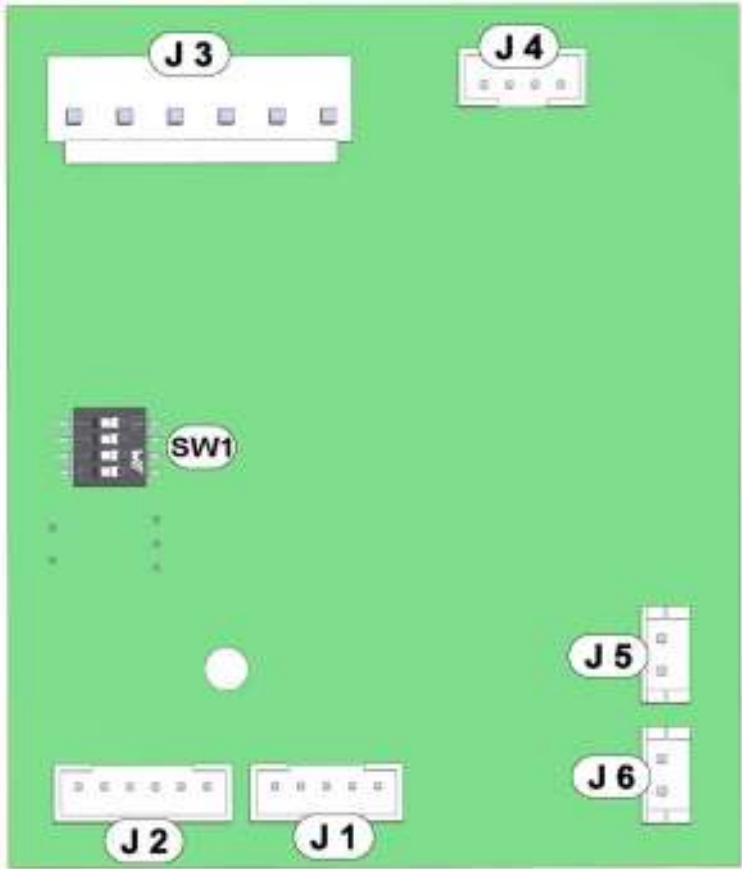


Figure 25–34: WBDP PCBA, front view

| Number |            | Connection |
|--------|------------|------------|
| J1     | Download   |            |
| J2     | Sensor IN  |            |
| J3     | PWR/CAN    |            |
| J4     | Pump motor |            |
| J5     | Valve 1    |            |
| J6     | Valve 2    |            |

Table 25–21: Connections of the WBDP PCBA

| Number |                      | Meaning/Function |
|--------|----------------------|------------------|
| SW 1   | Individual selection |                  |

Table 25–22: DIP-Switch of the WBDP PCBA



| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | OFF                                     |
|            | 2 | OFF                                     |
|            | 3 | OFF                                     |
|            | 4 | OFF                                     |

Table 25–23: DIP-Switch settings at WBDP PCBA

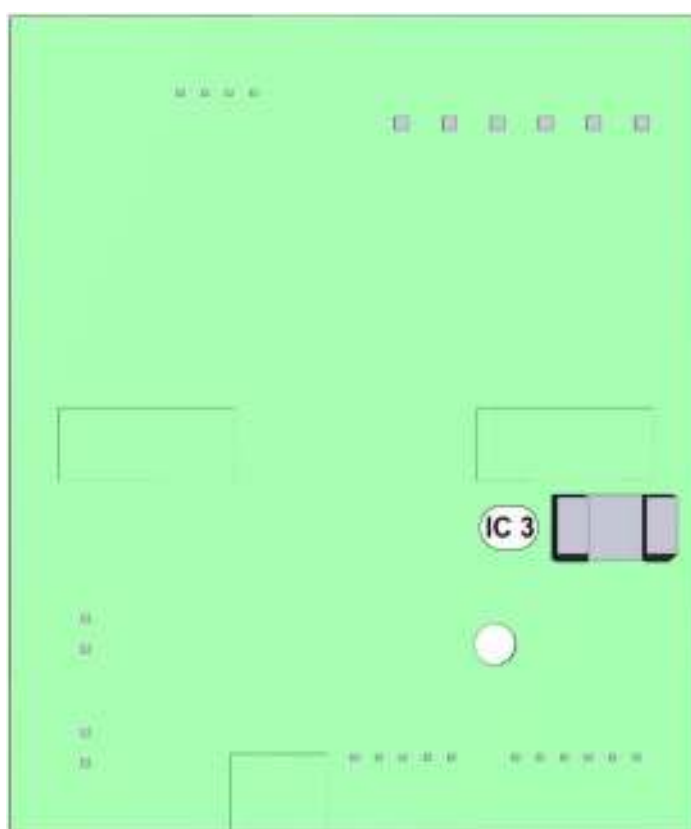


Figure 25–35: WBDP PCBA, rear view

| Number | Meaning/Function                      |
|--------|---------------------------------------|
| IC 3   | Cartridge piston home (light barrier) |

Table 25–24: IC of the WBDP PCBA

25.5.3 SDP PCBA

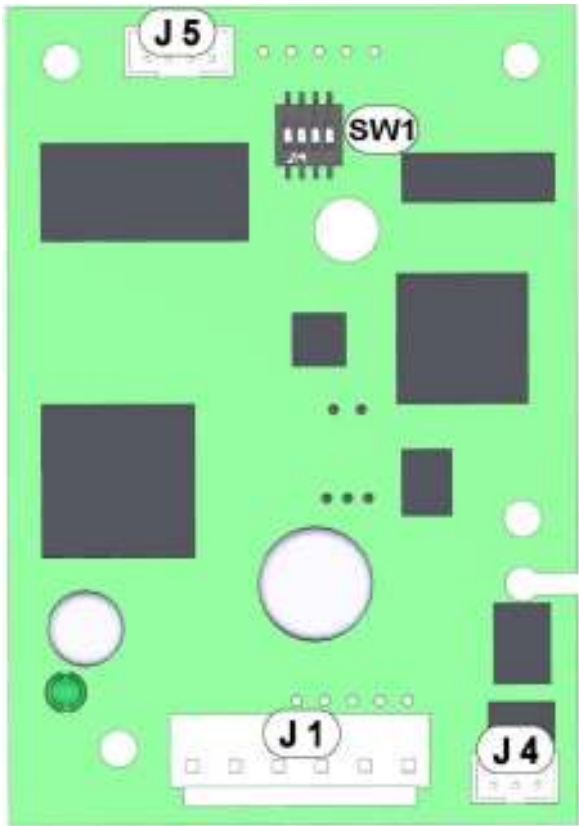


Figure 25–36: SDP PCBA, front view

| Number |  | Connection    |
|--------|--|---------------|
| J1     |  | Main PWR/CAN  |
| J4     |  | PWR out       |
| J5     |  | Stepper motor |

Table 25–25: Connections of the SDP PCBA

| Number |  | Meaning/Function     |
|--------|--|----------------------|
| SW 1   |  | Individual selection |

Table 25–26: Switch of the SDP PCBA

| DIP-Switch      Meaning/Function/Default Switch Setting - Starter 1 |   |     |
|---------------------------------------------------------------------|---|-----|
| SW1                                                                 | 1 | OFF |
|                                                                     | 2 | OFF |
|                                                                     | 3 | OFF |
|                                                                     | 4 | OFF |

Table 25–27: DIP-Switch settings at SDP1 PCBA

| DIP-Switch      Meaning/Function/Default Switch Setting - Starter 2 |   |     |
|---------------------------------------------------------------------|---|-----|
| SW1                                                                 | 1 | ON  |
|                                                                     | 2 | OFF |
|                                                                     | 3 | OFF |
|                                                                     | 4 | OFF |

Table 25–28: DIP-Switch settings at SDP2 PCBA

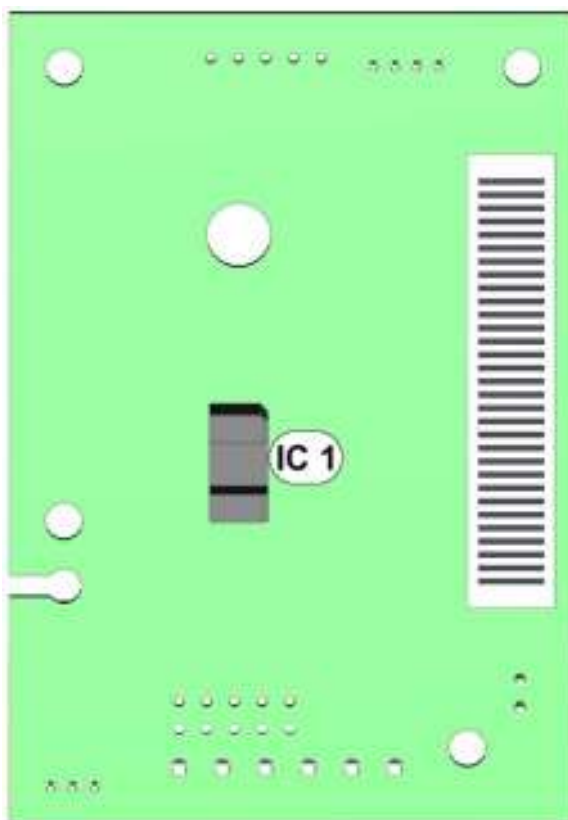


Figure 25–37: SDP PCBA, rear view

| Number | Meaning/Function     |
|--------|----------------------|
| IC 1   | Home (light barrier) |

Table 25–29: IC of the SDP PCBA



# 26 Pumps for external-tanks configuration

This chapter shows an overview of all pumps of the **LIAISON® XS** Analyzer.

In the first section of the chapter, “Safety” notes concerning pumps are specified, while the next section gives a brief overview of the pumps and where they are located inside the instrument.

This section is followed by a larger part containing informations about repair and calibration of all the pumps (sections “Replacement and Repair”, “Adjustment and Calibration”). In the last section, PCBA layout and connections are described.

## 26.1 Safety

### DANGER



#### Risk of infection!

The pumps and tubings, (parts with contact to potentially infectious liquid) shall be treated as potentially infectious. Improper handling of infectious parts will cause skin irritations, illnesses and possible death.

- Strictly follow the local and national provisions, legislation and laboratory regulations.
- Use appropriate gloves!
- Use an appropriate lab coat!
- Use an appropriate eye protection (e.g. protective glasses)!
- Avoid contact between skin/mucous membrane and samples/test reagents or parts of the instrument.
- Clean, disinfect and decontaminate the instrument immediately if potentially infectious material has been spilled.
- Do not use broken or chipped tubes or bottles.
- Observe the instructions in the package inserts for correct use of reagents.
- Observe the legal regulations for the handling of infectious material.
- Never use bio-hazardous liquids for testing the instrument!
- The instrument shall be cleaned, disinfected and decontaminated before servicing!

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 2.5 on page 2-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

## 26.2 Overview

This section shows an overview of the pumps and components located on the Left Side of the instrument and below the Cuvette - Tip Loading Module.

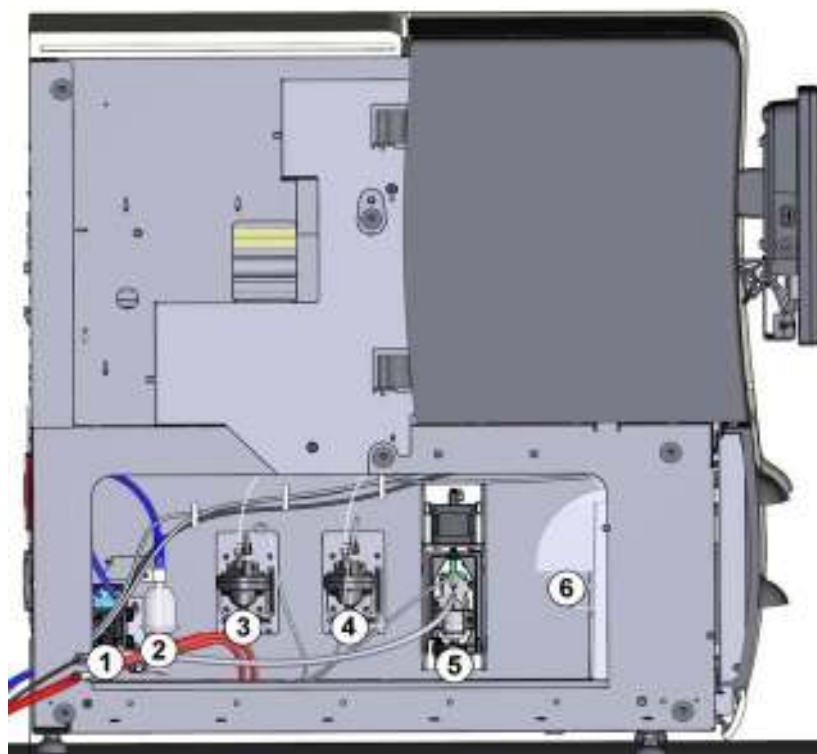


Figure 26–1: Instrument, left side view

| No. | Description                      |
|-----|----------------------------------|
| 1   | PRP (Probe Rinsing Pump)         |
| 2   | MANN filter                      |
| 3   | SDP1 (Starter Dispense Pump)     |
| 4   | SDP2 (Starter Dispense Pump)     |
| 5   | WBDP (Wash Buffer Dispense Pump) |
| 6   | Starter Reagent Module           |

Table 26–1: Pumps and components located behind the Left Side Cover of the instrument

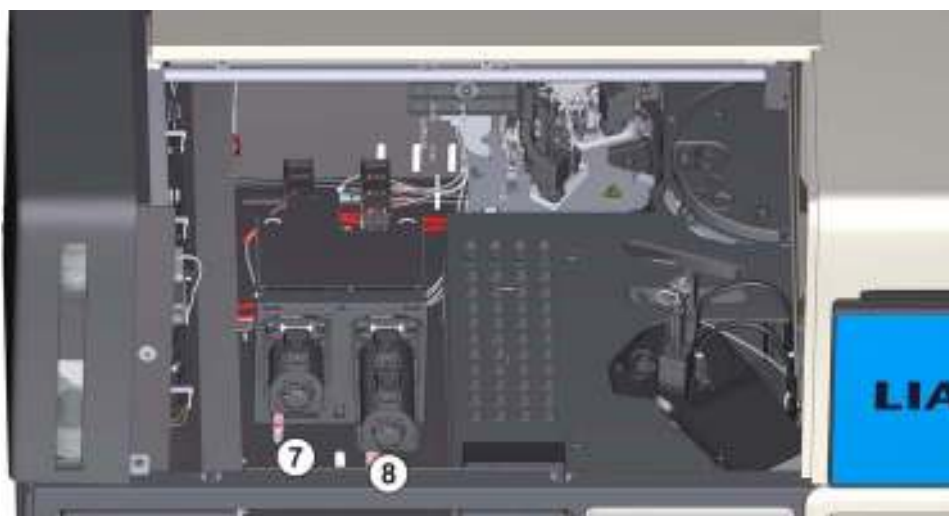


Figure 26-2: Instrument, with Left Top Cover and Cuvette - Tip drawer opened

| No. | Description                                                |
|-----|------------------------------------------------------------|
| 7   | LWP (Liquid Waste Pump) or 3-channels peristaltic pump     |
| 8   | WAP (Waste Aspiration Pump) or 5-channels peristaltic pump |

Table 26-2: Pumps located below the Cuvette - Tip Loading Module

### 26.2.1 Liquid Distribution Overview

#### NOTE

For an overview of the complete liquid distribution, see Chapter 4.1.7 on page 4-8.

## 26.3 Replacement of Pumps

### 26.3.1 Replacement of the LWP (Liquid Waste Pump, 3-channels peristaltic pump)

#### 26.3.1.1 Overview

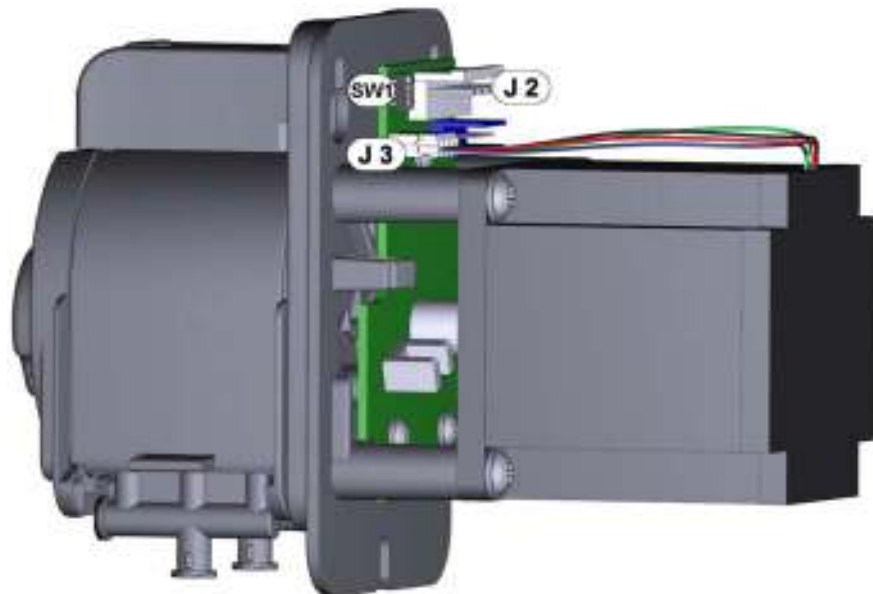


Figure 26–3: LWP

| Number |            | Connection |
|--------|------------|------------|
| J2     | PWR/CAN    |            |
| J3     | Pump motor |            |

Table 26–3: Connections of the LWP PCBA board

#### 26.3.1.2 Electrical Circuit Overview

#### NOTE

For the detail of the LWP connection and DIP-Switch setting, see Chapter 26.5.1 on page 26-36.



### 26.3.1.3 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.

### 26.3.1.4 Removal of the LWP (Liquid Waste Pump, 3-channels peristaltic pump)

1. Tilt peristaltic pumps rack up.
2. Unplug all tubing of LWP (A-B).

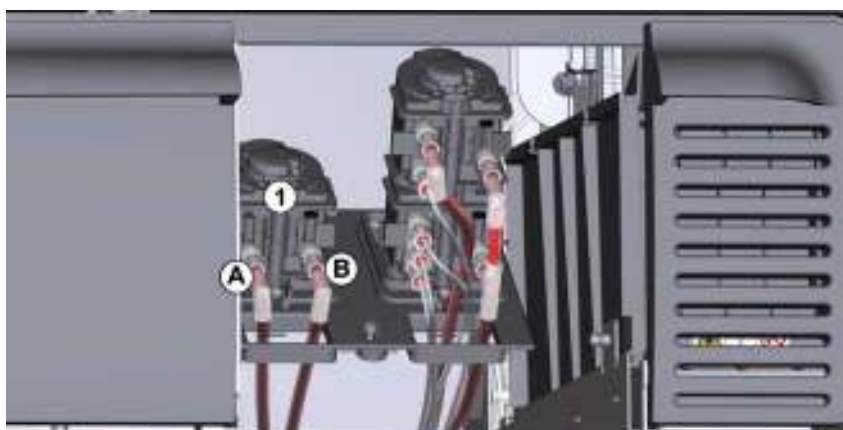


Figure 26-4: LWP, bottom view

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the LWP tubing.

3. Loosen three screws (2), lift the LWP (1) a bit up and carefully pull it out, paying attention that the PWR/CAN cable is still connected.

## 26. Pumps for external-tanks configuration

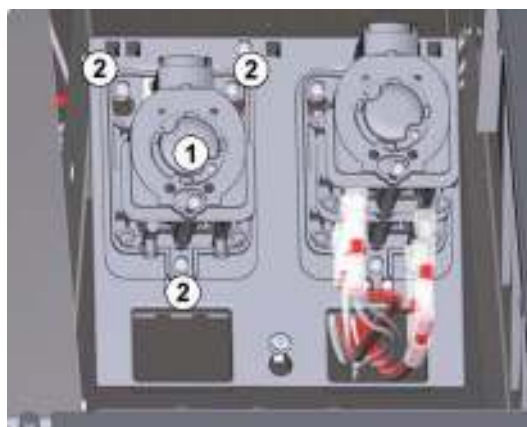


Figure 26-5: Peristaltic pumps rack, front view

4. Unplug PWR/CAN cable (J2) from LWP and remove the pump.



Figure 26-6: LWP, top view

### 26.3.1.5 Assembly of the LWP (Liquid Waste Pump, 3-channels peristaltic pump)

1. Check DIP-Switch on the LWP PCBA for correct setting, see Table 26-19.
2. Hold the LWP in front of the peristaltic pumps bracket and connect PWR/CAN cable onto LWP PCBA connector (J2).
3. Set keyholes into the screws and fasten three screws (2).
4. Connect following fitting onto LWP (1):

| No. | Connection at Pump | Tubing              | Connection to                    |
|-----|--------------------|---------------------|----------------------------------|
| A   | LWP                | Wash station-LWP    | Wash station                     |
| B   | LWP                | LWP-Waste interface | Waste interface<br>(Y-connector) |

Table 26–4: LWP, tube connection overview

5. Assemble the Cuvette - Tip drawer, see Chapter 15.3.1.3 on page 15-3.

#### 26.3.1.6 Set LWP

1. Run the Firmware Update of the complete system in case new LWP is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 26.3.2 Replacement of the WAP (Waste Aspiration Pump, 5-channels peristaltic pump)

#### 26.3.2.1 Overview

This section shows an overview of the connections of the Waste Aspiration Pump.

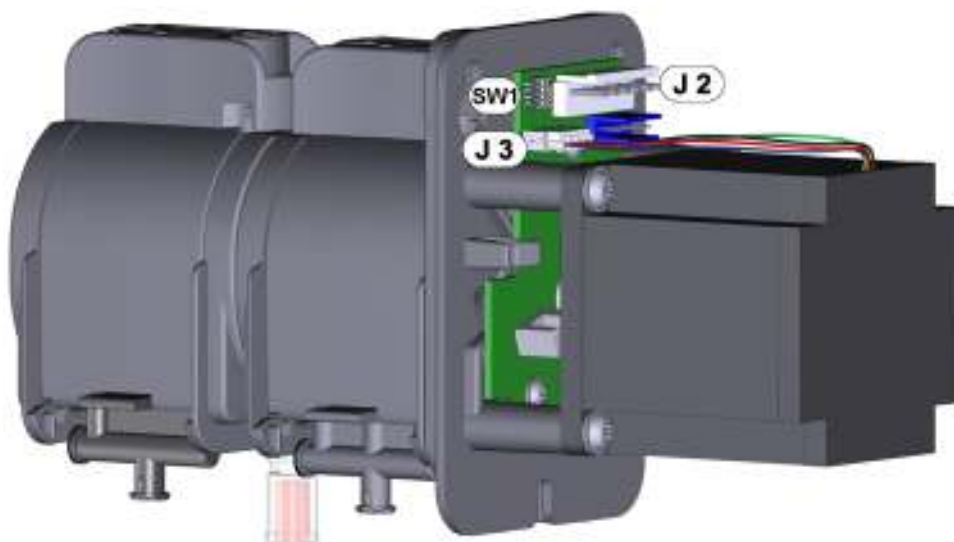


Figure 26–7: WAP, side view

| Number | Connection |
|--------|------------|
| J2     | PWR/CAN    |
| J3     | Pump motor |

Table 26–5: Connections of the WAP PCBA

### 26.3.2.2 Electrical Circuit Overview

**NOTE**

For the detail of the WAP connection and DIP-Switch setting, see Chapter 26.5.1 on page 26-36.

### 26.3.2.3 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.

### 26.3.2.4 Removal of the WAP (5-channels peristaltic pump)

1. Tilt peristaltic pumps rack up.
2. Unplug all tubing of the WAP (C-I).

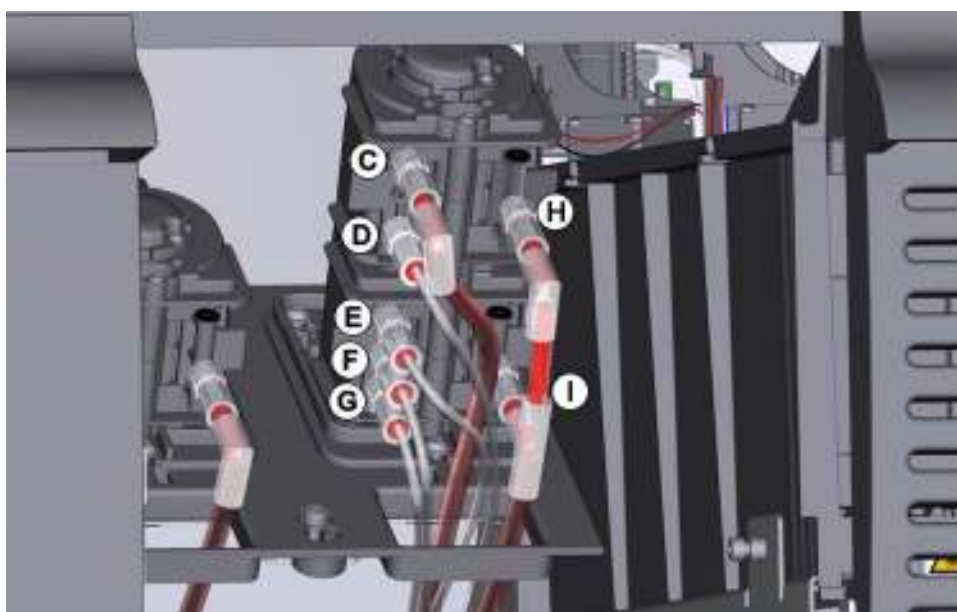


Figure 26–8: WAP, bottom view

**WARNING****Liquid Waste Spillage**

Please pay attention that liquid waste spillage is possible when unplugging the WAP tubing.

3. Loosen three screws (2) and lift the WAP (1) a bit up and carefully pull it out, paying attention that the PWR/CAN cable is still connected.

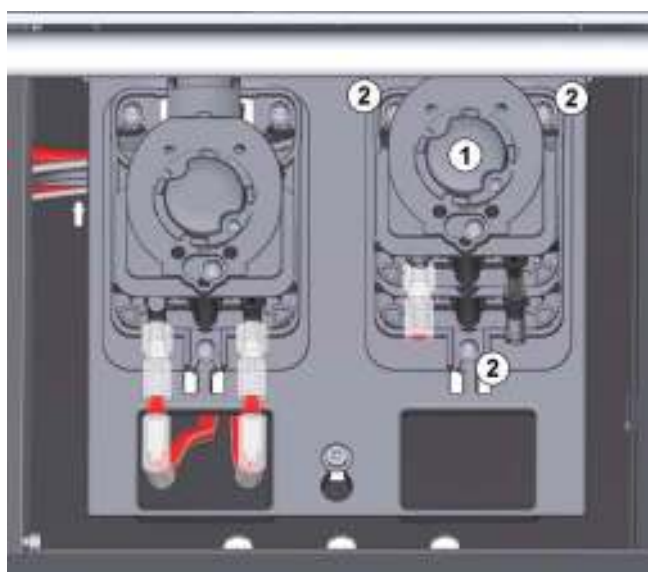


Figure 26–9: Peristaltic pumps rack, front view

## 26. Pumps for external-tanks configuration

---

4. Unplug PWR/CAN cable (J2) from WAP and remove the pump.



*Figure 26–10: WAP, top view*

### 26.3.2.5 Assembly of the WAP (Waste Aspiration Pump, 5-channels peristaltic pump)

1. Check DIP-Switch on the WAP PCBA for correct setting, see Table 26–20.
2. Hold the WAP in front of the peristaltic pump bracket and connect PWR/CAN cable onto WAP PCBA connector (J2).
3. Set keyholes into the screws and fasten three screws (2).
4. Connect following fitting onto the WAP (1):

| No. | Connection at Pump | Tubing              | Connection to                     |
|-----|--------------------|---------------------|-----------------------------------|
| C   | WAP                | Reagent Module-WAP  | Drain of the Reagent Module       |
| D   | WAP                | Reader-WAP          | Reader SIPAL                      |
| E   | WAP                | Washer-WAP          | Washer SIPAL 1                    |
| F   | WAP                | Washer-WAP          | Washer SIPAL 2                    |
| G   | WAP                | Washer-WAP          | Washer SIAPL 3 and Grounding pipe |
| H   | WAP                | WAP-Waste interface | Waste interface (Y-connector)     |
| I   | WAP                | WAP-Waste interface | Waste interface (Y-connector)     |

*Table 26–6: WAP tube connection overview*

5. Assemble the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.

### 26.3.2.6 Set WAP

1. Run the Firmware Update of the complete system in case new WAP is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 26.3.3 Replacement of the Peristaltic Pumps Tubings

#### 26.3.3.1 Overview



Figure 26-11: Peristaltic Pumps Tubings

| No. | Tubing for:                      |
|-----|----------------------------------|
| 1)  | LWP 3-channel Peristaltic Tubing |
| 2)  | WAP 3-channel Peristaltic Tubing |
| 3)  | WAP 2-channel Peristaltic Tubing |

Table 26-7: Peristaltic Pumps Tubings, overview



Figure 26-12: Prescribed position of Peristaltic Pumps Tubings



### 26.3.3.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Left Top Cover.
6. Move Sample pipettor to a safe location.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.

### 26.3.3.3 Removal of the Peristaltic Pumps Tubings

1. Unlock and remove the caps (1) of the Peristaltic Pumps (LWP and WAP) by pressing and turning the latches counterclockwise slightly less than 90°.



Figure 26–13: Peristaltic Pumps inside the instrument, top view

| No. | Description               |
|-----|---------------------------|
| 1)  | Caps                      |
| 2)  | Peristaltic Pumps Tubings |
| 3)  | Pump bracket              |

Table 26–8: Peristaltic Pumps, components

## 26. Pumps for external-tanks configuration

2. Tilt the pump bracket (3) upwards.
3. Unscrew the tube fittings (5) from the peristaltic tubings.
4. Pull the Peristaltic Pumps Tubings (4) out from each side of the pumps, and remove them.

**WARNING****Liquid Waste Spillage**

Please pay attention that liquid waste spillage is possible when unplugging the WAP tubing.



Figure 26-14: Pump bracket tilted upwards

| No. | Description               |
|-----|---------------------------|
| 4)  | Peristaltic Pumps Tubings |
| 5)  | Tube fittings             |

Table 26-9: Pump bracket tilted upwards

#### 26.3.3.4 Assembly of the Peristaltic Pumps Tubings

**NOTE**

**The tubes must not be twisted inside of the pump.**

If the tubes are twisted, then check them for dents or kinks.

Make sure that the tubes are properly seated in the tabs.

When dents or kinks are found, the tubes shall be replaced.

**WARNING**

**The Luer connectors of the tubes have different colors:**

- Connect the tubes with red Luer connectors always to the right side of the pumps.
- Connect the tubes with white Luer connectors always to the left side of the pumps.

1. Make sure the correct tube is mounted in the proper position:
  - a. Liquid Waste Pump is on the left and hosts 3 tubes with one inlet on the left and one outlet on the right.
  - b. Waste Aspiration Pump is on the right and host two brackets:
    - i. the bracket on the rear of the instrument hosts 3 tubes with three inlets on the left and one outlet on the right,
    - ii. the bracket on the front of the instrument hosts 2 tubes with 2 inlets on the left and one outlet on the right.
2. Snap the tubes into the Liquid Waste Pump, by sliding each side into the proper guiding, until the tube handle clicks in both the front and rear brackets at each side of the pump.
3. Remount the cap of the Liquid Waste Pump: insert the cap, and while pushing it down, rotate the latches clockwise slightly less than 90°, finishing with the latch clicking at 12 hour.
4. Connect the white and red tube fittings to the Liquid Waste Pump.
5. Snap the tubes into the Waste Aspiration Pump, by sliding each side into the proper guiding, until the tube handle clicks in both in the front and rear brackets at each side of the pump.
6. Remount the caps of the Waste Aspiration Pump: insert the cap, and while pushing it down, rotate the latches clockwise slightly less than 90°, finishing with the latch clicking at 12 hour.
7. Connect the white and red tube fittings to the Waste Aspiration Pump.
8. Push the pump bracket downwards.
9. Place the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3
10. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 26.3.4 Replacement of the SDP (Starter Dispense Pumps)

#### 26.3.4.1 Overview

This section shows an overview of the SDP (Starter Dispense Pump).

The SDPs are LCM pump 500uL.

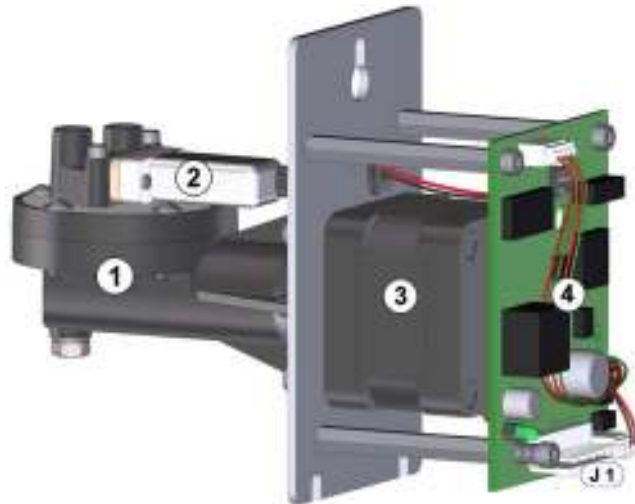


Figure 26-15: SDP

|   |                   |
|---|-------------------|
| 1 | Pump housing      |
| 2 | 2-way valve       |
| 3 | Pump motor        |
| 4 | SDP circuit board |

Table 26-10: SDP overview

#### NOTE

If the system is located at 2000mt. ASL or over you should install a Starter Pump shipped with a special package (custom vacuum package). For the detail, see Corp-TN-000529 (former LXS TN 028).

#### 26.3.4.2 Electrical Circuit Overview

#### NOTE

For the detail of the SDPs connection and DIP-Switch setting, see Chapter 26.5.3 on page 26-40.

### 26.3.4.3 Preliminary steps

#### NOTE

#### Priming in air

Before replacing the SDPs, remember to empty the Starter lines priming in air by **Service Software**, see Chapter 11.8.1.6 on page 11-65.



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.

### 26.3.4.4 Removal of the SDPs

1. Unscrew fittings of tubings:
  - SDP1-Starters (Q)
  - Reader-SDP1 (S)
  - SDP1-Starters (R)
  - Reader-SDP1 (T)

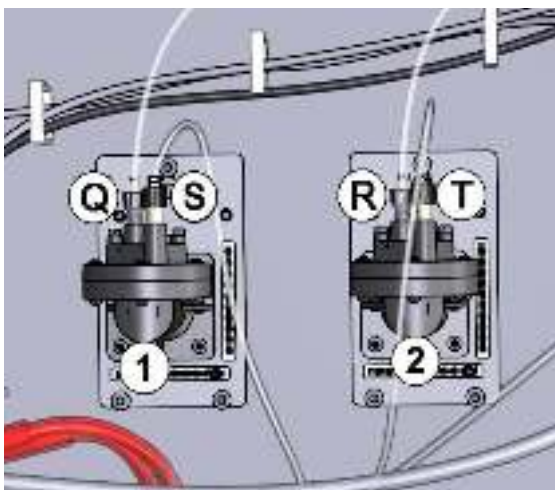


Figure 26–16: SDPs

## 26. Pumps for external-tanks configuration

2. Loosen three screws (2), lift the SDP (1) a bit up and carefully pull it out, paying attention that the cable is still connected to the pump.

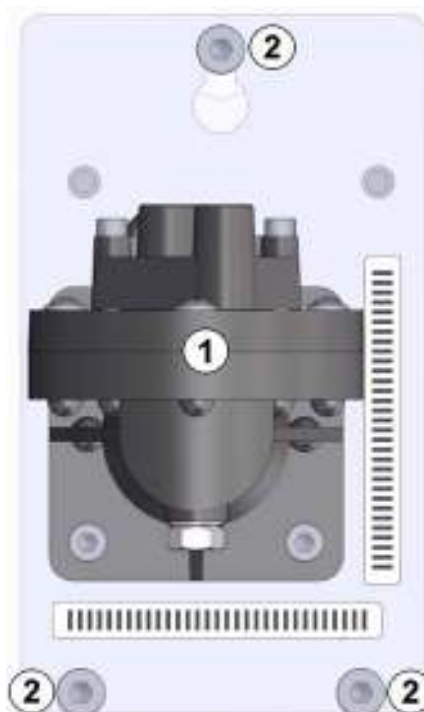


Figure 26-17: SDP, front view

3. Unplug PWR/CAN cable (J1) from the SDP and remove the pump.

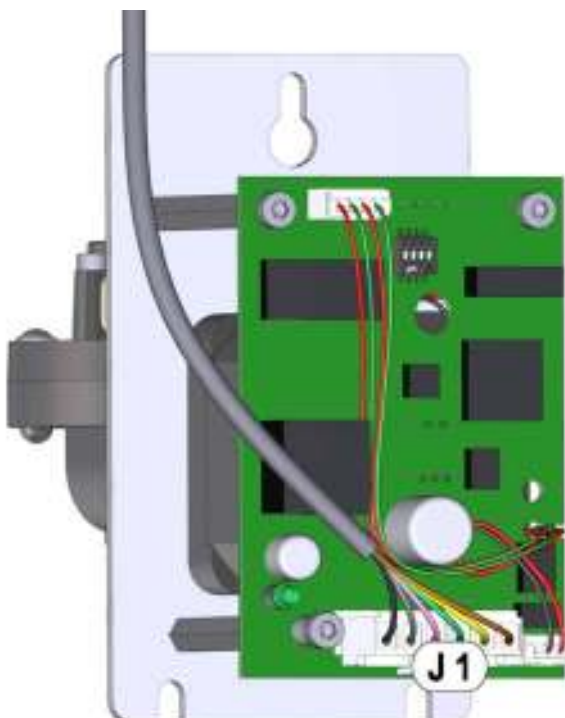


Figure 26-18: SDP, rear view

**26.3.4.5 Assembly of the SDPs**

1. Check DIP-Switch on the SDP PCBA for correct setting, see Table 26–27 and see Table 26–28.
2. Connect PWR/CAN cable onto pump connector (J1).
3. Hang the SDP into the three screws (2) and fasten screws.
4. Connect following fittings:

| No. | Connection at Pump | Tubing        | Connection to          |
|-----|--------------------|---------------|------------------------|
| Q   | SDP 1              | SDP1-Starters | Starter Reagent Module |
| S   | SDP 1              | Reader-SDP1   | Reader                 |
| R   | SDP 2              | SDP2-Starters | Starter Reagent Module |
| T   | SDP 2              | Reader-SDP2   | Reader                 |

*Table 26–11: SDP, tubings connections overview*

5. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**26.3.4.6 Set SDP**

1. Run the Firmware Update of the complete system in case new SDP is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the Starter Pump calibration, see Chapter 26.4.2.1 on page 26-35.
3. Perform Starter Reagents prime, see Chapter 12.8.1.6 on page 12-68.
4. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

**NOTE****Priming in liquid**

Please, be sure that the Starter lines are primed with Starter Reagent before you leave the instrument.

### 26.3.5 Replacement of the PRP (Probe Rinsing Pump)

#### 26.3.5.1 Overview

This section shows an overview of the PRP (Probe Rinsing Pump).

The PRP is a membrane pump.



Figure 26–19: PRP

|   |            |
|---|------------|
| 1 | Pump motor |
| 2 | Valve      |

Table 26–12: PRP overview

#### 26.3.5.2 Preliminary steps

##### NOTE

##### Priming in air

Before replacing the PRP, remove the System Liquid tank and remember to empty the lines priming in air by **Service Software**, see Chapter 12.8.1.4 on page 12-67.





1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
6. Unload the System liquid tank.

### 26.3.5.3 Removal of the PRP

1. Disconnect the tubings (N and P) of the PRP (1).

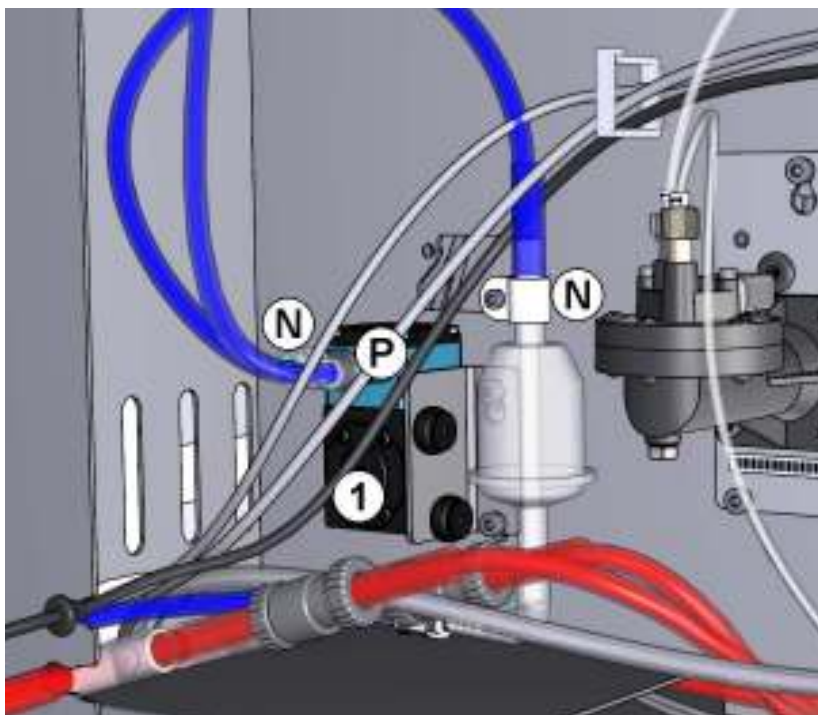


Figure 26–20: PRP inside the instrument

## 26. Pumps for external-tanks configuration

2. Loosen two screws (2) of PRP (1), lift it a bit up and carefully pull it out, paying attention that the cables are still connected to the pump.

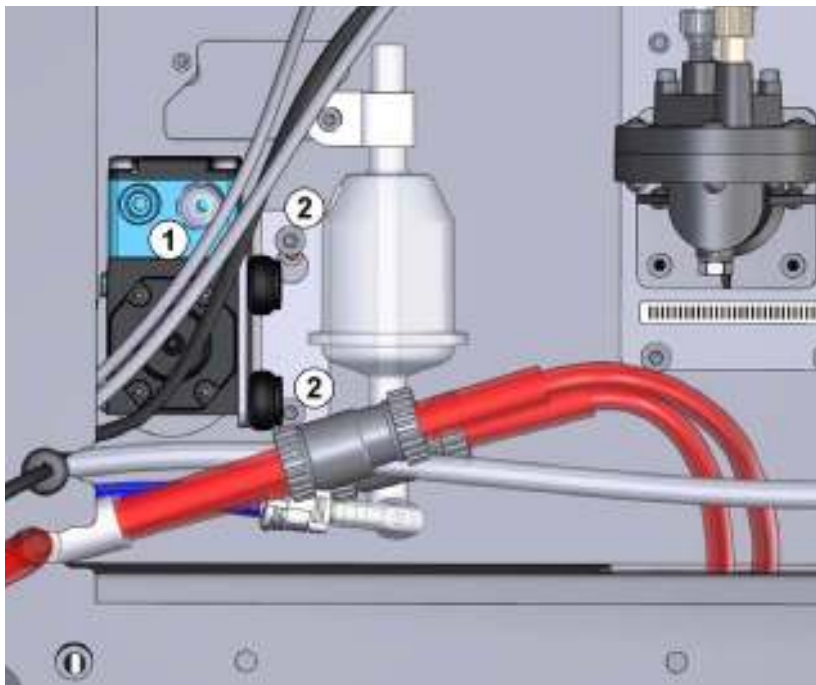


Figure 26-21: PRP inside the instrument, without connected tubings

3. Unplug cables of the PRP (1).



Figure 26-22: PRP, top view

**26.3.5.4 Assembly of the PRP**

1. Connect the brown cable onto plus cable lug and the black cable onto minus cable lug of the PRP (1).

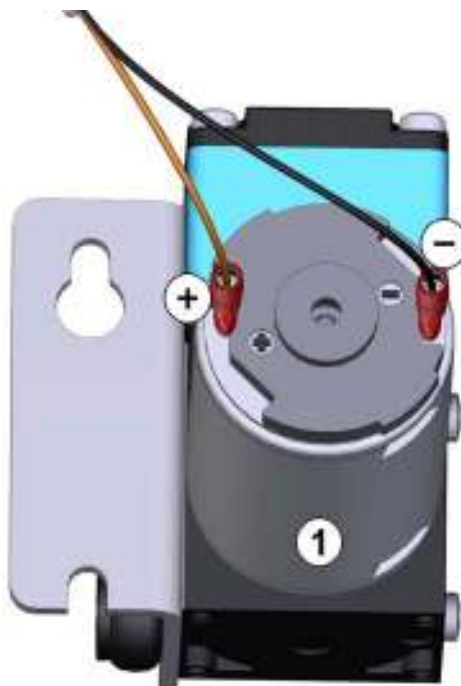


Figure 26-23: PRP, rear view

2. Hang the PRP into the two screws (2) and fasten screws.
3. Connect following fittings onto the PRP (1):

| No. | Connection at Pump | Tubing          | Connection to |
|-----|--------------------|-----------------|---------------|
| P   | PRP                | PRP-SPOLV       | SPOLV Module  |
| N   | PRP                | MANN filter-PRP | MANN filter   |

Table 26-13: PRP, tubings connections overview

4. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
5. Place back the System liquid tank.

**26.3.5.5 Set PRP**

1. Perform the SPOLV prime, see Chapter 12.8.1.4 on page 12-67.

**NOTE****Priming in liquid**

Please, be sure that the SPOLV line is primed before you leave the instrument.

## 26. Pumps for external-tanks configuration

---

**WARNING****Air inside the MANN filter**

While the system is priming the SPOLV, please observe the liquid flow from the System liquid tank through the MANN filter: pay attention that the filter is not full of air. If needed, tap on it to remove air as much as possible.

- 
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 26.3.6 Replacement of the WBDP (Wash Buffer Dispense Pump)

#### 26.3.6.1 Overview

This section shows an overview of the WBDP (Wash Buffer Dispense Pump).

The WBDP is a piston pump.

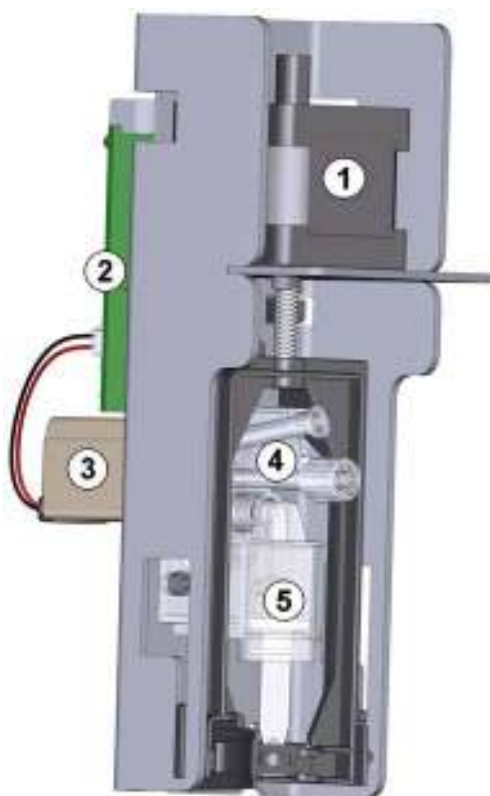


Figure 26–24: WBDP

|   |                                        |
|---|----------------------------------------|
| 1 | Pump motor                             |
| 2 | WBDP PCBA                              |
| 3 | 2 way valve                            |
| 4 | 3 way manifold                         |
| 5 | Cartridge (piston cylinder unit 2,5ml) |

Table 26–14: WBDP overview

## 26. Pumps for external-tanks configuration

### 26.3.6.2 Electrical Circuit Overview

#### NOTE

For the detail of the WBDP connection and DIP-Switch setting, see Chapter Table 26–19: on page 26-37.

### 26.3.6.3 Preliminary steps

#### NOTE

#### Priming in air

Before replacing the WBDP, remove the Wash Buffer tank and remember to empty the lines priming in air by **Service Software**, see Chapter 12.8.1.5 on page 12-67.



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
6. Remove the Wash Buffer tank.

### 26.3.6.4 Removal of the WBDP

1. Unscrew fittings of tubings of the WBDP:
  - WBDP-Washer (E-F-G)
  - WB TANK-WBDP (M)

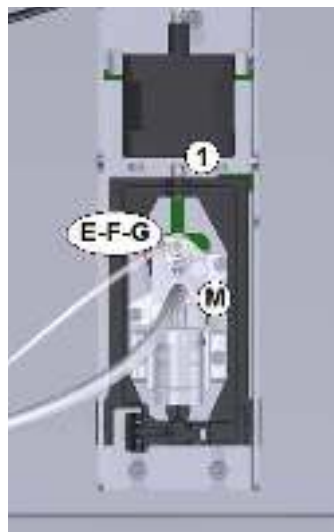


Figure 26–25: WBDP inside the instrument

**26. Pumps for external-tanks configuration**

2. Loosen three screws (2), lift the WBDP (1) a bit up and carefully pull it out, paying attention that the cable is still connected to the WBDP.

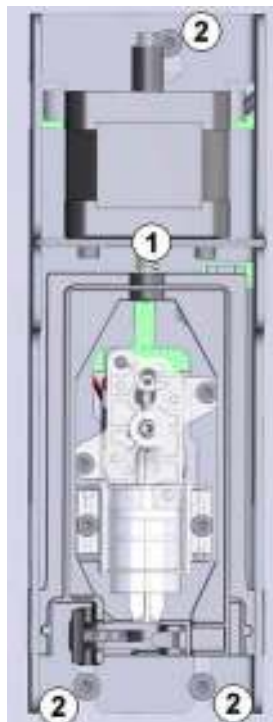


Figure 26–26: WBDP, front view

3. Unplug PWR/CAN cable (J3) from the WBDP and remove the pump.

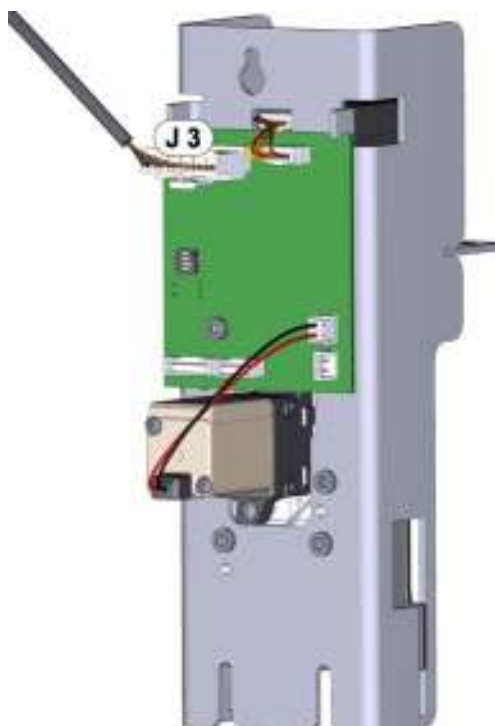


Figure 26–27: WBDP, rear view

## 26. Pumps for external-tanks configuration

### 26.3.6.5 Assembly of the WBDP

1. Connect PWR/CAN cable onto the WBDP connector (J3).
2. Hang the WBDP into the three screws (2) and fasten screws.
3. Connect following fittings onto the WBDP (1):

| No.   | Connection at Pump | Tubing       | Connection to    |
|-------|--------------------|--------------|------------------|
| E-F-G | WBDP               | WBDP-Washer  | Washer Module    |
| M     | WBDP               | WB TANK-WBDP | Wash Buffer tank |

Table 26–15: WBDP, tubings connections overview

4. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
5. Place back the Wash Buffer tank.

### 26.3.6.6 Set WBDP

1. Run the Firmware Update of the complete system in case new WBDP is installed, see Chapter 32.3.4 on page 32-6.
2. Perform the WBDP calibration, see Chapter 26.4.3.1 on page 26-35.
3. Perform the Washer prime, see Chapter 12.8.1.5 on page 12-67.

#### NOTE

#### Priming in liquid

Please, be sure that the Washer lines are primed with Wash Buffer before you leave the instrument.

4. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



## 26.3.7 Replacement of the WBDP Cartridge

### 26.3.7.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
6. Remove the Wash Buffer tank.

#### NOTE

#### Priming in air

Before replacing the WBDP, remove the Wash Buffer tank and remember to empty the lines priming in air by **Service Software**, see Chapter 12.8.1.5 on page 12-67.

---

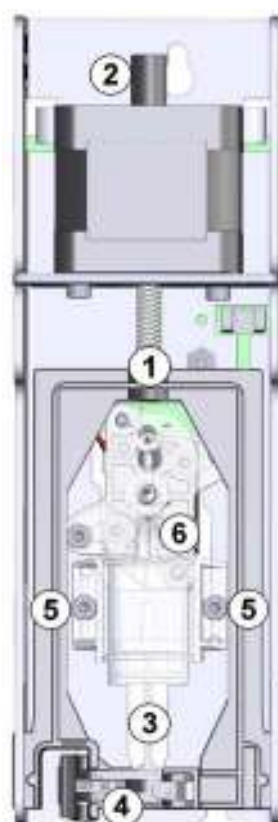
### 26.3.7.2 Removal of the WBDP Cartridge

1. Disconnect the WBDP tubes, see Chapter 26.3.6.4 on page 26-26.
2. Turn motor shaft (2) to extend the Cartridge piston (3) to the maximum down.
3. Open piston holder (4).
4. Remove screws (5) out of the WBDP Cartridge.

**26. Pumps for external-tanks configuration**

---

5. Pull the Cartridge up and out of the fitting connection (6).



*Figure 26-28: WBDP, front view*

### 26.3.7.3 Assembly of the WBDP Cartridge

1. Check for correct sit of the sealing inside of the fitting connection.

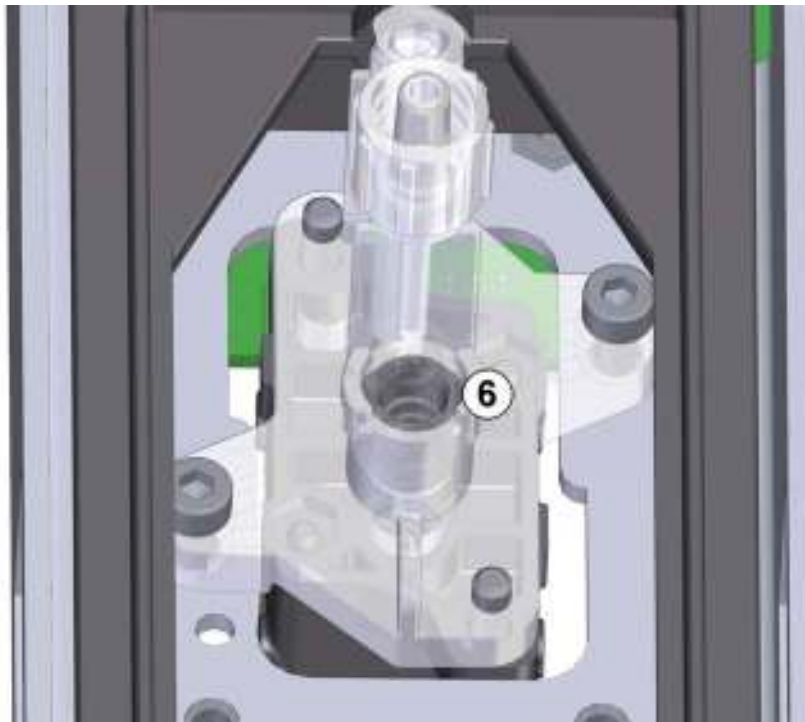


Figure 26-29: WBDP, 3-way manifold

2. Move the Cartridge fitting (7) carefully in the fitting connection (6) and be aware of the pins at the bottom of the cylinder housing that must sit correctly in the bore holes.

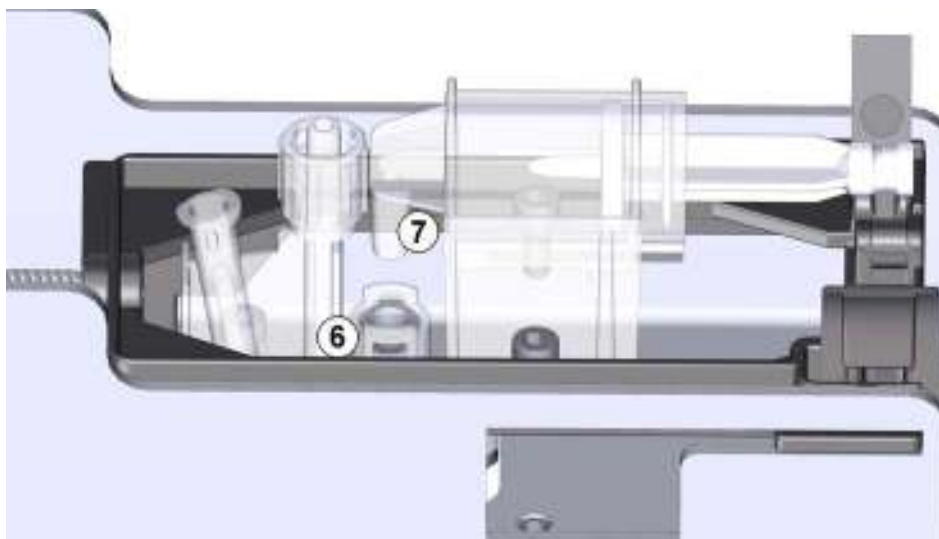


Figure 26-30: WBDP, side view

## 26. Pumps for external-tanks configuration

---

3. The piston shaft (8) shall sit on the bottom of the pusher frame (9).
4. Then close the piston shaft holder (10).

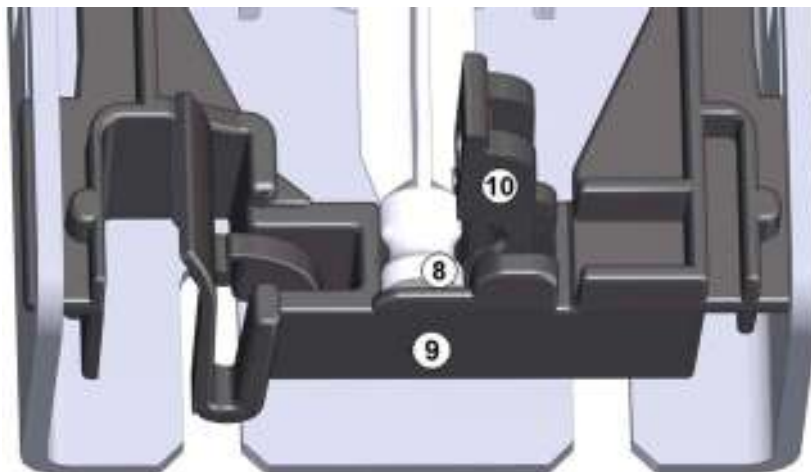


Figure 26–31: WBDP, piston bottom

5. Connect the WBDP tubes, see Chapter 26.3.6.5 on page 26-28.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
7. Place back the Wash Buffer tank.

### 26.3.7.4 Set WBDP cartridge

1. Perform the WBDP calibration, see Chapter 26.4.3.1 on page 26-35.
2. Perform the Washer prime, see Chapter 12.8.1.5 on page 12-67.

#### NOTE

#### Priming in liquid

Please, be sure that the Washer lines are primed with Wash Buffer before you leave the instrument.

3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

## 26.3.8 Replacement of the WBDP Cartridge valve unit

### 26.3.8.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
6. Remove the Wash Buffer tank.
7. Remove the WBDP, see Chapter 26.3.6.4 on page 26-26.
8. Remove the WBDP Cartridge, see Chapter 26.3.8.2 on page 26-33.

### 26.3.8.2 Removal of the WBDP Cartridge valve unit

#### NOTE

##### Priming in air

Before replacing the WBDP, remove the Wash Buffer tank and remember to empty the lines priming in air by **Service Software**, see Chapter 12.8.1.5 on page 12-67.

1. Unplug the WBDP Cartridge valve cable at the WBDP PCBA.
2. Unscrew two screws (2).

## 26. Pumps for external-tanks configuration

3. Turn the WBDP Cartridge valve unit (1) a bit counterclockwise and carefully lift it out.

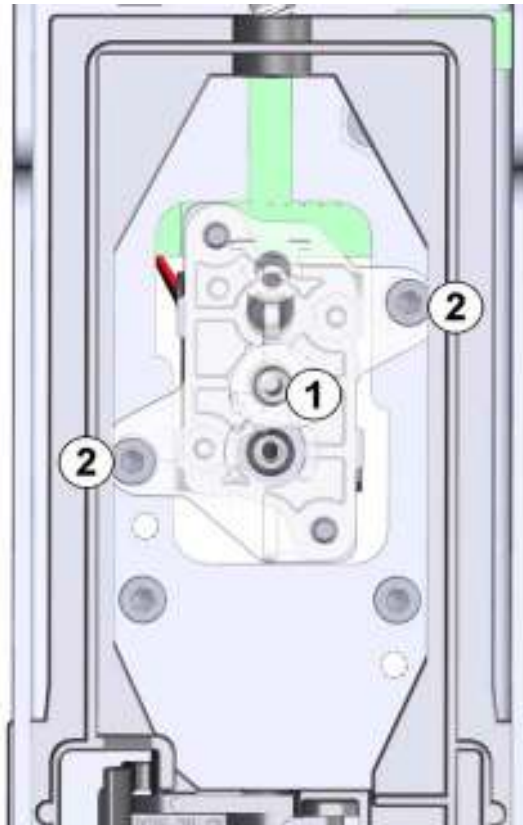


Figure 26–32: WBDP without Cartridge

### 26.3.8.3 Assembly of the WBDP Cartridge valve unit

1. Set the WBDP Cartridge valve unit (1) into the carrier for the pump and fasten with two screws (2).
2. Connect the WBDP Cartridge valve cable onto the WBDP PCBA.
3. Assemble the WBDP Cartridge, see Chapter 26.3.7.3 on page 26-31.
4. Assemble and set the WBDP, see Chapter 26.3.6.5 on page 26-28 and see Chapter 26.3.6.6 on page 26-28.
5. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
6. Place back the Wash Buffer tank.

### 26.3.8.4 Set WBDP Cartridge valve unit

1. Perform the WBDP calibration, see Chapter 26.4.3.1 on page 26-35.
2. Perform the Washer prime, see Chapter 12.8.1.5 on page 12-67.

#### NOTE

#### Priming in liquid

Please, be sure that the Washer lines are primed with Wash Buffer before you leave the instrument.

3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

## 26.4 Adjustment and Calibration

This section contains informations about the adjustment and calibration of the Pumps.

### 26.4.1 Initialization routines

- a. Close any open software and start the **Service Software**.
- b. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- c. Click on the **Instrument** button and choose the **Full initialization via COP** button to start the initialization sequence.
- d. The process is successful when the **Processing** window closes without failure message.

### 26.4.2 Calibration of the SDPs

#### 26.4.2.1 Calibration of the Starter Pump

In the **Service Software** at the navigation window choose **Reader** then **Teach & Calibration** bracket and select **Starter Pump**.

- Step 1**
- a. Click on the **Start** button to start the automatic calibration of the SDP1.
  - b. Check the results and proceed by pressing **Done** button.
- Step 2**
- c. Click on the **Start** button to start the automatic calibration of the SDP2.
  - d. Follow the instruction by pressing **Done** button.
- Step 3**
- e. Store the taught positions in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 4**
- f. The process is successful when, at the **Summarize** panel, all steps are in green.

### 26.4.3 Calibration of the WBDP

#### 26.4.3.1 Calibration of the WBDP

In the **Service Software** at the navigation window choose **Washer** then **Teach & Calibration** bracket and select **WBDP Calibration**.

- Step 1**
- a. Click on the **Start** button to start the automatic calibration of the WBDP.
  - b. Check the results and proceed by pressing **Done** button.
- Step 2**
- c. Store the taught positions in the EEPROM by selecting **Write ALL Parameters to NVM** button.
- Step 3**
- d. The process is successful when, at the **Summarize** panel, all steps are in green.

## 26.5 Electrical connections, switches and signals

This section shows an overview of the electronic boards of the pumps.

**NOTE**

see Chapter 4.1.8 on page 4-9 for an overview of the complete electric circuit of the system.

### 26.5.1 LWP and WAP PCBA

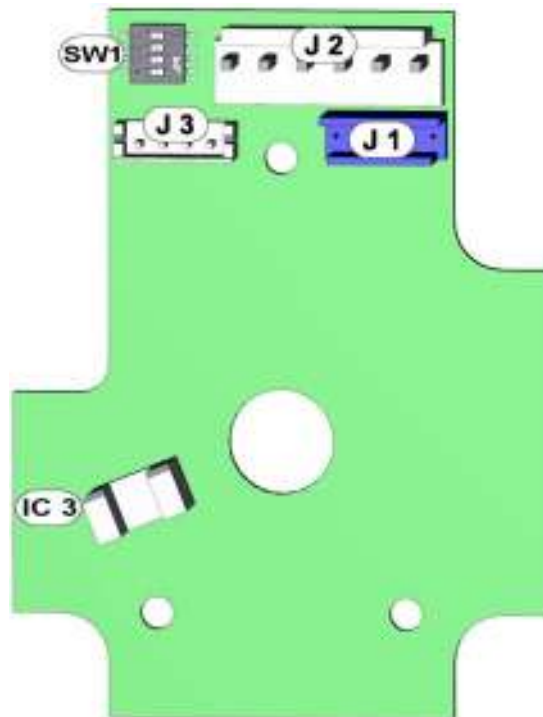


Figure 26–33: LWP and WAP PCBA

| Number | Connection |
|--------|------------|
| J1     | Download   |
| J2     | PWR/CAN    |
| J3     | Pump motor |

Table 26–16: Connections of the LWP and WAP PCBA

| Number | Meaning/Function     |
|--------|----------------------|
| IC 3   | Init (light barrier) |

Table 26–17: IC of the LWP and WAP PCBA



| Number | Meaning/Function     |
|--------|----------------------|
| SW 1   | Individual selection |

*Table 26–18: DIP-Switch of the LWP and WAP PCBA*

| DIP-Switch | Meaning/Function/Default Switch Setting |
|------------|-----------------------------------------|
| SW1 1      | ON                                      |
| 2          | OFF                                     |
| 3          | OFF                                     |
| 4          | OFF                                     |

*Table 26–19: DIP-Switch settings at LWP PCBA*

| DIP-Switch | Meaning/Function/Default Switch Setting |
|------------|-----------------------------------------|
| SW1 1      | OFF                                     |
| 2          | OFF                                     |
| 3          | OFF                                     |
| 4          | OFF                                     |

*Table 26–20: DIP-Switch settings at WAP PCBA*

26.5.2 WBDP PCBA

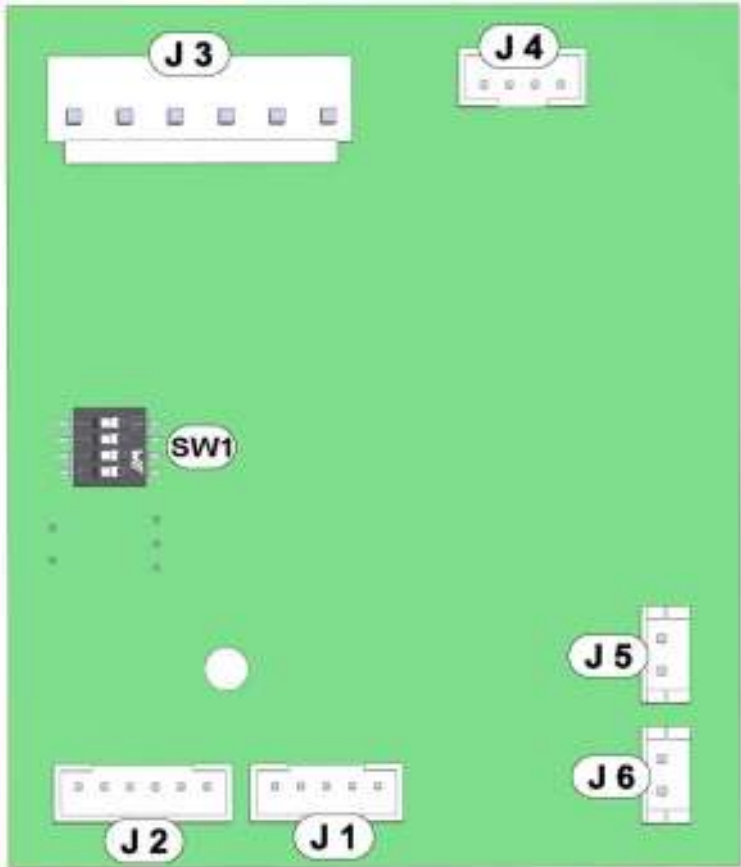


Figure 26–34: WBDP PCBA, front view

| Number |            | Connection |
|--------|------------|------------|
| J1     | Download   |            |
| J2     | Sensor IN  |            |
| J3     | PWR/CAN    |            |
| J4     | Pump motor |            |
| J5     | Valve 1    |            |
| J6     | Valve 2    |            |

Table 26–21: Connections of the WBDP PCBA

| Number |                      | Meaning/Function |
|--------|----------------------|------------------|
| SW 1   | Individual selection |                  |

Table 26–22: DIP-Switch of the WBDP PCBA

| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | OFF                                     |
|            | 2 | OFF                                     |
|            | 3 | OFF                                     |
|            | 4 | OFF                                     |

Table 26–23: DIP-Switch settings at WBDP PCBA

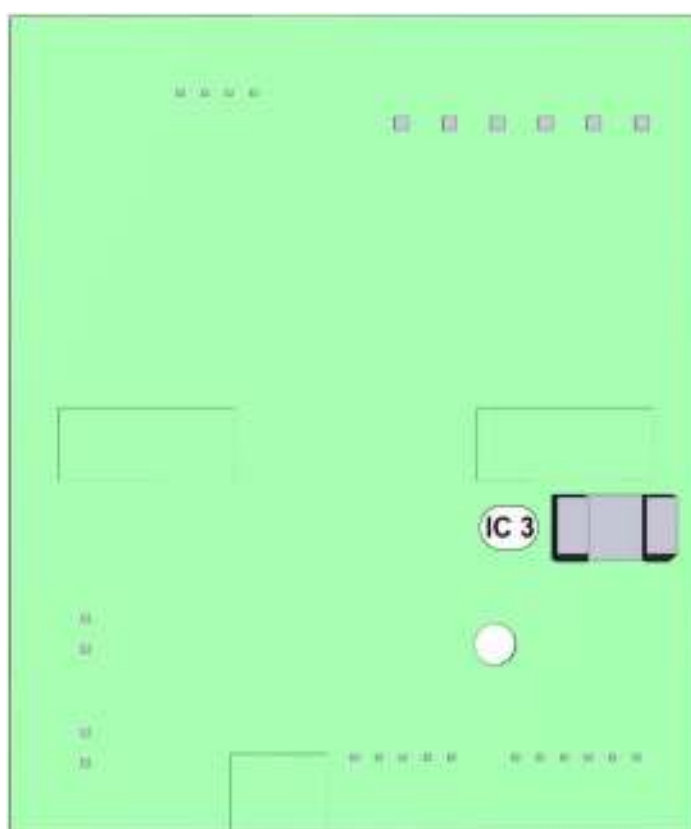


Figure 26–35: WBDP PCBA, rear view

| Number | Meaning/Function                      |
|--------|---------------------------------------|
| IC 3   | Cartridge piston home (light barrier) |

Table 26–24: IC of the WBDP PCBA

26.5.3 SDP PCBA

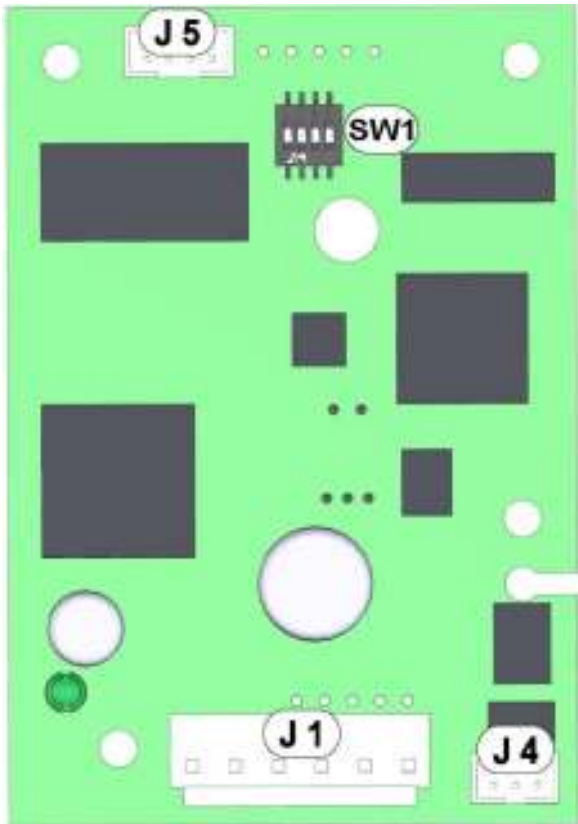


Figure 26–36: SDP PCBA, front view

| Number |  | Connection    |
|--------|--|---------------|
| J1     |  | Main PWR/CAN  |
| J4     |  | PWR out       |
| J5     |  | Stepper motor |

Table 26–25: Connections of the SDP PCBA

| Number |  | Meaning/Function     |
|--------|--|----------------------|
| SW 1   |  | Individual selection |

Table 26–26: Switch of the SDP PCBA

| DIP-Switch      Meaning/Function/Default Switch Setting - Starter 1 |   |     |
|---------------------------------------------------------------------|---|-----|
| SW1                                                                 | 1 | OFF |
|                                                                     | 2 | OFF |
|                                                                     | 3 | OFF |
|                                                                     | 4 | OFF |

Table 26–27: DIP-Switch settings at SDP1 PCBA

| DIP-Switch      Meaning/Function/Default Switch Setting - Starter 2 |   |     |
|---------------------------------------------------------------------|---|-----|
| SW1                                                                 | 1 | ON  |
|                                                                     | 2 | OFF |
|                                                                     | 3 | OFF |
|                                                                     | 4 | OFF |

Table 26–28: DIP-Switch settings at SDP2 PCBA

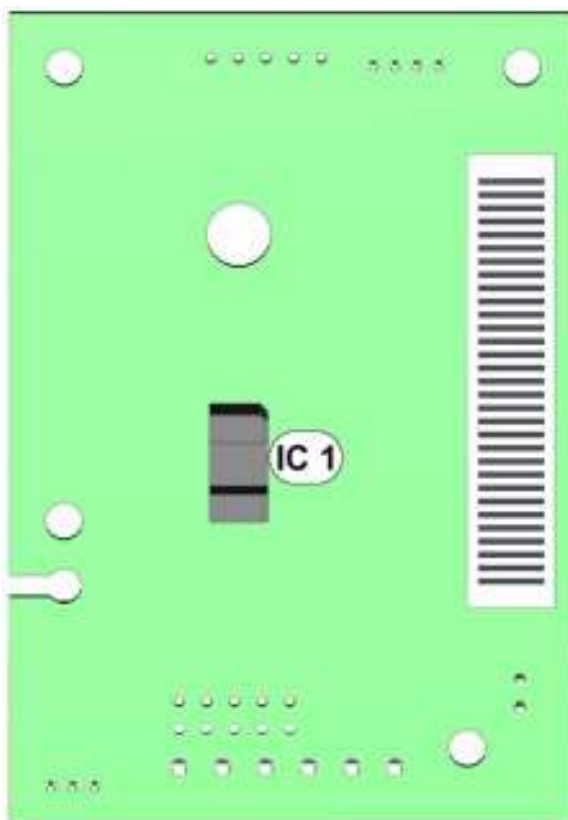


Figure 26–37: SDP PCBA, rear view

| Number | Meaning/Function     |
|--------|----------------------|
| IC 1   | Home (light barrier) |

Table 26–29: IC of the SDP PCBA



# 27 Front Module

This chapter describes the Front Module in detail. The first section of the chapter introduces “Safety” notes concerning the Front Module, followed by a brief overview of the main parts composing this module.

The following section contains information about repair and verification (“Replacement and Repair” and “Adjustment and Calibration” sections).

In the following section, Front Cover CU PCBA layout is described.

In the last section, the Magnetic Particles Resuspension Area is described.

## 27.1 Safety

---

**WARNING****Non-observance of safety instructions**

The non-observance of safety instructions may result in serious personal injury and material damage.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 Or see Chapter 2.5 on page 2-7.
  - Follow all warnings marked on the instrument.
  - Follow all warnings marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
- 

---

**WARNING****Sharp Edges!**

Sheet metal parts and PCBs located behind protective covers might have sharp edges. Contact might lead to injuries.

- Use caution at corners and edges!
-

## 27.2 Overview

This section contains an overview of the components of the Front Module.

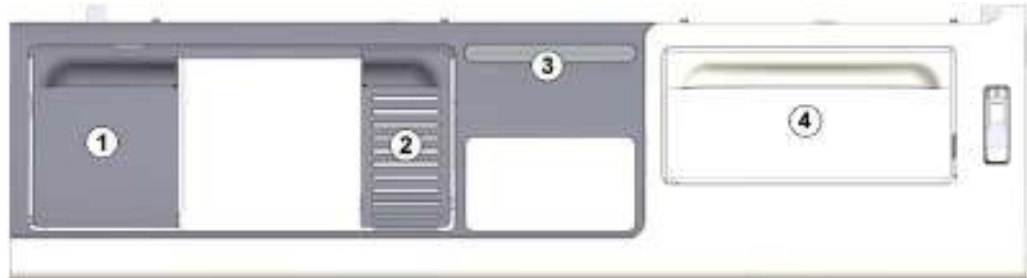


Figure 27-1: Front Module, front view

| No. | Description          |
|-----|----------------------|
| 1)  | Starter Reagent flap |
| 2)  | Sample Area flap     |
| 3)  | Status Light         |
| 4)  | Reagent Area flap    |

Table 27-1: Front Module, front view



Figure 27-2: Front Module, inside view

| No. | Description                          |
|-----|--------------------------------------|
| 5)  | Magnetic Particles Resuspension Tool |
| 6)  | Front Cover CU PCBA                  |

Table 27-2: Front Module, inside view

The Starter Reagent flap, Sample Area flap, Solid Waste Bin and Reagent Area flap include magnets to detect their status.

The position of the magnets is shown in the picture below.





Figure 27-3: Front Module, opened flaps (transparent cover plates)

| No. | Description                    |
|-----|--------------------------------|
| 7)  | Magnet of Starter Reagent flap |
| 8)  | Magnet of Sample Area flap     |
| 9)  | Magnet of Solid Waste Bin      |
| 10) | Magnet of Reagent Area flap    |

Table 27-3: Magnets of the Front Module

## 27.3 Replacement and repairs

This section contains instructions for the repair of individual components of the Front Module. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 27.3.1 Replacement of the Front Module

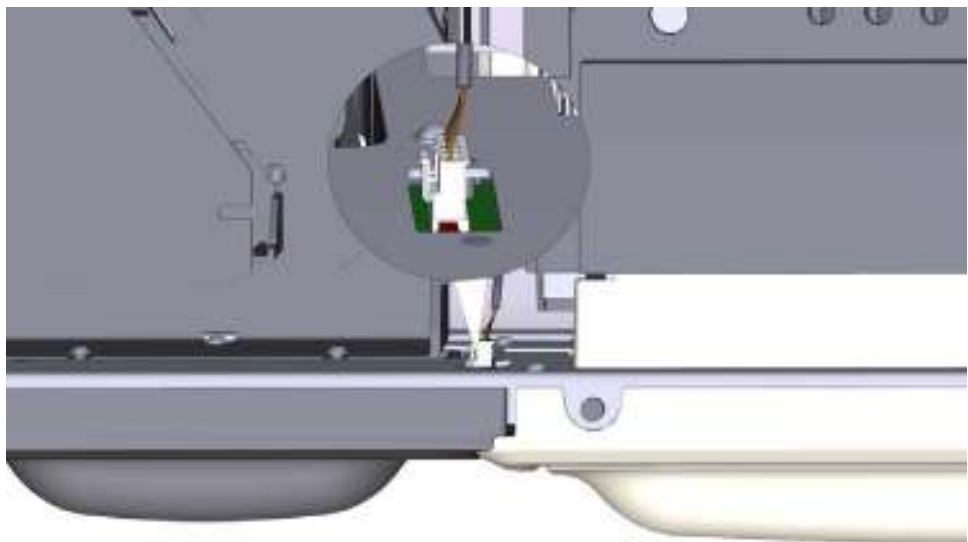
#### 27.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Solid Waste bin.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.

#### 27.3.1.2 Removal of the Front Module

1. Unplug the PWR/CAN cable connector from the Front Cover CU PCBA.



*Figure 27–4: Front Module PWR/CAN cable connector*

2. Unscrew the two screws (1).

3. Flip the Front Module to the front, lift it up and set it aside.



Figure 27-5: Front Module

### 27.3.1.3 Assembly of the Front Module

1. Set the Front Module into the notches (2).

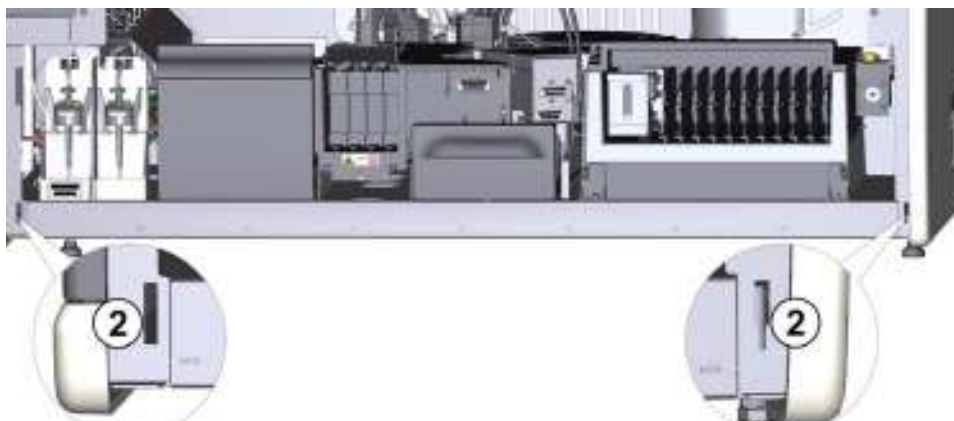


Figure 27-6: Chassis, notches for Front Module

2. Fasten the two screws (1).
3. Plug the PWR/CAN cable connector onto Front Cover CU PCBA.
4. Assemble the Cuvette-Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.
5. Insert the Solid Waste bin.

### 27.3.1.4 Set of the Front Module

1. In case a new module has been installed, proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. In case a new module has been installed, check the flap sensors, see Chapter 27.4.2.1 on page 27-24.
3. In case a new module has been installed, check the Beeper, see Chapter 27.4.2.2 on page 27-24.
4. In case a new module has been installed, check the Status light, see Chapter 27.4.2.3 on page 27-24.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 27.3.2 Replacement of the Starter Reagent flap

### 27.3.2.1 Spare part

The Spare part is composed by:

- Starter Reagent flap
- Washer
- Collar
- Axis



Figure 27-7: Starter Reagent flap spare parts

### 27.3.2.2 Preliminary step



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open Right and Left Top Cover.
6. Move Pipettor arms to a safe location.
7. Remove the solid waste bin.
8. Remove the Front module, see Chapter 27.3.1 on page 27-4

### 27.3.2.3 Removal of the Starter Reagent flap

1. Flip the Front cover and open the Starter reagent and Sample flap.
2. Unscrew the screws to remove the black cover part of the Front module.

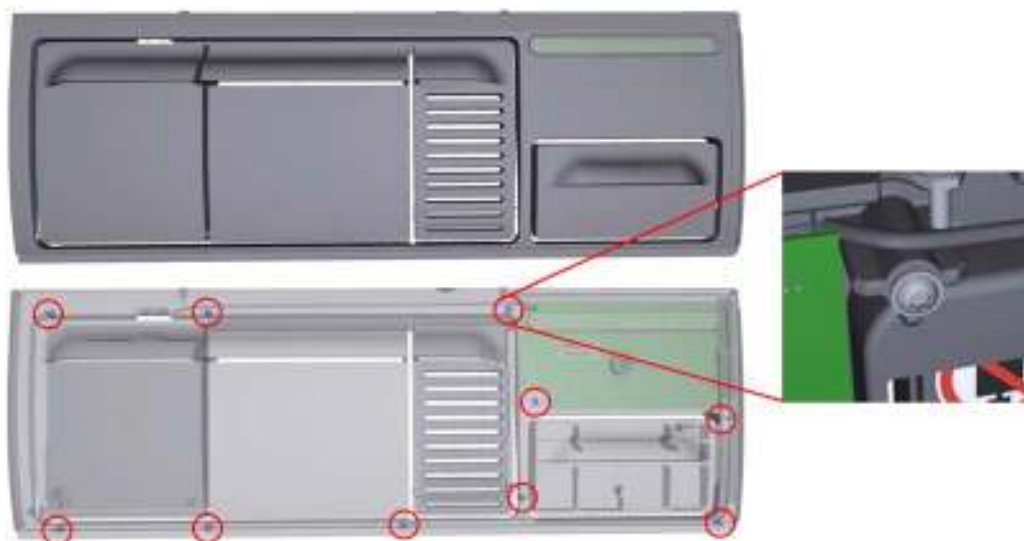


Figure 27–8: Screws location to remove the Front module black cover



Figure 27–9: Removed Front module black cover

3. Unlock and pull down the split pin of the gas pressure damper and remove the ball head. Then, remove the gas pressure damper.



Figure 27–10: Gas pressure damper of the Starter Reagent flap

4. Remove the left screw of the axis with the hexagon socket 2.5. Afterwards loosen the grub screw with the hexagon socket 1.5.



*Figure 27–11: Left screw and grub screw of the Starter Reagent flap*

5. Use a flat-blade screwdriver to lose the axis as far as possible and pull it out, subsequently.



*Figure 27–12: Use a flat-blade screwdriver to lose the axis as far as possible and pull it out*



*Figure 27-13: Removed Starter Reagent flap and used tools*

#### 27.3.2.4 Assembly of the Starter Reagent flap

1. Push the axis through the collar into the Starter reagent flap as far as it will go.



*Figure 27-14: Push the axis*

2. Fixate the axis with the flat-blade screwdriver.





*Figure 27-15: Fixate the axis*

3. Tighten the collar with the grub screw and put back the screw and the washer.



*Figure 27-16: Tighten the collar*

4. Put back the gas pressure damper and tighten it. Press on the ball head, then lock and push the splint pin upwards.





*Figure 27–17: Place back the gas pressure damper*

5. Put back the black covers on the Front module and tighten the screws from the back.



*Figure 27–18: Assembled Front module*

6. Assemble the Front module, see Chapter 27.3.1 on page 27-4.

### 27.3.3 Replacement of the Gas Spring for Starter Reagent Flap

#### 27.3.3.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.

#### 27.3.3.2 Removal of the Gas Spring for Starter Reagent Flap

1. Remove the securing spring (1) from the ball-hitch coupling (2).
2. Pull the ball-hitch coupling (2) out of the ball-head bolt (4).
3. Insert the securing spring (1) back into the ball hitch coupling (2).
4. Remove the screw (5) and nut (7) from the standoff (6) of the gas spring (3).

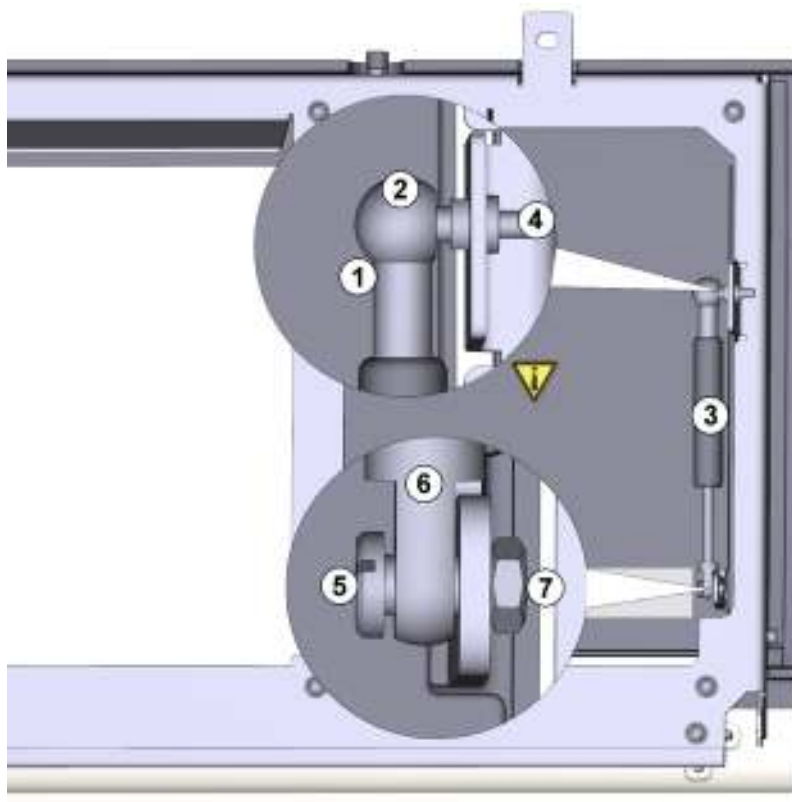


Figure 27–19: Gas Spring for Starter Reagent Flap

**27.3.3.3 Assembly of the Gas Spring for Starter Reagent Flap**

1. Set the standoff (6) of the gas spring (1) in, insert the screw (5) and fasten the nut (7).
2. Remove the securing spring (1) from the ball-hitch coupling (2).
3. Slide the ball-hitch coupling (2) over the ball-head bolt (4).
4. Secure the ball-hitch coupling (2) with the securing spring (1).
5. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.

## 27.3.4 Replacement of the Gas Spring for Sample Area Flap

### 27.3.4.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.

### 27.3.4.2 Removal of the Gas Spring for Sample Area Flap

1. Remove the securing spring (1) from the ball-hitch coupling (2).
2. Pull the ball-hitch coupling (2) out of the ball-head bolt (4).
3. Insert the securing spring (1) back into the ball hitch coupling (2).
4. Remove the screw (5) and nut (7) from the standoff (6) of the gas spring (3).

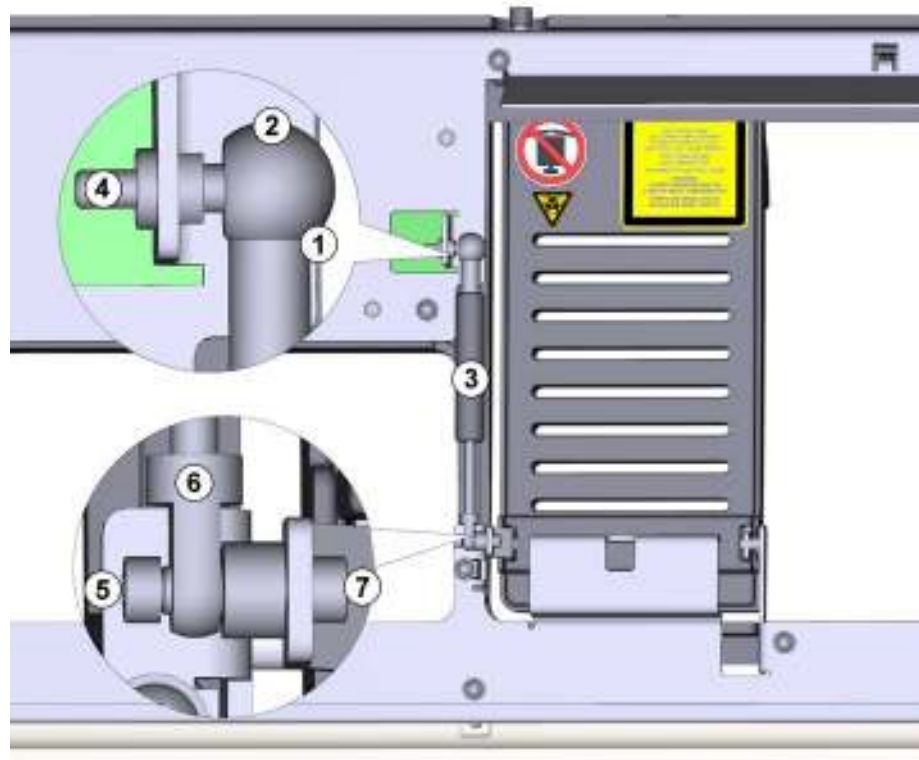


Figure 27–20: Gas Spring for Sample Area Flap

**27.3.4.3 Assembly of the Gas Spring for Sample Area Flap**

1. Set the standoff (6) of the gas spring (1) in, insert the screw (5) and fasten the nut (7).
2. Remove the securing spring (1) from the ball-hitch coupling (2).
3. Slide the ball-hitch coupling (2) over the ball-head bolt (4).
4. Secure the ball-hitch coupling (2) with the securing spring (1).
5. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
6. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 27.3.5 Replacement of the Gas Spring of Reagent Area flap

### 27.3.5.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.

### 27.3.5.2 Removal of the Gas Spring for Reagent Area Flap

1. Remove the securing spring (1) from the ball-hitch coupling (2) of the gas spring.
2. Pull the ball-hitch coupling (2) out of the ball-head bolt (4).
3. Insert the securing spring (1) back into the ball hitch coupling (2).
4. Remove the retaining ring (5) from the standoff (6) of the gas spring (3).

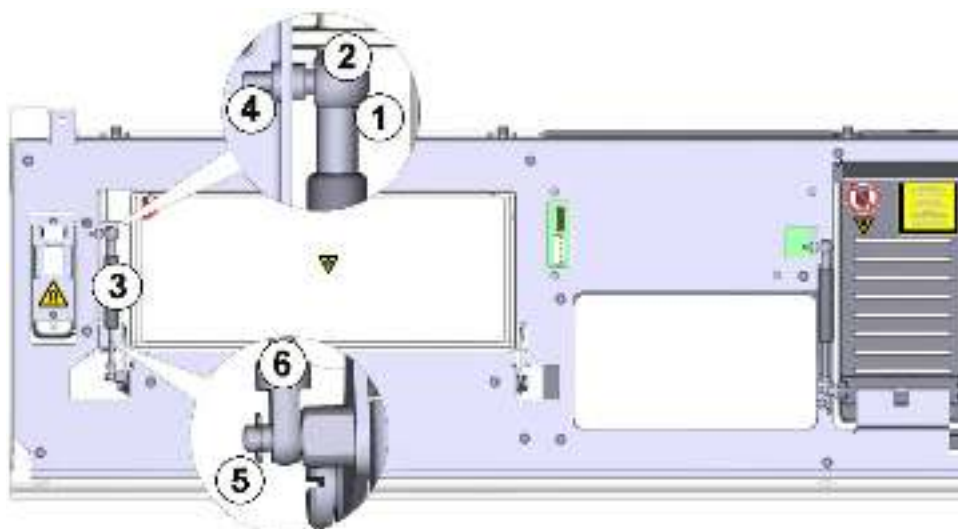


Figure 27–21: Gas spring for Reagent Area Flap

### 27.3.5.3 Assembly of the Gas Spring for Reagent Area Flap

1. Set the standoff (6) of the gas spring (1) in and secure it with the retaining ring (5).
2. Remove the securing spring (1) from the ball-hitch coupling (2).
3. Slide the ball-hitch coupling (2) over the ball-head bolt (4).
4. Secure the ball-hitch coupling (2) with the securing spring (1).
5. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
6. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 27.3.6 Replacement of the Front Cover CU PCB

### 27.3.6.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
7. Remove the Gas Spring for Reagent Area Flap, see Chapter 27.3.5.2 on page 27-16.

### 27.3.6.2 Removal of the Front Cover CU PCB

1. Remove two screws (2 and 3) and washers out of the Reagent Area Flap (1).
2. Lift the Reagent Area Flap out and set it aside.

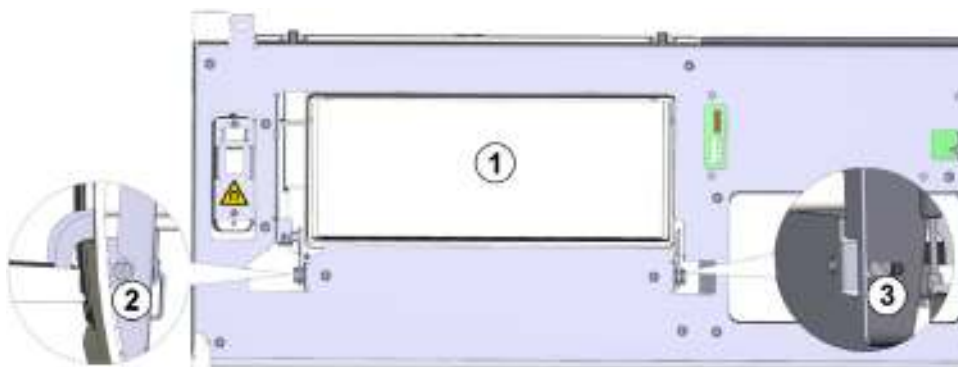
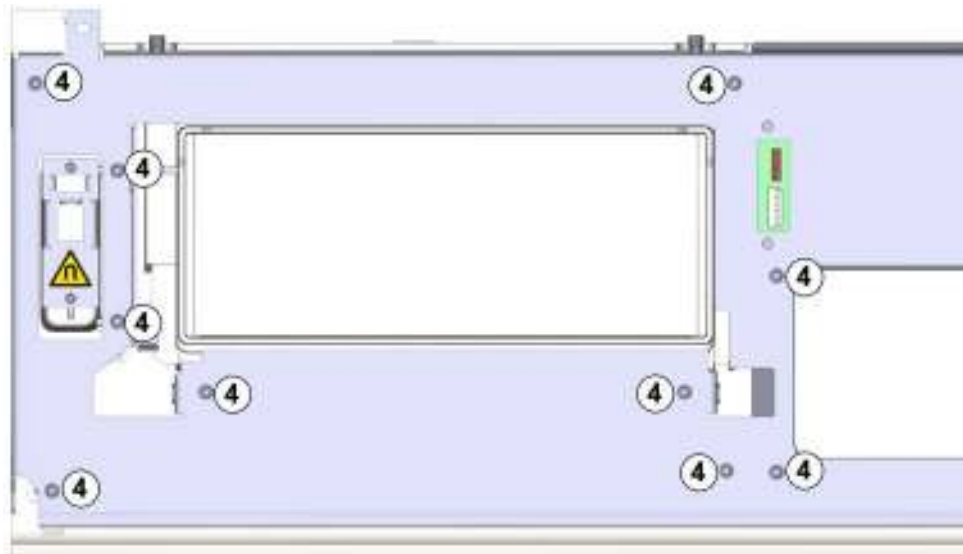
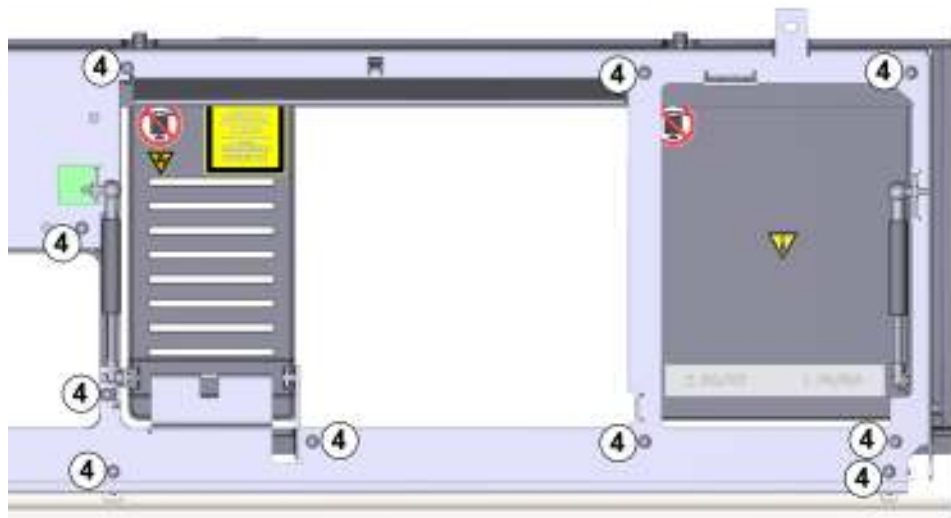


Figure 27–22: Front Module rear view, Reagent Area flap removal

3. Remove the twenty screws (4) of the Base Shell.



*Figure 27-23: Front Module, rear left side view*



*Figure 27-24: Front Module, rear right side view*

4. Lift the left and right parts of Front Module Cover and set them aside.



5. Unplug the Starter Flap cable connector (J2).
6. Unscrew the four screws (5) and lift the Front Cover CU PCB (6) out.

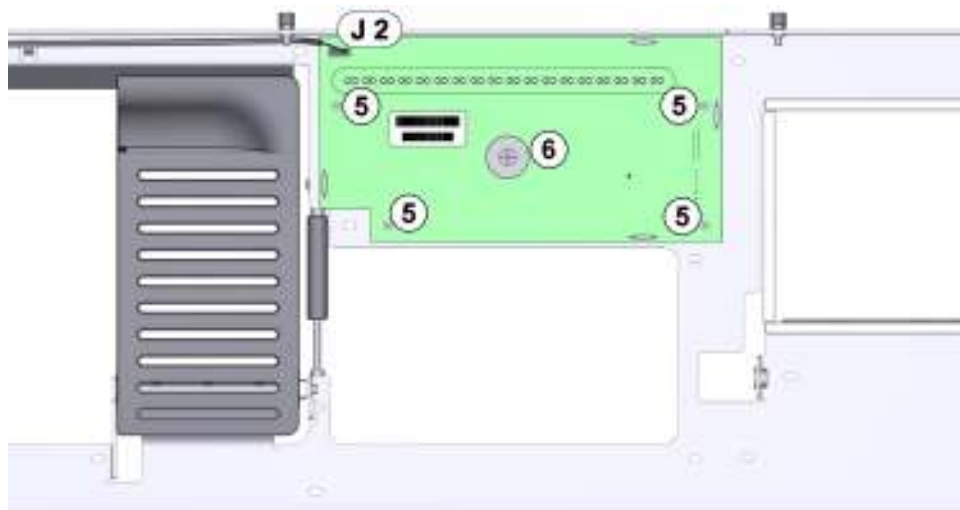


Figure 27–25: Front Module, view of Front Cover CU PCB

#### 27.3.6.3 Assembly of the Front Cover CU PCB

1. Insert the Front Cover CU PCB (6) and fasten with four screws (5).
2. Plug the Starter Flap cable connector (J2) onto the Front Cover CU PCB (6).
3. Lay the left Front Module cover onto the base shell and fasten with the ten screws (4).
4. Lay the right Front Module cover onto the base shell and fasten with the ten screws (4).
5. Insert the Reagent Area Flap (1), add the washers inside and fasten with two screws (2 and 3).
6. Assemble the Gas Spring for Reagent Area Flap, see Chapter 27.3.5.3 on page 27-16.
7. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
8. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 27.3.7 Replacement of the Reed Switch

### 27.3.7.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.

### 27.3.7.2 Removal of the Reed Switch

1. Unscrew the ten screws (1) of the Base Shell at the left Front Module Cover.
2. Lift the left Front Module Cover up and set it aside.

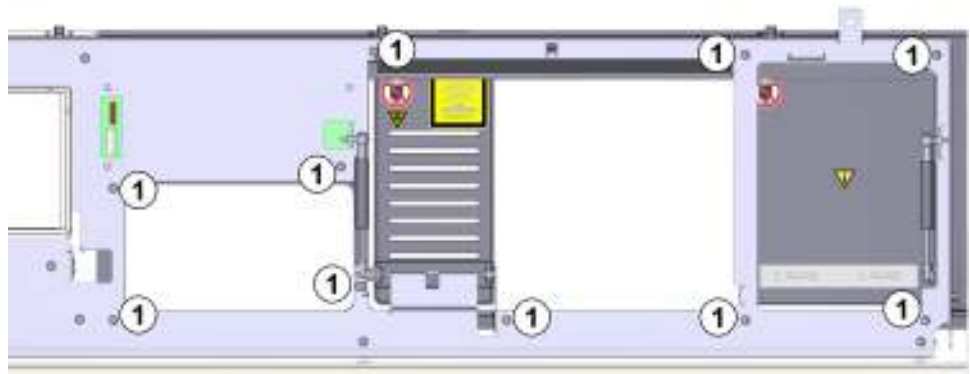


Figure 27–26: Removal of Base Shell, right side

3. Unplug the Reed Switch connector (J2) from the Front Cover CU PCB.
4. Remove two screws and washers (2) and remove the Reed Switch (1).



Figure 27–27: Front Cover left removed

**27.3.7.3 Assembly of the Reed Switch**

1. Insert the reed switch (1) and fasten with two screws and washers (2).
2. Connect the Reed Switch connector (J2) onto the Front Cover CU PCB.
3. Lay the left Front Module Cover onto the base shell and fasten ten screws (1).
4. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

### 27.3.8 Replacement of the Snap-in plate

#### 27.3.8.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.

#### 27.3.8.2 Removal of the Snap-in plate

1. Put the Front Module down to access the Magnetic Particles Resuspension Area with Snap-in plate.

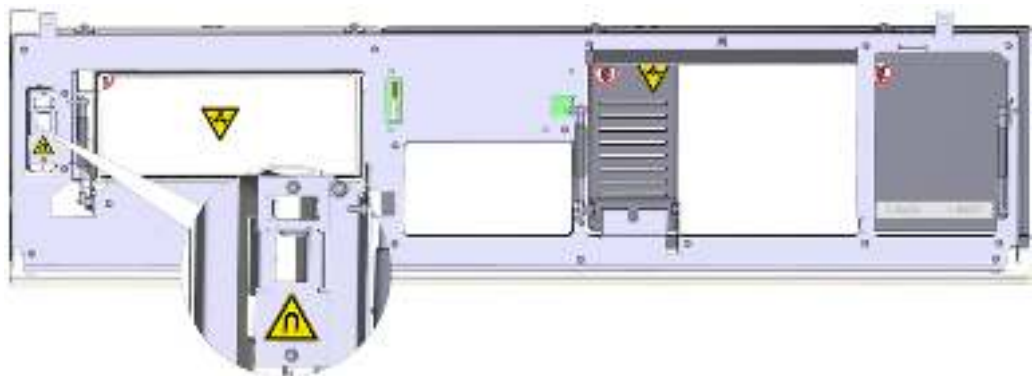


Figure 27–28: Inside view of the Front Module with enlarged Snap-in plate.

2. Unscrew the two screws (1).

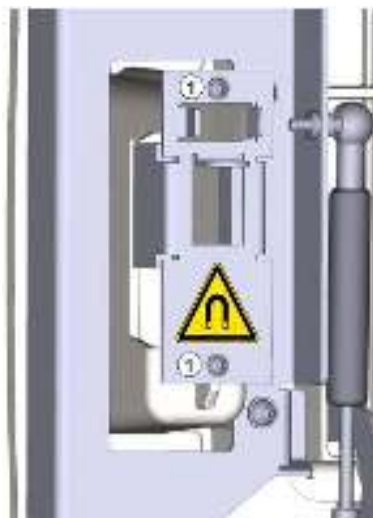


Figure 27–29: Snap-in plate

3. Lift the Snap-in plate carefully out of its position.

#### **27.3.8.3 Assembly of the Snap-in plate**

1. Place the Snap-in plate in position.
2. Tighten the two screws (1).
3. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.

#### **27.3.8.4 Set of the Snap-in plate**

1. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9 or see Chapter 10.4 on page 10-10.

## 27.4 Adjustment and calibration

This section contains information about check and verification of the Front Module.

### NOTE

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).

Do not run any of the software at the same time!

### 27.4.1 Initialization routine

- Close any open software and start the Service software from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **Front and Loading** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog controlled modules** to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 27.4.2 Adjustment and calibration of the Front Module

#### 27.4.2.1 Verifying of the Front Cover sensors

- In the **Service Software** at the navigation window, choose **Front and Loading** and the **General - Sensors** bracket.
- Click onto the **Front Cover** button.
- Check the Front Cover sensors with the **Auto-repeat** button:
  - open and close the following flaps: Reagent-, Sample-, Starter flap;
  - open and close the following: Left Top Cover, Solid Waste bin.

#### 27.4.2.2 Check of the Beeper

- In the **Service Software** at the navigation window, choose **Front and Loading** and the **General - Actors - Buzzer** bracket.
- Check the Beeper:
  - click onto the **On** button. The Beeper must be audible.
  - click onto the **Off** button.

#### 27.4.2.3 Check of the Status Light

- In the **Service Software** at the navigation window, choose **Front and Loading** and the **General - Actors - Status Light** bracket.
- Check the Status Light:
  - choose the **Yellow** bracket, and click onto the **On** button; then click onto **Flashing slow** button and onto **Flashing fast** button
  - choose the **Blue** bracket, and click onto the **On** button; then click onto **Flashing slow** button and onto **Flashing fast** button.

## 27.5 Electric connections, switches and signals

This section shows an overview of the electronic boards of the front module.

**NOTE**

see Chapter 3.1.7 on page 3-8 or see Chapter 4.1.8 on page 4-9 for an overview of the complete electric circuit of the system.

### 27.5.1 Front cover CU PCBA



Figure 27–30: Front cover CU PCBA, top view

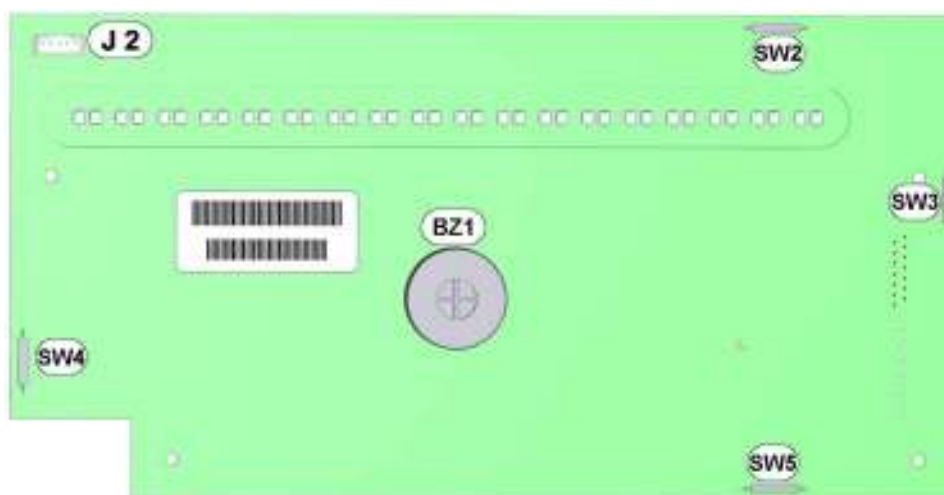


Figure 27–31: Front cover CU PCBA, rear view

| Number |                      | Connection |
|--------|----------------------|------------|
| J1     | Download             |            |
| J2     | Starter Reagent flap |            |
| J3     | PWR/CAN              |            |

Table 27–4: Connections of the front cover CU PCBA

| Number |                   | Meaning/Function |
|--------|-------------------|------------------|
| SW 1   | --                |                  |
| SW 2   | Top Cover         |                  |
| SW 3   | Reagent Area flap |                  |
| SW 4   | Sample Area flap  |                  |
| SW 5   | Solid Waste Bin   |                  |

Table 27–5: SW of the front cover CU PCBA

| Number |        | Meaning/Function |
|--------|--------|------------------|
| BZ 1   | Buzzer |                  |

Figure 27–32: Sound



## 27.6 Magnetic Particles Resuspension Area

In the resuspension area, the integral is pushed through the holes of the bracket, till it reaches the end-stop, see Figure 27–33.

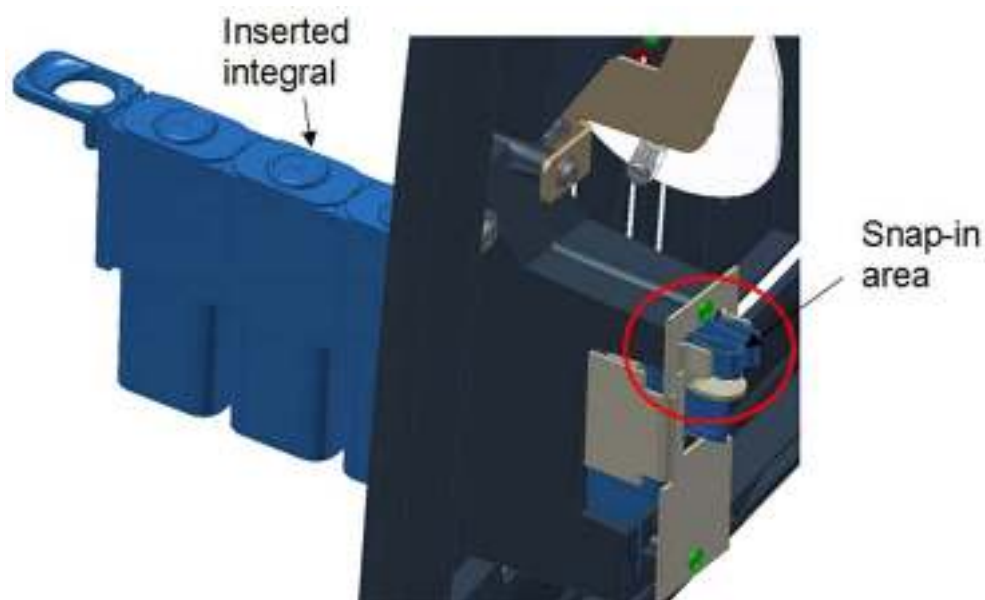


Figure 27–33: Resuspension area

Contours of the bracket provide a smooth snap-in effect during the insertion of the integral, using the latches of the integral to achieve the locking function (see Figure 27–34).

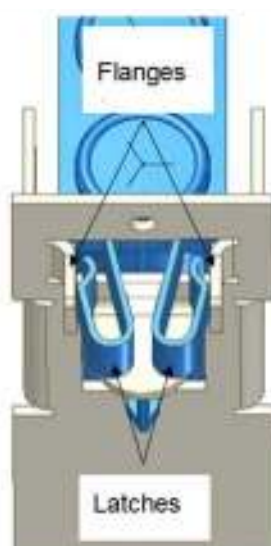


Figure 27–34: Snap-in function



# 28 Tubing for internal-tanks configuration

This chapter shows an overview of the tubing and the replacement of the tubings in the instrument.

## 28.1 Safety

### NOTE

The view of the instrument is a view from the bottom and some parts are transparent pictured.

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 1.5 on page 1-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

### WARNING

#### Pinched or bend tubing

- Do not bend, kink or damage any tubes while disassembling or assembling the tubing sets.
- At replaceable tank, canister or bottles the required minimal distance for laying out the hoses must be taken into account.

### WARNING



#### Liquid in the System

Liquid which gets into the system can cause illnesses with deadly consequences in case of contact. The system can be damaged by liquids.

- Switch off the system.
- Separate the system from the mains supply.
- Wear suitable protective clothing.
- Clean, disinfect and dry the system according to the applicable local and national provisions, legislation and laboratory procedures.
- Assure the system is completely dry before switch it on.

**Risk of infection!**

Parts, in contact to potentially infectious liquid, must be treated as potentially infectious. Improper handling of infectious parts can cause skin irritations, illnesses and possible death.

- Observe local and national provisions, legislation and laboratory regulations.
  - Use appropriate disposable gloves.
  - Use an appropriate lab coat.
  - Use an appropriate eye protection.
  - Avoid contact between skin/mucous membrane and samples/test reagents or parts of the instrument.
  - Do not pipette by mouth.
  - Do not eat, drink, smoke, apply cosmetics or handle contact lenses when handling human-sourced material or contaminated system components.
  - Clean and disinfect the system immediately if potentially infectious material has been spilled.
  - Any reagent spills should be washed with a chemical use sodium hypochlorite solution with 0.5% active chlorine (e.g. dilution 1:10 of a solution at 5% active chlorine) and disposed as though potentially infectious.
  - All samples, biological reagents and materials used in the assays must be considered potentially able to transmit infectious agents. They should therefore be disposed in accordance with the prevailing regulations and guidelines of the agencies holding jurisdiction over the laboratory, and the regulations of each Country.
  - Liquid waste must be decontaminated with a chemical use sodium hypochlorite solution with 0.1% active chlorine (e.g. dilution 1:50 of a solution at 5% active chlorine) for at least half a hour.
  - Do not use broken or chipped tubes or bottles.
  - Observe the instructions in the package inserts for a correct use of the reagents.
- 

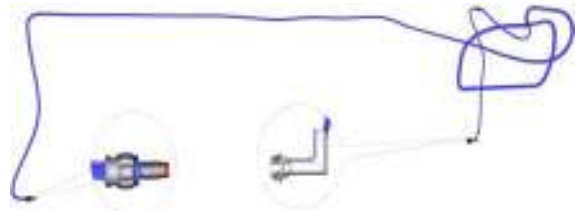
**28.1.1 Liquid Distribution Overview****NOTE**

For an overview of the complete liquid distribution, see Chapter 3.1.6 on page 3-7.

---

## 28.2 Overview

### 28.2.1 Overview tubing

| Chapter | Tubing description                              |                                                                                                                                                                                                                                                                                      |
|---------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 28.3.2  | LDV-PRP                                         |  A blue flexible tubing assembly with a 90-degree elbow at one end and a connector at the other. The elbow has a white label with the number '2'.                                                  |
| 28.3.3  | DI WATER TANK-LDV<br>(DI Water Docking Station) |  A blue flexible tubing assembly with a connector at one end and a docking station at the other. A circular inset shows a close-up of the connector, which has a white label with the number '1'. |
| 28.3.4  | PRP-SPOLV                                       |  A blue flexible tubing assembly with a connector at one end and a docking station at the other. Two circular insets show close-ups of the connector and the docking station.                    |
| 28.3.5  | WB TANK-WBDP-LDV<br>(WB Docking Station)        |  A blue flexible tubing assembly with a connector at one end and a docking station at the other.                                                                                                 |

## 28. Tubing for internal-tanks configuration

| Chapter | Tubing description                          |                                                                                      |
|---------|---------------------------------------------|--------------------------------------------------------------------------------------|
| 28.3.9  | Waste interface-Waste tank<br>Z             |    |
| 28.3.11 | WBDP-Washer Dispense Manifold<br>U, V and Z |   |
| 28.3.12 | SDP-Starters<br>(with labels)               |  |
| 28.3.13 | Reader-SDP                                  |  |

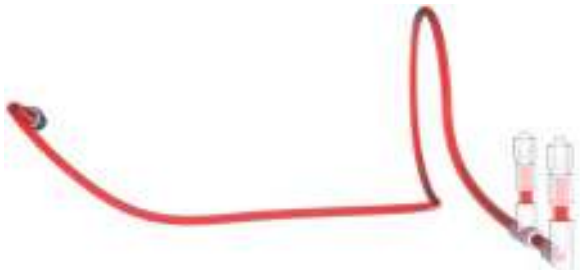
Table 28–1: Tubing overview

### 28.2.2 Overview Waste tubing at Left Energy Chain (LWP)

| Chapter | Tubing description       |                                                                                    |
|---------|--------------------------|------------------------------------------------------------------------------------|
| 28.3.7  | Wash station-LWP<br>A    |  |
| 28.3.7  | LWP-Waste interface<br>B |  |

Table 28–2: Waste tubing at Left Energy Chain (LWP)

### 28.2.3 Overview Waste tubing at Right Energy Chain (WAP)

| Chapter | Tubing description             |                                                                                      |
|---------|--------------------------------|--------------------------------------------------------------------------------------|
| 28.3.8  | WAP-Waste interface<br>H and I |  |
| 28.3.8  | Reagent Module-WAP<br>C        |  |

## 28. Tubing for internal-tanks configuration

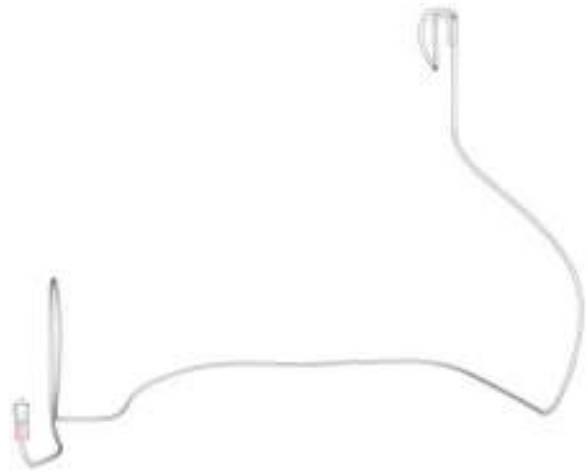
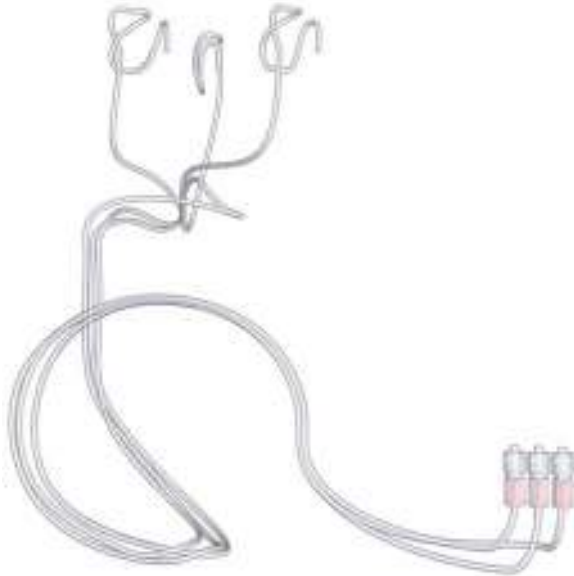
| Chapter | Tubing description                                                             |                                                                                     |
|---------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 28.3.8  | Reader-WAP<br>D                                                                |   |
| 28.3.8  | Washer-WAP:<br>E - SIPAL 1<br>F - SIPAL 2<br>G - Grounding pipe and<br>SIPAL 3 |  |

Table 28–3: Waste tubing at Right Energy Chain (WAP)



## 28.3 Replacements and repairs

This section contains instructions for the repair of individual components of the tubing and Wash Station. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 28.3.1 Replacement of the Liquid distribution valve (LDV)

#### 28.3.1.1 Overview of the LDV

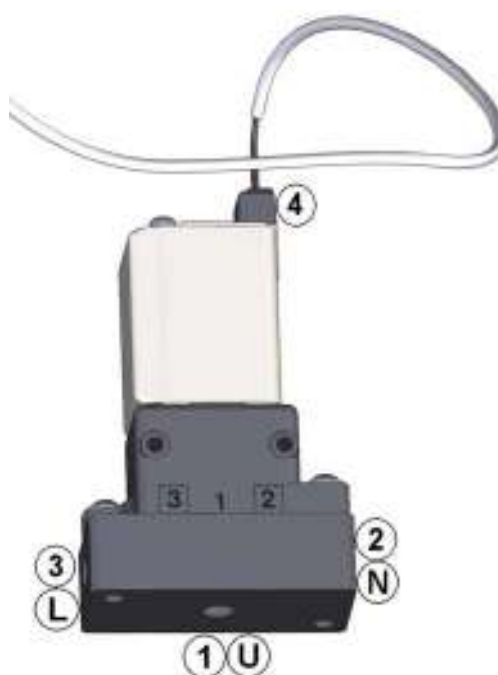


Figure 28–1: Liquid Distribution Valve

| No used on valve and tubing | Letters used in manual | Description                                         |
|-----------------------------|------------------------|-----------------------------------------------------|
| 1                           | U                      | Connected to DI water tank                          |
| 2                           | N                      | Connected to Probe Rinsing Pump (PRP)               |
| 3                           | L                      | Connected to WB tank                                |
| 4                           | --                     | Connected to PCB, PIPETTOR CU W/ JUMPER at slot J13 |

Table 28–4: Liquid Distribution Valve

## 28. Tubing for internal-tanks configuration

### 28.3.1.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the DI water tank and the Wash Buffer tank.

### 28.3.1.3 Removal of the LDV1

#### CAUTION



#### Possible flooding

Be aware that the removal of the tube can spill liquid into the instrument.

Take precaution measures:

- Keep a cleaning cloth and a small bucket close by.
- Wear suitable protective clothing and gloves.

1. Disconnect the valve cable at the PCB, Pipettor CU W/ Jumper at slot J13.

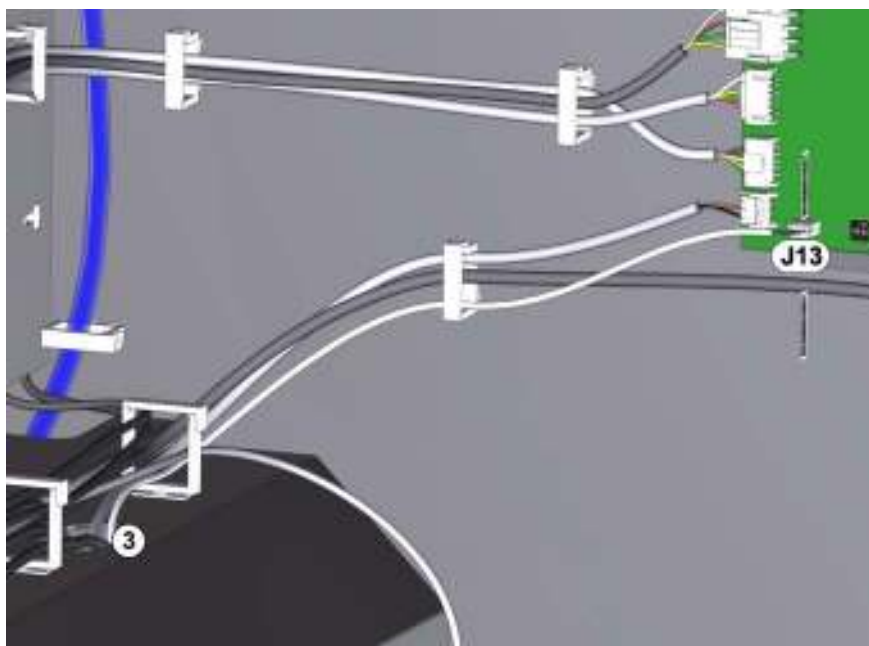


Figure 28–2: PCB, Pipettor CU W/ Jumper of the LDV

**NOTE**

Fittings connected to the tube can not be removed because they are glued. The tubing is glued into the docking station.

2. Remove the fitting (L, N and U) at the Liquid Distribution Valve (1).
3. Unscrew two screws (2).
4. Slide valve carefully out and thread the cable through chassis inner.

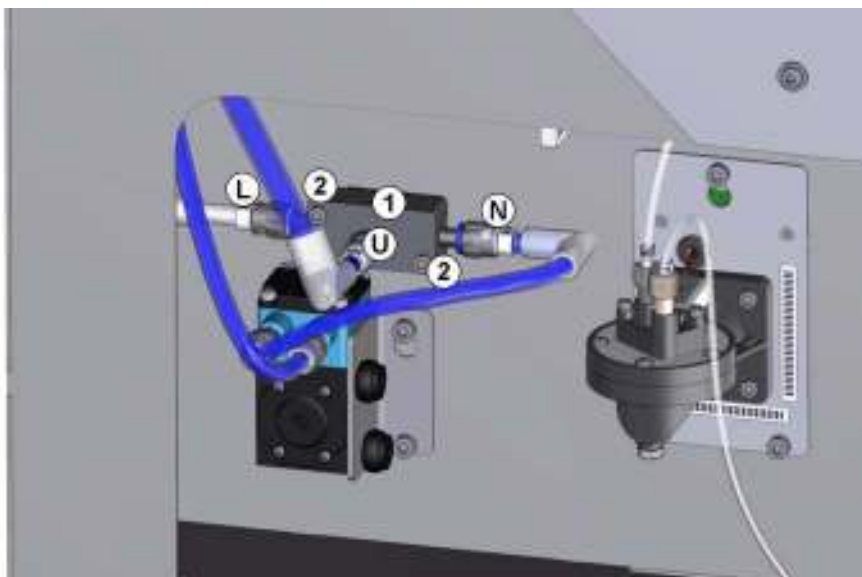


Figure 28-3: Liquid Distribution Valve

**28.3.1.4 Assembly of the LDV**

1. Slide the valve with the numbers face up in and fasten with two screws (2).
2. Draw the cable carefully through the chassis inner and connect the cable connector onto PCB, Pipettor CU W/ Jumper at slot J13.
3. Connect the fitting (L, N and U) onto the Liquid Distribution Valve (1).
4. Insert DI water tank and Wash Buffer tank.
5. Check the tube assembly for proper installation.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**28.3.1.5 Set the LDV**

1. Prime the SPOLV both with DI water and then with Wash Buffer, see Chapter 11.8.1.4 on page 11-64.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

## 28.3.2 Replacement of the LDV-PRP tubing

### 28.3.2.1 Tube routing of the LDV-PRP tubing

The specified tubing is connected from the Probe Rinse Pump (PRP) to Liquid Distribution Valve (LDV).

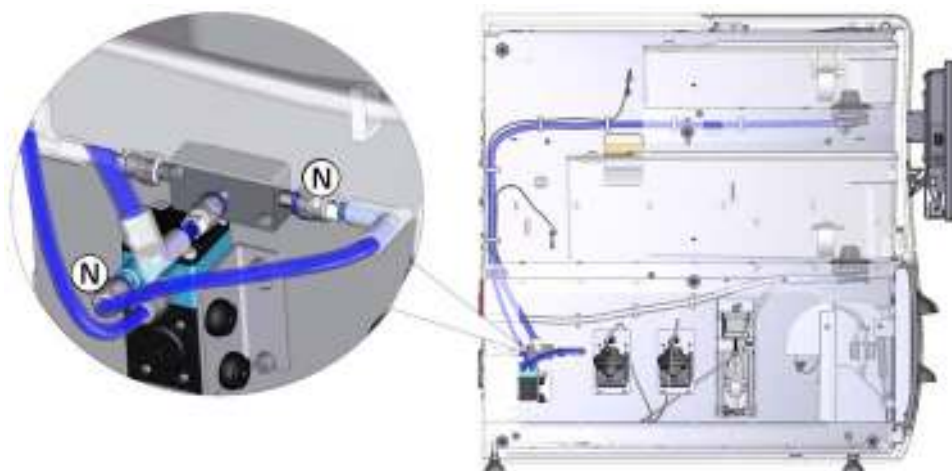


Figure 28-4: LDV-PRP tubing inside the instrument

### 28.3.2.2 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the DI water tank and the Wash Buffer tank.

### 28.3.2.3 Removal of the LDV-PRP tubing

**CAUTION****Possible flooding**

Be aware that the removal of the tube can spill liquid into the instrument.

Take precaution measures:

- Keep a cleaning cloth and a small bucket close by.
- Wear suitable protective clothing and gloves.

1. Disconnect the fittings (N) of the LDV-PRP tubing.

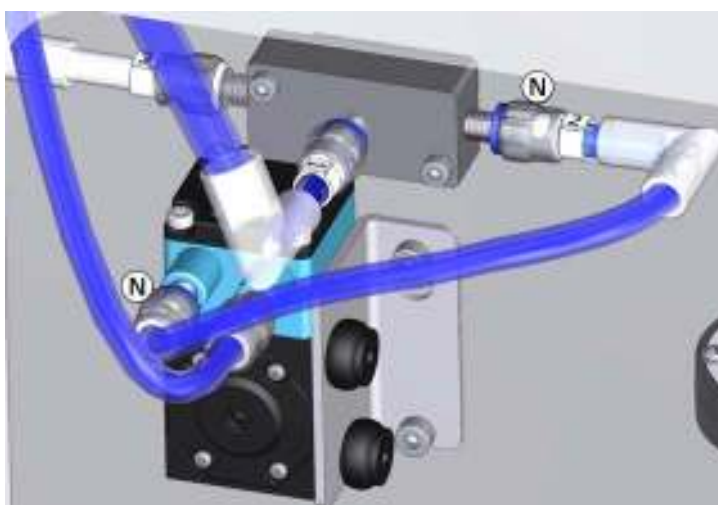


Figure 28–5: PRP and LDV

### 28.3.2.4 Assembly of the LDV-PRP tubing

1. Connect the fittings (N) of the LDV-PRP tubing.
2. Insert DI water tank and Wash Buffer tank.
3. Check the tube assembly for proper installation.
4. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 28.3.2.5 Set the LDV-PRP tubing

1. Prime the SPOLV both with DI water and then with Wash Buffer, see Chapter 11.8.1.4 on page 11-64.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 28.3.3 Replacement of the DI WATER TANK-LDV

#### 28.3.3.1 Tube routing of the DI WATER TANK-LDV

The DI WATER TANK-LDV tubing is routed from the DI water tank through the cable holders to the Liquid Distribution Valve (U).



Figure 28–6: DI WATER TANK-LVD tubing inside the instrument

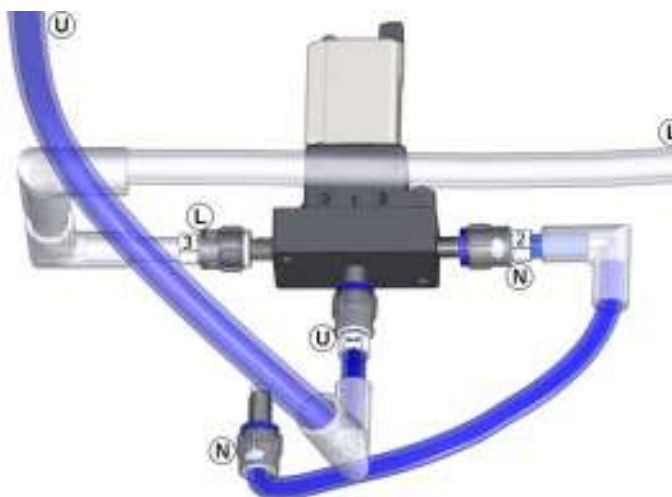


Figure 28–7: Connection to Liquid Distribution Valve

### 28.3.3.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the DI water tank and the Wash Buffer tank.

### 28.3.3.3 Removal of the DI WATER TANK-LDV tubing

---

**CAUTION****Possible flooding**

Be aware that the removal of the tube can spill liquid into the instrument.

Take precaution measures:

- Keep a cleaning cloth and a small bucket close by.
  - Wear suitable protective clothing and gloves.
- 

**NOTE**

Fittings connected to the tube can not be removed because they are glued.  
The tubing is glued into the docking station.

---

1. Remove the fitting (U) at the Liquid Distribution Valve (LDV).
  2. Lift the tubing out of the cable holders.
- 

**NOTE**

The hooks of the docking station are fragile and may brake.

---

3. Remove the DI water tank presence sensor (2) carefully out of the hooks (3).

## 28. Tubing for internal-tanks configuration

4. Remove the docking station (1) by turning it counter clockwise out of the tank support (4).

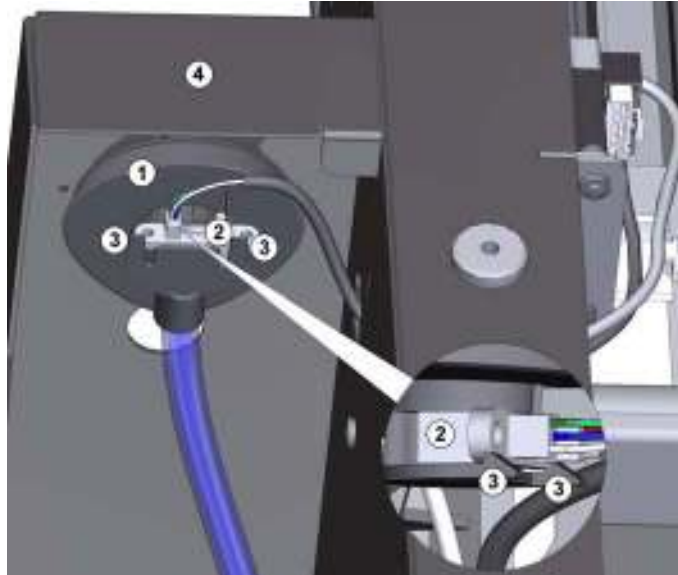


Figure 28-8: Docking station of the DI water tank, bottom view

5. Remove DI WATER TANK-LDV carefully out of the instrument.

### 28.3.3.4 Assembly of the DI WATER TANK-LDV tubing

1. Insert the docking station into the DI water tank support.
2. Insert the DI water tank presence sensor (2) carefully into the hooks (3).
3. Check that the presence sensor cable is properly routed and connected to the Liquid Support PCB (slot J21).
4. Insert the tubing into the cable holders.
5. Connect the fitting at the Liquid Distribution Valve (LDV).
6. Insert DI water tank and Wash Buffer tank.
7. Check the tube assembly for proper installation.
8. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 28.3.3.5 Set the DI WATER TANK-LDV tubing

1. Check the DI water tank presence sensor, see Chapter 28.4.2 on page 28-51.
2. Prime the SPOLV both with DI water and then with Wash Buffer, see Chapter 11.8.1.4 on page 11-64.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 28.3.4 Replacement of the PRP-SPOLV tubing

#### 28.3.4.1 Tube routing of the PRP-SPOLV tubing

The PRP-SPOLV tubing has a fitting connected to the right connector “exit” of the PRP (3). The tube (P) is routed through cable holders to the SPOLV and it is fixed with round cable holder to the Z-Bracket.



Figure 28–9: PRP-SPOLV tubing inside the instrument, side view

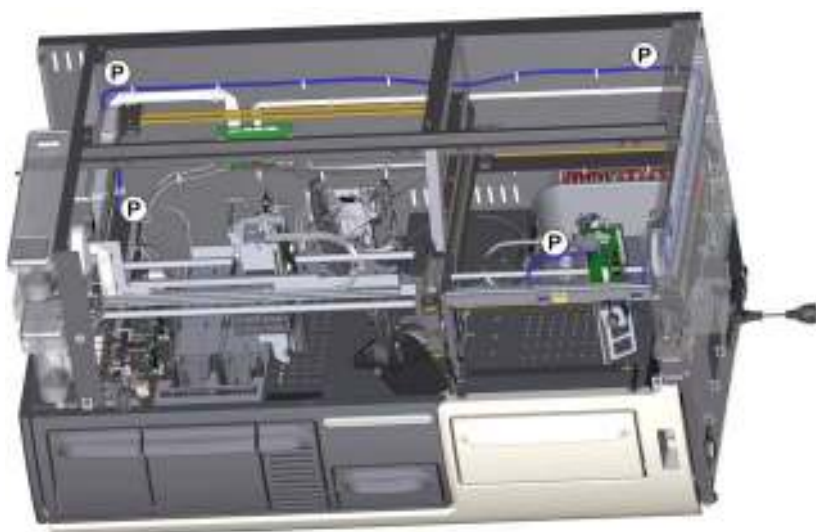


Figure 28–10: PRP-SPOLV tubing inside the instrument, front view

## 28. Tubing for internal-tanks configuration

---

### 28.3.4.2 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the DI water tank and the Wash Buffer tank.

### 28.3.4.3 Removal of the PRP-SPOLV tubing

1. Unscrew fitting at the PRP pump (3).
2. Thread the tube (P) through the left inner chassis opening.
3. Lift the tubing out of the cable holders at the chassis rear and the right side chassis.
4. Unplug the tubing at the SPOLV and remove it out of the holders.

### 28.3.4.4 Assembly of the PRP-SPOLV tubing

1. Plug the tubing into the SPOLV.
2. Insert the tubing into the cable holders at the right side chassis and the chassis rear.
3. Thread the tubing through the left inner chassis opening.
4. Connect the fitting onto the PRP pump (3).
5. Check the tube assembly for proper installation.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 28.3.4.5 Set the PRP-SPOLV tubing

1. Prime the SPOLV both with DI water and then with Wash Buffer, see Chapter 11.8.1.4 on page 11-64.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 28.3.5 Replacement of the WB TANK-WBDP-LDV

#### 28.3.5.1 Tube routing of the WB TANK-WBDP-LDV tubing

The WB TANK-WBDP-LDV tubing is routed from the Wash Buffer tank through the cable holders to the left side (L) of the Liquid Distribution Valve (2) and to the Wash Buffer Dispense Pump (4).

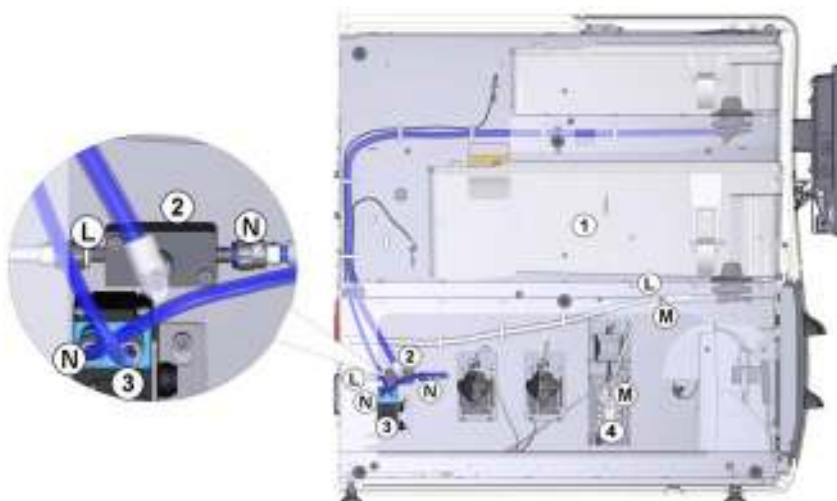


Figure 28-11: WB tank-WBDP-LDV tubing inside the instrument

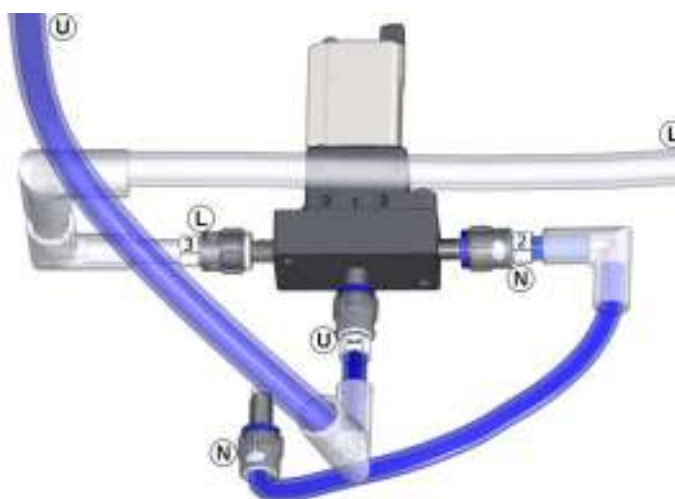


Figure 28-12: Connection to Liquid distribution Valve

## 28. Tubing for internal-tanks configuration

---

### 28.3.5.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the DI water tank and the Wash Buffer tank.

### 28.3.5.3 Removal of the WB TANK-WBDP-LDV tubing

#### CAUTION



#### Possible flooding.

Be aware that the removal of the tube can spill liquid into the instrument.

Take precaution measures:

- Keep a cleaning cloth and a small bucket close by.
- Wear suitable protective clothing and gloves.

#### NOTE

Fittings connected to the tube can not be removed because they are glued.  
The tubing is glued into the docking station.

1. Remove the fitting (L) at the Liquid Distribution Valve (LDV).
2. Remove the fitting (M) at the Wash Buffer Dispense Pump (WBDP)
3. Lift the tubing out of the cable holders.

4. Open the Starter Reagent flap.

**NOTE**

The hooks of the docking station are fragile and may brake.

5. Remove the WB tank presence sensor (2) carefully out of the hooks (3).
6. Remove the docking station (1) by turning it counter clockwise out of the tank support (4).

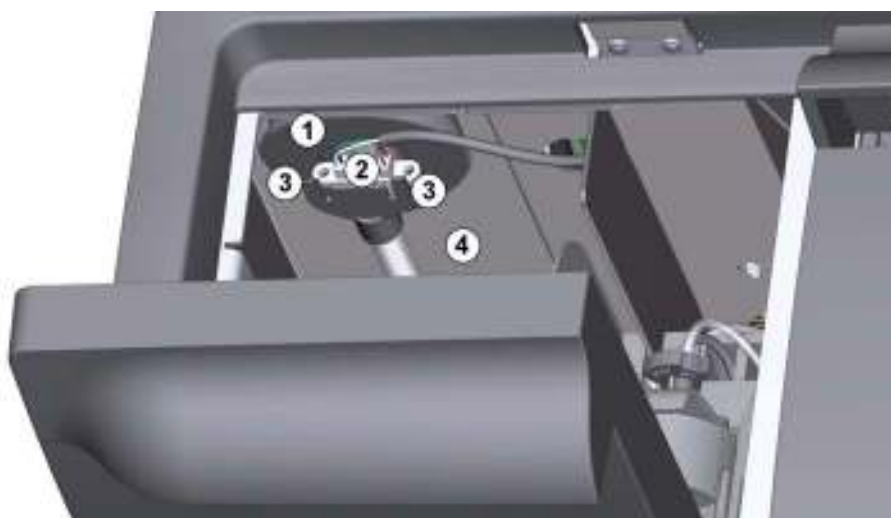


Figure 28-13: Wash Buffer tank docking station, bottom view

**28.3.5.4 Assembly of the WB TANK-WBDP-LDV tubing**

1. Insert the docking station into the tank support.
2. Insert the WB tank presence sensor (2) carefully into the hooks (3).
3. Check that the presence sensor cable is properly routed and connected to the Liquid Support PCB (slot J22).
4. Insert the tubing into the cable holders.
5. Connect the fitting (L) at the LDV.
6. Connect the fitting (M) at the WBDP.
7. Insert DI water tank and Wash Buffer tank.
8. Check the tube assembly for proper installation.
9. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**28.3.5.5 Set the WB TANK-WBDP-LDV tubing**

1. Check the Wash Buffer tank presence sensor, see Chapter 28.4.4 on page 28-52.
2. Prime the Washer, see Chapter 11.8.1.5 on page 11-65.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 28.3.6 Replacement of the Peristaltic pumps rack

#### 28.3.6.1 Tube connection of the Waste tubings connected to the Peristaltic pumps

This section shows an overview of the Waste tubings connected to the Peristaltic pumps.

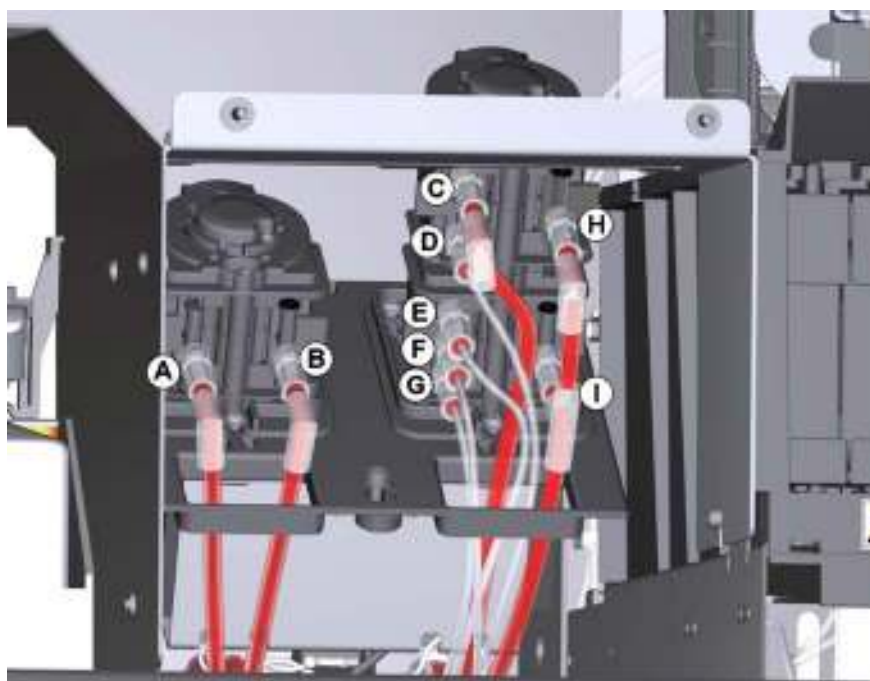


Figure 28–14: Waste tubings connected to the Peristaltic pumps overview

| No. | Connection at Pump | Tubing              | Connection to                        |
|-----|--------------------|---------------------|--------------------------------------|
| A   | LWP                | Wash station-LWP    | Wash station                         |
| B   | LWP                | LWP-Waste interface | Waste interface (Y-connector)        |
| C   | WAP                | Reagent Module-WAP  | TA Drain of the Reagent-Module       |
| D   | WAP                | Reader-WAP          | Assy-Sipal of the Measurement-Module |
| E   | WAP                | Washer-WAP          | SIPAL 1                              |
| F   | WAP                | Washer-WAP          | SIPAL 2                              |
| G   | WAP                | Washer-WAP          | Grounding pipe + SIPAL 3             |
| H   | WAP                | WAP-Waste interface | Waste interface (Y-connector)        |
| I   | WAP                | WAP-Waste interface | Waste interface (Y-connector)        |

Table 28–5: Pump block tube connection overview



### 28.3.6.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
9. Remove the Sample Module, see Chapter 21.3.2.2 on page 21-7.
10. Tilt Peristaltic pumps rack up.

### 28.3.6.3 Removal of the Peristaltic pumps rack

1. Unplug all fittings (A to I) of the Waste tubings connected to the Peristaltic pumps.

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

2. Loose screw (2) and slide the screw with the tube carrier down and push it to the back out.

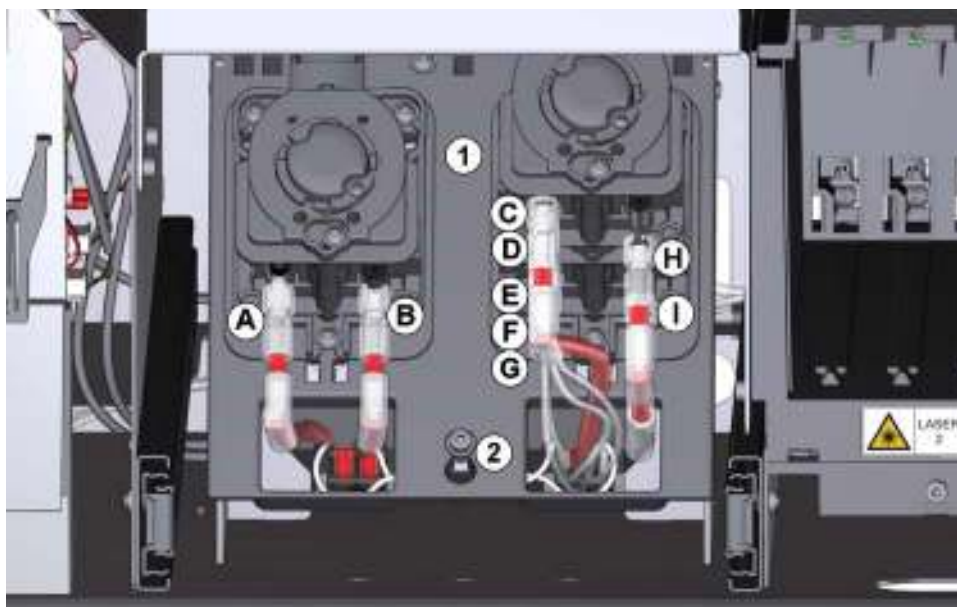


Figure 28–15: Peristaltic pumps, front view

## 28. Tubing for internal-tanks configuration

3. Remove the securing ring (3) of the pump bracket axis (4).
4. Pull the pump bracket axis to the right side out.
5. Unplug the cable connections at the pumps.
6. Remove the Peristaltic pumps rack out of the instrument.

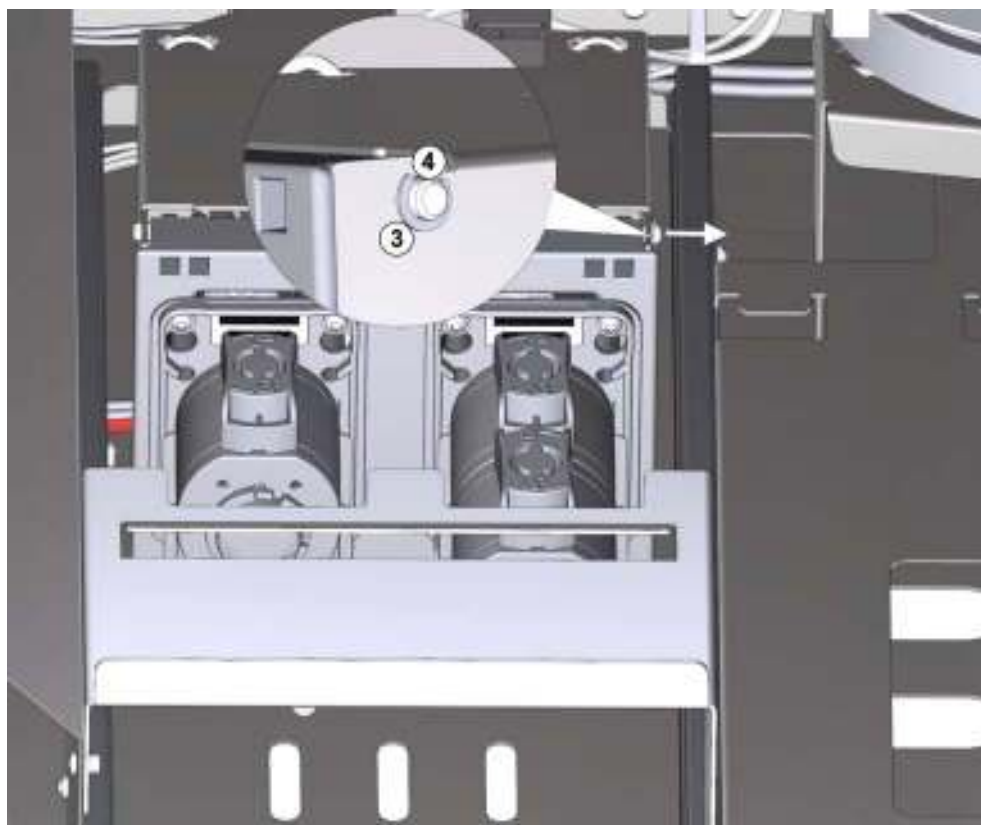


Figure 28-16: Peristaltic pumps rack

### 28.3.6.4 Assembly of the Peristaltic pumps rack

1. Insert the Peristaltic pumps rack (1) into the instrument and fix it with the pump bracket axis (4).
2. Secure the pump bracket axis (4) with a securing ring (3).
3. Connect the cable connections onto the pumps.
4. Slide the tube carrier with tubes in and set it into the screws of the Peristaltic pumps rack and fasten the fittings (A to I).
5. Close the Peristaltic pumps rack.
6. Assemble the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.
7. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
8. Assemble Sample Module, see Chapter 21.3.2.3 on page 21-8.
9. Open the Cuvette - Tip Loading Module and check the tube assembly for proper installation.
10. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 28.3.7 Replacement of the Waste tubing at Left Energy Chain (LWP)

**NOTE**

**The Waste tubings are an add-on of yearly Preventive Maintenance kit.**

The Field Service Engineer shall evaluate whether their replacement is needed or not, see Chapter 9.3.2 on page 9-8.

#### 28.3.7.1 Tube routing of the Waste tubing at Left Energy Chain (LWP)

The Waste tubing at Left Energy Chain are connected to the LWP:

- Wash station-LWP tubing (A) is routed through the Left Energy Chain and cable holders and connected to Wash Station.
- LWP-Waste interface tubing (B) is routed through the Left Energy Chain to the Waste interface.

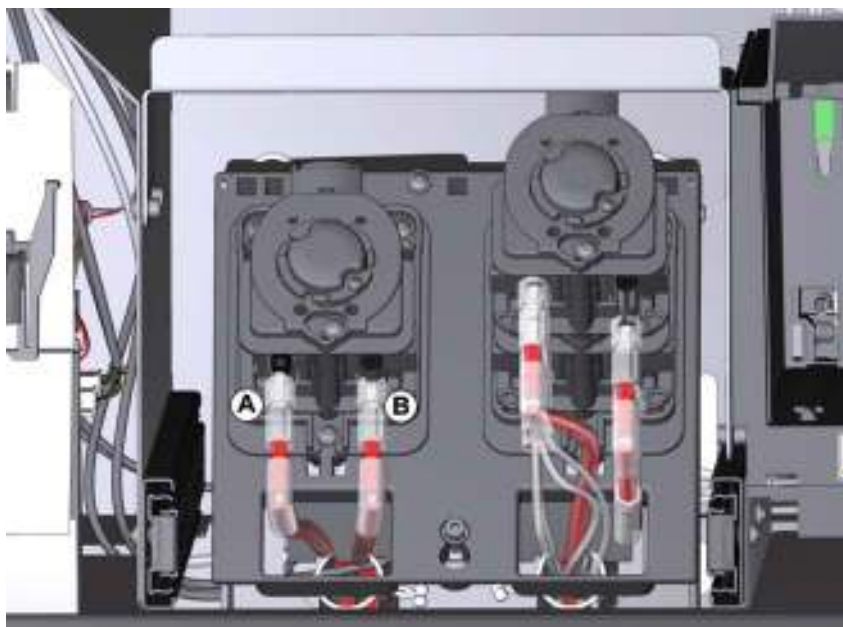


Figure 28–17: Peristaltic pumps, front view

## 28. Tubing for internal-tanks configuration

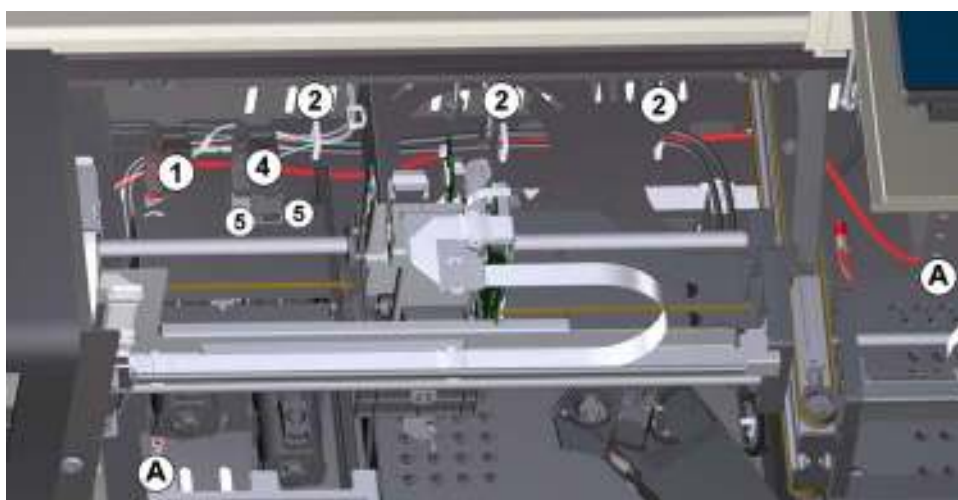


Figure 28-18: Wash station-LWP tubing, inside the instrument, without Incubator Module and Front Module

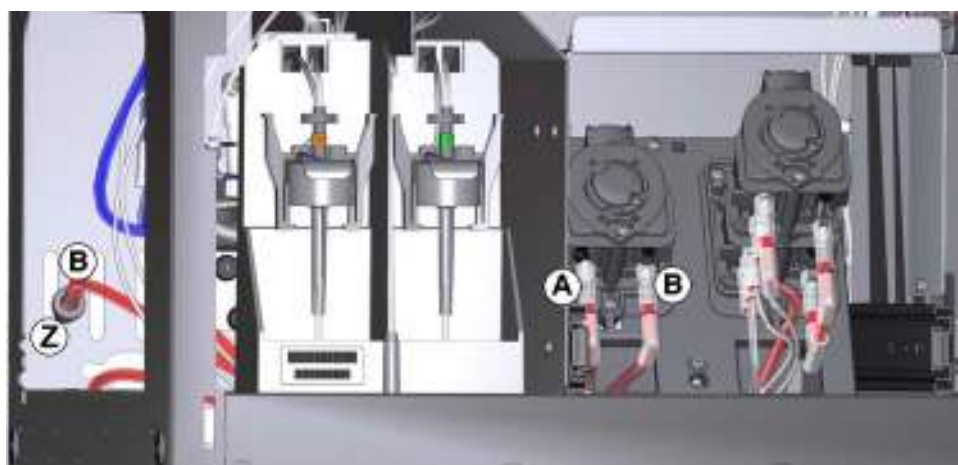


Figure 28-19: LWP-Waste interface tubing, inside the instrument

### 28.3.7.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
9. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.
10. Remove the Peristaltic pumps rack, see Chapter 28.3.6.3 on page 28-21.

### 28.3.7.3 Removal of the Waste tubing at Left Energy Chain (LWP)

1. Unplug the fitting of the Wash station-LWP (A) tubing at Wash station.

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

2. Unhook the tube at the cable holders (2).
3. Unplug the fitting of the LWP-Waste interface (B) tubing at the connection at the Waste interface-Waste tank (Z).
4. Unhook the tube at the cable holders.
5. Unhook the Left Energy Chain (1) at the last but one link (3) at the bottom.

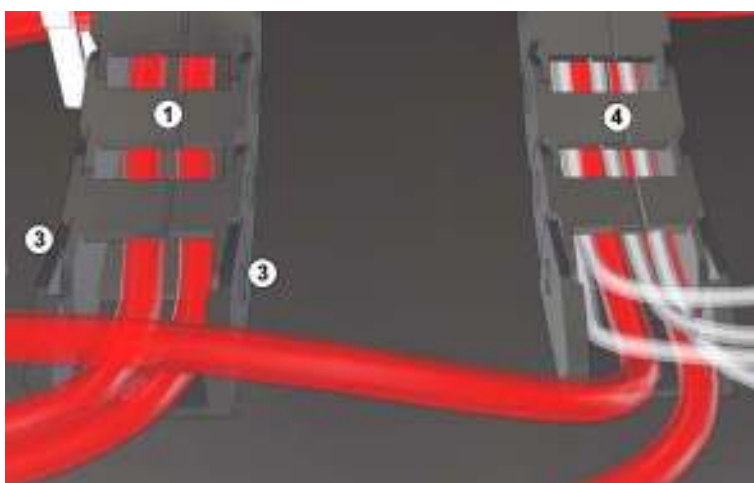


Figure 28–20: Energy chain mounted onto the bottom plate.

6. Open the frame stay and pull the tubing out of the Left Energy Chain.

## 28. Tubing for internal-tanks configuration

### 28.3.7.4 Assembly of the Waste tubing at Left Energy Chain (LWP)

1. Insert the tubing into the Left Energy Chain (1).
2. Connect the tube fitting of the Wash station-LWP (A) and the LWP-Waste interface (B) onto the LWP pump.
3. Close all frame stay of the Left Energy Chain (1).

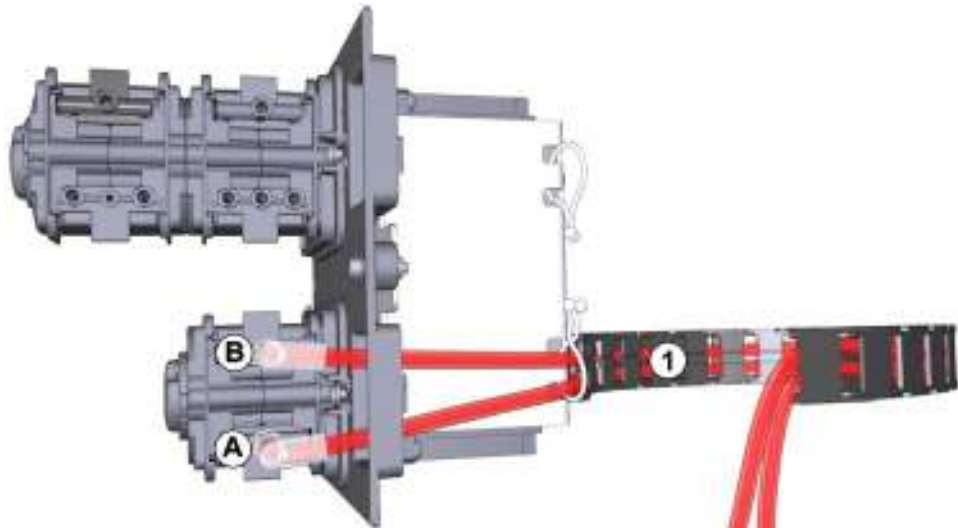


Figure 28-21: Pump block bottom view

4. Hook the Left Energy Chain onto the last but one link at the bottom.
5. Lay the Wash station-LWP (A) tubing out into the cable holders and close them.
6. Connect the tube fitting onto the Wash Station.
7. Lay the LWP-Waste interface (B) tubing underneath the bottom plate to the left side of the instrument.
8. Connect the fitting of the LWP-Waste interface (B) tubing onto the connection at the Waste interface-Waste tank (Z).
9. Assemble the Peristaltic pumps rack, see Chapter 28.3.6.4 on page 28-22.
10. Check the tube assembly for proper installation.
11. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
12. Assemble the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.
13. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
14. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 28.3.8 Replacement of the Waste tubing at Right Energy Chain (WAP)

**NOTE**

The Waste tubings are an add-on of yearly Preventive Maintenance kit.

The Field Service Engineer shall evaluate whether their replacement is needed or not, see Chapter 9.3.2 on page 9-8.

#### 28.3.8.1 Tube routing of the Waste tubing at Right Energy Chain (WAP)

The Waste tubing at Right Energy Chain are connected to the WAP:

- Reagent Module-WAP (C) tubing is routed through the Right Energy Chain and cable holders to drain of the Reagent Module (K).
- Reader-WAP (D) tubing is routed through Right Energy Chain and cable holders to the SIPAL of the Reader Module.
- Washer-WAP tubings are routed through Right Energy Chain to SIPAL 1, 2 and 3 of the Washer Module:
  - E = SIPAL 1,
  - F = SIPAL 2,
  - G = Grounding pipe + SIPAL 3.
- WAP-Waste interface (H) and (I) tubings are routed through the Right Energy Chain to the Waste interface-Waste tank.

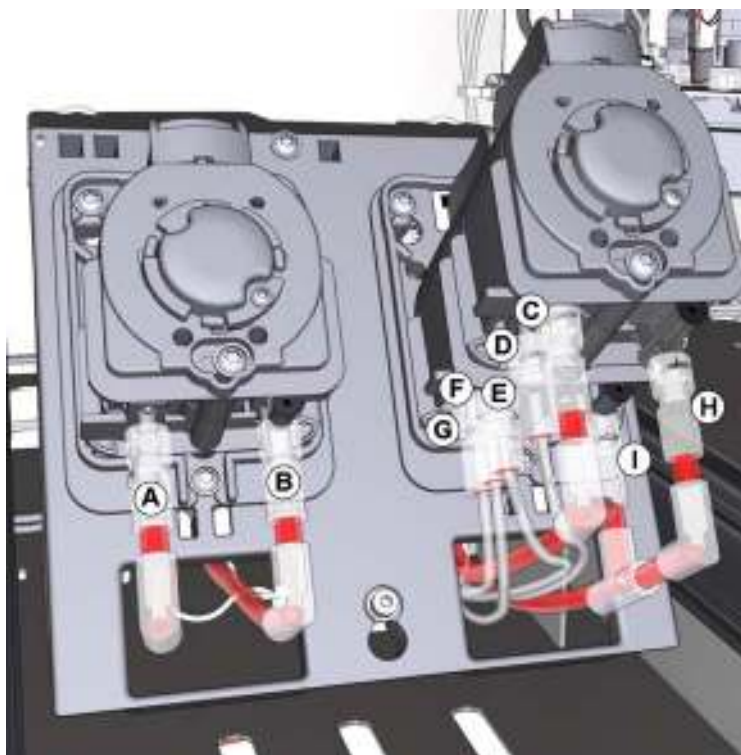


Figure 28–22: Peristaltic pump rack, front view



## 28. Tubing for internal-tanks configuration

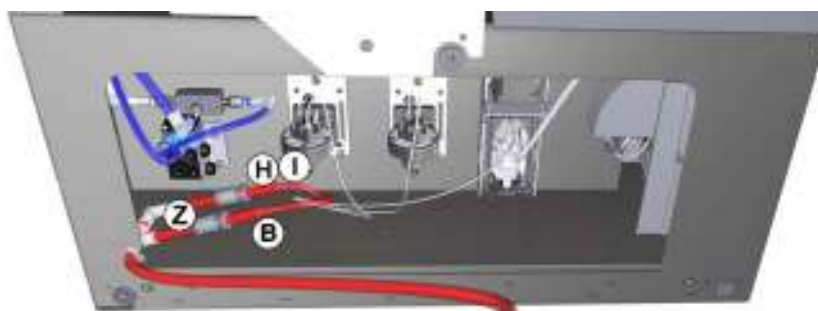


Figure 28-23: Instrument, left side view



Figure 28-24: Reagent module-WAP tubing inside the instrument, without Incubator and Washer modules

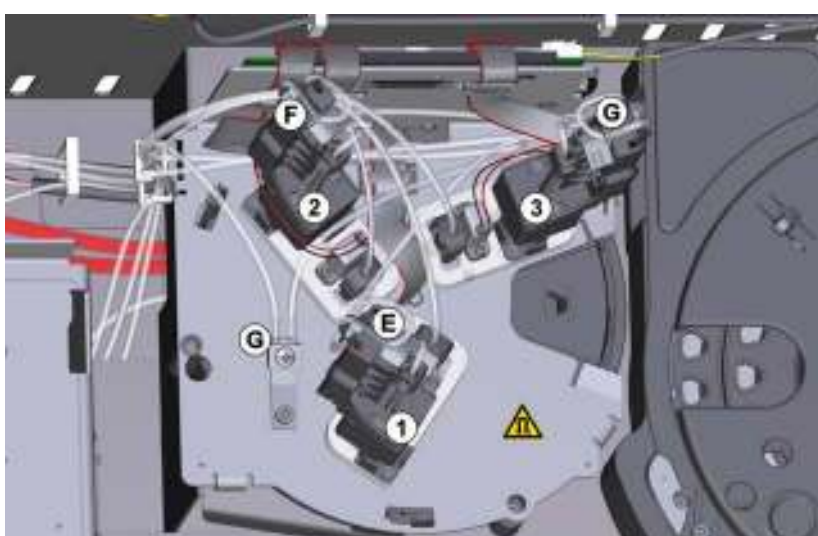


Figure 28-25: Washer Module, top view

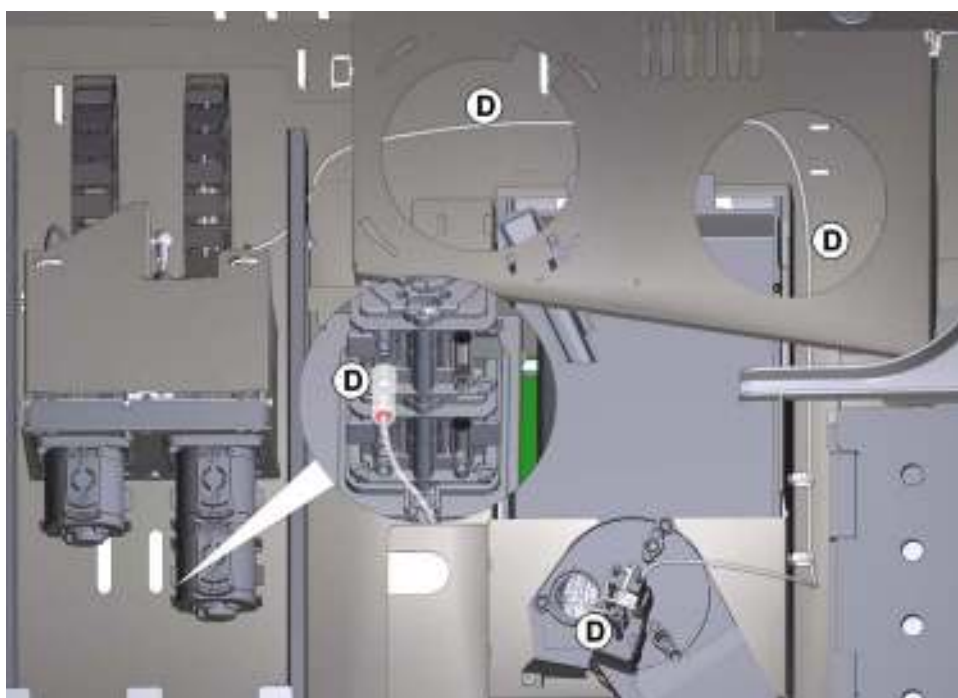


Figure 28-26: Reader-WAP tubing, inside the instrument, without Washer, Incubator and Sample modules

### 28.3.8.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
9. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.
10. Remove the Peristaltic pumps rack, see Chapter 28.3.6.3 on page 28-21.

## 28. Tubing for internal-tanks configuration

### 28.3.8.3 Removal of the Waste tubing at Right Energy Chain (WAP)

1. Unplug the fitting of the WAP-Waste interface at the pump (H-I) and at the connection at the Waste interface-Waste tank (Z).

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

2. Unplug the fitting of the Reagent Module-WAP at the pump (B) and at the drain of Reagent Module.
3. Unhook the tube at the cable holders (8).
4. Unplug the fitting of the Washer-WAP at the pump (E-F-G) and at the Washer Module. Disconnect the tubing at the three SIPAL of the Washer Module.
5. Unplug the fitting of the Reader-WAP at the pump (D) and at the Reader Module. Disconnect the tubing at the SIPAL of the Reader Module.
6. Unhook the tube at the cable holders.
7. Unhook the Right Energy Chain (4) at the last but one link (6) at the bottom.

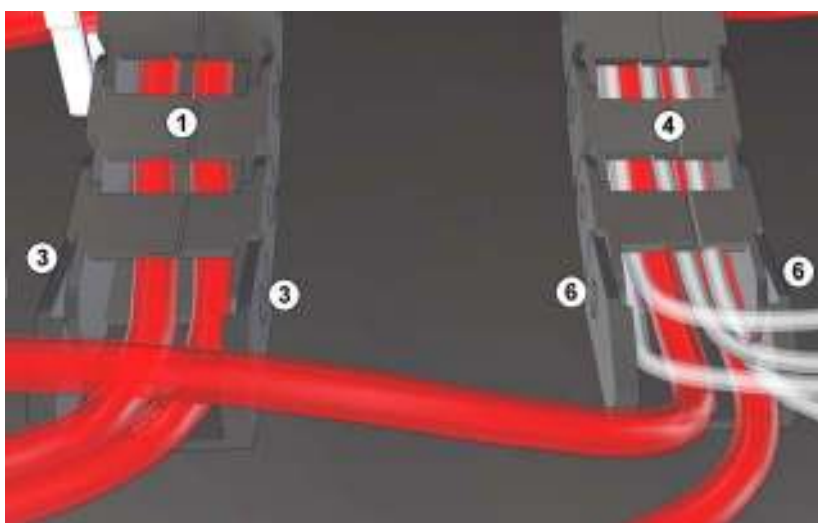


Figure 28-27: Energy chain mounted onto the bottom plate.

8. Open the frame stay and pull the tube out of the Right Energy Chain.



**28.3.8.4 Assembly of the Waste tubing at Right Energy Chain (WAP)**

1. Insert the tube into the Right Energy Chain (4).
2. Connect the tube fitting of the C, D, E, F, G, H, I connectors of the WAP onto the pump.
3. Close all frame stay of the Right Energy Chain (4).

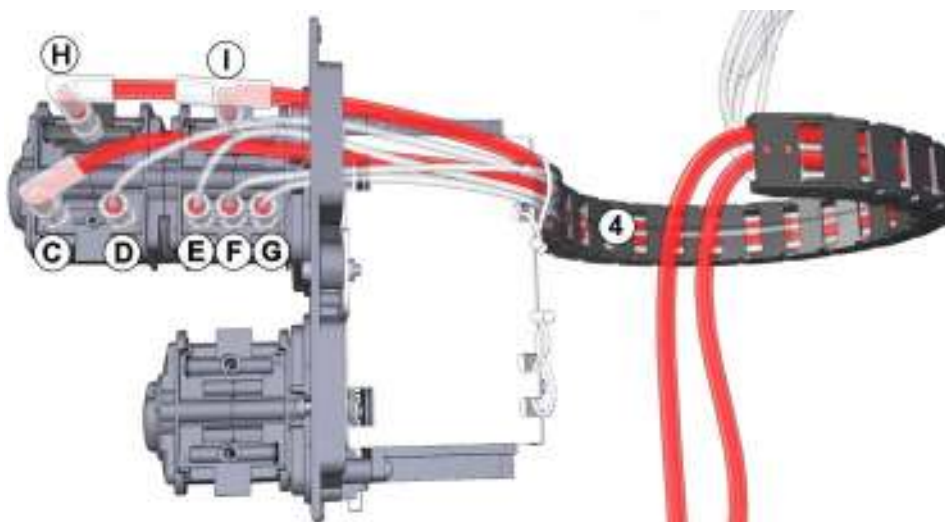


Figure 28–28: Peristaltic pumps and Right Energy Chain, bottom view

4. Hook the Right Energy Chain onto the last but one link at the bottom.
5. Lay the Reagent Module-WAP out into the cable holders and close them.
6. Lay the Washer-WAP onto the Washer Module.
7. Lay the Reader-WAP onto the Reader Module.
8. Lay the WAP-Waste interface (I-H) underneath the bottom plate to the left side of the instrument.
9. Connect the fitting of the WAP-Waste Interface (I-H) onto the connection at the Waste interface-Waste tank (Z).
10. Assemble the Peristaltic pumps rack, see Chapter 28.3.6.4 on page 28-22.
11. Check the tube assembly for proper installation.
12. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
13. Assemble the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.
14. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**28.3.8.5 Set the Waste tubing at Right Energy Chain (WAP)**

1. Perform the calibration of the Washer aspiration monitor, see Chapter 17.4.2.3 on page 17-46.
2. Perform the calibration of the Washer SIPAL lift, see Chapter 17.4.2.2 on page 17-46.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 28.3.9 Replacement of the Waste interface-Waste tank tubing

#### 28.3.9.1 Tube routing of the Waste interface-Waste tank tubing

The Waste interface-Waste tank tubing is connected from the Waste interface (Z) to the Liquid Waste tank.

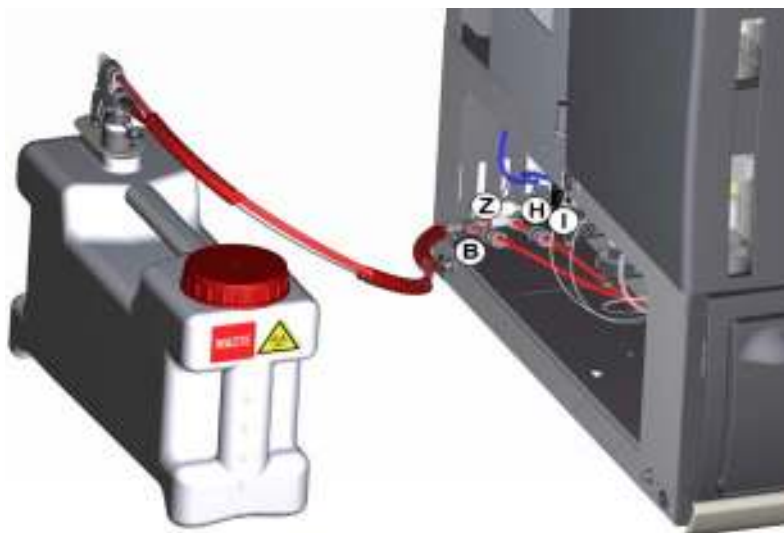


Figure 28–29: Liquid Waste tank connected to the system

#### 28.3.9.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the Left Top Cover.
6. Move the Sample Pipettor arm to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.

#### 28.3.9.3 Removal of the Waste interface-Waste tank tubing

1. Disconnect the Liquid Waste sensor cable from the Liquid Support PCB at slot J19,
2. Disassemble the routing of the sensor cable along the chassis.
3. Disconnect the fittings at the Waste interface (Z).
4. Disconnect the Liquid Waste tank.

**WARNING****Liquid Waste Spillage**

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

---

**28.3.9.4 Assembly of the Waste interface-Waste tank tubing**

1. Route the Liquid Waste sensor cable along the chassis.
2. Connect the cable to the Liquid Support PCB, at slot J19.
3. Connect the fittings at the Waste interface (Z) and clip the tubing into the chassis.
4. Connect the Liquid Waste tank.
5. Check the tube assembly for proper installation.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**28.3.9.5 Set the Waste interface-Waste tank tubing**

1. Open the **LIAISON® XS Main UI**.
2. Login and initialize the system.
3. Check the Liquid Waste presence sensor: open the **Status - Summary** toggle and verify the tank presence while connecting and dis-connecting the tank itself.
4. Check the Liquid Waste level sensor: open the **Status - Summary** toggle and verify the liquid level present in the tank while connecting the tank filled with different liquid level.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 28.3.10 Replacement of the Liquid Waste level sensor

#### 28.3.10.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.

#### 28.3.10.2 Removal of the Liquid Waste level sensor

1. Disconnect the Liquid Waste tank.

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

2. Decontaminate it according to paragraph “Decontamination of Liquid Waste tank” of the Service Manual, see Chapter 7.3.1 on page 7-3.
3. Close the dust cap of the sensor.

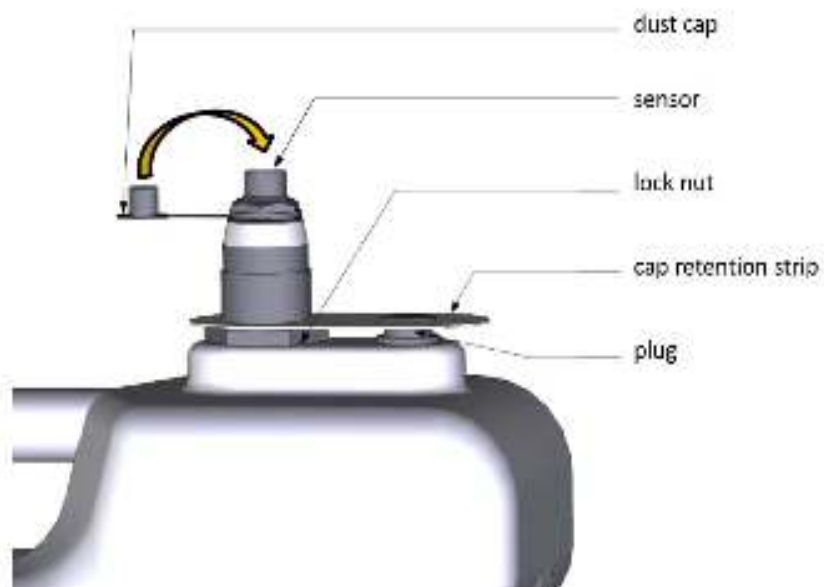
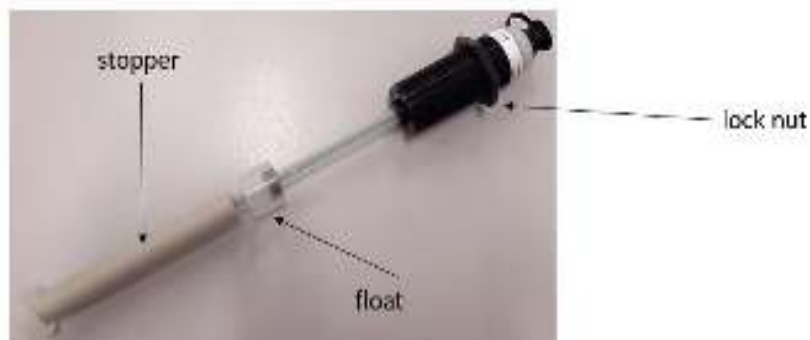


Figure 28–30: Sensor mounted in the tank.

4. Remove the plug and the cap retention strip.
5. Loosen the lock nut, unscrew the level sensor and pull it out.
6. Dispose the removed sensor.

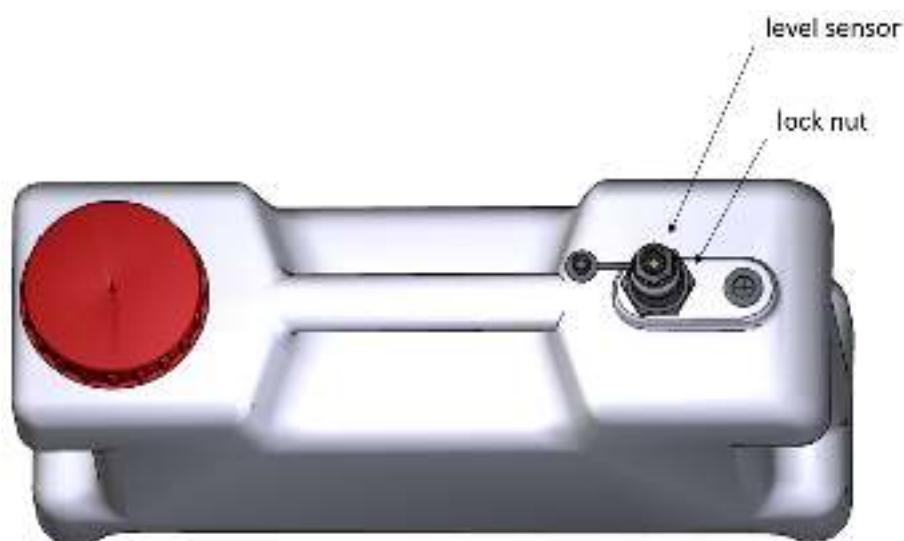
**28.3.10.3 Assembly of the Liquid Waste level sensor**

1. Unscrew the lock nut of the level sensor.



*Figure 28-31: New sensor.*

2. Screw the sensor into the thread of the tank, until the sensor almost touches the bottom of the tank.
3. Fix the sensor in the thread by tightening the lock nut.
4. Put back the cap retention strip on the outer thread of the level sensor. The plug must be above the cable entry and carefully pushed into it.
5. Check the correct alignment of the slot.
6. Connect the Liquid Waste tank.



*Figure 28-32: Tank with mounted level sensor.*

## 28. Tubing for internal-tanks configuration

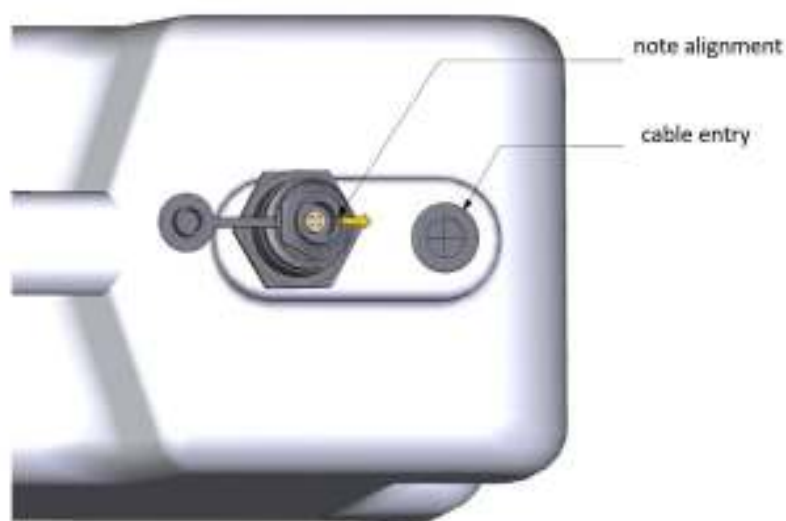


Figure 28–33: Slot alignment check.



Figure 28–34: Cap retention strip with plug for the cable entry.

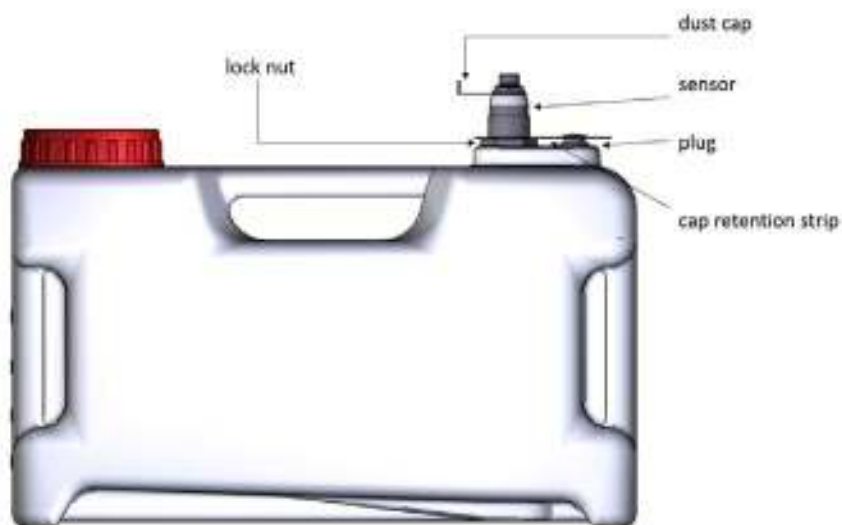


Figure 28–35: Completely mounted level sensor.

#### 28.3.10.4 Set the Liquid Waste level sensor

1. Open the **LIAISON<sup>®</sup> XS Main UI**.
2. Login and initialize the system.
3. Check the Liquid Waste presence sensor: open the **Status - Summary** toggle and verify the tank presence while connecting and dis-connecting the tank itself.
4. Check the Liquid Waste level sensor: open the **Status - Summary** toggle and verify the liquid level present in the tank while connecting the tank filled with different liquid level.
5. Requalification not needed.



### 28.3.11 Replacement of the WBDP-Washer tubings

#### 28.3.11.1 Tube routing of the WBDP-Washer tubings

The WBDP-Washer tubings are connected to the WBDP (4) connector (V-W-X) and they are routed through instrument to Washer in order to be connected to the Dispense Manifold of the three Washer SIPAL:

- V = SIPAL 1,
- W = SIPAL 2,
- X = SIPAL 3.

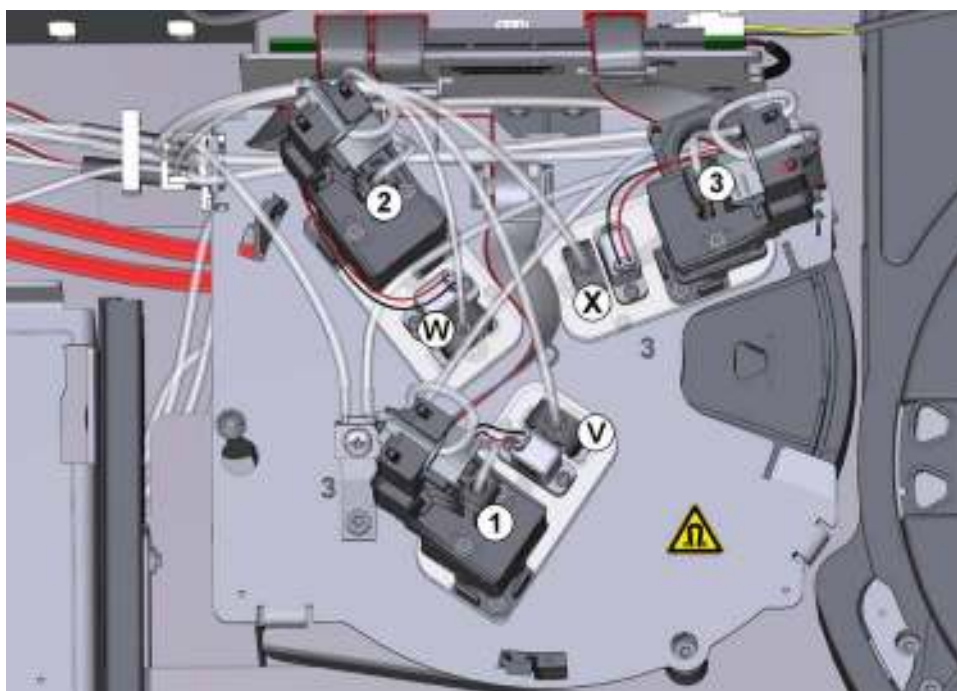


Figure 28–36: Washer Module, top view

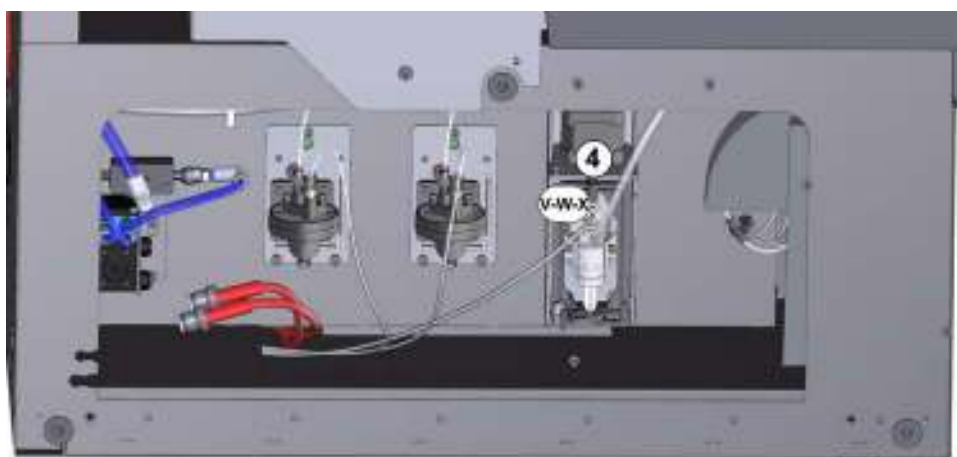


Figure 28–37: Instrument left side view



**28.3.11.2 Preliminary steps**

1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.

**28.3.11.3 Removal of the WBDP-Washer tubings**

1. Unplug the fitting of the WBDP-Washer tubings at the WBDP (V, W and X).
2. Unhook the tube at the cable holders.
3. Disconnect the tubing at the three Dispense Manifold of the SIPAL of the Washer Module.
4. Pull out the tubings from the instrument.

**28.3.11.4 Assembly of the WBDP-Washer tubings**

1. Connect the tube fittings (V, W and Z) onto the WBDP.
2. Connect the WBDP-Washer tubings to the three Dispense Manifold of the SIPAL of the Washer Module.
3. Check the tube assembly for proper installation.
4. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**28.3.11.5 Set the WBDP-Washer tubings**

1. Prime the Washer, see Chapter 11.8.1.5 on page 11-65.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 28.3.12 Replacement of the SDP-Starters tubings

#### 28.3.12.1 Tube routing of the SDP-Starters tubings

The SDP-Starter tubings are connected to the SDP pumps (fittings are marked with number tags):

- SDP-Starter1 is connected from the bottle of Starter Reagent 1 to the SDP1 left connector (Q).
- SDP-Starter2 is connected from the bottle of Starter Reagent 2 to the SDP2 left connector (R).

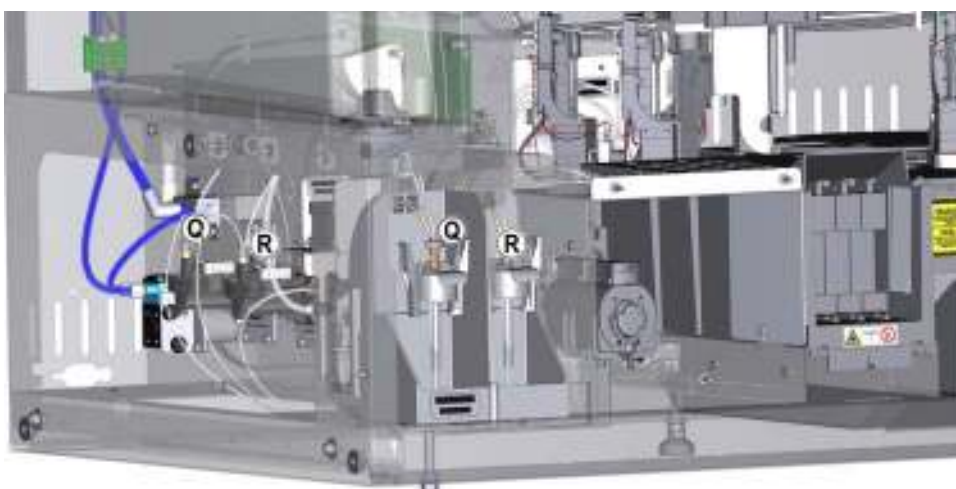


Figure 28-38: SDP-Starters tubings inside the instrument

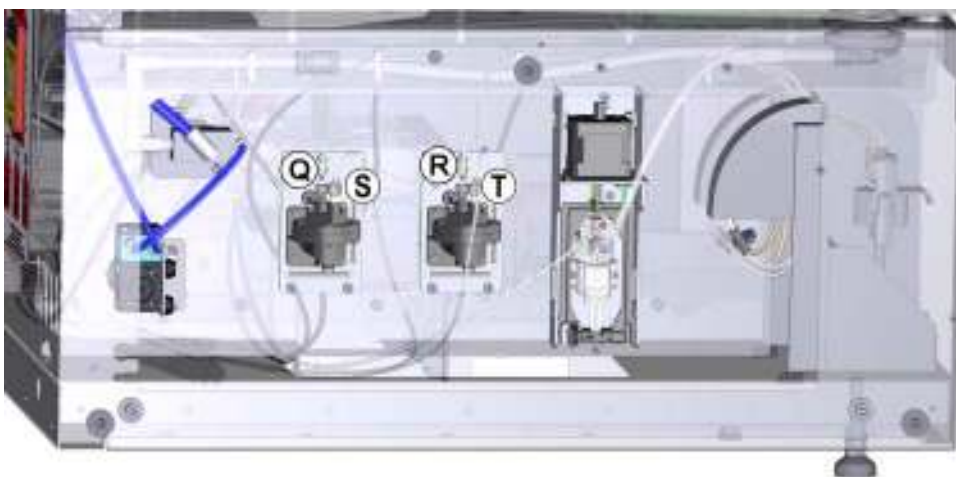


Figure 28-39: Instrument left side view

### 28.3.12.2 Preliminary steps



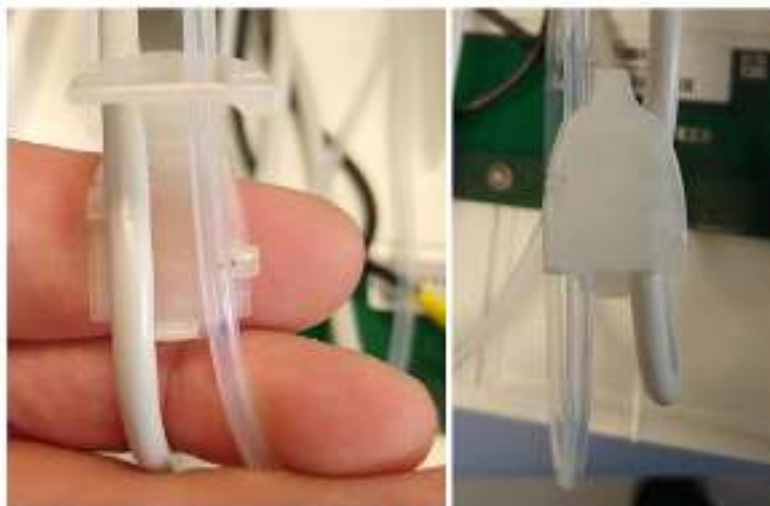
1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.

### 28.3.12.3 Removal of the SDP-Starter

1. Remove the Starter Reagent module, see Chapter 24.3.1.2 on page 24-4.
2. Open the hose clamp and the tubing protection. The hose clamp could be black or white according to the version of the Starter Reagent module.

### 28.3.12.4 Assembly of the SDP-Starters

1. Align the white hose clamps at around 22cm distance from Starter bottle and close them tightly.
2. Insert the Starter Reagent module back into the system while guiding the cables through the hole in the chassis.



*Figure 28–40: Close tightly the hose clamps.*

3. Assemble the Starter Reagent module, see Chapter 24.3.1.3 on page 24-6.
4. Check the tube for proper installation.

**28.3.12.5 Set the SDP-Starters**

1. Perform the Calibration of the Starter Bubble sensors, see Chapter 23.4.2.1 on page 23-16.
2. Prime the Reader, see Chapter 11.8.1.6 on page 11-65.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 28.3.13 Replacement of the Reader-SDP tubings

#### 28.3.13.1 Tube routing of the Reader-SDP tubings

The Reader-SDP tubings are connected to the SDP pumps (fittings are marked with number tags):

- Reader-SDP1 is connected from the SDP1 (S) to the injection of the Starter 1 on the Reader Module (1).
- Reader-SDP2 is connected from the SDP2 (T) to the injection of the Starter 2 on the Reader Module (2).

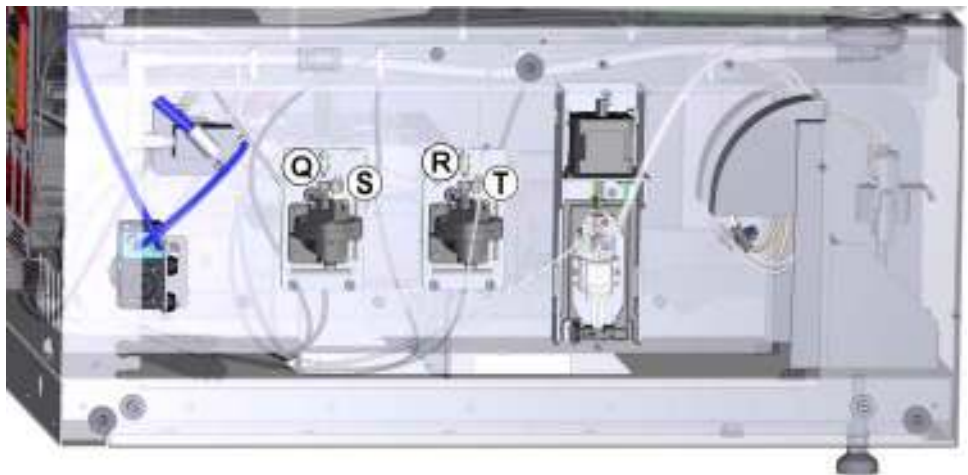


Figure 28-41: Instrument left side view

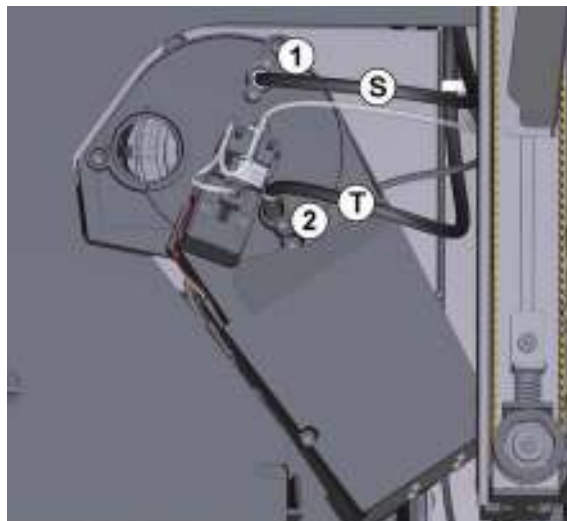


Figure 28-42: Reader Module, top view

## 28. Tubing for internal-tanks configuration

### 28.3.13.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.

### 28.3.13.3 Removal of the Reader-SDP tubings

1. Untwist at the Starter Dispense Pumps (1) and (2) the connected Reader-SDP tubings.

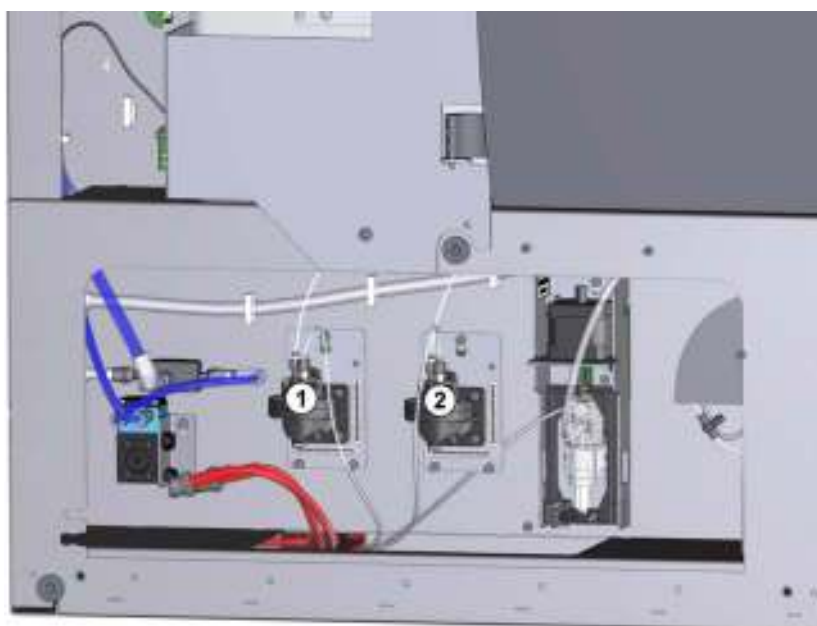
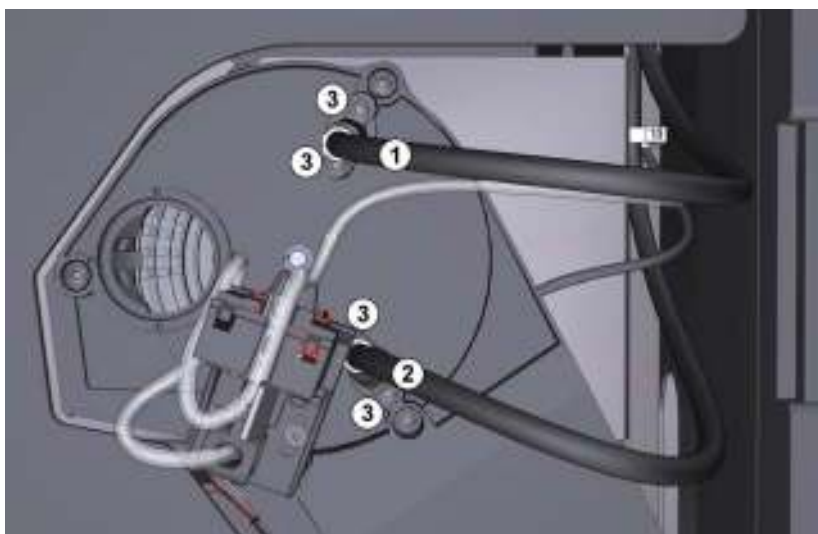


Figure 28–43: SDP overview

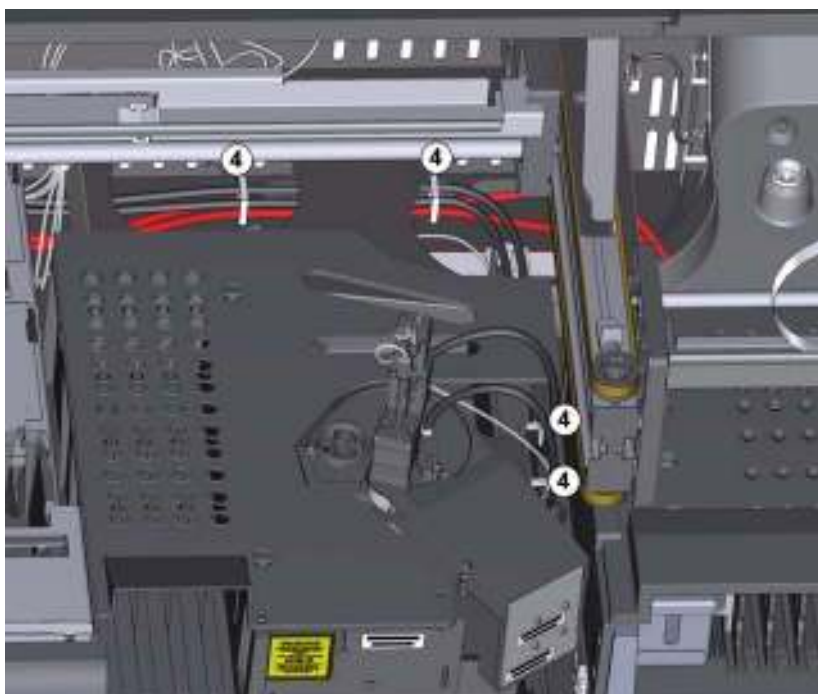
**28. Tubing for internal-tanks configuration**

2. Remove two screws (3) at Starters dispensing (1) and (2) of the Reader Module.



*Figure 28-44: Reader Module, top view*

3. At the Liquid Support CU PCBA, unplug the Starter heating cables, slots J4 and J5, see Chapter 30.4.4 on page 30-27 Or see Chapter 31.4.4 on page 31-27.
4. Remove the Reader-SDP tubings out of the cable holders (4) and pull out the tubings.



*Figure 28-45: Reader-SDP tubings overview*



## 28. Tubing for internal-tanks configuration

---

### 28.3.13.4 Assembly of the Reader-SDP tubings

1. Thread the Reader-SDP tubings from the left side of the instrument to the Reader Module.
2. Locate the Reader-SDP tubings with the four cable holder (4).

#### **NOTE**

---

**Before mounting the starter reagent tube.**

Be aware of the correct installation of the Starter 1 and Starter 2 lines.

---

3. Insert the Reader-SDP tubings (1) and (2) into the correct connection of the Reader Module and fasten each with two screws (3).
4. At the Liquid Support CU PCBA, plug the Starter heating cables, slots J4 and J5, see Chapter 30.4.4 on page 30-27 OR see Chapter 31.4.4 on page 31-27.
5. At the SDP (1) and (2), connect the Reader-SDP tubings.
6. Check the tube assembly for proper installation.
7. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
8. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 28.3.13.5 Set the Reader-SDP tubings

1. Prime the Reader, see Chapter 11.8.1.6 on page 11-65.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 28.3.14 Replacement of the Wash station cover

#### 28.3.14.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers. Move pipettor arms to a safe location.
6. Remove the Sample Module cover, see Chapter 21.3.1.2 on page 21-5.

#### 28.3.14.2 Removal of Wash station cover

1. Unscrew three screws (2) of Wash station cover (1).
2. Move the SPOLV pipettor arm to the rear left position.
3. Lift Wash station cover carefully out and set it aside.

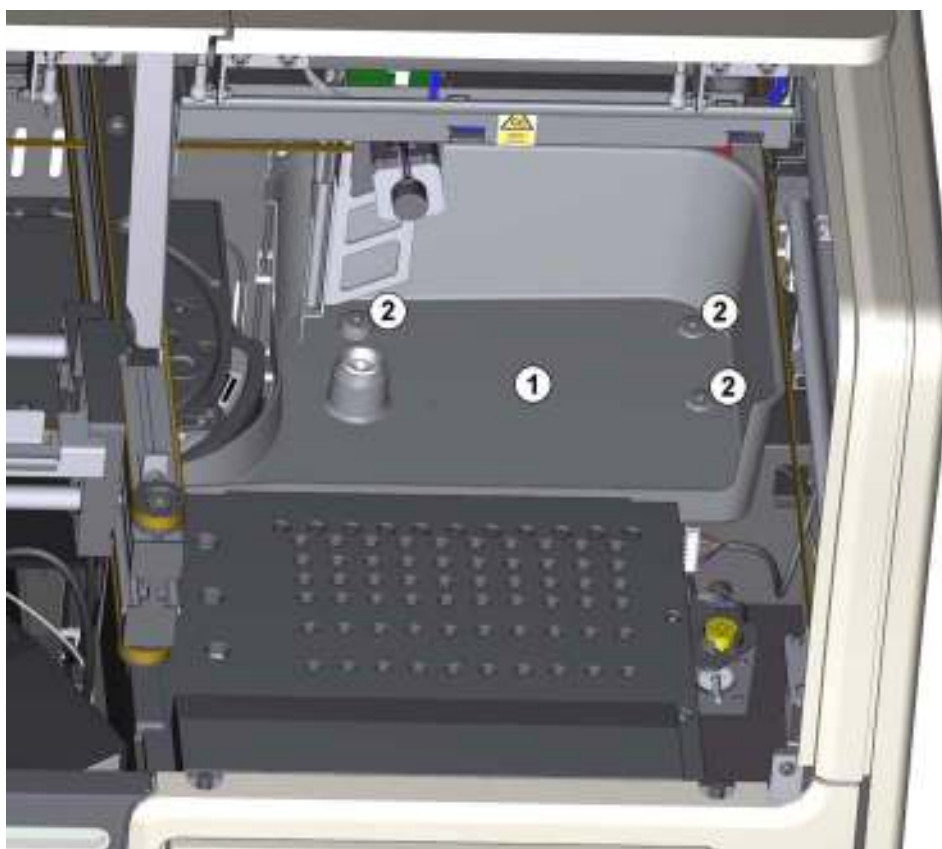


Figure 28–46: Wash Station cover

## 28. Tubing for internal-tanks configuration

---

### 28.3.14.3 Assembly of the Wash station cover

1. Set the bore of the Wash station cover over the Wash Station and slide the Wash Station carefully into the bore.

---

**CAUTION****Cover is fragile**

Be careful when fastening screws for the Wash station cover .  
The cover is fragile and could break or could get cracks.

---

2. Fasten carefully three screws (2).
3. Assemble the Sample Module cover, see Chapter 21.3.1.3 on page 21-6.
4. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 28.3.15 Replacement of the Wash station

#### 28.3.15.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers. Move pipettor arms to a safe location.
6. Remove the Wash station cover, see Chapter 28.3.14.2 on page 28-47.

#### 28.3.15.2 Removal of the Wash station

1. Lift the Wash station (1) carefully up by slightly turning left and right.

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

2. Unscrew fitting (2) at Wash station bottom and set Wash Station aside.

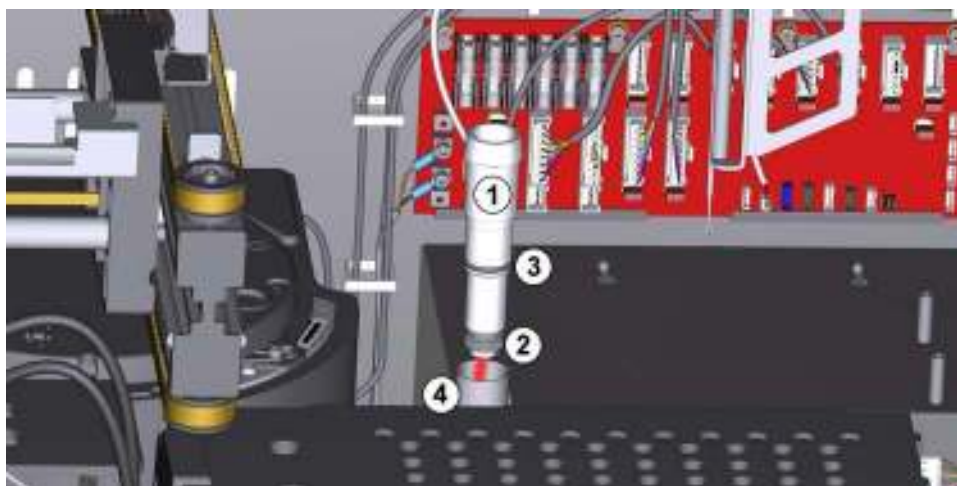


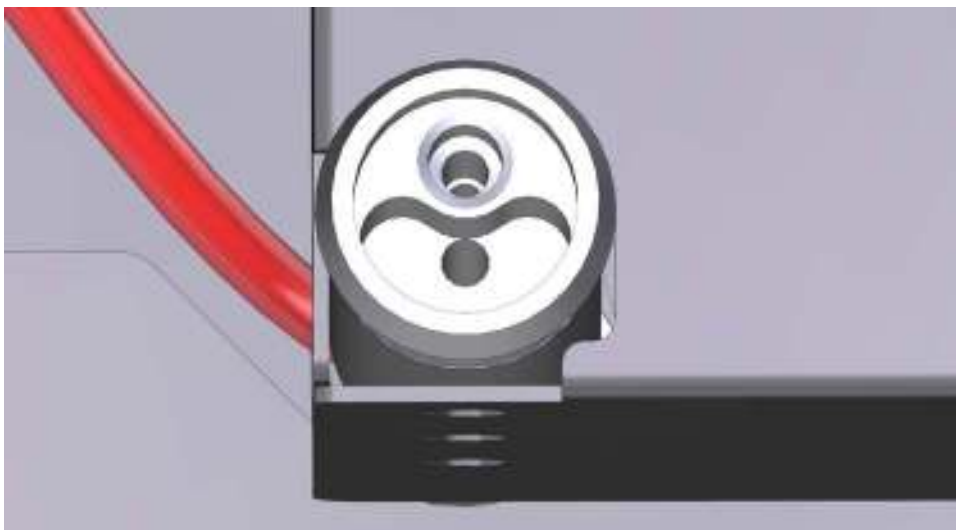
Figure 28-47: Wash station inside the instrument

**28. Tubing for internal-tanks configuration**

---

**28.3.15.3 Assembly of the Wash station**

1. Fasten the fitting (2) on the Wash station (1).
2. Moisten the seal (3) with a little bit grease.
3. Slide the Wash station carefully into the holding tube (4). Take care of the waste tubing, there shall be no nicks or not to be snapped off.
4. Check Wash station for correct orientation, see Figure 28–48: on page 28-50.



*Figure 28–48: Correct orientation of the Wash station, top view*

5. Check the tube assembly for proper installation.
6. Assemble the Wash station cover, see Chapter 28.3.14.3 on page 28-48.

**28.3.15.4 Set the Wash station**

1. Perform the teaching of the Right pipettor on the Wash station, see Chapter 13.15 on page 13-32.
2. Prime the SPOLV, see Chapter 11.8.1.4 on page 11-64.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

## 28.4 Verifying of tank presence sensors and level sensors

In this section information about the initialization and verifying of different level sensors and presence sensors can be found.

### NOTE

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).  
Do not run any of the software at the same time.

### 28.4.1 Initialization routine

- Close any open software and start the **Service Software** from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog Controlled Modules** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 28.4.2 Verifying of the DI water tank presence sensor

- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Sensors** button and then choose **Liquid State**.
- Click the **Auto** update button to play the function.
- Insert and remove the DI water tank, by checking on the display the correct status of the tank (Present or Absent).
- Click the **Auto** update button to stop the function.

### 28.4.3 Verifying of the DI water tank level sensor

- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Sensors** button and then choose **Liquid State**.
- Click the **Auto** update button to play the function.
- Insert and remove the DI water tank filled at different levels, by checking on the display the correct status of the tank (Empty or Full).
- Click the **Auto** update button to stop the function.

#### **28.4.4 Verifying of the Wash Buffer tank presence sensor**

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Sensors** button and then choose **Liquid State**.
- c. Click the **Auto** update button to play the function.
- d. Insert and remove the WB tank, by checking on the display the correct status of the tank (Present or Absent).
- e. Click the **Auto** update button to stop the function.

#### **28.4.5 Verifying of the Wash Buffer tank level sensor**

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Sensors** button and then choose **Liquid State**.
- c. Click the **Auto** update button to play the function.
- d. Insert and remove the WB tank filled at different levels, by checking on the display the correct status of the tank (Empty or Full).
- e. Click the **Auto** update button to stop the function.

#### **28.4.6 Verifying of the Liquid Waste tank presence sensor**

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Sensors** button and then choose **Liquid State**.
- c. Click the **Auto** update button to play the function.
- d. Insert and remove the Liquid Waste tank, by checking on the display the correct status of the tank (Present or Absent).
- e. Click the **Auto** update button to stop the function.

#### **28.4.7 Verifying of the Liquid Waste tank level sensor**

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Sensors** button and then choose **Liquid State**.
- c. Click the **Auto** update button to play the function.
- d. Insert and remove the Liquid Waste tank filled at different levels, by checking on the display the correct status of the tank (Empty or Full).
- e. Click the **Auto** update button to stop the function.

## 28.5 Direct Drain Configuration

LIAISON<sup>®</sup> XS analyzer supports also a direct drain configuration, where a waste tubing connected to the system allows to dispose liquid waste directly into a collector drain by-passing the liquid waste tank.

A dedicated direct drain kit shall be ordered to implement this configuration.

### NOTE

Direct Drain is an optional solution. All systems will be shipped with the liquid waste tank. Direct Drain configuration can be implemented during installation or at a later time.

### NOTE

The tubing provided with the direct drain kit has a length of 5 m.

The direct drain tubing features the same fitnut connectors of the standard waste tubing, to be screwed onto the internal waste lines of the instrument, and the same T-connector to be clipped to the chassis.

The open end of the direct drain tubing needs to be fixed to the collector drain.



Figure 28–49: Direct Drain tubing

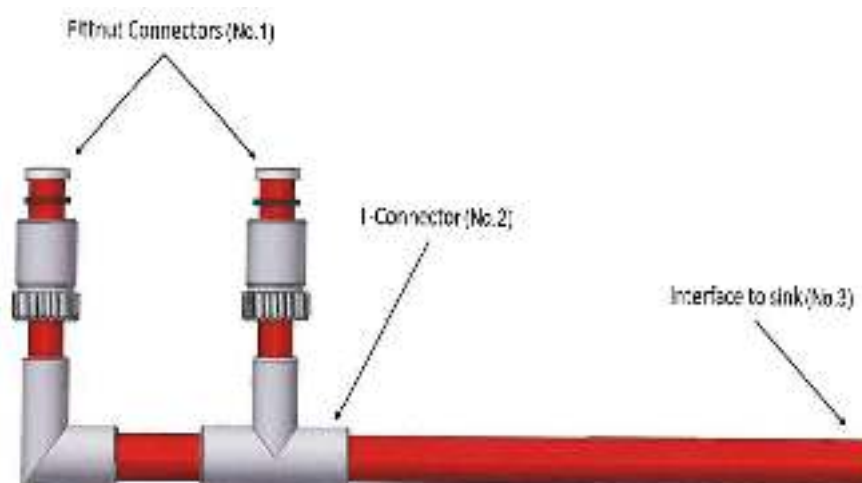


Figure 28–50: Direct Drain tubing layout

### 28.5.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the Left Top Cover.
6. Move the Sample Pipettor arm to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Waste Interface-Waste tank tubing, see Chapter 28.3.9.3 on page 28-32.

### 28.5.2 Installation

1. Screw the fitnut connectors of the Direct Drain tubing onto the internal waste lines of the instrument.
2. Clip the T-connector of the Direct Drain tubing to the chassis.
3. Verify that the Direct Drain tubing reaches the waste collector drain available at customer site.
4. Guide the open end of the Direct Drain tubing into the drain and securely fix it.

---

**CAUTION**

To avoid back flow into the instrument, the sink has to be at or below the bench where the instrument is placed on.

The routing of the Direct Drain tubing must not exceed 200 mm above the sink level.

---

### 28.5.3 Parameter setting

**NOTE**

Direct Drain can be used with software version 1.3.5 or following versions.

The FW provides an additional parameter for Direct Drain setting: if enabled, this parameter shows to the system the presence state of an empty liquid waste tank.

**For software version 1.3.5 only:**

Before starting the **Service Software**, ensure that file *ParameterWhiteListe-135-R01.txt* is copied in the "C:\LiaisonXS\Tools\ServiceSoftware\cfg\" folder and renamed to *ParameterWhiteListe.txt* (delete/replace the previous file).

- a. Start the **Service Software** from the Windows Administrator account.
- b. In the **Service Software**, at the navigation window, choose **System** and the **Parameter** bracket.



## 28. Tubing for internal-tanks configuration

- c. Change the Module to “Liquid Support” in the Selection Box.
- d. In the “Read Parameter(s)” field, enter the value “242” and press the **Update** button.
- e. Select the parameter and change its value according to:
  - 0: Use of Wast tank (default)
  - 1: Use of Direct Drain
- f. Confirm by pressing the **Write (Module RAM only)** button.
- g. Store the parameter by pressing the **Non-Volatile Memory** button.
- h. Confirm the pop-up dialog.

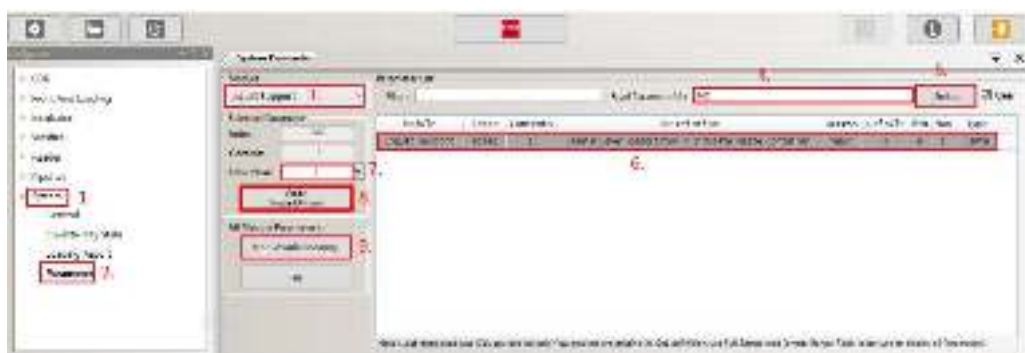


Figure 28–51: Direct Drain parameter setting

## 28.5.4 Requalification

- a. Insert all needed consumables and liquids.
- b. In the **Service Software**, at the navigation window, choose **System** and the **Cuvette Tray State** bracket.
- c. Set the cuvette trays.
- d. In the **Service Software**, at the navigation window, choose **System** and the **General** bracket. Click onto the **Liquid** button and choose the **Prime all** button to open the dedicated window.
- e. Customize on the screen the **Prime all** window:
  - Cycles Washer Port 1: 11
  - Cycles Washer Port 2: 5
  - Cycles Washer Port 3: 5
  - Cycles Measurement: 20
  - Cycles Pipettor: 11
  - Wash time: 18
  - OLV Calibration: Disabled
  - Liquid Control: Disabled
  - Liquid: WashBuffer
- f. Press the **Execute** button
- g. Check the connection to the Direct Drain tubing and to the sink, to ensure no leakage is present.



# 29 Tubing for external-tanks configuration

This chapter shows an overview of the tubing and the replacement of the tubings in the instrument.

## 29.1 Safety

### NOTE

The view of the instrument is a view from the bottom and some parts are transparent pictured.

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injuries and material damages.

- Follow all safety instructions, see Chapter 2.5 on page 2-7.
- Follow all working instructions included in this manual.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

### WARNING

#### Pinched or bend tubing

- Do not bend, kink or damage any tubes while disassembling or assembling the tubing sets.
- At replaceable tank, canister or bottles the required minimal distance for laying out the hoses must be taken into account.

### WARNING



#### Liquid in the System

Liquid which gets into the system can cause illnesses with deadly consequences in case of contact. The system can be damaged by liquids.

- Switch off the system.
- Separate the system from the mains supply.
- Wear suitable protective clothing.
- Clean, disinfect and dry the system according to the applicable local and national provisions, legislation and laboratory procedures.
- Assure the system is completely dry before switch it on.

**Risk of infection!**

Parts, in contact to potentially infectious liquid, must be treated as potentially infectious. Improper handling of infectious parts can cause skin irritations, illnesses and possible death.

- Observe local and national provisions, legislation and laboratory regulations.
  - Use appropriate disposable gloves.
  - Use an appropriate lab coat.
  - Use an appropriate eye protection.
  - Avoid contact between skin/mucous membrane and samples/test reagents or parts of the instrument.
  - Do not pipette by mouth.
  - Do not eat, drink, smoke, apply cosmetics or handle contact lenses when handling human-sourced material or contaminated system components.
  - Clean and disinfect the system immediately if potentially infectious material has been spilled.
  - Any reagent spills should be washed with a chemical use sodium hypochlorite solution with 0.5% active chlorine (e.g. dilution 1:10 of a solution at 5% active chlorine) and disposed as though potentially infectious.
  - All samples, biological reagents and materials used in the assays must be considered potentially able to transmit infectious agents. They should therefore be disposed in accordance with the prevailing regulations and guidelines of the agencies holding jurisdiction over the laboratory, and the regulations of each Country.
  - Liquid waste must be decontaminated with a chemical use sodium hypochlorite solution with 0.1% active chlorine (e.g. dilution 1:50 of a solution at 5% active chlorine) for at least half a hour.
  - Do not use broken or chipped tubes or bottles.
  - Observe the instructions in the package inserts for a correct use of the reagents.
- 

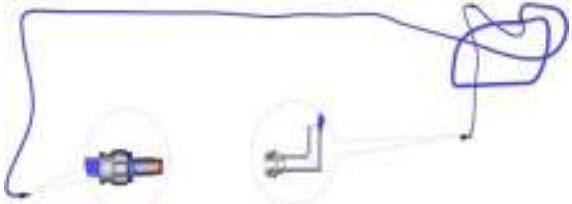
**29.1.1 Liquid Distribution Overview****NOTE**

For an overview of the complete liquid distribution, see Chapter 4.1.7 on page 4-8.

---

## 29.2 Overview

### 29.2.1 Overview tubing

| Chapter | Tubing description             |                                                                                      |
|---------|--------------------------------|--------------------------------------------------------------------------------------|
| 29.3.1  | MANN FILTER-PRP                |   |
| 29.3.3  | SYSTEM LIQUID TANK-MANN FILTER |  |
| 29.3.4  | PRP-SPOLV                      |  |
| 29.3.5  | WB TANK-WBDP                   |  |

## 29. Tubing for external-tanks configuration

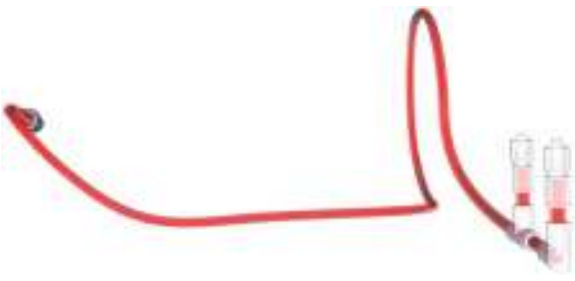
| Chapter | Tubing description                          |                                                                                      |
|---------|---------------------------------------------|--------------------------------------------------------------------------------------|
| 29.3.10 | Waste interface-Waste tank<br>Z             |    |
| 29.3.12 | WBDP-Washer Dispense Manifold<br>U, V and Z |   |
| 29.3.13 | SDP-Starters<br>(with labels)               |  |
| 29.3.14 | Reader-SDP                                  |  |

Table 29-1: Tubing overview

**29.2.2 Overview Waste tubing at Left Energy Chain (LWP)**

| Chapter | Tubing description       |                                                                                    |
|---------|--------------------------|------------------------------------------------------------------------------------|
| 29.3.8  | Wash station-LWP<br>A    |  |
| 29.3.8  | LWP-Waste interface<br>B |  |

*Table 29–2: Waste tubing at Left Energy Chain (LWP)***29.2.3 Overview Waste tubing at Right Energy Chain (WAP)**

| Chapter | Tubing description             |                                                                                      |
|---------|--------------------------------|--------------------------------------------------------------------------------------|
| 29.3.9  | WAP-Waste interface<br>H and I |  |
| 29.3.9  | Reagent Module-WAP<br>C        |  |

## 29. Tubing for external-tanks configuration

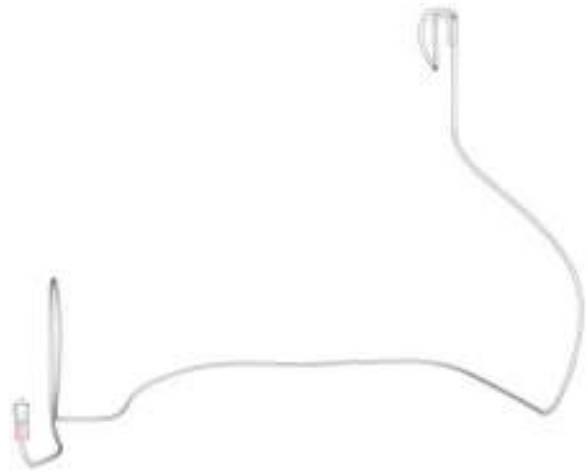
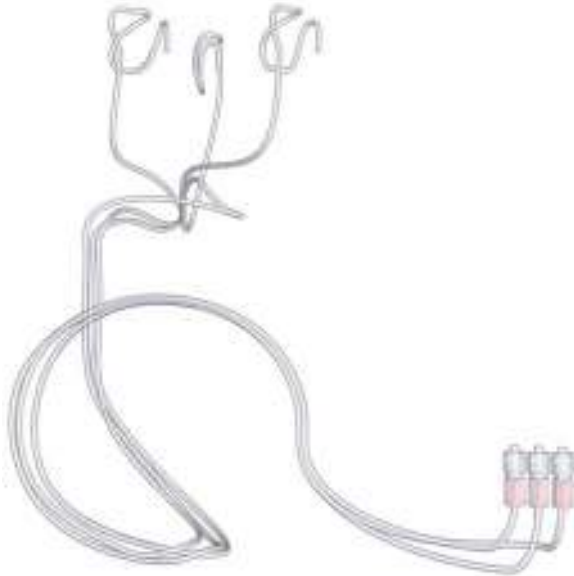
| Chapter | Tubing description                                                             |                                                                                     |
|---------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 29.3.9  | Reader-WAP<br>D                                                                |   |
| 29.3.9  | Washer-WAP:<br>E - SIPAL 1<br>F - SIPAL 2<br>G - Grounding pipe and<br>SIPAL 3 |  |

Table 29–3: Waste tubing at Right Energy Chain (WAP)



## 29.3 Replacements and repairs

This section contains instructions for the repair of individual components of the tubing and Wash Station. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

### 29.3.1 Replacement of the MANN filter

#### 29.3.1.1 Overview of the MANN filter

The MANN filter acts on the System Liquid supply to the SPOLV. It is placed upstream of the PRP: the filter inlet is connected to the tubing from the System Liquid tank, while the filter outlet is connected to the PRP inlet (black connector).

The MANN filter has a mesh size of 200 µm.

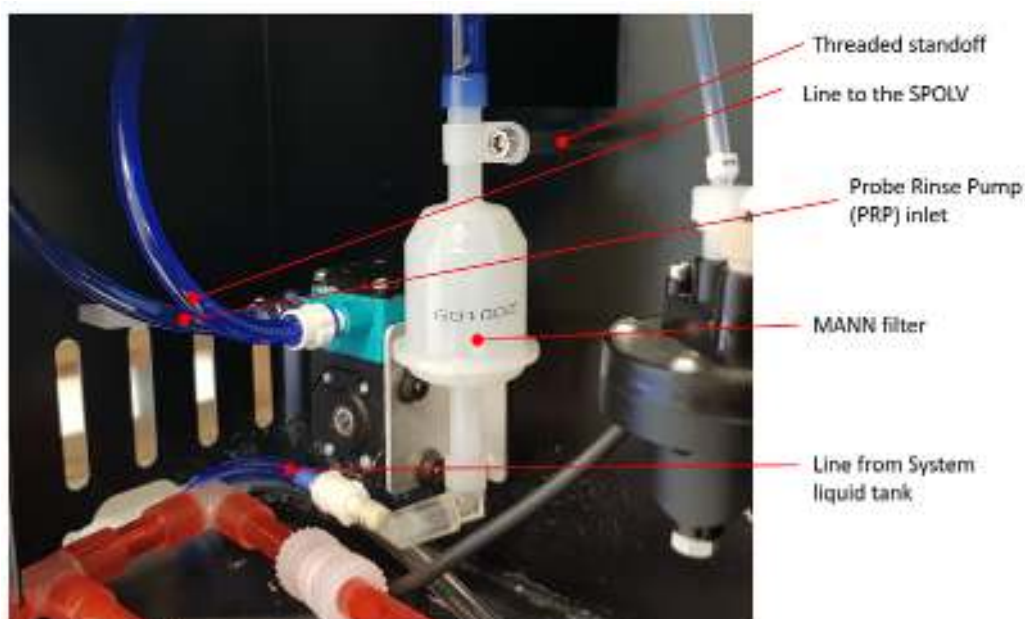


Figure 29–1: MANN filter inside the instrument

## 29. Tubing for external-tanks configuration

### 29.3.1.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Disconnect the System Liquid tank, unplugging both sensor cable and tubing.

### 29.3.1.3 Removal of the MANN filter

#### CAUTION



#### Possible flooding

Be aware that the removal of the MANN filter can spill liquid into the instrument.

Take precaution measures:

- Keep a cleaning cloth and a small bucket close by.
- Wear suitable protective clothing and gloves.

1. Remove the fitting (1) at the PRP inlet.
2. Unplug the fitting (2) at the System Liquid tubing.
3. Unscrew the nut (3) of the hose clamp.

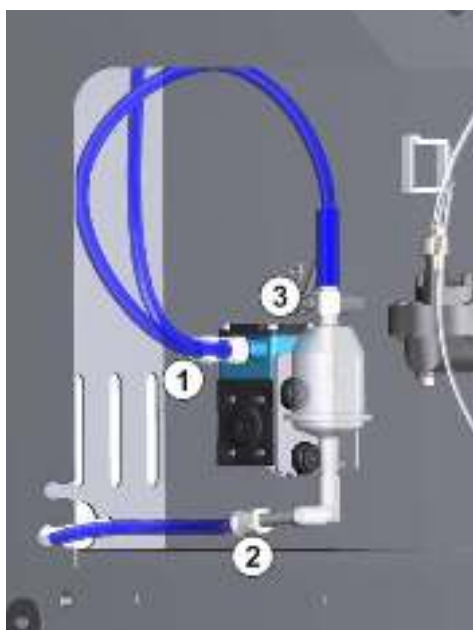


Figure 29–2: Removal of the MANN filter

4. Carefully remove the MANN filter from the instrument.

#### 29.3.1.4 Assembly of the MANN filter

1. Screw the nut (3) of the hose clamp in order to fix the MANN filter to the threaded standoff.
1. Screw the fitting (1) to the PRP inlet.
2. Plug the fitting (2) to the System Liquid tubing.
3. Check the tube assembly for proper installation.
4. Connect the System Liquid tank, plugging both sensor cable and tubing.
5. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

#### 29.3.1.5 Set the MANN filter

1. Prime the SPOLV, see Chapter 12.8.1.4 on page 12-67.

#### **WARNING**



---

#### **Air inside the MANN filter**

While the system is priming the SPOLV, please observe the liquid flow from the System liquid tank through the MANN filter: pay attention that the filter is not full of air. If needed, tap on it to remove air as much as possible.

---

2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

## 29.3.2 Replacement of the System Liquid tank sensor cable

### 29.3.2.1 Cable routing of the System Liquid tank sensor cable

The specified cable is connected from the Liquid Support PCBA (slots J15 and J21, see Chapter 31.4.4 on page 31-27) to the System Liquid tank.

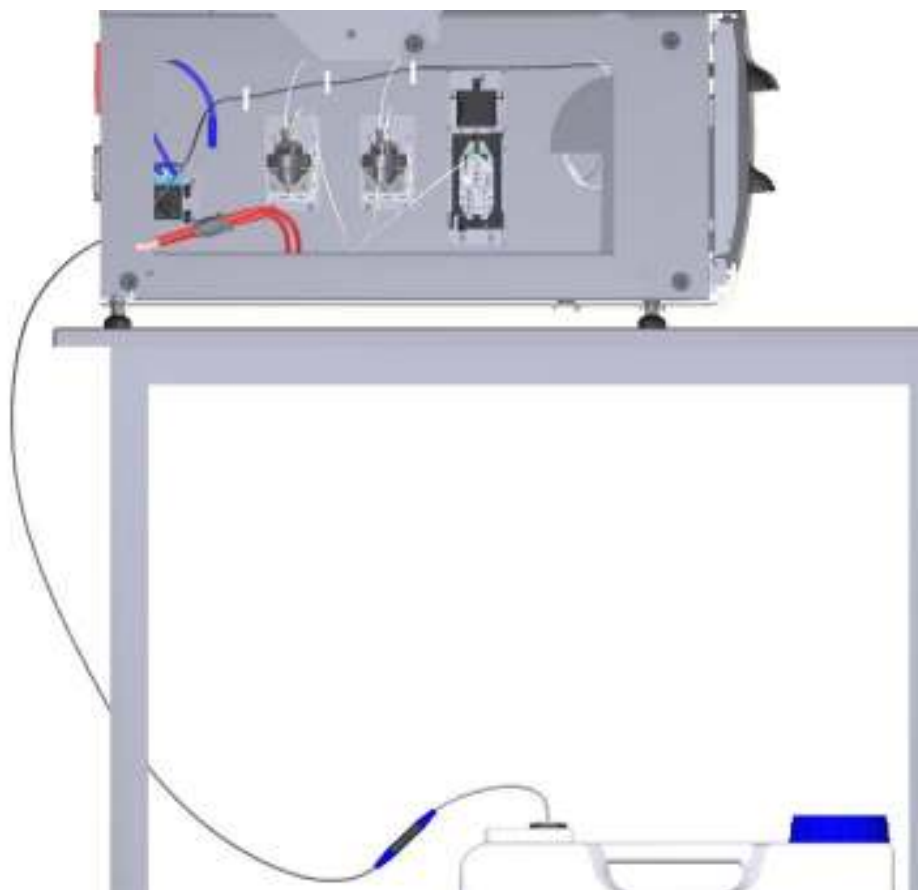


Figure 29–3: Routing of the System Liquid tank sensor cable

### 29.3.2.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Disconnect the System Liquid tank, unplugging both sensor cable and tubing.

### 29.3.2.3 Removal of the System Liquid tank sensor cable

1. At the Liquid Support PCBA, unplug the connectors J15 (System Liquid level switch) and J21 (System Liquid presence).

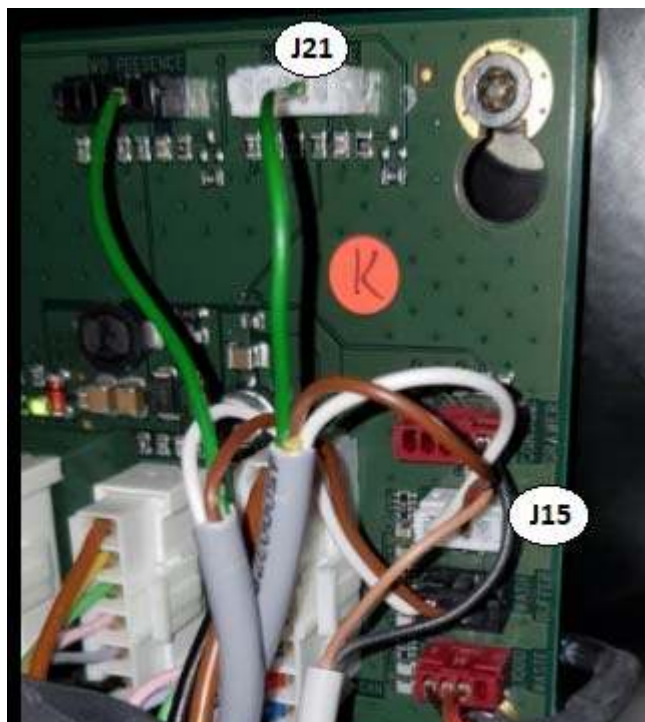


Figure 29-4: System Liquid tank sensor cable connection on the Liquid Support PCBA

2. Open the cable clips in the pumps area.
3. Disassemble the routing of the sensor cable along the chassis.
4. Remove the three blue cable hoses which hold together the System Liquid sensor cable and tubing.
5. Slide the sensor cable carefully out of the system from the left back hole in the chassis.

### 29.3.2.4 Assembly of the System Liquid tank presence sensor cable

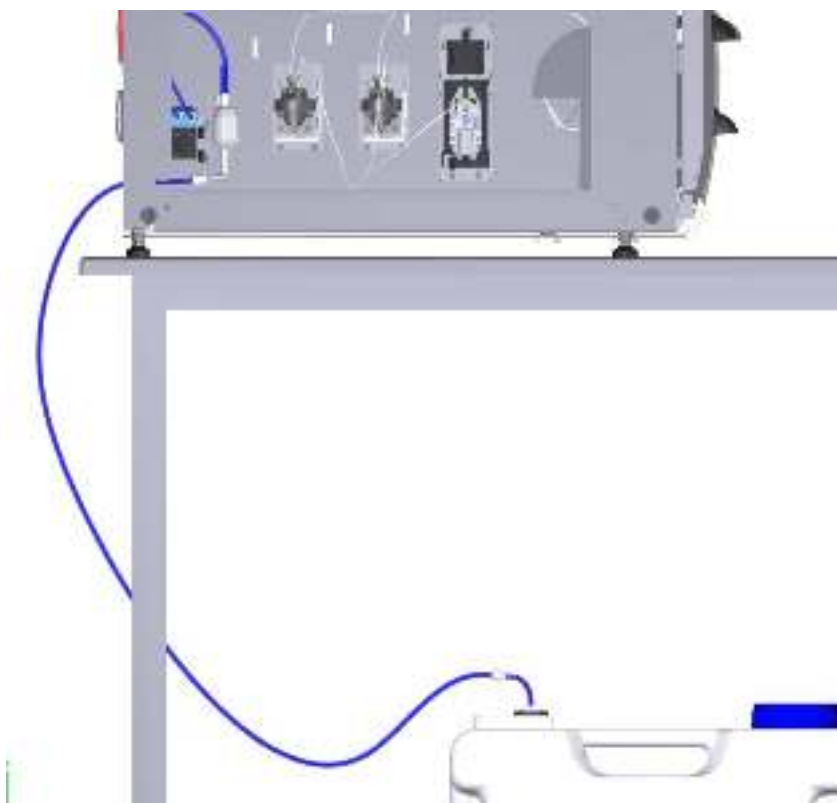
1. Insert the sensor cable through the left back hole in the chassis.
2. Carefully route the sensor cable along the chassis, through the cable clips in the pumps area and up to the Liquid Support PCBA.
3. Plug the connectors J15 (System Liquid level switch) and J21 (System Liquid presence) onto the Liquid Support PCBA.
4. Close the cable clips.
5. Connect the System Liquid tank, plugging both sensor cable and tubing.
6. Assemble the three blue cable hoses on the System Liquid sensor cable and tubing.
7. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**29.3.2.5 Set the System Liquid tank presence sensor cable**

1. Check the System Liquid tank presence sensor, see Chapter 29.4.2 on page 29-52.
2. Check the System Liquid tank level switch, see Chapter 29.4.3 on page 29-52.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

**29.3.3 Replacement of the SYSTEM LIQUID TANK-MANN FILTER tubing****29.3.3.1 Overview of the SYSTEM LIQUID TANK-MANN FILTER tubing**

The specified tubing is connected from the System Liquid tank to the Mann filter tubing.



*Figure 29–5: SYSTEM LIQUID TANK-MANN FILTER tubing*

### 29.3.3.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Disconnect the System Liquid tank, unplugging both sensor cable and tubing.

### 29.3.3.3 Removal of the SYSTEM LIQUID TANK-MANN FILTER tubing

#### CAUTION



#### Possible flooding

Be aware that the removal of the tube can spill liquid into the instrument.

Take precaution measures:

- Keep a cleaning cloth and a small bucket close by.
- Wear suitable protective clothing and gloves.

1. Remove the three blue cable hoses that hold together the System Liquid sensor cable and tubing.
2. Disconnect the System Liquid tubing (1) from the MANN filter tubing.

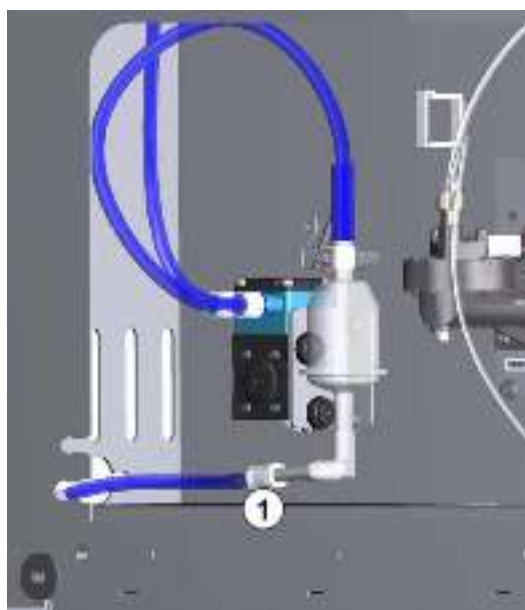


Figure 29–6: Disconnecting the System Liquid tubing from the Mann filter

## 29. Tubing for external-tanks configuration

---

3. Slide the System Liquid tubing carefully out from the left back hole in the chassis.

### 29.3.3.4 Assembly of the SYSTEM LIQUID TANK-MANN FILTER tubing

1. Insert the System Liquid tubing through the left back hole in the chassis.
2. Connect the System Liquid tubing (1) to the MANN filter tubing.
3. Connect the System Liquid tank, plugging both sensor cable and tubing.
4. Assemble the three blue cable hoses on the System Liquid sensor cable and tubing.
5. Check the tube assembly for proper installation.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 29.3.3.5 Set the SYSTEM LIQUID TANK-MANN FILTER tubing

1. Prime the SPOLV, see Chapter 12.8.1.4 on page 12-67.

#### **WARNING**



#### **Air inside the MANN filter**

While the system is priming the SPOLV, please observe the liquid flow from the System liquid tank through the MANN filter: pay attention that the filter is not full of air. If needed, tap on it to remove air as much as possible.

2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 29.3.4 Replacement of the PRP-SPOLV tubing

#### 29.3.4.1 Tube routing of the PRP-SPOLV tubing

The PRP-SPOLV tubing has a fitting connected to the outlet (right connector) of the PRP (1). The tube (P) is routed through cable holders to the SPOLV and it is fixed with round cable holder to the Z-Bracket.

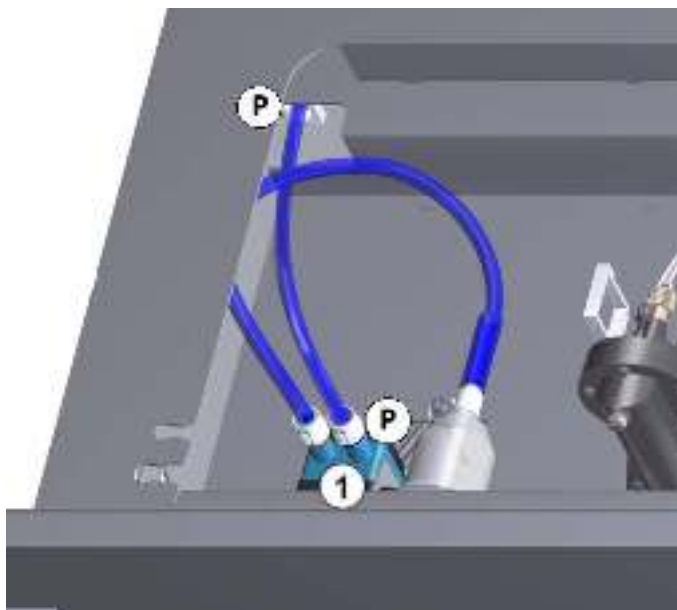


Figure 29–7: PRP-SPOLV tubing inside the instrument, side view

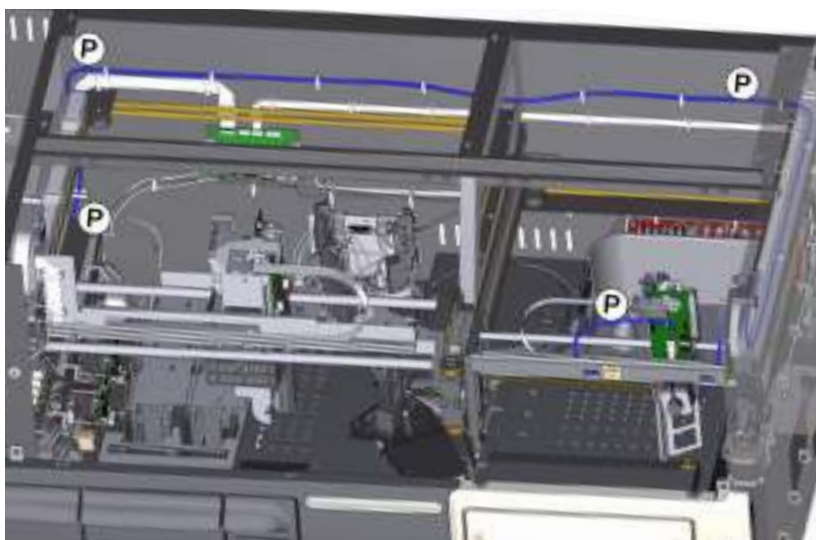


Figure 29–8: PRP-SPOLV tubing inside the instrument, front view

## 29. Tubing for external-tanks configuration

---

### 29.3.4.2 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Disconnect the System Liquid tank, unplugging both sensor cable and tubing.

### 29.3.4.3 Removal of the PRP-SPOLV tubing

1. Unscrew the fitting at the PRP pump (1).
2. Thread the tube (P) through the left inner chassis opening, inside the instrument.
3. Lift the tubing out of the cable holders along the rear and the right side of the chassis.
4. Unplug the tubing at the SPOLV and remove it out of the holders.

### 29.3.4.4 Assembly of the PRP-SPOLV tubing

1. Plug the tubing into the SPOLV.
2. Route the tubing along the Right Pipettor arm, the right side of the chassis and the rear of the chassis.
3. Thread the tubing through the left inner chassis opening, to the pump area.
4. Connect the fitting onto the PRP pump (1).
5. Check the tube assembly for proper installation.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 29.3.4.5 Set the PRP-SPOLV tubing

1. Prime the SPOLV, see Chapter 12.8.1.4 on page 12-67.

---

**WARNING****Air inside the MANN filter**

While the system is priming the SPOLV, please observe the liquid flow from the System liquid tank through the MANN filter: pay attention that the filter is not full of air. If needed, tap on it to remove air as much as possible.

---

2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 29.3.5 Replacement of the WB TANK-WBDP tubing

#### 29.3.5.1 Tube routing of the WB TANK-WBDP tubing

The WB TANK-WBDP tubing is routed from the Wash Buffer tank to the Wash Buffer Dispense Pump.



Figure 29–9: WB TANK-WBDP tubing inside the instrument

#### 29.3.5.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Disconnect the Wash Buffer tank, unplugging both sensor cable and tubing.

## 29. Tubing for external-tanks configuration

### 29.3.5.3 Removal of the WB TANK-WBDP tubing

#### CAUTION



#### Possible flooding.

Be aware that the removal of the tube can spill liquid into the instrument.

Take precaution measures:

- Keep a cleaning cloth and a small bucket close by.
- Wear suitable protective clothing and gloves.

1. Remove the three white cable hoses which hold together the Wash buffer sensor cable and tubing.
2. Unscrew the fitting (1) at the Wash Buffer Dispense Pump (WBDP).

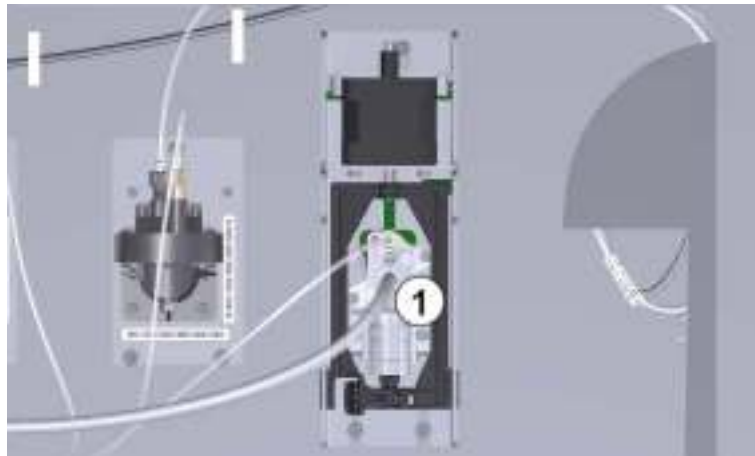


Figure 29-10:

3. Slide the Wash Buffer tubing carefully out of the instrument from the left back hole in the chassis.

### 29.3.5.4 Assembly of the WB TANK-WBDP tubing

1. Carefully slide the tubing into the system through the left back hole in the chassis.
2. Connect the Wash Buffer tubing (1) to the WBDP.
3. Connect the Wash Buffer tank, plugging both sensor cable and tubing.
4. Assemble the three white cable hoses on the sensor cable and tubing.
5. Check the tube assembly for proper installation.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

### 29.3.5.5 Set the WB TANK-WBDP tubing

1. Prime the Washer, see Chapter 12.8.1.5 on page 12-67.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 29.3.6 Replacement of the Wash Buffer tank sensor cable

#### 29.3.6.1 Cable routing of the Wash Buffer tank sensor cable

The specified cable is connected from the Liquid Support PCBA (slots J16 and J22, see Chapter 31.4.4 on page 31-27) to the Wash Buffer tank.

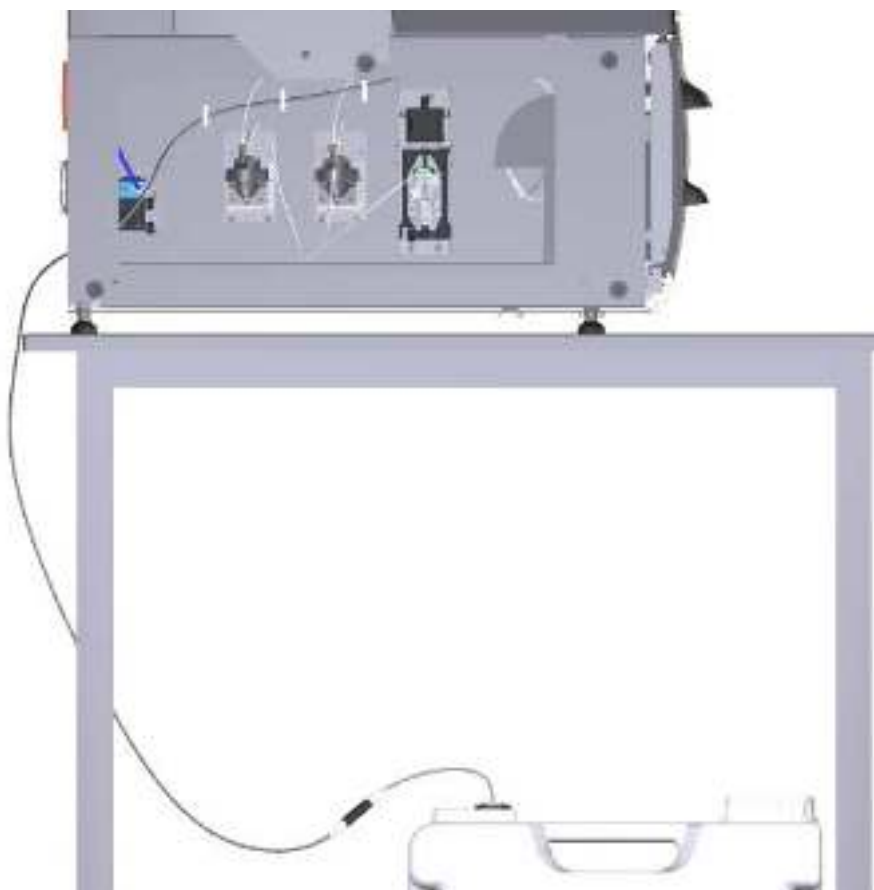


Figure 29–11: Routing of the Wash Buffer tank sensor cable

#### 29.3.6.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Disconnect the Wash Buffer tank, unplugging both sensor cable and tubing.

## 29. Tubing for external-tanks configuration

### 29.3.6.3 Removal of the Wash Buffer tank sensor cable

1. At the Liquid Support PCBA, unplug the connectors J16 (Wash Buffer level switch) and J22 (Wash Buffer presence).



Figure 29-12: Wash Buffer tank sensor cable connection on the Liquid Support PCBA

2. Open the cable clips in the pumps area.
3. Disassemble the routing of the sensor cable along the chassis.
4. Remove the three white cable hoses which hold together the Wash Buffer sensor cable and tubing.
5. Slide the sensor cable carefully out of the system from the left back hole in the chassis.

### 29.3.6.4 Assembly of the Wash Buffer tank presence sensor cable

1. Insert the sensor cable through the left back hole in the chassis.
2. Carefully route the sensor cable along the chassis, through the cable clips in the pumps area and up to the Liquid Support PCBA.
3. Plug the connectors J16 (Wash Buffer level switch) and J22 (Wash Buffer presence) onto the Liquid Support PCBA.
4. Close the cable clips.
5. Connect the Wash Buffer tank, plugging both sensor cable and tubing.
6. Assemble the three white cable hoses on the Wash Buffer sensor cable and tubing.
7. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**29.3.6.5 Set the Wash Buffer tank presence sensor cable**

1. Check the Wash Buffer tank presence sensor, see Chapter 29.4.4 on page 29-53.
2. Check the Wash Buffer tank level switch, see Chapter 29.4.5 on page 29-53.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 29.3.7 Replacement of the Peristaltic pumps rack

#### 29.3.7.1 Tube connection of the Waste tubings connected to the Peristaltic pumps

This section shows an overview of the Waste tubings connected to the Peristaltic pumps.

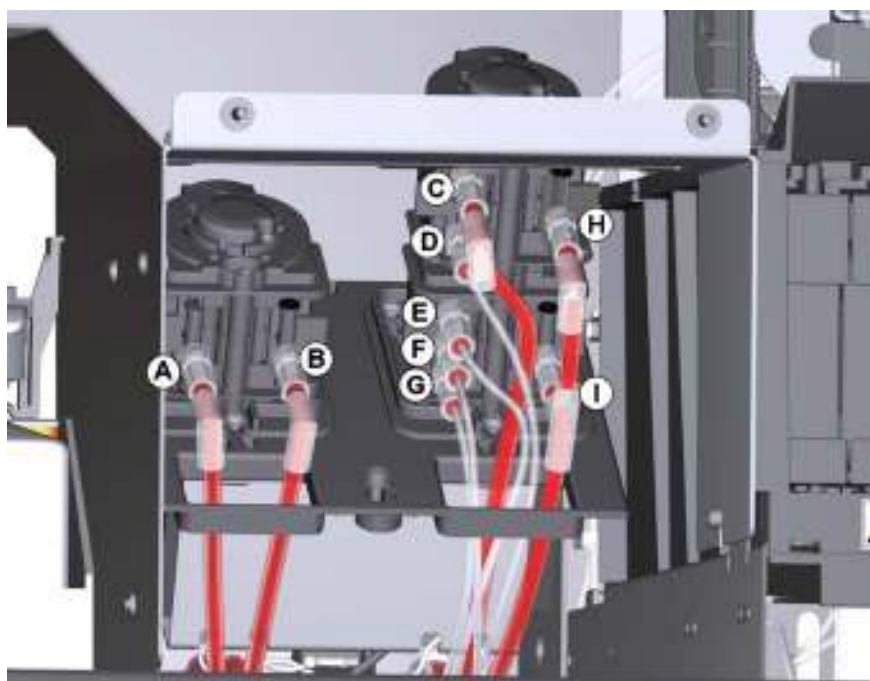


Figure 29–13: Waste tubings connected to the Peristaltic pumps overview

| No. | Connection at Pump | Tubing              | Connection to                        |
|-----|--------------------|---------------------|--------------------------------------|
| A   | LWP                | Wash station-LWP    | Wash station                         |
| B   | LWP                | LWP-Waste interface | Waste interface (Y-connector)        |
| C   | WAP                | Reagent Module-WAP  | TA Drain of the Reagent-Module       |
| D   | WAP                | Reader-WAP          | Assy-Sipal of the Measurement-Module |
| E   | WAP                | Washer-WAP          | SIPAL 1                              |
| F   | WAP                | Washer-WAP          | SIPAL 2                              |
| G   | WAP                | Washer-WAP          | Grounding pipe + SIPAL 3             |
| H   | WAP                | WAP-Waste interface | Waste interface (Y-connector)        |
| I   | WAP                | WAP-Waste interface | Waste interface (Y-connector)        |

Table 29–4: Pump block tube connection overview



### 29.3.7.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.
9. Remove the Sample Module, see Chapter 21.3.2.2 on page 21-7.
10. Tilt Peristaltic pumps rack up.

### 29.3.7.3 Removal of the Peristaltic pumps rack

1. Unplug all fittings (A to I) of the Waste tubings connected to the Peristaltic pumps.

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

2. Loose screw (2) and slide the screw with the tube carrier down and push it to the back out.

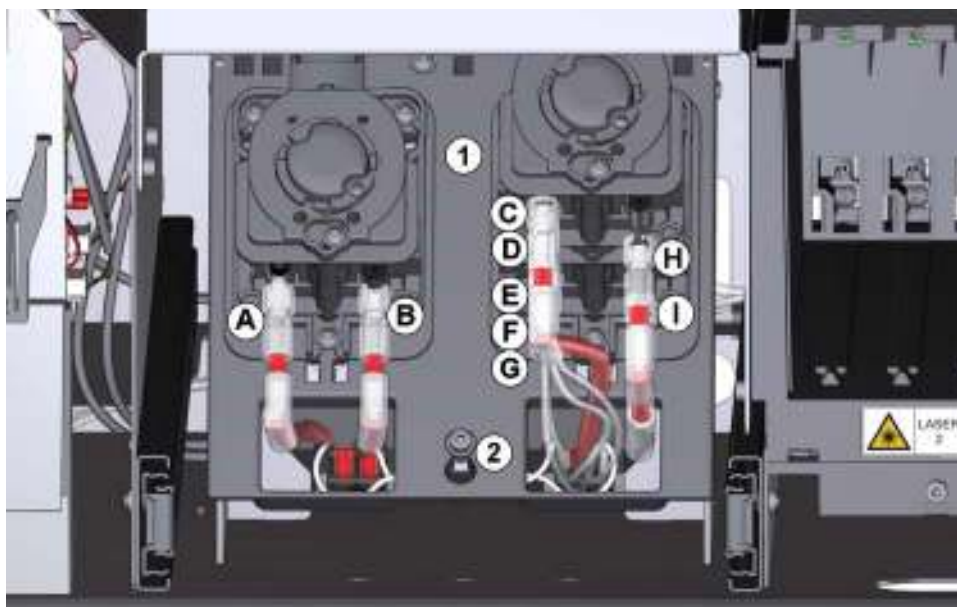


Figure 29-14: Peristaltic pumps, front view

## 29. Tubing for external-tanks configuration

3. Remove the securing ring (3) of the pump bracket axis (4).
4. Pull the pump bracket axis to the right side out.
5. Unplug the cable connections at the pumps.
6. Remove the Peristaltic pumps rack out of the instrument.

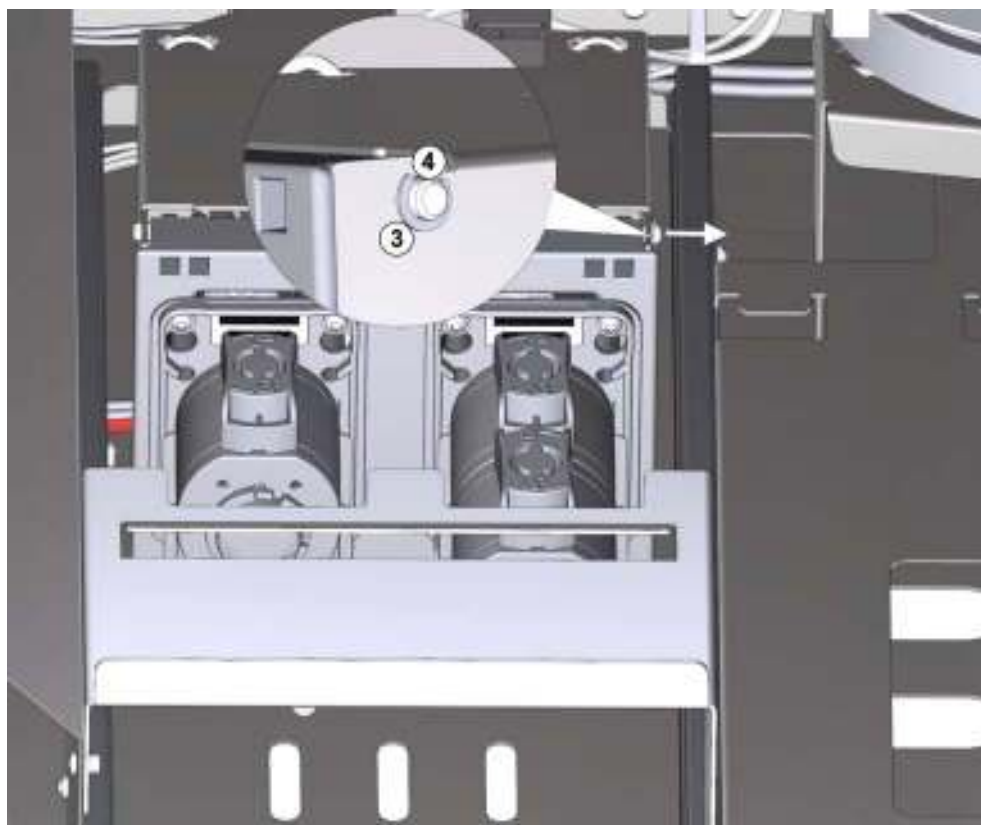


Figure 29-15: Peristaltic pumps rack

### 29.3.7.4 Assembly of the Peristaltic pumps rack

1. Insert the Peristaltic pumps rack (1) into the instrument and fix it with the pump bracket axis (4).
2. Secure the pump bracket axis (4) with a securing ring (3).
3. Connect the cable connections onto the pumps.
4. Slide the tube carrier with tubes in and set it into the screws of the Peristaltic pumps rack and fasten the fittings (A to I).
5. Close the Peristaltic pumps rack.
6. Assemble the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.
7. Assemble the Front Module, see Chapter 27.3.1.3 on page 27-5.
8. Assemble Sample Module, see Chapter 21.3.2.3 on page 21-8.
9. Open the Cuvette - Tip Loading Module and check the tube assembly for proper installation.
10. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 29.3.8 Replacement of the Waste tubing at Left Energy Chain (LWP)

**NOTE**

**The Waste tubings are an add-on of yearly Preventive Maintenance kit.**

The Field Service Engineer shall evaluate whether their replacement is needed or not, see Chapter 10.3.2 on page 10-9.

#### 29.3.8.1 Tube routing of the Waste tubing at Left Energy Chain (LWP)

The Waste tubing at Left Energy Chain are connected to the LWP:

- Wash station-LWP tubing (A) is routed through the Left Energy Chain and cable holders and connected to Wash Station.
- LWP-Waste interface tubing (B) is routed through the Left Energy Chain to the Waste interface.

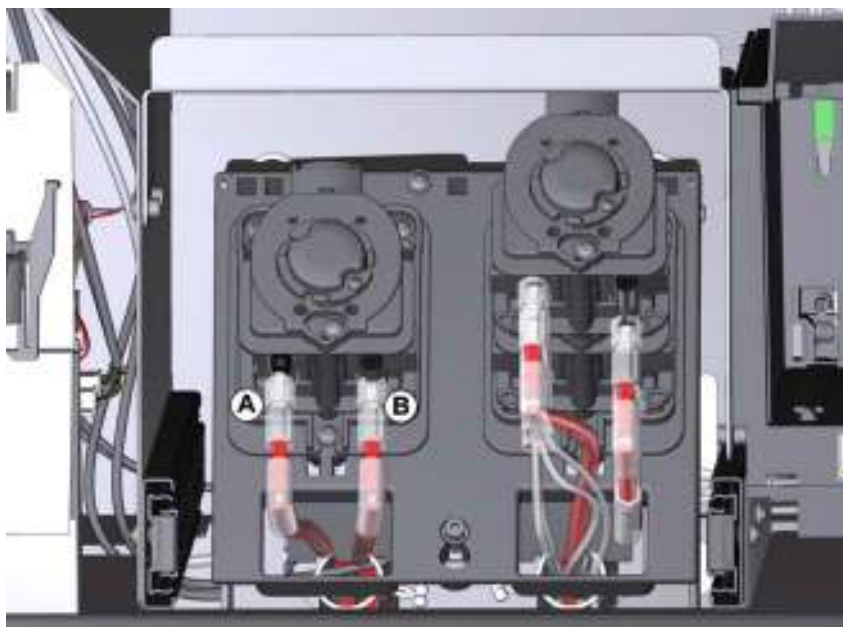


Figure 29–16: Peristaltic pumps, front view

## 29. Tubing for external-tanks configuration

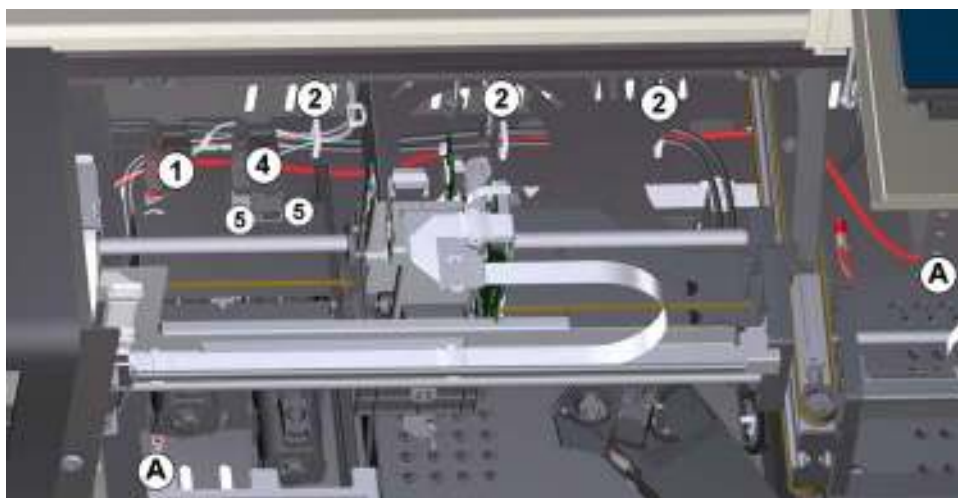


Figure 29-17: Wash station-LWP tubing, inside the instrument, without Incubator Module and Front Module

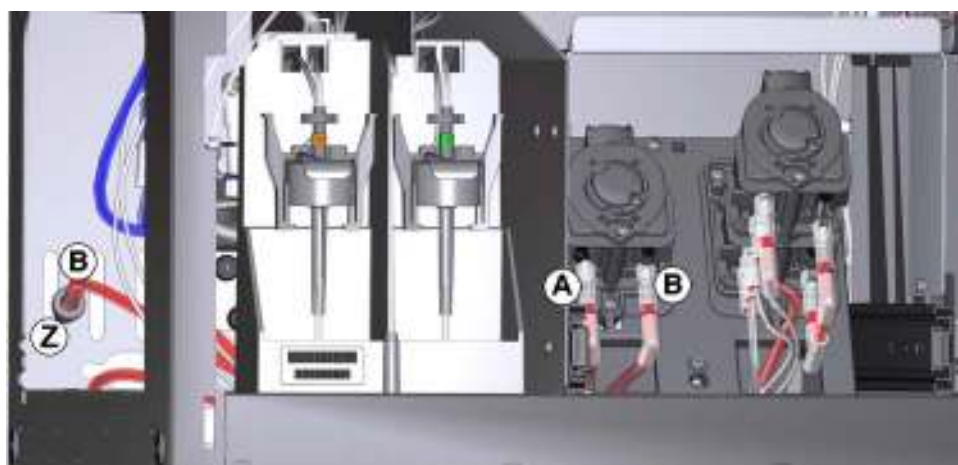


Figure 29-18: LWP-Waste interface tubing, inside the instrument

### 29.3.8.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
9. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.
10. Remove the Peristaltic pumps rack, see Chapter 29.3.7.3 on page 29-23.

### 29.3.8.3 Removal of the Waste tubing at Left Energy Chain (LWP)

1. Unplug the fitting of the Wash station-LWP (A) tubing at Wash station.

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

2. Unhook the tube at the cable holders (2).
3. Unplug the fitting of the LWP-Waste interface (B) tubing at the connection at the Waste interface-Waste tank (Z).
4. Unhook the tube at the cable holders.
5. Unhook the Left Energy Chain (1) at the last but one link (3) at the bottom.

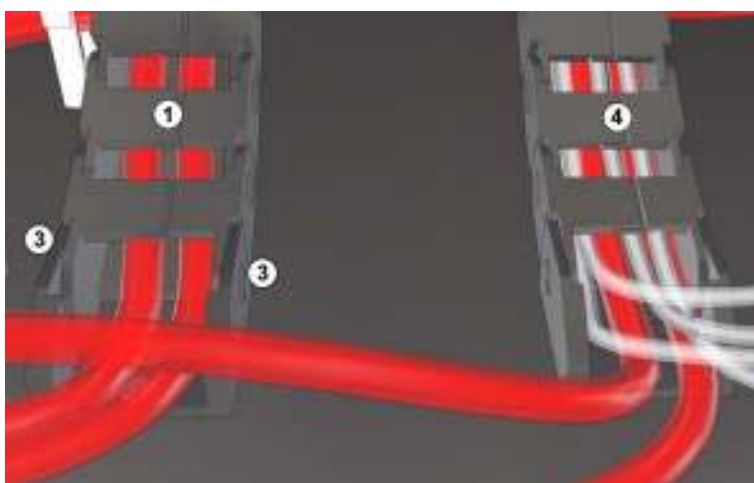


Figure 29-19: Energy chain mounted onto the bottom plate.

6. Open the frame stay and pull the tubing out of the Left Energy Chain.

## 29. Tubing for external-tanks configuration

### 29.3.8.4 Assembly of the Waste tubing at Left Energy Chain (LWP)

1. Insert the tubing into the Left Energy Chain (1).
2. Connect the tube fitting of the Wash station-LWP (A) and the LWP-Waste interface (B) onto the LWP pump.
3. Close all frame stay of the Left Energy Chain (1).

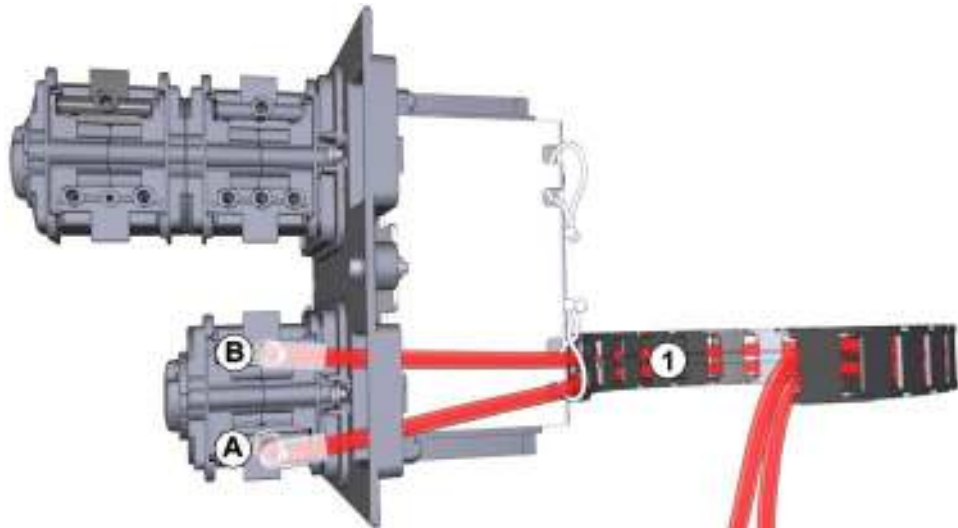


Figure 29–20: Pump block bottom view

4. Hook the Left Energy Chain onto the last but one link at the bottom.
5. Lay the Wash station-LWP (A) tubing out into the cable holders and close them.
6. Connect the tube fitting onto the Wash Station.
7. Lay the LWP-Waste interface (B) tubing underneath the bottom plate to the left side of the instrument.
8. Connect the fitting of the LWP-Waste interface (B) tubing onto the connection at the Waste interface-Waste tank (Z).
9. Assemble the Peristaltic pumps rack, see Chapter 29.3.7.4 on page 29-24.
10. Check the tube assembly for proper installation.
11. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
12. Assemble the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.
13. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.
14. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 29.3.9 Replacement of the Waste tubing at Right Energy Chain (WAP)

**NOTE**

The Waste tubings are an add-on of yearly Preventive Maintenance kit.

The Field Service Engineer shall evaluate whether their replacement is needed or not, see Chapter 10.3.2 on page 10-9.

#### 29.3.9.1 Tube routing of the Waste tubing at Right Energy Chain (WAP)

The Waste tubing at Right Energy Chain are connected to the WAP:

- Reagent Module-WAP (C) tubing is routed through the Right Energy Chain and cable holders to drain of the Reagent Module (K).
- Reader-WAP (D) tubing is routed through Right Energy Chain and cable holders to the SIPAL of the Reader Module.
- Washer-WAP tubings are routed through Right Energy Chain to SIPAL 1, 2 and 3 of the Washer Module:
  - E = SIPAL 1,
  - F = SIPAL 2,
  - G = Grounding pipe + SIPAL 3.
- WAP-Waste interface (H) and (I) tubings are routed through the Right Energy Chain to the Waste interface-Waste tank.

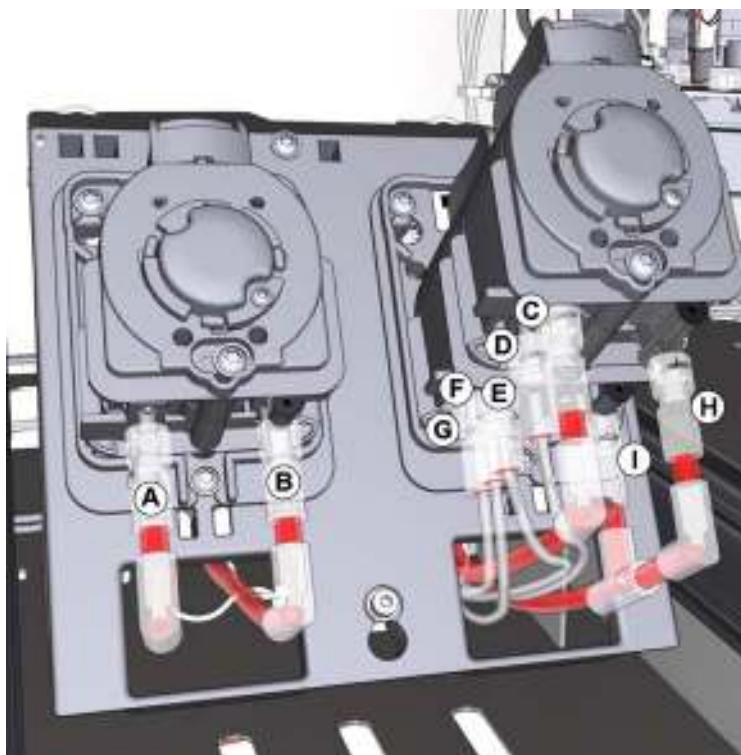


Figure 29–21: Peristaltic pump rack, front view

## 29. Tubing for external-tanks configuration

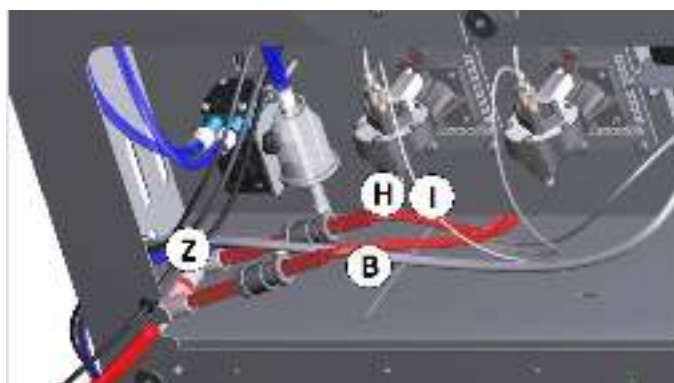


Figure 29-22: Instrument, left side view



Figure 29-23: Reagent module-WAP tubing inside the instrument, without Incubator and Washer modules

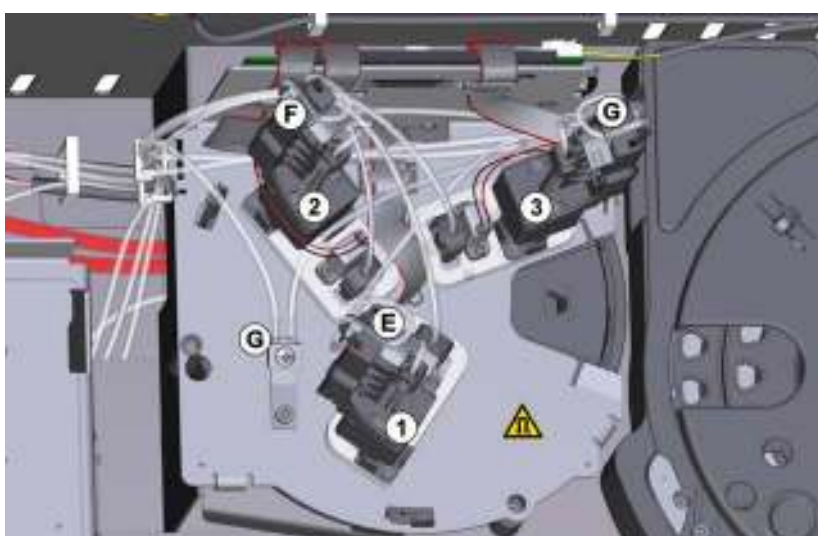


Figure 29-24: Washer Module, top view



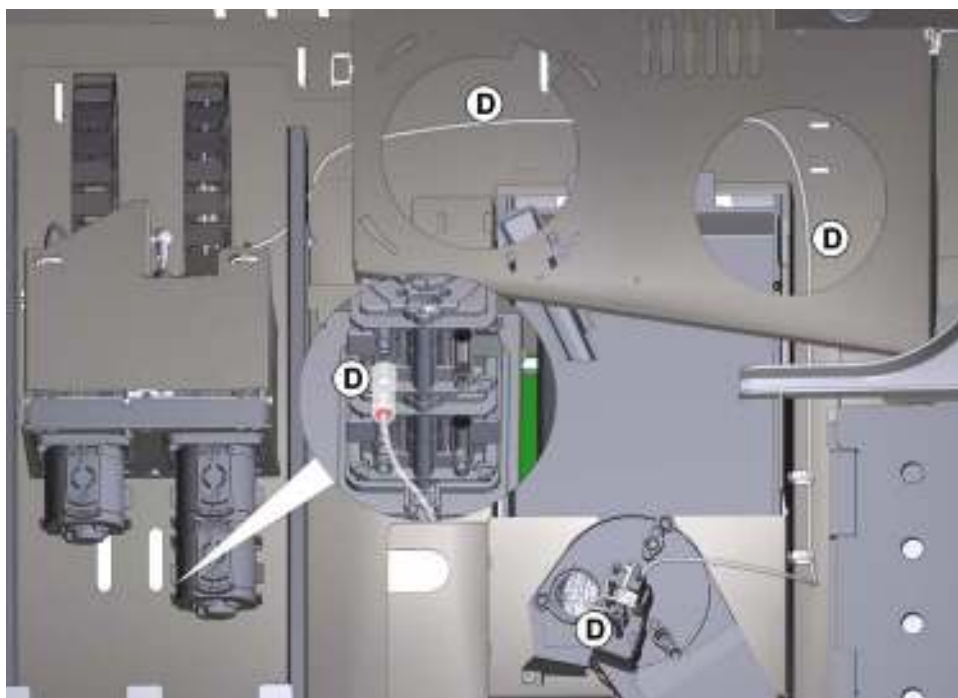


Figure 29-25: Reader-WAP tubing, inside the instrument, without Washer, Incubator and Sample modules

#### 29.3.9.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Cuvette - Tip Loading Module, see Chapter 15.3.1.2 on page 15-3.
9. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.
10. Remove the Peristaltic pumps rack, see Chapter 29.3.7.3 on page 29-23.

## 29. Tubing for external-tanks configuration

### 29.3.9.3 Removal of the Waste tubing at Right Energy Chain (WAP)

1. Unplug the fitting of the WAP-Waste interface at the pump (H-I) and at the connection at the Waste interface-Waste tank (Z).

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

2. Unplug the fitting of the Reagent Module-WAP at the pump (B) and at the drain of Reagent Module.
3. Unhook the tube at the cable holders (8).
4. Unplug the fitting of the Washer-WAP at the pump (E-F-G) and at the Washer Module. Disconnect the tubing at the three SIPAL of the Washer Module.
5. Unplug the fitting of the Reader-WAP at the pump (D) and at the Reader Module. Disconnect the tubing at the SIPAL of the Reader Module.
6. Unhook the tube at the cable holders.
7. Unhook the Right Energy Chain (4) at the last but one link (6) at the bottom.

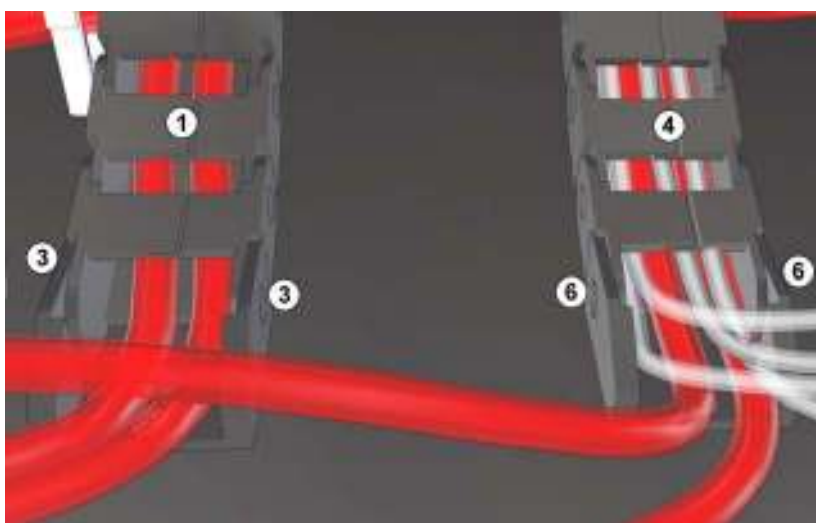


Figure 29-26: Energy chain mounted onto the bottom plate.

8. Open the frame stay and pull the tube out of the Right Energy Chain.

**29.3.9.4 Assembly of the Waste tubing at Right Energy Chain (WAP)**

1. Insert the tube into the Right Energy Chain (4).
2. Connect the tube fitting of the C, D, E, F, G, H, I connectors of the WAP onto the pump.
3. Close all frame stay of the Right Energy Chain (4).

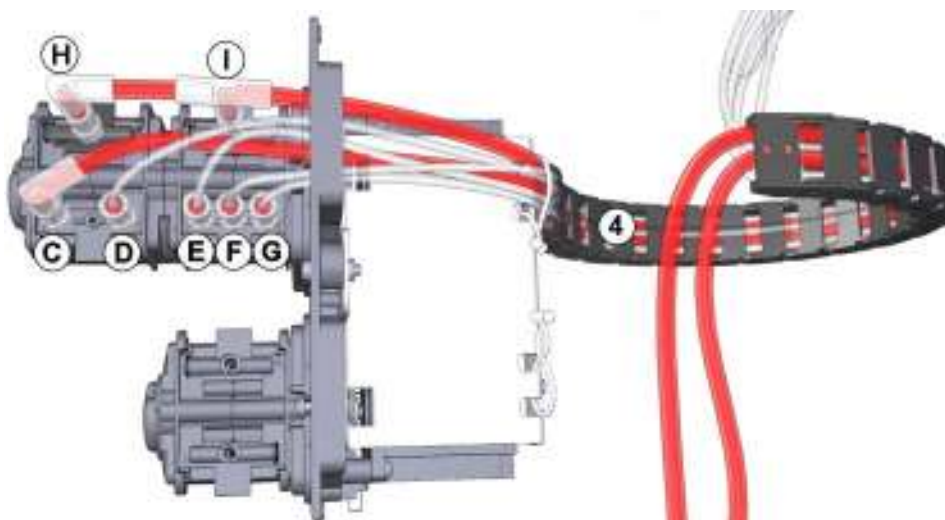


Figure 29–27: Peristaltic pumps and Right Energy Chain, bottom view

4. Hook the Right Energy Chain onto the last but one link at the bottom.
5. Lay the Reagent Module-WAP out into the cable holders and close them.
6. Lay the Washer-WAP onto the Washer Module.
7. Lay the Reader-WAP onto the Reader Module.
8. Lay the WAP-Waste interface (I-H) underneath the bottom plate to the left side of the instrument.
9. Connect the fitting of the WAP-Waste Interface (I-H) onto the connection at the Waste interface-Waste tank (Z).
10. Assemble the Peristaltic pumps rack, see Chapter 29.3.7.4 on page 29-24.
11. Check the tube assembly for proper installation.
12. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
13. Assemble the Cuvette - Tip Loading Module, see Chapter 15.3.1.3 on page 15-3.
14. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**29.3.9.5 Set the Waste tubing at Right Energy Chain (WAP)**

1. Perform the calibration of the Washer aspiration monitor, see Chapter 17.4.2.3 on page 17-46.
2. Perform the calibration of the Washer SIPAL lift, see Chapter 17.4.2.2 on page 17-46.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 29.3.10 Replacement of the Waste interface-Waste tank tubing

#### 29.3.10.1 Tube routing of the Waste interface-Waste tank tubing

The Waste interface-Waste tank tubing is connected from the Waste interface (Z) to the Liquid Waste tank.

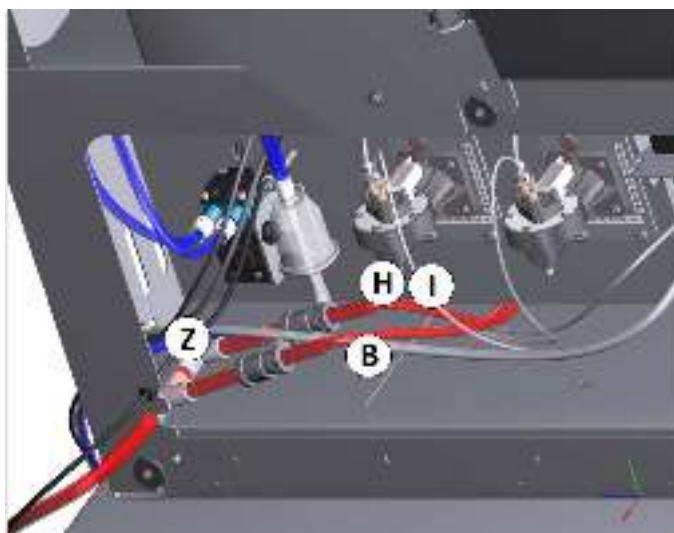


Figure 29–28: Waste interface

#### 29.3.10.2 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the Left Top Cover.
6. Move the Sample Pipettor arm to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.

#### 29.3.10.3 Removal of the Waste interface-Waste tank tubing

1. Disconnect the Liquid Waste sensor cable from the Liquid Support PCB, at slot J19, see Chapter 31.4.4 on page 31-27.
2. Disassemble the routing of the sensor cable along the chassis.
3. Disconnect the fittings at the Waste interface (Z).
4. Disconnect the Liquid Waste tank.

**WARNING****Liquid Waste Spillage**

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

---

**29.3.10.4 Assembly of the Waste interface-Waste tank tubing**

1. Route the Liquid Waste sensor cable along the chassis.
2. Connect the cable to the Liquid Support PCB, at slot J19.
3. Connect the fittings at the Waste interface (Z) and clip the tubing into the chassis.
4. Connect the Liquid Waste tank.
5. Check the tube assembly for proper installation.
6. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**29.3.10.5 Set the Waste interface-Waste tank tubing**

1. Open the **LIAISON<sup>®</sup> XS Main UI**.
2. Login and initialize the system.
3. Check the Liquid Waste presence sensor: open the **Status - Summary** toggle and verify the tank presence while connecting and dis-connecting the tank itself.
4. Check the Liquid Waste level sensor: open the **Status - Summary** toggle and verify the liquid level present in the tank while connecting the tank filled with different liquid level.
5. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 29.3.11 Replacement of the Liquid Waste level sensor

#### 29.3.11.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.

#### 29.3.11.2 Removal of the Liquid Waste level sensor

1. Disconnect the Liquid Waste tank.

#### WARNING



#### Liquid Waste Spillage

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

2. Decontaminate it according to paragraph “Decontamination of Liquid Waste tank” of the Service Manual, see Chapter 8.3.1 on page 8-3.
3. Close the dust cap of the sensor.

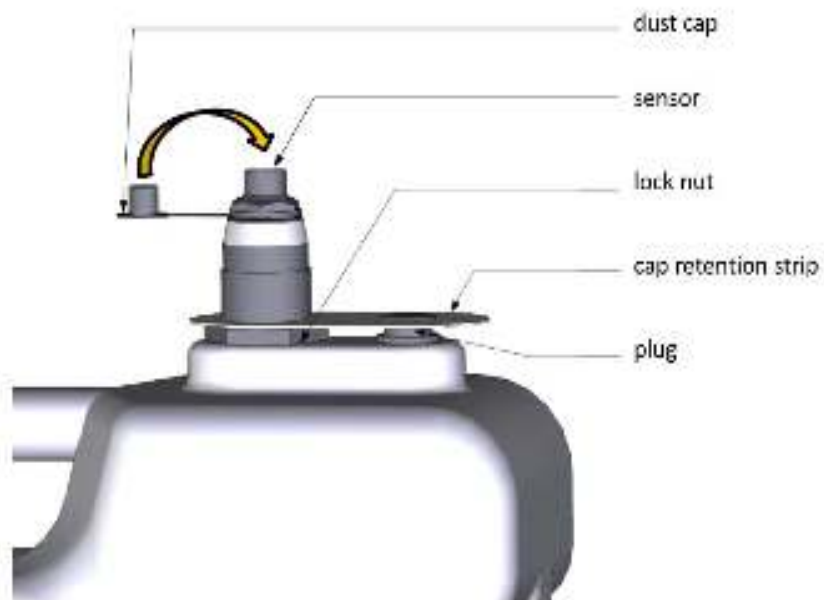


Figure 29–29: Sensor mounted in the tank.

4. Remove the plug and the cap retention strip.
5. Loosen the lock nut, unscrew the level sensor and pull it out.
6. Dispose the removed sensor.

**29.3.11.3 Assembly of the Liquid Waste level sensor**

1. Unscrew the lock nut of the level sensor.

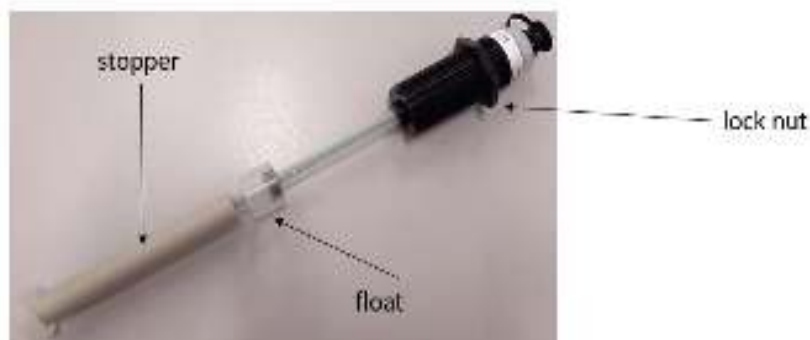


Figure 29-30: New sensor.

2. Screw the sensor into the thread of the tank, until the sensor almost touches the bottom of the tank.
3. Fix the sensor in the thread by tightening the lock nut.
4. Put back the cap retention strip on the outer thread of the level sensor. The plug must be above the cable entry and carefully pushed into it.
5. Check the correct alignment of the slot.
6. Connect the Liquid Waste tank.

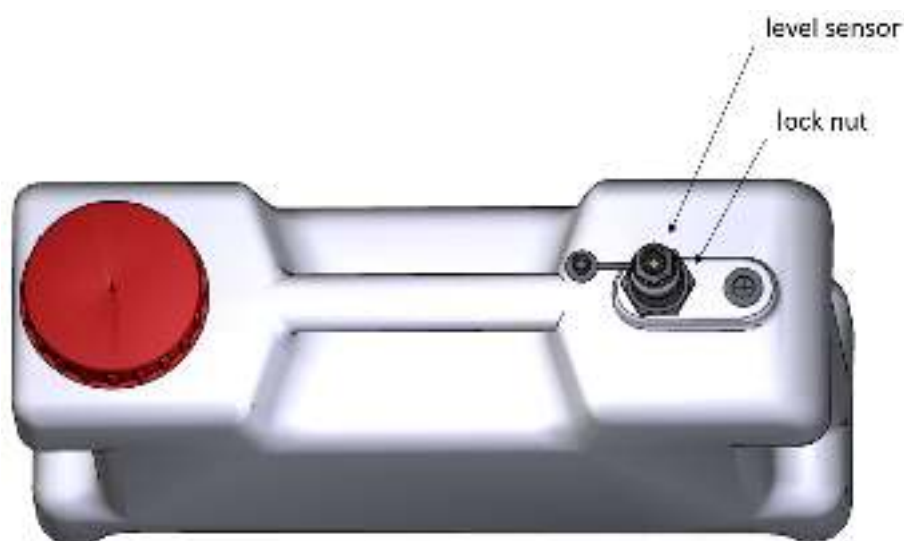


Figure 29-31: Tank with mounted level sensor.



## 29. Tubing for external-tanks configuration

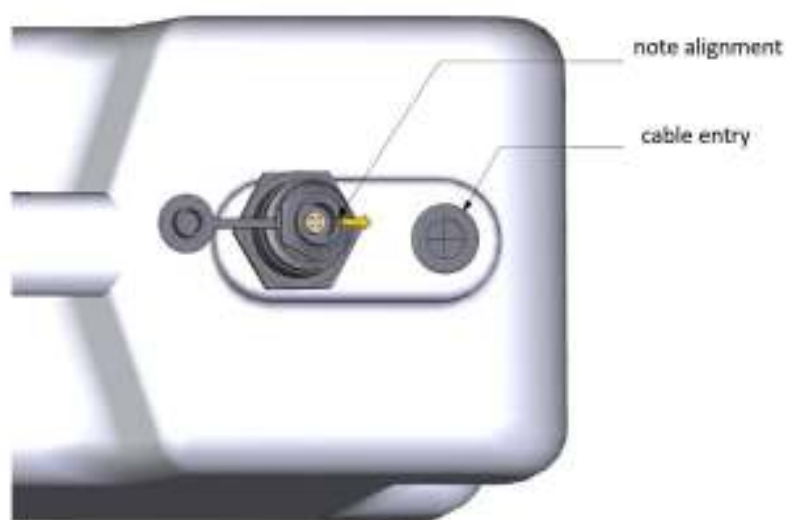


Figure 29–32: Slot alignment check.



Figure 29–33: Cap retention strip with plug for the cable entry.



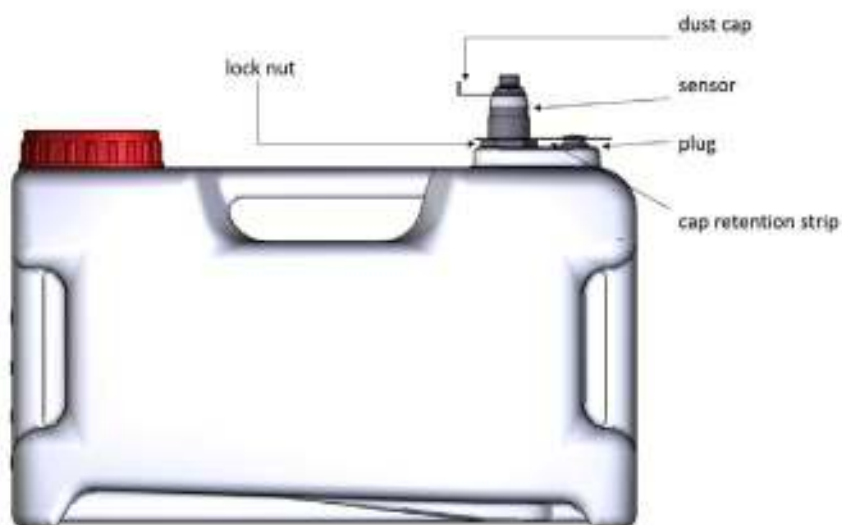


Figure 29–34: Complete mounted level sensor.

#### 29.3.11.4 Set the Liquid Waste level sensor

1. Open the **LIAISON® XS Main UI**.
2. Login and initialize the system.
3. Check the Liquid Waste presence sensor: open the **Status - Summary** toggle and verify the tank presence while connecting and dis-connecting the tank itself.
4. Check the Liquid Waste level sensor: open the **Status - Summary** toggle and verify the liquid level present in the tank while connecting the tank filled with different liquid level.
5. Requalification not needed.

### 29.3.12 Replacement of the WBDP-Washer tubings

#### 29.3.12.1 Tube routing of the WBDP-Washer tubings

The WBDP-Washer tubings are connected to the WBDP (4) connector (V-W-X) and they are routed through instrument to Washer in order to be connected to the Dispense Manifold of the three Washer SIPAL:

- V = SIPAL 1,
- W = SIPAL 2,
- X = SIPAL 3.

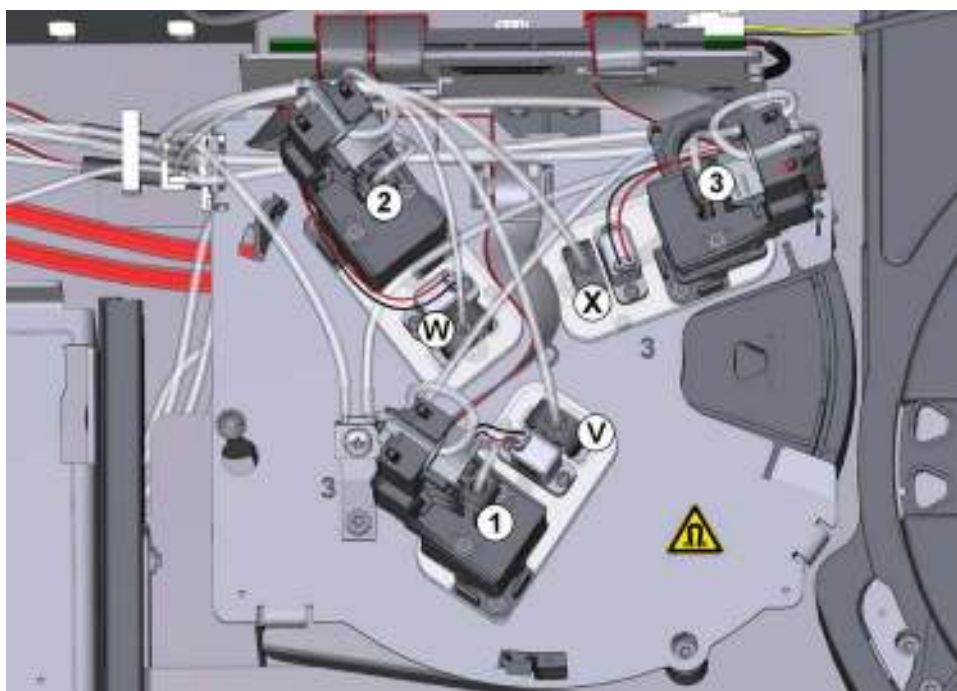


Figure 29–35: Washer Module, top view

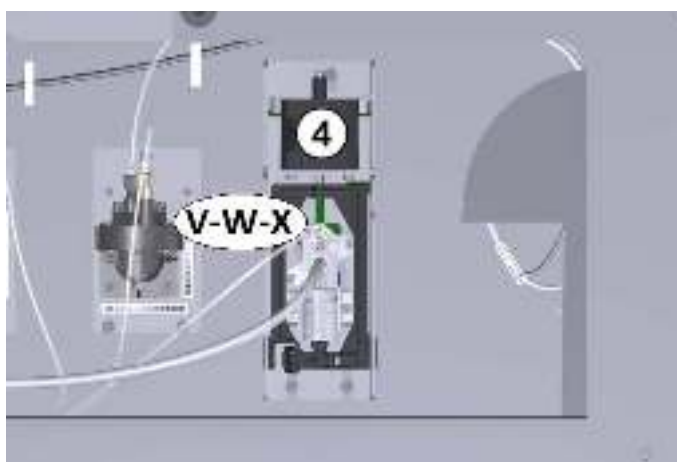


Figure 29–36: Instrument left side view, WBDP

**29.3.12.2 Preliminary steps**

1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.

**29.3.12.3 Removal of the WBDP-Washer tubings**

1. Unplug the fitting of WBDP-Washer tubings at the WBDP (V, W and X).
2. Unhook the tube at the cable holders.
3. Disconnect the tubing at the three Dispense Manifolds of Washer Module SIPALs.
4. Pull out the tubings from the instrument.

**29.3.12.4 Assembly of the WBDP-Washer tubings**

1. Connect the tube fittings (V, W and Z) onto the WBDP.
2. Route the tubings up to the Washer Module.
3. Connect the WBDP-Washer tubings to the three Dispense Manifolds of Washer Module SIPALs.
4. Check the tube assembly for proper installation.
5. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

**29.3.12.5 Set the WBDP-Washer tubings**

1. Prime the Washer, see Chapter 12.8.1.5 on page 12-67.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 29.3.13 Replacement of the SDP-Starters tubings

#### 29.3.13.1 Tube routing of the SDP-Starters tubings

The SDP-Starter tubings are connected to the SDP pumps (fittings are marked with number tags):

- SDP-Starter1 is connected from the bottle of Starter Reagent 1 to the SDP1 left connector (Q).
- SDP-Starter2 is connected from the bottle of Starter Reagent 2 to the SDP2 left connector (R).

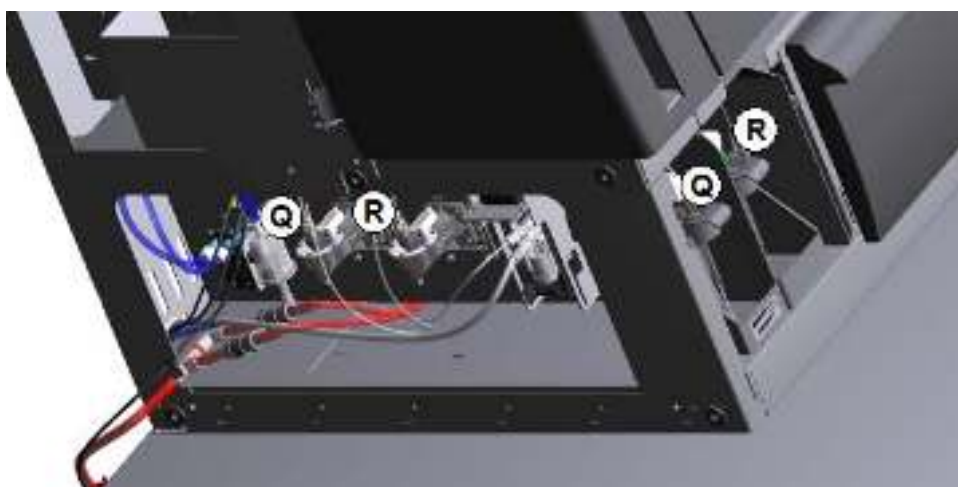


Figure 29-37: SDP-Starters tubings inside the instrument

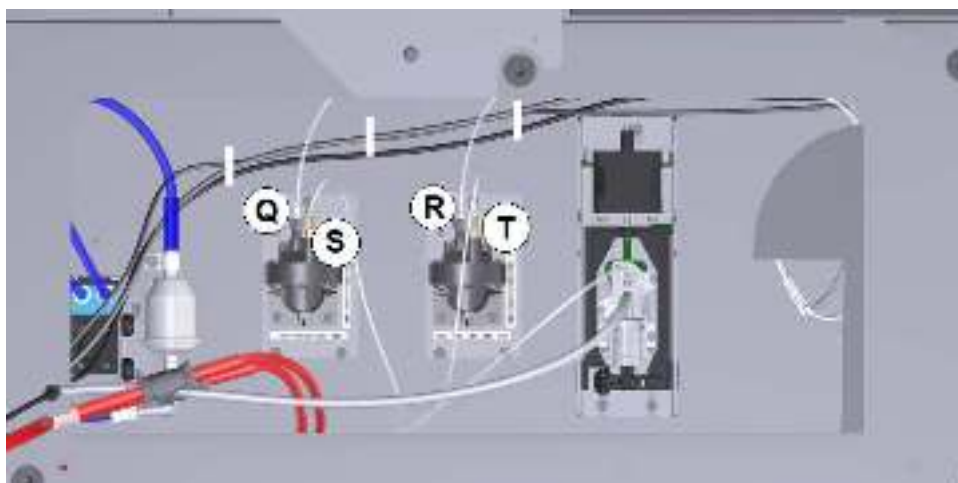


Figure 29-38: Instrument left side view

### 29.3.13.2 Preliminary steps



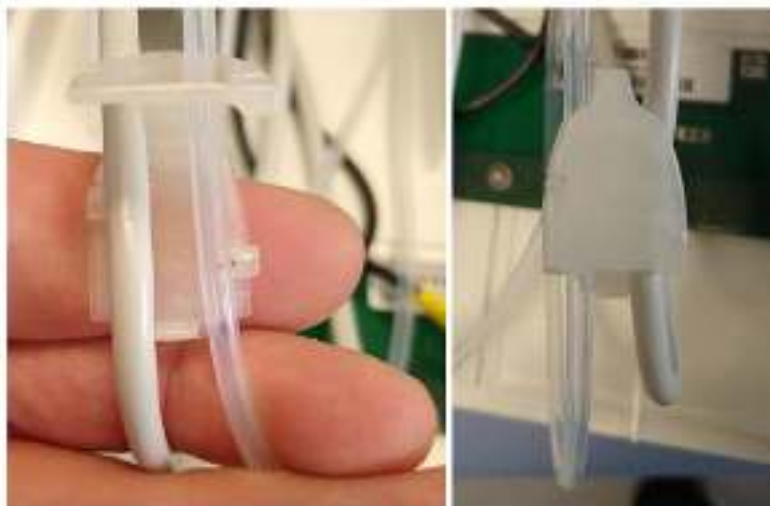
1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Front Module, see Chapter 27.3.1.2 on page 27-4.

### 29.3.13.3 Removal of the SDP-Starters

1. Remove the Starter Reagent module, see Chapter 24.3.1.2 on page 24-4.
2. Open the hose clamp and the tubing protection. The hose clamp could be black or white according to the version of the Starter Reagent module.

### 29.3.13.4 Assembly of the SDP-Starters

1. Align the white hose clamps at around 22cm distance from Starter bottle and close them tightly.
2. Insert the Starter Reagent module back into the system while guiding the cables through the hole in the chassis.



*Figure 29–39: Close tightly the hose clamps.*

3. Assemble the Starter Reagent module, see Chapter 24.3.1.3 on page 24-6.
4. Check the tube for proper installation.

**29.3.13.5 Set the SDP-Starters**

1. Perform the Calibration of the Starter Bubble sensors, see Chapter 24.4.2.1 on page 24-16
2. Prime the Reader, see Chapter 12.8.1.6 on page 12-68.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 29.3.14 Replacement of the Reader-SDP tubings

#### 29.3.14.1 Tube routing of the Reader-SDP tubings

The Reader-SDP tubings are connected to the SDP pumps (fittings are marked with number tags):

- Reader-SDP1 is connected from the SDP1 (S) to the injection of the Starter 1 on the Reader Module (1).
- Reader-SDP2 is connected from the SDP2 (T) to the injection of the Starter 2 on the Reader Module (2).

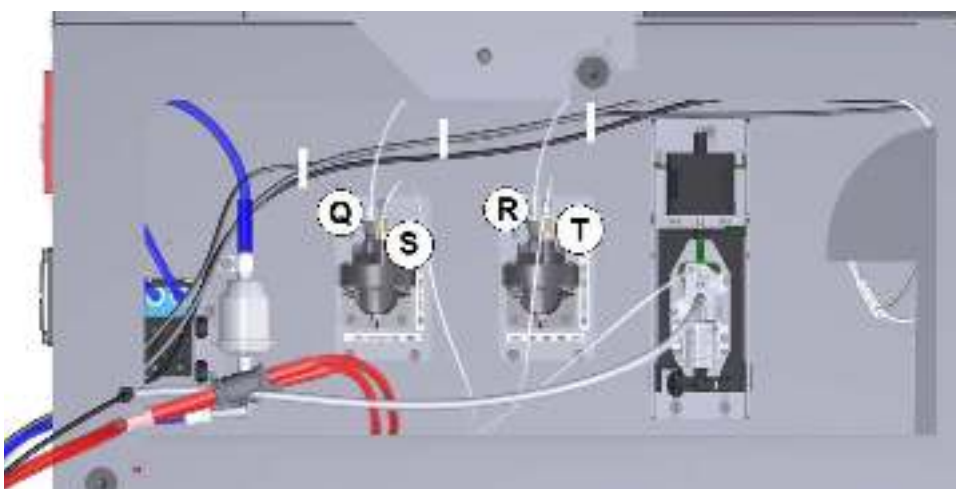


Figure 29-40: Instrument left side view

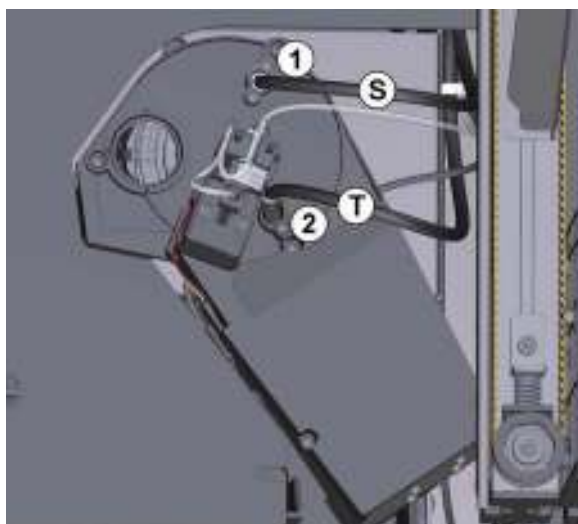


Figure 29-41: Reader Module, top view



## 29. Tubing for external-tanks configuration

### 29.3.14.2 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Incubator Module, see Chapter 16.3.2.2 on page 16-5.

### 29.3.14.3 Removal of the Reader-SDP tubings

1. At the Starter Dispense Pumps, untwist the connected Reader-SDP tubings.
2. Remove two screws (3) at Starter dispensing (1) and (2) of the Reader Module.

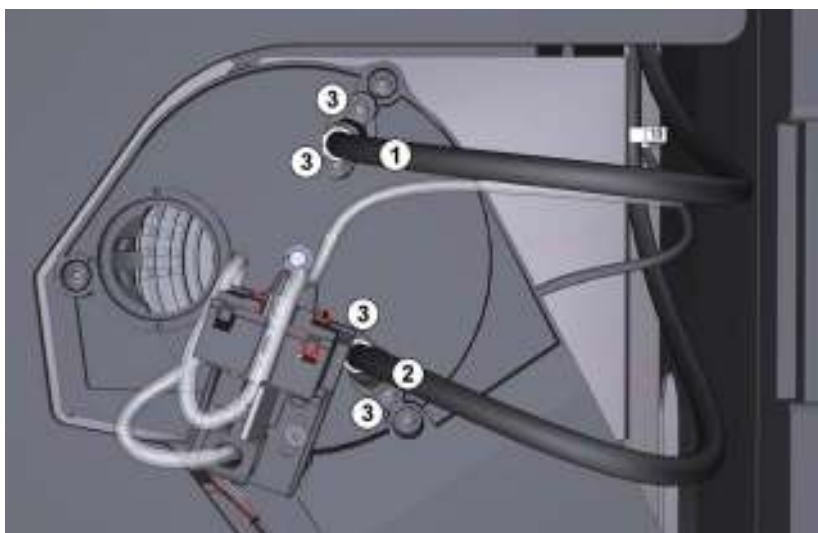


Figure 29–42: Reader Module, top view



3. Remove the Reader-SDP tubing out of the cable holders (4) and pull out the tubings.

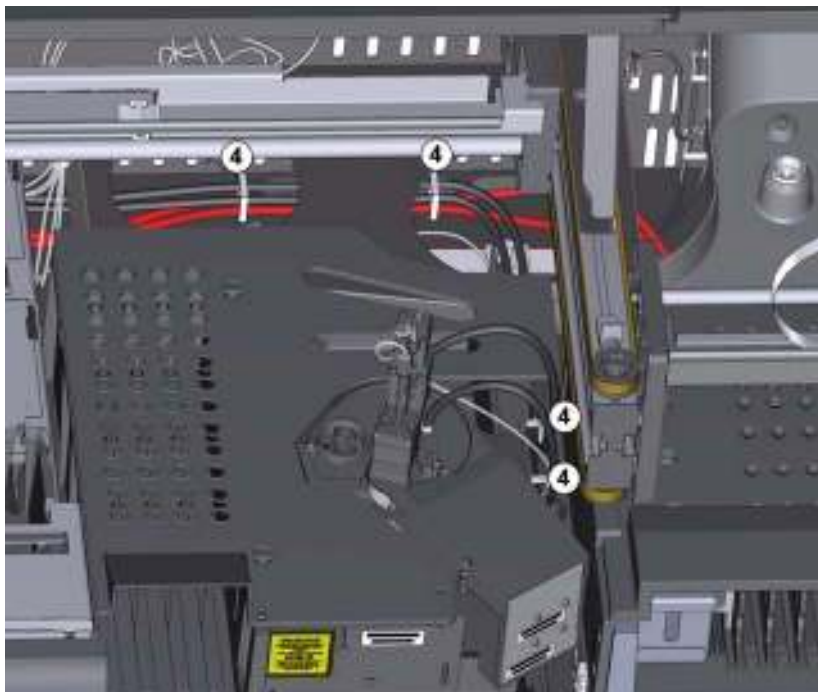


Figure 29-43: Reader-SDP tubings overview

#### 29.3.14.4 Assembly of the Reader-SDP tubings

1. Thread the Reader-SDP tubings from the left side of the instrument to the Reader Module.
2. Locate the Reader-SDP tubings with the four cable holders (4).

#### NOTE

##### Before mounting the starter reagent tube.

Be aware of the correct installation of the Starter 1 and Starter 2 lines.

3. Insert the Reader-SDP tubings (1) and (2) into the correct connection of the Reader Module and fasten each with two screws (3).
4. Connect the Reader-SDP tubings at the Starter Dispense Pumps.
5. Check the tube assembly for proper installation.
6. Assemble the Incubator Module, see Chapter 16.3.2.3 on page 16-6.
7. Assemble the Left Side Cover, see Chapter 14.3.1.4 on page 14-5.

#### 29.3.14.5 Set the Reader-SDP tubings

1. Prime the Reader, see Chapter 12.8.1.6 on page 12-68.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 29.3.15 Replacement of the Wash station cover

#### 29.3.15.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Sample Module cover, see Chapter 21.3.1.2 on page 21-5.

#### 29.3.15.2 Removal of Wash station cover

1. Unscrew three screws (2) of Wash station cover (1).
2. Move the SPOLV pipettor arm to the rear left position.
3. Lift Wash station cover carefully out and set it aside.

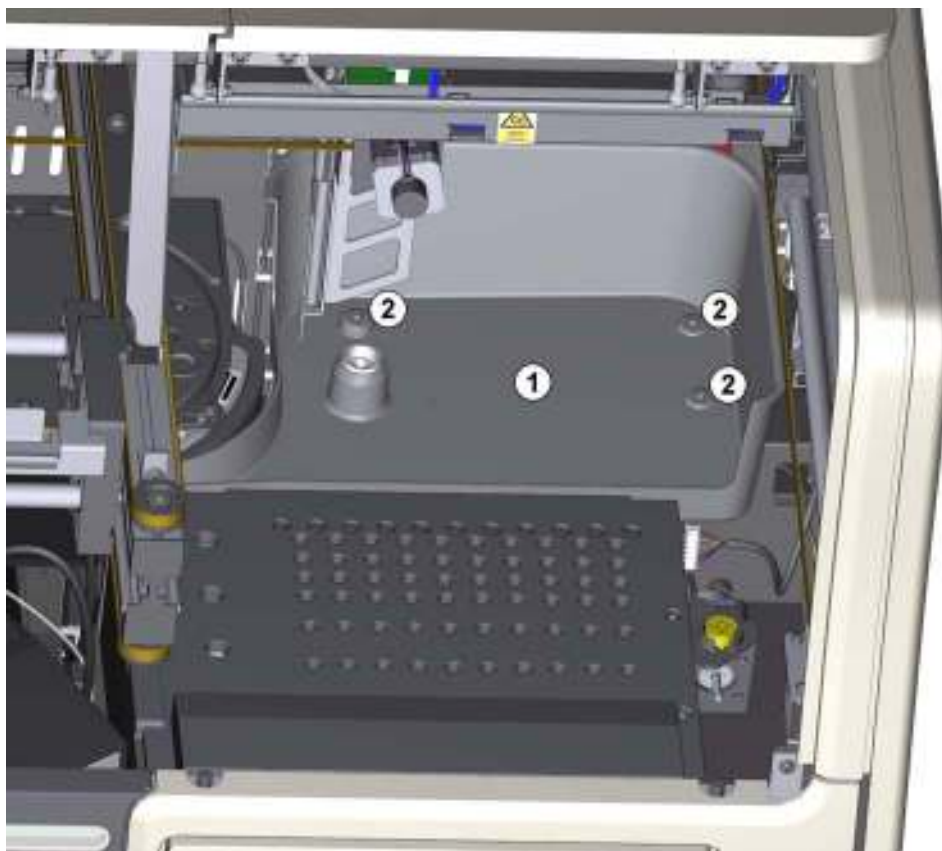


Figure 29–44: Wash Station cover

**29.3.15.3 Assembly of the Wash station cover**

1. Set the bore of the Wash station cover over the Wash Station and slide the Wash Station carefully into the bore.

**CAUTION****Cover is fragile**

Be careful when fastening screws for the Wash station cover .  
The cover is fragile and could break or could get cracks.

---

2. Fasten carefully three screws (2).
3. Assemble the Sample Module cover, see Chapter 21.3.1.3 on page 21-6.
4. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 29.3.16 Replacement of the Wash station

#### 29.3.16.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open top covers.
6. Move pipettor arms to a safe location.
7. Remove the Wash station cover, see Chapter 29.3.15.2 on page 29-48.

#### 29.3.16.2 Removal of the Wash station

1. Lift the Wash station (1) carefully up by slightly turning left and right.

---

**WARNING****Liquid Waste Spillage**

Please pay attention that liquid waste spillage is possible when unplugging the tubings.

---

2. Unscrew fitting (2) at Wash station bottom and set Wash Station aside.

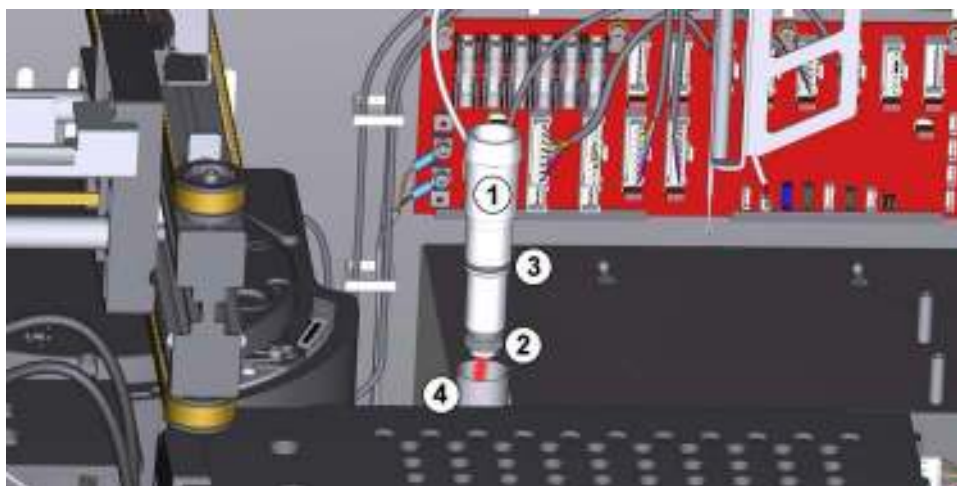
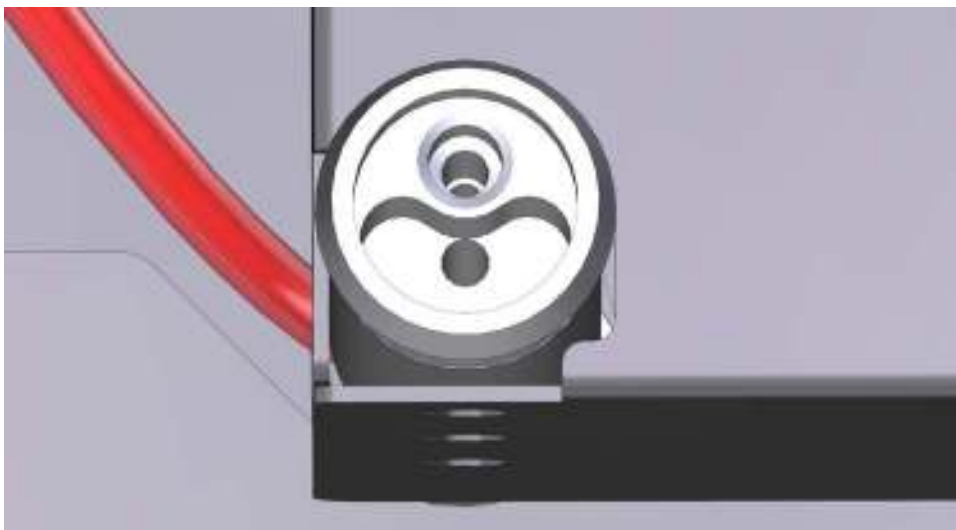


Figure 29-45: Wash station inside the instrument

**29.3.16.3 Assembly of the Wash station**

1. Fasten the fitting (2) on the Wash station (1).
2. Moisten the seal (3) with a little bit grease.
3. Slide the Wash station carefully into the holding tube (4). Take care of the waste tubing, there shall be no nicks or not to be snapped off.
4. Check Wash station for correct orientation, see Figure 29–46: on page 29-51.



*Figure 29–46: Correct orientation of the Wash station, top view*

5. Check the tube assembly for proper installation.
6. Assemble the Wash station cover, see Chapter 29.3.15.3 on page 29-49.

**29.3.16.4 Set the Wash station**

1. Perform the teaching of the Right pipettor on the Wash station, see Chapter 13.15 on page 13-32.
2. Prime the SPOLV, see Chapter 12.8.1.4 on page 12-67.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

## 29.4 Verifying of tank presence sensors and level sensors

In this section information about the initialization and verifying of different level sensors and presence sensors can be found.

### NOTE

#### Close software

Shut down previous used software (User-, Teaching- or Service Software).  
Do not run any of the software at the same time.

### 29.4.1 Initialization routine

- Close any open software and start the **Service Software** from the Windows Administrator account.
- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Initialize** button and choose the **All Dialog Controlled Modules** button to start the initialization sequence.
- The process is successful when the **Processing** window closes without failure message.

### 29.4.2 Verifying of the System Liquid tank presence sensor

- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Read** button and then choose **Liquid State**.
- Click the **Auto** update button to play the function.
- Connect and disconnect the System Liquid tank presence sensor cable, by checking on the display the correct status of the tank (Present or Absent).
- Click the **Auto** update button to stop the function.

### 29.4.3 Verifying of the System Liquid tank level sensor

- In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- Click onto the **Read** button and then choose **Liquid State**.
- Click the **Auto** update button to play the function.
- Insert and remove the System Liquid tank filled at different levels, by checking on the display the correct status of the tank (Empty or Full).
- Click the **Auto** update button to stop the function.

#### 29.4.4 Verifying of the Wash Buffer tank presence sensor

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Read** button and then choose **Liquid State**.
- c. Click the **Auto** update button to play the function.
- d. Insert and remove the WB tank, by checking on the display the correct status of the tank (Present or Absent).
- e. Click the **Auto** update button to stop the function.

#### 29.4.5 Verifying of the Wash Buffer tank level sensor

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Read** button and then choose **Liquid State**.
- c. Click the **Auto** update button to play the function.
- d. Insert and remove the WB tank filled at different levels, by checking on the display the correct status of the tank (Empty or Full).
- e. Click the **Auto** update button to stop the function.

#### 29.4.6 Verifying of the Liquid Waste tank presence sensor

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Read** button and then choose **Liquid State**.
- c. Click the **Auto** update button to play the function.
- d. Insert and remove the Liquid Waste tank, by checking on the display the correct status of the tank (Present or Absent).
- e. Click the **Auto** update button to stop the function.

#### 29.4.7 Verifying of the Liquid Waste tank level sensor

- a. In the **Service Software** at the navigation window, choose **System** and the **General** bracket.
- b. Click onto the **Read** button and then choose **Liquid State**.
- c. Click the **Auto** update button to play the function.
- d. Insert and remove the Liquid Waste tank filled at different levels, by checking on the display the correct status of the tank (Empty or Full).
- e. Click the **Auto** update button to stop the function.



## 29.5 Direct Drain Configuration

LIAISON® XS analyzer supports also a direct drain configuration, where a waste tubing connected to the system allows to dispose liquid waste directly into a collector drain by-passing the liquid waste tank.

A dedicated direct drain kit shall be ordered to implement this configuration.

**NOTE**

Direct Drain is an optional solution. All systems will be shipped with the liquid waste tank. Direct Drain configuration can be implemented during installation or at a later time.

**NOTE**

The tubing provided with the direct drain kit has a length of 5 m.

The direct drain tubing features the same fitnut connectors of the standard waste tubing, to be screwed onto the internal waste lines of the instrument, and the same T-connector to be clipped to the chassis.

The open end of the direct drain tubing needs to be fixed to the collector drain.



Figure 29-47: Direct Drain tubing

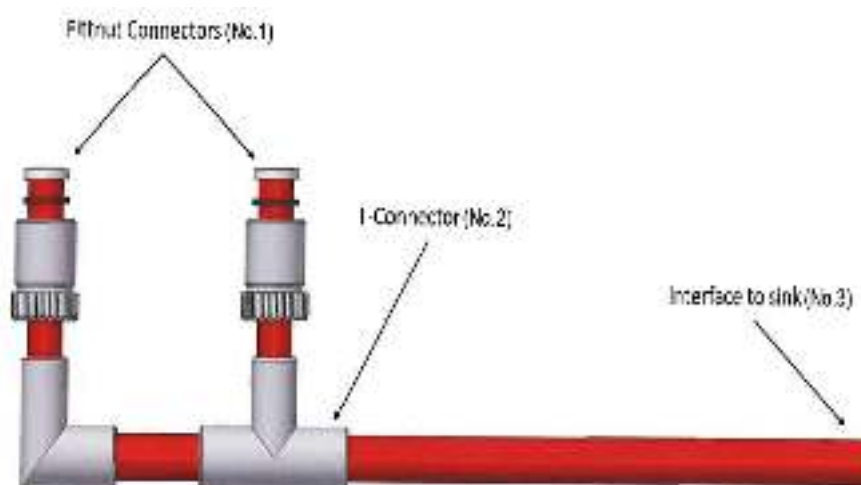


Figure 29-48: Direct Drain tubing layout



### 29.5.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shutdown is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the Left Top Cover.
6. Move the Sample Pipettor arm to a safe location.
7. Remove the Left Side Cover, see Chapter 14.3.1.3 on page 14-5.
8. Remove the Waste Interface-Waste tank tubing, see Chapter 29.3.10.3 on page 29-34.

### 29.5.2 Installation

1. Screw the fitnut connectors of the Direct Drain tubing onto the internal waste lines of the instrument.
2. Clip the T-connector of the Direct Drain tubing to the chassis.
3. Verify that the Direct Drain tubing reaches the waste collector drain available at customer site.
4. Guide the open end of the Direct Drain tubing into the drain and securely fix it.

---

**CAUTION**

To avoid back flow into the instrument, the sink has to be at or below the bench where the instrument is placed on.

The routing of the Direct Drain tubing must not exceed 200 mm above the sink level.

---

### 29.5.3 Parameter setting

**NOTE**

Direct Drain can be used with software version 1.3.5 or following versions.

The FW provides an additional parameter for Direct Drain setting: if enabled, this parameter shows to the system the presence state of an empty liquid waste tank.

- a. Start the **Service Software** from the Windows Administrator account.
- b. In the **Service Software**, at the navigation window, choose **System** and the **Parameter** bracket.
- c. Change the Module to "Liquid Support" in the Selection Box.
- d. In the "Read Parameter(s)" field, enter the value "242" and press the **Update** button.
- e. Select the parameter and change its value according to:

## 29. Tubing for external-tanks configuration

- 0: Use of Wast tank (default)
  - 1: Use of Direct Drain
- f. Confirm by pressing the **Write (Module RAM only)** button.
  - g. Store the parameter by pressing the **Non-Volatile Memory** button.
  - h. Confirm the pop-up dialog.

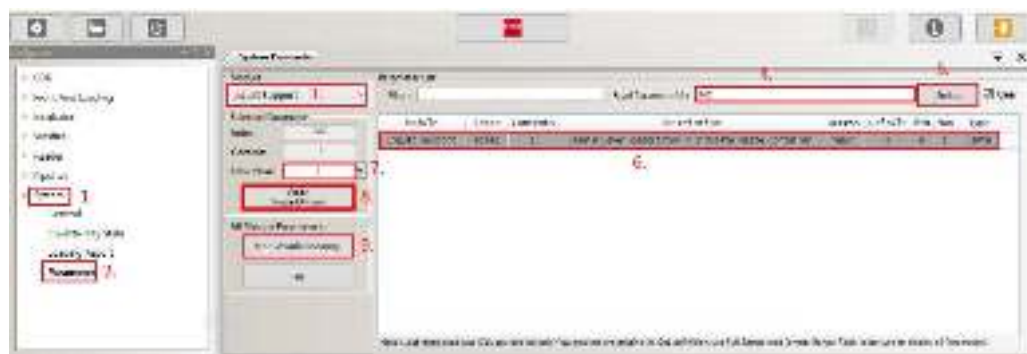


Figure 29–49: Direct Drain parameter setting

### 29.5.4 Requalification

- a. Insert all needed consumables and liquids.
- b. In the **Service Software**, at the navigation window, choose **System** and the **Cuvette Tray State** bracket.
- c. Set the cuvette trays.
- d. In the **Service Software**, at the navigation window, choose **System** and the **General** bracket. Click onto the **Liquid** button and choose the **Prime all** button to open the dedicated window.
- e. Customize on the screen the **Prime all** window:
  - Cycles Washer Port 1: 19
  - Cycles Washer Port 2: 5
  - Cycles Washer Port 3: 5
  - Cycles Measurement: 20
  - Cycles Pipettor: 12
  - Wash time: 18
  - OLV Calibration: Disabled
  - Liquid Control: Disabled
- f. Press the **Execute** button.
- g. Check the connection to the Direct Drain tubing and to the sink, to ensure no leakage is present.

# 30 Electrical Components for internal-tanks configuration

This chapter describes additional electrical components of **LIAISON<sup>®</sup> XS** system in detail. The first section of the chapter introduces “Safety” notes concerning these components, followed by a brief overview of the main circuit boards.

The following section contains information about repair and calibration (“Replacement and Repair” section) of electrical components.

In the last section, layout and connections of Liquid Support PCBA, Pipettor CU PCBA, COP EUS2CAN PCBA and PWR CAN EXT. CU PCBA are described.

## 30.1 Safety

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injury and material damage.

- Follow all safety instructions, see Chapter 1.5 on page 1-7.
- Follow all warnings marked on the instrument.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

### DANGER



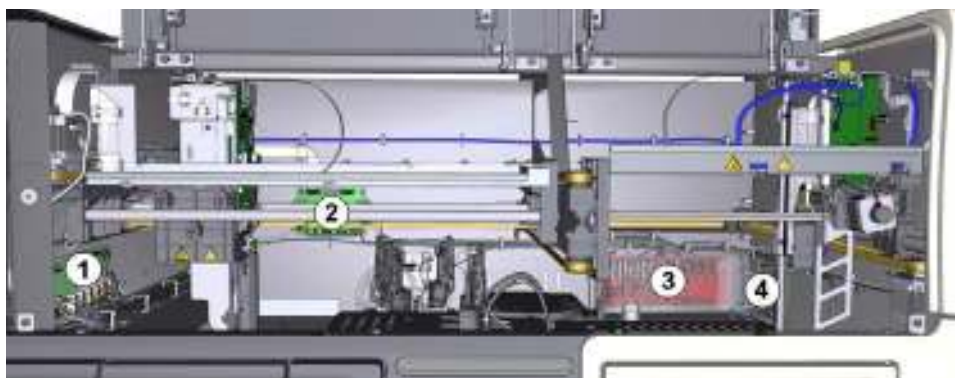
#### High Voltage!

The touching of high voltage parts or cables can result in serious personal injuries with deadly consequences.

- Never touch high voltage parts with instrument on.
- Separate the system from the mains supply before maintaining or cleaning it.
- Be careful! The following parts can be under high voltage during operation:
  - UPS device (100-240V, high current)
  - Main supply unit (100-240V, high current)
  - Enclosure metal part in Single Fault conditions (132V r.m.s., low current)
  - Photomultiplier (PMT) (~ 1000 Volts, low current)

## 30.2 Overview

This section shows an overview of the main circuit boards.



*Figure 30–1: Circuit boards overview*

|   |                      |
|---|----------------------|
| 1 | Liquid Support PCBA  |
| 2 | Pipettor CU PCBA     |
| 3 | PWR CAN EXT. CU PCBA |
| 4 | COP EUS2CAN PCBA     |

*Table 30–1: Circuit boards overview*

### 30.3 Replacement and repair

This section contains instructions for the repair of electrical components. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

#### 30.3.1 Replacement of the Liquid Support PCBA

##### 30.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the rearmost area of the instrument.

##### 30.3.1.2 Removal of the Liquid Support PCBA

1. Unplug all cables of Liquid Support PCB A (1).
2. Loosen the five screws (2), lift the PCB up and remove it.

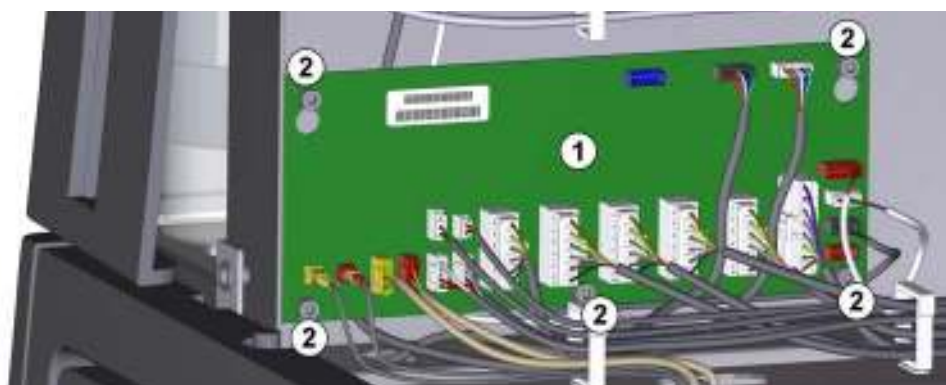


Figure 30-2: Liquid Support PCBA

##### 30.3.1.3 Assembly of the Liquid Support PCBA

1. Slide the PCB (1) into the keyholes and fasten the five screws (2).
2. Plug all cables onto the PCB, see Chapter 30.4.4 on page 30-27.

##### 30.3.1.4 Set of the Liquid Support PCBA

1. When the Liquid Support PCBA is installed and wired completely, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the calibration of Starter Bubble Sensors, see Chapter 11.6.3.4 on page 11-41.

### **30. Electrical Components for internal-tanks configuration**

---

3. Check the DI water tank level sensor, see Chapter 28.4.3 on page 28-51.
4. Check the Wash Buffer tank level sensor, see Chapter 28.4.5 on page 28-52.
5. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 30.3.2 Replacement of the Pipettor CU PCBA

#### 30.3.2.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the rightmost area of the instrument.

#### 30.3.2.2 Removal of the Pipettor CU PCBA

1. Unplug all cables of Pipettor CU PCBA (1).
2. Loosen the two screws (2) and lift the PCB out of the board holder (3).

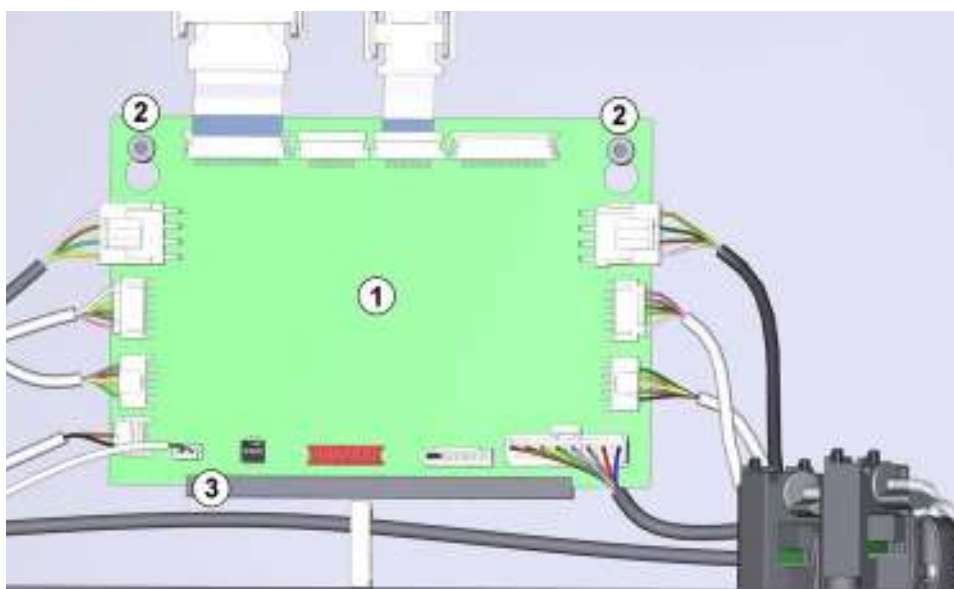


Figure 30–3: Pipettor CU PCBA

#### 30.3.2.3 Assembly of the Pipettor CU PCBA

1. Slide the PCB (1) into the board holder (3) and fasten the two screws (2).
2. Plug all cables onto the PCB, see Chapter 30.4.2 on page 30-23.

#### 30.3.2.4 Set of the Pipettor CU PCBA

1. When the Pipettor CU PCBA is installed and wired completely, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.

### 30.3.3 Replacement of PWR CAN EXT fuses



#### Electrocution/Fire Hazard!

Defective systems can result in serious injuries with deadly consequences and material damages (e.g. fire).

- Separate immediately the defective system from the mains supply, if a safe usage is no longer possible.
  - Secure the defective system against reconnection.
  - Label the defective system clearly as being defective.
- 



#### Danger of Electrocution

If the system is not separated from the mains supply before maintenance works, this can cause serious injuries with deadly consequences due to electrocution. Additionally, there is the danger that the system starts and causes injuries (e.g. contusion, cuts etc.) to the person working with the system.

- Determine and correct the cause of a blown fuse or thrown circuit-breaker before attempting to resume operations of the system.
  - Separate the system from the mains supply and secure it against restarting before replacing fuses.
  - Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.
  - Never repair or bridge blown fuses.
  - Never short-circuit the fuse holder.
- 

#### 30.3.3.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the Wash Station cover, see Chapter 28.3.14.2 on page 28-47.

#### NOTE

Although spare fuses are provided with the instrument and the replacement procedure is described, you should be aware that blown fuses are very often indicators of other malfunctions that may be affecting instrument modules, components or wires. If the fuses blow again shortly after their replacement and troubleshooting activities fail, please contact DiaSorin Corporate Service for assistance.

---



## 30. Electrical Components for internal-tanks configuration

## 30.3.3.2 Removal of the PWR CAN EXT blown fuse

1. Remove the faulty fuse from the PWR CAN EXT PCB, using an insulated plier.

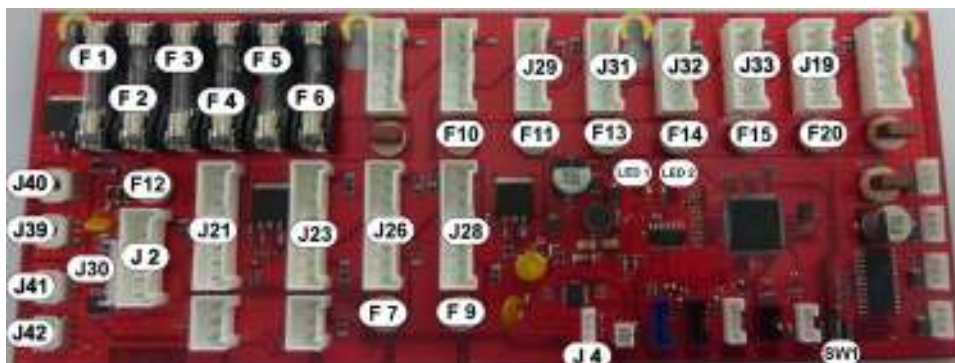


Figure 30–4: PWR CAN EXT SW CU PCBA

| Number | Connector | Circuit | Description               | Fuse type                                                               |
|--------|-----------|---------|---------------------------|-------------------------------------------------------------------------|
|        |           | 0       | COP digital supply        |                                                                         |
| F1     | J20       | 1       | not used                  | 250VAC, F5A                                                             |
| F2     | J21       | 1       |                           | 250VAC, F5A                                                             |
| F3     | J21       | 1       | Loading Bay               | 250VAC, F5A                                                             |
| F4     | J23       | 2       | Bridge                    | 250VAC, F5A                                                             |
| F5     | J23       | 2       | Bridge                    | 250VAC, F5A                                                             |
| F6     | J25       | 2       | not used                  | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F7     | J26       | 3       | RFID Liquid support<br>CU | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F8     | J3/J26    | 3       | Liquid support            | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F9     | J28       | 4       | Pipettor                  | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F10    | J27/J28   | 4       | Pipettor                  | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F11    | J29       | 5       | Incubator                 | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F12    | J2        | 0       | User interface            | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |

## 30. Electrical Components for internal-tanks configuration

| Number | Connector | Circuit | Description         | Fuse type                                                         |
|--------|-----------|---------|---------------------|-------------------------------------------------------------------|
| F13    | J31       | 6       | Measurement chamber | Temperature fuse or TE5 Time-Lag fuse, 392 series: 250VAC, T3.15A |
| F14    | J32       | 7       | Washer              | Temperature fuse or TE5 Time-Lag fuse, 392 series: 250VAC, T3.15A |
| F15    | J33       | 8       | Front cover         | Temperature fuse or TE5 Time-Lag fuse, 392 series: 250VAC, T3.15A |
| F20    | J19       | 9       | CAN termination.    | Temperature fuse or TE5 Time-Lag fuse, 392 series: 250VAC, T3.15A |

Table 30–2: Fuses with corresponding connectors and circuits

## 30.3.3.3 Assembly of the PWR CAN EXT new fuse

**NOTE****Replace fuses**

Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.

1. Insert the new fuse into the dedicated holder of PWR CAN EXT PCB.
2. Assemble the Wash Station cover, see Chapter 28.3.14.3 on page 28-48.
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 30.3.4 Replacement of the PWR CAN EXT. CU PCBA

#### 30.3.4.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the Wash Station cover, see Chapter 28.3.14.2 on page 28-47.

#### 30.3.4.2 Removal of the PWR CAN EXT. CU PCB

1. Unplug all the cables from the PWR CAN EXT. CU PCB (1).
2. Loosen the four screws (2) and lift the PCB board up and remove it.

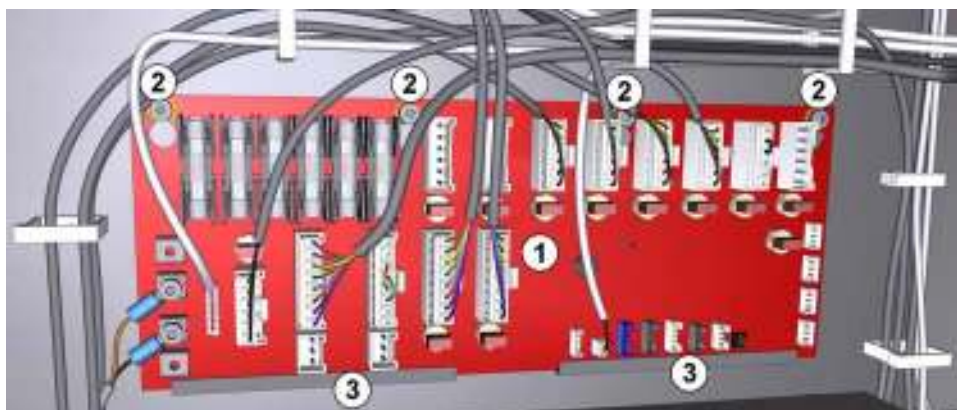


Figure 30-5: PCB, PWR CAN EXT. CU

#### 30.3.4.3 Assembly of the PWR CAN EXT. CU PCB

1. Slide the PWR CAN EXT. CU PCB (1) into the guide rails (3) and fasten the four screws (2).
2. Plug all the cables onto connectors of the circuit board, see Chapter 30.4.1 on page 30-21.
3. Assemble the Wash Station cover, see Chapter 28.3.14.3 on page 28-48.

#### 30.3.4.4 Set of the PWR CAN EXT. CU PCBA

1. When the PWR CAN EXT. CU PCB is installed and wired completely, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 30.3.5 Replacement of the COP EUS2CAN PCBA

#### 30.3.5.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the Wash Station cover, see Chapter 28.3.14.2 on page 28-47.

#### 30.3.5.2 Removal of the COP EUS2CAN PCB

1. Unplug all cables from the COP EUS2CAN PCB (1).
2. Loosen the two screws (2) and lift the PCB out.

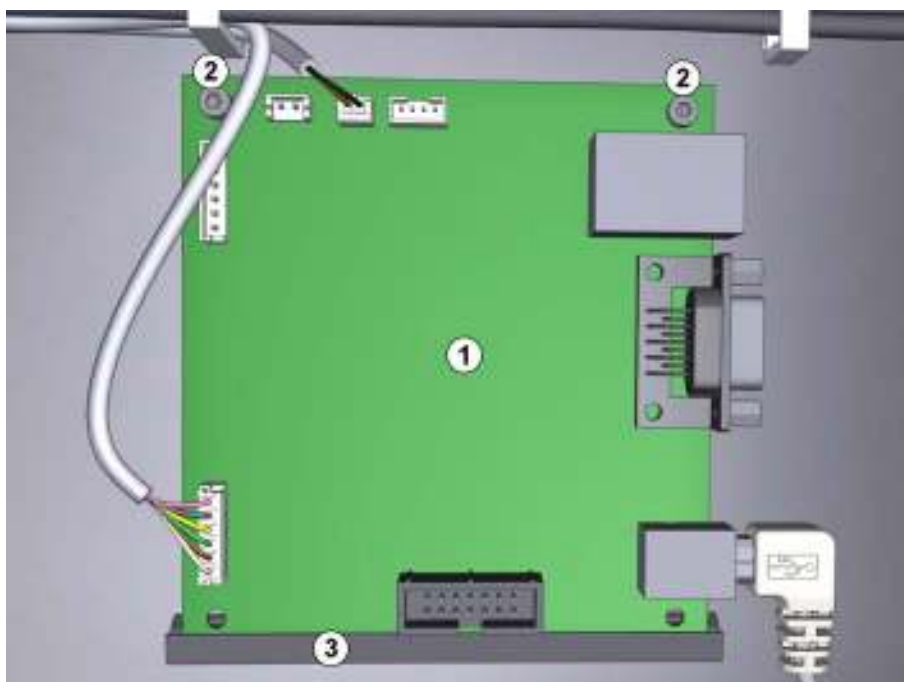


Figure 30–6: PCB, COP EUS2CAN

#### 30.3.5.3 Assembly of the COP EUS2CAN PCB

1. Slide the COP EUS2CAN PCB (1) into the guide rails (3) and fasten the two screws (2).
2. Plug all the cables onto the circuit board, see Chapter 30.4.3 on page 30-25.
3. Assemble the Wash Station cover, see Chapter 28.3.14.3 on page 28-48.

**30.3.5.4 Set of the COP EUS2CAN PCBA**

1. When the COP EUS2CAN PCB is installed and wired completely, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Open the **Service Software** from the Windows Administrator account.
3. At the navigation window, choose **COP** and the **Parameter** bracket.
4. In the **Read Parameter(s)** field, type **2** and press the **Update** button on the right.
5. Select the parameter 2 from the parameter list.
6. Type the system Serial Number in the **New Value** field on the left, and press the **Write** button.
7. Press the **Non-Volatile Memory** button and select **Overwrite**.
8. Check all the Pipettor teaching, see Chapter 13.4 on page 13-6.
9. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 30.3.6 Replacement of Fuses of the Main Switch

#### 30.3.6.1 Safety



---

##### Defective instrument

Any defective instrument will result in serious injuries with deadly consequences and material damage (e.g. fire).

- Immediately disconnect the defective instrument from the mains supply, if a safe usage is no longer possible.
  - Secure the defective instrument against reconnection.
  - Label the defective instrument clearly as being defective.
- 



---

##### Danger of Electrocutation

If the system is not separated from the mains supply before maintenance works, this can cause serious injuries with deadly consequences due to electrocution. Additionally, there is the danger that the system starts and causes injuries (e.g. contusion, cuts etc.) to the person working with the system.

- Determine and correct the cause of a blown fuse or thrown circuit-breaker before attempting to resume operations of the system.
  - Separate the system from the mains supply and secure it against restarting before replacing fuses.
  - Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.
  - Never repair or bridge blown fuses.
  - Never short-circuit the fuse holder.
- 

##### NOTE

Although spare fuses are provided with the instrument and the replacement procedure is described, you should be aware that blown fuses are very often indicators of other malfunctions that may be affecting instrument modules, components or wires. If the fuses blow again shortly after their replacement and troubleshooting activities fail, please contact DiaSorin Corporate Service for assistance.

---

#### 30.3.6.2 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.

### 30.3.6.3 Remove the Fuses of Main Switch

1. At the Main Switch (1), open the fuse box (2) with a screwdriver.
2. Remove the fuse box (3).
3. Remove defective fuses (4).



Figure 30-7: Right side view of instrument, main switch

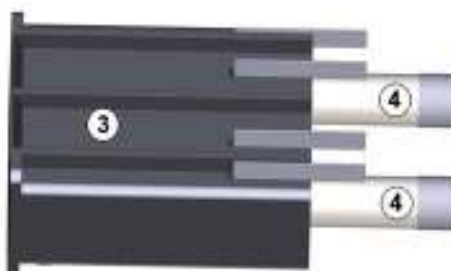


Figure 30-8: Fuse box

### 30.3.6.4 Assemble the Fuses of Main Switch

#### NOTE

#### Replace Fuses

Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.

Always replace blown fuses, do not try to repair them.

Never short-circuit the fuse holder.

1. Set the new fuses into the fuse box.
2. Slide the fuse box into the Main Switch and fasten the screws (2).
3. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.



### 30.3.7 Replacement of the User Access Panel CU PCB

#### 30.3.7.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.

#### 30.3.7.2 Removal of the User Access Panel CU PCB

1. Loosen the four screws (2) at the interface assembly (1).



Figure 30–9: Interface assembly



### 30. Electrical Components for internal-tanks configuration

2. Lift the interface assembly (1) a bit up, flip it out and unplug all the cables from the User Access Panel CU PCB.
3. Disconnect the network cable (A1) from the panel mount coupler (A2).

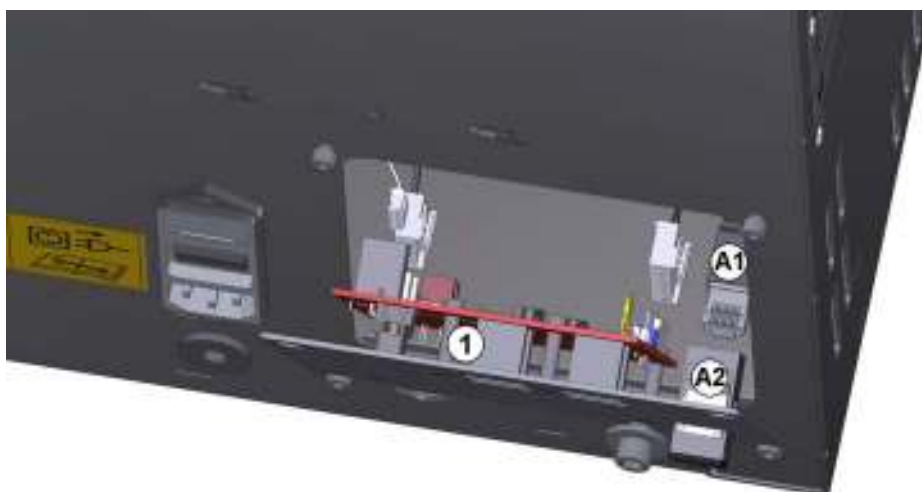


Figure 30-10: Interface assembly

4. Remove the four screws (3) of User Access Panel PCB (4).

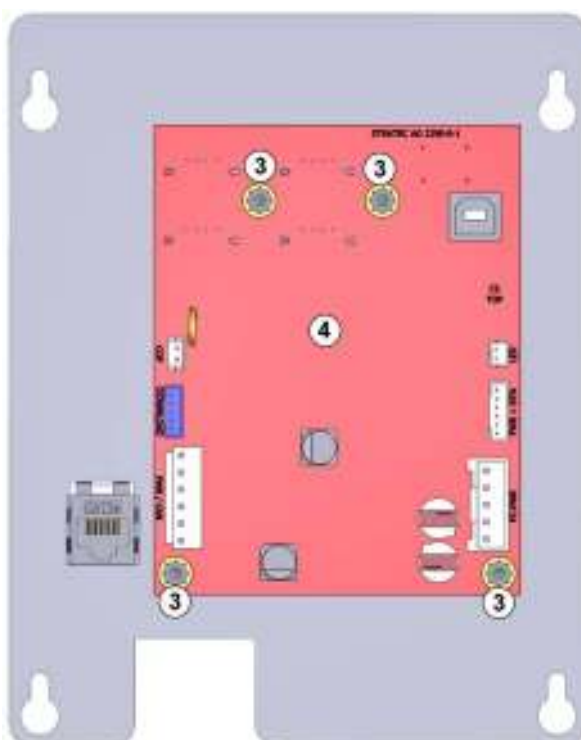


Figure 30-11: PCB, User Access Panel CU

**30.3.7.3 Assembly of the User Access Panel CU PCB**

1. Set the User Access Panel PCB (4) onto the interface assembly and fasten with four screws (3).
2. Plug all the cables onto the PCB User Access Panel CU, see Chapter 30.4.5 on page 30-28.
3. Plug the network cable (A1) onto the panel mount coupler (A2).
4. Assemble the interface assembly (1) and fasten with four screws (2).
5. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.

**30.3.7.4 Set of the User Access Panel CU PCB**

6. In case a new PCB has been installed, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
7. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### 30.3.8 Replacement of the Power Supply, 24V, 350W

The Power Supply, 24V, 350W is also called GCS350PS24-STF,PWRSPLY,24V,350W.

#### 30.3.8.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.
7. Remove the Wash Station cover, see Chapter 28.3.14.2 on page 28-47.
8. Remove the User Access Panel CU PCB, see Chapter 30.3.7.2 on page 30-14.

#### 30.3.8.2 Removal of the Power Supply, 24V, 350W

1. Loosen the two screws (3) of Power Supply.

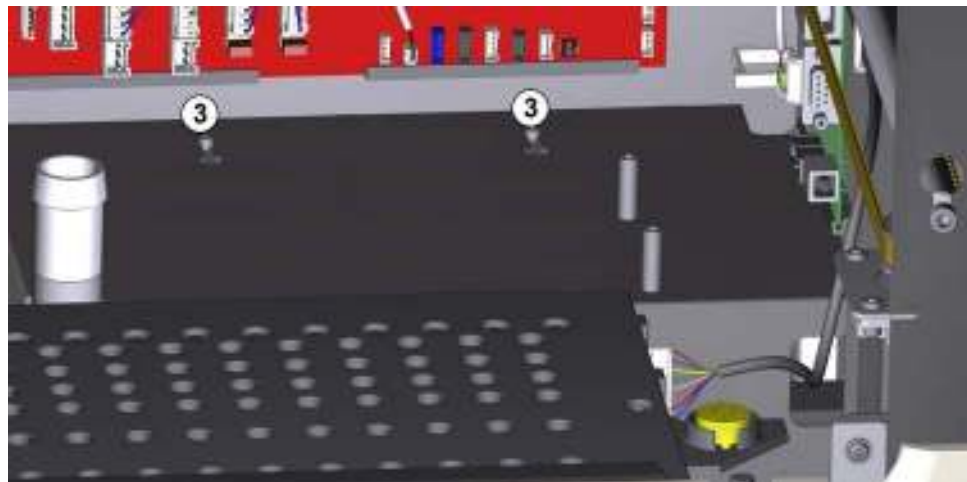


Figure 30–12: Power supply

### 30. Electrical Components for internal-tanks configuration

2. Hold on to the power supply (4), slide it out of the keyholes and set it at the instrument bottom.
3. Unplug the ground (6) and the power cable (5) from the Power Supply.

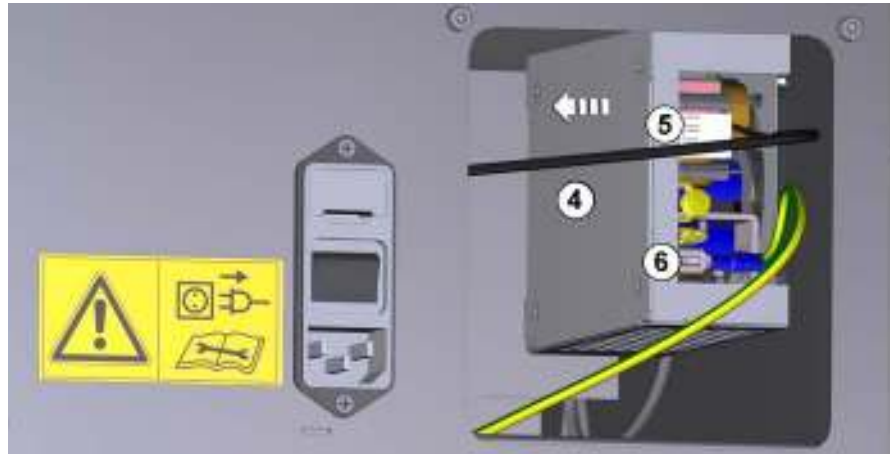


Figure 30–13: Power supply

4. Unscrew the plus (7) and minus (8) cables from the PWR CAN PCB (9).
5. Remove the two cables (7 and 8) from the cable clip (10).
6. Remove the Power Supply from the instrument.

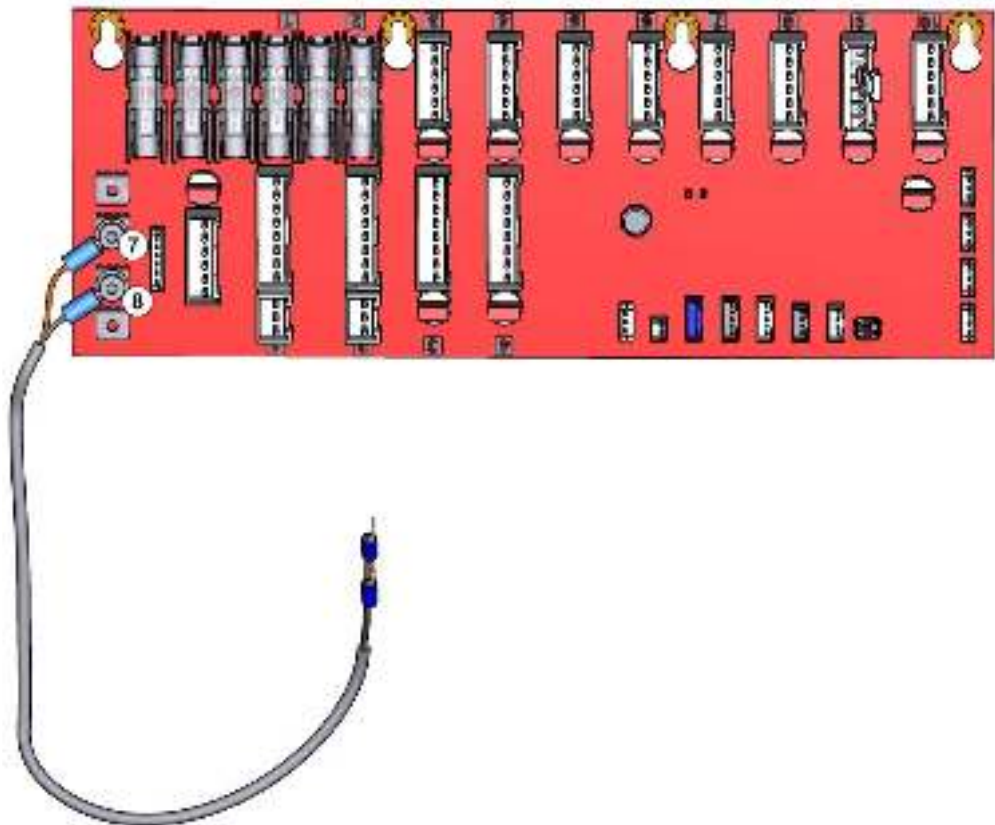


Figure 30–14: Unscrew the plus and minus cables from the PWR CAN PCB

### 30.3.8.3 Assembly of the Power Supply, 24V, 350W

1. If a new Power Supply has to be installed, then use the screws (3) of the old Power Supply and screw them into the new one.



Figure 30–15: Power supply

2. Set the Power Supply at the bottom of the instrument. While doing this, guide the plus (7, brown cable) and minus (8, black cable) cables into the opening of the chassis, making them arrive to the PWR CAN PCB (9).
3. Fasten the plus (7, brown cable) cable and the minus (8, black cable) cable onto the PWR CAN PCB (9).
4. Insert the two cables into the cable clip (10).
5. Connect the ground cable (6) and the power cable (5) to the Power Supply (4).
6. Slide the screws (3) into the keyholes and fasten the screws (3).
7. Assemble the User Access Panel CU PCBA, see Chapter 30.3.7.3 on page 30-16.
8. Assemble the Wash Station cover, see Chapter 28.3.14.3 on page 28-48.
9. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.
10. Perform the needed requalification after service visit, see Chapter 9.4 on page 9-9.

### **30.3.9 Replacement of the Remote Start Button PCB**

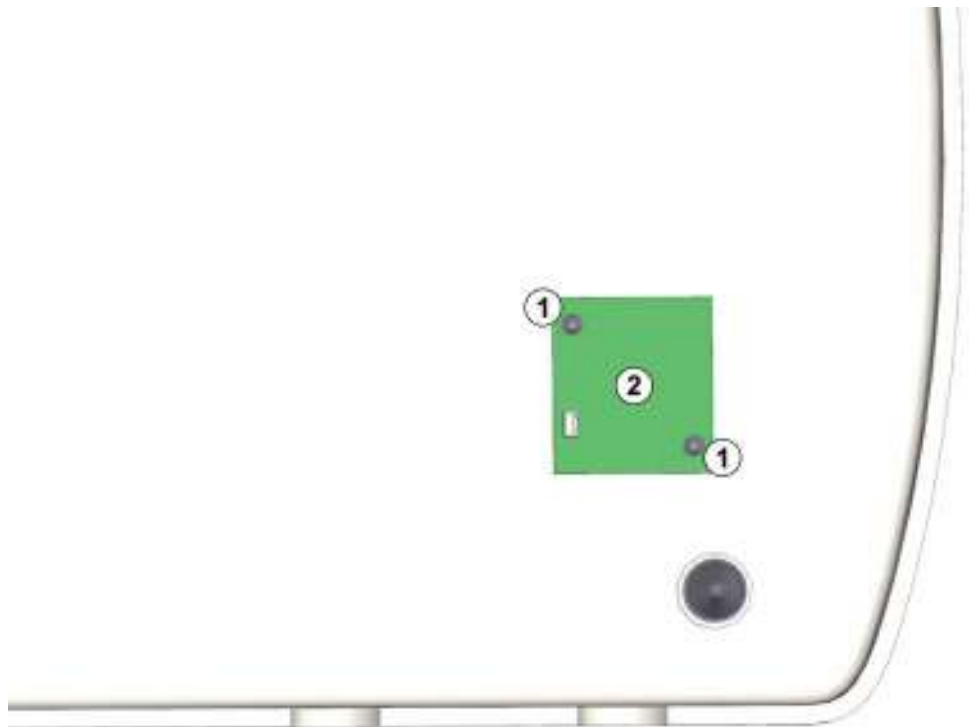
#### **30.3.9.1 Preliminary steps**



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.

#### **30.3.9.2 Removal of the Remote Start Button PCB**

1. On the inner side of the Right Side Cover, remove two screws (2) of the Remote Start Button PCB (2).
2. Remove the Remote Start Button PCB (2).



*Figure 30-16: Remote Start Button PCB*

#### **30.3.9.3 Assembly of the Remote Start Button PCB**

1. Set the Remote Start Button PCB (2) onto the Right Side Cover and fasten with two screws (1).
2. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.

## 30.4 Electric connections, switches and signals

This section shows an overview of the electronic boards of the module.

### 30.4.1 PWR CAN EXT. CU PCB

PWR CAN EXT. CU PCB is also PWR CAN EXT SW CU PCBA.

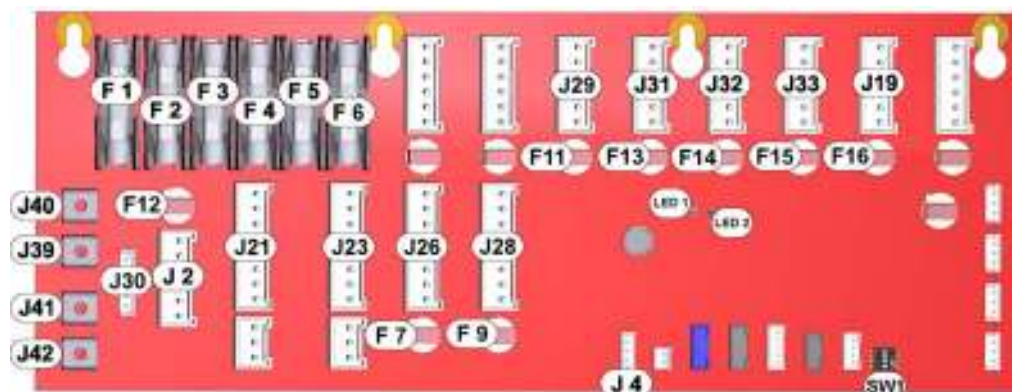


Figure 30–17: PCB, PWR CAN EXT. CU

| No. | Connection                    | Fuse       |
|-----|-------------------------------|------------|
| J2  | User Interface                | F12        |
| J4  | Shut down from COP            | no fuse    |
| J19 | CAN Termination               | F16        |
| J21 | Reagent Module                | F1, F2, F3 |
| J23 | Bridge                        |            |
| J26 | Liquid Support PCB            | F7         |
| J28 | Pipettor CU PCB               | F9         |
| J29 | Washer                        | F11        |
| J30 | Connection to COP EUS2CAN PCB |            |
| J31 | Reader                        | F13        |
| J32 | Incubator                     | F14        |
| J33 | Front Cover CU PCB            | F15        |
| J39 | GND (Power supply)            |            |
| J40 | GND (Power supply)            |            |
| J41 | +24 V (Power supply)          |            |
| J42 | +24 V (Power supply)          |            |

Figure 30–18: Connections of the PCB, PWR CAN EXT. CU



## 30. Electrical Components for internal-tanks configuration

| Number |         | Meaning/Function |
|--------|---------|------------------|
| LED 1  | Power   |                  |
| LED 2  | CAN bus |                  |

Table 30–3: LED of the PCB, PWR CAN EXT. CU

| Number |                                            | Meaning/Function |
|--------|--------------------------------------------|------------------|
| F 1    | Connector J20 and J21, 250VAC F5A,6.3x32mm |                  |
| F 2    | Connector J20 and J21, 250VAC F5A          |                  |
| F 3    | Connector J20 and J21, 250VAC F5A          |                  |
| F 4    | Connector J23 and J25, 250VAC F5A          |                  |
| F 5    | Connector J23 and J25, 250VAC F5A          |                  |
| F 6    | Connector J23 and J25, 250VAC F5A          |                  |
| F 7    | Liquid Support PCB (J26)                   |                  |
| F 9    | Pipettor (J28)                             |                  |
| F11    | Washer (J29)                               |                  |
| F12    | User Interface (J2)                        |                  |
| F13    | Reader (J31)                               |                  |
| F14    | Incubator (J32)                            |                  |
| F15    | CAN Termination (J33)                      |                  |

Table 30–4: Fuses of the PCB, PWR CAN EXT. CU

| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | ON/OFF                                  |
|            | 2 | ON/OFF                                  |
|            | 3 | ON/OFF                                  |
|            | 4 | ON/OFF                                  |

Table 30–5: DIP-Switches of the PCB, PWR CAN EXT. CU



### 30.4.2 Pipettor CU PCB

Pipettor CU PCB is also DUAL stepper PIP CU PCBA.

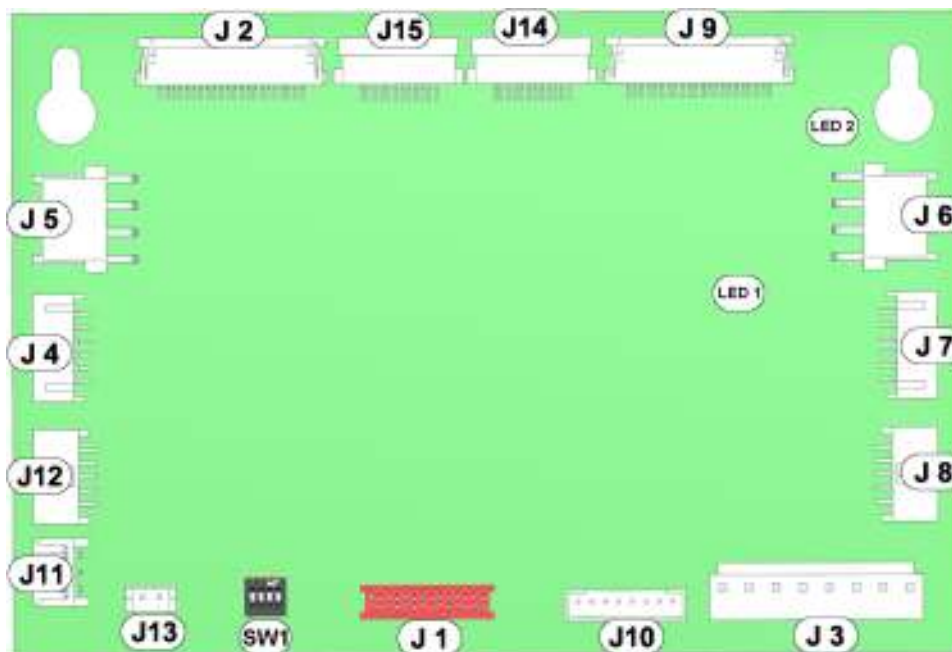


Figure 30–19: PCB, pipettor CU

| No. | Connection             | No. | Connection           |
|-----|------------------------|-----|----------------------|
| J 1 | Download (Emulator E1) | J 9 | N/A                  |
| J 2 | N/A                    | J10 | Safety circuit board |
| J 3 | Power / CAN            | J11 | PRP                  |
| J 4 | Encoder left           | J12 | Y-Home, Left Arm     |
| J 5 | Stepper drive left     | J13 | LDV                  |
| J 6 | Stepper drive right    | J14 | Right Arm            |
| J 7 | Encoder right          | J15 | Left Arm             |
| J 8 | Y-Home, Right Arm      |     |                      |

Table 30–6: Connections of the main PCB, pipettor CU

| Number | Meaning/Function |
|--------|------------------|
| LED 1  |                  |
| LED 2  |                  |

Table 30–7: LED of the PCB, pipettor CU

**30. Electrical Components for internal-tanks configuration**

---

| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | ON/OFF                                  |
|            | 2 | ON/OFF                                  |
|            | 3 | ON/OFF                                  |
|            | 4 | ON/OFF                                  |

*Table 30–8: DIP-Switches of the PCB, pipettor CU*

### 30.4.3 COP EUS2CAN PCB

COP EUS2CAN PCB is also COP EUS2CAN PCBA.

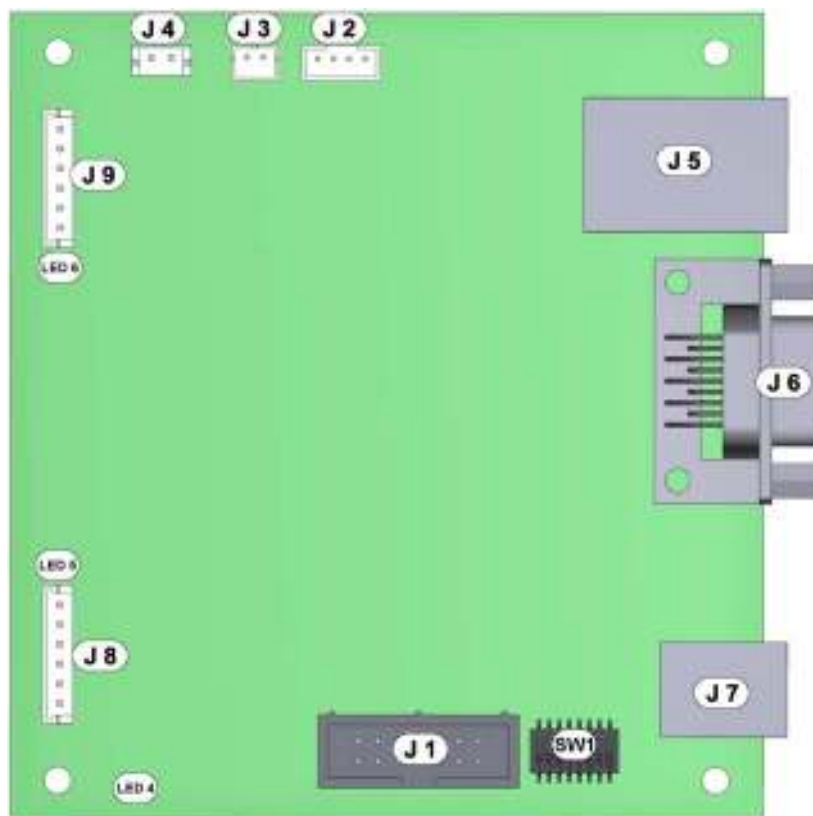


Figure 30–20: PCB, COP EUS2CAN

| No. | Connection       | No. | Connection |
|-----|------------------|-----|------------|
| J 1 | Download / Debug | J 6 | RS 232     |
| J 2 | PWR ON           | J 7 | USB        |
| J 3 | Shutdown         | J 8 | CAN-Bus 1  |
| J 4 | Standby          | J 9 | CAN-Bus 2  |
| J 5 | Ethernet         |     |            |

Table 30–9: Connections of the PCB, COP EUS2CAN

| Number | Meaning/Function | Number | Meaning/Function |
|--------|------------------|--------|------------------|
| LED 1  | System           | LED 4  | Power            |
| LED 2  | Main - PWR       | LED 5  | CAN 1 Activity   |
| LED 3  | Host COM         | LED 6  | CAN 2 Activity   |

Table 30–10: LED of the PCB, COP EUS2CAN

**30. Electrical Components for internal-tanks configuration**

---

| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | ON/OFF                                  |
|            | 2 | ON/OFF                                  |
|            | 3 | ON/OFF                                  |
|            | 4 | ON/OFF                                  |
|            | 5 | ON/OFF                                  |
|            | 6 | ON/OFF                                  |
|            | 7 | ON/OFF                                  |
|            | 8 | ON/OFF                                  |

*Table 30–11: DIP-Switches of the PCB, COP EUS2CAN*

### 30.4.4 Liquid Support PCB

LIQUID SUPPORT PCB is also RFID LIQUID SUPPORT CU PCBA.



Figure 30-21: PCB, Liquid support

| No.  | Connection              | No.  | Connection                |
|------|-------------------------|------|---------------------------|
| J 1  | Power / CAN             | J12  | EXT-OUT 2                 |
| J 2  | Download                | J13  | Antenna Starter 1         |
| J 3  | not used                | J14  | Antenna Starter 2         |
| J 4  | Tube heating, Starter 1 | J15  | DI water                  |
| J 5  | Tube heating, Starter 2 | J16  | Wash buffer               |
| J 6* | Waste Aspiration Pump   | J17* | Starter Dispense Pump 2   |
| J 7* | Liquid Waste Pump       | J18* | Wash Buffer Dispense Pump |
| J 8* | Starter Dispense Pump 1 | J19  | Liquid waste              |
| J 9  | BS Starter 1            | J20  | Consumable drawer         |
| J10  | BS Starter 2            | J21  | DI presence sensor        |
| J11  | EXT-OUT 1               | J22  | WB presence sensor        |

Table 30-12: Connections of the main PCB, Liquid support

\* connection sequence does not matter

| Number | Meaning/Function |
|--------|------------------|
| LED 1  | Power            |

Table 30-13: LED of the PCB, Liquid support

### 30.4.5 User access panel CU PCBA

USER ACCESS PANEL CU PCBA.

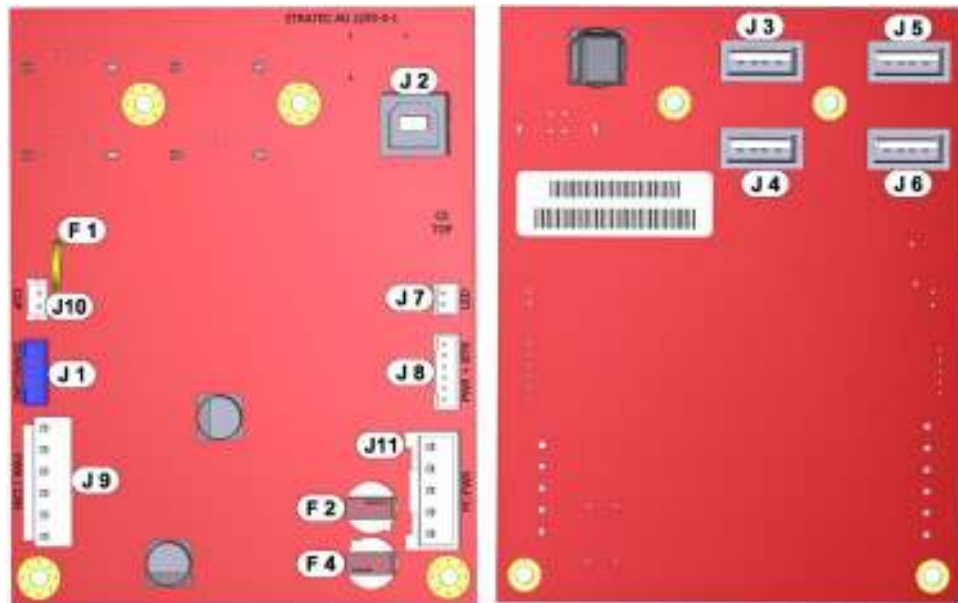


Figure 30-22: User access panel CU PCBA, front and rear view

| No. | Connection | No. | Connection    |
|-----|------------|-----|---------------|
| J 1 | Download   | J 7 | LED           |
| J 2 | PC-IN      | J 8 | Power-Button  |
| J 3 | USB 1      | J 9 | Power / CAN   |
| J 4 | USB 2      | J10 | COP (standby) |
| J 5 | USB 3      | J11 | PC-PWR        |
| J 6 | USB 4      |     |               |

Table 30-14: Connections of the user access panel CU PCBA

| Number | Meaning/Function |
|--------|------------------|
| F1     | Standby COP      |
| F2     | Fuse 2           |
| F4     | PC power         |

Table 30-15: Fuse of the user access panel CU PCBA

### 30.4.6 Remote Start Button PCB

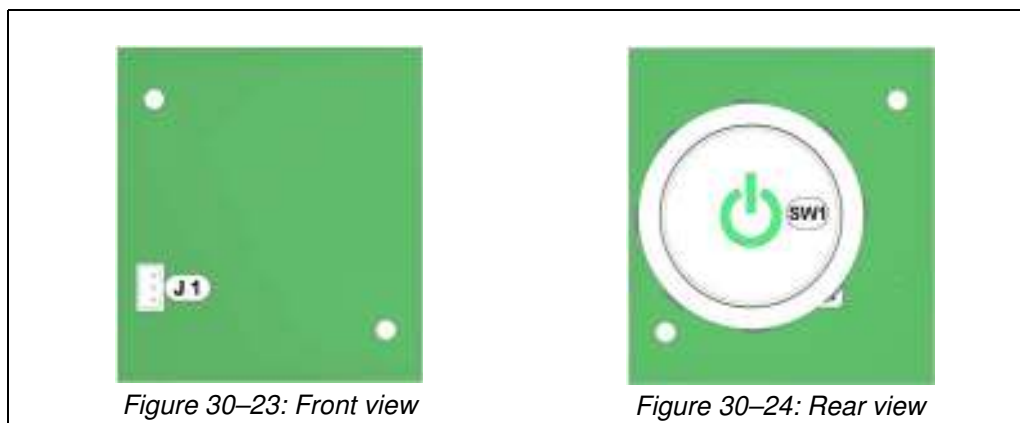


Figure 30-25: Remote Start Button PCB, front and rear view

| No. | Connection           |
|-----|----------------------|
| J 1 | User Access Panel CU |

Table 30-16: Connections of the Remote Start Button PCB

| Button | Meaning/Function/Default Switch Setting |
|--------|-----------------------------------------|
| SW1    | Start Button, ON/OFF                    |

Table 30-17: Switch of the Remote Start Button PCB





# 31 Electrical Components for external-tanks configuration

This chapter describes additional electrical components of **LIAISON<sup>®</sup> XS** system in detail. The first section of the chapter introduces “Safety” notes concerning these components, followed by a brief overview of the main circuit boards.

The following section contains information about repair and calibration (“Replacement and Repair” section) of electrical components.

In the last section, layout and connections of Liquid Support PCBA, Pipettor CU PCBA, COP EUS2CAN PCBA and PWR CAN EXT. CU PCBA are described.

## 31.1 Safety

### WARNING



#### Non-observance of safety instructions

The non-observance of safety instructions may result in serious personal injury and material damage.

- Follow all safety instructions, see Chapter 2.5 on page 2-7.
- Follow all warnings marked on the instrument.
- Follow all warnings marked on the instrument.
- Always wear personal protective equipment suitable for the activity.

### DANGER



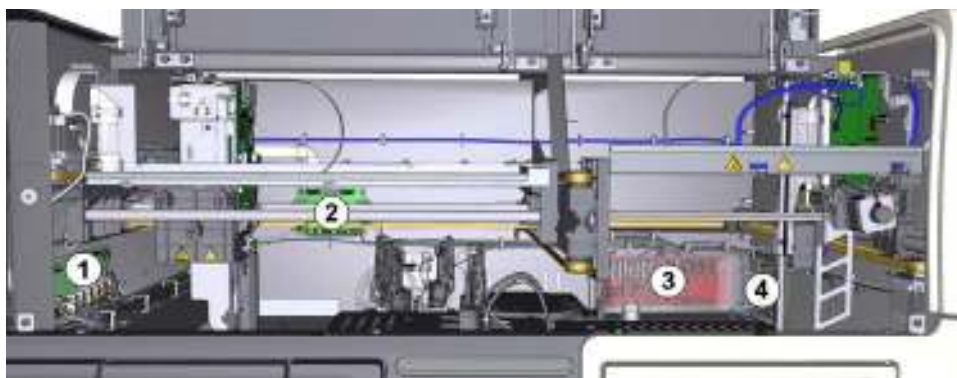
#### High Voltage!

The touching of high voltage parts or cables can result in serious personal injuries with deadly consequences.

- Never touch high voltage parts with instrument on.
- Separate the system from the mains supply before maintaining or cleaning it.
- Be careful! The following parts can be under high voltage during operation:
  - UPS device (100-240V, high current)
  - Main supply unit (100-240V, high current)
  - Enclosure metal part in Single Fault conditions (132V r.m.s., low current)
  - Photomultiplier (PMT) (~ 1000 Volts, low current)

## 31.2 Overview

This section shows an overview of the main circuit boards.



*Figure 31-1: Circuit boards overview*

|   |                      |
|---|----------------------|
| 1 | Liquid Support PCBA  |
| 2 | Pipettor CU PCBA     |
| 3 | PWR CAN EXT. CU PCBA |
| 4 | COP EUS2CAN PCBA     |

*Table 31-1: Circuit boards overview*

### 31.3 Replacement and repair

This section contains instructions for the repair of electrical components. After several repair works it is required that the individual components are readjusted. Follow the instructions in the individual sections.

#### 31.3.1 Replacement of the Liquid Support PCBA

##### 31.3.1.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the rearmost area of the instrument.

##### 31.3.1.2 Removal of the Liquid Support PCBA

1. Unplug all cables of Liquid Support PCB A (1).
2. Loosen the five screws (2), lift the PCB up and remove it.

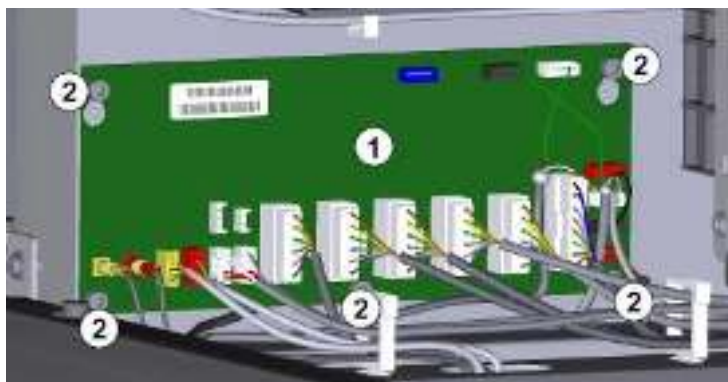


Figure 31–2: Liquid Support PCBA

##### 31.3.1.3 Assembly of the Liquid Support PCBA

1. Slide the PCB (1) into the keyholes and fasten the five screws (2).
2. Plug all cables onto the PCB, see Chapter 31.4.4 on page 31-27.

##### 31.3.1.4 Set of the Liquid Support PCBA

1. When the Liquid Support PCBA is installed and wired completely, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the calibration of Starter bubble sensors, see Chapter 12.6.3.4 on page 12-43.

### **31. Electrical Components for external-tanks configuration**

---

3. Check the System Liquid tank level sensor, see Chapter 29.4.3 on page 29-52.
4. Check the Wash Buffer tank level sensor, see Chapter 29.4.5 on page 29-53.
5. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 31.3.2 Replacement of the Pipettor CU PCBA

#### 31.3.2.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the left top cover.
6. Move the left pipettor arm to a safe location on the rightmost area of the instrument.

#### 31.3.2.2 Removal of the Pipettor CU PCBA

1. Unplug all cables of Pipettor CU PCBA (1).
2. Loosen the two screws (2) and lift the PCB out of the board holder (3).

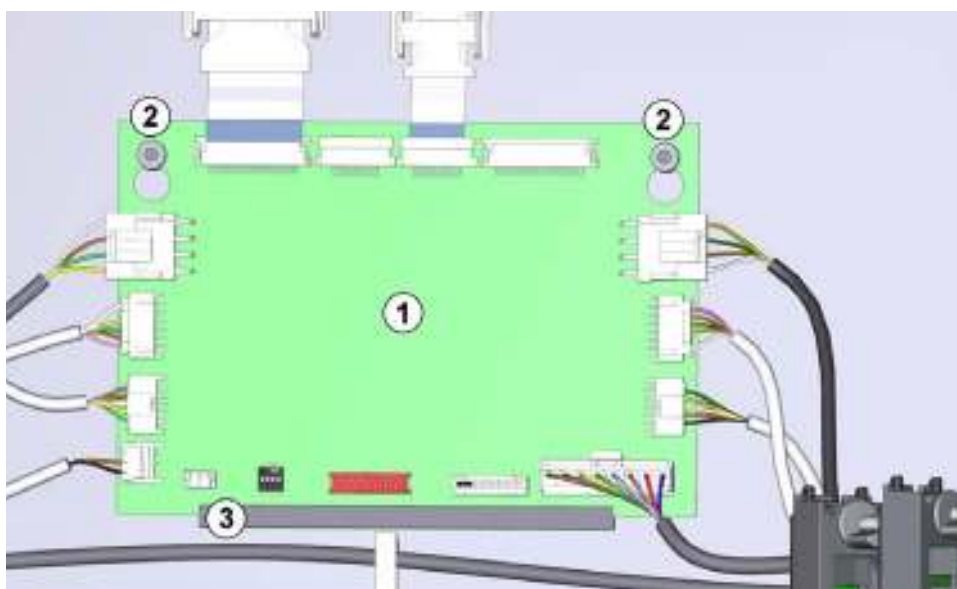


Figure 31–3: Pipettor CU PCBA

#### 31.3.2.3 Assembly of the Pipettor CU PCBA

1. Slide the PCB (1) into the board holder (3) and fasten the two screws (2).
2. Plug all cables onto the PCB, see Chapter 31.4.2 on page 31-23.

#### 31.3.2.4 Set of the Pipettor CU PCBA

1. When the Pipettor CU PCBA is installed and wired completely, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.

### 31.3.3 Replacement of PWR CAN EXT fuses



---

**Electrocution/Fire Hazard!**

Defective systems can result in serious injuries with deadly consequences and material damages (e.g. fire).

- Separate immediately the defective system from the mains supply, if a safe usage is no longer possible.
  - Secure the defective system against reconnection.
  - Label the defective system clearly as being defective.
- 



---

**Danger of Electrocution**

If the system is not separated from the mains supply before maintenance works, this can cause serious injuries with deadly consequences due to electrocution. Additionally, there is the danger that the system starts and causes injuries (e.g. contusion, cuts etc.) to the person working with the system.

- Determine and correct the cause of a blown fuse or thrown circuit-breaker before attempting to resume operations of the system.
  - Separate the system from the mains supply and secure it against restarting before replacing fuses.
  - Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.
  - Never repair or bridge blown fuses.
  - Never short-circuit the fuse holder.
- 

#### 31.3.3.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the Wash Station cover, see Chapter 29.3.15.2 on page 29-48.

---

**NOTE**

Although spare fuses are provided with the instrument and the replacement procedure is described, you should be aware that blown fuses are very often indicators of other malfunctions that may be affecting instrument modules, components or wires. If the fuses blow again shortly after their replacement and troubleshooting activities fail, please contact DiaSorin Corporate Service for assistance.

---

## 31. Electrical Components for external-tanks configuration

## 31.3.3.2 Removal of the PWR CAN EXT blown fuse

1. Remove the faulty fuse from the PWR CAN EXT PCB, using an insulated plier.

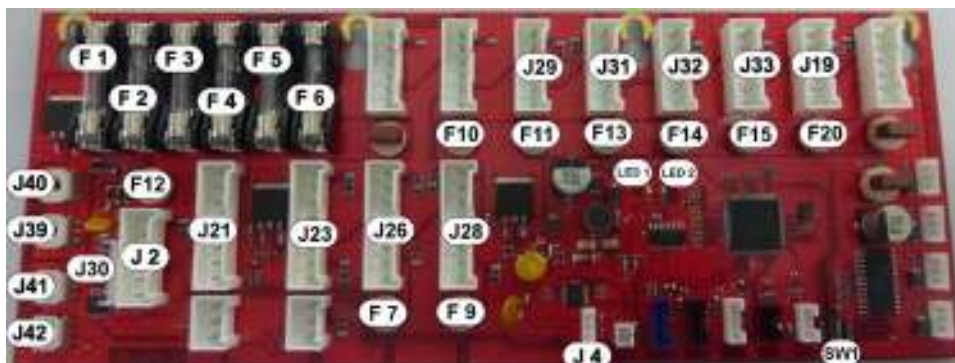


Figure 31–4: PWR CAN EXT SW CU PCBA

| Number | Connector | Circuit | Description               | Fuse type                                                               |
|--------|-----------|---------|---------------------------|-------------------------------------------------------------------------|
|        |           | 0       | COP digital supply        |                                                                         |
| F1     | J20       | 1       | not used                  | 250VAC, F5A                                                             |
| F2     | J21       | 1       |                           | 250VAC, F5A                                                             |
| F3     | J21       | 1       | Loading Bay               | 250VAC, F5A                                                             |
| F4     | J23       | 2       | Bridge                    | 250VAC, F5A                                                             |
| F5     | J23       | 2       | Bridge                    | 250VAC, F5A                                                             |
| F6     | J25       | 2       | not used                  | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F7     | J26       | 3       | RFID Liquid support<br>CU | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F8     | J3/J26    | 3       | Liquid support            | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F9     | J28       | 4       | Pipettor                  | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F10    | J27/J28   | 4       | Pipettor                  | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F11    | J29       | 5       | Incubator                 | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |
| F12    | J2        | 0       | User interface            | Temperature fuse or<br>TE5 Time-Lag fuse, 392<br>series: 250VAC, T3.15A |

## 31. Electrical Components for external-tanks configuration

| Number | Connector | Circuit | Description         | Fuse type                                                         |
|--------|-----------|---------|---------------------|-------------------------------------------------------------------|
| F13    | J31       | 6       | Measurement chamber | Temperature fuse or TE5 Time-Lag fuse, 392 series: 250VAC, T3.15A |
| F14    | J32       | 7       | Washer              | Temperature fuse or TE5 Time-Lag fuse, 392 series: 250VAC, T3.15A |
| F15    | J33       | 8       | Front cover         | Temperature fuse or TE5 Time-Lag fuse, 392 series: 250VAC, T3.15A |
| F20    | J19       | 9       | CAN termination.    | Temperature fuse or TE5 Time-Lag fuse, 392 series: 250VAC, T3.15A |

Table 31–2: Fuses with corresponding connectors and circuits

## 31.3.3.3 Assembly of the PWR CAN EXT new fuse

**NOTE****Replace fuses**

Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.

1. Insert the new fuse into the dedicated holder of PWR CAN EXT PCB.
2. Assemble the Wash Station cover, see Chapter 29.3.15.3 on page 29-49.
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 31.3.4 Replacement of the PWR CAN EXT. CU PCBA

#### 31.3.4.1 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the Wash Station cover, see Chapter 29.3.15.2 on page 29-48.

#### 31.3.4.2 Removal of the PWR CAN EXT. CU PCB

1. Unplug all the cables from the PWR CAN EXT. CU PCB (1).
2. Loosen the four screws (2) and lift the PCB board up and remove it.

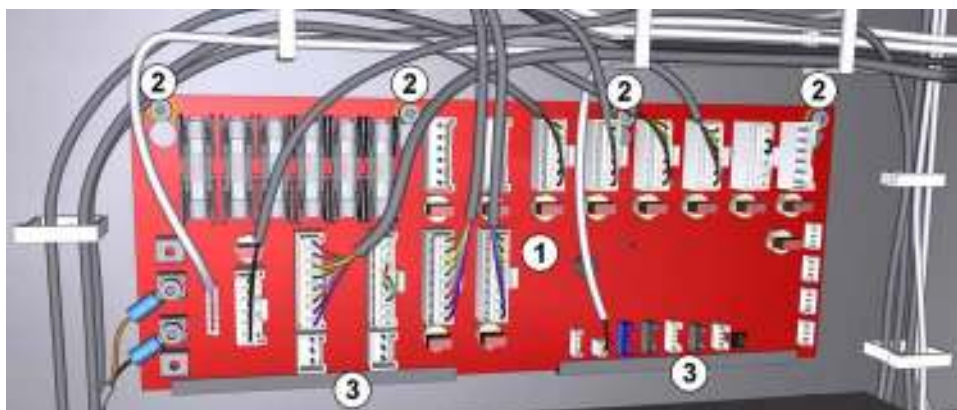


Figure 31-5: PCB, PWR CAN EXT. CU

#### 31.3.4.3 Assembly of the PWR CAN EXT. CU PCB

1. Slide the PWR CAN EXT. CU PCB (1) into the guide rails (3) and fasten the four screws (2).
2. Plug all the cables onto connectors of the circuit board, see Chapter 31.4.1 on page 31-21.
3. Assemble the Wash Station cover, see Chapter 29.3.15.3 on page 29-49.

#### 31.3.4.4 Set of the PWR CAN EXT. CU PCBA

1. When the PWR CAN EXT. CU PCB is installed and wired completely, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 31.3.5 Replacement of the COP EUS2CAN PCBA

#### 31.3.5.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Move the right pipettor arm to a safe location on the leftmost area of the instrument.
7. Remove the Wash Station cover, see Chapter 29.3.15.2 on page 29-48.

#### 31.3.5.2 Removal of the COP EUS2CAN PCB

1. Unplug all cables from the COP EUS2CAN PCB (1).
2. Loosen the two screws (2) and lift the PCB out.

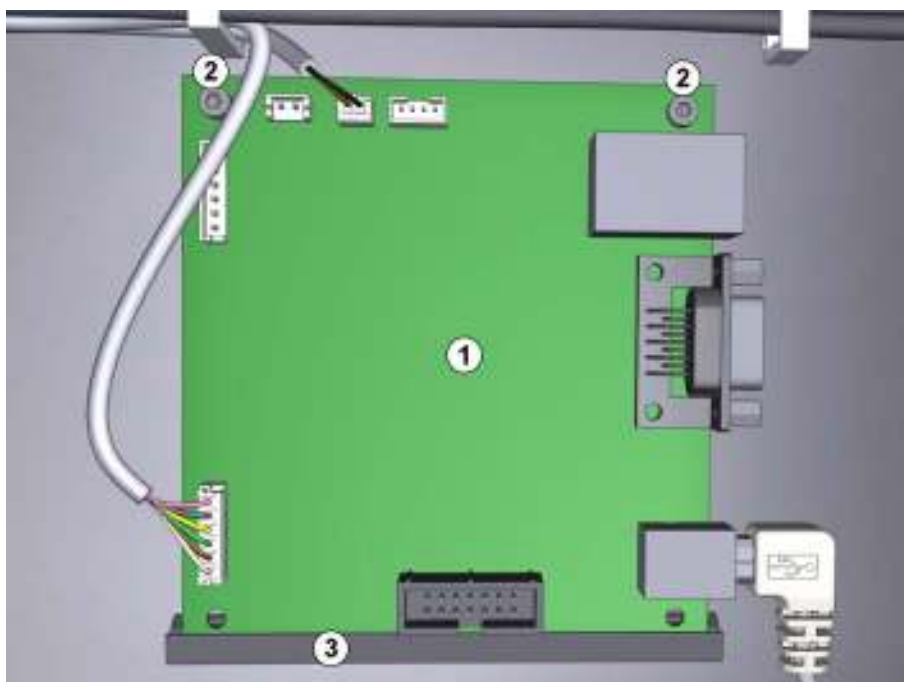


Figure 31–6: PCB, COP EUS2CAN

#### 31.3.5.3 Assembly of the COP EUS2CAN PCB

1. Slide the COP EUS2CAN PCB (1) into the guide rails (3) and fasten the two screws (2).
2. Plug all the cables onto the circuit board, see Chapter 31.4.3 on page 31-25.
3. Assemble the Wash Station cover, see Chapter 29.3.15.3 on page 29-49.

**31.3.5.4 Set of the COP EUS2CAN PCBA**

1. When the COP EUS2CAN PCB is installed and wired completely, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Open the **Service Software** from the Windows Administrator account.
3. At the navigation window, choose **COP** and the **Parameter** bracket.
4. In the **Read Parameter(s)** field, type **2** and press the **Update** button on the right.
5. Select the parameter 2 from the parameter list.
6. Type the system Serial Number in the **New Value** field on the left, and press the **Write** button.
7. Press the **Non-Volatile Memory** button and select **Overwrite**.
8. Check all the Pipettor teaching, see Chapter 13.15 on page 13-32.
9. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 31.3.6 Replacement of Fuses of the Main Switch

#### 31.3.6.1 Safety



---

##### Defective instrument

Any defective instrument will result in serious injuries with deadly consequences and material damage (e.g. fire).

- Immediately disconnect the defective instrument from the mains supply, if a safe usage is no longer possible.
  - Secure the defective instrument against reconnection.
  - Label the defective instrument clearly as being defective.
- 



---

##### Danger of Electrocutation

If the system is not separated from the mains supply before maintenance works, this can cause serious injuries with deadly consequences due to electrocution. Additionally, there is the danger that the system starts and causes injuries (e.g. contusion, cuts etc.) to the person working with the system.

- Determine and correct the cause of a blown fuse or thrown circuit-breaker before attempting to resume operations of the system.
  - Separate the system from the mains supply and secure it against restarting before replacing fuses.
  - Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.
  - Never repair or bridge blown fuses.
  - Never short-circuit the fuse holder.
- 

##### NOTE

Although spare fuses are provided with the instrument and the replacement procedure is described, you should be aware that blown fuses are very often indicators of other malfunctions that may be affecting instrument modules, components or wires. If the fuses blow again shortly after their replacement and troubleshooting activities fail, please contact DiaSorin Corporate Service for assistance.

---

#### 31.3.6.2 Preliminary steps



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.

### 31.3.6.3 Remove the Fuses of Main Switch

1. At the Main Switch (1), open the fuse box (2) with a screwdriver.
2. Remove the fuse box (3).
3. Remove defective fuses (4).



Figure 31-7: Right side view of instrument, main switch

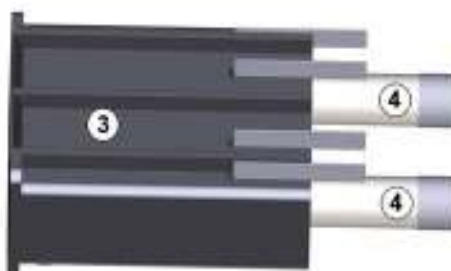


Figure 31-8: Fuse box

### 31.3.6.4 Assemble the Fuses of Main Switch

#### NOTE

#### Replace Fuses

Spare fuses must match the values (nominal voltage, nominal current and type) specified by the manufacturer.

Always replace blown fuses, do not try to repair them.

Never short-circuit the fuse holder.

1. Set the new fuses into the fuse box.
2. Slide the fuse box into the Main Switch and fasten the screws (2).
3. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### 31.3.7 Replacement of the User Access Panel CU PCB

#### 31.3.7.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.

#### 31.3.7.2 Removal of the User Access Panel CU PCB

1. Loosen the four screws (2) at the interface assembly (1).



Figure 31–9: Interface assembly

### 31. Electrical Components for external-tanks configuration

2. Lift the interface assembly (1) a bit up, flip it out and unplug all the cables from the User Access Panel CU PCB.
3. Disconnect the network cable (A1) from the panel mount coupler (A2).

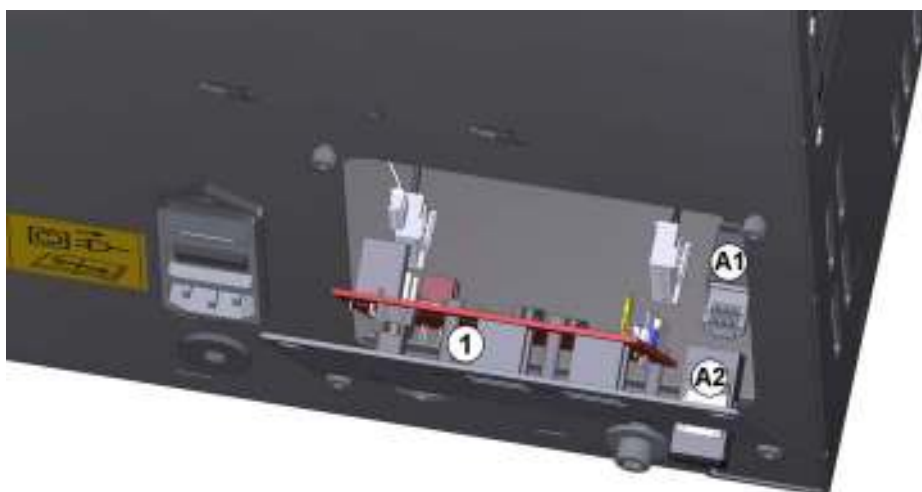


Figure 31-10: Interface assembly

4. Remove the four screws (3) of User Access Panel PCB (4).

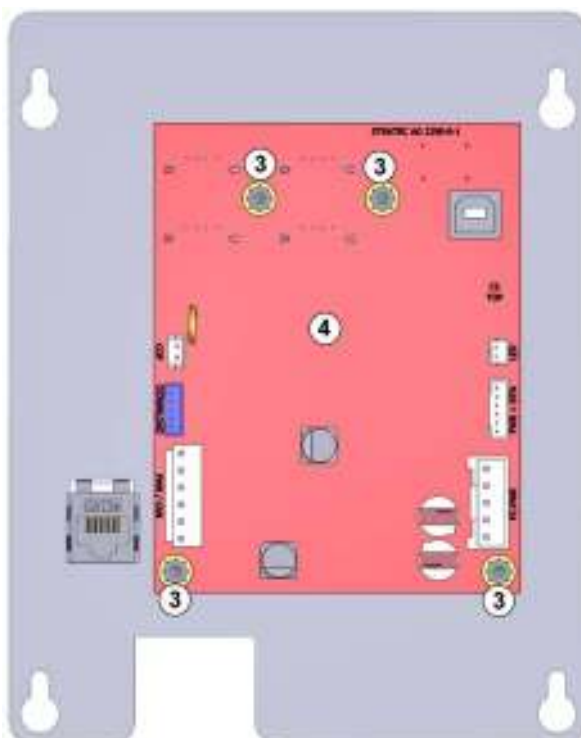


Figure 31-11: PCB, User Access Panel CU

**31.3.7.3 Assembly of the User Access Panel CU PCB**

1. Set the User Access Panel PCB (4) onto the interface assembly and fasten with four screws (3).
2. Plug all the cables onto the PCB User Access Panel CU, see Chapter 31.4.5 on page 31-28.
3. Plug the network cable (A1) onto the panel mount coupler (A2).
4. Assemble the interface assembly (1) and fasten with four screws (2).
5. Assemble the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.

**31.3.7.4 Set of the User Access Panel CU PCB**

1. In case a new PCB has been installed, then proceed with the firmware update, see Chapter 32.3.4 on page 32-6.
2. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.



### 31.3.8 Replacement of the Power Supply, 24V, 350W

The Power Supply, 24V, 350W is also called GCS350PS24-STF,PWRSPLY,24V,350W.

#### 31.3.8.1 Preliminary steps



1. Exit the **LIAISON<sup>®</sup> XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.
7. Remove the Wash Station cover, see Chapter 29.3.15.2 on page 29-48.
8. Remove the User Access Panel CU PCB, see Chapter 31.3.7.2 on page 31-14.

#### 31.3.8.2 Removal of the Power Supply, 24V, 350W

1. Loosen the two screws (3) of Power Supply.

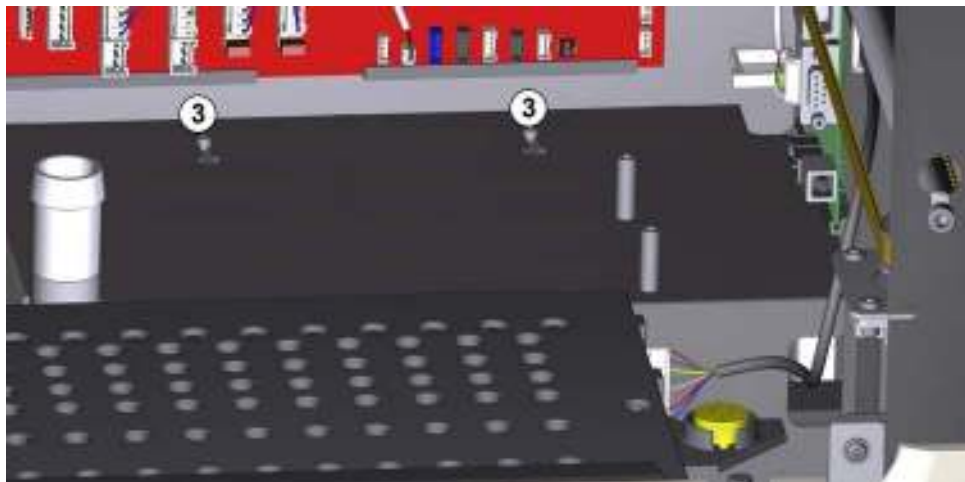


Figure 31–12: Power supply

### 31. Electrical Components for external-tanks configuration

2. Hold on to the power supply (4), slide it out of the keyholes and set it at the instrument bottom.
3. Unplug the ground (6) and the power cable (5) from the Power Supply.

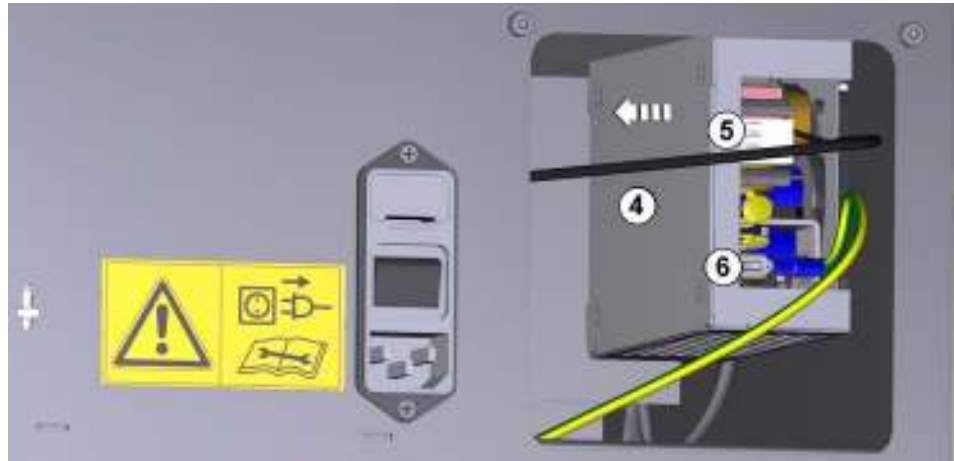


Figure 31–13: Power supply

4. Unscrew the plus (7) and minus (8) cables from the PWR CAN PCB (9).
5. Remove the two cables (7 and 8) from the cable clip (10).
6. Remove the Power Supply from the instrument.

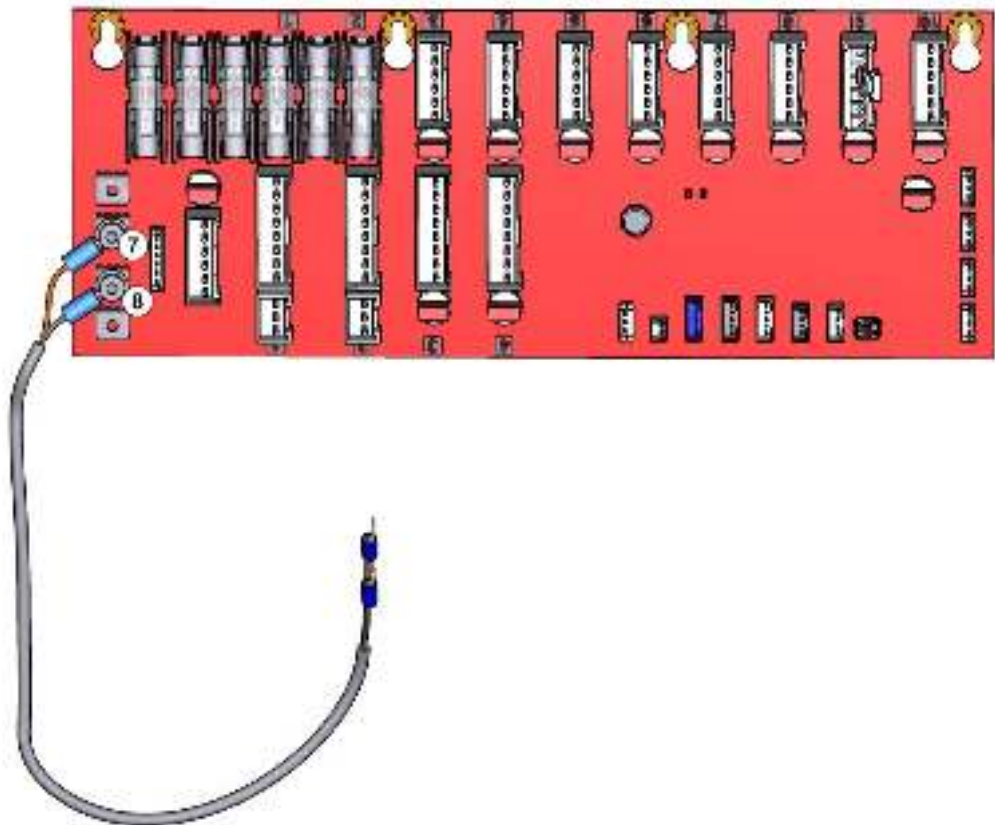


Figure 31–14: Unscrew the plus and minus cables from the PWR CAN PCB

**31.3.8.3 Assembly of the Power Supply, 24V, 350W**

1. If a new Power Supply has to be installed, then use the screws (3) of the old Power Supply and screw them into the new one.



*Figure 31-15: Power supply*

2. Set the Power Supply at the bottom of the instrument. While doing this, guide the plus (7, brown cable) and minus (8, black cable) cables into the opening of the chassis, making them arrive to the PWR CAN PCB (9).
3. Fasten the plus (7, brown cable) cable and the minus (8, black cable) cable onto the PWR CAN PCB (9).
4. Insert the two cables into the cable clip (10).
5. Connect the ground cable (6) and the power cable (5) to the Power Supply (4).
6. Slide the screws (3) into the keyholes and fasten the screws (3).
7. Assemble the User Access Panel CU PCBA, see Chapter 31.3.7.3 on page 31-16.
8. Assemble the Wash Station cover, see Chapter 29.3.15.3 on page 29-49.
9. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.
10. Perform the needed requalification after service visit, see Chapter 10.4 on page 10-10.

### **31.3.9 Replacement of the Remote Start Button PCB**

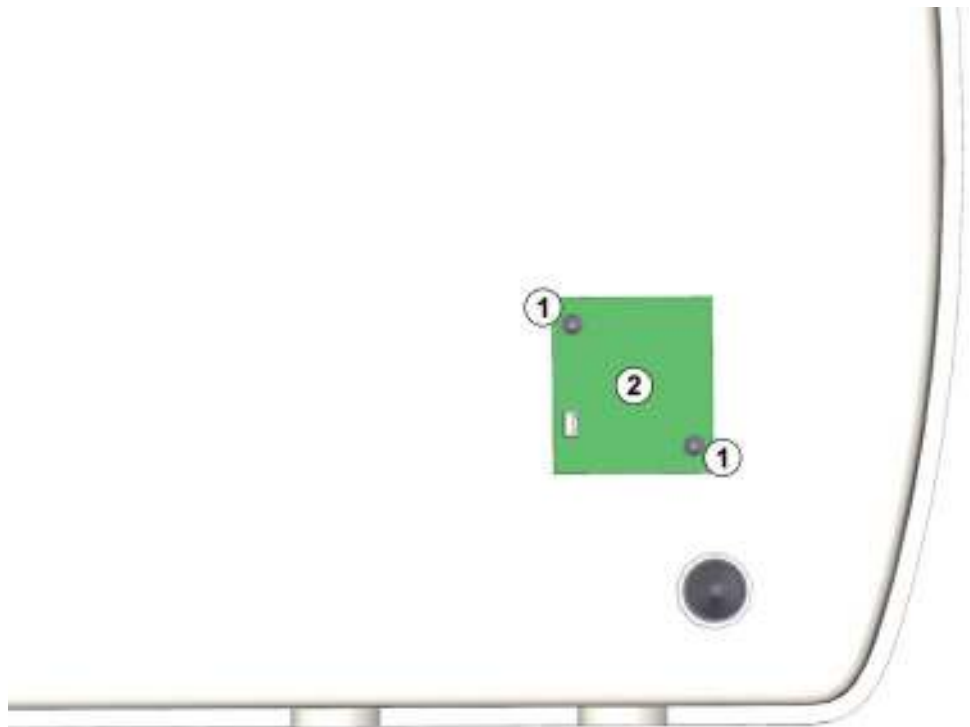
#### **31.3.9.1 Preliminary steps**



1. Exit the **LIAISON® XS** system software.
2. Shut down the PC, using the shut down button in Windows, wait until the shut down is complete.
3. On the right side of the system turn the power switch off.
4. Unplug the system power plug from the right side of the system.
5. Open the top covers.
6. Remove the Right Side Cover, see Chapter 14.3.3.3 on page 14-8.

#### **31.3.9.2 Removal of the Remote Start Button PCB**

1. On the inner side of the Right Side Cover, remove two screws (2) of the Remote Start Button PCB (2).
2. Remove the Remote Start Button PCB (2).



*Figure 31-16: Remote Start Button PCB*

#### **31.3.9.3 Assembly of the Remote Start Button PCB**

1. Set the Remote Start Button PCB (2) onto the Right Side Cover and fasten with two screws (1).
2. Assemble the Right Side Cover, see Chapter 14.3.3.4 on page 14-9.

## 31.4 Electric connections, switches and signals

This section shows an overview of the electronic boards of the module.

### 31.4.1 PWR CAN EXT. CU PCB

PWR CAN EXT. CU PCB is also PWR CAN EXT SW CU PCBA.

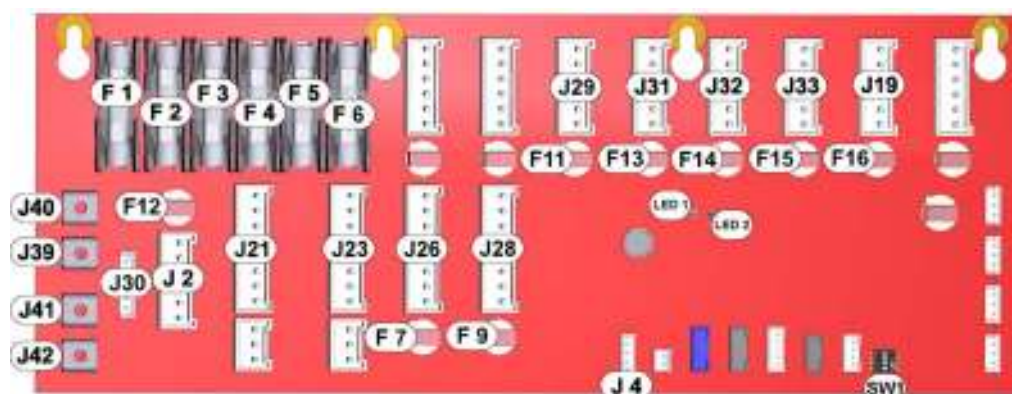


Figure 31–17: PCB, PWR CAN EXT. CU

| No. | Connection                    | Fuse       |
|-----|-------------------------------|------------|
| J2  | User Interface                | F12        |
| J4  | Shut down from COP            | no fuse    |
| J19 | CAN Termination               | F16        |
| J21 | Reagent Module                | F1, F2, F3 |
| J23 | Bridge                        |            |
| J26 | Liquid Support PCB            | F7         |
| J28 | Pipettor CU PCB               | F9         |
| J29 | Washer                        | F11        |
| J30 | Connection to COP EUS2CAN PCB |            |
| J31 | Reader                        | F13        |
| J32 | Incubator                     | F14        |
| J33 | Front Cover CU PCB            | F15        |
| J39 | GND (Power supply)            |            |
| J40 | GND (Power supply)            |            |
| J41 | +24 V (Power supply)          |            |
| J42 | +24 V (Power supply)          |            |

Table 31–3: Connections of the PCB, PWR CAN EXT. CU

## 31. Electrical Components for external-tanks configuration

| Number |         | Meaning/Function |
|--------|---------|------------------|
| LED 1  | Power   |                  |
| LED 2  | CAN bus |                  |

Table 31–4: LED of the PCB, PWR CAN EXT. CU

| Number |                                            | Meaning/Function |
|--------|--------------------------------------------|------------------|
| F 1    | Connector J20 and J21, 250VAC F5A,6.3x32mm |                  |
| F 2    | Connector J20 and J21, 250VAC F5A          |                  |
| F 3    | Connector J20 and J21, 250VAC F5A          |                  |
| F 4    | Connector J23 and J25, 250VAC F5A          |                  |
| F 5    | Connector J23 and J25, 250VAC F5A          |                  |
| F 6    | Connector J23 and J25, 250VAC F5A          |                  |
| F 7    | Liquid Support PCB (J26)                   |                  |
| F 9    | Pipettor (J28)                             |                  |
| F11    | Washer (J29)                               |                  |
| F12    | User Interface (J2)                        |                  |
| F13    | Reader (J31)                               |                  |
| F14    | Incubator (J32)                            |                  |
| F15    | CAN Termination (J33)                      |                  |

Table 31–5: Fuses of the PCB, PWR CAN EXT. CU

| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | ON/OFF                                  |
|            | 2 | ON/OFF                                  |
|            | 3 | ON/OFF                                  |
|            | 4 | ON/OFF                                  |

Table 31–6: DIP-Switches of the PCB, PWR CAN EXT. CU

### 31.4.2 Pipettor CU PCB

Pipettor CU PCB is also DUAL stepper PIP CU PCBA.

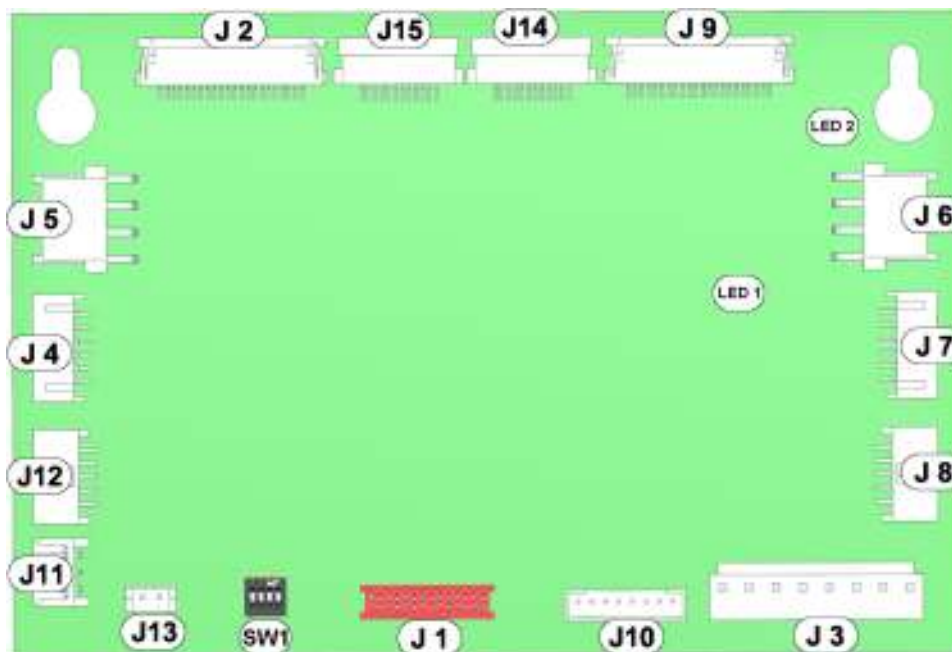


Figure 31–18: PCB, pipettor CU

| No. | Connection             | No. | Connection           |
|-----|------------------------|-----|----------------------|
| J 1 | Download (Emulator E1) | J 9 | N/A                  |
| J 2 | N/A                    | J10 | Safety circuit board |
| J 3 | Power / CAN            | J11 | PRP                  |
| J 4 | Encoder left           | J12 | Y-Home, Left Arm     |
| J 5 | Stepper drive left     | J13 | Not used             |
| J 6 | Stepper drive right    | J14 | Right Arm            |
| J 7 | Encoder right          | J15 | Left Arm             |
| J 8 | Y-Home, Right Arm      |     |                      |

Table 31–7: Connections of the main PCB, pipettor CU

| Number | Meaning/Function |
|--------|------------------|
| LED 1  |                  |
| LED 2  |                  |

Table 31–8: LED of the PCB, pipettor CU

**31. Electrical Components for external-tanks configuration**

---

| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | ON/OFF                                  |
|            | 2 | ON/OFF                                  |
|            | 3 | ON/OFF                                  |
|            | 4 | ON/OFF                                  |

*Table 31–9: DIP-Switches of the PCB, pipettor CU*



### 31.4.3 COP EUS2CAN PCB

COP EUS2CAN PCB is also COP EUS2CAN PCBA.

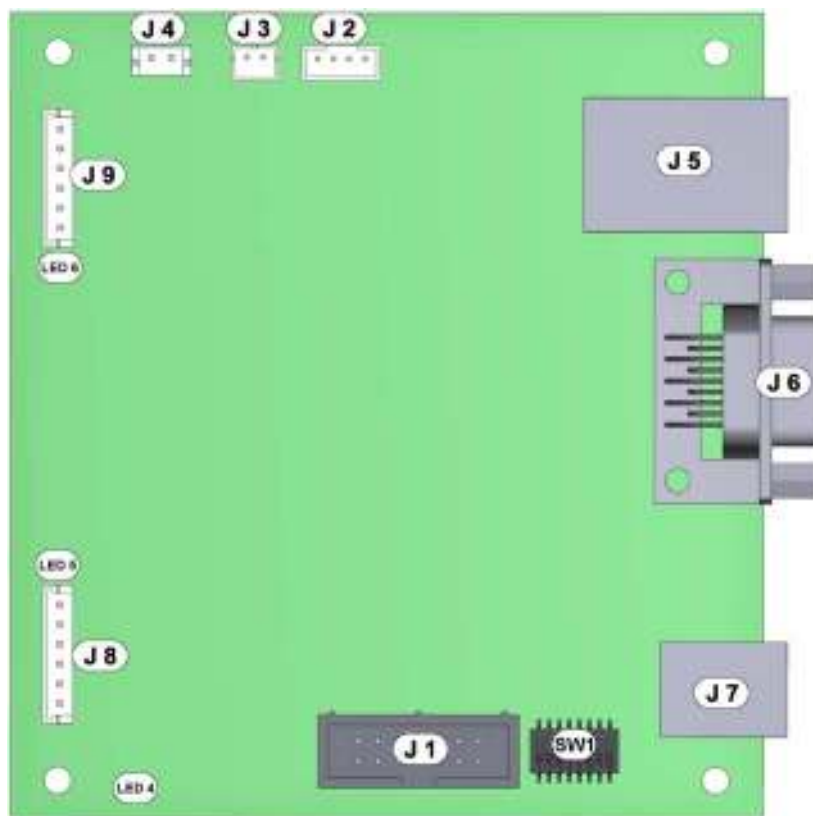


Figure 31–19: PCB, COP EUS2CAN

| No. | Connection       | No. | Connection |
|-----|------------------|-----|------------|
| J 1 | Download / Debug | J 6 | RS 232     |
| J 2 | PWR ON           | J 7 | USB        |
| J 3 | Shutdown         | J 8 | CAN-Bus 1  |
| J 4 | Standby          | J 9 | CAN-Bus 2  |
| J 5 | Ethernet         |     |            |

Table 31–10: Connections of the PCB, COP EUS2CAN

| Number | Meaning/Function | Number | Meaning/Function |
|--------|------------------|--------|------------------|
| LED 1  | System           | LED 4  | Power            |
| LED 2  | Main - PWR       | LED 5  | CAN 1 Activity   |
| LED 3  | Host COM         | LED 6  | CAN 2 Activity   |

Table 31–11: LED of the PCB, COP EUS2CAN

**31. Electrical Components for external-tanks configuration**

---

| DIP-Switch |   | Meaning/Function/Default Switch Setting |
|------------|---|-----------------------------------------|
| SW1        | 1 | ON/OFF                                  |
|            | 2 | ON/OFF                                  |
|            | 3 | ON/OFF                                  |
|            | 4 | ON/OFF                                  |
|            | 5 | ON/OFF                                  |
|            | 6 | ON/OFF                                  |
|            | 7 | ON/OFF                                  |
|            | 8 | ON/OFF                                  |

*Table 31–12: DIP-Switches of the PCB, COP EUS2CAN*

### 31.4.4 Liquid Support PCB

LIQUID SUPPORT PCB is also RFID LIQUID SUPPORT CU PCBA.



Figure 31–20: PCB, Liquid support

| No.  | Connection              | No.  | Connection                |
|------|-------------------------|------|---------------------------|
| J 1  | Power / CAN             | J12  | Not used                  |
| J 2  | Download                | J13  | Antenna Starter 1         |
| J 3  | not used                | J14  | Antenna Starter 2         |
| J 4  | Tube heating, Starter 1 | J15  | System liquid             |
| J 5  | Tube heating, Starter 2 | J16  | Wash buffer               |
| J 6* | Waste Aspiration Pump   | J17* | Starter Dispense Pump 2   |
| J 7* | Liquid Waste Pump       | J18* | Wash Buffer Dispense Pump |
| J 8* | Starter Dispense Pump 1 | J19  | Liquid waste              |
| J 9  | BS Starter 1            | J20  | Consumable drawer         |
| J10  | BS Starter 2            | J21  | SL presence sensor        |
| J11  | Not used                | J22  | WB presence sensor        |

Table 31–13: Connections of the main PCB, Liquid support

\* connection sequence does not matter

| Number | Meaning/Function |
|--------|------------------|
| LED 1  | Power            |

Table 31–14: LED of the PCB, Liquid support

### 31.4.5 User access panel CU PCBA

USER ACCESS PANEL CU PCBA.

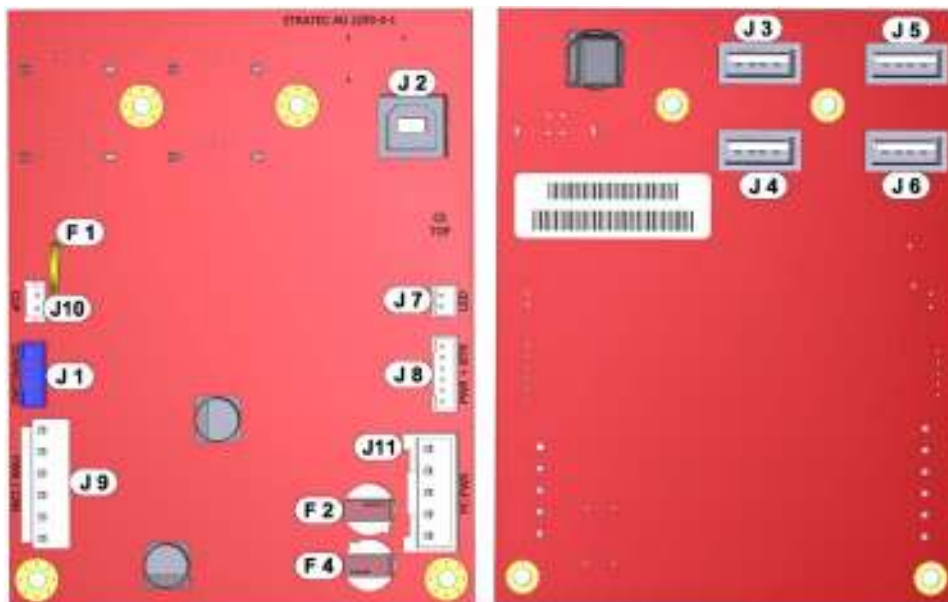


Figure 31-21: User access panel CU PCBA, front and rear view

| No. | Connection | No. | Connection    |
|-----|------------|-----|---------------|
| J 1 | Download   | J 7 | LED           |
| J 2 | PC-IN      | J 8 | Power-Button  |
| J 3 | USB 1      | J 9 | Power / CAN   |
| J 4 | USB 2      | J10 | COP (standby) |
| J 5 | USB 3      | J11 | PC-PWR        |
| J 6 | USB 4      |     |               |

Table 31-15: Connections of the user access panel CU PCBA

| Number | Meaning/Function |
|--------|------------------|
| F1     | Standby COP      |
| F2     | Fuse 2           |
| F4     | PC power         |

Table 31-16: Fuse of the user access panel CU PCBA

### 31.4.6 Remote Start Button PCB

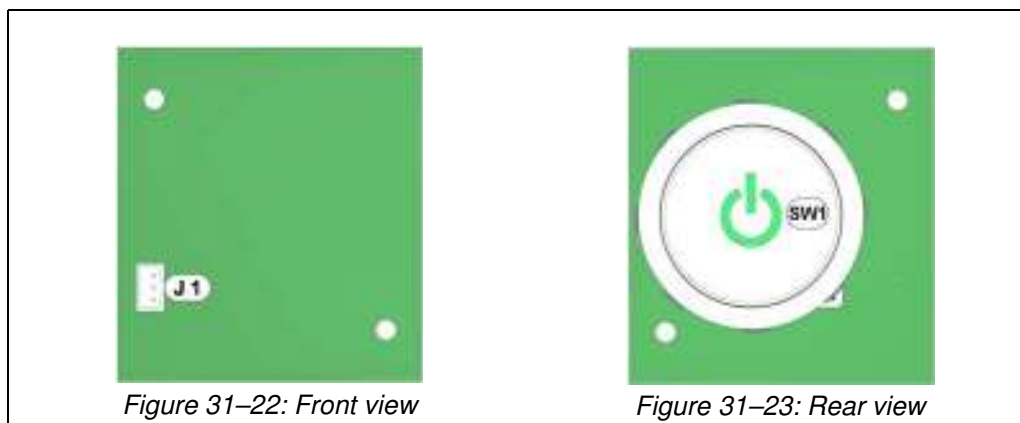


Figure 31-24: Remote Start Button PCB, front and rear view

| No. | Connection           |
|-----|----------------------|
| J 1 | User Access Panel CU |

Table 31-17: Connections of the Remote Start Button PCB

| Button | Meaning/Function/Default Switch Setting |
|--------|-----------------------------------------|
| SW1    | Start Button, ON/OFF                    |

Table 31-18: Switch of the Remote Start Button PCB



# 32 Software and Firmware Topics

The features and functionalities offered by **LIAISON<sup>®</sup> XS Main User Interface (Main UI)** are detailed in **LIAISON<sup>®</sup> XS** User Manual, so a Field Service Engineer (FSE) can use it to get information regarding the available buttons and the procedures for the execution of assays. Furthermore some topics relevant for technicians are not discussed in the User Manual because they don't give useful information to customers. The main purpose of this chapter is to fulfill this gap: the following sections detail some topics particularly useful for the everyday activities of a Service Engineer not presented in the User Manual.

## 32.1 Safety

---

**WARNING****Non-observance of safety instructions**

The non-observance of safety instructions may result in serious personal injury and material damage.

- Follow all safety instructions, see Chapter 1.5 on page 1-7 Or see Chapter 2.5 on page 2-7.
  - Follow all working instructions included in this manual.
  - Follow all warnings marked on the instrument.
  - Always wear personal protective equipment suitable for the activity.
- 

---

**CAUTION****Use of the Software**

Use the **LIAISON<sup>®</sup> XS** system software only to perform described functions.

---

---

**CAUTION**

It is absolutely forbidden to install softwares different from those specifically foreseen for **LIAISON<sup>®</sup> XS** system. The installation of not allowed softwares (e.g. not supported antivirus) brings the system to a not controlled and not validated status.

---

---

**CAUTION**

It is allowed to connect a **LIAISON<sup>®</sup> XS** system to a local network only for the following purposes:

- Printing on a network printer;
- Accessing to network paths for the functionalities specified in chapter 32.5.4;
- Remotely accessing to the system for troubleshooting purposes (i.e. through a remote connection, desktop sharing);
- IoT solution.

Any other network-based activity has not been validated and therefore is not allowed.

---

## 32.2 LIAISON® XS PC System

LIAISON® XS PC system includes the following components:

- Panel PC (Touch screen inclusive PC)
- Printer (optional)
- Handheld barcode scanner

---

**CAUTION**

The use of keyboard and mouse devices is allowed only to FSEs, they shall not be accessible to final users.

---

**NOTE**

The printer is not supplied directly with LIAISON® XS analyzer. A printer shall be purchased separately and installed, following the recommendations described in Corp-TN-000293 - former LXS TN 010.

---

### 32.2.1 Handheld Barcode Reader configuration

The following procedure shall be performed to complete the installation and the configuration of handheld barcode reader – PN A0179 or A0256:

1. Connect the handheld barcode reader to the LIAISON® XS PC.
2. Scan “Reset Default Settings” barcode, see Figure 32–1.



*Figure 32–1: Default settings*

3. Scan “Select USB Alternate Keyboard” barcode, see Figure 32–2.





## Select USB Alternate Keyboard

Figure 32–2: Select USB Alternate Keyboard

### 32.2.2 Touchscreen calibration

The following procedure shall be applied to perform the periodic recalibration of the touchscreen:

1. Push **Monitor** button of Desk Shield window.



Figure 32–3: Desk Shield

2. In Tablet PC Settings, push **Calibrate** button.



Figure 32-4: Table PC Settings

3. To proceed enter the Administrator password and follow the instructions visualized:

*“To provide calibration samples, tap the cross-hair each time that it appears on the screen. Right click anywhere on the screen to return to the last calibration point. Press the “Esc” button to close the tool. Do not change your screen orientation until you have completed the calibration process”*

## 32.3 LIAISON® XS Software

### 32.3.1 Recover after HDD loss (Manufacturer Default - all data lost)

You need to boot from the recover USB stick and to install the Software again.

#### 32.3.1.1 BIOS setting to start from USB Stick

The user has to set the boot configuration of the BIOS. BIOS has a built-in setup utility for initial computer configuration; to enter it, it is necessary to press a certain key combination (Esc, F9, Del, F1, Ctrl+Alt+Esc, Ctrl+Esc or some others, depending on the BIOS) during the POST (Power-On Self Test) sequence that starts immediately after the computer turns on. Usually the message with the required key combination is displayed during the startup test.

#### NOTE

The time interval during which it is possible to enter the setup utility of the BIOS is quite short. Be ready to press the proper key combination suddenly after the activation of the PC.

The pressure of the proper key combination opens the setup utility menu included in the BIOS, that allows booting operating systems not only from hard disks, but also from other devices.

#### 32.3.1.2 Change the Boot Option

1. Follow the BIOS setting.
2. Use the left and right arrow key to open the **Boot** screen.
3. Use the up and down arrow keys to move to the **Boot Option #1**.
4. Press the **Enter** key to start the change window to USB (depends on USB type). It is necessary to set the first booting device in the BIOS to USB stick.
5. Press the **F4** key and confirm the **Save & Exit Setup** screen with **Yes**.

#### 32.3.1.3 Install image from USB-Stick

1. With the next boot the PC will boot from the USB Stick.
2. In the prompt you are asked to confirm to apply the image to the PC. **This will delete all data on the PC** and set it back to manufacturer Default.
3. The PC will ask a confirmation before shutting down; when the PC is switched off, removed the USB-Stick.

#### 32.3.2 Change system Date and Time on Operating System

1. Log on as Administrator.
2. Right click at the bottom right onto the **Time and Date display**.
3. Click onto **Adjust date/time**.
4. The window for Settings **Date & Time** appears.
5. Change the Time zone accordingly to the local time.
6. Ensure that **Adjust for daylight saving time** is enabled.
7. Close the dialog and reboot the system.

#### 32.3.3 Software packages

The list of the items that can be installed on a **LIAISON<sup>®</sup> XS** PC is presented in the following table: the first installation of all these SWs (with the exclusion of the remote access tool) is completed by the manufacturer before shipping the instrument, so a FSE shall not perform any additional actions, unless if it is necessary to perform a reinstallation of the system or a SW/ FW update. In this case, it can be noticed that the greatest part of the items is provided in **Main UI** installation package, so it is only necessary to follow the related Installation Procedure to install/ update them all; once completed the installation/ update of the **Main UI** package, it is necessary to reinstall/ update separately **QC Software**, following the procedure described in the related Installation Procedure.

## 32. Software and Firmware Topics

In addition to the packages available by default on a **LIAISON® XS** PC, remote access tool can be also installed in the system: this configuration is optional and can be performed, where available, following the procedures described in Technical GOP 23.7210.

| Item                             | Provided in                                                       |
|----------------------------------|-------------------------------------------------------------------|
| <b>LIAISON® XS</b> Main Software | Main UI installation package                                      |
| QC software                      | QC SW installation package                                        |
| IFU of the Analyzer              | Available on DiaSorin online repository and not in the SW package |
| Service Software                 | Main UI installation package                                      |
| Teacher Software                 | Main UI installation package                                      |
| Firmware                         | Main UI installation package                                      |
| Antivirus                        | Main UI installation package                                      |
| Remote Access Tool (*)           | Remote access tool installation package                           |

Table 32–1: **LIAISON® XS** SW packages

(\*) Optional, not installed by default on a **LIAISON® XS** PC

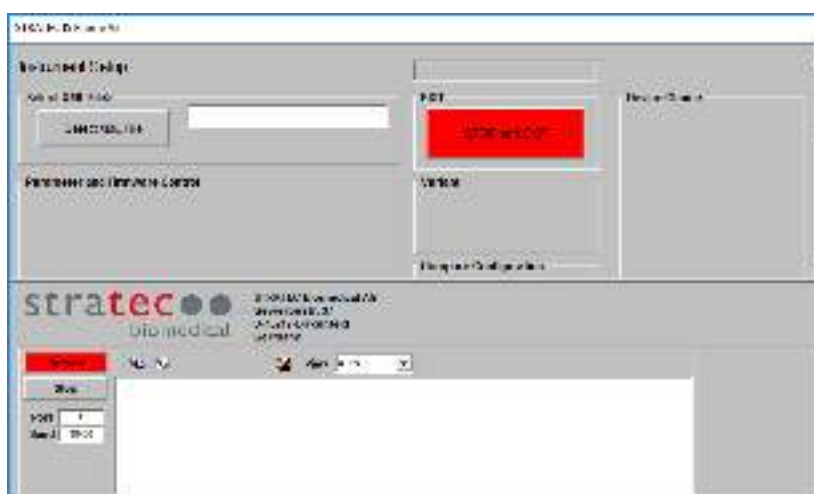
### 32.3.4 Software and Firmware Installation/Update

#### 32.3.4.1 Software installation and update

Refer to version specific Installation Procedure.

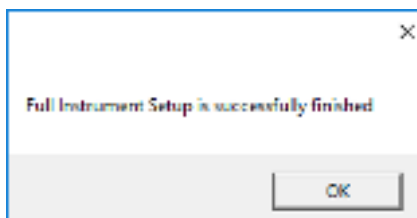
#### 32.3.4.2 Firmware download / Instrument Setup

1. Ensure to log in with the Windows account “Administrator”.
2. On the desktop, click on the IS-start.exe icon twice to start the program.



3. Click on the **Select XML File**, select in the prompted path “IS\_Config\_LiaisonXS.xml” file and press “Open”

4. Press **Full Instrument Setup**, acknowledge all pop-ups and wait until the following one is prompted:



5. Press **STOP and close**
6. Switch the system off and keep it unpowered for 1 minute.

### 32.3.5 Software structure

The list of system main files available in **LIAISON<sup>®</sup> XS** SW package is reported in the following table; for each file, a description of the contents is also presented.

| Path                                | File name              | Contents                                                                     |
|-------------------------------------|------------------------|------------------------------------------------------------------------------|
| C:\LiaisonXS\Backups                | *.zip                  | All backup folders                                                           |
| C:\LiaisonXS\BottleShapes           | BottleShape.xml        | Cross section definitions for reagent/ancillary/sample tubes                 |
| C:\LiaisonXS\Coordinates            | *.rac                  | Rack files                                                                   |
|                                     | Coordinate.coo         | Teaching coordinates collection                                              |
| C:\LiaisonXS\Databases              | *.db                   | Databases                                                                    |
| C:\LiaisonXS\Data\DiaSorin          | ExportAccessRights.txt | Access rights export file                                                    |
| C:\LiaisonXS\Data\DiaSorin\Logs     | FinishedJobs.txt       | List of measured jobs                                                        |
|                                     | DarkCounts.txt         | Dark count measures: average, single counts and detected spikes are reported |
|                                     | LightChecks.csv        | Background and Light Check results                                           |
| C:\LiaisonXS\Data\DiaSorin\Logs\LIS | ASTM1381Log_<Date>.log | ASTM 1382 communication in a human readable file                             |
|                                     | ASTM1394Log_<Date>.log | ASTM 1394 communication in a human readable file                             |

**32. Software and Firmware Topics**

---

| Path                           | File name | Contents                                                                                |
|--------------------------------|-----------|-----------------------------------------------------------------------------------------|
| C:\LiaisonXS\Data\ImportExport | *.ct3     | Controls definition files                                                               |
|                                | *.gr3     | Groups definition files                                                                 |
|                                | *.pr3     | Profiles definition files                                                               |
|                                | *.rr3     | Rerun rules definition files                                                            |
| C:\LiaisonXS\Data\OnlineHelp   | -         | All files (e.g. manuals, IFU, assay protocols) downloaded/acquired from the Online Help |
| C:\LiaisonXS\Data\Print        | *.pdf     | PDF printouts                                                                           |
| C:\LiaisonXS\Data\Results      | *.txt     | Export files                                                                            |
| C:\LiaisonXS\Data\SystemFiles  | *.zip     | Maintenance tasks and procedures folder                                                 |
|                                | *.di3     | Dilution table                                                                          |
|                                | *.gs3     | Generics                                                                                |
| C:\LiaisonXS\Data\Temp         | *.sil     | All log files generated by the system                                                   |
| C:\LiaisonXS\Data\Temp\archive | *.7z      | Archived log files generated by the system                                              |
| C:\LiaisonXS\QCStats\FromLXS   | *.xml     | QC results files                                                                        |

| Path                                                                                           | File name                | Contents                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C:\LiaisonXS\Tools\AV                                                                          | av.bat                   | AntiVirus command line                                                                                                                                                                |
|                                                                                                | clamlog.txt              | AntiVirus log file                                                                                                                                                                    |
| C:\LiaisonXS\Tools\Field GatewayServices\TelemetryData                                         | -                        | Instrument telemetry data stored in case the connection with the Vigilant Online server is not available; the data will be automatically transmitted once the connection is available |
| C:\LiaisonXS\Tools\InstrumentSetup                                                             | IS-Start.exe             | Instrument Setup application                                                                                                                                                          |
| C:\LiaisonXS\Tools\InstrumentSetup\Logs\Trace                                                  | -                        | Instrument Setup log files                                                                                                                                                            |
| C:\LiaisonXS\Tools\InstrumentSetup\Logs\IS_<IS_version>_LiaisonXS_<before/after>_<Date>_<Time> | -                        | Instrument Setup parameters export files                                                                                                                                              |
| C:\LiaisonXS\Tools\LogfileAnalyzer                                                             | LogfileAnalyser.exe      | Logfile Analyzer application                                                                                                                                                          |
| C:\LiaisonXS\Tools\ServiceSoftware                                                             | ServiceSoftware.exe      | Service Software application                                                                                                                                                          |
| C:\LiaisonXS\Tools\ServiceSoftware\log                                                         | -                        | Service Software log files                                                                                                                                                            |
| C:\LiaisonXS\Tools\Teacher                                                                     | LXSCoo.exe               | Teacher Software application                                                                                                                                                          |
|                                                                                                | CommunicationLogFile.txt | Teacher Software log file                                                                                                                                                             |

### 32.3.6 Files and information required to perform troubleshooting activities

The aim of this chapter is to list and detail the files to be downloaded from the analyzer and information required in order to perform troubleshooting activities on **LIAISON<sup>®</sup> XS**.

The list of files that shall be mandatory sent to Corporate Service are presented in Chapter 32.3.6.1 on page 32-9, see Chapter 32.3.6.2 on page 32-10 and Chapter 32.3.6.3 on page 32-10.

#### 32.3.6.1 Date and time

It is mandatory to send a detailed description of the issue and of the system behavior with date and time of the occurrences (especially in case of Software issue).

### 32.3.6.2 Database backup

The Database backup includes all databases for results, events, and system characteristics.

To know how to create and save a Database backup please refer to chapter “Sub Category Backup” of **LIAISON<sup>®</sup> XS** User Manual.

---

**NOTE**

Database are encrypted.

---

**NOTE**

In case of database restore, empty the incubator (from the Main SW) before performing any routine.

---

### 32.3.6.3 Troubleshooting Complete backup

Troubleshooting backup includes a selection of files useful for a general troubleshooting.

To know how to create and save a Database backup please refer to chapter “Sub Category Backup” of **LIAISON<sup>®</sup> XS** User Manual.

### 32.3.6.4 Other files and information to be provided

Besides the above files, other information have to be provided depending on the troubleshooting activity to be performed:

- In case of availability, movies, videos, pictures and screenshots have to be provided as well in order to help the troubleshooting activities.
- In particular cases Corporate Service can ask for other files (folders, info, other kinds of Backups etc.).

### 32.3.6.5 Contact address

To send to the Corporate Service the files, use DiaSorin file sharing system (see related Technical GOP).

Ensure that no patient demographic information is transmitted.

The official mail contact is:

**cs\_liaisonxs@diasorin.it**

## 32.4 LIAISON<sup>®</sup> XS Antivirus

### 32.4.1 Antivirus Installation

The installation of the Antivirus is performed automatically during SW installation.

### 32.4.2 Antivirus Update

The update of the Antivirus is performed automatically during SW installation.



**CAUTION**

It is forbidden to perform Internet updates.

### 32.4.3 Antivirus functioning and Virus detection

**LIAISON<sup>®</sup> XS** Antivirus shall be launched on-demand, using the dedicated **Virus Scan** button available on **Desk Shield**.

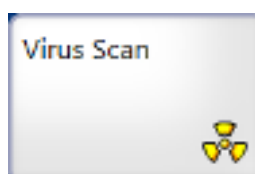


Figure 32–5: Virus Scan button

Once the routine starts, a DOS window, with the indication of the scanning progress, appears.

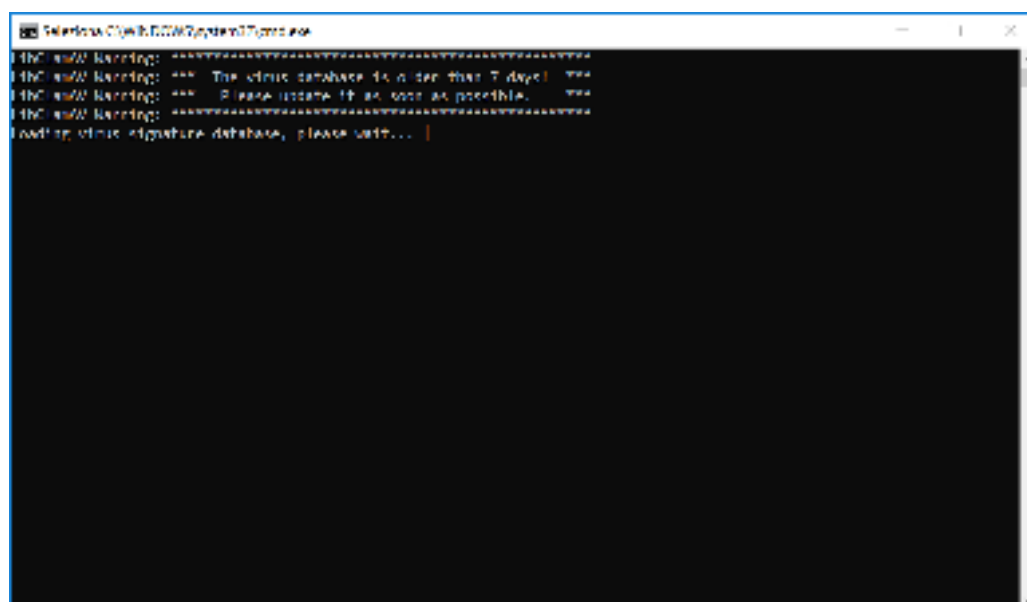


Figure 32–6: Scanning progress window

After the conclusion of the scan, the associated report is reported in the same DOS window; scan results are saved in `clamlog.txt` available at `C:\LiaisonXS\Tools\AV\` path.

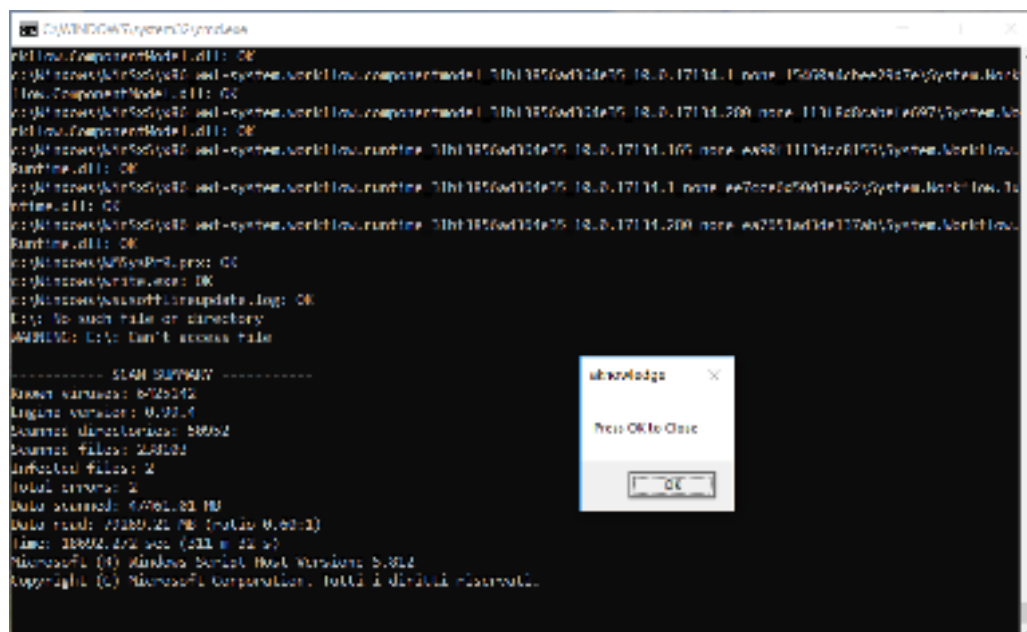


Figure 32-7: Scansion report window

When a virus is detected, contact DiaSorin Corporate Service.

## NOTE

No actions are automatically started by **LIAISON<sup>®</sup> XS** Antivirus when a virus is detected.

### 32.4.4 Antivirus management

Aim of this chapter is to:

- Provide a condensed description of the System Protection Measures suitable to customer explanations, tender requirements, and so on (all information are diffusively treated also in the User Manual);
- Give additional information to FSE on special procedures (scanning and update management, special care, ...).

As explained in **LIAISON<sup>®</sup> XS** User Manual, chapter "PC security measure", the system is protected against malware access with a list of system settings. These settings allow an "hardening" of the system structure to reduce the possibility that an unknown software enters the PC.

#### 32.4.4.1 Antivirus policy

As mentioned before, the antivirus is automatically installed on the **LIAISON<sup>®</sup> XS** system. The provided antivirus has been selected basing on software requirements and system restrictions and has been tested and validated during the software validation effort.

Due to the need of a dedicated validation procedure, **it is forbidden to install on the LIAISON<sup>®</sup> XS PC another antivirus and / or antimalware.**

For the same reason, it is forbidden to update the antivirus definitions without a specific official instruction from Corporate Service (i.e. via Technical Note).

#### 32.4.4.2 Antivirus management

The antivirus is not working on background, i.e. it is not active at every moment that the system is working, because of the risk associated to an unforeseen action of the antivirus itself.

It is indeed possible to start manually the antivirus from a specific icon on the DeskShield. By pressing the **Virus Scan** button the antivirus will be launched. It may take from few to many minutes depending on the quantity of files stored in the system.

#### 32.4.4.3 Scan management

**For Field Service Engineer.** The antivirus shall be launched during the Preventive maintenance activity.

**For final users.** There is no special indication for the final user to start the antivirus. It can be left to his/her decision whether to run the antivirus.

In case a malware is detected, the embedded antivirus will not automatically remove it. As stated in chapter 32.4.3, DiaSorin Corporate Service has to be contacted for removal operations.

Before contacting Corporate Service, it is recommended to retrieve the file `clamlog.txt` available at the path `C:\LiaisonXS\Tools\AV`.

#### 32.4.4.4 Update management

Antivirus updates are installed within each new released SW version.

### CAUTION



---

#### Special care when enabling Desktop.

When the Desktop is enabled, all the “hardening” feature explained in the User Manual may be impacted.

This would lead the system into a “weaker” status subject to malware introduction, for example from infected external devices such as USB drives.

For this reason, it is strongly recommended to field personnel to check their own external devices with the antivirus provided with the company laptop prior to the insertion into the **LIAISON® XS**.

In addition, before the release of the system to the final user, the PC shall be rebooted.

---

## 32.5 Software FSE functionalities

**LIAISON® XS** supports different privilege levels, that allow to access to different functionalities of the system; for this reason it is important that a FSE understands clearly what are the additional features foreseen for his profile and what are the system parameters he can access to and set. In the next paragraphs, an overview of the additional buttons available in **Main UI** and of the different system parameters that can be configured is presented.

### 32.5.1 Main UI FSE buttons

The list of the additional buttons available in Main UI for the FSE profile is reported in the following table, together with the indications of the page and the sub-page where each functionality can be found.

| Page               | Sub-Page         | Button           | Functionality description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------|------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Definitions</i> | <i>Assays</i>    | Delete           | Deletes one or more selected assays.<br><b>IMPORTANT:</b> <ul style="list-style-type: none"> <li>The assignments are checked before deleting assays. Assays that are still part of a profile, group or rerun rule cannot be deleted until they are removed from there. Assays that are part of a combi family can be deleted only after the combi father is deleted.</li> <li>Delete button can be used only if directly enforced by Corporate Service. Improper use of this button can create issues on the system. To update an assay it is necessary to overwrite it using Import button straight away.</li> </ul> |
| <i>Definitions</i> | <i>Assays</i>    | Import Generics  | Allows to import sequence for generic maintenance tasks: <ul style="list-style-type: none"> <li>Background;</li> <li>Light Check of both arms.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <i>Definitions</i> | <i>Assays</i>    | Import Dilutions | Allows to import all parameters necessary to perform dilutions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <i>Events</i>      | <i>Event Log</i> | Delete           | Deletes one or more selected events.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <i>Maintenance</i> | <i>Log</i>       | Delete           | Deletes one or more selected maintenance logs.<br><b>IMPORTANT:</b> it is no more possible to recover maintenance logs deleted from this window.                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <i>Maintenance</i> | <i>Tasks</i>     | Delete           | Deletes one or more selected maintenance tasks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <i>Maintenance</i> | <i>Tasks</i>     | Import           | Allows to import a maintenance package.<br><b>IMPORTANT:</b> all existing maintenance tasks will be deleted before the system will import new maintenance tasks.                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

| Page           | Sub-Page               | Button | Functionality description                                                                                                                                                                                                                                             |
|----------------|------------------------|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Results</i> | <i>Calibrations</i>    | Delete | Deletes one or more selected calibrations.<br><b>IMPORTANT:</b> it is not possible to recover calibrations deleted from this window. Calibrations are supposed to be never deleted, unless their number would be so high that scrolling the page becomes problematic. |
| <i>System</i>  | <i>Backups</i>         | Delete | Deletes one or more selected backups.<br><b>IMPORTANT:</b> once pressed the button, it is no more possible to recover the deleted backup.                                                                                                                             |
| <i>System</i>  | <i>Test Statistics</i> | Reset  | Resets all data reported in Test Statistics sub-page.<br><b>IMPORTANT:</b> once pressed the button, it is no more possible to recover the deleted data.                                                                                                               |

Table 32–2: FSE buttons

### 32.5.1.1 Configure button



Each Browser has a Button for configuration, clicking on it open a configuration dialog.

Figure 32–8: *Configure* column dialog

| Section                 | Description                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eligible Columns</b> | Columns not displayed in the table.                                                                                                                                                                                                                                                                                                              |
| <b>Visible Columns</b>  | Columns displayed in the table.                                                                                                                                                                                                                                                                                                                  |
| <b>Columns Setting</b>  | Enabled only an item in Visible Columns is selected: <ul style="list-style-type: none"> <li>• “Auto”: the column width is automatically adjusted to fit the column contents</li> <li>• “Fill”: the column width is set “x” times the default value</li> <li>• “Fix”: the column width is set according the the configured pixel value</li> </ul> |

Table 32–3: Sections of the **Configure** dialog

| Function           | Description                                                                                                                             |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>Restore all</b> | Restore default settings for all pages                                                                                                  |
| <b>Restore</b>     | Restore default settings for this page                                                                                                  |
| <b>Hide Column</b> | Enabled only if an entry in <b>Visible Columns</b> is selected. If clicked, this item moves to <b>Eligible Columns</b> .                |
| <b>Show Column</b> | Enabled only if an entry in <b>Eligible Columns</b> is selected. If clicked, this item moves to <b>Visible Columns</b> .                |
| <b>Move up</b>     | Enabled only if an entry in <b>Visible Columns</b> is selected (but not the first entry). If clicked, this item moves one position up.  |
| <b>Move down</b>   | Enabled only if an entry in <b>Visible Columns</b> is selected (but not the last entry). If clicked, this item moves one position down. |
| <b>Cancel</b>      | Closes the <b>Configure</b> dialog and returns without any changes to the previous display                                              |
| <b>Apply</b>       | Closes the <b>Configure</b> dialog and saves all changes                                                                                |

Figure 32–9: Functions of Configure

**NOTE**

In order to ensure a good visualization of the page, it is recommended not to exceed with the number of reported parameters.

Once completed the settings, it is always necessary to check the effective visualization of the enabled columns.

**32.5.2 LIS Details**

The system allows to specify a high number of parameters to accommodate the specific needs of the customer and the feature set of the software used as Host.

**NOTE**

Refer to User Manual for explanation of LIS Details main dialog and Option tab.

**NOTE**

The “Result Aspect” and “Replicates to Host” settings of the Definition - Assays page may also have an impact on the communication.

**Protocol**

Figure 32–10: Protocol

| Function                                                | Description                                                                                                                               |
|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Send Frame Timer</b>                                 | Overwrites the ASTM1381 'Send Frame Timer'                                                                                                |
| <b>Bad Send Frame &lt;NAK&gt; Timer (Interframe)</b>    | Overwrites the ASTM1381 'Bad Frame Send <NAK> Timer'                                                                                      |
| <b>Delimiters (Field, Component, Repeat and Escape)</b> | Delimiters are used for outgoing messages. Incoming messages may have different delimiters                                                |
| <b>Send Compressed Query</b>                            | The query for test orders is generated using repeat delimiters or not.                                                                    |
| <b>Send Compressed Comments</b>                         | The result comment record is generated using repeat delimiters or not.                                                                    |
| <b>Busy Timer</b>                                       | Overwrites the ASTM1381 'Busy Timer'                                                                                                      |
| <b>Patient ID</b>                                       | <b>P3</b> or <b>P4</b> , define the field used for the transmission of Patient ID. It applies to both incoming and outgoing transmissions |

Table 32–4: Functions of the Protocol tab

**NOTE**

The system can handle the PID both in P3 and P4. It will send the PID in the same field it was received if the test was requested from host.

In case of manually requested tests, the system will send the PID in the specified field.

## Serial



Figure 32–11: Serial

| Function         | Description                                                                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Baud Rate</b> | Specifies the speed (baud rate) used for transmissions between the <b>LIAISON<sup>®</sup> XS</b> and the host.<br>Overwrites the ASTM1381 requirements.<br>Value: 9600 or 19200. |

Table 32–5: Functions of the Serial tab

## TCP-IP



Figure 32–12: TCP/IP

| Function              | Description                                                |
|-----------------------|------------------------------------------------------------|
| <b>TCP-IP Address</b> | IP address of the host PC                                  |
| <b>TCP-Port</b>       | Port used in the communication between instrument and host |

Table 32–6: Functions of the TCP-IP tab



### 32.5.3 Sub Category System - Deleted

In System page, a Deleted sub-page is accessible only to FSEs; using this window, it is possible to see and to delete permanently the results moved to this window using Delete buttons available in the Results sub-pages. For the functionality of the buttons available in this sub-page, refer to User Manual (chapter “Software functions”).

Search:  Total Records: 60

| Sample ID | Assay  | Measured     | Status | Dilution | PLU | Dose  | User Unit | Label | Flags |
|-----------|--------|--------------|--------|----------|-----|-------|-----------|-------|-------|
| 6070      | TSP1   | ---          | Placed | 1        |     |       |           |       |       |
| 6070      | TSP2   | ---          | Placed | 1        |     |       |           |       |       |
| 740       | TSP4   | ---          | Placed | 1        |     |       |           |       |       |
| 740       | TSP4   | ---          | Placed | 1        |     |       |           |       |       |
| 100011    | 100011 | 001110010... | Done   | 1        | 311 | 0.6   | UI        |       |       |
| 100011    | 100012 | 001110010... | Done   | 1        | 320 | 0.64  | UI        |       |       |
| 100011    | 100011 | 001110010... | Done   | 1        |     | 1.34  | UI        |       |       |
| 100010    | 100011 | 001110010... | Done   | 1        | 302 | 0.6   | UI        |       |       |
| 100010    | 100012 | 001110010... | Done   | 1        | 319 | 0.640 | UI        |       |       |
| 100010    | 100011 | 001110010... | Done   | 1        |     | 1.34  | UI        |       |       |
| 10002     | TSP1   | 001110010... | Failed | 1        |     |       | WCD       |       | FI    |
| 10002     | TSP1   | 001110010... | Done   | 1        | 215 | 0.632 | WCD       |       |       |
| 0001      | TSP1   | 000000010... | Failed | 1        |     |       |           |       |       |

Accounts Applications Backup Connectivity Settings Test Statistics Versions Deleted Service

Loading Results Events Definitions System Diagnostics Maintenance Status

Figure 32–13: System - Deleted page

### 32.5.4 Sub Category System - Service

#### Sys. Settings

Some of the listed settings are only visible and not editable to the FSE.



Figure 32–14: Sub Category Service

| Function       | Description                                                                                                                  |
|----------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Default</b> | Sets all values to default and store them to the database.<br>A System Restart is needed to make sure that all changes apply |
| <b>Reload</b>  | Undoes all changes since the last store                                                                                      |
| <b>Store</b>   | Stores the values to the database. A system restart is needed to make sure that all changes apply                            |

Table 32–7: Functions of the Service sub category

In the following table only the editable options are described.

| Setting                                  | Description                                                                                                                                                                                                                                          |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Company Logo</b>                      | Set the Company logo within the print reports the <b>LIAISON<sup>®</sup> XS</b> Main SW will produce.<br>When the button with "ellipsis" is selected, the import file dialog opens to select the desired image                                       |
| <b>Company Name</b>                      | Set the Company name within the print reports the <b>LIAISON<sup>®</sup> XS</b> Main SW will produce and that will be transmitted to IoT                                                                                                             |
| <b>Country Code &amp; Country Others</b> | Set the country in which the instrument is installed, it is used by the system to download/import appropriate IFU for the specific country.<br>If the country is not present in the list, select "Other Country (OTHER)"                             |
| <b>Keyboard Language</b>                 | Set up the Main SW On-Screen Keyboard to match the standard layout for the chosen language (this setting is independent from the Main SW language).<br>Close and restart the <b>LIAISON<sup>®</sup> XS</b> application for the change to take effect |
| <b>Laboratory Address</b>                | Set the Laboratory address within the print reports the <b>LIAISON<sup>®</sup> XS</b> Main SW will produce and that will be transmitted to the QC SW and IoT                                                                                         |
| <b>Laboratory Name</b>                   | Set the Laboratory name within the print reports the <b>LIAISON<sup>®</sup> XS</b> Main SW will produce and that will be transmitted to the QC SW and IoT                                                                                            |
| <b>QCStat Start on Startup</b>           | If selected and the QCStat is installed, the QCStat is automatically started at Main SW startup                                                                                                                                                      |
| <b>LISEncoding</b>                       | Set the code page used during the LIS data transmission (default is Windows 1252)                                                                                                                                                                    |

**NOTE****Notes for print layout**

- The columns reported in the printout cannot be customized by a local technician
- In particular cases and for specific problems, it is possible to ask Corporate Functions for the implementation of customized printout report
- The image format must be either a PNG, JPEG or BMP; the recommended resolution is between 150 and 600 dpi. If visible artifacts are displayed in the printout, please consider to change the resolution. In case the problem persists, contact Corporate Functions
- Report layouts have been specifically developed to be printed using an A4 format. Other printing formats (e.g. Letter) can be used but can cause the partial deletion of the reported information
- Once completed the setting, it is always necessary to check the effective visualization of changed fields

## 32. Software and Firmware Topics

## Folders

In the Folder tab, the local and network path for the available actions are listed. The FSE cannot edit the Local paths defined.

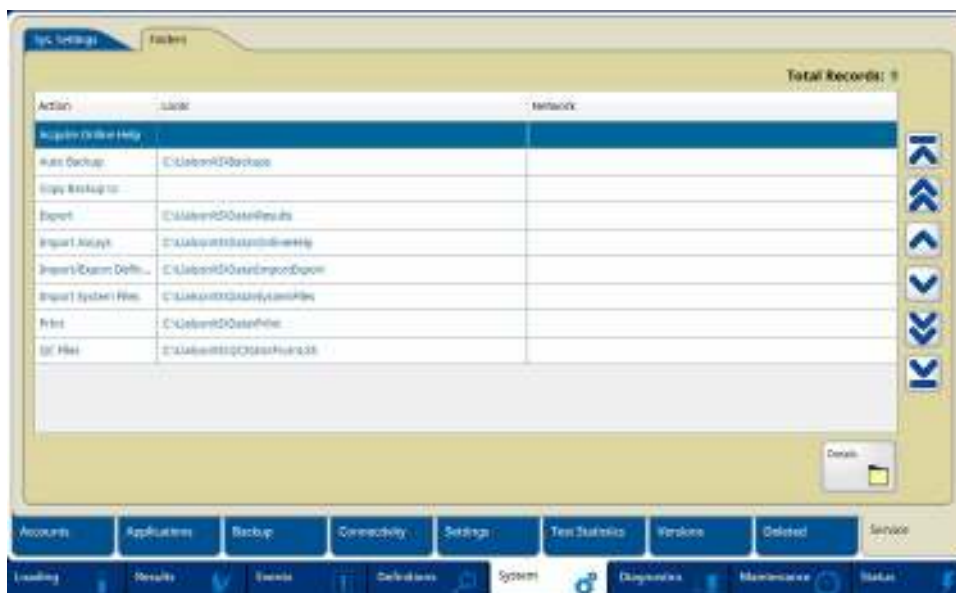


Figure 32–15: System Service- Folders tab

| Action                           | Description                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acquire Online Help</b>       | The defined path is added to the possible selection available in the Acquire dialog.                                                                                           |
| <b>Auto Backup</b>               | Path where the system stores a copy of performed Auto Backup.                                                                                                                  |
| <b>Copy Backup to</b>            | The defined path is added to the possible selection available in the Copy Backup dialog.                                                                                       |
| <b>Export</b>                    | The defined path is added to the possible selection available in the Export dialog.                                                                                            |
| <b>Import Assay</b>              | The defined path is added to the possible selection available in the Import dialog.                                                                                            |
| <b>Import/Export Definitions</b> | The defined path is added to the possible selection available in the Import/Export dialog for all files generated by the Main SW (e.g. control definitions, groups, profiles). |
| <b>Import System Files</b>       | The defined path is added to the possible selection available in the Import dialog for generics, dilution and maintenance files.                                               |
| <b>Print</b>                     | The defined path is added to the possible selection available in the Print dialog.                                                                                             |
| <b>QC Files</b>                  | Path where the system stores a copy of a produced control related file to be used for third party QC programs.                                                                 |

## Details



Figure 32–16: Edit Folders

| Column         | Description                           |
|----------------|---------------------------------------|
| <b>Action</b>  | The <b>Action</b> selected            |
| <b>Local</b>   | The local path defined (not editable) |
| <b>Network</b> | The network path defined              |

Table 32–8: Edit Folders

### 32.5.5 Sub Category Diagnostic - Options



Figure 32–17: Sub category Diagnostic - Options

| FSE Function                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Enable Windows Desktop</b>    | <p>Allows to enable the Windows tool-bar while the SW is running. After the desktop is enabled, the left header is displayed with blue stripes.</p> <p>If the <b>LIAISON® XS</b> software is closed, the desktop will appear.</p> <p><b>IMPORTANT:</b> Once completed the activity that requires the enabling of the Desktop, perform a PC reboot before allowing the customer to use the system.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Export Access Rights</b>      | <p>Exports the current settings with the following content:</p> <ul style="list-style-type: none"> <li>Access Rights List of Instrument <b>Serial Number</b>, create with Rights Level <b>Level of current</b> user on <b>Customer Formatted Date &amp; time</b>.</li> <li>The table of all <b>AccessProfileNames</b></li> <li>The table of all existing Users and their <b>AccessProfileName</b>.</li> <li>The table of all features and their level for <b>visible</b> and <b>editable</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Fast Resume</b>               | <p>Allows to shorten the resume procedure during service activities.</p> <p>The standard number of primes is used again as soon as any of the further conditions occurs:</p> <ul style="list-style-type: none"> <li>the feature is disabled</li> <li>the software is switched off</li> <li>the current user performs a log-out.</li> </ul> <p><b>IMPORTANT:</b> the software guarantees that end users cannot have access to this feature. Nevertheless, it is the sole responsibility of Service personnel to ensure that the system is returned to end users in the proper conditions, including an adequate number of primes.</p> <p>For this reason, in case the system was used with the “Fast Resume” check box checked, it is necessary to initialize the system before releasing to any users and anyway before performing any routine with real samples/reagents.</p> |
| <b>Allow open Cover</b>          | <p>If enabled, then the system does not halt even if the cover is opened while parts are moving.</p> <p>As soon as the current user performs a log-out or the software is switched off, the check box gets unchecked.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Pipettor Left and Gripper</b> | Initialize only left pipettor and gripper.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Pipettor Right</b>            | Initialize only right pipettor.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Incubator</b>                 | Initialize only incubator.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Washer</b>                    | Initialize only washer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Reader</b>                    | Initialize only reader.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Initialize</b>                | Initialize the selected assembly.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Table 32–9: Functions of the Options sub category

## 32.6 LIS connection

LIAISON<sup>®</sup> XS supports 2 different LIS settings:

- a **Complete Protocol**, to support all LIAISON<sup>®</sup> XS features;
- a **Reduced Protocol**, to be compatible with most of the existing Liaison connections.

The **Complete Protocol** features new data and options (refer to Main UI) in comparison to those offered by previous Liaison system; otherwise, the **Reduced Protocol** offers the same functionalities already available on Liaison, thus limiting the traceability measures foreseen for LIAISON<sup>®</sup> XS.

The choice between the two possible configurations depends on the features of the software used as host. The **Reduced Protocol** can be implemented using a specific combination of parameters and allows to install a LIAISON<sup>®</sup> XS in place of a Liaison analyzer reducing the possibility of introducing any modifications in the existing host interface.

### NOTE

During the creation of a LIS connection via serial, check the connection is set as COM1 and configured to match LIS requirements (e.g. baud rate/bits per second). Ensure that there are not duplicates of COM1.

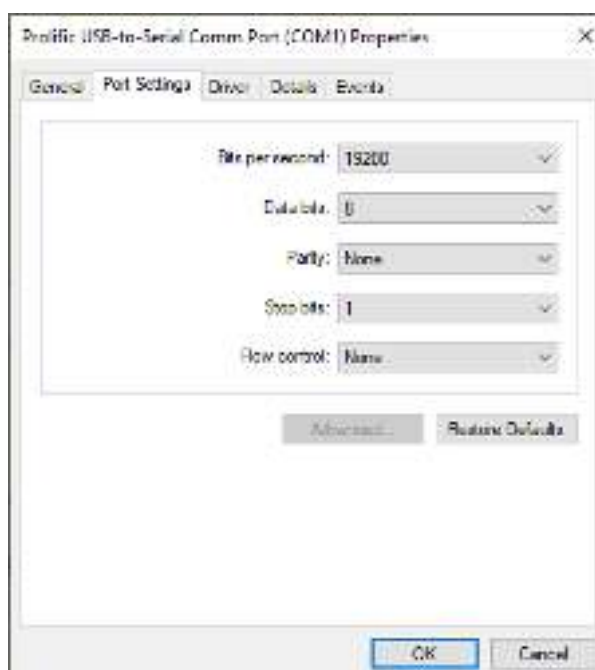


Figure 32–18: LIS COM port configuration

### NOTE

During the creation of a LIS connection via TCP/IP, always check the configuration of the IP address and related port. No special configuration for firewall are needed.



### 32.6.1 LIS configuration

To configure the instrument for the *Reduced Protocol*, set the following options as reported in the table.

| Page                                           | Option                                     | Description                                                                                                                                                                                                                                                                                                                       | Value       |
|------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <i>System-Connectivity-LIS Details-Options</i> | <i>Result flag style</i>                   | Allows to set the flag type, according to the number of digits accepted by the host.<br>Possible options:<br><ul style="list-style-type: none"> <li>• <b>Long</b>: result flags are transmitted as “full size” string</li> <li>• <b>Short</b>: results flags are transmitted by a 2 character abbreviation</li> </ul>             | <b>Long</b> |
| <i>System-Connectivity-LIS Details-Options</i> | <i>Enable Manufacturer record</i>          | If enabled, some additional information are sent to the host and shall be read on it:<br><ul style="list-style-type: none"> <li>• Integral Kit Number;</li> <li>• Integral Expiration Date;</li> <li>• Integral Lot Number;</li> <li>• Calibration ID;</li> <li>• Sample Lane Number;</li> <li>• Sample Tube Position.</li> </ul> | <b>Off</b>  |
| <i>Definitions-Assay-Details-Calculat</i>      | <i>Send Replicates to Host</i>             | Allows to chose if sending to the host the results associated to each replicate and the mean result ( <b>Yes</b> setting) or only the mean result ( <b>No</b> setting)                                                                                                                                                            | <b>No</b>   |
| <i>Definitions-Assay-Details-Calculat</i>      | <i>Result Aspect for Host Transmission</i> | Allows to define the aspect of the results sent to the host: it is possible to send only the dose ( <b>Dose</b> setting), only the qualitative label ( <b>Label</b> setting) or both dose and qualitative label ( <b>Both</b> setting)                                                                                            | <b>Dose</b> |

Table 32–10: Reduced Protocol Configuration (“Liaison compatible”)

### 32.6.2 Connection scheme

The LIS connection scheme supported by LIAISON<sup>®</sup> XS analyzer is represented in the following figure. Four different configurations can be deployed: setting the dedicated option available in System - Settings - Others subpage, it is possible to choose how the results of the tests shall be transferred to LIAISON<sup>®</sup> XS archive page (Automatically/Manually). In the same way the configuration of the dedicated option available in System - Connectivity



- LIS details - Options subpage allows to set how the archived results shall be sent to the host (automatically or not). This flexibility allows end-users to choose whether they want to perform a "validation" step directly on the LIAISON<sup>®</sup> XS interface or in any external device.

**NOTE**

The results can be sent to host only from the "Archived" page.

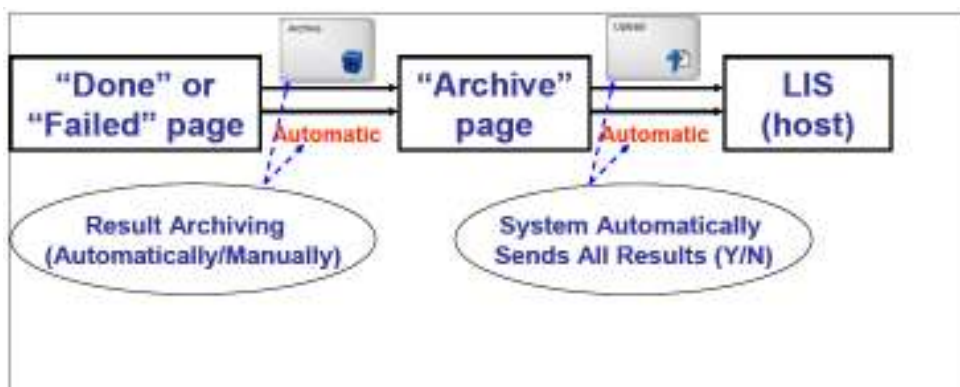


Figure 32–19: LIAISON<sup>®</sup> XS LIS connection scheme

### 32.6.3 Troubleshooting

Troubleshooting activities related to LIS connections are supported by two dedicated records, named ASTM1381 and ASTM1394.

All the log files related to ASTM1381 and ASTM1394 are stored in C:\LiaisonXS\Data\DiaSorin\Logs\Lis. In any case, the easiest way to access these data is to use the LIS dedicated backup from the Main UI or the troubleshooting backups.

## 32.7 Remote access tool

The purpose of this chapter is to describe the characteristics of the remote access tool available for the LIAISON<sup>®</sup> XS instruments that allows to connect remotely from a PC to the instrument PC.

### 32.7.1 Remote access tool setup

The Technical GOP 23.7210 describes everything that may concern the set up and the main characteristic of the remote access tool.

The document contains both the operations to be executed on the instrument and the operator's workstation.

### 32.7.2 Correct usage of the remote access tool and restrictions

The usage of the remote access tool application is only allowed to perform from remote:

- Product and instrument troubleshooting

## 32. Software and Firmware Topics

---

- Analytical data monitoring
- Software and firmware update

The usage of the remote access tool runs in parallel to **LIAISON® XS** application and is validated to allow to perform tests when a remote access is active.

**NOTE**

Once completed the activity using the remote access tool that required the enabling of the Desktop or the StoreLogs creation, perform a PC reboot before allowing the customer to use the system.

---

# 33 Service Check List for internal-tanks configuration

## 33.1 LIAISON® XS Installation Qualification (IQ)

### 1 General

- 1.1 Type \_\_\_\_\_
- 1.2 Serial No. \_\_\_\_\_
- 1.3 Software Version \_\_\_\_\_

### 2 Installation details

- 2.1 Installation date \_\_\_\_\_
- 2.2 Installed by \_\_\_\_\_

### 3 Customer details

- 3.1 Customer name \_\_\_\_\_
- 3.2 Contact person name \_\_\_\_\_
- 3.3 Company \_\_\_\_\_
- 3.4 Address \_\_\_\_\_
- 3.5 Phone \_\_\_\_\_
- 3.6 Email \_\_\_\_\_

**33. Service Check List for internal-tanks configuration**

| <b>4</b> | <b>Visual Inspection before unpacking</b> (see Chapter 5.3.1 on page 5-5)                  | <b>Flag</b> | <b>Note</b> |
|----------|--------------------------------------------------------------------------------------------|-------------|-------------|
| 4.1      | Transport box without damages.                                                             | [ ]         |             |
| 4.2      | Transport shock monitor and temperature indication.                                        | [ ]         |             |
| <b>5</b> | <b>Unpacking, visual inspection and inventory</b> (see Chapter 5.3.2 on page 5-6)          | <b>Flag</b> | <b>Note</b> |
| 5.1      | Unpack the instrument.                                                                     | [ ]         |             |
| 5.2      | Chassis without damages or gaps.                                                           | [ ]         |             |
| 5.3      | No visible cuts in the system or pipetting tubing.                                         | [ ]         |             |
| 5.4      | Covers without damages or scratches.                                                       | [ ]         |             |
| 5.5      | Check parts and accessories according to the packing lists, see Chapter 37.1 on page 37-1. | [ ]         |             |
| 5.6      | Transport tilt and shock monitor at the rear chassis.                                      | [ ]         |             |
| 5.7      | System leveling and stabilization                                                          | [ ]         |             |
| 5.8      | Unlock pipettors.                                                                          | [ ]         |             |
| <b>6</b> | <b>Manual inspection</b> (see Chapter 5.3.3 on page 5-14)                                  | <b>Flag</b> | <b>Note</b> |
| 6.1      | Both pipetting arms in x-axis are gliding smooth and without force.                        | [ ]         |             |
| 6.2      | Both pipetting arms in y-axis are gliding smooth and without force.                        | [ ]         |             |
| 6.3      | Both pipetting arms in z-axis are gliding smooth and without force.                        | [ ]         |             |
| <b>7</b> | <b>System set-up</b> (see Chapter 5.3.4 on page 5-15)                                      | <b>Flag</b> | <b>Note</b> |
| 7.1      | Place the Velcro strip for the USB stick.                                                  | [ ]         |             |
| 7.2      | Mount peristaltic tubes and caps of LWP and WAP pumps.                                     | [ ]         |             |
| 7.3      | Rinse the tanks.                                                                           | [ ]         |             |
| 7.4      | Fill DI water tank and put it into the instrument.                                         | [ ]         |             |
| 7.5      | Fill Wash Buffer tank and put it into the instrument.                                      | [ ]         |             |
| 7.6      | Liquid Waste Tank and basin set up.                                                        | [ ]         |             |
| 7.7      | Assemble covers, see Chapter 5.3.4.4 on page 5-19.                                         | [ ]         |             |
| 7.8      | Check the alignment of the Covers, see Chapter 14.4 on page 14-20.                         | [ ]         |             |
| 7.9      | Assemble Panel PC, see Chapter 14.3.4.3 on page 14-11.                                     | [ ]         |             |
| 7.10     | Check the system leveling.                                                                 | [ ]         |             |
| 7.11     | Re-tension the Y belt of Left and Right Pipettor arms.                                     | [ ]         |             |
| 7.12     | Insert the Ancillary tray in the Reagent Module.                                           | [ ]         |             |
| 7.13     | Install the Cuvette - Tip Loading Module handle and insert cuvette and tip trays.          | [ ]         |             |
| 7.14     | Place the Solid Waste bin in the instrument with the Waste box.                            | [ ]         |             |

### 33. Service Check List for internal-tanks

#### configuration

- |      |                                                              |     |
|------|--------------------------------------------------------------|-----|
| 7.15 | Insert the Starter reagent bottle into the dedicated module. | [ ] |
| 7.16 | Remove the protective foil from the Status light             | [ ] |
| 7.17 | Plug in and turn on <b>LIAISON® XS</b> system.               | [ ] |

#### 8 Rack configuration (see Chapter 5.3.5 on page 5-22 & see Chapter 5.3.6 on page 5-29) **Flag** **Note**

- |     |                                                                     |     |
|-----|---------------------------------------------------------------------|-----|
| 8.1 | Configure Sample racks, see Chapter 5.3.5.1 on page 5-23.           | [ ] |
| 8.2 | If needed, configure Control racks, see Chapter 5.3.6 on page 5-29. | [ ] |
| 8.3 | Barcode ID labels and Rack ID labels placed on all racks.           | [ ] |

#### 9 Pipettor and rack teaching (see Chapter 5.3.7 on page 5-30) **Flag** **Note**

- |     |                                                                                                   |     |
|-----|---------------------------------------------------------------------------------------------------|-----|
| 9.1 | Complete the teaching of the Pipettors via <b>Teacher Software</b> , see Chapter 13 on page 13-1. | [ ] |
| 9.2 | If required, perform the teaching of customized Sample racks (G, O, Y and Z racks).               | [ ] |

#### 11 Fluidic set-up (see Chapter 5.3.8 on page 5-30) **Flag** **Note**

- |      |                                                                                           |     |
|------|-------------------------------------------------------------------------------------------|-----|
| 11.1 | Perform fluidic prime via <b>Service Software</b> , see Chapter 11.8.1.3.1 on page 11-63. | [ ] |
|------|-------------------------------------------------------------------------------------------|-----|

#### Comments / Observations

---



---



---

\_\_\_\_\_  
Date

\_\_\_\_\_  
Signature

## **33.2 LIAISON<sup>®</sup> XS Operational Qualification (OQ)**

### **1 General**

- 1.1 Type \_\_\_\_\_
- 1.2 Serial No. \_\_\_\_\_
- 1.3 Software Version \_\_\_\_\_

### **2 Installation details**

- 2.1 Installation date \_\_\_\_\_
- 2.2 Installed by \_\_\_\_\_

### **3 Customer details**

- 3.1 Customer name \_\_\_\_\_
- 3.2 Contact person name \_\_\_\_\_
- 3.3 Company \_\_\_\_\_
- 3.4 Address \_\_\_\_\_
- 3.5 Phone \_\_\_\_\_
- 3.6 Email \_\_\_\_\_

## configuration

| 4   | Sample Bay                                                                                                                                               | Flag | Note |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|
| 4.1 | Sample Bay initialized successfully.                                                                                                                     | [ ]  |      |
| 4.2 | Check barcode scanner configuration.                                                                                                                     | [ ]  |      |
| 4.3 | Barcode reading tested successfully (with customer barcoded vials).                                                                                      | [ ]  |      |
| 4.4 | External handheld barcode reader plug-in and configuration.                                                                                              | [ ]  |      |
| 5   | System                                                                                                                                                   | Flag | Note |
| 5.1 | System initialized successfully, see Chapter 11.8.1.1 on page 11-58.                                                                                     | [ ]  |      |
| 5.2 | Perform Cuvette Endurance Run, see Chapter 11.8.1.2 on page 11-61.                                                                                       | [ ]  |      |
| 6   | Backup of system configuration                                                                                                                           | Flag | Note |
| 6.1 | Perform backup of <code>Coordinate.coo</code> file.                                                                                                      | [ ]  |      |
| 6.2 | Perform backup of “.rac” files.                                                                                                                          | [ ]  |      |
| 7   | Manuals download                                                                                                                                         | Flag | Note |
| 7.1 | Install the most updated version of the Instrument Manuals, corresponding to the Instrument SW version and customer’s language for the specific country. | [ ]  |      |
| 8   | System Verification                                                                                                                                      | Flag | Note |
| 8.1 | Initialize the instrument via <b>LIAISON® XS Main UI</b> .                                                                                               | [ ]  |      |
| 8.2 | Check Status toggle and assign Consumables.                                                                                                              | [ ]  |      |
| 8.3 | Load cuvette in the Incubator.                                                                                                                           | [ ]  |      |
| 9   | Installation Final Test                                                                                                                                  | Flag | Note |
| 9.1 | Max signal of Background Wash in range (3BGW - Max Signal $\leq 1000$ RLU).                                                                              | [ ]  |      |
| 9.2 | CV% of Background Wash by Series in range (3BGW - Max CV% $\leq 10,0\%$ ).                                                                               | [ ]  |      |
| 9.3 | CV% of Background Wash mean in range (3BGW - Mean CV% $\leq 6,5\%$ ).                                                                                    | [ ]  |      |
| 9.4 | Signal of Light Check Left in range (6LCL - $345000 \leq \text{Signal} \leq 520000$ RLU).                                                                | [ ]  |      |
| 9.5 | CV% of Light Check Left by Series in range (6LCL - Max CV% $\leq 4,0\%$ ).                                                                               | [ ]  |      |
| 9.6 | CV% of Light Check Left mean in range (6LCL - Mean CV% $\leq 2,5\%$ ).                                                                                   | [ ]  |      |
| 9.7 | Signal of Light Check Right in range (6LCR - $160000 \leq \text{Signal} \leq 245000$ RLU).                                                               | [ ]  |      |
| 9.8 | CV% of Light Check Right by Series in range (6LCR - Max CV% $\leq 4,0\%$ ).                                                                              | [ ]  |      |

**33. Service Check List for internal-tanks configuration**

|      |                                                                                                                               |     |
|------|-------------------------------------------------------------------------------------------------------------------------------|-----|
| 9.9  | CV% of Light Check Right mean in range (6LCR - Mean CV% $\leq$ 2,5%).                                                         | [ ] |
| 9.10 | Calibration of Assay is valid.                                                                                                | [ ] |
| 9.11 | 20 replicates Assay dose CV% in range.                                                                                        | [ ] |
| 9.12 | 20 replicates Assay dose average value in range.                                                                              | [ ] |
| 9.13 | Raw data and calibration export and backup.                                                                                   | [ ] |
| 9.14 | Verify that the last revision of<br>"Installation_Final_Test_Liaison_XS_[Assay].xls" file has been completed and<br>attached. | [ ] |

| 10   | Attachments                                                                | Flag | Note |
|------|----------------------------------------------------------------------------|------|------|
| 10.1 | BGW results attached to the SGAT Worksheet.                                | [ ]  |      |
| 10.2 | LCL results attached to the SGAT Worksheet.                                | [ ]  |      |
| 10.3 | LCR results attached to the SGAT Worksheet.                                | [ ]  |      |
| 10.4 | Calibration & Controls results attached to the SGAT Worksheet.             | [ ]  |      |
| 10.5 | Completed IFTP spreadsheet (last revision) attached to the SGAT Worksheet. | [ ]  |      |

| 11   | References                                                                            | Flag | Note |
|------|---------------------------------------------------------------------------------------|------|------|
| 11.1 | <b>LIAISON® XS</b> Service Manual version [____]<br>(Please fill the current version) | [ ]  |      |

**Comments / Observations**


---



---



---



---

Date

---

Signature



### 33.3 LIAISON® XS Uninstallation

#### 1 General

|     |                  |       |
|-----|------------------|-------|
| 1.1 | Type             | <hr/> |
| 1.2 | Serial No.       | <hr/> |
| 1.3 | Software Version | <hr/> |

#### 2 Uninstallation details

|     |                     |       |
|-----|---------------------|-------|
| 2.1 | Uninstallation date | <hr/> |
| 2.2 | Uninstalled by      | <hr/> |

#### 3 Customer details

|     |                     |       |
|-----|---------------------|-------|
| 3.1 | Customer name       | <hr/> |
| 3.2 | Contact person name | <hr/> |
| 3.3 | Company             | <hr/> |
| 3.4 | Address             | <hr/> |
| 3.5 | Phone               | <hr/> |
| 3.6 | Email               | <hr/> |

**33. Service Check List for internal-tanks configuration**

| <b>4</b> | <b>Data deleting</b> (see Chapter 7.3 on page 7-3)                                                                                                                             | <b>Flag</b> | <b>Note</b> |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|
| 4.1      | Delete patient database.                                                                                                                                                       | [ ]         |             |
| <b>5</b> | <b>Decontamination</b> (see Chapter 7.3 on page 7-3)                                                                                                                           | <b>Flag</b> | <b>Note</b> |
| 5.1      | Empty Liquid Waste tank, add hypochlorite solution (see chapter 5.4.6 "Connecting and separating the Liquid Waste" of the User Manual) and load it back into the system.       | [ ]         |             |
| 5.2      | Run <b>System Shutdown</b> task from the dedicated window of <b>Main User Interface</b> (Main category <b>Maintenance</b> , sub-category <b>Task</b> ) skipping the last step. | [ ]         |             |
| 5.5      | Remove all tanks from the system.                                                                                                                                              | [ ]         |             |
| 5.6      | If not disposed, decontaminate Liquid Waste tank.                                                                                                                              | [ ]         |             |
| 5.7      | Remove all cuvettes from the system.                                                                                                                                           | [ ]         |             |
| 5.8      | Remove all sample tubes out of the Sample racks.                                                                                                                               | [ ]         |             |
| 5.9      | Remove all integrals out of the Reagent Module.                                                                                                                                | [ ]         |             |
| 5.10     | Remove all ancillaries and holders out of the Ancillary rack.                                                                                                                  | [ ]         |             |
| 5.11     | Remove all disposable tips trays out of the Cuvette - Tip Loading Module.                                                                                                      | [ ]         |             |
| 5.12     | Remove all cuvette trays out of the Cuvette - Tip Loading Module.                                                                                                              | [ ]         |             |
| 5.13     | Close and dispose the Waste.                                                                                                                                                   | [ ]         |             |
| 5.14     | Perform the Cleaning procedure, see Chapter 7.3.3 on page 7-4.                                                                                                                 | [ ]         |             |
| 5.15     | Complete the certificate of the decontamination (included in DiaSorin Technical GOP 23.7201).                                                                                  | [ ]         |             |
| 5.16     | Add the certificate of decontamination to the transportation documents.                                                                                                        | [ ]         |             |
| <b>6</b> | <b>System Packing</b> (see Chapter 7.4 on page 7-5)                                                                                                                            | <b>Flag</b> | <b>Note</b> |
| 6.1      | Prepare the system for the packing, see Chapter 7.4.1 on page 7-5.                                                                                                             | [ ]         |             |
| 6.2      | Pack the accessories cardboard box, see Chapter 7.4.2 on page 7-21.                                                                                                            | [ ]         |             |
| 6.3      | Pack the instrument, see Chapter 7.4.3 on page 7-36.                                                                                                                           | [ ]         |             |
| 6.4      | Add shock watch and thermometer indicator.                                                                                                                                     | [ ]         |             |



Comments / Observations

---

---

---

---

Date

---

Signature

### **33.4 LIAISON<sup>®</sup> XS Yearly Maintenance**

**1 General**

- 1.1 Type \_\_\_\_\_
- 1.2 Serial No. \_\_\_\_\_
- 1.3 Software Version \_\_\_\_\_

**2 Maintenance details**

- 2.1 Maintenance date \_\_\_\_\_
- 2.2 Previous Maintenance date \_\_\_\_\_
- 2.2 Maintained by \_\_\_\_\_

**3 Customer details**

- 3.1 Customer name \_\_\_\_\_
- 3.2 Contact person name \_\_\_\_\_
- 3.3 Company \_\_\_\_\_
- 3.4 Address \_\_\_\_\_
- 3.5 Phone \_\_\_\_\_
- 3.6 Email \_\_\_\_\_

| 4    | System Cleaning (see Chapter 9.2.2 on page 9-3)                                                                                                                                                                                                                    | Flag | Note |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|
| 4.1  | Empty Liquid Waste tank, add hypochlorite solution (see chapter 5.4.6 “Connecting and separating the Liquid Waste” of the User Manual) and load it back into the system.                                                                                           | [ ]  |      |
| 4.2  | Remove the cuvette from the system in the Washer, Reader and Incubator performing the <b>Empty Incubator</b> from Main category <b>Diagnostics</b> .                                                                                                               | [ ]  |      |
| 4.3  | Run <b>Disinfection Procedure</b> task from the dedicated window of <b>Main User Interface</b> (Main category <b>Maintenance</b> , sub-category <b>Task</b> ), see chapter 10.5 “Disinfection procedure task” of the User Manual.                                  | [ ]  |      |
| 4.4  | Remove all sample tubes out of the Sample racks.                                                                                                                                                                                                                   | [ ]  |      |
| 4.5  | Remove all integrals out of the Reagent Module.                                                                                                                                                                                                                    | [ ]  |      |
| 4.6  | Remove all ancillaries and holders out of the Ancillary rack.                                                                                                                                                                                                      | [ ]  |      |
| 4.7  | Remove all disposable tips trays out of the Cuvette - Tip Loading Module.                                                                                                                                                                                          | [ ]  |      |
| 4.8  | Remove all cuvette trays out of the Cuvette - Tip Loading Module.                                                                                                                                                                                                  | [ ]  |      |
| 4.9  | Close and dispose the LIAISON® EASY Waste box.                                                                                                                                                                                                                     | [ ]  |      |
| 4.10 | Insert a new LIAISON® EASY Waste box in the solid waste bin.                                                                                                                                                                                                       | [ ]  |      |
| 5    | PC & Accessories (see Chapter 9.2.3 on page 9-3)                                                                                                                                                                                                                   | Flag | Note |
| 5.1  | Perform the <b>All_Databases</b> back-up task.                                                                                                                                                                                                                     | [ ]  |      |
| 5.2  | Check dimension of databases.                                                                                                                                                                                                                                      | [ ]  |      |
| 5.3  | Check C:\ partition of hard disk.                                                                                                                                                                                                                                  | [ ]  |      |
| 5.4  | Check printer functionalities (if present).                                                                                                                                                                                                                        | [ ]  |      |
| 5.5  | Check UPS functionalities (if present).                                                                                                                                                                                                                            | [ ]  |      |
| 5.6  | Check the presence of the most updated version of the Manuals, corresponding to the Instrument SW version and customer’s language for the specific country. If needed, update the Manual as written in the Installation procedure, see Chapter 5.4.4 on page 5-34. | [ ]  |      |
| 5.7  | Run Antivirus application, see Chapter 32.3.4 on page 32-6.                                                                                                                                                                                                        | [ ]  |      |
| 6    | Maintenance of the Reader and Washer Module (see Chapter 9.2.5 on page 9-4)                                                                                                                                                                                        | Flag | Note |
| 6.1  | Replace all Washer and Reader Aspiration needles, see Chapter 17.3.1 on page 17-6 and see Chapter 18.3.1 on page 18-4.                                                                                                                                             | [ ]  |      |
| 6.2  | Check Reader rotor for crystallization and clean it, if necessary.                                                                                                                                                                                                 | [ ]  |      |
| 6.3  | Check the Washer plate, rotor, dispense nozzles and clean them, if necessary.                                                                                                                                                                                      | [ ]  |      |
| 7    | Maintenance of the Pumps and Tubings (see Chapter 9.2.6 on page 9-5)                                                                                                                                                                                               | Flag | Note |

### 33. Service Check List for internal-tanks configuration

|           |                                                                                                                                                        |             |             |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|
| 7.1       | Replace the WBDP cartridge, see Chapter 25.3.7 on page 25-29.                                                                                          | [ ]         |             |
| 7.2       | Replace LWP and WAP Peristaltic tubings, see Chapter 25.3.3 on page 25-12 and see Chapter 25.3.3 on page 25-12.                                        | [ ]         |             |
| 7.3       | Inspect and, if necessary, replace Waste tubing at Right and Left Energy Chain, see Chapter 28.3.8 on page 28-27 and see Chapter 28.3.7 on page 28-23. | [ ]         |             |
| <b>8</b>  | <b>Maintenance of the Reagent Module</b> (see Chapter 9.2.7 on page 9-5)                                                                               | <b>Flag</b> | <b>Note</b> |
| 8.1       | Clean the Reagent Module (inner and outer, cover included).                                                                                            | [ ]         |             |
| 8.2       | If needed, clean the crank drive. Then, lubricate it, see Chapter 22.3.3.5 on page 22-10.                                                              | [ ]         |             |
| <b>9</b>  | <b>Maintenance of the Pipettor</b> (see Chapter 9.2.8 on page 9-5)                                                                                     | <b>Flag</b> | <b>Note</b> |
| 9.1       | Wipe the Pipettor rods with a lint free cloth soaked with isopropanol.                                                                                 | [ ]         |             |
| 9.2       | Wipe the Pipettor rods with a lint free cloth soaked with oil.                                                                                         | [ ]         |             |
| 9.3       | Verify the proper behavior moving both Pipettor modules 5 times in axial X-direction and 5 times in axial Y-direction.                                 | [ ]         |             |
| 9.4       | Inspect and, if necessary, replace the Gripper slide bearing, see Chapter 19.3.6 on page 19-25.                                                        | [ ]         |             |
| <b>10</b> | <b>Cleaning of other surfaces</b> (see Chapter 9.2.9 on page 9-6)                                                                                      | <b>Flag</b> | <b>Note</b> |
| 10.1      | Clean the Cuvette - Tip Loading Module.                                                                                                                | [ ]         |             |
| 10.2      | Clean inner parts of Left and Right Top Covers.                                                                                                        | [ ]         |             |
| <b>11</b> | <b>System Restore &amp; Periodic Recalibrations</b> (see Chapter 9.2.10 on page 9-6)                                                                   | <b>Flag</b> | <b>Note</b> |
| 11.1      | Power on the system, from <b>Service Software</b> , see Chapter 11.8.1 on page 11-58.                                                                  | [ ]         |             |
| 11.2      | Check Pipettor teaching, see Chapter 13.15 on page 13-32.                                                                                              | [ ]         |             |
| 11.3      | Backup of "Coordinates.coo".                                                                                                                           | [ ]         |             |
| 11.4      | Backup of ".rac." files.                                                                                                                               | [ ]         |             |
| 11.5      | Initialize the system from <b>Service Software</b> , see Chapter 11.8.1 on page 11-58.                                                                 | [ ]         |             |
| 11.6      | Perform the Washer SIPAL Lift Calibration, see Chapter 11.5.3.2 on page 11-32.                                                                         | [ ]         |             |
| 11.7      | Perform the Washer Aspiration Monitoring Calibration, see Chapter 11.5.3.3 on page 11-33.                                                              | [ ]         |             |
| 11.8      | Perform the WBDP Calibration, see Chapter 11.5.3.4 on page 11-34.                                                                                      | [ ]         |             |
| 11.9      | Perform the Reader SIPAL Lift Calibration, see Chapter 11.6.3.2 on page 11-39.                                                                         | [ ]         |             |
| 11.10     | Perform the Reader Aspiration Monitoring Calibration, see Chapter 11.6.3.3 on page 11-40.                                                              | [ ]         |             |

### 33. Service Check List for internal-tanks

#### configuration

|       |                                                                                                                                                                |     |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 11.11 | Insert at least two cuvette on trays in the Cuvette - Tip Loading Module and set them in the System - Cuvette Tray State, see Chapter 11.8.2 on page 11-66.    | [ ] |
| 11.12 | Perform the Cuvette Endurance Run, see Chapter 11.8.1.2.1 on page 11-61.                                                                                       | [ ] |
| 11.13 | Prime the fluidics, see Chapter 11.8.1.3.1 on page 11-63.                                                                                                      | [ ] |
| 11.14 | Check that the Antivirus scan has been completed.                                                                                                              | [ ] |
| 11.15 | Log into the <b>Main UI Software</b> and initialize the system.                                                                                                | [ ] |
| 11.16 | Reset the counter of the Solid Waste.                                                                                                                          | [ ] |
| 11.17 | Insert cuvette trays and disposable tips trays in the Cuvette - Tip Loading Module.                                                                            | [ ] |
| 11.18 | Check the <b>Status - Consumables</b> toggle and assign the Consumables (cuvettes and tips) in the <b>Status - Plastic</b> toggle in <b>Main UI Software</b> . | [ ] |
| 11.19 | Load the cuvette in the Incubator Module.                                                                                                                      | [ ] |

#### 12 Requalification Test (see Chapter 9.2.11 on page 9-7)

**Flag Note**

|      |                                                                                              |     |
|------|----------------------------------------------------------------------------------------------|-----|
| 12.1 | CV% of Background Wash by Series in range (3BGW - Max CV% ≤ 10,0% - RLU ≤ 1000).             | [ ] |
| 12.2 | CV% of Light Check Right by Series in range (3LCR - Max CV% ≤ 4,0% - 160000 < RLU < 245000). | [ ] |
| 12.3 | CV% of Light Check Left by Series in range (3LCL - Max CV% ≤ 4,0% - 345000 < RLU < 520000).  | [ ] |
| 12.4 | Calibration & Control of customer kit in range.                                              | [ ] |
| 12.5 | Raw data printout and backup.                                                                | [ ] |
| 12.6 | Inform the customer that all the assays in use must be recalibrated.                         | [ ] |

#### 10 Attachments

**Flag Note**

|      |                                                                |     |
|------|----------------------------------------------------------------|-----|
| 10.1 | BGW results attached to the SGAT Worksheet.                    | [ ] |
| 10.2 | LCL results attached to the SGAT Worksheet.                    | [ ] |
| 10.3 | LCR results attached to the SGAT Worksheet.                    | [ ] |
| 10.4 | Calibration & Controls results attached to the SGAT Worksheet. | [ ] |

#### Comments / Observations

---



---



---

**33. Service Check List for internal-tanks configuration**

---

---

Date

---

Signature



# 34 Service Check List for external-tanks configuration

## 34.1 LIAISON® XS Installation Qualification (IQ)

### 1 General

- 1.1 Type \_\_\_\_\_
- 1.2 Serial No. \_\_\_\_\_
- 1.3 Software Version \_\_\_\_\_

### 2 Installation details

- 2.1 Installation date \_\_\_\_\_
- 2.2 Installed by \_\_\_\_\_

### 3 Customer details

- 3.1 Customer name \_\_\_\_\_
- 3.2 Contact person name \_\_\_\_\_
- 3.3 Company \_\_\_\_\_
- 3.4 Address \_\_\_\_\_
- 3.5 Phone \_\_\_\_\_
- 3.6 Email \_\_\_\_\_

**34. Service Check List for external-tanks configuration**

| <b>4</b> | <b>Visual Inspection before unpacking</b> (see Chapter 6.3.1 on page 6-5)            | <b>Flag</b> | <b>Note</b> |
|----------|--------------------------------------------------------------------------------------|-------------|-------------|
| 4.1      | Transport box without damages.                                                       | [ ]         |             |
| 4.2      | Transport shock monitor and temperature indication.                                  | [ ]         |             |
| <b>5</b> | <b>Unpacking, visual inspection and inventory</b> (see Chapter 6.3.2 on page 6-6)    | <b>Flag</b> | <b>Note</b> |
| 5.1      | Unpack the instrument.                                                               | [ ]         |             |
| 5.2      | Chassis without damages or gaps.                                                     | [ ]         |             |
| 5.3      | No visible cuts in the system or pipetting tubing.                                   | [ ]         |             |
| 5.4      | Covers without damages or scratches.                                                 | [ ]         |             |
| 5.5      | Parts and accessories according to the packing lists, see Chapter 38.1 on page 38-1. | [ ]         |             |
| 5.6      | Transport tilt and shock monitor at the rear chassis.                                | [ ]         |             |
| 5.7      | System leveling and stabilization.                                                   | [ ]         |             |
| 5.8      | Unlock pipettors.                                                                    | [ ]         |             |
| <b>6</b> | <b>Manual inspection</b> (see Chapter 6.3.3 on page 6-15)                            | <b>Flag</b> | <b>Note</b> |
| 6.1      | Both pipetting arms in x-axis are gliding smooth and without force.                  | [ ]         |             |
| 6.2      | Both pipetting arms in y-axis are gliding smooth and without force.                  | [ ]         |             |
| 6.3      | Both pipetting arms in z-axis are gliding smooth and without force.                  | [ ]         |             |
| <b>7</b> | <b>System set-up</b> (see Chapter 6.3.4 on page 6-16)                                | <b>Flag</b> | <b>Note</b> |
| 7.1      | Place the Velcro strip for the USB stick.                                            | [ ]         |             |
| 7.2      | Mount peristaltic tubes and caps of LWP and WAP pumps.                               | [ ]         |             |
| 7.3      | Rinse the tanks.                                                                     | [ ]         |             |
| 7.4      | Fill System liquid tank and place it.                                                | [ ]         |             |
| 7.5      | Fill Wash Buffer tank and place it.                                                  | [ ]         |             |
| 7.6      | Set up System Liquid, Wash Buffer and Liquid Waste tubings and sensor cables.        | [ ]         |             |
| 7.7      | Set up the Liquid Waste Tank and its basin.                                          | [ ]         |             |
| 7.8      | Assemble covers, see Chapter 6.3.4.4 on page 6-21.                                   | [ ]         |             |
| 7.9      | Check the alignment of the Covers, see Chapter 14.4 on page 14-20.                   | [ ]         |             |
| 7.10     | Assemble Panel PC, see Chapter 14.3.4.3 on page 14-11.                               | [ ]         |             |
| 7.11     | Check the system leveling.                                                           | [ ]         |             |
| 7.12     | Re-tension the Y belt of Left and Right Pipettor arms.                               | [ ]         |             |
| 7.13     | Insert the Ancillary tray in the Reagent Module.                                     | [ ]         |             |

### 34. Service Check List for external-tanks

#### configuration

- |      |                                                                                   |     |
|------|-----------------------------------------------------------------------------------|-----|
| 7.14 | Install the Cuvette - Tip Loading Module handle and insert cuvette and tip trays. | [ ] |
| 7.15 | Place the Solid Waste bin in the instrument with the EASY Waste box.              | [ ] |
| 7.16 | Insert the Starter reagent bottle into the dedicated module.                      | [ ] |
| 7.17 | Remove the protective foil from the Status light                                  | [ ] |
| 7.18 | Plug in and turn on <b>LIAISON® XS</b> system.                                    | [ ] |

#### 8 Rack configuration (see Chapter 6.3.5 on page 6-23 & see Chapter 6.3.6 on page 6-30) **Flag** **Note**

- |     |                                                                     |     |
|-----|---------------------------------------------------------------------|-----|
| 8.1 | Configure Sample racks, see Chapter 6.3.5.1 on page 6-25.           | [ ] |
| 8.2 | If needed, configure Control racks, see Chapter 6.3.6 on page 6-30. | [ ] |
| 8.3 | Barcode ID labels and Rack ID labels placed on all racks.           | [ ] |

#### 9 Pipettor and rack teaching (see Chapter 6.3.7 on page 6-32) **Flag** **Note**

- |     |                                                                                                   |     |
|-----|---------------------------------------------------------------------------------------------------|-----|
| 9.1 | Complete the teaching of the Pipettors via <b>Teacher Software</b> , see Chapter 13 on page 13-1. | [ ] |
| 9.2 | If required, perform the teaching of customized Sample racks (G, O, Y and Z racks).               | [ ] |

#### 11 Fluidic set-up (see Chapter 6.3.8 on page 6-32) **Flag** **Note**

- |      |                                                                                           |     |
|------|-------------------------------------------------------------------------------------------|-----|
| 11.1 | Perform fluidic prime via <b>Service Software</b> , see Chapter 12.8.1.3.1 on page 12-66. | [ ] |
|------|-------------------------------------------------------------------------------------------|-----|

#### Comments / Observations

---



---



---

**34. Service Check List for external-tanks configuration**

---

\_\_\_\_\_  
Date

\_\_\_\_\_  
Signature

## 34.2 LIAISON® XS Operational Qualification (OQ)

### 1 General

- 1.1 Type \_\_\_\_\_
- 1.2 Serial No. \_\_\_\_\_
- 1.3 Software Version \_\_\_\_\_

### 2 Installation details

- 2.1 Installation date \_\_\_\_\_
- 2.2 Installed by \_\_\_\_\_

### 3 Customer details

- 3.1 Customer name \_\_\_\_\_
- 3.2 Contact person name \_\_\_\_\_
- 3.3 Company \_\_\_\_\_
- 3.4 Address \_\_\_\_\_
- 3.5 Phone \_\_\_\_\_
- 3.6 Email \_\_\_\_\_

## 34. Service Check List for external-tanks configuration

| 4   | Sample Bay (see Chapter 6.4.1 on page 6-35)                                                                                                              | Flag | Note |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|
| 4.1 | Sample Bay initialized successfully.                                                                                                                     | [ ]  |      |
| 4.2 | Check barcode scanner configuration.                                                                                                                     | [ ]  |      |
| 4.3 | Barcode reading tested successfully (with customer barcoded vials).                                                                                      | [ ]  |      |
| 4.4 | External handheld barcode reader plug-in and configuration.                                                                                              | [ ]  |      |
| 5   | System (see Chapter 6.4.2 on page 6-36)                                                                                                                  | Flag | Note |
| 5.1 | System initialized successfully, see Chapter 12.8.1.1 on page 12-60.                                                                                     | [ ]  |      |
| 5.2 | Perform Cuvette Endurance Run, see Chapter 12.8.1.1 on page 12-60.                                                                                       | [ ]  |      |
| 6   | Backup of system configuration (see Chapter 6.4.3 on page 6-36)                                                                                          | Flag | Note |
| 6.1 | Perform backup of <code>Coordinate.coo</code> file.                                                                                                      | [ ]  |      |
| 6.2 | Perform backup of ".rac" files.                                                                                                                          | [ ]  |      |
| 7   | Manuals download (see Chapter 6.4.4 on page 6-36)                                                                                                        | Flag | Note |
| 7.1 | Install the most updated version of the Instrument Manuals, corresponding to the Instrument SW version and customer's language for the specific country. | [ ]  |      |
| 8   | System Verification (see Chapter 6.4.5 on page 6-36)                                                                                                     | Flag | Note |
| 8.1 | Initialize the instrument via <b>LIAISON® XS Main UI</b> .                                                                                               | [ ]  |      |
| 8.2 | Check Status toggle and assign Consumables.                                                                                                              | [ ]  |      |
| 8.3 | Load cuvette in the Incubator.                                                                                                                           | [ ]  |      |
| 9   | Installation Final Test (see Chapter 6.4.6 on page 6-36)                                                                                                 | Flag | Note |
| 9.1 | Max signal of Background Wash in range (3BGW - Max Signal $\leq$ 1000 RLU).                                                                              | [ ]  |      |
| 9.2 | CV% of Background Wash by Series in range (3BGW - Max CV% $\leq$ 10,0%).                                                                                 | [ ]  |      |
| 9.3 | CV% of Background Wash mean in range (3BGW - Mean CV% $\leq$ 6,5%).                                                                                      | [ ]  |      |
| 9.4 | Signal of Light Check Left in range (6LCL - 345000 $\leq$ Signal $\leq$ 520000 RLU).                                                                     | [ ]  |      |
| 9.5 | CV% of Light Check Left by Series in range (6LCL - Max CV% $\leq$ 4,0%).                                                                                 | [ ]  |      |
| 9.6 | CV% of Light Check Left mean in range (6LCL - Mean CV% $\leq$ 2,5%).                                                                                     | [ ]  |      |
| 9.7 | Signal of Light Check Right in range (6LCR - 160000 $\leq$ Signal $\leq$ 245000 RLU).                                                                    | [ ]  |      |
| 9.8 | CV% of Light Check Right by Series in range (6LCR - Max CV% $\leq$ 4,0%).                                                                                | [ ]  |      |

### 34. Service Check List for external-tanks

#### configuration

|      |                                                                                                                         |     |
|------|-------------------------------------------------------------------------------------------------------------------------|-----|
| 9.9  | CV% of Light Check Right mean in range (6LCR - Mean CV% $\leq$ 2,5%).                                                   | [ ] |
| 9.10 | Calibration of Assay is valid.                                                                                          | [ ] |
| 9.11 | 20 replicates Assay dose CV% in range.                                                                                  | [ ] |
| 9.12 | 20 replicates Assay dose average value in range.                                                                        | [ ] |
| 9.13 | Raw data and calibration export and backup.                                                                             | [ ] |
| 9.14 | Verify that the last revision of "Installation_Final_Test_Liaison_XS_[Assay].xls" file has been completed and attached. | [ ] |

#### 10 Attachments Flag Note

|      |                                                                            |     |
|------|----------------------------------------------------------------------------|-----|
| 10.1 | BGW results attached to the SGAT Worksheet.                                | [ ] |
| 10.2 | LCL results attached to the SGAT Worksheet.                                | [ ] |
| 10.3 | LCR results attached to the SGAT Worksheet.                                | [ ] |
| 10.4 | Calibration & Controls results attached to the SGAT Worksheet.             | [ ] |
| 10.5 | Completed IFTP spreadsheet (last revision) attached to the SGAT Worksheet. | [ ] |

#### 11 References Flag Note

|      |                                                                                       |     |
|------|---------------------------------------------------------------------------------------|-----|
| 11.1 | <b>LIAISON® XS</b> Service Manual version [____]<br>(Please fill the current version) | [ ] |
|------|---------------------------------------------------------------------------------------|-----|

#### Comments / Observations

---



---



---

\_\_\_\_\_  
Date

\_\_\_\_\_  
Signature

### **34.3 LIAISON® XS Uninstallation**

**1 General**

- 1.1 Type \_\_\_\_\_
- 1.2 Serial No. \_\_\_\_\_
- 1.3 Software Version \_\_\_\_\_

**2 Uninstallation details**

- 2.1 Uninstallation date \_\_\_\_\_
- 2.2 Uninstalled by \_\_\_\_\_

**3 Customer details**

- 3.1 Customer name \_\_\_\_\_
- 3.2 Contact person name \_\_\_\_\_
- 3.3 Company \_\_\_\_\_
- 3.4 Address \_\_\_\_\_
- 3.5 Phone \_\_\_\_\_
- 3.6 Email \_\_\_\_\_



## configuration

| 4    | Data deleting (see Chapter 8.3 on page 8-3)                                                                                                                                    | Flag | Note |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|
| 4.1  | Delete patient database.                                                                                                                                                       | [ ]  |      |
| 5    | Decontamination (see Chapter 8.3 on page 8-3)                                                                                                                                  | Flag | Note |
| 5.1  | Empty Liquid Waste tank, add hypochlorite solution (see chapter 5.4.6 "Connecting and separating the Liquid Waste" of the User Manual) and load it back into the system.       | [ ]  |      |
| 5.2  | Run <b>System Shutdown</b> task from the dedicated window of <b>Main User Interface</b> (Main category <b>Maintenance</b> , sub-category <b>Task</b> ) skipping the last step. | [ ]  |      |
| 5.5  | Remove all tanks from the system.                                                                                                                                              | [ ]  |      |
| 5.6  | If not disposed, decontaminate Liquid Waste tank.                                                                                                                              | [ ]  |      |
| 5.7  | Remove all cuvettes from the system.                                                                                                                                           | [ ]  |      |
| 5.8  | Remove all sample tubes out of the Sample racks.                                                                                                                               | [ ]  |      |
| 5.9  | Remove all integrals out of the Reagent Module.                                                                                                                                | [ ]  |      |
| 5.10 | Remove all ancillaries and holders out of the Ancillary rack.                                                                                                                  | [ ]  |      |
| 5.11 | Remove all disposable tips trays out of the Cuvette - Tip Loading Module.                                                                                                      | [ ]  |      |
| 5.12 | Remove all cuvette trays out of the Cuvette - Tip Loading Module.                                                                                                              | [ ]  |      |
| 5.13 | Close and dispose the Waste.                                                                                                                                                   | [ ]  |      |
| 5.14 | Perform the Cleaning procedure, see Chapter 8.3.3 on page 8-4.                                                                                                                 | [ ]  |      |
| 5.15 | Complete the certificate of the decontamination (included in DiaSorin Technical GOP 23.7201).                                                                                  | [ ]  |      |
| 5.16 | Add the certificate of decontamination to the transportation documents.                                                                                                        | [ ]  |      |
| 6    | System Packing (see Chapter 8.4 on page 8-5)                                                                                                                                   | Flag | Note |
| 6.1  | Prepare the system for the packing, see Chapter 8.4.1 on page 8-5.                                                                                                             | [ ]  |      |
| 6.2  | Pack the accessories cardboard box, see Chapter 8.4.2 on page 8-21.                                                                                                            | [ ]  |      |
| 6.3  | Pack the instrument, see Chapter 8.4.3 on page 8-36.                                                                                                                           | [ ]  |      |
| 6.4  | Add shock watch and thermometer indicator.                                                                                                                                     | [ ]  |      |

**34. Service Check List for external-tanks configuration**

---

**Comments / Observations**

---

---

---

\_\_\_\_\_  
Date

\_\_\_\_\_  
Signature

## 34.4 LIAISON® XS Yearly Maintenance

### 1 General

- 1.1 Type \_\_\_\_\_
- 1.2 Serial No. \_\_\_\_\_
- 1.3 Software Version \_\_\_\_\_

### 2 Maintenance details

- 2.1 Maintenance date \_\_\_\_\_
- 2.2 Previous Maintenance date \_\_\_\_\_
- 2.2 Maintained by \_\_\_\_\_

### 3 Customer details

- 3.1 Customer name \_\_\_\_\_
- 3.2 Contact person name \_\_\_\_\_
- 3.3 Company \_\_\_\_\_
- 3.4 Address \_\_\_\_\_
- 3.5 Phone \_\_\_\_\_
- 3.6 Email \_\_\_\_\_

## 34. Service Check List for external-tanks configuration

| 4    | System Cleaning (see Chapter 10.2.2 on page 10-3)                                                                                                                                                                                                                  | Flag | Note |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|
| 4.1  | Empty Liquid Waste tank, add hypochlorite solution (see chapter 5.4.6 “Connecting and separating the Liquid Waste” of the User Manual) and load it back into the system.                                                                                           | [ ]  |      |
| 4.2  | Remove the cuvette from the system in the Washer, Reader and Incubator performing the <b>Empty Incubator</b> from Main category <b>Diagnostics</b> .                                                                                                               | [ ]  |      |
| 4.3  | Run <b>Disinfection Procedure</b> task from the dedicated window of <b>Main User Interface</b> (Main category <b>Maintenance</b> , sub-category <b>Task</b> ), see chapter 10.5 “Disinfection procedure task” of the User Manual.                                  | [ ]  |      |
| 4.4  | Remove all sample tubes out of the Sample racks.                                                                                                                                                                                                                   | [ ]  |      |
| 4.5  | Remove all integrals out of the Reagent Module.                                                                                                                                                                                                                    | [ ]  |      |
| 4.6  | Remove all ancillaries and holders out of the Ancillary rack.                                                                                                                                                                                                      | [ ]  |      |
| 4.7  | Remove all disposable tips trays out of the Cuvette - Tip Loading Module.                                                                                                                                                                                          | [ ]  |      |
| 4.8  | Remove all cuvette trays out of the Cuvette - Tip Loading Module.                                                                                                                                                                                                  | [ ]  |      |
| 4.9  | Close and dispose the LIAISON® EASY Waste box.                                                                                                                                                                                                                     | [ ]  |      |
| 4.10 | Insert a new LIAISON® EASY Waste box in the solid waste bin.                                                                                                                                                                                                       | [ ]  |      |
| 5    | PC & Accessories (see Chapter 10.2.3 on page 10-3)                                                                                                                                                                                                                 | Flag | Note |
| 5.1  | Perform the <b>All_Databases</b> back-up task.                                                                                                                                                                                                                     | [ ]  |      |
| 5.2  | Check dimension of databases.                                                                                                                                                                                                                                      | [ ]  |      |
| 5.3  | Check C:\ partition of hard disk.                                                                                                                                                                                                                                  | [ ]  |      |
| 5.4  | Check printer functionalities (if present).                                                                                                                                                                                                                        | [ ]  |      |
| 5.5  | Check UPS functionalities (if present).                                                                                                                                                                                                                            | [ ]  |      |
| 5.6  | Check the presence of the most updated version of the Manuals, corresponding to the Instrument SW version and customer's language for the specific country. If needed, update the Manual as written in the Installation procedure, see Chapter 6.4.4 on page 6-36. | [ ]  |      |
| 5.7  | Run Antivirus application, see Chapter 32.4 on page 32-10.                                                                                                                                                                                                         | [ ]  |      |
| 6    | Maintenance of the Reader and Washer Module (see Chapter 10.2.5 on page 10-5)                                                                                                                                                                                      | Flag | Note |
| 6.1  | Replace all Washer and Reader Aspiration needles, see Chapter 17.3.1 on page 17-6 and see Chapter 18.3.1 on page 18-4.                                                                                                                                             | [ ]  |      |
| 6.2  | Check Reader rotor for crystallization and clean it, if necessary.                                                                                                                                                                                                 | [ ]  |      |
| 6.3  | Check the Washer plate, rotor, dispense nozzles and clean them, if necessary.                                                                                                                                                                                      | [ ]  |      |

## 34. Service Check List for external-tanks

### configuration

| 7    | Maintenance of the Pumps and Tubings (see Chapter 10.2.6 on page 10-5)                                                                               | Flag | Note |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|
| 7.1  | Replace the WBDP cartridge, see Chapter 26.3.7 on page 26-29.                                                                                        | [ ]  |      |
| 7.2  | Replace LWP and WAP Peristaltic tubings, see Chapter 26.3.3 on page 26-12.                                                                           | [ ]  |      |
| 7.3  | Replace the MANN filter, see Chapter 29.3.1 on page 29-7.                                                                                            | [ ]  |      |
| 7.4  | Inspect and, if necessary, replace Waste tubing at Right and Left Energy Chain, see Chapter 29.2.3 on page 29-5 and see Chapter 29.2.2 on page 29-5. | [ ]  |      |
| 8    | Maintenance of the Reagent Module (see Chapter 10.2.7 on page 10-5)                                                                                  | Flag | Note |
| 8.1  | Clean the Reagent Module (inner and outer, cover included).                                                                                          | [ ]  |      |
| 8.2  | If needed, clean the crank drive. Then, lubricate it, see Chapter 22.3.3.5 on page 22-10.                                                            | [ ]  |      |
| 9    | Maintenance of the Pipettor (see Chapter 10.2.8 on page 10-6)                                                                                        | Flag | Note |
| 9.1  | Wipe the Pipettor rods with a lint free cloth soaked with isopropanol.                                                                               | [ ]  |      |
| 9.2  | Wipe the Pipettor rods with a lint free cloth soaked with oil.                                                                                       | [ ]  |      |
| 9.3  | Verify the proper behavior moving both Pipettor modules 5 times in axial X-direction and 5 times in axial Y-direction.                               | [ ]  |      |
| 9.4  | Inspect and, if necessary, replace the Gripper slide bearing, see Chapter 20.3.6 on page 20-25.                                                      | [ ]  |      |
| 10   | Cleaning of other surfaces (see Chapter 10.2.9 on page 10-6)                                                                                         | Flag | Note |
| 10.1 | Clean the Cuvette - Tip Loading Module.                                                                                                              | [ ]  |      |
| 10.2 | Clean inner parts of Left and Right Top Covers.                                                                                                      | [ ]  |      |
| 11   | System Restore & Periodic Recalibrations (see Chapter 10.2.10 on page 10-6)                                                                          | Flag | Note |
| 11.1 | Power on the system, from <b>Service Software</b> , see Chapter 12.8.1 on page 12-60.                                                                | [ ]  |      |
| 11.2 | Check Pipettor teaching, see Chapter 13 on page 13-1.                                                                                                | [ ]  |      |
| 11.3 | Backup of "Coordinates.coo".                                                                                                                         | [ ]  |      |
| 11.4 | Backup of ".rac." files.                                                                                                                             | [ ]  |      |
| 11.5 | Initialize the system from <b>Service Software</b> , see Chapter 12.8.1 on page 12-60.                                                               | [ ]  |      |
| 11.6 | Perform the Washer SIPAL Lift Calibration, see Chapter 12.5.3.2 on page 12-34.                                                                       | [ ]  |      |
| 11.7 | Perform the Washer Aspiration Monitoring Calibration, see Chapter 12.5.3.3 on page 12-35.                                                            | [ ]  |      |
| 11.8 | Perform the WBDP Calibration, see Chapter 12.5.3.4 on page 12-36.                                                                                    | [ ]  |      |
| 11.9 | Perform the Reader SIPAL Lift Calibration, see Chapter 12.6.3.2 on page 12-41.                                                                       | [ ]  |      |

### 34. Service Check List for external-tanks configuration

|           |                                                                                                                                                                |             |             |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-------------|
| 11.10     | Perform the Reader Aspiration Monitoring Calibration, see Chapter 12.6.3.3 on page 12-42.                                                                      | [ ]         |             |
| 11.11     | Insert at least two cuvette on trays in the Cuvette - Tip Loading Module and set them in the System - Cuvette Tray State, see Chapter 12.8.2 on page 12-69.    | [ ]         |             |
| 11.12     | Perform the Cuvette Endurance Run, see Chapter 12.8.1.2.1 on page 12-63.                                                                                       | [ ]         |             |
| 11.13     | Prime the fluidics, see Chapter 12.8.1.3.1 on page 12-66.                                                                                                      | [ ]         |             |
| 11.14     | Check that the Antivirus scan has been completed.                                                                                                              | [ ]         |             |
| 11.15     | Log into the <b>Main UI Software</b> and initialize the system.                                                                                                | [ ]         |             |
| 11.16     | Reset the counter of the Solid Waste.                                                                                                                          | [ ]         |             |
| 11.17     | Insert cuvette trays and disposable tips trays in the Cuvette - Tip Loading Module.                                                                            | [ ]         |             |
| 11.18     | Check the <b>Status - Consumables</b> toggle and assign the Consumables (cuvettes and tips) in the <b>Status - Plastic</b> toggle in <b>Main UI Software</b> . | [ ]         |             |
| 11.19     | Load the cuvette in the Incubator Module.                                                                                                                      | [ ]         |             |
| <b>12</b> | <b>Requalification Test</b> (see Chapter 10.2.11 on page 10-7)                                                                                                 | <b>Flag</b> | <b>Note</b> |
| 12.1      | CV% of Background Wash by Series in range (3BGW - Max CV% $\leq$ 10,0% - RLU $\leq$ 1000).                                                                     | [ ]         |             |
| 12.2      | CV% of Light Check Right by Series in range (3LCL - Max CV% $\leq$ 4,0% - 160000 < RLU < 245000).                                                              | [ ]         |             |
| 12.3      | CV% of Light Check Left by Series in range (3LCL - Max CV% $\leq$ 4,0% - 345000 < RLU < 520000).                                                               | [ ]         |             |
| 12.4      | Calibration & Control of customer kit in range.                                                                                                                | [ ]         |             |
| 12.5      | Raw data printout and backup.                                                                                                                                  | [ ]         |             |
| 12.6      | Inform the customer that all the assays in use must be recalibrated.                                                                                           | [ ]         |             |
| <b>10</b> | <b>Attachments</b>                                                                                                                                             | <b>Flag</b> | <b>Note</b> |
| 10.1      | BGW results attached to the SGAT Worksheet.                                                                                                                    | [ ]         |             |
| 10.2      | LCL results attached to the SGAT Worksheet.                                                                                                                    | [ ]         |             |
| 10.3      | LCR results attached to the SGAT Worksheet.                                                                                                                    | [ ]         |             |
| 10.4      | Calibration & Controls results attached to the SGAT Worksheet.                                                                                                 | [ ]         |             |

#### Comments / Observations

---



---



### 34. Service Check List for external-tanks

configuration

---

---

Date

---

Signature

#### 34. Service Check List for external-tanks configuration

---



## 35 Technical data for internal-tanks configuration

**NOTE****Specification**

Values are achieved under optimal conditions and can vary depending on environmental conditions, instrument status and processing conditions! Specifications are subject to change without notice.

### 35.1 Power requirements

|                      |                                                                 |
|----------------------|-----------------------------------------------------------------|
| Voltage:             | 100 V - 240 V $\pm$ 10 %                                        |
| Frequency:           | 50 - 60 Hz                                                      |
| Power consumption:   | mean: 200 W (peak: 300 W)<br>mean: 683 BTU/h (peak: 1024 BTU/h) |
| Input current range: | 4 - 1.7 A                                                       |
| Fuse:                | primary 250 VAC, T4AH                                           |

*Table 35–1: Power requirements*

## 35.2 Laser barcode scanner

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Class:                    | Class 2 laser product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Available bar-codes:      | <ul style="list-style-type: none"> <li>• Code 128/EAN 128 (set as default)</li> <li>• Codabar (set as default)</li> <li>• Code 39 (set as default)</li> <li>• 2/5 Interleaved (to be set)</li> </ul> <p>Please check with local service support the enabling status of the barcode types of interest.</p> <p>Barcode types not strictly compliant with this list are not allowed and must not be used on the system.</p> <p>Barcodes quality shall match category A or B (according to ANSI X3.182 standard) or category 4 and 3 (according to ISO/IEC 15416 standard).</p> <p>In addition:</p> <ul style="list-style-type: none"> <li>• <math>0.167 \text{ mm} \leq \text{width} \leq 0.5 \text{ mm}</math></li> <li>• <math>1:2.5 \leq \text{ratio} \leq 1:3</math> (if <math>0.167 \text{ mm} \leq \text{width} &lt; 0.2 \text{ mm}</math>) or</li> <li>• <math>1:2 \leq \text{ratio} \leq 1:3</math> (if <math>0.2 \text{ mm} \leq \text{width} \leq 0.5 \text{ mm}</math>).</li> </ul> |
| Maximal output radiation: | 85 $\mu\text{W}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Maximal pulse duration:   | 112 $\mu\text{s}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Emitted wave length:      | 660 nm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Standards:                | IEC 60825-1:2007; IEC 60825-1:2014, except otherwise noted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Reading distance:         | 45 - 270 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Table 35–2: Laser barcode scanner

### 35.2.1 Handheld barcode reader

Please, refer to handheld barcode reader user manual for technical information.

## 35.3 Computer and connections

|                                                                                       |                                                                                      |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Panel PC</b>                                                                       |                                                                                      |
| Processor type:                                                                       | Intel® (or compatible)                                                               |
| Touch screen type:                                                                    | Projected Capacitive Touch Panel                                                     |
| <b>User Access Panel</b> ( <i>right side of the instrument, next to Main Switch</i> ) |                                                                                      |
| Ports:                                                                                | <ul style="list-style-type: none"> <li>• 1 x RJ-45</li> <li>• 4 x USB 2.0</li> </ul> |

Table 35–3: Computer connections

|                   |                                               |
|-------------------|-----------------------------------------------|
| <b>Software:</b>  |                                               |
| Operating system: | Microsoft® Windows® 10 IoT Enterprise (64bit) |

*Table 35–4: Software*

## 35.4 Dimensions and weight

|             | <b>Weight</b>      | <b>Width</b>       | <b>Depth</b>        | <b>Height</b>     |
|-------------|--------------------|--------------------|---------------------|-------------------|
| Instrument: | 137 kg<br>(302 lb) | 1280 mm<br>(50 in) | 750 mm<br>(29.5 in) | 720 mm<br>(28 in) |

*Table 35–5: Dimensions and Weight (including covers)*

### 35.5 System clearance

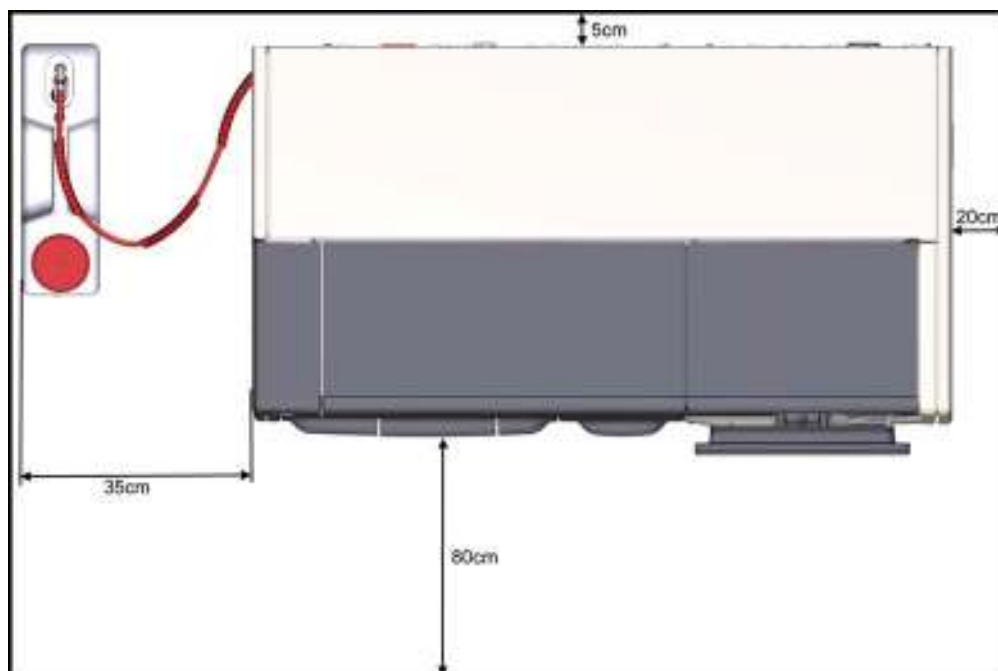


Figure 35-1: System clearance, top view

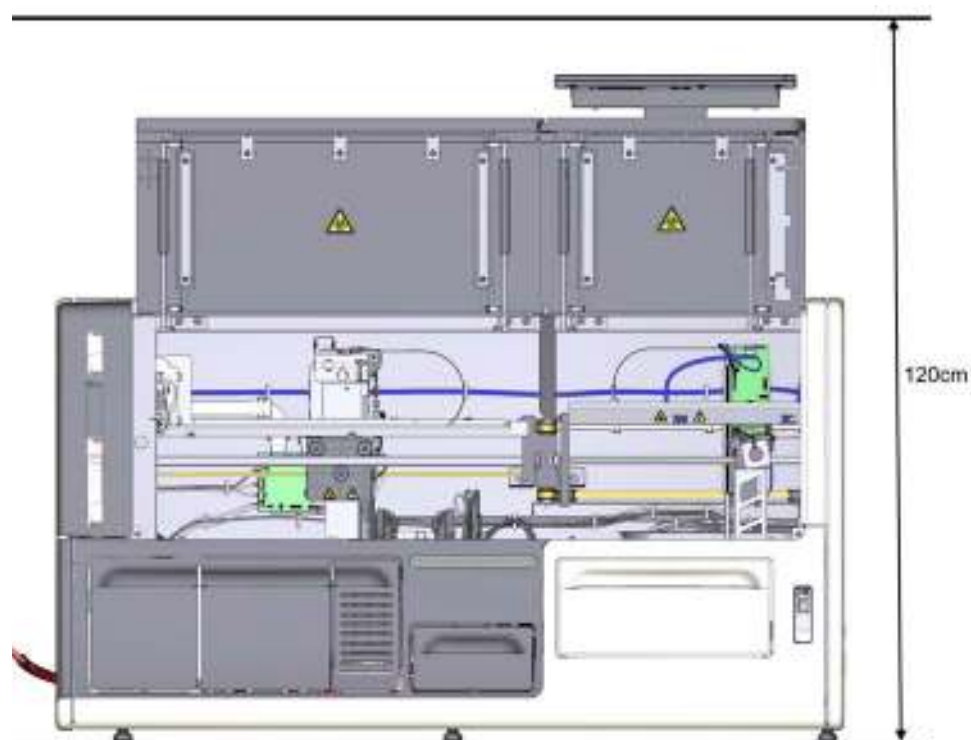


Figure 35-2: System clearance, front view

## 35.6 Environmental conditions

The following table shows the range of conditions needed to run the system safely.

|                           |                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental Conditions: | The system is made for Indoor use only.                                                                                                                  |
| Temperature:              | Operating: 15°C to 32°C (59°F to 89.6°F)<br>Storage: 5°C to 40°C (41°F to 104°F)<br>Transport: -20°C to 60°C (-4°F to 140°F)<br>(for no more than 48hrs) |
| Humidity:                 | Operating: 20 - 85 % non-condensing<br>Storage: 20 - 85 % non-condensing<br>Transport: 10 - 95 % non-condensing                                          |
| Pollution Degree:         | 2 (EN 61010-1:2001)                                                                                                                                      |
| Installation Class:       | 2 (EN 61010-1:2001)                                                                                                                                      |
| Limit Class:              | Class A<br>(For industrial use. Domestic use restricted)                                                                                                 |
| Sunlight:                 | No direct sunlight.<br>May mislead optical sensors and affect performance.                                                                               |
| Altitude:                 | Up to 3000 m (9843 ft) above mean sea level.                                                                                                             |
| Dust:                     | No excessive dust.                                                                                                                                       |

*Table 35–6: Environmental conditions*

## 35.7 Noise emission

Maximum value measured at normal user position, with closed cover hood at 1 m distance to enclosure (39.4 in) is 65 dB(A).

*Table 35–7: Noise emission*

## 35.8 Temperature Ranges

|                                  |                                              |
|----------------------------------|----------------------------------------------|
| Ambient Temperature:             | 15°C - 32°C (59°F - 89.6°F)                  |
| Incubating Temperature:          | 35°C - 37°C (95°F - 98.6°F), in liquid       |
| Temperature inside the system:   | Ambient temperature $\pm$ 5°C ( $\pm$ 9°F)   |
| Reagent Loading Bay Temperature: | 11°C - 15°C (51.8°F - 59°F)                  |
| Sample Loading Bay Temperature:  | Ambient temperature $\pm$ 3°C ( $\pm$ 5.4°F) |
| Starter tubes Temperature:       | 34°C - 36°C (93.2°F - 96.8°F)                |

*Table 35–8: Temperature Ranges*

### **35.9 Water quality**

The water to be used for the dilution and reconstitution of the Liaison XS reagents and consumables should comply with the definition of “Instrument Feed Water”, according to CLSI standard “Preparation and Testing of Reagent Water in the Clinical Laboratory; Approved Guideline – Fourth Edition”.

In particular, the water shall contain the following characteristics:

- pH: 5.0 – 8.0
- conductivity: <2  $\mu$ S/cm
- resistivity: > 0.5 M Ohm-cm
- TOC: < 500 ppb
- SiO<sub>2</sub>: <1.0 ppm
- Bacteria: <10 CFU/ml

## 35.10 Packaging

See enclosed delivery note for the contents, weights and dimensions.

|                          |   |
|--------------------------|---|
| Number of wooden pallet: | 1 |
|--------------------------|---|

*Table 35–9: Number of wooden pallet*

|                          |                                                             |
|--------------------------|-------------------------------------------------------------|
| <b>Card board box 1:</b> |                                                             |
| Contents:                | Instrument                                                  |
| Dimensions (WxDxH):      | 1380 mm x 800 mm x 900 mm<br>(54 in x 31.5 in x 35 in)      |
| Weight:                  | 125 kg<br>(276 lb),<br>total weight of instrument and crate |

*Table 35–10: Card board box 1*

|                          |                                                                     |
|--------------------------|---------------------------------------------------------------------|
| <b>Card board box 2:</b> |                                                                     |
| Contents:                | Covers and Accessories                                              |
| Dimensions (WxDxH):      | 1410 mm x 800 mm x 850 mm<br>(55.5 in x 31 in x 33 in)              |
| Weight:                  | 71 kg<br>(156 lb),<br>total weight of covers, accessories and crate |

*Table 35–11: Card board box 2*





## 36 Technical data for external-tanks configuration

**NOTE****Specification**

Values are achieved under optimal conditions and can vary depending on environmental conditions, instrument status and processing conditions! Specifications are subject to change without notice.

### 36.1 Power requirements

|                      |                                                                 |
|----------------------|-----------------------------------------------------------------|
| Voltage:             | 100 V - 240 V $\pm$ 10 %                                        |
| Frequency:           | 50 - 60 Hz                                                      |
| Power consumption:   | mean: 200 W (peak: 300 W)<br>mean: 683 BTU/h (peak: 1024 BTU/h) |
| Input current range: | 4 - 1.7 A                                                       |
| Fuse:                | primary 250 VAC, T4AH                                           |

Table 36–1: Power requirements

## 36.2 Laser barcode scanner

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Class:                    | Class 2 laser product                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Available bar-codes:      | <ul style="list-style-type: none"> <li>• Code 128/EAN 128 (set as default)</li> <li>• Codabar (set as default)</li> <li>• Code 39 (set as default)</li> <li>• 2/5 Interleaved (to be set)</li> </ul> <p>Please check with local service support the enabling status of the barcode types of interest.</p> <p>Barcode types not strictly compliant with this list are not allowed and must not be used on the system.</p> <p>Barcodes quality shall match category A or B (according to ANSI X3.182 standard) or category 4 and 3 (according to ISO/IEC 15416 standard).</p> <p>In addition:</p> <ul style="list-style-type: none"> <li>• <math>0.167 \text{ mm} \leq \text{width} \leq 0.5 \text{ mm}</math></li> <li>• <math>1:2.5 \leq \text{ratio} \leq 1:3</math> (if <math>0.167 \text{ mm} \leq \text{width} &lt; 0.2 \text{ mm}</math>) or</li> <li>• <math>1:2 \leq \text{ratio} \leq 1:3</math> (if <math>0.2 \text{ mm} \leq \text{width} \leq 0.5 \text{ mm}</math>).</li> </ul> |
| Maximal output radiation: | 85 $\mu\text{W}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Maximal pulse duration:   | 112 $\mu\text{s}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Emitted wave length:      | 660 nm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Standards:                | IEC 60825-1:2007; IEC 60825-1:2014, except otherwise noted                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Reading distance:         | 45 - 270 mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Table 36–2: Laser barcode scanner

### 36.2.1 Handheld barcode reader

Please, refer to handheld barcode reader user manual for technical information.

## 36.3 Computer and connections

|                                                                                       |                                                                                      |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Panel PC</b>                                                                       |                                                                                      |
| Processor type:                                                                       | Intel® (or compatible)                                                               |
| Touch screen type:                                                                    | Projected Capacitive Touch Panel                                                     |
| <b>User Access Panel</b> ( <i>right side of the instrument, next to Main Switch</i> ) |                                                                                      |
| Ports:                                                                                | <ul style="list-style-type: none"> <li>• 1 x RJ-45</li> <li>• 4 x USB 2.0</li> </ul> |

Table 36–3: Computer connections

|                   |                                               |
|-------------------|-----------------------------------------------|
| <b>Software:</b>  |                                               |
| Operating system: | Microsoft® Windows® 10 IoT Enterprise (64bit) |

*Table 36–4: Software*

## 36.4 Dimensions and weight

|             | <b>Weight</b>      | <b>Width</b>       | <b>Depth</b>        | <b>Height</b>     |
|-------------|--------------------|--------------------|---------------------|-------------------|
| Instrument: | 137 kg<br>(302 lb) | 1280 mm<br>(50 in) | 750 mm<br>(29.5 in) | 720 mm<br>(28 in) |

*Table 36–5: Dimensions and Weight (including covers)*

### 36.5 System clearance

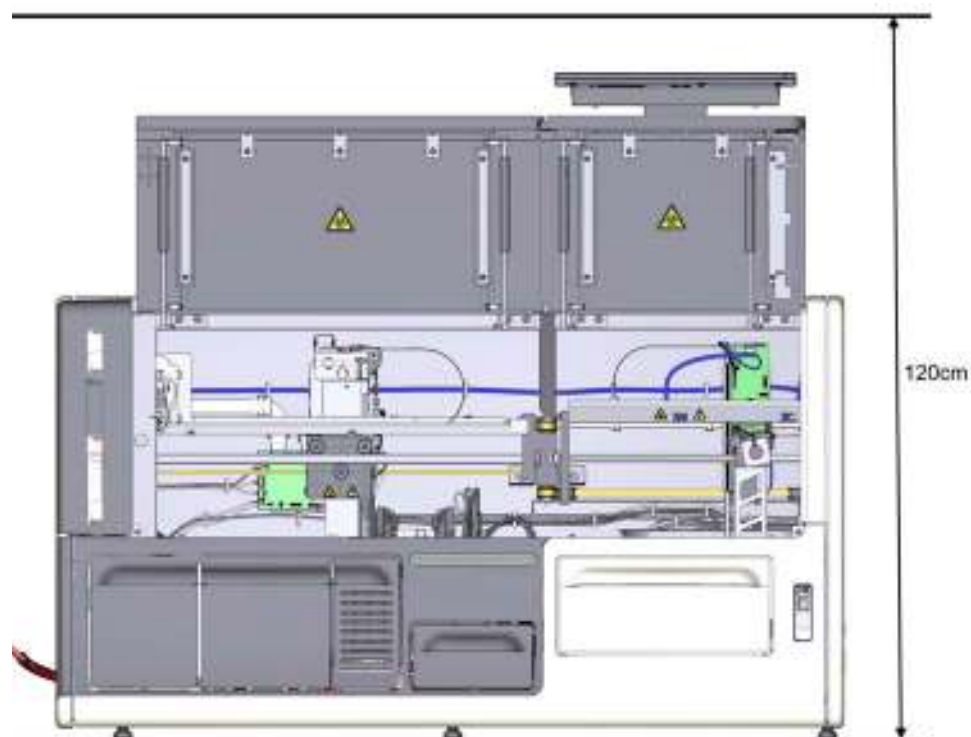


Figure 36-1: System clearance, front view



Figure 36-2: System clearance, side view

## 36. Technical data for external-tanks configuration

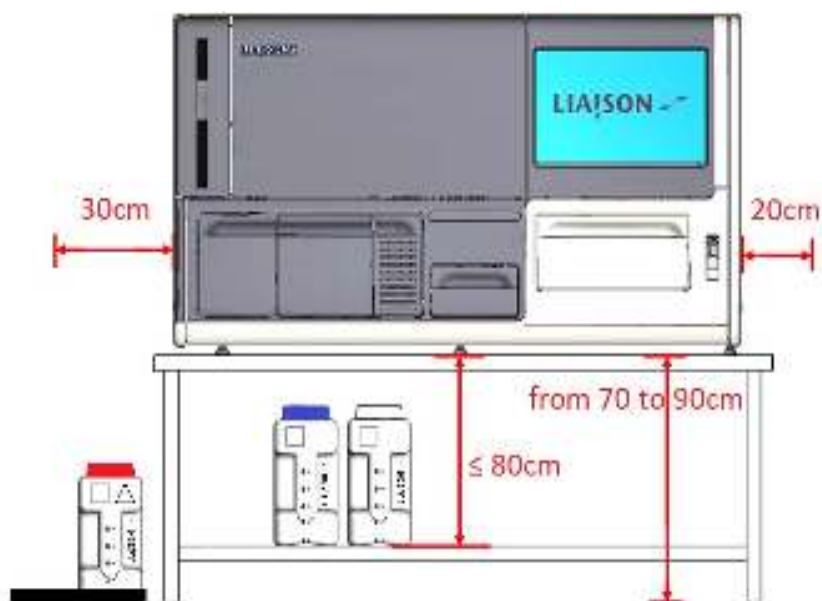


Figure 36-3: System clearance, front view, in case of tanks placed below the system

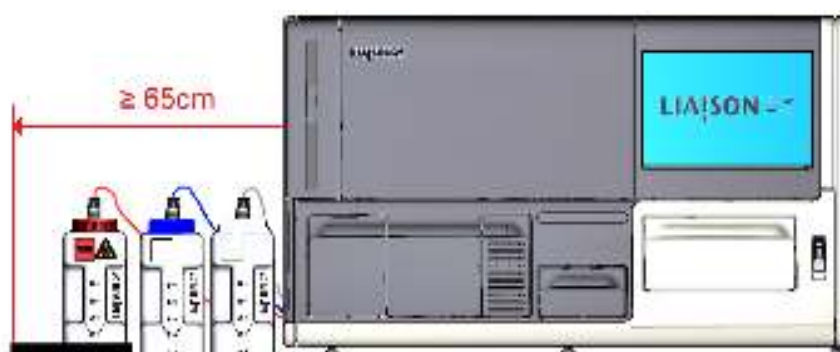


Figure 36-4: Left clearance, in case of tanks placed on the bench  
(right clearance unchanged)

## 36.6 Environmental conditions

The following table shows the range of conditions needed to run the system safely.

|                           |                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental Conditions: | The system is made for Indoor use only.                                                                                                                  |
| Temperature:              | Operating: 15°C to 32°C (59°F to 89.6°F)<br>Storage: 5°C to 40°C (41°F to 104°F)<br>Transport: -20°C to 60°C (-4°F to 140°F)<br>(for no more than 48hrs) |
| Humidity:                 | Operating: 20 - 85 % non-condensing<br>Storage: 20 - 85 % non-condensing<br>Transport: 10 - 95 % non-condensing                                          |
| Pollution Degree:         | 2 (EN 61010-1:2001)                                                                                                                                      |
| Installation Class:       | 2 (EN 61010-1:2001)                                                                                                                                      |
| Limit Class:              | Class A<br>(For industrial use. Domestic use restricted)                                                                                                 |
| Sunlight:                 | No direct sunlight.<br>May mislead optical sensors and affect performance.                                                                               |
| Altitude:                 | Up to 3000 m (9843 ft) above mean sea level.                                                                                                             |
| Dust:                     | No excessive dust.                                                                                                                                       |

Table 36–6: Environmental conditions

## 36.7 Noise emission

Maximum value measured at normal user position, with closed cover hood at 1 m distance to enclosure (39.4 in) is 65 dB(A).

Table 36–7: Noise emission

## 36.8 Temperature Ranges

|                                  |                                              |
|----------------------------------|----------------------------------------------|
| Ambient Temperature:             | 15°C - 32°C (59°F - 89.6°F)                  |
| Incubating Temperature:          | 35°C - 37°C (95°F - 98.6°F), in liquid       |
| Temperature inside the system:   | Ambient temperature $\pm$ 5°C ( $\pm$ 9°F)   |
| Reagent Loading Bay Temperature: | 11°C - 15°C (51.8°F - 59°F)                  |
| Sample Loading Bay Temperature:  | Ambient temperature $\pm$ 3°C ( $\pm$ 5.4°F) |
| Starter tubes Temperature:       | 34°C - 36°C (93.2°F - 96.8°F)                |

*Table 36–8: Temperature Ranges*

### 36.9 Water quality

The water to be used for the dilution and reconstitution of the Liaison XS reagents and consumables should comply with the definition of “Instrument Feed Water”, according to CLSI standard “Preparation and Testing of Reagent Water in the Clinical Laboratory; Approved Guideline – Fourth Edition”.

In particular, the water shall contain the following characteristics:

- pH: 5.0 – 8.0
- conductivity: <2 µS/cm
- resistivity: > 0.5 M Ohm-cm
- TOC: < 500 ppb
- SiO<sub>2</sub>: <1.0 ppm
- Bacteria: <10 CFU/ml

## 36.10 Packaging

See enclosed delivery note for the contents, weights and dimensions.

|                          |   |
|--------------------------|---|
| Number of wooden pallet: | 1 |
|--------------------------|---|

*Table 36–9: Number of wooden pallet*

|                          |                                                             |
|--------------------------|-------------------------------------------------------------|
| <b>Card board box 1:</b> |                                                             |
| Contents:                | Instrument                                                  |
| Dimensions (WxDxH):      | 1380 mm x 800 mm x 900 mm<br>(54 in x 31.5 in x 35 in)      |
| Weight:                  | 125 kg<br>(276 lb),<br>total weight of instrument and crate |

*Table 36–10: Card board box 1*

|                          |                                                                     |
|--------------------------|---------------------------------------------------------------------|
| <b>Card board box 2:</b> |                                                                     |
| Contents:                | Covers and Accessories                                              |
| Dimensions (WxDxH):      | 1410 mm x 800 mm x 850 mm<br>(55.5 in x 31 in x 33 in)              |
| Weight:                  | 71 kg<br>(156 lb),<br>total weight of covers, accessories and crate |

*Table 36–11: Card board box 2*



# 37 Packing List for internal-tanks configuration

## 37.1 Packing list

Check the scope of delivery for completeness by means of the following packing list.

### 37.1.1 Instrument cardboard box

1. Instrument with cables (schuko plug 220 Volt)
2. Sample Racks in Sample Module

### 37.1.2 Accessories cardboard box

1. Cover cardboard box
2. Left Top Cover
3. Small accessories cardboard box
4. Right Top Cover
5. Front Module
6. Top Cover
7. Panel PC cardboard box
8. Wash Buffer solution 5.5L tank (set of 2)
9. DI Water 1.5L tank
10. Cleaning solution 1.5L tank
11. **LIAISON<sup>®</sup> XS** accessories set
12. Liquid Waste tank
13. PN 10136280 Tank Filling Stand
14. Additional Sample rack
15. Sample rack Front Handle labels
16. Velcro strip for USB stick

### 37.1.3 Cover cardboard box

1. Right Side Cover
2. Left Side Cover

### 37.1.4 Small accessories cardboard box

1. Bulk Fluid Door
2. Solid Waste Bin
3. Cuvette - Tip Loading Module handle
4. Ancillary tray
5. Barcode Label Set for Sample Rack ID

**37. Packing List for internal-tanks configuration**

---

6. Caps for Peristaltic pumps

**37.1.5 Panel PC cardboard box**

1. Panel PC
2. Back plate
3. Cable carrier
4. Display holder
5. Accessories for Panel PC
  - Power cable
  - Ferrite

**37.1.6 LIAISON® XS accessories set**

1. A0055 SAMPLE TUBE TOOL (1 NU)
2. A0107 ANCILLARY ADAPTOR FOR LC (SET OF 2) (1 NC)
3. A0118 SAMPLE RACK ADAPTER - SET 60 (1 NC)
4. A0120 SAMPLE RACK HOLDER LIAISON XL/XS (1 NU)
5. A0128 INTEGRAL RACK HOLDER (1 NU)
6. A0179 EXTERNAL BARCODE SCANNER or A0256 EXTERNAL BARCODE SCANNER (1 NU)
7. A0243 XS CONTROL RACK 8 POSITIONS (QTY 2) (1 NC)
8. A0244 XS LIQUID WASTE BASIN (1 NU)
9. X0054 LIAISON® EASY Waste (QTY 10) (1 NC)
10. A0251 BACKUP USB STICK 16GB

## 38 Packing List for external-tanks configuration

### 38.1 Packing list

Check the scope of delivery for completeness by means of the following packing list.

#### 38.1.1 Instrument cardboard box

1. Instrument with cables (schuko plug 220 Volt)
2. Sample Racks in Sample Module

#### 38.1.2 Accessories cardboard box

1. Cover cardboard box
2. Left Top Cover
3. Small accessories cardboard box
4. Right Top Cover
5. Front Module
6. Top Cover
7. Panel PC cardboard box
8. Wash Buffer solution 10L tank (set of 2)
9. System liquid 10L tank
10. Cleaning solution 10L tank
11. **LIAISON<sup>®</sup> XS** accessories set
12. Liquid Waste tank
13. Additional Sample rack
14. Sample rack Front Handle labels
15. Velcro strip for USB stick

#### 38.1.3 Cover cardboard box

1. Right Side Cover
2. Left Side Cover

#### 38.1.4 Small accessories cardboard box

1. Bulk Fluid Door
2. Solid Waste Bin
3. Cuvette - Tip Loading Module handle
4. Ancillary tray
5. Barcode Label Set for Sample Rack ID
6. Caps for Peristaltic pumps

**38. Packing List for external-tanks configuration**

---

**38.1.5 Panel PC cardboard box**

1. Panel PC
2. Back plate
3. Cable carrier
4. Display holder
5. Accessories for Panel PC
  - Power cable
  - Ferrite

**38.1.6 LIAISON® XS accessories set**

1. A0055 SAMPLE TUBE TOOL (1 NU)
2. A0107 ANCILLARY ADAPTOR FOR LC (SET OF 2) (1 NC)
3. A0118 SAMPLE RACK ADAPTER - SET 60 (1 NC)
4. A0120 SAMPLE RACK HOLDER LIAISON XL/XS (1 NU)
5. A0128 INTEGRAL RACK HOLDER (1 NU)
6. A0179 EXTERNAL BARCODE SCANNER or A0256 EXTERNAL BARCODE SCANNER (1 NU)
7. A0243 XS CONTROL RACK 8 POSITIONS (QTY 2) (1 NC)
8. A0244 XS LIQUID WASTE BASIN (1 NU)
9. X0054 LIAISON® EASY Waste (QTY 10) (1 NC)
10. A0251 BACKUP USB STICK 16GB

# 39 LIS Interface Protocol

## 39.1 Purpose

This chapter is intended to describe the LIS interface for **LIAISON<sup>®</sup> XS** which is for communication to a host computer.

The specification shall provide sufficient detail so that a design and an implementation can be derived.

## 39.2 Related documents

For further information refer to the documents:

- CLSI LIS-1A (prior ASTM E1381). Specification for Low Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems.
- CLSI LIS-2A (prior ASTM E1394). Specification for Transferring Information Between Clinical Instruments and Computer Systems.

## 39.3 Overview

This description is intended to specify the interface for exchanging information between the **LIAISON<sup>®</sup> XS** instrument and a host computer. This includes both settings to be made to adapt the communication to lab specific properties as well as a description of message content and tools for troubleshooting.

## 39.4 General Information and Specification

Communication between **LIAISON<sup>®</sup> XS** and an external host computer consists of receiving test requests from the host and reporting results back to the host. This is accomplished via an RS232 connection (with a dedicated RS232 to USB adapter) or a TCP/IP connection and follows the CLSI LIS-1A (former ASTM1381, low level) and CLSI LIS-2A (former ASTM1394, high level) standards for communication.

The **LIAISON<sup>®</sup> XS** host interface consists of:

- ASTM 1381 low-level transfer protocol used to transmit or receive messages.
- Intermediate files used for temporarily storing data received (local to the **LIAISON<sup>®</sup> XS** application).
- Interpretation of data received from the intermediate files and entering into the **LIAISON<sup>®</sup> XS** database.
- Intermediate files used to temporarily store data to be sent to the host.

**NOTE**

There won't be specs for the lower level protocol, since this is described in sufficient detail in ASTM1381.

---

The general procedure related to the communication of the **LIAISON® XS** with an external host, as well as the specific implementation of the ASTM standards, are described here.

The ASTM 1394 defines how the data to be transmitted is represented as a structured message consisting of several records as described below. These messages are then translated into one or more frames that will actually be transmitted according to what described below and ASTM 1381.

### **39.4.1 Transfers initiated from **LIAISON® XS****

1. **LIAISON® XS** communication sessions with a host computer can be initiated on request of the **LIAISON® XS** operator.
  - An operator can upload all or a selection of test results on request through a **LIAISON® XS** menu selection (Results -> Archived -> Upload).
  - An operator can download all pending test orders from the host (Results -> All or Worklist -> Download Orders button). This is the QueryAll mode, available also when the system is in HostQuery mode.
2. **LIAISON® XS** can generate query messages during sample loading to query test orders for barcoded or manually entered samples tubes (See System Automatically Queries Host for Worklist in Table 29.6-1). This is the HostQuery mode.
3. The **LIAISON® XS** can upload patient results whenever results have become available.

### **39.4.2 LIS initiated transfers**

1. The LIS may send test orders spontaneously (Download mode). This can happen also in HostQuery Mode.
2. The LIS may send queries for results (see Table 39–19 for details).

## 39.5 Limitations

As a consequence of the allowed character set specified for ASTM1381 (it does not allow to transmit non-ASCII characters), for not-ASCII languages like Japanese, Chinese or Greek some information cannot be transmitted (e.g. patient names, demographic information or localized flags or qualitative result labels).

**LIAISON<sup>®</sup> XS** handles patient Sample IDs case-insensitive. Therefore sample IDs that are received from the host are automatically converted to uppercase.

Patient Sample IDs are sent to the host always in uppercase.

Allowed characters for Sample IDs are:

- A-Z (a-z is automatically converted to uppercase);
- 0-9;
- minus (-), dot (.), dollar sign (\$), plus (+), percent sign (%), pound sign (#), ampersand (&), equal (=), space ( ), slash (/);
- SIDs starting with a dollar sign (\$) denote calibrators and are rejected;
- SIDs starting with a pound sign (#) denote controls and are rejected.

Strict Parsing:

The LIS protocol validates the value of fields that are limited by the ASTM1394/LIS2-A standard to dedicated values. Even if the value of the fields are ignored by LXS (see "\$"Data Record Usage"), they must obey the values defined in the standard:

| Field Number | Field Name       | Allowed values |
|--------------|------------------|----------------|
| H12          | Processing ID    | P              |
| P9           | Patient Sex      | empty, M, F, U |
| O6           | Order Priority   | S, R           |
| O12          | Action Code      | empty, C       |
| O26          | Report Type      | A, X, F, Q, C  |
| L3           | Termination Code | N, I           |

Table 39–1: Data record usage

The LIS protocol does not perform a sequence number check.

### 39.5.1 Compatibility with LXL

The LXS LIS settings are fully compatible with LXL, except for the following:

- Serial connection always uses 8 databits (Data Bits no longer configurable)
- Serial connection always uses no parity check (Parity no longer configurable)

New features:

- LXS also supports TCP as connection protocol.
- The location of the patient ID in a ASTM1394 message is now configurable (P3 or P4)

**NOTE**

Ensure that the PID setting is defined before the transmission. A change during the transmission may result in wrong transmission of field P3/P4.

## 39.6 Result transmission configuration

These parameters are configurable in **Definition** -> **Assay** -> **Calculation** and the host software must satisfy their use.

| Parameter                           | Choice                  | Description                                                                      |
|-------------------------------------|-------------------------|----------------------------------------------------------------------------------|
| Send replicates to host             | Yes/No                  | Result Replicates to LIS: all Replicates are sent to LIS                         |
| Result Aspect for host transmission | Dose/<br>Label/<br>Both | Result Aspect for LIS transmission. This can be either "Dose", "Label" or "Both" |

Table 39–2: Parameters to satisfy

## 39.7 Logical Structure of the Message Level Protocol

The blocked stream of data sent between a host computer and the analyzer at a given time is called a message.

Messages consist of a hierarchy of records of various types:

| Level  | Segment Name                    | Identifier (Record TypeID) | Comments                                                |
|--------|---------------------------------|----------------------------|---------------------------------------------------------|
| 0      | Message Header Record           | 'H'                        |                                                         |
| 0      | Message Terminator Record       | 'L'                        |                                                         |
| 1      | Patient Information Record      | 'P'                        |                                                         |
| 1      | Query Record                    | 'Q'                        |                                                         |
| 1      | Scientific Record               | 'S'                        | Not allowed for <b>LIAISON® XS</b>                      |
| 2      | Test Order Record               | 'O'                        |                                                         |
| 3      | Result Record                   | 'R'                        |                                                         |
| common | Comment Record                  | 'C'                        |                                                         |
| 4      | Manufacturer Information Record | 'M'                        | Contains extended traceability information, if enabled. |

Table 39–3: Logical structure of the message level protocol

A record is identified by the first field of a record, the RecordTypeID.



**NOTE**

An unknown RecordTypeID causes an error and aborts the interpretation session. That means that this message is ignored by **LIAISON® XS** and the failed attempt is recorded in the Eventlog. The communication line is considered neutral and a new transmission can be initiated from either side.

Most of the various record types are related to each other in a definite hierarchy:

- A lower level record may never appear without the preceding higher level record.
- Order records must be preceded by a patient record, result records must be preceded by an order record etc.
- A sequence of records at one level is terminated by the appearance of a record of the same or higher level.

**NOTE**

A violation of the record hierarchy will cause an error and aborts the interpretation session. The LIS does not go off line, just that one message is lost. The Error will be logged that the hierarchy rule was ignored.

In case the host is switched off or the communication is temporarily interrupted, it may happen that the analyzer tries unsuccessfully to send results. As soon as the communication is restored, **LIAISON® XS** will try to send the results marked internally as "to be sent". After a successful transmission, **LIAISON® XS** stores the name of the current operator and the date and time in the result object.

If the system is working in Host-query and there is already a test order for a certain sample, when the operator unloads and reloads again the same rack, a new query shall be sent, to check if other new requests are pending for the same SID. If the **LIAISON® XS** receives again the same test (i.e. SID + assay) and the old one is either in Worklist or Ongoing, no new items shall be added; otherwise, if the test is either in Done or Archived or Failed, a new Work List Entry (WLE) shall be added.

## 39.8 Incoming transmission examples

Test orders downloading is used to request test orders from the host that are available for the **LIAISON® XS** instrument. The download session is initiated by the **LIAISON® XS** program only by using the **Query Record**.

### 39.8.1 Test orders

In case the query for test orders is generated manually by the operator through a **LIAISON® XS** menu option the query record contains „ALL“ in the „Starting Range ID“ and „Universal Test ID“ fields:

**LIAISON® XS** to Host: (request for test orders from host) manually

| Structure defined by ASTM 1394<br>(multiple records comprise a single message) | Structure defined by ASTM 1381<br>(each record is sent as one or more frames) |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Message Header Record                                                          | Frame 1                                                                       |
| Query Record                                                                   | :                                                                             |
| Message Terminator Record                                                      | Frame n                                                                       |

Table 39–4: Test orders - LIAISON<sup>®</sup> XS to Host manually

Example:

```
H|^&||| LiaisonXS |||||Host||P|1
Q|1|ALL||ALL|||||||O
L|1|N
```

LIAISON<sup>®</sup> XS to Host: (request for test orders from host) automatically

| Structure defined by ASTM 1394<br>(multiple records comprise a single message)              | Structure defined by ASTM 1381<br>(each record is sent as one or more frames) |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Message Header Record<br>Query Record 1<br>:<br>Query Record m<br>Message Terminator Record | Frame 1<br>:<br>Frame n                                                       |

Table 39–5: Test orders - LIAISON<sup>®</sup> XS to Host automatically

Example:

```
H|^&||| LiaisonXS |||||Host ||P|1
Q|1|SAMPLE01||ALL|||||||O
Q|2|SAMPLE02||ALL|||||||O
Q|3|BARCODE0815||ALL|||||||O
Q|4|12345||ALL|||||||O
L|1|N
```

If **CompressedOrderQuery** is enabled, the query for test orders is generated using repeat delimiters:

```
H|^&|||LiaisonXS|||||Host||P|1
Q|1|SAMPLE01\SAMPLE02\BARCODE0815\12345||ALL|||||||O
L|1|N
```

Host to LIAISON<sup>®</sup> XS: (response from host includes patient demographics, patient ID, sample ID, and test orders according to the following record hierarchy)

The response to requests for test orders is expected to be received within **<Timeout>** seconds after the request has been sent. **<Timeout>** is to be specified in the System -> Connectivity -> LIS Details.

| Structure defined by ASTM 1394<br>(multiple records comprise a single message)                                                                                                                                                | Structure defined by ASTM 1381<br>(each record is sent as one or more frames) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Message Header Record<br>Patient Information Record 1<br>Test Order Record 1<br>:<br>Test Order Record n<br>:<br>Patient Information Record n<br>Test Order Record 1<br>:<br>Test Order Record n<br>Message Terminator Record | Frame 1<br>:<br>Frame n                                                       |

Table 39–6: Test orders - Host to **LIAISON<sup>®</sup> XS**

In case there are no test orders available, the LIS should respond with an empty message containing header and terminator records only. The terminator record should contain an 'I' (no information available) flag in the Termination Code Field.

Example:

```
H|^&||||LIS||||LXS||P|1
L|1|I
```

It is also possible for the host to reply with a message with „empty“ P and O records:

```
H|^&||||LIS||||LXS||P|1
P|1
O|1
L|1|I
```

The comment record is used to return “bad test” etc messages back to the host after download session.

After the **LIAISON<sup>®</sup> XS** receives all test orders from the host, the records are interpreted. Valid test orders are entered into the work-list database, while invalid test orders are not. After the interpretation segment of the download session, the **LIAISON<sup>®</sup> XS** sends a message to the host containing information about all rejected or invalidated test orders as follows:

**LIAISON<sup>®</sup> XS** to Host: (information about rejected test orders)

| Structure defined by ASTM 1394<br>(multiple records comprise a single message)                                                                                                                                                                                                                                                                                | Structure defined by ASTM 1381<br>(each record is sent as one or more frames) |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Message Header Record<br>Patient Information Record 1<br>Test Order Record 1<br>Result Record 1<br>Comment 1<br>:<br>Test Order Record n<br>Result Record 1<br>Comment 1<br>:<br>Patient Information Record n<br>Test Order Record 1<br>Result Record 1<br>Comment 1<br>:<br>Test Order Record n<br>Result Record 1<br>Comment 1<br>Message Terminator Record | Frame 1<br>:<br>Frame n                                                       |

Table 39–7: Test orders - Host to LIAISON<sup>®</sup> XS

Example:

```

H|^&||LXS||||LaborEDV||1|19941115202738
P|1|PatID01||Meier^Anna|19741001|F||||MARTINEZ
O|1|SAMPLE01|^^^AFP^1:10|19980506||||||S||||||X
R|1|^^^AFP||||X||||Liaison
C|1||BAD_DILUTION|
L|1|N

```

Reasons for rejection of test orders are:

| Comment string | Description                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| BAD_TEST       | No match found for the testcode specified in TestOrderRecord, Field 5, Component 4.                                                             |
| QC_SAMPLE_ID   | If the SampleID in O3 matches the ID of a control (with or without leading #), the order is rejected. QC material can't be ordered by the host. |

| Comment string   | Description                                                                                                                                                                                                                                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| BAD_SAMPLE_ID    | SampleID field was either empty, too short or too long or it contained invalid characters. SampleID field was either empty or the SampleID had a leading '\$' character. Leading '\$' characters designate calibrator materials. Calibrators can't be ordered by the host. |
| BAD_DILUTION     | The dilution specified in TestOrderRecord, Field 5, Component 5 can not be applied to the test specified in TestOrderRecord, Field 5, Component 4.                                                                                                                         |
| WRONG_INSTRUMENT | The InstrumentID which is sent in the ReceiverID field of the Message Header Record doesn't match the InstrumentID in the System -> LIS/LAS -> LIS Dialogue.                                                                                                               |
| OTHER_ERROR      | Other error                                                                                                                                                                                                                                                                |

Table 39–8: Rejection of test orders

**Sample replicates**

The Host can require the desired number of replicates for any of the requested tests.

The value shall be inserted in field O5^6

If this field is empty, default value of the assay is ordered.

**NOTE**

It is not possible to ask for Combi Fathers with sons which have a different number of Replicates.

It is not possible to ask for less replicates than the minimum defined for the involved assay.

**Combi results**

If the Host requires only the combi father test order, the LIAISON<sup>®</sup> XS sends back only the combi father result to the Host.

In order to receive both the father and the sons results the Host shall request the complete family i.e. both the father and the involved sons within the same transmission (H-L).

**NOTE**

It is not possible to ask for Combi Fathers that is never run.

**39.8.2 Test Results**

Only the calculated result (concentration calculated from mean of RLU values) is transferred per test.

If the option "Send Replicates to host" is disabled only the mean result is transmitted.

If the option "Send Replicates to host" is enabled in assay definition, the mean result and all replicate results of a calculated result are transmitted, each in a separate result record.

LIAISON<sup>®</sup> XS to Host: (transmit sample information with corresponding tests and results)

| Structure defined by ASTM 1394<br>(multiple records comprise a single message)                                                                                                                                                                                                                                                                                                                                                                                                                                    | Structure defined by ASTM 1381<br>(each record is sent as one or more frames) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Message Header Record<br>Patient Information Record 1<br>Test Order Record 1<br>Result Record 1<br>Comment 1<br>:<br>Result Record n<br>Comment 1<br>:<br>Test Order Record n<br>Result Record 1<br>Comment 1<br>:<br>Result Record n<br>Comment 1<br>:<br>Patient Information Record n<br>Test Order Record 1<br>Result Record 1<br>Comment 1<br>:<br>Result Record n<br>Comment 1<br>:<br>Test Order Record n<br>Result Record 1<br>Comment 1<br>:<br>Result Record n<br>Comment 1<br>Message Terminator Record | Frame 1<br>:<br>Frame n                                                       |

Table 39–9: Test results

Example:

```

H|^&|||LiaisonXS|||LaborEDV|||1|19941115202738
P|1|PatID01|||Meier^Anna||19741001|F|||MARTINEZ
O|1|SAMPLE01||^AFP||19980506|||S|||F
R|1|^AFP^DOSE|13.1|IU/ml||H|F|||19980506123145|Liaison
C|1||RECALCULATED|I
L|1|N

```

A list of comments, which can detail the result in the preceding result record is given in [Result Comment Record](#).

Result transmission of mean and single replicates: as one 'R' record for the mean plus two 'R' records, one for each replicate

```
H|\^&|||LiaisonXS||||LaborEDV|||1|19941115202738
P|1|PatID01|||Meier^Anna||19741001|F||||MARTINEZ
O|1|SAMPLE01||^AFP||19980506|||||S|||||F
R|1|^AFP^DOSE|13.1|IU/ml||H||F|||19980506123145|Liaison
R|2|^AFP^REPL|13.0|IU/ml||H||S|||19980506123145|Liaison
R|3|^AFP^REPL|13.2|IU/ml||H||S|||19980506123145|Liaison
C|1||RECALCULATED||
L|1|N
```

In case more than one flag occurred, they are transmitted using repeat delimiters:

```
H|\^&|||LiaisonXS||||LIS||P|1
P|1|PatID01|||Meier^Anna||19741001|F||||MARTINEZ
O|1|SAMPLE01||^AFP||19980506|||||S|||||F
R|1|^AFP^DOSE|13.1|IU/ml||H||F|||19980506123145|Liaison
C|1||RECALCULATED\BAD_QC\IN_REFLEX_ZONE1|
L|1|N
```

In case results cannot be reported, the result is flagged („X“) in the 'ReportType' field (Field 26) of the [TestOrderRecord](#) and in the 'Status' field (Field 9) of the [ResultRecord](#). The reason is then appended in a [CommentRecord](#).

Example for transmission of failed results:

```
H|\^&|||LiaisonXS||||LaborEDV|||1|19941115202738
P|1|PatID01|||Meier^Anna||19741001|F||||MARTINEZ
O|1|SAMPLE01||^AFP||19980506|||||S|||||X
R|1|^AFP||||X|||19980506123145|Liaison
C|2||JOB_FAILED||
L|1|N
```

Queries for test result, which are still in process are transmitted with an „I“ flag („in instrument pending“) in the 'ReportType' field (Field 26) of the [TestOrderRecord](#) and in the 'Status' field (Field 9) of the [ResultRecord](#).

Example for transmission of pending results:

```
H|\^&|||LiaisonXS||||LaborEDV|||1|19941115202738
P|1|PatID01|||Meier^Anna||19741001|F||||MARTINEZ
O|1|SAMPLE01||^AFP||19980506|||||S|||||I
R|1|^AFP||||I|||Liaison
L|1|N
```

Corrections of test results are transmitted with a „C“ flag („correction of previously transformed result“) in the 'ReportType' field (Field 26) of the [TestOrderRecord](#) and in the 'Status' field (Field 9) of the [ResultRecord](#). This is the case if a result was changed using the "Change Result" function of the Results Journal.

Example for transmission of corrected results:

```
H|\^&|||LiaisonXS||||LaborEDV|||1|19941115202738
P|1|PatID01|||Meier^Anna||19741001|F||||MARTINEZ
```

```
O|1|SAMPLE01||^AFP||19980506|||||S|||||C
R|1|^AFP^DOSE|13.3|IU/ml||H||C|||19980506123145|Liaison
L|1|N
```

Results below (above) the calculation limits of the assay are transmitted with a numerical value of lower (upper) limit of the assay range and a '<' (>') flag in the 'Result Abnormal Flags' field (Field 7) of the [ResultRecord](#).

Example for transmission of off scale results ("< / > calculation limits):

```
H|\^&||Liaison||||LaborEDV|||1|19941115202738
P|1|PatID01|||Meier^Anna||19741001|F||||MARTINEZ
O|1|SAMPLE01||^AFP||19980506|||||S|||||F
R|1|^AFP^DOSE|< 0.20|IU/ml||<||F|||19980506123145
R|2|^AFP^INTR|neg|||<||F|||19980506123145
L|1|N
```

## 39.9 Data Record Usage

(refer to ASTM 1394, particularly sections 6 through 13)

Each record sent by the **LIAISON® XS** may not contain all fields possible for the record. An 'N' in the "Required" column indicates it is optional.

### 39.9.1 Message Header Record

The [Message Header Record](#) is always the first record in an application message. It contains general sender and receiver information. Its hierarchical level is 0.

The following tables describe the values each field can contain: they refer respectively to an "Outgoing" communication (from **LIAISON® XS** to host) or an "Incoming" communication (from host to **LIAISON® XS**).

| Field Number | Field Name               | Required | Value                                                                          |
|--------------|--------------------------|----------|--------------------------------------------------------------------------------|
| H1           | Record Type ID           | Y        | 'H'                                                                            |
| H2           | Delimiter Definition     | Y        | Default is  \^&                                                                |
| H5           | Sender Name              | N        | 1..10 characters InstrumentID as defined in the System -> LIS -> LIS dialogue. |
| H10          | Receiver ID              | N        | 1..10 characters HostID as defined in the System -> LIS -> LIS dialogue.       |
| H12          | Processing ID            | Y        | Always 1 character; always P                                                   |
| H13          | Version No.              | Y        | '1'                                                                            |
| H14          | Date and Time of Message | N        | Always empty, not sent.                                                        |

Table 39–10: Message Header Record - Outgoing



| Field Number | Field Name               | Required | Value                                                                                                                                                                                                     |
|--------------|--------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| H1           | Record Type ID           | Y        | 'H' or 'h'                                                                                                                                                                                                |
| H2           | Delimiter Definition     | Y        | Recommended using  ^&, but can use any 4 of the ASCII characters allowed                                                                                                                                  |
| H5           | Sender Name              | N        | Not used for incoming message processing.                                                                                                                                                                 |
| H10          | Receiver ID              | N        | 1..10 Characters Used to verify that the transmission is intended for the <b>LIAISON<sup>®</sup> XS</b> .<br>If sent, it must match the InstrumentID on System -> LIS -> LIS dialogue, otherwise ignored. |
| H12          | Processing ID            | N        | Not processed                                                                                                                                                                                             |
| H13          | Version No.              | N        | Not processed.                                                                                                                                                                                            |
| H14          | Date and Time of Message | N        | Not used for incoming message processing.                                                                                                                                                                 |

Table 39–11: Message Header Record - Incoming

### 39.9.2 Patient Information Record

The Patient Information Record is used to supply patient demographic information for order and result records that follow it. Its hierarchical level is 1.

The following tables describe the values each field can contain: they refer respectively to an “Outgoing” communication (from **LIAISON<sup>®</sup> XS** to host) or an “Incoming” communication (from host to **LIAISON<sup>®</sup> XS**).

| Field Number | Field Name      | Required | Value                                                                                                                                                                                                                                                                                                        |
|--------------|-----------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1           | Record Type ID  | Y        | 'P'                                                                                                                                                                                                                                                                                                          |
| P2           | Sequence Number | Y        | 1-5 characters. Formatted as a left justified number with a range from 1 through 65535. Equal to the numeric identifier of a patient record within the current outgoing application message. Used for validating the integrity of the application message by ensuring that the message contains all records. |

## 39. LIS Interface Protocol

| Field Number | Field Name                     | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------|--------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P3           | Practice Assigned Patient ID   | N        | 0..11 characters Set to the Patient ID (PID) field of an outgoing <b>LIAISON<sup>®</sup> XS</b> result entry.                                                                                                                                                                                                                                                                                                      |
| P4           | Laboratory Assigned Patient ID | N        | Sends the patient ID in the same field back, as it was received from Host (also when P3 was used, then LXS uses same field). If the host does not send any patient ID, the data can be entered in the LXS software (see patient Dialog)                                                                                                                                                                            |
| P6           | Patient Name                   | N        | 0-2 components. Total length of the field is 0-52 characters. Set to the patient Name field of the result entry. Each name component is separated by the Component Delimiter character (^), for example, Smith^John. Trailing empty name components and trailing Component Delimiter characters are not included. For example, Patient Name is set to Smith for a patient worklist entry with the last name Smith. |
| P6^1         | Patient Last Name              | N        | PatientName if available                                                                                                                                                                                                                                                                                                                                                                                           |
| P6^2         | Patient First Name             | N        | PatientFirstName if available                                                                                                                                                                                                                                                                                                                                                                                      |
| P6^3         | PatientInitials                | N        | PatientInitials if available                                                                                                                                                                                                                                                                                                                                                                                       |
| P8           | Birthdate                      | N        | 0 or 8 characters. Set to ANSI X3.30 format of patient date of birth (DOB) in result entry, for example, 19721005 for October 5, 1972. If the patient DOB field is not specified in the worklist or result entry, this field is empty.                                                                                                                                                                             |
| P9           | Patient Sex                    | N        | 1 character. Value set by the Patient Sex field in result entry. If sex is male, field is set to M. If sex is female, field is set to F. If the sex is unknown, the field is set to U. If the sample type is a control, field is blank.                                                                                                                                                                            |
| P14          | Attending Physician ID         | N        | 0-12 characters. Set to the Physician field of result entry.                                                                                                                                                                                                                                                                                                                                                       |

Table 39–12: Patient Information Record - Outgoing

| Field Number | Field Name                     | Required | Value                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|--------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P1           | Record Type ID                 | Y        | 'P' or 'p'                                                                                                                                                                                                                                                                                                                                                                    |
| P2           | Sequence Number                | Y        | 1 to any number of characters, with a default value of 0, which is always an invalid Sequence Number. Formatted as a string of ASCII numeric characters. Equal to the numeric identifier of a patient record within the current incoming application message. If the current Sequence Number does not equal the Sequence Number of the expected record, an error is reported. |
| P3           | Practice Assigned Patient ID   | N        | 0-11 characters. Written to the PID field of imported worklist entries. An empty field causes no change to the database field. A value of two double quotes ("" ) deletes the database field. This field is used only for user information. The <b>LIAISON<sup>®</sup> XS</b> system worklist allows the same PID field to exist in multiple worklist entries.                |
| P4           | Laboratory Assigned Patient ID | N        | Sends the patient ID in the same field back , as it was received from Host (also when P3 was used, then LXS uses same field). If the host does not send any patient ID, the data can be entered in the LXS software (see patient Dialog)                                                                                                                                      |
| P6           | Patient Name                   | N        | 0-3 components. (Components 3 and up are ignored.) Written to the patient Name field of the worklist entry. Total length of the field is 0-52 characters. An empty field causes no change to the database field. A value of two double quotes ("" ) causes the database field to be deleted.                                                                                  |
| P6^1         | Patient Last Name              | N        | 0..26 characters Becomes our PatientName                                                                                                                                                                                                                                                                                                                                      |
| P6^2         | Patient First Name             | N        | 0..26 characters Becomes our PatientFirstName                                                                                                                                                                                                                                                                                                                                 |

| Field Number | Field Name             | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P6^3         | PatientInitials        | N        | PatientInitials if available                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| P8           | Birthdate              | N        | 0 or 8 characters representing ANSI X3.30 date format. Written to date of birth (DOB) field in the worklist entry. An empty field causes no change to the database field. A value of two double quotes ("" ) deletes the database field. Invalid dates are treated as an empty field. This field is used for establishing normal ranges for the sample.                                                                                                                                 |
| P9           | Patient Sex            | N        | 0 or 1 character. Written to the Patient Sex field in worklist entry. If Patient Sex value is M, database field is set to male. If Patient Sex value is F, database field is set to female. If Patient Sex value is U, database field is set to unknown. An empty field causes no change to the database field. A value of two double quotes ("" ) causes the database field to be set to unknown sex. An invalid Patient Sex value causes the database field to be set to unknown sex. |
| P14          | Attending Physician ID | N        | 0-12 characters. Written to the Physician field of imported worklist entry. An empty field causes no change to the database field. A value of two double quotes ("" ) deletes the database field. This field is used only for user information.                                                                                                                                                                                                                                         |

Table 39–13: Patient Information Record - Incoming

### 39.9.3 Test Order Record

The Test **Order Record** is used to supply information on a specific sample's test requests. This information is necessary either for ordering tests on a specific sample or for reporting results for tests on a specific sample. An order record is associated with the previous patient record. Its hierarchical level is 2.

The following tables describe the allowable values for each field: they refer respectively to an "Outgoing" communication (from **LIAISON® XS** to host) or an "Incoming" communication (from host to **LIAISON® XS**).

| Field Number | Field Name             | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------|------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O1           | Record Type ID         | Y        | Always 'O'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| O2           | Sequence Number        | Y        | 1 to any number of characters; usually 1. Used for validating the integrity of the application message by ensuring that the message contains all records.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| O3           | Specimen ID            | Y        | 3-22 characters. Set to Sample ID (SID) field of result entry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| O5           | Universal Test ID      | Y        | List of tests, where each test is described by a multi-component field, separated by a Component Delimiter character (^). Each group of components is separated by a Repeat Delimiter character (\). If the current order record is for submitting a worklist request for a specific sample, this field describes the list of requested tests. If the current order record is for reporting single or multiple result values, this field describes the list of tests whose result values are reported in subsequent result records. The list of tests can contain from one up to the maximum number of tests the <b>LIAISON® XS</b> system can support including combi tests. Refer to individual components (field components O5^1 through O5^5) for data format information. |
| O5^1         | Universal Test ID      | N        | 0 characters; always empty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| O5^2         | Universal Test ID Name | N        | 0 characters; always empty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| O5^3         | Universal Test ID Type | N        | 0 characters; always empty.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

| Field Number | Field Name                       | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------|----------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O5^4         | Manufacturers Code               | Y        | 1-8 case-sensitive characters. The defined LIS alias name is used for the test scheduled for the sample.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| O5^5         | Dilution Ratio                   | N        | The dilution ratio, if the result was obtained from a diluted request: 2, 5, 10, 20, 50, 100, 200, 500, 1000, or 2500                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| O5^6         | Replicates                       | N        | Always empty                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| O6           | Priority                         | Y        | 1 character. Value determined by the stat flag in the result entry. If the stat flag is Yes, the Priority field is set to S (Stat). If the stat flag is No, the Priority field is set to R (Routine).                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| O7           | Requested/ Ordered Date and Time | N        | In YYYYMMDDHHMMSS format, will be assigned to the RequestDateTime property of the worklist entry.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| O12          | Action Code                      | N        | 0 or 1 character. If the result entry is for a control sample, the Action Code field is set to Q (QC Test Specimen). If the result entry is for a patient sample, the Action Code field is empty                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| O26          | Report Types                     | Y        | Zero or more repetitions of 1 character. <ul style="list-style-type: none"> <li>• If the outgoing message contains rejected incoming worklist data, the Report Type field contains X (Work Cannot Be Done).</li> <li>• If the outgoing message contains result data, the Report Type field contains F (Final Results).</li> <li>• If the outgoing message contains data in response to a query, the Report Type field contains Q (Response to a Query).</li> <li>• Corrections of test results, are transmitted with an „C“ flag. This is the case if a result was changed using the "Change Result" function of the Results Journal.</li> </ul> |

Table 39–14: Test Order Record - Outgoing

| Field Number | Field Name        | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------|-------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O1           | Record Type ID    | Y        | Can be 'O' or 'o'                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| O2           | Sequence Number   | Y        | 1 to any number of characters, with a default value of 0, which is always an invalid Sequence Number. Formatted as a string of ASCII numeric characters. Equal to the numeric identifier of an order record since the last patient record, which is considered the current patient record. If the current Sequence Number does not equal the Sequence Number of the expected record, an error is reported.                                                                                                                                                                                                                                                                                                                                                               |
| O3           | Specimen ID       | Y        | Written to the Sample ID (SID) field of imported worklist entry. Lowercase characters will be automatically converted into uppercase, before further processing.<br>3-22 characters.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| O5           | Universal Test ID | Y        | List of tests, where each test is described by a multi-component field, separated by a Component Delimiter character (^). Each group of components is separated by a Repeat Delimiter character (\). If a test is unknown to the <b>LIAISON<sup>®</sup> XS</b> system, the test order is rejected to the LIS and processing continues on the list of tests. The list of tests can contain any number of names. If a single Universal Test ID matches a combi test, all tests associated with the combi are automatically added to the list of tests. Refer to field O12, Action Code, for processing of updates and cancellations of the tests specified in this field. Refer to individual components (field components O5^1 through O5^5) for data format information. |

## 39. LIS Interface Protocol

| Field Number | Field Name                       | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|----------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O5^1         | Universal Test ID                | N        | 0 to any number of characters. This component is always ignored.                                                                                                                                                                                                                                                                                                                                        |
| O5^2         | Universal Test ID Name           | N        | 0 to any number of characters. This component is always ignored.                                                                                                                                                                                                                                                                                                                                        |
| O5^3         | Universal Test ID Type           | N        | 0 to any number of characters. This component is always.                                                                                                                                                                                                                                                                                                                                                |
| O5^4         | Manufacturers Code               | Y        | 1-8 case-sensitive characters. If a value exists, this value is used to search the LIAISON <sup>®</sup> XL test definition database. The LIAISON <sup>®</sup> XS system searches the test definition database for the matching LIS alias (LIS Name of the assay). If a match is found, that test is selected for the sample, otherwise the test order is rejected to the LIS (unknown test or profile). |
| O5^5         | Dilution Ratio                   | N        | Dilution ratio must be defined for the test.<br><br>(e.g. 2, 5, 10, 20, 50, 100, 200, 500, 1000, or 2500) If that dilution ratio is not defined then the order is rejected to the LIS.<br><br>The LIS must not use repeat delimiters in this field, multiple dilutions and neat work orders must be sent in individual O records.                                                                       |
| O5^6         | Replicates                       | N        | Number of replicates of the test specified in O5^4. If this field is empty, default value of the assay is ordered.<br><br><b>Note:</b> It is not possible to ask for CombiFathers with sons which have a different number of Replicates.                                                                                                                                                                |
| O6           | Priority                         | Y        | 1 character. Written to the stat flag in the worklist entry. If the Priority is S, the stat flag is set to Yes (STAT), otherwise to No (No STAT).                                                                                                                                                                                                                                                       |
| O7           | Requested/ Ordered Date and Time | N        | The RequestDateTime property of the worklist entry, in YYYYMMDDHHMMSS format.                                                                                                                                                                                                                                                                                                                           |



| Field Number | Field Name   | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------|--------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| O12          | Action Code  | N        | <p>1 character.</p> <p>If the order record is for an incoming worklist entry (refer to field O26, Report Type), this field is processed as follows:</p> <ul style="list-style-type: none"> <li>• If the Action Code is C (Cancel Request) and the worklist entry is in „to do“ or „placed“ status, the tests specified in Universal Test ID field O5 are canceled for the sample specified in field O3</li> <li>• If the Action Code does not specify Cancel Request, the Action Code is interpreted as a New Request or an Add Request, depending upon the existence of an entry in the <b>LIAISON<sup>®</sup> XS</b> system worklist for the Sample ID from field O3.</li> </ul> <p>Empty and invalid Action Code fields are interpreted as a New Request or as an Add Request, depending upon the existence of an entry in the <b>LIAISON<sup>®</sup> XS</b> system worklist for the Sample ID from field O3.</p> |
| O26          | Report Types | N        | <p>Zero or more repetitions of 1 character.</p> <ul style="list-style-type: none"> <li>• If the test order is not canceled (See Action Code (Field O12)) then the Report Type is assumed to be O (Order)</li> <li>• If the test order is canceled (See Action Code (Field O12)) then the Report Type is ignored.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

Table 39–15: Test Order Record – Incoming

### 39.9.4 Result Record

The **Result Record** supplies the final test result information for a single test of a specific sample. Results of multiple tests are specified in separate result records. The **LIAISON<sup>®</sup> XS** system ignores incoming result records from a remote system. The hierarchical level is 3. The result record is associated with the previous order record.

The following table describes the allowable values for each field: only the “Outgoing” communication (from **LIAISON<sup>®</sup> XS** to host) is in this case available.

| Field Number | Field Name             | Required | Value                                                                                                                                                                                                                                                                                                         |
|--------------|------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R1           | Record Type ID         | Y        | Always 'R'                                                                                                                                                                                                                                                                                                    |
| R2           | Sequence Number        | Y        | 1-5 characters, formatted as a left justified number, with a range from 1 through 65535. Equal to the numeric identifier of a result record since the last order record. Used to verify the integrity of an application message by ensuring that the message contains all records.                            |
| R3           | Universal Test ID      | Y        | A single test name described by 6 components, where each component is separated by a component delimiter character (^). The Universal Test ID specifies the name of the test associated with the result. Refer to the individual components (field components R3^1 through R3^5) for data format information. |
| R3^1         | Universal Test ID      | N        | 0 characters; always empty.                                                                                                                                                                                                                                                                                   |
| R3^2         | Universal Test ID Name | N        | 0 characters; always empty.                                                                                                                                                                                                                                                                                   |
| R3^3         | Universal Test ID type | N        | 0 characters; always empty.                                                                                                                                                                                                                                                                                   |
| R3^4         | Manufacturers Code     | Y        | 1-8 characters. The LIS alias of the test.                                                                                                                                                                                                                                                                    |
| R3^5         | Dilution Ratio         | Y        | The dilution ratio, if the result was obtained from a diluted request, otherwise empty (e.g. 2, 5, 10, 20, 50, 100, 200, 500, 1000, or 2500)                                                                                                                                                                  |

| Field Number | Field Name                | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------|---------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R3^6         | Result Aspect             | Y        | <p>0-4 characters. One of 3 possible text strings used to identify the specific aspect of the result being transmitted. The following are possible values and their interpretations:</p> <p><b>DOSE</b> - Result concentration mean value (in User Defined units, either from a regular test or a quantitative combi result) or the Result index value, from a qualitative test.</p> <p><b>REPL</b> - Replicate Result concentration value (in User Defined units) or the Replicate Result index value, from a qualitative test.</p> <p><b>INTR</b> - (Interpretation), qualitative result label or logical combi result derived from the mean result.</p> <p>The INTR aspect may be subject of localization and cannot be sent in non-ASCII languages.</p> |
| R4           | Data or Measurement Value | N        | <p>1-15 characters.</p> <p>If the Result Aspect in R3^6 is DOSE: the concentration value plus alphanumeric such as "&lt; value", or "&gt; value" . In case there is no assay range defined, the system shall send the result in decimal format, never in scientific notation.</p> <p>If the Result Aspect in R3^6 is INTR: the qualitative result label or the logical combi result.</p> <p>The decimal separator depends on the system setting.</p> <p>Example:<br/> English: 1.2.3 (dot, point)<br/> German (1,2,3) (comma)</p>                                                                                                                                                                                                                           |

39. LIS Interface Protocol

---

| Field Number | Field Name | Required | Value                                                                                                                                                                                                                                       |
|--------------|------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R5           | Units      | N        | 0-6 characters. If the Data Value field (R4) represents a quantitative result, this field is set to the Units field of the definition of the associated test.<br>If the result is not numerical but it is a label, the unit field is empty. |

| Field Number | Field Name            | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|-----------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R7           | Result Abnormal Flags | N        | <p>Zero or more repetitions of 1 character.</p> <p>The repetitions represent some of the <b>LIAISON<sup>®</sup> XS</b> system result flags and are one of the following:</p> <ul style="list-style-type: none"> <li>• If the result is flagged as low, this field contains L (Database flag LOW, under a defined normal range).</li> <li>• If the result is flagged as high, this field contains H (Database flag HIGH, over a defined normal range).</li> <li>• If the result is flagged as under assay range, this field contains &lt; (Database flag UNDER_RANGE, Under the conc. calc limit).</li> <li>• If the result is flagged as over assay range, this field contains &gt; (Database flag OVER_RANGE, over the conc. calc limit).</li> <li>• If the result is both within normal and assay range, i.e. no flags are given, , this field contains "N"</li> </ul> <p>A single abnormal flag represented by 1 character</p> <p>L: under normal range</p> <p>H: above normal range</p> <p>&lt;: under range/calc limit</p> <p>&gt;: above range/calc limit</p> <p>N: result within range and normal range</p> <p>Note: Refer to Result Comment Record (chapter 39.9.5), for other <b>LIAISON<sup>®</sup> XS</b> system result flags.</p> |

| Field Number | Field Name               | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------------|--------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| R9           | Result Status            | Y        | <p>One or more repetitions of 1 character.</p> <p>The Result Status field has a value of F (Final). Other values can occur, separated by the repeat delimiter (\):</p> <ul style="list-style-type: none"><li>• If the result is sent in response to a query, the field has a value of Q.</li><li>• If a previously sent result was changed by the system operator and not sent after the change, the result status field has a value of C to indicate correction of a previously transmitted result.</li><li>• If a result was sent previously, the result status field has a value of R. Values of R and C do not occur in combination.</li><li>• If the outgoing message contains rejected incoming worklist data, the Report Type field contains X (Work Cannot Be Done).</li><li>• If this is a replicate result, the result status field has a value of S (partial result).</li></ul> |
| R13          | Date/Time Test Completed | N        | <p>14 characters. Set to ANSI X3.43 format of date and time values. in YYYYMMDDHHMMSS form. Hours are in 24-hour form.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

Table 39–16: Result Record - Outgoing

### 39.9.5 Result Comment Record

The **LIAISON<sup>®</sup> XS** is used to communicate additional flags beyond those supported by the 'Result Abnormal Flags' field of the **Result Record**. The record is optional, i.e it is not sent when no flags are available for the preceeding result. It is used if a flag other than high (H), low (L), over assay range (>), or under assay range (<) is associated with the result. Each flag generates an additional result comment record.

The **LIAISON<sup>®</sup> XS** system ignores result data from a remote system. The system ignores incoming result comment records.

Its hierarchical level is 4. The result comment record always follows a result record.

The following table describes the allowable values for each field: only the "Outgoing" communication (from **LIAISON<sup>®</sup> XS** to host) is in this case available.

The order of flags follows the same priority as in the **LIAISON<sup>®</sup> XS** Main UI.

| Field Number | Field Name      | Required | Value                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------|-----------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1           | Record Type ID  | Y        | 1 character; always C.                                                                                                                                                                                                                                                                                                                                                                |
| C2           | Sequence Number | Y        | 1-5 characters, formatted as a left justified number, with a range from 1 through 65535. Equal to the numeric identifier of a result comment record within the current outgoing application message. Used to verify the integrity of an application message by ensuring that the message contains all records.                                                                        |
| C3           | Comment Source  | Y        | 1 character, always set to 'I'.                                                                                                                                                                                                                                                                                                                                                       |
| C4           | Comment         | N        | 0-60 characters. Result flags map to the values in the following table. If the flags do not exist for the result record, result comment records are not generated for transmission. This can also be the reason for work order rejection. Depending on the system parameter „SendCompressedComments“, this field can contain repeat delimiters and multiple comments in one C record. |
| C5           | Comment Type    | Y        | 1 character; always 'I'.                                                                                                                                                                                                                                                                                                                                                              |

Table 39–17: Result Comment Record – Outgoing

The following table shows the correspondence of text transmitted in the comment code field of the **Result Comment Record** to the result flags in the result database. Two different sets of strings are supported, according to a system parameter, either a two character abbreviation of a flag or a full size string.

| Database Flag                      | Commend Code    | Comment Code | Explanation                                                                                                                                                                                                             |
|------------------------------------|-----------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sample not present                 | JOB_FAILED      | S            | AM aborted job because of missing Sample (NoRack, Offline).                                                                                                                                                             |
| Starter Temperature out of range   | JOB_FAILED      | S            | Job aborted because Starter Reagent Temperature (SRT) was out-of-range                                                                                                                                                  |
| Starter depleted                   | JOB_FAILED      | S            |                                                                                                                                                                                                                         |
| Starter dispense failure           | JOB_FAILED      | S            |                                                                                                                                                                                                                         |
| Sample Integrity Error             | JOB_FAILED      | S            | AM aborted job because of Sample Integrity Error                                                                                                                                                                        |
| Sample depleted                    | JOB_FAILED      | S            |                                                                                                                                                                                                                         |
| Sample clotted                     | JOB_FAILED      | S            | AM aborted job due to clot detection                                                                                                                                                                                    |
| Ancillary not present              | JOB_FAILED      | R            | AM aborted job because of missing Ancillary (no ancillary, offline).                                                                                                                                                    |
| Integral not present               | JOB_FAILED      | R            | AM aborted job because of missing Integral (no integral, offline).                                                                                                                                                      |
| Reagent Integrity Error            | JOB_FAILED      | R            | AM aborted job because of Reagent Integrity Error                                                                                                                                                                       |
| Reagent depleted                   | JOB_FAILED      | R            |                                                                                                                                                                                                                         |
| Overdiluted                        | DILUTION_ERROR  | D            | The sample probe was overdiluted. (Only for Samples with Sample Specific Dilution)                                                                                                                                      |
| Reagent OBS Expired                | none            | E            | Any used Reagent (Integral, Ancillary) was expired (Onboard-Exiry)                                                                                                                                                      |
| No Calibration                     | BAD_CALIBRATION | C            | No usable calibration found                                                                                                                                                                                             |
| Control Expired                    | none            | QE           | The Control Material was expired when measured.                                                                                                                                                                         |
| Controls out of Manufacturer Range | BAD_QC          | Q            | <ul style="list-style-type: none"> <li>not generated if the range is empty, for "precision" controls</li> <li>based on last performance of any control name defined for that assay</li> <li>only for samples</li> </ul> |



| Database Flag               | Commend Code      | Comment Code | Explanation                                                                                                                                                                                                                          |
|-----------------------------|-------------------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QC Below Manufacturer Range | none              | QL           | this is for controls only <ul style="list-style-type: none"> <li>in case the control is below assay range, this flag is generated as well</li> <li>not generated if the control range is empty, for "precision" controls</li> </ul>  |
| QC Above Manufacturer Range | none              | QH           | this is for controls only <ul style="list-style-type: none"> <li>in case the control is above assay range, this flag is generated as well</li> <li>not generated if the control range is empty, for "precision" controls</li> </ul>  |
| QC Below User Range         | none              | QL           | this is for controls only <ul style="list-style-type: none"> <li>in case the control is below assay range, this flag is generated as well</li> <li>not generated if the control range is empty, for "precision" controls</li> </ul>  |
| QC Above User Range         | none              | QH           | this is for controls only <ul style="list-style-type: none"> <li>in case the control is above assay range, this flag is generated as well</li> <li>not generated if the control range is empty, for "precision" controls.</li> </ul> |
| RLU modified                | MODIFIED_MANUALLY | MM           | The RLU was manually modified.                                                                                                                                                                                                       |
| Recalculated                | RECALCULATED      | RM           | The result was recalculated.                                                                                                                                                                                                         |
| CausedRerun                 | IN_REFLEX_ZONE 1  | &            | This Result caused a rerun. (Sample Only).                                                                                                                                                                                           |
| Son CausedRerun             | none              | X&           |                                                                                                                                                                                                                                      |
| Son Above Assay range       | ABNORMAL_PARENTH  | X>           |                                                                                                                                                                                                                                      |
| Son Below Assay range       | ABNORMAL_PARENTH  | X<           |                                                                                                                                                                                                                                      |
| Son Above Normal Range      | ABNORMAL_PARENTH  | XH           |                                                                                                                                                                                                                                      |
| Son Below Normal Range      | ABNORMAL_PARENTH  | XL           |                                                                                                                                                                                                                                      |
| Below Normal Range          | none              | NL           |                                                                                                                                                                                                                                      |

| Database Flag       | Commend Code | Comment Code | Explanation |
|---------------------|--------------|--------------|-------------|
| Above Normal Range  | none         | NH           |             |
| Below Assay Range   | none         | <            |             |
| Above Normal Range  | none         | >            |             |
| All other flags are | JOB_FAILED   | *            |             |

Table 39–18: Database Flag explanation

### 39.9.6 Query Record

The **Query Record** is used to request data from host systems. Its hierarchical level is 1. The **LIAISON<sup>®</sup> XS** system processes only one received query at a time. If a second query is received while the first is processing, the system cancels the second query.

The following tables describe the allowable values for each field: they refer respectively to an “Outgoing” communication (from **LIAISON<sup>®</sup> XS** to host) or an “Incoming” communication (from host to **LIAISON<sup>®</sup> XS**).

| Field Number | Field Name        | Required | Value                                                                                                                                                                                                                                                                                                 |
|--------------|-------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1           | Record Type ID    | Y        | 1 character; always Q.                                                                                                                                                                                                                                                                                |
| Q2           | Sequence Number   | Y        | 1-5 characters, formatted as a left justified number, with a range from 1 through 65535. Equal to the numeric identifier of a query record within the current outgoing application message. Used to verify the integrity of an application message by ensuring that the message contains all records. |
| Q3           | Starting Range ID | Y        | If the operator requested a worklist, the Sample ID has no restrictions and the field is set to ALL.<br><br>If the <b>LIAISON<sup>®</sup> XS</b> system has initiated the worklist requests automatically, the field contains the Sample IDs, in the order from position 1 to 12 in the rack.         |
| Q4           | Ending Range ID   | N        | This field is always empty for all <b>LIAISON<sup>®</sup> XS</b> system-initiated query records.                                                                                                                                                                                                      |
| Q5           | Universal Test ID | Y        | Always set to ALL                                                                                                                                                                                                                                                                                     |

| Field Number | Field Name                      | Required | Value                                                                        |
|--------------|---------------------------------|----------|------------------------------------------------------------------------------|
| Q13          | Request Information Status Code | Y        | 1 character. Always 'O', which is a request for tests and demographics only. |

Table 39–19: Query Record – Outgoing

| Field Number | Field Name                      | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------|---------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Q1           | Record Type ID                  | Y        | Can be 'Q' or 'q'                                                                                                                                                                                                                                                                                                                                                                                      |
| Q2           | Sequence Number                 | Y        | 1 to any number of characters, with a default value of 0, which is always an invalid Sequence Number. Formatted as a string of ASCII numeric characters. Equal to the numeric identifier of a query record since the last header record, which is considered the current query record. If the current Sequence Number does not equal the Sequence Number of the expected record, an error is reported. |
| Q3           | Starting Range ID               | Y        | 3-22 characters. The Sample ID is used as a criterion for matching the SID field in worklist entries or results.<br>If ALL is sent in this field then LIAISON <sup>®</sup> XS will consider this as "match all" criterion, therefore will send all results from "Done" and "Failed" page (but not calibrators).                                                                                        |
| Q4           | Ending Range ID                 | N        | Not processed.                                                                                                                                                                                                                                                                                                                                                                                         |
| Q5           | Universal Test ID               | Y        | 0-8 characters. The LIS Alias code is used as a criterion for matching the test code field in worklist entries or results.                                                                                                                                                                                                                                                                             |
| Q13          | Request Information Status Code | Y        | When there is an 'X' in Q13, the query is ignored.                                                                                                                                                                                                                                                                                                                                                     |

Table 39–20: Query Record – Incoming

### 39.9.7 Message Terminator Record

The **Message Termination Record** is the last record of a message and closes the message. It provides an explanation for ending the message. Its hierarchical level is 0.

The following tables describe the allowable values for each field: they refer respectively to an “Outgoing” communication (from **LIAISON® XS** to host) or an “Incoming” communication (from host to **LIAISON® XS**).

| Field Number | Field Name       | Required | Value                                                                                                                                                                                                                                                        |
|--------------|------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1           | Record Type ID   | Y        | 1 character; always L.                                                                                                                                                                                                                                       |
| L2           | Sequence Number  | Y        | 1 character; always 1.                                                                                                                                                                                                                                       |
| L3           | Termination Code | Y        | 1 character, Can be 'N' or 'I'.<br>If the ending of the message is normal, this field is 'N' (Normal).<br>A Termination Code of 'I'. (No Information Available) is used to reply to a query if no information is available to send in response to the query. |

Table 39–21: Message Terminator Record – Outgoing

| Field Number | Field Name       | Required | Value                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------|------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| L1           | Record Type ID   | Y        | Can be 'L' or 'I'                                                                                                                                                                                                                                                                                                                                                                                                                    |
| L2           | Sequence         | Y        | 1 to any number of characters, with a default value of 0, which is always an invalid Sequence Number. Formatted as a string of ASCII numeric characters. If the current Sequence Number does not equal the Sequence Number of the expected record, a fatal parse error is posted to the event log. This value should always be 1 because multiple termination records not separated by a header record generate a fatal parse error. |
| L3           | Termination Code | Y        | 0 or 1 character, either 'N' or 'I'.                                                                                                                                                                                                                                                                                                                                                                                                 |

Table 39–22: Message Terminator Record – Incoming

### 39.9.8 Scientific Record

This record must not be send.

### 39.9.9 Manufacturer Information Record

The **Manufacturer Information Record** is used to send traceability information for a result record.

The **Manufacturer Information Record** is sent immediately after a Result Record, i.e. it always relates to the immediately preceding **Result Record**.

The following table describes the allowable values for each field: only the “Outgoing” communication (from LIAISON<sup>®</sup> XS to host) is in this case available.

| Field Number | Field Name               | Required | Value                                                                                                                                          |
|--------------|--------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------|
| M1           | Record Type ID           | Y        | 1 character; always M.                                                                                                                         |
| M2           | Sequence Number          | Y        | 1 character; always 1.                                                                                                                         |
| M3           | Integral Kit Number      | Y        | This is the Kit number of the reagent integral used to produce the result.<br>Kit Number is an integer ranging from 0 to 4.294.967.295         |
| M4           | Integral Expiration Date | Y        | This is the shelf expiration date of the reagent integral used to produce the result. In YYYYMMDD format.                                      |
| M5           | Integral LotNumber       | Y        | This is the reagent lot number of the reagent integral used to produce the result.<br>Lot Number is an integer ranging from 0 to 4.294.967.295 |
| M6           | Calibration ID           | N        | This is the internal ID of the calibration used to calculate the result. This is an integer value, 1 - 99.999.999.                             |
| M7^1         | Sample Lane Number       | Y        | The lane number from which the sample was aspirated for this test.<br>Lane number ranges from 1 to 4.                                          |
| M7^2         | Sample Tube Position     | Y        | The tube position within this lane from which the sample was aspirated for this test.<br>Ranges from 1 to 12.                                  |

## 39. LIS Interface Protocol

| Field Number | Field Name                            | Required | Value                                                                                        |
|--------------|---------------------------------------|----------|----------------------------------------------------------------------------------------------|
| M8           | Control Lot                           | N        | The lot number of the used control material<br>LotNumber are 1 - 19 characters, alaphnumeric |
| M9           | Control Name                          | N        | The name of the used control material, 1 - 19 characters, alphanumeric + " -/"               |
| M10          | Control Manufacturer                  | N        | The manufacturer of the control material, 1 - 19 characters, alphanumeric + " ."             |
| M11^1        | Control Manufacturer Range Low Value  | N        | The low value of the manufacturer range applied to the preceding control result.             |
| M11^2        | Control Manufacturer Range High Value | N        | The high value of the manufacturer range applied to the preceding control result.            |
| M12^1        | Control User Range Low Value          | N        | The low value of the user range applied to the preceding control result.                     |
| M12^2        | Control User Range High Value         | N        | The high value of the user range applied to the preceding control result.                    |

Table 39–23: Manufacturer Information Record – Outgoing

## 39.10 Tools for Troubleshooting

### 39.10.1 ASTM 1381 Layer

All bytes transmitted are recorded in "ASTM1381Log\_yyyymmdd.log" file (C:\LiaisonXS\Data\DiaSorin\Logs\LIS). The \*.log files have a plain text format and can be viewed with notepad.

### 39.10.2 ASTM 1394 Layer

All bytes transmitted are recorded in "ASTM1384Log\_yyyymmdd.log" file (C:\LiaisonXS\Data\DiaSorin\Logs\LIS). The \*.log files have a plain text format and can be viewed with notepad.

---

**NOTE**

In the ASTM logs no patient demographic data is reported.

---

## 39.11 Contention Case

Description:

With contention case it is intended a situation in which both the Host and the **LIAISON<sup>®</sup> XS** are requesting to communicate.

**LIAISON<sup>®</sup> XS** starts to send data, while the host is still transmitting, by sending an <EOT> as response to the header record from the host.

This is legal and explicitly defined in the ASTM standard (E1381 §6.3.5.1).

The host now has the option to ignore the <EOT> and continue its transmission, or to honor it and send an <EOT> as confirmation (E1381 §6.3.5.3 and §6.4).

In case the host does not follow the ASTM standard and just stops transmitting, i.e. it honors the <EOT> but it does not issue the <EOT> for confirmation, the communication can end in a "<ENQ><ENQ>" loop.

Example:

```

012-02-29, 10:18:52,936    <- <ENQ>
2012-02-29, 10:18:53,030    -> <ACK>
2012-02-29, 10:18:53,092    <-
                           <STX>1H|^&||||Lia3|||||HOST||P|1|<CR><ETX>BD<CR><L
                           F>
2012-02-29, 10:18:53,217    -> <ACK>
2012-02-29, 10:18:53,280    <-
                           <STX>2Q|1|0169035310||ALL|||||||O<CR><ETX>B8<CR><
                           LF>
2012-02-29, 10:18:53,373    -> <ACK>
2012-02-29, 10:18:53,436    <-
                           <STX>3Q|2|0806471110||ALL|||||||O<CR><ETX>BA<CR><
                           LF>
2012-02-29, 10:18:53,529    -> <ACK>
[...]
2012-02-29, 10:18:55,027    <-
                           <STX>5Q|12|0283584910||ALL|||||||O<CR><ETX>F9<CR>
                           <LF>
2012-02-29, 10:18:55,120    -> <ACK>
2012-02-29, 10:18:55,183    <- <STX>6L|1|N<CR><ETX>09<CR><LF>
2012-02-29, 10:18:55,276    -> <ACK>
2012-02-29, 10:18:55,339    <- <EOT>
2012-02-29, 10:18:55,432    -> <ENQ>
2012-02-29, 10:18:55,495    <- <ACK>
2012-02-29, 10:18:55,620    -> <STX>1H|^&|||||||P|1<CR><ETX>BA<CR><LF>
2012-02-29, 10:18:55,713    <- <EOT>
2012-02-29, 10:19:05,760    -> <ENQ>
2012-02-29, 10:19:25,774    -> <ENQ>
2012-02-29, 10:19:26,086    <- <ENQ>
2012-02-29, 10:19:36,164    -> <ENQ>
2012-02-29, 10:19:38,036    <- <ENQ>

```

**Workaround:**

As it turns out that the Host cannot always properly handle the **LIAISON<sup>®</sup> XS** request while it is transmitting data, then we suggest (as a temporary workaround before the Host is adapted to the ASTM standard) to disable the automatic results upload and to manually upload results.

The contention scenario may also appear when the user is loading several sample racks: here the Host sends the programmed tests for the already loaded tubes while the **LIAISON<sup>®</sup> XS** send a query for the newly loaded tubes.

To avoid such scenario, a solution could be to wait for the programmed tests of the first rack to be completely received before loading another rack.



### 39.12 Differences between LIAISON, LIAISON<sup>®</sup> XL and LIAISON<sup>®</sup> XS configuration parameters

The following table lists the differences between **LIAISON** and **LIAISON<sup>®</sup> XL** as far as configuration parameters are concerned and assessment of potential impact on the connection.

| Parameter                    | LIAISON                     | LIAISON XL          | LIAISON XS                  | Possible Impact                                                                                                                                                            |
|------------------------------|-----------------------------|---------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DataBits                     | 8<br>(cannot be changed)    | Parameter           | 8<br>(cannot be changed)    | None.<br>Reduces issues related to misconfiguration                                                                                                                        |
| StopBits                     | 1<br>(cannot be changed)    | Parameter           | 1<br>(cannot be changed)    | None.<br>Reduces issues related to misconfiguration                                                                                                                        |
| Parity                       | None<br>(cannot be changed) | Parameter           | None<br>(cannot be changed) | None.<br>Reduces issues related to misconfiguration                                                                                                                        |
| NumFormat                    | Parameter                   | Not a LIS parameter | Not a LIS parameter         | Values are stored as string in <b>LIAISON<sup>®</sup> XS</b> , depending on assay specific settings. It is not transformed for transmission to LIS.<br>No issues expected. |
| Send frame timeout           | 15 (cannot be changed)      | Parameter           | 15                          | None.<br>Reduces issues related to misconfiguration                                                                                                                        |
| Bad frame send (NAK) timeout | 30 (cannot be changed)      | Parameter           | 30                          | None.<br>Reduces issues related to misconfiguration                                                                                                                        |
| Busy (contention) timeout    | 10 (cannot be changed)      | Parameter           | 10                          | None.<br>Reduces issues related to misconfiguration                                                                                                                        |
| Send compressed comments     | Not present                 | Parameter           | Parameter                   | Added functionality.<br>No issues expected                                                                                                                                 |
| Upload frequency             | Parameter                   | Not present         | Not present                 | Automatic upload is triggered by auto-archiving in <b>LIAISON<sup>®</sup> XS</b> .<br>No issues expected.                                                                  |

| Parameter                      | LIAISON     | LIAISON XL      | LIAISON XS      | Possible Impact                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|-------------|-----------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Number of upload retries       | Parameter   | Not present     | Not present     | Results affected by transmission failure are transmitted with the next automatically transmitted results.<br>This is a change functionality in respect to <b>LIAISON<sup>®</sup></b> .<br>No issues expected.                                                                                                                                  |
| Host query frequency           | Parameter   | Not present     | Not present     | <b>LIAISON<sup>®</sup> XS</b> accepts all queries from the host, independently from any query frequency.<br>If <b>LIAISON<sup>®</sup> XS</b> is busy, the host cannot send the query, if <b>LIAISON<sup>®</sup> XS</b> is free, it answers.<br>This is a change functionality in respect to <b>LIAISON<sup>®</sup></b> .<br>No issue expected. |
| Enable/disable                 | Not present | Parameter       | Parameter       | Added functionality. No issue expected                                                                                                                                                                                                                                                                                                         |
| Result flag style              | Not present | Parameter       | Parameter       | Added functionality to keep compatibility with <b>LIAISON<sup>®</sup></b> .<br>No issue expected                                                                                                                                                                                                                                               |
| Enable manufacturer record     | Not present | Parameter       | Parameter       | Added functionality.<br>No issue expected                                                                                                                                                                                                                                                                                                      |
| Result Aspect                  | Not present | Assay Parameter | Assay Parameter | Added functionality.<br>No issue expected                                                                                                                                                                                                                                                                                                      |
| Send Replicate Results to Host | Not present | Assay Parameter | Assay Parameter | Added functionality.<br>No issue expected                                                                                                                                                                                                                                                                                                      |

## 40 Glossary

|                   |                                                                                                                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 00h; 0x00         | Shows hexadecimal (hex) code, used in trouble shooting error messages.                                                                                                                                                                                         |
| A                 | Amps, Ampere                                                                                                                                                                                                                                                   |
| Access level      | An integer number, which defines, which rights a user has, when he is logged in into the application.                                                                                                                                                          |
| Accuracy          | Accuracy is the nearness of a measurement to its true value or accepted value.                                                                                                                                                                                 |
| APE               | <b>Air Pipettor Module EC</b> -motor == short/symbol of FW which drives the ZSPip Module.                                                                                                                                                                      |
| APH               | <b>Air Pipettor Head</b> Module                                                                                                                                                                                                                                |
| APM               | <b>Aspirate pressure monitoring</b> system                                                                                                                                                                                                                     |
| APP               | <b>Air Pipettor Pump</b> Module                                                                                                                                                                                                                                |
| AM                | Analytical Module                                                                                                                                                                                                                                              |
| Analyte           | Substance of unknown concentration in a sample.                                                                                                                                                                                                                |
| Ancillary reagent | Additional reagents required for an assay.                                                                                                                                                                                                                     |
| Antibody          | Protein produced by the body in response to the presence of an antigen. Antibodies are used as part of the body's resistance to disease.                                                                                                                       |
| ASCII             | American Standard Code for Information Interchange (ASCII), pronounced is a character encoding based on the English alphabet.                                                                                                                                  |
| Asp. Mon.         | Aspiration monitor (SIPAL)                                                                                                                                                                                                                                     |
| Assay             | Generic term that refers to the chemical analysis for specific analyte in a sample. Each assay possesses a unique test protocol. Also called test.                                                                                                             |
| ASTM              | ASTM International, is an international standards organization that develops and publishes voluntary consensus technical standards.<br>ASTM 1381 and ASTM 1394 are determinations for the communication between computers.                                     |
| ASTM protocol     | Protocol for data transfer between Clinical Laboratory Instruments and Computer Systems (Host) proposed by the ASTM committee. It specifies the electrical standard and message contents for transferring data.                                                |
| Auto-dilution     | Auto-dilution: Function that features an automated rerun with a predefined predilution factor in case the sample dose result exceeds a given threshold.                                                                                                        |
| Background        | Electronic signal of the photomultiplier tube (PMT) in the absence of light.                                                                                                                                                                                   |
| Bisects           | Halved every step (To cut or divide into two parts, especially two equal parts.)                                                                                                                                                                               |
| Bleach            | Hydrogen peroxide (bleach, 3%)                                                                                                                                                                                                                                 |
| Bubble Sensor     | Sensor which detects air bubbles into a liquid line.                                                                                                                                                                                                           |
| C#                | Programming Language                                                                                                                                                                                                                                           |
| Calibration       | The use of two samples of know concentrations that you process on your system. The calibration data is then used to normalize your system against a stored master curve. The system calculates results from the normalized curve.                              |
| Calibration curve | Data that normalizes the system against stored Master Curve data for the tested analyte. The calibration data compensates for current running conditions and instrument variability. The system calculates the required sample dose from the normalized curve. |

## 40. Glossary

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Calibrator              | Solution that contains a known concentration or a known reactivity of one or more analytes and that provides a reference for converting measured signals into concentrations. Calibrators are analyzed with reagents to obtain calibration data points.                                                                                                                                                                              |
| CAN                     | Controller Area Network is a multicast shared serial bus standard for connecting electronic control units.                                                                                                                                                                                                                                                                                                                           |
| CCD                     | Charge Coupled Device                                                                                                                                                                                                                                                                                                                                                                                                                |
| CCW                     | Counter clock wise                                                                                                                                                                                                                                                                                                                                                                                                                   |
| CE                      | Conformité Européenne (European conformity)                                                                                                                                                                                                                                                                                                                                                                                          |
| Central Processing Unit | The unit of a computer that includes the circuits controlling the interpretation of program instructions and their execution. the CPU controls the entire computer. It receives and sends data through input-output channels, retrieves data and programmes from memory, and conducts mathematical and logical functions of a program.                                                                                               |
| CFR                     | Code of Federal Regulations (Codes of FCC, US government)                                                                                                                                                                                                                                                                                                                                                                            |
| CGS                     | Command CMD_GLOBAL_STATUS                                                                                                                                                                                                                                                                                                                                                                                                            |
| Checkpoint              | Checkpoint: In order to verify during calibration that the shape of a working curve is within reasonable limits, dose values at two selected points of the working curve candi-date, which are called checkpoints, are interpolated in order to obtain corresponding RLU values. The ratio of the RLU values corresponding to each checkpoint (imaginable as a kind of "slope" value) is checked against minimum and maximum limits. |
| Checksum                | (IEEE) A sum obtained by adding the digits in a numeral, or group of numerals [a file], usually without regard to meaning, position, or significance. See: check summation.                                                                                                                                                                                                                                                          |
| Chemiluminescence       | Chemiluminescence is the production of light photons by a chemical or electrochemical reaction. Chemiluminescence usually involves the oxidation of an organic compound such as luminol or acridinium esters by an oxidant such as hydrogen peroxide or hypochlorite. Chemiluminescent reactions occur in the presence of catalysts such as alkaline phosphatase, horseradish peroxidase, metal ions or metal complexes.             |
| CLR                     | A runtime environment that manages the execution of .NET program code, and provides services such as memory and exception management, debugging and profiling, and security. The CLR is a major component of the .NET Framework, and provides much of its functionality by following the rules defined in the Common Type System. Also known as the Virtual Execution System (VES).                                                  |
| CLSI                    | The Clinical and Laboratory Standards Institute (CLSI) is a global, nonprofit, standards - developing organization that promotes the development and use of voluntary consensus standards and guidelines within the healthcare community.                                                                                                                                                                                            |
| CM                      | Control Module is the PC with the GUI of the instrument.                                                                                                                                                                                                                                                                                                                                                                             |
| COM                     | COM is the original, yet still common, name of the serial port interface on PCs. It might not only refer to physical ports, but also to virtual ports, such as ports created by RS232 or USB adapters.                                                                                                                                                                                                                               |
| ComGen                  | Command Generator, which converts pipetting sequences into elementary pipetting commands.                                                                                                                                                                                                                                                                                                                                            |
| Concentration           | Amount of analyte measured.                                                                                                                                                                                                                                                                                                                                                                                                          |
| Control                 | A reference material used for proof of process control.                                                                                                                                                                                                                                                                                                                                                                              |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control ID                    | An integer number, which identifies a control as an element of a dialog window in a MS Windows environment. A control ID is always related to the ID of the dialog to which the control belongs.                                                                                                                                                                                                                                                                         |
| COP                           | Central Operating Processor is the master processor of the Analytical Module (AM).                                                                                                                                                                                                                                                                                                                                                                                       |
| COTS                          | Commercial Off-The-Shelf is a term for software or products that is developed for the general market, i.e. for a large number of customers, and that is delivered to many customers in identical format.                                                                                                                                                                                                                                                                 |
| CPS                           | Counts Per Second                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| CPU                           | Central Processing Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| CRC                           | Cyclical Redundancy Check is a mathematical method to ensure the integrity of data.                                                                                                                                                                                                                                                                                                                                                                                      |
| CSA                           | Canadian Standards Association                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| CU                            | Control Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Cut-off point                 | Medical decision point for a qualitative assay. Results above the cut-off point are reported with a different qualitative interpretation than those below the cut-off point.                                                                                                                                                                                                                                                                                             |
| Cuvette                       | Plastic, disposable container that holds the reaction mixture on the system. The chemiluminescent reaction occurs inside the cuvette.                                                                                                                                                                                                                                                                                                                                    |
| CV                            | Coefficient of variation, or CV, is a statistical measurement of the distribution of responses around the mean of those values. It is derived by dividing the standard deviation of the replicate adjusted or normalized response measurements by the mean of the replicate measurements (times 100 for percent). Since the measurement units are canceled out, the %CV is a relative value that is independent of the magnitude of the replicate response measurements. |
| Dark Counts                   | Electronic background signal of the photomultiplier tube (PMT) in the absence of light.                                                                                                                                                                                                                                                                                                                                                                                  |
| DC                            | Direct Current - In direct current, the electric charges flow in the same direction.                                                                                                                                                                                                                                                                                                                                                                                     |
| Diluent                       | Protein-based solution used to dilute a sample.                                                                                                                                                                                                                                                                                                                                                                                                                          |
| DITI                          | A disposable tip is a plastic tip which is used for one liquid transfer only, and then disposed afterwards.                                                                                                                                                                                                                                                                                                                                                              |
| Dose result                   | A result obtained from the measured RLU values by calculation.                                                                                                                                                                                                                                                                                                                                                                                                           |
| EEPROM or E <sup>2</sup> PROM | An Electrically Erasable Programmable Read-Only Memory, is a type of non-volatile memory used in computers and other electronic devices to store small amounts of data that must be saved when power is removed, e.g., calibration tables or device configuration.                                                                                                                                                                                                       |
| EIA                           | Enzyme Immuno Assay                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| ELISA                         | Enzyme Linked Immuno Sorbent Assay                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| EMC/EMI                       | Electromagnetic Compatibility (EMC) testing and Electromagnetic Interference testing (EMI).                                                                                                                                                                                                                                                                                                                                                                              |
| Error                         | Errors are special Events indicating that the operators intervention is needed or will be needed soon.                                                                                                                                                                                                                                                                                                                                                                   |
| Event                         | A notification by a program or operating system that "something has happened." An event may be fired (or raised) in response to the occurrence of a pre-defined action (e.g., a window getting focus, a user clicking a button, a timer indicating a specific interval of time has passed, or a program starting up or shutting down). In response to an event, an event handler is called.                                                                              |

## 40. Glossary

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Event Browser             | Browser which shows all the Events that are not yet deleted. The Browser supports searching, sorting and filtering functions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Event ID                  | Localization independent identifier of a special Event.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Event Log                 | Chronological list of all events, which are not directly sample related and result relevant.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Exception                 | A signal that is generated when an unplanned or unexpected event occurs. Exceptions are typically caught by an exception handler and dealt with in an appropriate way. A fatal exception (also called a critical or catastrophic error) is an event that cannot be properly handled to allow the application-or the operating system-to continue running.                                                                                                                                                                                                                                                                                            |
| Fatal Error               | An unexpected and non-recoverable termination of instrument operation including, but not limited to assay processing, maintenance, service, start-up, and shut-down.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Fatal Error Dialog        | The Fatal Error Dialog is a FIFO query that hides all new Fatal Errors until the first one is acknowledged.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| FCC                       | Federal Communications Commission, US government.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| FCM                       | The subassembly Front Cover Module.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Firmware                  | The combination of a hardware device; e.g., an IC; and computer instructions and data that reside as read only software on that device. Such software cannot be modified by the computer during processing.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Flag                      | A variable that is set to a prescribed state, often "true" or "false", based on the results of a process or the occurrence of a specified condition.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Floating Point Type       | Floating-point numbers use the IEEE (Institute of Electrical and Electronics Engineers) format. Single-precision values with float type have 4 bytes, consisting of a sign bit, an 8-bit excess-127 binary exponent, and a 23-bit mantissa. The mantissa represents a number between 1.0 and 2.0. Since the high-order bit of the mantissa is always 1, it is not stored in the number. This representation gives a range of approximately 3.4E-38 to 3.4E+38 for type float.                                                                                                                                                                        |
| FSE                       | Field Service Engineer trained on the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Full standardization mode | In this operating mode no masters Curve is needed. To receive a working curve, all standard points of a curve shall be measured by calibrators.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| FW                        | see Firmware                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Hardware                  | Physical components of the system.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Host                      | In computer networking a host is a main computer (e. g. server, central computer).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| HW                        | Hardware                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| I                         | Incubation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| IC                        | Integrated circuit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| ID                        | Identification. e.g. SID (sample identification).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| IFU                       | Instructions for use manual                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| Immunoassays              | An immunoassay is a ligand binding assay which uses an antibody and the antibody's corresponding antigen as the binder and ligand. The term immunoassay is often expanded to include any procedure in which the quantitation of an analyte depends on the progressive saturation of a specific binder or ligand, either of which may be the analyte, and the subsequent determination of the analyte's distribution between the bound and free fractions. This includes competitive protein binding assays, where the binder is a naturally occurring binding protein, and receptor assays, where the binder is a naturally occurring cell receptor. |

|                      |                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Integral             | Integral: An integrated kit containing a set of reagents used for one assay (or more).                                                                                                                                                                                                                                                                                     |
| Interface            | The set of properties, methods, and events exposed by an object that allow other objects to access its data and functionality. An object guarantees that it will support all of the elements of its interface by way of an interface contract.                                                                                                                             |
| IP                   | Integrated pipettor                                                                                                                                                                                                                                                                                                                                                        |
| IVD                  | In-vitro Diagnostic                                                                                                                                                                                                                                                                                                                                                        |
| Kit lot              | Kit lot: The production lot number of a reagent kit. It is supposed that reagent kits having the same lot number were produced together and have the same properties regarding calibration.                                                                                                                                                                                |
| L2                   | LIAISON 2; the system                                                                                                                                                                                                                                                                                                                                                      |
| LAN                  | A Local Area Network is a computer network.                                                                                                                                                                                                                                                                                                                                |
| LAS                  | Laboratory Automation System - System whereas several analyzers (of different or the same model) are linked together to a common device (work cell) to form an automated assembly line process used to optimize patient sample handling with minimum personnel.                                                                                                            |
| LC                   | Light check                                                                                                                                                                                                                                                                                                                                                                |
| LCMP                 | Liquid coupled membrane pump                                                                                                                                                                                                                                                                                                                                               |
| LED                  | Light Emitting Diode                                                                                                                                                                                                                                                                                                                                                       |
| LIMS                 | A Laboratory Information Management System (LIMS) is computer software that is used in laboratories for the management of samples, laboratory users, instruments, standards and other laboratory functions such as invoicing, plate management, and work flow automation.                                                                                                  |
| LIS                  | Laboratory Information System - System used for computerized management of labs dedicated to chemical-clinical or microbiological analysis. The LIS is an external program used in most clinical institutions for centralized information processing.                                                                                                                      |
| LIS01-A2             | Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems.                                                                                                                                                                                                                                                    |
| LIS02-A2             | Specification for Transferring Information Between Clinical Laboratory Instruments and Information Systems.                                                                                                                                                                                                                                                                |
| Logical Combi Assays | Logical combi assays are a (boolean) combination of up to four “partner” assays.                                                                                                                                                                                                                                                                                           |
| LLD                  | Liquid Level Detection<br>B-LLD - Barometric Liquid Level Detection<br>C-LLD - Capacitive Liquid Level Detection                                                                                                                                                                                                                                                           |
| LS                   | Liquid Sensor                                                                                                                                                                                                                                                                                                                                                              |
| LSP                  | The subassembly Left Side Panel                                                                                                                                                                                                                                                                                                                                            |
| LWP                  | Liquid Waste Pump                                                                                                                                                                                                                                                                                                                                                          |
| M                    | Measurement                                                                                                                                                                                                                                                                                                                                                                |
| Maintenance          | Maintenance - Activities such as adjusting, cleaning, modifying, overhauling equipment to assure performance in accordance with requirements. Maintenance to a software system includes correcting software errors, adapting software to a new environment, or making enhancements to software. See: adaptive maintenance, corrective maintenance, perfective maintenance. |



## 40. Glossary

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Master Curve      | The master Curve is the reference curve used to convert a measured signal into sample concentration. Calibration data normalizes the system against stored Master curve data for the tested analyte.                                                                                                                                                                                                                         |
| Master Curve Unit | Master Curve Unit (MCU): The unit of measure for the independent variable (x) of the master curve for a specific assay.                                                                                                                                                                                                                                                                                                      |
| MCU               | Master Curve Unit                                                                                                                                                                                                                                                                                                                                                                                                            |
| Message           | A special Event that is also shown in the Message Browser.                                                                                                                                                                                                                                                                                                                                                                   |
| Message Browser   | Those Events that are specified as Message will be shown in the Event Browser and the Message Browser. Messages are deleted after acknowledging but remaining the Event Browser.                                                                                                                                                                                                                                             |
| MP                | Magnetic Particles                                                                                                                                                                                                                                                                                                                                                                                                           |
| ms                | Milli seconds<br>(1000 ms = 1 s)                                                                                                                                                                                                                                                                                                                                                                                             |
| MSDS              | Material safety data sheet                                                                                                                                                                                                                                                                                                                                                                                                   |
| MSR               | Master Signal Range                                                                                                                                                                                                                                                                                                                                                                                                          |
| NA                | Not Applicable                                                                                                                                                                                                                                                                                                                                                                                                               |
| nm                | A nanometer is a unit of length in the metric system.<br>(1 nm = 0.000001 mm = $1 \times 10^{-9}$ m = $39.37 \times 10^{-9}$ inch)                                                                                                                                                                                                                                                                                           |
| NVM               | None volatile memory<br>is computer memory that can retain the stored information even when not powered.                                                                                                                                                                                                                                                                                                                     |
| .NET Framework    | A programming infrastructure created by Microsoft for building, deploying, and running applications and services that use .NET technologies, such as desktop applications.                                                                                                                                                                                                                                                   |
| OD                | Optical density                                                                                                                                                                                                                                                                                                                                                                                                              |
| Offline Status    | Condition, in which a certain consumable or module is not usable any more. The offline consumables/modules will come back online after a reinitialization or by unloading and reloading the consumable.                                                                                                                                                                                                                      |
| OLV               | Optical Liquid Verification                                                                                                                                                                                                                                                                                                                                                                                                  |
| Overdilution      | Overdilution: Condition of a sample whose dose determined after predilution is below a defined threshold; an overdiluted sample may result in a recalculated dose (multiplied by the dilution factor) that may be affected by inaccuracy or imprecision.                                                                                                                                                                     |
| P                 | Pipetting                                                                                                                                                                                                                                                                                                                                                                                                                    |
| PC                | Personal Computer                                                                                                                                                                                                                                                                                                                                                                                                            |
| PCB               | Printed Circuit Board                                                                                                                                                                                                                                                                                                                                                                                                        |
| Photon            | Unit of light; quantum of radiant energy.                                                                                                                                                                                                                                                                                                                                                                                    |
| PMT               | Photomultiplier Tube - Sensitive photo detector that captures individual photons of light. The photodetector consists of a vacuum-filled photo tube with a cathode (usually nine diodes) and an anode that system uses to generate a signal (pulse) during the measurement phase. The signal is directly proportional to the amount of illumination produced by the reaction and can be converted into sample concentration. |
| Power supply      | Electronic subsystem that converts the AC voltages in the power line to the DC voltages used inside the system.                                                                                                                                                                                                                                                                                                              |
| Precision         | Precision is a measurement of the reproducibility of replicate sample responses in an assay run. Precision is composed of both the random error and the systematic error of the measurement.                                                                                                                                                                                                                                 |



|                           |                                                                                                                                                                                                                                                                                           |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| prime                     | Function that introduces reagents into the fluidic system in preparation for analyzing samples.                                                                                                                                                                                           |
| PR1R2                     | PR1R2 = Representing R1 and R2 pipetting in one cycle, R1 is dispensed prior to R2.                                                                                                                                                                                                       |
| PR1SR2                    | PR1SR2 = Representing sample, R1 and R2 pipetting in one cycle, R1 is dispensed first, followed by sample and finally R2. Note: PR1SR2 is not be possible for internal calibrators.                                                                                                       |
| PRP                       | Probe Rinsing Pump                                                                                                                                                                                                                                                                        |
| PS                        | PS = Representing sample pipetting                                                                                                                                                                                                                                                        |
| PS2                       | The PS/2 (PS2) connector allows the connection of a keyboard or a mouse to a computer.                                                                                                                                                                                                    |
| PSR2                      | PSR2 = Representing sample and R2 pipetting in one cycle, Sample is dispensed prior to R2.                                                                                                                                                                                                |
| QA                        | Quality control Analysis                                                                                                                                                                                                                                                                  |
| Qualitative label         | Qualitative label: A text label assigned to a certain dose result for a certain range. Ranges and assorted labels can be defined.                                                                                                                                                         |
| Qualitative result        | Qualitative result: A numerical result obtained from a cut-off quotient, which can be considered as a value relative to a reference dose (i.e. a minimum dose value).                                                                                                                     |
| Quantitative Combi Assays | Quantitative combi assays are a (mathematical) combination of up to four “partner” assays.                                                                                                                                                                                                |
| Quantitative result       | Quantitative result: A numerical result obtained by two-point calibration of a master curve or by a “full standardization” working curve, giving an exact concentration.                                                                                                                  |
| Rack                      | Sample rack. Device used to present sample tubes to the system.                                                                                                                                                                                                                           |
| Rack ID                   | Special barcode at the end of every sample rack describing the geometrical layout of that rack.                                                                                                                                                                                           |
| RAM                       | Random access memory<br>When the computer is shut down, anything contained in RAM is lost.                                                                                                                                                                                                |
| Raw data                  | Total photon count that the system received for a flash. The raw data is not corrected.                                                                                                                                                                                                   |
| Reagent                   | A substance used to produce a specific chemical reaction. It may be used to detect, measure, or prepare other substances.                                                                                                                                                                 |
| Reliability               | The ability of a system or component to perform its required functions under stated conditions for a specified period of time.                                                                                                                                                            |
| Replicate                 | There is a one-to-one mapping to a cuvette. There may be one or more replicates requested for a test.                                                                                                                                                                                     |
| Re-schedule a replicate   | If any of the replicates of a test could not be aspirated in a run then when the next run is started the system will automatically reserve resources and aspiration position for that replicate.                                                                                          |
| Result Log                | List of all sample- and result relevant Events, is attached to individual sample result data block, accessible through detailed result view. List of all sample- and result relevant Events, is attached to individual sample result data block, accessible through detailed result view. |
| Resoak                    | A function of the pipettor after liquid dispense. If the Resoak steps are different from 0, immediately after the plunger has stopped dispensing, it moves in the opposite direction and picks up air the number of specified steps.                                                      |

## 40. Glossary

|                      |                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RFID                 | Radio Frequency Identification (RFID) is an automatic identification method, relying on storing and remotely retrieving data using devices called RFID tags or transponders. An RFID tag is an object that can be attached to or incorporated into a product, animal, or person for the purpose of identification using radio waves.                                                       |
| RLU                  | Relative Light unit. Counts generated when the PMT detects light during the chemiluminescent reaction. A relative light unit is calibrated against a tritium source and is unique to its measurement.                                                                                                                                                                                      |
| ROM                  | Read Only Memory                                                                                                                                                                                                                                                                                                                                                                           |
| RS232                | Serial bus standard to connect devices to a computer.                                                                                                                                                                                                                                                                                                                                      |
| RSP                  | The subassembly Right Side Panel.                                                                                                                                                                                                                                                                                                                                                          |
| Run                  | Analytical session, consisting of all actions related to scheduling and analysis of a group of samples, beginning with an operator-initiated start and ending when the analyzer returns to Ready state.                                                                                                                                                                                    |
| Sample               | A sample, or replicate, is an aliquot of a baseline standard specimen, standard specimen, control specimen, or unknown specimen, and all of the reagents which comprise an individual immunoassay incubation medium.                                                                                                                                                                       |
| Satellite assembly   | An assembly that contains only resources and no executable code. Satellite assemblies are typically used by .NET application to store localized data. Satellite assemblies can be added, modified, and loaded into a .NET application at runtime without the need to recompile the code. Satellite assemblies are created by compiling .resource files using the Assembly Linking Utility. |
| Schedule a replicate | The system automatically reserves resources to run requested replicates for a sample and decides when (within the scheme of the current run) the sample will be aspirated for that replicate.                                                                                                                                                                                              |
| SCTL                 | Servo Motor Controller (off the shelf).                                                                                                                                                                                                                                                                                                                                                    |
| SDP                  | Starter dispense pump                                                                                                                                                                                                                                                                                                                                                                      |
| Shared calibration   | Shared calibration: A calibration, which is shared among several reagent integrals having different kit number, but the same lot number.                                                                                                                                                                                                                                                   |
| SIPAL                | Single probe aspiration lift.                                                                                                                                                                                                                                                                                                                                                              |
| SM                   | Service Manual                                                                                                                                                                                                                                                                                                                                                                             |
| SP                   | Starter pump                                                                                                                                                                                                                                                                                                                                                                               |
| Spike                | Spike: Outliers within the signal (flash curve) due to events such as electro-static discharge.                                                                                                                                                                                                                                                                                            |
| SPP                  | Pipetting positions of sample pipettor.                                                                                                                                                                                                                                                                                                                                                    |
| SPOLV                | STRATEC Pipetting optical liquid verification                                                                                                                                                                                                                                                                                                                                              |
| SSW                  | Service Software                                                                                                                                                                                                                                                                                                                                                                           |
| Standard Deviation   | The standard deviation is the square root of the variance.                                                                                                                                                                                                                                                                                                                                 |
| Standby              | The system is functional, but no tests are running.                                                                                                                                                                                                                                                                                                                                        |
| STAT                 | Short turnaround time. The abbreviation STAT stands for "i Short Turnaround Time". STAT samples are used in emergency cases, were a result should be obtained as soon as possible, and have higher scheduling priority.                                                                                                                                                                    |
| SW                   | Software                                                                                                                                                                                                                                                                                                                                                                                   |
| System Liquid        | Solution present in the fluidic line of the reagent pipettor, used to wash the pipettor probe                                                                                                                                                                                                                                                                                              |
| TA                   | Tube assembly                                                                                                                                                                                                                                                                                                                                                                              |
| TAB                  | Tabulator                                                                                                                                                                                                                                                                                                                                                                                  |

|                                |                                                                                                                                                                                                                                                                                                                           |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TDS                            | Technical data sheet                                                                                                                                                                                                                                                                                                      |
| UI                             | The User Interface is the aggregate of means by which people (the users) interact with a particular machine, device, computer program or other complex tool (the system).                                                                                                                                                 |
| Unload                         | Download Tool                                                                                                                                                                                                                                                                                                             |
| USB                            | The Universal Serial Bus is a serial bus standard to connect devices to a computer.                                                                                                                                                                                                                                       |
| User                           | Any person, organization, or functional unit that uses the services of an information processing system. See: end user.                                                                                                                                                                                                   |
| V                              | Voltage                                                                                                                                                                                                                                                                                                                   |
| VC                             | Validation criteria                                                                                                                                                                                                                                                                                                       |
| VCD                            | Proximity coupling device (reader IC)                                                                                                                                                                                                                                                                                     |
| VGA                            | The Video Graphics Array is a standard interface to connect a screen to a computer.                                                                                                                                                                                                                                       |
| VICC                           | Proximity integrated circuit card (tag or smart label)                                                                                                                                                                                                                                                                    |
| WAP                            | Wash aspiration pump                                                                                                                                                                                                                                                                                                      |
| WB                             | Wash Buffer                                                                                                                                                                                                                                                                                                               |
| WBV                            | Washer Buffer Valve                                                                                                                                                                                                                                                                                                       |
| WCF                            | Windows Communication Foundation                                                                                                                                                                                                                                                                                          |
| WBDP                           | Wash Buffer Dispense Pump                                                                                                                                                                                                                                                                                                 |
| Work list                      | List of scheduled patient samples, controls, and calibrators, including SampleID and requested tests for each entry.                                                                                                                                                                                                      |
| Work order for patient samples | A job request for a sample which includes the sample ID and the tests to be run (as a minimum), it may also contain the patient name, Patient-ID, age, sex, dilution factor... may be generated manually on the system and/or Queried or downloaded from an LIS.                                                          |
| Working curve                  | Working curve: The working curve is derived from the master curve by "recalibration" in order to compensate environmental, reagent and consumable/kit variations at the customer site. After this recalibration has been performed, the working curve can be used to obtain dose values from RLU values by interpolation. |
| WS pump                        | Wash station pump                                                                                                                                                                                                                                                                                                         |
| WSR                            | Working Signal Range                                                                                                                                                                                                                                                                                                      |
| XML                            | A meta-markup language that provides a format for describing almost any type of structured data.                                                                                                                                                                                                                          |
| ZSPIP                          | Sample Pipettor                                                                                                                                                                                                                                                                                                           |
| μl                             | A microliter is a unit of volume in the metric system.<br>(1 μl = 0.001 ml = 1*10 <sup>-6</sup> l)                                                                                                                                                                                                                        |

Table 40–1: Glossary

